Morten Hjorth-Jensen

Quantum mechanics for Many-particle Systems; A computational perspective

From standard methods to quantum computing and machine learning

August 25, 2026

Springer

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1.1 Notations and conventions

Throughout these notes vectors, matrices and higher-order tensors are always set in boldface. Vectors are denoted by lower-case letters ($\boldsymbol{x}$, $\boldsymbol{v}$, ...) and matrices and tensors by upper-case letters ($\boldsymbol{A}$, $\boldsymbol{U}$, ...).

Unless stated otherwise the elements v_i of a vector $\boldsymbol{v}$ are assumed real. A real vector of length n satisfies $\boldsymbol{x} \in \mathbb{R}^n$; for complex vectors we write $\boldsymbol{x} \in \mathbb{C}^n$. A real $n \times n$ matrix satisfies $\boldsymbol{A} \in \mathbb{R}^{n\times n}$. Index counting starts at zero: matrix elements run from a_{00} to $a_{n-1,n-1}$.

We use the standard shorthand symbols: $\forall$ (for all), $\Rightarrow$ (implies), $\equiv$ (equivalent / defined as), with $\mathbb{R}$, $\mathbb{I}$, $\mathbb{C}$ for the real, integer and complex number fields.

1.2 Vectors

A column vector of length n and its transpose are

$$\boldsymbol{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix}, \qquad \boldsymbol{x}^T = \begin{bmatrix} x_0 \; x_1 \; \cdots \; x_{n-1} \end{bmatrix}.$$

For complex vectors the *Hermitian conjugate* (adjoint) is

$$\boldsymbol{x}^\dagger = \begin{bmatrix} x_0^* \; x_1^* \; \cdots \; x_{n-1}^* \end{bmatrix}.$$

The *inner (dot) product* is the scalar

$$\boldsymbol{x}^T \boldsymbol{x} = \sum_{i=0}^{n-1} x_i x_i = x_0^2 + x_1^2 + \cdots + x_{n-1}^2,$$

and for complex vectors $\boldsymbol{x}^\dagger \boldsymbol{x} = \sum_{i=0}^{n-1} x_i^* x_i = \|\boldsymbol{x}\|^2$.

The *outer product* of two real vectors $\boldsymbol{y}$ (length n) and $\boldsymbol{z}$ (length m) is the $n \times m$ matrix

$$\boldsymbol{y}\boldsymbol{z}^T \implies (\boldsymbol{y}\boldsymbol{z}^T)_{ij} = y_i z_j.$$

For complex vectors one writes $\boldsymbol{y}\boldsymbol{z}^\dagger$ with $(\boldsymbol{y}\boldsymbol{z}^\dagger)_{ij} = y_i z_j^*$.

Important vector operations:

$$\begin{aligned} \text{Addition / subtraction:} \quad & \boldsymbol{x} = \boldsymbol{y} \pm \boldsymbol{z} \;\Rightarrow\; x_i = y_i \pm z_i. \\ \text{Scalar multiplication:} \quad & \boldsymbol{x} = \gamma \boldsymbol{y} \;\Rightarrow\; x_i = \gamma y_i. \\ \text{Hadamard (element-wise) product:} \quad & \boldsymbol{x} = \boldsymbol{y} \circ \boldsymbol{z} \;\Rightarrow\; x_i = y_i z_i. \end{aligned}$$

1.3 Matrices: definitions and algebraic operations

A general $n \times n$ matrix and its column-vector representation:

$$\boldsymbol{A} = \begin{bmatrix} a_{00} & a_{01} & \cdots & a_{0,n-1} \\ a_{10} & a_{11} & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & \cdots & \cdots & a_{n-1,n-1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{a}_0 \; \boldsymbol{a}_1 \; \cdots \; \boldsymbol{a}_{n-1} \end{bmatrix}.$$

The fundamental algebraic operations are:

$$\begin{aligned} \text{Scalar mult.:} \quad & \boldsymbol{A} = \gamma \boldsymbol{B} \;\Rightarrow\; a_{ij} = \gamma b_{ij}. \\ \text{Addition:} \quad & \boldsymbol{A} = \boldsymbol{B} \pm \boldsymbol{C} \;\Rightarrow\; a_{ij} = b_{ij} \pm c_{ij}. \\ \text{Matrix-vector product:} \quad & \boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} \;\Rightarrow\; y_i = \sum_{j=0}^{n-1} a_{ij} x_j. \\ \text{Matrix-matrix product:} \quad & \boldsymbol{A} = \boldsymbol{B}\boldsymbol{C} \;\Rightarrow\; a_{ij} = \sum_{k=0}^{n-1} b_{ik} c_{kj}. \\ \text{Transpose:} \quad & \boldsymbol{A} = \boldsymbol{B}^T \;\Rightarrow\; a_{ij} = b_{ji}. \end{aligned}$$

Inverse.

If $\boldsymbol{A}$ is square and nonsingular, its inverse satisfies $\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$, where $\boldsymbol{I}$ is the identity matrix.

Hermitian conjugate.

The Hermitian conjugate $\boldsymbol{A}^\dagger$ is obtained by transposing $\boldsymbol{A}$ and complex-conjugating every element: $(A^\dagger)_{ij} = a^*_{ji}$.

Nonsingularity: equivalent characterisations.

For an $n \times n$ matrix $\boldsymbol{A}$ the following properties are all equivalent: (i) $\boldsymbol{A}^{-1}$ exists; (ii) $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$ implies $\boldsymbol{x} = \boldsymbol{0}$; (iii) the rows of $\boldsymbol{A}$ form a basis of $\mathbb{C}^n$; (iv) the columns of $\boldsymbol{A}$ form a basis of $\mathbb{C}^n$; (v) $\boldsymbol{A}$ is a product of elementary matrices; (vi) zero is not an eigenvalue of $\boldsymbol{A}$.

1.4 Special matrix types

Table 1.1 lists the matrix types that appear most frequently in quantum mechanics and many-body physics.

Name	Defining relation	Element condition
Symmetric	$\boldsymbol{A} = \boldsymbol{A}^T$	$a_{ij} = a_{ji}$
Real orthogonal	$\boldsymbol{A} = (\boldsymbol{A}^T)^{-1}$	$\sum_k a_{ik} a_{jk} = \delta_{ij}$
Hermitian	$\boldsymbol{A} = \boldsymbol{A}^\dagger$	$a_{ij} = a_{ji}^*$
Unitary	$\boldsymbol{A} = (\boldsymbol{A}^\dagger)^{-1}$	$\sum_k a_{ik} a_{jk}^* = \delta_{ij}$
Real matrix	$\boldsymbol{A} = \boldsymbol{A}^*$	$a_{ij} = a_{ij}^*$

Table 1.1 Important special matrix types and their defining properties.

Structured (sparse) matrices important for numerical many-body work:

- **Diagonal**: $a_{ij} = 0$ for $i \neq j$.
- **Upper/lower triangular**: $a_{ij} = 0$ for $i > j$ / $i < j$.
- **Upper/lower Hessenberg**: $a_{ij} = 0$ for $i > j+1$ / $i < j+1$.
- **Tridiagonal**: $a_{ij} = 0$ for $|i-j| > 1$.
- **Banded with bandwidth** p: $a_{ij} = 0$ for $|i-j| > p$.
- Block upper/lower triangular matrices, and more.

Many-body connection. Hamiltonian matrices in shell-model (full configuration-interaction, FCI) calculations are typically large, sparse, and often banded. Exploiting sparsity is essential to diagonalise them for systems with many single-particle states.

1.5 Dirac bra-ket notation

In quantum mechanics it is conventional to write state vectors using Dirac's bra-ket formalism. A ket $|x\rangle$ is a column vector

$$|x\rangle = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix},$$

and the corresponding bra $\langle x|$ is the row vector (dual)

$$\langle x| = \begin{bmatrix} x_0^* \; x_1^* \; \cdots \; x_{n-1}^* \end{bmatrix},$$

so $|x\rangle^\dagger = \langle x|$. The *inner product* (a complex number) is

$$\langle y|x\rangle = \sum_{i=0}^{n-1} y_i^* x_i.$$

Note that $\langle x|x\rangle \geq 0$ is always real and non-negative, and $\langle y|x\rangle^* = \langle x|y\rangle$.

The *outer product* $|x\rangle\langle y|$ is an $n \times n$ matrix with elements $(|x\rangle\langle y|)_{ij} = x_i y_j^*$:

$$|x\rangle\langle y| = \begin{bmatrix} x_0 y_0^* & x_0 y_1^* & \cdots & x_0 y_{n-1}^* \\ x_1 y_0^* & x_1 y_1^* & \cdots & x_1 y_{n-1}^* \\ \vdots & \vdots & \ddots & \vdots \\ x_{n-1} y_0^* & \cdots & \cdots & x_{n-1} y_{n-1}^* \end{bmatrix}.$$

Worked example.

Let $|x\rangle = \begin{bmatrix} 1-i \\ 2+i \end{bmatrix}$. Then

$$\langle x|x\rangle = (1+i)(1-i) + (2-i)(2+i) = 2+5 = 7 \in \mathbb{R}.$$

For two different vectors $|x\rangle = \begin{bmatrix} -1 \\ 2i \\ 1 \end{bmatrix}$, $|y\rangle = \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}$: $\langle x|y\rangle = -1+i \neq \langle y|x\rangle = -1-i$, confirming the rule $\langle x|y\rangle^* = \langle y|x\rangle$.

1.6 Computational basis states and superposition

A *two-level system* (qubit) is described by two orthonormal basis states, which we label

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \langle 0|1\rangle = 0, \quad \langle 0|0\rangle = \langle 1|1\rangle = 1.$$

These are the *computational basis* states. A general state is a *superposition*

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where $\alpha = \langle 0|\psi\rangle$ and $\beta = \langle 1|\psi\rangle$ are the probability amplitudes, satisfying $|\alpha|^2 + |\beta|^2 = 1$.

The identity operator decomposes as a sum of outer products:

$$\boldsymbol{I} = \sum_{i=0}^{1} |i\rangle\langle i| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Projection operators.

The outer products

$$\boldsymbol{P} = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \boldsymbol{Q} = |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

are *projection operators*. They are *idempotent* ($\boldsymbol{P}^2 = \boldsymbol{P}$, $\boldsymbol{Q}^2 = \boldsymbol{Q}$), commute with each other ($[\boldsymbol{P},\boldsymbol{Q}] = 0$), and satisfy $\boldsymbol{PQ} = 0$, $\boldsymbol{P}+\boldsymbol{Q} = \boldsymbol{I}$. Acting on a superposition state:

$$\boldsymbol{P}|\psi\rangle = \alpha|0\rangle, \qquad \boldsymbol{Q}|\psi\rangle = \beta|1\rangle.$$

The measurement postulate states that a measurement on $|\psi\rangle$ yields $|0\rangle$ with probability $|\alpha|^2$ and $|1\rangle$ with probability $|\beta|^2$.

Many-body connection. In many-body theory, projection operators $\hat{P}$ (onto occupied states) and $\hat{Q}$ (onto unoccupied / virtual states) define the particle-hole decomposition that is fundamental to perturbation theory, Hartree-Fock theory, and coupled-cluster methods.

Density matrix.

The density matrix (density operator) of a pure state $|\psi\rangle$ is the outer product

$$\boldsymbol{\rho} = |\psi\rangle\langle\psi| = \begin{bmatrix} \alpha\alpha^* & \alpha\beta^* \\ \beta\alpha^* & \beta\beta^* \end{bmatrix},$$

with $\operatorname{tr}\boldsymbol{\rho} = |\alpha|^2 + |\beta|^2 = 1$. Density matrices generalise to mixed states and subsystems; they are central to quantum information theory and to reduced-density-matrix many-body methods.

1.7 Pauli matrices

The three Pauli matrices are

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Key properties (easy to verify directly):

- **Involutory**: $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = \boldsymbol{I}$.
- **Unitary**: $\sigma_\alpha^\dagger = \sigma_\alpha^{-1}$ (Hermitian conjugate equals inverse).
- **Hermitian**: $\sigma_\alpha^\dagger = \sigma_\alpha$.
- $\det(\sigma_\alpha) = -1$ for each α.
- **Commutation**: $[\sigma_x, \sigma_y] = 2i\sigma_z$ (and cyclic permutations).

The Pauli matrices act on the computational basis as

$$\sigma_x|0\rangle = |1\rangle, \quad \sigma_x|1\rangle = |0\rangle; \qquad \sigma_z|0\rangle = +|0\rangle, \quad \sigma_z|1\rangle = -|1\rangle.$$

Acting on a superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:

$$\sigma_x|\psi\rangle = \beta|0\rangle + \alpha|1\rangle, \qquad \sigma_y|\psi\rangle = i\beta|0\rangle - i\alpha|1\rangle.$$

Quantum computing connection. The Pauli matrices are the elementary single-qubit gates X, Y, Z. Together with the Hadamard gate $H = (\sigma_x + \sigma_z)/\sqrt{2}$ and the phase gate T they form a universal single-qubit gate set.

Many-body connection. Via the Jordan-Wigner transformation, the creation/annihilation operators of second quantisation are mapped directly onto products (strings) of Pauli matrices, enabling the representation of fermionic Hamiltonians on qubit registers.

1.8 Orthonormal bases and the completeness relation

A set of states $\{|\phi_i\rangle\}_{i=0}^{n-1}$ is an *orthonormal basis* (ONB) of an n-dimensional Hilbert space if

$$\langle\phi_j|\phi_i\rangle = \delta_{ij} \quad \text{and} \quad \sum_{i=0}^{n-1} |\phi_i\rangle\langle\phi_i| = \boldsymbol{I} \quad \text{(completeness / resolution of the identity).}$$

Any state can then be expanded as $|\psi\rangle = \sum_i \langle\phi_i|\psi\rangle|\phi_i\rangle$, with $\langle\phi_i|\psi\rangle$ the projection (overlap) of $|\psi\rangle$ onto $|\phi_i\rangle$.

Many-body connection. The natural ONB in many-body physics is the set of single-particle eigenstates of the one-body Hamiltonian:

$$\hat{h}_0|\phi_i\rangle = \varepsilon_i|\phi_i\rangle, \qquad \phi_i(\mathbf{r}) = \langle \mathbf{r}|\phi_i\rangle.$$

This single-particle basis $\{|\phi_i\rangle\}$ is the starting point for virtually all many-body methods: Hartree-Fock, MBPT, FCI, coupled-cluster, and quantum Monte Carlo.

1.9 Unitary transformations and preservation of orthogonality

An $n \times n$ matrix $\boldsymbol{U}$ is *unitary* if $\boldsymbol{U}^\dagger \boldsymbol{U} = \boldsymbol{U}\boldsymbol{U}^\dagger = \boldsymbol{I}$, which means $\boldsymbol{U}^{-1} = \boldsymbol{U}^\dagger$.

Orthogonality is preserved.

If $\{\boldsymbol{v}_i\}$ is an orthonormal set, $\boldsymbol{v}_j^T \boldsymbol{v}_i = \delta_{ij}$, then the transformed set $\boldsymbol{w}_i = \boldsymbol{U}\boldsymbol{v}_i$ is also orthonormal:

$$\boldsymbol{w}_j^T \boldsymbol{w}_i = (\boldsymbol{U}\boldsymbol{v}_j)^T \boldsymbol{U}\boldsymbol{v}_i = \boldsymbol{v}_j^T \underbrace{\boldsymbol{U}^T \boldsymbol{U}}_{=\boldsymbol{I}} \boldsymbol{v}_i = \boldsymbol{v}_j^T \boldsymbol{v}_i = \delta_{ij}.$$

In Dirac notation, if $|\psi_i\rangle = \sum_j u_{ij}|\phi_j\rangle$ with (u_{ij}) unitary, then $\langle \psi_j|\psi_i\rangle = \delta_{ij}$.

Why only unitary transformations?

For the inner product (and hence the norm) to be preserved under $|\phi\rangle = \boldsymbol{U}|\psi\rangle$ we need

$$\langle \phi|\phi\rangle = \langle \psi|\boldsymbol{U}^\dagger \boldsymbol{U}|\psi\rangle = \langle \psi|\psi\rangle = 1,$$

which requires $\boldsymbol{U}^\dagger \boldsymbol{U} = \boldsymbol{I}$. Since probabilities must sum to unity, this is a fundamental physical requirement. Unitary transformations are rotations (and reflections) in Hilbert space. The fact that $\boldsymbol{U}^{-1} = \boldsymbol{U}^\dagger$ also means that every unitary transformation is perfectly reversible.

Example: two-state rotation.

Given $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and a unitary $\boldsymbol{U}$, the rotated state $|\phi\rangle = \gamma|0\rangle + \delta|1\rangle = \boldsymbol{U}|\psi\rangle$ satisfies

$$\begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

Since $|\alpha|^2 + |\beta|^2 = 1$ and $\boldsymbol{U}$ is unitary, $|\gamma|^2 + |\delta|^2 = 1$ automatically.

Many-body connection. In Hartree-Fock theory the self-consistent field (SCF) procedure generates a sequence of unitary rotations of the single-particle basis. The total HF energy is invariant under unitary rotations within the occupied or virtual subspace separately, which is the Thouless theorem.

Quantum computing connection. Every quantum gate is a unitary matrix. The evolution of a closed quantum system is described entirely by unitary operations; this is the quantum analogue of classical reversibility.

1.10 Tensor products

The *tensor (Kronecker) product* of two vectors $|x\rangle$ (length n) and $|y\rangle$ (length m) is the vector of length nm:

$$|x\rangle \otimes |y\rangle = |xy\rangle = \begin{bmatrix} x_0 y_0 \\ x_0 y_1 \\ x_1 y_0 \\ x_1 y_1 \end{bmatrix} \quad (\text{for } n = m = 2).$$

For the computational basis:

$$|0\rangle \otimes |0\rangle = |00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad |0\rangle \otimes |1\rangle = |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad |1\rangle \otimes |0\rangle = |10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad |1\rangle \otimes |1\rangle = |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

These four states form an ONB for the two-qubit Hilbert space $\mathbb{C}^4 = \mathbb{C}^2 \otimes \mathbb{C}^2$.

The most general state of N qubits is

$$|\Psi\rangle = \sum_{i_1 i_2 \cdots i_N} \Psi_{i_1 i_2 \cdots i_N} |i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_N\rangle,$$

requiring 2^N complex amplitudes. For $N \approx 300$ this exceeds the number of atoms in the observable universe — this is the *exponential complexity* at the heart of quantum many-body physics.

Many-body connection. The full N-particle Hilbert space is a tensor product of N single-particle spaces. The FCI method works in this exponentially large space; all approximate methods (HF, MBPT, coupled cluster, DMRG, ...) are strategies to identify the relevant low-dimensional submanifold.

Quantum computing connection. A quantum computer with N qubits *inherently* represents a 2^N-dimensional state vector. This is both the source of its potential power and the reason that quantum simulation of many-body systems is a leading application.

1.11 Spectral decomposition of Hermitian operators

A Hermitian operator $\boldsymbol{A} = \boldsymbol{A}^\dagger$ has the following key properties:

1. All eigenvalues λ_i are *real*.
2. Eigenvectors belonging to distinct eigenvalues are *orthogonal*.
3. $\boldsymbol{A}$ is diagonalisable (it is a *normal* operator, $[\boldsymbol{A}, \boldsymbol{A}^\dagger] = 0$).

The eigenvalue equation $\boldsymbol{A}|\psi\rangle = \lambda|\psi\rangle$ has non-trivial solutions only when

$$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0 \quad (\text{characteristic polynomial}).$$

Spectral decomposition.

Let $\{|i\rangle\}$ be the eigenbasis of $\boldsymbol{A}$ with eigenvalues λ_i. Any state expands as $|\psi\rangle = \sum_i \alpha_i |i\rangle$. Using projection operators $\boldsymbol{P}_i = |i\rangle\langle i|$ we obtain:

$$\boldsymbol{A}|\psi\rangle = \sum_i \alpha_i \lambda_i |i\rangle = \sum_i \lambda_i \boldsymbol{P}_i |\psi\rangle,$$

from which, since this holds for any $|\psi\rangle$,

$$\boxed{\boldsymbol{A} = \sum_{i=0}^{n-1} \lambda_i \, |i\rangle\langle i| = \sum_{i=0}^{n-1} \lambda_i \, \boldsymbol{P}_i.}$$

This is the *spectral decomposition*: the operator is expressed as a weighted sum of its projection operators.

Example for a 2×2 operator.

If $\boldsymbol{A}$ has eigenstates $|\psi_a\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$ and $|\psi_b\rangle = \beta_0|0\rangle + \beta_1|1\rangle$ with eigenvalues λ_a, λ_b, then

$$\boldsymbol{A} = \lambda_a \begin{bmatrix} |\alpha_0|^2 & \alpha_0\alpha_1^* \\ \alpha_1\alpha_0^* & |\alpha_1|^2 \end{bmatrix} + \lambda_b \begin{bmatrix} |\beta_0|^2 & \beta_0\beta_1^* \\ \beta_1\beta_0^* & |\beta_1|^2 \end{bmatrix}.$$

Many-body connection. The spectral decomposition is used every time one diagonalises a Hamiltonian matrix. In FCI the full N-body Hamiltonian is diagonalised in the many-determinant basis; in Hartree-Fock the Fock matrix (a single-particle Hermitian operator) is diagonalised to find the optimal orbital basis.

Quantum computing connection. Any unitary gate $\boldsymbol{U}$ can be written as $\boldsymbol{U} = e^{i\boldsymbol{H}}$ for a Hermitian generator $\boldsymbol{H}$. The spectral decomposition of $\boldsymbol{H}$ gives $\boldsymbol{U} = \sum_i e^{i\lambda_i}|i\rangle\langle i|$. This exponential relationship connects the physical Hamiltonian to the time-evolution operator $\hat{U} = e^{-i\hat{H}t/\hbar}$ and is the basis of the variational quantum eigensolver (VQE) and unitary coupled-cluster (UCC) ansatz.

1.12 From algebra to computation

Everything said so far is exact algebra. The moment we hand a matrix to a computer, however, two new questions arise. The first is one of *accuracy*: floating-point numbers carry a finite number of digits, and we need a way of measuring how large the resulting error is. The second is one of *cost*: a many-body Hamiltonian matrix may have a dimension of 10^6 or more, and an algorithm whose work grows as n^3 is then simply not available to us.

The remainder of this chapter is devoted to these two questions. We first introduce the norms with which errors are measured, then develop the direct (elimination) methods for linear systems, the iterative alternatives, and finally the algorithms for the algebraic eigenvalue problem $\boldsymbol{A}|x\rangle = \lambda|x\rangle$, which is the computational problem at the heart of essentially every method in this book. All programs are written in Python; the complete, runnable versions are collected in the `BookMaterial/Programs` directory.

Throughout we keep the convention of Section 1.1 that indices run from 0 to $n-1$, so that the mathematical formulae and the Python code carry the same indices.

1.13 Vector and matrix norms

A class of vector norms is given by the so-called p-norms

$$\|\boldsymbol{x}\|_p = (|x_0|^p + |x_1|^p + \cdots + |x_{n-1}|^p)^{1/p}, \qquad p \geq 1. \tag{1.1}$$

The three cases we shall need are $p = 1$, $p = 2$ and the limit $p \to \infty$,

$$\|\boldsymbol{x}\|_1 = |x_0| + |x_1| + \cdots + |x_{n-1}|, \tag{1.2}$$

$$\|\boldsymbol{x}\|_2 = \left(|x_0|^2 + \cdots + |x_{n-1}|^2\right)^{1/2} = \left(\boldsymbol{x}^\dagger \boldsymbol{x}\right)^{1/2}, \tag{1.3}$$

$$\|\boldsymbol{x}\|_\infty = \max_{0 \leq i \leq n-1} |x_i|. \tag{1.4}$$

The 2-norm is the one we have already been using implicitly: it is the square root of the inner product of Section 1.2, and in Dirac notation $\||x\rangle\|_2^2 = \langle x|x\rangle$. A quantum state is normalised precisely when its 2-norm equals unity.

Two inequalities follow from these definitions and will be used repeatedly. The first is the Cauchy-Schwarz inequality, valid for any two vectors in an inner product space,

$$\left|\boldsymbol{x}^\dagger \boldsymbol{y}\right| \leq \|\boldsymbol{x}\|_2 \, \|\boldsymbol{y}\|_2, \tag{1.5}$$

with equality only when $\boldsymbol{x}$ and $\boldsymbol{y}$ are linearly dependent. In Dirac notation this reads $|\langle x|y\rangle|^2 \leq \langle x|x\rangle\langle y|y\rangle$, and it is the inequality behind the variational principle: the overlap of a trial state with the exact ground state can never exceed the product of their norms. The second is the triangle inequality

$$\|\boldsymbol{x} + \boldsymbol{y}\|_2 \leq \|\boldsymbol{x}\|_2 + \|\boldsymbol{y}\|_2, \tag{1.6}$$

which follows from Eq. (1.5). Proofs may be found in Golub and Van Loan [1].

For matrices the most frequently used norms are the Frobenius norm

$$\|\boldsymbol{A}\|_F = \sqrt{\sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |a_{ij}|^2}, \tag{1.7}$$

which is simply the 2-norm of the matrix read as a long vector, and the induced or operator p-norms

$$\|\boldsymbol{A}\|_p = \max_{\boldsymbol{x} \neq 0} \frac{\|\boldsymbol{A}\boldsymbol{x}\|_p}{\|\boldsymbol{x}\|_p}. \tag{1.8}$$

The induced norm measures the largest factor by which $\boldsymbol{A}$ can stretch a vector. For a Hermitian matrix $\|\boldsymbol{A}\|_2$ is the largest absolute eigenvalue, a fact which follows immediately from the spectral decomposition of Section 1.11. A unitary matrix has $\|\boldsymbol{U}\|_2 = 1$; it rotates but never stretches, which is Section 1.9 restated in the language of norms.

1.13.1 The Frobenius norm as a trace

Definition (1.7) is a double sum over entries, which is convenient for a computer and inconvenient for a proof. The identity that makes the Frobenius norm usable in analysis is

$$\boxed{\|\boldsymbol{A}\|_F^2 = \operatorname{Tr}\left(\boldsymbol{A}^T \boldsymbol{A}\right), \qquad \|\boldsymbol{A}\|_F = \sqrt{\operatorname{Tr}\left(\boldsymbol{A}^T \boldsymbol{A}\right)}} \tag{1.9}$$

for a real $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, with $\boldsymbol{A}^T$ replaced by $\boldsymbol{A}^\dagger$ in the complex case. We give two proofs. The first is a line of index bookkeeping; the second explains *why* the identity has to hold, and it is the one worth remembering.

First proof: component by component.

The product $\boldsymbol{A}^T\boldsymbol{A}$ is an $n\times n$ matrix with entries

$$\left(\boldsymbol{A}^T\boldsymbol{A}\right)_{jk} = \sum_{i=0}^{m-1} a_{ij}a_{ik}, \tag{1.10}$$

so that on the diagonal, where $k = j$,

$$\left(\boldsymbol{A}^T\boldsymbol{A}\right)_{jj} = \sum_{i=0}^{m-1} a_{ij}^2,$$

which is the squared Euclidean length of column j of $\boldsymbol{A}$. The trace is the sum of the diagonal entries, so

$$\mathrm{Tr}\left(\boldsymbol{A}^T\boldsymbol{A}\right) = \sum_{j=0}^{n-1}\left(\boldsymbol{A}^T\boldsymbol{A}\right)_{jj} = \sum_{j=0}^{n-1}\sum_{i=0}^{m-1} a_{ij}^2 = \sum_{i=0}^{m-1}\sum_{j=0}^{n-1} a_{ij}^2,$$

the last step being nothing more than an interchange of two finite sums. The right-hand side is $\|\boldsymbol{A}\|_F^2$ by definition, which proves Eq. (1.9). Read the calculation once more and it says something slightly stronger: the Frobenius norm squared is the sum of the squared column lengths, and by running the same argument on $\boldsymbol{A}\boldsymbol{A}^T$ it is equally the sum of the squared row lengths. Hence $\mathrm{Tr}(\boldsymbol{A}^T\boldsymbol{A}) = \mathrm{Tr}(\boldsymbol{A}\boldsymbol{A}^T)$ even though the two matrices have different dimensions.

The Frobenius inner product.

The second proof begins by noticing that Eq. (1.9) is the norm statement of an inner product. For two matrices $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{m\times n}$ define

$$\langle \boldsymbol{A}, \boldsymbol{B}\rangle_F = \mathrm{Tr}\left(\boldsymbol{A}^T\boldsymbol{B}\right) = \sum_{i=0}^{m-1}\sum_{j=0}^{n-1} a_{ij}b_{ij}, \tag{1.11}$$

where the second equality follows by repeating Eq. (1.10) with b_{ik} in place of a_{ik} and taking the trace. This is bilinear, symmetric, and positive definite – $\langle \boldsymbol{A}, \boldsymbol{A}\rangle_F$ is a sum of squares and vanishes only when every entry vanishes – so it is an inner product on the vector space $\mathbb{R}^{m\times n}$, the *Frobenius* or Hilbert-Schmidt inner product. Every inner product induces a norm through $\|\boldsymbol{A}\| = \sqrt{\langle \boldsymbol{A}, \boldsymbol{A}\rangle}$, and for Eq. (1.11) that induced norm is precisely Eq. (1.7). Equation (1.9) is therefore not a coincidence about traces but the statement that the Frobenius norm is the norm belonging to the trace inner product.

Second proof: vectorisation.

The same point can be made constructively. Let $\mathrm{vec}: \mathbb{R}^{m\times n} \to \mathbb{R}^{mn}$ stack the columns of a matrix into one long vector,

$$\mathrm{vec}(\boldsymbol{A}) = \left(a_{00}\ a_{10}\ \ldots\ a_{m-1,0}\ a_{01}\ \ldots\ a_{m-1,n-1}\right)^T. \tag{1.12}$$

The particular ordering is irrelevant for what follows; all that matters is that every entry appears exactly once. Then

$$\|\mathrm{vec}(\boldsymbol{A})\|_2^2 = \sum_{i,j} a_{ij}^2 = \|\boldsymbol{A}\|_F^2,$$

by inspection, while for any two matrices of the same shape

$$\mathrm{vec}(\boldsymbol{A})^T \mathrm{vec}(\boldsymbol{B}) = \sum_{i,j} a_{ij} b_{ij} = \mathrm{Tr}\left(\boldsymbol{A}^T \boldsymbol{B}\right), \tag{1.13}$$

using Eq. (1.11). Setting $\boldsymbol{B} = \boldsymbol{A}$ gives Eq. (1.9) again. Geometrically, vectorisation is an isometry between $\mathbb{R}^{m\times n}$ equipped with the Frobenius norm and $\mathbb{R}^{mn}$ equipped with the ordinary Euclidean norm,

$$\|\boldsymbol{A}\|_F = \|\mathrm{vec}(\boldsymbol{A})\|_2, \tag{1.14}$$

so $\mathrm{Tr}(\boldsymbol{A}^T\boldsymbol{A})$ is simply the squared length of the matrix regarded as a point in mn dimensions. Matrix space is a Euclidean space, and the Frobenius norm is what makes it one.

Two consequences we shall use.

First, the Frobenius norm is unitarily invariant. If $\boldsymbol{U}$ and $\boldsymbol{V}$ are unitary of the appropriate sizes, then by the cyclic property of the trace

$$\|\boldsymbol{U}\boldsymbol{A}\boldsymbol{V}^\dagger\|_F^2 = \mathrm{Tr}\left(\boldsymbol{V}\boldsymbol{A}^\dagger\boldsymbol{U}^\dagger\boldsymbol{U}\boldsymbol{A}\boldsymbol{V}^\dagger\right) = \mathrm{Tr}\left(\boldsymbol{A}^\dagger\boldsymbol{A}\right) = \|\boldsymbol{A}\|_F^2.$$

Choosing $\boldsymbol{U}$ and $\boldsymbol{V}$ to be the singular vectors of Section 1.28 turns $\boldsymbol{A}$ into its diagonal matrix of singular values, so

$$\|\boldsymbol{A}\|_F^2 = \sum_k \sigma_k^2, \tag{1.15}$$

the Frobenius norm is the 2-norm of the vector of singular values, and the Eckart-Young theorem of Section 1.30 – that truncating the SVD gives the best low-rank approximation in the Frobenius norm – becomes a statement about discarding the smallest entries of that vector.

Second, because the Frobenius norm comes from an inner product, minimising it subject to linear constraints is an ordinary least-squares problem and can be solved with a Lagrange multiplier. That is exactly the calculation which produces Broyden's update and, through it, the BFGS method of Section 14.4: among all matrices satisfying a prescribed secant condition, one asks for the one closest to the current approximation in the Frobenius norm, and the answer is a rank-one correction. The trace identity (1.9) is what makes the differentiation $\partial\,\mathrm{Tr}(\boldsymbol{X}^T\boldsymbol{X})/\partial\boldsymbol{X} = 2\boldsymbol{X}$ available, and that single derivative is the whole of the derivation.

Relative error.

Let $fl(\boldsymbol{x})$ denote the machine representation of a vector $\boldsymbol{x} \neq 0$. The relative error is

$$\varepsilon = \frac{\|fl(\boldsymbol{x}) - \boldsymbol{x}\|}{\|\boldsymbol{x}\|}, \tag{1.16}$$

and if we use the ∞-norm the statement $\|fl(\boldsymbol{x}) - \boldsymbol{x}\|_\infty / \|\boldsymbol{x}\|_\infty \approx 10^{-l}$ says that the largest component of $fl(\boldsymbol{x})$ carries roughly l correct significant digits.

The condition number.

Suppose we solve $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ but, because of rounding, actually solve a slightly perturbed problem with right-hand side $\boldsymbol{b} + \delta\boldsymbol{b}$. The resulting error $\delta\boldsymbol{x}$ satisfies $\boldsymbol{A}\,\delta\boldsymbol{x} = \delta\boldsymbol{b}$, hence $\|\delta\boldsymbol{x}\| \leq \|\boldsymbol{A}^{-1}\|\,\|\delta\boldsymbol{b}\|$, while $\|\boldsymbol{b}\| \leq \|\boldsymbol{A}\|\,\|\boldsymbol{x}\|$. Combining the two gives

$$\frac{\|\delta\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \leq \kappa(\boldsymbol{A})\,\frac{\|\delta\boldsymbol{b}\|}{\|\boldsymbol{b}\|}, \qquad \kappa(\boldsymbol{A}) = \|\boldsymbol{A}\|\,\|\boldsymbol{A}^{-1}\|. \tag{1.17}$$

The quantity $\kappa(\boldsymbol{A})$ is the *condition number*. It tells us how much an input error can be amplified, and it is a property of the problem, not of the algorithm: no amount of cleverness can rescue an ill-conditioned system. In the 2-norm and for a Hermitian matrix

$$\kappa_2(\boldsymbol{A}) = \frac{\max_i |\lambda_i|}{\min_i |\lambda_i|}, \tag{1.18}$$

so a matrix is ill-conditioned exactly when it has eigenvalues of very different magnitude, that is, when it is close to being singular. As a rule of thumb, if $\kappa \approx 10^k$ one loses about k significant digits.

Many-body connection. Ill-conditioning is not an academic worry in many-body physics. Overlap matrices of non-orthogonal single-particle bases, such as the Gaussian basis sets of quantum chemistry, become nearly singular as the basis approaches linear dependence, and their condition numbers can reach 10^{10} and beyond. The standard cure is to diagonalise the overlap matrix and discard the eigenvectors with the smallest eigenvalues, which is the canonical orthogonalisation procedure met again in Hartree-Fock theory.

1.14 Linear systems and Gaussian elimination

We now turn to the solution of

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{w}, \tag{1.19}$$

with $\boldsymbol{A}$ non-singular. Before describing the algorithms it is worth seeing why such systems appear at all.

Where linear systems come from.

Consider the boundary value problem

$$-\frac{d^2u(x)}{dx^2} = f(x), \qquad x \in (a,b), \qquad u(a) = u(b) = 0. \tag{1.20}$$

We subdivide the interval into $n+1$ subintervals by setting $x_i = a + ih$ with step $h = (b-a)/(n+1)$, and replace the second derivative by the three-point formula

$$u''(x_i) \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} + \mathscr{O}(h^2). \tag{1.21}$$

For the interior points $i = 0, 1, \ldots, n-1$ this turns Eq. (1.20) into

$$\frac{-u_{i+1} + 2u_i - u_{i-1}}{h^2} = f_i, \tag{1.22}$$

where the boundary values u_{-1} and u_n are known and equal to zero. Collecting the coefficients we obtain $\boldsymbol{Au} = \boldsymbol{f}$ with the tridiagonal matrix

$$\boldsymbol{A} = \frac{1}{h^2}\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & \ddots & \\ & & \ddots & \ddots & -1 \\ & & & -1 & 2 \end{bmatrix}. \tag{1.23}$$

A differential equation has become a matrix equation. This is the single most important lesson of this chapter, and it will be repeated in a many-body setting in Section 1.27: discretising a continuous problem, or expanding a state in a finite basis, always produces a matrix problem.

The elimination.

The idea of Gaussian elimination is to use the first equation to eliminate x_0 from the remaining $n-1$ equations, then the new second equation to eliminate x_1 from the remaining $n-2$, and so on. After $n-1$ such steps we are left with an upper triangular system

$$\boldsymbol{Ux} = \boldsymbol{y}, \tag{1.24}$$

which is solved from the bottom upwards by *backward substitution*

$$x_m = \frac{1}{u_{mm}}\left(y_m - \sum_{k=m+1}^{n-1} u_{mk}x_k\right), \qquad m = n-1, n-2, \ldots, 0. \tag{1.25}$$

To eliminate x_0 we multiply the first equation by a_{j0}/a_{00} and subtract it from equation j, for $j = 1, \ldots, n-1$. The elements of the remaining $(n-1)\times(n-1)$ block are updated according to

$$a_{jk}^{(1)} = a_{jk} - \frac{a_{j0}\,a_{0k}}{a_{00}}, \qquad j,k = 1,\ldots,n-1, \tag{1.26}$$

and the same formula, applied to successively smaller blocks, defines the whole algorithm. Counting operations, the elimination requires $2n^3/3 + \mathscr{O}(n^2)$ floating point operations, while the backward substitution costs only $\mathscr{O}(n^2)$. The elimination dominates, and this cubic scaling is the reason why direct methods become unusable for the large matrices of Section 1.26.

Pivoting.

The derivation assumed $a_{00} \neq 0$, and more generally that no pivot element vanishes. Even a small pivot is dangerous, since dividing by it amplifies rounding errors. The cure is *partial pivoting*: before eliminating column k we search the column for the element of largest absolute value and interchange rows so that this element becomes the pivot. Row interchanges do not alter the solution, and they keep all multipliers bounded by unity. In practice one always pivots.

The following class implements the algorithm.

```
class GaussianElimination(DirectSolver):
    """Gaussian elimination with partial pivoting."""

    def solve(self, b):
```

```
        n = self.n
        M = self.A.copy()                 # the elimination destroys its input
        y = np.asarray(b, dtype=float).copy()

        for k in range(n - 1):            # forward elimination
            # partial pivoting: use the largest element in the column
            p = k + np.argmax(np.abs(M[k:, k]))
            if abs(M[p, k]) < 1.0e-14:
                raise np.linalg.LinAlgError("matrix is singular")
            if p != k:
                M[[k, p]] = M[[p, k]]
                y[k], y[p] = y[p], y[k]
            for i in range(k + 1, n):
                factor = M[i, k] / M[k, k]
                M[i, k:] -= factor * M[k, k:]
                y[i] -= factor * y[k]

        return self._back_substitute(M, y)

    @staticmethod
    def _back_substitute(U, y):
        """Solve U x = y for an upper triangular U."""
        n = U.shape[0]
        x = np.zeros(n)
        for m in range(n - 1, -1, -1):
            x[m] = (y[m] - U[m, m+1:] @ x[m+1:]) / U[m, m]
        return x
```

1.15 LU decomposition

Gaussian elimination as written above throws away the multipliers once it has used them. If we keep them instead, we obtain a factorisation of $\boldsymbol{A}$ itself, and this is what makes the method reusable. The LU decomposition writes

$$\boldsymbol{A} = \boldsymbol{L}\boldsymbol{U}, \tag{1.27}$$

with $\boldsymbol{L}$ unit lower triangular and $\boldsymbol{U}$ upper triangular; for a 4×4 matrix

$$\begin{bmatrix} a_{00} & a_{01} & a_{02} & a_{03} \\ a_{10} & a_{11} & a_{12} & a_{13} \\ a_{20} & a_{21} & a_{22} & a_{23} \\ a_{30} & a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ l_{10} & 1 & 0 & 0 \\ l_{20} & l_{21} & 1 & 0 \\ l_{30} & l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{00} & u_{01} & u_{02} & u_{03} \\ 0 & u_{11} & u_{12} & u_{13} \\ 0 & 0 & u_{22} & u_{23} \\ 0 & 0 & 0 & u_{33} \end{bmatrix}. \tag{1.28}$$

The factorisation exists whenever $\boldsymbol{A}$ is non-singular and, with the normalisation $l_{ii} = 1$, it is unique. Multiplying out the first column gives $a_{00} = u_{00}$ and $a_{j0} = l_{j0}u_{00}$, which determines u_{00} and the multipliers l_{j0}. The second column then gives u_{01}, u_{11} and l_{j1}, and so on: at every stage the unknowns are determined by quantities already computed. In practice $\boldsymbol{L}$ and $\boldsymbol{U}$ are stored in the same array as $\boldsymbol{A}$, since the unit diagonal of $\boldsymbol{L}$ need not be kept.

With partial pivoting the factorisation reads $\boldsymbol{PA} = \boldsymbol{LU}$ with $\boldsymbol{P}$ a permutation matrix, which in the program is recorded as a permutation array rather than as a matrix.

The determinant.

Because $\det(\boldsymbol{L}) = 1$ and the determinant of a triangular matrix is the product of its diagonal elements,

$$|\boldsymbol{A}| = \pm u_{00} u_{11} \cdots u_{n-1,n-1}, \tag{1.29}$$

where the sign is $(-1)^r$ with r the number of row interchanges. This is the only sensible way of computing a determinant numerically; expanding in minors would cost $\mathcal{O}(n!)$ operations.

Solving the system.

Writing $\boldsymbol{Ax} = \boldsymbol{LUx} = \boldsymbol{w}$ and introducing the intermediate vector $\boldsymbol{y} = \boldsymbol{Ux}$, the problem splits into two triangular systems,

$$\boldsymbol{Ly} = \boldsymbol{Pw} \quad \text{(forward substitution)}, \qquad \boldsymbol{Ux} = \boldsymbol{y} \quad \text{(backward substitution)}, \tag{1.30}$$

each costing only $\mathcal{O}(n^2)$ operations. This is the decisive advantage of LU over plain elimination: the expensive $\mathcal{O}(n^3)$ factorisation is performed once, after which any number of right-hand sides can be treated cheaply.

The inverse.

Column j of $\boldsymbol{A}^{-1}$ is the solution of $\boldsymbol{Ax} = \boldsymbol{e}_j$, where $\boldsymbol{e}_j$ is the jth unit vector. The inverse therefore costs one factorisation plus n substitutions. It is worth stressing that one should almost never compute an inverse: if the aim is to solve a linear system, solving it directly is both faster and more accurate than forming $\boldsymbol{A}^{-1}$ and multiplying.

```python
class LUDecomposition(DirectSolver):
    """Doolittle LU factorisation with partial pivoting, P A = L U."""

    def _factorize(self):
        n = self.n
        LU = self.A.copy()
        perm = np.arange(n)
        sign = 1.0

        for k in range(n):
            p = k + np.argmax(np.abs(LU[k:, k]))
            if abs(LU[p, k]) < 1.0e-14:
                raise np.linalg.LinAlgError("matrix is singular")
            if p != k:
                LU[[k, p]] = LU[[p, k]]
                perm[[k, p]] = perm[[p, k]]
                sign = -sign
            # the multipliers l_ik are stored in place, below the diagonal
            LU[k+1:, k] /= LU[k, k]
            # rank-one update of the trailing submatrix
            LU[k+1:, k+1:] -= np.outer(LU[k+1:, k], LU[k, k+1:])

        self.LU, self.perm, self.sign = LU, perm, sign

    def solve(self, b):
        """Solve A x = b in two steps: L y = P b, then U x = y."""
        y = np.asarray(b, dtype=float)[self.perm].copy()
        n = self.n
        for i in range(1, n):                    # forward, L has unit diagonal
            y[i] -= self.LU[i, :i] @ y[:i]
```

```
        x = np.zeros(n)
        for i in range(n - 1, -1, -1):            # backward
            x[i] = (y[i] - self.LU[i, i+1:] @ x[i+1:]) / self.LU[i, i]
        return x

    def determinant(self):
        return self.sign * np.prod(np.diag(self.LU))
```

Cholesky's factorisation.

If $\boldsymbol{A}$ is real, symmetric and positive definite it admits the special factorisation

$$\boldsymbol{A} = \boldsymbol{L}\boldsymbol{L}^T, \tag{1.31}$$

with $\boldsymbol{L}$ lower triangular. The elements follow from

$$L_{ii} = \left(A_{ii} - \sum_{k=0}^{i-1} L_{ik}^2 \right)^{1/2}, \tag{1.32}$$

$$L_{ji} = \frac{1}{L_{ii}} \left(A_{ji} - \sum_{k=0}^{i-1} L_{ik} L_{jk} \right), \qquad j = i+1, \ldots, n-1. \tag{1.33}$$

The algorithm costs half of a general LU factorisation, and since the pivots L_{ii} are guaranteed positive there is no need for pivoting at all. The square root in Eq. (1.32) also provides the cheapest available test of positive definiteness: if the argument turns negative, the matrix is not positive definite.

Many-body connection. Positive definite matrices are everywhere in many-body theory. Overlap matrices of linearly independent basis functions are positive definite by construction, and the Cholesky factor of the overlap matrix is the standard tool for transforming a generalised eigenvalue problem $\boldsymbol{H}\boldsymbol{c} = E\boldsymbol{S}\boldsymbol{c}$ into an ordinary one. The Hessian of the variance in variational Monte Carlo is likewise positive definite, which is what makes the conjugate gradient method of Section 1.18 applicable to the optimisation of trial wave functions.

1.16 Tridiagonal systems

The matrix of Eq. (1.23) is tridiagonal, and applying general Gaussian elimination to it would be an extravagant waste: almost all the operations would be spent on elements that are already zero. Writing the system as

$$a_i u_{i-1} + b_i u_i + c_i u_{i+1} = f_i, \qquad i = 0, 1, \ldots, n-1, \tag{1.34}$$

with a_0 and c_{n-1} not referenced, the elimination reduces to a single forward sweep followed by a single backward sweep. The cost is $\mathcal{O}(n)$ operations rather than $2n^3/3$, and the storage is three vectors rather than a full matrix. Whenever a problem produces a tridiagonal, banded or otherwise sparse matrix, exploiting that structure is not an optimisation but a necessity.

```
class TridiagonalSolver:
    """Solve a tridiagonal system in O(n) operations."""
```

```
    def solve(self, f):
        n = self.n
        f = np.asarray(f, dtype=float)
        temp = np.zeros(n)
        u = np.zeros(n)

        btemp = self.b[0]                       # forward substitution
        u[0] = f[0] / btemp
        for i in range(1, n):
            temp[i] = self.c[i-1] / btemp
            btemp = self.b[i] - self.a[i] * temp[i]
            u[i] = (f[i] - self.a[i] * u[i-1]) / btemp

        for i in range(n - 2, -1, -1):          # backward substitution
            u[i] -= temp[i+1] * u[i+1]
        return u
```

Applied to $-u'' = 2$ on $(0,1)$ with $u(0) = u(1) = 0$, whose exact solution is $u(x) = x(1-x)$, the solver reproduces the analytic answer to 8.6×10^{-16} with $n = 100$ interior points. The three-point formula is exact for polynomials of degree two, which is why the discretisation error vanishes here; for a general right-hand side the error would be $\mathscr{O}(h^2)$, as we shall see in Section 1.27.

1.17 Iterative methods for linear systems

The methods of the previous sections are *direct*: up to rounding they produce the exact answer in a finite, known number of operations. Their cost, however, grows as n^3, and they require the matrix to be stored. Iterative methods take the opposite view. We start from a guess and improve it until the change falls below a tolerance. The matrix enters only through the product $\boldsymbol{Ax}$, so a matrix that is sparse, or that is never formed at all but only applied, costs nothing extra. This is the reason iterative methods, and not the direct ones, are what one uses for large-scale many-body calculations.

We split the matrix as

$$\boldsymbol{A} = \boldsymbol{D} + \boldsymbol{L} + \boldsymbol{U}, \tag{1.35}$$

with $\boldsymbol{D}$ diagonal, $\boldsymbol{L}$ strictly lower and $\boldsymbol{U}$ strictly upper triangular.

Jacobi's method.

Solving row i for x_i and evaluating the right-hand side with the previous iterate gives

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j^{(k)} \right), \tag{1.36}$$

or, in matrix form,

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{D}^{-1} \left(\boldsymbol{b} - (\boldsymbol{L} + \boldsymbol{U}) \boldsymbol{x}^{(k)} \right). \tag{1.37}$$

Every component of the new iterate is computed from the old ones only, so all n updates are independent and the method parallelises trivially. It converges whenever $\boldsymbol{A}$ is strictly diagonally dominant or symmetric positive definite.

Gauss-Seidel.

Nothing forces us to wait: the components $x_0^{(k+1)},\dots,x_{i-1}^{(k+1)}$ are already available when we compute $x_i^{(k+1)}$. Using them immediately gives the Gauss-Seidel iteration

$$x_i^{(k+1)} = \frac{1}{a_{ii}}\left(b_i - \sum_{j<i} a_{ij}x_j^{(k+1)} - \sum_{j>i} a_{ij}x_j^{(k)}\right). \tag{1.38}$$

This is forward substitution applied to the splitting, and it typically halves the number of iterations. The price is that the sweep is now inherently sequential.

Successive over-relaxation.

Gauss-Seidel usually moves in the right direction but not far enough. Over-relaxation exaggerates the step by a factor ω,

$$x_i^{(k+1)} = (1-\omega)x_i^{(k)} + \frac{\omega}{a_{ii}}\left(b_i - \sum_{j<i} a_{ij}x_j^{(k+1)} - \sum_{j>i} a_{ij}x_j^{(k)}\right), \tag{1.39}$$

which in matrix form reads

$$\boldsymbol{x}^{(k+1)} = (\boldsymbol{D}+\omega\boldsymbol{L})^{-1}\left(\omega\boldsymbol{b} - [\omega\boldsymbol{U} + (\omega-1)\boldsymbol{D}]\,\boldsymbol{x}^{(k)}\right). \tag{1.40}$$

For symmetric positive definite matrices convergence is guaranteed for $0<\omega<2$, and $\omega=1$ returns Gauss-Seidel. A well chosen ω can reduce the iteration count by an order of magnitude, but the optimal value depends on the spectrum of $\boldsymbol{A}$ and is rarely known in advance.

1.18 The conjugate gradient method

The methods above treat the linear system as a fixed-point problem. The conjugate gradient method instead treats it as a minimisation. For a symmetric positive definite $\boldsymbol{A}$ the solution of $\boldsymbol{Ax}=\boldsymbol{b}$ is the unique minimum of the quadratic form

$$P(\boldsymbol{x}) = \tfrac{1}{2}\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x} - \boldsymbol{x}^T\boldsymbol{b}, \tag{1.41}$$

since $\nabla P = \boldsymbol{Ax}-\boldsymbol{b}$, which vanishes precisely at the solution. The residual

$$\boldsymbol{r} = \boldsymbol{b} - \boldsymbol{Ax} \tag{1.42}$$

is therefore the negative gradient of P, and steepest descent would move along it. Steepest descent is however notoriously slow, because successive steps undo part of the progress of their predecessors.

The remedy is to search along directions that are *conjugate*, meaning orthogonal with respect to the inner product defined by $\boldsymbol{A}$,

$$\boldsymbol{p}_i^T\boldsymbol{A}\boldsymbol{p}_j = 0, \qquad i\neq j. \tag{1.43}$$

The eigenvectors of $\boldsymbol{A}$ are an example, since $\boldsymbol{v}_i^T\boldsymbol{A}\boldsymbol{v}_j = \lambda_j\boldsymbol{v}_i^T\boldsymbol{v}_j$ vanishes for $i\neq j$; but the whole point of the method is that we shall construct conjugate directions without knowing any eigenvectors. A set of n mutually conjugate vectors forms a basis of $\mathbb{R}^n$, so we may expand

$$\boldsymbol{x} = \sum_{i=0}^{n-1} \alpha_i \boldsymbol{p}_i. \tag{1.44}$$

Multiplying $\boldsymbol{Ax} = \boldsymbol{b}$ from the left by $\boldsymbol{p}_k^T$ and using Eq. (1.43) collapses the sum to a single term,

$$\alpha_k = \frac{\boldsymbol{p}_k^T \boldsymbol{b}}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k}, \tag{1.45}$$

so each direction can be treated independently. In exact arithmetic the method therefore terminates after at most n steps, which makes it a direct method in disguise; what makes it useful is that a good approximation is normally reached in far fewer.

The directions are generated on the fly. Starting from $\boldsymbol{x}_0 = \boldsymbol{0}$ and $\boldsymbol{p}_0 = \boldsymbol{r}_0 = \boldsymbol{b}$, the new direction is the residual with its component along the previous direction projected out,

$$\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} - \frac{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{r}_{k+1}}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \boldsymbol{p}_k, \tag{1.46}$$

and the residual itself is updated recursively. From $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k$ we get

$$\boldsymbol{r}_{k+1} = \boldsymbol{b} - \boldsymbol{A}(\boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k) = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k, \tag{1.47}$$

which saves one matrix-vector product per iteration. Remarkably, the orthogonality relations that hold between the residuals allow Eq. (1.46) to be reduced to the two-term recurrence

$$\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k, \qquad \beta_k = \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k}, \tag{1.48}$$

so that only the current residual and the current direction need to be stored. This is what the implementation below uses.

```python
class ConjugateGradient(IterativeSolver):
    """Conjugate gradient for symmetric positive definite A.

    Only the matrix-vector product A p is required, so a `matvec` callable
    may be supplied instead of a dense matrix.
    """

    def solve(self, b, x0=None):
        b = np.asarray(b, dtype=float)
        x = self._initial_guess(x0)
        r = b - self.matvec(x)            # residual
        p = r.copy()                      # first search direction
        rs = r @ r

        for k in range(self.max_iter):
            Ap = self.matvec(p)
            alpha = rs / (p @ Ap)
            x += alpha * p                # x_{k+1} = x_k + alpha_k p_k
            r -= alpha * Ap               # r_{k+1} = r_k - alpha_k A p_k
            rs_new = r @ r
            if np.sqrt(rs_new) < self.tol:
                break
            p = r + (rs_new / rs) * p     # beta_k = |r_{k+1}|^2 / |r_k|^2
            rs = rs_new
        return x
```

Table 1.2 compares the four methods on the discretised second derivative of Eq. (1.23) with $n = 50$ interior points, for a residual tolerance of 10^{-8}. The ordering is the one to remember: Gauss-Seidel is roughly twice as fast as Jacobi, a well chosen relaxation parameter gains an order of magnitude, and the conjugate gradient method gains another.

Method	**Iterations**	**Error**
Jacobi	11049	1.0×10^{-9}
Gauss-Seidel	5526	1.0×10^{-9}
Successive over-relaxation ($\omega = 1.9$)	220	2.8×10^{-10}
Conjugate gradient	25	1.7×10^{-16}

Table 1.2 Iterations needed to solve $-u'' = 2$ on $(0,1)$ with $u(0) = u(1) = 0$, discretised on $n = 50$ interior points, to a residual tolerance of 10^{-8}. The error is measured against the analytic solution $u(x) = x(1-x)$.

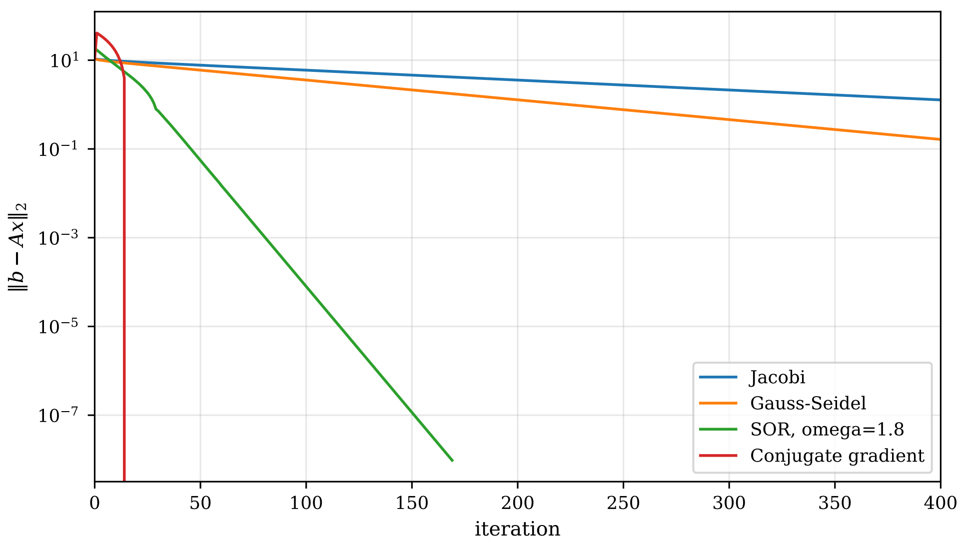

Fig. 1.1 Residual norm $\|\boldsymbol{b} - \boldsymbol{A}\boldsymbol{x}_k\|_2$ against the iteration index for the four solvers of Sections 1.17 and 1.18, applied to the discretised second derivative of Eq. (1.23) with $n = 30$ interior points, right-hand side $b_i = 2$ and starting vector $\boldsymbol{x}_0 = \boldsymbol{0}$. The ordinate is logarithmic, so the straight lines of Jacobi, Gauss-Seidel and successive over-relaxation display the geometric decay $\|\boldsymbol{r}_k\| \sim \rho^k$ characteristic of a stationary iteration, with ρ the spectral radius of its iteration matrix. Over-relaxation with $\omega = 1.8$ loses roughly one decade every fifteen sweeps and reaches the tolerance 10^{-8} after 170 iterations, whereas Jacobi needs 4030 and Gauss-Seidel 2016; the conjugate gradient curve falls vertically out of the frame after 16 steps, having reached the rounding level.

The same comparison is displayed as a function of the iteration index in Figure 1.1, for the slightly smaller problem $n = 30$. Three features of that figure repay attention. The first is that the Jacobi and Gauss-Seidel curves are straight and very nearly flat: after four hundred sweeps the residual has fallen by only one decade for Jacobi and by two for Gauss-Seidel, and the two lines are exactly parallel up to a factor of two in the abscissa, which is the graphical statement of the remark above that Gauss-Seidel halves the iteration count. Both are geometric with a rate ρ that approaches unity as the mesh is refined – for the discretised Laplacian $\rho_{\text{Jacobi}} = \cos(\pi h) \approx 1 - \pi^2 h^2/2$ – so that the number of iterations required grows as n^2 and the methods become useless long before the matrix does. The second is the qualitative change produced by over-relaxation. The choice $\omega = 1.8$ does not improve the constant in front of a slow decay, it changes the decay rate itself, and the resulting line descends nine decades in the space in which Gauss-Seidel manages two. The third, and the one that matters for the rest of the book, is the shape of the conjugate gradient curve. It is not a straight line at all. The residual first rises above its starting value, wanders for a dozen iterations while the Krylov space is still too small to contain a good approximation, and then collapses through twelve decades in two or three steps. This is the signature of a method that is building a basis rather than contracting a fixed map: nothing useful appears until the Krylov space of Eq. (1.44) has captured the dominant part of the spectrum, and once it has, the answer is essentially exact. The behaviour is the same one we shall meet in Figure 1.2 for the Lanczos algorithm, and for the same reason: both are Krylov methods, and both inherit their convergence from how well a polynomial of low degree can be made small on the spectrum of $\boldsymbol{A}$.

Many-body connection. The conjugate gradient method appears in two distinct roles in this book. As a linear solver it is used wherever a large sparse positive definite system arises. As a minimiser it is the workhorse of variational calculations: in variational Monte Carlo the parameters of a trial wave function are optimised by conjugate gradient steps on the energy or the variance, and the Hessian of the variance plays the role of $\boldsymbol{A}$. The non-linear generalisations of the method, discussed in the chapter on gradient methods, all rest on the linear theory developed here.

1.19 Cubic spline interpolation

Tridiagonal systems appear in a second, quite different context, and the solver of Section 1.16 can be reused verbatim. Suppose we are given $n+1$ knots $x_0 < x_1 < \cdots < x_n$ and the corresponding function values $y_i = f(x_i)$, and we want a smooth interpolant. A spline function s of degree k is a piecewise polynomial: on every subinterval $[x_i, x_{i+1})$ it is a polynomial of degree at most k, and it has $k-1$ continuous derivatives on the whole interval. For $k=1$ the spline is the broken line through the data points, with no continuous derivative at all. The useful choice is $k=3$, the cubic spline, which is twice continuously differentiable and therefore looks smooth to the eye.

On each of the n subintervals we have a cubic

$$s_i(x) = a_{i0} + a_{i1}x + a_{i2}x^2 + a_{i3}x^3, \tag{1.49}$$

giving $4n$ unknown coefficients. Interpolation at both ends of every subinterval provides $2n$ conditions, continuity of s' at the interior knots $n-1$ more, and continuity of s'' another $n-1$, for a total of $4n-2$. Two degrees of freedom remain, and they are fixed by a condition at the two ends; setting the second derivative to zero there gives the *natural* cubic spline.

The elegant way to construct the spline is to take the second derivatives $f_i = s''(x_i)$ as the unknowns. Since s'' is piecewise linear,

$$s_i''(x) = \frac{f_i}{x_{i+1}-x_i}(x_{i+1}-x) + \frac{f_{i+1}}{x_{i+1}-x_i}(x-x_i), \tag{1.50}$$

and integrating twice, fixing the integration constants by the interpolation conditions, gives $s_i(x)$ in terms of f_i and f_{i+1}. Demanding that s' be continuous at the interior knots then produces $n-1$ equations for the $n-1$ unknown interior second derivatives, and those equations are tridiagonal and diagonally dominant. A cubic spline through n points therefore costs $\mathscr{O}(n)$ operations, which is why splines rather than global polynomials are used to interpolate tabulated potentials, form factors and matrix elements.

1.20 Applications: integral equations and matrix inversion

Two further examples show how naturally physical problems turn into the matrix problems of this chapter.

A linear integral equation.

The scattering problem in quantum mechanics leads to the Lippmann-Schwinger equation, an integral equation of the form

$$R(k,k') = V(k,k') + \frac{2}{\pi}\mathscr{P}\int_0^\infty dq\, q^2 \frac{V(k,q)R(q,k')}{(k_0^2-q^2)/m}, \tag{1.51}$$

where $\mathscr{P}$ denotes the Cauchy principal value. If we approximate the integral by a quadrature rule with N points q_j and weights w_j, the unknown function R is replaced by the $N+1$ numbers $R(q_i,k_0)$, and Eq. (1.51) becomes

$$\sum_j A_{ij}R_j = V_i, \qquad A_{ij} = \delta_{ij} - \frac{2}{\pi}u_j V(k_i,q_j), \tag{1.52}$$

with the weights u_j containing the quadrature weights and the propagator. The principal value is handled by the standard subtraction trick, which removes the singularity at $q=k_0$ before the quadrature is applied. What began as an integral equation is now a linear system of the type treated in Section 1.15: the physics is entirely in the construction of the matrix, and the solution is one call to an LU solver.

An ill-conditioned inverse.

Not every matrix inversion is benign. Fitting a polynomial of degree $n-1$ through n points $x_0,\dots,x_{n-1}$ requires inverting the Vandermonde matrix

$$\boldsymbol{V} = \begin{bmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^{n-1} \\ 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n-1} & x_{n-1}^2 & \cdots & x_{n-1}^{n-1} \end{bmatrix}. \tag{1.53}$$

Its inverse is known in closed form, but its condition number grows exponentially with n: for equidistant points on $[0,1]$ and $n=20$ it exceeds 10^{20}, so that not a single significant digit survives in double precision. This is the numerical statement of a fact familiar from interpolation theory, namely that high-degree polynomial interpolation on a uniform grid is a bad idea, and it is the practical reason for preferring the piecewise construction of Section 1.19.

1.21 The algebraic eigenvalue problem

We return to the eigenvalue problem introduced in Section 1.11, this time asking how it is actually solved. The eigenvalues of a matrix $\boldsymbol{A}$ of dimension n are defined by

$$\boldsymbol{A}\boldsymbol{x}^{(\nu)} = \lambda^{(\nu)}\boldsymbol{x}^{(\nu)}, \tag{1.54}$$

where, unless stated otherwise, eigenvector means right eigenvector; the left eigenvectors satisfy $\boldsymbol{x}_L^{(\nu)}\boldsymbol{A} = \lambda^{(\nu)}\boldsymbol{x}_L^{(\nu)}$. Rewriting Eq. (1.54) as $(\boldsymbol{A} - \lambda^{(\nu)}\boldsymbol{I})\boldsymbol{x}^{(\nu)} = \boldsymbol{0}$, a non-trivial solution exists only if the matrix is singular, that is only if

$$P(\lambda) = \det(\lambda\boldsymbol{I} - \boldsymbol{A}) = 0. \tag{1.55}$$

The characteristic polynomial $P(\lambda)$ has degree n, and its roots

$$P(\lambda) = \prod_{i=0}^{n-1}(\lambda_i - \lambda) \tag{1.56}$$

form the *spectrum* $\lambda(\boldsymbol{A})$. Two useful identities follow at once from the factorisation,

$$|\boldsymbol{A}| = \lambda_0\lambda_1\cdots\lambda_{n-1}, \qquad \mathrm{Tr}(\boldsymbol{A}) = \lambda_0 + \lambda_1 + \cdots + \lambda_{n-1}. \tag{1.57}$$

It is tempting to conclude that one should find eigenvalues by computing the characteristic polynomial and looking for its roots. This is almost always a bad idea: the roots of a polynomial are extremely sensitive to its coefficients, so the detour through $P(\lambda)$ throws away accuracy that the original matrix still contained. The one case where the polynomial route is respectable is a tridiagonal matrix, for which the leading principal minors obey the recursion

$$P_k(\lambda) = (d_k - \lambda)P_{k-1}(\lambda) - e_{k-1}^2 P_{k-2}(\lambda), \tag{1.58}$$

with $P_0(\lambda) = d_0 - \lambda$ and $P_{-1} = 1$. The sequence $\{P_k(\lambda)\}$ is a Sturm sequence, and counting its sign changes tells us how many eigenvalues lie below λ, which is a robust way of isolating selected eigenvalues.

The standard approach is entirely different. We apply a sequence of similarity transformations that bring $\boldsymbol{A}$ either to diagonal form directly, or first to tridiagonal form and then to diagonal form. The second route is the one we follow, because diagonalising a tridiagonal matrix is vastly cheaper than diagonalising a full one.

1.22 Similarity transformations

Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be real and symmetric. Then there exists a real orthogonal matrix $\boldsymbol{S}$ such that

$$\boldsymbol{S}^T\boldsymbol{A}\boldsymbol{S} = \mathrm{diag}(\lambda_0, \lambda_1, \ldots, \lambda_{n-1}) \equiv \boldsymbol{D}, \tag{1.59}$$

and the jth column of $\boldsymbol{S}$ is the eigenvector belonging to λ_j [1]. We say that $\boldsymbol{B}$ is a similarity transform of $\boldsymbol{A}$ if

$$\boldsymbol{B} = \boldsymbol{S}^T\boldsymbol{A}\boldsymbol{S}, \qquad \boldsymbol{S}^T\boldsymbol{S} = \boldsymbol{S}^{-1}\boldsymbol{S} = \boldsymbol{I}. \tag{1.60}$$

The importance of such a transformation is that it leaves the eigenvalues unchanged while altering the eigenvectors in a controlled way. The proof is one line. Multiply $\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ from the left by $\boldsymbol{S}^T$ and insert $\boldsymbol{S}\boldsymbol{S}^T = \boldsymbol{I}$ between $\boldsymbol{A}$ and $\boldsymbol{x}$,

$$(\boldsymbol{S}^T\boldsymbol{A}\boldsymbol{S})(\boldsymbol{S}^T\boldsymbol{x}) = \lambda\,(\boldsymbol{S}^T\boldsymbol{x}), \tag{1.61}$$

that is $\boldsymbol{B}(\boldsymbol{S}^T\boldsymbol{x}) = \lambda(\boldsymbol{S}^T\boldsymbol{x})$. So λ is an eigenvalue of $\boldsymbol{B}$ as well, with eigenvector $\boldsymbol{S}^T\boldsymbol{x}$.

The strategy is therefore either to apply transformations until

$$\boldsymbol{S}_N^T\cdots\boldsymbol{S}_1^T\boldsymbol{A}\boldsymbol{S}_1\cdots\boldsymbol{S}_N = \boldsymbol{D}, \tag{1.62}$$

which is Jacobi's method of successive plane rotations, or to reduce $\boldsymbol{A}$ to tridiagonal form in a finite number of steps and then attack the tridiagonal matrix. We follow the second route, which is what all modern library routines do.

Many-body connection. Similarity transformations are not merely a numerical device. The whole of coupled-cluster theory rests on the similarity-transformed Hamiltonian $\bar{H} = e^{-T}\widehat{H}e^{T}$, which has the same eigenvalues as $\widehat{H}$ but is not Hermitian, since e^{T} is not unitary. The in-medium similarity renormalisation group instead uses a continuous sequence of *unitary* transformations to drive the Hamiltonian towards diagonal form, which is Eq. (1.62) carried out with infinitesimal steps.

1.23 The power method

The simplest of all eigenvalue algorithms is worth understanding, because everything that follows can be seen as a refinement of it. Assume $\boldsymbol{A}$ can be diagonalised, with eigenvalues $\lambda_0,\dots,\lambda_{n-1}$ and eigenvectors $\boldsymbol{v}_0,\dots,\boldsymbol{v}_{n-1}$, and assume there is a dominant eigenvalue, $|\lambda_0| > |\lambda_j|$ for $j > 0$. An arbitrary starting vector expands as

$$\boldsymbol{b}_0 = c_0\boldsymbol{v}_0 + c_1\boldsymbol{v}_1 + \cdots + c_{n-1}\boldsymbol{v}_{n-1}, \tag{1.63}$$

and applying $\boldsymbol{A}$ repeatedly gives

$$\boldsymbol{A}^k\boldsymbol{b}_0 = c_0\lambda_0^k\left(\boldsymbol{v}_0 + \sum_{j>0}\frac{c_j}{c_0}\left(\frac{\lambda_j}{\lambda_0}\right)^k\boldsymbol{v}_j\right). \tag{1.64}$$

Every ratio $|\lambda_j/\lambda_0|$ is smaller than one, so the bracket tends to $\boldsymbol{v}_0$: repeated multiplication filters out everything except the dominant eigenvector. Normalising at every step, $\boldsymbol{b}_k = \boldsymbol{A}^k\boldsymbol{b}_0/\|\boldsymbol{A}^k\boldsymbol{b}_0\|$, and estimating the eigenvalue by the Rayleigh quotient

$$\mu_k = \frac{\boldsymbol{b}_k^\dagger\boldsymbol{A}\boldsymbol{b}_k}{\boldsymbol{b}_k^\dagger\boldsymbol{b}_k}, \tag{1.65}$$

we obtain a method that converges geometrically with ratio $|\lambda_1/\lambda_0|$, where λ_1 is the second largest eigenvalue in magnitude. When the two are close, convergence is correspondingly slow.

The power method has two obvious weaknesses: it gives only one eigenpair, and it discards all the information contained in the intermediate vectors $\boldsymbol{b}_0, \boldsymbol{A}\boldsymbol{b}_0, \dots, \boldsymbol{A}^{k-1}\boldsymbol{b}_0$. Removing the second weakness by working in the whole space spanned by those vectors, the *Krylov space*, leads directly to the Lanczos algorithm of Section 1.26. Applying the power method to $\boldsymbol{A}^{-1}$ instead gives inverse iteration, which converges to the eigenvalue of smallest magnitude and, with a shift, to any eigenvalue one can bracket.

1.24 Householder's method for tridiagonalisation

The first of the two steps is to construct an orthogonal $\boldsymbol{S}$ that brings $\boldsymbol{A}$ to tridiagonal form. It is built as a product of $n-2$ elementary transformations,

$$\boldsymbol{S} = \boldsymbol{S}_0\boldsymbol{S}_1\cdots\boldsymbol{S}_{n-3}, \tag{1.66}$$

each of which clears one column and one row. Only $n-2$ steps are needed, since the last two rows are already tridiagonal. After the first transformation

$$
\boldsymbol{S}_0^T \boldsymbol{A} \boldsymbol{S}_0 = \begin{bmatrix} a_{00} & e_0 & 0 & \cdots & 0 \\ e_0 & a'_{11} & a'_{12} & \cdots & a'_{1,n-1} \\ 0 & a'_{21} & a'_{22} & \cdots & a'_{2,n-1} \\ \vdots & & & & \vdots \\ 0 & a'_{n-1,1} & \cdots & \cdots & a'_{n-1,n-1} \end{bmatrix}, \tag{1.67}
$$

where the primed block is a matrix of dimension $n-1$ which the next transformation will treat in the same way. Note that the effective size of the problem shrinks at every step; in Jacobi's method, by contrast, each rotation acts on the full matrix and previously created zeros are destroyed. After all $n-2$ steps we are left with the diagonal elements $a_{00}, a'_{11}, a''_{22}, \ldots$ and the off-diagonal elements $e_0, e_1, \ldots, e_{n-2}$ of a tridiagonal matrix.

It remains to find the transformations. We take

$$
\boldsymbol{S}_0 = \begin{bmatrix} 1 & \boldsymbol{0}^T \\ \boldsymbol{0} & \boldsymbol{P} \end{bmatrix}, \qquad \boldsymbol{P} = \boldsymbol{I} - 2\boldsymbol{u}\boldsymbol{u}^T, \qquad \boldsymbol{u}^T \boldsymbol{u} = 1, \tag{1.68}
$$

where $\boldsymbol{u}$ is a column vector of length $n-1$. The matrix $\boldsymbol{P}$, whose elements are $P_{ij} = \delta_{ij} - 2u_i u_j$, is a *Householder reflector*: it is symmetric and satisfies $\boldsymbol{P}^2 = \boldsymbol{I}$, hence $\boldsymbol{P}^T = \boldsymbol{P}^{-1}$, so it is orthogonal. Geometrically it reflects a vector in the hyperplane perpendicular to $\boldsymbol{u}$.

Applying $\boldsymbol{S}_0$ gives

$$
\boldsymbol{S}_0^T \boldsymbol{A} \boldsymbol{S}_0 = \begin{bmatrix} a_{00} & (\boldsymbol{P}\boldsymbol{v})^T \\ \boldsymbol{P}\boldsymbol{v} & \boldsymbol{A}' \end{bmatrix}, \qquad \boldsymbol{v}^T = \{a_{10}, a_{20}, \ldots, a_{n-1,0}\}, \tag{1.69}
$$

so we must choose $\boldsymbol{u}$ such that $\boldsymbol{P}\boldsymbol{v}$ has only one non-zero component,

$$
\boldsymbol{P}\boldsymbol{v} = \boldsymbol{v} - 2\boldsymbol{u}(\boldsymbol{u}^T \boldsymbol{v}) = \kappa \boldsymbol{e}, \qquad \boldsymbol{e}^T = \{1, 0, \ldots, 0\}. \tag{1.70}
$$

The constant κ follows from taking the scalar product of Eq. (1.70) with itself and using $\boldsymbol{P}^2 = \boldsymbol{I}$,

$$
(\boldsymbol{P}\boldsymbol{v})^T \boldsymbol{P}\boldsymbol{v} = \kappa^2 = \boldsymbol{v}^T \boldsymbol{v} = \sum_{i=1}^{n-1} a_{i0}^2, \tag{1.71}
$$

so $\kappa = \pm\|\boldsymbol{v}\|_2$. Rewriting Eq. (1.70) as $\boldsymbol{v} - \kappa\boldsymbol{e} = 2\boldsymbol{u}(\boldsymbol{u}^T \boldsymbol{v})$ shows that $\boldsymbol{u}$ is proportional to $\boldsymbol{v} - \kappa\boldsymbol{e}$, and normalising gives

$$
\boldsymbol{u} = \frac{\boldsymbol{v} - \kappa\boldsymbol{e}}{\|\boldsymbol{v} - \kappa\boldsymbol{e}\|_2}. \tag{1.72}
$$

The sign of κ is free, and here numerical care is essential: if $v_0 = a_{10}$ and κ have the same sign, the leading component of $\boldsymbol{v} - \kappa\boldsymbol{e}$ is a difference of nearly equal numbers and precision is lost. We therefore always choose

$$
\kappa = -\mathrm{sign}(v_0)\, \|\boldsymbol{v}\|_2, \tag{1.73}
$$

which makes the subtraction an addition.

Finally we need the transformed block. Writing $\boldsymbol{p} = \boldsymbol{A}'\boldsymbol{u}$ and $K = \boldsymbol{u}^T \boldsymbol{p}$, and introducing $\boldsymbol{z} = \boldsymbol{p} - K\boldsymbol{u}$, a short calculation gives the symmetric rank-two update

$$
\boldsymbol{P}\boldsymbol{A}'\boldsymbol{P} = \boldsymbol{A}' - 2\boldsymbol{z}\boldsymbol{u}^T - 2\boldsymbol{u}\boldsymbol{z}^T, \tag{1.74}
$$

which costs $\mathcal{O}(n^2)$ operations rather than the $\mathcal{O}(n^3)$ of forming the products explicitly. Summed over the $n-2$ steps, the whole reduction costs $\frac{4}{3}n^3$ operations, or twice that if the transformation matrix is accumulated.

```python
def tridiagonalize(self):
    """Return (d, e, S) with S^T A S tridiagonal."""
    n = self.n
    M = self.A.copy()
    S = np.eye(n)

    for k in range(n - 2):
        v = M[k+1:, k]                         # the column to be reduced
        norm_v = np.linalg.norm(v)
        if norm_v < self.tol:
            continue
        # the sign is chosen to avoid cancellation in v[0] - kappa
        kappa = -np.copysign(norm_v, v[0])
        w = v.copy()
        w[0] -= kappa
        u = w / np.linalg.norm(w)              # unit vector, u^T u = 1

        # similarity transformation of the trailing block,
        #   P A P = A - 2 z u^T - 2 u z^T,  z = A u - (u^T A u) u
        block = M[k+1:, k+1:]
        p = block @ u
        z = p - (u @ p) * u
        M[k+1:, k+1:] = block - 2.0*np.outer(z, u) - 2.0*np.outer(u, z)

        M[k+1:, k] = 0.0                       # the reduced column and row
        M[k+1, k] = kappa
        M[k, k+1:] = 0.0
        M[k, k+1] = kappa

        S[:, k+1:] -= 2.0 * np.outer(S[:, k+1:] @ u, u)   # S <- S P

    d = np.diag(M).copy()
    e = np.diag(M, -1).copy()
    return d, e, S
```

For a random symmetric 8×8 matrix the routine returns an $\boldsymbol{S}$ with $\|\boldsymbol{S}^T\boldsymbol{S}-\boldsymbol{I}\| = 1.1\times 10^{-15}$ and reproduces the original matrix, $\|\boldsymbol{S}\boldsymbol{T}\boldsymbol{S}^T-\boldsymbol{A}\| = 3.1\times 10^{-15}$. Orthogonality is preserved to machine precision, which is exactly why reflectors are preferred to the more obvious Gaussian-elimination-like schemes.

1.25 Diagonalising a tridiagonal matrix

The second step diagonalises the tridiagonal matrix

$$\boldsymbol{T} = \begin{bmatrix} d_0 & e_0 & & \\ e_0 & d_1 & e_1 & \\ & e_1 & d_2 & \ddots \\ & & \ddots & \ddots \end{bmatrix}. \tag{1.75}$$

The basic observation is that if any off-diagonal element e_i vanishes, the matrix splits into two independent blocks; in particular if $e_0 = 0$ then d_0 is already an eigenvalue. The algorithm therefore works by making the off-diagonal elements vanish one after another.

Francis' original scheme, and the QL and QR algorithms that grew out of it, apply a sequence of plane rotations

$$\boldsymbol{S}_i = \begin{bmatrix} \cos\theta & \cdots & \sin\theta \\ \vdots & & \vdots \\ -\sin\theta & \cdots & \cos\theta \end{bmatrix}, \tag{1.76}$$

choosing θ at each step so as to annihilate one off-diagonal element. Two refinements turn this idea into the algorithm actually used. The first is *shifting*: instead of working with $\boldsymbol{T}$ one works with $\boldsymbol{T} - \sigma\boldsymbol{I}$, where σ is an estimate of the eigenvalue being sought. The rate of convergence of the element e_{m-1} to zero is governed by the ratio of the distances of σ to the two nearest eigenvalues, so a good shift produces cubic convergence and typically two or three iterations per eigenvalue. The Wilkinson shift, obtained from the trailing 2×2 submatrix, is the standard choice. The second refinement is to apply the shift *implicitly*, by chasing an unwanted element down the subdiagonal with successive rotations, which avoids ever forming $\boldsymbol{T} - \sigma\boldsymbol{I}$ explicitly and so avoids the associated cancellation.

The resulting routine is traditionally called `tqli`. It takes the diagonal d and the off-diagonal e produced by the Householder step, together with the accumulated transformation matrix, and returns the eigenvalues in d and the eigenvectors as the columns of that matrix.

```
@staticmethod
def tqli(d, e, z=None, max_sweeps=50):
    """QL algorithm with implicit shifts for a symmetric tridiagonal matrix.

    Pass the S of the Householder step as z to obtain the eigenvectors of
    the original matrix A.
    """
    d = np.asarray(d, dtype=float).copy()
    n = len(d)
    e = np.concatenate([np.asarray(e, dtype=float), [0.0]])
    z = np.eye(n) if z is None else np.asarray(z, dtype=float).copy()
    eps = np.finfo(float).eps

    for l in range(n):
        for _ in range(max_sweeps):
            # look for a negligible off-diagonal element to split off
            m = n - 1
            for i in range(l, n - 1):
                if abs(e[i]) <= eps * (abs(d[i]) + abs(d[i+1])):
                    m = i
                    break
            if m == l:                        # d[l] has converged
                break

            # Wilkinson shift, computed without cancellation
            g = (d[l+1] - d[l]) / (2.0 * e[l])
            r = np.hypot(g, 1.0)
            g = d[m] - d[l] + e[l] / (g + np.copysign(r, g))

            s = c = 1.0
            p = 0.0
            for i in range(m - 1, l - 1, -1):
                f = s * e[i]
                b = c * e[i]
                r = np.hypot(f, g)
                e[i+1] = r
                s, c = f / r, g / r
                g = d[i+1] - p
                r = (d[i] - g) * s + 2.0 * c * b
                p = s * r
                d[i+1] = g + p
```

```
                g = c * r - b
                col = z[:, i+1].copy()       # accumulate the plane rotation
                z[:, i+1] = s * z[:, i] + c * col
                z[:, i] = c * z[:, i] - s * col
            d[l] -= p
            e[l] = g
            e[m] = 0.0
    return d, z
```

Together, the two steps constitute a complete eigenvalue solver. On a random symmetric 8×8 matrix it reproduces the eigenvalues returned by `numpy.linalg.eigvalsh` to 2.2×10^{-15}, with $\|\boldsymbol{A}\boldsymbol{V}-\boldsymbol{V}\boldsymbol{D}\| = 4.9\times 10^{-15}$ and $\|\boldsymbol{V}^T\boldsymbol{V}-\boldsymbol{I}\| = 2.7\times 10^{-15}$. The total cost is $\mathscr{O}(n^3)$, dominated by the Householder reduction.

Many-body connection. The Householder and QL combination is what lies behind every call to a library eigenvalue routine, and it is entirely adequate as long as the matrix fits in memory. For the Hartree-Fock and random-phase-approximation matrices of the following chapters, whose dimensions are set by the number of single-particle states, this is the method of choice. It becomes useless the moment the dimension is set by the number of Slater determinants, which is the situation of the next section.

1.26 Lanczos' algorithm

The methods of the previous two sections transform the matrix, and therefore need to have it. In large-scale many-body calculations we do not. A full configuration-interaction Hamiltonian for a realistic system may have a dimension of 10^9 or more; it is sparse, it is generated on the fly from the occupation numbers of the Slater determinants, and it is never stored. What we can always do is compute the product $\boldsymbol{A}\boldsymbol{q}$ for a given vector $\boldsymbol{q}$. And we rarely want the whole spectrum: the ground state and a handful of excited states are enough.

The Lanczos algorithm is built for exactly this situation. It applies to real symmetric matrices and has the following properties:

- it generates a sequence of real tridiagonal matrices $\boldsymbol{T}_k$ of dimension $k\times k$ with $k\leq n$, whose extremal eigenvalues are progressively better estimates of the extremal eigenvalues of $\boldsymbol{A}$;
- the matrix enters only through the product $\boldsymbol{A}\boldsymbol{q}$;
- the underlying transformation is the similarity transformation $\boldsymbol{T} = \boldsymbol{Q}^T\boldsymbol{A}\boldsymbol{Q}$, with the first Lanczos vector $\boldsymbol{Q}\boldsymbol{e}_0 = \boldsymbol{q}_0$ chosen at will.

The three-term recurrence.

Write $\boldsymbol{Q}$ in terms of its column vectors, $\boldsymbol{Q} = [\boldsymbol{q}_0\,\boldsymbol{q}_1\cdots\boldsymbol{q}_{n-1}]$, and let

$$\boldsymbol{T} = \begin{bmatrix} \alpha_0 & \beta_0 & & & \\ \beta_0 & \alpha_1 & \beta_1 & & \\ & \beta_1 & \alpha_2 & \ddots & \\ & & \ddots & \ddots & \beta_{n-2} \\ & & & \beta_{n-2} & \alpha_{n-1} \end{bmatrix}. \tag{1.77}$$

Using $\boldsymbol{Q}\boldsymbol{Q}^T = \boldsymbol{I}$ we may rewrite $\boldsymbol{T} = \boldsymbol{Q}^T\boldsymbol{A}\boldsymbol{Q}$ as $\boldsymbol{Q}\boldsymbol{T} = \boldsymbol{A}\boldsymbol{Q}$, and equating column k on both sides gives the three-term recurrence

$$A q_k = \beta_{k-1} q_{k-1} + \alpha_k q_k + \beta_k q_{k+1}, \tag{1.78}$$

with $\beta_{-1} q_{-1} = \mathbf{0}$. This single equation contains the whole algorithm. Since the Lanczos vectors are orthonormal, multiplying Eq. (1.78) from the left by q_k^T gives

$$\alpha_k = q_k^T A q_k, \tag{1.79}$$

and the remainder

$$r_k = (A - \alpha_k I) q_k - \beta_{k-1} q_{k-1} \tag{1.80}$$

determines the next vector. If $r_k \neq \mathbf{0}$ then

$$q_{k+1} = r_k / \beta_k, \qquad \beta_k = \pm \| r_k \|_2. \tag{1.81}$$

If r_k does vanish we have found an invariant subspace, and the eigenvalues of T_k are exact eigenvalues of A. The algorithm is therefore

$$\begin{array}{l} r_0 = q_0, \quad \beta_{-1} = 1, \quad q_{-1} = \mathbf{0}, \quad k = 0 \\ \textbf{while } \beta_{k-1} \neq 0 \\ \quad q_k = r_{k-1}/\beta_{k-1} \\ \quad \alpha_k = q_k^T A q_k \\ \quad r_k = (A - \alpha_k I) q_k - \beta_{k-1} q_{k-1} \\ \quad \beta_k = \| r_k \|_2, \quad k \leftarrow k+1 \\ \textbf{end while} \end{array} \tag{1.82}$$

Krylov spaces and Ritz values.

After m steps the vectors $q_0, \ldots, q_{m-1}$ span the Krylov space

$$\mathscr{K}_m(A, q_0) = \operatorname{span}\{q_0, A q_0, A^2 q_0, \ldots, A^{m-1} q_0\}, \tag{1.83}$$

which is precisely the sequence of vectors the power method of Section 1.23 generates and then throws away. Lanczos keeps them, orthonormalises them, and diagonalises A restricted to their span. The eigenvalues θ_i of the small matrix T_m are called *Ritz values* and the vectors $Q y_i$, with y_i the eigenvectors of T_m, are the *Ritz vectors*. They are the best approximations to eigenpairs of A obtainable within the Krylov space, in the variational sense: the lowest Ritz value is the minimum of the Rayleigh quotient over $\mathscr{K}_m$, and is therefore an upper bound on the exact ground-state energy. This is the same variational principle that underlies Hartree-Fock theory and the variational Monte Carlo method.

Because T_m is tridiagonal and small, it is diagonalised with the QL algorithm of Section 1.25. The two algorithms are thus used together: Lanczos reduces the enormous problem to a tiny tridiagonal one, and `tqli` finishes the job.

```
def run(self, m, q0=None, tol=1.0e-12):
    """Perform m Lanczos steps, returning the tridiagonal matrix elements."""
    n = self.n
    q = np.random.default_rng(2026).normal(size=n) if q0 is None \
        else np.asarray(q0, dtype=float).copy()
    q /= np.linalg.norm(q)

    Q = np.zeros((n, m))
    alpha = np.zeros(m)
    beta = np.zeros(max(m - 1, 0))
    q_prev = np.zeros(n)
    b_prev = 0.0
```

```
    for k in range(m):
        Q[:, k] = q
        w = self.matvec(q)                    # the only use made of A
        alpha[k] = q @ w                      # alpha_k = q_k^T A q_k
        w = w - alpha[k]*q - b_prev*q_prev    # r_k

        if self.reorthogonalize:
            for _ in range(2):                # modified Gram-Schmidt
                w -= Q[:, :k+1] @ (Q[:, :k+1].T @ w)

        if k == m - 1:
            break
        b = np.linalg.norm(w)
        if b < tol:                           # invariant subspace found
            alpha, beta, Q = alpha[:k+1], beta[:k], Q[:, :k+1]
            break
        beta[k] = b
        q_prev, b_prev = q, b
        q = w / b

    self.alpha, self.beta, self.Q = alpha, beta, Q
    return alpha, beta, Q
```

Loss of orthogonality.

In exact arithmetic the three-term recurrence produces orthogonal vectors automatically. In floating-point arithmetic it does not. As soon as one Ritz value converges, the corresponding direction starts to reappear in the remainder $\boldsymbol{r}_k$, orthogonality is lost, and the algorithm produces spurious extra copies of eigenvalues it has already found. These are the notorious *ghost* eigenvalues. The simplest cure, used in the code above, is full reorthogonalisation: at every step the new vector is explicitly orthogonalised against all previous ones. This requires storing all the Lanczos vectors and costs $\mathscr{O}(m^2 n)$ operations, which is acceptable when m is a few hundred. For very long runs one uses selective or periodic reorthogonalisation instead.

A many-body example.

Table 1.3 shows the algorithm at work on the pairing model that runs through this book: twelve doubly degenerate single-particle levels, six pairs, single-particle spacing $\delta = 1$ and pairing strength $g = 0.5$. In the seniority-zero space a basis state is a choice of which levels carry a pair, so the dimension is $\binom{12}{6} = 924$. The ground-state energy is converged to ten digits after thirty matrix-vector products and to machine precision after forty, in a space of dimension 924.

Figure 1.2 plots the same errors on a logarithmic scale, and the picture is worth more than the table. What one sees is a straight line: between $m = 10$ and $m = 30$ the error falls from 3×10^{-1} to 2.4×10^{-10}, which is nine decades in twenty matrix-vector products, or a gain of a little under a factor of three per step. The convergence of the Lanczos algorithm is thus geometric in m, and the rate is set not by the dimension of the matrix but by the relative gap between the lowest eigenvalue and the rest of the spectrum: the Ritz value at step m is the best that any polynomial of degree $m-1$ in $\boldsymbol{A}$ applied to the starting vector can do, and a well separated extremal eigenvalue is easy for such a polynomial to isolate. This is why a space of dimension 924 is exhausted, for the purpose of finding its lowest state, after forty products, and why the same algorithm remains usable when the dimension is 10^{10} and the

m	**Lowest Ritz value**	**Error**
5	32.865169933329	8.0×10^{0}
10	25.140140936891	3.0×10^{-1}
20	24.839267115133	9.4×10^{-5}
30	24.839172748689	2.4×10^{-10}
40	24.839172748451	5.7×10^{-14}
exact	24.839172748451	

Table 1.3 Convergence of the lowest Ritz value for the pairing model with twelve levels and six pairs, $\delta = 1$, $g = 0.5$, in a space of dimension 924. The exact value is obtained by full diagonalisation.

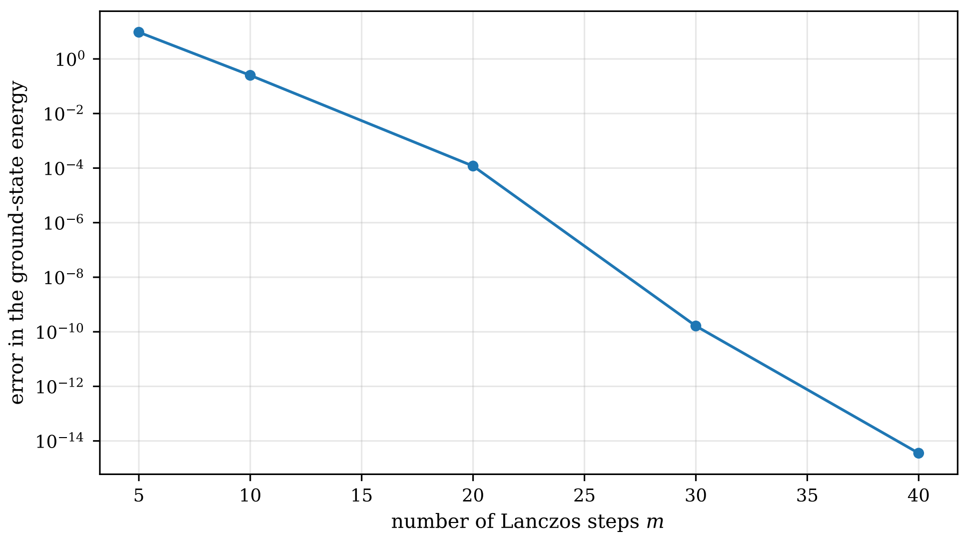

Fig. 1.2 Error of the lowest Ritz value with respect to the exact ground-state energy, plotted logarithmically against the number of Lanczos steps m, for the pairing model with twelve doubly degenerate levels, six pairs, $\delta = 1$ and $g = 0.5$, in a seniority-zero space of dimension 924. The five points correspond to the rows of Table 1.3. The error falls from 8 at $m = 5$ to 5.7×10^{-14} at $m = 40$, an almost perfectly geometric decay of about half a decade per matrix-vector product, and the slight flattening beyond $m = 30$ marks the point at which rounding, and not the size of the Krylov space, limits the accuracy.

matrix can never be written down. Two further observations follow from the figure. The curve flattens between $m = 30$ and $m = 40$ because the error has reached 10^{-14}, some fifteen orders of magnitude below the energy itself: we are then measuring the rounding of the arithmetic and not the quality of the Krylov space, and nothing is gained by continuing. And the first point, at $m = 5$, lies above the exact energy by 8 units – the lowest Ritz value is a variational upper bound at every step, since it is the minimum of the Rayleigh quotient over a subspace, so the curve approaches its limit monotonically from one side. The same monotonic, upper-bounding behaviour is what makes the Lanczos energy a legitimate variational estimate of the ground-state energy in the full configuration-interaction calculations of Chapter 2.

Run without reorthogonalisation for 120 steps on the same matrix, the Lanczos vectors lose orthogonality completely, $\|\boldsymbol{Q}^T\boldsymbol{Q} - \boldsymbol{I}\| = 4.9$, and three copies of the non-degenerate ground state appear among the Ritz values. With reorthogonalisation the deviation is 8.3×10^{-15} and the ground state appears exactly once.

Many-body connection. The Lanczos algorithm is the engine of large-scale shell-model and full configuration-interaction codes. Its combination of properties – only matrix-vector products are needed, the extremal eigenvalues converge first, and the lowest Ritz value is a variational upper bound – matches the structure of the many-body problem exactly: we want the ground state, the Hamiltonian is sparse, and the dimension is astronomical. The same Krylov idea reappears in quantum computing, where the quantum phase estimation and variational quantum eigensolver algorithms are, in different ways, attempts to extract the same information with quantum rather than classical resources.

1.27 Application: Schrödinger’s equation through diagonalisation

We close the chapter by putting the machinery to work on a genuine quantum mechanical problem, and one that will return in a many-body guise later: solving Schrödinger’s equation not as a differential equation, but as a matrix eigenvalue problem.

The radial equation for a particle in a central potential is

$$-\frac{\hbar^2}{2m}\left(\frac{1}{r^2}\frac{d}{dr}r^2\frac{d}{dr}-\frac{l(l+1)}{r^2}\right)R(r)+V(r)R(r)=ER(r). \tag{1.84}$$

Substituting $R(r)=u(r)/r$ removes the first-derivative term,

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2}+\left(V(r)+\frac{l(l+1)}{r^2}\frac{\hbar^2}{2m}\right)u(r)=Eu(r), \tag{1.85}$$

and introducing the dimensionless variable $\rho=r/\alpha$ with α a length makes the equation dimensionless. We specialise to one dimension, drop the centrifugal term, and take the harmonic oscillator potential $V(x)=\frac{1}{2}kx^2$. In units where $k=\hbar=m=\alpha=1$ the equation becomes

$$-\frac{d^2u}{dx^2}+x^2u(x)=2E\,u(x). \tag{1.86}$$

The factor two on the right is worth remembering: the program will return the values $2E_n=2n+1$, that is $1,3,5,7,\dots$, rather than $E_n=n+\frac{1}{2}$.

We now repeat the discretisation of Section 1.14. Choose $R_{\min}$ and $R_{\max}$, a number of mesh points N_{step}, and the step $h=(R_{\max}-R_{\min})/N_{\text{step}}$. With $x_i=R_{\min}+ih$ and the three-point formula (1.21), Eq. (1.86) becomes

$$-\frac{u_{i+1}-2u_i+u_{i-1}}{h^2}+V_iu_i=2E\,u_i, \tag{1.87}$$

that is the tridiagonal eigenvalue problem

$$\begin{bmatrix} \frac{2}{h^2}+V_1 & -\frac{1}{h^2} & & \\ -\frac{1}{h^2} & \frac{2}{h^2}+V_2 & -\frac{1}{h^2} & \\ & \ddots & \ddots & \ddots \\ & & -\frac{1}{h^2} & \frac{2}{h^2}+V_{N_{\text{step}}-1} \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N_{\text{step}}-1}\end{bmatrix}=2E\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N_{\text{step}}-1}\end{bmatrix}. \tag{1.88}$$

The boundary conditions $u(R_{\min})=u(R_{\max})=0$ are imposed simply by leaving the two endpoints out of the matrix, whose dimension is therefore $N_{\text{step}}-1$. The eigenvectors are the wave functions on the grid, normalised with the trapezoidal rule,

$$\int_{R_{\min}}^{R_{\max}} |u(x)|^2 dx \longrightarrow h\sum_i u_i^2 = 1. \tag{1.89}$$

Since the matrix is already tridiagonal, the Householder step is unnecessary and we go straight to `tqli`. Alternatively, and this is the point of the exercise, we may pretend that the matrix is far too large to store and use the Lanczos algorithm with a matrix-free operator.

```
class SchrodingerDiagonalization:
    """Discretise -u'' + V(x) u = 2E u and diagonalise the result."""

    def __init__(self, potential, rmin=-10.0, rmax=10.0, nsteps=400):
        self.potential = potential
        self.h = (rmax - rmin) / nsteps
        # interior mesh points only; u vanishes at the two endpoints
        self.x = rmin + self.h * np.arange(1, nsteps)
        self.n = len(self.x)

    @property
    def diagonal(self):
        return 2.0 / self.h**2 + self.potential(self.x)

    @property
    def offdiagonal(self):
        return np.full(self.n - 1, -1.0 / self.h**2)

    def matvec(self, v):
        """Apply the Hamiltonian without ever forming the matrix."""
        w = self.diagonal * v
        w[:-1] -= v[1:] / self.h**2
        w[1:] -= v[:-1] / self.h**2
        return w

    def solve(self, k=5):
        """The k lowest eigenvalues and normalised eigenfunctions."""
        values, vectors = SymmetricEigenSolver.tqli(self.diagonal,
                                                    self.offdiagonal)
        order = np.argsort(values)
        return (values[order][:k],
                self._normalise(vectors[:, order][:, :k]))

    def solve_lanczos(self, k=5, m=None):
        """The same eigenvalues from the Lanczos algorithm."""
        lan = Lanczos(matvec=self.matvec, n=self.n)
        values, vectors = lan.ritz(10*k if m is None else m, k=k)
        return values, self._normalise(vectors)
```

Table 1.4 shows the result on $x \in [-10, 10]$. The eigenvalues converge to the exact values $1, 3, 5, \ldots$ as h^2: doubling N_{step} divides the error by four, exactly as the truncation error of the three-point formula demands. Running the same problem through the Lanczos algorithm with $m = 200$ steps reproduces the three lowest eigenvalues to 6×10^{-13}, and the computed wave functions agree with the analytic $u_0(x) = \pi^{-1/4} e^{-x^2/2}$ to 7×10^{-5}.

The eigenvectors are as instructive as the eigenvalues, and they are shown in Figure 1.3. Nothing in the construction of the matrix (1.88) knows about Hermite polynomials; we imposed only a second-difference stencil, a parabolic potential and the requirement that u vanish at the two ends of the interval. Yet the eigenvectors come out with exactly the structure the analytic solution demands. The ground state is nodeless, even in x, and peaks at $u_0(0) = 0.7511$, which is $\pi^{-1/4}$ to four digits; the first excited state is odd with a single node at the origin, and the second is even with two nodes. The ordering of the eigenvalues and the number of nodes therefore track each other, which is the discrete version of the Sturm oscillation theorem and a useful check that a diagonalisation routine has not scrambled its output. Superimposing

N_{step}	$2E_0$	$2E_1$	$2E_2$	**Max error**
100	0.9974937026	2.9874430342	4.9672772610	1.0×10^{-1}
200	0.9993746086	2.9968714733	4.9918612670	2.6×10^{-2}
400	0.9998437256	2.9992185301	4.9979678946	6.4×10^{-3}
800	0.9999609360	2.9998046738	4.9994921341	1.6×10^{-3}
exact	1	3	5	

Table 1.4 The one-dimensional harmonic oscillator obtained by diagonalising Eq. (1.88) on $x \in [-10, 10]$. The error decreases as h^2.

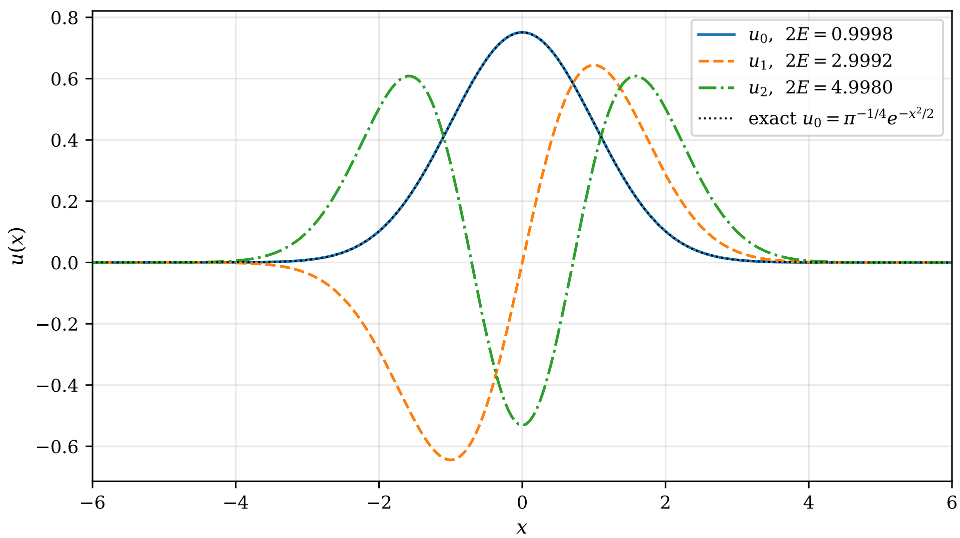

Fig. 1.3 The three lowest eigenfunctions $u_n(x)$ of the harmonic oscillator obtained by diagonalising the tridiagonal matrix of Eq. (1.88) on $x \in [-10, 10]$ with $N_{\text{step}} = 400$, normalised by the trapezoidal rule (1.89) and plotted over the interval where they are appreciably different from zero. The legend gives the computed eigenvalues $2E_n$, which should equal 1, 3 and 5. The dotted black curve is the analytic ground state $u_0 = \pi^{-1/4}e^{-x^2/2}$ and is indistinguishable from the computed u_0 on the scale of the plot; the maximum pointwise deviation is 7×10^{-5}.

the analytic $\pi^{-1/4}e^{-x^2/2}$ on the computed u_0 produces no visible second curve at all, and the largest discrepancy anywhere on the mesh is 7×10^{-5}, comparable with the 6.4×10^{-3} error in $2E_0$ of Table 1.4 once one remembers that the eigenvalue error is quadratic in the wavefunction error for a variational functional. The figure also settles a practical question that recurs whenever a continuum problem is put on a grid: how far out the box must extend. All three eigenfunctions have decayed into the line thickness by $|x| \approx 4$, well inside the boundary at $R_{\max} = 10$, so the boundary condition $u(R_{\max}) = 0$ costs us nothing here. Had we chosen $R_{\max} = 3$ the tails would have been amputated and the computed energies would have risen well above the exact ones, an error that no refinement of h could repair – the box size and the step length are two independent sources of truncation, and both must be checked.

Many-body connection. This calculation is the one-particle ancestor of everything that follows. Expanding a state in a finite basis and diagonalising the resulting matrix is exactly what full configuration interaction does, the only difference being that the basis consists of Slater determinants rather than grid points, and that the dimension grows exponentially with the number of particles rather than linearly with the number of mesh points. The eigenvectors computed here are the natural single-particle basis $\{|\phi_i\rangle\}$ of Section 1.8 from which those Slater determinants are built. Chapter 2 takes the first step in that direction.

1.28 The singular value decomposition

Every algorithm so far has assumed a square matrix, and most of them a symmetric one. Many of the objects we meet in many-body physics are neither. The coefficients of a state written across a partition of the system into two parts form a rectangular array; a set of sampled wave functions collected as columns forms a rectangular array; the two-body matrix elements arranged in a composite pair index form a matrix whose row and column dimensions have nothing to do with the dimension of the Hilbert space. For all of these we need a decomposition that does not require squareness, and preferably one that does not require diagonalisability either.

Recall that a square matrix can be diagonalised if and only if it is *normal*, that is if $\boldsymbol{X}\boldsymbol{X}^T = \boldsymbol{X}^T\boldsymbol{X}$ in the real case or $\boldsymbol{X}\boldsymbol{X}^\dagger = \boldsymbol{X}^\dagger\boldsymbol{X}$ in the complex one. Not every matrix satisfies this. The simplest counterexample is

$$\boldsymbol{X} = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, \tag{1.90}$$

whose determinant vanishes and for which $\boldsymbol{X}\boldsymbol{X}^T \neq \boldsymbol{X}^T\boldsymbol{X}$. It is *defective*: it cannot be diagonalised, and it has no inverse.

The singular value decomposition has no such restriction.

The theorem.

Any $m \times n$ matrix $\boldsymbol{X}$, real or complex, square or rectangular, can be written

$$\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T, \tag{1.91}$$

where $\boldsymbol{U}$ is an $m \times m$ orthogonal matrix, $\boldsymbol{U}\boldsymbol{U}^T = \boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}_m$, where $\boldsymbol{V}$ is an $n \times n$ orthogonal matrix, $\boldsymbol{V}\boldsymbol{V}^T = \boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}_n$, and where $\boldsymbol{\Sigma}$ is an $m \times n$ matrix which is zero except on the main diagonal, on which sit the *singular values*

$$\sigma_0 \geq \sigma_1 \geq \cdots \geq \sigma_{r-1} \geq 0, \qquad r = \min(m,n). \tag{1.92}$$

The columns of $\boldsymbol{U}$ are the *left* singular vectors and the columns of $\boldsymbol{V}$ the *right* singular vectors. The decomposition always exists. For the defective matrix of Eq. (1.90) it reads

$$\boldsymbol{X} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T, \tag{1.93}$$

with $\sigma_0 = 2$ and $\sigma_1 = 0$. A matrix that cannot be diagonalised at all is decomposed without difficulty, and the vanishing singular value states plainly what the determinant only hinted at.

Geometry.

Equation (1.91) says that every linear map is a rotation, followed by a scaling along the new axes, followed by another rotation. The singular values are the scale factors, and $\|\boldsymbol{X}\|_2 =$

σ_0, which is the statement of Section 1.13 that the induced 2-norm measures the largest stretching a matrix can produce.

Relation to the eigenvalue problem.

The SVD is not an independent construction; it is the symmetric eigenvalue problem of Section 1.21 in disguise. Using Eq. (1.91) and the orthogonality of $\boldsymbol{U}$,

$$\boldsymbol{X}^T\boldsymbol{X} = \boldsymbol{V}\boldsymbol{\Sigma}^T\boldsymbol{U}^T\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T = \boldsymbol{V}\boldsymbol{\Sigma}^T\boldsymbol{\Sigma}\boldsymbol{V}^T \equiv \boldsymbol{V}\tilde{\boldsymbol{\Sigma}}^2\boldsymbol{V}^T, \tag{1.94}$$

where $\tilde{\boldsymbol{\Sigma}}$ is the $n\times n$ diagonal matrix of singular values. Multiplying from the right by $\boldsymbol{V}$ gives

$$\left(\boldsymbol{X}^T\boldsymbol{X}\right)\boldsymbol{v}_i = \sigma_i^2\,\boldsymbol{v}_i, \tag{1.95}$$

so the right singular vectors are the eigenvectors of $\boldsymbol{X}^T\boldsymbol{X}$ and the squared singular values its eigenvalues. In exactly the same way

$$\boldsymbol{X}\boldsymbol{X}^T = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^T\boldsymbol{U}^T, \qquad \left(\boldsymbol{X}\boldsymbol{X}^T\right)\boldsymbol{u}_i = \sigma_i^2\,\boldsymbol{u}_i, \tag{1.96}$$

with the remaining $m-r$ eigenvalues equal to zero. Every singular value is therefore the non-negative square root of an eigenvalue of $\boldsymbol{X}^T\boldsymbol{X}$, and if $\boldsymbol{X}$ is itself symmetric the singular values are the absolute values of its eigenvalues. These two relations will do a great deal of work in the sections that follow: $\boldsymbol{X}\boldsymbol{X}^T$ and $\boldsymbol{X}^T\boldsymbol{X}$ will turn out to be the two reduced density matrices of a bipartite quantum state.

The economy-size decomposition.

If $m>n$ the last $m-n$ columns of $\boldsymbol{U}$ are multiplied by zeros in $\boldsymbol{\Sigma}$ and can never contribute. Removing them, and the corresponding rows of zeros in $\boldsymbol{\Sigma}$, gives the economy-size decomposition, in which $\boldsymbol{U}$ is $m\times n$ and $\boldsymbol{\Sigma}$ is $n\times n$. If $n>m$ one keeps instead the first m columns of $\boldsymbol{V}$. Nothing is lost and both storage and work are reduced; this is the default in most library routines.

How not to compute it.

Equation (1.95) suggests an algorithm: form $\boldsymbol{X}^T\boldsymbol{X}$, diagonalise it with the Householder and QL machinery of Sections 1.24 and 1.25, take square roots. This is mathematically correct and numerically poor. From Eq. (1.18) the condition number of the product is the square of that of $\boldsymbol{X}$,

$$\kappa_2(\boldsymbol{X}^T\boldsymbol{X}) = \kappa_2(\boldsymbol{X})^2, \tag{1.97}$$

so half of the available significant digits are destroyed before the eigenvalue solver is even called. Table 1.5 shows the damage for a 60×12 matrix constructed with singular values spread logarithmically between 1 and 10^{-7}: the direct SVD returns all twelve to machine precision, the detour through $\boldsymbol{X}^T\boldsymbol{X}$ loses seven digits on the smallest.

The stable algorithm never forms the product. It reduces $\boldsymbol{X}$ to bidiagonal form by Householder reflectors applied alternately from the left and the right – the same reflectors as in Section 1.24 – and then runs an implicitly shifted QR sweep on the bidiagonal matrix. This is the Golub-Kahan algorithm, and it is what `numpy.linalg.svd` calls through LAPACK. We use

Exact σ_i	From the SVD	From $X^T X$	Relative error
1.00000000×10^{0}	1.00000000×10^{0}	1.00000000×10^{0}	4.4×10^{-16}
$6.57933225 \times 10^{-4}$	$6.57933225 \times 10^{-4}$	$6.57933225 \times 10^{-4}$	2.0×10^{-11}
$1.87381742 \times 10^{-6}$	$1.87381742 \times 10^{-6}$	$1.87381889 \times 10^{-6}$	7.8×10^{-7}
$4.32876128 \times 10^{-7}$	$4.32876128 \times 10^{-7}$	$4.32893930 \times 10^{-7}$	4.1×10^{-5}
$1.00000000 \times 10^{-7}$	$1.00000000 \times 10^{-7}$	$9.99683051 \times 10^{-8}$	3.2×10^{-4}

Table 1.5 Singular values of a 60×12 matrix with $\kappa_2(\boldsymbol{X}) = 10^7$, computed directly and through the eigenvalues of $\boldsymbol{X}^T\boldsymbol{X}$, for which $\kappa_2 = 10^{14}$. Forming the product costs roughly half the significant digits.

the library routine in what follows: the interest here lies in what the decomposition is used for.

Many-body connection. Equations (1.95) and (1.96) are the reason the SVD keeps reappearing in many-body physics. Whenever a quantity is built as a product of a matrix with its own transpose – a density matrix, an overlap matrix, a two-body interaction written in a pair index – the SVD of the underlying rectangular object gives its spectrum directly, with better accuracy and often at lower cost than forming the product at all.

1.29 Rank, the pseudoinverse and ill-conditioned bases

The *rank* of a matrix is the number of linearly independent columns, and the SVD reads it off immediately: it is the number of non-zero singular values. In floating-point arithmetic one asks instead for the *numerical rank*, the number of singular values above a threshold, typically $\max(m,n)\,\varepsilon\,\sigma_0$ with ε the machine precision. A matrix is rank deficient when some columns are linear combinations of others, and near rank deficient – which is the practically dangerous case – when they nearly are.

The condition number of Section 1.13 is likewise a statement about singular values,

$$\kappa_2(\boldsymbol{X}) = \frac{\sigma_0}{\sigma_{r-1}}, \tag{1.98}$$

which is infinite exactly when $\boldsymbol{X}$ is singular. The SVD thus not only detects near-singularity, it quantifies it.

The Moore-Penrose pseudoinverse.

When $\boldsymbol{X}^{-1}$ does not exist, or exists but is useless because $\boldsymbol{X}$ is nearly singular, the SVD provides the natural substitute

$$\boldsymbol{X}^{+} = \boldsymbol{V}\boldsymbol{\Sigma}^{+}\boldsymbol{U}^{T}, \tag{1.99}$$

where $\boldsymbol{\Sigma}^{+}$ is obtained from $\boldsymbol{\Sigma}$ by transposing it and replacing every non-zero singular value by its reciprocal, *leaving the zeros as zeros*. In practice one treats as zero every singular value below some $\varepsilon_{\mathrm{cut}}\sigma_0$. If $\boldsymbol{X}$ is invertible and well conditioned, $\boldsymbol{X}^{+} = \boldsymbol{X}^{-1}$; otherwise $\boldsymbol{X}^{+}\boldsymbol{b}$ is the least-squares solution of smallest norm. Inverting a singular value of 10^{-13} would amplify the rounding noise in the corresponding direction by 10^{13}; discarding it simply declares that the problem contains no information about that direction, which is the truth.

Near-linearly-dependent basis sets.

This is not an abstract concern. Many-body calculations are routinely carried out in non-orthogonal single-particle bases – Gaussians in quantum chemistry, harmonic-oscillator states in different oscillator lengths, generator coordinates in nuclear physics – and such bases become linearly dependent long before one would like them to. The eigenvalue problem is then the generalised one,

$$\boldsymbol{H}\boldsymbol{c} = E\,\boldsymbol{S}\boldsymbol{c}, \qquad S_{ij} = \langle \phi_i | \phi_j \rangle, \tag{1.100}$$

with $\boldsymbol{S}$ the overlap matrix. Diagonalising $\boldsymbol{S} = \boldsymbol{V}\boldsymbol{s}\boldsymbol{V}^T$ and forming

$$\boldsymbol{X} = \boldsymbol{V}\boldsymbol{s}^{-1/2} \tag{1.101}$$

turns Eq. (1.100) into the ordinary problem $(\boldsymbol{X}^T\boldsymbol{H}\boldsymbol{X})\boldsymbol{c}' = E\boldsymbol{c}'$. The transformation involves $s_i^{-1/2}$, so a tiny overlap eigenvalue is precisely where the procedure breaks down. *Canonical orthogonalisation* is the fix: drop the eigenvectors whose eigenvalues lie below a threshold before forming Eq. (1.101).

Table 1.6 shows this for twelve Gaussians of width 1.4 placed on a uniform grid, a deliberately over-complete basis of the kind used in quantum chemistry. The overlap matrix has eigenvalues spanning thirteen orders of magnitude, from 6.9 down to 6.5×10^{-13}, and a condition number of 1.1×10^{13}: in double precision only three significant digits would survive. Discarding three basis directions removes six orders of magnitude of ill-conditioning at negligible cost in completeness.

Threshold	**Functions kept**	**Effective $\kappa_2(\boldsymbol{S})$**
none	12	1.1×10^{13}
10^{-12}	11	6.9×10^{10}
10^{-8}	9	2.1×10^{7}
10^{-5}	7	3.3×10^{4}

Table 1.6 Canonical orthogonalisation of a basis of twelve overlapping Gaussians. Eigenvectors of the overlap matrix with eigenvalue below the threshold, relative to the largest, are discarded.

```
class CanonicalOrthogonalization:
    """Orthogonalise a non-orthogonal basis with a near-singular overlap.

    The generalised problem H c = E S c is reduced to an ordinary one by
    X = V s^{-1/2}, keeping only the eigenvalues of S above a threshold.
    """

    def __init__(self, S, threshold=1.0e-8):
        self.S = np.asarray(S, dtype=float)
        w, V = np.linalg.eigh(self.S)
        order = np.argsort(w)[::-1]
        self.eigenvalues = w[order]
        self.eigenvectors = V[:, order]
        keep = self.eigenvalues > threshold * self.eigenvalues[0]
        self.kept = int(np.sum(keep))
        self.X = self.eigenvectors[:, keep] / np.sqrt(self.eigenvalues[keep])

    def solve_generalized(self, H):
        """Solve H c = E S c through the orthogonalised problem."""
        values, vectors = np.linalg.eigh(self.X.T @ H @ self.X)
        return values, self.X @ vectors
```

Many-body connection. The threshold is a physical choice, not a numerical detail. Setting it too low keeps directions that carry only rounding noise and poisons the whole calculation; setting it too high throws away genuine variational freedom and raises the computed energy. The same trade-off appears in the generator-coordinate method, where the overlap kernel of non-orthogonal mean-field states is routinely rank deficient, and in explicitly correlated Gaussian expansions, where it is the principal practical obstacle to enlarging the basis.

1.30 Low-rank approximation and the Schmidt decomposition

We now come to the property that makes the SVD indispensable in many-body physics. Truncating the sum

$$\boldsymbol{X} = \sum_{k=0}^{r-1} \sigma_k \boldsymbol{u}_k \boldsymbol{v}_k^T \tag{1.102}$$

after χ terms gives a matrix $\boldsymbol{X}_\chi$ of rank χ. The Eckart-Young theorem states that this is the *best* rank-χ approximation to $\boldsymbol{X}$, in both the 2-norm and the Frobenius norm, and that the error it makes is

$$\|\boldsymbol{X}-\boldsymbol{X}_\chi\|_2 = \sigma_\chi, \qquad \|\boldsymbol{X}-\boldsymbol{X}_\chi\|_F^2 = \sum_{k\geq\chi} \sigma_k^2. \tag{1.103}$$

No cleverer choice of χ vectors exists. If the singular values decay quickly, a matrix with mn entries is captured by $\chi(m+n+1)$ numbers.

The Schmidt decomposition.

Consider a quantum system divided into two parts, A and B, with orthonormal bases $\{|i\rangle_A\}$ and $\{|j\rangle_B\}$. Any state of the whole system can be written

$$|\Psi\rangle = \sum_{ij} C_{ij}\, |i\rangle_A \otimes |j\rangle_B, \tag{1.104}$$

and the coefficients form a matrix $\boldsymbol{C}$ which is rectangular whenever the two parts have different dimensions. Inserting its SVD $\boldsymbol{C} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ and absorbing $\boldsymbol{U}$ and $\boldsymbol{V}$ into new basis states $|u_k\rangle_A = \sum_i U_{ik}|i\rangle_A$ and $|v_k\rangle_B = \sum_j V_{jk}|j\rangle_B$, which are orthonormal because $\boldsymbol{U}$ and $\boldsymbol{V}$ are orthogonal, gives

$$\boxed{|\Psi\rangle = \sum_{k=0}^{\chi-1} \sigma_k\, |u_k\rangle_A \otimes |v_k\rangle_B} \tag{1.105}$$

This is the *Schmidt decomposition*. A double sum over the product basis has collapsed to a single sum, and the SVD has done all the work. The singular values σ_k are the Schmidt coefficients, and normalisation $\langle\Psi|\Psi\rangle = 1$ gives $\sum_k \sigma_k^2 = 1$.

The reduced density matrices follow at once. Tracing out B,

$$\boldsymbol{\rho}_A = \mathrm{Tr}_B\, |\Psi\rangle\langle\Psi| = \boldsymbol{C}\boldsymbol{C}^T = \boldsymbol{U}\boldsymbol{\Sigma}^2\boldsymbol{U}^T, \qquad \boldsymbol{\rho}_B = \boldsymbol{C}^T\boldsymbol{C} = \boldsymbol{V}\boldsymbol{\Sigma}^2\boldsymbol{V}^T, \tag{1.106}$$

which is Eqs. (1.95) and (1.96) read as physics. The two reduced density matrices have the *same* non-zero eigenvalues σ_k^2, however different the sizes of the two parts, and their eigenvectors are the Schmidt states. The number of non-zero σ_k is the *Schmidt rank*: it equals one precisely when $|\Psi\rangle$ is a product state, so a single singular value is the signature of an unentangled state. The *entanglement entropy*

$$S_A = -\mathrm{Tr}\,\boldsymbol{\rho}_A \ln \boldsymbol{\rho}_A = -\sum_k \sigma_k^2 \ln \sigma_k^2 \tag{1.107}$$

measures how far from a product the state is, and it is manifestly symmetric between A and B.

Truncation.

Keeping only the χ largest Schmidt coefficients gives, by Eq. (1.103), the best possible approximation to $|\Psi\rangle$ of that rank, with discarded weight

$$\varepsilon_\chi = \sum_{k \geq \chi} \sigma_k^2. \tag{1.108}$$

This single quantity is the truncation error of the density-matrix renormalisation group, and χ is its bond dimension. Whether a many-body state can be compressed is therefore a question about the decay of the Schmidt spectrum, and that decay is what area laws for the entanglement entropy assert: for ground states of local, gapped Hamiltonians the entropy across a cut grows with the *boundary* of the cut rather than with the volume of the subsystem, so a modest χ suffices no matter how large the system. Matrix product states are the systematic version of this observation, obtained by applying Eq. (1.105) at every bond of a one-dimensional lattice in turn.

The pairing model across a cut.

Take the ground state of the pairing model of Section 1.26 and split the twelve single-particle levels into the lowest six and the highest six. A seniority-zero basis state is a set of occupied levels, which factorises into its part in A and its part in B, so the coefficients form a 64×64 matrix. Its singular values are the Schmidt coefficients across the cut. Table 1.7 gives the discarded weight as a function of χ: three Schmidt states out of 64 already capture all but 5×10^{-3} of the state, and ten capture all but 8×10^{-7}. The entanglement entropy is $S_A = 0.9238$, well below the maximum $\ln 64 = 4.16$ that an arbitrary state of this size could carry.

Schmidt states kept, χ	**Discarded weight ε_χ**
1	4.831×10^{-1}
2	7.461×10^{-2}
3	5.448×10^{-3}
5	1.854×10^{-4}
7	7.363×10^{-6}
10	7.530×10^{-7}
15	4.225×10^{-9}

Table 1.7 Schmidt spectrum of the pairing-model ground state (12 levels, 6 pairs, $\delta = 1$, $g = 0.5$) across the cut separating the lowest six levels from the highest six. The coefficient matrix is 64×64 and the entanglement entropy is 0.9238.

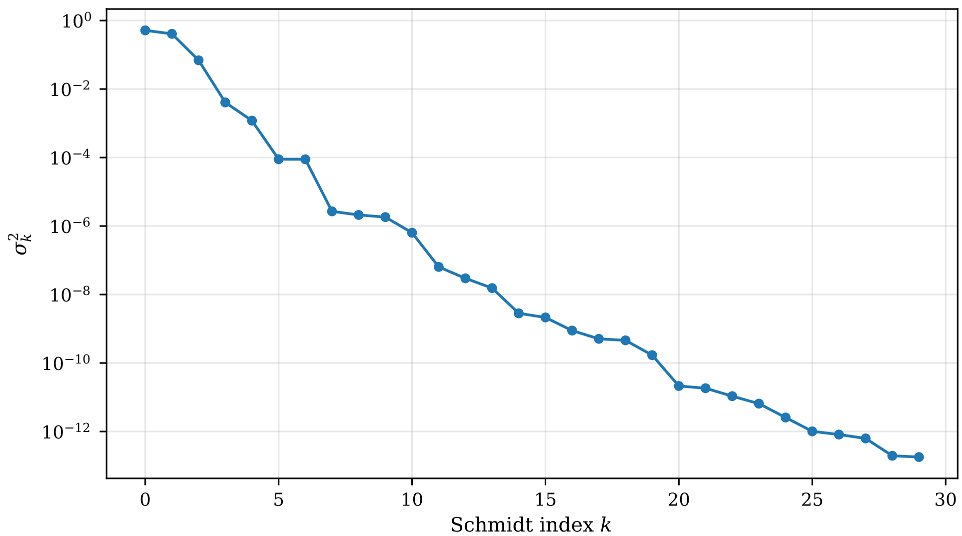

Fig. 1.4 Schmidt weights σ_k^2 of the pairing-model ground state (12 levels, 6 pairs, $\delta = 1$, $g = 0.5$) across the cut separating the lowest six levels from the highest six, plotted logarithmically against the Schmidt index k for the thirty largest of the 64 values. The first two weights, 0.5169 and 0.4085, are of the same order and together account for 92.5% of the norm; thereafter the spectrum falls by roughly half a decade per index, reaching 10^{-6} by $k = 8$ and 10^{-12} by $k = 25$. The small plateaux, for instance the pair of equal weights 8.9×10^{-5} at $k = 5$ and $k = 6$, are degeneracies imposed by the symmetry of the Hamiltonian.

Figure 1.4 shows the individual weights rather than the accumulated tail, and the two ways of looking at the same list of numbers are complementary. The discarded weight of Table 1.7 is what one needs in order to choose a bond dimension; the spectrum itself is what tells one whether such a choice will remain sensible as the system grows. Two things stand out. The first is that the leading two weights, 0.5169 and 0.4085, are of the same size. This state is emphatically not a product, and truncating at $\chi = 1$ – which is what a mean-field ansatz does across this cut – throws away 48% of the norm. The pairing interaction correlates the two halves of the level scheme strongly, and it does so in a way that a single determinant cannot represent. The second is that from $k = 2$ onwards the decay is close to exponential in the index, about half a decade per Schmidt state, and it is this, not the smallness of the first discarded weight, that makes compression work: an exponentially decaying spectrum means that the cost of an extra digit of accuracy is a constant number of additional Schmidt states, independent of how accurate one already is. Reading the figure at $\sigma_k^2 = 10^{-8}$ gives $k \approx 17$, so seventeen of the 64 available states suffice for eight digits. The entanglement entropy $S_A = 0.9238$ is the same information compressed into one number; that it lies so far below the maximum $\ln 64 = 4.16$ is the quantitative statement that the state occupies only a small corner of the 64×64-dimensional space of coefficient matrices. The visible plateaux, where two or three consecutive weights coincide to all displayed digits, are not numerical accidents: they are exact degeneracies of the reduced density matrix inherited from the symmetry of the pairing Hamiltonian, and a truncation that keeps one member of such a multiplet while discarding another will break that symmetry, which is why practical density-matrix renormalisation group codes truncate on multiplets rather than on individual states.

```
class SchmidtDecomposition:
    """The SVD of a bipartitioned state vector.

    For |Psi> = sum_ij C_ij |i>_A |j>_B the SVD C = U Sigma V^T gives
    |Psi> = sum_k sigma_k |u_k>_A |v_k>_B.  The eigenvalues of both reduced
    density matrices are sigma_k^2.
    """
```

```python
    def __init__(self, C):
        self.C = np.asarray(C, dtype=float)
        self.U, self.sigma, self.Vt = np.linalg.svd(C, full_matrices=False)
        self.weights = self.sigma**2        # normalised: sum_k sigma_k^2 = 1

    def schmidt_rank(self, tol=1.0e-10):
        return int(np.sum(self.sigma > tol * self.sigma[0]))

    def entanglement_entropy(self, base=np.e):
        w = self.weights[self.weights > 1.0e-16]
        entropy = -np.sum(w * np.log(w)) / np.log(base)
        return float(entropy) if entropy > 0.0 else 0.0

    def truncation_error(self, k):
        """Weight discarded when only k Schmidt states are kept."""
        return float(np.sum(self.weights[k:]))

    def truncate(self, k):
        """The best rank-k approximation to the state, renormalised."""
        C = (self.U[:, :k] * self.sigma[:k]) @ self.Vt[:k]
        return C / np.linalg.norm(C)
```

Many-body connection. The Schmidt decomposition is the bridge between linear algebra and the modern theory of many-body wave functions. The density-matrix renormalisation group truncates the Schmidt spectrum at every bond; matrix product states, projected entangled pair states and the whole family of tensor networks are built by iterating Eq. (1.105); and the entanglement entropy, computed from nothing more than a list of singular values, is what tells us in advance whether such a compression can work at all.

Quantum computing connection. The same numbers govern how hard a state is to simulate classically. A state of N qubits whose Schmidt rank across every cut is bounded by χ can be stored in $\mathcal{O}(N\chi^2)$ numbers and evolved classically; only states whose Schmidt rank grows exponentially require a quantum computer. Entanglement, quantified by Eq. (1.107), is thus simultaneously the resource that quantum algorithms exploit and the obstacle that classical ones must overcome.

1.31 Natural orbitals and occupation numbers

For two identical particles the Schmidt decomposition has a name of its own. Write the two-particle wave function in a single-particle basis $\{\varphi_p\}$,

$$\Psi(x_1,x_2) = \sum_{pq} C_{pq}\, \varphi_p(x_1)\varphi_q(x_2), \tag{1.109}$$

where symmetry under exchange makes $\boldsymbol{C}$ symmetric (or antisymmetric). The SVD of $\boldsymbol{C}$, which for a symmetric matrix is its eigen-decomposition up to signs, gives the *natural expansion*

$$\Psi(x_1,x_2) = \sum_{k} \sigma_k\, \chi_k(x_1)\chi_k(x_2), \tag{1.110}$$

a single sum in terms of the transformed orbitals $\chi_k = \sum_p U_{pk}\varphi_p$. These are Löwdin's *natural orbitals*, and they are the eigenvectors of the one-body density matrix

$$\rho_{pq} = \langle \Psi | a_p^\dagger a_q | \Psi \rangle, \qquad \boldsymbol{\rho} \propto \boldsymbol{C}\boldsymbol{C}^\dagger = \boldsymbol{U}\boldsymbol{\Sigma}^2\boldsymbol{U}^\dagger, \tag{1.111}$$

with eigenvalues $n_k \propto \sigma_k^2$: the *occupation numbers*. For a single Slater determinant exactly N occupation numbers equal one and the rest vanish; correlation depletes the occupied orbitals and populates the others, and the deviation of the largest n_k from unity is a direct measure of how far the state is from a mean-field one.

Table 1.8 illustrates this for two particles in a one-dimensional harmonic trap interacting through a softened Coulomb potential of strength g, solved on a grid of 40 points by full diagonalisation of the 1600-dimensional two-particle problem. At $g = 0$ the ground state is a product, there is a single natural orbital with $n_0 = 1$, and the entanglement entropy vanishes exactly. Switching on the interaction depletes n_0 and populates n_1, n_2 and so on, and the entropy rises.

g	E_0	S_A	n_0	n_1	χ for 10^{-8}
0	1.992536	0.000000	1.000000	0.000000	1
1	3.173866	0.040874	0.993528	0.006017	10
2	4.245117	0.139618	0.970732	0.027407	11
5	6.755062	0.507970	0.821806	0.168066	11

Table 1.8 Natural occupation numbers and entanglement entropy for two particles in a harmonic trap with a softened Coulomb repulsion of strength g, on a grid of 40 points. The last column gives the number of natural orbitals needed to recover all but 10^{-8} of the norm.

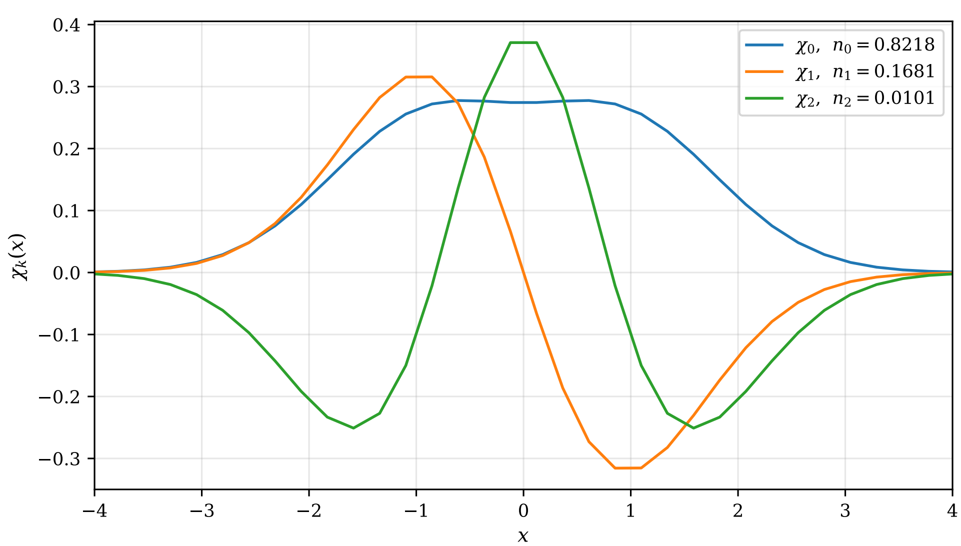

Fig. 1.5 The three leading natural orbitals $\chi_k(x)$ of two particles in a one-dimensional harmonic trap with a softened Coulomb repulsion of strength $g = 5$, obtained as the eigenvectors of the one-body density matrix (1.111) on a grid of 40 points. The legend gives the corresponding occupation numbers. The leading orbital carries $n_0 = 0.8218$ and is nodeless but visibly flattened at the centre; the second, with $n_1 = 0.1681$, is odd with one node; the third, with $n_2 = 0.0101$, is even with two. The orbitals are ordered by occupation, not by energy, and the sign of each is fixed by requiring its largest component to be positive.

The orbitals themselves, for the most strongly correlated case $g = 5$ of Table 1.8, are drawn in Figure 1.5, and they make visible what the occupation numbers only summarise. In shape the three of them are recognisably the ground state, the first and the second excited state of a one-dimensional oscillator: they have zero, one and two nodes, they alternate in parity, and they are mutually orthogonal. But they are not the oscillator states. The leading orbital has been pushed apart at the origin into a broad, almost flat-topped distribution with a shallow dip at $x = 0$, and the outer lobes of χ_1 and χ_2 sit further out than their oscillator counterparts would. This is the repulsion at work: the two particles avoid each other, and the single-particle description that best accommodates that avoidance is one built from orbitals that themselves keep away from the centre of the trap. The occupation numbers quantify how much of this the mean field is missing. At $g = 5$ we have $n_0 = 0.8218$, so eighteen per cent of the particle density has been promoted out of the leading orbital, of which $n_1 = 0.1681$ sits in the first excited natural orbital and only $n_2 = 0.0101$ in the second. A Hartree-Fock calculation on this system would keep the first orbital and discard the rest, and would therefore be in error at the level of tens of per cent in the density – this is a strongly correlated state and it needs a multireference treatment. Compare this with the weakly interacting case $g = 1$ of the same table, where $n_0 = 0.9935$ and a single determinant is an excellent starting point. The progression through the table, from $S_A = 0$ and a single occupied orbital at $g = 0$ to $S_A = 0.508$ and a spread of occupations at $g = 5$, is the cleanest illustration in this chapter of what correlation means in the language of linear algebra: the rank of the coefficient matrix $\boldsymbol{C}$ rises from one to many, and the entanglement entropy (1.107) rises with it.

Many-body connection. Natural orbitals are the optimal single-particle basis in a precise sense: for a given number of retained orbitals, no other choice gives a better representation of the state, which is Eq. (1.103) again. This is why natural-orbital truncation is the standard way of reducing a correlated basis in quantum chemistry and in nuclear structure, and why occupation numbers are reported as a diagnostic: a system whose largest occupation number is 0.99 is well described by a single determinant and Hartree-Fock theory of Chapter 2 will be a good starting point, while one whose occupation numbers are spread over several orbitals is strongly correlated and demands a multireference treatment.

1.32 Low-rank factorisation of the two-body interaction

The last application concerns not the wave function but the Hamiltonian. The two-body matrix elements V_{pqrs} form a four-index array with n^4 entries, where n is the number of single-particle states, and for n of a few hundred this is already the dominant storage cost of a many-body calculation. Reading the pairs (pq) and (rs) as composite indices turns the array into an ordinary $n^2 \times n^2$ matrix,

$$V_{(pq),(rs)} = \int\!\!\int dxdy\, \varphi_p(x)\varphi_q(x)\, K(x,y)\, \varphi_r(y)\varphi_s(y), \tag{1.112}$$

and for a Coulomb-like kernel K this matrix is symmetric and positive semidefinite. Its eigendecomposition, equivalently its Cholesky factorisation of Section 1.15, gives

$$V_{(pq),(rs)} = \sum_{\gamma=0}^{M-1} L^{\gamma}_{pq} L^{\gamma}_{rs}, \qquad L^{\gamma}_{pq} = \sqrt{\lambda_\gamma}\, U_{(pq),\gamma}. \tag{1.113}$$

The decisive fact is that the eigenvalues λ_γ decay rapidly, because the pair densities $\varphi_p\varphi_q$ are far from linearly independent as functions. Keeping only $M \ll n^2$ of them reduces the storage from n^4 to Mn^2 and turns the contraction of the interaction with a wave function into a sequence of matrix products.

Table 1.9 shows the decay for the twelve-Gaussian basis of Section 1.29, for which there are 144 pair indices. Nine vectors reproduce the interaction to one part in 10^6, and ten to one part in 10^8: a reduction of the rank by more than a factor of ten.

Target weight	**Vectors** M	$\|\boldsymbol{V}-\boldsymbol{L}^T\boldsymbol{L}\|/\|\boldsymbol{V}\|$
10^{-4}	7	2.1×10^{-5}
10^{-6}	9	1.7×10^{-7}
10^{-8}	10	1.2×10^{-8}

Table 1.9 Low-rank factorisation of the two-body matrix elements in a basis of twelve Gaussians, for which the pair index runs over 144 values. The leading eigenvalues of the pair matrix are 44.7, 15.6, 5.18, 1.38, 0.309, 0.055, ..., a geometric decay.

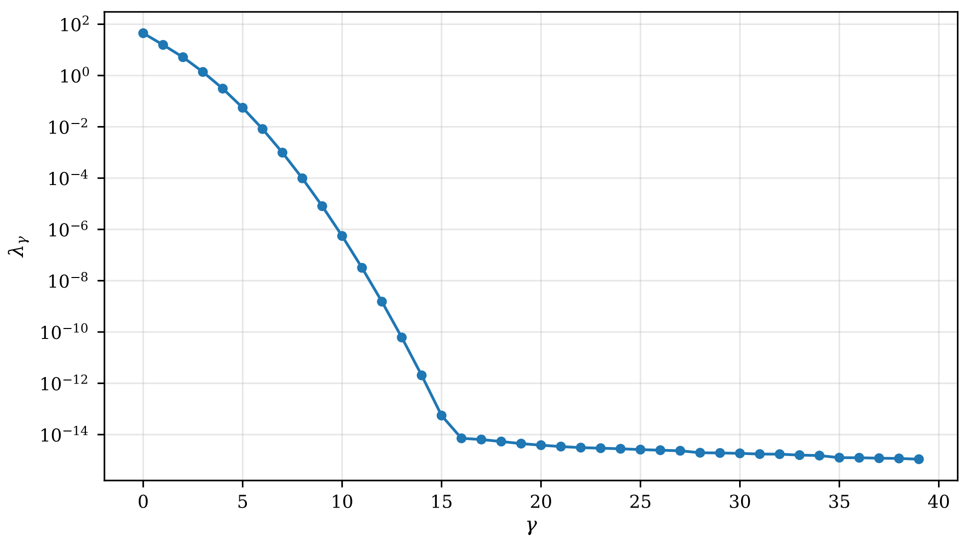

Fig. 1.6 The forty largest eigenvalues λ_γ of the two-body matrix (1.112) in the composite pair index, on a logarithmic scale, for the basis of twelve Gaussians of Section 1.29, for which the pair index runs over 144 values. The spectrum falls from $\lambda_0 = 44.7$ to 10^{-14} within sixteen indices and is flat thereafter: the tail beyond $\gamma \approx 16$ sits at the rounding level 10^{-14}, roughly λ_0 times the machine epsilon, and carries no information. The numerical rank of the interaction is therefore about sixteen, an order of magnitude below the 144 that naive counting of pair indices would suggest.

Figure 1.6 shows the whole spectrum rather than the three entries of Table 1.9, and the shape of it is the reason the method works. The first six eigenvalues, 44.7, 15.6, 5.18, 1.38, 0.309 and 0.055, fall by ratios that grow from about three to about six, so the decay is not merely geometric but accelerating; by $\gamma = 10$ we are at 6×10^{-7} and by $\gamma = 15$ at 2×10^{-14}. Then the curve turns over abruptly and runs flat out to $\gamma = 40$ at a level of 10^{-14}, which is λ_0 multiplied by something of the order of the machine epsilon. That flat tail is not physics. It is what is left of the positive semidefinite matrix after the true eigenvalues have fallen below the precision with which the matrix elements were computed, and its height is a useful diagnostic: an eigenvalue solver that returned a tail at $10^{-8}\lambda_0$ instead would be telling us that the integrals, and not the diagonalisation, were the limiting factor. The point at which the spectrum meets the noise floor is the numerical rank, here about sixteen out of 144, and it is what one should quote when asked how large M has to be. The reason for so drastic a collapse is stated in the text above: the 144 pair densities $\varphi_p \varphi_q$ are functions of a single variable x, and there is simply not room in that one-dimensional function space for 144 of them to be independent once the Gaussians overlap. The overlap matrix of the same basis, whose eigenvalues run from 6.93 down to 6.5×10^{-13}, was diagnosed in Section 1.29 as dangerously ill-conditioned; here the identical linear dependence appears not as a hazard but as an asset, since it is precisely what allows the four-index array to be replaced by sixteen matrices without measurable loss. The storage saving is a factor $n^2/M = 144/16 \approx 9$ in this small example, and it grows with the basis, because M tends to scale linearly with n while n^2 does not.

```python
class TwoBodyFactorization:
    """Low-rank factorisation of a two-body interaction.

    The pair-index matrix V_{(pq),(rs)} is positive semidefinite for a
    Coulomb-like interaction, so it factorises as a sum of outer products,
    V = sum_gamma L^gamma (L^gamma)^T, with few terms.
    """

    def __init__(self, V):
        self.V = np.asarray(V, dtype=float)
        w, U = np.linalg.eigh(0.5 * (self.V + self.V.T))
        order = np.argsort(w)[::-1]
        self.eigenvalues = w[order]
        self.vectors = U[:, order]

    def rank_for(self, tol):
        """Smallest M such that the discarded weight is below tol."""
        w = np.maximum(self.eigenvalues, 0.0)
        tail = np.cumsum(w[::-1])[::-1]
        below = np.where(tail <= tol * np.sum(w))[0]
        return int(below[0]) if len(below) else len(w)

    def cholesky_vectors(self, M):
        """The M leading vectors L^gamma."""
        w = np.maximum(self.eigenvalues[:M], 0.0)
        return (self.vectors[:, :M] * np.sqrt(w)).T
```

Many-body connection. Equation (1.113) is known as the Cholesky decomposition of the two-electron integrals, or, when the auxiliary vectors are chosen from a fitted basis rather than computed, as density fitting or the resolution of the identity. It is what makes large-basis coupled-cluster and many-body perturbation theory calculations feasible, since the rate-determining contractions can be performed on the factors L^γ rather than on the full four-index array.

Quantum computing connection. Applying the same idea twice gives the *double factorisation*: each matrix L^γ_{pq} is itself diagonalised, $\boldsymbol{L}^\gamma = \boldsymbol{W}^\gamma \boldsymbol{d}^\gamma (\boldsymbol{W}^\gamma)^T$, so that the Hamiltonian becomes a short sum of squares of one-body operators. Since a one-body operator maps under the Jordan-Wigner transformation of Section 1.7 onto a small number of commuting Pauli strings, this reduces the number of terms that must be measured in a variational quantum eigensolver, and the cost of block encoding the Hamiltonian for qubitisation, from $\mathscr{O}(n^4)$ to $\mathscr{O}(n^3)$ or better. The same singular values control the truncation in both the classical and the quantum setting.

1.33 Summary and the programs

The chapter has moved from notation to algorithms. The linear algebra of Sections 1.1-1.11 supplies the language in which quantum mechanics is written: states are vectors, observables are Hermitian matrices, symmetries and basis changes are unitary transformations, and the physical content of an operator is its spectral decomposition. The numerical sections supply the means of extracting numbers from that language.

Three dividing lines are worth keeping in mind. The first separates direct from iterative methods. Direct methods – Gaussian elimination, LU, Cholesky, Householder with QL – deliver the complete answer in a fixed $\mathcal{O}(n^3)$ number of operations, and require the matrix to be stored. Iterative methods – Jacobi, Gauss-Seidel, over-relaxation, conjugate gradient, Lanczos – deliver an approximation that improves with each step, and require only the action of the matrix on a vector. The second line separates dense from structured problems: a tridiagonal system costs $\mathcal{O}(n)$ rather than $\mathcal{O}(n^3)$, and recognising structure is usually worth more than any amount of tuning.

The third line is the one drawn by the singular value decomposition, and it is of a different kind. It separates the part of a problem that carries information from the part that does not. The same list of singular values tells us the numerical rank of a matrix, how far a basis is from linear dependence, how much of a wave function survives a truncation, how entangled a state is across a cut, which single-particle orbitals are actually occupied, and how few auxiliary vectors are needed to represent a two-body interaction. Sections 1.28-1.32 are five readings of one decomposition, and the thread connecting them – Eq. (1.103), that truncating the singular values is the best approximation of its rank – is the mathematical justification for a large part of modern many-body theory, from natural-orbital bases to the density-matrix renormalisation group.

The complete programs are collected in `doc/BookManybody/BookMaterial/Programs`:

- `direct_solvers.py` – Gaussian elimination, LU, Cholesky and the tridiagonal solver, with determinants and inverses.
- `iterative_solvers.py` – Jacobi, Gauss-Seidel, over-relaxation and the conjugate gradient method, in both a matrix-based and a matrix-free form.
- `householder.py` – Householder tridiagonalisation, the QL algorithm with implicit shifts, and the power method.
- `lanczos.py` – the Lanczos algorithm with optional full reorthogonalisation, together with the pairing-model Hamiltonian used in Table 1.3.
- `schrodinger_diagonalization.py` – Schrödinger's equation by diagonalisation, solved both directly and by Lanczos.
- `svd.py` – the singular value decomposition: rank, the pseudoinverse, canonical orthogonalisation, the Schmidt decomposition with entanglement entropies and natural occupation numbers, and the low-rank factorisation of a two-body interaction. It also contains the two physical models used in Sections 1.30-1.32: two particles in a trap, and the pairing model bipartitioned across a cut.

Each file runs as a script and reproduces the numbers quoted in this chapter. An executable version of the same material, in which the reader can vary the parameters, is available as a Jupyter notebook in the accompanying Jupyter-book.

1.34 Exercises

Warm-up exercises

1. **Inner and outer products.** Let $\boldsymbol{u} = \begin{bmatrix}1-i & 2+i\end{bmatrix}^T$ and $\boldsymbol{v} = \begin{bmatrix}1 & i\end{bmatrix}^T$. (a) Compute $\boldsymbol{u}^\dagger \boldsymbol{v}$ and $\boldsymbol{v}^\dagger \boldsymbol{u}$. (b) Verify $(\boldsymbol{u}^\dagger \boldsymbol{v})^* = \boldsymbol{v}^\dagger \boldsymbol{u}$. (c) Compute the outer product $\boldsymbol{u}\boldsymbol{v}^\dagger$ and show $\mathrm{tr}(\boldsymbol{u}\boldsymbol{v}^\dagger) = \boldsymbol{v}^\dagger \boldsymbol{u}$.
2. **Special matrix types.** For each matrix below, identify which type(s) from Table 1.1 it belongs to, and verify your answer using the element conditions in the table: (a) σ_x; (b) σ_z; (c) the Hadamard gate $H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$.
3. **Pauli algebra.** (a) Prove $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = \boldsymbol{I}$. (b) Compute all commutators $[\sigma_i, \sigma_j]$ for $i, j \in \{x, y, z\}$. (c) Which Pauli matrix has $|0\rangle$ and $|1\rangle$ as its eigenbasis? What are the eigenvalues?
4. **Projection operators.** (a) Verify that $\boldsymbol{P} = |0\rangle\langle 0|$ satisfies $\boldsymbol{P}^2 = \boldsymbol{P}$ and $\det(\boldsymbol{P}) = 0$. (b) Show $\boldsymbol{P} + \boldsymbol{Q} = \boldsymbol{I}$ and $\boldsymbol{P}\boldsymbol{Q} = 0$. (c) What does $\boldsymbol{P}|\psi\rangle$ equal for $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$?
5. **Commutator identities.** Prove the following for arbitrary operators $\hat{A}$, $\hat{B}$, $\hat{C}$: (a) $[\hat{A}+\hat{B}, \hat{C}] = [\hat{A}, \hat{C}] + [\hat{B}, \hat{C}]$; (b) $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$; (c) $[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0$ (Jacobi identity).
6. **Shared eigenstates.** Prove that if $[\hat{A}, \hat{B}] = 0$ then $\hat{A}$ and $\hat{B}$ share a common eigenbasis.
7. **Tensor products.** (a) Write out all four two-qubit basis states $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$ as column vectors of length 4 and verify they are mutually orthogonal. (b) Compute $C_X|00\rangle$, $C_X|01\rangle$, $C_X|10\rangle$, $C_X|11\rangle$. Explain the action of the CNOT gate in words.
8. **Unitary transformation.** Let $\boldsymbol{U}$ be a 2×2 unitary matrix and $\{|v_0\rangle, |v_1\rangle\}$ an ONB. Prove that $\{\boldsymbol{U}|v_0\rangle, \boldsymbol{U}|v_1\rangle\}$ is also an ONB.
9. **Spectral decomposition.** (a) Find the eigenvalues and normalised eigenvectors of σ_x. (b) Write the spectral decomposition $\sigma_x = \sum_i \lambda_i |i\rangle\langle i|$ explicitly. (c) Verify that $\sigma_x|\psi_a\rangle = \lambda_a|\psi_a\rangle$ for each eigenvector.

Exercise session, week 34: unitary transformations and determinants

The exercises of this session are mainly meant as reminders of the specific linear algebra elements that we shall use throughout the course.

Exercise 1: unitary transformations and orthogonality.

In many-body theories it is common to expand a new basis in terms of a known one. Well-known examples are the analytical solutions of the quantum mechanical harmonic oscillator, or the hydrogen atom with a Coulomb interaction between the electron and the nucleus. Varying the expansion coefficients leads to a variational minimum which is expected to lie closer to the true ground state of a given Hamiltonian. This is the background for mean-field theories, full configuration interaction theory and many other many-body methods, and the expansion coefficients are normally the matrix elements of a unitary or orthogonal matrix.

Assume that the matrix $\boldsymbol{U}$ is unitary with dimension $n \times n$ and matrix elements $u_{p\lambda}$. Using Dirac notation we represent a new vector $|\psi_p\rangle$ of length n in terms of another vector $|\phi_\lambda\rangle$ of length n through

$$|\psi_p\rangle = \boldsymbol{U}|\phi_\lambda\rangle. \tag{1.114}$$

It is common to assume that the basis vectors $|\phi_\lambda\rangle$ are normalised and orthogonal, and that they are the eigenvectors of an eigenvalue problem

$$\boldsymbol{O}|\phi_\lambda\rangle = o_\lambda|\phi_\lambda\rangle, \tag{1.115}$$

where $\boldsymbol{O}$ is a Hermitian matrix of dimension $n\times n$ and o_λ one of its n eigenvalues. We have thus defined our new basis by performing a unitary (or orthogonal) transformation.

a) Write down the properties of unitary and orthogonal matrices.
b) Write down the properties of Hermitian matrices.
c) Assuming that the coefficients $u_{p\lambda}$ belong to a unitary or orthogonal transformation, show that the new basis is orthogonal and normalised as well.
d) Show that the eigenvalues obtained in the new basis are the same as those of the original basis.

Exercise 2: determinants.

a) Consider the determinant

$$\begin{vmatrix}\alpha_1 b_{11}+\alpha_2 b_{12} & a_{12}\\ \alpha_1 b_{21}+\alpha_2 b_{22} & a_{22}\end{vmatrix} = \alpha_1\begin{vmatrix}b_{11} & a_{12}\\ b_{21} & a_{22}\end{vmatrix}+\alpha_2\begin{vmatrix}b_{12} & a_{12}\\ b_{22} & a_{22}\end{vmatrix},$$

which generalises to the linearity in a column expressed by Eq. (2.12). Use it to show that, with every element a_{ij} given as a linear combination $a_{ij}=\sum_{k=1}^n b_{ik}c_{kj}$,

$$\begin{vmatrix}\sum_k b_{1k}c_{k1} & \cdots & \sum_k b_{1k}c_{kn}\\ \vdots & & \vdots\\ \sum_k b_{nk}c_{k1} & \cdots & \sum_k b_{nk}c_{kn}\end{vmatrix} = |\boldsymbol{C}|\,|\boldsymbol{B}|,$$

where $|\boldsymbol{B}|$ and $|\boldsymbol{C}|$ are the determinants of the $n\times n$ matrices with elements b_{ij} and c_{ij}. This is the product rule (2.13), and we shall use it repeatedly in the mean-field theories of the following chapters.

b) With the new basis of the previous exercise, now specialised to a basis of single-particle functions,

$$\psi_p(x) = \sum_\lambda C_{p\lambda}\phi_\lambda(x),$$

show that the determinant in the new basis $\psi_p(x)$ can be written as (omitting the factor $1/\sqrt{N!}$, with N the number of particles)

$$\begin{vmatrix}\psi_p(x_1) & \cdots & \psi_p(x_N)\\ \vdots & \ddots & \vdots\\ \psi_t(x_1) & \cdots & \psi_t(x_N)\end{vmatrix} = \begin{vmatrix}\sum_\lambda C_{p\lambda}\phi_\lambda(x_1) & \cdots & \sum_\lambda C_{p\lambda}\phi_\lambda(x_N)\\ \vdots & \ddots & \vdots\\ \sum_\lambda C_{t\lambda}\phi_\lambda(x_1) & \cdots & \sum_\lambda C_{t\lambda}\phi_\lambda(x_N)\end{vmatrix},$$

which is nothing but $|\boldsymbol{C}||\boldsymbol{\Phi}|$, with $|\boldsymbol{\Phi}|$ the determinant built from the basis functions $\phi_\lambda(x)$.

c) Show that the new determinant differs from $|\boldsymbol{\Phi}|$ by a complex phase only.

Answers, week 34

Exercise 1a.

A matrix $\boldsymbol{U}\in\mathbb{C}^{n\times n}$ is *unitary* if

$$\boldsymbol{U}^\dagger \boldsymbol{U} = \boldsymbol{U}\boldsymbol{U}^\dagger = \boldsymbol{I}, \qquad \text{equivalently} \qquad \boldsymbol{U}^{-1} = \boldsymbol{U}^\dagger. \tag{1.116}$$

In terms of matrix elements this reads $\sum_k u_{ik}u^*_{jk} = \sum_k u^*_{ki}u_{kj} = \delta_{ij}$: the rows form an orthonormal set, and so do the columns. Three consequences follow at once and were used throughout Section 1.9. The inner product is preserved, $(\boldsymbol{U}\boldsymbol{x})^\dagger(\boldsymbol{U}\boldsymbol{y}) = \boldsymbol{x}^\dagger\boldsymbol{y}$, hence so is the norm; the eigenvalues all have modulus one, since $\boldsymbol{U}\boldsymbol{x} = \lambda\boldsymbol{x}$ with $\|\boldsymbol{x}\| = 1$ gives $|\lambda|^2 = \boldsymbol{x}^\dagger\boldsymbol{U}^\dagger\boldsymbol{U}\boldsymbol{x} = 1$; and $||\boldsymbol{U}|| = 1$, because $|\boldsymbol{U}|^*|\boldsymbol{U}| = |\boldsymbol{U}^\dagger\boldsymbol{U}| = 1$. A unitary matrix is thus a rotation, and it is perfectly reversible.

A *real* unitary matrix is called *orthogonal*: $\boldsymbol{O}^T\boldsymbol{O} = \boldsymbol{O}\boldsymbol{O}^T = \boldsymbol{I}$ and $\boldsymbol{O}^{-1} = \boldsymbol{O}^T$, with $|\boldsymbol{O}| = \pm 1$. These are the matrix types listed in Table 1.1.

Exercise 1b.

A matrix is *Hermitian* if $\boldsymbol{A} = \boldsymbol{A}^\dagger$, that is $a_{ij} = a^*_{ji}$; a real Hermitian matrix is symmetric. As shown in Section 1.11, the eigenvalues of a Hermitian matrix are real, eigenvectors belonging to distinct eigenvalues are orthogonal, and the matrix is diagonalisable by a unitary transformation, $\boldsymbol{U}^\dagger\boldsymbol{A}\boldsymbol{U} = \mathrm{diag}(\lambda_0, \ldots, \lambda_{n-1})$. It follows that expectation values $\langle\psi|\boldsymbol{A}|\psi\rangle$ are real, which is why observables are represented by Hermitian operators.

Exercise 1c.

Write the transformation in components, $|\psi_p\rangle = \sum_\lambda u_{p\lambda}|\phi_\lambda\rangle$, and use $\langle\phi_\lambda|\phi_\mu\rangle = \delta_{\lambda\mu}$:

$$\langle\psi_p|\psi_q\rangle = \sum_{\lambda\mu} u^*_{p\lambda}u_{q\mu}\langle\phi_\lambda|\phi_\mu\rangle = \sum_\lambda u^*_{p\lambda}u_{q\lambda} = \left(\boldsymbol{U}\boldsymbol{U}^\dagger\right)_{qp} = \delta_{pq}. \tag{1.117}$$

Orthonormality of the new basis is therefore *equivalent* to unitarity of the coefficient matrix; nothing else about $\boldsymbol{U}$ is needed, and nothing less would do. This is the statement proved geometrically in Eq. (2.19).

Exercise 1d.

In the new basis the operator $\boldsymbol{O}$ is represented by $\boldsymbol{O}' = \boldsymbol{U}\boldsymbol{O}\boldsymbol{U}^\dagger$. This is a similarity transformation of the kind studied in Section 1.22, and by Eq. (1.61) it leaves the spectrum unchanged: if $\boldsymbol{O}|\phi_\lambda\rangle = o_\lambda|\phi_\lambda\rangle$ then

$$\boldsymbol{O}'\left(\boldsymbol{U}|\phi_\lambda\rangle\right) = \boldsymbol{U}\boldsymbol{O}\boldsymbol{U}^\dagger\boldsymbol{U}|\phi_\lambda\rangle = o_\lambda\left(\boldsymbol{U}|\phi_\lambda\rangle\right). \tag{1.118}$$

The eigenvalues are the same and the eigenvectors are the rotated ones. Physically: the spectrum is a property of the operator, not of the basis we happen to write it in.

Exercise 2a.

Let $\boldsymbol{A} = \boldsymbol{B}\boldsymbol{C}$, so that column j of $\boldsymbol{A}$ is $\boldsymbol{a}_j = \sum_k c_{kj}\boldsymbol{b}_k$, a linear combination of the columns of $\boldsymbol{B}$. Applying the linearity (2.12) to the first column, then to the second, and so on through all n columns, gives

$$|\boldsymbol{A}| = \sum_{k_1=1}^{n}\sum_{k_2=1}^{n}\cdots\sum_{k_n=1}^{n} c_{k_1 1}c_{k_2 2}\cdots c_{k_n n}\left|\boldsymbol{b}_{k_1}\,\boldsymbol{b}_{k_2}\cdots\boldsymbol{b}_{k_n}\right|. \tag{1.119}$$

There are n^n terms, but a determinant with two equal columns vanishes, so every term in which two indices coincide drops out. Only the $n!$ terms in which $(k_1, \ldots, k_n)$ is a permutation

σ of $(1,\ldots,n)$ survive, and reordering the columns of such a term into their natural order produces the sign $(-1)^\sigma$ together with $|\boldsymbol{B}|$. Hence

$$|\boldsymbol{A}| = \left(\sum_\sigma (-1)^\sigma c_{\sigma(1)1} c_{\sigma(2)2} \cdots c_{\sigma(n)n}\right) |\boldsymbol{B}| = |\boldsymbol{C}|\,|\boldsymbol{B}|, \tag{1.120}$$

where the bracket is precisely the definition (2.10) of $|\boldsymbol{C}|$.

Two remarks are worth making. First, the derivation is combinatorial and says nothing about how to *compute* either determinant; for that one uses the LU factorisation of Section 1.15 and Eq. (1.29), at a cost of $\mathscr{O}(n^3)$ rather than the $\mathscr{O}(n!)$ of the sum above. Second, the result immediately gives $|\boldsymbol{A}^{-1}| = 1/|\boldsymbol{A}|$ and, for a unitary matrix, $||\boldsymbol{U}|| = 1$.

Exercise 2b.

The matrix whose determinant we want has elements

$$M_{pj} = \psi_p(x_j) = \sum_\lambda C_{p\lambda}\phi_\lambda(x_j) = (\boldsymbol{C}\boldsymbol{\Phi})_{pj}, \qquad \Phi_{\lambda j} = \phi_\lambda(x_j), \tag{1.121}$$

so $\boldsymbol{M} = \boldsymbol{C}\boldsymbol{\Phi}$ and part a) gives directly

$$|\boldsymbol{M}| = |\boldsymbol{C}|\,|\boldsymbol{\Phi}|. \tag{1.122}$$

The rows of $\boldsymbol{M}$ are labelled by the new single-particle states and its columns by the particle coordinates, exactly as in Eq. (2.47).

Exercise 2c.

If $\boldsymbol{C}$ is unitary then $|\boldsymbol{C}|^*|\boldsymbol{C}| = |\boldsymbol{C}^\dagger\boldsymbol{C}| = |\boldsymbol{I}| = 1$, so $|\boldsymbol{C}| = e^{i\theta}$ for some real θ and

$$|\boldsymbol{M}| = e^{i\theta}\,|\boldsymbol{\Phi}|. \tag{1.123}$$

The two determinants differ by a global phase. Since all physical predictions depend on $|\Psi|^2$ or on expectation values $\langle\Psi|\widehat{O}|\Psi\rangle$, in which the phase cancels, the two describe the *same* state. This is the result exploited in Section 2.5: one may rotate the occupied orbitals among themselves at will without changing any physics, which is the freedom underlying Hartree-Fock theory and Thouless' theorem.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter01/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/linearalgebra.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter01/`.

`direct_solvers.py` Gaussian elimination with partial pivoting, the Doolittle LU factorisation, Cholesky and the tridiagonal algorithm. Runs in a few seconds.

`iterative_solvers.py` Jacobi, Gauss–Seidel, successive over-relaxation and the conjugate gradient method. Runs in a few seconds.

`householder.py` Householder tridiagonalisation, the QL algorithm with implicit shifts (`tqli`) and the power method. Runs in a few seconds.

`lanczos.py` The Lanczos algorithm with reorthogonalisation, applied to the pairing model. Runs in a few seconds.

`svd.py` The singular value decomposition, the pseudoinverse, the Schmidt decomposition and the factorisation of a two-body matrix. Runs in a few seconds.

`schrodinger_diagonalization.py` The one-dimensional Schrödinger equation solved by diagonalisation on a grid. Runs in a few seconds.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 2
From Linear Algebra to Many-Body Physics

Chapter 1 developed a language and a set of algorithms. This chapter puts them to work. We shall see that the central object of fermionic many-body theory, the Slater determinant, is a determinant in exactly the sense of Section 2.3; that changing the single-particle basis is a unitary transformation of the kind studied in Section 1.9; that the Hartree-Fock energy is minimised by solving a symmetric eigenvalue problem with the machinery of Sections 1.24 and 1.25; and that the obstacle which makes the whole subject difficult – the exponential growth of the Hilbert space – is the tensor product of Section 1.10.

Nothing in this chapter is new mathematics. What is new is the physics that the mathematics is asked to carry.

2.1 The many-body Hamiltonian

Throughout these lectures we assume that the interacting part of the Hamiltonian can be approximated by a two-body interaction, so the full N-particle Hamiltonian takes the form

$$\hat{H} = \hat{H}_0 + \hat{H}_I = \sum_{i=1}^{N} \hat{h}_0(x_i) + \sum_{i<j}^{N} \hat{v}(r_{ij}), \tag{2.1}$$

where x_i denotes the coordinates – position and spin – of particle i. The one-body operator splits into a kinetic and a potential part,

$$\hat{h}_0(x_i) = \hat{t}(x_i) + \hat{u}_{\text{ext}}(x_i), \tag{2.2}$$

where $\hat{u}_{\text{ext}}$ is typically a harmonic oscillator potential, the Coulomb attraction of a nucleus, or the self-consistent Hartree-Fock potential derived later in this chapter.

We shall assume that the single-particle functions ψ_α are eigenfunctions of $\hat{h}_0$,

$$\hat{h}_0(x)\,\psi_\alpha(x) = \left(\hat{t}(x) + \hat{u}_{\text{ext}}(x)\right)\psi_\alpha(x) = \varepsilon_\alpha\,\psi_\alpha(x), \tag{2.3}$$

with ε_α the *unperturbed* or non-interacting single-particle energies. In the absence of $\hat{H}_I$ the total energy would simply be the sum of the occupied ε_α. Everything that makes many-body physics hard is contained in the correction to that statement.

Equation (2.3) is an eigenvalue problem, and in practice we solve it exactly as in Section 1.27: discretise on a grid, or expand in a finite basis, and diagonalise the resulting matrix with the algorithms of Sections 1.24 and 1.25. The harmonic oscillator eigenfunctions computed there are the single-particle basis used in the numerical examples of this chapter.

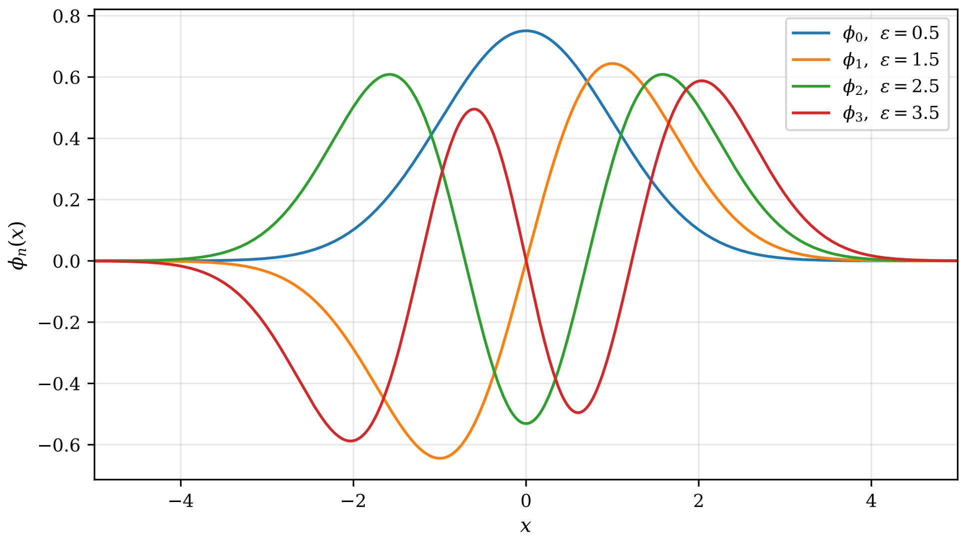

Fig. 2.1 The four lowest orbitals $\phi_n(x)$ of the one-dimensional harmonic oscillator basis used in the numerical examples of this chapter, drawn against the dimensionless coordinate x and labelled by the single-particle energies $\varepsilon_n = n + \frac{1}{2}$ of Eq. (2.3). The functions are generated from the Hermite recursion on a grid of 601 points covering $x \in [-8, 8]$ and normalised on that grid; the nth orbital carries exactly n nodes, and each successive orbital reaches a little further into the classically forbidden region while its peak amplitude drops. Orthonormality is here a numerical result rather than an assumption: the computed overlap matrix reproduces the identity to 1.3×10^{-15}.

The four lowest members of that basis are shown in Figure 2.1, and it is worth pausing over them, because every number quoted later in this chapter is built from these functions. Two features matter. The first is the node structure: ϕ_0 has none, ϕ_1 one, ϕ_2 two, and so on, so that a determinant built from the lowest N of them already oscillates rapidly in each coordinate before any interaction has been switched on. The second is the growth of the spatial extent with n. The ground-state orbital is essentially confined to $|x| \lesssim 2$ and peaks at 0.75, whereas the outermost maximum of ϕ_3 sits at $|x| \simeq 2.0$ and reaches only 0.59, the same normalisation having to be spread over a wider region. A local interaction $\hat{v}(r_{ij})$ therefore couples neighbouring orbitals strongly and distant ones weakly, which is the reason the two-body matrix of Section 2.10 is so far from having full rank. Because the ϕ_n are eigenfunctions of $\hat{h}_0$ the one-body matrix $\langle p|\hat{h}_0|q\rangle = \varepsilon_p \delta_{pq}$ is diagonal by construction and costs nothing; all the labour, and all the physics, sits in the two-body elements.

Finally, the Hamiltonian (2.1) is *symmetric* under the interchange of any two particles. Writing $\hat{P}_{ij}$ for the operator that permutes particles i and j,

$$[\hat{H}, \hat{P}_{ij}] = 0, \tag{2.4}$$

so the eigenfunctions of $\hat{H}$ can be chosen to be eigenfunctions of $\hat{P}_{ij}$ as well.

2.2 The Pauli exclusion principle and antisymmetry

Let Ψ_λ be such a simultaneous eigenfunction. Then

$$\hat{P}_{ij}\Psi_\lambda(x_1, \ldots, x_i, \ldots, x_j, \ldots, x_N) = \beta\, \Psi_\lambda(x_1, \ldots, x_i, \ldots, x_j, \ldots, x_N), \tag{2.5}$$

and since applying the permutation twice returns the original function, $\beta^2 = 1$ and $\beta = \pm 1$. For bosons $\beta = +1$; for fermions, which are our concern here, the Pauli principle requires

$$\beta = -1, \tag{2.6}$$

so the wave function changes sign under every transposition.

An arbitrary product of single-particle functions has no such symmetry, and must be projected onto the antisymmetric subspace. The projector is the *antisymmetrisation operator*

$$\hat{A} = \frac{1}{N!}\sum_{p}(-)^{p}\hat{P}, \tag{2.7}$$

where the sum runs over all $N!$ permutations $\hat{P}$ and p is the parity of each, that is the number of transpositions required to build it. Two properties of $\hat{A}$ will do all the work in Section 2.8. First, since $\hat{H}_0$ and $\hat{H}_I$ are invariant under every permutation,

$$[\hat{H}_0,\hat{A}] = [\hat{H}_I,\hat{A}] = 0. \tag{2.8}$$

Second, $\hat{A}$ is idempotent,

$$\hat{A}^2 = \hat{A}, \tag{2.9}$$

because permuting an already antisymmetrised function reproduces it up to the sign that the prefactor $(-)^p$ removes. A projector, in the sense of Section 1.6: $\hat{A}$ projects the full N-particle space onto its antisymmetric subspace, exactly as $\boldsymbol{P} = |0\rangle\langle 0|$ projects a qubit onto a computational basis state.

2.3 Determinants, and how they are actually computed

For an $n \times n$ matrix $\boldsymbol{A}$, the determinant is defined by summing over all $n!$ permutations of the column indices,

$$|\boldsymbol{A}| = |\boldsymbol{A}| = \sum_{i=1}^{n!}(-1)^{p_i}\hat{P}_i\, a_{11}a_{22}\cdots a_{nn}, \tag{2.10}$$

where $\hat{P}_i$ permutes the column indices and p_i is the number of transpositions needed to restore the natural ordering $a_{11}a_{22}\cdots a_{nn}$. For $n = 2$ the sum has two terms,

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = (-1)^0 a_{11}a_{22} + (-1)^1 a_{12}a_{21}, \tag{2.11}$$

the second obtained by interchanging the column indices 1 and 2.

The resemblance between Eq. (2.10) and the antisymmetriser (2.7) is not an accident, and it is the reason determinants appear in fermionic physics at all: both are the signed sum over the symmetric group, and the determinant is simply what the antisymmetriser produces when applied to a product of single-particle functions.

Two properties we shall need.

The determinant is linear in each column separately,

$$\begin{vmatrix} a_{11} & \cdots & \sum_{k=1}^{n} c_k b_{1k} & \cdots \\ a_{21} & \cdots & \sum_{k=1}^{n} c_k b_{2k} & \cdots \\ \vdots & & \vdots & \\ a_{n1} & \cdots & \sum_{k=1}^{n} c_k b_{nk} & \cdots \end{vmatrix} = \sum_{k=1}^{n} c_k \begin{vmatrix} a_{11} & \cdots & b_{1k} & \cdots \\ a_{21} & \cdots & b_{2k} & \cdots \\ \vdots & & \vdots & \\ a_{n1} & \cdots & b_{nk} & \cdots \end{vmatrix}, \tag{2.12}$$

and if every column of $\boldsymbol{A}$ is a linear combination of the columns of $\boldsymbol{B}$ with coefficient matrix $\boldsymbol{C}$, that is $a_{ij} = \sum_k b_{ik} c_{kj}$ or $\boldsymbol{A} = \boldsymbol{BC}$, then

$$|\boldsymbol{A}| = |\boldsymbol{B}| \cdot |\boldsymbol{C}|. \tag{2.13}$$

Equation (2.13) follows by applying Eq. (2.12) to each column in turn; the reader is invited to verify it for $n = 2$. It is the single most important algebraic fact in this chapter, and Section 2.5 is nothing but its physical interpretation.

A warning about Eq. (2.10).

The definition is a definition, not an algorithm. Evaluating a determinant by summing over permutations costs $\mathscr{O}(n!)$ operations, which for $n = 20$ is 2×10^{18} and for $n = 25$ exceeds the age of the universe on any machine. The LU decomposition of Section 1.15 computes the same number in $\mathscr{O}(n^3)$ operations as the product of the diagonal elements of $\boldsymbol{U}$, Eq. (1.29). Table 2.1 makes the comparison. Every Slater determinant evaluated in a Monte Carlo or configuration-interaction code is computed this way, never from Eq. (2.10).

n	**From Eq. (2.10),** $n!$	**From LU,** $\sim \frac{2}{3} n^3$
5	1.2×10^2	8.3×10^1
10	3.6×10^6	6.7×10^2
20	2.4×10^{18}	5.3×10^3
50	3.0×10^{64}	8.3×10^4

Table 2.1 Operation counts for evaluating an $n \times n$ determinant from the definition and from the LU decomposition of Section 1.15.

2.4 The Slater determinant

We now have everything needed to write down an antisymmetric N-particle wave function. Given N single-particle orbitals $\psi_\alpha, \psi_\beta, \ldots, \psi_\sigma$, the *Slater determinant* is

$$\Phi(x_1, \ldots, x_N; \alpha, \beta, \ldots, \sigma) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_\alpha(x_1) & \psi_\alpha(x_2) & \cdots & \psi_\alpha(x_N) \\ \psi_\beta(x_1) & \psi_\beta(x_2) & \cdots & \psi_\beta(x_N) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_\sigma(x_1) & \psi_\sigma(x_2) & \cdots & \psi_\sigma(x_N) \end{vmatrix}, \tag{2.14}$$

where $\alpha, \beta, \ldots, \sigma$ are the quantum numbers labelling the occupied orbitals. The two defining physical properties follow from two elementary properties of determinants:

- interchanging two particles interchanges two *columns*, which reverses the sign – the wave function is antisymmetric, as Eq. (2.6) demands;
- placing two particles in the same orbital gives two identical *rows*, and the determinant vanishes – no two fermions may occupy the same single-particle state.

The Pauli principle is thus not an extra postulate bolted onto the formalism; it is a property of determinants.

An equivalent form uses the antisymmetriser directly. Writing the unsymmetrised product, the *Hartree function*,

$$\Phi_H(x_1, \ldots, x_N; \alpha, \ldots, \nu) = \psi_\alpha(x_1)\psi_\beta(x_2)\cdots\psi_\nu(x_N), \tag{2.15}$$

we have

$$\Phi = \frac{1}{\sqrt{N!}}\sum_p (-)^p \hat{P}\,\Phi_H = \sqrt{N!}\hat{A}\,\Phi_H. \tag{2.16}$$

This is the form used in the derivations of Section 2.8.

Link to Chapter 1. The Hartree function (2.15) is precisely a tensor product, $|\alpha\rangle \otimes |\beta\rangle \otimes \cdots \otimes |\nu\rangle$, of the kind introduced in Section 1.10. The Slater determinant is its antisymmetric projection. A tensor product of N single-particle states is the least entangled state there is, and its Schmidt rank across any cut is one; antisymmetrisation introduces exactly the correlation demanded by statistics and no more. This is the precise sense in which a Slater determinant is an *uncorrelated* state, and it is made quantitative by the occupation numbers of Section 1.31: for a single Slater determinant they are exactly 1 for the N occupied orbitals and exactly 0 for all the rest. Any deviation from those values is a measure of genuine, dynamical correlation.

The numerical companion `slaterdeterminant.py` verifies both statements. Evaluating Eq. (2.14) for three particles at (x_1, x_2, x_3) and at (x_2, x_1, x_3) gives values that cancel to machine zero, and putting two particles in the same orbital gives $\Phi = -6 \times 10^{-18}$. The eigenvalues of the density matrix $\rho_{\lambda\mu} = \sum_i C^*_{i\lambda} C_{i\mu}$ come out as $(1, 1, 1, 0, 0, 0, 0, 0)$ to machine precision.

2.5 Single-particle bases and unitary transformations

In practice one rarely knows the right orbitals in advance. The standard strategy is to expand them in a fixed, known basis $\{\phi_\lambda\}$ – harmonic oscillator functions, hydrogen-like functions, plane waves, Gaussians – and to vary the expansion coefficients,

$$\psi_p(x) = \sum_\lambda C_{p\lambda}\,\phi_\lambda(x). \tag{2.17}$$

The basis $\{\phi_\lambda\}$ is normally chosen as the eigenfunctions of the one-body Hamiltonian, Eq. (2.3), and is orthonormal. In Dirac notation Eq. (2.17) reads

$$|p\rangle = \sum_\lambda C_{p\lambda}|\lambda\rangle, \qquad \phi_\lambda(\boldsymbol{r}) = \langle \boldsymbol{r}|\lambda\rangle. \tag{2.18}$$

Two questions arise immediately. Does the new basis remain orthonormal? And what happens to the Slater determinant? Chapter 1 has already answered both.

Orthonormality is preserved if and only if $\boldsymbol{C}$ is unitary.

This was proved in Section 1.9: for an orthonormal set $\boldsymbol{v}_j^T\boldsymbol{v}_i = \delta_{ij}$ and a transformation $\boldsymbol{w}_i = \boldsymbol{U}\boldsymbol{v}_i$,

$$\boldsymbol{w}_j^T\boldsymbol{w}_i = \boldsymbol{v}_j^T \underbrace{\boldsymbol{U}^T\boldsymbol{U}}_{=\boldsymbol{I}} \boldsymbol{v}_i = \delta_{ij}, \tag{2.19}$$

so $\langle p|q\rangle = \delta_{pq}$ follows whenever $(C_{p\lambda})$ is unitary. We shall use this constantly: it is what allows us to search over single-particle bases without ever having to re-orthogonalise.

The Slater determinant picks up a phase.

Substituting Eq. (2.17) into Eq. (2.14), every entry of the determinant becomes a sum $\sum_\lambda C_{p\lambda}\phi_\lambda(x_i)$, that is, the matrix of orbitals is the product of the coefficient matrix $\boldsymbol{C}$ with the matrix $\boldsymbol{\Phi}$ of basis functions. By the product rule (2.13),

$$\frac{1}{\sqrt{N!}}\begin{vmatrix} \sum_\lambda C_{p\lambda}\phi_\lambda(x_1) & \cdots & \sum_\lambda C_{p\lambda}\phi_\lambda(x_N) \\ \vdots & \ddots & \vdots \\ \sum_\lambda C_{t\lambda}\phi_\lambda(x_1) & \cdots & \sum_\lambda C_{t\lambda}\phi_\lambda(x_N) \end{vmatrix} = |\boldsymbol{C}|\cdot|\boldsymbol{\Phi}|. \tag{2.20}$$

If $\boldsymbol{C}$ is unitary then $||\boldsymbol{C}|| = 1$, so the rotated Slater determinant differs from the original only by an overall phase. It describes the *same* physical state.

Many-body connection. Equation (2.20) is the algebraic foundation of a fact used everywhere in this book: one may perform any unitary rotation among the occupied orbitals without changing the many-body state, the density, or the total energy. In Hartree-Fock theory this is the freedom that makes canonical and localised orbitals equally valid descriptions of the same determinant, and it is the content of Thouless' theorem. In configuration interaction and coupled-cluster theory it is why the choice of reference orbitals is a matter of convenience rather than principle. The numerical check in `slaterdeterminant.py` makes it concrete: a random orthogonal rotation of three occupied orbitals leaves the energy unchanged to 2.7×10^{-15}, while a rotation that mixes occupied and unoccupied orbitals changes it by more than a factor of two.

Truncation, and what happens when the basis is not orthogonal.

A complete basis $\{|\lambda\rangle\}$ contains infinitely many states, and any calculation must truncate the sum in Eq. (2.17). This is the first and often the largest approximation in a many-body calculation, and its convergence has to be studied explicitly.

Furthermore, some of the most useful bases are not orthogonal at all. Gaussian basis sets in quantum chemistry, oscillator states of different oscillator length, and generator-coordinate states all have non-trivial overlap matrices $S_{ij} = \langle\phi_i|\phi_j\rangle$, and Eq. (2.19) no longer applies. The eigenvalue problem then becomes the generalised one, $\boldsymbol{H}\boldsymbol{c} = E\boldsymbol{S}\boldsymbol{c}$, treated in Section 1.29: one diagonalises $\boldsymbol{S}$, discards the directions whose eigenvalues lie below a threshold, and transforms with $\boldsymbol{X} = \boldsymbol{V}\boldsymbol{s}^{-1/2}$. The threshold is a physical choice, and Table 1.6 shows how much ill-conditioning it removes.

2.6 Particle-hole notation and the Fermi level

Before deriving the energy it is worth fixing the index conventions that will be used for the rest of the book. Given a reference Slater determinant – usually the one obtained by filling the N lowest single-particle levels – the single-particle states divide into those that are occupied in the reference and those that are not. The dividing line is the *Fermi level*, denoted F. We write

$$\begin{array}{ll} i,j,k,l,\dots \le F & \text{occupied states, or } \textit{hole} \text{ states,} \\ a,b,c,d,\dots > F & \text{unoccupied states, or } \textit{particle} \text{ states,} \\ p,q,r,s,\dots & \text{general single-particle states.} \end{array} \tag{2.21}$$

The names come from the picture in which the reference determinant is treated as a new vacuum: exciting a particle from an occupied state i to an unoccupied state a creates a particle in a and leaves a hole in i. Chapter 3 develops this into the particle-hole formalism of second quantisation, where it becomes the natural language for perturbation theory, coupled-cluster theory and the shell model. For now it is simply a convention, but it is one worth adopting at once: every equation from here on is easier to read with it.

2.7 The variational principle

Let E_0 be the exact ground-state energy of $\hat{H}$. For any normalised trial function Φ,

$$E_0 \le E[\Phi] = \int \Phi^* \hat{H}\, \Phi\, d\boldsymbol{\tau}, \qquad \int \Phi^* \Phi\, d\boldsymbol{\tau} = 1, \tag{2.22}$$

with the shorthand $d\boldsymbol{\tau} = dx_1 dx_2 \cdots dx_N$. The proof is the spectral decomposition of Section 1.11 and nothing more. Expanding Φ in the exact eigenstates, $|\Phi\rangle = \sum_k c_k |\Psi_k\rangle$ with $\hat{H}|\Psi_k\rangle = E_k|\Psi_k\rangle$ and $\sum_k |c_k|^2 = 1$, gives

$$E[\Phi] = \sum_k |c_k|^2 E_k \;\ge\; E_0 \sum_k |c_k|^2 = E_0, \tag{2.23}$$

with equality only when Φ is the ground state itself.

This one inequality is the engine of most of what follows. Restricting the trial function to a single Slater determinant and minimising gives Hartree-Fock theory; allowing a linear combination of determinants gives configuration interaction; parametrising it by a quantum circuit gives the variational quantum eigensolver of Section 2.16. It is also the reason the Lanczos algorithm of Section 1.26 is so well suited to many-body problems: the lowest Ritz value is the minimum of the Rayleigh quotient over the Krylov space, and therefore an upper bound on E_0 that can only improve as the space grows.

2.8 Expectation values with a Slater determinant

We now compute $E[\Phi]$ for the Slater determinant (2.14). The calculation uses only Eqs. (2.8), (2.9) and the orthonormality of the single-particle functions, and it is worth following in full: the same manipulations recur throughout many-body theory, and second quantisation exists largely to make them automatic.

The one-body contribution.

Using the form (2.16) of the Slater determinant,

$$\int \Phi^* \hat{H}_0 \Phi \, d\boldsymbol{\tau} = N! \int \Phi_H^* \hat{A} \hat{H}_0 \hat{A} \, \Phi_H \, d\boldsymbol{\tau}. \tag{2.24}$$

Because $\hat{H}_0$ commutes with $\hat{A}$ we may move the left-hand antisymmetriser through it, and because $\hat{A}^2 = \hat{A}$ the two then collapse into one,

$$\int \Phi^* \hat{H}_0 \Phi \, d\boldsymbol{\tau} = N! \int \Phi_H^* \hat{H}_0 \hat{A} \, \Phi_H \, d\boldsymbol{\tau}. \tag{2.25}$$

Replacing $\hat{A}$ by its definition (2.7), which cancels the factor $N!$, and $\hat{H}_0$ by the sum of one-body operators,

$$\int \Phi^* \hat{H}_0 \Phi \, d\boldsymbol{\tau} = \sum_{i=1}^{N} \sum_{p} (-)^p \int \Phi_H^* \hat{h}_0(x_i) \hat{P} \, \Phi_H \, d\boldsymbol{\tau}. \tag{2.26}$$

Now the key step. The integrand factorises into single-particle integrals, and $\hat{h}_0(x_i)$ touches only one coordinate. If $\hat{P}$ permutes any two particles, at least one of the remaining factors is an overlap $\langle \mu | \nu \rangle$ between two *different* orbitals, which vanishes. Only the identity permutation survives,

$$\int \Phi^* \hat{H}_0 \Phi \, d\boldsymbol{\tau} = \sum_{i=1}^{N} \int \Phi_H^* \hat{h}_0(x_i) \Phi_H \, d\boldsymbol{\tau}, \tag{2.27}$$

and integrating out the $N-1$ spectator coordinates, each of which contributes unity, leaves

$$\int \Phi^* \hat{H}_0 \Phi \, d\boldsymbol{\tau} = \sum_{\mu=1}^{N} \int \psi_\mu^*(x) \, \hat{h}_0 \, \psi_\mu(x) dx. \tag{2.28}$$

Introducing the shorthand

$$\langle \mu | \hat{h}_0 | \mu \rangle = \int \psi_\mu^*(x) \hat{h}_0 \psi_\mu(x) dx, \tag{2.29}$$

this is simply

$$\int \Phi^* \hat{H}_0 \Phi \, d\boldsymbol{\tau} = \sum_{\mu=1}^{N} \langle \mu | \hat{h}_0 | \mu \rangle. \tag{2.30}$$

The one-body energy is the sum of the diagonal matrix elements over the occupied orbitals – the result one would have guessed, now derived.

The two-body contribution.

The same three steps apply,

$$\int \Phi^* \hat{H}_I \Phi \, d\boldsymbol{\tau} = N! \int \Phi_H^* \hat{A} \hat{H}_I \hat{A} \, \Phi_H \, d\boldsymbol{\tau} = \sum_{i<j}^{N} \sum_{p} (-)^p \int \Phi_H^* \hat{v}(r_{ij}) \hat{P} \, \Phi_H \, d\boldsymbol{\tau}, \tag{2.31}$$

but the conclusion is different. Because $\hat{v}(r_{ij})$ depends on *two* coordinates, the permutation that exchanges precisely those two particles no longer produces a vanishing overlap. Two permutations survive: the identity, and the transposition $\hat{P}_{ij}$. Hence

$$\int \Phi^* \hat{H}_I \Phi \, d\boldsymbol{\tau} = \sum_{i<j}^{N} \int \Phi_H^* \, \hat{v}(r_{ij}) \left(1 - \hat{P}_{ij}\right) \Phi_H \, d\boldsymbol{\tau}. \tag{2.32}$$

Writing out the two terms and integrating over the spectator coordinates,

$$\int \Phi^* \hat{H}_I \Phi \, d\boldsymbol{\tau} = \frac{1}{2} \sum_{\mu=1}^{N} \sum_{\nu=1}^{N} \left[\int \psi_\mu^*(x_i) \psi_\nu^*(x_j)\, \hat{v}(r_{ij})\, \psi_\mu(x_i) \psi_\nu(x_j)\, dx_i dx_j \right.$$
$$\left. - \int \psi_\mu^*(x_i) \psi_\nu^*(x_j)\, \hat{v}(r_{ij})\, \psi_\nu(x_i) \psi_\mu(x_j)\, dx_i dx_j \right], \tag{2.33}$$

where the factor $\frac{1}{2}$ compensates for the unrestricted double sum, which counts every pair twice. The first term is the *direct* or *Hartree* term: it is the classical electrostatic energy of one charge distribution in the field of another. The second is the *exchange* or *Fock* term, and it has no classical counterpart whatsoever; it exists solely because the wave function is antisymmetric, and it is the sole trace left in the energy by the transposition $\hat{P}_{ij}$ in Eq. (2.32).

Notation for the matrix elements.

Writing the single-particle functions as overlaps, $\psi_\alpha(x) = \langle x|\alpha\rangle$, we abbreviate the two integrals as

$$\langle \mu\nu|\hat{v}|\mu\nu\rangle = \int \psi_\mu^*(x_i) \psi_\nu^*(x_j) \hat{v}(r_{ij}) \psi_\mu(x_i) \psi_\nu(x_j)\, dx_i dx_j, \tag{2.34}$$

$$\langle \mu\nu|\hat{v}|\nu\mu\rangle = \int \psi_\mu^*(x_i) \psi_\nu^*(x_j) \hat{v}(r_{ij}) \psi_\nu(x_i) \psi_\mu(x_j)\, dx_i dx_j, \tag{2.35}$$

and define the *antisymmetrised* matrix element

$$\langle \mu\nu|\hat{v}|\mu\nu\rangle_{\mathrm{AS}} = \langle \mu\nu|\hat{v}|\mu\nu\rangle - \langle \mu\nu|\hat{v}|\nu\mu\rangle. \tag{2.36}$$

Note that $\langle \mu\mu|\hat{v}|\mu\mu\rangle_{\mathrm{AS}} = 0$: a particle does not interact with itself. The self-interaction that the Hartree term would otherwise contain is cancelled exactly by the exchange term, which is one of the reasons Hartree-Fock theory is better behaved than the Hartree theory that preceded it.

2.9 The energy as a functional of the coefficients

Collecting Eqs. (2.30) and (2.33), the expectation value of the Hamiltonian in a Slater determinant is

$$E[\Phi] = \sum_{\mu=1}^{N} \langle \mu|\hat{h}_0|\mu\rangle + \frac{1}{2} \sum_{\mu=1}^{N} \sum_{\nu=1}^{N} \langle \mu\nu|\hat{v}|\mu\nu\rangle_{\mathrm{AS}}, \tag{2.37}$$

or, in the particle-hole notation of Section 2.6,

$$E[\Phi] = \sum_{i=1}^{N} \langle i|\hat{h}_0|i\rangle + \frac{1}{2} \sum_{i,j=1}^{N} \langle ij|\hat{v}|ij\rangle_{\mathrm{AS}}. \tag{2.38}$$

This is a functional of the occupied orbitals. To turn it into something we can minimise we insert the expansion (2.17) and let the coefficients carry the variation,

$$E[\Psi] = \sum_{i=1}^{N} \sum_{\alpha\beta} C_{i\alpha}^* C_{i\beta} \langle \alpha|\hat{h}_0|\beta\rangle + \frac{1}{2} \sum_{i,j=1}^{N} \sum_{\alpha\beta\gamma\delta} C_{i\alpha}^* C_{j\beta}^* C_{i\gamma} C_{j\delta} \langle \alpha\beta|\hat{v}|\gamma\delta\rangle_{\mathrm{AS}}. \tag{2.39}$$

The Greek indices run over the full, fixed basis; the Latin indices run over the N occupied orbitals only. Everything on the right-hand side except the C's is computed once and stored.

Minimising Eq. (2.39) with respect to $C^*_{i\alpha}$ subject to the orthonormality constraint yields the Hartree-Fock equations. We defer the derivation, but the structure of the answer can be read off immediately and is worth stating now. Define the *density matrix* and the *Fock matrix*

$$\rho_{\gamma\delta} = \sum_{i=1}^{N} C^*_{i\gamma} C_{i\delta}, \qquad f_{\alpha\beta} = \langle\alpha|\hat{h}_0|\beta\rangle + \sum_{\gamma\delta} \rho_{\gamma\delta} \langle\alpha\gamma|\hat{v}|\beta\delta\rangle_{\mathrm{AS}}. \tag{2.40}$$

The condition for a stationary point is then

$$\sum_{\beta} f_{\alpha\beta}\, C_{i\beta} = \varepsilon_i\, C_{i\alpha}, \tag{2.41}$$

a Hermitian eigenvalue problem of exactly the kind solved in Sections 1.24 and 1.25 – with the single complication that the matrix $\boldsymbol{f}$ depends, through $\boldsymbol{\rho}$, on the eigenvectors we are trying to find. One therefore guesses, builds $\boldsymbol{f}$, diagonalises, rebuilds and repeats until nothing changes. This is the *self-consistent field* procedure, and it is an iteration in the sense of Section 1.17, with a matrix diagonalisation in place of a single sweep.

Summary: Hartree-Fock is Chapter 1 applied to fermions. Expand the states in an orthonormal basis (Section 1.8); change basis with a unitary transformation (Section 1.9), which leaves the determinant invariant up to a phase by the product rule (Section 2.3); minimise a quadratic form subject to a constraint, which produces a Hermitian eigenvalue problem (Section 1.11); solve it by Householder reduction followed by the QL algorithm (Sections 1.24 and 1.25); and iterate to self-consistency. Every step has already been developed.

The minimal implementation in `slaterdeterminant.py` does exactly this, using the eigenvalue solver written in Chapter 1. For spinless fermions in a one-dimensional trap with a softened Coulomb repulsion, and a basis of eight harmonic oscillator orbitals, it converges in nine iterations and lowers the energy below that of the naive determinant built from the N lowest oscillator states, as shown in Table 2.2.

N	E (lowest N orbitals)	E (Hartree-Fock)	**Gain**
2	2.69018342	2.66308878	-2.7×10^{-2}
3	6.46801021	6.39016133	-7.8×10^{-2}
4	11.77084712	11.62305385	-1.5×10^{-1}

Table 2.2 Self-consistent Hartree-Fock energies for N spinless fermions in a one-dimensional harmonic trap with a softened Coulomb interaction, in a basis of eight oscillator orbitals. The middle column is the energy of the determinant built from the N lowest oscillator states; the Hartree-Fock solution optimises the orbitals and always lies below it.

For $N=2$ the exact ground state in the same truncated basis can be obtained by diagonalising the Hamiltonian in the space of antisymmetric pairs, and the difference

$$E_{\mathrm{corr}} = E_{\mathrm{exact}} - E_{\mathrm{HF}} \tag{2.42}$$

defines the *correlation energy*. Here it is -3.86×10^{-3}: the system is weakly correlated, and the largest amplitude in the exact state, 0.9930, sits on the Hartree-Fock configuration. Re-covering the remaining 0.7% of the amplitude is what every method in the rest of this book is built to do.

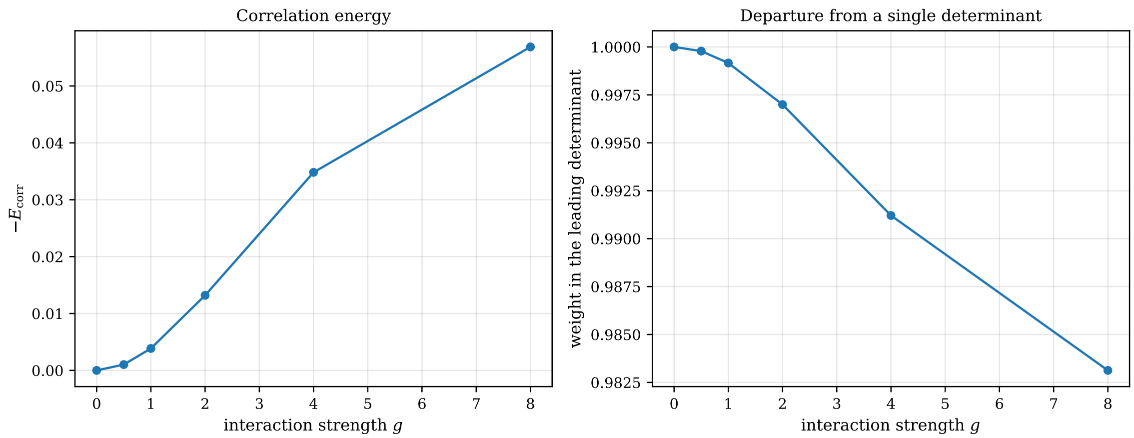

Fig. 2.2 Breakdown of the single-determinant description as the interaction is turned up, for two spinless fermions in the one-dimensional trap and the same basis of eight oscillator orbitals. The left panel shows the correlation energy of Eq. (2.42), plotted as $-E_{\mathrm{corr}}$ against the strength g of the softened Coulomb repulsion; the right panel shows the weight carried by the leading determinant in the exact state, that is the sum of the two largest natural occupation numbers of Section 1.31, which equals unity for a single Slater determinant. Both curves are flat at the origin and then depart quadratically: at $g = 1$ the correlation energy is -3.9×10^{-3} and the leading weight 0.9992, while at $g = 8$ they have grown to -5.7×10^{-2} and fallen to 0.9831.

How the single determinant fails, and how fast, is shown in Figure 2.2, where the whole calculation is repeated for interaction strengths g between 0 and 8. At $g = 0$ the Hartree-Fock determinant is the exact ground state, the correlation energy vanishes identically and the leading weight is exactly one; this is the free-fermion limit, and it is the only point at which the ansatz is exact. Away from it both panels leave their starting values quadratically: doubling g from 0.5 to 1 multiplies $-E_{\mathrm{corr}}$ by 3.8, and doubling it again to 2 multiplies it by a further 3.4, both close to the factor of four that second-order perturbation theory demands, since the leading correction to the energy is $\mathscr{O}(g^2)$ and the leading admixture of the doubly excited determinant is $\mathscr{O}(g)$. The same quadratic law governs the right panel: the deficit $1 - w$ grows from 2.2×10^{-4} at $g = 0.5$ to 8.4×10^{-4} at $g = 1$ and 3.0×10^{-3} at $g = 2$. Beyond $g \simeq 2$ the growth slackens markedly – from $g = 4$ to $g = 8$ the correlation energy rises by only 63% rather than the fourfold factor of the perturbative regime – and the reason is instructive. The reference is not held fixed as g is increased: the self-consistent field of Eq. (2.40) relaxes the orbitals so that the determinant keeps absorbing as much of the interaction as a product state can hold, and only what is left over counts as correlation. Notice finally the order of magnitude of the vertical scale on the left. Even at the strongest coupling shown, E_{corr} is -5.7×10^{-2} against a total energy of 6.13, less than one per cent, while the fraction of the wave function that is missing is not much larger. That is the awkward arithmetic of many-body theory: the quantity we are after is a small residue of a large number, and it must nevertheless be computed to a relative accuracy that a 1% error in the total energy would destroy entirely.

2.10 The price of the two-body matrix elements

Equation (2.39) contains a sum over four basis indices, so a calculation in a basis of n single-particle states requires n^4 two-body matrix elements. For $n = 100$ that is 10^8 numbers, for

$n = 1000$ it is 10^{12}, and long before the many-body problem becomes hard the mere storage of the interaction does.

Symmetries help. For a real, local interaction the matrix elements obey

$$\langle \alpha\beta|\hat{v}|\gamma\delta\rangle = \langle \beta\alpha|\hat{v}|\delta\gamma\rangle = \langle \gamma\delta|\hat{v}|\alpha\beta\rangle, \tag{2.43}$$

which reduces the count by roughly a factor of eight, and rotational or translational invariance can reduce it much further by making most elements vanish. But the scaling remains $\mathscr{O}(n^4)$.

The structural remedy is the one developed in Section 1.32. Reading the pairs $(\alpha\gamma)$ and $(\beta\delta)$ as composite indices turns the four-index array into an ordinary matrix, which for a Coulomb-like interaction is positive semidefinite with rapidly decaying eigenvalues, so that

$$\langle \alpha\beta|\hat{v}|\gamma\delta\rangle = \sum_{\eta=0}^{M-1} L^{\eta}_{\alpha\gamma} L^{\eta}_{\beta\delta}, \qquad M \ll n^2. \tag{2.44}$$

The storage drops from n^4 to Mn^2 and the contractions in Eq. (2.39) can be performed on the factors rather than on the full array. For the eight-orbital basis used above, the 64×64 pair matrix has numerical rank 15, not 64: less than a quarter of the nominal size. This is the same effect that Table 1.9 quantified, and it is what makes large-basis coupled-cluster and perturbation-theory calculations possible at all.

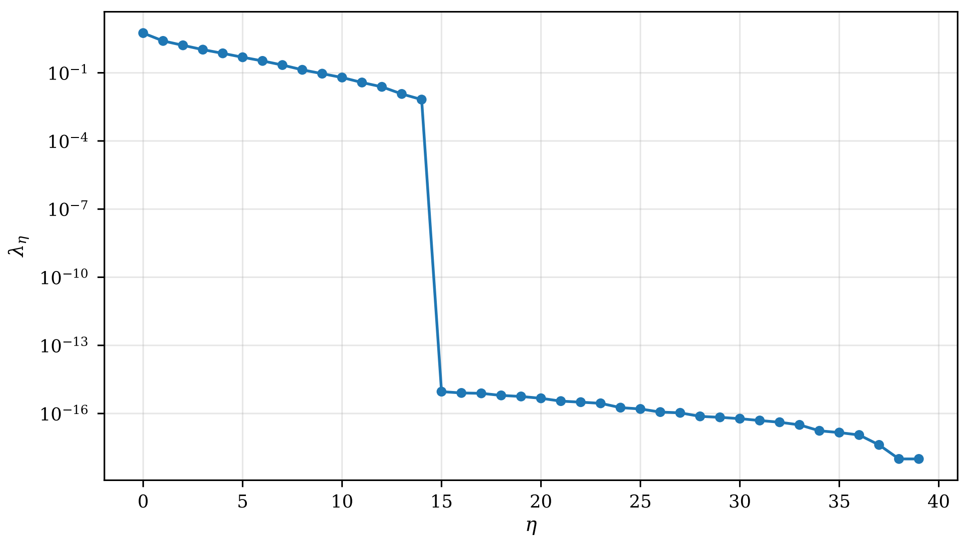

Fig. 2.3 Eigenvalues λ_η of the two-body interaction read as a matrix in the composite pair index $(\alpha\gamma), (\beta\delta)$, for the eight-orbital oscillator basis of Figure 2.1 with the softened Coulomb repulsion. The matrix is 64×64 and its eigenvalues are plotted in descending order on a logarithmic scale; the first forty are shown. The spectrum decays smoothly from $\lambda_0 = 5.63$ through $\lambda_1 = 2.54$ and $\lambda_7 = 0.22$ to $\lambda_{14} = 6.6 \times 10^{-3}$, and then falls off a cliff to the 10^{-15} level, so that the numerical rank is 15 rather than 64 and the sum in Eq. (2.44) may be truncated at $M = 15$ with no loss whatsoever.

Figure 2.3 shows where that rank comes from. The leading fifteen eigenvalues fall smoothly and almost geometrically, each roughly two thirds of its predecessor, spanning between them not quite three orders of magnitude; the sixteenth is 9×10^{-16}, close to thirteen orders of magnitude below the fifteenth, and everything after it hovers at the same level. A drop of that size in a single step is not physics but arithmetic: those directions are exactly null, and what is plotted for $\eta \geq 15$ is the rounding error of the eigenvalue solver rather than any property of the interaction. The origin of the null space is easy to identify, and it is worth identifying because it shows that the rank is a property of the basis rather than an accident. A local interaction $v(x,y)$ enters only through the pair densities $\rho_{\alpha\gamma}(x) = \phi_\alpha(x)\phi_\gamma(x)$, and a product of two Hermite functions is a polynomial of degree $\alpha+\gamma$ multiplied by e^{-x^2}. With eight orbitals the degree can only take the fifteen values $0,1,\ldots,14$, so however the 64 index pairs are arranged they span a space of just $2n-1=15$ independent functions, and the pair matrix – a Gram matrix of those functions in the metric supplied by the kernel – cannot have rank above that number. The practical consequence is Eq. (2.44) with $M=15$: storage falls from $n^4=4096$ numbers to $Mn^2=960$, a factor of 4.3 in a case small enough to make the counting transparent, and the factor grows with n because M is governed by the smoothness of the interaction rather than by the size of the basis. The reader should note how closely this repeats the Schmidt-spectrum argument of Section 1.30 and the Eckart-Young theorem: in both places a physically meaningful matrix turns out to be well approximated by one of much lower rank, and in both places the eigenvalue plot on a logarithmic axis is the diagnostic that tells us where to cut.

2.11 Evaluating Slater determinants in Monte Carlo simulations

The energy functional of Section 2.9 needs the Slater determinant once. A Monte Carlo calculation needs it several million times, and needs its gradient and its Laplacian as well. The variational and diffusion Monte Carlo methods developed later in this book propose a new configuration, accept or reject it, and repeat; the local energy requires

$$\frac{\nabla_i|\boldsymbol{D}|}{|\boldsymbol{D}|} \qquad \text{and} \qquad \frac{\nabla_i^2|\boldsymbol{D}|}{|\boldsymbol{D}|} \tag{2.45}$$

for every particle i at every step. Since the cost of a single determinant is $\mathscr{O}(N^3)$ by Section 2.3, and there are $N\cdot d$ coordinates to differentiate, brute-force evaluation costs

$$Nd\cdot\mathscr{O}(N^3)\sim\mathscr{O}(d\cdot N^4) \tag{2.46}$$

per sweep. This is prohibitive, and it is entirely avoidable. The technical material of this section and the next reduces it to $\mathscr{O}(d\cdot N^2)$. It is placed here because it is nothing but linear algebra applied to the object introduced in Section 2.4; the physics that makes it necessary comes later.

The Slater matrix.

Define the $N\times N$ matrix $\boldsymbol{D}$ with elements

$$d_{ij}=\phi_j(\boldsymbol{r}_i), \tag{2.47}$$

where ϕ_j is a single-particle wave function. Rows are labelled by particles and columns by the single-particle states. The Slater determinant of Eq. (2.14) is $|\boldsymbol{D}|/\sqrt{N!}$.

The observation that makes everything work is this: *if we move only one particle at a time, only one row of* $\boldsymbol{D}$ *changes*. Recomputing the whole determinant then throws away almost all of the work already done. The solution is to keep track of the *inverse* of the Slater matrix.

The ratio of two determinants.

Let $\boldsymbol{r}^{\text{old}}$ be the current configuration and $\boldsymbol{r}^{\text{new}}$ the proposed one, differing only in the position of particle i. The Metropolis test needs the ratio

$$R \equiv \frac{|\boldsymbol{D}(\boldsymbol{r}^{\text{new}})|}{|\boldsymbol{D}(\boldsymbol{r}^{\text{old}})|}. \tag{2.48}$$

Expanding each determinant along row i in terms of the cofactors C_{ij},

$$R = \frac{\sum_{j=1}^{N} d_{ij}(\boldsymbol{r}^{\text{new}})\, C_{ij}(\boldsymbol{r}^{\text{new}})}{\sum_{j=1}^{N} d_{ij}(\boldsymbol{r}^{\text{old}})\, C_{ij}(\boldsymbol{r}^{\text{old}})}. \tag{2.49}$$

Now recall that the cofactor C_{ij} is the signed minor obtained by *deleting* row i and column j, so the ith row of the cofactor matrix does not depend on the ith row of $\boldsymbol{D}$ at all. Since the two configurations differ only in that row,

$$C_{ij}(\boldsymbol{r}^{\text{new}}) = C_{ij}(\boldsymbol{r}^{\text{old}}) \qquad \forall\, j \in \{1,\ldots,N\}. \tag{2.50}$$

Next, the inverse of a matrix is the transposed cofactor matrix divided by the determinant,

$$d_{ij}^{-1} = \frac{C_{ji}}{|\boldsymbol{D}|}, \tag{2.51}$$

and by definition $\sum_k d_{ik} d_{kj}^{-1} = \delta_{ij}$. Substituting Eq. (2.50) into Eq. (2.49) and replacing the cofactors using Eq. (2.51), the denominator becomes exactly unity, and we are left with

$$\boxed{R = \sum_{j=1}^{N} d_{ij}(\boldsymbol{r}^{\text{new}})\, d_{ji}^{-1}(\boldsymbol{r}^{\text{old}}) = \sum_{j=1}^{N} \phi_j(\boldsymbol{r}_i^{\text{new}})\, d_{ji}^{-1}(\boldsymbol{r}^{\text{old}})} \tag{2.52}$$

The ratio of two $N \times N$ determinants is a single dot product between the vector of single-particle functions evaluated at the new position and the ith column of the old inverse. The cost is $\mathscr{O}(N)$ instead of $\mathscr{O}(N^3)$, and no determinant is computed at all.

Updating the inverse.

The price is that we must maintain $\boldsymbol{D}^{-1}$. If the move is accepted, the inverse is repaired in $\mathscr{O}(N^2)$ operations rather than recomputed in $\mathscr{O}(N^3)$. For every column $j \neq i$ form

$$S_j = \Big(\boldsymbol{D}(\boldsymbol{r}^{\text{new}})\, \boldsymbol{D}^{-1}(\boldsymbol{r}^{\text{old}})\Big)_{ij} = \sum_{l=1}^{N} d_{il}(\boldsymbol{r}^{\text{new}})\, d_{lj}^{-1}(\boldsymbol{r}^{\text{old}}), \tag{2.53}$$

and update

$$d_{kj}^{-1}(\boldsymbol{r}^{\text{new}}) = d_{kj}^{-1}(\boldsymbol{r}^{\text{old}}) - \frac{S_j}{R}\, d_{ki}^{-1}(\boldsymbol{r}^{\text{old}}), \qquad k = 1,\ldots,N, \quad j \neq i. \tag{2.54}$$

The ith column is simply rescaled,

$$d_{ki}^{-1}(\boldsymbol{r}^{\text{new}}) = \frac{1}{R}\, d_{ki}^{-1}(\boldsymbol{r}^{\text{old}}), \qquad k = 1,\ldots,N. \tag{2.55}$$

Equations (2.54) and (2.55) are the Sherman-Morrison formula specialised to a rank-one change of a single row. Note that both contain a division by R: the algorithm degrades precisely when the proposed determinant is small, a point we return to in Section 2.14.

```
def ratio(self, i, r_new):
    """R = |D(r_new)| / |D(r_old)|, in O(N) operations."""
    row = np.array([self.s.basis.value(j, r_new)[0]
                    for j in range(self.s.N)])
    return float(row @ self.s.Dinv[:, i]), row


def accept(self, i, r_new, R=None, row=None):
    """Update D and D^{-1} after the move of particle i is accepted."""
    if row is None:
        R, row = self.ratio(i, r_new)
    Dinv = self.s.Dinv

    # S_j = (D(new) D^{-1}(old))_{ij} for every column j
    S = row @ Dinv                          # length N, S[i] == R
    new_inv = Dinv.copy()
    for j in range(self.s.N):
        if j != i:
            new_inv[:, j] = Dinv[:, j] - (S[j] / R) * Dinv[:, i]
    new_inv[:, i] = Dinv[:, i] / R

    self.s.Dinv = new_inv
    self.s.D[i, :] = row
    self.s.positions[i] = r_new
```

For six particles in a two-dimensional oscillator basis the ratio computed from Eq. (2.52) agrees with the brute-force quotient of two determinants to 2×10^{-16}, as `slater_update.py` confirms.

2.12 Gradients, the Laplacian and the quantum force

The same argument applies to derivatives, and this is where the real saving lies. Differentiating the Slater determinant with respect to the coordinates of particle i changes only the ith row of $\boldsymbol{D}$, exactly as moving particle i did. The derivation leading to Eq. (2.52) therefore goes through unchanged with d_{ij} replaced by its derivative,

$$\frac{\nabla_i|\boldsymbol{D}(\boldsymbol{r})|}{|\boldsymbol{D}(\boldsymbol{r})|}=\sum_{j=1}^{N}\nabla_i d_{ij}(\boldsymbol{r})\,d_{ji}^{-1}(\boldsymbol{r})=\sum_{j=1}^{N}\nabla_i\phi_j(\boldsymbol{r}_i)\,d_{ji}^{-1}(\boldsymbol{r}), \tag{2.56}$$

$$\frac{\nabla_i^2|\boldsymbol{D}(\boldsymbol{r})|}{|\boldsymbol{D}(\boldsymbol{r})|}=\sum_{j=1}^{N}\nabla_i^2 d_{ij}(\boldsymbol{r})\,d_{ji}^{-1}(\boldsymbol{r})=\sum_{j=1}^{N}\nabla_i^2\phi_j(\boldsymbol{r}_i)\,d_{ji}^{-1}(\boldsymbol{r}). \tag{2.57}$$

To obtain every derivative of the Slater determinant we therefore need only the derivatives of the *single-particle* functions and the elements of the inverse Slater matrix. Each derivative is an $\mathscr{O}(N)$ dot product, and there are $d\cdot N$ of them, so the total evaluation costs $\mathscr{O}(d\cdot N^2)$. With the $\mathscr{O}(N^2)$ update of the inverse the overall scaling is no worse, which should be compared with the $\mathscr{O}(d\cdot N^4)$ of Eq. (2.46). Table 2.3 collects the counts.

Equation (2.56) is the *quantum force* (up to a factor of two) that drives the importance-sampled random walk of diffusion Monte Carlo, and Eq. (2.57) supplies the kinetic energy in the local energy. Both are therefore evaluated at every single step of the simulation.

Quantity	Brute force	With the inverse
Ratio R for one move	$\mathcal{O}(N^3)$	$\mathcal{O}(N)$
Update of $\boldsymbol{D}^{-1}$	$\mathcal{O}(N^3)$	$\mathcal{O}(N^2)$
All gradients and Laplacians	$\mathcal{O}(dN^4)$	$\mathcal{O}(dN^2)$

Table 2.3 Operation counts for the quantities a Monte Carlo step requires, with and without maintaining the inverse Slater matrix.

A practical remark.

In almost all cases one has closed-form expressions for the single-particle wave functions. It is then worth differentiating them analytically and writing separate functions for $\nabla_i \phi_j$ and $\nabla_i^2 \phi_j$. Finite differences would cost both accuracy and speed, and the derivatives are needed far too often to tolerate either. The check in `slater_update.py` compares the analytic Eqs. (2.56) and (2.57) against numerical differentiation of $\ln|\boldsymbol{D}|$ and finds agreement to 5×10^{-10} and 9×10^{-6} respectively – the residual being the error of the finite differences, not of the formulae.

2.13 Spin factorisation of the Slater determinant

A further and quite substantial saving follows from a property of the spin degree of freedom. Consider beryllium, four electrons in the $1s$ and $2s$ orbitals. The Slater determinant reads

$$\Phi(\boldsymbol{r}_1,\boldsymbol{r}_2,\boldsymbol{r}_3,\boldsymbol{r}_4) = \frac{1}{\sqrt{4!}} \begin{vmatrix} \psi_{100\uparrow}(\boldsymbol{r}_1) & \psi_{100\uparrow}(\boldsymbol{r}_2) & \psi_{100\uparrow}(\boldsymbol{r}_3) & \psi_{100\uparrow}(\boldsymbol{r}_4) \\ \psi_{100\downarrow}(\boldsymbol{r}_1) & \psi_{100\downarrow}(\boldsymbol{r}_2) & \psi_{100\downarrow}(\boldsymbol{r}_3) & \psi_{100\downarrow}(\boldsymbol{r}_4) \\ \psi_{200\uparrow}(\boldsymbol{r}_1) & \psi_{200\uparrow}(\boldsymbol{r}_2) & \psi_{200\uparrow}(\boldsymbol{r}_3) & \psi_{200\uparrow}(\boldsymbol{r}_4) \\ \psi_{200\downarrow}(\boldsymbol{r}_1) & \psi_{200\downarrow}(\boldsymbol{r}_2) & \psi_{200\downarrow}(\boldsymbol{r}_3) & \psi_{200\downarrow}(\boldsymbol{r}_4) \end{vmatrix}. \tag{2.58}$$

As written this determinant *vanishes*, because the spatial parts of the spin-up and spin-down rows are identical: rows one and two are the same function of the spatial coordinates, and so are rows three and four. The spin labels carry the distinction, and a determinant of spatial functions alone cannot see them.

Expanding Eq. (2.58) in terms of 2×2 determinants built separately from the spin-up and spin-down orbitals gives

$$\begin{aligned} \Phi \;\propto\; & \det\uparrow(1,2)\det\downarrow(3,4) - \det\uparrow(1,3)\det\downarrow(2,4) \\ & - \det\uparrow(1,4)\det\downarrow(3,2) + \det\uparrow(2,3)\det\downarrow(1,4) \\ & - \det\uparrow(2,4)\det\downarrow(1,3) + \det\uparrow(3,4)\det\downarrow(1,2), \end{aligned} \tag{2.59}$$

with

$$\det\uparrow(1,2) = \frac{1}{\sqrt{2}} \begin{vmatrix} \psi_{100\uparrow}(\boldsymbol{r}_1) & \psi_{100\uparrow}(\boldsymbol{r}_2) \\ \psi_{200\uparrow}(\boldsymbol{r}_1) & \psi_{200\uparrow}(\boldsymbol{r}_2) \end{vmatrix}, \qquad \det\downarrow(3,4) = \frac{1}{\sqrt{2}} \begin{vmatrix} \psi_{100\downarrow}(\boldsymbol{r}_3) & \psi_{100\downarrow}(\boldsymbol{r}_4) \\ \psi_{200\downarrow}(\boldsymbol{r}_3) & \psi_{200\downarrow}(\boldsymbol{r}_4) \end{vmatrix}. \tag{2.60}$$

The total is of course still zero; the expansion has merely reorganised the sum over spin assignments.

The Moskowitz-Kalos ansatz.

What we would like is to avoid summing over spin variables altogether, particularly when the interaction does not depend on spin. It can be shown, see Moskowitz and Kalos [2], that for the variational energy one may replace the full determinant by the single product

$$\Phi(\boldsymbol{r}_1,\ldots,\boldsymbol{r}_N) \;\propto\; \det\uparrow \cdot \det\downarrow, \tag{2.61}$$

in which $\det\uparrow$ involves only the electrons with spin up and $\det\downarrow$ only those with spin down. This ansatz is *not* antisymmetric under the exchange of two electrons of opposite spin, but as long as the Hamiltonian is spin independent it yields the same expectation value for the energy as the full determinant. The reason is visible in Section 2.8: the exchange term of Eq. (2.33) connects only orbitals with the same spin, so the missing antisymmetry never enters the energy.

The saving.

The full determinant factorises,

$$|\boldsymbol{D}| = |\boldsymbol{D}|_{\uparrow} \cdot |\boldsymbol{D}|_{\downarrow}, \tag{2.62}$$

and the combined dimensionality of the two smaller determinants equals that of the original. The ratio then factorises too,

$$\frac{|\boldsymbol{D}|^{\text{new}}}{|\boldsymbol{D}|^{\text{old}}} = \frac{|\boldsymbol{D}|^{\text{new}}_{\uparrow}}{|\boldsymbol{D}|^{\text{old}}_{\uparrow}} \cdot \frac{|\boldsymbol{D}|^{\text{new}}_{\downarrow}}{|\boldsymbol{D}|^{\text{old}}_{\downarrow}}, \tag{2.63}$$

and when a single particle moves, one of the two factors is unchanged and cancels. Only a determinant of size $N/2$ is involved in each ratio and each inverse update, so the cost of one move drops from

$$\mathscr{O}_R(N) + \mathscr{O}_{\text{inverse}}(N^2) \qquad \text{to} \qquad \mathscr{O}_R(N/2) + \mathscr{O}_{\text{inverse}}(N^2/4), \tag{2.64}$$

and the cost of a full sweep over all N particles from $\mathscr{O}_R(N^2) + \mathscr{O}_{\text{inverse}}(N^3)$ to

$$\mathscr{O}_R(N^2/2) + \mathscr{O}_{\text{inverse}}(N^3/4). \tag{2.65}$$

The reduction is a constant factor, but a large one, and it is greatest when the numbers of spin-up and spin-down particles are equal so that the two factorised determinants are exactly half the size of the original.

2.14 Determinants, inverses and numerical stability

The updating scheme of Section 2.11 still needs the inverse and the determinant *once*, at the start of the simulation and whenever the accumulated error has to be reset. This is where the machinery of Chapter 1 enters directly.

The determinant from LU.

Never use the definition (2.10). The LU factorisation of Section 1.15 gives $\boldsymbol{PD} = \boldsymbol{LU}$ in $\mathscr{O}(N^3)$ operations, and the determinant follows from Eq. (1.29) as the signed product of the diagonal elements of $\boldsymbol{U}$,

$$|\boldsymbol{D}| = \pm u_{00} u_{11} \cdots u_{N-1,N-1}. \tag{2.66}$$

The same factorisation delivers the inverse for N additional back substitutions, Eq. (1.30). One factorisation therefore supplies both objects the Monte Carlo run needs.

In practice one should not store $|\boldsymbol{D}|$ at all but its logarithm. A product of N orbital values underflows or overflows long before N becomes physically interesting, whereas

$$\ln||\boldsymbol{D}|| = \sum_{k=0}^{N-1} \ln|u_{kk}| \tag{2.67}$$

is perfectly well behaved, and the sign is tracked separately as the parity of the row interchanges. Since the Metropolis test uses only ratios, working with $\ln|\Phi|$ throughout costs nothing.

The determinant from the SVD.

The singular value decomposition of Section 1.28 gives the same number,

$$||\boldsymbol{D}|| = \prod_{k=0}^{N-1} \sigma_k, \tag{2.68}$$

since $\boldsymbol{D} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$ with $||\boldsymbol{U}|| = ||\boldsymbol{V}|| = 1$. It costs several times more than LU and is never the right way to compute a determinant on its own. It is, however, exactly the right tool when one wants to know *how close to singular* the Slater matrix is, because it returns the condition number of Eq. (1.98) in the same breath,

$$\kappa_2(\boldsymbol{D}) = \frac{\sigma_0}{\sigma_{N-1}}. \tag{2.69}$$

The nodal surface.

This matters because the Slater determinant vanishes on the *nodal surface*, the set of configurations at which $|\boldsymbol{D}| = 0$. As a configuration approaches it the matrix becomes singular, the smallest singular value goes to zero, and the condition number diverges. Table 2.4 shows what happens as two of six particles are brought together: the determinant and the smallest singular value fall in step, the condition number rises by the same factor, and the accuracy of the computed inverse degrades exactly as Eq. (1.17) predicts.

Separation	$\lvert\lvert\boldsymbol{D}\rvert\rvert$	$\kappa_2(\boldsymbol{D})$	$\lvert\lvert\boldsymbol{D}\boldsymbol{D}^{-1} - \boldsymbol{I}\rvert\rvert$
10^{0}	2.81×10^{0}	2.58×10^{1}	8.7×10^{-16}
10^{-2}	3.21×10^{-2}	1.51×10^{3}	5.5×10^{-14}
10^{-4}	3.20×10^{-4}	1.54×10^{5}	4.3×10^{-12}
10^{-6}	3.20×10^{-6}	1.54×10^{7}	4.7×10^{-10}

Table 2.4 Six particles in a two-dimensional oscillator basis, with two of them brought to the separation shown. Each factor of 100 in the separation costs two digits in the inverse.

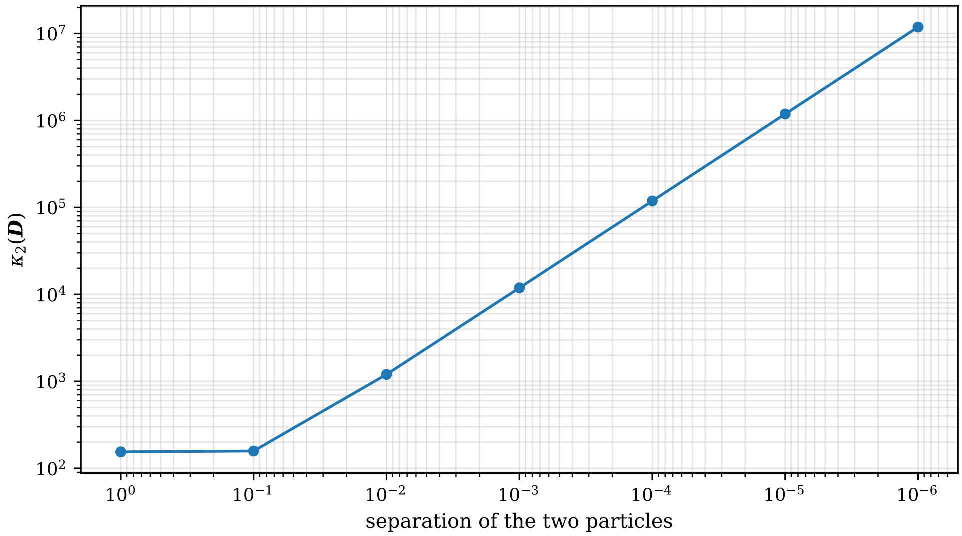

Fig. 2.4 Condition number $\kappa_2(\boldsymbol{D}) = \sigma_0/\sigma_{N-1}$ of the Slater matrix for six particles in a two-dimensional oscillator basis, as two of the six are brought together while the other four are held fixed. Both axes are logarithmic and the abscissa is reversed, so that the approach to the nodal surface runs from left to right. For separations above about 0.1 oscillator lengths the condition number is flat at $\kappa_2 \simeq 1.5 \times 10^2$, a value set by the shape of the basis and by the random placement of the spectator particles; below that it rises by exactly one decade for every decade of approach, reaching 1.2×10^7 at a separation of 10^{-6}. The straight line of unit slope is the signature of a simple zero of $|\boldsymbol{D}|$ on the nodal surface.

Figure 2.4 extends the same experiment over seven decades of separation, for a different random placement of the four spectator particles, and the shape of the curve says more than the individual entries of Table 2.4 do. Two regimes are visible. Far from the node the condition number is a constant of order 10^2: the Slater matrix is merely as well or as badly scaled as the oscillator orbitals themselves happen to make it, and nothing dramatic is going on. Once the two particles come within about one tenth of an oscillator length of each other the curve turns and becomes a straight line of slope -1 in the separation, $\kappa_2 \propto 1/s$. That exponent is worth a moment. Two rows of $\boldsymbol{D}$ that are evaluated at nearby points differ, to leading order, by the separation times the gradient of the orbitals, so the smallest singular value is linear in s while the largest is unaffected, and the ratio must diverge as s^{-1}. The determinant vanishes linearly for the same reason, as the second column of Table 2.4 confirms. Combining this with the error bound of Eq. (1.17) gives the practical rule: an inverse computed at a separation s carries a relative error of about $\kappa_2 \varepsilon_{\text{mach}} \sim 10^{-16}/s$, so we lose one significant digit for every factor of ten we approach the node, and by $s = 10^{-6}$ six of the sixteen available digits have already gone. Nothing in the algorithm can prevent this – the same figure would be obtained with any method of inversion – which is precisely the content of the statement that ill-conditioning is a property of the problem and not of the code. It is also the numerical face of a piece of physics we shall meet repeatedly: the nodal surface is where fermionic wave functions are hardest to handle, in diffusion Monte Carlo because the sign problem lives there, and here because the linear algebra loses its digits there.

When to rebuild.

Equations (2.54) and (2.55) divide by R, so every near-nodal move multiplies the error in $\boldsymbol{D}^{-1}$ by roughly $1/R$, and the damage is permanent: subsequent updates propagate it. Under Metropolis sampling of $|\Phi|^2$ this is rare, since the sampling density itself vanishes on the node

– in a run of 5000 moves the smallest accepted $|R|$ was 0.143 and the error in the inverse never exceeded 5×10^{-14}. But rare is not never. Forcing a sequence of near-nodal moves shows the mechanism plainly:

R **of the move**	$\|\boldsymbol{DD}^{-1} - \boldsymbol{I}\|$ **after**
(start)	2.2×10^{-15}
-7.0×10^{-3}	3.4×10^{-14}
1.2×10^{-4}	3.5×10^{-12}
1.2×10^{-6}	2.0×10^{-10}
1.2×10^{-8}	3.2×10^{-8}
after an LU refresh	1.7×10^{-15}

Table 2.5 Error accumulated in the updated inverse by a sequence of moves that pass close to the nodal surface, and its removal by one LU refactorisation.

The remedy is simple and universal: rebuild $\boldsymbol{D}^{-1}$ from an LU factorisation every few hundred accepted moves. The $\mathscr{O}(N^3)$ cost, amortised over that many $\mathscr{O}(N^2)$ updates, is negligible, and the accumulated error is reset to machine precision. A production Monte Carlo code should also monitor $\|\boldsymbol{DD}^{-1} - \boldsymbol{I}\|$, or the condition number of Eq. (2.69), and refresh whenever it exceeds a threshold.

Link to Chapter 1. Three separate results from the first chapter meet here. The LU decomposition of Section 1.15 supplies the determinant and the inverse at the start of the run; the condition number of Section 1.13, computed from the singular values of Section 1.28, tells us how many digits the inverse has lost; and the observation that the accuracy of a computed inverse is governed by κ_2 rather than by the algorithm, Eq. (1.17), explains why no cleverer updating formula could avoid the problem. The nodal surface of a fermionic wave function is, from the point of view of linear algebra, simply the locus where the Slater matrix becomes singular.

2.15 Quantum computing: the gate model and its linear-algebra foundation

A quantum computer is a well-controlled quantum many-body system, and the abstract *gate-based* model is defined entirely in the language of Chapter 1. Starting from N qubits initialised in $|\Psi\rangle^{(0)} = |00\cdots 0\rangle$, the state evolves by successive applications of unitary gates,

$$|\Psi\rangle^{(n+1)} = \hat{U}^{(n)}|\Psi\rangle^{(n)}, \qquad \hat{U}^{(n)} = T\exp\left(-i\int_{t_n}^{t_{n+1}} \hat{H}(t)\,dt\right).$$

A *single-qubit gate* acts on qubit a via a 2×2 unitary:

$$\Psi^{(n+1)}_{i_1\cdots i_N} = \sum_{i'_a} U^{(n)}_{i_a i'_a}\,\Psi^{(n)}_{i_1\cdots i_{a-1} i'_a i_{a+1}\cdots i_N}.$$

A *two-qubit gate* acts on qubits a, b via a 4×4 unitary. Standard examples are the Pauli gates X, Y, Z of Section 1.7, the Hadamard gate

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

the phase gate $T = \begin{pmatrix} e^{i\pi/8} & 0 \\ 0 & e^{-i\pi/8} \end{pmatrix}$, and the two-qubit CNOT (controlled-NOT):

$$C_X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

When qubit a is *measured*, the result $\alpha \in \{0,1\}$ occurs with probability

$$P_\alpha = \sum_{i_1 \cdots \hat{i}_a \cdots i_N} \left| \Psi^{(n)}_{i_1 \cdots \alpha \cdots i_N} \right|^2,$$

after which the state collapses to the corresponding projected state – the projection operators of Section 1.6 acting for real.

In a nutshell. Quantum computing is a restricted form of linear algebra: matrix-vector multiplications with special unitary matrices (gates) on exponentially large vectors (wave functions). The challenge – shared with classical many-body physics – is that only a tiny fraction of the 2^N-dimensional space is physically relevant; identifying that subspace is the core problem in both fields.

2.16 The exponential-dimension problem and why it matters

The state of N spin-$\frac{1}{2}$ particles requires 2^N complex numbers to describe, as the tensor-product construction of Section 1.10 already made clear. For $N \approx 300$, $2^{300} \gg 10^{82}$, the estimated number of atoms in the observable universe. The same counting applies to the fermionic problem: the number of Slater determinants that can be formed by distributing N particles over n single-particle states is $\binom{n}{N}$, which for $n = 40$ and $N = 20$ is already 1.4×10^{11}.

How do physicists actually compute?

1. **Clever approximations**: Bethe ansatz, perturbation theory, mean-field approximations such as the Hartree-Fock theory derived above.
2. **Computer simulations**: DFT, quantum Monte Carlo, FCI, coupled cluster, DMRG.
3. **Restricting to tractable subsystems**: systems where a single-particle basis is a good approximation.

Chapter 1 supplied two of the reasons this is not hopeless. The first is that we rarely want the whole spectrum: the Lanczos algorithm of Section 1.26 extracts the lowest eigenvalues of a matrix of dimension 10^9 using only matrix-vector products, and the pairing-model example there converged to machine precision in forty of them. The second is that physically relevant states are compressible: the Schmidt spectrum of a ground state decays rapidly (Section 1.30), and the Eckart-Young theorem, Eq. (1.103), guarantees that truncating it is the best approximation of its rank. Density-matrix renormalisation group and tensor-network methods are built on precisely that observation.

For a quantum circuit of depth D acting on N qubits, the accessible subspace has dimension $\mathcal{O}(DN)$, vastly smaller than 2^N. For $N = 100$ and $D = 10^6$, this subspace still has 10^{20} times fewer degrees of freedom than the full Hilbert space. The question is whether the *physically* *relevant* states lie in this reachable subspace – a central open problem in quantum many-body theory and quantum computing alike (see, e.g., LaRocca et al.).

The variational quantum eigensolver (VQE) tackles the problem by using a parametrised quantum circuit as an ansatz for the ground state and minimising $\langle\Psi(\boldsymbol{\theta})|\hat{H}|\Psi(\boldsymbol{\theta})\rangle$ classically over the circuit parameters. It is the quantum-computing form of the variational principle (2.22) that underlies Hartree-Fock and coupled-cluster theory: the bound $E(\boldsymbol{\theta}) \geq E_0$ holds for every choice of parameters, and the quality of the result is decided entirely by whether the ansatz can reach the true ground state.

2.17 Exercises

Warm-up exercises

1. **Determinant product rule.** For 2×2 matrices $\boldsymbol{B}$ and $\boldsymbol{C}$, define $a_{ij} = \sum_k b_{ik}c_{kj}$. Show by direct calculation that $\det(\boldsymbol{A}) = \det(\boldsymbol{B})\det(\boldsymbol{C})$.
2. **Slater determinant: $N = 2$.** Write the Slater determinant for two electrons in orbitals ψ_α and ψ_β. (a) Show it is antisymmetric under particle exchange. (b) Show it vanishes if $\psi_\alpha = \psi_\beta$.
3. **Antisymmetriser.** Verify for $N = 2$ that $\hat{A}^2 = \hat{A}$ and that $[\hat{H}_0, \hat{A}] = 0$ when $\hat{H}_0 = \hat{h}_0(x_1) + \hat{h}_0(x_2)$.
4. **Basis change of the Slater determinant.** With the expansion $\psi_p = \sum_\lambda C_{p\lambda}\phi_\lambda$ and a 2×2 unitary $\boldsymbol{C}$, show that the two-particle Slater determinant in the new basis equals $\det(\boldsymbol{C})\det(\boldsymbol{\Phi})$ and that $|\det(\boldsymbol{C})| = 1$ leaves the physics unchanged.
5. **Hartree-Fock energy.** For a two-electron system with orbitals ψ_α and ψ_β, write out the HF energy functional (2.37) explicitly, identifying the kinetic, direct (Hartree), and exchange (Fock) contributions.
6. **Variational principle.** Show that $E[\Phi] \geq E_0$ for any normalised trial state Φ, where E_0 is the exact ground state energy. Use the spectral decomposition of $\hat{H}$ in its eigenbasis.
7. **Exponential dimension.** How many complex numbers are needed to store the full N-particle wave function for $N = 10$, 20, 50? A modern GPU can perform roughly 10^{14} floating-point operations per second. Estimate how long a single matrix-vector multiplication in the $N = 50$ space would take.
8. **One-qubit gates numerically.** Write a short Python function that represents $|0\rangle$ and $|1\rangle$ as NumPy arrays and applies each Pauli matrix, the Hadamard gate H, and the phase gate T in turn. Verify the action of X (spin flip), Z (phase flip) and $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ (creation of a superposition state).
9. **Pauli gate as a quantum NOT.** Explain why σ_x is the quantum analogue of the classical NOT gate. What is the difference between the classical and quantum cases when the input is a superposition $\alpha|0\rangle + \beta|1\rangle$?
10. **VQE and the variational principle.** The variational quantum eigensolver minimises $E(\boldsymbol{\theta}) = \langle\Psi(\boldsymbol{\theta})|\hat{H}|\Psi(\boldsymbol{\theta})\rangle$ over circuit parameters $\boldsymbol{\theta}$. (a) Explain why $E(\boldsymbol{\theta}) \geq E_0$ for any $\boldsymbol{\theta}$. (b) Describe the connection to the Hartree-Fock variational principle. (c) What determines the quality of the approximation in VQE compared to FCI?
11. **The antisymmetriser as a projector.** Using Eq. (2.7), verify for $N = 2$ that $\hat{A}^2 = \hat{A}$ and that $[\hat{H}_0, \hat{A}] = 0$ when $\hat{H}_0 = \hat{h}_0(x_1) + \hat{h}_0(x_2)$. Explain why $\hat{A}$ is a projection operator in the sense of Section 1.6, and identify the subspace it projects onto.
12. **Why only two permutations survive.** In the derivation of Eq. (2.32) it was argued that only the identity and the transposition $\hat{P}_{ij}$ contribute to the two-body expectation value. Write out the case $N = 3$ explicitly, listing all six permutations, and show that four of them

give a vanishing integral. Then explain why the corresponding argument for the one-body operator leaves only one surviving permutation.

13. **Self-interaction.** Show from Eq. (2.36) that $\langle \mu\mu|\hat{v}|\mu\mu\rangle_{\mathrm{AS}} = 0$, and explain in words what would go wrong in Eq. (2.37) if the exchange term were omitted.
14. **Occupation numbers of a Slater determinant.** For the determinant (2.14) the density matrix is $\rho_{\lambda\mu} = \sum_i C^*_{i\lambda} C_{i\mu}$. (a) Show that $\boldsymbol{\rho}^2 = \boldsymbol{\rho}$ when the coefficients are orthonormal. (b) Conclude that the eigenvalues – the natural occupation numbers of Section 1.31 – can only be 0 or 1. (c) What does it mean physically when a computed occupation number comes out as 0.93?
15. **Cost of a determinant.** Reproduce Table 2.1. For $n = 20$, estimate how long the evaluation of a single determinant would take from Eq. (2.10) on a machine performing 10^{12} operations per second, and compare with the LU route of Section 1.15.
16. **Two-body storage.** How many two-body matrix elements $\langle \alpha\beta|\hat{v}|\gamma\delta\rangle$ are there for $n = 50$, $n = 200$ and $n = 1000$ single-particle states? How much memory does each case require in double precision? Using Eq. (2.44) with $M = 5n$, how much is saved?
17. **Hartree-Fock numerically.** Using `slaterdeterminant.py`, reproduce Table 2.2. Then vary the interaction strength and plot the correlation energy (2.42) and the largest natural occupation number against it. At what point would you no longer trust a single-determinant description?
18. **The ratio R by hand.** Take $N = 2$ and a Slater matrix $\boldsymbol{D}$ with elements $d_{ij} = \phi_j(\boldsymbol{r}_i)$. Move particle 1 and compute the ratio R both from the definition (2.48), by evaluating two 2×2 determinants, and from Eq. (2.52). Verify that they agree, and count the floating point operations each route requires.
19. **Why the cofactor row does not change.** Explain in your own words why $C_{ij}(\boldsymbol{r}^{\mathrm{new}}) = C_{ij}(\boldsymbol{r}^{\mathrm{old}})$ when only particle i has moved, Eq. (2.50). Where in the derivation of Eq. (2.52) would the argument fail if two particles were moved at once?
20. **Updating the inverse.** Implement Eqs. (2.53)–(2.55) for a 3×3 Slater matrix and check the result against `numpy.linalg.inv`. Confirm that the update costs $\mathcal{O}(N^2)$ operations by counting the loops.
21. **Derivatives of the determinant.** Using `slater_update.py`, compare the analytic gradient and Laplacian ratios of Eqs. (2.56) and (2.57) with numerical differentiation of $\ln|\boldsymbol{D}|$. Why does the Laplacian agree to fewer digits than the gradient?
22. **The nodal surface as an ill-conditioned matrix.** Reproduce Table 2.4. Plot $\ln\kappa_2(\boldsymbol{D})$ against $\ln(\text{separation})$ and explain the slope. Using Eq. (1.17), predict how many digits of $\boldsymbol{D}^{-1}$ survive at a separation of 10^{-8}, and check your prediction.
23. **When to refresh.** Design a rule for deciding when to rebuild $\boldsymbol{D}^{-1}$ from an LU factorisation. Estimate the fraction of the total run time that the refreshes consume for $N = 50$ if you rebuild every 500 accepted moves, and compare with the cost of doing so at every move.
24. **Spin factorisation.** For $N = 4$ with two spin-up and two spin-down particles, write out Eq. (2.59) and verify that the full determinant of Eq. (2.58) vanishes. Then show, using the structure of Eq. (2.33), that the factorised ansatz (2.61) gives the same energy for a spin-independent Hamiltonian.

Exercise session, week 35: Slater determinants and a first FCI model

Exercise 1: expectation values with a Slater determinant.

Consider the fermion Slater determinant used as an ansatz for a quantum-mechanical state,

$$\Phi_\lambda^{AS}(x_1x_2\ldots x_N;\alpha_1\alpha_2\ldots\alpha_N) = \frac{1}{\sqrt{N!}}\sum_p(-)^p\hat{P}\prod_{i=1}^N\psi_{\alpha_i}(x_i), \tag{2.70}$$

where $\hat{P}$ permutes the particles and the sum runs over all $N!$ permutations. The number of particles equals the number of available single-particle states $\alpha_1\alpha_2\ldots\alpha_N$, and the single-particle basis ψ_{α_i} is orthonormal.

a) Write out Φ^{AS} for $N=3$.
b) Show that

$$\int dx_1dx_2\cdots dx_N\,\left|\Phi_\lambda^{AS}(x_1x_2\ldots x_N;\alpha_1\alpha_2\ldots\alpha_N)\right|^2 = 1.$$

c) Define a general one-body operator $\hat{F} = \sum_i^N \hat{f}(x_i)$ and a general two-body operator $\hat{G} = \sum_{i>j}^N \hat{g}(x_i,x_j)$, with $\hat{g}$ invariant under the interchange of the coordinates of particles i and j. Calculate the matrix elements for a two-particle Slater determinant,

$$\langle\Phi^{AS}_{\alpha_1\alpha_2}|\hat{F}|\Phi^{AS}_{\alpha_1\alpha_2}\rangle \quad\text{and}\quad \langle\Phi^{AS}_{\alpha_1\alpha_2}|\hat{G}|\Phi^{AS}_{\alpha_1\alpha_2}\rangle.$$

Which properties do you expect these operators to have, in addition to permutation symmetry?
d) Repeat the calculation for a general N-particle Slater determinant.

Exercise 2: a three-level model.

We now consider a simple three-level problem, our first and very simple model of a many-fermion system and of what we shall later call full configuration interaction theory. The particles are fermions. The single-particle states are labelled by the quantum number p and each can accommodate up to two particles, so every single-particle state is doubly degenerate; one may think of the two as spin up and spin down. The spacing between the doubly degenerate states is constant with value d, the first state has energy d, and there are three available states $p=1,2,3$, as in Figure 2.5.

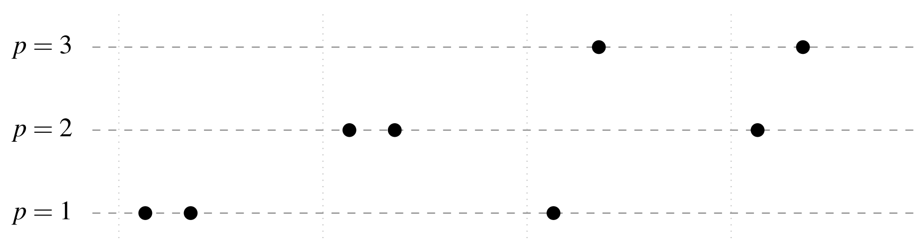

Fig. 2.5 Schematic plot of the possible single-particle levels with double degeneracy. The filled circles indicate occupied states and the spacing between the levels is constant. Four possible two-particle states are shown; the two leftmost are the configurations retained in parts b) and c), in which both particles occupy the same level.

a) How many two-particle Slater determinants can we construct in this space?
b) Limit the system to the two lowest orbits, $p=1$ and $p=2$, and two particles. Write the Hamiltonian as $\hat{H} = \hat{H}_0 + \hat{H}_I$, with the one-body part satisfying

$$\hat{h}_0\psi_{p\sigma} = p\,d\,\psi_{p\sigma},$$

where σ is the spin quantum number. Assume that the only two-particle states that can exist are those in which both particles occupy the same level p, and that all two-particle matrix elements of $\hat{H}_I$ have the constant value $-g$. Show that the Hamiltonian matrix can be written as

$$\begin{pmatrix} 2d-g & -g \\ -g & 4d-g \end{pmatrix},$$

and find its eigenvalues and eigenvectors. What is the admixture of the state with two particles in $p=2$ to the wave function with two particles in $p=1$? Discuss the result in terms of a linear combination of Slater determinants.

c) Add the possibility that the two particles occupy $p=3$, and find the Hamiltonian matrix, the eigenvalues and the eigenvectors. We still insist that only two-particle states with both particles in the same level are allowed. The 3×3 matrix may be diagonalised numerically.

This small model catches several birds with one stone. It shows how linear combinations of Slater determinants are built and how the coefficients are interpreted as admixtures to a given state. The two-particle states above $p=1$ are excitations of the ground-state configuration, and the reliability of the single-determinant ansatz depends on the ratio of the interaction strength g to the level spacing d. Finally, the model is a schematic ansatz for pairing correlations, and hence for superfluidity and superconductivity in fermionic systems; it is the two-level ancestor of the pairing model used in Section 1.26.

Answers, week 35

Exercise 1a.

For $N=3$ the six permutations give

$$\begin{aligned}\Phi_\lambda^{AS} = \frac{1}{\sqrt{6}}\Big(&\psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\psi_{\alpha_3}(x_3) - \psi_{\alpha_1}(x_2)\psi_{\alpha_2}(x_1)\psi_{\alpha_3}(x_3)\\ &+\psi_{\alpha_1}(x_2)\psi_{\alpha_2}(x_3)\psi_{\alpha_3}(x_1) - \psi_{\alpha_1}(x_3)\psi_{\alpha_2}(x_2)\psi_{\alpha_3}(x_1)\\ &+\psi_{\alpha_1}(x_3)\psi_{\alpha_2}(x_1)\psi_{\alpha_3}(x_2) - \psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_3)\psi_{\alpha_3}(x_2)\Big),\end{aligned} \tag{2.71}$$

which is the expansion of the 3×3 determinant of Eq. (2.14) along its rows.

Exercise 1b.

Introduce the antisymmetriser (2.7) and the Hartree function (2.15), so that $\Phi_\lambda^{AS} = \sqrt{N!}\hat{A}\Phi_H$. Then

$$\langle\Phi_\lambda^{AS}|\Phi_\lambda^{AS}\rangle = N!\int \Phi_H^*\hat{A}^\dagger\hat{A}\Phi_H\,d\boldsymbol{\tau} = N!\int \Phi_H^*\hat{A}\Phi_H\,d\boldsymbol{\tau}, \tag{2.72}$$

where we used that $\hat{A}$ is Hermitian and idempotent, Eq. (2.9). Writing out the antisymmetriser cancels the factor $N!$,

$$\langle\Phi_\lambda^{AS}|\Phi_\lambda^{AS}\rangle = \sum_p(-)^p\int \Phi_H^*\hat{P}\Phi_H\,d\boldsymbol{\tau}. \tag{2.73}$$

The permutation operator acts on only one of the two Hartree functions, so any $\hat{P}$ other than the identity produces at least one overlap $\langle\alpha_i|\alpha_j\rangle$ with $i\neq j$, which vanishes by orthonormality. Only the identity survives and $\langle\Phi_\lambda^{AS}|\Phi_\lambda^{AS}\rangle = \int\Phi_H^*\Phi_H d\boldsymbol{\tau} = 1$.

Exercise 1c.

For two particles the same manipulation gives

$$\langle \Phi^{AS}_{\alpha_1\alpha_2}|\hat{F}|\Phi^{AS}_{\alpha_1\alpha_2}\rangle = \sum_p (-)^p \int \psi^*_{\alpha_1}(x_1)\psi^*_{\alpha_2}(x_2)\left[\hat{f}(x_1)+\hat{f}(x_2)\right]\hat{P}\,\psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\,d\boldsymbol{\tau}. \tag{2.74}$$

As in b), a permuted Hartree function makes the integral vanish, so only the identity contributes. In each remaining term the one-body operator acts on a single coordinate; integrating out the other gives unity, and

$$\langle \Phi^{AS}_{\alpha_1\alpha_2}|\hat{F}|\Phi^{AS}_{\alpha_1\alpha_2}\rangle = \langle \alpha_1|\hat{f}|\alpha_1\rangle + \langle \alpha_2|\hat{f}|\alpha_2\rangle, \tag{2.75}$$

with the shorthand of Eq. (2.29).

For the two-body operator there is only one term, $\hat{G} = \hat{g}(x_1,x_2)$, and now the transposition survives as well, since $\hat{g}$ depends on both coordinates. The two permutations give

$$\begin{aligned}\langle \Phi^{AS}_{\alpha_1\alpha_2}|\hat{G}|\Phi^{AS}_{\alpha_1\alpha_2}\rangle &= \int \psi^*_{\alpha_1}(x_1)\psi^*_{\alpha_2}(x_2)\hat{g}(x_1,x_2)\psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\,d\boldsymbol{\tau} \\ &\quad - \int \psi^*_{\alpha_1}(x_1)\psi^*_{\alpha_2}(x_2)\hat{g}(x_1,x_2)\psi_{\alpha_1}(x_2)\psi_{\alpha_2}(x_1)\,d\boldsymbol{\tau} \\ &= \langle \alpha_1\alpha_2|\hat{g}|\alpha_1\alpha_2\rangle_{\mathrm{AS}}.\end{aligned} \tag{2.76}$$

The first term is the direct or Hartree term, the second the exchange or Fock term, exactly as in Eq. (2.33).

Beyond permutation symmetry we expect both operators to be *Hermitian*, since they represent physical observables and must have real eigenvalues. Permutation symmetry is forced on us because the particles are indistinguishable, and it implies $[\hat{A},\hat{F}] = [\hat{A},\hat{G}] = 0$, which is what allowed the manipulations above.

Exercise 1d.

For general N the steps are the same. For the one-body operator,

$$\langle \Phi^{AS}|\hat{F}|\Phi^{AS}\rangle = \sum_{i=1}^{N}\sum_p (-)^p \int \Phi^*_H \hat{f}(x_i)\hat{P}\Phi_H\,d\boldsymbol{\tau} = \sum_{i=1}^{N}\langle \alpha_i|\hat{f}|\alpha_i\rangle. \tag{2.77}$$

For the two-body operator, each pair (x_i,x_j) receives two contributing permutations with opposite signs, giving the factor $(1-\hat{P}_{ij})$ of Eq. (2.32). Replacing the restricted sum $i<j$ by an unrestricted double sum with a compensating $\frac{1}{2}$, and noting that the $i=j$ terms cancel identically, leaves

$$\langle \Phi^{AS}|\hat{G}|\Phi^{AS}\rangle = \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\langle \alpha_i\alpha_j|\hat{g}|\alpha_i\alpha_j\rangle_{\mathrm{AS}}. \tag{2.78}$$

Equations (2.77) and (2.78) together are Eq. (2.37), the Hartree-Fock energy functional.

Exercise 2a.

With three levels, each doubly degenerate, there are six single-particle states and hence $\binom{6}{2} = 15$ two-particle Slater determinants in general. If we count only the level labels and ignore spin, three states with two distinct levels and three with both particles in the same level give six configurations. Under the restriction of parts b) and c), that both particles occupy the *same* level, exactly three Slater determinants remain, one for each p.

Exercise 2b.

Denote by $|\Phi_p\rangle$ the state with both particles in level p. Since $\psi_{p\sigma}$ is an eigenstate of $\hat{h}_0$ with eigenvalue pd, the one-body part contributes $\langle\Phi_p|\hat{H}_0|\Phi_p\rangle = 2\langle p|\hat{h}_0|p\rangle = 2pd$, using Eq. (2.77) with $N=2$. Every matrix element of $\hat{H}_I$ equals $-g$. With $|\Phi_1\rangle$ and $|\Phi_2\rangle$ as basis,

$$\boldsymbol{H} = \begin{pmatrix} 2d-g & -g \\ -g & 4d-g \end{pmatrix}, \tag{2.79}$$

which is the requested matrix. This is the matrix representation of an operator in a chosen basis, exactly as in Section 1.21: the Slater determinants are not eigenstates of $\hat{H}$, but they form a basis in which the eigenstates can be expanded, $|\Psi_i\rangle = c_{i0}|\Phi_1\rangle + c_{i1}|\Phi_2\rangle$.

The characteristic polynomial follows from the trace $6d-2g$ and the determinant $(2d-g)(4d-g)-g^2 = 8d^2-6dg$,

$$\lambda^2 + (2g-6d)\lambda + 8d^2 - 6dg = 0, \tag{2.80}$$

and solving it gives

$$\lambda_\pm = 3d - g \pm \sqrt{d^2+g^2}. \tag{2.81}$$

Substituting back, the unnormalised eigenvectors are

$$|\Psi_\mp\rangle = \frac{d \pm \sqrt{d^2+g^2}}{g}|\Phi_1\rangle + |\Phi_2\rangle, \tag{2.82}$$

or, with the dimensionless ratio $\gamma \equiv d/g$,

$$|\Psi_\mp\rangle = \left(\gamma \pm \sqrt{1+\gamma^2}\right)|\Phi_1\rangle + |\Phi_2\rangle. \tag{2.83}$$

The single parameter γ measures the strength of the interaction against the level spacing. For $g \to 0$ ($\gamma \to \infty$) the ground state becomes pure $|\Phi_1\rangle$: the single Slater determinant is exact. As g grows the admixture of $|\Phi_2\rangle$ grows with it, and at $\gamma = 0$ the two determinants enter with equal weight. This is correlation, in its simplest possible form, and it is the same story told quantitatively by the occupation numbers of Section 1.31.

Exercise 2c.

Adding $p=3$ changes nothing structurally: the diagonal element is $2pd - g = 6d - g$ and the off-diagonal elements remain $-g$, so

$$\boldsymbol{H} = \begin{pmatrix} 2d-g & -g & -g \\ -g & 4d-g & -g \\ -g & -g & 6d-g \end{pmatrix} = g\begin{pmatrix} 2\gamma-1 & -1 & -1 \\ -1 & 4\gamma-1 & -1 \\ -1 & -1 & 6\gamma-1 \end{pmatrix}. \tag{2.84}$$

The eigenvalues may be found symbolically, or numerically with the Householder and QL solver of Sections 1.24 and 1.25 – three lines of Python. For $\gamma \gg 1$ the ground state is again dominated by $|\Phi_1\rangle$ with admixtures of order $1/\gamma$ from the two excited configurations. Continuing this construction to L levels and n pairs gives exactly the pairing model of Table 1.3, whose dimension $\binom{L}{n}$ grows fast enough that the Lanczos algorithm becomes the only practical route.

Exercise session, week 36: Condon-Slater rules and second quantisation

Exercise 1: Condon-Slater rules for expectation values.

Consider three N-particle Slater determinants $|SD\rangle$, $|SD_i^j\rangle$ and $|SD_{ij}^{kl}\rangle$, where $|SD_i^j\rangle$ differs from $|SD\rangle$ by a single single-particle state – the state ψ_i has been replaced by ψ_j – and $|SD_{ij}^{kl}\rangle$ differs by two. Define, as in week 35, a general one-body operator $\hat{F} = \sum_i^N \hat{f}(x_i)$ and a general two-body operator $\hat{G} = \sum_{i>j}^N \hat{g}(x_i, x_j)$ with $\hat{g}$ invariant under interchange of two coordinates. The single-particle states ψ_i are not necessarily eigenstates of $\hat{f}$.

a) Find $\langle SD|\hat{F}|SD\rangle$ and $\langle SD|\hat{G}|SD\rangle$.
b) Find $\langle SD|\hat{F}|SD_i^j\rangle$ and $\langle SD|\hat{G}|SD_i^j\rangle$.
c) Find $\langle SD|\hat{F}|SD_{ij}^{kl}\rangle$ and $\langle SD|\hat{G}|SD_{ij}^{kl}\rangle$. What happens to the two-body operator for a transition $\langle SD|\hat{G}|SD_{ijk}^{lmn}\rangle$, in which the determinant on the right differs by more than two single-particle states?

Exercise 2: a first encounter with second quantisation.

a) Show that the density of particles at $\boldsymbol{x}$,

$$n(\boldsymbol{x}) = N \int d\boldsymbol{x}_2 \cdots d\boldsymbol{x}_N \, |\Psi_{AS}(\boldsymbol{x}, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N)|^2,$$

can be written in terms of the single-particle states as

$$n(\boldsymbol{x}) = \sum_k |\psi_k(\boldsymbol{x})|^2.$$

b) Calculate the matrix elements $\langle \alpha_1 \alpha_2|\hat{F}|\alpha_1 \alpha_2\rangle$ and $\langle \alpha_1 \alpha_2|\hat{G}|\alpha_1 \alpha_2\rangle$ with

$$|\alpha_1 \alpha_2\rangle = a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger |0\rangle,$$
$$\hat{F} = \sum_{\alpha\beta} \langle \alpha|f|\beta\rangle a_\alpha^\dagger a_\beta,$$
$$\hat{G} = \frac{1}{2} \sum_{\alpha\beta\gamma\delta} \langle \alpha\beta|g|\gamma\delta\rangle a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma,$$

where the matrix elements are the integrals of Eqs. (2.29), (2.34) and (2.35). Compare with exercise 1c) of week 35.

Answers, week 36

Throughout we assume the two determinants have been brought to *maximum coincidence*: the common single-particle states appear in the same order in both, so that no additional sign from reordering rows is left over. If they have not, every result below acquires the factor $(-1)^\sigma$ of the permutation needed to bring them into that form.

Exercise 1a: the diagonal elements.

These are the results already derived in Section 2.8 and in exercise 1d) of week 35,

$$\langle SD|\hat{F}|SD\rangle = \sum_{i=1}^{N}\langle i|\hat{f}|i\rangle, \qquad \langle SD|\hat{G}|SD\rangle = \frac{1}{2}\sum_{i,j=1}^{N}\langle ij|\hat{g}|ij\rangle_{\mathrm{AS}}. \tag{2.85}$$

The one-body operator sees each occupied orbital once; the two-body operator sees each occupied pair once, with the exchange term subtracted.

Exercise 1b: one differing state.

Write $|SD_i^j\rangle$ for the determinant in which ψ_i has been replaced by ψ_j, with j not among the states occupied in $|SD\rangle$. Using $\Phi = \sqrt{N!}\hat{A}\Phi_H$ and the same argument as in Section 2.8, a term survives only if every orbital not acted on by the operator overlaps with an *identical* orbital on the other side. Since exactly one orbital differs, the one-body operator must be the one that connects ψ_i to ψ_j, and every other factor contributes unity:

$$\langle SD|\hat{F}|SD_i^j\rangle = \langle i|\hat{f}|j\rangle. \tag{2.86}$$

A single term, not a sum: the one-body operator can repair exactly one mismatch.

For the two-body operator, one of the two coordinates must carry the $i \to j$ replacement while the other runs over all the orbitals common to both determinants. Both the direct and the exchange contraction survive, so

$$\langle SD|\hat{G}|SD_i^j\rangle = \sum_{k=1}^{N}\langle ik|\hat{g}|jk\rangle_{\mathrm{AS}} = \sum_{k=1}^{N}\left[\langle ik|\hat{g}|jk\rangle - \langle ik|\hat{g}|kj\rangle\right], \tag{2.87}$$

where the sum runs over the states occupied in $|SD\rangle$. The term $k = i$ contributes nothing, since $\langle ii|\hat{g}|ji\rangle_{\mathrm{AS}} = 0$ by the antisymmetry of Eq. (2.36).

Equation (2.87) is worth pausing on. It says that a two-body interaction, evaluated between determinants differing by one orbital, behaves like an *effective one-body* operator obtained by summing over the occupied states. That sum is precisely the interaction part of the Fock matrix of Eq. (2.40); the Hartree-Fock condition $\langle SD|\hat{H}|SD_i^j\rangle = 0$, that no single excitation couples to the reference, is Brillouin's theorem.

Exercise 1c: two differing states, and beyond.

The one-body operator can repair only one mismatch at a time, so with two differing orbitals at least one overlap $\langle i|k\rangle$ between different states is always left over and

$$\langle SD|\hat{F}|SD_{ij}^{kl}\rangle = 0. \tag{2.88}$$

The two-body operator can repair exactly two, and both coordinates are used up doing so. Nothing is left to sum over, and

$$\langle SD|\hat{G}|SD_{ij}^{kl}\rangle = \langle ij|\hat{g}|kl\rangle_{\mathrm{AS}} = \langle ij|\hat{g}|kl\rangle - \langle ij|\hat{g}|lk\rangle. \tag{2.89}$$

Finally, if the determinants differ by three or more single-particle states,

$$\langle SD|\hat{G}|SD_{ijk}^{lmn}\rangle = 0, \tag{2.90}$$

because a two-body operator acts on at most two coordinates and therefore cannot bridge three mismatches; at least one vanishing overlap always survives.

Equations (2.85)–(2.90) are the *Condon-Slater rules*, and the pattern is simple enough to memorise: an n-body operator connects Slater determinants differing by at most n single-particle states. Their consequence is the reason large-scale diagonalisation is feasible at all. A configuration-interaction Hamiltonian matrix contains only one- and two-body operators, so all matrix elements between determinants differing by three or more orbitals vanish identically. The matrix is therefore extremely *sparse*: of the $\binom{n}{N}^2$ possible elements only a tiny fraction are non-zero, which is exactly the structure the Lanczos algorithm of Section 1.26 was designed to exploit, and the reason the Hamiltonian need never be stored.

Exercise 2a: the density.

Expand the Slater determinant along the row belonging to the coordinate $\boldsymbol{x}$,

$$\Psi_{AS}(\boldsymbol{x},\boldsymbol{x}_2,\ldots,\boldsymbol{x}_N) = \frac{1}{\sqrt{N}}\sum_{k=1}^{N}(-)^{k+1}\psi_k(\boldsymbol{x})\,\Psi_{AS}^{(k)}(\boldsymbol{x}_2,\ldots,\boldsymbol{x}_N), \tag{2.91}$$

where $\Psi_{AS}^{(k)}$ is the normalised $(N-1)$-particle Slater determinant built from all the occupied orbitals *except* ψ_k. Because two such determinants with $k \neq k'$ differ by one orbital, they are orthogonal,

$$\int d\boldsymbol{x}_2\cdots d\boldsymbol{x}_N\,\Psi_{AS}^{(k)*}\Psi_{AS}^{(k')} = \delta_{kk'}. \tag{2.92}$$

Squaring Eq. (2.91) and integrating therefore kills every cross term and leaves

$$\int d\boldsymbol{x}_2\cdots d\boldsymbol{x}_N\,|\Psi_{AS}|^2 = \frac{1}{N}\sum_{k=1}^{N}|\psi_k(\boldsymbol{x})|^2, \tag{2.93}$$

so that

$$n(\boldsymbol{x}) = N\cdot\frac{1}{N}\sum_{k=1}^{N}|\psi_k(\boldsymbol{x})|^2 = \sum_{k=1}^{N}|\psi_k(\boldsymbol{x})|^2. \tag{2.94}$$

The density of a Slater determinant is the sum of the densities of its occupied orbitals, with no interference between them. Integrating gives $\int n(\boldsymbol{x})d\boldsymbol{x} = N$, as it must. Note that Eq. (2.94) is the diagonal of the one-body density matrix of Eq. (1.111) in coordinate space, whose eigenvalues we already know to be 0 and 1 for a single determinant.

Exercise 2b: the same results from second quantisation.

Start from the one-body operator. Acting with a_β on the two-particle state and using $\{a_\alpha, a_\beta^\dagger\} = \delta_{\alpha\beta}$,

$$a_\beta\, a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger|0\rangle = \delta_{\beta\alpha_1}a_{\alpha_2}^\dagger|0\rangle - \delta_{\beta\alpha_2}a_{\alpha_1}^\dagger|0\rangle, \tag{2.95}$$

so that

$$a_\alpha^\dagger a_\beta|\alpha_1\alpha_2\rangle = \delta_{\beta\alpha_1}a_\alpha^\dagger a_{\alpha_2}^\dagger|0\rangle - \delta_{\beta\alpha_2}a_\alpha^\dagger a_{\alpha_1}^\dagger|0\rangle. \tag{2.96}$$

Projecting on $\langle\alpha_1\alpha_2| = \langle 0|a_{\alpha_2}a_{\alpha_1}$ and using $\langle\alpha_1\alpha_2|\alpha\alpha_2\rangle = \delta_{\alpha\alpha_1}$ and $\langle\alpha_1\alpha_2|\alpha\alpha_1\rangle = -\delta_{\alpha\alpha_2}$ gives

$$\langle\alpha_1\alpha_2|a_\alpha^\dagger a_\beta|\alpha_1\alpha_2\rangle = \delta_{\alpha\alpha_1}\delta_{\beta\alpha_1} + \delta_{\alpha\alpha_2}\delta_{\beta\alpha_2}, \tag{2.97}$$

and therefore

$$\langle \alpha_1\alpha_2|\hat{F}|\alpha_1\alpha_2\rangle = \langle \alpha_1|f|\alpha_1\rangle + \langle \alpha_2|f|\alpha_2\rangle. \tag{2.98}$$

For the two-body operator, $a_\delta a_\gamma$ annihilates the state unless $\{\gamma,\delta\} = \{\alpha_1,\alpha_2\}$, and likewise $a^\dagger_\alpha a^\dagger_\beta$ must recreate the same pair. The four surviving index assignments give, after collecting the signs from the anticommutators,

$$\langle \alpha_1\alpha_2|\hat{G}|\alpha_1\alpha_2\rangle = \langle \alpha_1\alpha_2|g|\alpha_1\alpha_2\rangle - \langle \alpha_1\alpha_2|g|\alpha_2\alpha_1\rangle = \langle \alpha_1\alpha_2|g|\alpha_1\alpha_2\rangle_{\mathrm{AS}}. \tag{2.99}$$

The factor $\frac{1}{2}$ in the definition of $\hat{G}$ is cancelled by the two equivalent assignments of the pair.

Comparison.

Equations (2.98) and (2.99) are *identical* to the results (2.75) and (2.76) obtained in week 35 by integrating over permutations of the Hartree function. This is the point of the exercise. The first-quantised route required us to write out the antisymmetriser, sort through $N!$ permutations, and argue term by term about which overlaps vanish; the second-quantised route required only the anticommutation relations $\{a_\alpha, a^\dagger_\beta\} = \delta_{\alpha\beta}$. Antisymmetry, which was an explicit and increasingly painful bookkeeping task in Section 2.8, has been built into the algebra of the operators themselves.

This is why the next chapter is devoted to second quantisation. The Slater determinant of Eq. (2.14) becomes a string of creation operators acting on the vacuum, the Condon-Slater rules of exercise 1 become elementary consequences of Wick's theorem, and calculations that are barely feasible by hand for $N = 3$ become mechanical for arbitrary N.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter02/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/manybodybasics.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter02/`.

`slaterdeterminant.py` Slater determinants, the energy functional and a minimal self-consistent field iteration. Runs in a few seconds.

`slater_update.py` The ratio R of determinants, Sherman–Morrison updating and the stability of the nodal surface. Runs in a few seconds.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 3
Second quantization

Chapter 2 ended with a warning. Every expectation value there was obtained by writing out the antisymmetriser, sorting through $N!$ permutations, and arguing term by term about which overlaps vanish. For two particles this is an exercise; for the eight-operator strings of Section 2.8 it is already unpleasant; for the excited determinants that perturbation theory and coupled-cluster theory require it is impossible. Exercise 2 of the week-36 session made the point sharply: the same matrix elements fall out of the anticommutation relations in three lines.

This chapter develops that observation into a formalism. Antisymmetry, which was an explicit and increasingly costly bookkeeping task in Chapter 2, is built once and for all into the algebra of the creation and annihilation operators. The Slater determinant of Eq. (2.14) becomes a string of creation operators acting on the vacuum, the Condon-Slater rules of Section 2.6 become consequences of Wick's theorem, and calculations that were barely feasible by hand for $N=3$ become mechanical for arbitrary N.

We introduce the time-independent operators $a_\alpha^\dagger$ and a_α which create and annihilate, respectively, a particle in the single-particle state φ_α. We define the fermion creation operator $a_\alpha^\dagger$

$$a_\alpha^\dagger|0\rangle \equiv |\alpha\rangle, \tag{3.1}$$

and

$$a_\alpha^\dagger|\alpha_1\dots\alpha_n\rangle_{\mathrm{AS}} \equiv |\alpha\alpha_1\dots\alpha_n\rangle_{\mathrm{AS}} \tag{3.2}$$

In Eq. (3.1) the operator $a_\alpha^\dagger$ acts on the vacuum state $|0\rangle$, which does not contain any particles. Alternatively, we could define a closed-shell nucleus or atom as our new vacuum, but then we need to introduce the particle-hole formalism, see the discussion to come.

In Eq. (3.2) $a_\alpha^\dagger$ acts on an antisymmetric n-particle state and creates an antisymmetric $(n+1)$-particle state, where the one-body state φ_α is occupied, under the condition that $\alpha \neq \alpha_1,\alpha_2,\dots,\alpha_n$. It follows that we can express an antisymmetric state as the product of the creation operators acting on the vacuum state.

$$|\alpha_1\dots\alpha_n\rangle_{\mathrm{AS}} = a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger\dots a_{\alpha_n}^\dagger|0\rangle \tag{3.3}$$

It is easy to derive the commutation and anticommutation rules for the fermionic creation operators $a_\alpha^\dagger$. Using the antisymmetry of the states (3.3)

$$|\alpha_1\dots\alpha_i\dots\alpha_k\dots\alpha_n\rangle_{\mathrm{AS}} = -|\alpha_1\dots\alpha_k\dots\alpha_i\dots\alpha_n\rangle_{\mathrm{AS}} \tag{3.4}$$

we obtain

$$a_{\alpha_i}^\dagger a_{\alpha_k}^\dagger = -a_{\alpha_k}^\dagger a_{\alpha_i}^\dagger \tag{3.5}$$

Using the Pauli principle

$$|\alpha_1 \dots \alpha_i \dots \alpha_i \dots \alpha_n\rangle_{\rm AS} = 0 \tag{3.6}$$

it follows that

$$a^\dagger_{\alpha_i} a^\dagger_{\alpha_i} = 0. \tag{3.7}$$

If we combine Eqs. (3.5) and (3.7), we obtain the well-known anti-commutation rule

$$a^\dagger_\alpha a^\dagger_\beta + a^\dagger_\beta a^\dagger_\alpha \equiv \{a^\dagger_\alpha, a^\dagger_\beta\} = 0 \tag{3.8}$$

The hermitian conjugate of $a^\dagger_\alpha$ is

$$a_\alpha = (a^\dagger_\alpha)^\dagger \tag{3.9}$$

If we take the hermitian conjugate of Eq. (3.8), we arrive at

$$\{a_\alpha, a_\beta\} = 0 \tag{3.10}$$

What is the physical interpretation of the operator a_α and what is the effect of a_α on a given state $|\alpha_1\alpha_2 \dots \alpha_n\rangle_{\rm AS}$? Consider the following matrix element

$$\langle \alpha_1\alpha_2 \dots \alpha_n | a_\alpha | \alpha'_1 \alpha'_2 \dots \alpha'_m\rangle \tag{3.11}$$

where both sides are antisymmetric. We distinguish between two cases. The first (1) is when $\alpha \in \{\alpha_i\}$. Using the Pauli principle of Eq. (3.6) it follows

$$\langle \alpha_1\alpha_2 \dots \alpha_n | a_\alpha = 0 \tag{3.12}$$

The second (2) case is when $\alpha \notin \{\alpha_i\}$. It follows that an hermitian conjugation

$$\langle \alpha_1\alpha_2 \dots \alpha_n | a_\alpha = \langle \alpha\alpha_1\alpha_2 \dots \alpha_n | \tag{3.13}$$

Eq. (3.13) holds for case (1) since the lefthand side is zero due to the Pauli principle. We write Eq. (3.11) as

$$\langle \alpha_1\alpha_2 \dots \alpha_n | a_\alpha | \alpha'_1 \alpha'_2 \dots \alpha'_m\rangle = \langle \alpha_1\alpha_2 \dots \alpha_n | \alpha\alpha'_1 \alpha'_2 \dots \alpha'_m\rangle \tag{3.14}$$

Here we must have $m = n+1$ if Eq. (3.14) has to be trivially different from zero.

For the last case, the minus and plus signs apply when the sequence $\alpha, \alpha_1, \alpha_2, \dots, \alpha_n$ and $\alpha'_1, \alpha'_2, \dots, \alpha'_{n+1}$ are related to each other via even and odd permutations. If we assume that $\alpha \notin \{\alpha_i\}$ we obtain

$$\langle \alpha_1\alpha_2 \dots \alpha_n | a_\alpha | \alpha'_1 \alpha'_2 \dots \alpha'_{n+1}\rangle = 0 \tag{3.15}$$

when $\alpha \in \{\alpha'_i\}$. If $\alpha \notin \{\alpha'_i\}$, we obtain

$$a_\alpha \underbrace{|\alpha'_1 \alpha'_2 \dots \alpha'_{n+1}\rangle}_{\neq\alpha} = 0 \tag{3.16}$$

and in particular

$$a_\alpha |0\rangle = 0 \tag{3.17}$$

If $\{\alpha\alpha_i\} = \{\alpha'_i\}$, performing the right permutations, the sequence $\alpha, \alpha_1, \alpha_2, \dots, \alpha_n$ is identical with the sequence $\alpha'_1, \alpha'_2, \dots, \alpha'_{n+1}$. This results in

$$\langle \alpha_1\alpha_2 \dots \alpha_n | a_\alpha | \alpha\alpha_1\alpha_2 \dots \alpha_n\rangle = 1 \tag{3.18}$$

and thus

$$a_\alpha |\alpha\alpha_1\alpha_2 \dots \alpha_n\rangle = |\alpha_1\alpha_2 \dots \alpha_n\rangle \tag{3.19}$$

The action of the operator a_α from the left on a state vector is to to remove one particle in the state α. If the state vector does not contain the single-particle state α, the outcome of the operation is zero. The operator a_α is normally called for a destruction or annihilation operator.

The next step is to establish the commutator algebra of $a_\alpha^\dagger$ and a_β.

The action of the anti-commutator $\{a_\alpha^\dagger, a_\alpha\}$ on a given n-particle state is

$$\begin{aligned} a_\alpha^\dagger a_\alpha \underbrace{|\alpha_1\alpha_2\dots\alpha_n\rangle}_{\neq\alpha} &= 0 \\ a_\alpha a_\alpha^\dagger \underbrace{|\alpha_1\alpha_2\dots\alpha_n\rangle}_{\neq\alpha} &= a_\alpha \underbrace{|\alpha\alpha_1\alpha_2\dots\alpha_n\rangle}_{\neq\alpha} = \underbrace{|\alpha_1\alpha_2\dots\alpha_n\rangle}_{\neq\alpha} \end{aligned} \tag{3.20}$$

if the single-particle state α is not contained in the state.

If it is present we arrive at

$$\begin{aligned} a_\alpha^\dagger a_\alpha |\alpha_1\alpha_2\dots\alpha_k\alpha\alpha_{k+1}\dots\alpha_{n-1}\rangle &= a_\alpha^\dagger a_\alpha (-1)^k |\alpha\alpha_1\alpha_2\dots\alpha_{n-1}\rangle \\ &= (-1)^k |\alpha\alpha_1\alpha_2\dots\alpha_{n-1}\rangle = |\alpha_1\alpha_2\dots\alpha_k\alpha\alpha_{k+1}\dots\alpha_{n-1}\rangle \\ a_\alpha a_\alpha^\dagger |\alpha_1\alpha_2\dots\alpha_k\alpha\alpha_{k+1}\dots\alpha_{n-1}\rangle &= 0 \end{aligned} \tag{3.21}$$

From Eqs. (3.20) and (3.21) we arrive at

$$\{a_\alpha^\dagger, a_\alpha\} = a_\alpha^\dagger a_\alpha + a_\alpha a_\alpha^\dagger = 1 \tag{3.22}$$

The action of $\left\{a_\alpha^\dagger, a_\beta\right\}$, with $\alpha \neq \beta$ on a given state yields three possibilities. The first case is a state vector which contains both α and β, then either α or β and finally none of them.

The first case results in

$$\begin{aligned} a_\alpha^\dagger a_\beta |\alpha\beta\alpha_1\alpha_2\dots\alpha_{n-2}\rangle &= 0 \\ a_\beta a_\alpha^\dagger |\alpha\beta\alpha_1\alpha_2\dots\alpha_{n-2}\rangle &= 0 \end{aligned} \tag{3.23}$$

while the second case gives

$$\begin{aligned} a_\alpha^\dagger a_\beta |\beta \underbrace{\alpha_1\alpha_2\dots\alpha_{n-1}}_{\neq\alpha}\rangle &= |\alpha \underbrace{\alpha_1\alpha_2\dots\alpha_{n-1}}_{\neq\alpha}\rangle \\ a_\beta a_\alpha^\dagger |\beta \underbrace{\alpha_1\alpha_2\dots\alpha_{n-1}}_{\neq\alpha}\rangle &= a_\beta |\alpha\beta \underbrace{\beta\alpha_1\alpha_2\dots\alpha_{n-1}}_{\neq\alpha}\rangle \\ &= -|\alpha \underbrace{\alpha_1\alpha_2\dots\alpha_{n-1}}_{\neq\alpha}\rangle \end{aligned} \tag{3.24}$$

Finally if the state vector does not contain α and β

$$\begin{aligned} a_\alpha^\dagger a_\beta |\underbrace{\alpha_1\alpha_2\dots\alpha_n}_{\neq\alpha,\beta}\rangle &= & 0 \\ a_\beta a_\alpha^\dagger |\underbrace{\alpha_1\alpha_2\dots\alpha_n}_{\neq\alpha,\beta}\rangle &= & a_\beta |\alpha \underbrace{\alpha_1\alpha_2\dots\alpha_n}_{\neq\alpha,\beta}\rangle = 0 \end{aligned} \tag{3.25}$$

For all three cases we have

$$\{a_\alpha^\dagger, a_\beta\} = a_\alpha^\dagger a_\beta + a_\beta a_\alpha^\dagger = 0, \quad \alpha \neq \beta \tag{3.26}$$

We can summarize our findings in Eqs. (3.22) and (3.26) as

$$\{a_\alpha^\dagger, a_\beta\} = \delta_{\alpha\beta} \tag{3.27}$$

with $\delta_{\alpha\beta}$ is the Kroenecker δ-symbol.

The properties of the creation and annihilation operators can be summarized as (for fermions)

$$a_\alpha^\dagger|0\rangle \equiv |\alpha\rangle,$$

and

$$a_\alpha^\dagger|\alpha_1 \dots \alpha_n\rangle_{\mathrm{AS}} \equiv |\alpha\alpha_1 \dots \alpha_n\rangle_{\mathrm{AS}}.$$

from which follows

$$|\alpha_1 \dots \alpha_n\rangle_{\mathrm{AS}} = a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger \dots a_{\alpha_n}^\dagger|0\rangle.$$

The hermitian conjugate has the folowing properties

$$a_\alpha = (a_\alpha^\dagger)^\dagger.$$

Finally we found

$$a_\alpha \underbrace{|\alpha_1'\alpha_2' \dots \alpha_{n+1}'}\rangle_{\neq\alpha} = 0, \quad \text{in particular } a_\alpha|0\rangle = 0,$$

and

$$a_\alpha|\alpha\alpha_1\alpha_2 \dots \alpha_n\rangle = |\alpha_1\alpha_2 \dots \alpha_n\rangle,$$

and the corresponding commutator algebra

$$\{a_\alpha^\dagger, a_\beta^\dagger\} = \{a_\alpha, a_\beta\} = 0 \qquad \{a_\alpha^\dagger, a_\beta\} = \delta_{\alpha\beta}.$$

3.1 One-body operators in second quantization

The result we are about to derive was obtained once already, in Section 2.8, by integrating over permutations of the Hartree function. It is worth comparing the two routes as we go: what took a page of careful argument there follows here from the anticommutation relations alone.

A very useful operator is the so-called number-operator. Most physics cases we will study in this text conserve the total number of particles. The number operator is therefore a useful quantity which allows us to test that our many-body formalism conserves the number of particles. In for example (d,p) or (p,d) reactions it is important to be able to describe quantum mechanical states where particles get added or removed. A creation operator $a_\alpha^\dagger$ adds one particle to the single-particle state α of a give many-body state vector, while an annihilation operator a_α removes a particle from a single-particle state α.

Let us consider an operator proportional with $a_\alpha^\dagger a_\beta$ and $\alpha = \beta$. It acts on an n-particle state resulting in

$$a_\alpha^\dagger a_\alpha|\alpha_1\alpha_2 \dots \alpha_n\rangle = \begin{cases} 0 & \alpha \notin \{\alpha_i\} \\ \\ |\alpha_1\alpha_2 \dots \alpha_n\rangle & \alpha \in \{\alpha_i\} \end{cases} \tag{3.28}$$

Summing over all possible one-particle states we arrive at

$$\left(\sum_\alpha a_\alpha^\dagger a_\alpha\right)|\alpha_1\alpha_2 \dots \alpha_n\rangle = n|\alpha_1\alpha_2 \dots \alpha_n\rangle \tag{3.29}$$

The operator

$$\hat{N} = \sum_{\alpha} a_{\alpha}^{\dagger} a_{\alpha} \tag{3.30}$$

is called the number operator since it counts the number of particles in a give state vector when it acts on the different single-particle states. It acts on one single-particle state at the time and falls therefore under category one-body operators. Next we look at another important one-body operator, namely $\hat{H}_0$ and study its operator form in the occupation number representation.

We want to obtain an expression for a one-body operator which conserves the number of particles. Here we study the one-body operator for the kinetic energy plus an eventual external one-body potential. The action of this operator on a particular n-body state with its pertinent expectation value has already been studied in coordinate space. In coordinate space the operator reads

$$\hat{H}_0 = \sum_i \hat{h}_0(x_i) \tag{3.31}$$

and the anti-symmetric n-particle Slater determinant is defined as

$$\Phi(x_1, x_2, \dots, x_n, \alpha_1, \alpha_2, \dots, \alpha_n) = \frac{1}{\sqrt{n!}} \sum_p (-1)^p \hat{P} \psi_{\alpha_1}(x_1) \psi_{\alpha_2}(x_2) \dots \psi_{\alpha_n}(x_n).$$

Defining

$$\hat{h}_0(x_i) \psi_{\alpha_i}(x_i) = \sum_{\alpha_k'} \psi_{\alpha_k'}(x_i) \langle \alpha_k' | \hat{h}_0 | \alpha_k \rangle \tag{3.32}$$

we can easily evaluate the action of $\hat{H}_0$ on each product of one-particle functions in Slater determinant. From Eq. (3.32) we obtain the following result without permuting any particle pair

$$\begin{aligned}
& \left(\sum_i \hat{h}_0(x_i)\right) \psi_{\alpha_1}(x_1) \psi_{\alpha_2}(x_2) \dots \psi_{\alpha_n}(x_n) \\
= & \sum_{\alpha_1'} \langle \alpha_1' | \hat{h}_0 | \alpha_1 \rangle \psi_{\alpha_1'}(x_1) \psi_{\alpha_2}(x_2) \dots \psi_{\alpha_n}(x_n) \\
+ & \sum_{\alpha_2'} \langle \alpha_2' | \hat{h}_0 | \alpha_2 \rangle \psi_{\alpha_1}(x_1) \psi_{\alpha_2'}(x_2) \dots \psi_{\alpha_n}(x_n) \\
+ & \dots \\
+ & \sum_{\alpha_n'} \langle \alpha_n' | \hat{h}_0 | \alpha_n \rangle \psi_{\alpha_1}(x_1) \psi_{\alpha_2}(x_2) \dots \psi_{\alpha_n'}(x_n)
\end{aligned} \tag{3.33}$$

If we interchange particles 1 and 2 we obtain

$$\begin{aligned}
& \left(\sum_i \hat{h}_0(x_i)\right) \psi_{\alpha_1}(x_2) \psi_{\alpha_1}(x_2) \dots \psi_{\alpha_n}(x_n) \\
= & \sum_{\alpha_2'} \langle \alpha_2' | \hat{h}_0 | \alpha_2 \rangle \psi_{\alpha_1}(x_2) \psi_{\alpha_2'}(x_1) \dots \psi_{\alpha_n}(x_n) \\
+ & \sum_{\alpha_1'} \langle \alpha_1' | \hat{h}_0 | \alpha_1 \rangle \psi_{\alpha_1'}(x_2) \psi_{\alpha_2}(x_1) \dots \psi_{\alpha_n}(x_n) \\
+ & \dots \\
+ & \sum_{\alpha_n'} \langle \alpha_n' | \hat{h}_0 | \alpha_n \rangle \psi_{\alpha_1}(x_2) \psi_{\alpha_1}(x_2) \dots \psi_{\alpha_n'}(x_n)
\end{aligned} \tag{3.34}$$

We can continue by computing all possible permutations. We rewrite also our Slater determinant in its second quantized form and skip the dependence on the quantum numbers x_i. Summing up all contributions and taking care of all phases $(-1)^p$ we arrive at

$$\begin{aligned}
\hat{H}_0|\alpha_1,\alpha_2,\dots,\alpha_n\rangle = & \sum_{\alpha_1'}\langle\alpha_1'|\hat{h}_0|\alpha_1\rangle|\alpha_1'\alpha_2\dots\alpha_n\rangle \\
+ & \sum_{\alpha_2'}\langle\alpha_2'|\hat{h}_0|\alpha_2\rangle|\alpha_1\alpha_2'\dots\alpha_n\rangle \\
+ & \dots \\
+ & \sum_{\alpha_n'}\langle\alpha_n'|\hat{h}_0|\alpha_n\rangle|\alpha_1\alpha_2\dots\alpha_n'\rangle
\end{aligned} \tag{3.35}$$

In Eq. (3.35) we have expressed the action of the one-body operator of Eq. (3.31) on the n-body state in its second quantized form. This equation can be further manipulated if we use the properties of the creation and annihilation operator on each primed quantum number, that is

$$|\alpha_1\alpha_2\dots\alpha_k'\dots\alpha_n\rangle = a^{\dagger}_{\alpha_k'}a_{\alpha_k}|\alpha_1\alpha_2\dots\alpha_k\dots\alpha_n\rangle \tag{3.36}$$

Inserting this in the right-hand side of Eq. (3.35) results in

$$\begin{aligned}
\hat{H}_0|\alpha_1\alpha_2\dots\alpha_n\rangle = & \sum_{\alpha_1'}\langle\alpha_1'|\hat{h}_0|\alpha_1\rangle a^{\dagger}_{\alpha_1'}a_{\alpha_1}|\alpha_1\alpha_2\dots\alpha_n\rangle \\
+ & \sum_{\alpha_2'}\langle\alpha_2'|\hat{h}_0|\alpha_2\rangle a^{\dagger}_{\alpha_2'}a_{\alpha_2}|\alpha_1\alpha_2\dots\alpha_n\rangle \\
+ & \dots \\
+ & \sum_{\alpha_n'}\langle\alpha_n'|\hat{h}_0|\alpha_n\rangle a^{\dagger}_{\alpha_n'}a_{\alpha_n}|\alpha_1\alpha_2\dots\alpha_n\rangle \\
= & \sum_{\alpha,\beta}\langle\alpha|\hat{h}_0|\beta\rangle a^{\dagger}_{\alpha}a_{\beta}|\alpha_1\alpha_2\dots\alpha_n\rangle
\end{aligned} \tag{3.37}$$

In the number occupation representation or second quantization we get the following expression for a one-body operator which conserves the number of particles

$$\hat{H}_0 = \sum_{\alpha\beta}\langle\alpha|\hat{h}_0|\beta\rangle a^{\dagger}_{\alpha}a_{\beta} \tag{3.38}$$

Obviously, $\hat{H}_0$ can be replaced by any other one-body operator which preserved the number of particles. The stucture of the operator is therefore not limited to say the kinetic or single-particle energy only.

The opearator $\hat{H}_0$ takes a particle from the single-particle state β to the single-particle state α with a probability for the transition given by the expectation value $\langle\alpha|\hat{h}_0|\beta\rangle$.

It is instructive to verify Eq. (3.38) by computing the expectation value of $\hat{H}_0$ between two single-particle states

$$\langle\alpha_1|\hat{h}_0|\alpha_2\rangle = \sum_{\alpha\beta}\langle\alpha|\hat{h}_0|\beta\rangle\langle 0|a_{\alpha_1}a^{\dagger}_{\alpha}a_{\beta}a^{\dagger}_{\alpha_2}|0\rangle \tag{3.39}$$

Using the commutation relations for the creation and annihilation operators we have

$$a_{\alpha_1}a^{\dagger}_{\alpha}a_{\beta}a^{\dagger}_{\alpha_2} = (\delta_{\alpha\alpha_1} - a^{\dagger}_{\alpha}a_{\alpha_1})(\delta_{\beta\alpha_2} - a^{\dagger}_{\alpha_2}a_{\beta}), \tag{3.40}$$

which results in

$$\langle 0|a_{\alpha_1}a^{\dagger}_{\alpha}a_{\beta}a^{\dagger}_{\alpha_2}|0\rangle = \delta_{\alpha\alpha_1}\delta_{\beta\alpha_2} \tag{3.41}$$

and

$$\langle\alpha_1|\hat{h}_0|\alpha_2\rangle = \sum_{\alpha\beta}\langle\alpha|\hat{h}_0|\beta\rangle\delta_{\alpha\alpha_1}\delta_{\beta\alpha_2} = \langle\alpha_1|\hat{h}_0|\alpha_2\rangle \tag{3.42}$$

3.2 Two-body operators in second quantization

The antisymmetrised matrix elements $\langle pq|\hat{v}|rs\rangle_{AS}$ that appear below are exactly those of Eq. (2.36), and the n^4 of them are the storage problem discussed in Section 2.10. Nothing in this chapter changes that count; what changes is the ease with which they are contracted.

Let us now derive the expression for our two-body interaction part, which also conserves the number of particles. We can proceed in exactly the same way as for the one-body operator. In the coordinate representation our two-body interaction part takes the following expression

$$\hat{H}_I = \sum_{i<j} V(x_i,x_j) \tag{3.43}$$

where the summation runs over distinct pairs. The term V can be an interaction model for the nucleon-nucleon interaction or the interaction between two electrons. It can also include additional two-body interaction terms.

The action of this operator on a product of two single-particle functions is defined as

$$V(x_i,x_j)\psi_{\alpha_k}(x_i)\psi_{\alpha_l}(x_j) = \sum_{\alpha_k'\alpha_l'}\psi'_{\alpha_k}(x_i)\psi'_{\alpha_l}(x_j)\langle\alpha_k'\alpha_l'|\hat{v}|\alpha_k\alpha_l\rangle \tag{3.44}$$

We can now let $\hat{H}_I$ act on all terms in the linear combination for $|\alpha_1\alpha_2\dots\alpha_n\rangle$. Without any permutations we have

$$\begin{aligned}
& \left(\sum_{i<j}V(x_i,x_j)\right)\psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\dots\psi_{\alpha_n}(x_n) \\
= & \sum_{\alpha_1'\alpha_2'}\langle\alpha_1'\alpha_2'|\hat{v}|\alpha_1\alpha_2\rangle\psi'_{\alpha_1}(x_1)\psi'_{\alpha_2}(x_2)\dots\psi_{\alpha_n}(x_n) \\
+ & \dots \\
+ & \sum_{\alpha_1'\alpha_n'}\langle\alpha_1'\alpha_n'|\hat{v}|\alpha_1\alpha_n\rangle\psi'_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\dots\psi'_{\alpha_n}(x_n) \\
+ & \dots \\
+ & \sum_{\alpha_2'\alpha_n'}\langle\alpha_2'\alpha_n'|\hat{v}|\alpha_2\alpha_n\rangle\psi_{\alpha_1}(x_1)\psi'_{\alpha_2}(x_2)\dots\psi'_{\alpha_n}(x_n) \\
+ & \dots
\end{aligned} \tag{3.45}$$

where on the rhs we have a term for each distinct pairs.

For the other terms on the rhs we obtain similar expressions and summing over all terms we obtain

$$
\begin{aligned}
H_I|\alpha_1\alpha_2\dots\alpha_n\rangle = & \sum_{\alpha_1',\alpha_2'} \langle\alpha_1'\alpha_2'|\hat{v}|\alpha_1\alpha_2\rangle|\alpha_1'\alpha_2'\dots\alpha_n\rangle \\
+ & \dots \\
+ & \sum_{\alpha_1',\alpha_n'} \langle\alpha_1'\alpha_n'|\hat{v}|\alpha_1\alpha_n\rangle|\alpha_1'\alpha_2\dots\alpha_n'\rangle \\
+ & \dots \\
+ & \sum_{\alpha_2',\alpha_n'} \langle\alpha_2'\alpha_n'|\hat{v}|\alpha_2\alpha_n\rangle|\alpha_1\alpha_2'\dots\alpha_n'\rangle \\
+ & \dots
\end{aligned}
\tag{3.46}
$$

We introduce second quantization via the relation

$$
\begin{aligned}
& a^\dagger_{\alpha_k'} a^\dagger_{\alpha_l'} a_{\alpha_l} a_{\alpha_k}|\alpha_1\alpha_2\dots\alpha_k\dots\alpha_l\dots\alpha_n\rangle \\
= & (-1)^{k-1}(-1)^{l-2} a^\dagger_{\alpha_k'} a^\dagger_{\alpha_l'} a_{\alpha_l} a_{\alpha_k}|\alpha_k\alpha_l\underbrace{\alpha_1\alpha_2\dots\alpha_n}_{\neq\alpha_k,\alpha_l}\rangle \\
= & (-1)^{k-1}(-1)^{l-2}|\alpha_k'\alpha_l'\underbrace{\alpha_1\alpha_2\dots\alpha_n}_{\neq\alpha_k',\alpha_l'}\rangle \\
= & |\alpha_1\alpha_2\dots\alpha_k'\dots\alpha_l'\dots\alpha_n\rangle
\end{aligned}
\tag{3.47}
$$

Inserting this in (3.46) gives

$$
\begin{aligned}
H_I|\alpha_1\alpha_2\dots\alpha_n\rangle = & \sum_{\alpha_1',\alpha_2'} \langle\alpha_1'\alpha_2'|\hat{v}|\alpha_1\alpha_2\rangle a^\dagger_{\alpha_1'} a^\dagger_{\alpha_2'} a_{\alpha_2} a_{\alpha_1}|\alpha_1\alpha_2\dots\alpha_n\rangle \\
+ & \dots \\
= & \sum_{\alpha_1',\alpha_n'} \langle\alpha_1'\alpha_n'|\hat{v}|\alpha_1\alpha_n\rangle a^\dagger_{\alpha_1'} a^\dagger_{\alpha_n'} a_{\alpha_n} a_{\alpha_1}|\alpha_1\alpha_2\dots\alpha_n\rangle \\
+ & \dots \\
= & \sum_{\alpha_2',\alpha_n'} \langle\alpha_2'\alpha_n'|\hat{v}|\alpha_2\alpha_n\rangle a^\dagger_{\alpha_2'} a^\dagger_{\alpha_n'} a_{\alpha_n} a_{\alpha_2}|\alpha_1\alpha_2\dots\alpha_n\rangle \\
+ & \dots \\
= & {\sum_{\alpha,\beta,\gamma,\delta}}' \langle\alpha\beta|\hat{v}|\gamma\delta\rangle a^\dagger_\alpha a^\dagger_\beta a_\delta a_\gamma|\alpha_1\alpha_2\dots\alpha_n\rangle
\end{aligned}
\tag{3.48}
$$

Here we let $\sum'$ indicate that the sums running over α and β run over all single-particle states, while the summations γ and δ run over all pairs of single-particle states. We wish to remove this restriction and since

$$
\langle\alpha\beta|\hat{v}|\gamma\delta\rangle = \langle\beta\alpha|\hat{v}|\delta\gamma\rangle \tag{3.49}
$$

we get

$$
\sum_{\alpha\beta}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle a^\dagger_\alpha a^\dagger_\beta a_\delta a_\gamma = \sum_{\alpha\beta}\langle\beta\alpha|\hat{v}|\delta\gamma\rangle a^\dagger_\alpha a^\dagger_\beta a_\delta a_\gamma \tag{3.50}
$$

$$
= \sum_{\alpha\beta}\langle\beta\alpha|\hat{v}|\delta\gamma\rangle a^\dagger_\beta a^\dagger_\alpha a_\gamma a_\delta \tag{3.51}
$$

where we have used the anti-commutation rules.

Changing the summation indices α and β in (3.51) we obtain

$$\sum_{\alpha\beta}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma = \sum_{\alpha\beta}\langle\alpha\beta|\hat{v}|\delta\gamma\rangle a_\alpha^\dagger a_\beta^\dagger a_\gamma a_\delta \tag{3.52}$$

From this it follows that the restriction on the summation over γ and δ can be removed if we multiply with a factor $\frac{1}{2}$, resulting in

$$\hat{H}_I = \frac{1}{2}\sum_{\alpha\beta\gamma\delta}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma \tag{3.53}$$

where we sum freely over all single-particle states α, β, γ og δ.

With this expression we can now verify that the second quantization form of $\hat{H}_I$ in Eq. (3.53) results in the same matrix between two anti-symmetrized two-particle states as its corresponding coordinate space representation. We have

$$\langle\alpha_1\alpha_2|\hat{H}_I|\beta_1\beta_2\rangle = \frac{1}{2}\sum_{\alpha\beta\gamma\delta}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle\langle 0|a_{\alpha_2}a_{\alpha_1}a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma a_{\beta_1}^\dagger a_{\beta_2}^\dagger|0\rangle. \tag{3.54}$$

Using the commutation relations we get

$$\begin{aligned}
& a_{\alpha_2}a_{\alpha_1}a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma a_{\beta_1}^\dagger a_{\beta_2}^\dagger \\
= {} & a_{\alpha_2}a_{\alpha_1}a_\alpha^\dagger a_\beta^\dagger(a_\delta\delta_{\gamma\beta_1}a_{\beta_2}^\dagger - a_\delta a_{\beta_1}^\dagger a_\gamma a_{\beta_2}^\dagger) \\
= {} & a_{\alpha_2}a_{\alpha_1}a_\alpha^\dagger a_\beta^\dagger(\delta_{\gamma\beta_1}\delta_{\delta\beta_2} - \delta_{\gamma\beta_1}a_{\beta_2}^\dagger a_\delta - a_\delta a_{\beta_1}^\dagger\delta_{\gamma\beta_2} + a_\delta a_{\beta_1}^\dagger a_{\beta_2}^\dagger a_\gamma) \\
= {} & a_{\alpha_2}a_{\alpha_1}a_\alpha^\dagger a_\beta^\dagger(\delta_{\gamma\beta_1}\delta_{\delta\beta_2} - \delta_{\gamma\beta_1}a_{\beta_2}^\dagger a_\delta \\
& -\delta_{\delta\beta_1}\delta_{\gamma\beta_2} + \delta_{\gamma\beta_2}a_{\beta_1}^\dagger a_\delta + a_\delta a_{\beta_1}^\dagger a_{\beta_2}^\dagger a_\gamma)
\end{aligned} \tag{3.55}$$

The vacuum expectation value of this product of operators becomes

$$\begin{aligned}
& \langle 0|a_{\alpha_2}a_{\alpha_1}a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma a_{\beta_1}^\dagger a_{\beta_2}^\dagger|0\rangle \\
= {} & (\delta_{\gamma\beta_1}\delta_{\delta\beta_2} - \delta_{\delta\beta_1}\delta_{\gamma\beta_2})\langle 0|a_{\alpha_2}a_{\alpha_1}a_\alpha^\dagger a_\beta^\dagger|0\rangle \\
= {} & (\delta_{\gamma\beta_1}\delta_{\delta\beta_2} - \delta_{\delta\beta_1}\delta_{\gamma\beta_2})(\delta_{\alpha\alpha_1}\delta_{\beta\alpha_2} - \delta_{\beta\alpha_1}\delta_{\alpha\alpha_2})
\end{aligned} \tag{3.56}$$

Insertion of Eq. (3.56) in Eq. (3.54) results in

$$\begin{aligned}
\langle\alpha_1\alpha_2|\hat{H}_I|\beta_1\beta_2\rangle = {} & \frac{1}{2}\big[\langle\alpha_1\alpha_2|\hat{v}|\beta_1\beta_2\rangle - \langle\alpha_1\alpha_2|\hat{v}|\beta_2\beta_1\rangle \\
& -\langle\alpha_2\alpha_1|\hat{v}|\beta_1\beta_2\rangle + \langle\alpha_2\alpha_1|\hat{v}|\beta_2\beta_1\rangle\big] \\
= {} & \langle\alpha_1\alpha_2|\hat{v}|\beta_1\beta_2\rangle - \langle\alpha_1\alpha_2|\hat{v}|\beta_2\beta_1\rangle \\
= {} & \langle\alpha_1\alpha_2|\hat{v}|\beta_1\beta_2\rangle_{\mathrm{AS}}.
\end{aligned} \tag{3.57}$$

The two-body operator can also be expressed in terms of the anti-symmetrized matrix elements we discussed previously as

$$\begin{aligned}
\hat{H}_I = {} & \frac{1}{2}\sum_{\alpha\beta\gamma\delta}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma \\
= {} & \frac{1}{4}\sum_{\alpha\beta\gamma\delta}[\langle\alpha\beta|\hat{v}|\gamma\delta\rangle - \langle\alpha\beta|\hat{v}|\delta\gamma\rangle]a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma \\
= {} & \frac{1}{4}\sum_{\alpha\beta\gamma\delta}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle_{\mathrm{AS}}a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma
\end{aligned} \tag{3.58}$$

The factors in front of the operator, either $\frac{1}{4}$ or $\frac{1}{2}$ tells whether we use antisymmetrized matrix elements or not.

We can now express the Hamiltonian operator for a many-fermion system in the occupation basis representation as

$$H = \sum_{\alpha,\beta} \langle \alpha | \hat{t} + \hat{u}_{\mathrm{ext}} | \beta \rangle a_\alpha^\dagger a_\beta + \frac{1}{4} \sum_{\alpha\beta\gamma\delta} \langle \alpha\beta | \hat{v} | \gamma\delta \rangle a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma. \tag{3.59}$$

This is the form we will use in the rest of these lectures, assuming that we work with anti-symmetrized two-body matrix elements.

3.3 Particle-hole formalism

The index conventions used here – i, j, k, l for hole states below the Fermi level, a, b, c, d for particle states above it, and p, q, r, s for either – were introduced in Section 2.6. We now give them their proper setting: the reference determinant becomes a new vacuum, and the operators are redefined with respect to it.

Second quantization is a useful and elegant formalism for constructing many-body states and quantum mechanical operators. One can express and translate many physical processes into simple pictures such as Feynman diagrams. Expecation values of many-body states are also easily calculated. However, although the equations are seemingly easy to set up, from a practical point of view, that is the solution of Schroedinger's equation, there is no particular gain. The many-body equation is equally hard to solve, irrespective of representation. The cliche that there is no free lunch brings us down to earth again. Note however that a transformation to a particular basis, for cases where the interaction obeys specific symmetries, can ease the solution of Schroedinger's equation.

But there is at least one important case where second quantization comes to our rescue. It is namely easy to introduce another reference state than the pure vacuum $|0\rangle$, where all single-particle states are active. With many particles present it is often useful to introduce another reference state than the vacuum state$|0\rangle$. We will label this state $|c\rangle$ (c for core) and as we will see it can reduce considerably the complexity and thereby the dimensionality of the many-body problem. It allows us to sum up to infinite order specific many-body correlations. The particle-hole representation is one of these handy representations.

In the original particle representation these states are products of the creation operators $a_{\alpha_i}^\dagger$ acting on the true vacuum $|0\rangle$. Following Eq. (3.3) we have

$$|\alpha_1\alpha_2 \dots \alpha_{n-1}\alpha_n\rangle = \quad a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger \dots a_{\alpha_{n-1}}^\dagger a_{\alpha_n}^\dagger |0\rangle \tag{3.60}$$

$$|\alpha_1\alpha_2 \dots \alpha_{n-1}\alpha_n\alpha_{n+1}\rangle = \quad a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger \dots a_{\alpha_{n-1}}^\dagger a_{\alpha_n}^\dagger a_{\alpha_{n+1}}^\dagger |0\rangle \tag{3.61}$$

$$|\alpha_1\alpha_2 \dots \alpha_{n-1}\rangle = \quad a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger \dots a_{\alpha_{n-1}}^\dagger |0\rangle \tag{3.62}$$

If we use Eq. (3.60) as our new reference state, we can simplify considerably the representation of this state

$$|c\rangle \equiv |\alpha_1\alpha_2 \dots \alpha_{n-1}\alpha_n\rangle = a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger \dots a_{\alpha_{n-1}}^\dagger a_{\alpha_n}^\dagger |0\rangle \tag{3.63}$$

The new reference states for the $n+1$ and $n-1$ states can then be written as

$$|\alpha_1\alpha_2 \dots \alpha_{n-1}\alpha_n\alpha_{n+1}\rangle = \quad (-1)^n a_{\alpha_{n+1}}^\dagger |c\rangle \equiv (-1)^n |\alpha_{n+1}\rangle_c \tag{3.64}$$

$$|\alpha_1\alpha_2 \dots \alpha_{n-1}\rangle = \quad (-1)^{n-1} a_{\alpha_n} |c\rangle \equiv (-1)^{n-1} |\alpha_{n-1}\rangle_c \tag{3.65}$$

The first state has one additional particle with respect to the new vacuum state $|c\rangle$ and is normally referred to as a one-particle state or one particle added to the many-body reference state. The second state has one particle less than the reference vacuum state $|c\rangle$ and is referred to as a one-hole state. When dealing with a new reference state it is often convenient to introduce new creation and annihilation operators since we have from Eq. (3.65)

$$a_\alpha|c\rangle \neq 0 \tag{3.66}$$

since α is contained in $|c\rangle$, while for the true vacuum we have $a_\alpha|0\rangle = 0$ for all α.

The new reference state leads to the definition of new creation and annihilation operators which satisfy the following relations

$$\begin{aligned} b_\alpha|c\rangle &= 0 & (3.67)\\ \{b_\alpha^\dagger, b_\beta^\dagger\} = \{b_\alpha, b_\beta\} &= 0 \\ \{b_\alpha^\dagger, b_\beta\} &= \delta_{\alpha\beta} & (3.68) \end{aligned}$$

We assume also that the new reference state is properly normalized

$$\langle c|c\rangle = 1 \tag{3.69}$$

The physical interpretation of these new operators is that of so-called quasiparticle states. This means that a state defined by the addition of one extra particle to a reference state $|c\rangle$ may not necesseraly be interpreted as one particle coupled to a core. We define now new creation operators that act on a state α creating a new quasiparticle state

$$b_\alpha^\dagger|c\rangle = \begin{cases} a_\alpha^\dagger|c\rangle = |\alpha\rangle, & \alpha > F \\ \\ a_\alpha|c\rangle = |\alpha^{-1}\rangle, & \alpha \leq F \end{cases} \tag{3.70}$$

where F is the Fermi level representing the last occupied single-particle orbit of the new reference state $|c\rangle$.

The annihilation is the hermitian conjugate of the creation operator

$$b_\alpha = (b_\alpha^\dagger)^\dagger,$$

resulting in

$$b_\alpha^\dagger = \begin{cases} a_\alpha^\dagger & \alpha > F \\ \\ a_\alpha & \alpha \leq F \end{cases} \qquad b_\alpha = \begin{cases} a_\alpha & \alpha > F \\ \\ a_\alpha^\dagger & \alpha \leq F \end{cases} \tag{3.71}$$

With the new creation and annihilation operator we can now construct many-body quasiparticle states, with one-particle-one-hole states, two-particle-two-hole states etc in the same fashion as we previously constructed many-particle states. We can write a general particle-hole state as

$$|\beta_1\beta_2\dots\beta_{n_p}\gamma_1^{-1}\gamma_2^{-1}\dots\gamma_{n_h}^{-1}\rangle \equiv \underbrace{b_{\beta_1}^\dagger b_{\beta_2}^\dagger \dots b_{\beta_{n_p}}^\dagger}_{>F} \underbrace{b_{\gamma_1}^\dagger b_{\gamma_2}^\dagger \dots b_{\gamma_{n_h}}^\dagger}_{\leq F} |c\rangle \tag{3.72}$$

We can now rewrite our one-body and two-body operators in terms of the new creation and annihilation operators. The number operator becomes

$$\hat{N} = \sum_\alpha a_\alpha^\dagger a_\alpha = \sum_{\alpha > F} b_\alpha^\dagger b_\alpha + n_c - \sum_{\alpha \leq F} b_\alpha^\dagger b_\alpha \tag{3.73}$$

where n_c is the number of particle in the new vacuum state $|c\rangle$. The action of $\hat{N}$ on a many-body state results in

$$N|\beta_1\beta_2\ldots\beta_{n_p}\gamma_1^{-1}\gamma_2^{-1}\ldots\gamma_{n_h}^{-1}\rangle = (n_p+n_c-n_h)|\beta_1\beta_2\ldots\beta_{n_p}\gamma_1^{-1}\gamma_2^{-1}\ldots\gamma_{n_h}^{-1}\rangle \tag{3.74}$$

Here $n = n_p+n_c-n_h$ is the total number of particles in the quasi-particle state of Eq. (3.72). Note that $\hat{N}$ counts the total number of particles present

$$N_{qp} = \sum_\alpha b_\alpha^\dagger b_\alpha, \tag{3.75}$$

gives us the number of quasi-particles as can be seen by computing

$$N_{qp} = |\beta_1\beta_2\ldots\beta_{n_p}\gamma_1^{-1}\gamma_2^{-1}\ldots\gamma_{n_h}^{-1}\rangle = (n_p+n_h)|\beta_1\beta_2\ldots\beta_{n_p}\gamma_1^{-1}\gamma_2^{-1}\ldots\gamma_{n_h}^{-1}\rangle \tag{3.76}$$

where $n_{qp} = n_p+n_h$ is the total number of quasi-particles.

We express the one-body operator $\hat{H}_0$ in terms of the quasi-particle creation and annihilation operators, resulting in

$$\begin{aligned}\hat{H}_0 = & \sum_{\alpha\beta>F}\langle\alpha|\hat{h}_0|\beta\rangle b_\alpha^\dagger b_\beta + \sum_{\alpha>F,\beta\leq F}\left[\langle\alpha|\hat{h}_0|\beta\rangle b_\alpha^\dagger b_\beta^\dagger + \langle\beta|\hat{h}_0|\alpha\rangle b_\beta b_\alpha\right] \\ + & \sum_{\alpha\leq F}\langle\alpha|\hat{h}_0|\alpha\rangle - \sum_{\alpha\beta\leq F}\langle\beta|\hat{h}_0|\alpha\rangle b_\alpha^\dagger b_\beta\end{aligned} \tag{3.77}$$

The first term gives contribution only for particle states, while the last one contributes only for holestates. The second term can create or destroy a set of quasi-particles and the third term is the contribution from the vacuum state $|c\rangle$.

Before we continue with the expressions for the two-body operator, we introduce a nomenclature we will use for the rest of this text. It is inspired by the notation used in quantum chemistry. We reserve the labels $i,j,k,\ldots$ for hole states and $a,b,c,\ldots$ for states above F, viz. particle states. This means also that we will skip the constraint $\leq F$ or $>F$ in the summation symbols. Our operator $\hat{H}_0$ reads now

$$\begin{aligned}\hat{H}_0 = & \sum_{ab}\langle a|\hat{h}|b\rangle b_a^\dagger b_b + \sum_{ai}\left[\langle a|\hat{h}|i\rangle b_a^\dagger b_i^\dagger + \langle i|\hat{h}|a\rangle b_i b_a\right] \\ + & \sum_i\langle i|\hat{h}|i\rangle - \sum_{ij}\langle j|\hat{h}|i\rangle b_i^\dagger b_j\end{aligned} \tag{3.78}$$

The two-particle operator in the particle-hole formalism is more complicated since we have to translate four indices $\alpha\beta\gamma\delta$ to the possible combinations of particle and hole states. When performing the commutator algebra we can regroup the operator in five different terms

$$\hat{H}_I = \hat{H}_I^{(a)} + \hat{H}_I^{(b)} + \hat{H}_I^{(c)} + \hat{H}_I^{(d)} + \hat{H}_I^{(e)} \tag{3.79}$$

Using anti-symmetrized matrix elements, the term $\hat{H}_I^{(a)}$ is

$$\hat{H}_I^{(a)} = \frac{1}{4}\sum_{abcd}\langle ab|\hat{V}|cd\rangle b_a^\dagger b_b^\dagger b_d b_c \tag{3.80}$$

The next term $\hat{H}_I^{(b)}$ reads

$$\hat{H}_I^{(b)} = \frac{1}{4}\sum_{abci}\left(\langle ab|\hat{V}|ci\rangle b_a^\dagger b_b^\dagger b_i^\dagger b_c + \langle ai|\hat{V}|cb\rangle b_a^\dagger b_i b_b b_c\right) \tag{3.81}$$

This term conserves the number of quasiparticles but creates or removes a three-particle-one-hole state. For $\hat{H}_I^{(c)}$ we have

$$\begin{aligned}\hat{H}_I^{(c)} = \quad & \frac{1}{4}\sum_{abij}\left(\langle ab|\hat{V}|ij\rangle b_a^\dagger b_b^\dagger b_j^\dagger b_i^\dagger + \langle ij|\hat{V}|ab\rangle b_a b_b b_j b_i\right) + \\ & \frac{1}{2}\sum_{abij}\langle ai|\hat{V}|bj\rangle b_a^\dagger b_j^\dagger b_b b_i + \frac{1}{2}\sum_{abi}\langle ai|\hat{V}|bi\rangle b_a^\dagger b_b. \end{aligned} \tag{3.82}$$

The first line stands for the creation of a two-particle-two-hole state, while the second line represents the creation to two one-particle-one-hole pairs while the last term represents a contribution to the particle single-particle energy from the hole states, that is an interaction between the particle states and the hole states within the new vacuum state. The fourth term reads

$$\begin{aligned}\hat{H}_I^{(d)} = \quad & \frac{1}{4}\sum_{aijk}\left(\langle ai|\hat{V}|jk\rangle b_a^\dagger b_k^\dagger b_j^\dagger b_i + \langle ji|\hat{V}|ak\rangle b_k^\dagger b_j b_i b_a\right) + \\ & \frac{1}{4}\sum_{aij}\left(\langle ai|\hat{V}|ji\rangle b_a^\dagger b_j^\dagger + \langle ji|\hat{V}|ai\rangle - \langle ji|\hat{V}|ia\rangle b_j b_a\right). \end{aligned} \tag{3.83}$$

The terms in the first line stand for the creation of a particle-hole state interacting with hole states, we will label this as a two-hole-one-particle contribution. The remaining terms are a particle-hole state interacting with the holes in the vacuum state. Finally we have

$$\hat{H}_I^{(e)} = \frac{1}{4}\sum_{ijkl}\langle kl|\hat{V}|ij\rangle b_i^\dagger b_j^\dagger b_l b_k + \frac{1}{2}\sum_{ijk}\langle ij|\hat{V}|kj\rangle b_k^\dagger b_i + \frac{1}{2}\sum_{ij}\langle ij|\hat{V}|ij\rangle \tag{3.84}$$

The first terms represents the interaction between two holes while the second stands for the interaction between a hole and the remaining holes in the vacuum state. It represents a contribution to single-hole energy to first order. The last term collects all contributions to the energy of the ground state of a closed-shell system arising from hole-hole correlations.

3.4 Summarizing and defining a normal-ordered Hamiltonian

Everything collected here has an exact counterpart in Chapter 2. The Slater determinant is Eq. (2.14); the antisymmetrised two-body matrix element is Eq. (2.36); and the energy functional that follows from the normal-ordered Hamiltonian is Eq. (2.37), whose minimisation gives the Hartree-Fock equations (2.41). The difference is that the derivations there ran over permutations, while here they will run over contractions.

$$\Phi_{AS}(\alpha_1,\dots,\alpha_A;x_1,\dots x_A) = \frac{1}{\sqrt{A}}\sum_{\hat{P}}(-1)^P\hat{P}\prod_{i=1}^{A}\psi_{\alpha_i}(x_i),$$

which is equivalent with $|\alpha_1\dots\alpha_A\rangle = a_{\alpha_1}^\dagger\dots a_{\alpha_A}^\dagger|0\rangle$. We have also

$$a_p^\dagger|0\rangle = |p\rangle, \quad a_p|q\rangle = \delta_{pq}|0\rangle$$

$$\delta_{pq} = \left\{a_p, a_q^\dagger\right\},$$

and

$$0 = \left\{a_p^\dagger, a_q\right\} = \left\{a_p, a_q\right\} = \left\{a_p^\dagger, a_q^\dagger\right\}$$

$$|\Phi_0\rangle = |\alpha_1 \dots \alpha_A\rangle, \quad \alpha_1,\dots,\alpha_A \leq \alpha_F$$

$$\left\{a_p^\dagger, a_q\right\} = \delta_{pq}, p,q \leq \alpha_F$$
$$\left\{a_p, a_q^\dagger\right\} = \delta_{pq}, p,q > \alpha_F$$

with $i,j,\dots \leq \alpha_F, \quad a,b,\dots > \alpha_F, \quad p,q,\dots -$ any

$$a_i|\Phi_0\rangle = |\Phi_i\rangle, \qquad a_a^\dagger|\Phi_0\rangle = |\Phi^a\rangle$$

and

$$a_i^\dagger|\Phi_0\rangle = 0 \qquad a_a|\Phi_0\rangle = 0$$

The one-body operator is defined as

$$\hat{F} = \sum_{pq} \langle p|\hat{f}|q\rangle a_p^\dagger a_q$$

while the two-body opreator is defined as

$$\hat{V} = \frac{1}{4}\sum_{pqrs} \langle pq|\hat{v}|rs\rangle_{AS} a_p^\dagger a_q^\dagger a_s a_r$$

where we have defined the antisymmetric matrix elements

$$\langle pq|\hat{v}|rs\rangle_{AS} = \langle pq|\hat{v}|rs\rangle - \langle pq|\hat{v}|sr\rangle.$$

We can also define a three-body operator

$$\hat{V}_3 = \frac{1}{36}\sum_{pqrstu} \langle pqr|\hat{v}_3|stu\rangle_{AS} a_p^\dagger a_q^\dagger a_r^\dagger a_u a_t a_s$$

with the antisymmetrized matrix element

$$\langle pqr|\hat{v}_3|stu\rangle_{AS} = \langle pqr|\hat{v}_3|stu\rangle + \langle pqr|\hat{v}_3|tus\rangle + \langle pqr|\hat{v}_3|ust\rangle - \langle pqr|\hat{v}_3|sut\rangle - \langle pqr|\hat{v}_3|tsu\rangle - \langle pqr|\hat{v}_3|uts\rangle. \tag{3.85}$$

3.5 Normal ordering

The manipulations of the previous sections all follow the same pattern. Take a product of creation and annihilation operators, push the annihilation operators to the right using the anticommutation relations until they meet the vacuum and give zero, and collect the Kronecker deltas that fall out along the way. For two operators this is immediate. For the four operators of an overlap between two-particle states it is already an exercise in care, and for the eight or more operators that arise in perturbation theory and coupled-cluster theory it is hopeless. Wick's theorem organises the whole calculation once and for all. We need two definitions first.

A product of creation and annihilation operators is in *normal-ordered* form when all creation operators stand to the left of all annihilation operators. For an arbitrary product we define the normal-ordered product

$$N[xyz\cdots w] = (-1)^{\kappa}\Big(\text{the same operators, all creation operators to the left,}\Big) \tag{3.86}$$

where κ counts the interchanges of neighbouring operators needed to bring the product into that form. Every interchange of two fermion operators costs a minus sign, and the factor $(-1)^\kappa$ records them. Two elementary cases are

$$N\left[a_p a_q^\dagger\right] = -a_q^\dagger a_p, \qquad N\left[a_p^\dagger a_q\right] = a_p^\dagger a_q, \tag{3.87}$$

the first requiring one interchange and the second none. The ordering of the creation operators among themselves, and of the annihilation operators among themselves, is a matter of convention as long as the same sign rule is applied consistently.

The one property that makes normal ordering useful follows immediately from $a_p|0\rangle = 0$ and $\langle 0|a_p^\dagger = 0$:

$$\boxed{\langle 0|N\left[xyz\cdots w\right]|0\rangle = 0} \qquad \text{for every non-empty product.} \tag{3.88}$$

Whatever else it contains, a normal-ordered product has an annihilation operator on the far right or a creation operator on the far left, and either one kills the vacuum. A normal-ordered product therefore contributes nothing to a vacuum expectation value. The whole problem of evaluating $\langle 0|xyz\cdots w|0\rangle$ reduces to finding the pieces that are *not* normal-ordered.

3.6 Contractions

The *contraction* of two operators is defined as the difference between their product and their normal-ordered product,

$$\overbracket{x\,y} \;\equiv\; xy - N\left[xy\right], \tag{3.89}$$

and we indicate it by a line joining the two operators. Since the normal-ordered piece has vanishing vacuum expectation value by Eq. (3.88), a contraction is just a number,

$$\overbracket{x\,y} = \langle 0|xy|0\rangle. \tag{3.90}$$

There are four possible pairs of operators, and only one of them survives:

$$\overbracket{a_p\,a_q^\dagger} = \langle 0|a_p a_q^\dagger|0\rangle = \delta_{pq}, \tag{3.91}$$

$$\overbracket{a_p\,a_q} = \langle 0|a_p a_q|0\rangle = 0, \tag{3.92}$$

$$\overbracket{a_q^\dagger\,a_p} = \langle 0|a_q^\dagger a_p|0\rangle = 0, \tag{3.93}$$

$$\overbracket{a_p^\dagger\,a_q^\dagger} = \langle 0|a_p^\dagger a_q^\dagger|0\rangle = 0. \tag{3.94}$$

Only an annihilation operator standing to the *left* of a creation operator gives a non-vanishing contraction. This single fact carries all the bookkeeping that follows: it is the statement that the vacuum contains no particles, and every zero in the calculations below can be traced back to it.

When the two contracted operators are not adjacent, the contraction carries the sign of the interchanges needed to bring them together. For a set of contractions the total sign is

$$(-1)^\nu, \qquad \nu = \text{number of crossings of the contraction lines,} \tag{3.95}$$

since each crossing corresponds to one interchange of neighbouring fermion operators. We shall use this rule constantly, and it is verified numerically for arbitrary strings in `wick.py`.

3.7 Two operators

We now build the theorem up from the simplest cases, using nothing more than the anticommutation relations. For two operators the content of the theorem is the definition (3.89) rearranged,

$$xy = N[xy] + \contraction{}{x}{}{y}xy. \tag{3.96}$$

Take $x = a_p$ and $y = a_q^\dagger$. The anticommutation relation $\{a_p, a_q^\dagger\} = \delta_{pq}$ gives

$$a_p a_q^\dagger = \delta_{pq} - a_q^\dagger a_p = \delta_{pq} + N\left[a_p a_q^\dagger\right], \tag{3.97}$$

where the second step is Eq. (3.87). Comparing with Eq. (3.96) identifies the contraction as δ_{pq}, which is Eq. (3.91). Taking the vacuum expectation value and using Eq. (3.88) recovers the overlap of two single-particle states,

$$\langle p|q\rangle = \langle 0|a_p a_q^\dagger|0\rangle = \delta_{pq}. \tag{3.98}$$

The three remaining combinations $a_p a_q$, $a_p^\dagger a_q$ and $a_p^\dagger a_q^\dagger$ are already normal-ordered up to a sign, so their contractions vanish and so do their vacuum expectation values.

3.8 Three operators

An odd number of operators can never be paired up completely. Whatever contractions we carry out, at least one operator is always left over, and a single uncontracted operator inside a vacuum expectation value gives

$$\langle 0|a_p|0\rangle = \langle 0|a_p^\dagger|0\rangle = 0. \tag{3.99}$$

Hence

$$\langle 0|xyz|0\rangle = 0 \tag{3.100}$$

for every choice of the three operators, as one may also check directly by anticommuting. Take $x = a_p$, $y = a_q^\dagger$, $z = a_r^\dagger$:

$$a_p a_q^\dagger a_r^\dagger = \left(\delta_{pq} - a_q^\dagger a_p\right) a_r^\dagger = \delta_{pq} a_r^\dagger - a_q^\dagger \left(\delta_{pr} - a_r^\dagger a_p\right), \tag{3.101}$$

and every surviving term contains a single unpaired creation operator acting on the vacuum, so the vacuum expectation value vanishes.

The result is worth stating explicitly because it is used constantly: *any* vacuum expectation value of an odd number of creation and annihilation operators is zero. The same argument reappears in the proof of the theorem, where it disposes of half the terms in the induction step.

3.9 Four operators

Four operators is the first genuinely interesting case, and it is the one that appears whenever we compute the overlap of two two-particle states,

$$|rs\rangle = a_r^\dagger a_s^\dagger|0\rangle, \qquad |pq\rangle = a_p^\dagger a_q^\dagger|0\rangle, \qquad \langle rs|pq\rangle = \langle 0|a_s a_r a_p^\dagger a_q^\dagger|0\rangle. \tag{3.102}$$

By anticommutation.

We evaluate it the elementary way first, so that the answer can be checked against the theorem afterwards. Anticommuting a_r past $a_p^\dagger$ with $a_r a_p^\dagger = \delta_{rp} - a_p^\dagger a_r$,

$$a_s a_r a_p^\dagger a_q^\dagger = a_s\left(\delta_{rp} - a_p^\dagger a_r\right)a_q^\dagger = \delta_{rp}\, a_s a_q^\dagger - a_s a_p^\dagger a_r a_q^\dagger. \tag{3.103}$$

The first term is handled by Eq. (3.97),

$$\delta_{rp}\, a_s a_q^\dagger = \delta_{rp}\left(\delta_{sq} - a_q^\dagger a_s\right) \;\longrightarrow\; \delta_{rp}\delta_{sq}, \tag{3.104}$$

the arrow denoting that we take the vacuum expectation value and drop the normal-ordered remainder. For the second term we anticommute twice more,

$$-a_s a_p^\dagger a_r a_q^\dagger = -a_s a_p^\dagger\left(\delta_{rq} - a_q^\dagger a_r\right) = -\delta_{rq}\, a_s a_p^\dagger + a_s a_p^\dagger a_q^\dagger a_r, \tag{3.105}$$

and the last term has an annihilation operator on the far right, so it dies on the vacuum. The surviving piece is

$$-\delta_{rq}\, a_s a_p^\dagger = -\delta_{rq}\left(\delta_{sp} - a_p^\dagger a_s\right) \;\longrightarrow\; -\delta_{rq}\delta_{sp}. \tag{3.106}$$

Collecting the two contributions,

$$\boxed{\langle rs|pq\rangle = \langle 0|a_s a_r a_p^\dagger a_q^\dagger|0\rangle = \delta_{rp}\delta_{sq} - \delta_{sp}\delta_{rq}} \tag{3.107}$$

Two-particle states are orthonormal up to precisely the antisymmetry expected of fermions: the overlap is the antisymmetrised product of single-particle overlaps.

By contractions.

The same answer can be read off without any algebra at all. Four operators can be paired in three ways. The first pairs the two annihilation operators with each other and the two creation operators with each other,

$$a_s\, a_r\, a_p^\dagger\, a_q^\dagger \;=\; 0, \tag{3.108}$$

which vanishes by Eqs. (3.92) and (3.94). The second is the nested pairing, with no crossings and hence a plus sign,

$$a_s\, a_r\, a_p^\dagger\, a_q^\dagger \;=\; +\delta_{sq}\delta_{rp}. \tag{3.109}$$

The third has the two lines crossing once, and therefore a minus sign by Eq. (3.95); we draw one line above and one below,

$$a_s\, a_r\, a_p^\dagger\, a_q^\dagger \;=\; -\delta_{sp}\delta_{rq}. \tag{3.110}$$

Adding the three gives Eq. (3.107) again.

Three pairings, one of which vanishes on inspection and two of which are read straight off: this is the economy the theorem buys. The program `wick.py` carries out both routes for arbitrary strings and confirms that they agree. For the eight-operator strings of Section 3.14 the anticommutation recursion takes 129 steps, while the theorem enumerates 105 pairings of which only a handful survive.

3.10 Wick's theorem

The pattern of the previous three sections generalises. *Wick's theorem* states that an arbitrary product of creation and annihilation operators can be written as the normal-ordered product plus the sum of all possible contractions,

$$\begin{aligned} xyz\cdots w = N\,[xyz\cdots w] & \\ +\sum_{(1)} N\,[xyz\cdots w] & \qquad \text{(all single contractions)} \\ +\sum_{(2)} N\,[xyz\cdots w] & \qquad \text{(all double contractions)} \\ +\cdots+\sum_{[M/2]} N\,[xyz\cdots w]\,, & \end{aligned} \tag{3.111}$$

where $\sum_{(m)}$ runs over all ways of choosing m disjoint pairs of operators to contract, each term carrying the sign (3.95), and $[M/2]$ is the largest number of disjoint pairs that M operators admit.

The theorem becomes a computational tool the moment we take a vacuum expectation value. Every term in Eq. (3.111) that still contains an uncontracted operator is normal-ordered and therefore vanishes by Eq. (3.88). Only the *fully* contracted terms survive:

$$\boxed{\langle 0|xyz\cdots w|0\rangle = \sum_{[M/2]} N\,[xyz\cdots w] \quad \text{(fully contracted terms only)}} \tag{3.112}$$

For M even there are $(M-1)!!$ such terms – one, three, fifteen and one hundred and five for $M = 2,4,6,8$ – and most of them vanish on inspection because one of their contractions is of the type (3.92)–(3.94). For M odd there are none, which is Eq. (3.100) again.

It is worth being precise about what this buys, and Figure 3.1 is drawn so that we can be. Three counts are plotted against the length of the string: the number of recursive calls the elementary anticommutation route of Sections 3.7–3.9 makes, the number $(2n-1)!!$ of pairings the theorem enumerates, and the number of those pairings that do not vanish on inspection. The last of the three is $n!$ exactly, since a non-zero contraction must join one of the n annihilation operators to one of the n creation operators and every such assignment is a permutation. The first thing to notice is that the plot does *not* show the theorem winning a race on raw operation count. For the eight operators of Eq. (3.132) the recursion takes 129 steps against 105 pairings, and from $2n = 10$ on the recursion is actually the cheaper of the two in this crude sense – 651 calls against 945 pairings at $2n = 10$, and 27 399 against 135 135 at $2n = 14$ – because every branch that begins with a creation operator or ends with an annihilation operator dies immediately.

What separates the two routes is therefore not the size of the count but what one can do with it. The recursion has no way of knowing its own structure until it has been carried out, one interchange at a time, and it delivers the answer as a single unsimplified expression

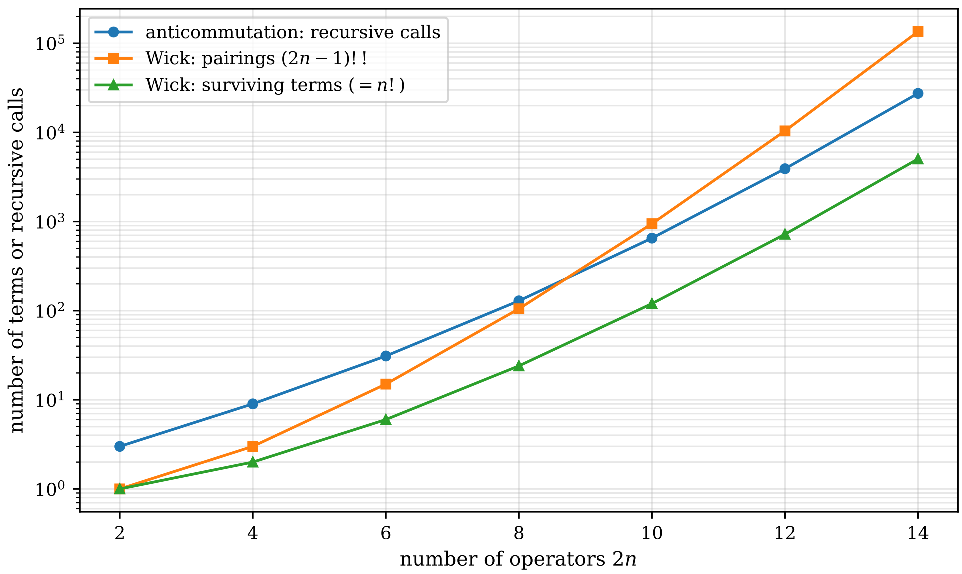

Fig. 3.1 Counting the work in a vacuum expectation value of $2n$ creation and annihilation operators, plotted logarithmically against the number of operators. The strings are the ones produced by `wick.py`, with all n annihilation operators standing to the left of all n creation operators. Circles give the number of recursive calls the brute-force anticommutation route actually makes, squares the number $(2n-1)!!$ of perfect matchings that Wick's theorem enumerates, and triangles the $n!$ of those matchings that survive the elementary contractions (3.92)–(3.94). The recursion prunes so hard on these particular strings that from $2n = 10$ onwards it makes fewer calls than there are pairings; the gap that matters is the one between either upper curve and the lowest, since the surviving terms are the only ones that ever have to be written down.

only at the very end; it cannot be stopped halfway, and it cannot be used as an ingredient in an algebraic argument. The pairings, by contrast, are known in advance: they are a purely combinatorial object, written down without any search at all, and each one can be inspected on its own. Most of them are then discarded at a glance, and the survivors – the lowest curve, 24 terms at $2n = 8$ and $5\,040$ at $2n = 14$ against the $135\,135$ pairings there – are read off directly with their signs taken from the crossings. It is the ratio between the middle curve and the bottom one, a factor of 27 already at $2n = 14$ and growing like $(2n-1)!!/n!$, that makes hand calculation possible, and it is the *a priori* character of the enumeration that makes the theorem an algebraic tool rather than merely an algorithm.

The practical consequence is the one anticipated in Section 2.16. Because an n-body operator can connect only Slater determinants differing by at most n single-particle states, a configuration-interaction Hamiltonian built from one- and two-body operators is extremely sparse: of the $\binom{n}{N}^2$ possible matrix elements only a vanishing fraction are non-zero. That sparsity is precisely what the Lanczos algorithm of Section 1.26 exploits, and it is why the Hamiltonian never has to be stored.

3.11 Proof of Wick's theorem

The proof is by induction on the number of operators. The case $M = 2$ is Eq. (3.96), established directly from the anticommutation relation. The induction step requires one preparatory result, which does all the real work.

A lemma: appending one operator on the right.

Let $x, y, z, \ldots, w$ be a set of operators and let Ω be one further creation or annihilation operator. Then

$$N\left[xyz\cdots w\right]\,\Omega = N\left[xyz\cdots w\,\Omega\right] + \sum_{(1)\Omega} N\left[xyz\cdots w\,\Omega\right], \tag{3.113}$$

where $\sum_{(1)\Omega}$ runs over all single contractions *involving* Ω, that is over the contractions of Ω with each of $x, y, z, \ldots, w$ in turn.

Proof of the lemma.

There are four cases to consider, and three of them are immediate.

(i) *Ω is an annihilation operator.* Then $N[xyz\cdots w]\,\Omega$ is already in normal-ordered form, since appending an annihilation operator on the far right cannot disturb the ordering, and so the left-hand side equals $N[xyz\cdots w\,\Omega]$. At the same time every contraction with Ω vanishes: a non-zero contraction requires an annihilation operator to the *left* of a creation operator, and here Ω is itself an annihilation operator standing to the right of everything. Both extra terms are absent and the lemma holds trivially.

(ii) *All of $x, y, \ldots, w$ are annihilation operators and Ω is an annihilation operator.* This is a special case of (i), and the lemma is again trivially satisfied.

(iii) *Ω is a creation operator.* Here we need only prove the lemma when all of $x, y, \ldots, w$ are annihilation operators. The reason is that a creation operator inside the normal-ordered product already stands to the left of Ω's eventual position, and its contraction with Ω vanishes by Eq. (3.94),

$$\overbracket{a_i^\dagger\, a_j^\dagger} = 0.$$

Creation operators in the string thus contribute neither to the reordering nor to the sum of contractions, and may be carried along untouched.

(iv) It remains to anticommute Ω leftwards through the annihilation operators $x, y, \ldots, w$, one at a time. Each step uses

$$a_j\,\Omega = \delta_{j\Omega} - \Omega\,a_j,$$

and produces two pieces: a Kronecker delta, which is precisely the contraction of a_j with Ω carrying the sign of the operators Ω has had to cross, and a term in which Ω has moved one place to the left. Iterating until Ω has passed all of them, the accumulated delta terms are exactly the sum $\sum_{(1)\Omega} N[xyz\cdots w\,\Omega]$, and the final term, in which Ω stands to the left of every annihilation operator, is exactly $N[xyz\cdots w\,\Omega]$. This is Eq. (3.113). □

The lemma at work.

For $M = 1$ the lemma reads

$$N\left[a_1\right] a_\ell^\dagger = N\left[a_1 a_\ell^\dagger\right] + N\left[\overbracket{a_1\, a_\ell^\dagger}\right] = -\langle 0|a_\ell^\dagger a_1|0\rangle + \delta_{1\ell}, \tag{3.114}$$

which is just Eq. (3.97) once more. For $M = 2$,

$$a_0 N\left[a_1\right] a_\ell^\dagger = N\left[a_0 a_1 a_\ell^\dagger\right] + N\left[a_0\,\overbracket{a_1\, a_\ell^\dagger}\right] + N\left[\contraction{}{a_0}{\,a_1\,}{a_\ell^\dagger}a_0\,a_1\,a_\ell^\dagger\right], \tag{3.115}$$

that is $N[a_0 a_1 a_\ell^\dagger] + \delta_{1\ell} N[a_0] + \delta_{0\ell} N[a_1]$ up to the signs of the crossings. Generally,

$$a_0 N\left[a_1 a_2 \cdots a_M\right] a_\ell^\dagger = \sum_{(1)\Omega} N\left[a_0 a_1 \cdots a_M a_\ell^\dagger\right] + N\left[a_0 a_1 \cdots a_M a_\ell^\dagger\right]. \tag{3.116}$$

The induction step.

Assume Wick's theorem, Eq. (3.111), holds for a product of M operators. Multiply both sides from the right by one further operator Ω and apply the lemma to every term. The first term gives

$$N\left[xyz\cdots w\right]\Omega = N\left[xyz\cdots w\,\Omega\right] + \sum_{(1)\Omega} N\left[xyz\cdots w\,\Omega\right], \tag{3.117}$$

the m-fold contracted terms give

$$\sum_{(m)} N\left[xyz\cdots w\right]\Omega = \sum_{(m)\neq\Omega} N\left[xyz\cdots w\,\Omega\right] + \sum_{(m)\Omega} N\left[xyz\cdots w\,\Omega\right], \tag{3.118}$$

where $\sum_{(m)\neq\Omega}$ collects the m-fold contractions that do not involve Ω and $\sum_{(m)\Omega}$ those in which one of the m contractions ends on Ω. But an m-fold contraction of the $M+1$ operators either involves Ω or it does not, so the two sums recombine,

$$\sum_{(m)\neq\Omega} + \sum_{(m)\Omega} = \sum_{(m)} \qquad \text{over all } M+1 \text{ operators.} \tag{3.119}$$

Collecting the terms by the number of contractions therefore reproduces Eq. (3.111) with M replaced by $M+1$,

$$xyz\cdots w\,\Omega = N\left[xyz\cdots w\,\Omega\right] + \sum_{(1)} N\left[\cdots\right] + \sum_{(2)} N\left[\cdots\right] + \cdots + \sum_{[(M+1)/2]} N\left[\cdots\right]. \tag{3.120}$$

This completes the induction. □

A remark on parity.

If $M+1$ is odd, the last sum $\sum_{[(M+1)/2]}$ still leaves one operator uncontracted, since an odd number of operators cannot be paired up completely. On taking the vacuum expectation value these terms vanish by Eq. (3.99), and we recover the result of Section 3.8: vacuum expectation values of odd strings are zero. For $M+1$ even the last sum consists of the fully contracted terms, and Eq. (3.112) follows.

3.12 Wick's generalised theorem

In practice we rarely meet a bare product of operators. What we meet is a product of groups, each of which is *already* in normal-ordered form: the bra, the operator, and the ket. Wick's generalised theorem states that for such a product

$$N\left[A_1 A_2 \cdots\right] N\left[B_1 B_2 \cdots\right] N\left[C_1 C_2 \cdots\right] = N\left[A_1 A_2 \cdots B_1 B_2 \cdots C_1 C_2 \cdots\right] + \sum_{\text{all}} N\left[A_1 A_2 \cdots C_1 C_2 \cdots\right], \tag{3.121}$$

where the sum runs over all contractions *between operators belonging to different normal-ordered groups*. Contractions within a single group are absent, precisely because that group is already normal-ordered and its internal contractions vanish by Eqs. (3.92)–(3.94).

This is a substantial saving, and it is worth putting a number on it before we prove anything. The one-body matrix element of the next section has three groups of two operators each. Its six operators admit $5!! = 15$ pairings, of which only 8 join two different groups. The two-body matrix element has groups of two, four and two operators; its eight operators admit $7!! = 105$ pairings, of which only 24 are between different groups. The saving grows quickly with the size of the groups: for three groups of four operators the counts are $10\,395$ and 1728. All three numbers are produced by `wick.py`.

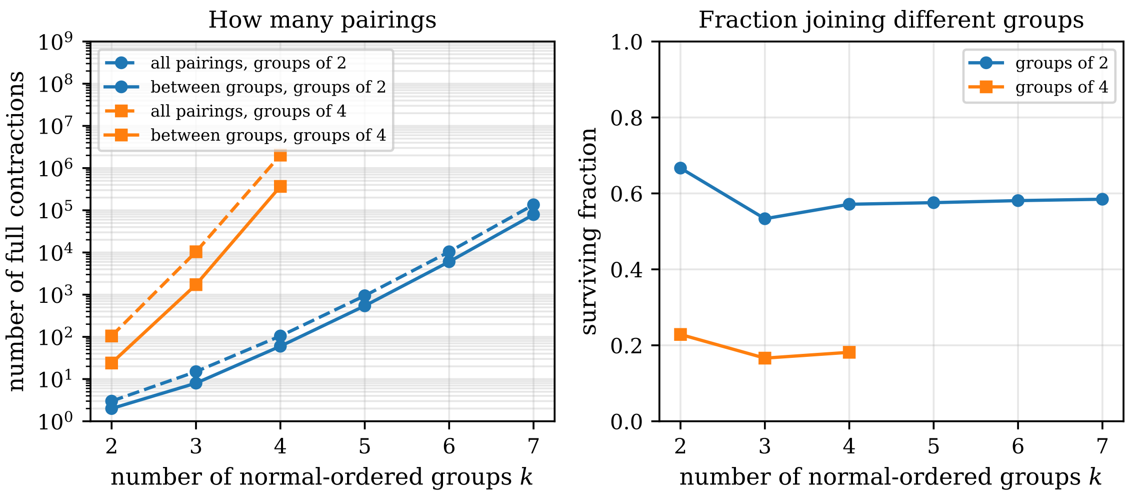

Fig. 3.2 Effect of the restriction to inter-group contractions, for a string cut into k normal-ordered groups all of the same size. In the left panel, plotted logarithmically, the dashed curves count every perfect matching of the $2n$ operators and the solid curves count only those matchings in which every line joins two different groups; circles are groups of two operators, the shape of a bra, a ket or a one-body operator, and squares are groups of four, the shape of a two-body operator. The right panel gives the ratio of the two, that is the fraction of pairings that the generalised theorem allows. Both counts grow factorially, but the surviving fraction settles at a value fixed by the group size, close to 0.58 for groups of two and near 0.18 for groups of four. Computed by `wick.py`.

The way those numbers behave as the string is cut into more and more pieces is worth a moment, and it is drawn in Figure 3.2. Take k normal-ordered groups all of the same size, count every perfect matching of the operators, and count again keeping only the matchings whose every line joins two different groups. The left panel shows that nothing stops either count from growing factorially: for groups of two the totals run $3, 15, 105, 945, 10\,395, 135\,135$ as k goes from two to seven, and the inter-group counts run $2, 8, 60, 544, 6040, 79\,008$ behind them. The generalised theorem does not tame the factorial, and it was never going to – the number of ways of pairing $2n$ objects is what it is. What the right panel shows is the more useful statement. The surviving fraction does not drift towards zero; it settles almost immediately on a number characteristic of the group *size*, a little under three fifths for groups of two and roughly a fifth for groups of four, and thereafter barely moves as k grows.

The practical reading of this is that large normal-ordered blocks are worth more than many small ones. A matrix element is not a string of $2n$ separate operators but a small number of fat groups, and it is the fatness that pays: going from groups of two to groups of four drops the allowed fraction from about 0.57 to about 0.18 at $k = 4$, a factor of three thrown away before any index has been looked at. This is exactly the structure of the calculations of Section 3.14. The two-body matrix element there is a $2+4+2$ string, and of its 105 pairings only 24 join different groups; the elementary contractions then cut those 24 down to the four terms that actually contribute. The same arithmetic, applied to the much longer strings of the coupled-cluster equations of Chapter 10, is what keeps those derivations finite, and it is the reason one always writes the interaction in normal-ordered form before starting rather than after.

3.13 Proof of the generalised theorem

The proof needs no machinery beyond what we already have. It rests on one observation about normal-ordered groups, and on the appending lemma (3.113) that carried the proof of the ordinary theorem in Section 3.11.

Throughout we take each group to be *written* in normal-ordered form, that is with its creation operators standing to the left of its annihilation operators. This costs nothing: by the definition (3.86) any group may be brought into that arrangement at the price of a sign $(-1)^{\kappa}$, and since $N[\cdots]$ is unchanged by the same reordering the factor multiplies both sides of Eq. (3.121) and cancels. With the groups written this way, $N[A_1A_2\cdots]$ *is* the product $A_1A_2\cdots$, and we may drop the brackets whenever it is convenient.

Step 0: a group has no internal contractions.

Let $A_1A_2\cdots A_n$ be normal-ordered and pick two of its operators, A_k and A_l with $k<l$. Their contraction vanishes. There are only two cases. If A_k is a creation operator, then A_l is either a creation operator, and the contraction vanishes by Eq. (3.94), or an annihilation operator standing to its right, and it vanishes by Eq. (3.93). If instead A_k is an annihilation operator, then everything to its right is an annihilation operator too, since the group is normal-ordered, and the contraction vanishes by Eq. (3.92). Hence

$$\contraction{}{A}{_k \cdots}{A}A_k \cdots A_l = 0 \qquad \text{for any two operators of the same normal-ordered group.} \tag{3.122}$$

This is the whole reason the internal contractions may be omitted from Eq. (3.121): they are not thrown away, they are zero. It is also the reason the arrangement matters. Written in the opposite order the pair $a_p^\dagger a_q$ becomes $a_q a_p^\dagger$, whose contraction is δ_{pq} and not zero at all, and the bookkeeping would have to be redone.

Step 1: two groups.

We first prove

$$N[A_1\cdots A_n]\,N[B_1\cdots B_m] = N[A_1\cdots A_nB_1\cdots B_m] + \sum_{A\text{–}B} N[A_1\cdots A_nB_1\cdots B_m], \tag{3.123}$$

where $\sum_{A\text{–}B}$ runs over all sets of disjoint contractions, each line joining an operator of the first group to an operator of the second. The proof is by induction on m, the number of operators in the second group.

For $m=0$ there is nothing to prove. For $m=1$ the statement is the appending lemma (3.113) verbatim, with $\Omega = B_1$: it produces $N[A_1\cdots A_nB_1]$ together with the sum of the single contractions of B_1 with each of $A_1,\ldots,A_n$, and every one of those lines joins the two groups.

Assume now that Eq. (3.123) holds for a second group of m operators, and append one more,

$$N[A_1\cdots A_n]\,B_1\cdots B_m B_{m+1} = \Big(N[A_1\cdots A_n B_1\cdots B_m] + \sum_{A\text{–}B} N[\cdots]\Big)B_{m+1}. \tag{3.124}$$

Every term on the right is a number – the product of the contractions it carries – multiplying a single normal-ordered product of the operators that are still uncontracted. The lemma (3.113) therefore applies to each of them separately, with $\Omega = B_{m+1}$, and each application yields two kinds of contribution:

(a) the term in which B_{m+1} is absorbed into the normal-ordered product, and
(b) the contractions of B_{m+1} with the uncontracted operators inside the bracket.

The partners available in (b) are of two sorts. A partner taken from the first group gives a legitimate A–B line. A partner taken from $B_1,\ldots,B_m$ gives a line internal to the second group, and it vanishes by Step 0, since $B_1\cdots B_{m+1}$ is normal-ordered and B_{m+1} stands to the right of all of them. Only the inter-group lines survive.

Collecting the survivors, the term (a) of the leading piece is $N[A_1\cdots A_n B_1\cdots B_{m+1}]$, and every set of A–B contractions of the $n+m+1$ operators appears exactly once: either it leaves B_{m+1} uncontracted, in which case it comes from case (a) applied to a contracted term of the induction hypothesis, or it contracts B_{m+1} with some A_k, in which case it comes from case (b). Each carries the sign (3.95) of the crossings of its lines, because the lemma appends Ω from the right and the sign it generates is precisely the number of operators the new line has to cross. This is Eq. (3.123) with $m+1$ in place of m, and the induction is complete.

Step 2: any number of groups.

The general case now follows by a second induction, this time on the number of groups. One group is trivial and two groups are Step 1. Suppose Eq. (3.121) holds for g groups, and multiply it from the right by one further normal-ordered group $N[H_1\cdots H_k]$. Each term on the left-hand side is again a number times a single normal-ordered product, so Step 1 applies to each of them with A standing for what is left of the first g groups and B for the new one. It produces the uncontracted term and all contractions between H and the remaining operators of the earlier groups – all of them inter-group lines – and no contractions internal to H, by Step 1.

Every admissible set of contractions of the $g+1$ groups is generated exactly once, because such a set splits uniquely into the lines that touch H and those that do not: the latter are generated by the induction hypothesis at stage g, the former by the application of Step 1. Hence Eq. (3.121) holds for $g+1$ groups. □

The corollary we shall actually use.

Taking the vacuum expectation value of Eq. (3.121) and using Eq. (3.88) once more, every term that still contains an uncontracted operator drops out and we are left with

$$\boxed{\langle 0|\,N[A_1A_2\cdots]\,N[B_1B_2\cdots]\,N[C_1C_2\cdots]\,|0\rangle = \sum_{\substack{\text{fully contracted,}\\ \text{no line inside a group}}} (-1)^{\nu}\prod \text{contractions}} \tag{3.125}$$

with ν the number of crossings as in Eq. (3.95). This is the form in which the theorem is used in the next section and, relentlessly, in Chapters 9 and 10.

A remark on the Fermi vacuum.

Nothing in the argument used the fact that $|0\rangle$ is the true vacuum. Replacing $|0\rangle$ by the reference determinant $|c\rangle$ of Section 3.3 and the operators a_p, $a_p^\dagger$ by the quasiparticle operators b_p, $b_p^\dagger$ of Eq. (3.67), the definitions of normal ordering and of a contraction carry over word for word, and so do Steps 0, 1 and 2. The only change is in the elementary contractions themselves: with respect to $|c\rangle$ the surviving contractions are

$$\contraction{}{a_a}{}{a_b^\dagger} a_a a_b^\dagger = \delta_{ab} \quad (a,b > F), \qquad \contraction{}{a_i^\dagger}{}{a_j} a_i^\dagger a_j = \delta_{ij} \quad (i,j \leq F), \tag{3.126}$$

the second of which has no counterpart above: a hole state below the Fermi level is occupied in $|c\rangle$, so removing and replacing a particle there gives unity. This particle-hole form of the generalised theorem is the one that generates the Hartree-Fock equations of Chapter 6, the perturbation expansion of Chapter 9 and the coupled-cluster amplitude equations of Chapter 10.

Numerical verification.

The program `wick.py` checks all of this. It confirms that the internal contractions of a normal-ordered group vanish, it counts the pairings quoted above, and it evaluates the one- and two-body matrix elements of the next section along three independent routes – brute-force anticommutation, the ordinary theorem, and the generalised theorem – and finds them identical. It then goes one step further. Equation (3.121) is an identity between *operators*, not merely between numbers, and the program verifies it as such by representing the creation and annihilation operators as matrices on the 2^n states of a small Fock space and comparing the two sides entry by entry. The residual is zero to machine precision, for products of two, three and four groups, and for the ordinary theorem recovered as the special case in which every group holds a single operator.

3.14 Applications: matrix elements between two-particle states

We close with the two calculations that motivated the whole construction. Both use the generalised theorem, and in both the bra, the operator and the ket are separately normal-ordered.

A one-body operator.

With

$$\hat{O}^{(1)} = \sum_{pq} \langle p|\hat{o}^{(1)}|q\rangle\, a_p^\dagger a_q, \tag{3.127}$$

and the two-particle states $|ij\rangle = a_i^\dagger a_j^\dagger|0\rangle$, $\langle ij| = \langle 0|a_j a_i$, we need

$$\langle ij|\hat{O}^{(1)}|k\ell\rangle = \sum_{pq} \langle p|\hat{o}^{(1)}|q\rangle\, \langle 0|\, \underbrace{a_j a_i}_{\text{bra}}\; \underbrace{a_p^\dagger a_q}_{\text{operator}}\; \underbrace{a_k^\dagger a_\ell^\dagger}_{\text{ket}}\, |0\rangle. \tag{3.128}$$

Only contractions between different groups contribute, and each must join an annihilation operator to a creation operator standing to its right. One such term is

$$a_j\, a_i\, a_p^\dagger\, a_q\, a_k^\dagger\, a_\ell^\dagger \;=\; \delta_{ip}\, \delta_{qk}\, \delta_{j\ell}, \tag{3.129}$$

which contributes $\delta_{j\ell}\,\langle i|\hat{o}^{(1)}|k\rangle$ after the sums over p and q are carried out. Collecting the four such terms gives the familiar result

$$\langle ij|\hat{O}^{(1)}|k\ell\rangle = \delta_{j\ell}\langle i|\hat{o}^{(1)}|k\rangle - \delta_{jk}\langle i|\hat{o}^{(1)}|\ell\rangle - \delta_{i\ell}\langle j|\hat{o}^{(1)}|k\rangle + \delta_{ik}\langle j|\hat{o}^{(1)}|\ell\rangle. \tag{3.130}$$

The matrix element vanishes whenever the two Slater determinants differ by more than one single-particle state, since then no assignment of the deltas can be satisfied. This is the Condon-Slater rule of Section 2.6, now obtained without a single explicit permutation.

The two-body interaction.

With the normal-ordered interaction

$$\hat{H}_I = \frac{1}{2}\sum_{pqrs}\langle pq|v|rs\rangle\, a_p^\dagger a_q^\dagger a_s a_r, \tag{3.131}$$

the diagonal matrix element in the state $|ij\rangle$ is

$$\langle ij|\hat{H}_I|ij\rangle = \frac{1}{2}\sum_{pqrs}\langle pq|v|rs\rangle\,\langle 0|\,\underbrace{a_j a_i}_{\text{bra}}\,\underbrace{a_p^\dagger a_q^\dagger a_s a_r}_{\text{operator}}\,\underbrace{a_i^\dagger a_j^\dagger}_{\text{ket}}\,|0\rangle. \tag{3.132}$$

This is an eight-operator string. Enumerating its 105 pairings by hand would be unpleasant, but the generalised theorem restricts us to contractions between the three groups, and of those only four survive. Two of them are

$$a_j\, a_i\, a_p^\dagger\, a_q^\dagger\, a_s\, a_r\, a_i^\dagger\, a_j^\dagger \;=\; \delta_{ip}\delta_{jq}\delta_{ri}\delta_{sj}, \tag{3.133}$$

giving $\langle ij|v|ij\rangle$, and the one obtained by exchanging the roles of r and s, which crosses one more line and therefore enters with the opposite sign, giving $-\langle ij|v|ji\rangle$. The two remaining terms are these two with i and j interchanged; they are equal to the first two and cancel the factor $\frac{1}{2}$. The result is

$$\boxed{\langle ij|\hat{H}_I|ij\rangle = \langle ij|v|ij\rangle - \langle ij|v|ji\rangle = \langle ij|v|ij\rangle_{\mathrm{AS}}} \tag{3.134}$$

which is exactly the antisymmetrised matrix element obtained in Chapter 2 by writing out the antisymmetriser and sorting through permutations. The two derivations agree, as they must; the point is that this one required no permutations at all, only the observation that a contraction is non-zero when an annihilation operator stands to the left of a creation operator.

The program `wick.py` performs every calculation of these sections in both ways – by brute-force anticommutation and by summing contractions with their signs – and verifies that they agree, for strings of two, four, six and eight operators. It also reproduces Eq. (3.134) by enumerating which of the (p,q,r,s) assignments survive.

3.15 Operators in second quantization

What follows is the bridge from the formalism to a running code. The Hamiltonian matrix built here is the sparse, never-stored matrix that the Lanczos algorithm of Section 1.26 was designed for: only the product $\boldsymbol{Hv}$ is needed, and the Condon-Slater rules of Section 3.14 say which of its elements can be non-zero. The bit representation below is how that product is evaluated in practice.

In the build-up of a shell-model or FCI code that is meant to tackle large dimensionalities is the action of the Hamiltonian $\hat{H}$ on a Slater determinant represented in second quantization as

$$|\alpha_1 \dots \alpha_n\rangle = a^\dagger_{\alpha_1} a^\dagger_{\alpha_2} \dots a^\dagger_{\alpha_n} |0\rangle.$$

The time consuming part stems from the action of the Hamiltonian on the above determinant,

$$\left(\sum_{\alpha\beta} \langle\alpha|t+u|\beta\rangle a^\dagger_\alpha a_\beta + \frac{1}{4}\sum_{\alpha\beta\gamma\delta} \langle\alpha\beta|\hat{v}|\gamma\delta\rangle a^\dagger_\alpha a^\dagger_\beta a_\delta a_\gamma\right) a^\dagger_{\alpha_1} a^\dagger_{\alpha_2} \dots a^\dagger_{\alpha_n} |0\rangle.$$

A practically useful way to implement this action is to encode a Slater determinant as a bit pattern.

Assume that we have at our disposal n different single-particle orbits $\alpha_0, \alpha_2, \dots, \alpha_{n-1}$ and that we can distribute among these orbits $N \le n$ particles.

A Slater determinant can then be coded as an integer of n bits. As an example, if we have $n = 16$ single-particle states $\alpha_0, \alpha_1, \dots, \alpha_{15}$ and $N = 4$ fermions occupying the states α_3, α_6, α_{10} and α_{13} we could write this Slater determinant as

$$\Phi_\Lambda = a^\dagger_{\alpha_3} a^\dagger_{\alpha_6} a^\dagger_{\alpha_{10}} a^\dagger_{\alpha_{13}} |0\rangle.$$

The unoccupied single-particle states have bit value 0 while the occupied ones are represented by bit state 1. In the binary notation we would write this 16 bits long integer as

$$\begin{array}{cccccccccccccccc} \alpha_0 & \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 & \alpha_5 & \alpha_6 & \alpha_7 & \alpha_8 & \alpha_9 & \alpha_{10} & \alpha_{11} & \alpha_{12} & \alpha_{13} & \alpha_{14} & \alpha_{15} \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \end{array}$$

which translates into the decimal number

$$2^3 + 2^6 + 2^{10} + 2^{13} = 9288.$$

We can thus encode a Slater determinant as a bit pattern.

With N particles that can be distributed over n single-particle states, the total number of Slater determinats (and defining thereby the dimensionality of the system) is

$$\dim(\mathscr{H}) = \binom{n}{N}.$$

The total number of bit patterns is 2^n.

We assume again that we have at our disposal n different single-particle orbits $\alpha_0, \alpha_2, \dots, \alpha_{n-1}$ and that we can distribute among these orbits $N \le n$ particles. The ordering among these states is important as it defines the order of the creation operators. We will write the determinant

$$\Phi_\Lambda = a^\dagger_{\alpha_3} a^\dagger_{\alpha_6} a^\dagger_{\alpha_{10}} a^\dagger_{\alpha_{13}} |0\rangle,$$

in a more compact way as

$$\Phi_{3,6,10,13} = |0001001000100100\rangle.$$

The action of a creation operator is thus

$$a^\dagger_{\alpha_4}\Phi_{3,6,10,13} = a^\dagger_{\alpha_4}|0001001000100100\rangle = a^\dagger_{\alpha_4}a^\dagger_{\alpha_3}a^\dagger_{\alpha_6}a^\dagger_{\alpha_{10}}a^\dagger_{\alpha_{13}}|0\rangle,$$

which becomes

$$-a^\dagger_{\alpha_3}a^\dagger_{\alpha_4}a^\dagger_{\alpha_6}a^\dagger_{\alpha_{10}}a^\dagger_{\alpha_{13}}|0\rangle = -|0001101000100100\rangle.$$

Similarly

$$a^\dagger_{\alpha_6}\Phi_{3,6,10,13} = a^\dagger_{\alpha_6}|0001001000100100\rangle = a^\dagger_{\alpha_6}a^\dagger_{\alpha_3}a^\dagger_{\alpha_6}a^\dagger_{\alpha_{10}}a^\dagger_{\alpha_{13}}|0\rangle,$$

which becomes

$$-a^\dagger_{\alpha_4}(a^\dagger_{\alpha_6})^2a^\dagger_{\alpha_{10}}a^\dagger_{\alpha_{13}}|0\rangle = 0!$$

This gives a simple recipe:

- If one of the bits b_j is 1 and we act with a creation operator on this bit, we return a null vector
- If $b_j = 0$, we set it to 1 and return a sign factor $(-1)^l$, where l is the number of bits set before bit j.

Consider the action of $a^\dagger_{\alpha_2}$ on various slater determinants:

$$\begin{array}{lll}
a^\dagger_{\alpha_2}\Phi_{00111} = a^\dagger_{\alpha_2}|00111\rangle & = 0\times|00111\rangle \\
a^\dagger_{\alpha_2}\Phi_{01011} = a^\dagger_{\alpha_2}|01011\rangle & = (-1)\times|01111\rangle \\
a^\dagger_{\alpha_2}\Phi_{01101} = a^\dagger_{\alpha_2}|01101\rangle & = 0\times|01101\rangle \\
a^\dagger_{\alpha_2}\Phi_{01110} = a^\dagger_{\alpha_2}|01110\rangle & = 0\times|01110\rangle \\
a^\dagger_{\alpha_2}\Phi_{10011} = a^\dagger_{\alpha_2}|10011\rangle & = (-1)\times|10111\rangle \\
a^\dagger_{\alpha_2}\Phi_{10101} = a^\dagger_{\alpha_2}|10101\rangle & = 0\times|10101\rangle \\
a^\dagger_{\alpha_2}\Phi_{10110} = a^\dagger_{\alpha_2}|10110\rangle & = 0\times|10110\rangle \\
a^\dagger_{\alpha_2}\Phi_{11001} = a^\dagger_{\alpha_2}|11001\rangle & = (+1)\times|11101\rangle \\
a^\dagger_{\alpha_2}\Phi_{11010} = a^\dagger_{\alpha_2}|11010\rangle & = (+1)\times|11110\rangle
\end{array}$$

What is the simplest way to obtain the phase when we act with one annihilation(creation) operator on the given Slater determinant representation?

We have an SD representation

$$\Phi_\Lambda = a^\dagger_{\alpha_0}a^\dagger_{\alpha_3}a^\dagger_{\alpha_6}a^\dagger_{\alpha_{10}}a^\dagger_{\alpha_{13}}|0\rangle,$$

in a more compact way as

$$\Phi_{0,3,6,10,13} = |1001001000100100\rangle.$$

The action of

$$a^\dagger_{\alpha_4}a_{\alpha_0}\Phi_{0,3,6,10,13} = a^\dagger_{\alpha_4}|0001001000100100\rangle = a^\dagger_{\alpha_4}a^\dagger_{\alpha_3}a^\dagger_{\alpha_6}a^\dagger_{\alpha_{10}}a^\dagger_{\alpha_{13}}|0\rangle,$$

which becomes

$$-a^\dagger_{\alpha_3}a^\dagger_{\alpha_4}a^\dagger_{\alpha_6}a^\dagger_{\alpha_{10}}a^\dagger_{\alpha_{13}}|0\rangle = -|0001101000100100\rangle.$$

The action

$$a_{\alpha_0}\Phi_{0,3,6,10,13} = |0001001000100100\rangle,$$

can be obtained by subtracting the logical sum (AND operation) of $\Phi_{0,3,6,10,13}$ and a word which represents only α_0, that is

$$|1000000000000000\rangle,$$

from $\Phi_{0,3,6,10,13} = |1001001000100100\rangle$.

This operation gives $|0001001000100100\rangle$.

Similarly, we can form $a^\dagger_{\alpha_4}a_{\alpha_0}\Phi_{0,3,6,10,13}$, say, by adding $|0000100000000000\rangle$ to $a_{\alpha_0}\Phi_{0,3,6,10,13}$, first checking that their logical sum is zero in order to make sure that orbital α_4 is not already occupied.

It is trickier however to get the phase $(-1)^l$. One possibility is as follows

- Let S_1 be a word that represents the $1-$bit to be removed and all others set to zero.

In the previous example $S_1 = |1000000000000000\rangle$

- Define S_2 as the similar word that represents the bit to be added, that is in our case

$S_2 = |0000100000000000\rangle$.

- Compute then $S = S_1 - S_2$, which here becomes

$$S = |0111000000000000\rangle$$

- Perform then the logical AND operation of S with the word containing

$$\Phi_{0,3,6,10,13} = |1001001000100100\rangle,$$

which results in $|0001000000000000\rangle$. Counting the number of $1-$bits gives the phase. Here you need however an algorithm for bitcounting. Several efficient ones available.

3.16 From fermions to qubits: the Jordan-Wigner transformation

The bit representation of the previous section was introduced as a programming device. It is more than that. A Slater determinant *is* a string of bits, and a register of qubits is a string of bits that a quantum computer can put into superposition. The two are the same object, and the transformation that makes the identification precise – the Jordan-Wigner transformation – is the last piece of formalism this chapter needs. It is what turns the Hamiltonians of Chapter 4 into something a quantum computer can be handed, and it is used again for the Trotter expansion of Section 6.9 and for the unitary coupled-cluster ansatz of Chapter 10.

States are easy; operators are not.

A fermionic Fock state is specified by its occupation numbers,

$$|n_0 n_1 \cdots n_{M-1}\rangle, \qquad n_p \in \{0,1\}, \tag{3.135}$$

and this is already a string of M bits. We may therefore declare

$$|n_0 n_1 \cdots n_{M-1}\rangle_{\text{Fock}} \longleftrightarrow |n_0 n_1 \cdots n_{M-1}\rangle_{\text{qubit}}, \tag{3.136}$$

with $|0\rangle$ an empty orbital and $|1\rangle$ an occupied one, and be done with the states. The work is entirely in the operators. What we need is a dictionary

$$a^\dagger_p,\ a_p,\ \hat{n}_p = a^\dagger_p a_p \longrightarrow \text{sums of products of } I, X, Y, Z, \tag{3.137}$$

because those Pauli operators, and their tensor products, are what a quantum computer implements. A product $P \in \{I, X, Y, Z\}^{\otimes M}$ is called a *Pauli string*, and the goal of the whole exercise is to write

$$\hat{H} = \sum_\alpha h_\alpha P_\alpha, \tag{3.138}$$

a weighted sum of Pauli strings. Once a Hamiltonian is in that form one can estimate its expectation value term by term on hardware, build a Trotterised time evolution from it, or feed it to a variational algorithm.

Why the obvious guess fails.

On a single qubit define

$$\sigma^+ = \frac{X - iY}{2} = |1\rangle\langle 0|, \qquad \sigma^- = \frac{X + iY}{2} = |0\rangle\langle 1|, \tag{3.139}$$

so that σ^+ fills an empty orbital and σ^- empties a full one. The obvious guess is $a_p^\dagger \sim \sigma_p^+$ and $a_p \sim \sigma_p^-$. It is wrong, and it is wrong for the reason this whole chapter has been about. Pauli operators on *different* qubits commute,

$$\sigma_p^- \sigma_q^- = \sigma_q^- \sigma_p^- \qquad (p \neq q),$$

whereas fermion operators on different orbitals *anti*commute, $a_p a_q = -a_q a_p$. The naive map would describe distinguishable spins, not fermions, and would lose the antisymmetry that Eq. (3.5) built in at the very start. The program `jordanwigner.py` quantifies the damage: with the naive map $\{\sigma_0^-, \sigma_1^-\}$ has norm 2 instead of 0.

The parity string.

The cure is to attach to each operator a factor that counts how many particles sit before it in the chosen ordering. Since

$$Z|0\rangle = +|0\rangle, \qquad Z|1\rangle = -|1\rangle,$$

the product $Z_0 Z_1 \cdots Z_{p-1}$ returns precisely

$$\prod_{j=0}^{p-1} Z_j \longrightarrow (-1)^{n_0 + n_1 + \cdots + n_{p-1}} \tag{3.140}$$

on a basis state. This is exactly the phase $(-1)^l$ that the bit-counting algorithm at the end of Section 3.15 computes by masking and counting ones: the number of occupied orbitals that the operator has to be carried past. The classical algorithm and the quantum operator are the same piece of bookkeeping.

With that factor in place we may define

$$\boxed{a_p^\dagger = \left(\prod_{j=0}^{p-1} Z_j\right) \frac{X_p - iY_p}{2}, \qquad a_p = \left(\prod_{j=0}^{p-1} Z_j\right) \frac{X_p + iY_p}{2}} \tag{3.141}$$

which is the Jordan-Wigner transformation. The string of Z's is called the *parity string*. Note that it depends on an *ordering* of the orbitals; different orderings give different Pauli strings for the same operator, all of them correct but not all of them equally convenient.

It is a legal transformation.

Everything rests on the claim that Eq. (3.141) reproduces the anticommutation relations (3.8), (3.10) and (3.27). It does. For $p=q$ the parity strings cancel, since $Z^2=I$, and

$$a_p^\dagger a_p = \frac{I-Z_p}{2}, \qquad a_p a_p^\dagger = \frac{I+Z_p}{2}, \qquad \{a_p, a_p^\dagger\} = I. \tag{3.142}$$

For $p<q$ the string attached to a_q contains a Z_p that the operator on site p does not commute with, and it is that single anticommuting factor which converts the commuting Pauli algebra into the anticommuting fermion algebra. The program `jordanwigner.py` builds the operators as explicit matrices and checks all three relations for up to five modes; the largest violation is zero to machine precision.

Number operators and one-body terms.

The first entry of Eq. (3.142) is the single most useful identity in the subject,

$$\hat{n}_p = a_p^\dagger a_p = \frac{I-Z_p}{2}, \tag{3.143}$$

and it says that a diagonal one-body operator becomes a sum of I and Z only,

$$\sum_p \varepsilon_p a_p^\dagger a_p \longrightarrow \frac{1}{2}\sum_p \varepsilon_p (I-Z_p). \tag{3.144}$$

A non-interacting Hamiltonian is therefore already diagonal on the qubit register: no circuit is needed to find its spectrum, only to prepare its eigenstates. The off-diagonal, hopping-like combination is where the parity string first shows itself. For $p<q$,

$$a_p^\dagger a_q + a_q^\dagger a_p \longrightarrow \frac{1}{2}\left(X_p Z_{p+1}\cdots Z_{q-1} X_q + Y_p Z_{p+1}\cdots Z_{q-1} Y_q\right), \tag{3.145}$$

two strings whose length grows with $|p-q|$. A hop has to know how many particles it jumps over, and the Z's are how it finds out.

Pair operators.

Creating two particles rather than moving one changes exactly one sign. For $p<q$,

$$a_p^\dagger a_q^\dagger = \frac{1}{4}(X_p - iY_p) Z_{p+1}\cdots Z_{q-1} (X_q - iY_q), \tag{3.146}$$

and the Hermitian combination that appears in every pairing Hamiltonian is

$$a_p^\dagger a_q^\dagger + a_q a_p \longrightarrow \frac{1}{2}\left(X_p Z_{p+1}\cdots Z_{q-1} X_q - Y_p Z_{p+1}\cdots Z_{q-1} Y_q\right). \tag{3.147}$$

Compare with Eq. (3.145): the two operators differ only in the relative sign of the XX and YY channels. That one sign is the difference between transporting a particle and creating a pair, and it is worth remembering, because it is the only thing that distinguishes the hopping term of the Hubbard model from the pair-transfer term of the pairing model once both are on a quantum computer.

The general two-body term.

The two-body operator of Eq. (3.59) is a sum of terms $a_p^\dagger a_q^\dagger a_s a_r$. When the four indices are distinct the Hermitian combination produces eight Pauli strings, each of weight four and each with coefficient $\frac{1}{8}$. For four consecutive orbitals $0 < 1 < 2 < 3$, so that all the intervening parity strings are empty, `jordanwigner.py` gives

$$a_0^\dagger a_1^\dagger a_3 a_2 + \text{h.c.} = \frac{1}{8}\Big(XXXX + XYXY + XYYX \\ + YXXY + YXYX + YYYY \\ - XXYY - YYXX\Big), \tag{3.148}$$

where $XYXY$ means $X_0Y_1X_2Y_3$ and so on. This particular operator will reappear in Chapter 4: read the four orbitals as $(p+, p-, q+, q-)$ and Eq. (3.148) is the pair-transfer term $\hat{P}_p^+\hat{P}_q^- + \text{h.c.}$ of the pairing model.

For a general two-body Hamiltonian on M orbitals the number of distinct Pauli strings grows as $\mathscr{O}(M^4)$, and the longest of them span the whole register. Table 3.1 gives the counts for a generic Hamiltonian, again computed rather than estimated.

M	4^M	Pauli strings	largest weight
2	16	6	2
3	64	19	3
4	256	55	4
5	1024	136	5
6	4096	292	6

Table 3.1 Pauli strings in the Jordan-Wigner image of a generic one- plus two-body Hamiltonian on M orbitals. The count is a small fraction of the 4^M available strings and grows as $\mathscr{O}(M^4)$, but the largest weight is M: the parity chains reach across the entire register. Computed by `jordanwigner.py`.

The weight spectrum of a structured Hamiltonian.

A generic Hamiltonian is the worst case, and the models we actually solve are not generic. It is therefore instructive to look not at the two summary numbers of Table 3.1 but at the whole distribution of weights for a Hamiltonian with structure in it. Figure 3.3 does this for the pairing Hamiltonian of Chapter 4 in the full $2L$-qubit encoding, with the decomposition taken straight from `jordanwigner.py`, every coefficient being obtained from the exact trace $c_P = \mathrm{Tr}(PH)/2^M$. That costs 4^M traces, which is why the figure stops at three levels and six qubits.

The shape in Figure 3.3 is the same for all three register sizes and every bar in it can be accounted for by hand. There is always exactly one string of weight zero, the identity, carrying the constant part of the Hamiltonian that Eq. (3.144) generated. The $2L$ strings of weight one – two, four and six bars in the figure – are the single Z_p's coming from the single-particle energies and from the diagonal part of the pairing interaction, again by Eq. (3.143). The L strings of weight two are the Z_pZ_q's produced by the $p = q$ pair-transfer terms, one for each level. Everything else has weight four: each of the $L(L-1)/2$ pairs of distinct levels contributes the eight strings of Eq. (3.148), which is why the weight-four bar jumps from

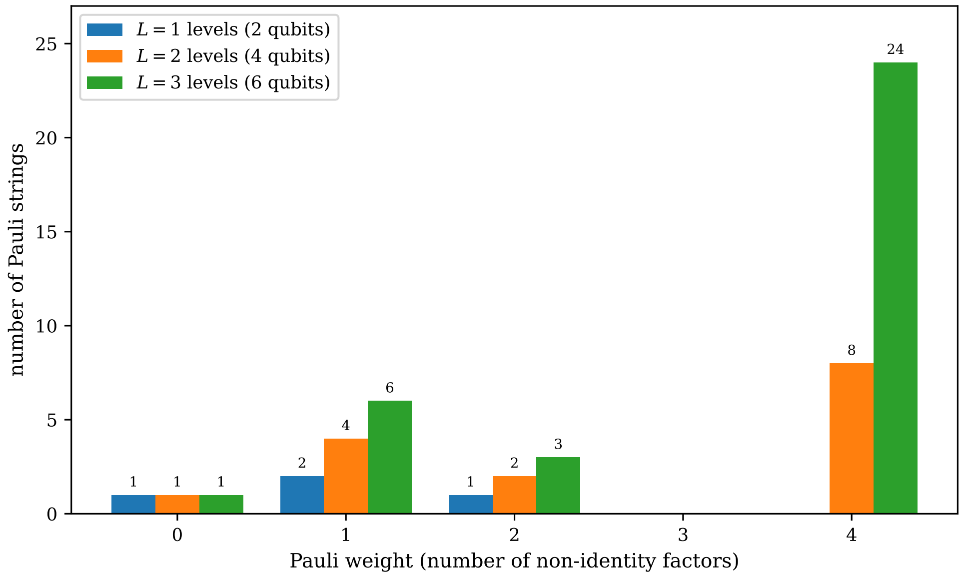

Fig. 3.3 Distribution of Pauli weights in the Jordan-Wigner image of the pairing Hamiltonian with pairing strength $g = 1$, in the full encoding that gives one qubit to each of the $2L$ spin-orbitals. The horizontal axis is the weight, that is the number of non-identity factors in a Pauli string, and the vertical axis counts the strings of that weight; the three sets of bars are $L = 1$, 2 and 3 levels, corresponding to registers of two, four and six qubits. Only the weights 0, 1, 2 and 4 occur, and the weight-four strings of the pair-transfer term dominate the count as soon as there is more than one level. Obtained by exact trace decomposition with `jordanwigner.py`.

nothing at $L = 1$ to 8 at $L = 2$ and 24 at $L = 3$ and dominates the totals of 4, 15 and 34 strings. The gap at weight three is not an accident of small numbers. The pair-transfer operator moves two particles at a time and therefore touches four spin-orbitals at once, while the one-body part touches one or two; nothing in the model touches exactly three, so no odd-weight string beyond the single Z's can appear. Reading the figure the other way round, the parity strings of Eq. (3.141) have done no damage at all here, even though the levels p and q may lie far apart in the register. The reason is that the operators arrive in pairs acting on two *neighbouring* spin-orbitals $2p$ and $2p+1$, so that the two parity strings differ by a single factor and almost everything between them cancels by $Z^2 = I$. The longest strings therefore stay at weight four however large L becomes, which is a first illustration of the remark below that the ordering of the orbitals may be chosen to keep the strings that matter short.

The price, and the alternatives.

The Jordan-Wigner transformation is exact and it is the easiest to derive, but the parity strings are a real cost. An operator that is local in the orbital index becomes an operator of weight $\mathscr{O}(M)$ on the qubit register, which means deeper circuits and noisier measurements. Two standard alternatives trade this differently. The *parity mapping* stores cumulative sums rather than occupations, moving the string from the annihilation operators to the number operators. The *Bravyi-Kitaev* mapping stores a balanced binary tree of partial sums and reduces the weight to $\mathscr{O}(\log M)$, at the cost of a considerably less transparent derivation. For the models of this book, which are small and highly structured, Jordan-Wigner is the right choice, and the ordering of the orbitals can be chosen so that the strings that matter are short – Section 4.7 does exactly that for the pairing model.

What imposing a symmetry first saves.

There is a cheaper move still, and it costs nothing in accuracy. If the pairing Hamiltonian is restricted at the outset to its seniority-zero sector, in which the spin-orbitals of a level are either both empty or both occupied, then a level is fully described by a single bit and the register needs L qubits rather than $2L$. The comparison between the two encodings of the same physics is a comparison of three quantities at once – how many qubits the register needs, how many Pauli strings must be measured, and how long the longest of them is – and Figure 3.4 puts all three together.

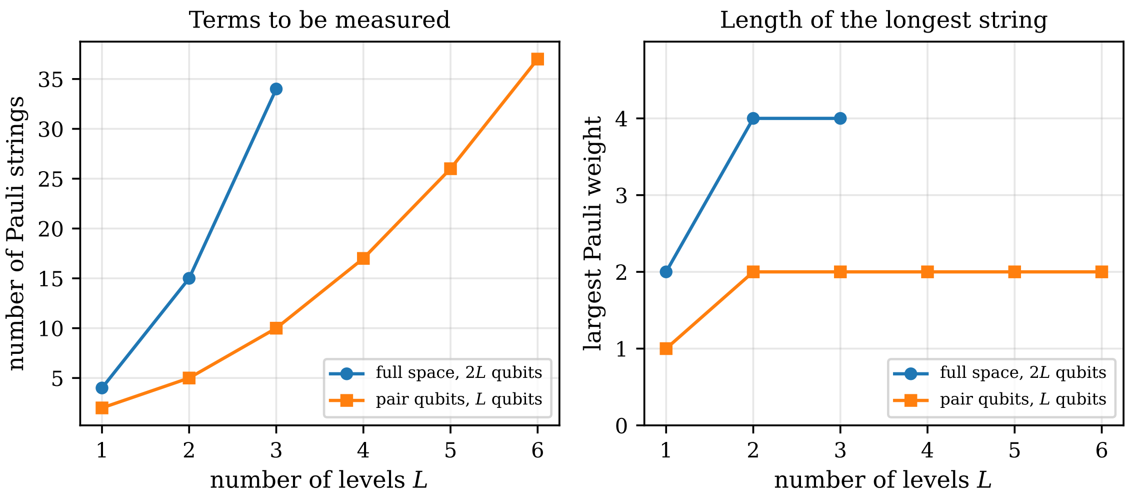

Fig. 3.4 Cost of the two Jordan-Wigner encodings of the same pairing Hamiltonian, plotted against the number of levels L. Circles are the full encoding, which gives one qubit to each of the $2L$ spin-orbitals, and squares the compact encoding, which uses one qubit per level and describes only the seniority-zero sector. The left panel counts the Pauli strings that have to be measured, $4L(L-1)+3L+1$ for the full encoding against L^2+1 for the compact one; the right panel gives the largest weight occurring, four against two. The full encoding stops at $L=3$ because its exact trace decomposition costs 4^{2L} traces. The compact encoding is not an approximation: its spectrum is exactly the seniority-zero part of the full one.

The compact encoding wins on all three counts, and it wins by a widening margin. It halves the register. Its string count is L^2+1 – one identity, L single Z's and the X_pX_q and Y_pY_q couplings of the XY form that Eq. (3.147) produced – against the $4L(L-1)+3L+1$ of the full encoding, so the leading term is smaller by a factor of four; at $L=3$ that is 10 strings against 34, and by $L=6$ the compact encoding still needs only 37, fewer than the full encoding already requires at three levels. The right panel is the more important of the two for hardware. The heaviest string in the full encoding has weight four, so each Trotter factor $\exp(-ih_\alpha P_\alpha \Delta t)$ of Section 6.9 is a four-qubit rotation, whereas in the compact encoding the heaviest weight is two and every factor is a two-qubit rotation. Both curves are flat in L, which is the cancellation of the parity strings noted above, but they are flat at different heights.

None of this is an approximation. The compact spectrum is exactly the seniority-zero part of the full spectrum, as `jordanwigner.py` verifies eigenvalue by eigenvalue, so the entire saving comes from having imposed the symmetry *before* choosing the encoding rather than asking the quantum computer to rediscover it variationally. That is a general lesson and not a peculiarity of the pairing model: the classical structure of the problem, of the kind this chapter has spent its length setting up, is worth more on a small quantum device than any amount of additional circuit depth.

Quantum-computing connection. Equation (3.138) is the input to essentially every quantum algorithm for many-body systems. In the variational quantum eigensolver of Chapter 10 the energy $\langle\Psi(\theta)|\hat{H}|\Psi(\theta)\rangle$ is evaluated by measuring each Pauli string separately and adding the results with the weights h_α, so the number of strings is the number of measurements. In a Trotterised time evolution, Section 6.9, each string contributes one factor $\exp(-ih_\alpha P_\alpha \Delta t)$ to the circuit, so the number of strings is the circuit depth and the weight of each is the number of entangling gates it costs. Both counts are what Table 3.1 and Figures 3.3 and 3.4 are really about, and both are the reason it pays to exploit a symmetry before choosing an encoding rather than after.

3.17 Exercises

The first four exercises drill the algebra of this chapter; the last two work through the Jordan-Wigner transformation by hand. Full solutions follow each one. Every numerical statement can be checked with `wick.py` or `jordanwigner.py`.

Exercise 1: normal ordering and contractions

a) Write $N\left[a_p a_q^\dagger a_r\right]$ out explicitly, including the sign, and verify that its vacuum expectation value vanishes.
b) Evaluate all four elementary contractions directly from the definition (3.89), without using Eq. (3.90).
c) Show that the contraction of two operators is unchanged if both are moved together through the same number of operators, and that interchanging them changes its sign. This is the content of the crossing rule (3.95).

Answer to a).

Normal ordering moves $a_q^\dagger$ to the front. Carrying it past a_p costs one interchange, so

$$N\left[a_p a_q^\dagger a_r\right] = -a_q^\dagger a_p a_r.$$

Its vacuum expectation value vanishes twice over: $a_r|0\rangle = 0$ kills it from the right, and $\langle 0|a_q^\dagger = 0$ from the left. This is Eq. (3.88), and it is the reason normal-ordered products are worth isolating in the first place.

Answer to b).

From $xy = N[xy] +$ contraction we read the contraction off as $xy - N[xy]$. For $a_p a_q^\dagger$ the anticommutator gives $a_p a_q^\dagger = \delta_{pq} - a_q^\dagger a_p$, and $N[a_p a_q^\dagger] = -a_q^\dagger a_p$, so the difference is δ_{pq}. The three other pairs are already in normal-ordered form up to a sign – $a_p a_q$ and $a_p^\dagger a_q^\dagger$ trivially, and $a_p^\dagger a_q$ because a creation operator standing to the left of an annihilation operator is what normal ordering means – so $xy = N[xy]$ and the contraction is zero. These are Eqs. (3.91)–(3.94).

Answer to c).

A contraction is a number, and the only thing that can change when the contracted pair is moved is the sign accumulated from anticommuting past other operators. Moving both members of the pair past one operator costs two interchanges, $(-1)^2 = +1$, so nothing happens. Interchanging the two contracted operators themselves costs one interchange and flips the sign. For a set of contractions, each crossing of two contraction lines is exactly one such interchange of neighbours, whence $(-1)^\nu$ with ν the number of crossings. The routine `matching_sign` in `wick.py` implements this rule, and `contraction_sign` shows that it agrees with the parity of the permutation that brings the contracted operators together.

Exercise 2: four operators the hard way and the easy way

a) Evaluate $\langle 0|a_s a_r a_p^\dagger a_q^\dagger|0\rangle$ by repeated use of the anticommutation relations.
b) Evaluate the same quantity with Wick's theorem, listing the three pairings and saying why two of them survive.
c) Interpret the result as an overlap between two-particle states.

Answer to a).

Push the annihilation operators to the right. With $a_r a_p^\dagger = \delta_{rp} - a_p^\dagger a_r$,

$$a_s a_r a_p^\dagger a_q^\dagger = \delta_{rp}\, a_s a_q^\dagger - a_s a_p^\dagger a_r a_q^\dagger,$$

and applying the same relation twice more to the second term gives, after taking the vacuum expectation value,

$$\langle 0|a_s a_r a_p^\dagger a_q^\dagger|0\rangle = \delta_{rp}\delta_{sq} - \delta_{sp}\delta_{rq}.$$

Every intermediate term with an operator left over dies against the vacuum.

Answer to b).

Four operators admit $3!! = 3$ pairings. The pairing that joins a_s to a_r must also join $a_p^\dagger$ to $a_q^\dagger$, and both contractions vanish by Eqs. (3.92) and (3.94). The two survivors join each annihilation operator to a creation operator standing to its right,

$$\langle 0|a_s a_r a_p^\dagger a_q^\dagger|0\rangle = \delta_{rp}\delta_{sq} - \delta_{sp}\delta_{rq},$$

the minus sign coming from the single crossing of the two contraction lines in the second term. Three terms written down, one discarded on sight, no permutations performed – against four applications of the anticommutator in part a). With eight operators the ratio is 105 terms against 129 recursive calls, and by twelve operators the direct route is hopeless.

Answer to c).

With $|pq\rangle = a_p^\dagger a_q^\dagger|0\rangle$ and $\langle rs| = \langle 0|a_s a_r$ the result is $\langle rs|pq\rangle = \delta_{rp}\delta_{sq} - \delta_{sp}\delta_{rq}$: two two-particle states overlap only if they contain the same pair of single-particle states, and the sign records whether the labels are in the same order. This is antisymmetry, obtained without writing a single determinant.

Exercise 3: the generalised theorem at work

a) Count the pairings of the eight operators in $\langle 0|\, a_j a_i\, a_p^\dagger a_q^\dagger a_s a_r\, a_k^\dagger a_\ell^\dagger\, |0\rangle$, first without and then with the restriction that no contraction may lie inside one of the three groups.
b) Explain why the restriction is legitimate here, and what would go wrong if the operator group were written as $a_p^\dagger a_s a_q^\dagger a_r$ instead.
c) Use the generalised theorem to show that a two-body operator has vanishing matrix elements between determinants differing by more than two single-particle states.

Answer to a).

Eight operators admit $7!! = 105$ pairings. Requiring every contraction line to join two different groups, with group sizes 2, 4 and 2, leaves 24. The counts are produced by `count_matchings` in `wick.py`, and of the 24 only four survive the elementary contractions – the four terms of Eq. (3.134).

Answer to b).

Each of the three groups is written in normal-ordered form: the bra is two annihilation operators, the ket two creation operators, and the operator has its creation operators to the left. By Eq. (3.122) all their internal contractions therefore vanish, and dropping them changes nothing. Written as $a_p^\dagger a_s a_q^\dagger a_r$ the group is no longer normal-ordered: the internal contraction of a_s with $a_q^\dagger$ is δ_{sq}, not zero, and those terms would have to be kept. One would first bring the group into normal order using Eq. (3.86), at the price of an overall sign.

Answer to c).

Let the bra and ket differ by n single-particle states. Every operator of the bra and of the ket must be contracted, and by the generalised theorem every contraction must reach a different group. The operator group supplies only four operators, two creation and two annihilation, so at most two of the bra's holes and two of the ket's particles can be accounted for. If $n > 2$ some operator of the bra has to contract with an operator of the ket, and those contractions vanish because the two determinants disagree there. Hence the matrix element is zero, which is the Condon-Slater rule of Section 3.14 and the reason the configuration-interaction matrix of Chapter 5 is sparse.

Exercise 4: normal ordering with respect to a Fermi vacuum

a) With $|c\rangle$ the reference determinant of Section 3.3, evaluate the four contractions of a_p and $a_p^\dagger$ with respect to $|c\rangle$ and say which survive.
b) Use them to compute $\langle c|\hat{H}_0|c\rangle$ for $\hat{H}_0 = \sum_{pq}\langle p|\hat{h}_0|q\rangle a_p^\dagger a_q$.
c) Explain why the same argument fails if $|c\rangle$ is replaced by a BCS state.

Answer to a).

With respect to $|c\rangle$ the quasiparticle operators are $b_a = a_a$ for $a > F$ and $b_i = a_i^\dagger$ for $i \le F$, and normal ordering means putting the $b^\dagger$'s to the left. Two contractions survive,

$$\langle c|a_a a_b^\dagger|c\rangle = \delta_{ab}\ (a,b > F), \qquad \langle c|a_i^\dagger a_j|c\rangle = \delta_{ij}\ (i,j \le F),$$

the second being new: a hole state is occupied in $|c\rangle$, so removing and replacing a particle there gives unity. All contractions mixing a particle index with a hole index vanish, as do $\langle c|a_p a_q|c\rangle$ and $\langle c|a_p^\dagger a_q^\dagger|c\rangle$.

Answer to b).

Only the fully contracted terms survive, and the only non-vanishing contraction of $a_p^\dagger a_q$ in the reference state is the hole one, so

$$\langle c|\hat{H}_0|c\rangle = \sum_{pq}\langle p|\hat{h}_0|q\rangle\, \delta_{pq}\, \theta(F-p) = \sum_{i\le F}\langle i|\hat{h}_0|i\rangle,$$

the sum of the occupied single-particle energies. This is Eq. (3.84) obtained in one line.

Answer to c).

The whole construction rests on the reference state being a single determinant, so that $\langle c|a_p^\dagger a_q|c\rangle$ is a Kronecker delta and every other contraction vanishes. A BCS state is a superposition of determinants with different particle numbers, and $\langle \mathrm{BCS}|a_p a_{\bar{p}}|\mathrm{BCS}\rangle$ – the pairing tensor – is *not* zero. Wick's theorem still applies, but with an extra kind of contraction, and the bookkeeping becomes the generalised Wick theorem of Bogoliubov quasiparticles used in Chapter 7.

Exercise 5: the Jordan-Wigner transformation by hand

a) Verify from Eq. (3.141) that $\hat{n}_p = (I - Z_p)/2$ and that $\{a_p, a_p^\dagger\} = I$.
b) Show that a_0 and a_1 anticommute, and identify precisely where the parity string does the work.
c) Derive the hopping identity (3.145) for adjacent orbitals $q = p+1$.
d) Derive the pair identity (3.147) for the same pair and compare the two.

Answer to a).

The parity strings of $a_p^\dagger$ and a_p are identical and $Z^2 = I$, so they cancel in any product of the two. Hence $\hat{n}_p = \sigma_p^+\sigma_p^- = |1\rangle\langle 0|\,|0\rangle\langle 1| = |1\rangle\langle 1| = (I - Z_p)/2$, using $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$. Likewise $a_p a_p^\dagger = |0\rangle\langle 0| = (I + Z_p)/2$, and the two add to I.

Answer to b).

Write $a_0 = \sigma_0^-$ and $a_1 = Z_0\sigma_1^-$. Then

$$a_0 a_1 = \sigma_0^- Z_0 \sigma_1^-, \qquad a_1 a_0 = Z_0 \sigma_0^- \sigma_1^-,$$

and the two differ only in the order of σ_0^- and Z_0 on the first qubit. Since $\sigma^- = |0\rangle\langle 1|$ and $Z|0\rangle\langle 1| = |0\rangle\langle 1|$ while $|0\rangle\langle 1|Z = -|0\rangle\langle 1|$, we have $\sigma_0^- Z_0 = -Z_0\sigma_0^-$, so $a_0 a_1 = -a_1 a_0$. The anticommutation comes entirely from the single factor Z_0 in the string of a_1; without it the two operators would act on different qubits and simply commute.

Answer to c).

For $q = p+1$ the strings cancel except for one factor, $a_p^\dagger a_{p+1} = \sigma_p^+ Z_p \sigma_{p+1}^- = \sigma_p^+ \sigma_{p+1}^-$, using $\sigma^+ Z = \sigma^+$. With $\sigma^\pm = (X \mp iY)/2$,

$$\sigma_p^+ \sigma_{p+1}^- = \frac{1}{4}(X_p - iY_p)(X_{p+1} + iY_{p+1}) = \frac{1}{4}(X_p X_{p+1} + Y_p Y_{p+1} + iX_p Y_{p+1} - iY_p X_{p+1}).$$

Adding the Hermitian conjugate cancels the two imaginary terms and doubles the others, leaving

$$\tfrac{1}{2}(X_p X_{p+1} + Y_p Y_{p+1}),$$

which is Eq. (3.145) with an empty parity string.

Answer to d).

The only change is that the second operator is a creation rather than an annihilation operator, so $\sigma_{p+1}^- \to \sigma_{p+1}^+$ and the sign of the iY_{p+1} term flips:

$$\sigma_p^+ \sigma_{p+1}^+ = \frac{1}{4}(X_p X_{p+1} - Y_p Y_{p+1} - iX_p Y_{p+1} - iY_p X_{p+1}),$$

and adding the Hermitian conjugate leaves $\frac{1}{2}(X_p X_{p+1} - Y_p Y_{p+1})$. The two operators are identical except for the relative sign of the YY term. On a quantum computer they cost exactly the same to measure and to exponentiate, which is a useful thing to know when one is comparing a hopping model with a pairing model.

Exercise 6: a Hamiltonian on two qubits

Consider two orbitals, $p = 0$ and $p = 1$, with

$$\hat{H} = \varepsilon_0 \hat{n}_0 + \varepsilon_1 \hat{n}_1 + U\hat{n}_0\hat{n}_1 + t\left(a_0^\dagger a_1 + a_1^\dagger a_0\right).$$

a) Map $\hat{H}$ to Pauli strings.
b) Write the 4×4 matrix in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ and identify the block structure.
c) Diagonalise the one-particle block and check the answer against the Pauli form.

Answer to a).

Using $\hat{n}_p = (I - Z_p)/2$ and Eq. (3.145),

$$\hat{H} = \frac{\varepsilon_0+\varepsilon_1}{2}I + \frac{U}{4}I - \frac{2\varepsilon_0+U}{4}Z_0 - \frac{2\varepsilon_1+U}{4}Z_1 + \frac{U}{4}Z_0Z_1 + \frac{t}{2}\left(X_0X_1+Y_0Y_1\right),$$

where the U term has been expanded as $\hat{n}_0\hat{n}_1 = (I-Z_0)(I-Z_1)/4$. Six Pauli strings, none of weight greater than two – which is the $M=2$ entry of Table 3.1.

Answer to b).

The I and Z terms are diagonal, and the $XX+YY$ term connects $|01\rangle$ and $|10\rangle$ only, since $\frac{1}{2}(X_0X_1+Y_0Y_1) = |01\rangle\langle 10| + |10\rangle\langle 01|$. Hence

$$\hat{H} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \varepsilon_1 & t & 0 \\ 0 & t & \varepsilon_0 & 0 \\ 0 & 0 & 0 & \varepsilon_0+\varepsilon_1+U \end{pmatrix},$$

block diagonal in the particle number, as it must be: every term of $\hat{H}$ conserves $\hat{N}$, so no Pauli string can connect sectors with different numbers of $|1\rangle$'s.

Answer to c).

The one-particle block has eigenvalues

$$E_\pm = \frac{\varepsilon_0+\varepsilon_1}{2} \pm \sqrt{\left(\frac{\varepsilon_0-\varepsilon_1}{2}\right)^2 + t^2},$$

the familiar two-level result of Section 2.1. For $\varepsilon_0 = \varepsilon_1$ the eigenvectors are $(|01\rangle \pm |10\rangle)/\sqrt{2}$, maximally entangled states of the two qubits: a delocalised particle and an entangled register are the same thing seen from two sides. The program `jordanwigner.py` builds this Hamiltonian, decomposes it and diagonalises it, and the three answers agree.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter03/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebooks `LectureNotes/wicktheorem.ipynb` and `LectureNotes/jordanwigner.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter03/`.

`wick.py` Vacuum expectation values evaluated three ways — by repeated anticommutation, by Wick contractions and by the generalised theorem — with the generalised theorem also checked as an operator identity in a small Fock space. Runs in a few seconds.

`jordanwigner.py` The Jordan–Wigner operators as matrices, the exact Pauli decomposition by traces, the anticommutator check, the hopping, pair and two-body identities, and the pairing and pairing-plus-particle-hole Hamiltonians in both the $2L$-qubit and the compact L-qubit encodings. Runs in a few seconds.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 4
Physical systems and models

The three preceding chapters built a formalism. This one puts it to work on a handful of systems.

The models collected here share a property that makes them invaluable: each is simple enough to be solved exactly, at least for small numbers of particles, yet each contains a genuine many-body mechanism. The Lipkin model has a collective mode driven by a two-body interaction that scatters pairs across a gap; the pairing model has superfluid correlations, and a small pair-breaking addition to it turns the particle-hole channel on and gives us the model that carries the whole of Chapter 7; the Fermi-Hubbard model has the competition between kinetic delocalisation and on-site repulsion that drives the Mott transition; its own strong-coupling limit is the quantum Heisenberg model, the simplest interacting spin system there is; and the Calogero-Sutherland model is exactly solvable for arbitrary particle number, with a ground state that is a pure Jastrow function. Because the exact answer is available, these models are the benchmarks against which every approximate method in the rest of this book is measured.

They also share a structure. Every one of them is written as

$$\hat{H} = \sum_{pq} \langle p|\hat{h}_0|q\rangle a_p^\dagger a_q + \frac{1}{4}\sum_{pqrs} \langle pq|\hat{v}|rs\rangle_{\mathrm{AS}} a_p^\dagger a_q^\dagger a_s a_r, \tag{4.1}$$

the general second-quantised Hamiltonian of Chapter 3, with a particular and very restricted choice of matrix elements. What differs between them is which single-particle states there are and which matrix elements are allowed to be non-zero. The symmetries that follow from those restrictions are what make the models tractable, and identifying them is the first step in every case. The one exception is the Heisenberg model of Section 4.5, which is written in spin operators rather than in fermion operators; the Jordan-Wigner transformation of Section 3.16 maps it back onto Eq. (4.1), and the mapping is instructive in its own right. Section 4.7 runs the same transformation the other way and puts all the models of this chapter on a qubit register.

Throughout, the detailed algebra is relegated to the exercises at the end of the chapter, where full solutions are given. The main text states each model, identifies its symmetries, and reports what the exact solution looks like. All numbers quoted are produced by `models.py`.

4.1 The Lipkin model

The Lipkin-Meshkov-Glick model [3] describes N fermions distributed over two levels, each of degeneracy Ω. The levels carry the quantum number $\sigma = \pm 1$: the upper has $\sigma = +1$ and energy $+\varepsilon/2$, the lower has $\sigma=-1$ and energy $-\varepsilon/2$. The substates of each level are labelled $p=1,\dots,\Omega$, so a single-particle state is $|p\sigma\rangle = a^\dagger_{p\sigma}|0\rangle$.

The Hamiltonian is

$$\hat{H} = \hat{H}_0 + \hat{H}_1 + \hat{H}_2, \tag{4.2}$$

with

$$\hat{H}_0 = \frac{1}{2}\varepsilon \sum_{p\sigma} \sigma\, a^\dagger_{p\sigma} a_{p\sigma}, \tag{4.3}$$

$$\hat{H}_1 = \frac{1}{2}V \sum_{pp'\sigma} a^\dagger_{p\sigma} a^\dagger_{p'\sigma} a_{p'-\sigma} a_{p-\sigma}, \tag{4.4}$$

$$\hat{H}_2 = \frac{1}{2}W \sum_{pp'\sigma} a^\dagger_{p\sigma} a^\dagger_{p'-\sigma} a_{p'\sigma} a_{p-\sigma}. \tag{4.5}$$

The operator $\hat{H}_1$ moves a *pair* of particles from one level to the other, with strength V; $\hat{H}_2$ is a spin-exchange term of strength W, which moves a pair from $(p\sigma, p'-\sigma)$ to $(p-\sigma, p'\sigma)$. Neither changes the substate labels p, and this is the crucial restriction: the interaction is completely degenerate in p.

Quasispin.

The degeneracy in p means the model has an $SU(2)$ symmetry. Define the *quasispin* operators

$$\hat{J}_\pm = \sum_p a^\dagger_{p\pm} a_{p\mp}, \qquad \hat{J}_z = \frac{1}{2}\sum_{p\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma}, \qquad \hat{J}^2 = \hat{J}_+\hat{J}_- + \hat{J}_z^2 - \hat{J}_z. \tag{4.6}$$

Exercise 4.9 shows, using only the anticommutation relations of Chapter 3, that these satisfy the angular-momentum algebra

$$[\hat{J}_z, \hat{J}_\pm] = \pm\hat{J}_\pm, \qquad [\hat{J}_+, \hat{J}_-] = 2\hat{J}_z, \qquad [\hat{J}^2, \hat{J}_\pm] = [\hat{J}^2, \hat{J}_z] = 0, \tag{4.7}$$

and that the Hamiltonian collapses to

$$\boxed{\hat{H} = \varepsilon\hat{J}_z + \frac{1}{2}V\left(\hat{J}_+^2 + \hat{J}_-^2\right) + \frac{1}{2}W\left(-\hat{N} + \hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+\right)} \tag{4.8}$$

with $\hat{N} = \sum_{p\sigma} a^\dagger_{p\sigma} a_{p\sigma}$ the number operator. Every trace of the substate index p has disappeared.

The consequence is decisive. Since $[\hat{H}, \hat{J}^2] = 0$ and $[\hat{H}, \hat{N}] = 0$, both J and N are good quantum numbers, and the Hamiltonian is block diagonal in J. A Hilbert space of dimension $\binom{2\Omega}{N}$ collapses into blocks of dimension $2J+1$. For $N=4$ particles at half filling the largest block is $J=2$, of dimension five: a 70-dimensional problem has become a 5×5 matrix.

The $J=2$ block.

The states are labelled $|J, J_z\rangle$ with $J_z = -J, \dots, J$. The lowest, with all four spins down, is

$$|2,-2\rangle = a^\dagger_{1-} a^\dagger_{2-} a^\dagger_{3-} a^\dagger_{4-}|0\rangle, \tag{4.9}$$

and the remaining four follow by applying $\hat{J}_+$ with the standard normalisations

$$\hat{J}_\pm |J, J_z\rangle = \sqrt{J(J+1) - J_z(J_z \pm 1)}\,|J, J_z \pm 1\rangle. \tag{4.10}$$

Working out the matrix elements of Eq. (4.8) – again Exercise 4.9 – gives for $\varepsilon = 2$, $V = -1/3$, $W = -1/4$

$$\boldsymbol{H}^{(1)}_{J=2} = \begin{bmatrix} -4 & 0 & -\sqrt{6}/3 & 0 & 0 \\ 0 & -11/4 & 0 & -1 & 0 \\ -\sqrt{6}/3 & 0 & -1 & 0 & -\sqrt{6}/3 \\ 0 & -1 & 0 & 5/4 & 0 \\ 0 & 0 & -\sqrt{6}/3 & 0 & 4 \end{bmatrix}, \tag{4.11}$$

whose eigenvalues are -4.21288, -2.98607, -0.91914, 1.48607 and 4.13201. The ground state is

$$|\psi_0\rangle = 0.96735|2,-2\rangle + 0.25221|2,0\rangle + 0.02507|2,2\rangle, \qquad E_0 = -4.21288. \tag{4.12}$$

Note the structure: the interaction connects only states differing by two units of J_z, because $\hat{J}^2_\pm$ moves a pair. The matrix therefore splits further into even and odd J_z.

Increasing the interaction to $V = -4/3$, $W = -1$ gives eigenvalues -7.75122, -7.47214, -1.55581, 1.47214, 5.30704 and the ground state

$$|\psi_0\rangle = 0.64268|2,-2\rangle + 0.73816|2,0\rangle + 0.20515|2,2\rangle, \qquad E_0 = -7.75122. \tag{4.13}$$

In the first case the unperturbed configuration $|2,-2\rangle$ carries 94% of the probability and a single Slater determinant is a good approximation. In the second the interaction matrix elements are comparable to the single-particle spacing, $|2,0\rangle$ has become the dominant component, and no single determinant will do. This is the same story the occupation numbers of Section 1.31 told, in a model small enough to see the whole of it.

The two couplings just quoted are two points on a curve, and it is worth looking at the whole of it. Figure 4.1 follows the $J = 2$ block as V is turned on continuously with $\varepsilon = 2$ and $W = -1/4$ held fixed. The left-hand panel is the ground-state energy. It leaves the unperturbed value $E_0 = -4$ quadratically, and we can say exactly why: the only state connected to $|2,-2\rangle$ is $|2,0\rangle$, through the matrix element $\sqrt{6}V$ of Eq. (4.11), and the two unperturbed energies differ by three, so second-order perturbation theory gives $E_0 \simeq -4 - 2V^2$. At $V = -1/3$ that is -4.222 against the exact -4.213, an error of two parts in a thousand. By $-V = 1$ the same estimate is -6 against -5.536 and by $-V = 2$ it is -12 against -8.396: the expansion has run away, while the exact curve has settled into an almost linear descent of slope -3.2 per unit of $-V$. A descent linear in the coupling is not something a power series in V reproduces at any finite order, and it is the first warning that a perturbative treatment of this model has a limited range.

The right-hand panel says where that range ends. It plots the probability $|\langle 2,-2|\psi_0\rangle|^2$ that the exact ground state is still the unperturbed configuration. The curve falls monotonically from unity, is 0.936 at $V = -1/3$ – the 94% quoted above – reaches 0.705 at $-V = 1$ and passes one half at $-V \simeq 2.16$, beyond which the reference determinant is the minority component of its own ground state. Two things deserve emphasis. The first is that nothing whatever happens to the exact solution at that point: both panels are smooth, there is no level crossing and no non-analyticity, because a five-dimensional matrix with entries analytic in V cannot produce one. Any sharp transition reported for this system at a finite coupling – and Hartree-Fock theory does report one, as the notebox below recalls – is an artefact of the approximation and not a property of the model. The second is that the erosion of the reference weight is gradual and begins at once, so there is no coupling at which the single-determinant picture is safe and beyond which it is not; there is only a slow handover, and each method fails somewhere along it. This is the concrete version of the occupation-number argument of Section 1.31, and the whole of the Lipkin model's usefulness as a testing ground lies in the fact that we can watch the handover happen along a single axis.

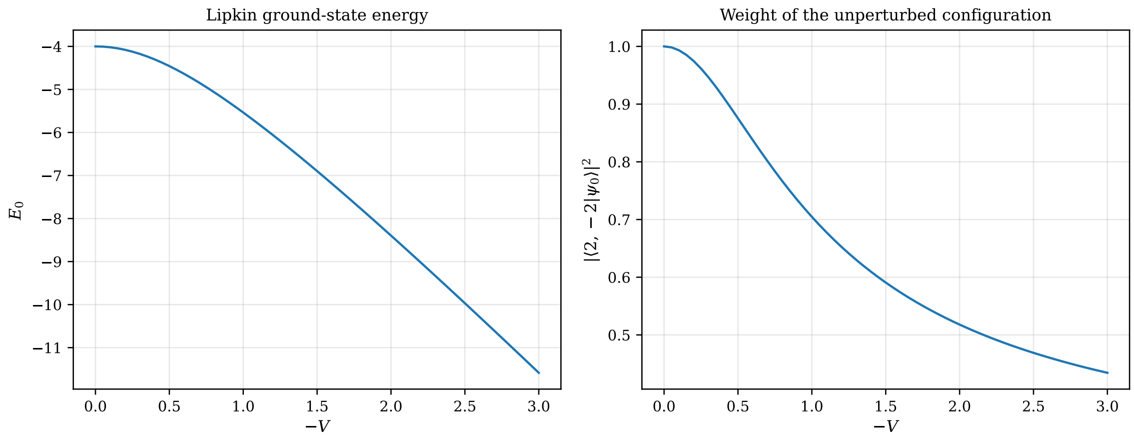

Fig. 4.1 The Lipkin model solved exactly in the $J = 2$ block for $N = 4$ particles, with $\varepsilon = 2$ and $W = -1/4$ fixed and the pair-transfer strength V varied along the negative axis. Left: the ground-state energy E_0, which starts from the unperturbed value -4, falls quadratically as $-4-2V^2$ for small $|V|$ and then almost linearly. Right: the weight $|\langle 2,-2|\psi_0\rangle|^2$ of the unperturbed configuration in the exact ground state, which decreases monotonically from unity and drops below one half at $-V \simeq 2.16$. The value $V = -1/3$ of Eq. (4.12) sits at 0.936 on the right-hand curve; the stronger case of Eq. (4.13) changes W as well and therefore does not lie on it. Both curves are smooth: the model has no phase transition at finite N.

Why this model matters. The Lipkin model was invented to test many-body methods, and it still is. Because the exact answer is a 5×5 eigenvalue problem while the interaction can be tuned from weak to strong, it exposes exactly where an approximation breaks down: Hartree-Fock theory undergoes a spurious phase transition at a critical coupling, the random-phase approximation predicts an excitation energy that goes imaginary there, and coupled-cluster theory tracks the exact result much further. Every one of these statements can be checked against Eq. (4.12) and its analogues.

Mapping to qubits.

The Lipkin model is a natural first target for quantum algorithms, and how one maps it onto qubits is worth a moment. There are three obvious schemes.

The *occupation mapping* assigns one qubit to each single-particle state (p,σ), the qubit being $|1\rangle$ or $|0\rangle$ according as that state is occupied or empty. It costs 2Ω qubits.

The *level mapping* exploits the fact that there are only two levels. Restricting to half filling, $N = \Omega$, each doublet $((p,+1),(p,-1))$ holds exactly one particle, so one qubit per doublet suffices: $|0\rangle$ for the particle in the upper state and $|1\rangle$ for the lower. This costs Ω qubits, half of the occupation mapping. It has a second advantage: moving a pair changes the state of two doublets, so any operator in the Hamiltonian is at most a two-qubit gate.

The *spin mapping* goes further and assigns a qubit to each state $|J,J_z\rangle$. For a given J there are only $2J+1$ of these, so with $J_{\max} = N/2$ the largest register is $N+1$ qubits. The price is that one must run a separate circuit for each J and take the lowest of the resulting energies.

Under the level mapping the quasispin operators become sums of Pauli matrices. Writing $\hat{J}_z = \sum_p j_z^{(p)}$ and $\hat{J}_\pm = \sum_p j_\pm^{(p)}$ with

$$j_z^{(p)} = \frac{1}{2}\sum_\sigma \sigma a^\dagger_{p\sigma}a_{p\sigma}, \qquad j_\pm^{(p)} = a^\dagger_{p\pm}a_{p\mp}, \tag{4.14}$$

each doublet operator is a single-qubit operator, and the correspondence with the Pauli matrices of Section 1.7 is

$$j_z^{(p)} \longleftrightarrow \tfrac{1}{2} Z_p, \qquad j_\pm^{(p)} \longleftrightarrow \tfrac{1}{2}\left(X_p \pm iY_p\right). \tag{4.15}$$

The Hamiltonian (4.8) therefore becomes a sum of Pauli strings of weight at most two – exactly the form a variational quantum eigensolver requires. The circuits themselves belong to the quantum-computing part of this book; what matters here is that the mapping costs Ω qubits and two-qubit interactions only, which is a direct consequence of the $SU(2)$ structure identified above.

4.2 The pairing model

The pairing model has appeared repeatedly already – as the test case for the Lanczos algorithm in Section 1.26, for the Schmidt decomposition in Section 1.30, and as the three-level exercise of week 35. We now set it out properly.

The single-particle space consists of L doubly degenerate, equally spaced levels labelled $p = 1, 2, \ldots, L$ with spin $\sigma = \pm 1$. The Hamiltonian is

$$\hat{H} = \hat{H}_0 + \hat{V}, \qquad \hat{H}_0 = \xi \sum_{p\sigma} (p-1)\, a^\dagger_{p\sigma} a_{p\sigma}, \qquad \hat{V} = -\frac{1}{2} g \sum_{pq} a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+}. \tag{4.16}$$

The spacing ξ is set to unity without loss of generality. The interaction has a single term and a single strength g: it destroys a pair of particles in level q with opposite spins and creates one in level p. Nothing else happens. In the language of Eq. (4.1), only the matrix elements $\langle p+\, p-|\hat{v}|q+\, q-\rangle$ are non-zero, and they are all equal.

Symmetries.

Exercise 4.9 shows that both $\hat{H}_0$ and $\hat{V}$ commute with the spin projection and the total spin,

$$\hat{S}_z = \frac{1}{2} \sum_{p\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma}, \qquad \hat{S}^2 = \hat{S}_z^2 + \frac{1}{2}\left(\hat{S}_+ \hat{S}_- + \hat{S}_- \hat{S}_+\right), \qquad \hat{S}_\pm = \sum_p a^\dagger_{p\pm} a_{p\mp}, \tag{4.17}$$

so the Hamiltonian is block diagonal in S and S_z. We work in the $S = 0$ block. There it is natural to introduce the pair operators

$$\hat{P}_p^+ = a^\dagger_{p+} a^\dagger_{p-}, \qquad \hat{P}_p^- = a_{p-} a_{p+}, \tag{4.18}$$

in terms of which

$$\hat{H} = \sum_{p\sigma} (p-1) a^\dagger_{p\sigma} a_{p\sigma} - \frac{1}{2} g \sum_{pq} \hat{P}_p^+ \hat{P}_q^-. \tag{4.19}$$

The pair creation operators commute among themselves, $[\hat{P}_p^+, \hat{P}_q^+] = 0$, which is why they behave like bosons and why the model describes a superfluid. They are *not* bosons, however: $(\hat{P}_p^+)^2 = 0$, the Pauli principle asserting itself.

The seniority-zero space.

Because the interaction only moves pairs, it never breaks one. If we start from a state in which every occupied level carries a full pair, we stay in that space forever. A basis state is then simply a choice of which n of the L levels carry a pair, and the dimension is $\binom{L}{n}$ rather than $\binom{2L}{2n}$. For $L=4$ and $n=2$ this is 6 instead of 28; for $L=12$ and $n=6$ it is 924 instead of 2704156. This is the space in which Table 1.3 was computed.

In that basis the Hamiltonian is easy to write down: the diagonal is $2\sum_{p\in\mathrm{occ}}(p-1)-\frac{1}{2}gn$, and any two configurations differing by the position of a single pair are connected by $-\frac{1}{2}g$.

Table 4.1 gives the exact ground-state energy for four particles in four levels as a function of g, together with the energy of the unperturbed configuration. At $g=0$ the reference determinant is exact. For small $|g|$ the correlation energy grows quadratically, as second-order perturbation theory requires, and for $g=1$ it has reached 36% of the reference energy.

g	E_{ref}	E_0 (exact)	E_{corr}
−1.00	3.000000	2.77987014	−0.22012986
−0.50	2.500000	2.43688426	−0.06311574
0.00	2.000000	2.00000000	0.00000000
0.50	1.500000	1.41677428	−0.08322572
1.00	1.000000	0.63554847	−0.36445153

Table 4.1 The pairing model with four doubly degenerate levels and two pairs, $\xi=1$. E_{ref} is the energy of the configuration with both pairs in the two lowest levels; E_0 is the exact ground-state energy in the six-dimensional seniority-zero space.

The five rows of the table are five points on the topmost curve of Figure 4.2, which scans the coupling continuously and adds two larger systems at the same filling: $L=6$ with three pairs and $L=8$ with four, of dimensions twenty and seventy. Three features are worth drawing out.

The first is the behaviour at the origin. Every curve is stationary at $g=0$ and curves downward on both sides, which is second-order perturbation theory made visible: the reference is exact at $g=0$, the first-order correction is already contained in E_{ref}, and the leading correction is therefore $O(g^2)$ and necessarily negative. For $L=4$ and $n=2$ we can even put a number on it without diagonalising anything. Four configurations are reachable from the reference by moving one pair, obtained by lifting a pair from level 2 or level 1 to level 3 or level 4; the matrix element is $-g/2$ in each case and the excitation energies are 2, 4, 4 and 6, so

$$E^{(2)}=-\frac{g^2}{4}\left(\frac{1}{2}+\frac{1}{4}+\frac{1}{4}+\frac{1}{6}\right)=-0.2917\,g^2. \tag{4.20}$$

At $|g|=1/2$ this predicts -0.0729, and the two exact values are -0.0832 for $g=+1/2$ and -0.0631 for $g=-1/2$, whose mean is -0.0732. The agreement of the mean is not a coincidence: the even orders of the expansion are the same for both signs, so the average of $E_{\mathrm{corr}}(g)$ and $E_{\mathrm{corr}}(-g)$ removes the odd orders and leaves Eq. (4.20) plus $O(g^4)$.

The second is the asymmetry itself. The curves are visibly steeper for positive g, where the interaction is attractive, than for negative g: at $|g|=1$ the $L=4$ system has -0.364 against -0.220 and the $L=8$ system -1.111 against -0.423, a factor of 2.6. Attraction lets the amplitudes of all the pair-transfer processes add coherently, which is the beginning of the superfluid state; repulsion makes them interfere and the system stays close to its reference. This asymmetry is entirely third and higher order, so it is also a measure of how far a low-order expansion can be trusted.

The third is what happens as the system grows. At $g=1$ the correlation energy per pair is -0.182, -0.233 and -0.278 for $n=2$, 3 and 4: it increases rather than settling down. The reason is that the pairing interaction of Eq. (4.16) is of infinite range in the level index – every level is coupled to every other with the same strength g – so the number of pair-transfer channels available to the reference grows like $n(L-n)$ and the correlation energy is not extensive. A method whose error per particle is constant will therefore look worse and worse on this model as L grows at fixed g, which is precisely the size-extensivity problem that truncated configuration interaction suffers from and coupled-cluster theory does not.

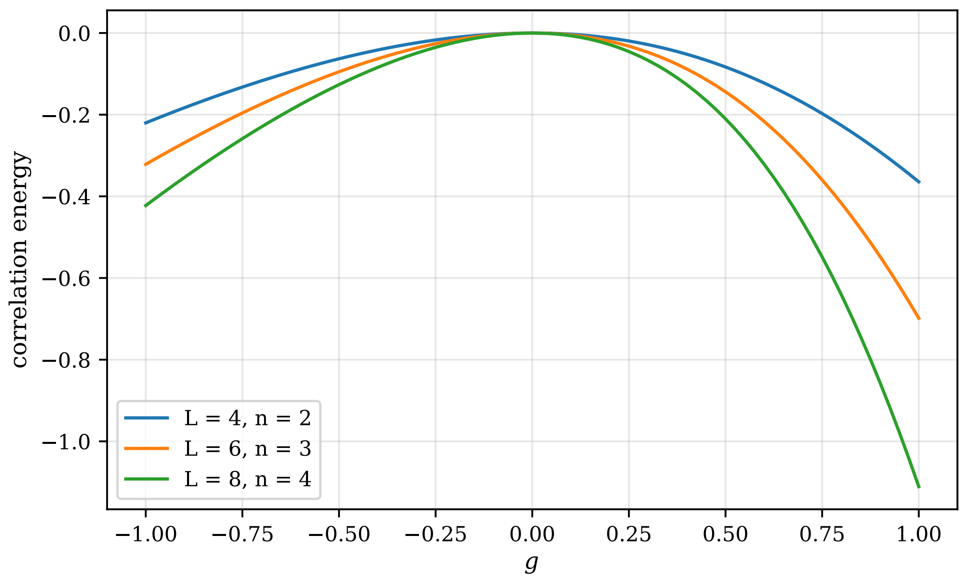

Fig. 4.2 Correlation energy $E_{\text{corr}} = E_0 - E_{\text{ref}}$ of the pairing model in the seniority-zero space, against the coupling g, for three half-filled systems with level spacing $\xi = 1$: $L = 4$ levels with $n = 2$ pairs, $L = 6$ with $n = 3$ and $L = 8$ with $n = 4$. The reference is in each case the configuration with the n lowest levels doubly occupied. All three curves are negative for either sign of the coupling and vanish quadratically at $g = 0$, as second-order perturbation theory requires (4.20); the attractive side $g > 0$ gains substantially more correlation energy than the repulsive side, and the gain grows with the number of levels. The $g = \pm 1$ values of the uppermost curve are the last and first rows of Table 4.1.

Why this model matters. The pairing model is the schematic version of superfluidity and superconductivity, and it is the standard proving ground for the hierarchy of many-body methods. Because the exact answer is available for any g, one can plot Hartree-Fock, second- and third-order perturbation theory, coupled cluster with doubles, and full configuration interaction on the same axes and watch each one fail at a different coupling. The second midterm of FYS4480 does exactly this, and the exercises at the end of this chapter set up the ingredients.

4.3 Breaking the pairs: a particle-hole interaction

The pairing model is almost too well behaved. Its interaction moves pairs and does nothing else, so the seniority – the number of particles that are not part of a complete pair – is a constant of the motion, and the whole calculation collapses into the $\binom{L}{n}$-dimensional seniority-zero space. That is a blessing when one wants exact answers cheaply, but it is a liability when one wants to test methods, because an entire class of excitations is simply absent. In particular the *particle-hole* channel, in which a single particle is lifted across the Fermi level leaving one hole behind, is empty: the pairing interaction has no matrix element between the reference determinant and any one-particle-one-hole state. The Tamm-Dancoff and random-phase approximations of Chapter 7 are built entirely on that channel, and on the pure pairing model they have nothing to say.

The cure is one extra term. We keep the level scheme of Eq. (4.16) and add an interaction that creates a pair while removing two particles which need not be partners,

$$\hat{H} = \hat{H}_0 + \hat{V}_{\text{pair}} + \hat{V}_{\text{ph}}, \qquad \hat{V}_{\text{ph}} = -\frac{f}{2}\sum_{pqr}\left(a^\dagger_{p+}a^\dagger_{p-}a_{q-}a_{r+} + \text{h.c.}\right), \tag{4.21}$$

with $\hat{H}_0$ and $\hat{V}_{\text{pair}}$ exactly as in Eq. (4.16). The new strength f is a second knob. At $f = 0$ we recover the pairing model and every number of Table 4.1; at $g = 0$ only pair breaking survives; and in between the two mechanisms compete. This is the model of Chapter 7, introduced here so that it sits alongside the others rather than arriving unannounced.

What the extra term actually does.

The triple sum in Eq. (4.21) hides three quite different processes, and it is worth separating them.

- When $q = r$ the operator is $a^\dagger_{p+}a^\dagger_{p-}a_{q-}a_{q+}$, which is the pairing term again. Together with its Hermitian conjugate, which contributes the same sum over again, these pieces merely renormalise $g \to g + 2f$; they move whole pairs and change nothing structurally.
- When $q = p$ and $r \neq p$ – more transparently, in the Hermitian conjugate, $a^\dagger_{r+}a^\dagger_{p-}a_{p-}a_{p+} = a^\dagger_{r+}\hat{n}_{p-}a_{p+}$ – a *single* spin-up particle is carried from level p to level r while its former partner stays put. This is a one-particle-one-hole excitation, and it is the process the pairing interaction cannot perform.
- When q, r and p are all different, two particles are removed from two different levels and a pair is deposited in a third. The result is a two-particle-two-hole determinant with two unpaired particles.

Figure 4.3 shows the level scheme and one representative of each excitation. Panel (a) is the reference determinant $|c\rangle$ with the two lowest levels filled; panel (b) is the only kind of excitation the pairing term can produce, a whole pair moved across the Fermi level; panels (c) and (d) are the one-particle-one-hole and pair-breaking two-particle-two-hole determinants that only $\hat{V}_{\text{ph}}$ can reach.

The price: the seniority-zero space is no longer closed.

Because $\hat{V}_{\text{ph}}$ breaks pairs, we can no longer work with the six pair configurations of Section 4.2. What survives is the weaker statement that both terms conserve N_+ and N_- separately, so $S_z = (N_+ - N_-)/2$ is still a good quantum number. For $L = 4$ and $N = 4$ the full space has $\binom{8}{4} = 70$ determinants and the balanced sector $S_z = 0$ has 36 – six times the seniority-zero dimension, and the price of one extra term.

Table 4.2 makes the content of Figure 4.3 quantitative. It lists every determinant that the Hamiltonian connects directly to the reference, classified by excitation rank and seniority, together with the matrix element. At $f = 0$ there are four such determinants and they are all of type (b). At $f \neq 0$ two further families appear, exactly as the figure suggests, and the seniority-zero matrix element picks up the promised $g \to g + 2f$.

The effect on the ground state is shown in Table 4.3. The first row is the pairing model, and reproduces the $g = 1$ entry of Table 4.1 exactly – a check worth making, since the two rows are

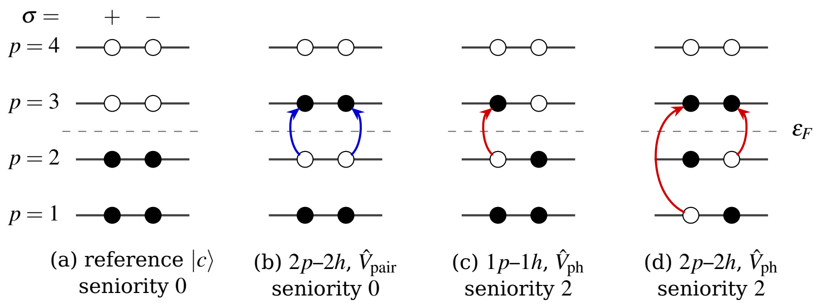

Fig. 4.3 Single-particle levels of the pairing model with $L = 4$ doubly degenerate levels, $\varepsilon_p = (p-1)\xi$, and $N = 4$ particles. Each level carries two slots, $\sigma = +$ and $\sigma = -$; a filled circle is an occupied spin-orbital and an open circle an empty one, so an open circle below ε_F is a hole. (a) The reference determinant. (b) A two-particle-two-hole excitation produced by the pairing term $\hat{V}_{\rm pair}$: a complete pair is carried from level 2 to level 3 and the seniority stays zero. (c) A one-particle-one-hole excitation and (d) a pair-breaking two-particle-two-hole excitation, both of which require the particle-hole term $\hat{V}_{\rm ph}$ of Eq. (4.21) and both of which have seniority two. In the pure pairing model panels (c) and (d) are unreachable from (a).

Channel	**Seniority**	**Number**	$\langle\Phi\|\hat{H}\|c\rangle$
$g = 1,\ f = 0$: *the pairing model*			
$2p$–$2h$	0	4	$-g/2$
$g = 1,\ f = 0.05$			
$1p$–$1h$	2	8	$\pm f/2$
$2p$–$2h$	0	4	$-(g+2f)/2$
$2p$–$2h$	2	8	$\pm f/2$

Table 4.2 The determinants connected to the reference by one action of the Hamiltonian, for $L = 4$ and $N = 4$ in the $S_z = 0$ sector. With $f = 0$ only the seniority-zero two-particle-two-hole states are reached, which is why the particle-hole channel of the pairing model is empty. Computed by `models.py`.

computed in spaces of dimension six and thirty-six respectively. As f grows the correlation energy grows much faster than the reference energy falls, which is the signature of a second, competing mean field; the whole of Chapter 7 is about deciding which of the two to build the fluctuations on.

g	f	$E_{\rm ref}$	E_0 (exact)	$E_{\rm corr}$
1.00	0.00	1.0000	0.63554847	−0.36445153
1.00	0.05	0.9000	0.45058234	−0.44941766
1.00	0.20	0.6000	−0.18348455	−0.78348455
1.00	0.50	0.0000	−1.69173670	−1.69173670

Table 4.3 The pairing model with a particle-hole term, $L = 4$, $N = 4$, $\xi = 1$, diagonalised in the 36-dimensional $S_z = 0$ sector. The first row is the pairing model and agrees with Table 4.1 to all digits shown.

Figure 4.4 turns the four rows of that table into a continuous scan. The dashed line is the reference energy and it is exactly straight, $E_{\rm ref} = 1 - 2f$, reaching zero at $f = 1/2$. That is the first of the three processes listed above and nothing else: the reference determinant has both pairs intact, the particle-hole pieces of $\hat{V}_{\rm ph}$ have no diagonal matrix element in it, and all that survives is the renormalisation $g \to g + 2f$ acting on the two pairs. The exact energy, the solid curve, starts 0.364 below it and falls away much faster, ending at -2.238 when $f = 0.6$, where the reference is still at -0.200. The correlation energy has grown from -0.364 to -2.038, a factor of 5.6, while the reference energy has moved by 1.2.

The initial slope of the exact curve is worth a moment, because it can be had exactly. At $f = 0$ the derivative of the ground-state energy with respect to f is, by the Hellmann-Feynman theorem, the expectation value of $\partial\hat{H}/\partial f$ in the pairing ground state; the pair-breaking operators have vanishing expectation value there, and the remainder is twice the derivative with respect to g. Numerically $dE_0/df = -3.548$ at $f = 0$ against $dE_0/dg = -1.774$, and the factor of two is exact to every digit computed. So the exact curve leaves the origin 1.77 times steeper than the reference line, and all of that steepening is still the old pairing mechanism in disguise. It is only further out that the new channels take over: by $f = 0.3$ the slope has grown to -4.913, which the $g \to g+2f$ renormalisation cannot account for, and the extra descent comes from the eight one-particle-one-hole and eight seniority-two two-particle-two-hole determinants of Table 4.2, each connected to the reference with strength $\pm f/2$. The exact energy passes through zero at $f \simeq 0.159$, well before the reference does at $f = 0.5$. For the mean-field methods of Chapter 7 this widening gap is the whole story: the dashed line is what a pairing-based mean field knows about, and everything between the two curves has to be recovered by the fluctuations built on top of it.

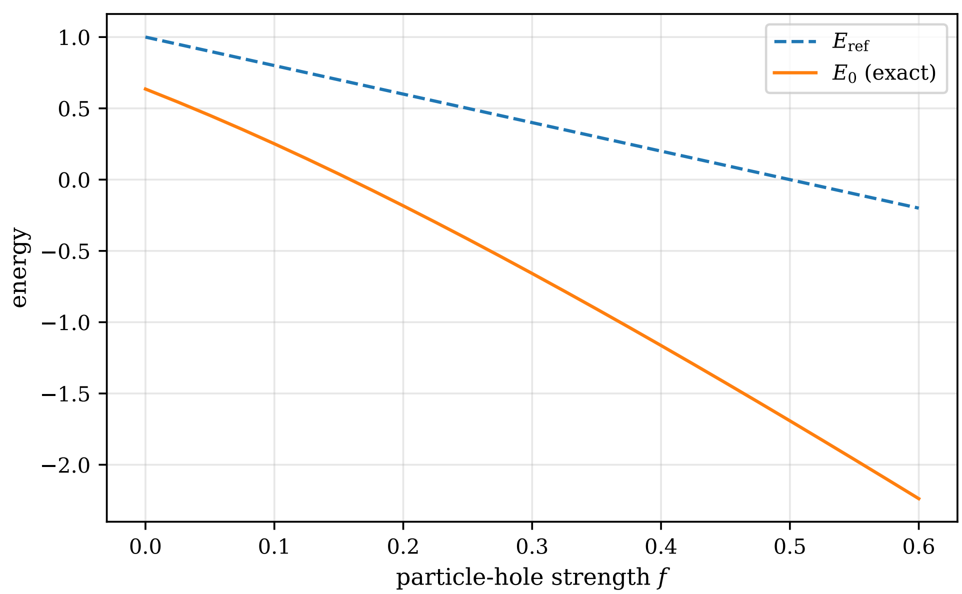

Fig. 4.4 The pairing model with the particle-hole term of Eq. (4.21) for $L = 4$ levels, $N = 4$ particles, $\xi = 1$ and $g = 1$, as the pair-breaking strength f is increased. The dashed line is the energy $E_{\mathrm{ref}} = 1-2f$ of the reference determinant with the two lowest levels filled, which feels f only through the renormalisation $g \to g+2f$; the solid line is the exact ground-state energy obtained by diagonalising the Hamiltonian in the 36-dimensional $S_z = 0$ sector. The two curves separate steadily, from 0.364 at $f = 0$ to 2.038 at $f = 0.6$, as the one-particle-one-hole and pair-breaking channels opened by $\hat{V}_{\mathrm{ph}}$ take up an ever larger share of the binding. The values at $f = 0$, 0.05, 0.2 and 0.5 are the four rows of Table 4.3.

Why this model matters. A model with one interaction tests one approximation. With two competing interactions and two strengths, the same system can be driven from a regime where a pairing mean field is right to one where a particle-hole mean field is, and one can watch each approximation fail where it should. That is precisely what Chapter 7 does with the Tamm-Dancoff approximation, the random-phase approximation, BCS and the quasiparticle RPA, all measured against the exact diagonalisation of this section.

4.4 The Fermi-Hubbard model

The two models so far live in an abstract single-particle space with no geometry. The Fermi-Hubbard model introduces a lattice, and with it the competition that dominates the physics of correlated electrons: kinetic energy wants to delocalise the particles, the interaction wants to keep them apart.

Electrons hop between neighbouring sites of a lattice and pay an energy U whenever two of them, necessarily of opposite spin, occupy the same site,

$$\boxed{\hat{H} = -\sum_{\langle ij\rangle\sigma} t_{ij}\left(c_{i\sigma}^{\dagger}c_{j\sigma} + \text{h.c.}\right) + U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}} \tag{4.22}$$

with $\hat{n}_{i\sigma} = c_{i\sigma}^{\dagger}c_{i\sigma}$. This is again Eq. (4.1): the one-body part is the hopping matrix, and the only non-vanishing two-body matrix element is the on-site one.

The single-particle basis is now the set of lattice sites, so the creation operators carry a position label. Fermionic antisymmetry is enforced by the Jordan-Wigner string: $c_i^{\dagger}c_j$ acting on an occupation-number state picks up the parity of the occupied sites strictly between i and j, which is exactly the phase computed by the bit-counting algorithm at the end of Chapter 3.

What the Hamiltonian actually needs from the lattice is very little. The symbol $\langle ij\rangle$ in Eq. (4.22) is nothing but a list of pairs of sites, each carrying an amplitude t_{ij}, and everything the geometry contributes is contained in that list. Figure 4.5 shows one such list drawn as a graph: sixteen sites on a 4×4 lattice, numbered $i = 4x + y$ with x the column and y the row, joined by the solid nearest-neighbour bonds and, on alternate plaquettes, by the dashed diagonals that a next-nearest-neighbour hopping would add. The lattice is periodic in both directions, so every site has four nearest neighbours and there are 32 nearest-neighbour bonds in all; the eight of them that wrap round the boundary cannot be drawn without clutter and have been left out of the picture, which is why the sites at the edges look under-connected. The diagonals are placed on the eight plaquettes whose lower-left corner has $x+y$ even, and five of those eight lie wholly inside the drawn region.

Two consequences of the drawing matter for what follows. The first is that the nearest-neighbour lattice on its own is bipartite – colour the sites with $x+y$ even black and the rest white, and every solid bond joins a black site to a white one – and it is this property that gives the half-filled Hubbard model its particle-hole symmetry and, through the Marshall sign rule of Section 4.5, makes its strong-coupling limit free of the quantum Monte Carlo sign problem. Adding the dashed bonds destroys it: a diagonal joins two sites of the same colour, the lattice becomes frustrated and the sign problem returns. A single line in the bond list is the difference between a model one can simulate and one one cannot. The second is the sheer size of what is drawn. Sixteen sites at half filling give $\binom{16}{8}^2 = 1.66\times 10^8$ determinants with $N_\uparrow = N_\downarrow = 8$, so the innocent-looking graph of Figure 4.5 is already beyond exact diagonalisation on a laptop, and the calculations reported below use rings of four and six sites instead.

Symmetries and sector dimensions.

The Hamiltonian conserves $N_\uparrow$ and $N_\downarrow$ separately, so we may diagonalise in a fixed sector of dimension

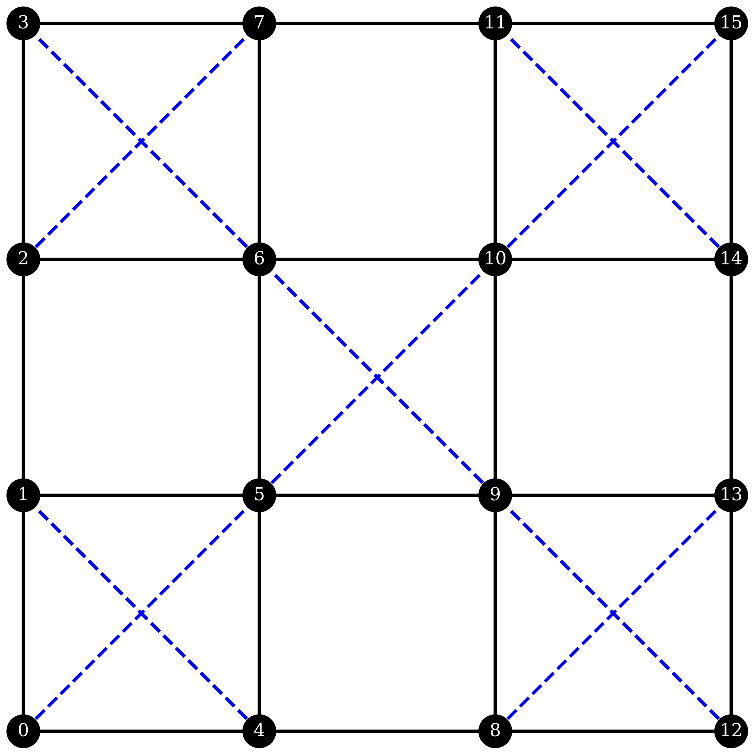

Fig. 4.5 The bond list of a two-dimensional Hubbard lattice drawn as a graph: a 4×4 square lattice with periodic boundary conditions in both directions, the sites numbered $i = 4x + y$. Solid lines are the nearest-neighbour bonds carrying the hopping t of Eq. (4.22), dashed lines the two diagonals of every plaquette whose lower-left corner has $x + y$ even, which is how a next-nearest-neighbour hopping is added to the model. The eight bonds that wrap round the periodic boundary are not drawn. The nearest-neighbour lattice by itself is bipartite; the diagonals connect sites of the same sublattice and frustrate it.

$$\dim = \binom{N}{N_\uparrow}\binom{N}{N_\downarrow}. \tag{4.23}$$

Better still, the hopping term acts on the two spin species independently, so the sector factorises as a tensor product and the kinetic part of the Hamiltonian is

$$\boldsymbol{K}_\uparrow \otimes \boldsymbol{I}_\downarrow + \boldsymbol{I}_\uparrow \otimes \boldsymbol{K}_\downarrow, \tag{4.24}$$

the tensor-product construction of Section 1.10 used in earnest. The interaction is diagonal in the occupation basis. On a periodic chain translational invariance gives a further reduction into momentum sectors.

From metal to Mott insulator.

Table 4.4 follows the half-filled four-site ring as U grows. At $U = 0$ the model is exact free fermions: the single-particle energies are $\varepsilon_k = -2t\cos k$ with $k = 0, \pm\pi/2, \pi$, and filling the two lowest levels for each spin gives $E_0 = -4t$, which the diagonalisation reproduces. As U increases, the double occupancy falls steadily – the electrons avoid each other – and the ground-state energy rises towards zero.

U/t	E_0/t	$E_0/(Nt)$	$\sum_i \langle n_{i\uparrow} n_{i\downarrow} \rangle$
0	-4.00000000	-1.000000	1.000000
2	-2.82842712	-0.707107	0.453082
4	-2.10274848	-0.525687	0.287325
8	-1.32023496	-0.330059	0.129992
16	-0.72286432	-0.180716	0.042033

Table 4.4 The half-filled four-site Hubbard ring, $N_\uparrow = N_\downarrow = 2$, sector dimension 36. The last column is the total double occupancy in the ground state.

The strong-coupling limit.

When $U \gg t$ the doubly occupied states are pushed far up in energy and can be eliminated perturbatively. What remains is a model of localised spins,

$$\hat{H}_{tJ} = \hat{P}_G \left[-\sum_{\langle ij \rangle \sigma} t_{ij} \left(c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.} \right) \right] \hat{P}_G + \sum_{\langle ij \rangle} J_{ij} \left[\boldsymbol{S}_i \cdot \boldsymbol{S}_j - \tfrac{1}{4} \hat{P}_G \hat{n}_i \hat{n}_j \hat{P}_G \right], \qquad J_{ij} = \frac{4t_{ij}^2}{U}, \tag{4.25}$$

where $\hat{P}_G$ is the Gutzwiller projector removing all doubly occupied sites. At half filling there is no empty site to hop into, the kinetic term is frozen, and the t-J model reduces to the Heisenberg antiferromagnet.

This is a prediction we can check. For the four-site ring the Heisenberg part contributes $-2J$ and the constant $-\frac{1}{4} n_i n_j$ term on four bonds contributes $-J$, so $E_0 \to -3J = -12t^2/U$. Table 4.5 shows the ratio E_0/J converging to -3.

U/t	E_0/t	$E_0/J, \quad J = 4t^2/U$
8	-1.3202349583	-2.64047
16	-0.7228643224	-2.89146
32	-0.3714104223	-2.97128
64	-0.1870445461	-2.99271
128	-0.0936928521	-2.99817

Table 4.5 The half-filled four-site Hubbard ring at strong coupling. The ratio converges to -3, confirming the mapping onto the Heisenberg model with $J = 4t^2/U$.

Figure 4.6 plots that ratio against U/t on a logarithmic axis, with the predicted value drawn in. The convergence is unmistakable but it is also slow, and the rate is informative. The deviations from -3 are 0.360, 0.109, 0.029, 0.0073 and 0.0018 at $U/t = 8$, 16, 32, 64 and 128, and the successive ratios are 3.3, 3.8, 3.9 and 4.0: each doubling of U divides the error by four. That is exactly what one should expect. The elimination of the doubly occupied states is a perturbation expansion in t/U whose leading term is the exchange $J = 4t^2/U$; the next term is of order t^4/U^3, and the ratio of the two is $(t/U)^2$, which quarters when U doubles. The figure therefore does more than confirm the mapping: it measures the size of the first term the mapping throws away, and tells us that at $U = 8t$ – a perfectly realistic value for a transition-metal oxide – the discarded physics is still 12% of the answer. Not until $U = 64t$ has the error fallen below a quarter of a per cent.

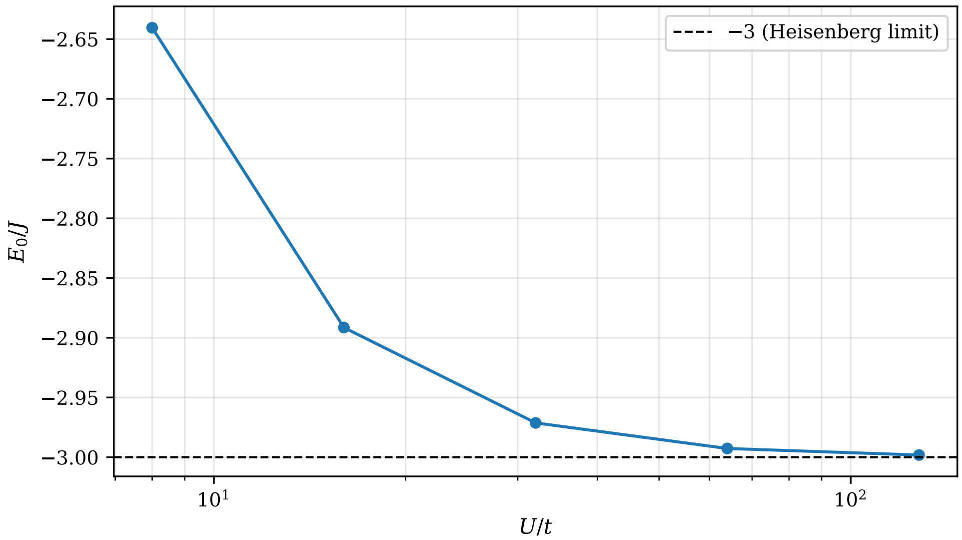

Fig. 4.6 The half-filled four-site Hubbard ring at strong coupling. Plotted is the exact ground-state energy in units of the effective exchange $J = 4t^2/U$, against U/t on a logarithmic scale; the dashed line is the value -3 predicted by the mapping onto the Heisenberg antiferromagnet, $-2J$ from the exchange on the four bonds and $-J$ from the constant $-\frac{1}{4}\hat{n}_i\hat{n}_j$ term. The remaining deviation falls by a factor of four for every doubling of U, the signature of a leading correction of order t^4/U^3. The five points are the rows of Table 4.5.

What the t-J model leaves out.

The four-site ring at half filling is a special case, because the Gutzwiller projector freezes the kinetic term completely and the t-J Hamiltonian (4.25) is the Heisenberg model and nothing else. It is worth repeating the comparison on a larger ring and against the full t-J model, and Figure 4.7 does so for six sites with three electrons of each spin. On the left the two ground-state energies per site are plotted together over $4 \le U/t \le 128$. At half filling the t-J energy is proportional to J, so a single diagonalisation fixes the whole curve: the six-site Heisenberg ring has $E_0 = -2.8028J$ and the six bonds contribute a further $-1.5J$, giving $E_0^{tJ} = -4.3028J = -17.211\,t^2/U$. The two curves are indistinguishable to the eye above $U \simeq 20t$; at $U = 4t$ they differ by 0.106 per site, the t-J estimate lying too low by 17%.

The right-hand panel is the interesting one. It plots the absolute difference of the two total energies on logarithmic axes, and the points fall on a straight line of slope -3: the residual is $0.634t$ at $U = 4t$, $0.0219t$ at $U \simeq 14t$ and $3.2 \times 10^{-5}t$ at $U = 128t$, and the local slope between the last two points is 3.00 to three digits. A power law with exponent three is a statement about which term of the strong-coupling expansion is missing. The t-J model keeps the second-order superexchange $4t^2/U$ and stops; the fourth-order processes, in which a particle hops round a plaquette or through an intermediate doubly occupied site and back by a different route, generate ring-exchange couplings of order t^4/U^3, and those are precisely what the truncation discards. The figure measures them. It is also a warning about the ubiquitous habit of replacing the Hubbard model by the t-J model at intermediate coupling: at $U = 8t$ the two differ in the total energy by $0.103t$ on six sites, or 5%, which is comparable to the energy scales that distinguish competing ordered states in the cuprate problem.

Why this model matters. The Hubbard model is the minimal model of strong correlation. It is believed to contain the essential physics of the cuprate superconductors, it is the standard benchmark for quantum Monte Carlo, tensor-network and machine-learning

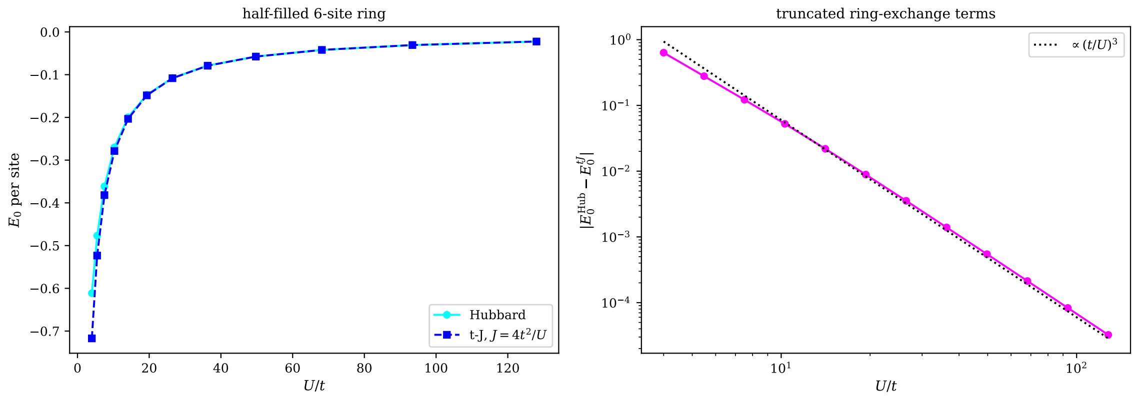

Fig. 4.7 The t-J model as the strong-coupling limit of the Hubbard model, for the half-filled six-site ring with periodic boundary conditions and $N_\uparrow = N_\downarrow = 3$. Left: the exact Hubbard ground-state energy per site (circles) and the t-J energy per site at $J = 4t^2/U$ (squares), against U/t; at half filling the latter is the Heisenberg ring energy $-4.3028J$ and scales exactly as $1/U$. Right: the absolute difference of the two total energies on logarithmic axes, together with a line $\propto (t/U)^3$. The residual follows that power law over more than four decades, identifying it as the fourth-order ring-exchange physics of order t^4/U^3 that the t-J truncation omits.

wave functions, and it is among the first targets for quantum simulation with cold atoms and with quantum computers. It is also the model in which the exponential wall of Section 2.16 bites hardest: the sector dimension (4.23) for a 4×4 lattice at half filling is already 1.66×10^8, and the two-dimensional model remains unsolved.

4.5 The quantum Heisenberg model

The strong-coupling limit of the previous section produced a model of localised spins, and that model deserves a section of its own. At half filling and $U \gg t$ there is no room for an electron to hop, the charge degrees of freedom are frozen out, and all that is left on each site is a spin one-half. What remains is the *quantum Heisenberg model*,

$$\boxed{\hat{H} = J \sum_{\langle ij \rangle} \boldsymbol{S}_i \cdot \boldsymbol{S}_j - h \sum_i \hat{S}_i^z} \tag{4.26}$$

with $\boldsymbol{S}_i = \frac{1}{2}\boldsymbol{\sigma}_i$ built from the Pauli matrices of Section 1.7, and h an external field which we shall usually set to zero. For $J > 0$ neighbouring spins prefer to be antiparallel and the model is antiferromagnetic; for $J < 0$ it is ferromagnetic.

Written out with $\hat{S}^\pm = \hat{S}^x \pm i\hat{S}^y$ the exchange term is

$$\boldsymbol{S}_i \cdot \boldsymbol{S}_j = \hat{S}_i^z \hat{S}_j^z + \frac{1}{2}\left(\hat{S}_i^+ \hat{S}_j^- + \hat{S}_i^- \hat{S}_j^+\right), \tag{4.27}$$

which is how one implements it. In the basis of $|\uparrow\downarrow\uparrow\cdots\rangle$ configurations the first term is diagonal, contributing $\pm J/4$ per bond according to whether the two spins are parallel or not, and the second – the flip-flop term – exchanges an antiparallel pair. It is worth pausing on what is *absent* here. Spin operators on different sites commute, so unlike every fermionic model in this chapter there is no Jordan-Wigner string and no sign to track: the matrix elements of Eq. (4.27) are $+J/2$ and nothing else. That single fact is why spin models are the natural first target for quantum simulation.

Symmetries.

The flip-flop term moves one unit of S^z from one site to another, so the total magnetisation $\hat{S}^z_{\text{tot}} = \sum_i \hat{S}^z_i$ commutes with $\hat{H}$ and the matrix is block diagonal in it; the ground state of the antiferromagnet lies in the $S^z_{\text{tot}} = 0$ sector, of dimension $\binom{N}{N/2}$. The full $SU(2)$ is a symmetry as well, $[\hat{H}, \boldsymbol{S}^2_{\text{tot}}] = 0$, which is the same structure met in the Lipkin model of Section 4.1 and in the S, S_z classification of the pairing model. On a ring, translations give a further decomposition into momentum sectors, and on a bipartite lattice the Marshall sign rule fixes the sign of every amplitude in the ground state, which is what makes the antiferromagnetic Heisenberg model free of the sign problem in quantum Monte Carlo.

Checking the mapping from the Hubbard model.

Section 4.4 predicted that the half-filled Hubbard ring at strong coupling becomes the Heisenberg antiferromagnet with $J = 4t^2/U$, and that the four-site ring should approach $-3J$: $-2J$ from the exchange and $-J$ from the constant $-\frac{1}{4}\hat{n}_i\hat{n}_j$ term on four bonds. We can now verify both halves separately. Diagonalising Eq. (4.26) directly on the four-site ring gives

$$E_0 = -2J, \qquad \langle \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} \rangle = -\tfrac{1}{2}, \qquad \langle \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+2} \rangle = +\tfrac{1}{4}, \tag{4.28}$$

the four nearest-neighbour bonds accounting for the whole energy. And the Hubbard energies of Table 4.5, with the constant $-J$ removed, converge on that number:

U/t	E_0^{Hubbard}/J	$E_0^{\text{Hubbard}}/J + 1$
8	-2.640470	-1.640470
16	-2.891457	-1.891457
32	-2.971283	-1.971283
64	-2.992713	-1.992713
128	-2.998171	-1.998171

The last column tends to -2, which is Eq. (4.28). The two models really are the same at low energy, and we have now checked the statement from both sides.

The Bethe ansatz and finite-size scaling.

The one-dimensional spin-$\frac{1}{2}$ chain is one of the very few interacting many-body systems solved exactly in the thermodynamic limit. Bethe's ansatz gives the ground-state energy per site of the infinite antiferromagnetic chain as

$$\frac{E_0}{NJ} \longrightarrow \frac{1}{4} - \ln 2 = -0.4431471806\ldots, \tag{4.29}$$

a number that no perturbative expansion in J produces, since J is the only scale in the problem. Table 4.6 compares it with exact diagonalisation of finite rings. The convergence is instructive: the finite chains lie *below* the Bethe value and the deviation falls off as $1/N^2$, so that even sixteen sites – a matrix of dimension 12 870 in the $S^z_{\text{tot}} = 0$ sector – is still 0.7% away. Extracting the thermodynamic limit from small systems is hard, and this is the cheapest illustration of why.

The right way to look at those numbers is Figure 4.8, which plots $E_0/(NJ)$ for $N = 4$ to 12 not against N but against $1/N^2$, with the Bethe value drawn as a horizontal line. On that axis

N	dimension	E_0/J	$E_0/(NJ)$	deviation
4	6	-2.00000000	-0.50000000	-0.056853
6	20	-2.80277564	-0.46712927	-0.023982
8	70	-3.65109341	-0.45638668	-0.013239
10	252	-4.51544635	-0.45154464	-0.008397
12	924	-5.38739092	-0.44894924	-0.005802
14	3432	-6.26354953	-0.44739640	-0.004249
16	12870	-7.14229636	-0.44639352	-0.003246
∞	—	—	-0.44314718	0

Table 4.6 The spin-$\frac{1}{2}$ antiferromagnetic Heisenberg ring, $J = 1$, $h = 0$, diagonalised in the $S^z_{\text{tot}} = 0$ sector. The last row is the Bethe-ansatz result (4.29) and the last column is $E_0/(NJ) - (\frac{1}{4} - \ln 2)$. Beyond twelve sites the matrix is stored sparsely and the lowest eigenvalue found by the Lanczos algorithm of Section 1.26.

the points lie very nearly on a straight line, and the whole content of the $1/N^2$ law is in that straightness. It converts a slowly converging sequence into something we can extrapolate: a straight line through the two largest rings, $N = 10$ and $N = 12$, hits the vertical axis at -0.44305, which is 9.7×10^{-5} from the exact $\frac{1}{4} - \ln 2 = -0.4431472$. The largest chain on its own is 5.8×10^{-3} away. Two points and a ruler have therefore gained about sixty times in accuracy over the raw number, and this is the pattern of every finite-size study in the book: one does not compute the thermodynamic limit, one computes a sequence and extrapolates it, and knowing the correct power of $1/N$ is worth more than another factor of ten in the dimension of the matrix.

The slope of that line is not arbitrary either. Fitting the three largest rings gives -0.857, and the two-point value is -0.849. Conformal field theory predicts $-\pi c v/6$ with central charge $c = 1$ and spin-wave velocity $v = \pi J/2$ for the antiferromagnetic chain, that is $-\pi^2/12 = -0.8225$; the residual discrepancy is the well-known logarithmic correction to scaling, which is what makes the points in the figure bend very slightly away from the straight line at small N. Rings of a dozen spins, diagonalised on a laptop, thus reproduce a prediction of two-dimensional conformal field theory to a few per cent, which is a rather good advertisement for exact diagonalisation of small systems.

Back to fermions.

Equation (4.26) is not of the form (4.1), but it can be put there. The Jordan-Wigner transformation writes a spin-$\frac{1}{2}$ chain as spinless fermions,

$$\hat{S}^+_j = c^\dagger_j \prod_{k<j} (1 - 2\hat{n}_k), \qquad \hat{S}^z_j = \hat{n}_j - \tfrac{1}{2}, \tag{4.30}$$

read backwards. The string of $(1 - 2\hat{n}_k) = (-1)^{\hat{n}_k}$ is the parity string (3.140) of Section 3.16, supplying exactly the sign that the bit-counting algorithm of Section 3.15 computes. On a chain with nearest-neighbour couplings the strings cancel between the two sites of a bond, and Eq. (4.26) becomes

$$\hat{H} = \frac{J}{2} \sum_j \left(c^\dagger_j c_{j+1} + \text{h.c.} \right) + J \sum_j \left(\hat{n}_j - \tfrac{1}{2} \right) \left(\hat{n}_{j+1} - \tfrac{1}{2} \right), \tag{4.31}$$

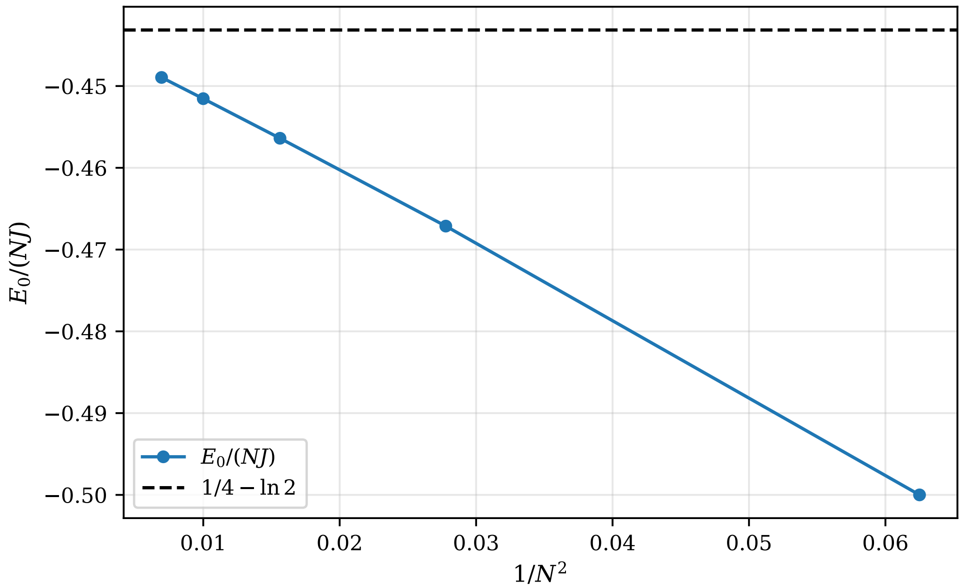

Fig. 4.8 Finite-size scaling of the spin-$\frac{1}{2}$ antiferromagnetic Heisenberg ring with $J = 1$ and $h = 0$. The ground-state energy per site $E_0/(NJ)$, obtained by exact diagonalisation in the $S^z_{\rm tot} = 0$ sector for $N = 4, 6, 8, 10$ and 12, is plotted against $1/N^2$; the dashed line is the Bethe-ansatz value $\frac{1}{4} - \ln 2 = -0.4431472$ for the infinite chain, which corresponds to the intercept at the left-hand edge of the plot. The points are close to collinear, so a linear extrapolation in $1/N^2$ is possible: through the two largest rings it gives -0.44305, in error by 1×10^{-4}, where the $N = 12$ ring alone is in error by 5.8×10^{-3}. The slope, -0.85, is close to the value $-\pi^2/12$ that conformal field theory predicts for a critical chain with central charge one.

a hopping term plus a nearest-neighbour density-density interaction: the Hubbard model of the previous section with the on-site U replaced by an inter-site one. Dropping the $\hat{S}^z\hat{S}^z$ piece leaves the XY model, which is free fermions and diagonal in momentum space. Everything in this book applies to Eq. (4.31) without modification.

Why this model matters. The Heisenberg model is the standard model of quantum magnetism, and the reference system for essentially every numerical method invented for correlated systems – the density-matrix renormalisation group was developed on it, and it is the benchmark for tensor networks, quantum Monte Carlo and neural-network wave functions alike. It is also the model closest to a quantum computer: written in Pauli matrices, Eq. (4.26) is already a sum of tensor products of one- and two-qubit operators, with none of the Jordan-Wigner overhead that fermionic Hamiltonians incur, so the Trotter circuits of Chapter 6 and the variational quantum eigensolver apply to it directly.

4.6 The Calogero-Sutherland model

The models above are all solvable because a symmetry cuts the Hilbert space down to a manageable size. The Calogero-Sutherland model [4, 5] is solvable for a different and much rarer reason: it is *integrable*, and its ground state is known in closed form for any number of particles.

The Calogero model describes N particles on a line in a harmonic trap, interacting through an inverse-square potential,

$$\hat{H} = -\frac{\hbar^2}{2m}\sum_{i=1}^{N}\frac{\partial^2}{\partial x_i^2} + \frac{m\omega^2}{2}\sum_{i=1}^{N} x_i^2 + \sum_{i<j}\frac{g}{(x_i - x_j)^2}. \tag{4.32}$$

It is conventional to parametrise the coupling as

$$g = \frac{\hbar^2}{m}\lambda(\lambda - 1), \tag{4.33}$$

so that λ measures the interaction strength. The Sutherland model is the same system on a ring of circumference L, with the inverse square replaced by the chord distance,

$$\hat{H} = -\frac{\hbar^2}{2m}\sum_{i}\frac{\partial^2}{\partial x_i^2} + \frac{\hbar^2}{m}\left(\frac{\pi}{L}\right)^2\sum_{i<j}\frac{\lambda(\lambda-1)}{\sin^2(\pi(x_i - x_j)/L)}. \tag{4.34}$$

The exact ground state.

In units $\hbar = m = 1$ the ground state of Eq. (4.32) is

$$\boxed{\Psi_0(x_1,\ldots,x_N) = \prod_{i<j}|x_i - x_j|^{\lambda}\,\exp\left(-\frac{\omega}{2}\sum_i x_i^2\right)} \tag{4.35}$$

with energy

$$E_0 = \omega\left[\frac{N}{2} + \frac{\lambda N(N-1)}{2}\right]. \tag{4.36}$$

Exercise 4.9 verifies this by direct substitution; the inverse-square form of the potential is precisely what is needed for the cross terms generated by the Laplacian to cancel.

The factor $\prod_{i<j}|x_i - x_j|^{\lambda}$ is a *Jastrow factor*, and it is the reason this model deserves a place in a book about many-body methods. It is a correlated wave function of exactly the type that variational Monte Carlo optimises numerically – but here it is exact, for every N and every λ. The model is therefore the sharpest available test of a Jastrow ansatz: any method that cannot reproduce Eq. (4.35) is missing something structural.

Two limits are instructive. At $\lambda = 0$ the coupling g vanishes and the particles are free bosons in a trap, $E_0 = \omega N/2$. At $\lambda = 1$ the coupling vanishes again, but now

$$\prod_{i<j}|x_i - x_j| \tag{4.37}$$

is the absolute value of the Vandermonde determinant of Section 1.20, which is precisely the Slater determinant of N free fermions in a harmonic trap. The Calogero model interpolates continuously between bosons and fermions, and for non-integer λ it describes particles obeying *fractional statistics*.

A numerical check.

For $N = 2$ the centre-of-mass and relative motions separate. With $x = x_1 - x_2$ the relative Hamiltonian is

$$\hat{h}_{\text{rel}} = -\frac{d^2}{dx^2} + \frac{\omega^2}{4}x^2 + \frac{\lambda(\lambda-1)}{x^2}, \tag{4.38}$$

whose exact ground-state energy is $\omega(\lambda + \frac{1}{2})$. Discretising on a grid and diagonalising with the machinery of Section 1.27 reproduces this to better than 10^{-5} for λ between 1 and 3, as `models.py` confirms. The inverse-square barrier keeps the wave function away from the origin, so the boundary condition $\Psi \to 0$ as $x \to 0$ is automatic for $\lambda > 0$.

Why this model matters. The Calogero-Sutherland model sits at the junction of several subjects. Its ground state is a Laughlin-type wave function, $\prod_{i<j}(z_i - z_j)^m e^{-\sum|z_i|^2/4}$, connecting it to the fractional quantum Hall effect; its excitation spectrum is that of free particles with fractional statistics; and the probability density $|\Psi_0|^2 \propto \prod_{i<j}|x_i - x_j|^{2\lambda} e^{-\omega\sum x_i^2}$ is exactly the eigenvalue distribution of a random matrix ensemble, with $\beta = 2\lambda$ the Dyson index. For our purposes it is above all a benchmark with a known answer for arbitrary N – something none of the other models in this chapter can offer.

4.7 The models on a quantum computer

The models of this chapter are small, exactly solvable and highly structured, which is precisely why they are the ones people run on quantum hardware. This section carries them across. The tool is the Jordan-Wigner transformation of Section 3.16; the goal is the Pauli form (3.138), $\hat{H} = \sum_\alpha h_\alpha P_\alpha$; and the lesson – which is the same lesson as the seniority – is that what one gets out depends on how much symmetry one is willing to exploit before starting.

The general two-body Hamiltonian.

Everything follows from four identities established in Section 3.16. A diagonal one-body term becomes a sum of I and Z by Eq. (3.143); an off-diagonal one-body term becomes the two strings of Eq. (3.145); a pair-creation term becomes the two strings of Eq. (3.147), differing from the previous case only in one sign; and a two-body term with four distinct indices becomes the eight weight-four strings of Eq. (3.148). The general Hamiltonian (4.1) is a sum of such terms, so its qubit image is a sum of $\mathscr{O}(M^4)$ Pauli strings on M qubits, one qubit per spin-orbital. Table 3.1 gave the counts.

That is the generic situation, and it is not encouraging: the parity strings reach across the whole register, so the longest Pauli strings have weight M. Every one of the models below does better than generic, and each does so for a reason we have already met.

The pairing model, one qubit per spin-orbital.

Order the spin-orbitals so that the two partners of a level are adjacent, $(1+, 1-, 2+, 2-, \dots)$, which keeps the parity strings inside a level empty. With L levels this needs $2L$ qubits. The one-body part and the diagonal $p = q$ part of the interaction are then products of I and Z alone,

$$\hat{H}_0 + \hat{V}_{\text{diag}} = \sum_p \left\{ \xi(p-1)\left[(I - Z_{p+}) + (I - Z_{p-})\right] - \frac{g}{8}(I - Z_{p+})(I - Z_{p-}) \right\}, \tag{4.39}$$

using $\hat{n}_{p\sigma} = (I - Z_{p\sigma})/2$ twice and $\hat{P}_p^+ \hat{P}_p^- = \hat{n}_{p+}\hat{n}_{p-}$. The pair-transfer terms with $p \neq q$ are the eight-string operators of Eq. (3.148), one set for each pair of levels. For $L = 2$, $\xi = 1$ and $g = 1$ the whole Hamiltonian is fifteen Pauli strings on four qubits,

$$\hat{H} = \tfrac{3}{4}\,IIII - \tfrac{3}{8}\,(IIZI + IIIZ) + \tfrac{1}{8}\,(ZIII + IZII) - \tfrac{1}{8}\,(ZZII + IIZZ) - \tfrac{1}{16}\Big(XXXX + XYXY + XYYX + YXXY + YXYX + YYYY - XXYY - YYXX\Big), \tag{4.40}$$

where the qubits are ordered $(1+,1-,2+,2-)$ and $XYXY$ means $X_{1+}Y_{1-}X_{2+}Y_{2-}$. Seven diagonal strings, eight of weight four. The program `jordanwigner.py` produces this decomposition by taking traces against all $4^4 = 256$ Pauli strings, so it is exact and not a hand calculation to be trusted.

The pairing model, one qubit per level.

We can do much better, and the reason is the seniority. In the seniority-zero space of Section 4.2 a level is either empty or holds a complete pair, so its state is one bit, not two. Make that precise with the local quasispin operators

$$\hat{S}^+_p = \hat{P}^+_p = a^\dagger_{p+}a^\dagger_{p-}, \qquad \hat{S}^-_p = \hat{P}^-_p = a_{p-}a_{p+}, \qquad \hat{S}^z_p = \tfrac{1}{2}\left(\hat{N}_p - 1\right), \tag{4.41}$$

with $\hat{N}_p = \hat{n}_{p+} + \hat{n}_{p-}$ the number of particles in level p. Exercise 4.9 shows from the anticommutation relations alone that these obey

$$\left[\hat{S}^z_p, \hat{S}^\pm_p\right] = \pm\hat{S}^\pm_p, \qquad \left[\hat{S}^+_p, \hat{S}^-_p\right] = 2\hat{S}^z_p, \qquad \left[\hat{S}^\alpha_p, \hat{S}^\beta_q\right] = 0 \quad (p \neq q), \tag{4.42}$$

so each level carries its own $su(2)$ and the levels are independent. This is the same algebra that organised the Lipkin model in Section 4.1, and it is why the two models look so alike on a quantum computer.

A spin one-half is a qubit. Identifying $|0\rangle_p$ with an empty level and $|1\rangle_p$ with an occupied one,

$$\hat{S}^+_p = \frac{X_p - iY_p}{2}, \qquad \hat{S}^-_p = \frac{X_p + iY_p}{2}, \qquad \hat{S}^z_p = -\frac{Z_p}{2}, \qquad \hat{N}_p = I - Z_p, \tag{4.43}$$

and the pairing Hamiltonian (4.19) collapses to

$$\boxed{\hat{H} = \sum_{p=1}^{L}\left[\xi(p-1) - \frac{g}{4}\right](I - Z_p) - \frac{g}{4}\sum_{p<q}(X_pX_q + Y_pY_q)} \tag{4.44}$$

on L qubits. The derivation is two lines: the $p = q$ part of the interaction is $-\frac{g}{2}\sum_p \hat{S}^+_p\hat{S}^-_p = -\frac{g}{4}\sum_p(I - Z_p)$, which merges with the one-body term, and the $p \neq q$ part uses

$$\hat{S}^+_p\hat{S}^-_q + \hat{S}^+_q\hat{S}^-_p = \frac{1}{2}(X_pX_q + Y_pY_q)\,. \tag{4.45}$$

For two levels with $\xi = 1$ and $g = 1$ this is five strings on two qubits,

$$\hat{H}^{(L=2)} = \tfrac{1}{2}\,II + \tfrac{1}{4}\,ZI - \tfrac{3}{4}\,IZ - \tfrac{1}{4}\,(XX + YY)\,, \tag{4.46}$$

against fifteen strings on four qubits in Eq. (4.40). Equation (4.45) also identifies what kind of spin model the pairing Hamiltonian is: an XY model with infinite-range couplings, in an inhomogeneous field. Compare Eq. (4.26) – the XY model is the Heisenberg model with the $\hat{S}^z\hat{S}^z$ term removed, so the pairing model and the antiferromagnet are close relatives once both are on qubits.

The compact form is a *restriction*, not an approximation. Its spectrum is exactly the seniority-zero part of the full spectrum at fixed particle number, level by level, and `jordanwigner.py` verifies this for $L = 2$ and $L = 3$ and every pair number. Nothing is lost as long as the physics stays in the seniority-zero space.

What the particle-hole term costs.

And that is exactly what the particle-hole term of Section 4.3 destroys. Once $f \neq 0$ the seniority-zero space is no longer invariant – Figure 4.3(c) and (d) are the states that leak out – so the one-qubit-per-level encoding is simply not available. One has to go back to $2L$ qubits. The leakage is easy to quantify: the largest matrix element connecting the seniority-zero block to the rest is $f/2$, exactly the $1p$–$1h$ matrix element of Table 4.2, and it is zero if and only if $f = 0$.

The bill is presented in Table 4.7. Adding the particle-hole term to $L = 3$ takes the Hamiltonian from 34 Pauli strings of weight at most four to 118 strings of weight up to six, on the same six qubits, while the compact encoding that would have needed three qubits and ten strings is no longer legitimate at all. Every one of those numbers is a direct cost on hardware: strings are measurements in a variational calculation, and weights are entangling gates in a Trotter step.

Hamiltonian	**Encoding**	**Qubits**	**Pauli strings**	**Largest weight**
Pairing, $L = 2$	one per level	2	5	2
Pairing, $L = 2$	one per orbital	4	15	4
Pairing + p-h, $L = 2$	one per orbital	4	27	4
Pairing, $L = 3$	one per level	3	10	2
Pairing, $L = 3$	one per orbital	6	34	4
Pairing + p-h, $L = 3$	one per orbital	6	118	6

Table 4.7 The cost of putting the pairing model on a quantum computer. The compact encoding halves the number of qubits and caps the Pauli weight at two, but it requires the seniority to be conserved. The particle-hole term breaks that conservation, so the middle option becomes the only option and the count grows sharply. Produced by `jordanwigner.py`.

The other models.

The same reading applies to the rest of the chapter. The Lipkin model of Section 4.1 is already written in quasispin operators, so it maps onto a small number of qubits directly, and the compressed encoding that uses $\lceil \log_2(2J+1) \rceil$ qubits for one J block is the analogue of the pair-qubit trick here. The Hubbard model of Section 4.4 needs one qubit per site and spin, and its hopping terms are the two strings of Eq. (3.145), which is why orbital ordering matters so much for lattice models: a hop between distant sites in the chosen ordering drags a long parity chain behind it. The Heisenberg model of Section 4.5 is the easiest of all, because Eq. (4.26) is *already* a sum of Pauli strings of weight two – there is no transformation to perform, and no parity strings to pay for. That is the point made in passing in Section 4.5, now with the cost of the alternative in view.

Quantum-computing connection. All of this is preparation for the variational quantum eigensolver. There one prepares a parametrised state $|\Psi(\theta)\rangle$ on the register, measures $\langle\Psi(\theta)|\hat{H}|\Psi(\theta)\rangle$ by measuring each Pauli string of the decomposition separately, and minimises over θ classically. The unitary coupled-cluster ansatz of Section 10.14 supplies the state, the Trotter expansion of Section 6.9 turns it into a circuit, and Eq. (4.44) supplies the observable. The pairing model with $L = 2$ is then a genuine two-qubit calculation which can be run on hardware; with the particle-hole term switched on it is a four-qubit calculation, which is a rather different proposition.

4.8 What the models have in common

Table 4.8 collects them. The pattern is worth stating explicitly, because it is the pattern of the whole subject.

Model	**One-body space**	**Interaction**	**Symmetry used**	**Exactly solvable**
Lipkin	two levels, degeneracy Ω	pair transfer, exchange	$SU(2)$ quasispin	yes, block by block
Pairing	L doubly degenerate levels	pair transfer	S, S_z, seniority	yes, small L
Pairing + p-h	L doubly degenerate levels	+ pair breaking	S_z only	yes, small L
Hubbard	lattice sites	on-site U	$N_\uparrow$, $N_\downarrow$, momentum	1D only
Heisenberg	spins on a lattice	exchange $\boldsymbol{S}_i \cdot \boldsymbol{S}_j$	S^z_{tot}, $SU(2)$	1D only
Calogero	continuum, N particles	inverse square	integrability	yes, any N

Table 4.8 The models of this chapter. Each is the general Hamiltonian (4.1) with a drastically restricted set of two-body matrix elements – the Heisenberg model after the Jordan-Wigner transformation (4.30) – and in each case a symmetry reduces the dimension of the problem. The third row is the second row with one term added, and the price of that term is visible in the fourth column: the seniority is gone, and with it the reduction from thirty-six dimensions to six.

In every case the route is the same. Write the Hamiltonian in second quantisation; identify the operators that commute with it; use them to block diagonalise; and solve the blocks. The first step is Chapter 3, the second is a commutator calculation of the kind Wick's theorem makes routine, the third is the statement that a symmetry gives a basis in which the matrix is block diagonal (Section 1.11), and the fourth is the eigenvalue problem of Sections 1.24 and 1.26.

What separates these models from a realistic system is only the last step of the reduction. In a real nucleus or a real molecule the symmetries reduce the dimension by orders of magnitude but not to five; the matrix is sparse but enormous; and one must fall back on the approximate methods that occupy the rest of this book. The value of the models in this chapter is that they let us watch those approximations succeed and fail against an answer we already know.

4.9 Exercises

Exercise 1: the Lipkin model in the quasispin basis

a) Show that the quasispin operators (4.6) obey the angular-momentum algebra (4.7), using only the anticommutation relations of Chapter 3.
b) Show that the Lipkin Hamiltonian (4.2) can be rewritten as Eq. (4.8).
c) Show that $[\hat{H},\hat{J}^2]=0$ and $[\hat{H},\hat{N}]=0$, and explain what this implies for the structure of the Hamiltonian matrix.
d) Construct the five states $|2,J_z\rangle$ for four particles, and compute the matrix (4.11). Diagonalise it and compare with Eq. (4.12).

Answer to a).

Insert the definitions and use $\{a_l,a_k\}=\{a_l^\dagger,a_k^\dagger\}=0$ and $\{a_l^\dagger,a_k\}=\delta_{lk}$. For the first commutator,

$$[\hat{J}_z,\hat{J}_\pm]=\frac{1}{2}\sum_{pp'\sigma}\sigma\left(a^\dagger_{p\sigma}a_{p\sigma}a^\dagger_{p'\pm}a_{p'\mp}-a^\dagger_{p'\pm}a_{p'\mp}a^\dagger_{p\sigma}a_{p\sigma}\right). \tag{4.47}$$

Anticommute the operators of the second product into the order of the first. Using $a_{p'\mp}a^\dagger_{p\sigma}=\delta_{pp'}\delta_{\mp\sigma}-a^\dagger_{p\sigma}a_{p'\mp}$ and then $a_{p\sigma}a^\dagger_{p'\pm}=\delta_{pp'}\delta_{\pm\sigma}-a^\dagger_{p'\pm}a_{p\sigma}$, the four-operator terms cancel in pairs and what survives is

$$[\hat{J}_z,\hat{J}_\pm]=\frac{1}{2}\sum_{p\sigma}\sigma\left(\delta_{\pm\sigma}a^\dagger_{p\sigma}a_{p\mp}-\delta_{\mp\sigma}a^\dagger_{p\pm}a_{p\sigma}\right)=\pm\frac{1}{2}\sum_{p}\left(a^\dagger_{p\pm}a_{p\mp}+a^\dagger_{p\pm}a_{p\mp}\right)=\pm\hat{J}_\pm, \tag{4.48}$$

where the two Kronecker deltas each pick out one value of σ, and the factor $\sigma=\pm1$ supplies the sign. The same manipulation gives $[\hat{J}_+,\hat{J}_-]=2\hat{J}_z$, and $[\hat{J}^2,\hat{J}_\pm]=[\hat{J}^2,\hat{J}_z]=0$ then follows from the definition $\hat{J}^2=\hat{J}_+\hat{J}_-+\hat{J}_z^2-\hat{J}_z$ exactly as for ordinary angular momentum.

Answer to b).

The one-body part is immediate: comparing Eqs. (4.3) and (4.6) gives $\hat{H}_0=\varepsilon\hat{J}_z$.

For $\hat{H}_1$, first anticommute the two annihilation operators, $a_{p'-\sigma}a_{p-\sigma}=-a_{p-\sigma}a_{p'-\sigma}$, and then move $a_{p-\sigma}$ past $a^\dagger_{p'\sigma}$ using $a_{p-\sigma}a^\dagger_{p'\sigma}=\delta_{pp'}\delta_{-\sigma\sigma}-a^\dagger_{p'\sigma}a_{p-\sigma}$. Since $\delta_{-\sigma\sigma}=0$, the delta term drops out and

$$\hat{H}_1=\frac{1}{2}V\sum_{pp'\sigma}a^\dagger_{p\sigma}a_{p-\sigma}\,a^\dagger_{p'\sigma}a_{p'-\sigma}. \tag{4.49}$$

The two factors are now independent sums over p and p'. Writing out $\sigma=\pm$ separately,

$$\hat{H}_1=\frac{1}{2}V\left[\left(\sum_p a^\dagger_{p+}a_{p-}\right)\left(\sum_{p'}a^\dagger_{p'+}a_{p'-}\right)+\left(\sum_p a^\dagger_{p-}a_{p+}\right)\left(\sum_{p'}a^\dagger_{p'-}a_{p'+}\right)\right]=\frac{1}{2}V\left(\hat{J}_+^2+\hat{J}_-^2\right). \tag{4.50}$$

For $\hat{H}_2$ the same two steps give

$$\hat{H}_2=\frac{1}{2}W\left(-\sum_{p\sigma}a^\dagger_{p\sigma}a_{p\sigma}+\sum_{pp'\sigma}a^\dagger_{p\sigma}a_{p-\sigma}a^\dagger_{p'-\sigma}a_{p'\sigma}\right), \tag{4.51}$$

where this time $\delta_{-\sigma-\sigma} = 1$ and the delta term survives as $-\hat{N}$. Splitting the remaining sum over σ as before identifies it as $\hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+$, so $\hat{H}_2 = \frac{1}{2}W(-\hat{N} + \hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+)$. Adding the three pieces gives Eq. (4.8).

Answer to c).

Written as in Eq. (4.8), $\hat{H}$ is built entirely from $\hat{J}_z$, $\hat{J}_\pm$ and $\hat{N}$. Since $\hat{J}^2$ commutes with each of $\hat{J}_z$ and $\hat{J}_\pm$ by part a), and with $\hat{N}$ trivially, it commutes with $\hat{H}$. For $\hat{N}$: the one-body term conserves particle number, and every term of the interaction contains two creation and two annihilation operators, so $[\hat{H}, \hat{N}] = 0$.

Both J and N are therefore good quantum numbers. In a basis $|J, J_z\rangle$ the Hamiltonian matrix is block diagonal, one block for each J, of dimension $2J+1$. Within a block, $\hat{J}_z$ is diagonal and $\hat{J}_\pm^2$ connects J_z to $J_z \pm 2$, so the block splits once more into even and odd J_z. A problem of nominal dimension $\binom{2\Omega}{N}$ has been reduced to a set of matrices of dimension at most $N/2+1$.

Answer to d).

Start from $|2,-2\rangle$, Eq. (4.9), and apply Eq. (4.10). With $J = 2$ and $J_z = -2$ the factor is $\sqrt{6-2} = 2$, so

$$|2,-1\rangle = \frac{1}{2}\hat{J}_+|2,-2\rangle = \frac{1}{2}\Big(a^\dagger_{1+}a^\dagger_{2-}a^\dagger_{3-}a^\dagger_{4-} + a^\dagger_{1-}a^\dagger_{2+}a^\dagger_{3-}a^\dagger_{4-} + a^\dagger_{1-}a^\dagger_{2-}a^\dagger_{3+}a^\dagger_{4-} + a^\dagger_{1-}a^\dagger_{2-}a^\dagger_{3-}a^\dagger_{4+}\Big)|0\rangle, \tag{4.52}$$

one term for each substate that can be flipped. Applying $\hat{J}_+$ again with the factor $\sqrt{6-0} = \sqrt{6}$ gives $|2,0\rangle$ as the normalised sum of the six ways of flipping two of the four substates, and so on up to $|2,2\rangle$ with all four spins up.

The matrix elements follow from Eq. (4.8). The diagonal is

$$\langle 2, J_z|\hat{H}|2, J_z\rangle = \varepsilon J_z + \frac{W}{2}\Big(-N + 2\big[J(J+1) - J_z^2\big]\Big), \tag{4.53}$$

since $\hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+ = 2(\hat{J}^2 - \hat{J}_z^2)$. The off-diagonal elements come from $\frac{1}{2}V(\hat{J}_+^2 + \hat{J}_-^2)$,

$$\langle 2, J_z+2|\hat{H}|2, J_z\rangle = \frac{V}{2}\sqrt{J(J+1) - J_z(J_z+1)}\,\sqrt{J(J+1) - (J_z+1)(J_z+2)}. \tag{4.54}$$

With $\varepsilon = 2$, $V = -1/3$, $W = -1/4$ and $N = 4$ this gives Eq. (4.11); for instance the $(-2,0)$ element is $\frac{1}{2}V\sqrt{4}\sqrt{6} = -\sqrt{6}/3$, and the $J_z = -1$ diagonal element is $-2 + \frac{1}{2}(-\frac{1}{4})(-4 + 2[6-1]) = -2 - \frac{3}{4} = -\frac{11}{4}$. Diagonalising reproduces Eq. (4.12).

Exercise 2: the pairing model

a) Show that $\hat{H}_0$ and $\hat{V}$ of Eq. (4.16) commute with $\hat{S}_z$ and $\hat{S}^2$ of Eq. (4.17).
b) Show that the Hamiltonian can be written in the pair form (4.19), and that the pair creation operators commute among themselves while $(\hat{P}_p^+)^2 = 0$.
c) Construct the Hamiltonian matrix for four particles in the four lowest levels, with no broken pairs and $S = 0$, and find its eigenvalues as a function of g.

Answer to a).

$\hat{H}_0$ is a sum of number operators $a^\dagger_{p\sigma}a_{p\sigma}$, and $\hat{S}_z$ is a weighted sum of the same operators, so the two commute term by term. For $\hat{V}$, note that $a^\dagger_{p+}a^\dagger_{p-}$ creates one particle with $\sigma=+1$ and one with $\sigma=-1$, changing S_z by $\frac{1}{2}(+1)+\frac{1}{2}(-1)=0$; likewise $a_{q-}a_{q+}$ destroys such a pair. Hence $[\hat{V},\hat{S}_z]=0$.

For $\hat{S}^2$ it is enough to show $[\hat{V},\hat{S}_\pm]=0$. The operator $\hat{S}_+=\sum_p a^\dagger_{p+}a_{p-}$ acting on a fully paired level gives zero, because the state $|p+\rangle$ is already occupied and $(a^\dagger_{p+})^2=0$; acting on an empty level it also gives zero. Since $\hat{V}$ maps the fully paired space into itself, $\hat{S}_+\hat{V}$ and $\hat{V}\hat{S}_+$ both annihilate it, and the commutator vanishes. Both S and S_z are therefore good quantum numbers and the Hamiltonian is block diagonal in them.

Answer to b).

Insert the definitions (4.18) into Eq. (4.16):

$$-\frac{1}{2}g\sum_{pq}a^\dagger_{p+}a^\dagger_{p-}a_{q-}a_{q+}=-\frac{1}{2}g\sum_{pq}\hat{P}^+_p\hat{P}^-_q, \tag{4.55}$$

which is Eq. (4.19).

For the commutator, $\hat{P}^+_p\hat{P}^+_q=a^\dagger_{p+}a^\dagger_{p-}a^\dagger_{q+}a^\dagger_{q-}$. Moving $a^\dagger_{q+}a^\dagger_{q-}$ to the front requires four interchanges of neighbouring creation operators, each contributing a minus sign, so the total sign is $(-1)^4=+1$ and $[\hat{P}^+_p,\hat{P}^+_q]=0$. The pair operators therefore behave like bosons under exchange.

They are not bosons, however. Since $(a^\dagger_{p+})^2=0$ by the Pauli principle,

$$\left(\hat{P}^+_p\right)^2=a^\dagger_{p+}a^\dagger_{p-}a^\dagger_{p+}a^\dagger_{p-}=-\left(a^\dagger_{p+}\right)^2\left(a^\dagger_{p-}\right)^2=0. \tag{4.56}$$

A level can hold one pair but not two. This is the hard-core constraint that distinguishes the pairing model from a gas of free bosons, and it is what makes the model non-trivial.

Answer to c).

With four levels and two pairs there are $\binom{4}{2}=6$ basis states, which we label by the occupied levels: (12), (13), (14), (23), (24), (34). The diagonal element of a configuration is $2\sum_{p\in\mathrm{occ}}(p-1)-g$, the last term coming from the $p=q$ terms of the interaction with two pairs present. Every pair of configurations differing by the position of one pair is connected by $-g/2$; configurations differing in both pairs are not connected. Ordering the basis as above,

$$\boldsymbol{H}=\begin{bmatrix}2-g & -g/2 & -g/2 & -g/2 & -g/2 & 0\\ -g/2 & 4-g & -g/2 & -g/2 & 0 & -g/2\\ -g/2 & -g/2 & 6-g & 0 & -g/2 & -g/2\\ -g/2 & -g/2 & 0 & 6-g & -g/2 & -g/2\\ -g/2 & 0 & -g/2 & -g/2 & 8-g & -g/2\\ 0 & -g/2 & -g/2 & -g/2 & -g/2 & 10-g\end{bmatrix}. \tag{4.57}$$

Diagonalising for a range of g gives Table 4.1. Note that the (13) and (23) configurations are degenerate at $g=0$ – both have unperturbed energy 6 – and that the correlation energy is negative for both signs of g, since the interaction can always lower the energy by mixing configurations.

Exercise 3: the Hubbard model

a) Show that the Hubbard Hamiltonian (4.22) conserves $N_\uparrow$ and $N_\downarrow$ separately, and hence S_z.
b) Diagonalise the $U = 0$ Hubbard ring analytically and verify the entry $E_0 = -4t$ of Table 4.4.
c) Derive the effective spin Hamiltonian in the limit $U \gg t$ at half filling, and predict the ratio E_0/J for the four-site ring.

Answer to a).

The hopping term contains $c^\dagger_{i\sigma} c_{j\sigma}$ with the *same* σ on both operators, so it conserves the number of particles of each spin separately. The interaction is a product of number operators and conserves everything. Hence $[\hat{H}, \hat{N}_\uparrow] = [\hat{H}, \hat{N}_\downarrow] = 0$, and since $\hat{S}_z = \frac{1}{2}(\hat{N}_\uparrow - \hat{N}_\downarrow)$, also $[\hat{H}, \hat{S}_z] = 0$. This is what allows the diagonalisation to be carried out in a fixed sector of dimension (4.23).

Answer to b).

At $U = 0$ the Hamiltonian is a sum of independent one-body problems. On a ring of N sites with periodic boundary conditions the hopping matrix is diagonalised by plane waves $c^\dagger_{k\sigma} = N^{-1/2} \sum_j e^{ikj} c^\dagger_{j\sigma}$ with $k = 2\pi n/N$, giving

$$\varepsilon_k = -2t \cos k. \tag{4.58}$$

For $N = 4$ the allowed momenta are $k = 0, \pm\pi/2, \pi$ with energies $-2t$, 0, 0, $+2t$. Filling the two lowest for each spin gives $-2t + 0 = -2t$ per spin species, hence $E_0 = -4t$, which is the first line of Table 4.4. The degeneracy at $k = \pm\pi/2$ means the $U = 0$ ground state is not unique; switching on U lifts the degeneracy.

Answer to c).

At half filling and $U \gg t$ every site carries exactly one electron in the ground state, and states with a doubly occupied site lie an energy U higher. Treat the hopping as a perturbation. A single hop creates a doubly occupied site and an empty one, costing U; hopping back restores a singly occupied configuration. Second-order perturbation theory therefore gives an energy gain $-2t^2/U$ for each such virtual excursion, and the process is possible only if the two spins involved are antiparallel, since the Pauli principle forbids two electrons of the same spin on one site.

Collecting the terms gives the Heisenberg Hamiltonian

$$\hat{H}_{\text{eff}} = J \sum_{\langle ij \rangle} \left(\boldsymbol{S}_i \cdot \boldsymbol{S}_j - \frac{\hat{n}_i \hat{n}_j}{4} \right), \qquad J = \frac{4t^2}{U}, \tag{4.59}$$

which is the half-filled limit of the t-J model, Eq. (4.25). For the four-site ring the Heisenberg ground-state energy is $-2J$, and the constant $-\frac{1}{4} n_i n_j$ contributes $-\frac{1}{4}$ per bond on four bonds, that is $-J$. Hence

$$E_0 \longrightarrow -3J = -\frac{12t^2}{U}, \qquad \text{so} \qquad \frac{E_0}{J} \longrightarrow -3, \tag{4.60}$$

which is what Table 4.5 confirms numerically.

Exercise 4: the Calogero ground state

a) Verify by direct substitution that Eq. (4.35) is an eigenstate of Eq. (4.32) with energy (4.36).
b) Show that at $\lambda = 1$ the ground state is the Slater determinant of N free fermions in a harmonic trap.
c) Separate the $N = 2$ problem and verify the relative energy $\omega(\lambda + \frac{1}{2})$.

Answer to a).

Work in units $\hbar = m = 1$ and write $\Psi_0 = e^{-\Phi}$ with

$$\Phi = \frac{\omega}{2}\sum_i x_i^2 - \lambda \sum_{i<j} \ln|x_i - x_j|. \tag{4.61}$$

For any such form,

$$-\frac{1}{2}\sum_i \frac{\partial^2 \Psi_0}{\partial x_i^2} = \frac{1}{2}\left[\sum_i \frac{\partial^2 \Phi}{\partial x_i^2} - \sum_i \left(\frac{\partial \Phi}{\partial x_i}\right)^2\right]\Psi_0. \tag{4.62}$$

The derivatives are

$$\frac{\partial \Phi}{\partial x_i} = \omega x_i - \lambda \sum_{j\neq i} \frac{1}{x_i - x_j}, \qquad \frac{\partial^2 \Phi}{\partial x_i^2} = \omega + \lambda \sum_{j\neq i} \frac{1}{(x_i - x_j)^2}. \tag{4.63}$$

Squaring the first,

$$\sum_i \left(\frac{\partial \Phi}{\partial x_i}\right)^2 = \omega^2 \sum_i x_i^2 - 2\lambda\omega \sum_i \sum_{j\neq i} \frac{x_i}{x_i - x_j} + \lambda^2 \sum_i \sum_{j\neq i} \sum_{k\neq i} \frac{1}{(x_i - x_j)(x_i - x_k)}. \tag{4.64}$$

Two simplifications now do all the work. In the second term, pairing (i,j) with (j,i) gives $x_i/(x_i - x_j) + x_j/(x_j - x_i) = 1$, so the double sum equals $N(N-1)/2$ times two, that is $-2\lambda\omega \cdot N(N-1)/2$. In the third term the pieces with $j = k$ give $\lambda^2 \sum_{i\neq j}(x_i - x_j)^{-2}$, while the pieces with $j \neq k$ cancel identically by the three-term identity

$$\frac{1}{(x_i - x_j)(x_i - x_k)} + \frac{1}{(x_j - x_i)(x_j - x_k)} + \frac{1}{(x_k - x_i)(x_k - x_j)} = 0. \tag{4.65}$$

Assembling everything, the inverse-square terms combine as

$$\frac{1}{2}\left[\lambda \sum_{i\neq j} \frac{1}{(x_i - x_j)^2} - \lambda^2 \sum_{i\neq j} \frac{1}{(x_i - x_j)^2}\right] = -\lambda(\lambda - 1)\sum_{i<j} \frac{1}{(x_i - x_j)^2}, \tag{4.66}$$

which cancels the interaction term of the Hamiltonian exactly; the $\omega^2 x_i^2$ terms cancel the trap; and what is left is the constant

$$E_0 = \frac{1}{2}\left[N\omega + \lambda\omega N(N-1)\right] = \omega\left[\frac{N}{2} + \frac{\lambda N(N-1)}{2}\right], \tag{4.67}$$

which is Eq. (4.36). The cancellation in Eq. (4.66) works only because the interaction falls off as the inverse *square*; this is the sense in which the model is fine-tuned to be solvable.

Answer to b).

At $\lambda = 1$ the coupling $g = \lambda(\lambda - 1)$ vanishes and the particles are free, but the wave function

$$\Psi_0 = \prod_{i<j} |x_i - x_j| \, e^{-\omega \sum_i x_i^2/2} \tag{4.68}$$

is not the bosonic ground state. Up to an overall sign it is the Vandermonde determinant of Section 1.20,

$$\prod_{i<j}(x_j - x_i) = \det\left[x_i^{j-1}\right], \tag{4.69}$$

and multiplying each column by the Gaussian and taking suitable linear combinations turns $x_i^{j-1} e^{-\omega x_i^2/2}$ into the Hermite functions of Section 1.27. The result is precisely the Slater determinant of the N lowest harmonic oscillator orbitals. The energy check is immediate: Eq. (4.36) with $\lambda = 1$ gives $E_0 = \omega N^2/2 = \omega \sum_{n=0}^{N-1}(n + \frac{1}{2})$, the sum of the N lowest oscillator energies.

For non-integer λ the wave function is neither symmetric nor antisymmetric, and the particles obey fractional statistics interpolating between the two.

Answer to c).

Introduce $X = (x_1 + x_2)/2$ and $x = x_1 - x_2$. The kinetic energy separates as $-\frac{1}{4}\partial_X^2 - \partial_x^2$ and the trap as $\omega^2 X^2 + \frac{1}{4}\omega^2 x^2$, so with $\hbar = m = 1$

$$\hat{H} = \left[-\frac{1}{4}\frac{\partial^2}{\partial X^2} + \omega^2 X^2\right] + \left[-\frac{\partial^2}{\partial x^2} + \frac{\omega^2}{4}x^2 + \frac{\lambda(\lambda - 1)}{x^2}\right]. \tag{4.70}$$

The centre-of-mass part is a harmonic oscillator with ground-state energy $\omega/2$. The relative part is Eq. (4.38); its ground state is $x^\lambda e^{-\omega x^2/4}$ with energy $\omega(\lambda + \frac{1}{2})$, as substitution confirms. The total is $\omega/2 + \omega(\lambda + \frac{1}{2}) = \omega(1 + \lambda)$, which agrees with Eq. (4.36) for $N = 2$. Solving the relative problem numerically on a grid, as in Section 1.27, reproduces $\omega(\lambda + \frac{1}{2})$ to better than 10^{-5}; see `models.py`.

Exercise 5: the particle-hole term

a) Show that $\hat{V}_{\mathrm{ph}}$ of Eq. (4.21) conserves N_+ and N_- separately, and hence that S_z remains a good quantum number.
b) Isolate the $q = r$ part of the sum and show that, together with its Hermitian conjugate, it is the pairing interaction with g replaced by $2f$. Deduce the diagonal matrix element of $\hat{H}$ in the reference determinant of Figure 4.3(a) for $L = 4$, $N = 4$.
c) Let $|c\rangle$ be that reference determinant and let $|\Phi_1\rangle$ be the one-particle-one-hole determinant of Figure 4.3(c). Show that $\langle\Phi_1|\hat{V}_{\mathrm{pair}}|c\rangle = 0$ and evaluate $\langle\Phi_1|\hat{V}_{\mathrm{ph}}|c\rangle$.
d) Show that the seniority is not conserved once $f \neq 0$, by exhibiting a state of seniority two that $\hat{H}$ reaches from $|c\rangle$.

Answer to a).

Every term of $\hat{V}_{\rm ph}$ contains $a^\dagger_{p+}a^\dagger_{p-}a_{q-}a_{r+}$: one spin-up operator is created and one destroyed, and likewise for spin down. So $[\hat{V}_{\rm ph},\hat{N}_+] = [\hat{V}_{\rm ph},\hat{N}_-] = 0$, and since $S_z = (\hat{N}_+ - \hat{N}_-)/2$ it commutes with S_z as well. Note that this is weaker than the pairing case: there $\hat{V}_{\rm pair}$ commutes with $\hat{S}^2$ too, and with the seniority operator; here neither holds.

Answer to b).

Setting $q = r$ in Eq. (4.21) gives

$$-\frac{f}{2}\sum_{pq} a^\dagger_{p+}a^\dagger_{p-}a_{q-}a_{q+},$$

which is $\hat{V}_{\rm pair}$ with $g \to f$. Its Hermitian conjugate is $-\frac{f}{2}\sum_{pq} a^\dagger_{q+}a^\dagger_{q-}a_{p-}a_{p+}$, and relabelling $p \leftrightarrow q$ shows it to be the same sum again. The two together therefore contribute $\hat{V}_{\rm pair}$ with $g \to 2f$, so in every seniority-conserving matrix element g is replaced by $g+2f$. For $L = 4$, $N = 4$ the reference has complete pairs in levels 1 and 2, so

$$\langle c|\hat{H}|c\rangle = \underbrace{2\xi(0) + 2\xi(1)}_{\hat{H}_0} - \frac{g+2f}{2} \times 2 = 2\xi - (g+2f),$$

which is 0.9 for $\xi = 1$, $g = 1$, $f = 0.05$ – the first entry of Table 4.3.

Answer to c).

$|\Phi_1\rangle$ differs from $|c\rangle$ in a single spin-orbital: the spin-up particle of level 2 has moved to level 3. A two-body operator can connect determinants differing by one orbital only through its "$\hat{n}$-contracted" pieces, and $\hat{V}_{\rm pair}$ has none: every term $a^\dagger_{p+}a^\dagger_{p-}a_{q-}a_{q+}$ changes the occupation of level q by two and of level p by two, so the two determinants must differ by an even number of orbitals in *each* spin channel. Hence $\langle\Phi_1|\hat{V}_{\rm pair}|c\rangle = 0$, and the particle-hole channel of the pure pairing model is empty.

For $\hat{V}_{\rm ph}$ take the Hermitian-conjugate term with $r = 3$, $p = q = 2$,

$$-\frac{f}{2}a^\dagger_{3+}a^\dagger_{2-}a_{2-}a_{2+} = -\frac{f}{2}a^\dagger_{3+}\hat{n}_{2-}a_{2+},$$

which acting on $|c\rangle$ gives exactly $|\Phi_1\rangle$, since level 2 is full. Collecting the signs from the anticommutations, $\langle\Phi_1|\hat{V}_{\rm ph}|c\rangle = \pm f/2$, the sign depending on the ordering convention for the spin-orbitals. This is the entry $\pm f/2$ of Table 4.2, and `models.py` confirms that all eight one-particle-one-hole determinants have matrix elements of this magnitude.

Answer to d).

Take $p = 3$, $q = 2$, $r = 1$ in Eq. (4.21): $a^\dagger_{3+}a^\dagger_{3-}a_{2-}a_{1+}$ removes the spin-down particle from level 2 and the spin-up particle from level 1 and deposits a complete pair in level 3. The result is Figure 4.3(d): levels 1 and 2 each retain a single, unpaired particle, so the seniority is two, while $|c\rangle$ had seniority zero. A single application of $\hat{H}$ therefore changes the seniority, and the seniority-zero space of Section 4.2 is no longer invariant.

Exercise 6: the Heisenberg model

a) Verify Eq. (4.27) from the definitions $\hat{S}^{\pm} = \hat{S}^x \pm i\hat{S}^y$, and show that $[\hat{H}, \hat{S}^z_{\rm tot}] = 0$.
b) Solve the two-site problem exactly, and identify the singlet and the triplet.
c) For the four-site ring, use the value $E_0 = -2J$ quoted in Eq. (4.28) together with translational invariance to deduce $\langle \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} \rangle$, and then obtain $\langle \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+2} \rangle$ from the fact that the ground state is a total spin singlet.
d) Verify the Jordan-Wigner transformation (4.30): show that the strings cancel for a nearest-neighbour bond and that Eq. (4.31) follows.

Answer to a).

$\hat{S}^+_i \hat{S}^-_j + \hat{S}^-_i \hat{S}^+_j = 2(\hat{S}^x_i \hat{S}^x_j + \hat{S}^y_i \hat{S}^y_j)$, since the cross terms $i(\hat{S}^y_i \hat{S}^x_j - \hat{S}^x_i \hat{S}^y_j)$ cancel between the two products. Adding $\hat{S}^z_i \hat{S}^z_j$ gives Eq. (4.27). For the commutator, note that $\hat{S}^z_i \hat{S}^z_j$ is diagonal and that $\hat{S}^+_i \hat{S}^-_j$ raises S^z on site i and lowers it on site j, leaving the sum unchanged.

Answer to b).

With one bond, $\hat{H} = J\boldsymbol{S}_1 \cdot \boldsymbol{S}_2 = \frac{J}{2}\left(\boldsymbol{S}^2_{\rm tot} - \boldsymbol{S}^2_1 - \boldsymbol{S}^2_2\right) = \frac{J}{2}\left(S(S+1) - \frac{3}{2}\right)$. The singlet $S = 0$ has $E = -3J/4$ and the triplet $S = 1$ has $E = +J/4$: the familiar splitting, and the reason a bond wants to be a singlet. The gap is J.

Answer to c).

Translational invariance makes all four nearest-neighbour bonds equivalent, so $E_0 = 4J\langle \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} \rangle = -2J$ gives $\langle \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} \rangle = -\frac{1}{2}$. For the second, use

$$0 = \langle \boldsymbol{S}^2_{\rm tot} \rangle = \sum_{ij} \langle \boldsymbol{S}_i \cdot \boldsymbol{S}_j \rangle = 4 \cdot \tfrac{3}{4} + 8\langle \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} \rangle + 4\langle \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+2} \rangle,$$

where the singlet ground state has $\boldsymbol{S}^2_{\rm tot} = 0$, each site contributes $\boldsymbol{S}^2_i = \frac{3}{4}$, there are eight ordered nearest-neighbour pairs and four ordered next-nearest ones. Solving, $\langle \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+2} \rangle = -\frac{1}{4}(3-4) = +\frac{1}{4}$. Both values are confirmed by `models.py`. The positive next-nearest-neighbour correlation is the antiferromagnet: sites two apart are on the same sublattice.

Answer to d).

Write $K_j = \prod_{k<j}(1 - 2\hat{n}_k)$, so that $\hat{S}^+_j = c^\dagger_j K_j$ and $\hat{S}^-_j = K_j c_j$. For j and $j+1$, $K_j K_{j+1} = (1 - 2\hat{n}_j)$, since every factor with $k < j$ appears twice and $(1 - 2\hat{n}_k)^2 = 1$. Then

$$\hat{S}^+_j \hat{S}^-_{j+1} = c^\dagger_j (1 - 2\hat{n}_j) c_{j+1} = c^\dagger_j c_{j+1},$$

because $c^\dagger_j$ acts only on states with $\hat{n}_j = 0$, where the factor is unity. The strings have cancelled. With $\hat{S}^z_j = \hat{n}_j - \frac{1}{2}$ the diagonal term is $J(\hat{n}_j - \frac{1}{2})(\hat{n}_{j+1} - \frac{1}{2})$, and collecting both gives Eq. (4.31). The function `jordan_wigner_chain` in `models.py` builds the fermionic form directly and confirms that the two Hamiltonians have identical spectra, not merely identical ground-state energies, for open chains of four to ten sites.

Exercise 7: quasispin and the qubit encodings

a) Derive the quasispin algebra (4.42) from the fermion anticommutation relations alone.
b) Verify Eq. (4.45) and deduce the compact form (4.44).
c) Write out the one-level pairing Hamiltonian $\hat{H} = \varepsilon_p \hat{N}_p - g\,\hat{n}_{p+}\hat{n}_{p-}$ in the full two-qubit space and show that it is diagonal.
d) Explain, without computing anything, why the compact encoding cannot be used once $f \neq 0$.

Answer to a).

For the first commutator use $[\hat{N}_p, a^\dagger_{p\sigma}] = a^\dagger_{p\sigma}$, which follows from $\{a_{p\sigma}, a^\dagger_{p\sigma}\} = 1$. Applying it twice, $[\hat{N}_p, a^\dagger_{p+}a^\dagger_{p-}] = 2a^\dagger_{p+}a^\dagger_{p-}$, so

$$\left[\hat{S}^z_p, \hat{S}^+_p\right] = \tfrac{1}{2}\left[\hat{N}_p, \hat{S}^+_p\right] = \hat{S}^+_p,$$

and the Hermitian conjugate gives the $-$ case. For the second, expand

$$\hat{S}^+_p \hat{S}^-_p = a^\dagger_{p+}a^\dagger_{p-}a_{p-}a_{p+} = a^\dagger_{p+}\hat{n}_{p-}a_{p+} = \hat{n}_{p+}\hat{n}_{p-},$$

and, using $a_{p+}a^\dagger_{p+} = 1 - \hat{n}_{p+}$ twice,

$$\hat{S}^-_p \hat{S}^+_p = a_{p-}a_{p+}a^\dagger_{p+}a^\dagger_{p-} = (1-\hat{n}_{p+})\,(1-\hat{n}_{p-})\,.$$

Subtracting, $[\hat{S}^+_p, \hat{S}^-_p] = \hat{n}_{p+} + \hat{n}_{p-} - 1 = \hat{N}_p - 1 = 2\hat{S}^z_p$. Operators belonging to different levels involve disjoint sets of orbitals and each contains an even number of fermion operators, so they commute.

Answer to b).

With $\hat{S}^\pm = (X \mp iY)/2$ on each qubit,

$$\hat{S}^+_p \hat{S}^-_q = \tfrac{1}{4}\left(X_pX_q + Y_pY_q + iX_pY_q - iY_pX_q\right),$$

and adding the term with p and q interchanged cancels the imaginary parts, leaving $\frac{1}{2}(X_pX_q + Y_pY_q)$. Hence $-\frac{g}{2}\sum_{p\neq q}\hat{S}^+_p\hat{S}^-_q = -\frac{g}{4}\sum_{p<q}(X_pX_q + Y_pY_q)$. The diagonal $p = q$ terms give $-\frac{g}{2}\sum_p \hat{n}_{p+}\hat{n}_{p-} = -\frac{g}{4}\sum_p(I - Z_p)$ in the paired subspace, and the one-body term contributes $\xi(p-1)$ per particle, that is $2\xi(p-1)$ per pair, or $\xi(p-1)(I - Z_p)$. Collecting gives Eq. (4.44).

Answer to c).

Both terms are built from number operators, and $\hat{n}_{p\sigma} = (I - Z_{p\sigma})/2$, so

$$\hat{H} = \frac{\varepsilon_p}{2}\,(2I - Z_{p+} - Z_{p-}) - \frac{g}{4}\,(I - Z_{p+})\,(I - Z_{p-}) = \left(\varepsilon_p - \frac{g}{4}\right)I - \left(\frac{\varepsilon_p}{2} - \frac{g}{4}\right)(Z_{p+} + Z_{p-}) - \frac{g}{4}Z_{p+}Z_{p-}.$$

Only I and Z appear, so the Hamiltonian is diagonal in the computational basis and its eigenvalues can be read off: 0, ε_p, ε_p and $2\varepsilon_p - g$ for zero, one and two particles. A single pair level needs no quantum computer.

Answer to d).

The compact encoding assigns one qubit to a whole level, which presupposes that the level is either empty or fully occupied – that is, that the seniority is zero. The particle-hole term produces the states of Figure 4.3(c) and (d), in which a level carries a single unpaired particle, and those states have no representative in the compact Hilbert space. Restricting to it would not be an approximation but a projection onto a subspace the Hamiltonian does not leave invariant, and the resulting energies would be wrong even qualitatively.

Exercise 8: numerical explorations

a) Using `models.py`, plot the Lipkin ground-state energy and the weight of $|2,-2\rangle$ in the ground state as functions of V at fixed ε and W. At roughly what value of V/ε does the single-configuration picture break down?
b) Repeat the pairing calculation of Table 4.1 for six and eight levels. How does the correlation energy at $g=1$ scale with the number of levels?
c) Compute the double occupancy of the Hubbard ring as a function of U/t and identify the crossover. Compare a six-site ring with the four-site one.
d) For the pairing model with twelve levels and six pairs, compare the exact ground-state energy from full diagonalisation with the Lanczos result of Table 1.3, and with the Schmidt truncation of Table 1.7. How many Schmidt states are needed for the energy to be correct to six digits?

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter04/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebooks `LectureNotes/models.ipynb` and `LectureNotes/Hubbard.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter04/`.

`models.py` The Lipkin, pairing, pairing-plus-particle-hole, Hubbard, Heisenberg and Calogero models: matrices, spectra, exact limits and the Jordan–Wigner check. Runs in a few seconds.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Part II
Wave function based methods

Chapter 5
Full configuration interaction theory

We will start our discussions of many-body methods with what is called full configuration interaction theory, or just FCI. The reason for this is simple. If the effective degrees of freedom which are relevant for describing a physical system give a sufficiently good reproduction of the static properties (and later we will add dynamical properties) of a many-body system, FCI gives us all relevant eigenpairs (eigenvalues and eigenstates) and allows us to compute other expectation values than the energy levels.

Although it would have been natural to start with mean-field methods (FCI is often labelled as a post Hartree-Fock method), we have chosen a slightly different route. The reason for this is that FCI, within a restricted effective space, provides us with the exact eigenvalues and can thus serve as a benchmark for the approximative methods discussed in the coming chapters, from mean field methods to the Tamm-Dancoff (TDA) and random-phase (RPA) approximations, variants of many-body perturbation theory and coupled-cluster theory, Greens' function approaches and other. We will use FCI to illustrate how these other methods can be interpreeted in terms of specific truncations/approximations of FCI.

We start thus with, what this authour likes to claim as the simplest possible approach to many-body problems, an approach which involves us being able to set up a many-body Hamiltonian matrix (within an effective and reduced Hilbert space), and then simply diagonalize. The overarching principles and the underlying mathematics are straightforward, although the actual implementations may easily lead to endless frustrations and sleepless nights. Let us get started.

5.1 Slater determinants as a basis

The simplest many-body wave function is a product,

$$\Psi(x_1,x_2,\dots,x_N) \approx \phi_1(x_1)\phi_2(x_2)\cdots\phi_N(x_N), \tag{5.1}$$

because we are really only good at thinking about one particle at a time. Product functions are easy to work with: if the single-particle states are orthonormal the products are too, and matrix elements factorise into one-dimensional integrals. The price is the complete absence of correlations, which must then be built up by superposing many, many products. This is the trade-off that runs through the whole subject – a compact representation with difficult integrals, or easy integrals and very many states.

Antisymmetry turns the product into a determinant, and Chapter 2 showed that the two elementary properties of determinants – a sign change under the interchange of two columns, and vanishing when two rows coincide – are exactly the Pauli principle:

- no two particles at the same place (two equal columns);
- no two particles in the same state (two equal rows).

Beyond $N = 4$ the coordinate-space determinant becomes unwieldy, which is why Chapter 3 replaced it by the occupation representation. There a Slater determinant is a string of creation operators acting on the vacuum, each index appearing at most once because $a_i^\dagger a_i^\dagger = 0$, and antisymmetry is carried by the algebra rather than by the notation.

We take that formalism as given. Writing the reference determinant as

$$|\Phi_0\rangle = \left(\prod_{i\le F} \hat{a}_i^\dagger\right)|0\rangle, \tag{5.2}$$

the excited determinants are

$$|\Phi_i^a\rangle = \hat{a}_a^\dagger \hat{a}_i|\Phi_0\rangle, \qquad |\Phi_{ij}^{ab}\rangle = \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i|\Phi_0\rangle, \tag{5.3}$$

and in general

$$|\Phi_{ijk\dots}^{abc\dots}\rangle = \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_c^\dagger \cdots \hat{a}_k \hat{a}_j \hat{a}_i|\Phi_0\rangle, \tag{5.4}$$

with the hole indices i, j, k below the Fermi level and the particle indices a, b, c above it, following the convention of Section 2.6. We call such a determinant an np-nh excitation, or a *configuration*.

5.2 The configuration-interaction expansion

The exact ground state is an arbitrary superposition of all of these,

$$|\Psi_0\rangle = C_0|\Phi_0\rangle + \sum_{ai} C_i^a|\Phi_i^a\rangle + \sum_{abij} C_{ij}^{ab}|\Phi_{ij}^{ab}\rangle + \cdots = \left(C_0 + \hat{C}\right)|\Phi_0\rangle, \tag{5.5}$$

where we have introduced the *correlation operator*

$$\hat{C} = \sum_{ai} C_i^a \hat{a}_a^\dagger \hat{a}_i + \sum_{abij} C_{ij}^{ab} \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i + \cdots \tag{5.6}$$

The normalisation of $|\Psi_0\rangle$ is at our disposal, and C_0 is by hypothesis non-zero, so we may set $C_0 = 1$ with proportional changes in all the other coefficients. With this *intermediate normalisation*,

$$\langle\Psi_0|\Phi_0\rangle = \langle\Phi_0|\Phi_0\rangle = 1, \qquad |\Psi_0\rangle = \left(1 + \hat{C}\right)|\Phi_0\rangle. \tag{5.7}$$

It is convenient to compress the notation. Writing H for a set of $0, 1, \dots, n$ hole indices and P for a set of $0, 1, \dots, n$ particle indices,

$$|\Psi_0\rangle = \sum_{PH} C_H^P|\Phi_H^P\rangle = \left(\sum_{PH} C_H^P \hat{A}_H^P\right)|\Phi_0\rangle, \tag{5.8}$$

with $\hat{A}_H^P$ the operator that creates the corresponding excitation. Unit normalisation reads $\sum_{PH} |C_H^P|^2 = 1$, and the energy is

$$E = \langle\Psi_0|\hat{H}|\Psi_0\rangle = \sum_{PP'HH'} C_H^{*P} \langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'}. \tag{5.9}$$

If the expansion runs over *all* determinants that can be built from the chosen single-particle basis, this is *full configuration interaction*. The only approximation left is the truncation of the single-particle space itself.

5.3 The eigenvalue problem

Equation (5.9) is solved by diagonalisation, and it is worth seeing why the diagonalisation is not merely convenient but is the variational answer. We minimise the energy subject to normalisation,

$$\delta\left[\langle\Psi_0|\hat{H}|\Psi_0\rangle - \lambda\langle\Psi_0|\Psi_0\rangle\right] = 0, \tag{5.10}$$

with λ a Lagrange multiplier. Carrying out the variation,

$$\sum_{P'H'}\left\{\delta[C_H^{*P}]\langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'} + C_H^{*P}\langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle\delta[C_{H'}^{P'}] - \lambda\left(\delta[C_H^{*P}]C_{H'}^{P'} + C_H^{*P}\delta[C_{H'}^{P'}]\right)\right\} = 0. \tag{5.11}$$

Since $\delta[C_H^{*P}]$ and $\delta[C_{H'}^{P'}]$ are complex conjugates, it is necessary and sufficient that the quantities multiplying $\delta[C_H^{*P}]$ vanish, giving

$$\sum_{P'H'}\langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'} = \lambda\, C_H^P \tag{5.12}$$

for every set PH. Multiplying by C_H^{*P} and summing identifies $\lambda = E$, so the multiplier is the energy and

$$\boxed{\sum_{P'H'}\langle\Phi_H^P|\hat{H}-E|\Phi_{H'}^{P'}\rangle C_{H'}^{P'} = 0} \tag{5.13}$$

This is the full CI equation. The same result follows more directly by writing $(\hat{H}-E)|\Psi_0\rangle = 0$ and projecting successively onto every $\langle\Phi_H^P|$.

Equation (5.13) is a Hermitian eigenvalue problem, and the diagonalisation is a Rayleigh-Ritz variational calculation in the space spanned by the chosen determinants: the lowest eigenvalue is the best upper bound to E_0 obtainable in that space, and it can only improve as the space grows. This is exactly the property that made the Lanczos algorithm of Section 1.26 so well suited to the problem – the lowest Ritz value there is the same variational bound, computed in a Krylov space instead of a space of determinants.

Everything now depends on being able to build and diagonalise the matrix $\langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle$. The matrix elements are supplied by the Condon-Slater rules of Section 3.14, and the diagonalisation by the algorithms of Chapter 1: Householder and QL for small matrices, Lanczos for large ones.

5.4 The structure of the Hamiltonian matrix

Suppose we have six fermions below the Fermi level, so that at most $6p$-$6h$ excitations are possible. Ordering the determinants by excitation level, the Hamiltonian matrix has the block structure

$$\begin{array}{c|ccccccc}
 & 0p0h & 1p1h & 2p2h & 3p3h & 4p4h & 5p5h & 6p6h \\
\hline
0p0h & \times & \times & \times & 0 & 0 & 0 & 0 \\
1p1h & \times & \times & \times & \times & 0 & 0 & 0 \\
2p2h & \times & \times & \times & \times & \times & 0 & 0 \\
3p3h & 0 & \times & \times & \times & \times & \times & 0 \\
4p4h & 0 & 0 & \times & \times & \times & \times & \times \\
5p5h & 0 & 0 & 0 & \times & \times & \times & \times \\
6p6h & 0 & 0 & 0 & 0 & \times & \times & \times
\end{array} \tag{5.14}$$

The zeros are not accidental. A two-body operator connects Slater determinants differing by at most two single-particle states, which is the last of the Condon-Slater rules of Section 3.14. The matrix is therefore *banded* in the excitation level, with bandwidth two, and overwhelmingly sparse: for six particles the fraction of non-zero blocks is already small, and it falls as the space grows. Chapter 1 called this the structure that Lanczos exploits; here we see where it comes from.

In a Hartree-Fock basis.

If the single-particle states are the solutions of the Hartree-Fock equations rather than the eigenstates of $\hat{h}_0$, one more set of elements vanishes. Brillouin's theorem, met in Section 3.14, states that

$$\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = 0, \tag{5.15}$$

so the reference determinant does not couple to the singles at all,

$$\begin{array}{c|ccccccc}
 & 0p0h & 1p1h & 2p2h & 3p3h & 4p4h & 5p5h & 6p6h \\
\hline
0p0h & \tilde{\times} & 0 & \tilde{\times} & 0 & 0 & 0 & 0 \\
1p1h & 0 & \tilde{\times} & \tilde{\times} & \tilde{\times} & 0 & 0 & 0 \\
2p2h & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} & 0 & 0 \\
3p3h & 0 & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} & 0 \\
4p4h & 0 & 0 & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} \\
5p5h & 0 & 0 & 0 & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} \\
6p6h & 0 & 0 & 0 & 0 & \tilde{\times} & \tilde{\times} & \tilde{\times}
\end{array} \tag{5.16}$$

where the tildes are a reminder that a unitary transformation of the single-particle basis has changed every matrix element. The correlation energy is then carried entirely by the doubles at leading order, which is the starting point for both many-body perturbation theory and coupled-cluster theory.

The programme `fci.py` builds this structure explicitly. For the pairing model with four levels and four particles the full space has $\binom{8}{4} = 70$ determinants, distributed as 1, 16, 36, 16, 1 over the excitation levels 0 to 4, and the computed block table is

$$\begin{array}{c|ccccc}
 & 0p0h & 1p1h & 2p2h & 3p3h & 4p4h \\
\hline
0p0h & \times & 0 & \times & 0 & 0 \\
1p1h & 0 & \times & 0 & \times & 0 \\
2p2h & \times & 0 & \times & 0 & \times \\
3p3h & 0 & \times & 0 & \times & 0 \\
4p4h & 0 & 0 & \times & 0 & \times
\end{array} \tag{5.17}$$

Even the odd blocks are empty here, because the pairing interaction moves particles two at a time and can never change the number of broken pairs. The symmetry has done more work than Brillouin's theorem would.

5.5 The exponential wall

Full configuration interaction is exact, and that is precisely its problem. With N particles distributed over n single-particle states the number of Slater determinants is

$$\dim \mathscr{H} = \binom{n}{N}, \tag{5.18}$$

which is the count of bit patterns with N bits set out of n, in the representation of Section 3.15. This is a combinatorial expression, but combinatorial growth turns into exponential growth as soon as the particle number scales with the size of the basis. At half filling, $N = n/2$, Stirling's formula gives

$$\binom{n}{n/2} \sim \frac{2^n}{\sqrt{\pi n/2}} \sim \frac{2^n}{\sqrt{n}}, \tag{5.19}$$

so the dimension doubles every time one single-particle state is added. The program `fci.py` tabulates both sides:

n	$\binom{n}{n/2}$	$2^n/\sqrt{n}$
10	2.520×10^2	3.238×10^2
20	1.848×10^5	2.345×10^5
40	1.378×10^{11}	1.738×10^{11}
80	1.075×10^{23}	1.352×10^{23}
160	9.205×10^{46}	1.155×10^{47}

The estimate is accurate to within a factor of order one over five decades of n and twenty-five orders of magnitude of dimension.

A realistic example.

Consider oxygen-16 and let the neutrons occupy the four lowest major oscillator shells, the $0s$, $0p$, $1s0d$ and $1p0f$ shells, which together hold $2+6+12+20 = 40$ single-particle states for each species. The eight neutrons can be distributed in

$$\binom{40}{8} = 7.690 \times 10^7 \tag{5.20}$$

ways, and since the protons can be distributed independently the full proton-neutron space has

$$\binom{40}{8}^2 = 5.914 \times 10^{15} \tag{5.21}$$

Slater determinants. Rotational and parity symmetry reduce this by orders of magnitude, but adding a single further major shell, or relaxing the truncation, restores the growth immediately. Direct diagonalisation by Householder and QL, Sections 1.24 and 1.25, is limited to matrices of dimension around 10^5 because it stores and modifies the whole matrix. Lanczos, Section 1.26, needs only the product $\boldsymbol{Hv}$ and reaches perhaps 10^{10} on a large machine. Beyond that no rearrangement of the linear algebra helps: the expansion itself must be truncated.

Figure 5.1 collects all of this on a single plot, and it repays a careful look, because the fixed-N curves and the half-filled curve say rather different things. At fixed particle number $\binom{n}{N} \sim n^N/N!$ is a polynomial in n, and a polynomial drawn on a logarithmic vertical scale bends over: the $N = 2$ curve has not reached 10^4 by $n = 80$, and even $N = 8$ arrives at $\binom{80}{8} =$

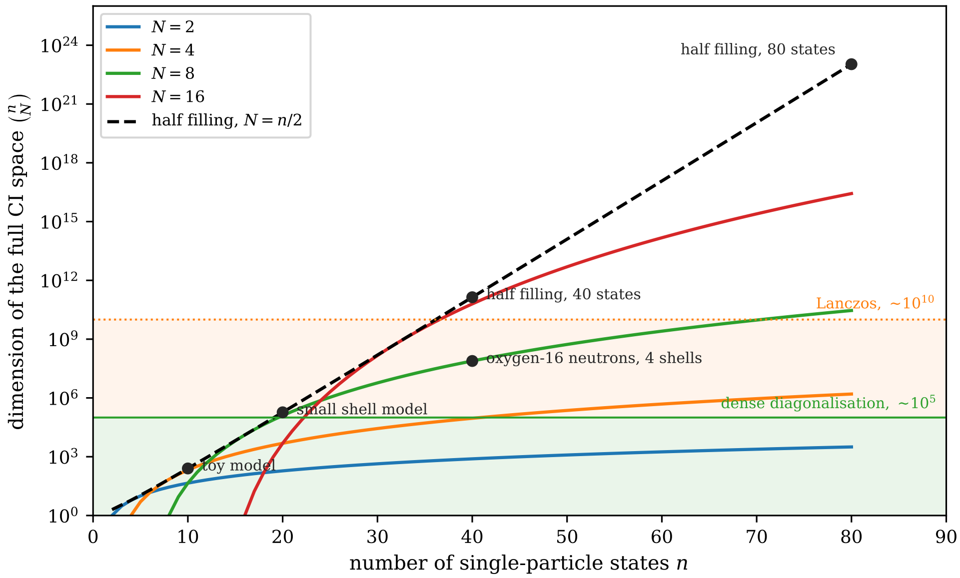

Fig. 5.1 Dimension $\binom{n}{N}$ of the full configuration-interaction space as a function of the number of single-particle states n, on a logarithmic vertical scale. The four solid curves hold the particle number fixed at $N = 2, 4, 8, 16$, while the dashed curve follows the half-filled line $N = n/2$. Filled circles mark the landmarks tabulated by `fci.py`: a toy model ($n = 10$, $N = 5$), a small shell-model space ($n = 20$, $N = 10$), the neutrons of oxygen-16 in four major oscillator shells ($n = 40$, $N = 8$), and half filling at forty and at eighty single-particle states. The two shaded bands are the practical limits of the linear algebra of Chapter 1: dense Householder-QL diagonalisation reaches a dimension of about 10^5, and Lanczos, which needs only matrix-vector products, about 10^{10}. Everything above the upper band is territory that no exact diagonalisation will enter.

2.9×10^{10} only at the right-hand edge of the plot. Adding single-particle states to a fixed and small number of valence particles is therefore comparatively cheap, which is precisely why calculations near a closed shell are so much easier than calculations in the middle of an open one. The dashed curve, by contrast, is a straight line, and a straight line on a semi-logarithmic plot is an exponential: at half filling every new single-particle state multiplies the dimension by two, a factor of 10^3 for every ten states, exactly as Eq. (5.19) says. It crosses the dense-diagonalisation band at $n = 20$, where $\binom{20}{10} = 1.85 \times 10^5$, and the Lanczos band between $n = 36$ and $n = 38$, since $\binom{36}{18} = 9.08 \times 10^9$ while $\binom{38}{19} = 3.49 \times 10^{10}$. Each fixed-$N$ curve touches the dashed one at $n = 2N$ and falls away from it on both sides, so the half-filled line is the envelope of the whole family: whatever the filling, the worst case is the middle of the shell.

The landmarks say the same thing in physical language. The eight neutrons of oxygen-16 in four major shells sit at 7.69×10^7, comfortably inside the Lanczos band, and it is only the independence of the protons that lifts the combined space to the 5.9×10^{15} of Eq. (5.21), five orders of magnitude above it. Half filling with forty single-particle states already stands at 1.38×10^{11}, just beyond reach, and with eighty states at 1.08×10^{23}, of the order of Avogadro's number of complex amplitudes for a single state of a single nucleus. No improvement in the linear algebra will close a gap of thirteen orders of magnitude, and no foreseeable machine will store 10^{23} doubles. The wall is not a statement about our algorithms; it is a statement about the size of the space, and the only way past it is to stop asking for all of the amplitudes.

The same wall in the models of Chapter 4.

Each of the models of the previous chapter meets Eq. (5.19) in its own way.

- The *Lipkin model* with N particles in two N-fold degenerate levels has $\binom{2N}{N}$ determinants in total and $\binom{N}{N/2} \sim 2^N/\sqrt{N}$ in the maximal-quasispin multiplet. That the SU(2) structure of Section 4.1 collapses the problem to an $(N+1)$-dimensional matrix is a symmetry accident, not a general escape.
- The *pairing model* of Section 4.2 has $\binom{2n}{N}$ determinants for n doubly degenerate levels, of which only $\binom{n}{N/2}$ have seniority zero. Again a symmetry – here the conservation of the number of broken pairs – takes the square root of the dimension, and again it is special to the model.
- The *Hubbard model* of Section 4.4 has a local Hilbert space of dimension four, so L sites give 4^L states in total. At fixed particle number this becomes $\binom{L}{N_\uparrow}\binom{L}{N_\downarrow}$, Eq. (4.23), which is again exponential at any extensive filling.
- A *spin chain*, the limit of the Hubbard model at large U, has no combinatorial disguise at all: $\mathscr{H} = \bigotimes_{i=1}^{L} \mathbb{C}^2$ has dimension 2^L by construction.

The pattern is worth stating plainly. Combinatorial counting and tensor-product structure are two faces of the same growth; the tensor product is fundamental, the binomial coefficient is what survives after a conservation law has been imposed. A conserved quantity can remove a factor, but never the exponential.

Why the wall is not the end of the story. The dimension of $\mathscr{H}$ counts the coefficients an *arbitrary* state would need. Physical ground states are not arbitrary. For local Hamiltonians with a gap, the entanglement entropy across a cut grows with the area of the cut rather than with the volume it encloses, and states obeying such an area law are described to high accuracy by a number of parameters polynomial in L. The Schmidt decomposition of Section 1.30 is the one-cut version of this statement, and the singular values decaying rapidly is exactly what makes the truncation work. Applying it at every bond of a chain gives a matrix product state,

$$|\psi\rangle = \sum_{s_1 \ldots s_L} A_1^{s_1} A_2^{s_2} \cdots A_L^{s_L} |s_1 s_2 \ldots s_L\rangle, \tag{5.22}$$

whose cost is polynomial in the bond dimension D; this is the density-matrix renormalisation group. The same idea appears elsewhere in this book in different clothing: a neural-network quantum state replaces the exponentially many amplitudes by a polynomially parameterised function of the occupation pattern, and quantum computing replaces them by n qubits whose Hilbert space is exponential by construction. Full CI is the reference against which all of these compressions are measured, which is why we treat it first.

5.6 Full CI for the pairing model

The pairing Hamiltonian of Section 4.2,

$$\hat{H} = \xi \sum_{p\sigma} (p-1) \hat{a}^\dagger_{p\sigma} \hat{a}_{p\sigma} - \frac{1}{2} g \sum_{pq} \hat{a}^\dagger_{p+} \hat{a}^\dagger_{p-} \hat{a}_{q-} \hat{a}_{q+}, \tag{5.23}$$

is the ideal first test. It is small enough to diagonalise in the full space, and Chapter 4 already gave the exact answer in the seniority-zero subspace, so the two calculations can be checked against each other.

Take four doubly degenerate levels and four particles, $\xi = 1$. The full space has $\binom{8}{4} = 70$ determinants, distributed over the excitation levels as 1, 16, 36, 16, 1. The seniority-zero subspace used in Chapter 4 has only $\binom{4}{2} = 6$ states. Diagonalising the 70×70 matrix at $g = 1$

gives $E_0 = 0.63554847$, identical to the value in Table 4.1 to all digits shown. The determinants with broken pairs are present in the basis but carry no amplitude in the ground state, because the interaction cannot create a broken pair from a paired configuration. The symmetry that we imposed by hand in Chapter 4 is recovered automatically by the diagonalisation.

This is the general situation. A symmetry of $\hat{H}$ block-diagonalises the FCI matrix, and one may either exploit it, gaining a smaller matrix and a labelled spectrum, or ignore it, gaining a simpler code at the cost of diagonalising a larger matrix. Production codes exploit it; the program accompanying this chapter does not, precisely so that the two answers can be compared.

5.7 Full CI for the Hubbard model

The Hubbard model of Section 4.4 is the harder test, and the more instructive one. We take a ring of $L = 6$ sites at half filling, $N_\uparrow = N_\downarrow = 3$, and work in the momentum basis, where the hopping term is diagonal,

$$\hat{H} = \sum_{k\sigma} \varepsilon_k \hat{c}^\dagger_{k\sigma} \hat{c}_{k\sigma} + \frac{U}{L} \sum_{kk'q} \hat{c}^\dagger_{k+q\uparrow} \hat{c}^\dagger_{k'-q\downarrow} \hat{c}_{k'\downarrow} \hat{c}_{k\uparrow}, \qquad \varepsilon_k = -2t\cos k, \tag{5.24}$$

with $k = 2\pi m/L$. In this basis the non-interacting ground state is a single determinant – the Fermi sea – so the machinery of Section 5.2 applies directly with $|\Phi_0\rangle$ the filled Fermi sea. The dimension is $\binom{6}{3}^2 = 400$, with the excitation levels populated as 1, 18, 99, 164, 99, 18, 1.

The interesting quantity is how much of the correlation energy survives truncation as U/t grows:

U/t	E_{ref}	CIS	CID	CISD	FCI
1	−6.500000	−6.50000000	−6.59980159	−6.59989778	−6.60115829
2	−5.000000	−5.00000000	−5.38877703	−5.39026458	−5.40945685
4	−2.000000	−2.00000000	−3.41192866	−3.43018837	−3.66870618
8	4.000000	2.48338852	−0.36779217	−1.05692183	−2.04813089

The doubles recover 98.66% of the correlation energy at $U/t = 1$ but only 72.22% at $U/t = 8$. Two features deserve comment.

First, the reference does not couple to the singles at all: the interaction conserves total momentum, and a single excitation changes it, so $\langle\Phi_0|\hat{H}|\Phi_i^a\rangle = 0$ exactly, for every U. This is Brillouin's theorem, Eq. (5.15), enforced here by a symmetry rather than by a variational condition – the plane-wave determinant is the Hartree-Fock solution of a translationally invariant problem. The CIS matrix is therefore block diagonal, and its lowest eigenvalue is the smaller of the reference energy and the lowest eigenvalue of the singles block alone. For $U/t \leq 4$ the reference wins and CIS reproduces it exactly. At $U/t = 8$ it does not: the singles block, strongly mixed by the large interaction, has an eigenvalue at 2.483, below the reference at 4.000. The number in the table is that eigenvalue, and the state it belongs to is orthogonal to $|\Phi_0\rangle$. What CIS reports as a ground state is by then not a correlated version of the reference at all.

Second, the deterioration with U/t is the signature of *strong correlation*. At small U the ground state is a single determinant with small corrections and any method built on that picture works. At large U the ground state is a superposition of very many determinants with comparable weight, the reference is a poor starting point, and a truncation at any fixed excitation level fails. This is where the tensor-network and quantum-computing approaches of the later chapters become attractive, and it is why the Hubbard model at intermediate U/t remains a benchmark problem.

5.8 Truncating the expansion

Since the full expansion is out of reach for all but the smallest systems, we truncate it. The natural truncation is by excitation level: keep all determinants up to n particle-hole pairs and discard the rest. This gives the familiar hierarchy

$$|\Psi_{\text{CIS}}\rangle = \left(1+\hat{C}_1\right)|\Phi_0\rangle, \tag{5.25}$$

$$|\Psi_{\text{CID}}\rangle = \left(1+\hat{C}_2\right)|\Phi_0\rangle, \tag{5.26}$$

$$|\Psi_{\text{CISD}}\rangle = \left(1+\hat{C}_1+\hat{C}_2\right)|\Phi_0\rangle, \tag{5.27}$$

and so on to CISDT, CISDTQ, until at $\hat{C}_N$ we are back at full CI. Each of these is again a variational calculation in a subspace, so each lowest eigenvalue is an upper bound to E_0 and the bounds decrease monotonically as the truncation is relaxed. The cost, for a basis of n states and N particles, is roughly $\binom{N}{k}\binom{n-N}{k}$ determinants at excitation level k, that is polynomial of degree $2k$ – which is why CISD, at $\mathscr{O}(n^6)$, is affordable and CISDTQ, at $\mathscr{O}(n^{10})$, is not.

For the pairing model with four levels and four particles the truncated spaces have dimensions 1, 17, 53, 69, 70, and the energies are

g	E_{ref}	CIS	CID	CISD	FCI
0.25	1.750000	1.75000000	1.73050700	1.73050700	1.73045985
0.50	1.500000	1.50000000	1.41759645	1.41759645	1.41677428
1.00	1.000000	1.00000000	0.64890655	0.64890655	0.63554847
2.00	0.000000	0.00000000	−1.35208670	−1.35208670	−1.48965216

The singles change nothing at all, and CISD coincides with CID: the pairing interaction moves pairs and cannot break them, so it has no matrix element between the reference and a $1p$-$1h$ state, nor between a $1p$-$1h$ state and the reference through any intermediate. The doubles carry essentially the whole correlation energy – 99.8% at $g=0.25$, still 96.3% at $g=1$, but only 90.8% at $g=2$. The pattern is the same as in the Hubbard model: truncation at fixed excitation level works while the reference is good and degrades as the coupling grows.

Figure 5.2 follows that degradation continuously, and in a space large enough that the higher truncations mean something: six levels and six particles rather than four and four, so that CISDTQ keeps 887 of the 924 determinants instead of coinciding with full CI. The left panel is the reassuring one and the right panel is the honest one. On the left, CIS lies flat on zero at every coupling, for the reason already given – the pairing interaction cannot break a pair, so the reference is decoupled from the entire singles block – while CISD and CISDTQ both track the full CI curve down to a correlation energy of -3.19 at $g=2$, and CISDTQ is indistinguishable from it by eye across the whole plot. Only above $g\approx 1$ does the CISD curve visibly detach. The right panel dismantles that impression. At $g=0.05$ the doubles are wrong by 2.5×10^{-7} and the quadruples by 6.5×10^{-12}; by $g=2$ the doubles are wrong by 0.73 and the quadruples by 2.9×10^{-2}. The error of a fixed truncation therefore grows by six and a half orders of magnitude for CISD, and by nine and a half for CISDTQ, while nothing about the calculation has changed except the strength of the interaction.

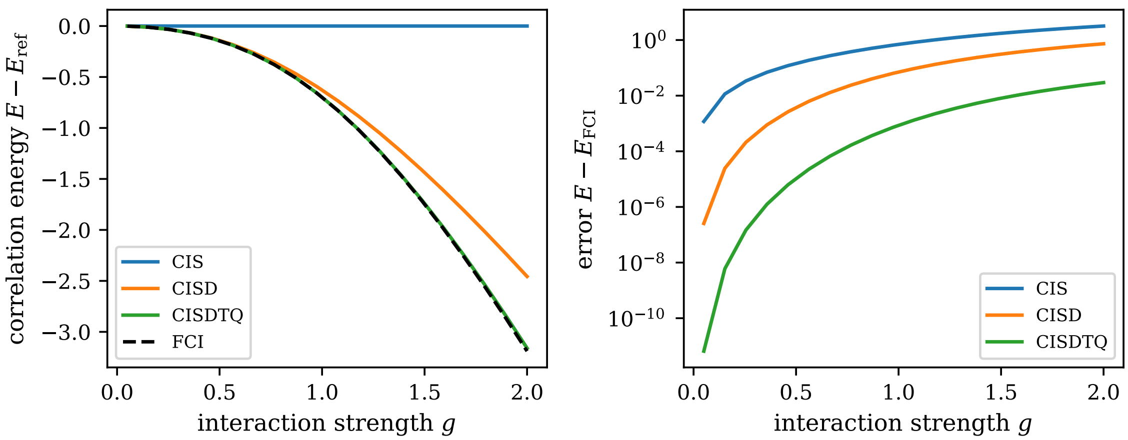

Fig. 5.2 Truncated configuration interaction against full CI for the pairing Hamiltonian of Eq. (5.23) with six doubly degenerate levels, six particles and $\xi = 1$, over the range $0.05 \leq g \leq 2$. The left panel shows the correlation energy $E - E_{\text{ref}}$ recovered by CIS, CISD and CISDTQ, together with the full CI result as a dashed line; the right panel shows what each truncation misses, $E - E_{\text{FCI}}$, on a logarithmic scale. The full space holds $\binom{12}{6} = 924$ determinants, with excitations up to $6p$-$6h$, and the truncated spaces 37, 262 and 887 of them, so that the quadruples are a genuine truncation and not the whole space in disguise. The errors climb by six to ten orders of magnitude across the range while the truncation level stays fixed.

Two further features of the right panel are worth drawing out. The first is that the curves converge as g grows: at $g = 0.05$ the quadruples improve on the doubles by a factor of 4×10^4, at $g = 1$ by a factor of 82, and at $g = 2$ by a factor of only 25. Adding a whole excitation level buys less and less the harder the problem becomes, which is the least comfortable property a hierarchy of approximations can have – the correction is largest exactly where the estimate of the correction is worst. The second is a comparison with the four-level table above. At $g = 2$, CISD there recovered 90.8% of the correlation energy, whereas here, with six levels and six particles, the same truncation recovers only 77.0%. The relative error has grown although the truncation has not, simply because the system is larger; the missing configurations are the higher products of doubles, and there are more of them to miss. That observation is not an accident of the pairing model but the size-consistency defect of Section 5.9, seen from the side.

5.9 Size consistency

There is a defect of truncated CI that is more serious than the loss of accuracy, because it does not go away as the basis is improved. Consider two identical systems A and B, infinitely far apart and non-interacting. The Hamiltonian is $\hat{H}_A + \hat{H}_B$ and the exact ground state is the product $|\Psi_A\rangle|\Psi_B\rangle$, so the exact energy must satisfy

$$E(AB) = E(A) + E(B) = 2E(A). \tag{5.28}$$

A method that reproduces this is called *size consistent*.

Full CI is size consistent, since the product state lies in the full space. Truncated CI is not. The reason is immediate: if A and B are each doubly excited, the combined state is a *quadruple* excitation of the combined reference, and CID has no room for it. The product of two doubles is a quadruple, and truncating at a fixed excitation level is not a multiplicatively closed operation.

The program `fci.py` demonstrates this on two non-interacting copies of a two-level, two-particle pairing system:

g	$2E(A)$	$E(AB)$, FCI	$E(AB)$, CID	CID error
0.25	-0.26556444	-0.26556444	-0.26550480	5.96×10^{-5}
0.50	-0.56155281	-0.56155281	-0.56066017	8.93×10^{-4}
1.00	-1.23606798	-1.23606798	-1.22474487	1.13×10^{-2}

Full CI passes to machine precision; CID fails, and the failure grows with the coupling. With M subsystems instead of two the error grows with M, so the energy per particle computed with truncated CI drifts as the system is made larger. For an extended system – nuclear matter, the electron gas, a large molecule – this is fatal.

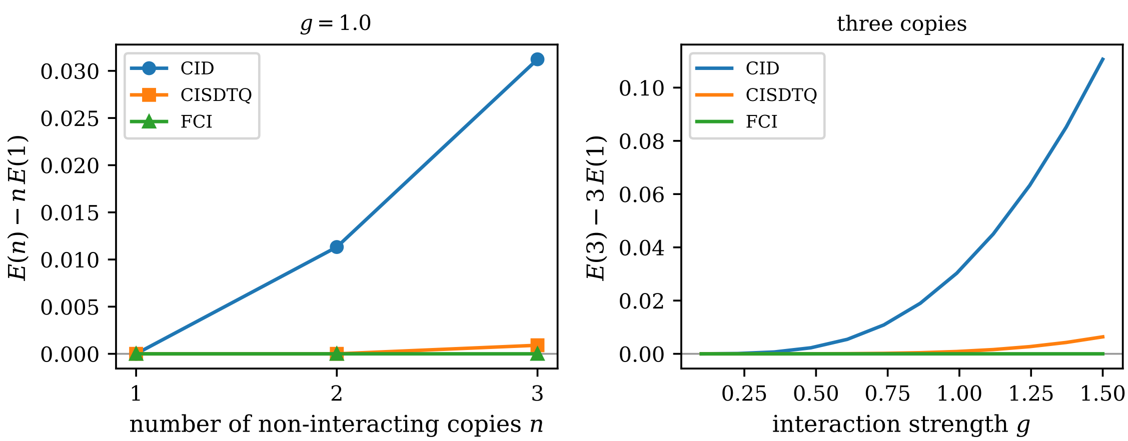

Fig. 5.3 Size-consistency error of truncated CI for n identical, non-interacting copies of a two-level, two-particle pairing system, built by the class `NonInteractingCopies`. The left panel shows $E(n) - nE(1)$ at fixed coupling $g = 1$ for CID, CISDTQ and full CI, each scheme being compared with its own one-copy energy, which is the only fair comparison. The right panel shows the same quantity for three copies as a function of g. Full CI lies on zero to machine precision, as it must, while CID drifts upwards both with the number of subsystems and with the coupling; CISDTQ is exact for two copies, where nothing beyond a quadruple excitation is missing, and fails again at three.

Figure 5.3 turns that last remark into a picture, and it is the cleanest numerical statement of the defect in this chapter. The three copies of the two-level system are genuinely independent – the combined Hamiltonian is a sum with no cross terms – so the exact energy is additive by construction, and the full CI curve sits at -1.6×10^{-15}, which is zero to the precision of the diagonalisation. CID does not. At $g = 1$ its error is $+1.13 \times 10^{-2}$ for two copies, the same number that appears in the table above, and $+3.12 \times 10^{-2}$ for three, a growth of 2.8 against the naive counting factor of three that one gets from the three distinct pairs of subsystems, each of which contributes one missing quadruple excitation. The sign is always positive: the truncated space is smaller than the exact one, the variational bound is looser for the assembly than for the pieces, and the assembly is therefore always under-bound relative to n times one copy. This is worth stating carefully, because a variational method that is exact for one subsystem and variational for n of them does not simply get worse, it gets worse in a way that is proportional to the size of the system, so the error per subsystem does not vanish as the calculation is refined in any other respect.

The CISDTQ curve shows the same mechanism at the next order. For two copies the error is -2.2×10^{-16}, that is, exact: the largest excitation the product state requires is the simultaneous double excitation of both subsystems, a quadruple of the combined reference, and quadruples are kept. For three copies a sextuple is needed and CISDTQ is not exact any more, with an error of $+9.1\times10^{-4}$ at $g=1$ – smaller than CID by a factor of 34, but no longer zero, and no longer improvable by anything short of raising the truncation level again. Every subsystem added demands one more excitation level, so no fixed truncation survives the thermodynamic limit. The right panel adds the second axis of the failure: for three copies the CID error grows from essentially nothing at $g=0.1$ to 2.6×10^{-3} at $g=0.5$ and 0.11 at $g=1.5$, faster than g^2, while CISDTQ reaches 6.4×10^{-3} over the same range and full CI stays at the 10^{-15} level throughout. Strong coupling and large systems are two ways of saying the same thing to a truncated CI calculation, and it fails on both counts at once.

The repair is to make the ansatz multiplicative rather than additive. Replacing $1+\hat{C}$ by $\exp(\hat{T})$ with $\hat{T}=\hat{T}_1+\hat{T}_2$ gives

$$|\Psi_{\mathrm{CCSD}}\rangle = e^{\hat{T}_1+\hat{T}_2}|\Phi_0\rangle = \left(1+\hat{T}_1+\hat{T}_2+\tfrac{1}{2}\hat{T}_2^2+\cdots\right)|\Phi_0\rangle, \tag{5.29}$$

in which the quadruple excitation $\frac{1}{2}\hat{T}_2^2$ appears automatically, with its coefficient fixed by the doubles rather than determined independently. For two non-interacting subsystems $e^{\hat{T}}=e^{\hat{T}_A}e^{\hat{T}_B}$ and size consistency is exact. This is coupled-cluster theory, and Eq. (5.29) is the reason it, rather than truncated CI, is the workhorse of modern many-body physics. We shall develop it properly in a later chapter; the point here is that its central idea is a response to a defect that full CI lets us see clearly.

5.10 Truncations and the other many-body methods

We can now do what the introduction to this chapter promised, and read the standard many-body methods as truncations of Eq. (5.13). Write the projected equations explicitly. Multiplying $(\hat{H}-E)|\Psi_0\rangle=0$ from the left by $\langle\Phi_0|$ and by $\langle\Phi_H^P|$ gives

$$E = \langle\Phi_0|\hat{H}|\Phi_0\rangle + \sum_{P'H'\neq 0}\langle\Phi_0|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'}, \tag{5.30}$$

$$\left(E-\langle\Phi_H^P|\hat{H}|\Phi_H^P\rangle\right)C_H^P = \sum_{P'H'\neq PH}\langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'}. \tag{5.31}$$

The first equation says that the correlation energy comes entirely from the coupling of the reference to the doubles, since only those have a non-zero matrix element with $|\Phi_0\rangle$. The second is a set of coupled equations for the coefficients, which can be solved by iteration: guess the C's, compute the right-hand side, update, repeat. Every method in the remaining chapters is a prescription for which terms of Eq. (5.31) to keep and how to solve it.

- **Hartree-Fock theory** keeps only C_0 and instead varies the single-particle basis so that $\langle\Phi_0|\hat{H}|\Phi_i^a\rangle=0$ – the Brillouin condition of Eq. (5.15). It is the optimisation of the reference rather than of the expansion, and it fixes the basis in which every subsequent truncation is made. In the language of Eq. (5.16), Hartree-Fock chooses the unitary transformation that empties the $0p0h$-$1p1h$ block.
- **The Tamm-Dancoff approximation** restricts the space to the $1p$-$1h$ block alone and diagonalises it. It is CIS applied to excited states, and it gives the excitation spectrum of the reference at the cheapest non-trivial level.
- **The random-phase approximation** enlarges TDA by allowing the ground state to contain $2p$-$2h$ correlations, so that the excitation operator carries both $1p$-$1h$ and $1h$-$1p$ pieces. It is a partial summation of the doubles block, performed to all orders in the coupling but only for a selected class of terms.

- **Many-body perturbation theory** solves Eq. (5.31) by expanding in powers of the interaction rather than by iterating to convergence. Setting E on the left-hand side to the reference energy and $C_{H'}^{P'}$ on the right to zero gives the first-order coefficients; substituting these back gives the second-order energy, and so on. The order in perturbation theory is the number of times the right-hand side has been substituted into itself.
- **Coupled-cluster theory** keeps the same truncation in excitation level as CISD but writes the wave function multiplicatively, Eq. (5.29). The equations look like Eq. (5.31) with additional nonlinear terms, and it is those nonlinear terms that restore size consistency.

Seen this way the methods are not a list of unrelated techniques but a set of answers to one question: given that we cannot keep all of Eq. (5.5), which parts do we keep, and do we keep them additively or multiplicatively? Full CI is the fixed point that lets us measure each answer. Whenever a model is small enough for Eq. (5.13) to be solved exactly – the pairing model, the Lipkin model, a short Hubbard ring – we should solve it, and use the result to calibrate the approximation before applying it to a system where no calibration is possible.

5.11 Practical considerations

Three remarks on implementation, which anticipate the code discussed in the companion notebook.

Never store the matrix.

The FCI matrix is sparse, with the band structure of Eq. (5.14), and for interesting dimensions it cannot be stored. The Lanczos algorithm of Section 1.26 needs only the product $\boldsymbol{Hv}$, and that product is computed by looping over the determinants and over the non-zero matrix elements of $\hat{H}$, using the bit representation of Section 3.15 to find which determinant each term produces and with which sign. This is the single most important structural choice in an FCI code.

Reorthogonalise.

The Lanczos vectors lose orthogonality as soon as an eigenvalue has converged, and spurious copies of converged eigenvalues appear. Section 1.26 discussed the remedies; for FCI the usual choice is full or selective reorthogonalisation, at the price of storing the Lanczos vectors.

Use the symmetries.

Angular momentum, parity, particle number, total momentum and spin projection all commute with $\hat{H}$ and block-diagonalise the matrix. Working in a single symmetry block reduces the dimension by orders of magnitude and, just as importantly, labels the states. The pairing calculation of Section 5.6 illustrated both options – 70 determinants without the symmetry, 6 with it, the same ground-state energy.

5.12 Summary

Full configuration interaction is the statement that a many-body state is a superposition of all Slater determinants that the single-particle basis allows, together with the observation that finding the best superposition is a Hermitian eigenvalue problem, Eq. (5.13). Everything else in this chapter follows from that. The Condon-Slater rules make the matrix banded in the excitation level; a Hartree-Fock basis empties one more block; the combinatorics of Eq. (5.18) makes the matrix too large to build beyond a few tens of single-particle states; and the truncations that this forces upon us are precisely the approximations that the following chapters elevate into methods.

The program `fci.py` in `BookMaterial/Programs/` carries out every calculation quoted above: the pairing model in the full 70-dimensional space and in the seniority-zero subspace, the block structure of the Hamiltonian, the CIS/CID/CISD/FCI comparison for pairing and for the Hubbard ring, the size-consistency test, and the dimension tables of Section 5.5. The accompanying notebook `fci.ipynb` runs the same code cell by cell.

5.13 Exercises

Exercise 1: counting determinants

a) Show that the number of Slater determinants with N particles in n single-particle states that are k particle-hole excitations of a given reference is $\binom{N}{k}\binom{n-N}{k}$, and verify that $\sum_k \binom{N}{k}\binom{n-N}{k} = \binom{n}{N}$.
b) Evaluate this for $n=8$, $N=4$ and compare with the distribution $1,16,36,16,1$ quoted in Section 5.4.
c) Use Stirling's formula to derive Eq. (5.19), and determine the largest n for which $\binom{n}{n/2}$ stays below 10^{10}.

Answer to a).

Choosing which k of the N occupied states to empty can be done in $\binom{N}{k}$ ways, and which k of the $n-N$ empty states to fill in $\binom{n-N}{k}$ ways; the two choices are independent. Every determinant with N particles arises in this way exactly once, for exactly one value of k, namely the number of occupied states it does not share with the reference. Summing over k must therefore give the total, which is Vandermonde's identity

$$\sum_k \binom{N}{k}\binom{n-N}{k} = \sum_k \binom{N}{N-k}\binom{n-N}{k} = \binom{n}{N}. \tag{5.32}$$

Answer to b).

With $n=8$ and $N=4$ the products are $\binom{4}{k}^2 = 1,16,36,16,1$ for $k=0,\dots,4$, summing to $70=\binom{8}{4}$. This is exactly the distribution that `fci.py` finds by classifying the 70 determinants of the pairing model by excitation level.

Answer to c).

Stirling's formula $m! \sim \sqrt{2\pi m}\,(m/e)^m$ gives

$$\binom{n}{n/2} = \frac{n!}{[(n/2)!]^2} \sim \frac{\sqrt{2\pi n}\,(n/e)^n}{2\pi(n/2)\,\left[(n/2e)^{n/2}\right]^2} = \frac{2^n}{\sqrt{\pi n/2}}, \tag{5.33}$$

which is Eq. (5.19) up to the constant $\sqrt{\pi/2} \approx 1.25$. Solving $\binom{n}{n/2} = 10^{10}$ numerically: $\binom{36}{18} = 9.08 \times 10^9$ and $\binom{38}{19} = 3.49 \times 10^{10}$, so $n = 36$ is the last even value below the limit. Every further pair of single-particle states multiplies the dimension by about four.

Exercise 2: which blocks vanish

Let $\hat{H} = \hat{H}_0 + \hat{H}_I$ with $\hat{H}_I$ a two-body operator.

a) Show that $\langle \Phi_H^P | \hat{H} | \Phi_{H'}^{P'} \rangle = 0$ whenever the two determinants differ by more than two single-particle states, and deduce the band structure of Eq. (5.14).
b) Suppose $\hat{H}$ also contains a genuine three-body force. How does the band structure change, and what does that do to the cost of a matrix-vector product?
c) For the pairing Hamiltonian, Eq. (5.23), show that every block connecting an even to an odd excitation level vanishes, as in Eq. (5.17).

Answer to a).

Write both determinants in the occupation representation of Section 3.15. A two-body operator is a product of two creation and two annihilation operators, so it can change at most two occupation numbers from one to zero and two from zero to one. If the two determinants differ in more than four occupation numbers – that is, by more than two single-particle states – then in the Wick expansion of Section 3.14 at least one operator is left uncontracted in every term, and the matrix element is zero. Since an *np*-*nh* and an *mp*-*mh* determinant differ by at least $|n-m|$ single-particle states, the element vanishes for $|n-m| > 2$. The matrix is therefore banded with bandwidth two in the excitation level, which is Eq. (5.14).

Answer to b).

A three-body operator changes up to three occupations of each kind, so the bandwidth becomes three and the $0p0h$ row now reaches the $3p3h$ block. The number of non-zero elements per row grows from $\mathscr{O}(N^2(n-N)^2)$ to $\mathscr{O}(N^3(n-N)^3)$, so a matrix-vector product costs a further factor $N(n-N)$ – typically two to three orders of magnitude. This is why three-body forces, though physically important in nuclear physics, are usually normal-ordered with respect to the reference determinant and then truncated at the two-body level, a procedure whose justification rests on Section 3.5.

Answer to c).

The interaction $-\frac{1}{2} g \sum_{pq} \hat{a}^\dagger_{p+} \hat{a}^\dagger_{p-} \hat{a}_{q-} \hat{a}_{q+}$ removes a complete pair from level q and creates a complete pair in level p. Define the seniority s as the number of unpaired particles. Both the one-body term and the interaction leave s unchanged, so $[\hat{H},\hat{s}]=0$ and $\hat{H}$ is block diagonal in seniority. An excitation of odd order relative to the paired reference must leave at least two particles unpaired, so it has $s\geq 2$, whereas the reference and all even excitations reached from it by moving whole pairs have $s=0$. Every even-odd matrix element therefore connects different seniorities and vanishes. Numerically the 70×70 matrix splits into a six-dimensional $s=0$ block, a $s=2$ block and so on, and the ground state lies in the first.

Exercise 3: full CI for the pairing model

Take four doubly degenerate levels with single-particle energies $\varepsilon_p=(p-1)\xi$, $p=1,\ldots,4$, four particles, $\xi=1$, and the Hamiltonian of Eq. (5.23).

a) Set up the six seniority-zero configurations and write down the 6×6 Hamiltonian matrix.
b) Diagonalise it for $g=-1,-0.5,0,0.5,1$ and reproduce Table 4.1.
c) Repeat the calculation in the full 70-dimensional space and confirm that the ground-state energy is unchanged. Inspect the eigenvector and verify that every determinant with a broken pair has zero amplitude.
d) Plot the correlation energy against g and show that it is quadratic for small $|g|$, as second-order perturbation theory requires.

Answer to a).

Label a seniority-zero configuration by the pair of levels that are occupied. With four levels and two pairs there are $\binom{4}{2}=6$ of them,

$$|12\rangle,\ |13\rangle,\ |14\rangle,\ |23\rangle,\ |24\rangle,\ |34\rangle, \tag{5.34}$$

where $|pq\rangle$ has pairs in levels p and q. The diagonal element of $|pq\rangle$ is $2\xi[(p-1)+(q-1)]-g$: the factor two because each level holds two particles, and $-g$ because the $p=q$ terms of the interaction give $-\frac{1}{2}g$ for each of the two occupied pairs. A single application of the interaction moves *one* pair, so two configurations are connected by $-\frac{1}{2}g$ if and only if they share exactly one occupied level; the three pairs of configurations that share no level, $|12\rangle$–$|34\rangle$, $|13\rangle$–$|24\rangle$ and $|14\rangle$–$|23\rangle$, are not connected at all. With $\xi=1$ and the ordering of Eq. (5.34),

$$\boldsymbol{H}=\begin{pmatrix} 2-g & -g/2 & -g/2 & -g/2 & -g/2 & 0 \\ -g/2 & 4-g & -g/2 & -g/2 & 0 & -g/2 \\ -g/2 & -g/2 & 6-g & 0 & -g/2 & -g/2 \\ -g/2 & -g/2 & 0 & 6-g & -g/2 & -g/2 \\ -g/2 & 0 & -g/2 & -g/2 & 8-g & -g/2 \\ 0 & -g/2 & -g/2 & -g/2 & -g/2 & 10-g \end{pmatrix}. \tag{5.35}$$

Answer to b).

Diagonalising Eq. (5.35) reproduces Table 4.1; at $g=1$ the lowest eigenvalue is 0.63554847 and the reference energy is $E_{\mathrm{ref}}=H_{11}=1$.

Answer to c).

The class `PairingFCI` in `fci.py` builds the full space with no seniority restriction. Its dimension is $\binom{8}{4} = 70$, the excitation levels are populated as $1, 16, 36, 16, 1$, and the lowest eigenvalue at $g = 1$ is 0.63554847, identical to b) to every digit. Projecting the ground-state eigenvector onto the 64 determinants with at least one broken pair gives zero to machine precision, as Exercise 2c) predicts.

Answer to d).

The reference is $|12\rangle$. Second-order perturbation theory in the interaction gives

$$E^{(2)} = \sum_{\Phi \neq \Phi_0} \frac{|\langle \Phi_0 | \hat{H}_I | \Phi \rangle|^2}{E_0^{(0)} - E_{\Phi}^{(0)}} = \frac{g^2}{4} \left(\frac{1}{-2} + \frac{1}{-4} + \frac{1}{-4} + \frac{1}{-6} \right) = -\frac{7}{24} g^2, \tag{5.36}$$

using the four configurations reachable from $|12\rangle$ by moving one pair, with unperturbed excitation energies $2, 4, 4, 6$. A log-log plot of $|E_{\text{corr}}|$ against g has slope 2 for $|g| \lesssim 0.2$ and bends away above that, where the higher orders take over. At $g = 0.25$, Eq. (5.36) gives -0.01823 against the exact -0.01954; at $g = 0.5$ it gives -0.0729 against -0.0832, and at $g = 1$ only -0.2917 against -0.3645.

Exercise 4: truncations and size consistency

a) Using the code of the previous exercise, restrict the basis to excitation levels ≤ 1 and ≤ 2 and reproduce the CIS and CID columns of Section 5.8. Explain why CIS returns exactly the reference energy.
b) Build a system of two identical, non-interacting two-level pairing systems. Show that full CI gives exactly twice the energy of one subsystem while CID does not, and tabulate the error for $g = 0.25, 0.5, 1$.
c) Identify the configuration that CID is missing, and show that including quadruple excitations restores the exact result.
d) Show that the ansatz $\exp(\hat{T}_2)|\Phi_0\rangle$, with the same number of independent parameters as CID, is size consistent for this system.

Answer to a).

In a truncated space the eigenvalue problem is Eq. (5.13) restricted to the corresponding rows and columns. The dimensions for the cuts at $0, 1, 2, 3, 4$ are $1, 17, 53, 69, 70$. CIS returns the reference energy because $\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = 0$: a single excitation breaks a pair and so changes the seniority, which the pairing Hamiltonian conserves (Exercise 2c). The reference is therefore decoupled from the entire singles block and remains an exact eigenstate of the truncated matrix. Its energy is also the lowest diagonal element, and the singles block lies far above it for the couplings considered, so it is the CIS ground state.

Answer to b).

Let each subsystem have two levels at 0 and ξ, holding one pair. The combined system then has four levels with energies $0,\xi,0,\xi$ and two pairs, one per subsystem, with the interaction acting only within a subsystem. The class `NonInteractingCopies` gives

$$\begin{array}{ccccc} g & 2E(A) & E(AB)_{\mathrm{FCI}} & E(AB)_{\mathrm{CID}} & \text{CID error} \\ 0.25 & -0.26556444 & -0.26556444 & -0.26550480 & 5.96\times10^{-5} \\ 0.50 & -0.56155281 & -0.56155281 & -0.56066017 & 8.93\times10^{-4} \\ 1.00 & -1.23606798 & -1.23606798 & -1.22474487 & 1.13\times10^{-2} \end{array} \tag{5.37}$$

Full CI is exact to machine precision; the CID error grows by roughly an order of magnitude each time g is doubled.

Answer to c).

Writing $\hat{A}_2^X$ for the operator that excites the pair of subsystem X, the exact product state is

$$\left(1+c\hat{A}_2^A\right)\left(1+c\hat{A}_2^B\right)|\Phi_0\rangle = \left(1+c\hat{A}_2^A+c\hat{A}_2^B+c^2\hat{A}_2^A\hat{A}_2^B\right)|\Phi_0\rangle. \tag{5.38}$$

The last term excites both subsystems at once and is a $4p$-$4h$ excitation of the combined reference; it is absent from CID. Adding the quadruples – which for this small system means going to the full space – recovers the exact energy, and the coefficient of the quadruple comes out equal to c^2, the product of the two double amplitudes.

Answer to d).

With $\hat{T}_2 = t\left(\hat{A}_2^A+\hat{A}_2^B\right)$ the two terms commute, since they act on disjoint sets of orbitals, so

$$e^{\hat{T}_2} = e^{t\hat{A}_2^A}e^{t\hat{A}_2^B} = \left(1+t\hat{A}_2^A\right)\left(1+t\hat{A}_2^B\right), \tag{5.39}$$

the last step because $(\hat{A}_2^X)^2=0$ – one cannot excite the same pair twice. The exponential ansatz therefore generates exactly the product state of Eq. (5.38) with one parameter per subsystem, the same count as CID, and the energy is additive by construction. This is coupled cluster with doubles in miniature, and it shows that the cure for size inconsistency costs nothing in parameters – only a change from a linear to a multiplicative ansatz.

Exercise 5: the Hubbard ring by full CI

Take a ring of $L=6$ sites at half filling with $t=1$, in the momentum basis of Eq. (5.24).

a) Verify that the dimension is $\binom{6}{3}^2=400$ and that the excitation levels are populated as 1, 18, 99, 164, 99, 18, 1.
b) Compute the exact ground-state energy for $U/t=1,2,4,8$ and compare with the CID result.
c) Show that $\langle\Phi_0|\hat{H}|\Phi_i^a\rangle=0$ for every single excitation, and explain this using conservation of total momentum.
d) At $U/t=8$ the CIS energy lies *below* the reference energy even though the reference does not couple to the singles. Explain how this is possible and what it says about the reliability of a variational bound.

e) Compare the exact energies at large U with the Heisenberg limit of Section 4.4.

Answer to a).

There are $\binom{6}{3} = 20$ ways to place three electrons of each spin, and the two species are independent, so the dimension is 400. Counting for each determinant the number of occupied momenta outside the Fermi sea, separately for each spin, and adding the two counts gives $1, 18, 99, 164, 99, 18, 1$, which sums to 400. The distribution is symmetric because a kp-kh excitation of the Fermi sea is a $(6-k)p$-$(6-k)h$ excitation of the completely inverted determinant.

Answer to b).

The values are those of Section 5.7. Relative to the reference, the doubles recover 98.66%, 94.95%, 84.61% and 72.22% of the correlation energy at $U/t = 1, 2, 4, 8$.

Answer to c).

The interaction is $\frac{U}{L}\sum_{kk'q} \hat{c}^{\dagger}_{k+q\uparrow}\hat{c}^{\dagger}_{k'-q\downarrow}\hat{c}_{k'\downarrow}\hat{c}_{k\uparrow}$, in which the momenta of the two created operators sum to those of the two annihilated ones; total crystal momentum is therefore conserved. A single excitation $i \to a$ changes the total momentum by $k_a - k_i \neq 0$ and so cannot be reached from the Fermi sea. The one-body part is diagonal in k and cannot connect them either. Evaluating the couplings numerically gives zero to machine precision for every U. Note that this is Brillouin's theorem, Eq. (5.15), obtained here from a symmetry rather than from a variational condition: for a translationally invariant problem the plane-wave determinant *is* the Hartree-Fock solution.

Answer to d).

Because the coupling vanishes, the CIS matrix is block diagonal: a 1×1 block holding E_{ref} and an 18×18 singles block. Its lowest eigenvalue is the smaller of the two numbers. At $U/t \leq 4$ the reference wins and CIS reproduces it exactly. At $U/t = 8$ the singles block, strongly mixed by the large interaction, has an eigenvalue at 2.483, which lies below $E_{\text{ref}} = 4.000$, and the variational principle selects the corresponding state – one that is exactly orthogonal to $|\Phi_0\rangle$. The number is a legitimate upper bound to E_0, but it is not a correlated correction to the reference; the calculation has silently changed which state it describes. The moral is that a lower variational energy does not by itself mean a better description of the state one set out to compute, and that the overlap with the reference should always be monitored alongside the energy.

Answer to e).

For $U/t \gg 1$ the half-filled Hubbard model maps onto a Heisenberg chain with $J = 4t^2/U$, Eq. (4.25), whose ground-state energy per site tends to $-J(\ln 2 - \frac{1}{4})$ in the thermodynamic limit. At $U/t = 8$ and $L = 6$ this estimate gives about -1.3 against the exact -2.05. The discrepancy is expected: the mapping neglects terms of order t^4/U^3, the ring is short, and $U/t = 8$ is not yet deep in the strong-coupling regime. Repeating the comparison at $U/t = 20$ and 40 shows the two converging.

Exercise 6: the exponential wall in practice

a) A shell-model code stores the Hamiltonian matrix in double precision. For a dimension of 10^6 with on average 10^3 non-zero elements per row, and a 4-byte column index per element, how much memory does the sparse matrix need? Compare with storing the full matrix.
b) How large a single-particle space can be treated at half filling if a Lanczos code can handle 10^{10} determinants?
c) A matrix product state with bond dimension D on L sites has about LD^2d parameters, with d the local dimension. For a spin chain with $d = 2$, $L = 100$ and $D = 200$, compare this with 2^L.
d) Discuss what must be true of the physical state for the comparison in c) to be a fair one.

Answer to a).

The sparse matrix holds $10^6 \times 10^3 = 10^9$ elements at $8 + 4 = 12$ bytes each, or 12 GB – large but routine. The full matrix would need 8×10^{12} bytes, 8 TB, and diagonalising it by Householder and QL would take $\mathscr{O}(10^{18})$ floating-point operations. This is the practical content of Section 5.11: the dense algorithms of Sections 1.24 and 1.25 are unusable here, and even the sparse matrix is best not stored at all but regenerated on the fly inside the matrix-vector product.

Answer to b).

From Eq. (5.19), $2^n/\sqrt{n} = 10^{10}$ gives $n \approx 38$, and the exact condition $\binom{n}{n/2} \leq 10^{10}$ gives $n = 36$. A Lanczos calculation at the limit of current hardware therefore reaches roughly 36–40 single-particle states at half filling. Away from half filling the binomial coefficient is much smaller and the reach is greater, which is why calculations near closed shells are so much easier than calculations in the middle of a shell.

Answer to c).

The matrix product state has about $100 \times 200^2 \times 2 = 8 \times 10^6$ parameters, against $2^{100} = 1.27 \times 10^{30}$ amplitudes for a general state – a compression by twenty-four orders of magnitude.

Answer to d).

The comparison is fair only if the state of interest is actually representable at that bond dimension. The Schmidt spectrum across every bond must decay fast enough that keeping D singular values costs little, which is the truncation error of the Eckart-Young theorem in Section 1.30 applied bond by bond. This holds for the ground states of gapped one-dimensional local Hamiltonians, which obey an area law. For critical chains the required D grows polynomially with L; for volume-law states – generic excited states, or the state produced by a quantum quench after a time proportional to L – no polynomial D suffices and the exponential wall stands undiminished. The same caveat applies to every other compressed representation, including the neural-network and quantum-circuit ansaetze of the later chapters: each buys its efficiency by restricting the class of states it can describe, and full configuration interaction remains the only method that makes no such restriction at all.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter05/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/fullci.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter05/`.

`fci.py` Determinants as bit strings, full configuration interaction for the pairing and Hubbard models, truncations, size consistency and the growth of the Hilbert space. Runs in a few seconds.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 6
Mean-field approaches

Chapter 5 showed that the exact answer is a superposition of an exponentially large number of Slater determinants, and that we cannot afford it. This chapter takes the opposite tack. Instead of adding determinants we keep exactly one and ask which one is best. The answer is Hartree-Fock theory, and it is the foundation on which every method in the remaining chapters is built: perturbation theory expands about it, coupled cluster exponentiates corrections to it, and the truncated configuration interaction of Section 5.8 is measured against it.

Chapter 2 already wrote down the Hartree-Fock equations, Eq. (2.41), and deferred their derivation. We supply it here, in two independent ways – by varying the expansion coefficients of the orbitals, and by varying the determinant itself in second quantization. The second route requires Thouless' theorem, which characterises what a "nearby determinant" means, and Thouless' theorem in turn rests on the Baker-Campbell-Hausdorff expansion. That expansion is worth a section of its own, because the same algebra reappears in the Trotter splitting used for time evolution and for quantum simulation later in the book. Armed with it we can ask not merely whether the Hartree-Fock solution is stationary but whether it is a minimum, and we close with the one many-body system for which the whole programme can be carried through analytically: the infinite homogeneous electron gas.

6.1 Why a mean field?

Hartree-Fock theory is an algorithm for finding an approximate ground state of a given Hamiltonian. It has two ingredients. The first is a single-particle basis defined by its own eigenvalue problem,

$$\hat{h}^{\mathrm{HF}}\psi_\alpha = \varepsilon_\alpha \psi_\alpha, \qquad \hat{h}^{\mathrm{HF}} = \hat{t} + \hat{u}_{\mathrm{ext}} + \hat{u}^{\mathrm{HF}}, \tag{6.1}$$

in which $\hat{u}^{\mathrm{HF}}$ is a one-body potential still to be determined. The second is the prescription for determining it: choose $\hat{u}^{\mathrm{HF}}$ so that

$$E^{\mathrm{HF}} = \langle \Phi_0 | \hat{H} | \Phi_0 \rangle \tag{6.2}$$

is stationary, with $|\Phi_0\rangle$ a single Slater determinant built from the N lowest solutions of Eq. (6.1). The variational principle of Section 2.7 then guarantees $E^{\mathrm{HF}} \geq E_0$, so whatever we obtain is an upper bound on the exact ground-state energy.

We shall see that the resulting one-body operator is precisely the operator $\hat{f}$ that appears when the Hamiltonian is normal-ordered with respect to the reference determinant. For a specific matrix element,

$$\langle p|\hat{h}^{\mathrm{HF}}|q\rangle = \langle p|\hat{f}|q\rangle = \langle p|\hat{t} + \hat{u}_{\mathrm{ext}}|q\rangle + \sum_{i\leq F} \langle pi|\hat{v}|qi\rangle_{\mathrm{AS}}, \tag{6.3}$$

so that

$$\langle p|\hat{u}^{\mathrm{HF}}|q\rangle = \sum_{i\leq F}\langle pi|\hat{v}|qi\rangle_{\mathrm{AS}}. \tag{6.4}$$

The sum runs over all single-particle states below the Fermi level, and this is the whole content of the mean-field idea: the two-body interaction, summed over the other fermions, creates an effective one-body field in which a given fermion moves.

Two features of Eq. (6.4) deserve emphasis. It brings an explicit *medium dependence*, since the potential felt by a particle depends on which states are occupied; and it brings an explicit dependence on the interaction, so different interactions give different mean fields even at the same density. For atoms and molecules there is an external potential $\hat{u}_{\mathrm{ext}}$ – the attraction of the nuclei. For nuclei there is none: a nucleus is a self-bound system, held together by the interaction between its own constituents, and the entire one-body field must be generated by Eq. (6.4).

6.2 Variational calculus and Lagrange multipliers

Before deriving the Hartree-Fock equations we recall the machinery, in its simplest setting. The calculus of variations deals with problems in which the quantity to be made stationary is an integral,

$$E[\Phi] = \int_a^b f\left(\Phi(x), \frac{\partial\Phi}{\partial x}, x\right) dx. \tag{6.5}$$

The difficulty is that although f is a known function of Φ, $\partial\Phi/\partial x$ and x, the dependence of Φ on x is not known: the limits are fixed but the path is not. In our case the unknowns are the single-particle wave functions, and the constraints are the integrals expressing their orthonormality. Constraints of this kind are handled with Lagrange multipliers.

Take one particle in three dimensions, with

$$E = \int \psi^* \hat{H}\psi \, dxdydz, \qquad \int \psi^*\psi \, dxdydz = 1, \qquad \hat{H} = -\frac{1}{2}\nabla^2 + V. \tag{6.6}$$

Integrating the kinetic term by parts, with ψ vanishing at the boundaries,

$$\int \psi^* \left(-\frac{1}{2}\nabla^2\right)\psi \, dxdydz = \int \frac{1}{2}\nabla\psi^*\cdot\nabla\psi \, dxdydz, \tag{6.7}$$

so that the constrained variation reads

$$\delta\left\{\int\left(\frac{1}{2}\nabla\psi^*\cdot\nabla\psi + V\psi^*\psi - \lambda\psi^*\psi\right) dxdydz\right\} = 0. \tag{6.8}$$

Writing the integrand as

$$f = \frac{1}{2}\left(\psi_x^*\psi_x + \psi_y^*\psi_y + \psi_z^*\psi_z\right) + V\psi^*\psi - \lambda\psi^*\psi, \tag{6.9}$$

with subscripts denoting partial derivatives, the Euler-Lagrange equation for ψ^*,

$$\frac{\partial f}{\partial\psi^*} - \frac{\partial}{\partial x}\frac{\partial f}{\partial\psi_x^*} - \frac{\partial}{\partial y}\frac{\partial f}{\partial\psi_y^*} - \frac{\partial}{\partial z}\frac{\partial f}{\partial\psi_z^*} = 0, \tag{6.10}$$

gives

$$-\frac{1}{2}\nabla^2\psi + V\psi = \lambda\psi. \tag{6.11}$$

The multiplier is the energy, and the stationarity condition is nothing but the Schrödinger equation. This is the pattern we shall meet again: a constrained variation produces an eigenvalue problem, and the multiplier enforcing normalisation turns out to be the eigenvalue. It happened already in Section 5.3, where the same manipulation produced the full configuration-interaction equation, and it will happen once more below.

6.3 Determinants under a linear change of basis

One property of determinants is needed before we can vary the coefficients, and it is worth isolating because it explains why the whole construction is consistent.

A determinant is linear in each column separately. For a 2×2 example,

$$\begin{vmatrix} \alpha_1 b_{11}+\alpha_2 b_{12} & a_{12} \\ \alpha_1 b_{21}+\alpha_2 b_{22} & a_{22} \end{vmatrix} = \alpha_1 \begin{vmatrix} b_{11} & a_{12} \\ b_{21} & a_{22} \end{vmatrix} + \alpha_2 \begin{vmatrix} b_{12} & a_{12} \\ b_{22} & a_{22} \end{vmatrix}, \tag{6.12}$$

and in general, for an $n\times n$ matrix in which one column is a linear combination $\sum_k c_k b_{jk}$,

$$\det\left(\ldots\ \textstyle\sum_k c_k b_{jk}\ \ldots\right) = \sum_{k=1}^{n} c_k \det\left(\ldots\ b_{jk}\ \ldots\right), \tag{6.13}$$

where the omitted columns are the same on both sides. Applying this to *every* column in turn gives the product rule

$$\det(\boldsymbol{BC}) = \det(\boldsymbol{B})\,\det(\boldsymbol{C}), \tag{6.14}$$

which we already used in Section 2.3. The reader is encouraged to check the case $n=2$ explicitly.

Now let the new orbitals be expanded in a fixed orthonormal basis, as in Eq. (2.17),

$$\psi_p(x) = \sum_\lambda C_{p\lambda}\,\phi_\lambda(x). \tag{6.15}$$

The Slater determinant built from the ψ's is

$$\frac{1}{\sqrt{N!}} \begin{vmatrix} \sum_\lambda C_{p\lambda}\phi_\lambda(x_1) & \ldots & \sum_\lambda C_{p\lambda}\phi_\lambda(x_N) \\ \vdots & & \vdots \\ \sum_\lambda C_{t\lambda}\phi_\lambda(x_1) & \ldots & \sum_\lambda C_{t\lambda}\phi_\lambda(x_N) \end{vmatrix} = \det(\boldsymbol{C})\ \Phi(\phi), \tag{6.16}$$

with $\Phi(\phi)$ the determinant of the original basis functions. If $\boldsymbol{C}$ is unitary then $|\det(\boldsymbol{C})| = 1$ and the two determinants differ by a phase only. Two consequences follow, and both are used constantly.

First, orthonormality is preserved. If $\boldsymbol{v}_j^T\boldsymbol{v}_i = \delta_{ij}$ and $\boldsymbol{w}_i = \boldsymbol{U}\boldsymbol{v}_i$ with $\boldsymbol{U}$ unitary, then

$$\boldsymbol{w}_j^\dagger\boldsymbol{w}_i = \boldsymbol{v}_j^\dagger\boldsymbol{U}^\dagger\boldsymbol{U}\boldsymbol{v}_i = \boldsymbol{v}_j^\dagger\boldsymbol{v}_i = \delta_{ij}, \tag{6.17}$$

which is Section 1.9 restated. So if $\langle\alpha|\beta\rangle = \delta_{\alpha\beta}$ then $\langle p|q\rangle = \delta_{pq}$, and a unitary change of coefficients never spoils the basis we are working in.

Second, a Slater determinant is invariant, up to a phase, under any unitary mixing of its occupied orbitals. The determinant does not know which particular orbitals were used to build it, only which N-dimensional subspace they span. This is why the Hartree-Fock solution is properly a statement about a subspace, and why the individual orbitals are determined only up to such a mixing – the freedom that is fixed, conventionally, by demanding that the Fock matrix be diagonal.

Where this comes from. Every ingredient of this section already appeared in Chapter 1: multilinearity and the product rule in Section 2.3, unitary transformations in Section 1.9, and the practical evaluation of $\det(\boldsymbol{C})$ by LU decomposition in Section 1.15. The last is not an idle remark: in variational Monte Carlo the ratio of two such determinants is evaluated millions of times, which is why Section 2.11 developed the Sherman-Morrison update instead.

6.4 The Hartree-Fock equations by varying the coefficients

We can now carry out the derivation that Chapter 2 postponed. The energy functional of a single determinant, Eq. (2.37), is

$$E[\Phi] = \sum_{\mu=1}^{N} \langle \mu|\hat{h}_0|\mu\rangle + \frac{1}{2}\sum_{\mu=1}^{N}\sum_{\nu=1}^{N} \langle \mu\nu|\hat{v}|\mu\nu\rangle_{\mathrm{AS}}, \tag{6.18}$$

and inserting Eq. (6.15) turns it into a function of the coefficients,

$$E[\Phi^{\mathrm{HF}}] = \sum_{i=1}^{N}\sum_{\alpha\beta} C^*_{i\alpha}C_{i\beta}\langle\alpha|\hat{h}_0|\beta\rangle + \frac{1}{2}\sum_{i,j=1}^{N}\sum_{\alpha\beta\gamma\delta} C^*_{i\alpha}C^*_{j\beta}C_{i\gamma}C_{j\delta}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle_{\mathrm{AS}}. \tag{6.19}$$

Greek indices run over the whole fixed basis, Latin indices over the N occupied orbitals.

The coefficients are not free: orthonormality of the new orbitals requires

$$\langle i|j\rangle = \delta_{ij} = \sum_{\alpha\beta} C^*_{i\alpha}C_{i\beta}\langle\alpha|\beta\rangle = \sum_{\alpha} C^*_{i\alpha}C_{i\alpha}, \tag{6.20}$$

so we introduce one multiplier ε_i per occupied orbital and minimise

$$F[\Phi^{\mathrm{HF}}] = E[\Phi^{\mathrm{HF}}] - \sum_{i=1}^{N}\varepsilon_i\sum_{\alpha} C^*_{i\alpha}C_{i\alpha}. \tag{6.21}$$

The coefficients and their complex conjugates may be varied independently – the real and imaginary parts are two independent real variations, and it is equivalent and more convenient to use C and C^* – so we impose

$$\frac{\partial}{\partial C^*_{i\alpha}}\left[E[\Phi^{\mathrm{HF}}] - \sum_{j}\varepsilon_j\sum_{\beta} C^*_{j\beta}C_{j\beta}\right] = 0. \tag{6.22}$$

Differentiating Eq. (6.19) is straightforward. The one-body term contributes $\sum_\beta C_{i\beta}\langle\alpha|\hat{h}_0|\beta\rangle$. In the two-body term $C^*_{i\alpha}$ appears twice, once as $C^*_{i\alpha}$ and once as $C^*_{j\beta}$ with $j=i$; the two contributions are equal by the antisymmetry of $\langle\alpha\beta|\hat{v}|\gamma\delta\rangle_{\mathrm{AS}}$ under the simultaneous interchange of both pairs, and they cancel the factor $\frac{1}{2}$. The result is

$$\sum_{\beta} C_{i\beta}\langle\alpha|\hat{h}_0|\beta\rangle + \sum_{j=1}^{N}\sum_{\beta\gamma\delta} C^*_{j\beta}C_{j\delta}C_{i\gamma}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle_{\mathrm{AS}} = \varepsilon_i^{\mathrm{HF}}C_{i\alpha}. \tag{6.23}$$

Relabelling the dummy indices so that the coefficient $C_{i\beta}$ can be factored out of both terms,

$$\sum_{\beta}\left\{\langle\alpha|\hat{h}_0|\beta\rangle+\sum_{j=1}^{N}\sum_{\gamma\delta}C^*_{j\gamma}C_{j\delta}\langle\alpha\gamma|\hat{v}|\beta\delta\rangle_{\mathrm{AS}}\right\}C_{i\beta}=\varepsilon_i^{\mathrm{HF}}C_{i\alpha}. \tag{6.24}$$

Defining the Hartree-Fock, or Fock, matrix

$$h^{\mathrm{HF}}_{\alpha\beta}=\langle\alpha|\hat{h}_0|\beta\rangle+\sum_{j=1}^{N}\sum_{\gamma\delta}C^*_{j\gamma}C_{j\delta}\langle\alpha\gamma|\hat{v}|\beta\delta\rangle_{\mathrm{AS}}, \tag{6.25}$$

the condition becomes

$$\boxed{\sum_{\beta}h^{\mathrm{HF}}_{\alpha\beta}C_{i\beta}=\varepsilon_i^{\mathrm{HF}}C_{i\alpha}} \tag{6.26}$$

This is Eq. (2.41), now derived. It is a standard Hermitian eigenvalue problem, solvable by the Householder reduction and QL iteration of Sections 1.24 and 1.25, or by Lanczos, Section 1.26, when the basis is large – with the one complication that the matrix depends on its own eigenvectors.

If the fixed basis is chosen as the eigenbasis of $\hat{h}_0$, which is the usual choice, the first term is diagonal,

$$h^{\mathrm{HF}}_{\alpha\beta}=\varepsilon_\alpha\delta_{\alpha\beta}+\sum_{j=1}^{N}\sum_{\gamma\delta}C^*_{j\gamma}C_{j\delta}\langle\alpha\gamma|\hat{v}|\beta\delta\rangle_{\mathrm{AS}}, \tag{6.27}$$

and Eq. (6.26) can be written compactly as $\boldsymbol{h}^{\mathrm{HF}}\boldsymbol{C}=\boldsymbol{\varepsilon}^{\mathrm{HF}}\boldsymbol{C}$.

An important practical point is visible in Eq. (6.27): no integrals need to be recomputed during the iteration. The matrix elements $\langle\alpha|\hat{h}_0|\beta\rangle$ and $\langle\alpha\gamma|\hat{v}|\beta\delta\rangle_{\mathrm{AS}}$ are tabulated once and for all; every subsequent iteration is a contraction of those tables with the current coefficients, followed by a diagonalisation. The cost of storing them is the n^4 scaling discussed in Section 2.10, and it is the reason the factorisations of Section 1.32 matter.

6.5 The density matrix and the self-consistent field algorithm

The double sum over the occupied orbitals in Eq. (6.27) depends on the coefficients only through the combination

$$\rho_{\gamma\delta}=\sum_{i=1}^{N}\langle\gamma|i\rangle\langle i|\delta\rangle=\sum_{i=1}^{N}C_{i\gamma}C^*_{i\delta}, \tag{6.28}$$

the one-body *density matrix*, which is precisely the object introduced in Section 1.31. It is the projector onto the occupied subspace, $\boldsymbol{\rho}^2=\boldsymbol{\rho}$ and $\mathrm{tr}\,\boldsymbol{\rho}=N$, which is the formal statement of the invariance found in Section 6.3: the Fock matrix depends on the occupied subspace, not on the particular orbitals chosen to span it. With it,

$$h^{\mathrm{HF}}_{\alpha\beta}=\varepsilon_\alpha\delta_{\alpha\beta}+\sum_{\gamma\delta}\rho_{\gamma\delta}\langle\alpha\gamma|\hat{v}|\beta\delta\rangle_{\mathrm{AS}}. \tag{6.29}$$

Building $\boldsymbol{\rho}$ once per iteration and contracting it with the stored two-body table is the efficient way to organise the calculation.

The algorithm is then an iteration of the kind discussed in Section 1.17, with a diagonalisation replacing a single sweep.

1. Start from a guess, conventionally $C^{(0)}_{i\alpha}=\delta_{i\alpha}$, which makes the reference determinant the one built from the N lowest eigenstates of $\hat{h}_0$. Random orthonormal vectors work too, and

are sometimes preferable because they avoid convergence onto a symmetry-constrained saddle point.
2. Build $\boldsymbol{\rho}$ from Eq. (6.28) and $\boldsymbol{h}^{\mathrm{HF}}$ from Eq. (6.29).
3. Diagonalise, obtaining $C^{(1)}_{i\alpha}$ and $\varepsilon_i^{\mathrm{HF}(1)}$.
4. Repeat from step 2 until the single-particle energies stop moving,

$$\frac{1}{m}\sum_p \left|\varepsilon_p^{(n)} - \varepsilon_p^{(n-1)}\right| \leq \lambda, \tag{6.30}$$

with m the number of single-particle states and λ a prescribed tolerance, typically 10^{-8} or smaller.

The program `hartreefock.py` implements exactly this loop. For the system of Section 2.9 – spinless fermions in a one-dimensional oscillator trap with a softened Coulomb repulsion, in a basis of eight oscillator orbitals – it reproduces Table 2.2, and the gain over the unoptimised determinant grows with the particle number:

N	iterations	E(reference)	E(Hartree-Fock)	gain
2	17	2.69018342	2.66308878	-2.71×10^{-2}
3	17	6.46801021	6.39016133	-7.78×10^{-2}
4	17	11.77084712	11.62305385	-1.48×10^{-1}

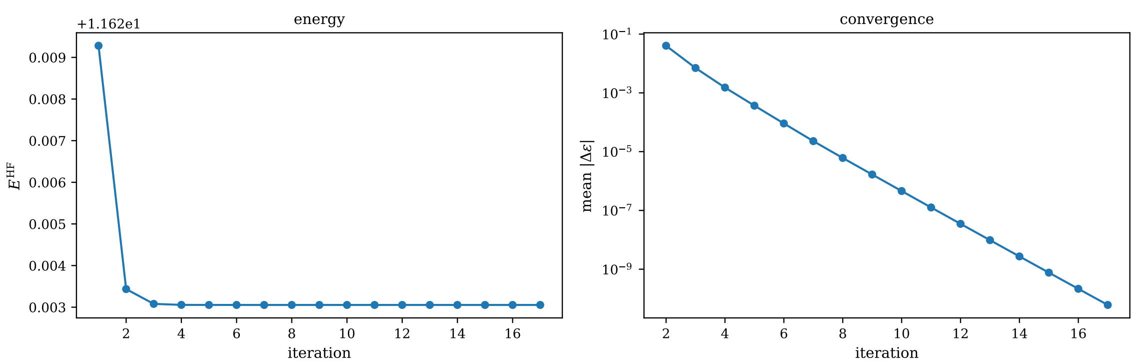

Fig. 6.1 The self-consistent field loop for four spinless fermions in a one-dimensional oscillator trap with a softened Coulomb repulsion, in a basis of eight oscillator orbitals. The left panel gives the Hartree-Fock energy E^{HF} obtained after each diagonalisation, on a vertical scale offset by 11.62; the right panel gives the mean drift of the single-particle energies, $m^{-1}\sum_p|\varepsilon_p^{(n)}-\varepsilon_p^{(n-1)}|$ of Eq. (6.30), on a logarithmic scale. The energy has settled to the width of the plot after three iterations, while the orbitals need all seventeen to bring the drift below 10^{-9}.

The two panels of Figure 6.1 say rather different things about the same $N=4$ calculation, and the difference is worth dwelling on. The very first diagonalisation already brings the energy from the reference value 11.77085 down to 11.62925, which is 96% of the total gain of -1.48×10^{-1}; by the third iteration the curve in the left panel is flat on any scale a reader can resolve. The right panel, plotted logarithmically, shows that nothing of the sort has happened to the orbitals: the mean drift falls from about 4×10^{-2} at the second iteration to some 10^{-10} at the seventeenth along a straight line, that is by a fixed factor of roughly four per iteration. This is linear convergence of exactly the kind met for the stationary iterative solvers of Section 1.17 – the self-consistency loop is a fixed-point iteration, and its rate is set by the spectral radius of the map, not by any property of the energy. The reason the energy looks so much better behaved is the variational principle itself: at a stationary point the energy is quadratic in the displacement of the orbitals, so a drift of 10^{-5} in the single-particle energies shows up as something of order 10^{-10} in E^{HF}. The practical lesson is that one must never use the energy as the convergence criterion. It is converged long before the density matrix is, and quantities that depend on the orbitals rather than on the energy – the Fock matrix elements f_{ai} below, transition matrix elements, anything fed into a subsequent coupled-cluster or perturbation calculation – are only as accurate as the right-hand panel says they are.

Brillouin's theorem.

At convergence the Fock matrix is diagonal in its own eigenbasis, so its occupied-unoccupied block vanishes,

$$f_{ai} = 0 \qquad \Longleftrightarrow \qquad \langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = 0. \tag{6.31}$$

The program reports $\max|f_{ai}| = 8.7 \times 10^{-11}$ for $N = 4$, at the prescribed tolerance. This is Brillouin's theorem, met first in Section 3.14 and used in Section 5.4 to empty the $0p0h$–$1p1h$ block of the configuration-interaction matrix. We can now see what it means: the Hartree-Fock solution is the choice of single-particle basis that decouples the reference from all single excitations. Everything left over is carried by the doubles and higher, which is why Eq. (5.30) contained only doubles and why both perturbation theory and coupled-cluster theory start there.

6.6 Hartree-Fock in second quantization

The derivation of Section 6.4 varied the orbitals. We now vary the determinant, which is more abstract but far better suited to the question of stability, and which is the natural language given Chapter 3.

Our ansatz is the determinant

$$|\Phi_0\rangle = |c\rangle = \hat{a}_i^\dagger \hat{a}_j^\dagger \cdots \hat{a}_l^\dagger |0\rangle, \tag{6.32}$$

and we seek $\hat{u}^{\mathrm{HF}}$ such that $E_0^{\mathrm{HF}} = \langle c|\hat{H}|c\rangle$ is a local minimum. The stationarity condition is that the energy does not change to first order under any variation of the determinant,

$$\langle \delta c|\hat{H}|c\rangle = 0 \tag{6.33}$$

for every admissible $|\delta c\rangle$. To make this precise we must know what the admissible variations are – that is, what the neighbourhood of a Slater determinant looks like inside the space of Slater determinants. That is the content of Thouless' theorem.

6.7 Thouless' theorem

Theorem 6.1 (Thouless). *Any Slater determinant $|c'\rangle$ that is not orthogonal to a given determinant $|c\rangle = \prod_{i=1}^{n} \hat{a}_{\alpha_i}^\dagger |0\rangle$ can be written as*

$$|c'\rangle = \exp\left\{ \sum_{a>F} \sum_{i\le F} C_{ai} \hat{a}_a^\dagger \hat{a}_i \right\} |c\rangle. \tag{6.34}$$

In words: every determinant in the neighbourhood of $|c\rangle$ is obtained from it by an exponentiated linear combination of one-particle-one-hole excitations. This is a strong statement. It says that the manifold of Slater determinants is parametrised, near any point, by the $(n-N)\times N$ amplitudes C_{ai} and by nothing else – so the Hartree-Fock variation has exactly that many degrees of freedom.

Step one: the exponential collapses.

The exponential of a sum of operators factorises into a product of exponentials only when the operators commute. When they do not, the Baker-Campbell-Hausdorff formula of Section 6.8 supplies the correction. Here we are lucky: for particle indices $a,b>F$ and hole indices $i,j\leq F$, the two index sets are disjoint, and a short calculation with the anticommutation relations of Chapter 3 gives

$$\left[\hat{a}_a^\dagger \hat{a}_i,\, \hat{a}_b^\dagger \hat{a}_j\right] = 0. \tag{6.35}$$

Indeed, since $i\neq b$ we may write $\hat{a}_i\hat{a}_b^\dagger = -\hat{a}_b^\dagger\hat{a}_i$, and likewise $\hat{a}_j\hat{a}_a^\dagger = -\hat{a}_a^\dagger\hat{a}_j$; both orderings then reduce to $-\hat{a}_a^\dagger\hat{a}_b^\dagger\hat{a}_i\hat{a}_j$ and the commutator vanishes. Writing

$$\hat{T} = \sum_{i\leq F}\hat{T}_i, \qquad \hat{T}_i = \sum_{a>F} C_{ai}\hat{a}_a^\dagger\hat{a}_i, \tag{6.36}$$

we therefore have $e^{\hat{T}} = \prod_i e^{\hat{T}_i}$. Moreover each factor truncates, because

$$\hat{T}_i^2 = \sum_{ab>F} C_{ai}C_{bi}\,\hat{a}_a^\dagger\hat{a}_i\hat{a}_b^\dagger\hat{a}_i = 0, \tag{6.37}$$

since the rightmost $\hat{a}_i$ empties the hole i and the second one then annihilates the state – an instance of $(\hat{a}_i)^n = 0$ for $n>1$. Hence

$$|c'\rangle = \prod_{i\leq F}\left\{1+\sum_{a>F} C_{ai}\hat{a}_a^\dagger\hat{a}_i\right\}|c\rangle. \tag{6.38}$$

Note carefully that the product in Eq. (6.38) is *not* the same as $1+\hat{T}$: the cross terms $\hat{T}_i\hat{T}_j$ with $i\neq j$ survive and describe $2p$-$2h$ admixtures. Dropping them is legitimate only for infinitesimal amplitudes, which is exactly the situation in the stability analysis below.

Step two: the product is a determinant.

Insert $|c\rangle = \hat{a}_{i_1}^\dagger\hat{a}_{i_2}^\dagger\cdots\hat{a}_{i_n}^\dagger|0\rangle$ into Eq. (6.38). The factor labelled i_1 acts only on $\hat{a}_{i_1}^\dagger$, the factor labelled i_2 only on $\hat{a}_{i_2}^\dagger$, and so on, so the product reorganises into

$$\begin{aligned}|c'\rangle &= \left(1+\sum_{a>F} C_{ai_1}\hat{a}_a^\dagger\hat{a}_{i_1}\right)\hat{a}_{i_1}^\dagger\left(1+\sum_{a>F} C_{ai_2}\hat{a}_a^\dagger\hat{a}_{i_2}\right)\hat{a}_{i_2}^\dagger\cdots|0\rangle\\ &= \prod_{i\leq F}\left(\hat{a}_i^\dagger+\sum_{a>F} C_{ai}\hat{a}_a^\dagger\right)|0\rangle.\end{aligned} \tag{6.39}$$

Defining a new creation operator

$$\hat{b}_i^\dagger = \hat{a}_i^\dagger + \sum_{a>F} C_{ai}\hat{a}_a^\dagger, \tag{6.40}$$

we have $|c'\rangle = \prod_i \hat{b}_i^\dagger |0\rangle$, which has exactly the form of a Slater determinant. The rotated state is a single determinant, not a superposition: this is the whole point.

Step three: it is the general determinant.

Equation (6.40) looks less general than it should, because the sum runs only over states above the Fermi level. Consider therefore the most general one,

$$\tilde{b}_i^\dagger = \sum_p f_{ip}\hat{a}_p^\dagger, \qquad |\tilde{c}\rangle = \prod_i \tilde{b}_i^\dagger |0\rangle, \tag{6.41}$$

with f unitary, and require only that $\langle c|\tilde{c}\rangle \neq 0$. Expanding,

$$\langle c|\tilde{c}\rangle = \langle 0|\hat{a}_{i_n}\cdots\hat{a}_{i_1}\left(\sum_p f_{i_1 p}\hat{a}_p^\dagger\right)\cdots\left(\sum_t f_{i_n t}\hat{a}_t^\dagger\right)|0\rangle = \det\left(f_{ip}\right)_{i,p\leq F}, \tag{6.42}$$

the determinant of the occupied block of f: only the terms with p below the Fermi level survive the vacuum expectation value, and summing over the permutations reconstructs the determinant. The hypothesis $\langle c|\tilde{c}\rangle \neq 0$ says exactly that this determinant is non-zero, so the occupied block is invertible, and intermediate normalisation lets us rescale to $\det(f_{ip}) = 1$. With f^{-1} available we can form

$$\sum_i f_{ki}^{-1}\tilde{b}_i^\dagger = \sum_i\sum_p f_{ki}^{-1} f_{ip}\hat{a}_p^\dagger = \hat{a}_k^\dagger + \sum_{p>F}\left(\sum_{i\leq F} f_{ki}^{-1} f_{ip}\right)\hat{a}_p^\dagger, \tag{6.43}$$

where the terms with $p \leq F$ have collapsed to δ_{kp}. Setting $C_{pk} = \sum_{i\leq F} f_{ki}^{-1} f_{ip}$ returns Eq. (6.40) exactly. Since a unitary mixing of the $\tilde{b}^\dagger$ changes the determinant only by a factor, as Section 6.3 showed, $|\tilde{c}\rangle$ and $|c'\rangle$ are the same state. The theorem is proved.

Numerical check.

The program `hartreefock.py` verifies each step explicitly for six orbitals and three particles, with random amplitudes of size 0.3: $\max|[\hat{T}_i,\hat{T}_j]| = 2.8\times10^{-17}$, $\max|\hat{T}_i^2| = 0$, $|c'\rangle$ agrees with $\prod_i \hat{b}_i^\dagger|0\rangle$ to 5.6×10^{-17}, the overlap $\langle c|c'\rangle$ is exactly one, and $\det(f_{ip}) = 1$. The difference between $e^{\hat{T}}|c\rangle$ and $(1+\hat{T})|c\rangle$ falls by a factor of four each time the amplitudes are halved, confirming that the discarded piece is second order.

What it buys us.

Thouless' theorem lets us write any nearby determinant as

$$|c'\rangle = |c\rangle + |\delta c\rangle, \qquad |c'\rangle \simeq \left(1 + \sum_{ai}\delta C_{ai}\hat{a}_a^\dagger\hat{a}_i\right)|c\rangle \tag{6.44}$$

to first order in the amplitudes. The stationarity condition (6.33) then reads

$$\sum_{ai}\delta C_{ai}^*\langle c|\hat{a}_i^\dagger\hat{a}_a\hat{H}|c\rangle = 0 \quad \text{for arbitrary } \delta C \qquad \Longrightarrow \qquad \langle\Phi_i^a|\hat{H}|\Phi_0\rangle = 0, \tag{6.45}$$

which is Brillouin's theorem again, Eq. (6.31). The two derivations therefore agree: varying the orbitals and varying the determinant give the same equations. What the second route adds is the ability to go to second order, which is the subject of Section 6.10.

6.8 The Baker-Campbell-Hausdorff expansion

Thouless' theorem needed only the trivial case of the following result, but the general case is used repeatedly later – in the similarity transformation of coupled-cluster theory, in the interaction picture, and in the Trotter splitting of the next section – so we develop it here.

For two operators X and Y that do not commute, $e^X e^Y \neq e^{X+Y}$. The Baker-Campbell-Hausdorff formula states that the product is nevertheless a single exponential, $e^X e^Y = e^Z$, with

$$Z = X + Y + \frac{1}{2}[X,Y] + \frac{1}{12}\Big([X,[X,Y]] + [Y,[Y,X]]\Big) - \frac{1}{24}\big[Y,[X,[X,Y]]\big] + \cdots \tag{6.46}$$

Every term beyond the first is a nested commutator of X and Y. No ordinary products appear, which is the essential structural statement: if X and Y belong to a Lie algebra, so does Z. The coefficients $\frac{1}{2}, \frac{1}{12}, \frac{1}{24}, \ldots$ involve Bernoulli numbers and were first given in general by Dynkin.

Derivation of the first terms.

Expand both sides in powers and match order by order. With $Z = X + Y + A_2 + A_3 + \cdots$, where A_n collects the terms of total degree n,

$$e^X e^Y = I + (X+Y) + \frac{1}{2}\left(X^2 + 2XY + Y^2\right) + \mathscr{O}(3), \qquad e^Z = I + Z + \frac{1}{2}Z^2 + \mathscr{O}(3). \tag{6.47}$$

At first order, A_1 is absent and $Z^{(1)} = X + Y$. At second order the left-hand side supplies $\frac{1}{2}(X^2 + 2XY + Y^2)$ while $\frac{1}{2}(X+Y)^2 = \frac{1}{2}(X^2 + XY + YX + Y^2)$; the difference is $\frac{1}{2}(XY - YX) = \frac{1}{2}[X,Y]$, which must therefore be supplied by A_2,

$$A_2 = \frac{1}{2}[X,Y]. \tag{6.48}$$

The same procedure at third order, using the Jacobi identity $[X,[Y,W]] + [Y,[W,X]] + [W,[X,Y]] = 0$ to reduce the independent structures, gives

$$A_3 = \frac{1}{12}\big[X,[X,Y]\big] - \frac{1}{12}\big[Y,[X,Y]\big] = \frac{1}{12}\Big([X,[X,Y]] + [Y,[Y,X]]\Big). \tag{6.49}$$

When the series terminates.

If $[X,Y]$ commutes with both X and Y – in particular if it is a multiple of the identity, $[X,Y] = cI$ – then every higher commutator vanishes and

$$e^X e^Y = \exp\left(X + Y + \tfrac{1}{2}[X,Y]\right) \tag{6.50}$$

is exact. This is the case for canonically conjugate operators, $[\hat{x},\hat{p}] = i\hbar$, from which the Weyl form of the displacement operator follows, and for the ladder operators of the harmonic oscillator, $[\hat{a},\hat{a}^\dagger] = 1$. It is also, trivially, the case in Thouless' theorem, where by Eq. (6.35) the commutator is zero and the series stops at the first term.

How good is a truncation?

For the two nilpotent matrices $X = \begin{pmatrix} 0 & 0.1 \\ 0 & 0 \end{pmatrix}$ and $Y = \begin{pmatrix} 0 & 0 \\ 0.1 & 0 \end{pmatrix}$, `hartreefock.py` gives

order	terms kept	$\lVert e^X e^Y - e^{Z_n} \rVert$
1	$X+Y$	7.071×10^{-3}
2	$+\,\frac{1}{2}[X,Y]$	2.379×10^{-4}
3	$+\,\frac{1}{12}\big([X,[X,Y]]+[Y,[Y,X]]\big)$	1.190×10^{-5}
4	$-\,\frac{1}{24}\big[Y,[X,[X,Y]]\big]$	4.757×10^{-7}

Each additional commutator gains between one and two orders of magnitude for amplitudes of this size. The series is asymptotic rather than convergent in general, and the gain per order shrinks as the operators grow; the practical lesson is that BCH is a small-parameter expansion, and the small parameter is the size of the commutator relative to the operators themselves.

6.9 The Trotter-Suzuki splitting

The most common use of Eq. (6.46) is in reverse. Suppose we can exponentiate H_1 and H_2 separately but not their sum – the standard situation, with H_1 the kinetic energy, diagonal in momentum space, and H_2 the potential, diagonal in coordinate space. Setting $X = -iH_1\Delta$ and $Y = -iH_2\Delta$ in Eq. (6.46),

$$e^{-iH_1\Delta} e^{-iH_2\Delta} = \exp\left(-i(H_1+H_2)\Delta - \frac{1}{2}[H_1,H_2]\Delta^2 + \mathscr{O}(\Delta^3)\right), \tag{6.51}$$

so a single split step is correct to first order in Δ, with an error governed by the commutator. Dividing the evolution into N steps of length t/N gives the Lie-Trotter product formula

$$e^{-i(H_1+H_2)t} = \lim_{N\to\infty} \left(e^{-iH_1 t/N} e^{-iH_2 t/N}\right)^N, \tag{6.52}$$

whose error at finite N is $\mathscr{O}(t^2/N)$.

Symmetrising the step cancels the leading commutator. The second-order, or Strang, splitting

$$S_2(\Delta) = e^{-iH_1\Delta/2}\, e^{-iH_2\Delta}\, e^{-iH_1\Delta/2} = e^{-i(H_1+H_2)\Delta} + \mathscr{O}(\Delta^3) \tag{6.53}$$

has a local error governed by the double commutators $[H_1,[H_1,H_2]]$ and $[H_2,[H_1,H_2]]$, hence a global error $\mathscr{O}(t^3/N^2)$. Suzuki's recursive construction generalises this to any even order, at the cost of more exponentials per step and, from fourth order on, of steps with negative coefficients.

For a single qubit with $\hat{H} = \sigma_x + \sigma_z$, so that $[\sigma_x, \sigma_z] = -2i\sigma_y \neq 0$, and $t = 1$, `hartreefock.py` finds

N	first order	ratio	second order	ratio
1	1.130×10^{0}	—	4.436×10^{-1}	—
2	5.125×10^{-1}	2.21	1.006×10^{-1}	4.41
4	2.493×10^{-1}	2.06	2.443×10^{-2}	4.12
8	1.238×10^{-1}	2.01	6.061×10^{-3}	4.03
16	6.177×10^{-2}	2.00	1.513×10^{-3}	4.01
32	3.087×10^{-2}	2.00	3.779×10^{-4}	4.00
64	1.543×10^{-2}	2.00	9.448×10^{-5}	4.00

Doubling N halves the first-order error and quarters the second-order one, exactly as the leading terms of Eq. (6.46) predict. The ratios approach their asymptotic values from above, which is the usual signature of higher-order terms still contributing at small N.

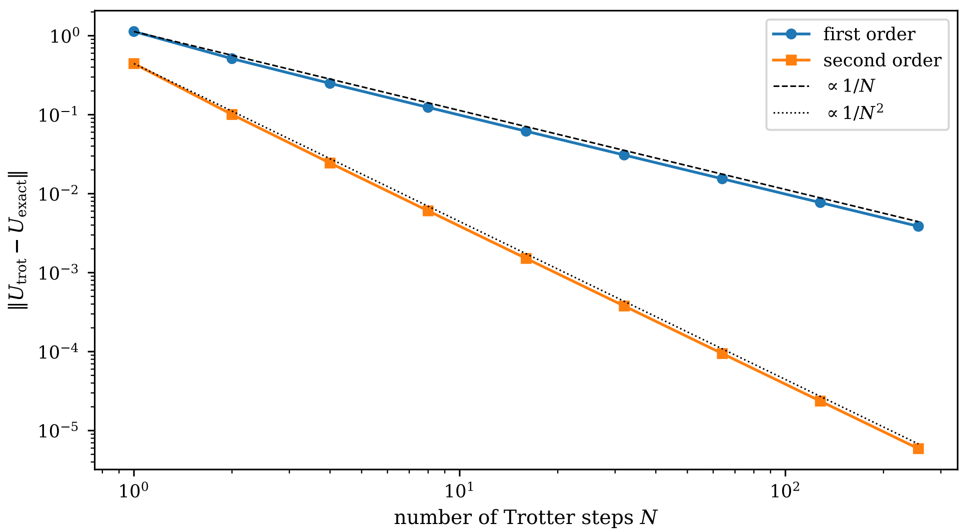

Fig. 6.2 Spectral-norm error $\|U_{\text{trot}} - U_{\text{exact}}\|$ of the Lie-Trotter product (6.52) and of the Strang splitting (6.53) for a single qubit with $\hat{H} = \sigma_x + \sigma_z$ propagated to $t = 1$, against the number of Trotter steps N, on logarithmic scales. The two straight reference lines are $e_1(1)/N$ and $e_2(1)/N^2$, anchored on the respective one-step errors. Both sequences settle onto their asymptotic slopes of -1 and -2 after a couple of halvings of the step, and at $N = 256$ the symmetrised scheme is nearly three orders of magnitude more accurate than the unsymmetrised one.

Figure 6.2 carries the same experiment two decades further, to $N = 256$, and puts it on logarithmic axes where the orders of the two schemes become slopes. Both sets of points fall on straight lines, of slope -1 for the Lie-Trotter product and -2 for the Strang splitting, and both lie slightly *below* the reference lines drawn through their own one-step errors – by some 12% at $N = 256$ – which is the graphical version of the ratios in the table approaching 2 and 4 from above. The vertical gap between the two curves is the whole point. At $N = 1$ the symmetrised scheme is better by a factor of 2.5, which is barely worth the extra exponential; at $N = 256$ the errors are 3.9×10^{-3} and 5.8×10^{-6}, a factor of almost 700. Read the other way round, an accuracy of 10^{-3} costs about $N \simeq 10^3$ first-order steps but only $N \simeq 20$ second-order ones, and since a Strang step contains three exponentials rather than two, the symmetrised scheme reaches that accuracy with roughly thirty times fewer gates. That ratio grows without bound as the tolerance is tightened, which is why no serious real-time or quantum-simulation calculation is ever done with the first-order formula, and why the search for higher-order and randomised product formulas is an active subject in its own right.

Why this belongs in a chapter on mean fields. The immediate reason is that Thouless' theorem is a statement about an exponential of one-body operators, and the reason it is so simple is that those operators commute. The longer-range reason is that the same expansion controls three later developments. In coupled-cluster theory the similarity-transformed Hamiltonian $e^{-\hat{T}}\hat{H}e^{\hat{T}}$ is expanded by the Baker-Hausdorff lemma $e^{-X}Ye^{X} = Y - [X,Y] + \frac{1}{2!}[X,[X,Y]] - \cdots$, and terminates exactly at the fourth commutator because $\hat{H}$ contains at most two-body terms. In time-dependent Hartree-Fock and in real-time methods generally, the propagator is built by splitting. And in quantum computing the Trotter formula is how a Hamiltonian is turned into a circuit: each exponential $e^{-iH_j\Delta}$ becomes a gate, Eq. (6.52) fixes how many of them are needed, and the trade-off between circuit depth and accuracy is precisely the trade-off between N and the error above. A first-order scheme needs $N = \mathscr{O}(t^2/\varepsilon)$ steps for an accuracy ε; a second-order scheme needs only $N = \mathscr{O}(t^{3/2}/\sqrt{\varepsilon})$.

The pieces H_j that are split apart in that last application are the Pauli strings of the Jordan-Wigner transformation of Section 3.16. Writing $\hat{H} = \sum_\alpha h_\alpha P_\alpha$ as in Eq. (3.138), each factor $e^{-ih_\alpha P_\alpha \Delta}$ is a rotation generated by a single Pauli string and has a standard circuit whose depth is set by the weight of P_α. The number of strings is therefore the number of factors per Trotter step, and their weights are the entangling cost of each – which is why Table 3.1 and Table 4.7 were worth computing. Strings that commute with one another can be exponentiated together with no splitting error at all, and grouping them is one of the main practical optimisations.

6.10 Stability of the Hartree-Fock solution

The variational condition (6.45) guarantees only that $\langle c|\hat{H}|c\rangle$ is *stationary*. A stationary point may be a minimum, a maximum or a saddle, and which one it is matters: a Hartree-Fock solution sitting on a saddle point is a poor reference for perturbation theory or coupled cluster, because the corrections it generates are organised about the wrong state. Thouless' theorem lets us settle the question by going to second order.

We ask whether

$$\frac{\langle c'|\hat{H}|c'\rangle}{\langle c'|c'\rangle} \;\geq\; \langle c|\hat{H}|c\rangle = E_0 \tag{6.54}$$

for every nearby determinant $|c'\rangle = |c\rangle + |\delta c\rangle$. By Thouless' theorem,

$$|c'\rangle = \exp\left\{\sum_{a>F}\sum_{i\leq F}\delta C_{ai}\hat{a}_a^\dagger\hat{a}_i\right\}|c\rangle = \left\{1 + \sum_{ai}\delta C_{ai}\hat{a}_a^\dagger\hat{a}_i + \frac{1}{2!}\sum_{abij}\delta C_{ai}\delta C_{bj}\hat{a}_a^\dagger\hat{a}_i\hat{a}_b^\dagger\hat{a}_j + \cdots\right\}|c\rangle, \tag{6.55}$$

with the amplitudes δC small. With intermediate normalisation $\langle c'|c\rangle = 1$ the norm is

$$\langle c'|c'\rangle = 1 + \sum_{a>F}\sum_{i\leq F}|\delta C_{ai}|^2 + \mathscr{O}(\delta C^3). \tag{6.56}$$

The energy to second order.

Expanding the numerator and using the Hartree-Fock condition (6.45) to kill the terms linear in δC,

$$\begin{aligned}\langle c'|\hat{H}|c'\rangle &= \langle c|\hat{H}|c\rangle + \sum_{ab>F}\sum_{ij\le F}\delta C_{ai}^{*}\delta C_{bj}\langle c|\hat{a}_i^{\dagger}\hat{a}_a\hat{H}\hat{a}_b^{\dagger}\hat{a}_j|c\rangle \\ &\quad + \frac{1}{2!}\sum_{ab>F}\sum_{ij\le F}\delta C_{ai}\delta C_{bj}\langle c|\hat{H}\hat{a}_a^{\dagger}\hat{a}_i\hat{a}_b^{\dagger}\hat{a}_j|c\rangle \\ &\quad + \frac{1}{2!}\sum_{ab>F}\sum_{ij\le F}\delta C_{ai}^{*}\delta C_{bj}^{*}\langle c|\hat{a}_j^{\dagger}\hat{a}_b\hat{a}_i^{\dagger}\hat{a}_a\hat{H}|c\rangle + \cdots \end{aligned} \tag{6.57}$$

Each of these expectation values is evaluated with Wick's theorem, Section 3.10. The first of them is the $1p$-$1h$ matrix element computed in Section 3.14,

$$\begin{aligned}\langle c|\{\hat{a}_i^{\dagger}\hat{a}_a\}\hat{H}\{\hat{a}_b^{\dagger}\hat{a}_j\}|c\rangle &= \sum_{pq}\langle p|\hat{h}_0|q\rangle\langle c|\{\hat{a}_i^{\dagger}\hat{a}_a\}\{\hat{a}_p^{\dagger}\hat{a}_q\}\{\hat{a}_b^{\dagger}\hat{a}_j\}|c\rangle \\ &\quad + \frac{1}{4}\sum_{pqrs}\langle pq|\hat{v}|rs\rangle\langle c|\{\hat{a}_i^{\dagger}\hat{a}_a\}\{\hat{a}_p^{\dagger}\hat{a}_q^{\dagger}\hat{a}_s\hat{a}_r\}\{\hat{a}_b^{\dagger}\hat{a}_j\}|c\rangle, \end{aligned} \tag{6.58}$$

and collecting the surviving contractions gives

$$E_0\sum_{ai}|\delta C_{ai}|^2 + \sum_{ai}|\delta C_{ai}|^2(\varepsilon_a-\varepsilon_i) - \sum_{ijab}\langle aj|\hat{v}|bi\rangle\,\delta C_{ai}^{*}\delta C_{bj}. \tag{6.59}$$

The third line of Eq. (6.57) is the Hermitian conjugate of the second, since $\langle a|\hat{A}|b\rangle = \left(\langle b|\hat{A}^{\dagger}|a\rangle\right)^{*}$ and $\hat{H}$ is Hermitian, and the second evaluates to

$$\frac{1}{2}\sum_{ijab}\langle ab|\hat{v}|ij\rangle\,\delta C_{ai}^{*}\delta C_{bj}^{*}. \tag{6.60}$$

The stability matrix.

Define the two antisymmetrised blocks

$$A_{ai,bj} = -\langle aj|\hat{v}|bi\rangle_{\mathrm{AS}}, \qquad B_{ai,bj} = \langle ab|\hat{v}|ij\rangle_{\mathrm{AS}}, \tag{6.61}$$

together with the diagonal matrix of unperturbed excitation energies

$$\Delta_{ai,bj} = (\varepsilon_a-\varepsilon_i)\,\delta_{ab}\delta_{ij}. \tag{6.62}$$

Then Eqs. (6.59) and (6.60) combine into

$$\begin{aligned}\langle c'|\hat{H}|c'\rangle &= \left(1+\sum_{ai}|\delta C_{ai}|^2\right)\langle c|\hat{H}|c\rangle + \sum_{ai}|\delta C_{ai}|^2\left(\varepsilon_a^{\mathrm{HF}}-\varepsilon_i^{\mathrm{HF}}\right) + \sum_{ijab}A_{ai,bj}\,\delta C_{ai}^{*}\delta C_{bj} \\ &\quad + \frac{1}{2}\sum_{ijab}B_{ai,bj}^{*}\,\delta C_{ai}\delta C_{bj} + \frac{1}{2}\sum_{ijab}B_{ai,bj}\,\delta C_{ai}^{*}\delta C_{bj}^{*} + \mathscr{O}(\delta C^3), \end{aligned} \tag{6.63}$$

so that, dividing by the norm (6.56),

$$\frac{\langle c'|\hat{H}|c'\rangle}{\langle c'|c'\rangle} = E_0 + \frac{\Delta E}{1+\sum_{ai}|\delta C_{ai}|^2}, \qquad \Delta E = \frac{1}{2}\langle\chi|\hat{M}|\chi\rangle, \tag{6.64}$$

with the composite vector and matrix

$$\chi = \left[\delta C\ \delta C^{*}\right]^{T}, \qquad \hat{M} = \begin{pmatrix}\Delta + A & B \\ B^{*} & \Delta + A^{*}\end{pmatrix}. \tag{6.65}$$

The conclusion is now immediate. The denominator in Eq. (6.64) is positive, so the Hartree-Fock solution is a local minimum if and only if

$$\boxed{\Delta E = \frac{1}{2}\langle\chi|\hat{M}|\chi\rangle \ \geq\ 0 \quad \text{for every } \chi} \tag{6.66}$$

that is, if and only if every eigenvalue of $\hat{M}$ is non-negative. A necessary but not sufficient condition, obtained by restricting χ to the upper block, is that

$$(\varepsilon_a - \varepsilon_i)\,\delta_{ab}\delta_{ij} + A_{ai,bj} \ \geq\ 0 \tag{6.67}$$

as a matrix. This is cheap to test and is the usual first check.

The stability matrix is the RPA matrix. The upper-left block $\Delta + A$ is exactly the matrix diagonalised in the Tamm-Dancoff approximation: it is the Hamiltonian restricted to the $1p$-$1h$ space, measured from the reference energy. The full $\hat{M}$, with its off-diagonal B blocks, is the matrix of the random-phase approximation. So the statement “the Hartree-Fock solution is stable” and the statement “the RPA frequencies are real” are one and the same, and an imaginary RPA frequency is not a numerical accident but a signal that the mean field itself is wrong. This is why Chapter 5 could describe TDA and RPA as truncations of full configuration interaction and why they belong immediately after this chapter: they are the second derivative of the Hartree-Fock energy surface.

6.11 An explicit instability: the Lipkin model

The Lipkin model of Section 4.1 is the standard illustration, because everything can be computed in closed form. Take $W = 0$, so that

$$\hat{H} = \varepsilon\hat{J}_z + \frac{V}{2}\left(\hat{J}_+^2 + \hat{J}_-^2\right), \tag{6.68}$$

with N particles in two N-fold degenerate levels separated by ε. The natural mean-field ansatz rotates every single-particle state by the same angle α in the two-level space,

$$\hat{b}_p^\dagger = \cos\left(\tfrac{\alpha}{2}\right)\hat{a}_{p-}^\dagger + \sin\left(\tfrac{\alpha}{2}\right)\hat{a}_{p+}^\dagger, \tag{6.69}$$

which is a Thouless rotation with $C_{ai} = \tan(\alpha/2)\,\delta_{pq}$. The resulting energy is the classic result

$$\frac{E(\alpha)}{N} = -\frac{\varepsilon}{2}\cos\alpha - \frac{\varepsilon\chi}{4}\sin^2\alpha, \qquad \chi = \frac{|V|(N-1)}{\varepsilon}, \tag{6.70}$$

where χ is the dimensionless coupling. Differentiating,

$$\frac{1}{N}\frac{dE}{d\alpha} = \frac{\varepsilon}{2}\sin\alpha\,(1 - \chi\cos\alpha), \tag{6.71}$$

so there are two stationary points: the *spherical* solution $\alpha = 0$, which exists always, and a *deformed* solution $\cos\alpha = 1/\chi$, which exists only for $\chi \geq 1$. The second derivative at $\alpha = 0$ is

$$\frac{1}{N}\left.\frac{d^2E}{d\alpha^2}\right|_{\alpha=0} = \frac{\varepsilon}{2}(1-\chi), \tag{6.72}$$

which changes sign at $\chi = 1$. The spherical solution turns from a minimum into a maximum exactly where the deformed solution appears, which is the signature of a continuous symmetry-breaking transition.

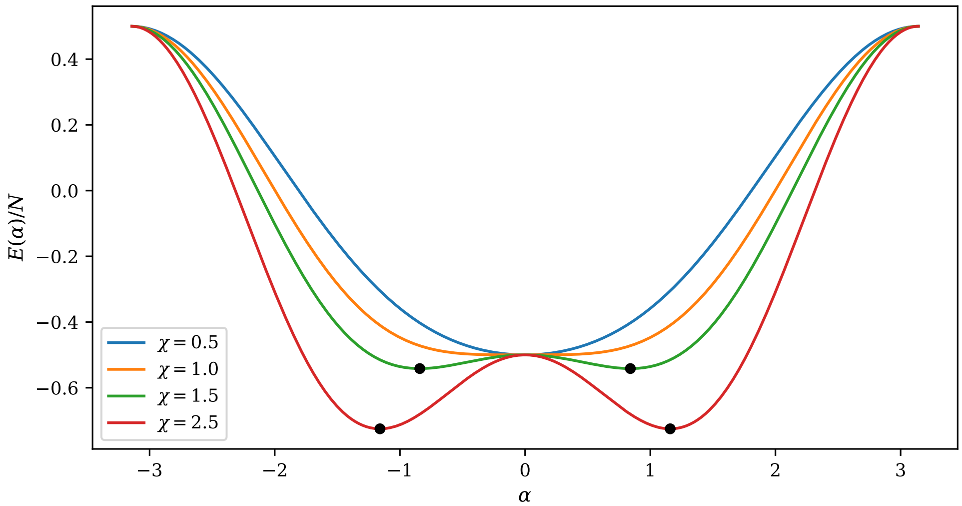

Fig. 6.3 The Lipkin mean-field energy per particle $E(\alpha)/N$ of Eq. (6.70) against the Thouless rotation angle α, for $N = 4$, $\varepsilon = 1$, $W = 0$ and four values of the dimensionless coupling $\chi = |V|(N-1)/\varepsilon$. All four curves pass through $E/N = -\varepsilon/2$ at $\alpha = 0$, the spherical solution. For $\chi < 1$ that point is the global minimum; at $\chi = 1$ the curvature vanishes and the surface is quartic there; for $\chi > 1$ it has become a local maximum and two degenerate deformed minima, marked by the filled circles, have moved out to $\cos\alpha = 1/\chi$.

Figure 6.3 draws the energy surface on either side of the transition, and it is worth reading slowly, because almost everything the stability analysis of Section 6.10 does algebraically can be seen in it directly. The spherical solution sits at $\alpha = 0$ with $E/N = -\varepsilon/2$ for every coupling: the interaction, which changes J_z by two units, contributes nothing at all to the expectation value in the undeformed determinant, so the reference energy is completely insensitive to χ. What changes is the curvature. For $\chi = 0.5$ the surface is a simple well; at $\chi = 1$ it has flattened, the quadratic term of Eq. (6.72) having vanished and the leading behaviour become quartic; and by $\chi = 1.5$ and $\chi = 2.5$ the point $\alpha = 0$ is a local maximum with a pair of minima either side of it, at $\alpha = \pm 0.841$ and $\alpha = \pm 1.159$ respectively. The energy at those minima is $-\frac{\varepsilon}{4}(\chi + 1/\chi)$, which gives -0.542 at $\chi = 1.5$ and -0.725 at $\chi = 2.5$ against the -0.5 of the spherical state. Two features of the deformed branch matter beyond this model. The first is that the minima come in pairs, $\pm\alpha$, because the Hamiltonian is invariant under the rotation that flips the sign of the deformation while the individual determinants are not: the mean field has broken a symmetry of $\hat{H}$, and the two solutions are degenerate for that reason and for no other. The second is that the gain is bought slowly. At $\chi = 1.5$, half again beyond the critical coupling, the deformed state is only 0.042ε per particle below the spherical one, so a calculation that stopped at the spherical solution would not look obviously wrong from its energy alone – which is precisely why one needs a stability criterion and not a plausibility argument.

The stability matrix says the same thing without any reference to the parametrisation (6.69). Because the interaction in Eq. (6.68) changes J_z by two units, it has no matrix element within the $1p$-$1h$ space, so $A = 0$ and $\Delta + A = \varepsilon \boldsymbol{I}$. The block B has matrix elements $\pm V$ connecting different pairs, and its largest eigenvalue is $|V|(N-1)$. The lowest eigenvalue of $\hat{M}$ is therefore

$$\lambda_{\min} = \varepsilon - |V|(N-1) = \varepsilon\,(1-\chi)\,, \tag{6.73}$$

which is exactly Eq. (6.72) up to the factor $N/2$. The program `hartreefock.py` builds $\hat{M}$ numerically from the Hamiltonian in the determinant basis, for $N = 4$ and $\varepsilon = 1$:

χ	$\lambda_{\min}(\hat{M})$	$\alpha_{\min}$	E_{mf}	E_{exact}	solution
0.25	0.75	0.000000	−2.000000	−2.020726	spherical
0.50	0.50	0.000000	−2.000000	−2.081666	spherical
0.90	0.10	0.000000	−2.000000	−2.253886	spherical
1.00	0.00	0.000000	−2.000000	−2.309401	critical
1.50	−0.50	0.841069	−2.166667	−2.645751	deformed
2.00	−1.00	1.047198	−2.500000	−3.055050	deformed

Table 6.1 Stability of the spherical Hartree-Fock solution of the Lipkin model with $N = 4$, $\varepsilon = 1$ and $W = 0$. The lowest eigenvalue of the stability matrix crosses zero exactly at $\chi = 1$, where the deformed minimum appears. E_{mf} is the mean-field energy at the minimum and E_{exact} the exact ground-state energy.

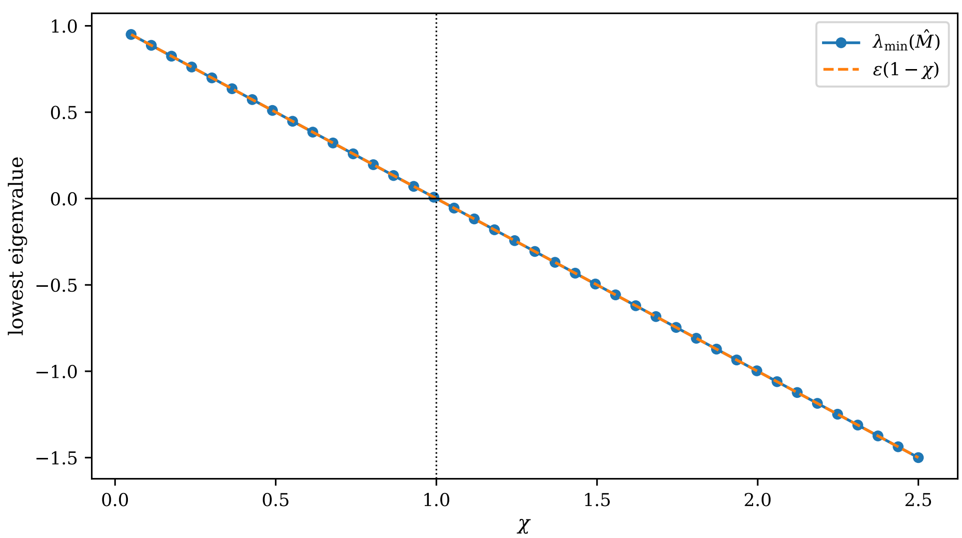

Fig. 6.4 Lowest eigenvalue of the Hartree-Fock stability matrix (6.65) for the Lipkin model with $N = 4$, $\varepsilon = 1$ and $W = 0$, built numerically from the Hamiltonian in the determinant basis at forty values of the coupling χ, compared with the analytic result $\varepsilon(1-\chi)$ of Eq. (6.73). The horizontal line marks zero and the dotted vertical line the critical coupling $\chi = 1$. The numerical eigenvalues lie on the analytic straight line to plotting accuracy and cross zero exactly where the deformed minimum of Figure 6.3 appears.

Figure 6.4 extends the six rows of Table 6.1 into a continuous scan and makes the agreement between the numerical stability matrix and the closed form (6.73) complete: the computed eigenvalues sit on the line $\varepsilon(1-\chi)$ over the whole range $0.05 \le \chi \le 2.5$, with a slope of exactly $-\varepsilon$ and a zero at $\chi = 1$. The exactness is not an accident of small N. It follows from the two structural facts noted above – that the interaction cannot connect the reference to a $1p$-$1h$ state, so $A = 0$ and $\Delta + A = \varepsilon \boldsymbol{I}$, and that the pair-scattering block B is a rank-structured matrix whose largest eigenvalue is $|V|(N-1)$ – and both are consequences of the quasi-spin symmetry of the model rather than of its size. What the figure adds to the table is the behaviour on the far side of the transition: $\lambda_{\min}$ does not level off after the instability sets in but continues to fall linearly, reaching -1.5ε at $\chi = 2.5$. The spherical determinant becomes ever more strongly unstable, and the curvature of the surface at $\alpha = 0$ in Figure 6.3 steepens in step with it. A negative eigenvalue of $\hat{M}$ is thus not a marginal warning to be weighed against other considerations; it is a direction in the space of determinants along which the energy decreases without limit until the nonlinearity of the parametrisation catches it, and a perturbation expansion built on such a reference is an expansion about a saddle, with small energy denominators of the wrong sign in the very channel that is unstable.

Two lessons are worth drawing. The first is that a solution can satisfy the Hartree-Fock equations perfectly and still be useless: for $\chi > 1$ the spherical state is a genuine stationary point, obeys Brillouin's theorem, and is a saddle. The second is that the deformed solution buys a lower energy by breaking a symmetry the exact ground state does not break – at $\chi = 2$ it gains 0.5 but is still 0.55 above the exact answer. Symmetry breaking in a mean field is a device for capturing correlations cheaply, and the symmetry must eventually be restored, by projection or by a method that never broke it.

6.12 When the mean field does nothing

It is instructive to see the opposite extreme. Apply the same machinery to the pairing model of Section 4.2, with four doubly degenerate levels and four particles. The program reports

g	E(reference)	E(Hartree-Fock)	E(exact)	iterations
0.25	1.75000000	1.75000000	1.73045985	2
0.50	1.50000000	1.50000000	1.41677428	2
1.00	1.00000000	1.00000000	0.63554847	2

Hartree-Fock gains nothing at all. The reason is the seniority conservation of Section 5.6: the pairing interaction moves complete pairs and cannot connect the reference to a $1p$-$1h$ state, so $\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = 0$ identically, the Fock matrix is diagonal from the first iteration, and the self-consistency loop terminates without having moved. The entire correlation energy – 36% of the reference energy at $g = 1$ – must come from the doubles, which is precisely what Table 4.1 and Section 5.8 showed.

The moral is that a mean field is only as useful as the one-body physics the interaction actually contains. Where an interaction can be well approximated by an average field – atoms, molecules, closed-shell nuclei – Hartree-Fock does most of the work. Where the interaction is intrinsically a pair-correlation effect – pairing, superconductivity, the strongly coupled Hubbard model of Section 5.7 – it does none, and one must either generalise the mean field, as Bardeen, Cooper and Schrieffer did by allowing the number of particles to fluctuate, or go beyond it.

6.13 The infinite homogeneous electron gas

The electron gas is very likely the only realistic model of an interacting many-particle system for which the Hartree-Fock equations can be solved in closed form. Better still, the total energy and several other properties follow analytically to first order in the interaction, so the whole chapter can be tested against an exact mean-field answer.

The model is defined by three assumptions:

- the electrons feel no external force except the attraction of a uniform, stationary background of positive charge;

- the system as a whole is neutral;
- there are N electrons in a cubic box of side L and volume $\Omega = L^3$, containing a uniform positive charge of density Ne/Ω.

It is a very good approximation for the valence electrons of a metal, and the three-dimensional gas is the cornerstone of the local-density approximation of density-functional theory. Two-dimensional electron fluids occur on metal and liquid-helium surfaces and at metal-oxide-semiconductor interfaces. For all its simplicity, the correlations of this system have resisted a satisfactory first-principles description by anything except quantum Monte Carlo; the diffusion Monte Carlo calculations of Ceperley, and of Ceperley and Tanatar, remain the benchmark in three and in two dimensions. At low density the electrons localise into a lattice – Wigner crystallisation, a direct consequence of the long range of the repulsion – while at high density the system behaves as a liquid. The infinite range of the Coulomb interaction also generates finite-size effects that are absent in nuclear or neutron matter, and that any finite calculation must handle with care.

6.13.1 Plane waves and periodic boundary conditions

The system is homogeneous, so the single-particle states are plane waves normalised in the box,

$$\psi_{\boldsymbol{k}\sigma}(\boldsymbol{r}) = \frac{1}{\sqrt{\Omega}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\,\xi_\sigma, \qquad \xi_{+1/2} = \begin{pmatrix}1\\0\end{pmatrix}, \quad \xi_{-1/2} = \begin{pmatrix}0\\1\end{pmatrix}, \tag{6.74}$$

and periodic boundary conditions restrict the wave numbers to

$$k_i = \frac{2\pi n_i}{L}, \qquad i = x, y, z, \qquad n_i = 0, \pm 1, \pm 2, \ldots \tag{6.75}$$

The limit $L \to \infty$ is taken after the expectation values have been computed. The kinetic energy operator is diagonal,

$$\hat{T} = \sum_{\boldsymbol{k}\sigma} \frac{\hbar^2 k^2}{2m} \hat{a}^\dagger_{\boldsymbol{k}\sigma} \hat{a}_{\boldsymbol{k}\sigma}, \tag{6.76}$$

which is the same structure as the momentum-space Hubbard model of Section 5.7: the plane-wave determinant is automatically the Hartree-Fock solution of a translationally invariant problem, since momentum conservation forces $f_{ai} = 0$. Everything in this section is therefore a calculation *in* the Hartree-Fock basis, not a search for it.

The full Hamiltonian has three pieces,

$$\hat{H} = \hat{H}_{\mathrm{el}} + \hat{H}_b + \hat{H}_{\mathrm{el}-b}, \tag{6.77}$$

namely the electrons,

$$\hat{H}_{\mathrm{el}} = \sum_{i=1}^{N} \frac{p_i^2}{2m} + \frac{e^2}{2} \sum_{i \neq j} \frac{e^{-\mu|\boldsymbol{r}_i - \boldsymbol{r}_j|}}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}, \tag{6.78}$$

the background with itself,

$$\hat{H}_b = \frac{e^2}{2} \iint d\boldsymbol{r}\, d\boldsymbol{r}' \frac{n(\boldsymbol{r}) n(\boldsymbol{r}') e^{-\mu|\boldsymbol{r} - \boldsymbol{r}'|}}{|\boldsymbol{r} - \boldsymbol{r}'|}, \tag{6.79}$$

with $n(\boldsymbol{r}) = N/\Omega$, and the electrons with the background,

$$\hat{H}_{\text{el}-b} = -\frac{e^2}{2}\sum_{i=1}^{N}\int d\boldsymbol{r}\frac{n(\boldsymbol{r})e^{-\mu|\boldsymbol{r}-\boldsymbol{x}_i|}}{|\boldsymbol{r}-\boldsymbol{x}_i|}. \tag{6.80}$$

The factor $e^{-\mu r}$ is a convergence device: each integral is computed at finite μ, and the limit $\mu \to 0$ is taken at the end. Individually the three terms diverge as μ^{-2},

$$\hat{H}_b = \frac{e^2}{2}\frac{N^2}{\Omega}\frac{4\pi}{\mu^2}, \qquad \hat{H}_{\text{el}-b} = -e^2\frac{N^2}{\Omega}\frac{4\pi}{\mu^2}, \tag{6.81}$$

and the direct term of the electron-electron repulsion supplies exactly the missing $+\frac{1}{2}$. The three divergences cancel, and what survives is the kinetic energy and the exchange term. This cancellation is special to the Coulomb interaction and to a neutral system; for a short-ranged interaction, nuclear forces for example, both direct and exchange terms must be kept. What remains is

$$\hat{H} = \hat{H}_0 + \hat{H}_I, \qquad \hat{H}_0 = \sum_{\boldsymbol{k}\sigma}\frac{\hbar^2k^2}{2m}\hat{a}^\dagger_{\boldsymbol{k}\sigma}\hat{a}_{\boldsymbol{k}\sigma}, \tag{6.82}$$

$$\hat{H}_I = \frac{e^2}{2\Omega}\sum_{\sigma_1\sigma_2}\sum_{\boldsymbol{q}\neq 0,\boldsymbol{k},\boldsymbol{p}}\frac{4\pi}{q^2}\hat{a}^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma_1}\hat{a}^\dagger_{\boldsymbol{p}-\boldsymbol{q},\sigma_2}\hat{a}_{\boldsymbol{p}\sigma_2}\hat{a}_{\boldsymbol{k}\sigma_1}, \tag{6.83}$$

with the $\boldsymbol{q} = 0$ term – the one that diverged – removed by the background.

6.13.2 The interaction in momentum space

Equation (6.83) deserves a derivation, since the same steps apply to any interaction depending only on the relative distance. Starting from

$$\hat{V} = \frac{1}{2}\sum_{\substack{\sigma_p\sigma_q\\ \sigma_r\sigma_s}}\sum_{\substack{\boldsymbol{k}_p\boldsymbol{k}_q\\ \boldsymbol{k}_r\boldsymbol{k}_s}}\langle \boldsymbol{k}_p\sigma_p,\boldsymbol{k}_q\sigma_q|\hat{v}|\boldsymbol{k}_r\sigma_r,\boldsymbol{k}_s\sigma_s\rangle\,\hat{a}^\dagger_{\boldsymbol{k}_p\sigma_p}\hat{a}^\dagger_{\boldsymbol{k}_q\sigma_q}\hat{a}_{\boldsymbol{k}_s\sigma_s}\hat{a}_{\boldsymbol{k}_r\sigma_r}, \tag{6.84}$$

insert the plane waves (6.74). The spin functions are orthonormal and $\hat{v}$ is spin independent, so

$$\langle \boldsymbol{k}_p\sigma_p,\boldsymbol{k}_q\sigma_q|\hat{v}|\boldsymbol{k}_r\sigma_r,\boldsymbol{k}_s\sigma_s\rangle = \frac{\delta_{\sigma_p\sigma_r}\delta_{\sigma_q\sigma_s}}{\Omega^2}\iint e^{-i\boldsymbol{k}_p\cdot\boldsymbol{r}_p}e^{-i\boldsymbol{k}_q\cdot\boldsymbol{r}_q}\,v(r)\,e^{i\boldsymbol{k}_r\cdot\boldsymbol{r}_p}e^{i\boldsymbol{k}_s\cdot\boldsymbol{r}_q}d\boldsymbol{r}_pd\boldsymbol{r}_q. \tag{6.85}$$

Change variables to $\boldsymbol{r} = \boldsymbol{r}_p - \boldsymbol{r}_q$ at fixed $\boldsymbol{r}_q$; the limits are unchanged, and the exponentials separate,

$$= \frac{\delta_{\sigma_p\sigma_r}\delta_{\sigma_q\sigma_s}}{\Omega^2}\int v(r)e^{i(\boldsymbol{k}_r-\boldsymbol{k}_p)\cdot\boldsymbol{r}}d\boldsymbol{r}\int e^{i(\boldsymbol{k}_s-\boldsymbol{k}_q+\boldsymbol{k}_r-\boldsymbol{k}_p)\cdot\boldsymbol{r}_q}d\boldsymbol{r}_q. \tag{6.86}$$

The second integral is a Kronecker delta enforcing momentum conservation,

$$\langle \boldsymbol{k}_p\sigma_p,\boldsymbol{k}_q\sigma_q|\hat{v}|\boldsymbol{k}_r\sigma_r,\boldsymbol{k}_s\sigma_s\rangle = \frac{\delta_{\sigma_p\sigma_r}\delta_{\sigma_q\sigma_s}}{\Omega}\,\delta_{\boldsymbol{k}_p+\boldsymbol{k}_q,\boldsymbol{k}_r+\boldsymbol{k}_s}\int v(r)e^{i(\boldsymbol{k}_r-\boldsymbol{k}_p)\cdot\boldsymbol{r}}d\boldsymbol{r}. \tag{6.87}$$

The matrix element vanishes unless $\boldsymbol{k}_p + \boldsymbol{k}_q = \boldsymbol{k}_r + \boldsymbol{k}_s$. Using this to eliminate one summation variable and writing $\boldsymbol{p} = \boldsymbol{k}_p + \boldsymbol{k}_q - \boldsymbol{k}_r$, $\boldsymbol{k} = \boldsymbol{k}_r$ and $\boldsymbol{q} = \boldsymbol{k}_p - \boldsymbol{k}_r$,

$$\hat{V} = \frac{1}{2\Omega}\sum_{\sigma\sigma'}\sum_{\boldsymbol{kpq}}\left[\int v(r)e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}d\boldsymbol{r}\right]\hat{a}^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma}\hat{a}^\dagger_{\boldsymbol{p}-\boldsymbol{q},\sigma'}\hat{a}_{\boldsymbol{p}\sigma'}\hat{a}_{\boldsymbol{k}\sigma}. \tag{6.88}$$

For the screened Coulomb interaction the bracket is $4\pi e^2/(\mu^2+q^2)$, which becomes $4\pi e^2/q^2$ as $\mu\to 0$, giving Eq. (6.83). The structure – a momentum transfer $\boldsymbol{q}$ shared between two particles – is exactly the Hubbard interaction of Eq. (5.24), with $4\pi e^2/q^2$ in place of the constant U.

6.13.3 The reference energy

With $|\Phi_0\rangle$ the filled Fermi sea, the kinetic energy follows from the contractions of Section 3.14,

$$\langle\Phi_0|\hat{T}|\Phi_0\rangle = \sum_{i\le F}\frac{\hbar^2 k_i^2}{2m}\cdot 2 = \frac{2\Omega}{(2\pi)^3}\frac{\hbar^2}{2m}\int_0^{k_F} 4\pi k^4 dk = \frac{\hbar^2\Omega}{10\pi^2 m}k_F^5, \tag{6.89}$$

the factor two being the spin degeneracy. The same density of states fixes the particle number,

$$N = \frac{2\Omega}{(2\pi)^3}\cdot\frac{4}{3}\pi k_F^3 = \frac{\Omega}{3\pi^2}k_F^3 \qquad \Longrightarrow \qquad k_F = \left(\frac{3\pi^2 N}{\Omega}\right)^{1/3}, \tag{6.90}$$

so that

$$\frac{\langle\Phi_0|\hat{T}|\Phi_0\rangle}{N} = \frac{3}{5}\frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2(3\pi^2)^{5/3}}{10\pi^2 m}\rho^{2/3}, \qquad \rho = \frac{N}{\Omega}. \tag{6.91}$$

For the interaction, only two classes of contraction survive: either $\boldsymbol{k}+\boldsymbol{q}=\boldsymbol{p}$ with $\sigma=\sigma'$, or $\boldsymbol{q}=0$. The first is the exchange term, the second the direct one, and

$$\langle\Phi_0|\hat{V}|\Phi_0\rangle = \frac{1}{2\Omega}\left(N^2\int v(r)d\boldsymbol{r} - N\sum_{\boldsymbol{q}}\int v(r)e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}d\boldsymbol{r}\right), \tag{6.92}$$

valid for any interaction depending only on r. For the electron gas the first term is cancelled by the background, as we saw, and only the exchange term remains.

6.13.4 The single-particle energy

The Hartree-Fock single-particle energy is the kinetic energy plus the exchange self-energy,

$$\varepsilon_k^{\mathrm{HF}} = \frac{\hbar^2 k^2}{2m} - \frac{e^2}{\Omega^2}\sum_{p\le k_F}\int d\boldsymbol{r}\, e^{i(\boldsymbol{p}-\boldsymbol{k})\cdot\boldsymbol{r}}\int d\boldsymbol{r}'\,\frac{e^{i(\boldsymbol{k}-\boldsymbol{p})\cdot\boldsymbol{r}'}}{|\boldsymbol{r}-\boldsymbol{r}'|}, \tag{6.93}$$

and we shall show that it evaluates to

$$\boxed{\varepsilon_k^{\mathrm{HF}} = \frac{\hbar^2 k^2}{2m} - \frac{e^2 k_F}{2\pi}\left[2 + \frac{k_F^2-k^2}{kk_F}\ln\left|\frac{k+k_F}{k-k_F}\right|\right].} \tag{6.94}$$

Insert the convergence factor $e^{-\mu|\boldsymbol{r}-\boldsymbol{r}'|}$ and replace the sum by an integral, $\sum_{\boldsymbol{p}}\to\Omega(2\pi)^{-3}\int d\boldsymbol{p}$,

$$\lim_{\mu\to 0}\frac{e^2}{\Omega(2\pi)^3}\iint d\boldsymbol{r}d\boldsymbol{r}'\,\frac{e^{-\mu|\boldsymbol{r}-\boldsymbol{r}'|}}{|\boldsymbol{r}-\boldsymbol{r}'|}\int d\boldsymbol{p}\, e^{i(\boldsymbol{p}-\boldsymbol{k})\cdot(\boldsymbol{r}-\boldsymbol{r}')}. \tag{6.95}$$

Change variables to $\boldsymbol{x}=\boldsymbol{r}-\boldsymbol{r}'$ and $\boldsymbol{y}=\boldsymbol{r}'$. The integrand no longer depends on $\boldsymbol{y}$, whose integral gives Ω, and the $\boldsymbol{x}$ integral in spherical coordinates is

$$\int d\boldsymbol{x}\, e^{i(\boldsymbol{p}-\boldsymbol{k})\cdot\boldsymbol{x}}\frac{e^{-\mu x}}{x} = 4\pi\int_0^\infty \frac{\sin(|\boldsymbol{p}-\boldsymbol{k}|x)}{|\boldsymbol{p}-\boldsymbol{k}|}e^{-\mu x}dx = \frac{4\pi}{\mu^2+|\boldsymbol{p}-\boldsymbol{k}|^2}. \tag{6.96}$$

The limit $\mu \to 0$ is now harmless, and we are left with

$$\frac{e^2}{2\pi^2}\int_{p\le k_F}\frac{d\boldsymbol{p}}{|\boldsymbol{p}-\boldsymbol{k}|^2} = \frac{e^2}{\pi}\int_0^{k_F} p^2 dp\int_{-1}^1 \frac{du}{p^2+k^2-2pku}, \tag{6.97}$$

having set $u = \cos\theta$. The angular integral is elementary,

$$\int_{-1}^1 \frac{du}{p^2+k^2-2pku} = \frac{1}{pk}\ln\left|\frac{p+k}{p-k}\right|, \tag{6.98}$$

so the exchange term becomes

$$\frac{e^2}{k\pi}\int_0^{k_F} p\ln\left|\frac{p+k}{p-k}\right|dp = \frac{e^2}{k\pi}\left[kk_F + \frac{k_F^2-k^2}{2}\ln\left|\frac{k+k_F}{k-k_F}\right|\right], \tag{6.99}$$

where the remaining integral is done by substituting $x = p+k$ and $y = p-k$. Rearranging gives Eq. (6.94).

6.13.5 Dimensionless form, band width and effective mass

Introduce the radius r_0 of the sphere whose volume is the volume per electron,

$$\rho = \frac{k_F^3}{3\pi^2} = \frac{3}{4\pi r_0^3} \qquad \Longrightarrow \qquad k_F = \frac{(9\pi/4)^{1/3}}{r_0} = \frac{1.9192}{r_0}, \tag{6.100}$$

the Bohr radius $a_0 = \hbar^2/me^2$ and the dimensionless density parameter $r_s = r_0/a_0$, which lies between about 2 and 6 for most metals. With $x = k/k_F$ and

$$F(x) = \frac{1}{2}+\frac{1-x^2}{4x}\ln\left|\frac{1+x}{1-x}\right|, \tag{6.101}$$

Equation (6.94) becomes

$$\varepsilon_k^{\mathrm{HF}} = \frac{\hbar^2k^2}{2m} - \frac{2e^2k_F}{\pi}F(x), \qquad \frac{\varepsilon_k^{\mathrm{HF}}}{\varepsilon_0^F} = x^2 - \frac{4}{\pi k_F a_0}F(x) = x^2 - 0.663\, r_s F(x), \tag{6.102}$$

where $\varepsilon_0^F = \hbar^2k_F^2/2m$ is the free energy at the Fermi surface. The function F is smooth except at $x=1$, where it takes the value $\frac{1}{2}$ but has an infinite derivative; at $x=0$ it equals 1. The program `hartreefock.py` tabulates the result for $r_s = 4$:

Band width.

The width of the occupied band is

$$\Delta\varepsilon^{\mathrm{HF}} = \varepsilon_{k_F}^{\mathrm{HF}} - \varepsilon_0^{\mathrm{HF}}. \tag{6.103}$$

At $r_s = 4$ the two entries of Table 6.2 give -0.326 and -2.652, hence $\Delta\varepsilon^{\mathrm{HF}} = 2.326$ in units of ε_0^F. In general, using $F(1) = \frac{1}{2}$ and $F(0) = 1$,

$x = k/k_F$	$F(x)$	$\varepsilon_k^{\mathrm{HF}}/\varepsilon_0^F$	$\varepsilon_k^{(0)}/\varepsilon_0^F$
0.00	1.000000	−2.652000	0.0000
0.25	0.978899	−2.533540	0.0625
0.50	0.911980	−2.168570	0.2500
0.75	0.783779	−1.516081	0.5625
1.00	0.500000	−0.326000	1.0000
1.25	0.252812	0.892042	1.5625
1.50	0.164700	1.813214	2.2500
2.00	0.088020	3.766570	4.0000

Table 6.2 The Hartree-Fock single-particle energy of the three-dimensional electron gas at $r_s = 4$, compared with the free-particle energy. The exchange term lowers every state and lowers the bottom of the band far more than the top.

$$\Delta\varepsilon^{\mathrm{HF}} = 1 + \frac{0.663}{2} r_s = 1 + 0.3315\, r_s, \tag{6.104}$$

which reproduces 2.326 at $r_s = 4$. The free gas has a band width of exactly one, so exchange more than doubles it – an effect of the right sign but, as experiment shows, considerably too large.

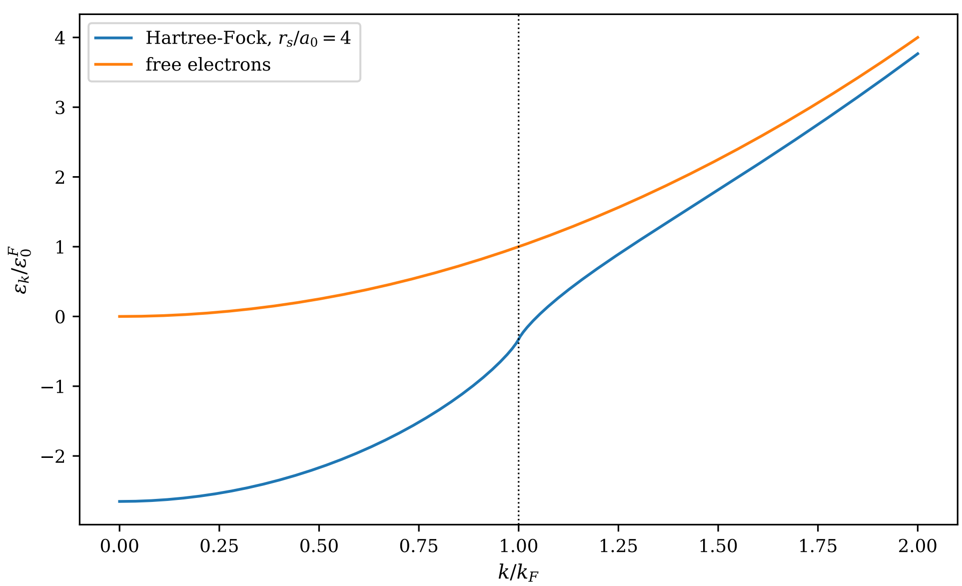

Fig. 6.5 Hartree-Fock single-particle energy of the three-dimensional electron gas, Eq. (6.102), at $r_s = 4$, in units of the free Fermi energy $\varepsilon_0^F = \hbar^2 k_F^2/2m$ and against $x = k/k_F$, together with the free-particle parabola x^2. The dotted vertical line marks the Fermi surface. Exchange lowers every level, by 2.652 at the bottom of the band and by only 1.326 at the Fermi surface, so the occupied band is stretched from a width of 1 to a width of 2.326; above k_F the two curves converge again as $F(x)$ falls off, the gap being 0.233 at $x = 2$.

Figure 6.5 plots the two columns of Table 6.2 as functions of x, and three things about it deserve comment. The first is that the exchange shift is not a constant: it is largest at the bottom of the band, where a state overlaps with every occupied state in the Fermi sea, and smallest at and above k_F, where the phase-space for exchange with the occupied states is smaller. It is this k-dependence, and not the overall shift, that has physical consequences, since a rigid shift of all levels would merely redefine the zero of energy. The second is that the whole occupied band lies below the free-particle curve and indeed below zero out to $x \simeq 1.05$: the exchange energy of a state at the Fermi surface, $-0.663\, r_s F(1) \varepsilon_0^F = -1.326\, \varepsilon_0^F$ at $r_s = 4$, is larger in magnitude than its kinetic energy, which is what makes the gas bound at all. The third is the feature at $x = 1$. The Hartree-Fock curve is visibly steeper there than anywhere else, and the steepening is confined to a narrow neighbourhood of the Fermi surface; on either side of it the curve is perfectly smooth. That local anomaly is not a numerical artefact but the logarithmic singularity of F' announced under Eq. (6.101), and it is the subject of the next paragraph. Notice also that the size of the whole effect is governed by the single combination $0.663\, r_s$: at metallic densities, r_s between 2 and 6, the exchange shift at the bottom of the band ranges from 1.3 to 4.0 in units of ε_0^F, so exchange is never a small perturbation on the kinetic energy.

Effective mass and level density.

Expanding about the Fermi surface,

$$\varepsilon_k^{\mathrm{HF}} = \varepsilon_{k_F}^{\mathrm{HF}} + \left(\frac{\partial \varepsilon_k^{\mathrm{HF}}}{\partial k} \right)_{k_F} (k - k_F) + \cdots \tag{6.105}$$

and comparing with the free-particle expansion $\varepsilon_k^{(0)} = \hbar^2 k_F^2/2m + (\hbar^2 k_F/m)(k - k_F) + \cdots$ defines the effective mass

$$m_{\mathrm{HF}}^* \equiv \hbar^2 k_F \left(\frac{\partial \varepsilon_k^{\mathrm{HF}}}{\partial k} \right)_{k_F}^{-1} = \frac{2m}{2 - 0.663\, r_s F'(1)}. \tag{6.106}$$

Here is the difficulty: $F'(x)$ diverges logarithmically as $x \to 1$, so the slope of $\varepsilon_k^{\mathrm{HF}}$ is infinite at the Fermi surface and $m_{\mathrm{HF}}^* \to 0$. Approaching from below, the program finds $m^*/m = 0.256, 0.185, 0.144, 0.118, 0.100$ at $1 - x = 10^{-2}, \ldots, 10^{-6}$: the ratio creeps towards zero like $1/\ln(1-x)$. Since the level density is

$$n(\varepsilon) = \frac{\Omega k^2}{2\pi^2} \left(\frac{\partial \varepsilon}{\partial k} \right)^{-1}, \tag{6.107}$$

it vanishes at the Fermi surface.

Figure 6.6 shows how slowly the catastrophe arrives. The x-axis has been reversed so that the Fermi surface is on the right, and it covers eight decades; over that whole range m_{HF}^*/m falls only from 0.404 at $1 - x = 10^{-1}$ to 0.076 at $1 - x = 10^{-8}$, passing through the tabulated values 0.256 and 0.100 at 10^{-2} and 10^{-6}. There is no sign of a plateau: the curve keeps bending downwards, and the product $m^* \ln(1/(1-x))$ stays within a factor of two of unity across the plot, which is the numerical signature of the $1/\ln$ law implied by the divergence of F'. Two consequences follow, and they pull in opposite directions. The first is that Hartree-Fock is unambiguously wrong at the Fermi surface: the limit is zero, the level density (6.107) goes with it, and the linear specific heat of a metal is lost. The second is that the pathology is so weak, logarithmically weak, that at any momentum a finite-basis calculation can actually resolve – and the discrete $\boldsymbol{k}$-meshes of Section 6.13.1 have a spacing of order $1/L$, hence $1 - x$ of order $N^{-1/3}$ – one sees not a divergence but a merely rather small effective mass, around 0.3 to 0.4. This is the uncomfortable position such calculations occupy: the number one computes is finite, plausible and basis-dependent, and only the analytic result tells us that it has no limit.

This is qualitatively wrong. Metals have a finite density of states at the Fermi level – it is what gives them a linear specific heat and a finite Pauli susceptibility. The culprit is the

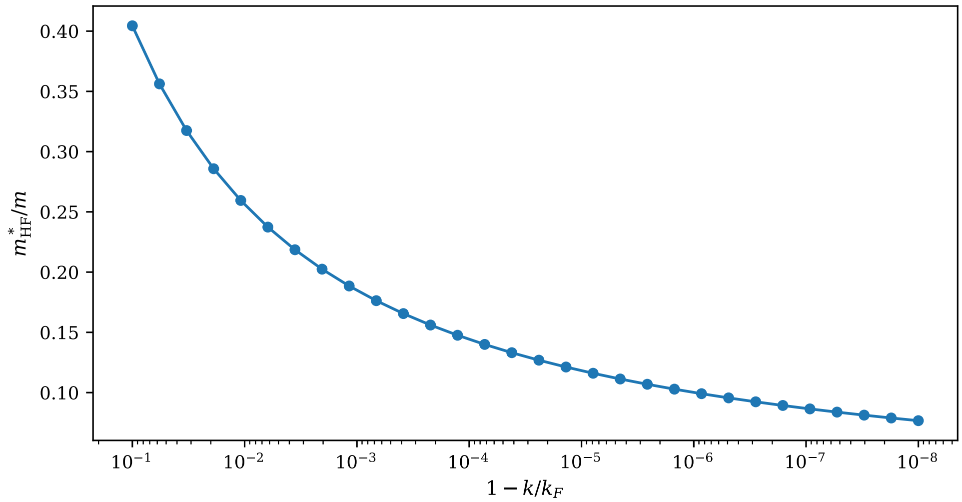

Fig. 6.6 Hartree-Fock effective mass (6.106) of the electron gas at $r_s = 4$, in units of the bare mass, evaluated at thirty momenta approaching the Fermi surface from below and plotted against $1 - k/k_F$ on a logarithmic scale that runs from 10^{-1} on the left to 10^{-8} on the right, so that the Fermi surface lies at the right-hand edge. The ratio falls monotonically from 0.40 to 0.076, and it does so like the reciprocal of $\ln(1 - k/k_F)$: eight decades in the distance to the Fermi surface cost only a factor of five in m^*.

$1/q^2$ singularity of the bare Coulomb interaction at small momentum transfer, which Hartree-Fock treats without any screening. In a real metal the other electrons rearrange to screen it, and the effective interaction falls off exponentially beyond the screening length. Summing the diagrams that produce that screening is precisely what the random-phase approximation does, and it restores a finite effective mass. So the electron gas illustrates both what a mean field does well and where it must fail, and it points directly to the next method.

6.13.6 The energy per electron

Collecting the kinetic energy (6.91) and the exchange term, and measuring energies in Rydberg, $e^2/2a_0 = 13.606$ eV, the Hartree-Fock energy per electron is

$$\boxed{\frac{E_0}{N} = \frac{e^2}{2a_0}\left[\frac{2.21}{r_s^2} - \frac{0.916}{r_s}\right].} \tag{6.108}$$

The two coefficients are not empirical:

$$\frac{3}{5}\left(\frac{9\pi}{4}\right)^{2/3} = 2.2099, \qquad \frac{3}{2\pi}\left(\frac{9\pi}{4}\right)^{1/3} = 0.9163, \tag{6.109}$$

as `hartreefock.py` confirms. The first term is the kinetic energy, which scales as $\rho^{2/3} \propto r_s^{-2}$ and dominates at high density; the second is exchange, which scales as $\rho^{1/3} \propto r_s^{-1}$ and wins at low density. Because they carry opposite signs and different powers, their sum has a minimum:

$$\frac{d}{dr_s}\left[\frac{2.21}{r_s^2} - \frac{0.916}{r_s}\right] = 0 \qquad \Longrightarrow \qquad r_s = \frac{2 \times 2.21}{0.916} = 4.83, \tag{6.110}$$

at which $E_0/N = -0.0949$ Ry $= -1.29$ eV. The values along the curve are

r_s	E_0/N [Ry]
1.00	1.294000
2.00	0.094500
3.00	−0.059778
4.00	−0.090875
4.83	−0.094916
6.00	−0.091278
8.00	−0.079969

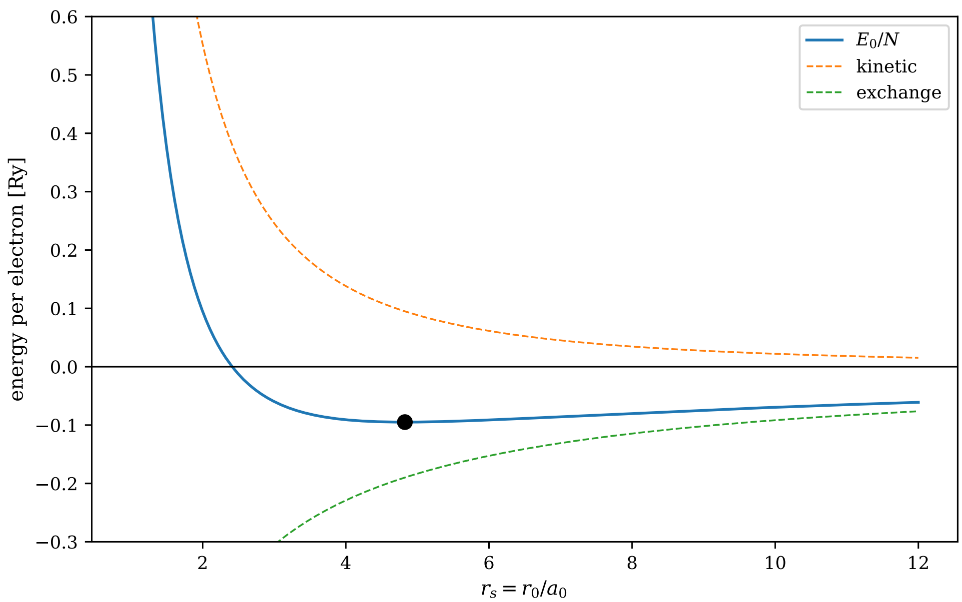

Fig. 6.7 Hartree-Fock energy per electron of the three-dimensional electron gas, Eq. (6.108), in Rydberg and against the density parameter $r_s = r_0/a_0$, decomposed into the kinetic term $2.21/r_s^2$ and the exchange term $-0.916/r_s$. The two contributions carry opposite signs and different powers of the density, so their sum turns over; the filled circle marks the minimum at $r_s = 4.83$, where $E_0/N = -0.0949$ Ry $= -1.29$ eV. The total crosses zero at $r_s = 2.41$ and the well is very shallow, the energy staying within 5% of its minimum over the whole interval $4 \leq r_s \leq 6$.

Figure 6.7 draws the competition of Eq. (6.108) with the two terms shown separately, and the shape of the result repays attention. At high density, small r_s, the kinetic term wins because it grows as r_s^{-2} against the r_s^{-1} of exchange: the gas is unbound, and E_0/N crosses zero at $r_s = 2.21/0.916 = 2.41$. At low density exchange wins, but both terms are then going to zero, so the binding per electron dies away as r_s^{-1} and the curve creeps back towards the axis from below. In between lies the minimum at $r_s = 4.83$. What the figure shows and the table only hints at is how flat that minimum is: E_0/N is -0.0909 Ry at $r_s = 4$, -0.0949 at 4.83 and -0.0913 at 6, so a 50% change in the interparticle spacing – a factor of more than three in density – costs less than 5% of the binding energy. Two things follow. The first is physical: a system this soft against uniform compression has a small bulk modulus, and the alkali metals, whose observed r_s values of 3.93 for sodium and 5.62 for caesium sit right in the flat region, are indeed among the most compressible of solids. The second is methodological: because the well is so shallow, the correlation energy of about -0.06 Ry – comparable to the entire Hartree-Fock binding and only weakly dependent on r_s in this range – shifts the predicted equilibrium density substantially even though it changes the physics not at all. A mean-field theory can therefore get the mechanism of cohesion right, as this one does, and still be quantitatively unreliable for the equilibrium lattice constant, which is one of the reasons the electron gas remained a testing ground for many-body methods long after its Hartree-Fock solution was known. at a density lying squarely in the range observed in the alkali metals. This is a real success: exchange alone, with no correlation whatsoever, explains why a metal is stable against expansion. The binding energy is too small by roughly a factor of two – the missing part is the correlation energy, which for this system is about -0.06 Ry near equilibrium – and the equilibrium density is somewhat too low, but the mechanism is right.

6.14 Summary

Hartree-Fock theory is the variational principle of Section 2.7 applied to the smallest interesting trial space: a single Slater determinant, with the orbitals free. The stationarity condition is a nonlinear eigenvalue problem, Eq. (6.26), solved by iterating a linear one – the self-consistent field. Its solution is characterised by Brillouin's theorem, $\langle \Phi_0|\hat{H}|\Phi_i^a\rangle = 0$, which is exactly the statement that the reference decouples from all single excitations and which is why every post-Hartree-Fock method begins at the doubles.

Thouless' theorem makes the geometry precise: the neighbourhood of a determinant, inside the manifold of determinants, is parametrised by $e^{\hat{T}}$ with $\hat{T}$ a linear combination of one-particle-one-hole operators. Because those operators commute, the exponential collapses, and the Baker-Campbell-Hausdorff series – which controls the general case and reappears in the Trotter splitting and in coupled-cluster theory – reduces to its first term. Going to second order in $\hat{T}$ produces the stability matrix (6.65), whose positivity decides whether the stationary point is a minimum, and which is the same matrix that the random-phase approximation diagonalises.

The two model calculations bracket the range of behaviour. For the pairing model the mean field does nothing at all, since the interaction conserves seniority. For the Lipkin model it does too much: beyond $\chi = 1$ the symmetric solution goes unstable and a symmetry-broken one takes over, buying energy at the cost of a broken symmetry that must later be restored. Between them lies the electron gas, where the whole programme goes through analytically, predicts a bound system at a realistic density, and yet gets the effective mass qualitatively wrong for want of screening.

The program `hartreefock.py` carries out every calculation of this chapter, and the notebook `hartreefock.ipynb` runs it cell by cell.

6.15 Exercises

Exercise 1: determinants and unitary transformations

a) Verify Eq. (6.12) by direct expansion, and then prove Eq. (6.13) for a general n.
b) Use a) to prove the product rule $\det(\boldsymbol{BC}) = \det(\boldsymbol{B})\det(\boldsymbol{C})$ for $n = 2$, and convince yourself that the argument generalises.
c) Show that a Slater determinant is invariant, up to a phase, under any unitary mixing of its occupied orbitals, and that the one-body density matrix (6.28) is strictly invariant.

d) Show that $\boldsymbol{\rho}^2 = \boldsymbol{\rho}$ and $\operatorname{tr}\boldsymbol{\rho} = N$, and explain what these two statements say about the occupied subspace.

Answer to a).

Expanding the left-hand side of Eq. (6.12),

$$(\alpha_1 b_{11} + \alpha_2 b_{12})a_{22} - (\alpha_1 b_{21} + \alpha_2 b_{22})a_{12} = \alpha_1(b_{11}a_{22} - b_{21}a_{12}) + \alpha_2(b_{12}a_{22} - b_{22}a_{12}), \tag{6.111}$$

which is the right-hand side. For general n, use the Leibniz formula $\det(\boldsymbol{A}) = \sum_\pi \operatorname{sgn}(\pi)\prod_j a_{\pi(j)j}$. Exactly one factor in each product comes from column j; substituting $a_{ij} \to \sum_k c_k b_{ik}$ in that column and interchanging the two sums gives $\sum_k c_k \sum_\pi \operatorname{sgn}(\pi) b_{\pi(j)k} \prod_{l\neq j} a_{\pi(l)l}$, which is Eq. (6.13).

Answer to b).

Write $\boldsymbol{A} = \boldsymbol{BC}$, so that column j of $\boldsymbol{A}$ is $\sum_k C_{kj}\boldsymbol{b}_k$ with $\boldsymbol{b}_k$ the columns of $\boldsymbol{B}$. Applying Eq. (6.13) to both columns of a 2×2 matrix,

$$\det(\boldsymbol{A}) = \sum_{k_1 k_2} C_{k_1 1} C_{k_2 2} \det\left(\boldsymbol{b}_{k_1}\ \boldsymbol{b}_{k_2}\right). \tag{6.112}$$

Terms with $k_1 = k_2$ vanish, and the two surviving terms combine to $(C_{11}C_{22} - C_{21}C_{12})\det(\boldsymbol{b}_1\ \boldsymbol{b}_2) = \det(\boldsymbol{C})\det(\boldsymbol{B})$. For general n the same expansion leaves only the $n!$ terms with distinct k's, each carrying the sign of the corresponding permutation, and the sum is again $\det(\boldsymbol{C})\det(\boldsymbol{B})$.

Answer to c).

Let $\psi_i' = \sum_j U_{ij}\psi_j$ with $\boldsymbol{U}$ unitary and i, j running over the occupied orbitals. By Eq. (6.16) the new determinant is $\det(\boldsymbol{U})$ times the old one, and $|\det(\boldsymbol{U})| = 1$ for a unitary matrix, so the two states differ by a phase and describe the same physics. The density matrix is $\boldsymbol{\rho}' = \boldsymbol{U}^\dagger\boldsymbol{\rho}\boldsymbol{U}$ evaluated on the occupied block; since $\boldsymbol{\rho}$ is the projector onto the span of the occupied orbitals and $\boldsymbol{U}$ merely rotates within that span, $\boldsymbol{\rho}$ is unchanged exactly, not merely up to a phase.

Answer to d).

With orthonormal orbitals, $\rho_{\gamma\delta} = \sum_i C_{i\gamma}C_{i\delta}^*$ gives

$$(\boldsymbol{\rho}^2)_{\gamma\delta} = \sum_{ij\lambda} C_{i\gamma}C_{i\lambda}^* C_{j\lambda}C_{j\delta}^* = \sum_{ij} \delta_{ij} C_{i\gamma}C_{j\delta}^* = \rho_{\gamma\delta}, \tag{6.113}$$

and $\operatorname{tr}\boldsymbol{\rho} = \sum_i\sum_\gamma |C_{i\gamma}|^2 = N$. A Hermitian idempotent is a projector, and its trace is the dimension of the space it projects onto. So $\boldsymbol{\rho}$ is the projector onto the N-dimensional occupied subspace, and its eigenvalues are N ones and the rest zeros. Together these say that a Hartree-Fock solution is a choice of subspace, and that only for a Slater determinant are the occupation numbers exactly zero and one – for a correlated state the eigenvalues of $\boldsymbol{\rho}$ are the fractional natural occupations of Section 1.31.

Exercise 2: the Hartree-Fock equations

a) Carry out the differentiation in Eq. (6.22) in full, and confirm that the factor $\frac{1}{2}$ in the two-body term is cancelled by the two ways in which $C^*_{i\alpha}$ can appear.
b) Show that the Hartree-Fock energy is *not* the sum of the single-particle energies, and derive the correct relation

$$E^{\mathrm{HF}} = \sum_{i=1}^{N} \varepsilon_i^{\mathrm{HF}} - \frac{1}{2} \sum_{i,j=1}^{N} \langle ij|\hat{v}|ij\rangle_{\mathrm{AS}}.$$

c) Interpret $\varepsilon_i^{\mathrm{HF}}$ physically by computing the energy needed to remove a particle from orbital i, assuming the remaining orbitals do not relax. (This is Koopmans' theorem.)
d) Show directly from Eq. (6.25) that at self-consistency $f_{ai} = 0$, and hence that $\langle \Phi_0|\hat{H}|\Phi_i^a\rangle = 0$.

Answer to a).

The one-body term of Eq. (6.19) is linear in $C^*_{i\alpha}$ and differentiates to $\sum_\beta C_{i\beta}\langle\alpha|\hat{h}_0|\beta\rangle$. In the two-body term $C^*_{i\alpha}$ can be either the first starred factor, with the sum over j free, or the second, with the roles of the index pairs exchanged. Relabelling the dummy indices of the second contribution and using

$$\langle\alpha\beta|\hat{v}|\gamma\delta\rangle_{\mathrm{AS}} = \langle\beta\alpha|\hat{v}|\delta\gamma\rangle_{\mathrm{AS}} \tag{6.114}$$

shows the two are identical, so together they cancel the $\frac{1}{2}$ and give the second term of Eq. (6.23).

Answer to b).

Multiply Eq. (6.26) by $C^*_{i\alpha}$ and sum over α and over the occupied i:

$$\sum_{i=1}^{N} \varepsilon_i^{\mathrm{HF}} = \sum_{i=1}^{N} \langle i|\hat{h}_0|i\rangle + \sum_{i,j=1}^{N} \langle ij|\hat{v}|ij\rangle_{\mathrm{AS}}. \tag{6.115}$$

Comparing with Eq. (6.18), which carries a factor $\frac{1}{2}$ on the two-body term, gives the stated relation. The interaction energy is counted twice in the sum of single-particle energies – once for each partner – and half of it must be removed. This is the standard double-counting correction, and forgetting it is one of the commonest errors in a first Hartree-Fock code.

Answer to c).

Removing a particle from orbital i without relaxing the others changes the energy by

$$E^{\mathrm{HF}}(N) - E^{\mathrm{HF}}(N-1) = \langle i|\hat{h}_0|i\rangle + \sum_{j\neq i} \langle ij|\hat{v}|ij\rangle_{\mathrm{AS}} = \varepsilon_i^{\mathrm{HF}}, \tag{6.116}$$

using $\langle ii|\hat{v}|ii\rangle_{\mathrm{AS}} = 0$. So each Hartree-Fock eigenvalue is, in the frozen-orbital approximation, minus the ionisation energy from that orbital. This is Koopmans' theorem. It is only an approximation, because in reality the remaining orbitals relax – an effect that lowers the energy of the $(N-1)$-particle system and therefore makes the true ionisation energy smaller than

$-\varepsilon_i^{\mathrm{HF}}$. Correlation works in the opposite direction, and for atoms the two errors partly cancel, which is why Koopmans' theorem is more accurate than it has any right to be.

Answer to d).

At self-consistency the coefficients diagonalise the Fock matrix, so in the Hartree-Fock basis $f_{pq} = \varepsilon_p \delta_{pq}$ and in particular $f_{ai} = 0$ for $a > F \geq i$. By the Condon-Slater rules of Section 3.14, the matrix element of $\hat{H}$ between the reference and a single excitation is

$$\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = \langle a | \hat{h}_0 | i \rangle + \sum_{j \leq F} \langle a j | \hat{v} | i j \rangle_{\mathrm{AS}} = f_{ai} = 0. \tag{6.117}$$

This is the identity that makes the Hartree-Fock determinant the natural reference for everything that follows.

Exercise 3: Thouless' theorem

a) Prove Eq. (6.35) from the anticommutation relations, being explicit about where the disjointness of the particle and hole index sets is used.
b) Show that $\hat{T}_i^2 = 0$ and hence that $e^{\hat{T}} = \prod_i (1 + \hat{T}_i)$. Explain why this is *not* the same as $1 + \hat{T}$, and identify what the difference describes.
c) Verify all of this numerically with `hartreefock.py` for six orbitals and three particles, and check that the discarded piece scales as the square of the amplitudes.
d) Show that $\langle c | \tilde{c} \rangle = \det(f_{ip})$ over the occupied block, and explain why the hypothesis $\langle c | \tilde{c} \rangle \neq 0$ is exactly what is needed for the proof to go through.

Answer to a).

Take $a, b > F$ and $i, j \leq F$, so that all four indices in a given product are distinct except possibly $a = b$ or $i = j$. Since $i \neq b$, $\hat{a}_i \hat{a}_b^\dagger = -\hat{a}_b^\dagger \hat{a}_i$, whence

$$\hat{a}_a^\dagger \hat{a}_i \hat{a}_b^\dagger \hat{a}_j = -\hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_i \hat{a}_j. \tag{6.118}$$

Since $j \neq a$, the same step on the reversed product gives

$$\hat{a}_b^\dagger \hat{a}_j \hat{a}_a^\dagger \hat{a}_i = -\hat{a}_b^\dagger \hat{a}_a^\dagger \hat{a}_j \hat{a}_i = -\hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_i \hat{a}_j, \tag{6.119}$$

where the second equality interchanges both the two creation operators and the two annihilation operators, contributing two signs that cancel. The two products are equal, so the commutator vanishes. The disjointness of the index sets is what allowed both anticommutations to be performed without generating a delta function; this is why the result is special to particle-hole operators and fails for general $\hat{a}_p^\dagger \hat{a}_q$.

Answer to b).

$\hat{T}_i^2$ contains $\hat{a}_i \cdots \hat{a}_i$, and by Eq. (6.118) the two annihilation operators can be brought adjacent; $\hat{a}_i^2 = 0$, so $\hat{T}_i^2 = 0$ and $e^{\hat{T}_i} = 1 + \hat{T}_i$. Combining with a) gives $e^{\hat{T}} = \prod_i e^{\hat{T}_i} = \prod_i (1 + \hat{T}_i)$. Expanding the product,

$$\prod_i (1+\hat{T}_i) = 1 + \sum_i \hat{T}_i + \sum_{i<j} \hat{T}_i \hat{T}_j + \cdots = 1 + \hat{T} + \mathscr{O}(\hat{T}^2), \tag{6.120}$$

so the difference from $1+\hat{T}$ consists of the cross terms $\hat{T}_i\hat{T}_j$ with $i \neq j$, which are $2p$-$2h$, $3p$-$3h$, ... excitations. They are second and higher order in the amplitudes, which is why the linearisation is exact for an infinitesimal variation and why the stability analysis of Section 6.10, which works to second order in the energy but first order in the state, is consistent.

Answer to c).

The class `ThoulessRotation` reports $\max|[\hat{T}_i, \hat{T}_j]| = 2.8 \times 10^{-17}$, $\max|\hat{T}_i^2| = 0$ exactly, agreement between $e^{\hat{T}}|c\rangle$ and $\prod_i \hat{b}_i^\dagger|0\rangle$ to 5.6×10^{-17}, an overlap $\langle c|c'\rangle = 1$ and $\det(f) = 1$ over the occupied block. Scaling the amplitudes by s and comparing $e^{s\hat{T}}|c\rangle$ with $(1+s\hat{T})|c\rangle$ gives errors 3.68×10^{-2}, 9.20×10^{-3}, 2.30×10^{-3}, 5.75×10^{-4} and 1.44×10^{-4} for $s = 0.4, 0.2, 0.1, 0.05, 0.025$: each halving of s divides the error by four, confirming the $\mathscr{O}(s^2)$ scaling.

Answer to d).

Expanding Eq. (6.41) and taking the overlap with $\langle c| = \langle 0|\hat{a}_{i_n}\cdots\hat{a}_{i_1}$, only terms in which every $\hat{a}_p^\dagger$ carries an index below the Fermi level survive, and each must be contracted with a distinct $\hat{a}_{i_k}$. Summing over the assignments with their signs reconstructs $\sum_\pi \operatorname{sgn}(\pi)\prod_k f_{i_k\pi(i_k)} = \det(f_{ip})$ over the occupied block. The proof needs f restricted to that block to be invertible, since f^{-1} appears in Eq. (6.43); a non-vanishing determinant is precisely invertibility. Geometrically, if $\langle c|\tilde{c}\rangle = 0$ the two determinants have no overlap and $|\tilde{c}\rangle$ is not in the neighbourhood of $|c\rangle$ at all – one cannot reach it by an infinitesimal rotation, and Thouless' theorem makes no claim about it.

Exercise 4: BCH and Trotter

a) Derive Eq. (6.48) by matching the power series of $e^X e^Y$ and e^Z to second order.
b) Show that if $[X,Y] = cI$ then Eq. (6.50) is exact, and use it to derive the Weyl form $e^{ia\hat{x}/\hbar}e^{ib\hat{p}/\hbar} = e^{-iab/2\hbar}e^{i(a\hat{x}+b\hat{p})/\hbar}$.
c) Use Eq. (6.46) to show that the leading error of the first-order splitting is $-\frac{1}{2}[H_1,H_2]\Delta^2$, and verify that the symmetric formula (6.53) cancels it. Which commutators govern the leading error of S_2?
d) Prove the Baker-Hausdorff lemma $e^{-X}Ye^X = Y - [X,Y] + \frac{1}{2!}[X,[X,Y]] - \cdots$ and explain why the corresponding series in coupled-cluster theory terminates.
e) Reproduce the two tables of Sections 6.8 and 6.9 with `hartreefock.py`, and estimate how many Trotter steps a first- and a second-order scheme need for an accuracy $\varepsilon = 10^{-3}$ at $t = 1$.

Answer to a).

See Eq. (6.47) and the discussion following it. The essential observation is that $\frac{1}{2}(X+Y)^2$ symmetrises the cross term into $\frac{1}{2}(XY+YX)$, whereas $e^X e^Y$ produces XY with unit weight; the deficit is $\frac{1}{2}(XY-YX)$.

Answer to b).

If $[X,Y]=cI$ then $[X,[X,Y]]=[Y,[X,Y]]=0$ and every term of Eq. (6.46) beyond $\frac{1}{2}[X,Y]$ contains at least one such double commutator. For $X=ia\hat{x}/\hbar$ and $Y=ib\hat{p}/\hbar$ we have $[X,Y]=-(ab/\hbar^2)[\hat{x},\hat{p}]=-iab/\hbar$, so

$$e^X e^Y=\exp\left(\frac{i}{\hbar}(a\hat{x}+b\hat{p})-\frac{iab}{2\hbar}\right)=e^{-iab/2\hbar}\,e^{i(a\hat{x}+b\hat{p})/\hbar}. \tag{6.121}$$

The same argument applied to $[\hat{a},\hat{a}^\dagger]=1$ gives the normal-ordered form of the coherent-state displacement operator.

Answer to c).

Setting $X=-iH_1\Delta$, $Y=-iH_2\Delta$ in Eq. (6.46), the second term is $\frac{1}{2}[X,Y]=-\frac{1}{2}[H_1,H_2]\Delta^2$, which is Eq. (6.51). For the symmetric product, write $S_2(\Delta)=e^{X/2}e^Ye^{X/2}$; applying BCH twice, the $[X,Y]$ contributions from the two half-steps enter with opposite signs and cancel exactly. What survives at the next order is a combination of $[H_1,[H_1,H_2]]$ and $[H_2,[H_1,H_2]]$ multiplying Δ^3. This is also visible from symmetry alone: $S_2(-\Delta)=S_2(\Delta)^{-1}$, so the error must be an odd function of Δ and cannot contain a Δ^2 term.

Answer to d).

Define $G(\lambda)=e^{-\lambda X}Ye^{\lambda X}$. Then $G'(\lambda)=e^{-\lambda X}[Y,X]e^{\lambda X}=-e^{-\lambda X}[X,Y]e^{\lambda X}$, and iterating, $G^{(n)}(0)=(-1)^n\,\mathrm{ad}_X^n(Y)$ with $\mathrm{ad}_X(Y)=[X,Y]$. Taylor-expanding about $\lambda=0$ and setting $\lambda=1$ gives the lemma. In coupled-cluster theory $X=\hat{T}$ is a sum of excitation operators, all of which commute with one another, and $Y=\hat{H}$ contains at most two-body terms. Each commutator with $\hat{T}$ must contract at least one annihilation operator of $\hat{H}$, and $\hat{H}$ has only four, so the series terminates exactly after the fourth commutator – which is why coupled cluster is a finite, not an asymptotic, expansion.

Answer to e).

The tables are reproduced by `demo_bch()` and `demo_trotter()`. Reading off the first-order column, the error is about $1.0/N$, so $\varepsilon=10^{-3}$ requires $N\approx10^3$ steps. The second-order error is about $0.39/N^2$, giving $N\approx20$. The second-order scheme costs about 50% more exponentials per step, so it is cheaper by more than an order of magnitude here – and the advantage grows as the accuracy demand tightens, which is why practical quantum-simulation circuits are never built at first order.

Exercise 5: stability and the Lipkin model

a) Derive Eq. (6.56) for the norm of the rotated determinant, keeping track of which terms are second order.
b) Starting from Eq. (6.63), obtain Eq. (6.64) and the stability matrix (6.65). Why does χ contain both δC and δC^*?

c) For the Lipkin model with $W = 0$, show that $A = 0$ and that the largest eigenvalue of B is $|V|(N-1)$, and hence recover Eq. (6.73).
d) Derive the mean-field energy (6.70) from the rotation (6.69), and locate both stationary points.
e) Reproduce Table 6.1 and comment on the gap between E_{mf} and E_{exact} on either side of $\chi = 1$.

Answer to a).

Expanding Eq. (6.55) and taking the overlap with itself, the zeroth-order term gives 1; the cross terms between the constant and the linear piece vanish, since $\langle c|\hat{a}_a^\dagger \hat{a}_i|c\rangle = 0$ for $a > F \geq i$; and the linear piece with itself gives $\sum_{abij} \delta C_{ai}^* \delta C_{bj} \langle \Phi_i^a | \Phi_j^b \rangle = \sum_{ai} |\delta C_{ai}|^2$ by the orthonormality of the excited determinants. Terms involving the quadratic piece of the expansion are of third order or higher when paired with the linear one, and of fourth order with themselves.

Answer to b).

Subtract $(1 + \sum |\delta C|^2) E_0$ from Eq. (6.63) and divide by the norm; what is left over the norm is ΔE. Collecting the terms, those proportional to $\delta C^* \delta C$ assemble into $\chi^\dagger$ times the diagonal blocks $\Delta + A$, and those proportional to $\delta C \delta C$ and $\delta C^* \delta C^*$ into the off-diagonal blocks B and B^*. The doubled vector χ is needed precisely because the quadratic form is not Hermitian in δC alone: the terms $B\,\delta C^* \delta C^*$ pair two conjugated amplitudes, and only by treating δC and δC^* as independent components of a single vector can the whole expression be written as a Hermitian form. This doubling is exactly the structure of the RPA equations, where it produces the pairs of solutions $\pm\omega$.

Answer to c).

With $W = 0$ the interaction is $(V/2)(\hat{J}_+^2 + \hat{J}_-^2)$, which changes J_z by ± 2, so it has no matrix element within the $1p$-$1h$ space and $A_{ai,bj} = 0$; the diagonal block is $\Delta = \varepsilon \boldsymbol{I}$, since every particle-hole excitation costs ε. For B, the operator $\hat{J}_-^2$ connects the reference to a $2p$-$2h$ state in which levels p and q are both excited, giving $B_{ai,bj} = V$ for $p \neq q$ and zero for $p = q$; that is, $\boldsymbol{B} = V(\boldsymbol{J} - \boldsymbol{I})$ within each spin block, with $\boldsymbol{J}$ the matrix of ones. Its eigenvalues are $V(N-1)$, once, and $-V$, $N-1$ times. Because $\hat{M}$ has the form $\left(\begin{smallmatrix} \varepsilon\boldsymbol{I} & \boldsymbol{B} \\ \boldsymbol{B} & \varepsilon\boldsymbol{I} \end{smallmatrix}\right)$, its eigenvalues are $\varepsilon \pm b$ with b an eigenvalue of $\boldsymbol{B}$, and the smallest is $\varepsilon - |V|(N-1) = \varepsilon(1-\chi)$.

Answer to d).

Under the rotation (6.69) the quasispin vector rotates rigidly, $\langle \hat{J}_z \rangle = -\frac{N}{2}\cos\alpha$ and $\langle \hat{J}_x \rangle = \frac{N}{2}\sin\alpha$. The one-body term gives $-\frac{N\varepsilon}{2}\cos\alpha$ directly. For the interaction, $\hat{J}_+^2 + \hat{J}_-^2 = 2(\hat{J}_x^2 - \hat{J}_y^2)$, and evaluating this in the rotated determinant – where the N particles are uncorrelated, so that $\langle \hat{J}_x^2 \rangle = \langle \hat{J}_x \rangle^2 + \mathscr{O}(N)$ – gives $-\frac{N\varepsilon\chi}{4}\sin^2\alpha$ with $\chi = |V|(N-1)/\varepsilon$, the factor $N-1$ rather than N arising because a particle does not interact with itself. Setting the derivative (6.71) to zero gives $\sin\alpha = 0$ or $\cos\alpha = 1/\chi$, the second having a solution only when $\chi \geq 1$.

Answer to e).

The table is produced by `demo_stability()`. Below $\chi = 1$ the mean-field energy is flat at $-N\varepsilon/2 = -2$ while the exact energy falls steadily, so the error grows: the mean field is stable but increasingly inadequate, and the missing energy is pure correlation. At $\chi = 1.5$ and 2.0 the deformed solution recovers part of it – 0.167 and 0.5 respectively – but remains 0.48 and 0.56 above the exact answer. The deformed state breaks the symmetry of $\hat{H}$ under the rotation generated by $\hat{J}_z$ parity, which the exact ground state respects, so part of the remaining error is spurious and would be removed by projecting the broken symmetry back out.

Exercise 6: the electron gas

a) Derive Eq. (6.88) for a general interaction $v(r)$, and specialise it to the screened Coulomb interaction.
b) Show that $\hat{H}_b$ and $\hat{H}_{\mathrm{el}-b}$ take the forms (6.81), and that together with the direct term of the electron-electron repulsion they cancel in the thermodynamic limit.
c) Carry out the steps from Eq. (6.93) to Eq. (6.94).
d) Compute the Hartree-Fock effective mass m^*_{HF} and the level density $n(\varepsilon_F^{\mathrm{HF}})$, and comment on the result.
e) Show that the energy per electron is Eq. (6.108), plot it, and explain why the system is bound. Compute the pressure $P = -(\partial E/\partial\Omega)_N$ and the bulk modulus $B = -\Omega(\partial P/\partial\Omega)_N$, and comment.

Answer to a).

See Eqs. (6.85)–(6.88). For the screened Coulomb interaction the Fourier transform is

$$\int \frac{e^2 e^{-\mu r}}{r} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}} d\boldsymbol{r} = \frac{4\pi e^2}{\mu^2 + q^2} \xrightarrow[\mu\to 0]{} \frac{4\pi e^2}{q^2}, \tag{6.122}$$

which is Eq. (6.96) again. The q^{-2} divergence at small momentum transfer is the signature of the infinite range of the interaction, and it is the source of every difficulty in this section.

Answer to b).

Both are elementary integrals of $e^{-\mu r}/r$ over the box with a constant density $n = N/\Omega$: $\int d\boldsymbol{r}\, e^{-\mu r}/r = 4\pi/\mu^2$, whence $\hat{H}_b = \frac{e^2}{2} n^2 \Omega \cdot 4\pi/\mu^2 = \frac{e^2}{2}\frac{N^2}{\Omega}\frac{4\pi}{\mu^2}$ and $\hat{H}_{\mathrm{el}-b} = -e^2 \frac{N^2}{\Omega}\frac{4\pi}{\mu^2}$, twice as large and of opposite sign. The direct term of Eq. (6.92) is $\frac{N^2}{2\Omega}\int v(r) d\boldsymbol{r} = \frac{e^2}{2}\frac{N^2}{\Omega}\frac{4\pi}{\mu^2}$, and the three sum to zero. Physically: a neutral system has no monopole, so the $\boldsymbol{q} = 0$ component of the interaction must drop out, and it is only the separation into three individually divergent pieces that made the cancellation look accidental.

Answer to c).

See Eqs. (6.95)–(6.99). The last integral is

$$\int_0^{k_F} p \ln\left|\frac{p+k}{p-k}\right| dp = kk_F + \frac{k_F^2 - k^2}{2} \ln\left|\frac{k+k_F}{k-k_F}\right|, \tag{6.123}$$

which may be checked by differentiating with respect to k_F, or numerically.

Answer to d).

Differentiating Eq. (6.102),

$$\frac{m^*_{\rm HF}}{m} = \frac{2}{2 - 0.663\, r_s F'(1)}, \qquad F'(x) \sim -\frac{1}{2}\ln\left|\frac{2}{1-x}\right| \;\text{ as } x \to 1. \tag{6.124}$$

Since $F'(1) = -\infty$, the denominator diverges and $m^*_{\rm HF} \to 0$; the program gives 0.256, 0.185, 0.144, 0.118 and 0.100 at $1-x = 10^{-2}, \ldots, 10^{-6}$, decreasing like the reciprocal logarithm. By Eq. (6.107) the level density is proportional to m^* and therefore vanishes at the Fermi surface. This contradicts experiment: the electronic specific heat of a metal is linear in T with a coefficient proportional to $n(\varepsilon_F)$, and it is measured to be finite. The failure is traceable to the unscreened $1/q^2$, and it is repaired by the random-phase approximation, which replaces it by a screened interaction with a finite range.

Answer to e).

The kinetic term is Eq. (6.91); writing $k_F = (9\pi/4)^{1/3}/r_0$ and measuring in Rydberg,

$$\frac{\langle \hat{T} \rangle}{N} = \frac{3}{5}\frac{\hbar^2 k_F^2}{2m} = \frac{e^2}{2a_0}\frac{3}{5}\left(\frac{9\pi}{4}\right)^{2/3}\frac{1}{r_s^2} = \frac{e^2}{2a_0}\frac{2.21}{r_s^2}, \tag{6.125}$$

and the exchange term evaluates similarly to $-0.916/r_s$ in the same units. The system is bound because the two terms carry opposite signs and different powers of r_s: the kinetic repulsion, $\propto r_s^{-2}$, dominates at high density and the exchange attraction, $\propto r_s^{-1}$, at low density, so their sum has a minimum, at $r_s = 4.83$ with $E_0/N = -0.0949$ Ry. For the pressure, with $\Omega \propto N r_0^3$ so that $\partial/\partial\Omega = -(r_s/3\Omega)\,\partial/\partial r_s$,

$$P = -\left(\frac{\partial E}{\partial \Omega}\right)_N = \frac{\rho\, r_s}{3}\frac{d}{dr_s}\left(\frac{E_0}{N}\right) = \frac{e^2\rho}{2a_0}\frac{1}{3}\left[-\frac{4.42}{r_s^2} + \frac{0.916}{r_s}\right], \tag{6.126}$$

which vanishes exactly at the equilibrium r_s, as it must, is negative (the gas would contract) at higher density and positive at lower. The bulk modulus $B = -\Omega(\partial P/\partial\Omega)_N$ is positive at the minimum, confirming mechanical stability, and its measured values in the alkali metals are of the right order but consistently smaller than the Hartree-Fock prediction – correlation softens the gas.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter06/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/hartreefock.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter06/`.

`hartreefock.py` The self-consistent field with the density matrix, Thouless rotations, the stability matrix, the Baker–Campbell–Hausdorff and Trotter errors, and the infinite electron gas. Runs in a few seconds; needs `scipy.linalg` for `expm` and `logm`.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 7
Mean-field applications: TDA and RPA

Chapter 6 ended with a matrix. Expanding the Hartree-Fock energy to second order in the amplitudes of a Thouless rotation produced the stability matrix (6.65), and we remarked in passing that it is the matrix of the random-phase approximation. This chapter makes good on that remark. We shall see that the same two blocks A and B, arranged Hermitian, decide whether the mean field is a minimum, and arranged non-Hermitian, give the excitation energies of the system. Stability and spectroscopy are two readings of one calculation.

The vehicle is a schematic model with two competing interactions – a pairing term that scatters whole pairs between levels, and a particle-hole term that breaks them. At zero particle-hole strength it reduces exactly to the pairing model of Section 4.2, so every number can be checked against Table 4.1; switching the particle-hole term on makes the particle-hole channel non-trivial and lets us watch two mean fields compete. We solve the model exactly by the methods of Chapter 5, then in the Tamm-Dancoff approximation, then in the random-phase approximation on the Hartree-Fock state, and finally in the quasiparticle random-phase approximation built on a BCS vacuum. Comparing four approximations against one exact answer, in a system small enough that nothing is hidden, is the point of the exercise.

7.1 The pairing plus particle-hole model

We take L doubly degenerate, equally spaced single-particle levels $p = 1,\ldots,L$, each carrying a spin label $\sigma = \pm$, occupied by N spin-$\frac{1}{2}$ fermions. The Hamiltonian is $\hat{H} = \hat{H}_0 + \hat{V}_{\text{pair}} + \hat{V}_{\text{ph}}$ with

$$\hat{H}_0 = \xi \sum_{p\sigma} (p-1)\, \hat{a}^\dagger_{p\sigma} \hat{a}_{p\sigma}, \tag{7.1}$$

$$\hat{V}_{\text{pair}} = -\frac{g}{2} \sum_{pq} \hat{a}^\dagger_{p+} \hat{a}^\dagger_{p-} \hat{a}_{q-} \hat{a}_{q+}, \tag{7.2}$$

$$\hat{V}_{\text{ph}} = -\frac{f}{2} \sum_{pqr} \left(\hat{a}^\dagger_{p+} \hat{a}^\dagger_{p-} \hat{a}_{q-} \hat{a}_{r+} + \text{h.c.} \right). \tag{7.3}$$

The one-body term gives level p the energy $\xi(p-1)$; we set $\xi = 1$ throughout. The pairing term, met already in Section 4.2, moves a complete pair – both spin components of one level – from level q to level p. The particle-hole term removes a spin-down particle from level q and a spin-up particle from level r and deposits a pair at level p; when $q \neq r$ it therefore *breaks* a pair, which is exactly what the pairing term cannot do.

Both terms conserve N_+ and N_- separately, so $S_z = (N_+ - N_-)/2$ is a good quantum number and the low-lying spectrum lives in the balanced sector $S_z = 0$. For $N = L = 4$ the full space has $\binom{8}{4} = 70$ determinants, of which 36 have $S_z = 0$.

Two limits organise everything that follows.

At $f = 0$.

The model is the pairing model of Chapter 4. The seniority – the number of unpaired particles – is conserved, the ground state has seniority zero, and the exact energies are those of Table 4.1. The program `rpa.py` reproduces them:

g	E_{ref}	E_0 (this chapter)	Table 4.1
−1.00	3.000000	2.77987014	2.77987014
−0.50	2.500000	2.43688426	2.43688426
0.00	2.000000	2.00000000	2.00000000
0.50	1.500000	1.41677428	1.41677428
1.00	1.000000	0.63554847	0.63554847

At $g = 0$.

Only the pair-breaking term survives, and the seniority is maximally violated. For intermediate f/g the two mechanisms compete, and the interesting question – the one the whole chapter is about – is which mean field to build the fluctuations on.

7.2 Exact diagonalisation

The exact solution is Chapter 5 applied without modification. Determinants are bit strings, as in Section 3.15; the Hamiltonian is built by applying each term of Eqs. (7.1)–(7.3) to every basis state and routing the result back to its index with the accumulated fermion sign; and the lowest eigenvalues follow from the algorithms of Chapter 1 – a dense solve when the matrix is small enough, Lanczos when it is not.

Restricting to $S_z = 0$ already cuts the dimension from 70 to 36, an instance of the general remark of Section 5.11 that symmetries are worth exploiting. For $g = 0.5$ and $f = 0.05g$ the lowest exact eigenvalues are

$$E_0 = 1.347133, \quad E_1 = 2.710867, \quad E_2 = 2.711098, \quad E_3 = 3.413340, \tag{7.4}$$

with the near-degeneracy of E_1 and E_2 split only by the small particle-hole term. These are the numbers every approximation in this chapter will be measured against.

Figure 7.1 follows the same eigenvalues continuously as the couplings are switched on, and it repays a look before any approximation is attempted, because most of what the later sections struggle with is already visible in it. Both panels are pinned at the origin to the unperturbed spectrum $2, 3, 3, 4, 4, 4$ of $\hat{H}_0$ alone, with a non-degenerate ground state and two degenerate multiplets above it. The ground state leaves that point smoothly and falls away at a rate close to $-(g+2f)$, which is what first-order perturbation theory gives for the filled Fermi sea: the diagonal part of $\hat{V}_{\text{pair}}$ contributes $-g/2$ for each of the two occupied pairs, and the $p = q = r$ part of $\hat{V}_{\text{ph}}$ contributes $-f$ for each of them. The combination $g + 2f$ is not an accident of the first order; it is the same combination through which the BCS mean field of Section 7.10 sees the whole interaction. The excited curves, by contrast, have a visible corner at $g = 0$. That is degenerate perturbation theory at work: a degenerate multiplet splits at first

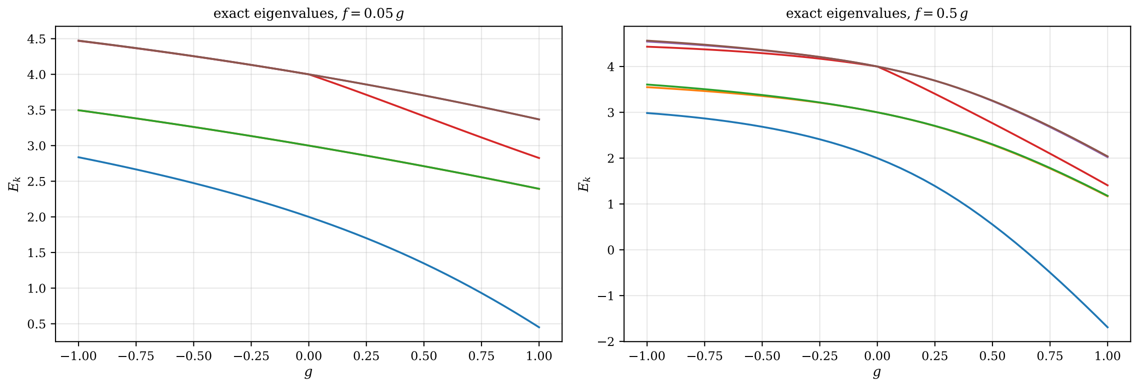

Fig. 7.1 The six lowest exact eigenvalues of the pairing plus particle-hole Hamiltonian (7.1)–(7.3) for $N = L = 4$ and $\xi = 1$, plotted against the pairing strength g over $-1 \le g \le 1$ with the particle-hole strength tied to it as $f = 0.05\,g$ (left panel) and $f = 0.5\,g$ (right panel). Every curve passes through the unperturbed spectrum $2,3,3,4,4,4$ at $g = 0$, where both interactions vanish. With the weak particle-hole term the degenerate multiplets stay together and the curves are close to straight lines; with the strong one they fan out and the ground state plunges, reaching -1.6917 at $g = 1$ against 0.4506 in the left panel. All eigenvalues were obtained by dense diagonalisation in the 36-dimensional $S_z = 0$ sector.

order in the coupling, and because we drive g through zero with f tied to it, the splitting grows like $|g|$ on both sides and the eigenvalues meet in a wedge rather than smoothly.

The two panels differ in degree, not in kind, and the degree is what matters. In the left panel, with $f = 0.05\,g$, the members of each multiplet stay within a line width of one another all the way to $g = 1$ – the pair $E_1 = 2.3931$ and $E_2 = 2.3941$ is drawn as a single curve – which is the graphical version of the near-degeneracy in Eq. (7.4) and of Exercise 7.14d). In the right panel, with $f = 0.5\,g$, every multiplet has separated by $g \approx 0.5$ and the ground state has been pushed down to -1.6917 at $g = 1$, so that the first excitation energy is $1.1678 - (-1.6917) = 2.8595$, the number in the last row of Table 7.2. Two features are worth carrying forward. The ground state never crosses another level anywhere in the window, on either side of $g = 0$, so a single-reference method has at least a well-defined state to aim at. And the gap above it *grows* with the pairing strength rather than closing: from 1.94 at $g = 1,\ f = 0.05g$ to 2.86 at $g = 1,\ f = 0.5g$. Any approximation whose lowest excitation flattens out with increasing g is therefore going wrong in a way that no amount of numerical care will repair, and Section 7.9 will show both TDA and RPA doing exactly that.

7.3 The Hartree-Fock reference

Both approximations to come are built on a stationary mean field, so we solve the Hartree-Fock equations of Section 6.4 first. The antisymmetrised two-body matrix elements are read off the two-particle sector of $\hat{V} = \hat{H} - \hat{H}_0$,

$$\bar{v}_{abcd} = \left(\hat{a}_a^\dagger \hat{a}_b^\dagger |0\rangle\right)^\dagger \hat{V} \left(\hat{a}_c^\dagger \hat{a}_d^\dagger |0\rangle\right), \tag{7.5}$$

and the self-consistent field of Section 6.5 does the rest:

g	f	E_{HF}	iterations	Hartree-Fock energies
0.70	0.00	1.30000	2	$-0.350,\ -0.350,\ 0.650,\ 0.650,\dots$
0.70	0.05	1.19709	20	$-0.404,\ -0.404,\ 0.600,\ 0.600,\dots$
1.00	0.50	-0.26976	40	$-1.505,\ -1.505,\ 0.119,\ 0.119,\dots$

The first row is the case of Section 6.12 again: with pure pairing the filled Fermi sea is already the Hartree-Fock solution, the iteration converges immediately, and the only effect of the interaction is to lower each occupied level by $g/2$ while leaving the empty ones alone. The particle-hole term does connect the reference to a $1p$-$1h$ state, so the mean field must rotate the orbitals until Brillouin's theorem, Eq. (6.31), is restored, and the iteration becomes non-trivial.

7.4 The equations of motion

Rather than derive TDA and RPA separately, we obtain both from a single principle. Suppose the exact eigenstates satisfy $\hat{H}|\nu\rangle = E_\nu|\nu\rangle$ and define, for each excited state, an operator $\hat{Q}_\nu^\dagger$ with

$$|\nu\rangle = \hat{Q}_\nu^\dagger|0\rangle, \qquad \hat{Q}_\nu|0\rangle = 0, \tag{7.6}$$

where $|0\rangle$ is the exact ground state. Such an operator always exists – $\hat{Q}_\nu^\dagger = |\nu\rangle\langle 0|$ will do – and it satisfies

$$\left[\hat{H},\hat{Q}_\nu^\dagger\right]|0\rangle = \omega_\nu \hat{Q}_\nu^\dagger|0\rangle, \qquad \omega_\nu = E_\nu - E_0. \tag{7.7}$$

Taking the matrix element of Eq. (7.7) with an arbitrary variation $\delta\hat{Q}$ and using $\hat{Q}_\nu|0\rangle = 0$ to symmetrise, one arrives at the *equation-of-motion* form

$$\boxed{\langle 0|\left[\delta\hat{Q},\left[\hat{H},\hat{Q}_\nu^\dagger\right]\right]|0\rangle = \omega_\nu\,\langle 0|\left[\delta\hat{Q},\hat{Q}_\nu^\dagger\right]|0\rangle.} \tag{7.8}$$

This is still exact. It becomes an approximation as soon as we (i) restrict the operators $\hat{Q}^\dagger$ to a manageable family, and (ii) replace the unknown exact ground state $|0\rangle$ by something we can compute. Every method in this chapter is a choice of those two things.

The double commutator on the left has a virtue worth noticing: it is symmetric in the two operators, so each of them can be evaluated with fewer contractions than the raw product would need. This is why the RPA matrices are traditionally written as double commutators, and it is how the program computes them – literally, as expectation values of $[\hat{O}_K^\dagger,[\hat{H},\hat{O}_L]]$ in the chosen reference, with nothing transcribed by hand.

7.5 The Tamm-Dancoff approximation

The simplest choice is to build the excitation from one particle-hole pair only, and to take the Hartree-Fock determinant for the ground state:

$$\hat{Q}_\nu^\dagger = \sum_{mi} X_{mi}^\nu \hat{a}_m^\dagger \hat{a}_i, \qquad |0\rangle \longrightarrow |\mathrm{HF}\rangle, \tag{7.9}$$

with $i,j,\dots$ labelling occupied and $m,n,\dots$ unoccupied Hartree-Fock orbitals. Since $\hat{a}_m^\dagger\hat{a}_i$ does annihilate the determinant when acting to the left, condition (7.6) holds exactly, and the right-hand side of Eq. (7.8) reduces to $\delta_{mn}\delta_{ij}$. What remains is an ordinary Hermitian eigenvalue problem,

$$\sum_{nj} A_{mi,nj} X^{\nu}_{nj} = \omega_\nu X^{\nu}_{mi}, \tag{7.10}$$

with

$$A_{mi,nj} = \langle \mathrm{HF}| \left[\hat{a}^\dagger_i \hat{a}_m, \left[\hat{H}, \hat{a}^\dagger_n \hat{a}_j\right]\right] |\mathrm{HF}\rangle = (\varepsilon_m - \varepsilon_i)\,\delta_{mn}\delta_{ij} + \bar{v}_{mjin}. \tag{7.11}$$

There is a plainer way to say what this is. The matrix A is nothing but the Hamiltonian restricted to the space of $1p$-$1h$ determinants, measured from the reference energy,

$$A_{mi,nj} = \langle \Phi^m_i | \hat{H} | \Phi^n_j \rangle - E_{\mathrm{HF}}\,\delta_{mn}\delta_{ij}. \tag{7.12}$$

The program checks this identity directly and finds agreement to 3×10^{-15}. So the Tamm-Dancoff approximation is exactly the configuration-interaction truncation of Section 5.8 applied to excited states: keep the reference and all single excitations, diagonalise, and read off the excitation energies. It is CIS, no more and no less, and it inherits both the variational property – every TDA root is an upper bound to the corresponding exact excitation energy within that space – and the defect, that the ground state is left uncorrelated.

7.6 The random-phase approximation

The Tamm-Dancoff ground state is a single determinant, which we know is wrong. The random-phase approximation improves it by allowing the phonon to *de*-excite as well as excite,

$$\hat{Q}^\dagger_\nu = \sum_{mi} \left(X^{\nu}_{mi}\, \hat{a}^\dagger_m \hat{a}_i - Y^{\nu}_{mi}\, \hat{a}^\dagger_i \hat{a}_m \right). \tag{7.13}$$

The second term does not annihilate a Slater determinant acting to the right, so the condition $\hat{Q}_\nu|0\rangle = 0$ can no longer be satisfied by the Hartree-Fock state. It defines instead a new, correlated ground state $|\mathrm{RPA}\rangle$ – one that we never construct explicitly, but whose existence is the whole content of the approximation.

To make progress we need the norm on the right of Eq. (7.8), which requires the commutator of two particle-hole operators,

$$\left[\hat{a}^\dagger_i \hat{a}_m, \hat{a}^\dagger_n \hat{a}_j\right] = \delta_{mn} \hat{a}^\dagger_i \hat{a}_j - \delta_{ij} \hat{a}^\dagger_n \hat{a}_m. \tag{7.14}$$

This is an operator, not a number. The *quasiboson approximation* replaces it by its expectation value in the reference determinant,

$$\left[\hat{a}^\dagger_i \hat{a}_m, \hat{a}^\dagger_n \hat{a}_j\right] \;\longrightarrow\; \langle \mathrm{HF}| \left[\hat{a}^\dagger_i \hat{a}_m, \hat{a}^\dagger_n \hat{a}_j\right] |\mathrm{HF}\rangle = \delta_{mn}\delta_{ij}, \tag{7.15}$$

which is to say that particle-hole pairs are treated as bosons. It is exact when the occupations are exactly zero and one and nothing else is going on; it fails to the extent that the true ground state has particles above the Fermi level, and it is the origin of every pathology of the method.

With Eq. (7.15), Eq. (7.8) becomes the *RPA eigenvalue problem*

$$\boxed{\begin{pmatrix} A & B \\ -B^* & -A^* \end{pmatrix} \begin{pmatrix} X^\nu \\ Y^\nu \end{pmatrix} = \omega_\nu \begin{pmatrix} X^\nu \\ Y^\nu \end{pmatrix},} \tag{7.16}$$

with A as before and

$$B_{mi,nj} = -\langle \mathrm{HF}| \left[\hat{a}^\dagger_i \hat{a}_m, \left[\hat{H}, \hat{a}^\dagger_j \hat{a}_n\right]\right] |\mathrm{HF}\rangle = \bar{v}_{mnij}. \tag{7.17}$$

The program verifies both closed forms against the double commutators to better than 10^{-13}. Setting $B=0$ discards the Y amplitudes and returns the Tamm-Dancoff equation, so TDA is RPA with the ground-state correlations switched off.

Equation (7.16) is not Hermitian. Its eigenvalues come in pairs $\pm\omega_\nu$ – if (X,Y) solves it with ω, then (Y^*,X^*) solves it with $-\omega$ – and they need not be real. The physical solutions are the positive ones, normalised by

$$\sum_{mi}\left(|X^\nu_{mi}|^2-|Y^\nu_{mi}|^2\right)=1, \tag{7.18}$$

a metric that is indefinite rather than positive, which is precisely why real eigenvalues are not guaranteed.

7.7 RPA and the stability of the mean field

We can now close the loop with Chapter 6. The blocks A and B of Eqs. (7.11) and (7.17) are, term for term, the blocks of the stability matrix (6.65); only the arrangement differs,

$$\hat{M}_{\text{stab}}=\begin{pmatrix}A & B\\ B^* & A^*\end{pmatrix}\ \text{(Hermitian)},\qquad \hat{M}_{\text{RPA}}=\begin{pmatrix}A & B\\ -B^* & -A^*\end{pmatrix}\ \text{(not Hermitian)}. \tag{7.19}$$

The relation between them is exact and easy to state. Writing $\boldsymbol{\eta}=\mathrm{diag}(\mathbf{1},-\mathbf{1})$ for the metric of Eq. (7.18), we have $\hat{M}_{\text{RPA}}=\boldsymbol{\eta}\,\hat{M}_{\text{stab}}$. If $\hat{M}_{\text{stab}}$ is positive definite it has a Hermitian square root, and

$$\hat{M}^{1/2}_{\text{stab}}\,\boldsymbol{\eta}\,\hat{M}^{1/2}_{\text{stab}} \tag{7.20}$$

is Hermitian and similar to $\hat{M}_{\text{RPA}}$, so the RPA frequencies are real. Conversely, if $\hat{M}_{\text{stab}}$ has a negative eigenvalue the argument fails and imaginary roots appear. Hence

$$\boxed{\text{the RPA frequencies are real} \iff \hat{M}_{\text{stab}}\geq 0 \iff \text{the mean field is a local minimum.}} \tag{7.21}$$

The Lipkin model of Sections 4.1 and 6.11 shows this happening. Feeding the same blocks into both arrangements:

χ	$\lambda_{\min}(\hat{M}_{\text{stab}})$	imaginary roots	lowest real RPA root
0.25	0.75	0	0.968246
0.50	0.50	0	0.866025
0.90	0.10	0	0.435890
1.00	0.00	0	0.000000
1.50	−0.50	2	—
2.00	−1.00	2	—

Table 7.1 The Lipkin model with $N=4$, $\varepsilon=1$, $W=0$. The lowest eigenvalue of the Hermitian stability matrix and the RPA spectrum built from the same blocks cross zero together at $\chi=1$. Beyond that the collective root is imaginary and no excitation energy can be quoted.

The collective root goes *soft* – falls to zero – exactly at $\chi=1$, where the deformed solution of Section 6.11 appears, and becomes imaginary beyond it. This is the general pattern, and it is one of the most useful things RPA does: a vanishing frequency announces a mean-field phase transition, and an imaginary one says the reference is a saddle point. The excitation spectrum and the stability analysis are the same calculation read two ways.

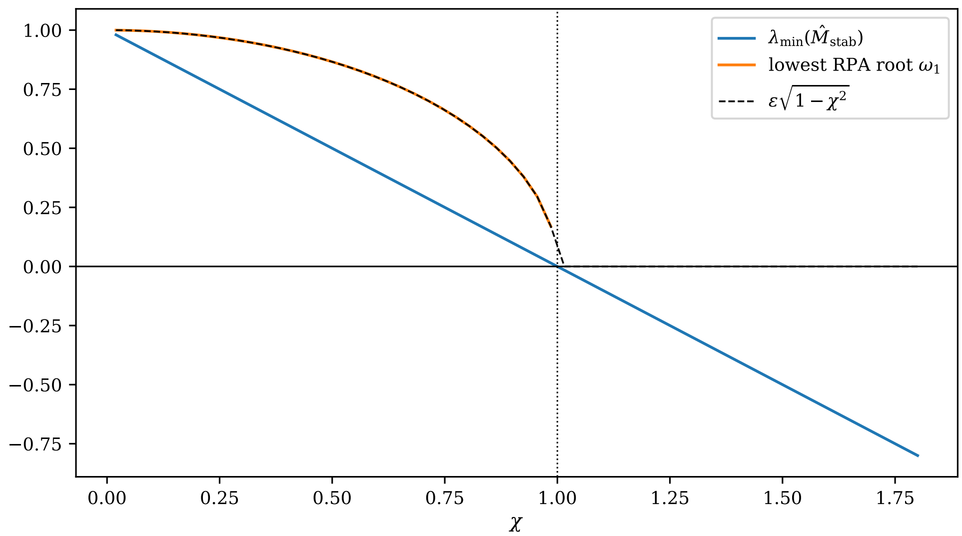

Fig. 7.2 The Lipkin model with $N = 4$, $\varepsilon = 1$ and $W = 0$ as the dimensionless interaction strength χ is varied. The lowest eigenvalue of the Hermitian stability matrix falls along the straight line $\lambda_{\min} = \varepsilon(1-\chi)$ and changes sign at $\chi = 1$, marked by the dotted vertical line, while the lowest root of the non-Hermitian RPA matrix built from the same two blocks follows $\varepsilon\sqrt{1-\chi^2}$ (dashed) and reaches zero at the same point with infinite slope. The RPA curve stops there because beyond $\chi = 1$ the root has become imaginary and there is no excitation energy to plot. The figure is the continuous version of Table 7.1.

Figure 7.2 shows the two readings side by side over a continuous range of couplings, and the difference in the *manner* of their vanishing is worth dwelling on. Both quantities cross zero at $\chi = 1$, as they must, but the stability eigenvalue crosses linearly while the frequency comes down as a square root. The relation between the two makes this inevitable. In the collective channel the Lipkin blocks reduce to the numbers $A = \varepsilon$ and $B = \varepsilon\chi$, so that $\omega_1 = \sqrt{A^2 - B^2} = \sqrt{(A-B)(A+B)}$ with $A - B = \varepsilon(1-\chi) = \lambda_{\min}$, and hence $\omega_1 = \sqrt{\lambda_{\min}\,\varepsilon(1+\chi)}$: the frequency is literally the square root of the stability eigenvalue, up to a factor that stays finite. A linear zero in the Hermitian problem therefore always appears as a square-root zero in the frequency, and this is general, not a peculiarity of Lipkin.

The practical consequence is that the frequency is a poor early warning. At $\chi = 0.9$ the mean field has lost nine tenths of its rigidity, $\lambda_{\min} = 0.1$, but the collective mode still sits at $\omega_1 = 0.4359$, four times higher, and on the scale of the figure looks perfectly healthy. By the time ω_1 has fallen to half its non-interacting value, at $\chi = 0.866$, the stability eigenvalue is already down to 0.134. If one is hunting for the point at which a Hartree-Fock solution ceases to be a minimum – and after Chapter 6 that is a question one should always ask – it is the eigenvalues of $\hat{M}_{\text{stab}}$ that should be monitored, because they approach zero at a finite rate and can be extrapolated. The reader should also keep in mind that none of this happens to the *exact* Lipkin spectrum, which passes through $\chi = 1$ perfectly smoothly for $N = 4$; the transition drawn here belongs to the mean field, and Section 7.10 will produce another example of the same illusion in the pairing channel.

For the pairing plus particle-hole model the particle-hole stability matrix never goes negative – $\lambda_{\min}$ stays at or above 1 for every coupling we consider, including $g = 4$, $f = 2$. The particle-hole channel is simply not where this model becomes collective. Its instability is in the pairing channel, and reaching it requires a different mean field, which is the subject of Sections 7.10 and 7.11.

7.8 The RPA ground state and its correlation energy

Because $\hat{Q}_\nu|\mathrm{RPA}\rangle = 0$ with $\hat{Q}$ containing the de-excitation piece, the RPA ground state contains particle-hole pairs. One can show, and Exercise 7.14 asks for the argument, that $|\mathrm{RPA}\rangle$ has the form

$$|\mathrm{RPA}\rangle = \mathscr{N} \exp\left\{ \frac{1}{2} \sum_{mnij} Z_{mi,nj}\, \hat{a}_m^\dagger \hat{a}_i \hat{a}_n^\dagger \hat{a}_j \right\} |\mathrm{HF}\rangle, \qquad \boldsymbol{Z} = \boldsymbol{Y}^* \left(\boldsymbol{X}^*\right)^{-1}, \tag{7.22}$$

an exponential of $2p$-$2h$ operators acting on the Hartree-Fock determinant. The reader will recognise the structure of Thouless' theorem, Eq. (6.34), with two-body operators in place of one-body ones, and the structure of the coupled-cluster ansatz, Eq. (5.29), with the doubles amplitude fixed by the RPA amplitudes rather than solved for independently. RPA is a doubles theory in disguise.

Its energy relative to Hartree-Fock is

$$E_{\mathrm{corr}}^{\mathrm{RPA}} = \frac{1}{2} \left(\sum_\nu \omega_\nu - \mathrm{Tr} A \right), \tag{7.23}$$

a remarkably compact result: the correlation energy is half the difference between the sum of the RPA frequencies and the sum of the TDA ones. Since each RPA root lies below the corresponding TDA root, the correction is always negative – RPA always lowers the energy. For $g = 0.7$, $f = 0.05$ the program gives $\mathrm{Tr}A = 38.423313$ and $\sum_\nu \omega_\nu = 37.824314$, hence $E_{\mathrm{corr}} = -0.299500$.

7.9 Results in the particle-hole channel

We can now compare. The table gives, for four combinations of the two couplings, the Hartree-Fock energy, the RPA-corrected ground-state energy, the exact ground-state energy, and the lowest excitation energy in the three treatments.

g	f	E_{HF}	$E_{\mathrm{HF}} + E_{\mathrm{corr}}$	E_{exact}	ω_1^{TDA}	ω_1^{RPA}	$\omega_1^{\mathrm{exact}}$
0.50	0.025	1.44927	1.29898	1.34713	1.2748	1.2448	1.3637
0.70	0.050	1.19709	0.89759	0.96976	1.3992	1.3412	1.5972
1.00	0.050	0.89709	0.36862	0.45058	1.5492	1.4488	1.9425
1.00	0.500	−0.26976	−1.73461	−1.69174	1.8111	1.6445	2.8596

Table 7.2 The pairing plus particle-hole model with $N = L = 4$, $\xi = 1$. Hartree-Fock, RPA and exact ground-state energies, and the lowest excitation energy in the Tamm-Dancoff and random-phase approximations compared with the exact one.

Three things are worth reading off.

First, RPA always lowers the energy below Hartree-Fock, by construction, and here it *overshoots* the exact answer in every row. This is the standard behaviour: the quasiboson approximation, by treating particle-hole pairs as bosons that can be created without limit, over-counts the ground-state correlations. The overshoot grows with the coupling, from 0.048 at $g = 0.5$ to 0.043 at $g = 1$, $f = 0.5$ – modest here, but in a strongly collective system RPA can overbind badly and eventually collapse altogether.

Second, $\omega_1^{\mathrm{RPA}} < \omega_1^{\mathrm{TDA}}$ in every row. The B block always pushes the collective root down, for the same reason it lowers the ground state: it adds correlations to the ground state, and the excitation energy is measured from a lower floor.

Third, both approximations put the lowest excitation well below the exact one, and the gap widens as the pairing strength grows – at $g = 1$, $f = 0.5$ the exact first excitation is 2.86 against an RPA value of 1.64. This is not a failure of arithmetic but of the choice of channel. The particle-hole phonon knows nothing about pairing correlations, and in this model the low-lying spectrum is shaped by them.

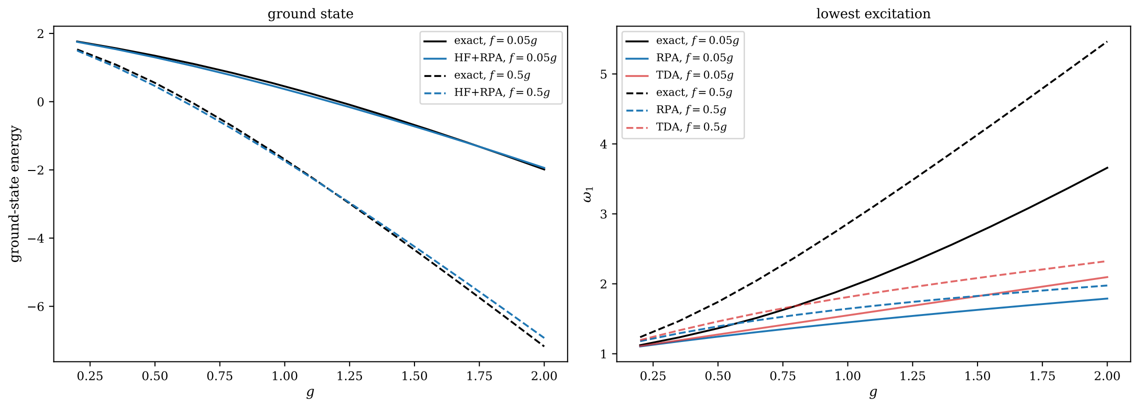

Fig. 7.3 Left: the ground-state energy of the pairing plus particle-hole model with $N = L = 4$ and $\xi = 1$ as a function of g, exactly (black) and as Hartree-Fock plus the RPA correlation energy of Eq. (7.23) (blue), for $f = 0.05\,g$ (solid) and $f = 0.5\,g$ (dashed). Right: the lowest excitation energy ω_1 for the same two cases, exactly and in the random-phase and Tamm-Dancoff approximations. The ground-state curves are barely distinguishable on this scale, the RPA one running just below the exact one at weak and moderate coupling and crossing it near $g \approx 1.75$ for $f = 0.05\,g$ and near $g \approx 1.2$ for $f = 0.5\,g$. The excitation energies are another matter: both approximations flatten out towards 2 while the exact ω_1 climbs almost linearly to 3.66 and 5.46 at $g = 2$.

Figure 7.3 turns the four rows of Table 7.2 into two continuous curves each, and the contrast between its two panels is the most instructive thing in this section. On the left, a theory that is right to about one part in a hundred; on the right, the same theory wrong by a factor of nearly three. The reason is that the two quantities are extracted from the calculation in quite different ways. The correlation energy (7.23) is a sum over *all* the roots, and a sum of sixteen numbers is remarkably forgiving of an error in any one of them; the excitation energy is a single root, with nothing to hide behind. It is a general moral, and not confined to RPA: a method can produce a good total energy while getting the spectrum badly wrong, and the total energy alone is a weak test.

The flattening of the two lower curves in the right panel has a definite cause, and Eq. (7.24) below names it. The particle-hole phonon energy grows like $\sqrt{\xi^2 + g\xi}$, that is like $\sqrt{g}$ at large coupling, whereas the exact first excitation of this model costs the breaking of a pair and therefore grows linearly in g. A square root cannot chase a straight line, and the two curves separate without limit. Notice also how little the approximations care which panel they are in: multiplying f by ten moves the RPA curve at $g = 2$ only from 1.789 to 1.976, and the TDA curve from 2.096 to 2.325, while the exact ω_1 moves from 3.66 to 5.46. The extra particle-hole strength is absorbed almost entirely into the mean field – it rotates the orbitals and lowers E_{HF} – and almost none of it reaches the phonon. That is the clearest statement of the diagnosis given above: the channel is wrong, and refining the treatment within it will not help.

A closed form.

For pure pairing the particle-hole problem collapses. All sixteen particle-hole pairs are degenerate, the matrices reduce to multiples of simple structures, and one finds

$$\omega^{\mathrm{TDA}} = \xi + \frac{g}{2}, \qquad \omega^{\mathrm{RPA}} = \sqrt{\xi^2 + g\xi} = \sqrt{\left(\xi + \tfrac{g}{2}\right)^2 - \left(\tfrac{g}{2}\right)^2}, \tag{7.24}$$

the second being the classic $\sqrt{A^2 - B^2}$ of a two-level RPA problem with $A = \xi + g/2$ and $B = g/2$. The program confirms both to eight digits:

g	TDA	$\xi + g/2$	RPA	$\sqrt{1+g}$
0.5	1.25000000	1.25000000	1.22474487	1.22474487
1.0	1.50000000	1.50000000	1.41421356	1.41421356
2.0	2.00000000	2.00000000	1.73205081	1.73205081
3.0	2.50000000	2.50000000	2.00000000	2.00000000
5.0	3.50000000	3.50000000	2.44948974	2.44948974

Equation (7.24) is worth staring at. The RPA root is real for all $g > -\xi$, so the particle-hole channel of the pure pairing model never goes unstable, however strong the pairing. The instability is elsewhere, and finding it requires a mean field that can break a different symmetry.

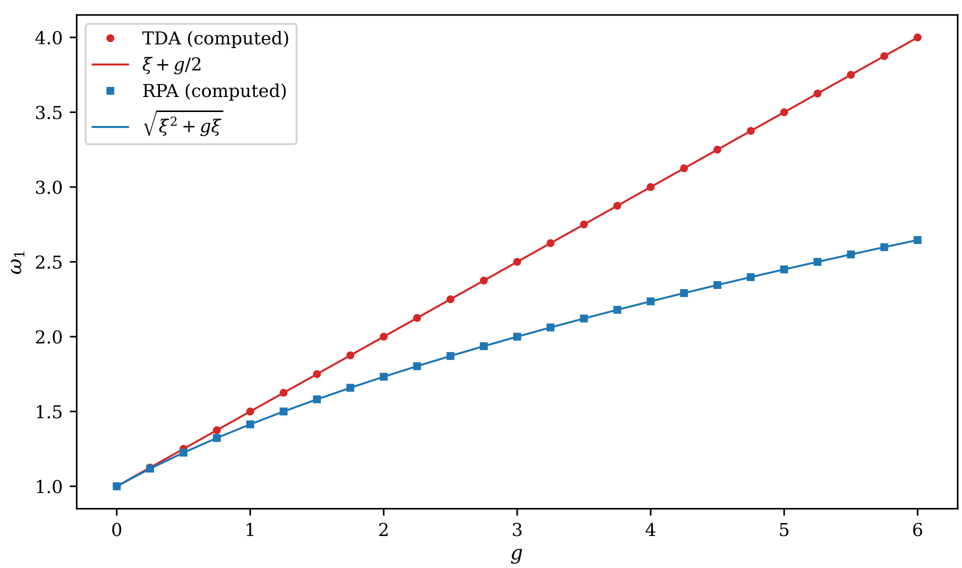

Fig. 7.4 The lowest particle-hole phonon of the pure pairing model ($f = 0$, $N = L = 4$, $\xi = 1$) over $0 \le g \le 6$. The markers are the roots obtained by diagonalising the Tamm-Dancoff matrix A (circles) and the full non-Hermitian RPA matrix of Eq. (7.16) (squares); the lines are the closed forms $\xi + g/2$ and $\sqrt{\xi^2 + g\xi}$ of Eq. (7.24). Computed roots and closed forms agree to the width of the symbols over the whole range. The two branches leave $\omega_1 = \xi = 1$ together at $g = 0$, separate quadratically, and then diverge: at $g = 6$ the Tamm-Dancoff root is exactly 4 while the RPA root is only 2.6458.

Figure 7.4 draws the two closed forms out to $g = 6$, far beyond any coupling for which the model is physically sensible, precisely because the asymptotics are the point. At small g the branches are hard to tell apart: expanding the square root gives $\omega^{\mathrm{RPA}} = \xi + g/2 - g^2/(8\xi) + \mathscr{O}(g^3)$, so the gap between them opens only at second order, and at $g = 1$ it is

$1.5 - 1.4142 = 0.0858$, close to the $1/8$ that the expansion predicts. At large g they part company completely, the Tamm-Dancoff root growing linearly and the RPA root only as $\sqrt{g\xi}$, so that their ratio grows without bound – already 1.51 at $g = 6$. Whatever one thinks of the quasiboson approximation, the correlations it puts into the ground state are not a small correction here, and any statement that TDA and RPA "agree to within a few per cent" is a statement about weak coupling only.

The figure also makes the stability question sharp. For this problem $A = \xi + g/2$ and $B = g/2$, so that $\omega^{\mathrm{RPA}} = \sqrt{(A-B)(A+B)}$ with $A - B = \xi$ *independent of the coupling*. The quantity whose vanishing would signal an instability is therefore frozen at the single-particle level spacing and can never be driven to zero, however large B becomes. This is why the RPA curve, though it bends over, never turns down towards the axis: the particle-hole channel of the pairing model is stable by construction, not by accident. Contrast this with the Lipkin model of Section 7.7, where $A - B = \varepsilon(1-\chi)$ does depend on the coupling and does reach zero, and the difference between a model that becomes collective in the particle-hole channel and one that does not is laid bare in a single algebraic line.

7.10 BCS: the pairing mean field

The pairing interaction is $-G\sum_{pq}\hat{P}_p^\dagger\hat{P}_q$ with $\hat{P}_p^\dagger = \hat{a}_{p+}^\dagger\hat{a}_{p-}^\dagger$ and $G = g/2$. Its natural mean field is not of Hartree-Fock type at all, because the object that acquires an expectation value is $\langle\hat{P}_p^\dagger\rangle$, which does not conserve particle number. The trial state is the BCS wave function

$$|\mathrm{BCS}\rangle = \prod_p \left(u_p + v_p\,\hat{a}_{p+}^\dagger\hat{a}_{p-}^\dagger\right)|0\rangle, \qquad u_p^2 + v_p^2 = 1, \tag{7.25}$$

a product over levels in which each level is empty with amplitude u_p and occupied by a pair with amplitude v_p. It is a superposition of states with different particle numbers, so the number is fixed only on average, by minimising $\langle\mathrm{BCS}|\hat{H} - \lambda\hat{N}|\mathrm{BCS}\rangle$ with λ adjusted until $\langle\hat{N}\rangle = N$.

Because the trial state is a product over levels, every expectation value is available in closed form. Writing $u_p = \cos\theta_p$, $v_p = \sin\theta_p$,

$$\langle\hat{N}\rangle = 2\sum_p v_p^2, \qquad \langle\hat{H}_0\rangle = 2\sum_p \varepsilon_p v_p^2, \tag{7.26}$$

and, using $\langle\hat{P}_p^\dagger\hat{P}_q\rangle = u_p v_p u_q v_q$ for $p \neq q$ and v_p^2 for $p = q$,

$$\langle\hat{H}\rangle = 2\sum_p \varepsilon_p v_p^2 - S\left[\sum_p v_p^2 + \sum_{p\neq q} u_p v_p u_q v_q\right], \qquad S = \frac{g}{2} + f. \tag{7.27}$$

The program checks Eq. (7.27) against the expectation value computed directly in the 2^{2L}-dimensional Fock space and finds agreement to 5×10^{-16}.

The particle-hole term as extra pairing.

Equation (7.27) contains a small surprise. Only the $q = r$ part of $\hat{V}_{\mathrm{ph}}$ survives in the BCS state – for $q \neq r$ the operator $\hat{a}_{q-}\hat{a}_{r+}$ leaves two broken pairs, and the BCS state has none – and $\hat{a}_{q-}\hat{a}_{q+} = \hat{P}_q$ exactly. Seen from the BCS vacuum, therefore, the particle-hole term *is* a pairing term, and the whole problem depends on g and f only through the combination $g+2f$. The program confirms it:

g	f	$g+2f$	E_{BCS}	gap Δ
1.4	0.0	1.4	0.515684	0.789218
1.2	0.1	1.4	0.515684	0.789218
0.8	0.3	1.4	0.515684	0.789218
0.0	0.7	1.4	0.515684	0.789218

This is a clean illustration of a general point: a mean field sees only that part of the interaction which can be written as a one-body field in its own channel. Hartree-Fock and BCS decompose the same $\hat{V}$ differently, and each is blind to what the other sees.

The pairing transition.

With $\Delta = S\sum_p u_p v_p$ the pairing gap, the solution of the BCS equations for $f=0$ is

g	Δ	λ	v_p^2
0.40	0.00000	1.2500	1.000, 1.000, 0.000, 0.000
0.80	0.00000	1.2500	1.000, 1.000, 0.000, 0.000
1.00	0.00000	1.2500	1.000, 1.000, 0.000, 0.000
1.20	0.46716	1.2000	0.984, 0.925, 0.075, 0.016
1.60	1.05608	1.1000	0.934, 0.783, 0.217, 0.066
2.00	1.54088	1.0000	0.887, 0.709, 0.291, 0.113

Table 7.3 The BCS solution of the pairing model with $L=4$ levels and $N=4$ particles. Below $g_c = 1.07037$ the only solution is the normal Fermi sea; above it a gap opens and the occupations become fractional.

Below a critical coupling the only solution is the normal Fermi sea, with $\Delta = 0$ and sharp occupations; above it a gap opens and the occupation numbers spread. The condition for a non-trivial solution, obtained by linearising the gap equation about $\Delta = 0$, is

$$1 = 2S\left[\frac{1}{3\xi+S} + \frac{1}{\xi+S}\right], \tag{7.28}$$

whose root is $S_c = 0.53519$, so that $g_c = 1.07037$ at $f=0$ – exactly the value the program locates by bisection.

Two comments. The transition is *sharp* even though the system is finite, but that sharpness is an artefact of the mean field: the exact ground-state energy of Table 4.1 is perfectly smooth through g_c, and nothing whatever happens there. Mean-field phase transitions in finite systems are approximations to smooth crossovers. And above g_c the reference has stopped being an eigenstate of $\hat{N}$, which is a broken symmetry, with all that this implies for the fluctuations built on top of it.

Figure 7.5 puts the two halves of that statement side by side. The left panel is a textbook order parameter: identically zero on one side of a critical point and rising with the characteristic mean-field exponent on the other, $\Delta \propto (g-g_c)^{1/2}$ with a coefficient close to 1.3 over the whole range shown, so that $\Delta = 0.2168$ at $g = 1.1$ and $\Delta = 0.4672$ at $g = 1.2$. Nothing in the underlying problem is singular at g_c; the non-analyticity is manufactured entirely by insisting

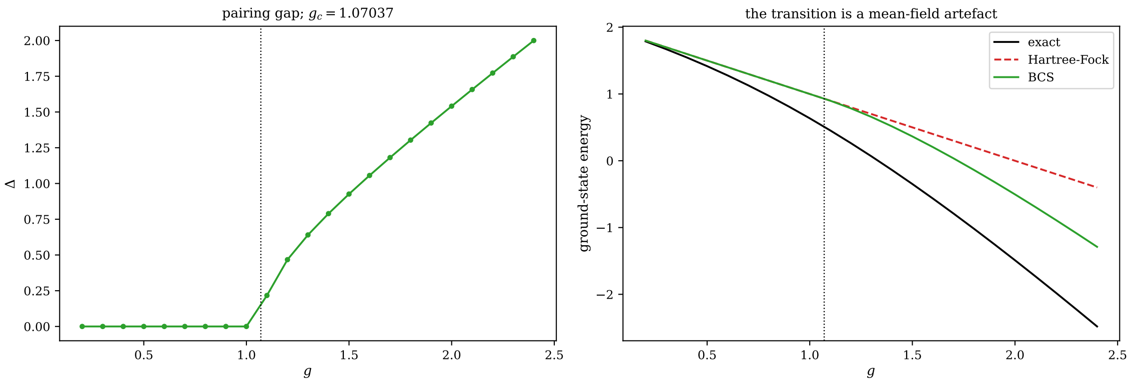

Fig. 7.5 Left: the BCS pairing gap Δ of the pure pairing model with $L = 4$ levels and $N = 4$ particles, obtained by solving the gap and number equations self-consistently at each value of g. It is identically zero below the critical coupling $g_c = 1.07037$, marked by the dotted line, and rises above it like $\Delta \approx 1.3\sqrt{g - g_c}$, reaching $\Delta = 2.0$ at $g = 2.4$. Right: the ground-state energy over the same range, exact, in Hartree-Fock and in BCS. The Hartree-Fock line $2 - g$ is hidden under the BCS curve below g_c, where the two coincide exactly, and separates from it above. The exact curve passes through g_c without a trace of a feature.

that the wave function be a product of the form (7.25). Below g_c that form is flexible enough only to reproduce the sharp Fermi sea, and the variational minimum sits at the boundary $\Delta = 0$ of the parameter space; above g_c the minimum moves into the interior and Δ starts to grow. A constrained minimisation whose solution leaves a boundary is the whole mechanism, and it is why mean-field transitions occur at all in systems of four particles.

The right panel says how little is bought by it. Just above the transition the BCS energy is lower than the Hartree-Fock one by 0.0148 at $g = 1.2$, a gain so small that the two curves cannot be told apart there; by $g = 2.4$ it has grown to 0.889, but the exact energy has by then fallen to -2.4812 against -0.4 for Hartree-Fock, so BCS recovers 43 per cent of a correlation energy of -2.081 and leaves the rest. Below g_c it recovers nothing whatever. The exact curve, meanwhile, bends downwards smoothly and crosses zero near $g = 4/3$ with no hint that anything is supposed to be happening at 1.07. It is the same lesson that Figure 7.2 taught in the particle-hole channel, transposed into the pairing channel: the transition is a statement about the trial wave function, not about the system. What is genuinely new here is that the broken symmetry is $U(1)$ rather than a discrete one, and a broken continuous symmetry leaves a Goldstone mode behind, which is the first thing Section 7.11 has to deal with.

7.11 Quasiparticle RPA

The BCS state is a vacuum, not for particles, but for the Bogoliubov quasiparticles

$$\hat{\alpha}^\dagger_{p+} = u_p \hat{a}^\dagger_{p+} - v_p \hat{a}_{p-}, \qquad \hat{\alpha}^\dagger_{p-} = u_p \hat{a}^\dagger_{p-} + v_p \hat{a}_{p+}, \tag{7.29}$$

which satisfy $\hat{\alpha}_{p\sigma}|\mathrm{BCS}\rangle = 0$. The program verifies this to 3×10^{-17}. These are the natural elementary excitations of the paired system, and the quasiparticle random-phase approximation builds collective modes out of *pairs* of them,

$$\hat{Q}^\dagger_\nu = \sum_{a<b} \left(X^\nu_{ab}\, \hat{\alpha}^\dagger_a \hat{\alpha}^\dagger_b - Y^\nu_{ab}\, \hat{\alpha}_b \hat{\alpha}_a \right). \tag{7.30}$$

The structure of Eq. (7.8) is unchanged; only the reference and the elementary operators are different, so the same routine produces

$$A_{ab,cd} = \langle \mathrm{BCS}| \left[\hat{\alpha}_b\hat{\alpha}_a, \left[\hat{H}', \hat{\alpha}_c^\dagger\hat{\alpha}_d^\dagger\right]\right] |\mathrm{BCS}\rangle, \qquad B_{ab,cd} = -\langle \mathrm{BCS}| \left[\hat{\alpha}_b\hat{\alpha}_a, \left[\hat{H}', \hat{\alpha}_d\hat{\alpha}_c\right]\right] |\mathrm{BCS}\rangle, \quad (7.31)$$

and the same eigenvalue problem (7.16). For $L = 4$ there are $2L = 8$ quasiparticle states and hence $2L(2L-1)/2 = 28$ two-quasiparticle operators.

Two features require attention, and they are the physics of this section.

7.11.1 The spurious mode

The BCS state breaks the $U(1)$ symmetry generated by $\hat{N}$. Whenever a mean field breaks a continuous symmetry of the Hamiltonian, the generator of that symmetry produces an exact zero mode of the RPA equations – a Goldstone mode. The argument is short: since $[\hat{H},\hat{N}] = 0$, the amplitudes

$$X_K^{\mathrm{sp}} = \langle \mathrm{BCS}| \left[\hat{O}_K, \hat{N}\right] |\mathrm{BCS}\rangle, \qquad Y_K^{\mathrm{sp}} = \langle \mathrm{BCS}| \left[\hat{O}_K^\dagger, \hat{N}\right] |\mathrm{BCS}\rangle \quad (7.32)$$

satisfy $\hat{M}_{\mathrm{RPA}}\,(X^{\mathrm{sp}}, Y^{\mathrm{sp}})^T = 0$. Physically the mode is not an excitation at all: it is the rotation of the BCS phase, which costs nothing and merely moves the system to a state with a different average particle number.

The standard device is to build the QRPA with the *cranked* Hamiltonian

$$\hat{H}' = \hat{H} - \lambda\hat{N}, \quad (7.33)$$

using the same λ that the BCS equations extremise, which places the spurious root at zero where it can be discarded. In finite arithmetic it does not land exactly on zero, and because the RPA metric is singular there it may emerge as a small *imaginary* pair rather than a small real one. This is worth emphasising, since it is easily mistaken for an instability:

g	Δ	$\lvert\omega_{\mathrm{sp}}\rvert$	overlap with $\hat{N}$	imaginary roots
1.20	0.46716	1.14×10^{-4}	1.000000	0
1.40	0.78922	1.83×10^{-4}	1.000000	0
1.60	1.05608	2.59×10^{-4}	1.000000	0
1.80	1.30338	1.64×10^{-4}	1.000000	0
2.00	1.54088	2.77×10^{-4}	1.000000	0

Table 7.4 The spurious mode of the QRPA for the pairing model, $f = 0$. The root nearest zero has unit overlap with the amplitudes (7.32) built from the number operator, identifying it beyond doubt. Once it is removed, no imaginary eigenvalues remain at any coupling.

The overlap with the number-operator amplitudes is one to six decimal places, which identifies the near-zero root beyond any doubt. Once it is removed, the QRPA matrix has no imaginary eigenvalues anywhere in the superfluid regime. The lesson generalises: before declaring a mean field unstable on the strength of an imaginary RPA root, check whether the root is a spurious mode in disguise. The test is cheap – compute the amplitudes (7.32) for each broken symmetry and take overlaps.

7.11.2 The mode content

The second feature is that quasiparticles mix particle number, so a two-quasiparticle phonon need not conserve it. Building the phonon operator explicitly, applying it to the BCS vacuum and taking the expectation value of $\hat{N}$ separates the modes cleanly:

ω_ν	$\langle \Delta\hat{N} \rangle$	character
$g = 2.0$, $f = 0$; *exact excitations* 3.3274, 3.3274, 3.4897, 4.2536		
2.6015	0.0000	excitation of the N-particle system
2.9212	0.0000	excitation of the N-particle system (fourfold)
3.7879	−0.3566	pairing vibration
3.7879	+0.3566	pairing vibration

Table 7.5 QRPA phonons at $g = 2$, classified by the average particle number they carry. Modes with $\langle \Delta\hat{N} \rangle = 0$ are excitations within the N-particle system; the others connect towards $N \pm 2$ and are the pairing vibrations. The shift is not quantised because the BCS vacuum is not an eigenstate of $\hat{N}$.

Only the $\langle \Delta\hat{N} \rangle = 0$ modes are directly comparable with the exact spectrum of the N-particle system; the others describe transitions to the neighbouring even systems and would be compared with $E_0(N \pm 2)$. Notice that even the comparable ones lie well below the exact excitations – 2.60 and 2.92 against 3.33 and 3.49 – the same downward bias RPA showed in the particle-hole channel.

7.12 Everything against everything

We can now put the four treatments side by side. For pure pairing, where the exact answers are those of Chapter 4:

g	E_{exact}	E_{HF}	$E_{\text{HF}} + E_{\text{corr}}^{\text{RPA}}$	E_{BCS}	$E_{\text{BCS}} + E_{\text{corr}}^{\text{QRPA}}$
0.40	1.548001	1.60000	1.51748	1.60000	1.46366
0.60	1.277449	1.40000	1.22390	1.40000	1.08964
0.80	0.973552	1.20000	0.90179	1.20000	0.62540
1.00	0.635548	1.00000	0.55459	1.00000	−0.01157
1.20	0.264504	0.80000	0.18506	0.78522	−0.40860
1.40	−0.137035	0.60000	−0.20453	0.51568	−0.67091
1.60	−0.565660	0.40000	−0.61230	0.20452	−1.04351
2.00	−1.489652	0.00000	−1.47622	−0.50353	−1.94223

Table 7.6 Ground-state energies for the pure pairing model, $N = L = 4$, $\xi = 1$. The Hartree-Fock energy is the unperturbed reference, since the pairing interaction cannot couple it to a $1p$-$1h$ state. The BCS energy departs from it only above $g_c = 1.07$.

The table repays reading across.

Hartree-Fock does nothing at all, exactly as Section 6.12 predicted: $E_{\mathrm{HF}} = 2 - g$ is the unperturbed reference energy for every g. The whole correlation energy has to come from the fluctuations.

Particle-hole RPA supplies almost all of it, and tracks the exact curve remarkably closely – -0.61230 against -0.565660 at $g = 1.6$ – crossing it between $g = 1.6$ and $g = 2$. It is slightly over-bound below the crossing and slightly under-bound above it. That a particle-hole theory should do this well on a pure pairing interaction is a reminder that the labels "particle-hole" and "pairing" refer to how a calculation is organised, not to a partition of the physics.

BCS improves on Hartree-Fock only above g_c, and then only modestly. A mean field that merely smears the occupation numbers cannot reproduce a genuinely correlated ground state; at $g = 2$ it recovers 0.50 of the required 1.49.

QRPA on top of BCS overshoots badly, and Section 7.11.2 says why: the two-quasiparticle space contains pairing vibrations reaching towards $N \pm 2$, and the standard formula (7.23) counts every root. Restoring particle number by projection is necessary before BCS plus QRPA becomes quantitative for a system this small. In a heavy nucleus, with many levels and a small relative number fluctuation, the same machinery is far better behaved – which is why it remains a standard tool there.

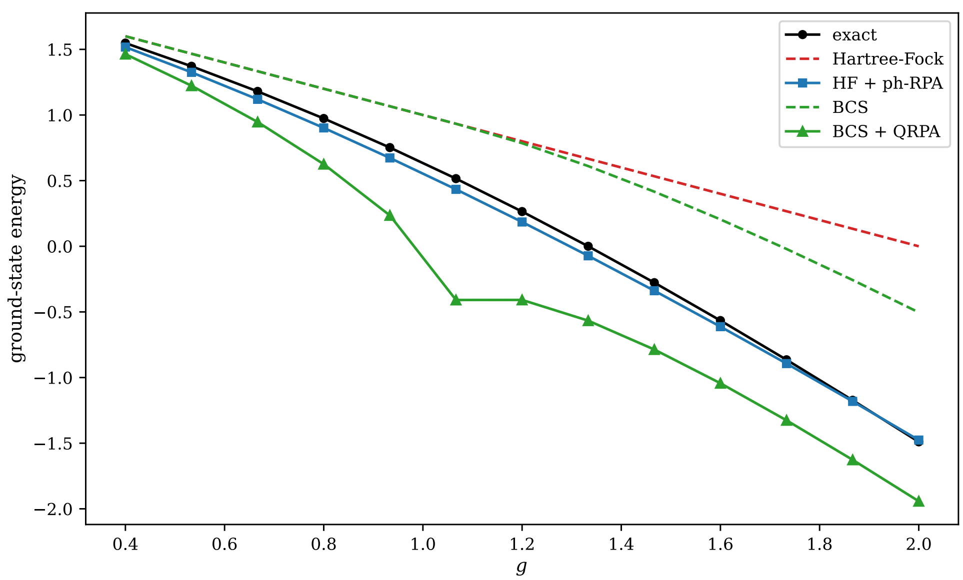

Fig. 7.6 Ground-state energy of the pure pairing model, $N = L = 4$ and $\xi = 1$, in the four treatments of this chapter over $0.4 \le g \le 2.0$: exact, Hartree-Fock, Hartree-Fock plus particle-hole RPA, BCS, and BCS plus QRPA. Hartree-Fock is the straight line $2 - g$, and the BCS curve, which lies on top of it below $g_c = 1.07$, leaves it there. The particle-hole RPA curve is almost indistinguishable from the exact one across the whole range, whereas BCS plus QRPA lies far below it everywhere, by 0.45 at $g = 2$. The kink in the QRPA curve just above $g = 1$ is the pairing instability of the normal vacuum. The figure is the graphical form of Table 7.6.

Figure 7.6 shows all five curves at once, and it is the summary picture of the chapter. The first thing one notices is how close the particle-hole RPA curve stays to the exact one: the largest discrepancy anywhere in the range is 0.082, at $g \approx 1.07$, and the two cross near $g \approx 1.95$. Given that Hartree-Fock is a straight line missing the exact energy by 1.49 at $g = 2$, the fluctuations are doing essentially all of the work, and doing it well. The second thing is that BCS, the mean field supposedly built for this interaction, is the worse of the two references throughout: it lies on top of the Hartree-Fock line until g_c and then recovers only a third of what is missing. Yet the correlations built on it overshoot by a wide margin. A poor reference and a large correlation energy are the same statement seen twice, and the arithmetic that repairs the one tends to overcorrect the other.

The kink in the QRPA curve near $g \approx 1.1$ deserves a sentence of its own, because it is not a numerical accident but the very instability that the particle-hole channel could not see. Below g_c the QRPA is built on the normal Fermi sea, and as g approaches g_c from below the lowest two-quasiparticle root of that calculation goes soft in exactly the manner of Section 7.7: it is 0.3745 at $g = 1$, 0.0512 at $g = 1.07$ and 0.0116 at $g = 1.0704$, one ten-thousandth of the level spacing away from the critical coupling. By Eq. (7.23) the correlation energy dips with it, reaching -1.4265 at the transition before relaxing back to -1.1938 at $g = 1.2$ once the BCS vacuum has taken over and the soft root has hardened again – it is back up to 0.7681 at $g = 1.2$ and 2.6015 at $g = 2$, the value of Table 7.5. So the vanishing frequency does what Eq. (7.21) says it must: it announces the pairing transition, from the side of the phase that is about to lose its stability, and it does so in the channel where Figure 7.4 showed that $A - B = \xi$ could never go soft at all. The two figures together are the cleanest statement in this chapter of what it means to choose a channel.

What to take away. Both approximations of this chapter are the *same* approximation applied to different reference states and different elementary operators: linearise the equations of motion, replace the commutator of two elementary excitations by its expectation value in the reference, and diagonalise. Everything that follows – the $\pm\omega$ pairing of the roots, the indefinite metric, the possibility of imaginary frequencies, the correlation-energy formula, the appearance of Goldstone modes whenever the reference breaks a symmetry – follows from that one step, and is independent of which channel one is working in. The quality of the answer, however, is not: it depends entirely on whether the reference has already captured the physics, and the two halves of Table 7.6 show the same interaction treated well and badly by that criterion.

7.13 Summary

The Tamm-Dancoff approximation diagonalises the Hamiltonian in the $1p$-$1h$ space; it is the excited-state version of the CIS truncation of Chapter 5, and its matrix A is the Hamiltonian restricted to that space, measured from the reference. The random-phase approximation adds the de-excitation amplitudes Y, which correlates the ground state, lowers every root, and turns the Hermitian eigenvalue problem into the non-Hermitian one of Eq. (7.16). The blocks A and B that appear there are precisely the blocks of Chapter 6's stability matrix, and the RPA frequencies are real exactly when the mean field is a local minimum: excitation spectrum and stability analysis are one calculation.

Applied to a model with competing pairing and particle-hole interactions, the approximations behave as their construction demands. Particle-hole RPA is stable at every coupling and does surprisingly well on the ground-state energy, while systematically underestimating the excitation energies. BCS finds the pairing instability that the particle-hole channel cannot see, at the price of breaking particle-number conservation, and QRPA built on it carries a spurious Goldstone mode which must be identified and removed – by its overlap with the number-operator amplitudes, not by inspecting the spectrum alone.

The program `rpa.py` performs every calculation quoted here, and the notebook `tdarpa.ipynb` runs it cell by cell. All the matrices are built as literal double commutators, so the same routine serves the particle-hole and the quasiparticle case, and the textbook formulas (7.11) and (7.17) are checked against them rather than assumed.

7.14 Exercises

Exercise 1: the model and its limits

a) Show that $\hat{V}_{\text{pair}}$ and $\hat{V}_{\text{ph}}$ both conserve N_+ and N_- separately, and hence that S_z is a good quantum number. What is the dimension of the $S_z = 0$ sector for $N = L = 4$?
b) Show that at $f = 0$ the seniority is conserved, and that at $f \neq 0$ it is not. Which matrix elements does the particle-hole term supply that the pairing term does not?
c) Verify with `rpa.py` that the $f = 0$ spectrum reproduces Table 4.1, and that the ground state at $f = 0$ lies entirely in the seniority-zero subspace.
d) For $g = 0.5$, $f = 0.05g$ the two lowest excited states are split by only 2×10^{-4}. Explain the near-degeneracy.

Answer to a).

Every term of Eq. (7.2) contains one $\hat{a}^\dagger_{p+}$ and one $\hat{a}_{q+}$, so N_+ is unchanged, and likewise for N_-. In Eq. (7.3) the operator $\hat{a}^\dagger_{p+}\hat{a}^\dagger_{p-}\hat{a}_{q-}\hat{a}_{r+}$ again creates and destroys one particle of each spin. Hence $[\hat{H}, \hat{N}_\pm] = 0$ and $S_z = (N_+ - N_-)/2$ is conserved. For $N = L = 4$ the balanced sector requires two spin-up and two spin-down particles among four levels each, so its dimension is $\binom{4}{2}^2 = 36$, against $\binom{8}{4} = 70$ for the full space.

Answer to b).

The pairing term moves both members of a pair together, so the number of unpaired particles – the seniority – is unchanged, as Exercise 5.13c) showed. The particle-hole term with $q \neq r$ removes a spin-down from level q and a spin-up from a *different* level r, leaving unpaired particles behind at both, and deposits a pair at p. It therefore changes the seniority by two. These are exactly the matrix elements that connect the seniority-zero subspace to the rest of the $S_z = 0$ space, and they are why the full 36-dimensional problem must be solved once $f \neq 0$.

Answer to c).

Running `demo_model()` prints the $f = 0$ energies alongside Table 4.1; they agree to all eight digits shown. Projecting the ground-state eigenvector onto the determinants with broken pairs gives zero to machine precision.

Answer to d).

At $f = 0$ the first excited state of the seniority-zero problem is doubly degenerate by the symmetry of the level scheme. The particle-hole term lifts the degeneracy, but only at order f, and $f = 0.025$ here; the splitting $2.711098 - 2.710867 = 2.3 \times 10^{-4}$ is of order f^2/ξ, consistent with the degeneracy being lifted at second order.

Exercise 2: from the equations of motion to TDA and RPA

a) Derive Eq. (7.8) from Eq. (7.7), stating where $\hat{Q}_\nu|0\rangle = 0$ is used.
b) Show that with the operator (7.9) and a Slater determinant reference, Eq. (7.8) reduces to the Hermitian problem (7.10), and evaluate $A_{mi,nj}$ using Wick's theorem.
c) Show that the Tamm-Dancoff matrix is the Hamiltonian restricted to the $1p$-$1h$ space, Eq. (7.12), and hence that TDA is CIS.
d) With the operator (7.13), derive Eq. (7.16) and identify precisely where the quasiboson approximation enters.
e) Show that if (X,Y) solves Eq. (7.16) with frequency ω, then (Y^*,X^*) solves it with $-\omega$.

Answer to a).

Take the matrix element of Eq. (7.7) with $\langle 0|\delta\hat{Q}\cdot|\rangle$: $\langle 0|\delta\hat{Q}[\hat{H},\hat{Q}_\nu^\dagger]|0\rangle = \omega_\nu\langle 0|\delta\hat{Q}\hat{Q}_\nu^\dagger|0\rangle$. Now use $\hat{Q}_\nu|0\rangle = 0$, and its adjoint $\langle 0|\hat{Q}_\nu^\dagger|\rangle = 0$, to add for free the terms $\langle 0|[\hat{H},\hat{Q}_\nu^\dagger]\delta\hat{Q}|0\rangle$ and $\omega_\nu\langle 0|\hat{Q}_\nu^\dagger\delta\hat{Q}|0\rangle$, which vanish because $\delta\hat{Q}$ is itself a de-excitation operator. Subtracting turns both sides into commutators and gives Eq. (7.8). The symmetrised form is preferable because the double commutator can be evaluated with fewer contractions, and because it is manifestly Hermitian.

Answer to b).

With $\delta\hat{Q} = \hat{a}_i^\dagger\hat{a}_m$ and $\hat{Q}^\dagger = \sum_{nj} X_{nj}\hat{a}_n^\dagger\hat{a}_j$ the right-hand side of Eq. (7.8) is $\sum_{nj} X_{nj}\langle\mathrm{HF}|[\hat{a}_i^\dagger\hat{a}_m,\hat{a}_n^\dagger\hat{a}_j]|\mathrm{HF}\rangle = X_{mi}$ by Eq. (7.14), exactly and without approximation, because the expectation value of $\hat{a}_i^\dagger\hat{a}_j$ in a determinant is δ_{ij} for occupied indices and zero otherwise. The left-hand side is $\sum_{nj} A_{mi,nj}X_{nj}$ with A given by Eq. (7.11); evaluating the double commutator with the generalised Wick theorem of Section 3.12 gives the one-body piece $(\varepsilon_m-\varepsilon_i)\delta_{mn}\delta_{ij}$ from $\hat{H}_0$ and the single surviving contraction $\bar{v}_{mjin}$ from the interaction.

Answer to c).

The $1p$-$1h$ determinants are orthonormal, and the Condon-Slater rules of Section 3.14 give

$$\langle\Phi_i^m|\hat{H}|\Phi_j^n\rangle = E_{\mathrm{HF}}\delta_{mn}\delta_{ij} + (\varepsilon_m-\varepsilon_i)\,\delta_{mn}\delta_{ij} + \bar{v}_{mjin}, \tag{7.34}$$

which is $E_{\mathrm{HF}}\delta + A_{mi,nj}$. Diagonalising A is therefore diagonalising $\hat{H}$ in the space spanned by $|\mathrm{HF}\rangle$ and all singles – CIS – and reading the eigenvalues relative to the reference. The reference itself decouples by Brillouin's theorem, which is why it does not appear in A. The program checks $\max|A-(\boldsymbol{H}_{1p1h}-E_{\mathrm{HF}})| = 3\times 10^{-15}$.

Answer to d).

Now $\delta\hat{Q}$ ranges over both $\hat{a}_i^\dagger\hat{a}_m$ and $\hat{a}_m^\dagger\hat{a}_i$, so Eq. (7.8) becomes a pair of coupled equations for X and Y. The left-hand sides give A and B as in Eqs. (7.11) and (7.17). The right-hand sides require $\langle\mathrm{RPA}|[\hat{a}_i^\dagger\hat{a}_m,\hat{a}_n^\dagger\hat{a}_j]|\mathrm{RPA}\rangle$ in the *correlated* ground state, which we do not have. Replacing it by its value in the Hartree-Fock determinant, Eq. (7.15), is the quasiboson approximation, and it is the only approximation made. It produces the norm matrix $\boldsymbol{\eta} = \mathrm{diag}(\mathbf{1}, -\mathbf{1})$; the minus sign, from the second row, is what makes Eq. (7.16) non-Hermitian.

Answer to e).

Take the complex conjugate of Eq. (7.16) and interchange the two blocks of rows: $A^*X^* + B^*Y^* = \omega^* X^*$ and $-BY^* - AX^* = \omega^* Y^*$, that is

$$\begin{pmatrix} A & B \\ -B^* & -A^* \end{pmatrix} \begin{pmatrix} Y^* \\ X^* \end{pmatrix} = -\omega^* \begin{pmatrix} Y^* \\ X^* \end{pmatrix}. \tag{7.35}$$

So the spectrum is symmetric about zero. If ω is real this is the partner $-\omega$; if it is complex the four roots $\pm\omega, \pm\omega^*$ appear together, which is why imaginary frequencies arrive in pairs.

Exercise 3: RPA, stability and Thouless

a) Show that $\hat{M}_{\mathrm{RPA}} = \boldsymbol{\eta}\hat{M}_{\mathrm{stab}}$ with $\boldsymbol{\eta} = \mathrm{diag}(\mathbf{1}, -\mathbf{1})$, and deduce Eq. (7.21).
b) Reproduce Table 7.1 for the Lipkin model and show that the collective root behaves as $\omega = \varepsilon\sqrt{1-\chi^2}$ near the transition.
c) Verify Eq. (7.24) for the pure pairing model by computing A and B analytically in the particle-hole channel.
d) Show that $\hat{Q}_\nu|\mathrm{RPA}\rangle = 0$ is solved by Eq. (7.22), and compare its structure with Thouless' theorem and with the coupled-cluster ansatz.

Answer to a).

Multiplying $\hat{M}_{\mathrm{stab}}$ on the left by $\boldsymbol{\eta}$ flips the sign of its second block row, turning (B^*, A^*) into $(-B^*, -A^*)$, which is $\hat{M}_{\mathrm{RPA}}$. If $\hat{M}_{\mathrm{stab}} > 0$ it has a positive-definite Hermitian square root $\boldsymbol{S}$, and $\boldsymbol{S}^{-1}\hat{M}_{\mathrm{RPA}}\boldsymbol{S} = \boldsymbol{S}^{-1}\boldsymbol{\eta}\boldsymbol{S}^2\boldsymbol{S} = \boldsymbol{S}\boldsymbol{\eta}\boldsymbol{S}$, which is Hermitian. Similar matrices have the same spectrum, so the RPA eigenvalues are real. If $\hat{M}_{\mathrm{stab}}$ has a negative eigenvalue the square root is no longer Hermitian and the argument collapses; the roots may then be imaginary, and for a single negative direction they are.

Answer to b).

Section 6.11 gives $\Delta + A = \varepsilon\boldsymbol{I}$ and $\boldsymbol{B} = V(\boldsymbol{J} - \boldsymbol{I})$, whose largest eigenvalue is $V(N-1) = \varepsilon\chi$. Since A and B commute in this model, the RPA roots are $\omega = \sqrt{A^2 - B^2}$ evaluated eigenvalue by eigenvalue, and the collective one is

$$\omega = \sqrt{\varepsilon^2 - \varepsilon^2\chi^2} = \varepsilon\sqrt{1-\chi^2}. \tag{7.36}$$

This vanishes at $\chi = 1$ and is imaginary beyond, reproducing Table 7.1: at $\chi = 0.9$ it gives $\sqrt{1-0.81} = 0.43589$, exactly the tabulated value. Note also the square root singularity: the mode softens with infinite slope as the transition is approached, which is the mean-field signature of a second-order transition.

Answer to c).

With pure pairing the Hartree-Fock orbitals are the original ones, the occupied levels shifted down by $g/2$, so $\varepsilon_m - \varepsilon_i$ for the lowest particle-hole pair is $\xi + g/2$. The interaction contributes $\bar{v}_{mjin}$ to A and $\bar{v}_{mnij}$ to B; for the pairing force only the terms in which m,n and i,j are spin partners of the same level survive, giving a diagonal $A = \xi + g/2$ and $B = g/2$ in the collective channel. Then $\omega = \sqrt{A^2 - B^2} = \sqrt{(\xi + g/2)^2 - (g/2)^2} = \sqrt{\xi^2 + g\xi}$, which is Eq. (7.24). Since $\xi^2 + g\xi > 0$ for all $g > -\xi$, the particle-hole channel never becomes unstable.

Answer to d).

Write $|\mathrm{RPA}\rangle = \mathscr{N} e^{\hat{S}}|\mathrm{HF}\rangle$ with $\hat{S} = \frac{1}{2}\sum Z_{mi,nj}\hat{a}_m^\dagger \hat{a}_i \hat{a}_n^\dagger \hat{a}_j$. Since the particle-hole operators commute among themselves in the quasiboson approximation, $[\hat{a}_i^\dagger \hat{a}_m, \hat{S}] = \sum_{nj} Z_{mi,nj}\hat{a}_n^\dagger \hat{a}_j$, and imposing $\hat{Q}_\nu e^{\hat{S}}|\mathrm{HF}\rangle = 0$ gives $Y^\nu = \boldsymbol{Z}\boldsymbol{X}^\nu$ in matrix form, hence $\boldsymbol{Z} = \boldsymbol{Y}^*(\boldsymbol{X}^*)^{-1}$. The structure is the same as Thouless' theorem, Eq. (6.34), with $2p$-$2h$ operators in place of the $1p$-$1h$ ones – so the RPA ground state stands to the doubles as a Slater determinant does to the singles. It is also the coupled-cluster doubles ansatz $e^{\hat{T}_2}|\Phi_0\rangle$ of Eq. (5.29), with the amplitudes fixed by the RPA equations rather than determined by their own nonlinear equations. RPA is thus an approximation to coupled-cluster doubles, and the comparison between the two is a standard way of exposing what the quasiboson approximation costs.

Exercise 4: BCS

a) Derive Eq. (7.27), being explicit about the $p = q$ and $p \neq q$ contributions.
b) Show that only the $q = r$ part of $\hat{V}_{\mathrm{ph}}$ has a non-zero expectation value in the BCS state, and hence that the problem depends on g and f only through $g + 2f$.
c) Derive the critical condition (7.28) and solve it for S_c and g_c.
d) Verify that the Bogoliubov quasiparticles (7.29) annihilate the BCS state and obey the fermion anticommutation relations.
e) Compare the BCS ground-state energy with the exact one through the transition and comment on the sharpness of the mean-field transition.

Answer to a).

$|\mathrm{BCS}\rangle$ is a product over levels, so $\langle \hat{N}\rangle = 2\sum_p v_p^2$ and $\langle \hat{H}_0\rangle = 2\sum_p \varepsilon_p v_p^2$ immediately. For the interaction, $\hat{P}_q$ acting on the product picks out the v_q term and leaves level q empty; $\hat{P}_p^\dagger$ in the bra does the same at p. For $p = q$ the two operations are on the same level and the overlap is v_p^2; for $p \neq q$ the surviving overlap has u_p at p and u_q at q, giving $u_p v_p u_q v_q$. Assembling, $\sum_{pq}\langle \hat{P}_p^\dagger \hat{P}_q\rangle = \sum_p v_p^2 + \sum_{p\neq q} u_p v_p u_q v_q$, and multiplying by $-S$ gives Eq. (7.27).

Answer to b).

$\hat{a}_{q-}\hat{a}_{r+}$ with $q \neq r$ leaves an unpaired particle at each of q and r; the resulting state has two broken pairs and is orthogonal to anything $\hat{P}_p^\dagger$ can produce from $|\mathrm{BCS}\rangle$, which contains only paired levels. For $q = r$, $\hat{a}_{q-}\hat{a}_{q+} = \hat{P}_q$ exactly, so the term reduces to $-\frac{f}{2}\sum_{pq}\hat{P}_p^\dagger \hat{P}_q$ and its Hermitian conjugate contributes an equal amount. The total is $-f\sum_{pq}\hat{P}_p^\dagger \hat{P}_q$, which added to $-\frac{g}{2}\sum_{pq}\hat{P}_p^\dagger \hat{P}_q$ gives the single strength $S = g/2 + f$, that is $g \to g + 2f$. The program confirms that $(g, f) = (1.4, 0)$, $(1.2, 0.1)$, $(0.8, 0.3)$ and $(0, 0.7)$ give identical energies and gaps.

Answer to c).

Linearise about $\Delta = 0$. There $v_p^2 = 1$ for the two lowest levels and zero above, so the self-energy shifts them to $\tilde{\varepsilon}_p = \varepsilon_p - S$ for $p \leq 2$ and leaves $\tilde{\varepsilon}_p = \varepsilon_p$ above. With $\varepsilon_p = 0, 1, 2, 3$ this gives $(-S, 1-S, 2, 3)$, and the chemical potential sits midway between the highest occupied and lowest empty level, $\lambda = (3-S)/2$. The linearised gap equation $1 = (S/2)\sum_p |\tilde{\varepsilon}_p - \lambda|^{-1}$ becomes

$$1 = \frac{S}{2}\left[\frac{2}{(3+S)/2}\cdot 1 + \frac{2}{(1+S)/2}\cdot 1\right]\cdot\frac{1}{2}\cdot 2 = 2S\left[\frac{1}{3+S} + \frac{1}{1+S}\right], \tag{7.37}$$

which is Eq. (7.28) with $\xi = 1$. Solving numerically gives $S_c = 0.53519$ and $g_c = 2S_c = 1.07037$ at $f = 0$, matching the bisection in the program.

Answer to d).

$\hat{\alpha}_{p+}|\mathrm{BCS}\rangle = (u_p\hat{a}_{p+} - v_p\hat{a}^\dagger_{p-})\prod_q(u_q + v_q\hat{P}^\dagger_q)|0\rangle$. Only the factor at level p matters; acting on $(u_p + v_p\hat{a}^\dagger_{p+}\hat{a}^\dagger_{p-})$ gives $u_pv_p\hat{a}^\dagger_{p-} - v_pu_p\hat{a}^\dagger_{p-} = 0$. For the anticommutators,

$$\left\{\hat{\alpha}_{p+}, \hat{\alpha}^\dagger_{p+}\right\} = u_p^2\left\{\hat{a}_{p+}, \hat{a}^\dagger_{p+}\right\} + v_p^2\left\{\hat{a}^\dagger_{p-}, \hat{a}_{p-}\right\} = u_p^2 + v_p^2 = 1, \tag{7.38}$$

the cross terms vanishing because they pair two creation or two annihilation operators. The normalisation $u_p^2 + v_p^2 = 1$ is exactly what makes the transformation canonical. The program checks $\max\|\hat{\alpha}_a|\mathrm{BCS}\rangle\| = 3\times 10^{-17}$.

Answer to e).

From Table 7.6, $E_{\mathrm{BCS}} = E_{\mathrm{HF}}$ exactly below $g_c = 1.07$ and departs from it above. The exact energy, by contrast, runs smoothly through g_c with no feature of any kind: at $g = 1.0$, 1.2 and 1.4 it is 0.6355, 0.2645 and -0.1370, a perfectly regular sequence. The mean-field transition is therefore an artefact of restricting the trial state to a product form. What is physical is the crossover – the gradual build-up of pair correlations – and what is spurious is its sharpness, and with it the broken symmetry. For a large system the crossover narrows and the mean-field picture becomes accurate; for four particles it does not.

Exercise 5: QRPA and the spurious mode

a) Show that if $[\hat{H}, \hat{F}] = 0$ while the reference is not an eigenstate of $\hat{F}$, then the amplitudes (7.32) built from $\hat{F}$ give an exact zero mode of the RPA equations.
b) Explain why cranking with $\hat{H}' = \hat{H} - \lambda\hat{N}$ places the spurious root at zero, and why the root may come out as a small *imaginary* pair rather than a small real one.
c) Reproduce Table 7.4 and confirm the unit overlap. What would you conclude if the overlap were small?
d) Classify the QRPA phonons at $g = 2$ by $\langle\Delta\hat{N}\rangle$, and explain why the shift is not quantised at ± 2.

e) Explain why $E_{\mathrm{corr}}^{\mathrm{QRPA}}$ overshoots so badly in Table 7.6, and suggest what would fix it.

Answer to a).

Because $[\hat{H},\hat{F}] = 0$, the operator $\hat{F}$ generates a family of states $e^{i\theta\hat{F}}|\mathrm{ref}\rangle$ all with the same energy. If the reference is not an eigenstate of $\hat{F}$ these are genuinely different states, so the energy surface has a flat direction, and the second derivative – which is what $\hat{M}_{\mathrm{stab}}$ is – has a zero eigenvalue along it. Algebraically, substituting the amplitudes (7.32) into the left-hand side of Eq. (7.8) gives $\langle\mathrm{ref}|[\delta\hat{Q},[\hat{H},\hat{F}]]|\mathrm{ref}\rangle = 0$ identically, so $\omega = 0$.

Answer to b).

The BCS state extremises $\hat{H} - \lambda\hat{N}$, not $\hat{H}$, so it is $\hat{H}'$ whose energy surface is flat at the BCS point and whose second derivative therefore has the exact zero eigenvalue. Building the QRPA from $\hat{H}$ instead displaces the spurious root to a small non-zero value and contaminates the physical ones. The root may emerge imaginary because the RPA metric $\boldsymbol{\eta}$ is singular in the spurious direction: there $\sum(|X|^2 - |Y|^2) = 0$, the mode cannot be normalised, and rounding errors of either sign are amplified into a small $\pm i\varepsilon$ pair. This is a property of the numerics, not of the physics.

Answer to c).

`demo_qrpa()` prints the table. The overlap between the near-zero eigenvector and the amplitudes (7.32) is 1.000000 throughout, so the root is unambiguously the number mode. Were the overlap small, the conclusion would be the opposite and far more interesting: a genuine soft mode, signalling that the BCS solution is about to become unstable against some other deformation, exactly as the Lipkin collective root does at $\chi = 1$. The two cases look identical in the spectrum alone, which is why the overlap test matters.

Answer to d).

Running `classify_modes` gives $\langle\Delta\hat{N}\rangle = 0$ to machine precision for the roots at 2.6015 and 2.9212, and ± 0.357 for the pair at 3.7879. The shift is not ± 2 because $|\mathrm{BCS}\rangle$ is not an eigenstate of $\hat{N}$: it is a wave packet in particle number, and $\langle\hat{N}\rangle$ in the phonon state measures the *mean* of a shifted packet, not a quantum number. What is exact and meaningful is the vanishing of the shift for the modes that stay within the N-particle system, which is enforced by symmetry. Number projection would restore the quantisation.

Answer to e).

Two effects compound. First, Eq. (7.23) sums over *all* roots, and the two-quasiparticle space contains pairing vibrations that describe the neighbouring $N \pm 2$ systems; their zero-point energy has no business in the correlation energy of the N-particle system. Second, the quasiboson approximation is worse here than in the particle-hole channel, because the BCS occupations are fractional and the commutator of two two-quasiparticle operators is correspondingly further from a c-number. The cure for the first is particle-number projection before or after variation; the cure for the second is a renormalised QRPA that keeps the leading correction to Eq. (7.15). Both are standard, and both are essential for a system with only four particles; in a heavy nucleus the relative number fluctuation $\Delta N/N$ is small and plain QRPA is adequate.

Exercise 6: numerical explorations

a) Using `rpa.py`, plot the exact, TDA and RPA lowest excitation energies against g for $f = 0.05g$ and for $f = 0.5g$, and comment on where the approximations fail.
b) Repeat Table 7.6 for $L = 6$ and $N = 6$. Does RPA improve or deteriorate as the system grows?
c) Compute the RPA correlation energy from Eq. (7.23) and, separately, as the difference between the exact and Hartree-Fock energies. Plot the ratio against g.
d) Find, by sweeping f at fixed g, the coupling at which the particle-hole stability matrix of this model first acquires a negative eigenvalue – or show that it never does within the range you can reach.
e) Restrict the QRPA correlation energy to the $\langle \Delta\hat{N} \rangle = 0$ modes and compare with Table 7.6. How much of the overshoot does this remove?

Answer.

These are open-ended; a few pointers. In b) the RPA correlation energy is an extensive quantity while the quasiboson error per phonon is not, so RPA generally improves in relative terms as the level space grows – the opposite of truncated configuration interaction, whose size-consistency error grows (Section 5.9). In c) the ratio starts near one at small g and drifts past it as RPA begins to overshoot; the crossing point is a useful single number for characterising where the method turns over. In d) the answer for this model is that it never does within any reasonable range, since Eq. (7.24) is real for all $g > -\xi$ and the particle-hole term only stiffens the matrix; the instability is in the pairing channel, which requires the BCS reference to see. In e) removing the pairing-vibration roots takes out a large part of the overshoot but not all of it, since the quasiboson approximation remains; the residual is a fair estimate of what number projection alone cannot fix.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter07/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/tdarpa.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter07/`.

`rpa.py` The pairing-plus-particle-hole model solved by exact diagonalization, TDA, RPA, BCS and QRPA, with every matrix built as a double commutator. Runs in about twenty seconds; needs `scipy.linalg`, `scipy.sparse` and `scipy.optimize`.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 8
Density functional theory

Every method so far has been a method for the wave function. Full configuration interaction stores its coefficients, Hartree-Fock optimises the orbitals it is built from, and the random-phase approximation describes the fluctuations around that. All of them run into the same wall: the object being computed lives in a space whose dimension grows exponentially with the number of particles, Section 5.5.

Density functional theory takes a different route. It replaces the wave function of N electrons – a function of $3N$ coordinates – by the electronic density, a function of three. This is not an approximation but, remarkably, an exact reformulation: Hohenberg and Kohn proved that the ground-state density determines the external potential, hence the Hamiltonian, hence everything. The catch, which we shall be careful to state precisely, is that the theorem asserts the existence of a universal functional without telling us its form.

The chapter has three parts. We first assemble the tools – the Born-Oppenheimer Hamiltonian and the reduced density matrices of Sections 1.31 and 2.9, now in coordinate space. We then prove the Hohenberg-Kohn theorems and examine what they do and do not deliver. Finally we build the Kohn-Sham scheme, which recovers most of what the mean-field methods of Chapter 6 do well while replacing the expensive part by a local functional, and we construct that functional explicitly from the electron gas of Section 6.13.

Density functional theory has been an extraordinary practical success. Most crystalline materials and a very large fraction of molecular structures have been described with it; Walter Kohn shared the 1998 Nobel Prize in Chemistry for it; and it is now standard in quantum chemistry, materials science, condensed-matter physics and, increasingly, low-energy nuclear physics. It is also, as we shall see, an approximation whose errors are of a quite different character from those of the wave-function methods, and understanding that difference is the point of putting it here.

8.1 The electronic Hamiltonian

Any piece of matter – a crystal, a molecule, a living cell – is a collection of nuclei and electrons held together by the Coulomb interaction, and is governed by

$$i\hbar\frac{\partial}{\partial t}\Phi(\boldsymbol{r};t) = \left(-\sum_{i}^{N}\frac{\hbar^2}{2m_i}\frac{\partial^2}{\partial \boldsymbol{r}_i^2} + \sum_{i<j}^{N}\frac{e^2 Z_i Z_j}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}\right)\Phi(\boldsymbol{r};t). \tag{8.1}$$

Nothing else is needed in principle, and nothing is soluble in practice.

The first simplification is universal. Even the lightest nucleus is nearly two thousand times heavier than an electron, so on the time scale of electronic motion the nuclei may be treated as fixed. This is the *Born-Oppenheimer approximation*, and it turns Eq. (8.1) into a time-independent problem for the electrons alone, moving in the static field of the nuclei,

$$\left[\sum_{i}^{N}\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial \boldsymbol{r}_i^2}+v_{\text{ext}}(\boldsymbol{r}_i)\right)+\sum_{i<j}^{N}v(r_{ij})\right]\Psi(\boldsymbol{r})=E_0\Psi(\boldsymbol{r}), \tag{8.2}$$

with $r_{ij}=|\boldsymbol{r}_i-\boldsymbol{r}_j|$ and the external potential supplied by the nuclei,

$$v_{\text{ext}}(\boldsymbol{r}_i)=-\sum_j \frac{e^2 Z_j}{|\boldsymbol{r}_i-\boldsymbol{R}_j|}. \tag{8.3}$$

We have written a generic two-body interaction $v(r_{ij})$ rather than the bare Coulomb repulsion, because everything below applies unchanged to any interaction depending only on the relative distance – nuclear forces included, where v_{ext} is absent and the system is self-bound, as Section 6.1 noted.

Equation (8.2) is the Hamiltonian of Eq. (4.1) written in coordinate space. Only one thing about it will matter in what follows: the kinetic energy and the interaction are the *same for every system*, and the only thing that distinguishes one material from another is v_{ext}.

8.2 Reduced density matrices

The central object of this chapter is the electronic density. It is the diagonal of the one-body reduced density matrix, which we met in Section 1.31 as a matrix in a single-particle basis and now write in coordinate space.

8.2.1 The one-body density

Define the density operator

$$\hat{n}(\boldsymbol{r})=\sum_{i=1}^{N}\delta(\boldsymbol{r}-\boldsymbol{r}_i), \qquad n(\boldsymbol{r})=\frac{\langle\Psi|\hat{n}(\boldsymbol{r})|\Psi\rangle}{\langle\Psi|\Psi\rangle}, \tag{8.4}$$

for any normalisable state Ψ. Equivalently, integrating out all coordinates but one,

$$n(\boldsymbol{r})=N\int d^3\boldsymbol{r}_2\cdots d^3\boldsymbol{r}_N\,|\Psi(\boldsymbol{r},\boldsymbol{r}_2,\ldots,\boldsymbol{r}_N)|^2. \tag{8.5}$$

For a single Slater determinant built from orbitals ψ_i this collapses to

$$n(\boldsymbol{r})=\sum_{i=1}^{N}|\psi_i(\boldsymbol{r})|^2, \tag{8.6}$$

a result already used in Section 6.5.

In second quantization the operator reads

$$\hat{n}(\boldsymbol{r})=\sum_{\alpha\beta}\psi_\alpha^*(\boldsymbol{r})\psi_\beta(\boldsymbol{r})\,\hat{a}_\alpha^\dagger\hat{a}_\beta, \tag{8.7}$$

so that its expectation value in *any* state – a full configuration-interaction ground state $|\Psi_0\rangle = \sum_i C_{0i}|\Phi_i\rangle$ included – is

$$n(\boldsymbol{r}) = \sum_{ij} C_{0i}^* C_{0j} \sum_{\alpha\beta} \psi_\alpha^*(\boldsymbol{r})\psi_\beta(\boldsymbol{r})\langle\Phi_i|\hat{a}_\alpha^\dagger\hat{a}_\beta|\Phi_j\rangle. \tag{8.8}$$

Integrating over $\boldsymbol{r}$ and using orthonormality gives $\int n(\boldsymbol{r})d^3\boldsymbol{r} = N$, as it must.

The more general object, needed below, is the one-body reduced density matrix off the diagonal,

$$\rho^{(1)}(x';x) = N\int dx_2\cdots dx_N\,\Psi^*(x',x_2,\dots,x_N)\,\Psi(x,x_2,\dots,x_N), \tag{8.9}$$

with x the combined space and spin coordinate. It is Hermitian, its diagonal is the density, $n(x) = \rho^{(1)}(x;x)$, and its eigenvalues are the occupation numbers of Section 1.31: at most one for fermions, and exactly N ones and the rest zeros when – and only when – Ψ is a single Slater determinant, in which case $\rho^{(1)}$ is the projector onto the occupied subspace.

8.2.2 The two-body density

Two-body operators need the pair density,

$$\rho^{(2)}(x_1,x_2;y_1,y_2) = \frac{N(N-1)}{2}\int dx_3\cdots dx_N\,\Psi^*(y_1,y_2,x_3,\dots,x_N)\,\Psi(x_1,x_2,x_3,\dots,x_N), \tag{8.10}$$

the prefactor counting the $N(N-1)/2$ distinct pairs. Its diagonal $\rho^{(2)}(x_1,x_2)$ is the joint probability of finding one particle at x_1 and another at x_2, and it contains everything needed for the expectation value of any two-body operator – which is to say, everything needed for the energy. In second quantization,

$$\hat{\rho}^{(2)}(\boldsymbol{r}_1,\boldsymbol{r}_2) = \sum_{\alpha\beta\gamma\delta}\rho_{\alpha\gamma}(\boldsymbol{r}_1)\rho_{\beta\delta}(\boldsymbol{r}_2)\,\hat{a}_\alpha^\dagger\hat{a}_\beta^\dagger\hat{a}_\delta\hat{a}_\gamma, \qquad \rho_{\alpha\beta}(\boldsymbol{r}) = \psi_\alpha^*(\boldsymbol{r})\psi_\beta(\boldsymbol{r}). \tag{8.11}$$

8.2.3 Antisymmetry, and when the pair density factorises

Fermionic wave functions change sign under the exchange of any two particles,

$$\Psi(\dots,x_i,\dots,x_j,\dots) = -\Psi(\dots,x_j,\dots,x_i,\dots), \tag{8.12}$$

and this has a direct consequence for the pair density: it must vanish when $x_1 = x_2$, since two identical fermions cannot be at the same point in the same spin state. This is the Pauli principle expressed as a statement about $\rho^{(2)}$, and any approximate treatment that violates it is in trouble.

For a single Slater determinant the pair density is completely determined by the one-body density matrix. Applying Wick's theorem, Section 3.10, to $\langle\Phi_0|\hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a}|\Phi_0\rangle$ leaves exactly two contractions, and

$$\boxed{\rho^{(2)}(x_1,x_2;y_1,y_2) = \rho^{(1)}(x_1;y_1)\rho^{(1)}(x_2;y_2) - \rho^{(1)}(x_1;y_2)\rho^{(1)}(x_2;y_1).} \tag{8.13}$$

The first term is the naive product – what one would write for independent particles – and the second is the exchange term that enforces antisymmetry. On the diagonal,

$$\rho^{(2)}(x_1,x_2) = n(x_1)n(x_2) - \left|\rho^{(1)}(x_1;x_2)\right|^2, \tag{8.14}$$

and setting $x_1 = x_2$ gives $n(x_1)^2 - n(x_1)^2 = 0$ exactly, as required.

Equation (8.13) is worth dwelling on, because it marks precisely the boundary of mean-field theory. For a Hartree product – a plain product of orbitals with no antisymmetry – one would have $\rho^{(2)} = n(x_1)n(x_2)$, but such a state is not a legitimate fermionic wave function. The closest legitimate object is a single determinant, for which Eq. (8.13) holds *exactly*, with the exchange term as the only two-particle correlation. Beyond that – for any genuinely correlated state – the pair density cannot be reduced to one-body quantities at all, and the difference is what we have been calling correlation throughout this book.

Where this leaves us. The energy of Eq. (8.2) needs only $\rho^{(1)}$ and $\rho^{(2)}$; the full wave function is superfluous. That observation is the origin of reduced-density-matrix methods, which try to determine $\rho^{(2)}$ directly. The difficulty is the *N-representability problem*: not every Hermitian, antisymmetric, correctly normalised $\rho^{(2)}$ comes from an actual N-fermion wave function, and characterising those that do is extremely hard. Density functional theory sidesteps this by going all the way down to $n(\boldsymbol{r})$, at the price of an unknown functional. The two difficulties – an unknown functional here, an unknown representability condition there – are two faces of the same problem, and neither has been solved.

8.3 The Hohenberg-Kohn theorems

We now come to the two results on which everything rests. Consider the family of Hamiltonians

$$\hat{H} = \hat{T} + \hat{V}_{\text{ext}} + \hat{V}, \qquad \hat{V}_{\text{ext}} \in \mathscr{V}_{\text{ext}}, \tag{8.15}$$

in which $\hat{T}$ and $\hat{V}$ are fixed once and for all – they are properties of electrons, not of materials – and only the external potential varies. Assume for now that the ground state is non-degenerate.

Theorem 8.1 (Hohenberg-Kohn I). *The external potential $v_{\text{ext}}(\boldsymbol{r})$ is determined, up to an additive constant, by the ground-state density $n_0(\boldsymbol{r})$.*

Theorem 8.2 (Hohenberg-Kohn II). *There exists a universal functional $F[n]$, the same for every external potential, such that*

$$E[n] = F[n] + \int d^3\boldsymbol{r}\, v_{\text{ext}}(\boldsymbol{r})n(\boldsymbol{r}), \qquad F[n] = T[n] + V_{\text{int}}[n], \tag{8.16}$$

is minimised, over densities integrating to N, by the true ground-state density, and its minimum value is the ground-state energy E_0.

Together these say something startling. The density – three variables – determines the potential, hence the Hamiltonian, hence the full many-body wave function and every ground-state observable. A quantity that is measured routinely by X-ray diffraction contains, in principle, all the information in the $3N$-dimensional Ψ.

8.3.1 Proof of the first theorem

The proof is a two-line *reductio ad absurdum* and is worth giving in full, because its simplicity is itself informative.

Suppose two external potentials $v^{(1)}$ and $v^{(2)}$, differing by more than a constant, produced the same ground-state density $n(\boldsymbol{r})$. They have different Hamiltonians $\hat{H}^{(1)},\hat{H}^{(2)}$ and, by the unique continuation theorem, different ground states $\Psi^{(1)} \neq \Psi^{(2)}$ with energies $E^{(1)},E^{(2)}$.

Use $\Psi^{(2)}$ as a trial function for $\hat{H}^{(1)}$. The variational principle of Section 2.7 is strict for a non-degenerate ground state, so

$$E^{(1)} < \langle\Psi^{(2)}|\hat{H}^{(1)}|\Psi^{(2)}\rangle = \langle\Psi^{(2)}|\hat{H}^{(2)}|\Psi^{(2)}\rangle + \int \left[v^{(1)} - v^{(2)}\right] n(\boldsymbol{r})d^3\boldsymbol{r} = E^{(2)} + \int \left[v^{(1)} - v^{(2)}\right] n. \quad (8.17)$$

The two Hamiltonians differ only in the external potential, and the density is by hypothesis the same, which is what allows the middle step. Interchanging the labels,

$$E^{(2)} < E^{(1)} + \int \left[v^{(2)} - v^{(1)}\right] n. \quad (8.18)$$

Adding Eqs. (8.17) and (8.18), the integrals cancel and we are left with

$$E^{(1)} + E^{(2)} < E^{(1)} + E^{(2)}, \quad (8.19)$$

which is absurd. Hence no two potentials differing by more than a constant can share a ground-state density.

The second theorem follows immediately. Since n_0 fixes v_{ext}, it fixes $\hat{H}$ and therefore $\Psi[n]$, so

$$F_{\mathrm{HK}}[n] \equiv \langle\Psi[n]|\hat{T} + \hat{V}|\Psi[n]\rangle \quad (8.20)$$

is a well-defined functional of the density alone, and it is universal because $\hat{T}$ and $\hat{V}$ contain no reference to v_{ext}. The Rayleigh-Ritz principle applied within this family gives $E_0 = \min_n E[n]$.

8.3.2 What the theorems do not give

Three caveats deserve stating plainly, because they are the difference between a beautiful theorem and a working method.

They are existence theorems.

Nothing in the proof constructs $F[n]$, and nothing suggests it is simple. All the many-body difficulty of the preceding chapters is still there; it has been moved inside $F[n]$. This is why the practical history of density functional theory is a history of *approximate functionals*.

v-representability.

The argument assumes that the density under consideration is the ground-state density of *some* potential in the family. Not every well-behaved, positive, correctly normalised $n(\boldsymbol{r})$ is, and characterising those that are is unsolved. Levy and Lieb removed the difficulty by re-defining the functional as a constrained search over wave functions,

$$F_{\mathrm{LL}}[n] = \min_{\Psi \to n} \langle \Psi | \hat{T} + \hat{V} | \Psi \rangle, \tag{8.21}$$

the minimum running over all antisymmetric Ψ that yield the density n. This is defined for any N-representable density, agrees with F_{HK} where the latter exists, and extends the theory to degenerate ground states. It is also, of course, no more computable.

Ground states only.

The functional delivers E_0 and n_0 and nothing else. Excitation energies – the subject of Chapter 7 – are not obtained from the ground-state functional, and reaching them requires a separate construction such as time-dependent density functional theory. The Kohn-Sham eigenvalues introduced below are *not* excitation energies, a point we return to in Section 8.5.1.

8.3.3 The variational equation

Given $F[n]$, the ground state follows from minimising $E[n]$ subject to $\int n = N$. Introducing a Lagrange multiplier exactly as in Section 6.2,

$$\frac{\delta}{\delta n(\boldsymbol{r})} \left\{ E[n] - \mu \left(\int n \, d^3\boldsymbol{r} - N \right) \right\} = 0 \qquad \Longrightarrow \qquad \left. \frac{\delta E[n]}{\delta n(\boldsymbol{r})} \right|_{n=n_0} = \mu, \tag{8.22}$$

so that the functional derivative of the energy is constant throughout the system. That constant is the chemical potential, the Fermi level. The pattern is by now familiar: a constrained variation produces a stationarity condition, and the multiplier enforcing normalisation is the energy conjugate to the particle number.

8.4 Building a functional from the electron gas

The theorems tell us $F[n]$ exists. To make progress we need an approximation, and the oldest and still the most instructive strategy is to take a system we *can* solve exactly and use it locally.

That system is the homogeneous electron gas of Section 6.13. There we computed, for a uniform density n, both the kinetic energy per unit volume,

$$t_0[n] = \frac{3\hbar^2(3\pi^2)^{2/3}}{10m} n^{5/3}, \tag{8.23}$$

which is Eq. (6.91) divided by the volume, and the exchange energy per particle. If the density varies slowly enough, we may pretend that each small region behaves like a uniform gas at the local density and integrate,

$$\mathscr{T}_0[n] = \frac{3\hbar^2(3\pi^2)^{2/3}}{10m} \int d^3\boldsymbol{r} \, (n(\boldsymbol{r}))^{5/3}. \tag{8.24}$$

This is the *Thomas-Fermi* kinetic functional, and with the classical electrostatic (Hartree) energy

$$\mathscr{U}_{\mathrm{H}}[n] = \frac{e^2}{2} \iint d^3\boldsymbol{r} d^3\boldsymbol{r}' \, \frac{n(\boldsymbol{r})n(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}, \tag{8.25}$$

it gives a complete, if crude, theory. Everything not accounted for is collected into a single remainder, the *exchange-correlation energy* $E_{\rm xc}[n]$, and the Hohenberg-Kohn functional is written

$$\mathscr{E}_{\rm HK}[n] = \mathscr{T}_0[n] + \int d^3\boldsymbol{r}\, v_{\rm ext}(\boldsymbol{r})n(\boldsymbol{r}) + \mathscr{U}_{\rm H}[n] + E_{\rm xc}[n]. \tag{8.26}$$

The decomposition is a definition, not an approximation: whatever is missing is in $E_{\rm xc}$ by construction. It is useful only if that remainder is small, which is the whole gamble of the subject.

Why Thomas-Fermi is not enough.

Equation (8.24) has no orbitals in it, and minimising Eq. (8.26) directly over n gives an algebraic equation. It is enormously cheaper than anything in the preceding chapters. It also fails, and the failure is instructive rather than marginal. The program `dft.py` compares the exact kinetic energy of N non-interacting fermions in a one-dimensional trap with the local estimate:

N	T (exact)	T (Thomas-Fermi)	ratio
1	0.250000	0.302300	1.2092
2	1.000000	1.074844	1.0748
3	2.250000	2.340025	1.0400
4	4.000000	4.101805	1.0255
6	9.000000	9.120038	1.0133
8	16.000000	16.134286	1.0084

The error falls steadily as the particle number grows, which is exactly what the construction promises – the density becomes smoother on the scale of the local Fermi wavelength – but a one per cent error in the *largest* term of the energy is fatal when chemical accuracy is wanted. Worse, the Thomas-Fermi density has no shell structure at all: it is smooth and featureless, with none of the nodal structure that orbitals supply, and Teller proved that within the model no molecule is bound. Something has to be done about the kinetic energy specifically.

Figure 8.1 makes both halves of that verdict visible at once. The left panel plots the ratio of the table against N, and the approach to unity is unmistakable but slow: the curve falls steeply from 1.2092 at $N = 1$ to 1.0748 at $N = 2$, then flattens into a long tail that is still 0.8% high with all eight orbitals filled. What deserves attention is that the *absolute* error hardly moves at all. Subtracting the two columns of the table gives 0.052, 0.075, 0.090, 0.102, 0.120 and 0.134, so that the error grows by rather less than a factor of three – it is creeping up logarithmically – while the kinetic energy itself grows from 0.25 to 16, a factor of sixty-four. The improving ratio is therefore not the sign of a functional converging on the right answer; it is the sign of a roughly constant error being divided by a rapidly growing denominator. For a fixed, finite system, which is what atoms, molecules and nuclei are, an error of a tenth of a Hartree in the largest single term of the energy is simply not available, and no amount of taking N to infinity will make it so.

The right panel shows the failure that cannot be repaired by any adjustment of coefficients. The exact density of the eight-particle system carries eight separate maxima, at $x \simeq \pm 0.40, \pm 1.16, \pm 2.00$ and ± 2.92, which are nothing other than the nodal structure of the occupied orbitals surviving the sum over them; the density rises to 1.308 at the innermost maxima and dips to 1.234 between them, and the whole distribution ends abruptly near the

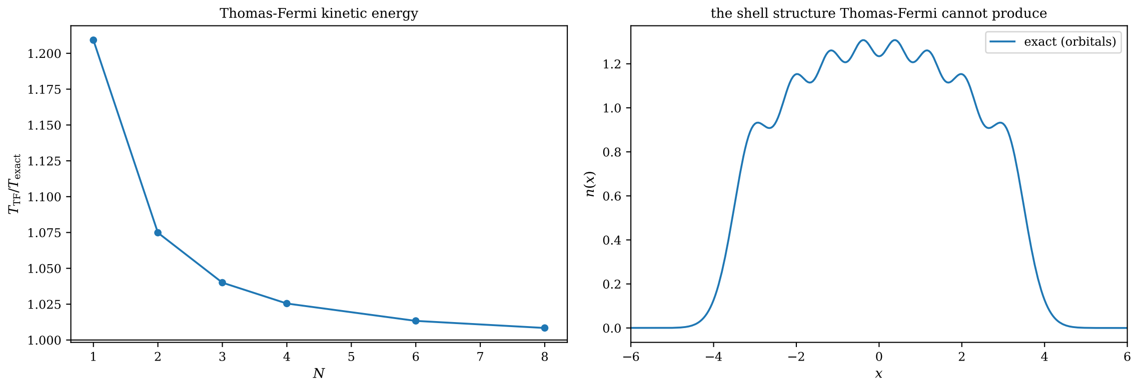

Fig. 8.1 Left: the ratio of the Thomas-Fermi kinetic functional (8.24) to the exact kinetic energy of N non-interacting spinless fermions in the one-dimensional harmonic trap, both evaluated on the exact orbital density on a grid of 401 points covering $x \in [-8, 8]$; the horizontal line marks the exact result. The local estimate is always too large, by 21% for a single particle and still by 0.8% when all eight oscillator orbitals are occupied. Right: the density $n(x) = \sum_{k<8} |\phi_k(x)|^2$ of that filled eight-orbital configuration, which oscillates between 1.234 and 1.308 near the centre and carries eight distinct maxima, one per occupied orbital. A functional of the density alone reproduces the smooth envelope and none of the oscillation.

classical turning point of the highest occupied level, $x = \sqrt{2\varepsilon_7} = \sqrt{15} \simeq 3.87$. Thomas-Fermi, having no orbitals, produces instead a smooth dome that follows the envelope and has no oscillation whatever, and the shell effects that organise the whole of atomic and nuclear structure are lost with it. This is the observation the Kohn-Sham construction is built on: the offending quantity is the kinetic energy, the offending feature is the shell structure, and both can be recovered exactly at the modest price of putting the orbitals back in for that one term.

8.5 The Kohn-Sham construction

Kohn and Sham's idea was to keep orbitals for the one term that cannot be approximated locally, and only that term.

Introduce an auxiliary system of N *non-interacting* fermions moving in some effective potential $v_{\rm KS}(\boldsymbol{r})$, chosen so that its ground-state density equals the density of the real, interacting system. Its kinetic energy,

$$T_s[n] = -\frac{\hbar^2}{2m}\sum_{i=1}^{N}\int d^3\boldsymbol{r}\,\psi_i^*(\boldsymbol{r})\nabla^2\psi_i(\boldsymbol{r}), \tag{8.27}$$

is computed exactly from orbitals rather than approximated from the density, and it captures the shell structure that Thomas-Fermi threw away. It is not the true kinetic energy $T[n]$ of the interacting system, but the difference is small, and we simply redefine the remainder to absorb it,

$$E_{\rm xc}[n] \equiv (T[n] - T_s[n]) + (V_{\rm int}[n] - \mathscr{U}_{\rm H}[n]). \tag{8.28}$$

The total energy is then

$$E[n] = T_s[n] + \int d^3\boldsymbol{r}\, v_{\rm ext} n + \mathscr{U}_{\rm H}[n] + E_{\rm xc}[n]. \tag{8.29}$$

Varying Eq. (8.29) with respect to the orbitals, subject to orthonormality, and following exactly the steps of Section 6.4, gives the *Kohn-Sham equations*

$$\boxed{\left(-\frac{\hbar^2}{2m}\nabla^2 + v_{\mathrm{KS}}(\boldsymbol{r})\right)\psi_i(\boldsymbol{r}) = \varepsilon_i\psi_i(\boldsymbol{r}),} \tag{8.30}$$

with the Kohn-Sham potential

$$v_{\mathrm{KS}}(\boldsymbol{r}) = v_{\mathrm{ext}}(\boldsymbol{r}) + \int d^3\boldsymbol{r}'\,\frac{e^2 n(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|} + v_{\mathrm{xc}}(\boldsymbol{r}), \qquad v_{\mathrm{xc}}(\boldsymbol{r}) = \frac{\delta E_{\mathrm{xc}}[n]}{\delta n(\boldsymbol{r})}, \tag{8.31}$$

and the density reconstructed from the occupied orbitals,

$$n(\boldsymbol{r}) = \sum_{i=1}^{N} |\psi_i(\boldsymbol{r})|^2. \tag{8.32}$$

Since v_{KS} depends on n and n on the orbitals, the equations are nonlinear and must be iterated to self-consistency – the same self-consistent field loop as Section 6.5, and for the same reason.

The total energy.

The sum of the Kohn-Sham eigenvalues is *not* the total energy, exactly as in Hartree-Fock, Exercise 6.15b). Since $\sum_i \varepsilon_i = T_s + \int v_{\mathrm{KS}} n$,

$$E = \sum_{i=1}^{N} \varepsilon_i - \mathscr{U}_{\mathrm{H}}[n] + E_{\mathrm{xc}}[n] - \int d^3\boldsymbol{r}\, v_{\mathrm{xc}}(\boldsymbol{r}) n(\boldsymbol{r}), \tag{8.33}$$

the last three terms correcting the double counting of the Hartree and exchange-correlation contributions.

8.5.1 Kohn-Sham against Hartree-Fock

It is worth setting the two schemes side by side, since they look almost identical and differ in one decisive respect. The Hartree-Fock equation of Chapter 6 reads, in coordinate space,

$$\begin{aligned}\left(-\frac{\hbar^2}{2m}\nabla^2 + v_{\mathrm{ext}}(\boldsymbol{r}) + \int d^3\boldsymbol{r}'\, w(\boldsymbol{r},\boldsymbol{r}') n(\boldsymbol{r}')\right)\phi_k(\boldsymbol{r})& \\ \underbrace{-\sum_{l=1}^{N}\int d^3\boldsymbol{r}'\,\phi_l^*(\boldsymbol{r}') w(\boldsymbol{r},\boldsymbol{r}')\phi_k(\boldsymbol{r}')\phi_l(\boldsymbol{r})}_{\text{exchange}} &= \varepsilon_k\phi_k(\boldsymbol{r}),\end{aligned} \tag{8.34}$$

whereas the Kohn-Sham equation is

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + v_{\mathrm{ext}}(\boldsymbol{r}) + \int d^3\boldsymbol{r}'\, w(\boldsymbol{r},\boldsymbol{r}') n(\boldsymbol{r}') + \underbrace{v_{\mathrm{xc}}([n];\boldsymbol{r})}_{\text{exchange \textit{and} correlation}}\right)\phi_k(\boldsymbol{r}) = \varepsilon_k\phi_k(\boldsymbol{r}). \tag{8.35}$$

The difference is the last term. Hartree-Fock's exchange is *non-local* – an integral operator, different for each orbital – and it is exact. Kohn-Sham's v_{xc} is a *local multiplicative potential*, the same for every orbital, and it is approximate; but it also contains correlation, which Hartree-Fock does not have at all. The trade is non-locality and exactness against locality, cheapness and a bit of correlation. Both scale formally as the cube of the number of basis functions, but the Kohn-Sham prefactor is far smaller because no four-index integrals are needed for v_{xc}.

What the Kohn-Sham orbitals are. They are the orbitals of a fictitious non-interacting system chosen to reproduce the right density. They are not approximations to the true one-electron wave functions, and the eigenvalues ε_i are not ionisation energies – Koopmans' theorem, Exercise 6.15c), applies to Hartree-Fock, not here. The one exception is the highest occupied level, which for the exact functional equals minus the exact ionisation energy. The systematic underestimation of band gaps by Kohn-Sham eigenvalue differences is not a numerical deficiency but a consequence of asking the orbitals a question they were not built to answer.

8.6 The local density approximation

Everything now rests on $E_{\mathrm{xc}}[n]$. The oldest approximation, and still the reference point for all the others, is the *local density approximation*: assume that the exchange-correlation energy density at a point depends only on the density at that point, and take that dependence from the uniform electron gas,

$$E_{\mathrm{xc}}^{\mathrm{LDA}}[n] = \int d^3\boldsymbol{r}\, n(\boldsymbol{r})\, \varepsilon_{\mathrm{xc}}(n(\boldsymbol{r}))\,, \tag{8.36}$$

where $\varepsilon_{\mathrm{xc}}(n)$ is the exchange-correlation energy per particle of a uniform gas of density n. The corresponding potential follows by differentiation,

$$v_{\mathrm{xc}}(\boldsymbol{r}) = \frac{d}{dn}\left[n\, \varepsilon_{\mathrm{xc}}(n)\right]_{n=n(\boldsymbol{r})}. \tag{8.37}$$

The exchange part is available in closed form. It is the exchange energy of Section 6.13.4 expressed per particle rather than per state,

$$\varepsilon_x(n) = -\frac{3}{4}\left(\frac{3}{\pi}\right)^{1/3} n^{1/3}, \qquad v_x(n) = \frac{4}{3}\varepsilon_x(n), \tag{8.38}$$

in Hartree atomic units. That this must reproduce the electron-gas result of Chapter 6 is a consistency check the program performs:

r_s	ε_x [Ha]	ε_x [Ry]	$-0.916/r_s$ [Ry]
1.0000	−0.458165	−0.916331	−0.916000
2.0000	−0.229083	−0.458165	−0.458000
4.0000	−0.114541	−0.229083	−0.229000
4.8253	−0.094951	−0.189901	−0.189833
6.0000	−0.076361	−0.152722	−0.152667

The coefficient comes out as 0.916331 against the 0.9163 of Eq. (6.109). This agreement is a tautology – the local approximation is *defined* to be exact for the uniform gas – and the interesting question is not whether it holds but why the approximation remains useful when the density is far from uniform.

Figure 8.2 draws the same comparison over a continuous range of densities rather than the five points of the table, and the two curves cannot be told apart anywhere: the solid line computed from $\varepsilon_x(n)$ and the dashed $-0.916/r_s$ overlap to within the width of the pen. It is

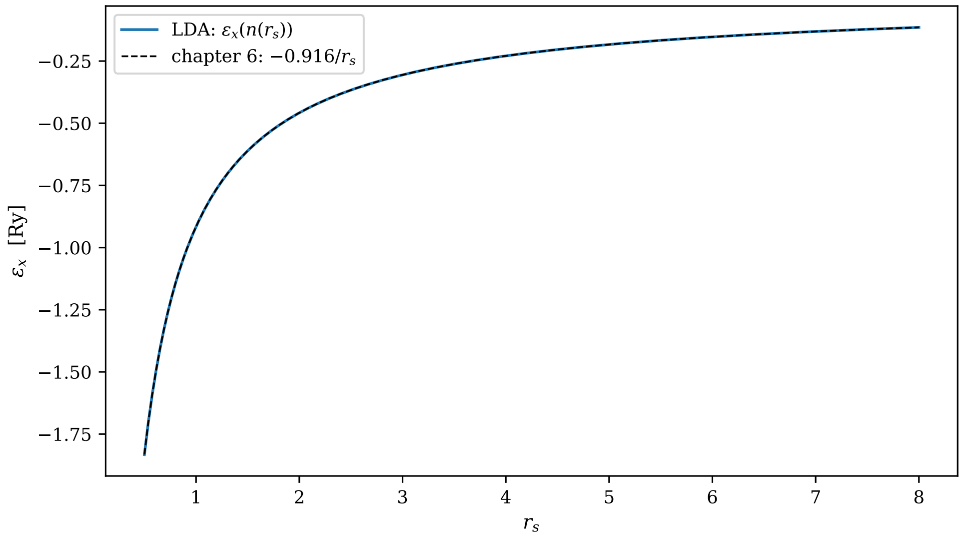

Fig. 8.2 The local exchange energy per particle $\varepsilon_x(n) = -\frac{3}{4}(3/\pi)^{1/3}n^{1/3}$ of Eq. (8.38), converted to Rydberg and plotted against the Wigner-Seitz radius r_s through $n = 3/4\pi r_s^3$, together with the $-0.916/r_s$ obtained from the filled Fermi sea in Chapter 6. The two curves are drawn separately over $0.5 \le r_s \le 8$ and are indistinguishable at any point, the coefficients differing by 3.6×10^{-4} in relative terms. The agreement is exact by construction and the figure is a check on the algebra rather than on the physics.

worth being clear about what has and has not been demonstrated. The figure verifies that the closed-form local functional agrees with the electron-gas exchange energy of Section 6.13.6, which is a statement about our algebra, not about nature; the two expressions are the same expression written in different variables. What the figure does convey is the shape of the dependence, and that shape is the reason the approximation is used at all. Exchange per particle scales as $n^{1/3}$, so it falls off only as the cube root of the density: over the range shown, a sixteenfold decrease in n from $r_s = 0.5$ to $r_s = 2$ costs a factor of four in ε_x, and beyond $r_s \simeq 4$ the curve is so flat that a substantial local error in n translates into a small error in ε_x. A functional this insensitive to its argument can afford to be evaluated on a density that is only roughly uniform, and that is a good deal of what makes the local approximation survive in systems where its stated assumption is plainly false.

The correlation part $\varepsilon_c(n)$ has no closed form. It is obtained from quantum Monte Carlo calculations of the uniform gas, the diffusion Monte Carlo results of Ceperley and Alder being the standard source, and then fitted to a convenient analytic form. This is worth noticing: the input to the most widely used approximation in computational materials science is a wave-function calculation of the kind developed in the preceding chapters.

8.6.1 The exchange hole

The reason a local approximation works at all becomes clearer if one looks at what exchange actually does. Divide the pair density (8.14) by the product of densities to define the pair correlation function g; for the uniform gas at the Hartree-Fock level it depends only on $y = k_F s$ and is known exactly,

$$g(y) = 1 - \frac{9}{2}\left[\frac{\sin y - y\cos y}{y^3}\right]^2. \tag{8.39}$$

The program tabulates it:

$k_F s$	$g(s)$	
0.00	0.500000	same spins never coincide
0.50	0.524471	
1.00	0.591838	
2.00	0.786732	
4.00	0.996208	
8.00	0.999920	the hole has healed

At coincidence $g(0) = \frac{1}{2}$: electrons of the same spin are completely excluded, those of opposite spin are uncorrelated at this level. Each electron therefore moves inside a hole in the density of its neighbours, and that hole obeys an exact sum rule – it contains precisely one missing electron,

$$n \int d^3\boldsymbol{s}\,[g(s)-1] = \frac{4}{3\pi}\int_0^\infty dy\, y^2\,[g(y)-1] = -1, \tag{8.40}$$

which the program verifies numerically to four digits.

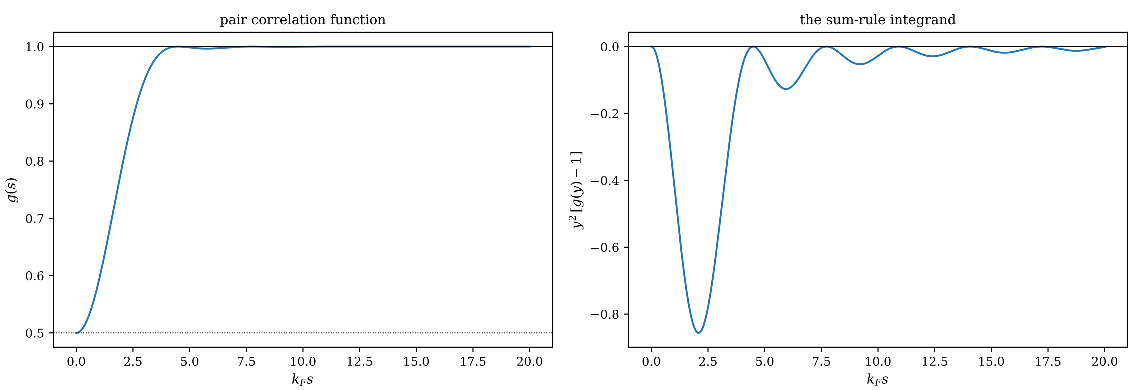

Fig. 8.3 Left: the Hartree-Fock pair correlation function (8.39) of the unpolarised electron gas against the scaled separation $y = k_F s$, with the uncorrelated value $g = 1$ and the coincidence value $g(0) = \frac{1}{2}$ marked by the solid and dotted horizontal lines. Right: the integrand $y^2[g(y)-1]$ of the sum rule (8.40), whose integral over all y must be $-3\pi/4$. The hole is deepest at coincidence and has essentially healed by $y \simeq 4$, where $g = 0.9962$, but the integrand carries a long oscillatory tail decaying only as y^{-2}: the range plotted here accounts for 95% of the sum rule, and the remaining 5% lies beyond $k_F s = 20$.

Figure 8.3 puts the two statements side by side, and the contrast between the panels is the whole point. In the left panel the correlation function starts at exactly one half, rises monotonically through $g = 0.5918$ at $y = 1$ and $g = 0.7867$ at $y = 2$, and reaches 0.9962 by $y = 4$; after that it is unity to three decimals, with only the faintest Friedel-like ripples left over from the sharp edge of the Fermi sea. Read on its own the left panel says that the exchange hole is a short-ranged object, of radius roughly k_F^{-1}, and that beyond a few interparticle spacings an electron knows nothing of its neighbours at this level of approximation. The right panel says something rather different. Weighting by the volume element y^2 turns those residual ripples into contributions that are not negligible at all: the integrand has a deep principal minimum of -0.856 at $y = 2.08$, but its subsequent oscillations decay only as y^{-2}, because $g(y) - 1 \to -\frac{9}{2}\cos^2 y/y^4$ at large y. Integrating out to the edge of the plot recovers -0.953 of the required -1; one must go to $k_F s \simeq 400$ before the quadrature returns the -0.9976 quoted in Exercise 8.9e). The sum rule is exact, but it is satisfied slowly, and by a tail rather than by the visible hole.

This is the deeper reason the local approximation succeeds. The exchange-correlation energy is, to a good approximation, the electrostatic interaction of each electron with its own hole. That energy depends mostly on the *total charge* of the hole – which the sum rule fixes exactly – and only weakly on its shape. A functional that gets the shape badly wrong but the normalisation right can still give good energies, and this is what the local density approximation does.

8.6.2 Beyond LDA

The density in a real atom or molecule is anything but slowly varying, and the natural first correction is to let the functional see the gradient as well,

$$E_{\rm xc}^{\rm GGA}[n] = \int d^3\boldsymbol{r}\, n(\boldsymbol{r})\, \varepsilon_{\rm xc}(n(\boldsymbol{r}), |\nabla n(\boldsymbol{r})|)\,, \tag{8.41}$$

the *generalised gradient approximation*. A naive gradient expansion of the uniform gas actually makes things worse; what works is to construct the gradient dependence so as to preserve the exact constraints – the sum rule above, scaling relations, known limits – and this is how the widely used functionals were built. Beyond that lie meta-GGAs, which add the kinetic energy density, and hybrids, which mix in a fraction of exact Hartree-Fock exchange, Eq. (8.34), to repair the self-interaction error discussed next.

8.7 Kohn-Sham in practice

To see the machinery work, and fail, we apply it to a system we have solved twice already: N spinless fermions in a one-dimensional harmonic trap with a softened Coulomb repulsion, in a basis of eight oscillator orbitals. This is the system of Section 2.9 and of the Hartree-Fock calculation of Section 6.5, so the three treatments can be compared for the same Hamiltonian.

There is no tabulated exchange functional for a one-dimensional gas with this interaction, so the program builds one, which is instructive in itself. For a spinless one-dimensional gas the Fermi momentum is $k_F = \pi n$ and the one-body density matrix is $\rho^{(1)}(s) = \sin(k_F s)/\pi s$, so

$$\varepsilon_x(n) = -\frac{1}{n}\int_0^\infty ds\, v(s)\left[\frac{\sin(k_F s)}{\pi s}\right]^2, \tag{8.42}$$

which is evaluated numerically on a grid of densities, tabulated and interpolated. This is precisely how real functionals are made. The result:

n	$\varepsilon_x(n)$	$v_x(n)$
0.05	−0.17352608	−0.29740316
0.10	−0.27855522	−0.45906496
0.20	−0.42332628	−0.65789106
0.40	−0.59557914	−0.84878595
0.80	−0.75884371	−0.96587633
1.50	−0.86548980	−0.99703512
2.50	−0.91895548	−0.99989688

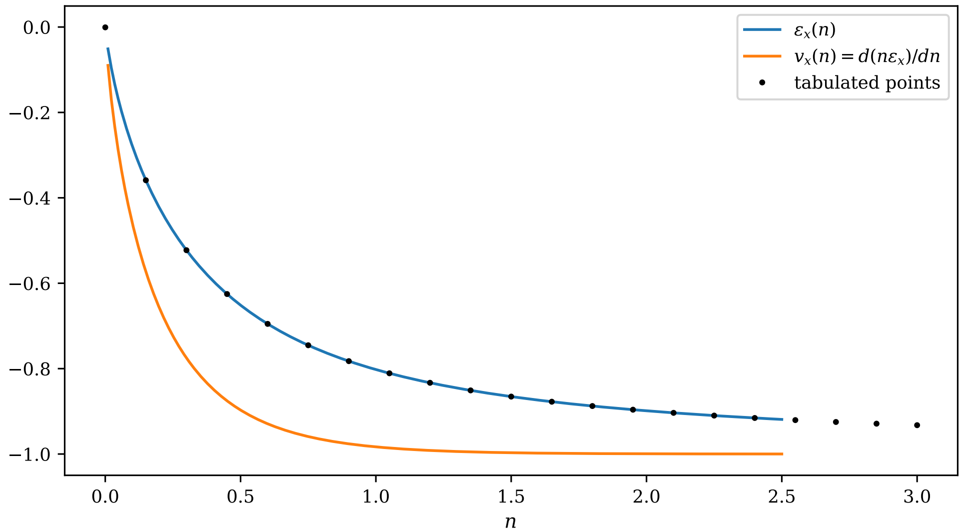

Fig. 8.4 The one-dimensional exchange functional built numerically from Eq. (8.42) for the softened Coulomb interaction $v(s) = 1/\sqrt{s^2+0.25}$ of Chapters 2 and 6: the exchange energy per particle $\varepsilon_x(n)$, the corresponding potential $v_x(n) = d(n\varepsilon_x)/dn$ of Eq. (8.37), and the 241 tabulated densities on which the cubic spline is built, shown every twelfth point. Both curves fall steeply out of the origin and then saturate, v_x approaching -1 and ε_x still climbing slowly towards it; the spline passes through the tabulated values by construction, and the points continue to $n = 3$ so that the interpolation is never extrapolated inside the self-consistent loop.

The functional so constructed is drawn in Figure 8.4, and it repays a moment's attention because it behaves quite unlike its three-dimensional counterpart. There $\varepsilon_x \propto n^{1/3}$ grows without bound; here both ε_x and v_x saturate, the potential having already reached -0.9833 at $n = 1$ and -0.99990 at $n = 2.5$. The limit is not an accident of the numerics. At high density the Fermi momentum $k_F = \pi n$ is large, the one-body density matrix $\sin(k_F s)/\pi s$ collapses onto a spike of width k_F^{-1} and height n, and the integral in Eq. (8.42) samples the interaction only at contact, giving $\varepsilon_x \to -v(0)/2 = -1$ for the softening 0.5 used throughout. Softening the Coulomb singularity has thus put a floor under the exchange energy per particle, and the local functional inherits it. The practical consequence is visible in the vertical gap between the two curves: v_x lies below ε_x everywhere, by 0.25 at $n = 0.5$ and by 0.18 at $n = 1$, which is the one-dimensional version of the factor $\frac{4}{3}$ of Eq. (8.38) and of Exercise 8.9c). It is the potential, not the energy per particle, that enters the Kohn-Sham equation, and the difference between the two is exactly what the double-counting correction of Eq. (8.33) has to undo afterwards.

Solving Eq. (8.30) self-consistently in the oscillator basis then gives

N	E (Kohn-Sham)	E (Hartree-Fock)	E (exact)
1	0.54752674	0.50000000	0.50000000
2	2.72022768	2.66308878	2.65923049
3	6.45108052	6.39016133	—
4	11.68602229	11.62305385	—

Table 8.1 Kohn-Sham with a local exchange functional, Hartree-Fock and the exact answer for N spinless fermions in a one-dimensional trap with a softened Coulomb repulsion, in a basis of eight oscillator orbitals. The Hartree-Fock column reproduces Table 2.2; the exact value for $N = 2$ comes from full diagonalisation in the same basis.

Note first that the exact two-particle energy, 2.65923049, lies -3.86×10^{-3} below the Hartree-Fock value, which is the correlation energy quoted in Eq. (2.42) – the three chapters are consistent. Note next that Kohn-Sham lies *above* both. This is not a failure of the variational principle, because we have used exchange only and put no correlation functional in at all; it is the error of treating exchange locally, compounded by a second defect that we now isolate.

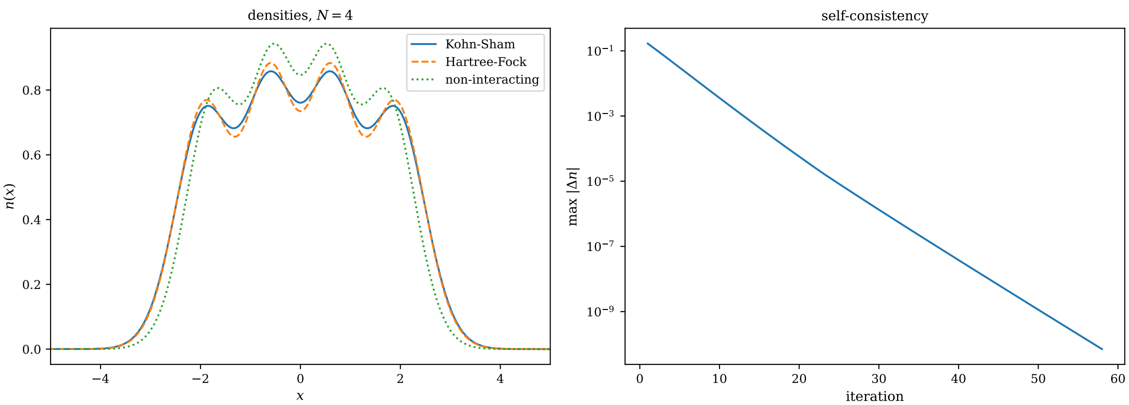

Fig. 8.5 Left: the density of $N = 4$ spinless fermions in the one-dimensional trap, computed self-consistently from the Kohn-Sham equations (8.30) with the local exchange functional of Figure 8.4, from Hartree-Fock in the same eight-orbital basis, and from the non-interacting determinant. All three carry four maxima, one per occupied orbital, but the repulsion pushes the outer pair from $x = \pm 1.64$ out to ± 1.84 and lowers the central maxima from 0.943 to 0.858. Right: the largest change in the density from one iteration to the next, on a logarithmic scale, for the same $N = 4$ calculation with linear mixing at 30%; the decay is a clean straight line, a factor 0.685 per iteration, reaching 7×10^{-11} after 58 iterations.

Before doing so it is worth looking at what the self-consistent solution actually produces, and Figure 8.5 shows it for $N = 4$. The left panel confirms first of all that the Kohn-Sham density is a perfectly sensible density: the four maxima that the four occupied orbitals generate are all present, so the shell structure that Thomas-Fermi destroyed in Figure 8.1 has indeed been recovered by putting the orbitals back. Turning the interaction on then does what one expects of a repulsion, pushing the distribution outwards and flattening it: the outer maxima move from $x = \pm 1.64$ to ± 1.84 and drop from 0.806 to 0.751, the inner pair from 0.943 to 0.858, and the weight beyond $|x| = 3$ grows from 0.024 particles to 0.062. The Kohn-Sham and Hartree-Fock densities lie almost on top of each other, as they should for two mean fields built from the same Hamiltonian, and their largest absolute difference is only 0.028, at $x \simeq \pm 1.24$. Where they part company is instructive. The Hartree-Fock density is the more strongly modulated of the two, dipping to 0.735 at the origin against the Kohn-Sham 0.761, so that a non-local exchange operator *sharpens* the shell structure while the smooth multiplicative v_{xc} of Eq. (8.35) washes a little of it out; and in the tails, where the density is small and the local approximation is at its worst, the Kohn-Sham solution carries some seven per cent more weight beyond $|x| = 3$ than Hartree-Fock does. That excess diffuseness is the fingerprint of the defect analysed in the next subsection, an electron being spuriously repelled by its own charge and spreading out to escape it.

The right panel is about the numerics rather than the physics, and it is the same picture we saw for the Hartree-Fock loop of Section 6.5. The largest pointwise change in the density falls from 0.167 at the first iteration to 7×10^{-11} at the fifty-eighth, and on the logarithmic axis the descent is a straight line, which is to say that the iteration converges linearly with a fixed contraction factor of 0.685 per step – some three iterations per decimal digit. Linear rather than quadratic convergence is what a fixed-point iteration with simple linear mixing of the density delivers, and the slope is set by the mixing parameter, here 0.3: mixing too little makes the loop unstable, mixing too much makes this line shallower. The absence of any plateau or kink is itself worth noting, since it says that the tabulated functional of Figure 8.4 is smooth enough that the spline introduces no noise floor down to 10^{-10}, and that the self-consistent density is a genuine attracting fixed point of Eqs. (8.30)–(8.32) rather than something the convergence criterion has merely stopped short of.

8.7.1 The self-interaction error

Take a single particle. There is nothing for it to interact with, so the exact energy is the oscillator ground state, 0.5. Hartree-Fock gives exactly that: in Eq. (8.34) the direct term with $l = k$ is cancelled term by term by the exchange term with $l = k$, and a Slater determinant is rigorously free of self-interaction.

A local functional cannot do this, because it sees only the density, and the density of one electron is indistinguishable from the density of half of two. The program reports, for $N = 1$:

$$\mathscr{U}_{\mathrm{H}} = 0.60855522, \qquad E_x^{\mathrm{LDA}} = -0.56166291, \qquad \mathscr{U}_{\mathrm{H}} + E_x^{\mathrm{LDA}} = 0.04689232, \tag{8.43}$$

where the sum should vanish identically. The residue 0.047 is the *self-interaction error*, and it accounts for almost all of the 0.0475 by which Kohn-Sham misses the exact one-particle energy.

This single number explains a long list of known LDA failures: anions are overbound, reaction barriers too low, band gaps too small, and charge is over-delocalised, because an electron is spuriously repelled by itself and spreads out to reduce that repulsion. Self-interaction corrections remove it explicitly, orbital by orbital; hybrid functionals reduce it by mixing in a fraction of exact exchange, which is self-interaction free by construction.

8.7.2 How local is too local?

We can isolate the error of *locality* alone, cleanly, by evaluating on one and the same density both the exact exchange energy of the determinant and the local estimate:

N	E_x (exact)	E_x (local)	ratio
1	-0.60855522	-0.56166291	0.9229
2	-1.33077771	-1.27749456	0.9600
3	-2.11087395	-2.05548416	0.9738
4	-2.92523775	-2.86866697	0.9807

The local approximation recovers 92% of the exchange energy for one particle and 98% for four, improving steadily as the density smooths out on the scale of the local Fermi wavelength. This is the behaviour the construction promises, and the corresponding figure for real atoms in three dimensions is close to 90% – an error of the order of ten per cent in a large term, which is exactly why gradient corrections were needed before density functional theory became quantitative for chemistry.

Figure 8.6 plots the last column of that table, and the regularity of it is worth extracting. Multiplying the shortfall $1 - r$ by the particle number gives 0.077, 0.080, 0.079 and 0.077 for $N = 1, \ldots, 4$ – constant to within a couple of per cent – so that the relative error of the local

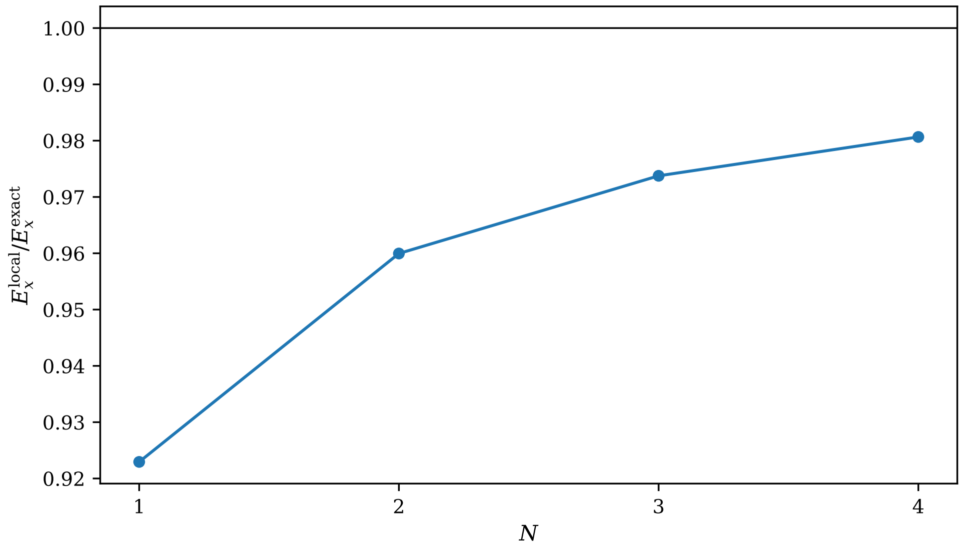

Fig. 8.6 The ratio of the local exchange energy $\int n\,\varepsilon_x(n)$ to the exact exchange energy of the same Slater determinant, for $N = 1$ to 4 spinless fermions in the one-dimensional trap. Both quantities are evaluated on one and the same self-consistent Kohn-Sham density, so that the only error being measured is that of locality, with no self-consistency and no basis-truncation error mixed in; the horizontal line marks the exact result. The local functional is too small throughout, by 7.7% for one particle and by 1.9% for four, the shortfall falling essentially as $1/N$.

approximation behaves as $1 - r \simeq 0.078/N$ over the whole range. That is precisely the scaling one should expect if the error lives in the surface of the density rather than in its interior. The local functional is exact for a uniform gas and wrong only where the density varies on the scale of k_F^{-1}, which in a trapped system means the two edges and, to a lesser extent, the shell minima of Figure 8.5; the number of such regions does not grow with N while the bulk contribution does, and the ratio of the two is $1/N$. Extrapolating the fit, one would need of the order of eighty particles before locality alone cost less than a tenth of a per cent of the exchange energy, which is a sobering estimate for a quantity that is itself several Hartree. Two further points should be kept in view. The error is one-signed – the curve approaches unity strictly from below, and the local functional never overbinds – so it does not average away in an energy difference between two systems of different size, which is exactly the situation in a dissociation or reaction energy. And it is a smooth, slowly varying function of N, which is what makes a gradient correction of the form (8.41) a plausible repair in the first place: a single additional term sensitive to $|\nabla n|$ has some chance of capturing a systematic edge effect, though as Exercise 8.9e) shows, one fitted parameter will not do it for every N at once.

8.8 Summary

Density functional theory rests on an exact statement – the ground-state density determines the Hamiltonian – and on an approximation that the theorem itself does not supply. The Hohenberg-Kohn theorems guarantee a universal functional $F[n]$ and a variational principle for it, but they are existence theorems; all the many-body difficulty of the preceding chapters survives, relocated inside $F[n]$.

The Kohn-Sham construction is what makes the theory workable. By reintroducing orbitals for the kinetic energy – the one contribution that a local approximation handles badly, as

Thomas-Fermi showed – it keeps shell structure exactly and leaves only a small remainder to be approximated. What is left is a self-consistent one-body problem of exactly the same form as Hartree-Fock, Eq. (6.26), differing in a single term: a local multiplicative v_{xc} in place of the non-local exchange operator.

The local density approximation takes that remainder from the electron gas of Chapter 6, and its success rests less on the density being slowly varying – it never is – than on the exchange-correlation hole sum rule, which fixes the quantity the energy is most sensitive to. Its two characteristic failures are visible in a system with four particles: exchange treated locally is wrong by a few per cent, and the self-interaction that Hartree-Fock cancels identically survives.

It is worth being clear about where this method sits relative to the others in this book. Configuration interaction, coupled cluster and the random-phase approximation are systematically improvable: there is a hierarchy, and one can in principle climb it towards the exact answer. Density functional theory is not of that kind. There is no sequence of functionals converging to $F[n]$, and no internal estimate of the error. What it offers instead is a cost that scales gently enough to treat thousands of atoms, and an accuracy that – for reasons that are now partly understood – is far better than the crudeness of the approximations would suggest.

The program `dft.py` performs every calculation of this chapter, and the notebook `dft.ipynb` runs it cell by cell.

8.9 Exercises

Exercise 1: densities and density matrices

a) Show that Eq. (8.4) and Eq. (8.5) define the same quantity, and that $\int n(\boldsymbol{r})d^3\boldsymbol{r} = N$.
b) Derive Eq. (8.6) for a single Slater determinant, and show that $\rho^{(1)}$ is then the projector onto the occupied orbitals.
c) Derive the factorisation (8.13) from Wick's theorem, and verify that the diagonal vanishes at $x_1 = x_2$.
d) Show that for a correlated state the eigenvalues of $\rho^{(1)}$ are strictly between zero and one, and relate this to the occupation numbers of Section 1.31.

Answer to a).

Insert Eq. (8.4) into the expectation value and use the fact that Ψ is antisymmetric, so that every term of $\sum_i \delta(\boldsymbol{r}-\boldsymbol{r}_i)$ contributes equally:

$$\langle\Psi|\hat{n}(\boldsymbol{r})|\Psi\rangle = N\int d^3\boldsymbol{r}_2\cdots d^3\boldsymbol{r}_N\,|\Psi(\boldsymbol{r},\boldsymbol{r}_2,\dots)|^2\,, \tag{8.44}$$

which is Eq. (8.5). Integrating over $\boldsymbol{r}$ restores the full normalisation integral, giving $N\cdot 1 = N$.

Answer to b).

For a determinant $\rho^{(1)}(x';x) = \sum_{i=1}^{N}\psi_i(x')\psi_i^*(x)$, which follows from Eq. (8.9) by expanding the determinant and noting that only terms pairing the same orbital in bra and ket survive the integration over the remaining $N-1$ coordinates. Setting $x' = x$ gives Eq. (8.6). Regarded as an operator, $\rho^{(1)} = \sum_i |\psi_i\rangle\langle\psi_i|$, so $(\rho^{(1)})^2 = \rho^{(1)}$ by orthonormality and $\mathrm{tr}\,\rho^{(1)} = N$: a projector of rank N, which is Exercise 6.15d) in coordinate language.

Answer to c).

Write the pair density as $\langle\Phi_0|\hat{\psi}^\dagger(y_1)\hat{\psi}^\dagger(y_2)\hat{\psi}(x_2)\hat{\psi}(x_1)|\Phi_0\rangle$ and apply Wick's theorem. Only contractions pairing a creation with an annihilation operator survive, and there are exactly two ways to do it: $(y_1x_1)(y_2x_2)$ and $(y_1x_2)(y_2x_1)$, the second differing from the first by one interchange and hence carrying a minus sign. Each contraction is a factor $\rho^{(1)}$, giving Eq. (8.13). On the diagonal with $x_1 = x_2$ the two terms become $n(x_1)^2$ and $|\rho^{(1)}(x_1;x_1)|^2 = n(x_1)^2$, which cancel.

Answer to d).

$\rho^{(1)}$ is Hermitian and positive semi-definite, so its eigenvalues n_k are real and non-negative; they sum to N. For fermions no eigenvalue can exceed one, since $n_k = \langle\Psi|\hat{a}_k^\dagger\hat{a}_k|\Psi\rangle \leq 1$ by the Pauli principle. Equality for exactly N of them forces $\rho^{(1)}$ to be idempotent, hence forces Ψ to be a single determinant. Any correlated state therefore has all n_k strictly inside $[0,1]$, and the departure of the largest from unity is the diagnostic used in Table 1.8.

Exercise 2: the Hohenberg-Kohn theorems

a) Reproduce the proof of the first theorem, stating exactly where non-degeneracy is used.
b) Show that the theorem determines v_{ext} only up to an additive constant, and explain why that is harmless.
c) Prove the second theorem from the first, and state what "universal" means.
d) Explain the v-representability problem and how the Levy-Lieb constrained search, Eq. (8.21), removes it.
e) The theorems say the density determines everything. Why, then, can we not obtain excitation energies from $E[n]$?

Answer to a).

See Eqs. (8.17)–(8.19). Non-degeneracy enters in making the variational inequality *strict*: if the ground state were degenerate, $\Psi^{(2)}$ might be another ground state of $\hat{H}^{(1)}$ and the inequality would become an equality, so that adding the two relations would give $E^{(1)} + E^{(2)} \leq E^{(1)} + E^{(2)}$ – true, and no contradiction. This is exactly the loophole that the Levy-Lieb formulation closes.

Answer to b).

Adding a constant c to v_{ext} adds Nc to every eigenvalue and changes no eigenstate, hence no density. The map from densities to potentials therefore cannot possibly resolve the constant, and does not need to: energy differences, densities and all observables are unaffected. It is the same ambiguity as the zero of a potential in elementary mechanics.

Answer to c).

By the first theorem n_0 determines v_{ext}, hence $\hat{H}$, hence its ground state $\Psi[n]$. So $F_{\text{HK}}[n] = \langle\Psi[n]|\hat{T}+\hat{V}|\Psi[n]\rangle$ is a functional of n alone. "Universal" means that F contains no reference to v_{ext}: the very same functional applies to a hydrogen atom, a copper crystal and a stretch of DNA, since $\hat{T}$ and $\hat{V}$ are properties of electrons. For any trial n, the Rayleigh-Ritz principle gives $E[n] = F[n] + \int v_{\text{ext}} n \geq E_0$, with equality only at $n = n_0$.

Answer to d).

The construction of F_{HK} requires that n be the ground-state density of *some* potential in the admissible family – that it be v-representable. Densities are known that are not, and no useful characterisation of the v-representable ones exists. Levy and Lieb sidestep the issue by defining F through a minimisation over all antisymmetric wave functions yielding the given density, Eq. (8.21). This requires only N-representability, which is easy to characterise for a one-body density – positivity, correct normalisation and finite $\int|\nabla n^{1/2}|^2$ suffice – and it handles degenerate ground states.

Answer to e).

The theorems concern the ground state. The variational principle $E[n] \geq E_0$ singles out the minimum and says nothing about stationary points corresponding to excited states, and the functional $F[n]$ was constructed from ground-state wave functions only. Excitations require extra structure: time-dependent density functional theory obtains them from the linear response of the density, which is formally the density-functional analogue of the random-phase approximation of Chapter 7, with the exchange-correlation kernel $\delta v_{\text{xc}}/\delta n$ playing the part of the residual interaction in A and B.

Exercise 3: from the electron gas to a functional

a) Derive the Thomas-Fermi kinetic density (8.23) from the kinetic energy of Chapter 6, and confirm it numerically.
b) Derive the local exchange energy (8.38) and show that it reproduces the $-0.916/r_s$ of Section 6.13.6.
c) Show that $v_x = \frac{4}{3}\varepsilon_x$ for the local exchange functional, and explain why the factor is not one.
d) Derive the one-dimensional analogue (8.42) and the kinetic density $t[n] = (\pi^2/6)n^3$.
e) Verify the exchange-hole sum rule (8.40) analytically.

Answer to a).

Equation (6.91) gives the kinetic energy per particle as $\frac{3}{5}\hbar^2 k_F^2/2m$; multiplying by n and using $k_F = (3\pi^2 n)^{1/3}$ gives the energy per unit volume

$$t_0[n] = \frac{3}{10}\frac{\hbar^2}{m}\left(3\pi^2\right)^{2/3} n^{5/3}, \tag{8.45}$$

which is Eq. (8.23). The program compares the two expressions at three densities and finds them identical to eight digits.

Answer to b).

The exchange energy is $E_x = -\frac{1}{2}\iint v(|\boldsymbol{r}-\boldsymbol{r}'|)|\rho^{(1)}(\boldsymbol{r};\boldsymbol{r}')|^2$, the second term of Eq. (8.14) integrated against the interaction. For the uniform gas $\rho^{(1)}$ depends only on $s = |\boldsymbol{r}-\boldsymbol{r}'|$ and equals $3n[\sin y - y\cos y]/y^3$ with $y = k_F s$; doing the integral with $v = e^2/s$ gives $E_x/V = -\frac{3}{4}(3/\pi)^{1/3}n^{4/3}$, hence Eq. (8.38). Evaluating at $n = 3/(4\pi r_0^3)$,

$$\varepsilon_x = -\frac{3}{4}\left(\frac{3}{\pi}\right)^{1/3}\left(\frac{3}{4\pi}\right)^{1/3}\frac{1}{r_s} = -\frac{0.4582}{r_s}\ \text{Ha} = -\frac{0.9163}{r_s}\ \text{Ry}, \tag{8.46}$$

which is the exchange term of Eq. (6.108).

Answer to c).

$E_x = \int n\varepsilon_x(n)$ with $\varepsilon_x \propto n^{1/3}$, so the integrand is $\propto n^{4/3}$ and

$$v_x = \frac{d}{dn}\left[n\varepsilon_x(n)\right] = \frac{4}{3}\varepsilon_x(n). \tag{8.47}$$

The factor is not one because v_x is a functional derivative, not an energy per particle: changing the density at a point changes both the number of electrons there and the energy each of them carries. The distinction is the same as that between $\sum_i \varepsilon_i$ and the total energy in Eq. (8.33), and forgetting it is a standard first error.

Answer to d).

For a spinless one-dimensional gas, $n = \int_{-k_F}^{k_F} dk/2\pi = k_F/\pi$, so $k_F = \pi n$, and $\rho^{(1)}(s) = \int_{-k_F}^{k_F}(dk/2\pi)e^{iks} = \sin(k_F s)/\pi s$. Inserting into the exchange integral and using the symmetry of the two half-lines to cancel the factor $\frac{1}{2}$ gives Eq. (8.42). For the kinetic energy, $T/L = \int_{-k_F}^{k_F}(dk/2\pi)(k^2/2) = k_F^3/6\pi = (\pi^2/6)n^3$.

Answer to e).

Substituting Eq. (8.39),

$$\frac{4}{3\pi}\int_0^\infty dy\, y^2\left[g(y)-1\right] = -\frac{4}{3\pi}\cdot\frac{9}{2}\int_0^\infty dy\,\frac{(\sin y - y\cos y)^2}{y^4} = -\frac{6}{\pi}\cdot\frac{\pi}{6} = -1, \tag{8.48}$$

using the standard integral $\int_0^\infty(\sin y - y\cos y)^2 y^{-4}dy = \pi/6$. The program obtains -0.9976 by direct quadrature, the residue being the slowly decaying oscillatory tail.

Exercise 4: the Kohn-Sham equations

a) Derive Eq. (8.30) by varying Eq. (8.29) with respect to the orbitals subject to orthonormality.
b) Derive the total-energy expression (8.33) and explain each correction term.
c) Compare Eqs. (8.34) and (8.35) term by term. Which is variational, and with respect to what?
d) Why is $T_s[n]$, and not $T[n]$, computed from the orbitals? What is the difference, and where does it go?
e) Show that for one electron the exact E_{xc} must satisfy $E_{\text{xc}}[n] = -\mathscr{U}_{\text{H}}[n]$.

Answer to a).

Write T_s and n in terms of the orbitals and impose $\int \psi_i^* \psi_j = \delta_{ij}$ with multipliers ε_{ij}. Varying with respect to ψ_i^*,

$$-\frac{\hbar^2}{2m}\nabla^2\psi_i + \left[v_{\text{ext}} + \frac{\delta \mathscr{U}_{\text{H}}}{\delta n} + \frac{\delta E_{\text{xc}}}{\delta n}\right] \frac{\delta n}{\delta|\psi_i|^2}\psi_i = \sum_j \varepsilon_{ij}\psi_j, \tag{8.49}$$

and since $\delta n/\delta|\psi_i|^2 = 1$ the bracket is exactly v_{KS}. A unitary rotation among the occupied orbitals diagonalises ε_{ij} without changing the density – the same freedom as in Section 6.3 – giving Eq. (8.30).

Answer to b).

Multiplying Eq. (8.30) by ψ_i^*, integrating and summing over the occupied orbitals gives $\sum_i \varepsilon_i = T_s + \int v_{\text{KS}} n$. Substituting v_{KS} from Eq. (8.31) and comparing with Eq. (8.29),

$$E = \sum_i \varepsilon_i - \underbrace{\int v_{\text{H}} n}_{=2\mathscr{U}_{\text{H}}} - \int v_{\text{xc}} n + \mathscr{U}_{\text{H}} + E_{\text{xc}}, \tag{8.50}$$

which is Eq. (8.33). The band sum counts the Hartree energy twice, once for each partner, so one copy is removed; and it contains $\int v_{\text{xc}} n$, which is not E_{xc}, so the former is subtracted and the latter added. This is the same double-counting correction as Exercise 6.15b).

Answer to c).

The kinetic, external and Hartree terms are identical. The difference is the last term: Hartree-Fock has a non-local exchange *operator*, acting differently on each orbital, exact for a determinant and containing no correlation; Kohn-Sham has a local multiplicative *potential*, the same for all orbitals, approximate, but containing correlation as well. Hartree-Fock is variational with respect to the true energy – it is a genuine upper bound on E_0. Kohn-Sham is variational only with respect to the energy of the model functional it is given; with an approximate E_{xc} the result may lie above or below the exact energy, as Table 8.1 shows.

Answer to d).

$T[n]$, the true kinetic energy of the interacting system, is not accessible from the density; that was the problem in the first place. $T_s[n]$, the kinetic energy of the non-interacting system with the same density, is computed exactly from its orbitals. The difference $T[n]-T_s[n]$ is small – it is a correlation effect – and it is folded into $E_{\rm xc}$ by the definition (8.28). The whole construction is the observation that this is a far better place to put the difficulty than the kinetic energy itself.

Answer to e).

For $N=1$ the exact energy contains no electron-electron interaction at all. But Eq. (8.29) contains $\mathscr{U}_{\rm H}[n]$, which is non-zero for any density. The exact functional must therefore cancel it exactly, $E_{\rm xc}[n]=-\mathscr{U}_{\rm H}[n]$ for any one-electron density. The local density approximation violates this: the program finds a residue 0.0469 for the harmonic trap, Eq. (8.43). Imposing the one-electron condition exactly is the content of the Perdew-Zunger self-interaction correction.

Exercise 5: numerical explorations

a) Using `dft.py`, reproduce Table 8.1 and plot the Kohn-Sham, Hartree-Fock and non-interacting densities for $N=4$. Where do they differ most?
b) Vary the interaction strength and follow the ratio $E_x^{\rm local}/E_x^{\rm exact}$. Does the local approximation improve or deteriorate as the interaction is made stronger?
c) Implement the Perdew-Zunger self-interaction correction for $N=1$ – subtract $\mathscr{U}_{\rm H}[n_i]+E_x[n_i]$ orbital by orbital – and check that the one-particle energy returns to 0.5.
d) Solve the Thomas-Fermi equation for the trap directly by minimising over $n(x)$ on a grid, and compare the resulting density with the Kohn-Sham one. Is the shell structure visible?
e) Add a crude gradient correction, $E_x \to E_x - \beta\int|\nabla n|^2/n^{4/3}$, and tune β to fit the exact exchange energies of Section 8.7.2. Does one value work for all N?

Answer.

Some pointers. In a) the densities differ most in the tails, where the Kohn-Sham density is too diffuse – the signature of self-interaction, since a spuriously self-repelled electron spreads out. In b) the ratio deteriorates as the interaction strengthens, because exchange grows relative to the kinetic energy and the errors in it become more visible; this is the same trend that made the mean field fail at strong coupling in Section 5.7. In c) the correction is exact by construction for one particle and the energy returns to 0.5 to machine precision; the interesting question is what it does for $N>1$, where it is no longer exact and can over-correct. In d) the Thomas-Fermi density is a smooth dome with no oscillations, in contrast to the N visible maxima of the Kohn-Sham density – the clearest single illustration of what orbitals buy. In e) a single β does not fit all N; the failure of a one-parameter gradient correction is precisely why the construction of gradient functionals is done by imposing exact constraints instead of fitting.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter08/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/dft.ipynb`, so that the numbers on this page and the numbers on the Jupyterbook site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter08/`.

`dft.py` The local density approximation built from the uniform gas, the exchange hole, a Kohn–Sham solver in the chapter-6 basis, and the self-interaction and locality errors. Runs in a few seconds; needs `scipy.linalg` and `scipy.interpolate`.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 9
Many-body perturbation theory

Chapter 5 showed that the exact ground state is a superposition of all determinants, and that we cannot afford them. Chapter 6 found the best single determinant, and Chapter 7 described the fluctuations about it. Perturbation theory takes yet another route: it accepts the mean-field determinant as a starting point and generates corrections to it *order by order* in the interaction, each order being a finite, well-defined sum that can be written down and evaluated.

The appeal is obvious. There is no matrix to diagonalise, no iteration to converge, and no variational parameters – only sums over intermediate states, which for a two-body Hamiltonian are limited to particle-hole excitations of low rank. The price is equally clear and we shall be blunt about it: the method is not variational, so nothing guarantees that a higher order is a better answer; and the series need not converge at all.

This chapter develops the formalism algebraically. There is a beautiful diagrammatic language for organising these sums, and it is indispensable beyond third order, but it is a separate subject and we leave it out here in order to keep the algebra visible. We derive the two classical schemes – Brillouin-Wigner and Rayleigh-Schrödinger – write out the first, second and third-order terms, and then apply them to the four models that have run through the book: the pairing model of Section 4.2, the Lipkin model of Section 4.1, the Hubbard model of Section 4.4, and the pairing plus particle-hole model of Section 7.1. Every number is compared with the exact answer and with the methods of the preceding chapters.

9.1 An exact starting point

We are interested in the ground state, and we assume it is dominated by a single determinant $|\Phi_0\rangle$ that solves an unperturbed problem,

$$\hat{H}_0|\Phi_0\rangle = W_0|\Phi_0\rangle, \qquad \hat{H} = \hat{H}_0 + \hat{H}_I. \tag{9.1}$$

The exact state is expanded in the eigenstates of $\hat{H}_0$,

$$|\Psi_0\rangle = |\Phi_0\rangle + \sum_{m=1}^{\infty} C_m|\Phi_m\rangle, \tag{9.2}$$

with intermediate normalisation $\langle\Phi_0|\Psi_0\rangle = 1$, exactly as in Section 5.2. Note that $|\Psi_0\rangle$ is therefore not normalised to unity; this is a convenience, not an approximation.

Project the Schrödinger equation $\hat{H}|\Psi_0\rangle = E|\Psi_0\rangle$ onto the reference,

$$\langle\Phi_0|\hat{H}|\Psi_0\rangle = E\langle\Phi_0|\Psi_0\rangle = E, \tag{9.3}$$

and subtract the corresponding statement for $\hat{H}_0$,

$$\langle\Psi_0|\hat{H}_0|\Phi_0\rangle = W_0\langle\Psi_0|\Phi_0\rangle = W_0. \tag{9.4}$$

Both operators are Hermitian, so the two may be subtracted directly, giving

$$\boxed{\Delta E = E - W_0 = \langle\Phi_0|\hat{H}_I|\Psi_0\rangle.} \tag{9.5}$$

This is exact, and we call ΔE the correlation energy. Every perturbative derivation in this chapter starts from it.

As it stands, however, Eq. (9.5) is a rewriting of the Schrödinger equation and nothing more: the right-hand side contains the unknown exact state. To make progress we must expand $|\Psi_0\rangle$ in powers of $\hat{H}_I$.

9.2 Projection operators and the resolvent

Define the projector onto the reference and its complement,

$$\hat{P} = |\Phi_0\rangle\langle\Phi_0|, \qquad \hat{Q} = \sum_{m=1}^{\infty} |\Phi_m\rangle\langle\Phi_m| = 1 - \hat{P}, \tag{9.6}$$

so that our model space is one-dimensional. Both are idempotent, $\hat{P}^2 = \hat{P}$ and $\hat{Q}^2 = \hat{Q}$, both commute with $\hat{H}_0$ – they are built from its eigenstates – and $\hat{P}\hat{Q} = 0$. With intermediate normalisation,

$$|\Psi_0\rangle = \left(\hat{P} + \hat{Q}\right)|\Psi_0\rangle = |\Phi_0\rangle + \hat{Q}|\Psi_0\rangle. \tag{9.7}$$

Now introduce an energy parameter ω, to be fixed later, and rewrite the Schrödinger equation by adding and subtracting $\omega|\Psi_0\rangle$,

$$\left(\omega - \hat{H}_0\right)|\Psi_0\rangle = \left(\omega - E + \hat{H}_I\right)|\Psi_0\rangle. \tag{9.8}$$

Provided the operator on the left has an inverse – the *resolvent*, or unperturbed Green's function,

$$\left(\omega - \hat{H}_0\right)^{-1} = \frac{1}{\omega - \hat{H}_0}, \tag{9.9}$$

we may divide by it. Multiplying from the left by $\hat{Q}$ and using

$$\hat{Q}\frac{1}{\omega - \hat{H}_0}\hat{Q} = \hat{Q}\frac{1}{\omega - \hat{H}_0} = \frac{\hat{Q}}{\omega - \hat{H}_0}, \tag{9.10}$$

which holds because $\hat{Q}$ commutes with $\hat{H}_0$ and is idempotent, we obtain

$$|\Psi_0\rangle = |\Phi_0\rangle + \frac{\hat{Q}}{\omega - \hat{H}_0}\left(\omega - E + \hat{H}_I\right)|\Psi_0\rangle. \tag{9.11}$$

Equation (9.11) is still exact and still useless: it is a nonlinear equation for two unknowns, E and $|\Psi_0\rangle$. But it invites iteration. Substituting the whole right-hand side into itself repeatedly, starting from the natural guess $|\Psi_0\rangle \to |\Phi_0\rangle$, gives

$$|\Psi_0\rangle = \sum_{i=0}^{\infty}\left\{\frac{\hat{Q}}{\omega - \hat{H}_0}\left(\omega - E + \hat{H}_I\right)\right\}^i |\Phi_0\rangle, \tag{9.12}$$

and, inserted into Eq. (9.5),

$$\Delta E = \sum_{i=0}^{\infty} \langle \Phi_0 | \hat{H}_I \left\{ \frac{\hat{Q}}{\omega - \hat{H}_0} \left(\omega - E + \hat{H}_I \right) \right\}^i | \Phi_0 \rangle. \tag{9.13}$$

This is a genuine expansion in powers of $\hat{H}_I$. Two things still spoil it: the undetermined parameter ω, and the appearance of the exact energy E. The two classical schemes are the two natural ways of disposing of them.

9.3 Brillouin-Wigner perturbation theory

The first choice is $\omega = E$, which annihilates the awkward factor $(\omega - E)$ and leaves

$$\Delta E = \langle \Phi_0 | \hat{H}_I + \hat{H}_I \frac{\hat{Q}}{E - \hat{H}_0} \hat{H}_I + \hat{H}_I \frac{\hat{Q}}{E - \hat{H}_0} \hat{H}_I \frac{\hat{Q}}{E - \hat{H}_0} \hat{H}_I + \cdots | \Phi_0 \rangle. \tag{9.14}$$

Every term is explicit and the structure is simple: alternating interactions and resolvents. The price is that E appears in every denominator, so the equation must be solved self-consistently – guess E, evaluate the right-hand side, update, repeat.

There is more to be said. Write $\hat{e} = E - \hat{H}_0$ and recall that $\hat{Q}$ commutes with $\hat{H}_0$ and is idempotent. The geometric series

$$\hat{Q} \frac{1}{\hat{e} - \hat{Q}\hat{H}_I\hat{Q}} = \hat{Q} \left[\frac{1}{\hat{e}} + \frac{1}{\hat{e}} \hat{Q}\hat{H}_I\hat{Q} \frac{1}{\hat{e}} + \frac{1}{\hat{e}} \hat{Q}\hat{H}_I\hat{Q} \frac{1}{\hat{e}} \hat{Q}\hat{H}_I\hat{Q} \frac{1}{\hat{e}} + \cdots \right] \hat{Q} \tag{9.15}$$

resums Eq. (9.14) completely, giving the closed form

$$\boxed{\Delta E = \langle \Phi_0 | \hat{H}_I + \hat{H}_I \hat{Q} \frac{1}{E - \hat{H}_0 - \hat{Q}\hat{H}_I\hat{Q}} \hat{Q}\hat{H}_I | \Phi_0 \rangle.} \tag{9.16}$$

This is not an approximation. Solved self-consistently it returns the exact ground-state energy, and the program `mbpt.py` verifies that it does, for the pairing model, to eight digits at every coupling:

g	BW(2)	BW(3)	BW resummed	FCI
0.25	1.73328927	1.73091635	1.73045985	1.73045985
0.50	1.43866884	1.42314335	1.41677428	1.41677428
1.00	0.79140813	0.70800963	0.63554847	0.63554847
1.50	0.09433209	−0.10095126	−0.34819051	−0.34819051
2.00	−0.63318105	−0.96772735	−1.48965216	−1.48965216

Equation (9.16) is, in other words, one more disguise of the Schrödinger equation. Its truncations are simple to write down and, for a single small system, converge quickly. They also have a fatal defect, which Section 9.8 exhibits.

9.4 Rayleigh-Schrödinger perturbation theory

The second choice is $\omega = W_0$, the *unperturbed* reference energy, which removes the exact energy from the denominators at the cost of leaving ΔE in the numerators,

$$\Delta E = \langle \Phi_0 | \hat{H}_I + \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \left(\hat{H}_I - \Delta E \right) + \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \left(\hat{H}_I - \Delta E \right) \frac{\hat{Q}}{W_0 - \hat{H}_0} \left(\hat{H}_I - \Delta E \right) + \cdots | \Phi_0 \rangle. \quad (9.17)$$

One simplification is immediate. Since ΔE is a number and $\hat{Q}$ annihilates the reference,

$$\hat{Q}\, \Delta E | \Phi_0 \rangle = \Delta E\, \hat{Q} | \Phi_0 \rangle = 0, \quad (9.18)$$

so the rightmost ΔE in each term drops out and

$$\Delta E = \langle \Phi_0 | \hat{H}_I + \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \hat{H}_I + \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \left(\hat{H}_I - \Delta E \right) \frac{\hat{Q}}{W_0 - \hat{H}_0} \hat{H}_I + \cdots | \Phi_0 \rangle. \quad (9.19)$$

Equation (9.19) is still implicit in ΔE, and the resolution is to expand it too. Writing

$$\Delta E = \sum_{i=1}^{\infty} \Delta E^{(i)}, \quad (9.20)$$

and collecting powers of $\hat{H}_I$, we obtain the first three orders:

$$\Delta E^{(1)} = \langle \Phi_0 | \hat{H}_I | \Phi_0 \rangle, \quad (9.21)$$

$$\Delta E^{(2)} = \langle \Phi_0 | \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \hat{H}_I | \Phi_0 \rangle, \quad (9.22)$$

$$\Delta E^{(3)} = \langle \Phi_0 | \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \hat{H}_I | \Phi_0 \rangle - \langle \Phi_0 | \hat{H}_I | \Phi_0 \rangle \langle \Phi_0 | \hat{H}_I \frac{\hat{Q}}{\left(W_0 - \hat{H}_0 \right)^2} \hat{H}_I | \Phi_0 \rangle. \quad (9.23)$$

In a basis of unperturbed determinants, writing V_{mn} for the matrix elements of $\hat{H}_I$ and $D_m = W_0 - W_m$ for the energy denominators, these read

$$\Delta E^{(1)} = V_{00}, \quad (9.24)$$

$$\Delta E^{(2)} = \sum_{m \neq 0} \frac{V_{0m} V_{m0}}{D_m}, \quad (9.25)$$

$$\Delta E^{(3)} = \sum_{m,n \neq 0} \frac{V_{0m} V_{mn} V_{n0}}{D_m D_n} - V_{00} \sum_{m \neq 0} \frac{V_{0m} V_{m0}}{D_m^2}. \quad (9.26)$$

The second term of Eq. (9.26) is the one to pay attention to. It is the price of having removed E from the denominators, and it appears for the first time at third order. It is often called the *renormalisation* term, and it is not small: for the pairing model at $g = 0.5$ the program finds

$$\underbrace{-0.02300347}_{\text{principal}} - \underbrace{(-0.01258681)}_{\text{renormalisation}} = \underbrace{-0.01041667}_{\Delta E^{(3)}}, \quad (9.27)$$

so the correction is larger than the answer. Its function, as Section 9.8 shows, is to cancel exactly those contributions that would otherwise make the energy scale wrongly with the size of the system. This is the algebraic shadow of the linked-diagram theorem.

Generating the series to any order.

Rather than extend Eqs. (9.21)–(9.23) by hand, one uses the standard recursion. With $\hat{R} = \hat{Q}/(W_0 - \hat{H}_0)$,

$$|\Psi^{(n)}\rangle = \hat{R}\left[\hat{H}_I|\Psi^{(n-1)}\rangle - \sum_{k=1}^{n-1}\Delta E^{(k)}|\Psi^{(n-k)}\rangle\right], \qquad \Delta E^{(n+1)} = \langle\Phi_0|\hat{H}_I|\Psi^{(n)}\rangle, \tag{9.28}$$

starting from $|\Psi^{(0)}\rangle = |\Phi_0\rangle$. Unwinding two steps reproduces Eqs. (9.22) and (9.23) exactly, renormalisation term included, and the program checks the agreement numerically for all four models. We use the recursion in Section 9.7 to look at what the series does at high order.

9.5 The wave operator, and the link to configuration interaction

Before turning to numbers it is worth seeing what perturbation theory is doing to the wave function, because it connects directly to Chapter 5.

Recall the correlation operator of Eq. (5.6),

$$|\Psi_0\rangle = \left(1+\hat{C}\right)|\Phi_0\rangle, \qquad \hat{C} = \sum_{ai} C_i^a \hat{a}_a^\dagger \hat{a}_i + \sum_{abij} C_{ij}^{ab} \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i + \cdots \tag{9.29}$$

In a configuration-interaction calculation the coefficients come out of a diagonalisation. Perturbation theory produces them instead as an expansion, and it is instructive to see the two agree at lowest order.

Take a Hartree-Fock basis, so that $\langle\Phi_0|\hat{H}_I|\Phi_i^a\rangle = 0$ by Brillouin's theorem, Eq. (6.31). A two-body interaction can connect the reference only to $2p$-$2h$ states, so Eq. (9.22) collapses to the single well-known expression

$$\Delta E^{(2)} = \frac{1}{4}\sum_{abij}\frac{\langle ij|\hat{v}|ab\rangle\langle ab|\hat{v}|ij\rangle}{\varepsilon_i+\varepsilon_j-\varepsilon_a-\varepsilon_b}. \tag{9.30}$$

Compare this with the exact correlation energy in the same basis, Eq. (5.30), which for a Hartree-Fock reference reduces to

$$\Delta E = \sum_{abij}\langle ij|\hat{v}|ab\rangle C_{ij}^{ab}. \tag{9.31}$$

The two agree provided

$$C_{ij}^{ab} = \frac{1}{4}\frac{\langle ab|\hat{v}|ij\rangle}{\varepsilon_i+\varepsilon_j-\varepsilon_a-\varepsilon_b}, \tag{9.32}$$

which is precisely the first-order perturbative estimate of the doubles amplitudes. Second-order perturbation theory is thus configuration interaction with the doubles amplitudes taken to first order instead of being solved for.

The difference matters. Full configuration interaction determines C_{ij}^{ab} to *all* orders in the interaction, whereas Eq. (9.32) is the leading term of an expansion. Going to higher order in perturbation theory brings in successively more complicated correlations – and, at fourth order and beyond, genuinely new excitation classes. This is the sense in which the two hierarchies are complementary: configuration interaction truncates the *excitation rank* and keeps all orders in the interaction; perturbation theory truncates the *order in the interaction* and keeps all excitation ranks that can contribute. Coupled-cluster theory, which we come to later, is designed to keep the best of both.

Choosing the partition. Nothing in the derivation fixed $\hat{H}_0$; only that it be soluble and that $|\Phi_0\rangle$ be one of its eigenstates. Two choices are standard. Taking $\hat{H}_0$ to be the one-body part of $\hat{H}$ – in a Hartree-Fock basis, the Fock operator – is the Møller-Plesset partition, and it is what we use below. Taking $\hat{H}_0$ to be the entire *diagonal* of $\hat{H}$ in the determinant basis is the Epstein-Nesbet partition; then $\Delta E^{(1)} = 0$ by construction and the denominators contain the diagonal interaction as well. The two sum to the same exact answer and disagree at every finite order. Section 9.12 compares them, and finds that neither is uniformly better – which is why a perturbative result quoted without its partition means nothing.

9.6 The pairing model

We now apply all of this. Take the pairing model of Section 4.2 with four doubly degenerate levels and four particles, $\xi = 1$, and let $\hat{H}_0$ be the one-body part. The reference is the determinant filling the two lowest levels, with $W_0 = 2$, and the first-order energy is

$$\Delta E^{(1)} = \langle \Phi_0 | \hat{H}_I | \Phi_0 \rangle = -g, \tag{9.33}$$

one factor $-g/2$ for each of the two occupied pairs. The energy through first order is therefore $2-g$, which is exactly the Hartree-Fock energy of Section 6.12: for this interaction the mean field does nothing, and first-order perturbation theory reproduces it. Every subsequent order is genuine correlation energy.

g	E_{ref}	MBPT2	MBPT3	FCI	CID	HF + RPA
0.25	1.75000	1.73177083	1.73046875	1.73045985	1.73050700	1.71634735
0.50	1.50000	1.42708333	1.41666667	1.41677428	1.41759645	1.37450235
1.00	1.00000	0.70833333	0.62500000	0.63554847	0.64890655	0.55458933
1.50	0.50000	−0.15625000	−0.43750000	−0.34819051	−0.29113175	−0.40624787
2.00	0.00000	−1.16666667	−1.83333333	−1.48965216	−1.35208670	−1.47622319

Table 9.1 Rayleigh-Schrödinger perturbation theory for the pairing model with four levels and four particles, compared with full configuration interaction, with doubles configuration interaction (Section 5.8) and with the particle-hole random-phase approximation (Section 7.9).

At $g = 0.25$ the series is superb: second order is accurate to 1.3×10^{-3} and third order to 9×10^{-6}. By $g = 1$ third order has crossed the exact answer, and at $g = 2$ it lies 0.34 below it while second order lies 0.32 above. Nothing prevents this. Perturbation theory is not variational, and neither the sign nor the magnitude of the next correction is under any control.

Contrast the behaviour of the two other approximations in the table. Doubles configuration interaction is variational, so it stays above the exact answer at every coupling and degrades smoothly. The random-phase approximation sums a restricted class of contributions to all orders; it is the worst of the three at $g = 0.25$ and the best at $g = 2$.

9.7 How far can the series be trusted?

The recursion (9.28) lets us stop guessing and look. The following table gives the size of the n-th term for the same model:

g	$\lvert E^{(2)}\rvert$	$\lvert E^{(5)}\rvert$	$\lvert E^{(10)}\rvert$	$\lvert E^{(20)}\rvert$	$\lvert E^{(40)}\rvert$	through 20	exact
0.50	7.3×10^{-2}	3.6×10^{-4}	5.7×10^{-8}	2.6×10^{-12}	1.9×10^{-22}	1.4167743	1.4167743
1.00	2.9×10^{-1}	1.2×10^{-2}	5.9×10^{-5}	2.8×10^{-6}	2.1×10^{-10}	0.6355478	0.6355485
1.50	6.6×10^{-1}	8.8×10^{-2}	3.4×10^{-3}	9.2×10^{-3}	2.3×10^{-3}	−0.3532205	−0.3481905
2.00	1.2×10^{0}	3.7×10^{-1}	6.0×10^{-2}	2.9×10^{0}	2.3×10^{2}	−3.7973859	−1.4896522
2.50	1.8×10^{0}	1.1×10^{0}	5.6×10^{-1}	2.5×10^{2}	1.7×10^{6}	−240.36	−2.7371418

Table 9.2 The magnitude of successive Rayleigh-Schrödinger terms for the pairing model. At $g=0.5$ and $g=1$ the series converges to the exact energy; at $g=2$ and beyond the terms grow without limit.

At $g=0.5$ and $g=1$ the terms fall away steadily and the partial sums reproduce the exact energy to the digits shown. At $g=2$ and $g=2.5$ the terms grow by ten and by twelve orders of magnitude between the tenth and the fortieth, and the partial sums are worthless. The radius of convergence lies somewhere between $g=1$ and $g=2$; $g=1.5$ sits close to it, and no finite number of terms settles the question there.

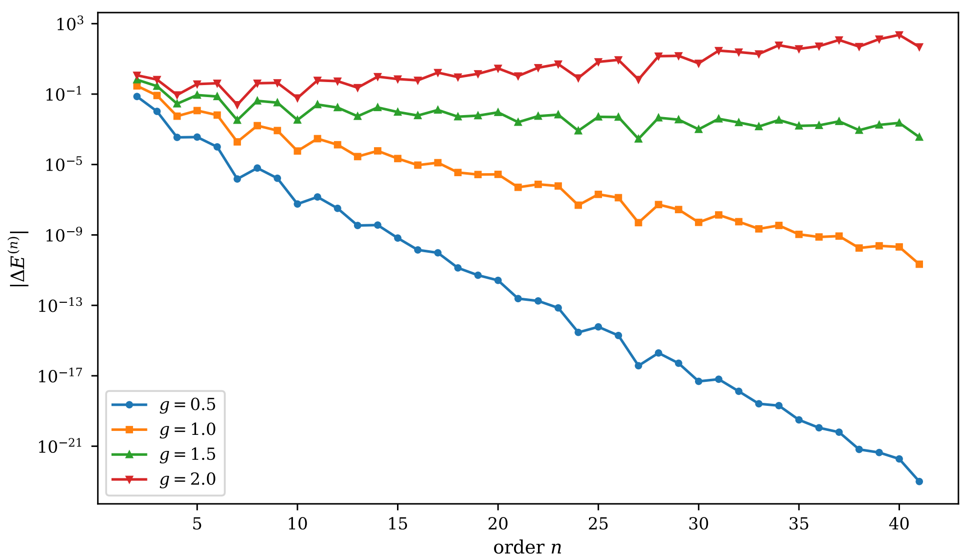

Fig. 9.1 Magnitude of the individual Rayleigh-Schrödinger corrections $|\Delta E^{(n)}|$ for the pairing model with four doubly degenerate levels, four particles and $\xi=1$, plotted logarithmically against the order n for four values of the pairing strength. The series was generated to $n=41$ with the recursion (9.28), and the four curves contain the same information as Table 9.2 at every order rather than at five of them. At $g=0.5$ the terms fall by half a decade per order, at $g=1$ by a fifth of a decade, at $g=1.5$ they are almost flat, and at $g=2$ they grow. Note that no curve is smooth: individual orders dip one or two decades below the local trend and recover.

Figure 9.1 shows the same experiment at every order, and it makes the situation far more definite than the table can. Each of the four curves is, apart from a jitter we return to in a moment, a straight line on a logarithmic scale, which is to say that the terms behave geometrically, $|\Delta E^{(n)}|\sim r^n$. Fitting the slope over the range $20\le n\le 60$ gives $r=0.317$ at $g=0.5$, $r=0.634$ at $g=1$, $r=0.951$ at $g=1.5$ and $r=1.268$ at $g=2$. Those four numbers are $0.634\,g$ to three digits, and that proportionality is the whole story: the ratio of successive terms is linear in the coupling, so the series converges precisely when $0.634\,g<1$, and the radius of convergence of the pairing series is $g_c\simeq1.58$. This is why $g=1.5$ was the awkward case in the table. It lies four per cent below g_c; the terms shrink by a factor 0.951 per order, so that between $n=10$ and $n=41$ they fall by less than one decade, from 3.4×10^{-3} to 3.6×10^{-4}, and forty terms of a convergent series look exactly like forty terms of a divergent one. At $g=2$, by contrast, the growth is unmistakable once one looks far enough: the terms bottom out around $n = 10$ at 6.0×10^{-2} and are back up to 4.8×10^{1} by $n = 41$.

The jitter deserves a comment of its own, because it is the practical obstacle to doing any of this from a handful of terms. The curves are not monotone. At $g = 0.5$ the term at $n = 7$ is 1.5×10^{-6} and the one at $n = 8$ is 6.3×10^{-6}, four times larger; at $g = 1.5$ the terms wander between 2.8×10^{-4} and 2.5×10^{-2} over the last thirty orders without settling. The origin is that the corrections do not simply alternate in sign but arrive in blocks of two or three of the same sign, with a period of roughly five orders, so that a near-cancellation at the boundary between two blocks produces an anomalously small term followed by an entirely ordinary one. A ratio test applied to two consecutive terms is therefore worthless, and even a ratio test applied to ten of them is unreliable unless they are taken well beyond the region where the sign pattern is still settling down. This is the difficulty referred to in Exercise 9.15b), and it is not an artefact of this small model: it is the generic behaviour of a many-body perturbation series, and it is the reason that the convergence of such a series is almost never established in practice. What one does instead is quote the low orders and hope, which brings us to the next point.

This deserves to be taken seriously rather than noted in passing, because of how it looks from the inside. At $g = 1.5$ the first three orders are

$$E_{\text{ref}} = 0.50000, \qquad \text{through second} = -0.15625, \qquad \text{through third} = -0.43750, \tag{9.34}$$

against an exact -0.34819. Those numbers look entirely reasonable. Nothing in them warns that the series behind them may not converge, and in a realistic calculation one rarely has the exact answer to compare against. Truncating a perturbation series at low order is an act of faith, and the faith is sometimes misplaced.

The underlying reason is the same one we have met twice already. The expansion is about the reference determinant, and once the true ground state has stopped resembling it, the expansion has nothing left to say. In Chapter 6 the same coupling made the mean field unstable; in Chapter 5 it made truncated configuration interaction deteriorate. Here it makes the series diverge.

9.8 Size extensivity, and why Brillouin-Wigner is abandoned

Section 5.9 showed that truncated configuration interaction fails a basic test: for two non-interacting subsystems it does not give twice the energy of one. We now apply the same test to the two perturbation schemes, and the outcome is the sharpest distinction between them.

Take M identical, non-interacting copies of a small pairing system – three levels holding one pair, with the interaction acting only within a subsystem. Any acceptable method must give exactly M times the one-subsystem energy. The program finds, for $M = 2$:

Rayleigh-Schrödinger is exact *at every order separately*, to machine precision. This is the linked-diagram theorem in its most concrete form. The unlinked contributions – terms in which the intermediate state contains excitations in both subsystems with no interaction connecting them – would individually scale as M^2, and they are cancelled exactly by the renormalisation term of Eq. (9.26) and by its analogues at higher order. That is what the term is for.

Brillouin-Wigner has no such term. Its denominators contain the exact energy, which is itself extensive, so the denominators of a two-subsystem calculation differ from those of a one-subsystem calculation and the two can never match. The error is 7.8×10^{-3} at $g = 0.5$ and grows to 4.7×10^{-2} at $g = 1$, and it would grow linearly with the number of subsystems. For a

order	2× one copy	two copies	error
$g = 0.5$			
first	-0.500000000	-0.500000000	0
second	-0.093750000	-0.093750000	0
third	-0.007812500	-0.007812500	0
fourth	0.001098633	0.001098633	0
BW(2)	-0.583666572	-0.575845121	7.8×10^{-3}
$g = 1.0$			
first	-1.000000000	-1.000000000	0
second	-0.375000000	-0.375000000	0
third	-0.062500000	-0.062500000	0
fourth	0.017578125	0.017578125	0
BW(2)	-1.296377095	-1.249140538	4.7×10^{-2}

Table 9.3 Size extensivity for two non-interacting copies of a three-level pairing system. Rayleigh-Schrödinger is exact at every order, to machine precision; Brillouin-Wigner truncated at second order is not, and the error grows with the coupling.

calculation on bulk matter this is fatal, and it is the reason Brillouin-Wigner theory, for all its formal elegance and rapid convergence on small systems, is not used in many-body physics.

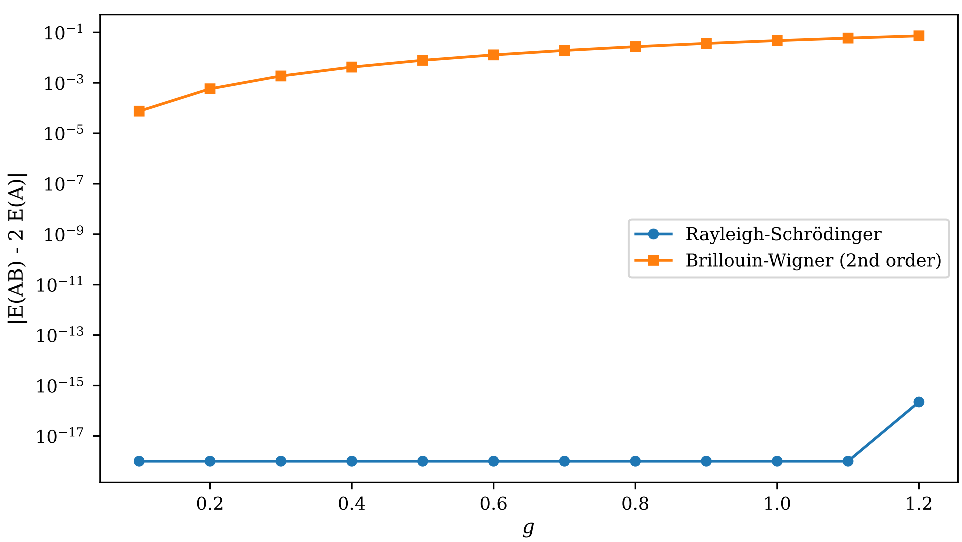

Fig. 9.2 Failure of size extensivity, measured as $|E(AB) - 2E(A)|$ for two non-interacting copies of a three-level pairing system holding one pair each, plotted logarithmically against the pairing strength g. Circles give Rayleigh-Schrödinger theory summed through third order, squares Brillouin-Wigner theory truncated at second order. The Rayleigh-Schrödinger points are exact zeros and lie on the floor of the plot at 10^{-18}, where the figure clamps them; only at $g = 1.2$ does the value become non-zero at all, at 2.2×10^{-16}, which is one unit in the last place of a number of order unity. The Brillouin-Wigner error is a smooth curve rising by three decades over the range, from 7.6×10^{-5} at $g = 0.1$ to 7.3×10^{-2} at $g = 1.2$. The two entries of Table 9.3 are the points at $g = 0.5$ and $g = 1$.

Figure 9.2 repeats the test at every coupling between $g = 0.1$ and $g = 1.2$, and the two curves could hardly be further apart. The Rayleigh-Schrödinger result is not merely small;

it is zero, in the strict sense that the difference between the two-subsystem sum and twice the one-subsystem sum is a floating-point zero at eleven of the twelve couplings and a single rounding unit at the twelfth. There is no trend, no growth with g, and nothing that could be reduced by working harder: the cancellation of the unlinked terms is an algebraic identity, and identities do not degrade. The Brillouin-Wigner curve, sitting thirteen to sixteen decades above it, is what the same test looks like when no such identity is available.

The shape of that curve is worth extracting, because it identifies the missing piece precisely. Over the small-g end of the range the Brillouin-Wigner error is accurately cubic, $|E(AB)-2E(A)| \propto g^3$ with a fitted exponent of 2.98, and cubic is exactly the order at which the renormalisation term of Eq. (9.26) first appears. The reason is visible in the structure of the two schemes. Doubling the system doubles ΔE, and ΔE sits inside every Brillouin-Wigner denominator $E - W_m = \Delta E - (W_m - W_0)$; expanding that denominator in ΔE multiplies the second-order sum, which is $\mathscr{O}(g^2)$, by the first-order energy, which is $\mathscr{O}(g)$. The leading non-extensive error is therefore the product $V_{00}\sum_m V_{0m}V_{m0}/D_m^2$ – the renormalisation term itself, uncancelled. Rayleigh-Schrödinger subtracts that product by construction and Brillouin-Wigner does not, and Figure 9.2 is the difference between the two, measured. By $g = 1.2$ the Brillouin-Wigner error has reached 4.5 per cent of the total energy of the two-subsystem calculation. Read as an energy per subsystem the same failure is a drift: at $g = 0.5$ one copy gives -0.291833 and two copies give -0.287923 each, and adding further copies moves the number again, without limit and in the same direction. A quantity that is supposed to be an intensive property of the bulk instead depends on how much of the bulk one chose to put in the calculation. That is the property the linked-diagram theorem guarantees, and it is the property coupled-cluster theory will inherit through the exponential ansatz of Eq. (5.29).

9.9 The Lipkin model

The Lipkin model of Section 4.1, with $W = 0$, is instructive because almost everything can be done in closed form. The interaction $(V/2)(\hat{J}_+^2 + \hat{J}_-^2)$ changes J_z by exactly two units, so

$$\Delta E^{(1)} = \langle \Phi_0 | \hat{H}_I | \Phi_0 \rangle = 0 \tag{9.35}$$

identically, and the reference energy is the unperturbed $-N\varepsilon/2$. The reference couples to exactly one state, $|J, -J+2\rangle$, at an excitation energy 2ε, with matrix element $\frac{V}{2}\sqrt{2N(N-1)}$, so

$$\Delta E^{(2)} = -\frac{N(N-1)V^2}{4\varepsilon}, \tag{9.36}$$

which the program confirms to ten digits.

The third order vanishes identically, and for a reason worth stating. The principal term of Eq. (9.26) requires a matrix element V_{mn} between two states both reachable from the reference; but every such state has $J_z = -J+2$, and $\hat{H}_I$ changes J_z by two units, so $V_{mn} = 0$. The renormalisation term carries a factor $V_{00} = 0$. Hence $\Delta E^{(3)} = 0$ exactly.

χ	E_{ref}	through $\Delta E^{(2)}$	through $\Delta E^{(3)}$	exact	HF + RPA
0.25	-2.00000	-2.02083333	-2.02083333	-2.02072594	-2.02109449
0.50	-2.00000	-2.08333333	-2.08333333	-2.08166600	-2.08796735
0.75	-2.00000	-2.18750000	-2.18750000	-2.17944947	-2.21691233
0.90	-2.00000	-2.27000000	-2.27000000	-2.25388553	-2.35114625
1.00	-2.00000	-2.33333333	-2.33333333	-2.30940108	-2.58578643

Here perturbation theory and the random-phase approximation may be compared directly, and the comparison is revealing. Both are close to the exact answer at small χ, with perturbation theory slightly the better of the two. As χ grows the two part company. The RPA collective frequency $\omega = \varepsilon\sqrt{1-\chi^2}$ of Section 7.7 goes soft as $\chi \to 1$, and since the correlation energy is $\frac{1}{2}(\sum_\nu \omega_\nu - \mathrm{Tr}A)$ the softening drives it steadily more negative: at $\chi = 1$ RPA gives -2.586 against an exact -2.309, an overshoot of 0.28. Beyond $\chi = 1$ the frequencies turn imaginary and the method stops altogether.

Perturbation theory does much better numerically – -2.333 at $\chi = 1$ – but it does so silently. It produces a perfectly ordinary number at the coupling where the mean field becomes unstable, and would go on producing ordinary numbers beyond it. RPA is less accurate here and more informative: its collapse is a diagnosis. Whether one prefers a method that fails loudly or one that fails quietly depends on whether an exact answer is available for comparison, and in a real calculation it is not.

9.10 The Hubbard model

For the Hubbard model of Section 4.4 we work in the momentum basis, where the plane-wave determinant is the Hartree-Fock solution (Section 5.7) and the first-order energy is the Hartree shift $UN_\uparrow N_\downarrow/L$. Six sites at half filling give $\Delta E^{(1)} = 3U/2$.

U/t	E_{ref}	MBPT2	MBPT3	MBPT4	FCI	CID
1	−6.50000	−6.60069444	−6.60069444	−6.60117251	−6.60115829	−6.59980159
2	−5.00000	−5.40277778	−5.40277778	−5.41042685	−5.40945685	−5.38877703
4	−2.00000	−3.61111111	−3.61111111	−3.73349623	−3.66870618	−3.41192866
8	4.00000	−2.44444444	−2.44444444	−4.40260631	−2.04813089	−0.36779217

Two features stand out. The third-order energy vanishes identically at this filling – the two middle columns agree to every digit, for every U – so we have included fourth order as well. This is special to the half-filled ring with a momentum-independent interaction; the pairing model of Section 9.6 has a perfectly good third-order term, and one should not expect the cancellation to survive a change of filling or of interaction.

The second feature is the familiar one. At $U/t = 1$ second order is already within 5×10^{-4} of the exact energy. At $U/t = 8$ fourth order overshoots it by more than a factor of two. This is the strong-correlation regime of Section 5.7 seen through the perturbation series: the reference determinant has ceased to be a good starting point, and the expansion about it fails.

9.11 The pairing plus particle-hole model

The model of Chapter 7 is the most interesting of the four, because its particle-hole term breaks pairs and therefore couples the reference to $1p$-$1h$ states. The second-order sum splits accordingly, and we may look at the two pieces separately:

The singles contribution is small but non-zero, and it is entirely an artefact of our choice of basis: in a Hartree-Fock basis it would vanish identically by Brillouin's theorem. This is one more reason – beyond the variational one of Chapter 6 – for doing perturbation theory on a Hartree-Fock reference.

For the three weakly coupled rows the series behaves well, third order sitting within a few parts in a thousand of the exact answer. The last row is the interesting one. With $f = 0.5g$ the particle-hole term is strong, and third order overshoots by 0.56. Chapter 7 found RPA

g	f	E_{ref}	$E^{(2)}_{1p1h}$	$E^{(2)}_{2p2h}$	MBPT2	MBPT3	exact
0.50	0.025	1.45000	−0.00072917	−0.08856250	1.36070833	1.34684681	1.34713265
0.70	0.050	1.20000	−0.00291667	−0.18800000	1.00908333	0.96643444	0.96975878
1.00	0.050	0.90000	−0.00291667	−0.35425000	0.54283333	0.43194111	0.45058234
1.00	0.500	0.00000	−0.29166667	−1.30000000	−1.59166667	−2.25388889	−1.69173670

Table 9.4 Perturbation theory for the pairing plus particle-hole model, with the second-order energy split by the excitation rank of the intermediate state.

overshooting the same row in the same direction, −1.73461 against −1.69174, but by only 0.04. Summing a restricted class of contributions to all orders is, here, an order of magnitude safer than summing all of them to third order.

9.12 Does the partition matter?

Nothing forced our choice of $\hat{H}_0$, and it is worth checking how much the answer depends on it. Taking the whole diagonal of $\hat{H}$ instead of its one-body part gives the Epstein-Nesbet partition.

For three of our four models the two partitions give *identical* results, and the reason is worth seeing. The diagonal of $\hat{H}_I$ is the same on every determinant the reference can reach, so the two choices of $\hat{H}_0$ differ by a constant, which cancels in every energy denominator:

model	spread of $\mathrm{diag}(\hat{H}_I)$
pairing	1.000000
Lipkin	0.000000
Hubbard	0.000000
pairing + particle-hole	2.000000

For the pairing model the spread is non-zero over the whole space, but the interaction conserves seniority, so every determinant the reference reaches has the same number of complete pairs and hence the same diagonal. Only the pairing plus particle-hole model, whose particle-hole term breaks pairs and contributes $-f$ per complete pair, has a genuinely varying diagonal within the active space. There the two partitions do disagree:

g	f	one-body		Epstein-Nesbet		exact
		$+\Delta E^{(2)}$	$+\Delta E^{(3)}$	$+\Delta E^{(2)}$	$+\Delta E^{(3)}$	
0.50	0.25	0.60208333	0.51930556	0.62435065	0.51885823	0.55119157
1.00	0.50	−1.59166667	−2.25388889	−1.44791667	−2.25694444	−1.69173670
1.50	0.75	−4.58125000	−6.81625000	−4.17225275	−6.82273352	−4.34336785

Both partitions sum to the same exact answer, but at any finite order they disagree, and neither is uniformly better: at second order Epstein-Nesbet wins the last row and loses the first two, and at third order the two are almost indistinguishable and both are very poor. There is no theorem saying one partition is better than another. The choice is part of the approximation, and a perturbative number quoted without its partition means nothing.

9.13 Everything against everything

The pairing model is the one system every chapter has now treated, so we collect the results.

g	HF	MBPT2	MBPT3	CID	HF + RPA	FCI
0.25	1.75000	1.731771	1.730469	1.730507	1.716347	1.730460
0.50	1.50000	1.427083	1.416667	1.417596	1.374502	1.416774
1.00	1.00000	0.708333	0.625000	0.648907	0.554589	0.635548
1.50	0.50000	−0.156250	−0.437500	−0.291132	−0.406248	−0.348191
2.00	0.00000	−1.166667	−1.833333	−1.352087	−1.476223	−1.489652
errors						
0.25		1.3×10^{-3}	8.9×10^{-6}	4.7×10^{-5}	-1.4×10^{-2}	
0.50		1.0×10^{-2}	-1.1×10^{-4}	8.2×10^{-4}	-4.2×10^{-2}	
1.00		7.3×10^{-2}	-1.1×10^{-2}	1.3×10^{-2}	-8.1×10^{-2}	
1.50		1.9×10^{-1}	-8.9×10^{-2}	5.7×10^{-2}	-5.8×10^{-2}	
2.00		3.2×10^{-1}	-3.4×10^{-1}	1.4×10^{-1}	1.3×10^{-2}	

Table 9.5 The pairing model treated by every method of the book so far. Hartree-Fock from Chapter 6, doubles configuration interaction from Chapter 5, the random-phase approximation from Chapter 7, and full configuration interaction as the exact reference.

The errors are where the interest lies. Hartree-Fock does nothing at all, since the pairing interaction cannot couple the reference to a $1p$-$1h$ state. Beyond that, the ordering changes with the coupling and no method wins everywhere. At $g = 0.25$ third-order perturbation theory is best by three orders of magnitude, and the random-phase approximation is the worst of the four. At $g = 2$ the ordering has reversed completely: RPA is best, and the perturbation series – which by then is diverging – is the worst. Doubles configuration interaction is the only one of the four that is variational, so it alone is guaranteed to stay above the exact answer, and it alone degrades gracefully.

Figure 9.3 plots the whole of Table 9.5 as a function of the coupling, and the right-hand panel is the one to dwell on, because it turns a statement about five numbers into a statement about a continuum. Read from left to right, the four error curves cross one another three times. Below $g \simeq 1.15$ third-order perturbation theory is the most accurate of the four, by a wide margin at the weak-coupling end: at $g = 0.1$ its error is 4.3×10^{-7} against 1.1×10^{-6} for doubles configuration interaction, 8.4×10^{-5} for second order and 2.6×10^{-3} for the random-phase approximation, a spread of four decades between the best and the worst method at a coupling where all four energies agree with the exact one to three decimals. At $g \simeq 1.15$ doubles configuration interaction overtakes third order; at $g \simeq 1.51$ the random-phase approximation overtakes doubles configuration interaction and becomes the best of the four; and at $g \simeq 1.96$ third order finally falls behind second order, so that the extra term one has worked to compute is by then making the answer worse. A method's ranking, in other words, is not a property of the method. It is a property of the method and the coupling together, and the interval over which any one of these curves is the lowest is at most a factor of two wide in g.

Two features of the right-hand panel are worth separating from the general trend, since both are easy to misread. The first is the sharp downward spike in the third-order curve at $g = 0.4$, where the error drops to 1.3×10^{-6}, a decade below its neighbours on either side.

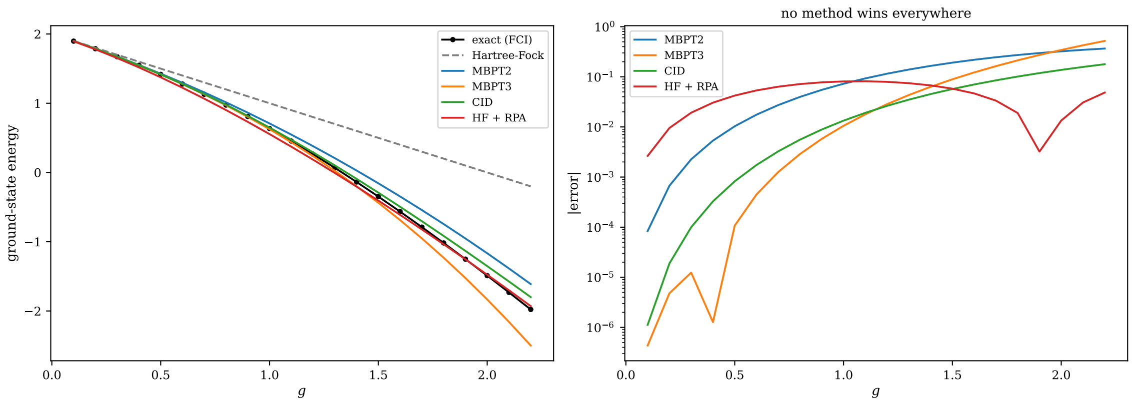

Fig. 9.3 Every method of the book so far applied to the pairing model with four doubly degenerate levels, four particles and $\xi = 1$, over $0.1 \leq g \leq 2.2$. Left: the ground-state energy, with the exact diagonalisation shown as black circles, the Hartree-Fock energy $2 - g$ as a grey dashed line, and second- and third-order Rayleigh-Schrödinger theory, doubles configuration interaction and the random-phase approximation on a Hartree-Fock reference as solid curves. Right: the absolute error of the four correlated methods with respect to the exact result, on a logarithmic scale. The five rows of Table 9.5 are five vertical cuts through this pair of panels; the right-hand panel shows that the ordering of the four methods changes three times as the coupling is increased.

Nothing good is happening there. The third-order energy simply crosses the exact energy at that coupling on its way from above to below, and the spike is the zero of a quantity that is changing sign. The second is the much deeper spike in the random-phase curve at $g = 1.9$, where the error falls to 3.2×10^{-3}, and it has the same origin: the RPA energy crosses the exact one there and continues past it. Both spikes are accidents of a particular model at a particular coupling, and a calculation that happened to be run at $g = 1.9$ would report a spectacular agreement that means nothing whatever. This is the practical argument for scanning a parameter rather than quoting a point, and it is an argument that applies with equal force to the realistic calculations of the later chapters, where the parameter is a model space or a resolution scale rather than a pairing strength.

The left-hand panel makes a quieter point. The grey dashed Hartree-Fock line separates from the exact curve almost immediately and never comes back: at $g = 2.2$ it sits at -0.2 against an exact -1.978, so the correlation energy it is missing, 1.78, is comparable with the total. Yet all four correlated curves lie on top of the exact one for $g \lesssim 0.5$ and are still visually close at $g = 1$. The mean field is thus badly wrong long before any of the correlated methods is, which is the reason a chapter on perturbation theory can be written at all. What ends the story is not the failure of the mean field but the failure of the *expansion about* it: beyond $g \simeq 1.58$, the radius of convergence found in Section 9.7, the perturbation series has nothing left to offer, while the random-phase approximation – which sums a restricted class of contributions to all orders and is therefore not a power series in g at all – is still tracking the exact curve closely at $g = 2.2$, with an error of 4.9×10^{-2} against 5.2×10^{-1} for third order.

None of this is an argument for one method over another. It is an argument for knowing which regime one is in, and for having more than one method available.

9.14 Summary

Perturbation theory begins from the exact identity $\Delta E = \langle \Phi_0 | \hat{H}_I | \Psi_0 \rangle$ and turns it into a series by iterating the resolvent equation (9.11). The free parameter ω chooses the scheme. Setting $\omega = E$ gives Brillouin-Wigner, whose denominators contain the exact energy, which must therefore be found self-consistently, and whose full resummation is exact. Setting $\omega = W_0$ gives Rayleigh-Schrödinger, whose denominators are known in advance but whose higher orders carry renormalisation terms.

Those renormalisation terms are the whole point. They cancel the unlinked contributions exactly, which makes Rayleigh-Schrödinger size extensive at every order – verified here to machine precision – while Brillouin-Wigner is not. That single property decides the matter for many-body physics.

Applied to four models, the series behaves as its construction demands: very accurate when the reference determinant is good, and worthless when it is not. It is not variational, so it may approach the exact answer from either side and cross it; and it need not converge at all, as the pairing model demonstrates beyond $g \approx 1.5$, with no warning visible in the first three orders.

What perturbation theory offers in exchange is a systematic, order-by-order recipe requiring no diagonalisation, applicable where configuration interaction is far out of reach, and – crucially – correct in its scaling with system size. The next step is to keep that property while summing selected contributions to all orders, which is what coupled-cluster theory does.

The program `mbpt.py` performs every calculation of this chapter. The accompanying notebook runs it cell by cell.

9.15 Exercises

Exercise 1: the exact starting point

a) Derive Eq. (9.5), stating where intermediate normalisation and the hermiticity of $\hat{H}$ and $\hat{H}_0$ are used.
b) Show that $\hat{P}$ and $\hat{Q}$ of Eq. (9.6) are idempotent, orthogonal and commute with $\hat{H}_0$, and prove Eq. (9.10).
c) Obtain Eq. (9.11) from Eq. (9.8), and explain why it is exact but useless as it stands.
d) Why does the iteration (9.12) start from $|\Phi_0\rangle$ rather than from any other determinant?

Answer to a).

Projecting $\hat{H}|\Psi_0\rangle = E|\Psi_0\rangle$ onto $\langle\Phi_0|$ gives $\langle\Phi_0|\hat{H}|\Psi_0\rangle = E\langle\Phi_0|\Psi_0\rangle$, and intermediate normalisation sets the overlap to one, so the right-hand side is E. Similarly $\langle\Psi_0|\hat{H}_0|\Phi_0\rangle = W_0$. Since both operators are Hermitian we may read the second as $\langle\Phi_0|\hat{H}_0|\Psi_0\rangle = W_0$ and subtract: $\langle\Phi_0|\hat{H} - \hat{H}_0|\Psi_0\rangle = E - W_0$, which is Eq. (9.5). Without hermiticity the two matrix elements could not be identified and the derivation fails – which is a real restriction for non-Hermitian formulations such as coupled-cluster theory.

Answer to b).

$\hat{P}^2 = |\Phi_0\rangle\langle\Phi_0||\Phi_0\rangle\langle\Phi_0| = \hat{P}$ by normalisation, and $\hat{Q}^2 = \hat{Q}$ by orthonormality of the Φ_m; $\hat{P}\hat{Q} = 0$ since the reference is excluded from the sum. Both are built from eigenstates of $\hat{H}_0$, so $\hat{H}_0\hat{Q} = \sum_m W_m|\Phi_m\rangle\langle\Phi_m| = \hat{Q}\hat{H}_0$. Any function of $\hat{H}_0$ is therefore also diagonal in the same basis, and $\hat{Q}f(\hat{H}_0)\hat{Q} = \hat{Q}^2 f(\hat{H}_0) = \hat{Q}f(\hat{H}_0)$, which is Eq. (9.10).

Answer to c).

Divide Eq. (9.8) by $(\omega - \hat{H}_0)$, multiply from the left by $\hat{Q}$, use Eq. (9.10), and add $|\Phi_0\rangle = \hat{P}|\Psi_0\rangle$ from Eq. (9.7). It is exact because no term has been dropped. It is useless because it is one equation in two unknowns, $|\Psi_0\rangle$ and E, with the unknown state appearing on both sides.

Answer to d).

Any starting vector would generate a formally correct series, but the iteration converges usefully only if the starting vector already has a large overlap with the target. $|\Phi_0\rangle$ is chosen precisely because it is assumed to dominate $|\Psi_0\rangle$ – and when that assumption fails, as at strong coupling in Section 9.7, the series fails with it.

Exercise 2: the two schemes

a) Set $\omega = E$ in Eq. (9.13) and obtain the Brillouin-Wigner series (9.14).
b) Derive the resummation (9.16) using the geometric series (9.15), and explain why it is exact.
c) Set $\omega = W_0$ and obtain Eq. (9.17); then show that Eq. (9.18) reduces it to Eq. (9.19).
d) Expand Eq. (9.19) order by order and derive Eqs. (9.21)–(9.23). At which order does the renormalisation term first appear, and why not earlier?
e) Verify that the recursion (9.28) reproduces $\Delta E^{(2)}$ and $\Delta E^{(3)}$.

Answer to a).

With $\omega = E$ the factor $(\omega - E + \hat{H}_I)$ collapses to $\hat{H}_I$, and the i-th term of Eq. (9.13) becomes i factors of $\hat{H}_I\hat{Q}/(E - \hat{H}_0)$ acting on $\hat{H}_I|\Phi_0\rangle$, which is Eq. (9.14).

Answer to b).

Identify the bracketed sum in Eq. (9.15) as the expansion of $(\hat{e} - \hat{Q}\hat{H}_I\hat{Q})^{-1}$ in powers of $\hat{Q}\hat{H}_I\hat{Q}$, with $\hat{e} = E - \hat{H}_0$. Substituting it into Eq. (9.14) collapses the whole series into Eq. (9.16). It is exact because the geometric series has been resummed, not truncated: no term of Eq. (9.14) has been discarded. Equivalently, Eq. (9.16) is what one gets by solving the Schrödinger equation exactly in the $\hat{Q}$ space and substituting the result back, which is a Feshbach projection and involves no approximation at all.

Answer to c).

With $\omega = W_0$ the factor becomes $(W_0 - E + \hat{H}_I) = (\hat{H}_I - \Delta E)$ since $\Delta E = E - W_0$, giving Eq. (9.17). The rightmost factor always acts directly on $|\Phi_0\rangle$; there ΔE is a number and $\hat{Q}|\Phi_0\rangle = 0$, so that piece vanishes and each term loses its final ΔE, leaving Eq. (9.19).

Answer to d).

Count powers of $\hat{H}_I$, remembering that ΔE is itself of first order and higher. At order one only $\langle \Phi_0|\hat{H}_I|\Phi_0\rangle$ survives. At order two only the second term of Eq. (9.19) contributes, since the ΔE in the third term would make it third order. At order three the third term contributes both with $\hat{H}_I$ in the middle – the principal term – and with $-\Delta E^{(1)}$ in the middle, which gives $-V_{00}\sum_m V_{0m}V_{m0}/D_m^2$. The renormalisation term cannot appear earlier because it needs at least two resolvents, hence at least three interactions.

Answer to e).

With $|\Psi^{(0)}\rangle = |\Phi_0\rangle$ the recursion gives $|\Psi^{(1)}\rangle = \hat{R}\hat{H}_I|\Phi_0\rangle$, whence $\Delta E^{(2)} = \langle \Phi_0|\hat{H}_I\hat{R}\hat{H}_I|\Phi_0\rangle$, which is Eq. (9.22). At the next step

$$|\Psi^{(2)}\rangle = \hat{R}\left[\hat{H}_I\hat{R}\hat{H}_I - \Delta E^{(1)}\hat{R}\hat{H}_I\right]|\Phi_0\rangle, \tag{9.37}$$

and projecting with $\langle \Phi_0|\hat{H}_I$ gives exactly the two terms of Eq. (9.23), the second carrying $\hat{R}^2$ as required. The program checks the agreement numerically for all four models.

Exercise 3: the models to third order

a) For the pairing model show that $\Delta E^{(1)} = -g$ and that the energy through first order is the Hartree-Fock energy of Section 6.12.
b) For the pairing model with four levels and two pairs, evaluate $\Delta E^{(2)}$ by hand and compare with $-\frac{7}{24}g^2$, obtained in Exercise 5.13d).
c) For the Lipkin model derive Eq. (9.36) and prove that $\Delta E^{(3)} = 0$ identically.
d) Explain why the Hubbard model at half filling has $\Delta E^{(3)} = 0$ but $\Delta E^{(4)} \neq 0$, and why this should not be expected in general.
e) For the pairing plus particle-hole model, show that the $1p$-$1h$ part of $\Delta E^{(2)}$ vanishes in a Hartree-Fock basis.

Answer to a).

The pairing interaction $-\frac{g}{2}\sum_{pq}\hat{a}^\dagger_{p+}\hat{a}^\dagger_{p-}\hat{a}_{q-}\hat{a}_{q+}$ has a diagonal contribution only from $p = q$, giving $-\frac{g}{2}$ for each occupied pair. The reference has two, so $\Delta E^{(1)} = -g$ and $W_0 + \Delta E^{(1)} = 2 - g$. Section 6.12 found the Hartree-Fock energy to be exactly the reference energy, because $\langle \Phi_0|\hat{H}|\Phi_i^a\rangle = 0$ by seniority conservation. The two agree, as they must: first-order perturbation theory *is* the mean-field energy when the reference is the Hartree-Fock determinant.

Answer to b).

Only the four seniority-zero determinants reachable by moving one pair contribute. Their unperturbed excitation energies relative to the reference are $2,4,4,6$ and each matrix element is $-g/2$, so

$$\Delta E^{(2)} = \frac{g^2}{4}\left(\frac{1}{-2} + \frac{1}{-4} + \frac{1}{-4} + \frac{1}{-6}\right) = -\frac{7}{24}g^2, \tag{9.38}$$

which is Eq. (5.36). At $g = 0.5$ this gives -0.0729, and the program's numerical value is -0.07291667.

Answer to c).

The reference is $|J,-J\rangle$ with $J = N/2$. The operator $(V/2)\hat{J}_+^2$ takes it to $|J,-J+2\rangle$ with matrix element

$$\frac{V}{2}\sqrt{J(J+1)-(-J)(-J+1)}\,\sqrt{J(J+1)-(-J+1)(-J+2)} = \frac{V}{2}\sqrt{2N(N-1)}, \tag{9.39}$$

and the unperturbed excitation energy is 2ε. Hence

$$\Delta E^{(2)} = -\frac{V^2\,2N(N-1)/4}{2\varepsilon} = -\frac{N(N-1)V^2}{4\varepsilon}. \tag{9.40}$$

For the third order, every intermediate state reachable from the reference has $J_z = -J+2$; but $\hat{H}_I$ changes J_z by exactly two units, so $V_{mn} = 0$ between any two of them and the principal term vanishes. The renormalisation term carries $V_{00} = 0$. Hence $\Delta E^{(3)} = 0$ exactly, and the vanishing is a consequence of the selection rule, not an accident of the parameters.

Answer to d).

The interaction is momentum-independent and couples only opposite spins, and at half filling the ring is particle-hole symmetric. Under these conditions the third-order contributions cancel among themselves, and the program finds $\Delta E^{(3)}$ to be zero to machine precision for every U. The cancellation is a property of this filling and this interaction: change either – or take the pairing model, where the third order is a perfectly ordinary non-zero number – and it disappears. The fourth order is not protected by the same argument and is not zero.

Answer to e).

The $1p$-$1h$ part of $\Delta E^{(2)}$ is $\sum_{ai} |\langle \Phi_0|\hat{H}_I|\Phi_i^a\rangle|^2/D$, and by Brillouin's theorem, Eq. (6.31), every numerator vanishes in the Hartree-Fock basis. In the unperturbed oscillator basis used in Table 9.4 it does not, and the residue is a measure of how far the reference is from the Hartree-Fock solution. This is the practical argument for building perturbation theory on a Hartree-Fock reference: it removes an entire class of contributions at every order.

Exercise 4: size extensivity

a) For two non-interacting subsystems, show that the second-order energy is exactly additive.
b) Show that the principal term of $\Delta E^{(3)}$ alone is *not* additive, and that adding the renormalisation term makes it so.
c) Explain why Brillouin-Wigner theory fails this test at every truncation order.
d) Reproduce Table 9.3 and extend it to three and four subsystems. How does the Brillouin-Wigner error scale?
e) Relate all of this to the size-consistency failure of truncated configuration interaction in Section 5.9.

Answer to a).

With $\hat{H} = \hat{H}_A + \hat{H}_B$ and no coupling, the reference is a product and any singly excited intermediate state lives entirely in A or entirely in B. The sum in Eq. (9.25) therefore splits into two disjoint sums, and the denominators of the A states involve only A excitation energies. Hence $\Delta E^{(2)}(AB) = \Delta E^{(2)}(A) + \Delta E^{(2)}(B)$.

Answer to b).

At third order the intermediate states m and n may lie in different subsystems: m an excitation of A and n an excitation of B. Such a term is *unlinked* – no interaction connects the two – and it contributes V_{00} times a product of A and B second-order-like factors, which scales as the square of the system size. Written out, these are exactly the terms $V_{00}\sum_m V_{0m}V_{m0}/D_m^2$, and the renormalisation term subtracts them identically. What survives is linked and additive. The program verifies additivity to machine precision through fourth order.

Answer to c).

Brillouin-Wigner denominators contain the exact energy $E = W_0 + \Delta E$, and ΔE is extensive. Doubling the system therefore changes every denominator, so the two-subsystem sum cannot reduce to twice the one-subsystem sum, at any truncation order. There is no renormalisation term available to repair this because the scheme has none: that was the whole point of setting $\omega = E$.

Answer to d).

The extension is a one-line change to the demonstration. The Brillouin-Wigner error grows roughly linearly with the number of subsystems for weak coupling, so the *energy per subsystem* drifts – which is the practical statement of the failure. For a calculation on bulk matter, where the number of subsystems is effectively infinite, no useful answer survives.

Answer to e).

They are the same defect wearing two hats. Truncated configuration interaction fails because a product of two double excitations is a quadruple, which the truncation discards; Brillouin-Wigner fails because its denominators do not factorise. In both cases the symptom is an energy that does not scale linearly with the system size, and in both cases the cure is to make the ansatz multiplicative – $e^{\hat{T}}$ instead of $1 + \hat{C}$, Eq. (5.29) – which is coupled-cluster theory. Rayleigh-Schrödinger achieves the same scaling order by order through the cancellation of unlinked terms, which is why it, and not Brillouin-Wigner, is the perturbative starting point for many-body theory.

Exercise 5: numerical explorations

a) Using `mbpt.py`, reproduce Table 9.1 and extend it to sixth order. Does every additional order help?

b) Reproduce Table 9.2 and locate the radius of convergence of the pairing series as accurately as you can. Why is this harder than it looks?
c) Compare the two partitions of Section 9.12 for the pairing plus particle-hole model over a range of f/g. Is there a regime where one is clearly preferable?
d) For the Hubbard model, verify that $\Delta E^{(3)} = 0$ and check whether it survives a change of filling or the addition of a next-nearest-neighbour hopping.
e) Implement a Padé resummation of the pairing series and see whether it extends the useful range beyond the radius of convergence.

Answer.

A few pointers. In a) the answer is no: at $g = 1$ the fourth-order term overshoots and the fifth brings it back, the partial sums oscillating about the exact value – convergence is not monotone and no individual order is guaranteed to improve matters. In b) the difficulty is that near the radius of convergence the terms decrease so slowly that any finite number of them is consistent with either behaviour; the ratio test needs terms of very high order, and those are contaminated by round-off once they have fallen many orders of magnitude below the leading ones. In c) neither partition wins uniformly, but Epstein-Nesbet is the more stable at second order when the diagonal interaction is large, which is the regime it was designed for. In d) the cancellation is fragile: it does not survive a filling that breaks particle-hole symmetry. In e) Padé approximants do extend the useful range, sometimes dramatically, because they can represent the singularity that limits the Taylor series – which is a reminder that a divergent series is not necessarily a useless one.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter09/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/mbptheory.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter09/`.

`mbpt.py` Rayleigh–Schrödinger perturbation theory to any order by recursion, Brillouin–Wigner solved self-consistently, four model systems, size extensivity and convergence. Imports `fci.py`, `hartreefock.py` and `rpa.py` and reuses their model builders. Runs in a few seconds.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 10
Coupled cluster theory

Three defects have followed us through the last five chapters. Truncated configuration interaction, Section 5.9, is variational but not size consistent, so it becomes useless for large systems. Perturbation theory, Chapter 9, is size extensive but not variational, and its series need not converge. The random-phase approximation, Chapter 7, sums one class of contributions to all orders but collapses when the mean field goes unstable.

Coupled-cluster theory repairs the first two at once. It is size extensive by construction, it sums infinite classes of contributions, it is systematically improvable by raising the excitation rank, and its cost grows as a modest power of the basis size rather than exponentially. It is not variational – that is the price – but in practice this matters far less than one might fear. It is, by a wide margin, the most successful many-body method in use: nuclei up to lead have been computed with it, and in quantum chemistry it is the standard against which everything else is measured.

The idea is a single change of ansatz. Where configuration interaction writes $|\Psi\rangle = (1 + \hat{C})|\Phi_0\rangle$, coupled cluster writes $|\Psi\rangle = e^{\hat{T}}|\Phi_0\rangle$. Everything follows from that exponential, and this chapter follows it through: the similarity-transformed Hamiltonian, the reason its expansion terminates exactly, the projected equations, and then full derivations of the doubles and the singles-and-doubles amplitude equations. We work algebraically throughout, as in Chapter 9; the diagrammatic language is a separate subject. The pairing model of Section 4.2 serves as the worked example, and we compare against every method the book has developed.

10.1 The exponential ansatz

We start, as always, from a reference determinant

$$|\Phi_0\rangle = \prod_{i\leq F} \hat{a}_i^\dagger |0\rangle, \tag{10.1}$$

usually but not necessarily the Hartree-Fock state of Chapter 6. The exact state is written

$$\boxed{|\Psi\rangle = e^{\hat{T}}|\Phi_0\rangle,} \tag{10.2}$$

with the *cluster operator*

$$\hat{T} = \hat{T}_1 + \hat{T}_2 + \cdots + \hat{T}_N, \tag{10.3}$$

$$\hat{T}_1 = \sum_{ia} t_i^a \hat{a}_a^\dagger \hat{a}_i, \tag{10.4}$$

$$\hat{T}_2 = \frac{1}{4} \sum_{ijab} t_{ij}^{ab} \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i, \tag{10.5}$$

$$\hat{T}_3 = \frac{1}{36} \sum_{ijkabc} t_{ijk}^{abc} \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_c^\dagger \hat{a}_k \hat{a}_j \hat{a}_i, \tag{10.6}$$

and so on. The factors $1/4$ and $1/36$ compensate the repeated counting of antisymmetric index pairs, $(2!)^2$ and $(3!)^2$. The amplitudes are antisymmetric under interchange of the occupied or of the virtual indices.

Two features of $\hat{T}$ decide everything that follows.

The cluster operators commute.

Every term of $\hat{T}$ consists of particle creation and hole annihilation operators only – $\hat{a}_a^\dagger$ with $a > F$ and $\hat{a}_i$ with $i \leq F$ – and never the reverse. The same argument as in Eq. (6.35) therefore gives

$$\left[\hat{T}_m, \hat{T}_n\right] = 0 \tag{10.7}$$

for all m, n. This is what makes the exponential tractable.

Excitations cannot be undone.

Because $\hat{T}$ only excites, $\hat{T}^n$ produces at least n particle-hole pairs, and acting on a state with N particles it must vanish for large enough n. The exponential is therefore a finite sum in any given matrix element.

Why exponentiate at all?

Expanding Eq. (10.2),

$$e^{\hat{T}} = 1 + \hat{T}_1 + \left(\hat{T}_2 + \tfrac{1}{2}\hat{T}_1^2\right) + \left(\hat{T}_3 + \hat{T}_1\hat{T}_2 + \tfrac{1}{6}\hat{T}_1^3\right) + \cdots, \tag{10.8}$$

we see that the coefficient of a $2p$-$2h$ determinant is $t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a$, and so on. Comparing with the configuration-interaction expansion $|\Psi\rangle = (1 + \hat{C}_1 + \hat{C}_2 + \cdots)|\Phi_0\rangle$ of Eq. (5.5), the two are related by

$$\hat{C}_1 = \hat{T}_1, \qquad \hat{C}_2 = \hat{T}_2 + \tfrac{1}{2}\hat{T}_1^2, \qquad \hat{C}_3 = \hat{T}_3 + \hat{T}_1\hat{T}_2 + \tfrac{1}{6}\hat{T}_1^3, \tag{10.9}$$

and so forth. The exponential does not add anything the linear expansion cannot represent; the two are equivalent if both are complete. What it does is *reorganise*: the high-rank coefficients are built from products of low-rank ones rather than treated as independent. When $\hat{T}$ is truncated, that reorganisation is exactly what survives.

This is the point. Truncating $\hat{C}$ at doubles discards the quadruple excitations entirely, and Section 5.9 showed that this destroys size consistency because a product of two doubles on separate subsystems *is* a quadruple. Truncating $\hat{T}$ at doubles keeps that quadruple, as $\frac{1}{2}\hat{T}_2^2$, with its amplitude fixed by the doubles rather than free. Nothing else needs to be said: the disease of Chapter 5 is cured by construction.

10.2 The similarity-transformed Hamiltonian

Insert Eq. (10.2) into the Schrödinger equation $\hat{H}|\Psi\rangle = E|\Psi\rangle$ and multiply from the left by $e^{-\hat{T}}$:

$$\boxed{\overline{H}|\Phi_0\rangle = E|\Phi_0\rangle, \qquad \overline{H} \equiv e^{-\hat{T}}\hat{H}e^{\hat{T}}.} \tag{10.10}$$

This is worth pausing on. A similarity transformation does not change the spectrum – if $\hat{H}|E\rangle = E|E\rangle$ then $\overline{H}\,e^{-\hat{T}}|E\rangle = E\,e^{-\hat{T}}|E\rangle$ – so $\overline{H}$ has exactly the same eigenvalues as $\hat{H}$. What the transformation does is move the correlation out of the wave function and into the operator. Coupled cluster is the search for a transformation that makes the *uncorrelated* reference an exact eigenstate.

Because $e^{\hat{T}}$ is not unitary – $\hat{T}$ is not anti-Hermitian – $\overline{H}$ is not Hermitian. That is a genuine cost: left and right eigenvectors differ, and expectation values require both. It also brings a large benefit, as we are about to see.

Normal-ordering with respect to the reference, Section 3.5, splits the Hamiltonian as $\hat{H} = E_{\text{ref}} + \hat{H}_N$ with

$$\hat{H}_N = \sum_{pq} f_{pq}\left\{\hat{a}_p^\dagger \hat{a}_q\right\} + \frac{1}{4}\sum_{pqrs}\langle pq\|rs\rangle\left\{\hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r\right\}, \tag{10.11}$$

where f_{pq} is the Fock matrix of Eq. (6.25) and $\langle pq\|rs\rangle$ the antisymmetrised two-body element. Since E_{ref} is a number it commutes with $\hat{T}$, so

$$\overline{H} = E_{\text{ref}} + \overline{H}_N, \qquad \overline{H}_N = e^{-\hat{T}}\hat{H}_N e^{\hat{T}}, \tag{10.12}$$

and Eq. (10.10) becomes $\overline{H}_N|\Phi_0\rangle = \Delta E|\Phi_0\rangle$ with $\Delta E = E - E_{\text{ref}}$ the correlation energy.

10.3 Why the expansion terminates

The Baker-Hausdorff lemma of Exercise 6.15d) gives

$$\overline{H}_N = \hat{H}_N + \left[\hat{H}_N,\hat{T}\right] + \frac{1}{2!}\left[\left[\hat{H}_N,\hat{T}\right],\hat{T}\right] + \frac{1}{3!}\left[\left[\left[\hat{H}_N,\hat{T}\right],\hat{T}\right],\hat{T}\right] + \cdots \tag{10.13}$$

For a general operator this is an infinite series. Here it is not, and the reason is Eq. (10.7).

Because the cluster operators commute with one another, no commutator in Eq. (10.13) can be non-zero by a $\hat{T}$ acting on another $\hat{T}$: every $\hat{T}$ must contract with $\hat{H}_N$ directly. In the language of Section 3.6, each $\hat{T}$ must be *connected* to $\hat{H}_N$. But $\hat{H}_N$ contains at most four operators that can be contracted – two annihilation operators and two creation operators in its two-body part – so at most four cluster operators can attach to it. Hence

$$\boxed{\overline{H}_N = \hat{H}_N + \left[\hat{H}_N,\hat{T}\right] + \frac{1}{2!}\left[\left[\hat{H}_N,\hat{T}\right],\hat{T}\right] + \frac{1}{3!}[\cdots] + \frac{1}{4!}[\cdots],} \tag{10.14}$$

and the series stops *exactly* at the fourfold commutator. Every term is connected, which is why one writes $\overline{H}_N = (\hat{H}_N e^{\hat{T}})_c$ with the subscript denoting connected terms only.

This is the technical heart of the method, and it is worth appreciating what has been bought. The non-Hermiticity of $\overline{H}_N$, which looked like a defect, is precisely what makes the expansion finite: a unitary transformation with anti-Hermitian $\hat{T}$ would contain de-excitation operators, those would not commute among themselves, and the series would never terminate. Unitary coupled-cluster theory, which we meet again in the quantum-computing chapters, pays exactly that price.

10.4 The coupled-cluster equations

Equation (10.10) says $|\Phi_0\rangle$ is an eigenstate of $\overline{H}$. Projecting it onto the reference and onto each excited determinant turns that statement into a computable set of conditions:

$$\Delta E = \langle\Phi_0|\overline{H}_N|\Phi_0\rangle, \tag{10.15}$$
$$0 = \langle\Phi_i^a|\overline{H}_N|\Phi_0\rangle, \tag{10.16}$$
$$0 = \langle\Phi_{ij}^{ab}|\overline{H}_N|\Phi_0\rangle, \tag{10.17}$$

and so on up the excitation ladder. The first line gives the energy once the amplitudes are known; the rest determine them.

Read physically, Eqs. (10.16) and (10.17) say that the similarity-transformed Hamiltonian has *no* matrix elements connecting the reference to $1p$-$1h$ or $2p$-$2h$ states. The reference has been decoupled from the simple excitations. It has not been decoupled from everything: with $\hat{T} = \hat{T}_1 + \hat{T}_2$, the operator $\overline{H}_N$ still connects $|\Phi_0\rangle$ to $3p$-$3h$ and $4p$-$4h$ states, and that residual coupling is the entire error of CCSD.

Why is CCD not exact? A similarity transformation preserves all eigenvalues, so one might expect the transformed problem to be exactly soluble. The resolution is that we are not solving the transformed problem exactly either. With $\hat{T} = \hat{T}_2$ the full $\overline{H}_N$ contains up to six-body terms – $\hat{H}_N$ has four contractible operators, and each $\hat{T}_2$ adds two more – and it is that full operator which would reproduce the exact energy. What we impose is only that the $2p$-$2h$ block of $\overline{H}_N$ vanishes. Coupled cluster at any finite truncation is therefore a similarity transformation *plus* a projection, and the projection is where the approximation lives.

10.5 The correlation energy

Evaluating Eq. (10.15) is the easiest part of the calculation. Only terms of $\overline{H}_N$ that de-excite by exactly the amount $\hat{T}$ excites can contribute, so at most two cluster operators can attach, and

$$\Delta E = \langle\Phi_0|\hat{H}_N\left(1+\hat{T}_1+\hat{T}_2+\tfrac{1}{2}\hat{T}_1^2\right)|\Phi_0\rangle_c. \tag{10.18}$$

The one-body part of $\hat{H}_N$ contracts with $\hat{T}_1$, the two-body part with $\hat{T}_2$ and with $\frac{1}{2}\hat{T}_1^2$, and nothing else survives:

$$\boxed{\Delta E = \sum_{ia} f_{ia}t_i^a + \frac{1}{4}\sum_{ijab}\langle ij\|ab\rangle t_{ij}^{ab} + \frac{1}{2}\sum_{ijab}\langle ij\|ab\rangle t_i^a t_j^b.} \tag{10.19}$$

In a Hartree-Fock basis $f_{ia} = 0$ by Brillouin's theorem, Eq. (6.31), and the first term disappears. The singles then enter the energy only through the last term, quadratically – which is why CCSD energies are often close to CCD energies even when the singles amplitudes are not small.

Notice also that Eq. (10.19) has exactly the form of the configuration-interaction correlation energy, Eq. (9.31), with C_{ij}^{ab} replaced by $t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a$. That combination will appear so often that it is given a name,

$$\tau_{ij}^{ab} = t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a, \tag{10.20}$$

and its half-weighted cousin

$$\tilde{\tau}_{ij}^{ab} = t_{ij}^{ab} + \tfrac{1}{2}\left(t_i^a t_j^b - t_i^b t_j^a\right). \tag{10.21}$$

10.6 Deriving the CCD equations

We derive the doubles equation first, with $\hat{T} = \hat{T}_2$ alone. This is the *coupled-cluster doubles* approximation, and it is exact for any system whose interaction cannot produce a $1p$-$1h$ excitation – the pairing model, nuclear matter on a momentum grid – and the leading approximation in a Hartree-Fock basis for everything else.

The equation to satisfy is $\langle \Phi_{ij}^{ab}|\overline{H}_N|\Phi_0\rangle = 0$ with

$$\overline{H}_N = \left(\hat{H}_N e^{\hat{T}_2}\right)_c = \left(\hat{H}_N\left[1 + \hat{T}_2 + \tfrac{1}{2}\hat{T}_2^2\right]\right)_c. \tag{10.22}$$

The series stops at $\hat{T}_2^2$: a third $\hat{T}_2$ would need two more contractible operators in $\hat{H}_N$ than it has, once the two free particle and two free hole lines of the $2p$-$2h$ projection are accounted for.

We take the three terms in turn. Throughout we use the permutation operators

$$P(ij)X_{ij} = X_{ij} - X_{ji}, \qquad P(ab)X^{ab} = X^{ab} - X^{ba}, \tag{10.23}$$

and their product $P(ij)P(ab)X_{ij}^{ab} = X_{ij}^{ab} - X_{ji}^{ab} - X_{ij}^{ba} + X_{ji}^{ba}$, which enforce the antisymmetry of the amplitudes under interchange of the two occupied or the two virtual labels. Repeated indices are summed.

No cluster operator.

$\langle \Phi_{ij}^{ab}|\hat{H}_N|\Phi_0\rangle$ is the Condon-Slater matrix element between the reference and a double excitation, Section 3.14, which is simply

$$\langle ab\|ij\rangle. \tag{10.24}$$

This is the driving term: without it all amplitudes would vanish.

One cluster operator.

$\langle \Phi_{ij}^{ab}|\left(\hat{H}_N\hat{T}_2\right)_c|\Phi_0\rangle$ requires all four external lines of the projection to be accounted for, with $\hat{T}_2$ supplying two particle and two hole lines and $\hat{H}_N$ connecting to the rest. Enumerating the ways to contract, the one-body part of $\hat{H}_N$ can attach to a particle line or to a hole line,

$$P(ab)\sum_c f_{bc}\, t_{ij}^{ac} - P(ij)\sum_k f_{kj}\, t_{ik}^{ab}, \tag{10.25}$$

and the two-body part in three distinct ways – both particle lines, both hole lines, or one of each:

$$\frac{1}{2}\sum_{cd}\langle ab\|cd\rangle\, t_{ij}^{cd} + \frac{1}{2}\sum_{kl}\langle kl\|ij\rangle\, t_{kl}^{ab} + P(ij)P(ab)\sum_{kc}\langle kb\|cj\rangle\, t_{ik}^{ac}. \tag{10.26}$$

These three are the *particle-particle ladder*, the *hole-hole ladder* and the *ring* term. The factors $\frac{1}{2}$ compensate the double counting of the two equivalent contractions in each ladder; the ring has no such factor because the two lines are inequivalent, and instead carries the full permutation $P(ij)P(ab)$.

Two cluster operators.

$\frac{1}{2}\langle\Phi_{ij}^{ab}|\left(\hat{H}_N\hat{T}_2^2\right)_c|\Phi_0\rangle$ needs both $\hat{T}_2$ connected to $\hat{H}_N$, which forces the two-body part and uses all four of its contractible operators. Only $\langle kl\|cd\rangle$ with two hole and two particle indices can do this. There are four distinct topologies, according to how the four contractions are distributed between the two amplitudes:

$$
\begin{aligned}
&+\frac{1}{4}\sum_{klcd}\langle kl\|cd\rangle\, t_{ij}^{cd}\, t_{kl}^{ab} \\
&+P(ij)P(ab)\,\frac{1}{2}\sum_{klcd}\langle kl\|cd\rangle\, t_{ik}^{ac}\, t_{jl}^{bd} \\
&-P(ij)\,\frac{1}{2}\sum_{klcd}\langle kl\|cd\rangle\, t_{ik}^{cd}\, t_{lj}^{ab} \\
&-P(ab)\,\frac{1}{2}\sum_{klcd}\langle kl\|cd\rangle\, t_{kl}^{ac}\, t_{ij}^{db}.
\end{aligned}
\tag{10.27}
$$

The first is the product of a hole-hole and a particle-particle ladder; the second is a ring on each side; the third and fourth attach both indices of one amplitude to the interaction and leave the other amplitude as a spectator.

The result.

Collecting Eqs. (10.24)–(10.27) and writing the diagonal Fock elements on the left, the CCD amplitude equation is

$$
\begin{aligned}
D_{ij}^{ab}\, t_{ij}^{ab} = {}& \langle ab\|ij\rangle + P(ab)\sum_c f'_{bc}\, t_{ij}^{ac} - P(ij)\sum_k f'_{kj}\, t_{ik}^{ab} \\
&+\frac{1}{2}\sum_{cd}\langle ab\|cd\rangle\, t_{ij}^{cd} + \frac{1}{2}\sum_{kl}\langle kl\|ij\rangle\, t_{kl}^{ab} + P(ij)P(ab)\sum_{kc}\langle kb\|cj\rangle\, t_{ik}^{ac} \\
&+\frac{1}{4}\sum_{klcd}\langle kl\|cd\rangle\, t_{ij}^{cd} t_{kl}^{ab} + \frac{1}{2}P(ij)P(ab)\sum_{klcd}\langle kl\|cd\rangle\, t_{ik}^{ac} t_{jl}^{bd} \\
&-\frac{1}{2}P(ij)\sum_{klcd}\langle kl\|cd\rangle\, t_{ik}^{cd} t_{lj}^{ab} - \frac{1}{2}P(ab)\sum_{klcd}\langle kl\|cd\rangle\, t_{kl}^{ac} t_{ij}^{db},
\end{aligned}
\tag{10.28}
$$

with the energy denominator

$$
D_{ij}^{ab} = f_{ii} + f_{jj} - f_{aa} - f_{bb} = \varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b
\tag{10.29}
$$

and f' the off-diagonal part of the Fock matrix, which vanishes in a canonical Hartree-Fock basis.

Equation (10.28) is a set of coupled quadratic equations, one for each amplitude, and it is solved by iteration: start from $t_{ij}^{ab} = \langle ab\|ij\rangle/D_{ij}^{ab}$, which is the second-order perturbation amplitude of Eq. (9.32), evaluate the right-hand side, divide by D, and repeat.

Intermediates.

Evaluated naively, the quadratic terms cost $\mathscr{O}(n^8)$. Factoring them through intermediates that are built once per iteration reduces this to $\mathscr{O}(n^6)$, which is what makes the method usable. Defining

$$\chi_{kj} = \frac{1}{2}\sum_{lcd}\langle kl\|cd\rangle\, t_{jl}^{cd}, \qquad \chi_{cb} = -\frac{1}{2}\sum_{kld}\langle kl\|cd\rangle\, t_{kl}^{bd},$$

$$\chi_{klij} = \frac{1}{2}\sum_{cd}\langle kl\|cd\rangle\, t_{ij}^{cd}, \qquad \chi_{bkcj} = \frac{1}{2}\sum_{ld}\langle kl\|cd\rangle\, t_{lj}^{db}, \tag{10.30}$$

the last four lines of Eq. (10.28) become

$$-P(ij)\sum_k t_{ik}^{ab}\chi_{kj} + P(ab)\sum_c t_{ij}^{ac}\chi_{cb} + \frac{1}{2}\sum_{kl} t_{kl}^{ab}\chi_{klij} + P(ij)P(ab)\sum_{kc} t_{ik}^{ac}\chi_{bkcj}, \tag{10.31}$$

each a single matrix multiplication. This is how the program implements it, and how every production code does.

10.7 Deriving the CCSD equations

With $\hat{T} = \hat{T}_1 + \hat{T}_2$ the derivation is the same in principle and far longer in practice, because $\hat{T}_1$ can appear up to four times. We give the result in the standard factored form, which is both the clearest statement and the one that is implemented.

The organising trick is to collect the recurring combinations into intermediates. Three carry two indices,

$$F_{ae} = f'_{ae} - \frac{1}{2}\sum_m f_{me} t_m^a + \sum_{mf} t_m^f \langle ma\|fe\rangle - \frac{1}{2}\sum_{mnf} \tilde{\tau}_{mn}^{af}\langle mn\|ef\rangle, \tag{10.32}$$

$$F_{mi} = f'_{mi} + \frac{1}{2}\sum_e t_i^e f_{me} + \sum_{en} t_n^e \langle mn\|ie\rangle + \frac{1}{2}\sum_{nef} \tilde{\tau}_{in}^{ef}\langle mn\|ef\rangle, \tag{10.33}$$

$$F_{me} = f_{me} + \sum_{nf} t_n^f \langle mn\|ef\rangle, \tag{10.34}$$

and three carry four,

$$W_{mnij} = \langle mn\|ij\rangle + P(ij)\sum_e t_j^e \langle mn\|ie\rangle + \frac{1}{4}\sum_{ef} \tau_{ij}^{ef}\langle mn\|ef\rangle, \tag{10.35}$$

$$W_{abef} = \langle ab\|ef\rangle - P(ab)\sum_m t_m^b \langle am\|ef\rangle + \frac{1}{4}\sum_{mn} \tau_{mn}^{ab}\langle mn\|ef\rangle, \tag{10.36}$$

$$\begin{aligned} W_{mbej} = {} & \langle mb\|ej\rangle + \sum_f t_j^f \langle mb\|ef\rangle - \sum_n t_n^b \langle mn\|ej\rangle \\ & - \sum_{nf}\left(\tfrac{1}{2}t_{jn}^{fb} + t_j^f t_n^b\right)\langle mn\|ef\rangle. \end{aligned} \tag{10.37}$$

These are the intermediates of Stanton, Gauss, Watts and Bartlett, and their structure repays a moment's attention: each is a bare integral, plus corrections in which the amplitudes dress it. F_{ae} and F_{mi} are the Fock matrix dressed by correlation; W_{mnij} and W_{abef} are the hole-hole and particle-particle ladders dressed likewise; and W_{mbej} is the dressed ring.

With them, the amplitude equations take a compact form. The singles:

$$\begin{aligned} D_i^a t_i^a = {} & f_{ia} + \sum_e t_i^e F_{ae} - \sum_m t_m^a F_{mi} + \sum_{me} t_{im}^{ae} F_{me} \\ & - \sum_{nf} t_n^f \langle na\|if\rangle - \frac{1}{2}\sum_{mef} t_{im}^{ef}\langle ma\|ef\rangle - \frac{1}{2}\sum_{men} t_{mn}^{ae}\langle nm\|ei\rangle, \end{aligned} \tag{10.38}$$

with $D_i^a = \varepsilon_i - \varepsilon_a$. The doubles:

$$
\begin{aligned}
D_{ij}^{ab} t_{ij}^{ab} = \langle ij \| ab \rangle + P(ab) \sum_e t_{ij}^{ae} \left(F_{be} - \tfrac{1}{2} \sum_m t_m^b F_{me} \right) \\
- P(ij) \sum_m t_{im}^{ab} \left(F_{mj} + \tfrac{1}{2} \sum_e t_j^e F_{me} \right) \\
+ \frac{1}{2} \sum_{mn} \tau_{mn}^{ab} W_{mnij} + \frac{1}{2} \sum_{ef} \tau_{ij}^{ef} W_{abef} \\
+ P(ij)P(ab) \sum_{me} \left(t_{im}^{ae} W_{mbej} - t_i^e t_m^a \langle mb \| ej \rangle \right) \\
+ P(ij) \sum_e t_i^e \langle ab \| ej \rangle - P(ab) \sum_m t_m^a \langle mb \| ij \rangle.
\end{aligned} \tag{10.39}
$$

Reading the equations.

Several structural features are worth noting, and each connects to an earlier chapter.

Setting all amplitudes on the right to zero leaves $D_{ij}^{ab} t_{ij}^{ab} = \langle ij \| ab \rangle$, which is the first-order perturbation amplitude, and $D_i^a t_i^a = f_{ia}$, which vanishes in a Hartree-Fock basis. The first iteration of a coupled-cluster calculation is therefore an MP2 calculation, exactly as Eq. (9.32) said.

Setting $\hat{T}_1 = 0$ makes $\tau = \tilde{\tau} = t_2$, kills the last two lines of Eq. (10.39) and reduces the intermediates to those of Eq. (10.30); Eq. (10.28) is recovered.

The singles equation is not driven in a Hartree-Fock basis: with $f_{ia} = 0$ the leading surviving term is $-\frac{1}{2} \sum_{mef} t_{im}^{ef} \langle ma \| ef \rangle$, which is first order in t_2 and hence second order in the interaction. Singles therefore start one order later than doubles, which is Brillouin's theorem showing up again.

The equations are nonlinear in both t_1 and t_2, and it is the nonlinearity that performs the infinite resummations. The Section 10.8 makes that statement quantitative.

Cost. The most expensive term in Eq. (10.39) is the particle-particle ladder, $\frac{1}{2} \tau_{ij}^{ef} W_{abef}$, which costs $\mathcal{O}(n_o^2 n_v^4)$. With n_o occupied and n_v virtual orbitals the total scales as n^6, against the exponential scaling of full configuration interaction, Section 5.5. This single fact is why coupled cluster reaches systems that configuration interaction cannot approach: for a hundred orbitals, $n^6 = 10^{12}$ operations is a large but finite calculation, while $\binom{100}{50} \approx 10^{29}$ determinants is not. Adding triples, CCSDT, costs n^8; the perturbative CCSD(T) approximation costs n^7 and is the standard compromise.

10.8 Coupled cluster as a resummation of perturbation theory

Chapter 9 built the correlation energy as a series in the interaction and found that it may converge slowly or not at all. Coupled cluster contains the same series, but does not sum it term by term. The relation can be made exact and, for the pairing model, checked numerically to every digit.

The CCD amplitude equation (10.28) is *quadratic* in the amplitudes – nothing of higher degree appears – so it may be written

$$
Dt = C + L[t] + Q[t,t], \tag{10.40}
$$

with C the driving integral, L the linear terms and Q the bilinear ones. Expanding $t = \sum_k t^{(k)}$ with $t^{(k)}$ of order k in the interaction and collecting powers,

$$Dt^{(1)} = C, \qquad Dt^{(k)} = L\left[t^{(k-1)}\right] + \sum_{i+j=k-1} Q\left[t^{(i)}, t^{(j)}\right], \tag{10.41}$$

since L and Q each carry one power of the interaction. The energy contribution of $t^{(k)}$ is of order $k+1$.

This is a perturbation expansion, and it is the Møller-Plesset one: the first amplitude is $\langle ab\|ij\rangle/D_{ij}^{ab}$ and the first energy is MP2. The program checks the identification against the Rayleigh-Schrödinger machinery of Chapter 9, computed independently in the determinant basis:

g	order	from CCD	from MBPT
0.50	2	-0.0623931624	-0.0623931624
	3	-0.0165183724	-0.0165183724
	4	-0.0038210766	-0.0038210766
1.00	2	-0.2190476190	-0.2190476190
	3	-0.1004988662	-0.1004988662
	4	-0.0403096858	-0.0403096858

They agree to every digit through fourth order. That CCD reproduces MP2 and MP3 exactly is general – those orders contain only doubles. That it also gets MP4 right here is special to the pairing model, where the interaction produces neither singles nor triples and the quadruple excitations enter only through the disconnected $\frac{1}{2}\hat{T}_2^2$ that the exponential already supplies. For a general interaction, MP4 contains connected triples that CCD does not have, and the agreement stops at third order.

A warning about partitions.

The numbers above are *not* those of Chapter 9, and the difference is instructive. There we took $\hat{H}_0$ to be the bare one-body operator; here the denominators contain the Fock matrix, whose occupied elements are shifted down by $g/2$. For the pairing model the second-order energy in the two partitions is

$$\Delta E^{(2)}_{\text{one-body}} = -\frac{7}{24}g^2, \qquad \Delta E^{(2)}_{\text{Møller-Plesset}} = -\frac{g^2}{4}\left[\frac{1}{2+g} + \frac{2}{4+g} + \frac{1}{6+g}\right], \tag{10.42}$$

which differ already at order g^3: at $g = 1$ they give -0.2917 and -0.2190. Section 9.12 warned that a perturbative number without its partition is meaningless, and here is the warning in force. Coupled cluster uses the Møller-Plesset partition because the Fock operator is what makes the singles vanish at first order.

Summing the series.

The point of the exponential is that the expansion need not be summed term by term. For the pairing model:

The series does converge here – but slowly, and non-monotonically, changing sign at seventh order. Solving Eq. (10.28) by iteration reaches the same number in twenty-odd steps,

order	$g = 0.5$	$g = 1.0$	$g = 1.5$
2	-6.24×10^{-2}	-2.19×10^{-1}	-4.40×10^{-1}
3	-1.65×10^{-2}	-1.00×10^{-1}	-2.68×10^{-1}
4	-3.82×10^{-3}	-4.03×10^{-2}	-1.43×10^{-1}
5	-6.58×10^{-4}	-1.23×10^{-2}	-5.86×10^{-2}
6	-2.98×10^{-5}	-1.31×10^{-3}	-9.80×10^{-3}
7	$+3.49 \times 10^{-5}$	$+1.62 \times 10^{-3}$	$+1.16 \times 10^{-2}$
sum of 16	-0.0833623	-0.3695742	-0.8857526
CCD	-0.0833623	-0.3695572	-0.8841653

Table 10.1 The order-by-order content of coupled-cluster doubles for the pairing model, and its resummation. The expansion converges back onto the CCD energy; solving the nonlinear equation reaches the same answer directly and without the intermediate oscillation.

with no reference to the order structure at all. That is the practical content of "infinite resummation": not that new physics appears, but that the physics already present in the series is obtained without having to sum it.

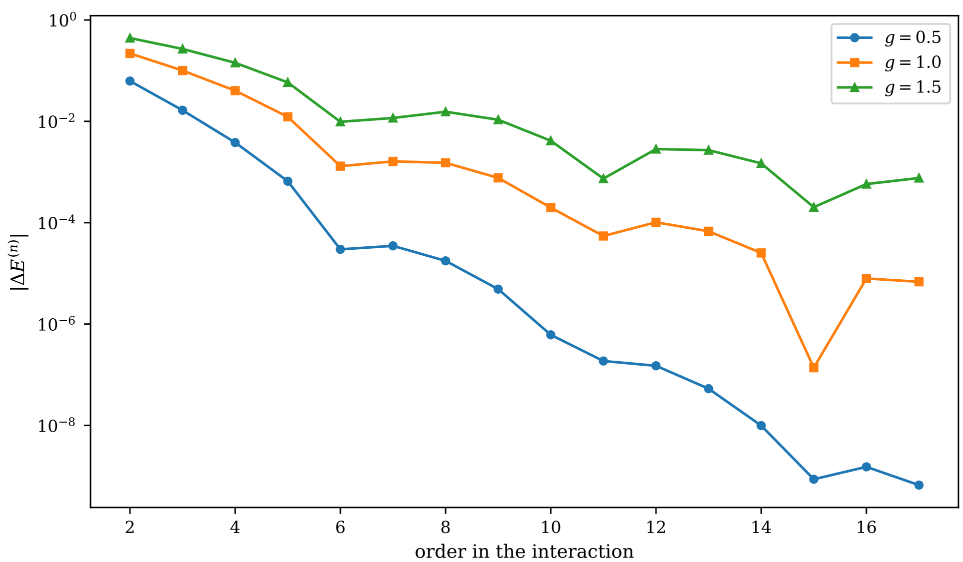

Fig. 10.1 Magnitude of the order-by-order energy increment $|\Delta E^{(n)}|$ generated by the recursion (10.41) from the coupled-cluster doubles equations, plotted logarithmically against the order n in the interaction for three couplings of the four-level pairing model with four particles and $\xi = 1$. Each curve carries the sixteen orders $n = 2, \ldots, 17$, so that its first point is the Møller-Plesset second-order energy and the first three points of the $g = 0.5$ and $g = 1.0$ curves are the entries of Table 10.1. The decay is geometric with a ratio that degrades sharply as the coupling grows, and the downward spikes – at $n = 15$ for $g = 1.0$, at $n = 11$ and $n = 15$ for $g = 1.5$ – are orders at which the increment passes through zero as it changes sign.

Figure 10.1 puts the same statement on a logarithmic scale and over sixteen orders rather than six, and it repays a careful look. All three curves start at much the same place, between 6×10^{-2} and 4×10^{-1}, and what separates them is not where they begin but how fast they fall. At $g = 0.5$ the increments drop by roughly one decade every two orders and have reached 10^{-9} by the seventeenth, which is why the sum of sixteen orders in Table 10.1 reproduces the converged CCD energy -0.0833623 to every digit printed. At $g = 1.0$ the same decay is about a decade every four orders, the seventeenth term is still of order 7×10^{-6}, and the truncated sum consequently misses CCD by 1.7×10^{-5}. At $g = 1.5$ the curve has almost stopped falling: between the sixth and the seventeenth order the increments wander between 10^{-2} and 10^{-4} without ever settling into a clean geometric decay, the last term computed is still of order 7×10^{-4}, and the sixteen-order sum is left 1.6×10^{-3} away from the CCD answer – a discrepancy of exactly the size of the terms one has stopped at, which is the signature of a series being truncated rather than converged. The non-monotonic structure is real and not numerical noise: the spikes are single orders at which $\Delta E^{(n)}$ crosses zero, and the flat stretch in the $g = 1.5$ curve around 10^{-2} marks the region where positive and negative contributions of similar magnitude are cancelling against one another. The lesson is the one Chapter 9 arrived at from the other direction. Summing the series is a bad way to obtain the CCD energy, and it becomes a worse way the more strongly correlated the system is, whereas iterating Eq. (10.28) converges in thirty steps at every coupling shown here and does not care about the order structure at all.

The classes summed are those the truncation can represent. The particle-particle ladder term of Eq. (10.28), iterated, sums the ladder series to all orders; the ring term sums the ring series, which is what the random-phase approximation of Chapter 7 does in its own channel; and the quadratic terms couple them. CCD is neither a ladder theory nor a ring theory but a self-consistent combination of both.

10.9 The pairing model

We now do the calculation. The pairing Hamiltonian of Section 4.2,

$$\hat{H} = \xi \sum_{p\sigma} (p-1)\, \hat{a}^\dagger_{p\sigma} \hat{a}_{p\sigma} - \frac{g}{2} \sum_{pq} \hat{a}^\dagger_{p+} \hat{a}^\dagger_{p-} \hat{a}_{q-} \hat{a}_{q+}, \tag{10.43}$$

has $\langle p{+}\, p{-} \| q{+}\, q{-} \rangle = -g/2$ and nothing else. Two consequences follow immediately, and both simplify the calculation enormously.

First, the interaction moves complete pairs, so it has no matrix element between the reference and a $1p$-$1h$ state. Hence $t_i^a = 0$ identically and CCSD reduces to CCD. The program verifies this: the singles amplitudes come out as exactly zero, and the two energies agree to every digit.

Second, only three of the interaction channels are non-zero – $\langle ab \| cd \rangle$, $\langle ab \| ij \rangle$ and $\langle ij \| kl \rangle$. There is no $\langle ab \| id \rangle$ or $\langle ai \| jb \rangle$, because those would require breaking a pair. The ring term of Eq. (10.28) therefore vanishes, and only the two ladders and the quadratic terms survive.

The Fock matrix is diagonal, with

$$\varepsilon_i = \xi(p_i - 1) - \frac{g}{2}, \qquad \varepsilon_a = \xi(p_a - 1), \tag{10.44}$$

the occupied levels shifted down by $g/2$ and the empty ones untouched – which is the Hartree-Fock solution found in Section 6.12. Taking four levels and four particles, the results are

At weak coupling CCD is very nearly exact – four parts in a million at $g = 0.25$, where MP2 is already wrong in the third decimal. It over-correlates, and the error grows with the coupling, reaching 0.12 at $g = 2$. What is missing is the *connected* quadruple excitations: the exponential supplies the $4p$-$4h$ amplitude as $\frac{1}{2}\hat{T}_2^2$, the disconnected part only, and for four particles in four levels the connected part is not negligible once the interaction is strong. Including $\hat{T}_4$ would make the calculation exact, since the exact state has no components beyond $4p$-$4h$.

Figure 10.2 shows the whole coupling range rather than the five points of the table, and the two panels say different things. In the left-hand panel the Hartree-Fock line is the exact energy only at $g = 0$ and thereafter peels away linearly, since $E_{\text{ref}} = 2 - g$ misses the entire

g	E_{ref}	MP2	CCD	FCI	CCD error	iterations
0.25	1.75000	1.73320261	1.73045621	1.73045985	-3.6×10^{-6}	13
0.50	1.50000	1.43760684	1.41663766	1.41677428	-1.4×10^{-4}	18
1.00	1.00000	0.78095238	0.63044275	0.63554847	-5.1×10^{-3}	26
1.50	0.50000	0.05974026	-0.38416526	-0.34819051	-3.6×10^{-2}	30
2.00	0.00000	-0.70833333	-1.60959440	-1.48965216	-1.2×10^{-1}	31

Table 10.2 Coupled-cluster doubles for the pairing model with four doubly degenerate levels and four particles, $\xi = 1$. The reference energy is the Hartree-Fock energy $2-g$ of Chapter 6, and the exact column is the full configuration interaction of Chapter 5.

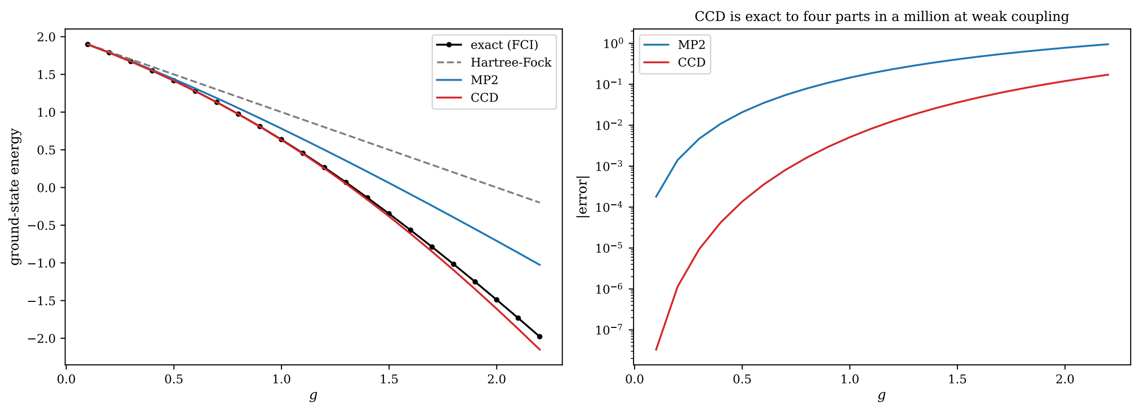

Fig. 10.2 Ground-state energy of the pairing model with four doubly degenerate levels, four particles and $\xi = 1$, as a function of the pairing strength g over twenty-two values between 0.1 and 2.2. Left: the exact diagonalisation of Chapter 5 (black circles), the Hartree-Fock reference energy $2-g$ (grey dashed), second-order Møller-Plesset theory (blue) and coupled-cluster doubles (red); the five couplings of Table 10.2 are points on these curves. Right: the absolute deviation of MP2 and of CCD from the exact energy on a logarithmic ordinate. The CCD curve lies three to four decades below the MP2 curve at weak coupling and the gap closes to less than one decade by $g = 2.2$.

correlation energy; by $g = 2.2$ it is high by almost two units. MP2 follows the exact curve to about $g = 0.6$ and then falls behind, its error growing as the square of the coupling because a second-order theory recovers only the leading term of a series whose higher terms are no longer small. The CCD curve is visually indistinguishable from the exact one over the whole range plotted, which is already the main result: a theory with four amplitudes tracks a full diagonalisation across a factor of twenty in the interaction strength. The right-hand panel is where the numbers live. At $g = 0.1$ the CCD error is 3×10^{-8} against an MP2 error of 2×10^{-4}, a ratio of some five thousand, and at $g = 0.25$ CCD is the four parts in a million quoted above. Both curves rise steeply and then flatten, but they rise at different rates: by $g = 1$ the ratio has fallen to about thirty (5.1×10^{-3} against 1.5×10^{-1}) and by $g = 2.2$ to about six. That the advantage of CCD over MP2 shrinks monotonically with g is the graphical form of Figure 10.1: resumming the series to all orders helps most when the series converges quickly, and when it barely converges the resummation is left carrying an error of its own – here the missing connected quadruples, which grow with the coupling for the reason given above.

10.10 When singles matter

To see CCSD do something CCD cannot, we need an interaction that breaks pairs. The pairing plus particle-hole model of Section 7.1 is exactly that: its particle-hole term

$$\hat{V}_{\text{ph}} = -\frac{f}{2}\sum_{pqr}\left(\hat{a}^\dagger_{p+}\hat{a}^\dagger_{p-}\hat{a}_{q-}\hat{a}_{r+} + \text{h.c.}\right) \tag{10.45}$$

connects the reference to $1p$-$1h$ states and the singles amplitudes are no longer zero. We run the calculation twice, once on the bare oscillator reference and once on the Hartree-Fock reference of Chapter 6, where Brillouin's theorem holds.

g	f	CCD	CCSD	CCSD (HF)	FCI	$\max\lvert t_1\rvert$	$\max\lvert t_1\rvert$ (HF)
0.50	0.025	1.3476038	1.3469075	1.3469088	1.3471327	1.1×10^{-2}	5.4×10^{-4}
0.70	0.050	0.9707634	0.9681476	0.9681644	0.9697588	1.9×10^{-2}	1.5×10^{-3}
1.00	0.050	0.4447429	0.4423659	0.4424119	0.4505823	1.7×10^{-2}	1.8×10^{-3}
0.50	0.250	0.6033126	0.5467210	0.5472745	0.5511916	8.4×10^{-2}	7.3×10^{-3}
1.00	0.500	−1.6764657	−1.7957374	−1.7805416	−1.6917367	1.0×10^{-1}	1.3×10^{-2}

Table 10.3 CCD and CCSD for the pairing plus particle-hole model of Chapter 7, on the bare oscillator reference and on the Hartree-Fock reference.

Three things to read off.

The singles matter. At $g = 0.5$, $f = 0.25$ they move the energy from 0.6033 to 0.5467 against an exact 0.5512, turning a nine per cent error into one of under one per cent.

Brillouin's theorem is visible. The singles amplitudes are an order of magnitude smaller on the Hartree-Fock reference, because there they start at second order in the interaction rather than first, Eq. (10.38). This is the practical reason for working in a Hartree-Fock basis: not that the singles vanish, but that they are small.

Coupled cluster is not invariant under a change of reference. The two CCSD columns are close but not identical. A truncated cluster operator depends on the determinant it is built from, and only the untruncated theory is reference-independent. In a Hartree-Fock basis, where $\hat{T}_1$ has little to do, the dependence is weak – which is the usual situation and the usual justification.

Note also the last row. At $g = 1$, $f = 0.5$ the particle-hole term is strong, CCSD overshoots the exact energy by 0.11, and CCD – which cannot see the singles at all – happens to be closer. This is not a reason to prefer CCD. It is a reminder that an uncontrolled error can have either sign, and that agreement at a single point proves nothing.

Figure 10.3 turns the five rows of Table 10.3 into a continuous scan and makes the mechanism visible. On the right, the largest singles amplitude leaves the origin with a finite slope of about 0.4 – at $f = 0.05$ it is already 2.1×10^{-2} – and this linearity is Eq. (10.38) with its driving term restored: on the bare oscillator reference f_{ia} no longer vanishes, so the singles are first order in the pair-breaking interaction rather than second. The curve then bends over and reaches 0.13 at $f = 0.5$, the flattening being the back-reaction of the doubles on the singles through the intermediates F_{ae} and F_{mi}. On the left the consequence is unambiguous. The CCD and CCSD curves are on top of each other out to $f \approx 0.1$, where $\max\lvert t_1\rvert$ is still below 0.04 and the energy contribution $\frac{1}{2}\langle ij\|ab\rangle t_i^a t_j^b$ of Eq. (10.19) is quadratically small, and they separate steadily thereafter, the CCD curve floating above the exact one and CCSD staying glued to it. At $f = 0.25$ the separation is the 0.6033 against 0.5467 of the table, an error of nine per cent reduced to under one; at $f = 0.5$ CCD is high by 0.14 while CCSD is low by 0.02. The comparison is worth holding beside Figure 10.2, where CCD was essentially exact: nothing about the doubles has changed, and the entire difference is that the Hamiltonian now has a channel the doubles cannot reach. Note also that the red curve does not approach the black one from one side. It runs just *below* it for every $f > 0$ – by 4×10^{-4} at $f = 0.025$ and by 2×10^{-2} at $f = 0.5$ – so CCSD is over-correlating here while CCD, on the same points, is under-correlating. There

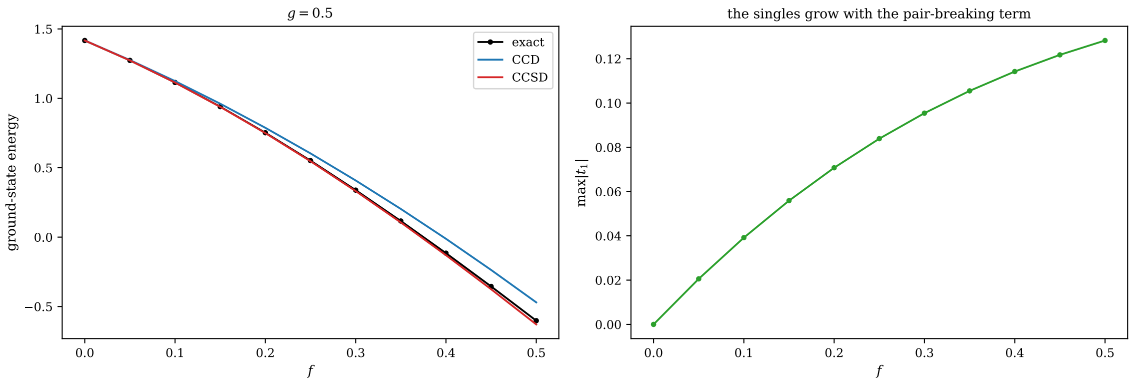

Fig. 10.3 The pairing plus particle-hole model of Section 7.1 at fixed pairing strength $g = 0.5$, on the bare oscillator reference, as the pair-breaking strength f of Eq. (10.45) is raised from 0 to 0.5 in eleven steps. Left: the exact ground-state energy (black circles) against coupled-cluster doubles (blue) and coupled-cluster singles and doubles (red). Right: the largest singles amplitude $\max|t_1|$ of the converged CCSD solution. At $f = 0$ the interaction moves only complete pairs, the singles vanish identically and the two coupled-cluster curves coincide; the gap that opens between them is the work the singles are doing.

is no contradiction: neither energy is an expectation value, and Section 10.14 is where we ask what it would take to put a sign on the error.

10.11 Size extensivity

The property that motivated the whole construction can now be verified. For two non-interacting subsystems A and B, the cluster operator separates, $\hat{T} = \hat{T}_A + \hat{T}_B$, the two pieces commute, and therefore

$$e^{\hat{T}}|\Phi_0\rangle = e^{\hat{T}_A} e^{\hat{T}_B}|\Phi_A\rangle|\Phi_B\rangle = \left(e^{\hat{T}_A}|\Phi_A\rangle\right)\left(e^{\hat{T}_B}|\Phi_B\rangle\right). \tag{10.46}$$

The wave function factorises exactly, and the energy is therefore additive – at any truncation level, since $\hat{T}_A$ and $\hat{T}_B$ are truncated independently. This is the whole argument; it is one line, and it is the reason coupled cluster exists.

The program confirms it for identical copies of a three-level pairing system:

g	$E(1)$	$E(2)-2E(1)$	$E(3)-3E(1)$
0.25	-0.012167069342	0	0
0.50	-0.050059379847	-2.1×10^{-13}	-3.2×10^{-13}
1.00	-0.205303400091	5.6×10^{-17}	-3.0×10^{-13}

Exact to machine precision. Compare Section 5.9, where doubles configuration interaction – the *linear* theory with the same excitation rank – had errors growing from 6×10^{-5} to 1.1×10^{-2} over the same range of couplings. The two methods keep the same excitations; only the ansatz differs; and that difference decides whether the method can be used on a solid.

10.12 Everything against everything

The pairing model has now been treated by every method in this book.

g	HF	MBPT2	MBPT3	CID	HF + RPA	CCD	FCI
0.25	1.75000	1.73177	1.73047	1.73051	1.71635	1.7304562	1.7304598
0.50	1.50000	1.42708	1.41667	1.41760	1.37450	1.4166377	1.4167743
1.00	1.00000	0.70833	0.62500	0.64891	0.55459	0.6304428	0.6355485
1.50	0.50000	−0.15625	−0.43750	−0.29113	−0.40625	−0.3841653	−0.3481905
2.00	0.00000	−1.16667	−1.83333	−1.35209	−1.47622	−1.6095944	−1.4896522
errors							
0.25		1.3×10^{-3}	8.9×10^{-6}	4.7×10^{-5}	-1.4×10^{-2}	-3.6×10^{-6}	
0.50		1.0×10^{-2}	-1.1×10^{-4}	8.2×10^{-4}	-4.2×10^{-2}	-1.4×10^{-4}	
1.00		7.3×10^{-2}	-1.1×10^{-2}	1.3×10^{-2}	-8.1×10^{-2}	-5.1×10^{-3}	
1.50		1.9×10^{-1}	-8.9×10^{-2}	5.7×10^{-2}	-5.8×10^{-2}	-3.6×10^{-2}	
2.00		3.2×10^{-1}	-3.4×10^{-1}	1.4×10^{-1}	1.3×10^{-2}	-1.2×10^{-1}	

Table 10.4 The pairing model treated by every method of this book: Hartree-Fock (Chapter 6), second- and third-order perturbation theory and doubles configuration interaction (Chapters 9 and 5), the random-phase approximation (Chapter 7), coupled-cluster doubles, and full configuration interaction as the exact reference.

At weak and moderate coupling coupled cluster is the best of the five, by two to three orders of magnitude at $g = 0.25$ and by a factor of two over third-order perturbation theory at $g = 1$. Unlike the perturbation series it does not fall apart as the coupling grows: at $g = 2$, where MBPT3 is wrong by 0.34, CCD is wrong by 0.12.

It does not win everywhere. At $g = 2$ the random-phase approximation happens to be closer, which is a reminder that a method summing a different class of contributions can prevail in the regime it was designed for. And CCD is not variational – the errors change sign across the table – so a single number carries no guarantee.

What distinguishes it is that no other entry in the table is simultaneously size extensive, systematically improvable and accurate across the whole range. Configuration interaction is variational but not extensive; perturbation theory is extensive but divergent; the random-phase approximation is neither systematically improvable nor stable. Coupled cluster is extensive by construction, improvable by raising the rank of $\hat{T}$, and polynomial in cost. That combination is why it, and not the others, became the standard.

10.13 Beyond CCSD

The hierarchy continues. Including $\hat{T}_3$ gives CCSDT, with a third amplitude equation

$$0 = \langle \Phi_{ijk}^{abc} | \overline{H}_N | \Phi_0 \rangle, \tag{10.47}$$

whose structure is the same – a driving term, linear terms, and nonlinear couplings – and whose length is formidable. The cost rises to n^8.

Empirically, CCSD recovers around 90% of the correlation energy of a well-behaved system and CCSDT around 99%. Since the triples are expensive and their effect modest, the standard compromise is to compute them perturbatively rather than iteratively. Approximating

$$t_{ijk}^{abc} \approx \frac{N_{ijk}^{abc}(t_1, t_2; \langle pq \| rs \rangle)}{D_{ijk}^{abc}}, \qquad D_{ijk}^{abc} = \varepsilon_i + \varepsilon_j + \varepsilon_k - \varepsilon_a - \varepsilon_b - \varepsilon_c, \tag{10.48}$$

with a numerator built from the converged singles and doubles, gives CCSD(T) at a cost of n^7 and with no iteration. This method is accurate enough, and cheap enough, to have earned the name "the gold standard of quantum chemistry".

Two further directions deserve mention. Because $\overline{H}_N$ is not Hermitian, its left and right eigenvectors differ, and computing properties other than the energy requires both – this is the price of the exponential noted in Section 10.2. And diagonalising $\overline{H}_N$ in the space of excited determinants, rather than merely demanding that its $1p$-$1h$ and $2p$-$2h$ blocks vanish, gives excitation energies: this is equation-of-motion coupled cluster, and it stands to the present chapter as the random-phase approximation of Chapter 7 stands to Hartree-Fock.

10.14 Unitary coupled cluster

Everything so far has rested on the exponential $e^{\hat{T}}$ being *non*-unitary. That was what allowed the Baker-Campbell-Hausdorff series to terminate, Section 10.3, and it is what makes the classical algorithm cheap. It also cost us the variational principle: nothing in Section 10.4 guarantees that the coupled-cluster energy lies above the exact one, and Table 10.2 shows that it does not.

There is an alternative. Take

$$\boxed{|\Psi(\boldsymbol{t})\rangle = e^{\hat{\sigma}}|\Phi_0\rangle, \qquad \hat{\sigma} = \hat{T} - \hat{T}^{\dagger}.} \tag{10.49}$$

Since $\hat{\sigma}^{\dagger} = -\hat{\sigma}$, the operator $e^{\hat{\sigma}}$ is unitary, and therefore

$$\langle\Psi|\Psi\rangle = \langle\Phi_0|e^{-\hat{\sigma}}e^{\hat{\sigma}}|\Phi_0\rangle = 1. \tag{10.50}$$

The energy

$$E(\boldsymbol{t}) = \langle\Psi(\boldsymbol{t})|\hat{H}|\Psi(\boldsymbol{t})\rangle = \langle\Phi_0|e^{-\hat{\sigma}}\hat{H}e^{\hat{\sigma}}|\Phi_0\rangle \tag{10.51}$$

is then an honest expectation value in a normalised state, so the variational principle of Section 2.7 applies in full:

$$E(\boldsymbol{t}) \;\geq\; E_0 \qquad \text{for every choice of amplitudes.} \tag{10.52}$$

This is *unitary coupled cluster*, and the amplitudes are now determined not by projection but by minimisation.

10.14.1 What is gained and what is lost

The gain is Eq. (10.52). The loss is the finite expansion, and it is worth seeing exactly why.

In the standard theory every term of $\hat{T}$ consists of particle creation and hole annihilation operators, so the cluster operators commute, Eq. (10.7), and every $\hat{T}$ in a commutator must attach to $\hat{H}_N$ directly. In the unitary theory $\hat{\sigma}$ contains $\hat{T}^{\dagger}$ as well, which *de*-excites, and

$$\left[\hat{T}, \hat{T}^{\dagger}\right] \neq 0. \tag{10.53}$$

The argument of Section 10.3 therefore collapses: successive commutators can be built from $\hat{\sigma}$ acting on $\hat{\sigma}$, no operator runs out of contractible partners, and

$$e^{-\hat{\sigma}}\hat{H}e^{\hat{\sigma}} = \hat{H} + \left[\hat{H}, \hat{\sigma}\right] + \frac{1}{2!}\left[\left[\hat{H}, \hat{\sigma}\right], \hat{\sigma}\right] + \cdots \tag{10.54}$$

is an infinite series. Truncating it is possible – the commonly used *unitary CCSD with a truncated BCH expansion* keeps terms through the second or third commutator – but it is an uncontrolled approximation, and it destroys the variational property that motivated the whole construction.

That is the whole story of unitary coupled cluster on a classical computer: it is more expensive, it is not systematically truncatable, and its one advantage is a bound. Usually that is a bad bargain, and the standard theory wins.

The calculus changes completely on a quantum computer. There the exponential of an anti-Hermitian operator is not something to be expanded but something to be *applied*: it is precisely a unitary, precisely what quantum gates implement. The terminating BCH expansion was never the point, and the variational bound is exactly what an optimisation algorithm wants. This is why unitary coupled cluster, a curiosity in classical quantum chemistry, is one of the standard ansatz families for the variational quantum eigensolver.

10.14.2 The generators

Write $\hat{\sigma}$ as a sum over individual excitations with one amplitude each,

$$\hat{\sigma}(\boldsymbol{t}) = \sum_k t_k \left(\hat{A}_k - \hat{A}_k^\dagger \right), \tag{10.55}$$

where the $\hat{A}_k$ run over the singles and doubles of Eqs. (10.4) and (10.5),

$$\hat{A}_i^a = \hat{a}_a^\dagger \hat{a}_i, \qquad \hat{A}_{ij}^{ab} = \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i, \tag{10.56}$$

with $i < j$ and $a < b$ so that each independent amplitude appears once. Each $\hat{A}_k - \hat{A}_k^\dagger$ is anti-Hermitian by construction. For n_o occupied and n_v virtual spin-orbitals there are $n_o n_v$ singles and $\binom{n_o}{2}\binom{n_v}{2}$ doubles, so the number of variational parameters grows only polynomially – which is the point.

The pairing model.

For the Hamiltonian of Eq. (10.43) the interaction moves complete pairs, so the only generators that can do anything are the pair excitations

$$\hat{A}_i^a = \hat{a}_{a+}^\dagger \hat{a}_{a-}^\dagger \hat{a}_{i-} \hat{a}_{i+}, \tag{10.57}$$

one for each occupied level i and empty level a. With four levels and two pairs that is four amplitudes, against thirty-six general doubles and fifty-two singles-and-doubles. The program checks that all three give *identical* energies:

generators	amplitudes	E(UCC)
pairs only	4	0.6369869810
doubles	36	0.6369869810
singles and doubles	52	0.6369869810

The extra generators are not wrong; they are simply idle, because the Hamiltonian cannot use them. Recognising such redundancies is important in practice, since every superfluous amplitude is a parameter the optimiser must handle and, on a quantum computer, a block of gates that must be executed.

10.14.3 UCCD and UCCSD for the pairing model

Minimising Eq. (10.51) over the amplitudes gives

g	E_{ref}	CCD	UCCD	FCI	CCD error	UCCD error
0.25	1.75000	1.73045621	1.73046012	1.73045985	-3.6×10^{-6}	$+2.8\times10^{-7}$
0.50	1.50000	1.41663766	1.41679562	1.41677428	-1.4×10^{-4}	$+2.1\times10^{-5}$
1.00	1.00000	0.63044275	0.63698698	0.63554847	-5.1×10^{-3}	$+1.4\times10^{-3}$
1.50	0.50000	−0.38416526	−0.33675149	−0.34819051	-3.6×10^{-2}	$+1.1\times10^{-2}$
2.00	0.00000	−1.60959440	−1.45347633	−1.48965216	-1.2×10^{-1}	$+3.6\times10^{-2}$

Table 10.5 Standard and unitary coupled cluster with doubles, for the pairing model with four levels and four particles. Every unitary error is positive, as Eq. (10.52) requires; every standard error is negative.

Every UCCD error is positive and every CCD error is negative, exactly as the variational property demands. For this model the unitary theory is also more accurate at every coupling, by a factor between three and thirteen. One should not generalise too freely from a four-particle model – the variational bound guarantees a sign, not a magnitude – but the pattern is a real one, and it becomes more pronounced as the coupling grows and the standard theory begins to over-correlate.

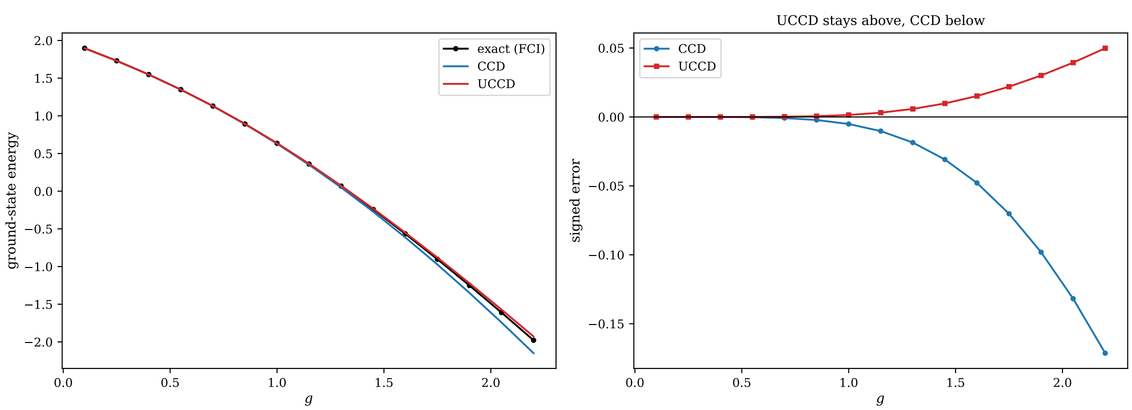

Fig. 10.4 Standard and unitary coupled-cluster doubles for the pairing model with four levels and four particles, over fifteen couplings between $g=0.1$ and $g=2.2$. Left: the exact energy (black circles), CCD obtained by solving the projected equations (10.28) (blue) and UCCD obtained by minimising the expectation value (10.51) over the four pair amplitudes of Eq. (10.57) (red). Right: the same two results plotted as *signed* errors against the exact energy, with the zero line drawn in. The unitary curve never crosses that line and the standard curve never touches it from above.

The right-hand panel of Figure 10.4 is the picture the variational principle predicts, and it is worth dwelling on because Eq. (10.52) is a statement one can check by eye. The red curve stays on or above the zero line at every one of the fifteen couplings, rising smoothly to $+0.048$ at $g=2.2$; the blue curve stays on or below it, falling to -0.168 at the same point. Neither ever strays onto the wrong side, and there is no reason it should: the unitary energy is an honest expectation value in a normalised state, and no expectation value can lie beneath the ground-state eigenvalue, while the projected energy of Eq. (10.15) is not an expectation value of anything and carries no such protection. The two errors are also of visibly different size. Out to $g \approx 0.7$ both are indistinguishable from zero on this linear scale – the model is nearly uncorrelated there and every method works – and beyond $g \approx 1$ they separate, the CCD error growing roughly three and a half times faster than the UCCD one all the way to the right-hand edge, in agreement with the factor between three and thirteen in Table 10.5. Two remarks are in order. First, the sign is guaranteed but the magnitude is not, and a system in which the unitary theory was the less accurate of the two would violate nothing; here the two happen to run together because the same missing connected quadruples are responsible for both errors, and the variational constraint merely stops UCCD from running past the exact answer. Second, the left-hand panel is a reminder of how little of this is visible in the energies themselves – all three curves lie within the thickness of the line over most of the range – which is why the signed error, and not the energy, is the quantity to plot when a bound is what one wants to demonstrate.

For the pairing plus particle-hole model of Section 7.1, where the singles are active, the same comparison at the singles-and-doubles level gives

g	f	UCCD	UCCSD	CCSD	FCI	UCCSD error	CCSD error
0.50	0.025	1.3478676	1.3471722	1.3469075	1.3471327	$+4.0 \times 10^{-5}$	-2.3×10^{-4}
1.00	0.050	0.4553916	0.4530194	0.4423659	0.4505823	$+2.4 \times 10^{-3}$	-8.2×10^{-3}
0.50	0.250	0.6093216	0.5526429	0.5467210	0.5511916	$+1.5 \times 10^{-3}$	-4.5×10^{-3}
1.00	0.500	-1.5302687	-1.6608653	-1.7957374	-1.6917367	$+3.1 \times 10^{-2}$	-1.0×10^{-1}

Table 10.6 Unitary and standard coupled cluster with singles and doubles for the pairing plus particle-hole model of Chapter 7.

The singles are worth having: at $g = 0.5$, $f = 0.25$ they take the energy from 0.6093 to 0.5526 against an exact 0.5512. And in the last row, where the particle-hole term is strong and CCSD overshoots by 0.10, UCCSD is wrong by 0.031 – three times better, and on the side the variational principle promises.

10.15 Trotterisation

A quantum computer cannot apply $e^{\hat{\sigma}}$ in one step. What it can do is apply the exponential of each generator separately, since each $e^{t_k(\hat{A}_k - \hat{A}_k^\dagger)}$ maps onto a short, known gate sequence after the Jordan-Wigner transformation of Section 3.16: the generator becomes a sum of Pauli strings by Eq. (3.148), and the exponential of each string is a standard rotation circuit. The two are not the same, because the generators do not commute, and the difference is exactly the Trotter splitting of Section 6.9:

$$\boxed{e^{\hat{\sigma}(\boldsymbol{t})} \approx \left[\prod_k e^{t_k(\hat{A}_k - \hat{A}_k^\dagger)/n}\right]^n.} \tag{10.58}$$

This is Eq. (6.52) with the cluster generators in place of the pieces of a Hamiltonian, and it inherits the same first-order error $\mathscr{O}(1/n)$ in the operator, governed by the commutators $[\hat{A}_k - \hat{A}_k^\dagger, \hat{A}_l - \hat{A}_l^\dagger]$.

The obvious question is how large n must be. The answer is surprising, and it is the most useful thing in this section.

10.15.1 Re-optimisation absorbs the splitting error

Optimising the amplitudes separately for each Trotter number gives

n	E(UCCD)	error against FCI
exact exponential	0.6369869810	1.44×10^{-3}
1	0.6369869810	1.44×10^{-3}
2	0.6369869810	1.44×10^{-3}
4	0.6369869810	1.44×10^{-3}
8	0.6369869810	1.44×10^{-3}

Every Trotter number gives the same energy to ten digits – including $n=1$, the crudest possible splitting. This is not because the splitting error is small. It is because the Trotterised ansatz is a *different parametrisation of a similar family of states*, and a variational search simply finds whichever amplitudes are best for the parametrisation it is given. The optimiser absorbs the splitting error into the amplitudes.

To see the error one must hold the amplitudes fixed. Taking those optimised for the exact exponential and evaluating the Trotterised state at the same amplitudes:

n	$\|\Psi_n-\Psi\|$	ratio	energy error	ratio
1	3.56×10^{-3}	—	6.78×10^{-5}	—
2	1.14×10^{-3}	3.13	6.38×10^{-6}	10.63
4	4.56×10^{-4}	2.50	9.37×10^{-7}	6.81
8	2.11×10^{-4}	2.16	1.93×10^{-7}	4.85
16	1.03×10^{-4}	2.04	4.57×10^{-8}	4.23
32	5.14×10^{-5}	2.01	1.13×10^{-8}	4.06

Table 10.7 The Trotter error of the UCCD ansatz for the pairing model at $g=1$, with the amplitudes frozen at their exact-exponential values. The state converges as $1/n$ and the energy as $1/n^2$.

Two things are visible. The state converges as $1/n$, which is the first-order Trotter behaviour of Section 6.9. The energy converges as $1/n^2$ – one power faster.

Figure 10.5 extends the table to $n=64$ and makes both slopes unmistakable. Read on a log-log plot a power law is a straight line, and the state error is straight over the whole range with the slope of the dashed $1/n$ guide: the last three points, 1.03×10^{-4}, 5.14×10^{-5} and 2.57×10^{-5}, halve exactly as n doubles. The energy error is also straight, but steeper by a factor of two in slope, and its last three points, 4.57×10^{-8}, 1.13×10^{-8} and 2.8×10^{-9}, fall by a factor of four per doubling. Both curves in fact sit somewhat below their guides at large n – the state by a factor of two at $n=64$, the energy by a factor of six – because the guides are anchored at $n=1$, where the splitting is crude and the leading behaviour has not yet set in; the useful information is the slope and not the offset. The vertical separation of the two curves is the practically important quantity. At $n=1$ the state is already wrong by 3.6×10^{-3} while the energy is wrong by only 6.8×10^{-5}, and by $n=64$ that gap of a factor of

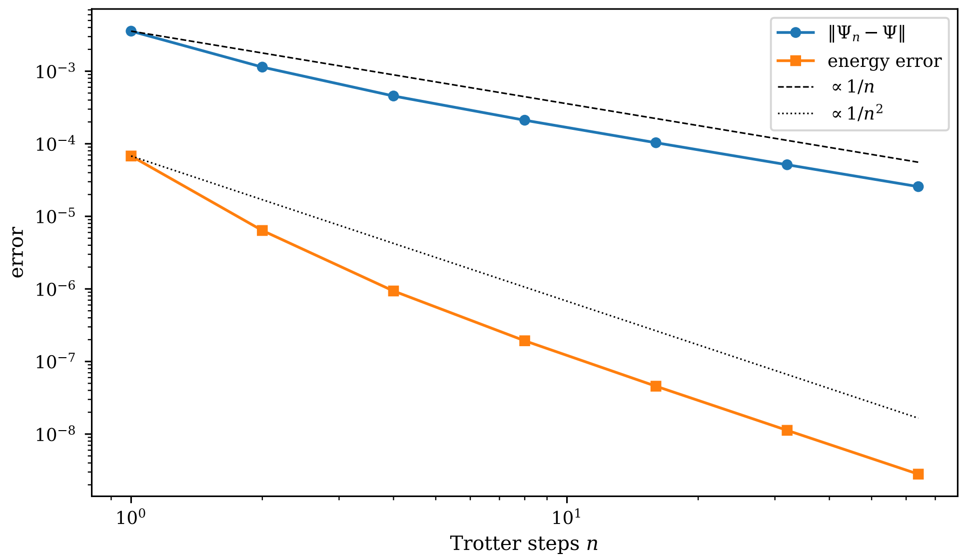

Fig. 10.5 Convergence of the Trotterised UCCD ansatz (10.58) for the pairing model at $g = 1$, with the four pair amplitudes held fixed at the values obtained by optimising the exact exponential. Plotted logarithmically against the number of Trotter steps n are the norm $\|\Psi_n - \Psi\|$ of the difference between the split and unsplit states, after fixing the overall phase, and the corresponding energy error $E_n - E$. The two straight guides are $1/n$ and $1/n^2$, each anchored at the $n = 1$ value of the quantity it accompanies. The tabulated points of Table 10.7 are the first six of the seven shown here.

fifty has widened to four orders of magnitude. This is Eq. (10.59) seen directly: the first-order response of a stationary energy to a perturbation of its state vanishes, so an error of size ε in the wave function costs only ε^2 in the energy. For a quantum computation the consequence is a pleasant one. If the observable of interest is the energy – and in a variational quantum eigensolver it is – then one buys two decades of accuracy for every decade of circuit depth, and even the single-step circuit of $n = 1$, which is all that near-term hardware is likely to afford, is already accurate to 7×10^{-5} before the optimiser has been allowed to absorb any of the remaining error.

The reason is worth spelling out, because it is a general and useful fact. Write the Trotterised state as $|\Psi_n\rangle = |\Psi\rangle + |\delta\rangle$ with $\|\delta\| = \mathscr{O}(1/n)$. Then

$$E_n - E = 2\,\mathrm{Re}\langle\Psi|\hat{H} - E|\delta\rangle + \langle\delta|\hat{H} - E|\delta\rangle + \cdots \tag{10.59}$$

The amplitudes were chosen to make E stationary, so the first-order term is suppressed and only the quadratic one survives. The program confirms this directly: the linear term is smaller than the quadratic one by three orders of magnitude already at $n = 1$.

Both observations point the same way. A variational algorithm is far more forgiving of Trotter error than a simulation algorithm, because it can re-optimise around it, and because the energy responds only quadratically to whatever error remains. This is one of the practical arguments for the variational quantum eigensolver over straightforward time evolution on near-term hardware, where circuit depth is the binding constraint and $n = 1$ is often all one can afford.

Towards the variational quantum eigensolver. What we have just described *is* a VQE, run on a classical simulator. The ingredients are a parametrised trial state – here $e^{\hat{\sigma}(\boldsymbol{t})}|\Phi_0\rangle$, Trotterised into a sequence of gates – an energy evaluated on it, and a classical optimiser adjusting $\boldsymbol{t}$. On real hardware the second step is replaced by repeated measurement of the qubit register in bases chosen so that the measured averages reconstruct $\langle\Psi|\hat{H}|\Psi\rangle$, which is where the Pauli matrices of Section 1.7, the gate model of Section 2.15 and the Jordan-Wigner mapping of Section 3.16 enter. The Hamiltonian is written as the weighted sum of Pauli strings (3.138), each string is measured separately, and the results are added with their weights: the number of strings is the number of measurements. For the pairing model used throughout this chapter those counts are in Table 4.7, and the compact encoding (4.44) makes the $L = 2$ case a five-term measurement on two qubits. Nothing else changes: the ansatz, the variational principle and the optimisation loop are the ones in this section. Chapter 10 therefore leads directly into the quantum-computing chapters, and the code written here is reused there.

10.16 Summary

Coupled-cluster theory replaces the linear ansatz $(1+\hat{C})|\Phi_0\rangle$ by the exponential $e^{\hat{T}}|\Phi_0\rangle$. The two are equivalent when complete, and they differ decisively when truncated: the exponential builds high-rank excitations as products of low-rank ones, so truncating $\hat{T}$ at doubles retains the disconnected quadruples that truncating $\hat{C}$ discards. That single reorganisation makes the method size extensive at every truncation level, which configuration interaction is not.

Requiring $e^{-\hat{T}}\hat{H}e^{\hat{T}}$ to have the reference as an eigenstate turns the Schrödinger equation into the projected equations (10.15)–(10.17). Because the cluster operators commute among themselves, every $\hat{T}$ in the Baker-Campbell-Hausdorff series must connect to $\hat{H}_N$, which has only four contractible operators; the expansion therefore terminates exactly at the fourfold commutator. The non-Hermiticity that this exploits is the same non-Hermiticity that complicates properties – one cannot have the one without the other.

The resulting equations, (10.28) for doubles and (10.38)–(10.39) for singles and doubles, are nonlinear and are solved by iteration. Their first iteration is MP2; their order-by-order expansion is Møller-Plesset perturbation theory, verified here through fourth order; and solving them to convergence sums that expansion within the retained excitation classes. For the pairing model CCD is accurate to four parts in a million at weak coupling and still the best method available at strong coupling, while remaining exactly size extensive.

The unitary variant makes the opposite trade. Taking $\hat{\sigma} = \hat{T} - \hat{T}^\dagger$ gives a unitary exponential, hence a normalised state and a variational energy, at the cost of a Baker-Campbell-Hausdorff series that no longer terminates. For the pairing model UCCD is above the exact energy at every coupling where CCD is below it, and here also closer. Trotterising the exponential, as any quantum circuit must, introduces an error that falls as $1/n$ in the state but only $1/n^2$ in the energy – and which a variational optimiser absorbs almost entirely by re-optimising the amplitudes. That is the structure of the variational quantum eigensolver, and it is where Chapter 10 hands over to the quantum-computing chapters.

The program `coupledcluster.py` performs every calculation of this chapter, including the unitary ones, and the class `UnitaryCC` is the one reused later.

10.17 Exercises

Exercise 1: the exponential ansatz

a) Prove Eq. (10.7), that the cluster operators commute, and identify exactly where the particle-hole structure is used.

b) Derive the relations (10.9) between the configuration-interaction and cluster amplitudes, and extend them to $\hat{C}_4$.
c) Show that for two non-interacting subsystems the exponential ansatz factorises, Eq. (10.46), and that the linear ansatz truncated at doubles does not.
d) Why does the exponential series terminate when acting on an N-particle determinant?

Answer to a).

Take two terms of $\hat{T}$, each a string of particle creation and hole annihilation operators. Since particle and hole indices are disjoint, every annihilation operator in one string anticommutes with every creation operator in the other, with no delta function generated. Moving one string past the other therefore produces a sign $(-1)^{n_1 n_2}$ from the interchange, and the same sign appears in the reverse ordering, so the two products are equal. This is Eq. (6.35) generalised, and it is exactly the disjointness of the index sets that makes it work.

Answer to b).

Expand $e^{\hat{T}}$ and collect terms by excitation rank. Rank one gives $\hat{C}_1 = \hat{T}_1$; rank two, $\hat{C}_2 = \hat{T}_2 + \frac{1}{2}\hat{T}_1^2$; rank three, $\hat{C}_3 = \hat{T}_3 + \hat{T}_1\hat{T}_2 + \frac{1}{6}\hat{T}_1^3$; and rank four,

$$\hat{C}_4 = \hat{T}_4 + \hat{T}_1\hat{T}_3 + \frac{1}{2}\hat{T}_2^2 + \frac{1}{2}\hat{T}_1^2\hat{T}_2 + \frac{1}{24}\hat{T}_1^4. \tag{10.60}$$

The term $\frac{1}{2}\hat{T}_2^2$ is the one that matters: it is present in CCD, where $\hat{T}_3 = \hat{T}_4 = \hat{T}_1 = 0$, and absent from CID.

Answer to c).

With no interaction between the subsystems, no amplitude can carry indices from both, so $\hat{T} = \hat{T}_A + \hat{T}_B$ with $[\hat{T}_A, \hat{T}_B] = 0$. Hence $e^{\hat{T}} = e^{\hat{T}_A} e^{\hat{T}_B}$ and, since the operators act on disjoint orbitals, the state factorises. The energy of a product state under $\hat{H}_A + \hat{H}_B$ is the sum of the two energies. For the linear ansatz, $(1 + \hat{C}_2^A + \hat{C}_2^B)$ is *not* the product $(1 + \hat{C}_2^A)(1 + \hat{C}_2^B)$: the cross term $\hat{C}_2^A\hat{C}_2^B$ is a quadruple excitation, which CID has thrown away. This is Section 5.9 in one line.

Answer to d).

Each application of $\hat{T}$ annihilates at least one hole, and there are only N holes. Hence $\hat{T}^n|\Phi_0\rangle = 0$ for $n > N$, and more sharply for a truncated $\hat{T}$: with $\hat{T} = \hat{T}_2$ the series stops at $\hat{T}_2^{N/2}$.

Exercise 2: the similarity transformation

a) Show that $\overline{H} = e^{-\hat{T}}\hat{H}e^{\hat{T}}$ has the same eigenvalues as $\hat{H}$ for any $\hat{T}$.
b) What condition on $\hat{T}$ would make $\overline{H}$ Hermitian? Why does coupled cluster not impose it?
c) Prove that the Baker-Campbell-Hausdorff series terminates at the fourfold commutator, and say precisely which property of $\hat{H}_N$ is used.
d) If a similarity transformation preserves all eigenvalues, why is the CCD energy not exact?

Answer to a).

If $\hat{H}|E\rangle = E|E\rangle$ then

$$\overline{H}\left(e^{-\hat{T}}|E\rangle\right) = e^{-\hat{T}}\hat{H}e^{\hat{T}}e^{-\hat{T}}|E\rangle = e^{-\hat{T}}\hat{H}|E\rangle = E\,e^{-\hat{T}}|E\rangle, \tag{10.61}$$

so $e^{-\hat{T}}|E\rangle$ is an eigenstate of $\overline{H}$ with the same eigenvalue. The map $|E\rangle \mapsto e^{-\hat{T}}|E\rangle$ is invertible, so the spectra coincide. Note that eigen*values* are preserved but eigen*vectors* are not, and neither are norms: $\overline{H}$ is not Hermitian and its left and right eigenvectors differ, satisfying $\langle L_\alpha|R_\beta\rangle = \delta_{\alpha\beta}$.

Answer to b).

$\overline{H}$ is Hermitian if $e^{\hat{T}}$ is unitary, that is if $\hat{T}^\dagger = -\hat{T}$. Coupled cluster does not impose it because an anti-Hermitian $\hat{T}$ contains de-excitation operators $\hat{a}_i^\dagger\hat{a}_a$ as well as excitations, those do not commute with the excitations, and the Baker-Campbell-Hausdorff series then never terminates. One would gain Hermiticity and lose the finite expansion. Unitary coupled cluster makes precisely that trade, and is used where the exponential can be realised directly rather than expanded – on a quantum computer, for instance.

Answer to c).

By Eq. (10.7) no commutator can be made non-zero by one $\hat{T}$ acting on another, so in $[[\cdots[\hat{H}_N,\hat{T}],\ldots],\hat{T}]$ every $\hat{T}$ must be contracted with $\hat{H}_N$ itself. Each such contraction consumes one of $\hat{H}_N$'s operators that can be contracted to the right – and its two-body part has exactly four, $\hat{a}_p^\dagger\hat{a}_q^\dagger$ and $\hat{a}_s\hat{a}_r$ in normal-ordered form. A fifth $\hat{T}$ has nothing left to attach to, so the term vanishes. The property used is that $\hat{H}_N$ is *at most two-body*; a three-body force would extend the series to the sixfold commutator.

Answer to d).

Because we do not solve the transformed problem exactly either. With $\hat{T} = \hat{T}_2$ the full $\overline{H}_N$ contains up to six-body terms, and it is that operator whose spectrum equals the exact one. What CCD imposes is only that the $2p$-$2h$ block of $\overline{H}_N$ vanish – the reference is decoupled from doubles but not from triples and quadruples. Coupled cluster at finite truncation is a similarity transformation *plus* a projection, and the error lives in the projection.

Exercise 3: deriving the CCD equations

a) Show that the expansion (10.22) stops at $\hat{T}_2^2$ when projected onto a $2p$-$2h$ state.
b) Derive the linear terms (10.25) and (10.26) and explain the factors $\frac{1}{2}$ and the permutation operators.
c) Derive the four quadratic terms (10.27).
d) Verify that setting all amplitudes on the right-hand side to zero gives the MP2 amplitude.
e) Show that the intermediates (10.30) reduce the cost of the quadratic terms from $\mathcal{O}(n^8)$ to $\mathcal{O}(n^6)$.

Answer to a).

A $2p$-$2h$ projection supplies two hole and two particle external lines. Each $\hat{T}_2$ must connect to $\hat{H}_N$, consuming two of its contractible operators, and $\hat{H}_N$ has four. Two cluster operators therefore exhaust it, and a third has nothing to attach to.

Answer to b).

Use the generalised Wick theorem of Section 3.12 on $\langle \Phi_{ij}^{ab}|\hat{H}_N\hat{T}_2|\Phi_0\rangle$. The one-body part can contract with either a particle or a hole line of the amplitude, giving the two terms of Eq. (10.25); the permutation operators appear because the external labels a,b and i,j must be antisymmetrised. For the two-body part, contracting both of its creation operators with the amplitude's particle lines gives the particle-particle ladder, both annihilation operators with the hole lines gives the hole-hole ladder, and one of each gives the ring. The factor $\frac{1}{2}$ in each ladder compensates the two equivalent ways of assigning the summed pair of indices; the ring has inequivalent lines and instead carries the full $P(ij)P(ab)$.

Answer to c).

Now both $\hat{T}_2$ connect to $\hat{H}_N$, which forces the two-body part and uses all four of its operators, so only $\langle kl\|cd\rangle$ contributes. The four topologies are fixed by how the four contractions are shared: two with each amplitude in the "ladder times ladder" pattern (first term), one particle and one hole with each (second term, the ring-ring), or both particle and both hole indices with one amplitude leaving the other a spectator (third and fourth). The combinatorial factors follow by counting equivalent assignments as in b).

Answer to d).

Setting $t=0$ on the right leaves $D_{ij}^{ab}t_{ij}^{ab} = \langle ab\|ij\rangle$, so

$$t_{ij}^{ab} = \frac{\langle ab\|ij\rangle}{\varepsilon_i+\varepsilon_j-\varepsilon_a-\varepsilon_b}, \tag{10.62}$$

which is Eq. (9.32). Substituting into Eq. (10.19) gives the MP2 energy. The program checks this against the closed form $-\frac{g^2}{4}[1/(2+g)+2/(4+g)+1/(6+g)]$ for the pairing model.

Answer to e).

Written out, a term such as $\sum_{klcd}\langle kl\|cd\rangle t_{ik}^{ac}t_{jl}^{bd}$ has four free external indices and four summed ones, hence $\mathscr{O}(n^8)$. Building $\chi_{bkcj} = \frac{1}{2}\sum_{ld}\langle kl\|cd\rangle t_{lj}^{db}$ first costs $\mathscr{O}(n^6)$ – four free, two summed – and contracting it with the remaining amplitude costs $\mathscr{O}(n^6)$ again. The saving is the standard one of never forming an object with more indices than necessary, and it is what makes the n^6 scaling of CCSD achievable.

Exercise 4: CCSD and its perturbative content

a) Derive the energy expression (10.19) and explain why no term with three or more cluster operators appears.
b) Show that Eqs. (10.38) and (10.39) reduce to the CCD equation when $\hat{T}_1 = 0$.
c) Show that in a Hartree-Fock basis the singles amplitudes are of second order in the interaction while the doubles are of first order.
d) Establish the order-by-order recursion (10.41) and show that the first two energies are MP2 and MP3.
e) Why does CCD reproduce MP4 for the pairing model but not in general?

Answer to a).

$\langle \Phi_0 | \overline{H}_N | \Phi_0 \rangle$ requires that the excitation produced by the cluster operators be exactly undone by $\hat{H}_N$, which can de-excite by at most two ranks. Hence at most $\hat{T}_2$ or $\frac{1}{2}\hat{T}_1^2$ can appear, plus $\hat{T}_1$ against the one-body part. Three cluster operators would produce at least a $3p$-$3h$ excitation, which $\hat{H}_N$ cannot annihilate.

Answer to b).

With $\hat{T}_1 = 0$ we have $\tau = \tilde{\tau} = t_2$; the last two lines of Eq. (10.39) vanish, as does the second half of the ring term; and the intermediates collapse to $F_{ae} \to f'_{ae} - \frac{1}{2} t_2 \cdot v$, $W_{mnij} \to \langle mn \| ij \rangle + \frac{1}{4} t_2 \cdot v$ and so on, which are exactly the combinations of Eq. (10.30). What remains is Eq. (10.28).

Answer to c).

In a Hartree-Fock basis $f_{ia} = 0$, so the driving term of Eq. (10.38) vanishes. The leading surviving term is $-\frac{1}{2} \sum_{mef} t_{im}^{ef} \langle ma \| ef \rangle$, first order in t_2; since t_2 is itself first order in the interaction, t_1 is second order. The doubles, by contrast, are driven directly by $\langle ij \| ab \rangle$ and are first order. The program shows this concretely: the singles amplitudes of the pairing plus particle-hole model drop by an order of magnitude when the reference is rotated to the Hartree-Fock one.

Answer to d).

The CCD equation is quadratic, so $Dt = C + L[t] + Q[t,t]$ exactly. Substituting $t = \sum_k t^{(k)}$ and noting that C, L and Q each carry one power of the interaction, the terms of order k are $L[t^{(k-1)}]$ and $Q[t^{(i)}, t^{(j)}]$ with $i + j = k - 1$, which is Eq. (10.41). The first amplitude is the MP2 one by d) of the previous exercise, and its energy is MP2. The second satisfies $Dt^{(2)} = L[t^{(1)}]$, whose energy contribution is the sum of the particle-particle ladder, hole-hole ladder and ring contractions of two MP2 amplitudes – which is precisely the third-order energy.

Answer to e).

At fourth order the exact energy receives contributions from singles, doubles, triples and quadruples. CCD has the doubles, and it has the quadruples in their disconnected form $\frac{1}{2}\hat{T}_2^2$, which is the whole of the fourth-order quadruple contribution. It lacks the singles and the connected triples. For the pairing model those are absent anyway – the interaction cannot produce a $1p$-$1h$ or a $3p$-$3h$ excitation – so CCD is complete at fourth order and the program finds exact agreement. For a general interaction the connected triples are present and CCD misses them, so agreement stops at third order. This is exactly why CCSD(T) exists.

Exercise 6: unitary coupled cluster

a) Show that $e^{\hat{\sigma}}$ with $\hat{\sigma} = \hat{T} - \hat{T}^\dagger$ is unitary, and hence that the energy (10.51) is variational.
b) Show that $[\hat{T}, \hat{T}^\dagger] \neq 0$ for a single doubles amplitude, and explain why this destroys the argument of Section 10.3.
c) For the pairing model, show that only the pair generators of Eq. (10.57) can contribute, and count the amplitudes for L levels and $N/2$ pairs.
d) Derive Eq. (10.59) and explain why the Trotter error in the energy falls one power of n faster than the error in the state.
e) Why does re-optimising the amplitudes at each Trotter number give the same energy? Is this a general phenomenon or an accident of this model?

Answer to a).

$\hat{\sigma}^\dagger = (\hat{T} - \hat{T}^\dagger)^\dagger = \hat{T}^\dagger - \hat{T} = -\hat{\sigma}$, so $\hat{\sigma}$ is anti-Hermitian and $(e^{\hat{\sigma}})^\dagger e^{\hat{\sigma}} = e^{-\hat{\sigma}} e^{\hat{\sigma}} = 1$. The state therefore stays normalised for any amplitudes, $E(\boldsymbol{t}) = \langle \Psi | \hat{H} | \Psi \rangle$ is a Rayleigh quotient with unit denominator, and Section 2.7 gives $E(\boldsymbol{t}) \geq E_0$. Note where the standard theory fails this test: $e^{\hat{T}}$ is not unitary, $|\Psi\rangle$ is not normalised, and the coupled-cluster energy is obtained by projection rather than as an expectation value.

Answer to b).

Take $\hat{A} = \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i$. Then $\hat{A}^\dagger \hat{A}$ acting on the reference gives back the reference (both hole states are occupied and both particle states empty), while $\hat{A}\hat{A}^\dagger$ annihilates it. Hence $[\hat{A}, \hat{A}^\dagger]|\Phi_0\rangle = -|\Phi_0\rangle \neq 0$. The argument of Section 10.3 required that every cluster operator in a nested commutator attach to $\hat{H}_N$, which followed from the cluster operators commuting among themselves. With $\hat{T}^\dagger$ present they do not, commutators of $\hat{\sigma}$ with itself are non-zero, and nothing limits the length of the series.

Answer to c).

The pairing interaction moves complete pairs, so it has no matrix element between the reference and any state with a broken pair. A generator that breaks a pair therefore produces a component of the trial state on which $\hat{H}$ has vanishing matrix elements with everything else that is populated, and the optimiser sets its amplitude to zero. Only $\hat{A}_i^a = \hat{a}_{a+}^\dagger \hat{a}_{a-}^\dagger \hat{a}_{i-} \hat{a}_{i+}$ survives, one for each occupied level i and empty level a, giving $(N/2)(L - N/2)$ amplitudes – four for the case $L = 4$, $N = 4$. The program confirms that the general doubles and the singles-and-doubles ansatz give exactly the same energy.

Answer to d).

Write $|\Psi_n\rangle = |\Psi\rangle + |\delta\rangle$ and expand the normalised expectation value. To second order in δ,

$$E_n - E = 2\,\mathrm{Re}\langle\Psi|\hat{H} - E|\delta\rangle + \langle\delta|\hat{H} - E|\delta\rangle + \mathscr{O}(\delta^3). \tag{10.63}$$

The amplitudes were obtained by minimising E, so the state is at a stationary point and the first-order term is suppressed. With $\|\delta\| = \mathscr{O}(1/n)$ from the first-order Trotter formula, the surviving quadratic term is $\mathscr{O}(1/n^2)$. Table 10.7 shows both scalings, and the program checks that the linear term is three orders of magnitude below the quadratic one at $n = 1$.

Answer to e).

It is general, though not exact. Trotterising changes the ansatz, not the variational principle: for each n one is minimising the same energy over a slightly different family of states, and both families contain states very close to the true minimum of the untrotterised one. The optimiser therefore compensates by shifting the amplitudes, and the residual difference is second order in the shift. The effect is not exact – with a coarser optimiser or a harder problem one would see small differences – but it is why practical VQE implementations use $n = 1$ almost universally. What it does *not* mean is that the Trotter error is unimportant in general: for time evolution, where the amplitudes are fixed by the physics and cannot be re-optimised, the full $\mathscr{O}(1/n)$ error is felt, as Section 6.9 showed.

Exercise 5: numerical explorations

a) Using `coupledcluster.py`, reproduce Table 10.2 and extend it to six and eight levels. Does the CCD error grow or shrink with the size of the space?
b) Verify the size-extensivity result for four and five subsystems, and compare with doubles configuration interaction over the same range.
c) Add $\hat{T}_4$ to the pairing calculation, or equivalently solve the seniority-zero problem exactly, and confirm that the remaining CCD error is entirely due to connected quadruples.
d) Study the convergence of the CCD iteration as a function of g. At what coupling does it stop converging, and how does that compare with the radius of convergence of the perturbation series in Section 9.7?
e) Implement the CCSD(T) correction of Eq. (10.48) for the pairing plus particle-hole model and see how much of the remaining error it removes.
f) Using the class `UnitaryCC`, reproduce Tables 10.5 and 10.7, and check that the UCCD energy stays above the exact one for every coupling you can reach.
g) Count the gates a Trotterised UCCSD circuit would need for L levels, assuming each generator costs $\mathscr{O}(1)$ two-qubit gates after a Jordan-Wigner mapping. How does the count grow, and how does it compare with the n^6 of classical CCSD?

Answer.

Some pointers. In a) the relative error shrinks as the space grows, because the neglected connected quadruples become a smaller fraction of a larger correlation energy – the opposite of the trend for truncated configuration interaction, whose size-consistency error grows. In b) the coupled-cluster error stays at machine precision for any number of subsystems, while the CID error grows roughly linearly. In c) the pairing model with four levels has no $6p$-$6h$ states, so CCD plus connected quadruples is exact, and the difference from CCD is precisely the number in the error column of Table 10.2. In d) the CCD iteration continues to converge well past the point where the perturbation series diverges, which is the practical meaning of resummation; it eventually fails too, and the failure is associated with the same physics that made the mean field unstable in Chapter 6. In e) the perturbative triples remove a large part of the residual error at weak coupling and much less of it at strong coupling, where the expansion underlying them is itself unreliable.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter10/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/coupledcluster.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter10/`.

`coupledcluster.py` General spin-orbital CCSD with intermediates, CCD, a validating full CI in the same basis, the order-by-order expansion, and `UnitaryCC` (UCCD and UCCSD, exact and Trotterised). Runs in a few seconds.

`ucc.py` An earlier standalone unitary coupled cluster implementation, kept for reference. Runs in a few seconds.

`CCD_PairingModel.py` The original CCD solver for the pairing model. Runs in a few seconds.

`NeutronMatterCCD_Ladders.py` Coupled cluster doubles with ladder diagrams for infinite neutron matter. Runs in minutes.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 11
Applications and more realistic Hamiltonians

Every system in this book so far has been a model. The pairing model has one interaction and one strength; the Lipkin model has two; the Hubbard model has an on-site U and nothing else. That was the point: with a single parameter one can turn correlation on and off at will and watch each approximation break where it should. But a model interaction also hides the two things that make real many-body calculations hard. The matrix elements were given to us in closed form and there were only a handful of them, and the single-particle space was finite by construction.

This chapter removes both crutches. The system is a *quantum dot*: N electrons confined to two dimensions by a parabolic trap and interacting through the bare Coulomb repulsion. Nothing is schematic. The interaction is the real one, every matrix element has to be computed, the single-particle basis is infinite and has to be truncated, and the truncation error has to be controlled alongside the many-body error. Quantum dots are also real: they are fabricated in semiconductor heterostructures, their level structure is measured, and they are among the candidate architectures for quantum computing.

The plan is to take the same hierarchy through this system that Chapters 6, 9 and 10 took through the pairing model – Hartree-Fock, second-order perturbation theory, coupled cluster with singles and doubles, and the unitary variant – and to measure all of them against exact diagonalisation wherever that is affordable. The programs are the same too: `quantumdot.py` builds the one- and two-body matrices, and everything after that is imported unchanged from `coupledcluster.py`. That reuse is deliberate. A method that works only on the model it was developed on is of no use.

The Hamiltonian of this chapter returns in the stochastic part of the book, where variational and diffusion Monte Carlo are applied to the same dots and compared with the results obtained here.

11.1 The system

We consider N electrons moving in two dimensions in a harmonic trap. In natural units, $\hbar = m_e = e = 1$, the Hamiltonian splits in the usual way into a one-body and a two-body part,

$$\boxed{\hat{H} = \hat{H}_0 + \hat{V}, \qquad \hat{H}_0 = \sum_{i=1}^{N} \left(-\tfrac{1}{2}\nabla_i^2 + \tfrac{1}{2}\omega^2 r_i^2\right), \qquad \hat{V} = \sum_{i<j}^{N} \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}} \tag{11.1}$$

Lengths are measured in the oscillator length $a_0^* = \sqrt{\hbar/(m_e\omega)}$. The single parameter of the problem is the trap frequency ω, and it does more work than it looks. The kinetic and confining energies scale as ω while the Coulomb energy scales as $\sqrt{\omega}$, so a shallow trap is a strongly correlated system and a steep one is nearly free. We shall see that dependence explicitly in Table 11.6 and Figure 11.2.

Single-particle states and magic numbers.

The eigenstates of $\hat{H}_0$ for one particle in two dimensions are labelled by a radial quantum number $n = 0, 1, 2, \ldots$ and an angular-momentum projection $m = 0, \pm 1, \pm 2, \ldots$, with energies

$$\varepsilon_{nm} = \hbar\omega\,(2n + |m| + 1)\,. \tag{11.2}$$

States with the same $N_s = 2n + |m|$ are degenerate, so the spectrum falls into shells of degeneracy $2(N_s + 1)$ including spin. Filling shells completely gives the closed-shell electron numbers

$$N = 2,\ 6,\ 12,\ 20,\ 30,\ 42,\ldots \tag{11.3}$$

the *magic numbers* of the quantum dot, listed shell by shell in Table 11.1. They play exactly the role that the closed shells of atomic and nuclear physics do: adding or removing an electron at a magic number costs more energy than elsewhere, and a closed-shell dot has a non-degenerate ground state which a single Slater determinant can describe reasonably. All the machinery of Chapters 6 to 10 assumes such a reference, so we work at $N = 2$ and $N = 6$.

N_s	states (n,m)	degeneracy	cumulative N
0	$(0,0)$	2	**2**
1	$(0,-1),\ (0,+1)$	4	**6**
2	$(0,-2),\ (1,0),\ (0,+2)$	6	**12**
3	$(0,\pm 3),\ (1,\pm 1)$	8	**20**
4	$(0,\pm 4),\ (1,\pm 2),\ (2,0)$	10	**30**
5	$(0,\pm 5),\ (1,\pm 3),\ (2,\pm 1)$	12	**42**

Table 11.1 Shell structure of the two-dimensional harmonic oscillator. Shell $N_s = 2n + |m|$ has $N_s + 1$ spatial states and twice that many spin-orbitals. The cumulative column is the sequence of magic numbers (11.3). The calculations of this chapter use all six shells, that is 42 spin-orbitals.

11.2 The single-particle basis

In Cartesian coordinates the two-dimensional oscillator separates into two one-dimensional ones and the eigenfunctions are products of Hermite polynomials. That is correct but inconvenient: the rotational symmetry of the trap is then invisible, and it is precisely that symmetry which will cut the number of non-zero Coulomb matrix elements by two thirds. In polar coordinates $x = r\cos\theta$, $y = r\sin\theta$ the angular momentum is manifest.

Writing $\psi(r,\theta) = R(r)\,Y(\theta)$, the angular factor is

$$Y(\theta) = \frac{1}{\sqrt{2\pi}}\,e^{im\theta}, \tag{11.4}$$

and single-valuedness, $\psi(r,\theta+2\pi)=\psi(r,\theta)$, forces m to be an integer. The radial equation is solved by associated Laguerre polynomials,

$$R_{nm}(r)=\sqrt{\frac{2n!}{(n+|m|)!}}\,\beta^{\frac{1}{2}(|m|+1)}\,r^{|m|}\,e^{-\beta r^2/2}\,L_n^{|m|}\left(\beta r^2\right),\qquad \beta=\frac{m_e\omega}{\hbar},\tag{11.5}$$

so that the complete single-particle eigenfunction is

$$\boxed{\psi_{nm}(r,\theta)=\sqrt{\frac{n!}{\pi\,(n+|m|)!}}\,\beta^{\frac{1}{2}(|m|+1)}\,r^{|m|}\,e^{-\beta r^2/2}\,L_n^{|m|}\left(\beta r^2\right)e^{im\theta}}\tag{11.6}$$

with the eigenvalue (11.2). The Cartesian and polar labellings agree, $n_x+n_y=2n+|m|$, as they must.

This basis is not the only possible one, and it is not obviously the best. It is the eigenbasis of the non-interacting problem, which makes $\hat{H}_0$ diagonal and gives a natural ordering by shell, but it is built from Gaussians and therefore cannot reproduce the cusp that the Coulomb interaction imposes on the exact wave function at $r_{12}=0$. That single fact governs how the results below converge, and it is worth keeping in mind when reading Table 11.4 and the convergence curve of Figure 11.1.

11.3 The Coulomb matrix element

Everything now hinges on

$$\langle pq|\hat{v}|rs\rangle=\left\langle (n_pm_p)(n_qm_q)\left|\frac{1}{|\boldsymbol{r}_1-\boldsymbol{r}_2|}\right|(n_rm_r)(n_sm_s)\right\rangle,\tag{11.7}$$

a four-dimensional integral over two planes. It can be done in closed form. The derivation is due to Anisimova and Matulis and consists of expanding the Laguerre polynomials binomially, doing the angular integrals with the selection rule below, and recognising the radial integrals as Gamma functions; the result is a finite sum of products of factorials, which `quantumdot.py` evaluates in logarithms so that nothing overflows.

Two exact statements about Eq. (11.7) do more for us than the closed form itself.

Angular momentum is conserved.

The Coulomb interaction depends only on $|\boldsymbol{r}_1-\boldsymbol{r}_2|$ and is therefore invariant under a simultaneous rotation of both particles. Hence

$$\boxed{m_p+m_q=M=m_r+m_s}\tag{11.8}$$

and the matrix element vanishes identically otherwise. This is the single most valuable fact in the implementation: of the M_s^4 index combinations for M_s spatial states, the great majority are eliminated before any work is done. For the six-shell basis, $21^4=194\,481$ combinations reduce to $14\,703$ non-zero ones.

The frequency dependence is a scale factor.

Lengths scale as $\omega^{-1/2}$ and the interaction as one over a length, so

$$\langle pq|\hat{v}|rs\rangle_{\omega} = \sqrt{\hbar\omega}\,\langle pq|\hat{v}|rs\rangle_{\omega=1}. \tag{11.9}$$

The expensive integrals need therefore be computed only once, and the whole ω dependence of Table 11.6 costs nothing extra. It also explains the physics: since $\hat{H}_0$ scales as ω and $\hat{V}$ as $\sqrt{\omega}$, the ratio of interaction to kinetic energy goes as $\omega^{-1/2}$, and correlation is a small-ω phenomenon.

Spin.

The closed form is a spatial integral; the spin has to be put in by hand. The antisymmetrised matrix element that all our machinery expects is

$$\langle pq|\hat{v}|rs\rangle_{\mathrm{AS}} = \langle pq|\hat{v}|rs\rangle\,\delta_{\sigma_p\sigma_r}\delta_{\sigma_q\sigma_s} - \langle pq|\hat{v}|sr\rangle\,\delta_{\sigma_p\sigma_s}\delta_{\sigma_q\sigma_r}, \tag{11.10}$$

the direct term requiring each particle to keep its spin and the exchange term requiring the two to swap. Both together conserve S_z. It is easy to get the second delta pair wrong – note that the roles of r and s are interchanged in the exchange term – and the resulting error is subtle, since the calculation still runs and still converges. The checks below exist for exactly that reason.

Verification.

Three tests catch almost every implementation error. The lowest element is known analytically,

$$\langle 00;00|\hat{v}|00;00\rangle = \sqrt{\frac{\pi}{2}}\,\sqrt{\hbar\omega} = 1.253314137316\,\sqrt{\hbar\omega}, \tag{11.11}$$

which `quantumdot.py` reproduces to twelve digits. The scaling (11.9) can be checked by computing the same element at several frequencies and dividing. And the element must satisfy

$$\langle pq|\hat{v}|rs\rangle = \langle qp|\hat{v}|sr\rangle = \langle rs|\hat{v}|pq\rangle, \tag{11.12}$$

particle exchange and hermiticity, both of which hold to machine precision. Equation (11.11) is also the first-order perturbative estimate of the two-electron ground-state energy, $E \approx 2\hbar\omega + \sqrt{\pi/2}\sqrt{\hbar\omega}$, which at $\hbar\omega = 1$ gives 3.253314 – the minimal-basis Hartree-Fock energy of the next section.

11.4 Hartree-Fock

The Hartree-Fock equations of Section 6.4 are used here without change. We expand the Hartree-Fock orbitals in the oscillator basis,

$$\psi_\alpha = \sum_p C_{p\alpha}\,\phi_p, \tag{11.13}$$

and the variational condition becomes the Roothaan eigenvalue problem $\boldsymbol{FC} = \boldsymbol{C\varepsilon}$ with

$$F_{pq} = \langle p|\hat{h}_0|q\rangle + \sum_{rs} \rho_{rs} \langle pr|\hat{v}|qs\rangle_{\mathrm{AS}}, \qquad \rho_{rs} = \sum_{a=1}^{N} C_{ra} C_{sa}, \tag{11.14}$$

solved by the self-consistent iteration of Section 6.5. In the oscillator basis $\langle p|\hat{h}_0|q\rangle = \varepsilon_p^{(0)}\delta_{pq}$ with $\varepsilon_p^{(0)}$ from Eq. (11.2), so the only non-trivial ingredient is the two-body matrix built in the previous section. The total energy is

$$E_{\mathrm{HF}} = \sum_{a=1}^{N} \varepsilon_a^{(0)} + \frac{1}{2} \sum_{a,b=1}^{N} \langle ab|\hat{v}|ab\rangle_{\mathrm{AS}}, \tag{11.15}$$

the factor $\frac{1}{2}$ removing the double counting of the interaction.

Convergence with the basis.

Table 11.2 follows the energy as shells are added. The sequence is monotone, as it must be: enlarging the basis gives the Hartree-Fock state more variational freedom and can only lower the energy. Two features are worth noting. The energy is unchanged when certain shells are added – from one shell to two for $N = 2$, and again from three to four – for a reason worked out in Exercise 11.10. And convergence is slow at the end: going from 30 to 42 orbitals still moves the six-electron energy by 0.028. That is the Gaussian basis struggling with the Coulomb cusp, and it is a much larger effect than anything the many-body method will contribute.

shells	orbitals	$N = 2$: E_{HF}	ΔE	$N = 6$: E_{HF}	ΔE
0	2	3.25331414	—	—	—
1	6	3.25331414	+0.000000	22.21981284	—
2	12	3.16269135	−0.090623	21.59319848	−0.626614
3	20	3.16269135	+0.000000	20.76691943	−0.826279
4	30	3.16192140	−0.000770	20.74840225	−0.018517
5	42	3.16192140	+0.000000	20.72025707	−0.028145

Table 11.2 Hartree-Fock energy as a function of the number of oscillator shells, $\hbar\omega = 1$. ΔE is the change from the previous row. Computed by `quantumdot.py`.

The single-particle spectrum.

The converged Hartree-Fock energies for six electrons in the full basis are

$$\varepsilon_{1,2} = 4.54225444, \qquad \varepsilon_{3,4,5,6} = 5.30056289, \tag{11.16}$$

against the bare oscillator values 1 and 2. The mean field has pushed every level up by more than three units – the Coulomb repulsion of the other electrons – and the degeneracies of Table 11.1 survive untouched, which is a useful check that the self-consistent solution has not broken the rotational symmetry. By Koopmans' theorem the ionisation energy of the dot is $-\varepsilon_6 = -5.30056289$; for two electrons the corresponding number is $-\varepsilon_2 = -2.12246475$.

11.5 Many-body perturbation theory

With a Hartree-Fock reference in hand, the natural partition is the Møller-Plesset one of Chapter 9: the unperturbed operator is the sum of Fock operators and the perturbation is the *fluctuation potential*, the difference between the true interaction and the mean field it generates,

$$\hat{H}_0^{\mathrm{MP}} = \sum_{i=1}^{N} \hat{f}(i), \qquad \hat{V}' = \hat{V} - \sum_{i=1}^{N} \hat{u}^{\mathrm{HF}}(i). \tag{11.17}$$

The first-order energy then vanishes by construction, $E^{(0)} + E^{(1)} = E_{\mathrm{HF}}$, so the leading correction is second order. Brillouin's theorem removes the singly excited determinants, $\langle \Phi_a^r | \hat{V}' | \Phi_0 \rangle = 0$, and only doubles survive:

$$\boxed{E_{\mathrm{MP2}} = \frac{1}{4} \sum_{ab}^{\mathrm{occ}} \sum_{rs}^{\mathrm{virt}} \frac{|\langle ab|\hat{v}|rs\rangle_{\mathrm{AS}}|^2}{\varepsilon_a + \varepsilon_b - \varepsilon_r - \varepsilon_s}} \tag{11.18}$$

The denominator is negative, so MP2 always lowers the energy. The matrix elements are needed in the Hartree-Fock basis, obtained from the oscillator ones by the four-index transformation

$$\langle \alpha\beta|\hat{v}|\gamma\delta\rangle = \sum_{pqrs} C_{p\alpha} C_{q\beta} \langle pq|\hat{v}|rs\rangle C_{r\gamma} C_{s\delta}, \tag{11.19}$$

carried out in two half-steps so that the cost is $\mathscr{O}(M^5)$ rather than $\mathscr{O}(M^8)$.

For the full basis at $\hbar\omega = 1$ the result is

$$E_{\mathrm{MP2}}(N=2) = -0.13488329, \qquad E_{\mathrm{MP2}}(N=6) = -0.41769581, \tag{11.20}$$

that is 11.6% and 3.9% of the Hartree-Fock interaction energy $E_{\mathrm{HF}} - E_0$. The smaller fraction for six electrons is not a statement that the six-electron dot is less correlated; it is that the mean field is doing more of the work, since each electron feels five others rather than one.

11.6 Coupled cluster

The coupled-cluster equations of Chapter 10 are applied here verbatim: the same amplitude equations, the same Stanton-Gauss-Watts-Bartlett intermediates, the same solver. Only the matrices change. That is the whole point of writing the equations in a spin-orbital basis with an antisymmetrised interaction – the method never learns what the interaction is.

A warning about the singles.

It is often said that for a closed-shell system in a canonical Hartree-Fock basis the singles amplitudes vanish by Brillouin's theorem, so that CCSD reduces to CCD. This is not true, and it is worth being explicit about why. Brillouin's theorem states that $f_{ai} = 0$, and that is what kills the first-order singles contribution in perturbation theory. But the CCSD singles equation is not homogeneous in t_1: it contains the terms

$$\tfrac{1}{2} \sum_{mef} t_{im}^{ef} \langle ma|\hat{v}|ef\rangle_{\mathrm{AS}} \qquad \text{and} \qquad -\tfrac{1}{2} \sum_{men} t_{mn}^{ae} \langle nm|\hat{v}|ei\rangle_{\mathrm{AS}}, \tag{11.21}$$

which are built from the *doubles* and survive at $t_1 = 0$. The singles are therefore driven by the doubles, they are small but not zero, and they describe the relaxation of the orbitals in response to the correlation the doubles have introduced. For the quantum dots computed here $\max|t_i^a| = 5.7 \times 10^{-3}$ at $N = 2$ and 1.0×10^{-2} at $N = 6$, and CCSD lies correspondingly a little below CCD. A calculation that reports $t_1 = 0$ identically has a bug in its singles equation, and Exercise 11.10 gives a cheap test that finds it.

Results.

Table 11.3 collects the whole hierarchy in the full 42-orbital basis. The ordering

$$E_{\mathrm{HF}} > E_{\mathrm{HF}} + E_{\mathrm{MP2}} > E_{\mathrm{HF}} + E_{\mathrm{CCD}} > E_{\mathrm{HF}} + E_{\mathrm{CCSD}} \tag{11.22}$$

holds for both electron numbers, and each step has a clear meaning: MP2 is the leading pair correlation, CCD resums the ladders and rings to all orders in $\hat{T}_2$, and CCSD adds the orbital relaxation just discussed.

quantity	$N = 2$	$N = 6$
non-interacting E_0	2.00000000	10.00000000
Hartree-Fock E_{HF}	3.16192140	20.72025707
interaction $E_{\mathrm{HF}} - E_0$	1.16192140	10.72025707
E_{MP2}	−0.13488329	−0.41769581
total	3.02703812	20.30256126
E_{CCD}	−0.14799908	−0.44624450
total	3.01392232	20.27401257
E_{CCSD}	−0.14829527	−0.44701044
total	3.01362613	20.27324663

Table 11.3 The hierarchy of methods for the two- and six-electron quantum dot at $\hbar\omega = 1$, in the basis of six oscillator shells (42 spin-orbitals). The self-consistent field converges in 15 and 24 iterations, the coupled-cluster equations in 23 and 28. Computed by `quantumdot.py`.

11.7 Comparison with the exact answer

Tables of energies produced by approximate methods are worth little without something to compare them with. For this system we have two benchmarks.

Two electrons: exact within the basis, and exact in the limit.

With $N = 2$ the determinant space has dimension $\binom{M}{2}$, which is 861 for 42 spin-orbitals and can be diagonalised directly. Moreover singles and doubles *exhaust* the excitations available to two particles, so CCSD must be exact in any basis. Table 11.4 confirms both statements: CCSD and full diagonalisation agree to twelve digits at every basis size. This is a stronger test of the coupled-cluster solver of Chapter 10 than anything the pairing model could provide, because here the interaction has no special structure to hide behind.

The second benchmark is analytic. Taut showed that the relative motion of two electrons in a parabolic trap is exactly solvable for a discrete set of frequencies, and at $\hbar\omega = 1$ in our units the ground state is exactly $E = 3$. The last column of Table 11.4 approaches that value from above, but slowly: at 42 orbitals the remaining error is 0.0136, comparable with the entire difference between MP2 and CCSD. The many-body problem has been solved exactly; the basis-set problem has not. For this system the truncation of the single-particle space, not the truncation of the cluster operator, is the dominant error – which is the opposite of the situation in the pairing model, and a lesson worth taking from a realistic Hamiltonian.

shells	orb.	dim	E_{HF}	$+E_{\mathrm{MP2}}$	$+E_{\mathrm{CCD}}$	$+E_{\mathrm{CCSD}}$	E_{FCI}
1	6	15	3.25331414	3.17856150	3.15232801	3.15232801	3.15232801
2	12	66	3.16269135	3.05797643	3.03904782	3.03860458	3.03860458
3	20	190	3.16269135	3.04440370	3.02527309	3.02523058	3.02523058
4	30	435	3.16192140	3.03341846	3.01794371	3.01760623	3.01760623
5	42	861	3.16192140	3.02703812	3.01392232	3.01362613	3.01362613
∞	—	—	—	—	—	—	3.00000000

Table 11.4 Two electrons at $\hbar\omega = 1$. CCSD agrees with full diagonalisation to 3×10^{-13} in every basis, as it must for two particles. The last row is Taut's analytic result for the complete basis. Computed by `quantumdot.py`.

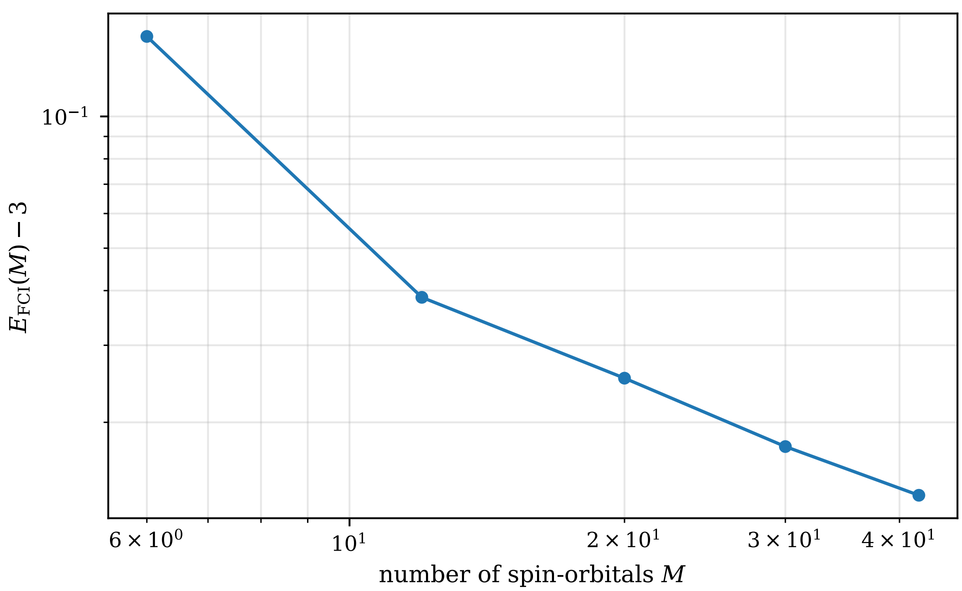

Fig. 11.1 Basis-set error of the two-electron ground-state energy at $\hbar\omega = 1$, $E_{\mathrm{FCI}}(M) - 3$, plotted logarithmically against the number of spin-orbitals M. The five points are the last column of Table 11.4 measured against Taut's analytic value $E = 3$, for one, two, three, four and five shells, that is $M = 6$, 12, 20, 30 and 42. The first step is steep – the error falls by a factor of four between $M = 6$ and $M = 12$ – but the remaining points lie on a straight line of slope close to -0.8, so that enlarging the basis by a further factor of 3.5 buys less than a factor of three in accuracy. The curve is a pure measure of the truncation of the single-particle space: for two electrons the many-body problem is solved exactly at every point.

Figure 11.1 is worth more than the table it is drawn from, because on logarithmic axes the *rate* of convergence becomes visible and the rate is the whole story. What is plotted is the distance from the exact answer, and since CCSD and full diagonalisation agree here to 3×10^{-13}, none of that distance is a many-body error; every bit of it is the basis. The first segment is steep, a slope of -1.98 between $M=6$ and $M=12$, and that is the last good news. From twelve orbitals onwards the points fall on a line of slope -0.83, -0.89 and -0.76 over the three remaining intervals, an algebraic decay a little slower than $1/M$. Fitting the four largest bases to the two-term form $a/M+b/M^2$ suggested in Exercise 5 gives $a=0.58$ and $b=-1.44$ and reproduces the computed errors to a few times 10^{-4}; the negative b is exactly what makes the local slope shallower than the asymptotic -1. The extrapolation is sobering. To push the remaining error below 10^{-3} one needs $M\approx600$ spin-orbitals, some twenty-four shells, and to reach 10^{-4} one needs of order six thousand. For two electrons that is merely expensive – the determinant space grows as $M^2/2$ and would still be manageable – but for six it is out of the question, and no improvement in the many-body method repairs it.

The reason for the slow decay was anticipated in Section 11.2. The exact wave function has a cusp at $r_{12}=0$, imposed by the requirement that the kinetic energy cancel the divergence of $1/r_{12}$, and a finite sum of products of Gaussians has a vanishing derivative there and cannot produce one. A basis that is wrong in a small region converges algebraically rather than exponentially, and the exponent is set by the dimensionality and by the order of the defect, not by anything one can tune. This is the same partial-wave convergence problem that quantum chemistry meets in its correlation-consistent basis sets and that nuclear physics meets in its oscillator model spaces, and the standard remedies – extrapolation, as above, or an explicitly correlated ansatz containing r_{12} directly – are the same too. It is also the sharpest contrast with the model Hamiltonians of the earlier chapters, where the single-particle space was finite by construction and Figure 11.1 would have had nothing to plot. The stochastic methods of the later chapters take the second remedy: a variational Monte Carlo trial function multiplies the determinant by a Jastrow factor that carries the cusp explicitly, so the error of Figure 11.1 simply does not arise there, and the price is paid instead in statistical noise.

Six electrons: exact in a small basis.

For six electrons the full space is out of reach at 42 orbitals – $\binom{42}{6}=5\,245\,786$ determinants – but in the two-shell basis it is only $\binom{12}{6}=924$, and there the truncation of the cluster operator genuinely matters. Table 11.5 is therefore the one comparison in this book where every method is measured against the exact answer on a realistic interaction in a system that is not trivially solvable. The last column is the fraction of the exact correlation energy recovered.

method	energy	error	% of E_{corr}
Hartree-Fock	21.59319848	1.7×10^{-1}	0.00
MP2	21.43344056	1.3×10^{-2}	92.55
CCD	21.42380972	3.2×10^{-3}	98.13
CCSD	21.42323158	2.6×10^{-3}	98.47
UCCSD	21.42196510	1.4×10^{-3}	99.20
exact diagonalisation	21.42058830	—	100.00

Table 11.5 Six electrons at $\hbar\omega=1$ in the two-shell basis, 12 spin-orbitals, determinant space of dimension 924. The correlation energy is $E_{\mathrm{corr}}=-0.17261018$. The unitary result is evaluated at the converged CCSD amplitudes rather than at its own variational optimum, and is still the best of the four. Computed by `quantumdot.py`.

Dependence on the confinement.

Table 11.6 varies the trap frequency. The last column, the correlation energy as a fraction of the Hartree-Fock interaction energy, falls by a factor of three between $\hbar\omega = 0.25$ and $\hbar\omega = 4$, which is the $\omega^{-1/2}$ scaling of Eq. (11.9) doing its work. A shallow trap is a strongly correlated system: the electrons have room to avoid one another, the mean field describes them badly, and the correlation methods have more to do. In the opposite limit the dot becomes a set of nearly free particles in a very steep well and Hartree-Fock is almost exact. This is the same story the pairing model told through g in Table 4.1, told now by a physical parameter that an experimentalist can actually tune.

$\hbar\omega$	E_0	E_{HF}	E_{MP2}	E_{CCSD}	ratio
$N = 2$					
0.25	0.5	1.04628424	−0.10380945	−0.11288484	−0.2066
0.50	1.0	1.79974794	−0.12036392	−0.13249076	−0.1657
1.00	2.0	3.16192140	−0.13488329	−0.14829527	−0.1276
2.00	4.0	5.67728232	−0.14693248	−0.15989645	−0.0953
4.00	8.0	10.40864632	−0.15651039	−0.16789538	−0.0697
$N = 6$					
0.25	2.5	7.39178633	−0.32952382	−0.35507942	−0.0726
0.50	5.0	12.27149922	−0.37861963	−0.40804475	−0.0561
1.00	10.0	20.72025707	−0.41769581	−0.44701044	−0.0417
2.00	20.0	35.67233283	−0.44700367	−0.47274020	−0.0302
4.00	40.0	62.74116274	−0.46768316	−0.48840005	−0.0215

Table 11.6 Dependence on the trap frequency, in the full 42-orbital basis. The last column is $E_{\mathrm{CCSD}}/(E_{\mathrm{HF}} - E_0)$. For $N = 2$ the CCSD column is exact within the basis, so $E_{\mathrm{HF}} + E_{\mathrm{CCSD}}$ is the full configuration-interaction energy at every frequency. Computed by `quantumdot.py`.

Figure 11.2 draws the last column of Table 11.6 and adds two things the table does not make obvious. The first is how nearly the two curves are parallel. Sixteenfold steepening of the trap costs the two-electron dot a factor $0.202/0.062 = 3.3$ of its relative correlation energy and the six-electron dot a factor $0.051/0.016 = 3.1$, which on logarithmic axes are slopes of -0.43 and -0.41. The naive scaling argument of Eq. (11.9) predicts $-1/2$ exactly: the interaction goes as $\sqrt{\omega}$ against a one-body part going as ω, so the ratio of the two should go as $\omega^{-1/2}$ and a factor of 16 in $\hbar\omega$ should give a factor of 4 in the ratio. That it gives 3.3 instead is not a failure of the argument but a reminder of what the denominator contains. The Hartree-Fock interaction energy $E_{\mathrm{HF}} - E_0$ is itself not a pure $\sqrt{\omega}$: the self-consistent orbitals relax differently at each frequency, spreading out in a shallow trap and contracting in a steep one, so both numerator and denominator carry a residual ω dependence and only their leading behaviour is the clean power. The deviation from $-1/2$ measures exactly that relaxation, and it is largest at the shallow end where the mean field is working hardest.

The second point is the vertical separation. At every frequency the six-electron dot has proportionally less correlation than the two-electron one – 0.032 against 0.118 at $\hbar\omega = 1$, a factor of 3.8 – and it would be a serious misreading to conclude from this that six electrons are easier to correlate than two. The absolute correlation energies run the other way, -0.339 against -0.137 in the same basis. What has changed is the denominator: each electron in the six-electron dot feels five neighbours rather than one, the mean field is a much larger and much better approximation to what it feels, and the fraction left over for the correlated methods shrinks even as its absolute size grows. This is the quantum-dot form of a statement

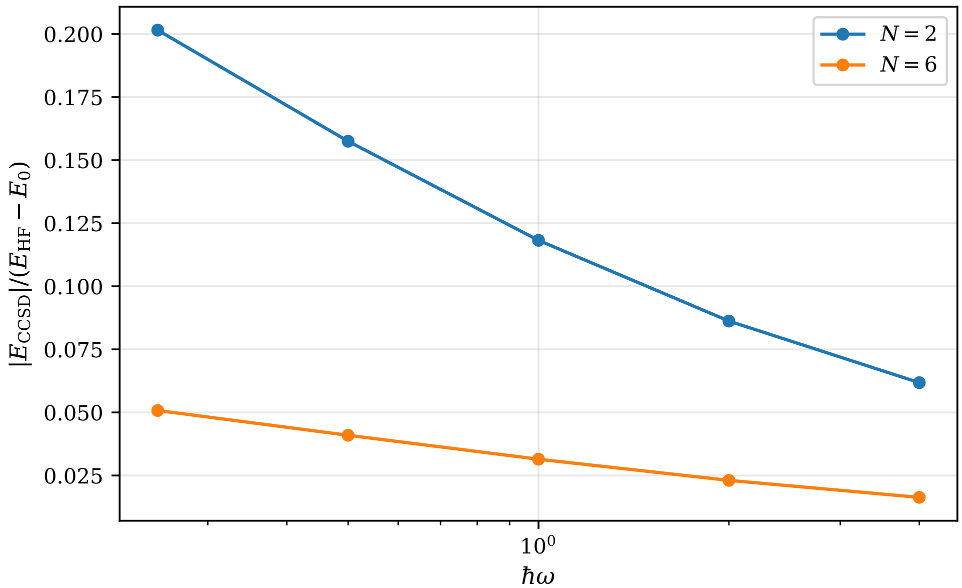

Fig. 11.2 Correlation energy as a fraction of the Hartree-Fock interaction energy, $|E_{\mathrm{CCSD}}|/(E_{\mathrm{HF}}-E_0)$, against the trap frequency $\hbar\omega$ on a logarithmic axis, for two electrons (upper curve) and six electrons (lower curve). The five frequencies are $\hbar\omega = 0.25$, 0.5, 1, 2 and 4 and the basis is the smaller one of shells $N_s = 0$ to 3, twenty spin-orbitals, so the numbers are slightly below the full-basis values of Table 11.6 while the trend is the same. Both curves fall monotonically as the trap is steepened, from 0.202 to 0.062 for $N = 2$ and from 0.051 to 0.016 for $N = 6$, and the six-electron curve lies a factor of about four below the two-electron one at every frequency.

that recurs throughout many-body physics, that mean-field theory improves as the number of interacting partners grows, and it is the reason nuclear and electronic-structure calculations of large systems can begin from a single determinant at all. Read together with Figure 11.1 the two figures divide the labour cleanly: the frequency controls how much correlation there is to capture, the basis size controls how well one can capture it, and for this Hamiltonian at $\hbar\omega = 1$ the second is by far the larger worry.

11.8 Unitary coupled cluster

The unitary ansatz of Section 10.14 carries over unchanged. With

$$\hat{\sigma} = \hat{T} - \hat{T}^{\dagger}, \qquad |\Psi(\boldsymbol{t})\rangle = e^{\hat{\sigma}(\boldsymbol{t})}|\Phi_0\rangle, \qquad E(\boldsymbol{t}) = \langle\Psi(\boldsymbol{t})|\hat{H}|\Psi(\boldsymbol{t})\rangle, \tag{11.23}$$

the energy is a genuine expectation value in a normalised state and is therefore variational: it lies above the exact energy for any amplitudes, which the projected CCSD energy does not. The price is that the Baker-Campbell-Hausdorff expansion no longer terminates, so there are no closed amplitude equations and the energy has to be evaluated by exponentiating $\hat{\sigma}$.

Two electrons: everything collapses.

Table 11.7 runs the unitary calculation for two electrons. Two things happen, both instructive. UCCSD, CCSD and full diagonalisation coincide, which is expected: with two particles singles and doubles span the whole space and any ansatz that is exact there must give the exact energy. Less obviously, UCCD coincides with CCD. The reason is that for two particles $\hat{T}_2\hat{T}_2$

annihilates the reference and so does $\hat{T}_2^\dagger$, so $e^{\hat{T}_2-\hat{T}_2^\dagger}|\Phi_0\rangle$ and $(1+\hat{T}_2)|\Phi_0\rangle$ sweep out the same two-dimensional manifold; the variational minimum over it is the lowest eigenvalue in the space spanned by the reference and the doubles, which is what the CCD equations return. The unitary and the standard ansatz differ only when the truncation genuinely bites.

shells	orb.	ampl.	CCD	UCCD	CCSD	UCCSD	FCI
1	6	14	3.15232801	3.15232801	3.15232801	3.15232801	3.15232801
2	12	65	3.03904782	3.03904782	3.03860458	3.03860458	3.03860458
3	20	189	3.02527309	3.02527309	3.02523058	3.02523058	3.02523058

Table 11.7 Unitary and standard coupled cluster for two electrons at $\hbar\omega = 1$. The unitary energies are obtained by minimising Eq. (11.23) over all amplitudes, the standard ones by solving the projected equations. Computed by `quantumdot.py`.

Six electrons: the unitary ansatz wins.

The six-electron row of Table 11.5 is where the difference appears. There the truncation to singles and doubles is a real approximation – the exact state contains triples and quadruples – and the unitary ansatz recovers 99.2% of the correlation energy against CCSD's 98.5%, while remaining above the exact energy as the variational principle requires. What makes this worth emphasising is that the unitary number was obtained *without optimising*: the converged CCSD amplitudes were simply substituted into Eq. (11.23). Even at borrowed amplitudes the unitary ansatz is the better wave function, because $e^{\hat{\sigma}}$ generates the de-excitations that $e^{\hat{T}}$ cannot.

The cost of doing better is what pushes one towards a quantum computer. A proper variational optimisation of the six-electron case means minimising over 261 amplitudes, each energy evaluation requiring the exponential of a 924×924 matrix, and both numbers grow exponentially with the size of the system. On a quantum computer the state $e^{\hat{\sigma}}|\Phi_0\rangle$ is prepared by a circuit whose depth grows polynomially, and the energy is measured rather than computed.

Trotterisation.

A circuit cannot apply $e^{\hat{\sigma}}$ in one step; it applies the generators one at a time, and the resulting splitting error is the Trotter error of Section 6.9. Table 11.8 measures it for the two-electron dot at the CCSD amplitudes. The error halves when the number of Trotter steps doubles – the $\mathscr{O}(1/n)$ rate of Eq. (6.52) – and n is the circuit depth. On hardware the Hamiltonian would first be mapped to Pauli strings by the Jordan-Wigner transformation of Section 3.16, each string contributing one rotation to the circuit; here the matrices are simply exponentiated.

Figure 11.3 is drawn on logarithmic axes precisely so that the reference line can be put next to the data, and the comparison is more interesting than a clean agreement would have been. The measured points do not lie on the $1/n$ line; they fall below it, with an average slope of -1.20 against the ideal -1, so that at sixteen steps the error is 3.77×10^{-7} where the line predicts 6.5×10^{-7}. The ratios in Table 11.8 tell the same story from the other side: they begin at 2.38, well above the asymptotic 2, and decrease steadily through 2.40, 2.26 and 2.15. Both observations say that we are not yet in the asymptotic regime. The first-order Lie-Trotter bound of Eq. (6.52) is a statement about the leading term in a series in $1/n$, and at $n = 1$ or $n = 2$ the higher terms are not negligible; here they happen to have the sign that

Trotter steps n	energy	error	ratio
1	3.03863183	1.04×10^{-5}	—
2	3.03863789	4.38×10^{-6}	2.38
4	3.03864044	1.83×10^{-6}	2.40
8	3.03864146	8.10×10^{-7}	2.26
16	3.03864189	3.77×10^{-7}	2.15
∞	3.03864227	—	—

Table 11.8 First-order Trotterisation of the unitary ansatz, two electrons in 12 orbitals at the converged CCSD amplitudes. The error is measured against the untrotterised exponential and falls off as $1/n$, the ratio approaching 2 from above. Computed by `quantumdot.py`.

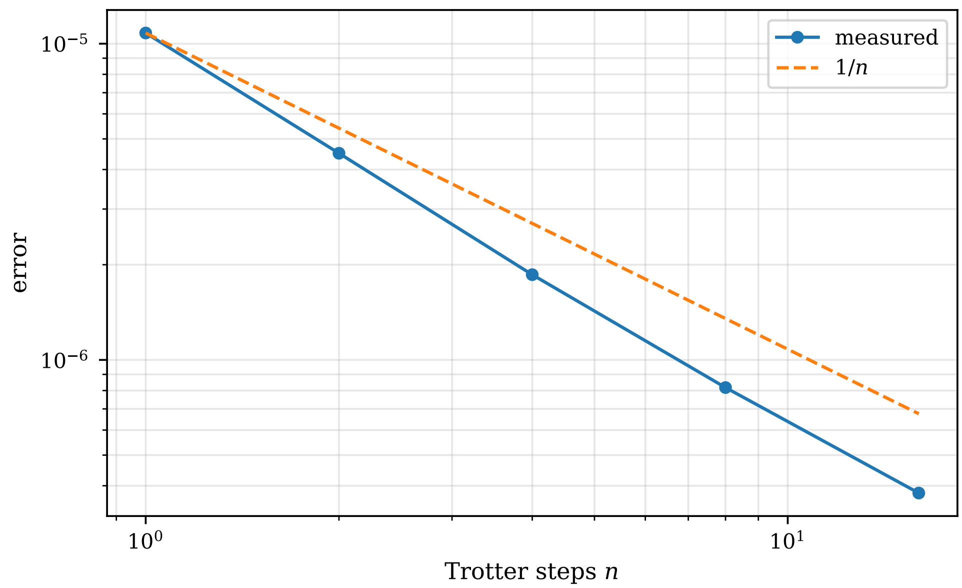

Fig. 11.3 Error of the first-order Trotter splitting of $e^{\hat{\sigma}}$ against the number of Trotter steps n, on logarithmic axes, for two electrons in 12 orbitals at the converged CCSD amplitudes. The measured error (solid, circles) is the deviation from the untrotterised energy 3.03864227 and is the third column of Table 11.8; the dashed line is the ideal $\mathcal{O}(1/n)$ rate, anchored at the $n = 1$ point. The measured curve is slightly steeper than $1/n$ over this range, falling from 1.04×10^{-5} at one step to 3.77×10^{-7} at sixteen, a factor of 28 where $1/n$ would give 16.

helps. Extending the study to a hundred steps would bring the slope up to -1 and the ratio down to 2 from above, and Exercise 5 e) asks for the same plot with the symmetric splitting of Eq. (6.53), where the line to compare against is $1/n^2$ and the same over-performance at small n reappears.

What deserves more emphasis than the exponent, however, is the vertical scale. Even a single Trotter step – the shallowest circuit one could write – misses the exact exponential by only 1.04×10^{-5}. Set that against the two other errors present in the same calculation. Evaluating the unitary energy at the borrowed CCSD amplitudes rather than at its own variational minimum costs $3.03864227 - 3.03860458 = 3.77 \times 10^{-5}$, four times more than the crudest Trotterisation; and the twelve-orbital basis itself sits 0.0386 above Taut's exact $E = 3$, a thousand times more again. The hierarchy of errors is therefore basis $\gg$ amplitudes $\gg$ circuit depth, and it is the reverse of the order in which one is tempted to worry about them. For a system of this size there is no point at all in spending circuit depth to drive the Trotter error below 10^{-5}: it is already invisible beneath the basis-set error of Figure 11.1. The lesson generalises.

The Trotter number should be chosen so that the splitting error matches the accuracy the rest of the calculation can support, and on a device with a finite coherence time – or a finite number of shots, since the statistical error of measuring $\langle\Psi|\hat{H}|\Psi\rangle$ falls only as the inverse square root of the sample size – every step spent beyond that point is a step wasted.

Why this system matters. The quantum dot is the smallest realistic many-fermion system on which the whole apparatus of this book can be run and checked. It is a genuine Coulomb problem, so the matrix elements and the basis-set convergence are those of quantum chemistry and nuclear physics; it is small enough that exact diagonalisation is available for two electrons in any basis and for six in a small one; and it has an analytic answer at special frequencies. It is also the system on which the stochastic methods of the later chapters will be tested, so the numbers computed here are the reference against which variational and diffusion Monte Carlo will be measured.

11.9 What the realistic system taught us

The methods survived the transition from a one-parameter model to a real interaction without modification. The same Hartree-Fock routine, the same MP2 sum, the same coupled-cluster solver and the same unitary ansatz that were developed and validated on the pairing model of Chapter 4 produce Tables 11.3 to 11.8 when handed a different pair of matrices. That portability is not an accident; it is what the spin-orbital formulation of Chapter 3 and the antisymmetrised matrix element were for.

Three things did change, and they are the reasons for doing a realistic calculation at all.

- **The basis is now an approximation.** In the pairing model the single-particle space was finite and the only error was the many-body method. Here the oscillator basis is truncated, and for two electrons the truncation error at 42 orbitals is 0.0136 against a CCSD-minus-MP2 difference of 0.0134. The two errors are comparable, and reporting one without the other is meaningless. Figure 11.1 shows how slowly the first of them goes away.
- **The hierarchy is quantitative, not qualitative.** Table 11.5 puts numbers on what each method buys: MP2 recovers 92.6% of the correlation energy, CCD 98.1%, CCSD 98.5% and the unitary ansatz 99.2%. The steps get smaller and the costs get larger, which is the shape of the whole subject.
- **Folklore does not survive contact with a computation.** The singles amplitudes do not vanish for a closed-shell Hartree-Fock reference, CCSD is not equal to CCD, and a calculation reporting otherwise is reporting a bug. The check that catches it is cheap: for two electrons CCSD must equal full diagonalisation, and it does, to twelve digits.

The Hamiltonian (11.1) returns later in the book. In the stochastic chapters it is treated by variational and diffusion Monte Carlo, which sample the many-body wave function directly rather than expanding it in determinants, and which therefore have no basis-set error at all – their error is statistical, and of a completely different nature. Comparing the two families on the same system, with the numbers of this chapter as the reference, is the cleanest way to see what each is good for.

11.10 Exercises

Exercise 1: the shell structure

a) Show that the number of spatial states with $2n+|m|=N_s$ is N_s+1, and hence that the magic numbers are $N=(N_s+1)(N_s+2)$.
b) Verify that the Cartesian and polar labellings give the same spectrum.
c) Why is $N=4$ not a magic number, and what would go wrong if we attempted a Hartree-Fock calculation there?

Answer to a).

For fixed N_s the allowed radial quantum numbers are $n=0,1,\ldots,\lfloor N_s/2\rfloor$ and $|m|=N_s-2n$. Each n with $|m|>0$ contributes two states, $m=\pm|m|$, and the single $n=N_s/2$ with $m=0$ (present only for even N_s) contributes one. Counting gives N_s+1 spatial states, or $2(N_s+1)$ with spin. The cumulative sum is $\sum_{k=0}^{N_s}2(k+1)=(N_s+1)(N_s+2)$, giving $2,6,12,20,30,42$.

Answer to b).

The Cartesian energy is $\hbar\omega(n_x+n_y+1)$ and the polar one is $\hbar\omega(2n+|m|+1)$, so the two agree whenever $n_x+n_y=2n+|m|$. For a given shell the Cartesian count is the number of pairs (n_x,n_y) with $n_x+n_y=N_s$, which is N_s+1 – the same as in a). The two bases are related by a unitary transformation within each shell, and either may be used; the polar one is chosen because it diagonalises $\hat{L}_z$.

Answer to c).

$N=4$ falls inside the $N_s=1$ shell, which holds four spin-orbitals of which only two would be occupied. The reference determinant is then not unique – any choice of two of the four degenerate orbitals gives the same unperturbed energy – and the ground state of the interacting system is degenerate or nearly so. A single-determinant reference is a poor starting point, the self-consistent iteration may fail to converge, and the perturbation denominators $\varepsilon_a+\varepsilon_b-\varepsilon_r-\varepsilon_s$ can vanish. This is the open-shell problem, and it needs multireference methods.

Exercise 2: the Coulomb matrix element

a) Prove the selection rule (11.8) from the angular integrals alone.
b) Derive the scaling (11.9).
c) Estimate what fraction of the M_s^4 index combinations survives the selection rule for a basis of M_s spatial states, and compare with the measured 14 703 out of 194 481 for six shells.

Answer to a).

Write $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ in polar coordinates. The interaction $1/|\boldsymbol{r}_1 - \boldsymbol{r}_2|$ depends on θ_1 and θ_2 only through $\theta_1 - \theta_2$, so a Fourier expansion in that difference gives terms $e^{ik(\theta_1-\theta_2)}$. The angular part of the matrix element is then

$$\int_0^{2\pi}\int_0^{2\pi} e^{-im_p\theta_1} e^{-im_q\theta_2} e^{ik(\theta_1-\theta_2)} e^{im_r\theta_1} e^{im_s\theta_2}\, d\theta_1\, d\theta_2,$$

which is non-zero only when $m_r - m_p + k = 0$ and $m_s - m_q - k = 0$ simultaneously. Adding the two conditions eliminates k and leaves $m_p + m_q = m_r + m_s$.

Answer to b).

The oscillator eigenfunctions (11.6) depend on r only through βr^2 with $\beta = m_e\omega/\hbar$, so the natural length in the problem is $\beta^{-1/2} \propto \omega^{-1/2}$. The matrix element of $1/r_{12}$ is an inverse length times a dimensionless number that depends only on the quantum numbers, and therefore scales as $\omega^{1/2}$. Numerically, dividing the element by $\sqrt{\hbar\omega}$ gives 1.253314137316 at every frequency, as `quantumdot.py` confirms.

Answer to c).

Fix m_p, m_q and m_r; then m_s is determined, and the combination is allowed only if that value lies in the basis. Roughly one index in four is fixed by the other three, so of order M_s^3 of the M_s^4 combinations survive, a fraction $\mathscr{O}(1/M_s)$. For $M_s = 21$ this suggests about 5%; the measured fraction is $14\,703/194\,481 = 7.6\%$, larger because the determined m_s often falls inside the basis rather than outside it, particularly for small $|m|$.

Exercise 3: checking a Hartree-Fock code

a) Show that for $N = 2$ in the minimal basis the Hartree-Fock energy is $2\hbar\omega + \sqrt{\pi/2}\sqrt{\hbar\omega}$, and check it against Table 11.2.
b) Explain why the $N = 2$ energy does not change from one shell to two, nor from three to four.
c) What would you conclude if the converged single-particle energies within a shell were not degenerate?

Answer to a).

With only the $(0,0)$ spatial orbital available, both electrons occupy it with opposite spins and no variational freedom remains: $\boldsymbol{C}$ is the identity. The energy is then the unperturbed $2\hbar\omega$ plus the single direct matrix element $\langle 00;00|\hat{v}|00;00\rangle$ – the exchange term vanishes because the two electrons have opposite spins – which is Eq. (11.11). At $\hbar\omega = 1$ this is $2 + 1.25331414 = 3.25331414$, the first row of the table.

Answer to b).

The Hartree-Fock orbitals carry good angular momentum, since $\hat{f}$ commutes with $\hat{L}_z$, so the occupied $m = 0$ orbital can mix only with other $m = 0$ states. Shell $N_s = 1$ contains none: its states are $(0, \pm 1)$. Adding it therefore changes nothing. The first new $m = 0$ state is $(1, 0)$ in shell 2, which is why the energy drops there, and the next is $(2, 0)$ in shell 4, which is the second drop. Shells 3 and 5 contain no $m = 0$ states at all. The same argument explains why the two-electron energy is converged to six digits long before the six-electron one.

Answer to c).

Either the calculation has not converged, or the antisymmetrised matrix elements are wrong in a way that breaks rotational invariance – the most common cause being the spin deltas of Eq. (11.10), where the exchange term has r and s interchanged. A broken degeneracy is a far more sensitive diagnostic than the total energy, which can be wrong by a little without looking wrong at all.

Exercise 4: the singles amplitudes

a) State precisely what Brillouin's theorem says, and what it does not say.
b) Show that $t_1 = 0$ is *not* a solution of the CCSD singles equation for an interacting system, using Eq. (11.21).
c) Design a numerical test that would catch a CCSD implementation in which the two driver terms have been omitted.

Answer to a).

Brillouin's theorem says that the Hartree-Fock determinant does not couple to singly excited determinants through the Hamiltonian, $\langle \Phi_i^a | \hat{H} | \Phi_0 \rangle = f_{ai} = 0$. That is a statement about a single matrix element. It implies that singles do not contribute at first order in perturbation theory, and hence that MP2 involves only doubles. It says nothing about the coupled-cluster amplitudes, which solve a non-linear system in which the singles are coupled to the doubles.

Answer to b).

Set $t_1 = 0$ in the singles equation. Every term containing t_1 vanishes and so does f_{ai}, but the two terms of Eq. (11.21) are built from t_2 alone and are generally non-zero. The residual is therefore not zero and $t_1 = 0$ is not a fixed point. Iterating from $t_1 = 0$ generates singles of the order of the doubles times a matrix element over an energy denominator, which is why they come out small – about 10^{-2} here.

Answer to c).

Run the code on a two-electron system. Singles and doubles exhaust the excitation space for two particles, so CCSD must equal full configuration interaction exactly. A solver missing the driver terms will converge to something close but not equal: the discrepancy in Table 11.4 would be of order 10^{-4} rather than 10^{-13}, and no amount of tightening the convergence threshold will remove it. The same test distinguishes a genuine convergence problem, which tightening does fix, from a wrong equation, which it does not.

Exercise 5: numerical explorations

a) Using `quantumdot.py`, reproduce Table 11.4 and extend it to seven shells. Fit the remaining error to $a/M + b/M^2$ and extrapolate to the complete basis. How close do you get to 3?
b) Compute the twelve-electron dot at $\hbar\omega = 1$ in the full basis. How does the fraction $E_{\mathrm{CCSD}}/(E_{\mathrm{HF}} - E_0)$ compare with the two- and six-electron values, and why?
c) Plot the Hartree-Fock and bare oscillator single-particle spectra for $N = 6$ and comment on Koopmans' theorem.
d) For the six-electron dot in the two-shell basis, optimise the unitary amplitudes properly rather than borrowing them from CCSD. How much further does the energy fall, and how many function evaluations does it take?
e) Repeat the Trotter study of Table 11.8 with the symmetric second-order splitting of Eq. (6.53) and confirm that the error now falls as $1/n^2$.
f) Estimate the number of qubits and Pauli strings needed for the six-electron dot in the two-shell basis, using Section 3.16. Compare with the pairing model of Table 4.7.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter11/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/quantumdots.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter11/`.

`quantumdot.py` The two-dimensional quantum dot: Coulomb integrals in closed form, the antisymmetrised two-body matrix, a fast two-electron full CI, and the whole chapter-10 hierarchy applied to it. Runs in a few minutes.
`ho1dim.py` Harmonic oscillator matrix elements in one dimension. Runs in seconds.
`ho2dim.py` Harmonic oscillator matrix elements in two dimensions. Runs in seconds.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Part III
Stochastic methods

Chapter 12
Elements from statistics

Every method so far has expanded the wave function in a basis. Hartree-Fock picked the best single determinant in that basis, perturbation theory and coupled cluster added determinants systematically, and full configuration interaction used all of them. The error was always a truncation – of the cluster operator, of the perturbation series, of the single-particle space – and Chapter 11 showed that for a realistic system the last of these can be the largest.

The stochastic methods of the coming chapters do something different. They never write down a basis at all. A variational Monte Carlo calculation takes a trial wave function, evaluates it at points drawn at random from $|\Psi_T|^2$, and averages; a diffusion Monte Carlo calculation lets an ensemble of walkers evolve under an equation that is the Schrödinger equation in imaginary time. Neither has a basis-set error. What they have instead is a *statistical* error, and controlling it needs a set of tools that has not appeared in this book yet.

That is what this chapter is for. We need probability distributions and their moments, the central limit theorem and the $1/\sqrt{N}$ law that follows from it, the correlation time that tells us how many of our samples were really independent, random walks and their relation to the diffusion equation, Markov chains and detailed balance, and finally the Metropolis algorithm – which is how one samples a distribution that can only be evaluated up to an unknown constant, the situation every many-body problem presents.

The examples are all Python and all in `montecarlo.py`, and wherever it is possible the running example is the quantum dot of Chapter 11. The chapter ends with a first variational Monte Carlo calculation on that system. The check that ties the two chapters together is worth stating in advance: with the variational parameter set to one, the trial function of Section 12.9 *is* the minimal-basis Hartree-Fock determinant, so the Monte Carlo energy must come out at 3.25331414, the first row of Table 11.2. It does.

12.1 Stochastic variables and probability distributions

A *stochastic variable* is characterised by two things: a *domain* $\mathbb{D} = \{x\}$ of the values it can take, and a *probability distribution function* (PDF) $p(x)$ on that domain. We write stochastic variables as capitals, $X, Y, \ldots$, and their values as lower case.

For a discrete domain the PDF is a probability outright, $p(x) = \mathrm{Prob}(X = x)$; the tossing of two dice has the domain $\{2, 3, \ldots, 12\}$ and the PDF $\{1/36, 2/36, \ldots, 1/36\}$. For a continuous domain $p(x)$ is a *density*: the probability of finding X in an infinitesimal interval about x is $p(x)\,dx$, and

$$\mathrm{Prob}(a \le X \le b) = \int_a^b p(x)\,dx. \tag{12.1}$$

The *cumulative distribution function* (CDF) is

$$P(x) = \mathrm{Prob}(X \le x) = \int_{-\infty}^{x} p(x')\,dx', \qquad p(x) = \frac{d}{dx}P(x), \tag{12.2}$$

and it will turn out to be the single most useful object in the chapter: it is what converts one distribution into another.

Every PDF must be positive and normalised,

$$p(x) \ge 0, \qquad \sum_{x_i \in \mathbb{D}} p(x_i) = 1 \quad \text{or} \quad \int_{\mathbb{D}} p(x)\,dx = 1. \tag{12.3}$$

Three continuous distributions will be used throughout: the uniform distribution on $[a,b]$, $p(x) = 1/(b-a)$; the exponential, $p(x) = \lambda e^{-\lambda x}$ on $[0,\infty)$, which is the PDF of radioactive decay; and the Gaussian or normal distribution

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \tag{12.4}$$

which the central limit theorem will single out.

Expectation values and moments.

For any function h on the domain,

$$\langle h \rangle = \int h(x)p(x)\,dx. \tag{12.5}$$

Taking $h(x) = x^n$ gives the *moments* $\langle x^n \rangle$. The zeroth moment is the normalisation, the first is the *mean* $\mu = \langle x \rangle$, and the second *central* moment is the *variance*,

$$\sigma^2 = \langle (x - \langle x \rangle)^2 \rangle = \langle x^2 \rangle - \langle x \rangle^2, \tag{12.6}$$

whose square root σ is the standard deviation, the “spread” of p about its mean. Two PDFs are equal if and only if all their moments are.

Transforming a distribution.

Let $Y = h(X)$ with h strictly monotonic, so that h^{-1} exists. Then $P_Y(y) = P_X(h^{-1}(y))$ for increasing h, and differentiating,

$$\boxed{p_Y(y) = p_X\left(h^{-1}(y)\right)\left|\frac{d}{dy}h^{-1}(y)\right|} \tag{12.7}$$

This is the formula that turns a uniform random number generator into a generator for anything we like, and Section 12.5 uses it repeatedly.

Several variables, and independence.

A PDF of one variable is *univariate*, of several *multivariate*. The variables $\{X_i\}$ are *independent* – equivalently *uncorrelated* – when the joint PDF factorises,

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} p_i(x_i). \tag{12.8}$$

Qualitatively: the events tell each other nothing. Independence is what makes the next two sections work, and its failure is what Section 12.3 is about.

12.2 The central limit theorem and the square-root law

Suppose we draw m values from a PDF $p(x)$ with mean μ and variance σ^2 and form their average $z=(x_1+\cdots+x_m)/m$. What is the PDF of z? Writing the constraint as a δ-function and using its integral representation,

$$\tilde{p}(z) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dq\, e^{iq(z-\mu)} \left[\int_{-\infty}^{\infty} dx\, p(x)\, e^{iq(\mu-x)/m}\right]^m . \tag{12.9}$$

Expanding the inner integral, the term linear in $(\mu-x)$ vanishes because μ is the mean, and what is left is

$$\int_{-\infty}^{\infty} dx\, p(x)\, e^{iq(\mu-x)/m} = 1 - \frac{q^2\sigma^2}{2m^2} + \cdots , \tag{12.10}$$

so that in the limit $m\to\infty$

$$\boxed{\tilde{p}(z) = \frac{1}{\sqrt{2\pi}\,(\sigma/\sqrt{m})}\exp\left(-\frac{(z-\mu)^2}{2(\sigma/\sqrt{m})^2}\right)} \tag{12.11}$$

The average of m samples from *any* distribution with a finite variance is normally distributed, with the same mean and with variance σ^2/m. Nothing about p survives except its first two moments.

Two consequences run through everything that follows. First, the error on a Monte Carlo average is

$$\sigma_m = \frac{\sigma}{\sqrt{m}} \qquad \text{or, for a measured variance,} \qquad \sigma_m \approx \frac{\sigma}{\sqrt{m-1}}, \tag{12.12}$$

independent of the dimensionality of the problem. A deterministic quadrature on a grid needs N^d points to reach a given accuracy in d dimensions; Monte Carlo needs N, whatever d is. That is the entire reason for using it on a many-body system. Second, the convergence is slow: a factor of ten in accuracy costs a factor of a hundred in samples. Everything called “variance reduction” is an attempt to shrink σ rather than to increase m.

Table 12.1 checks the theorem numerically on the least Gaussian distribution imaginable, the uniform one.

m	mean of averages	variance	σ^2/m
1	0.503888	0.084564	0.083333
2	0.498682	0.041927	0.041667
5	0.499093	0.016636	0.016667
10	0.500667	0.008348	0.008333
50	0.500089	0.001689	0.001667

Table 12.1 The central limit theorem for the uniform distribution on $[0,1)$, which has $\mu=1/2$ and $\sigma^2=1/12$. Each row is 20 000 averages of m samples. Computed by `montecarlo.py`.

Figure 12.1 is the same experiment drawn rather than tabulated, and it is worth looking at closely because the speed of the convergence is surprising. The parent distribution could

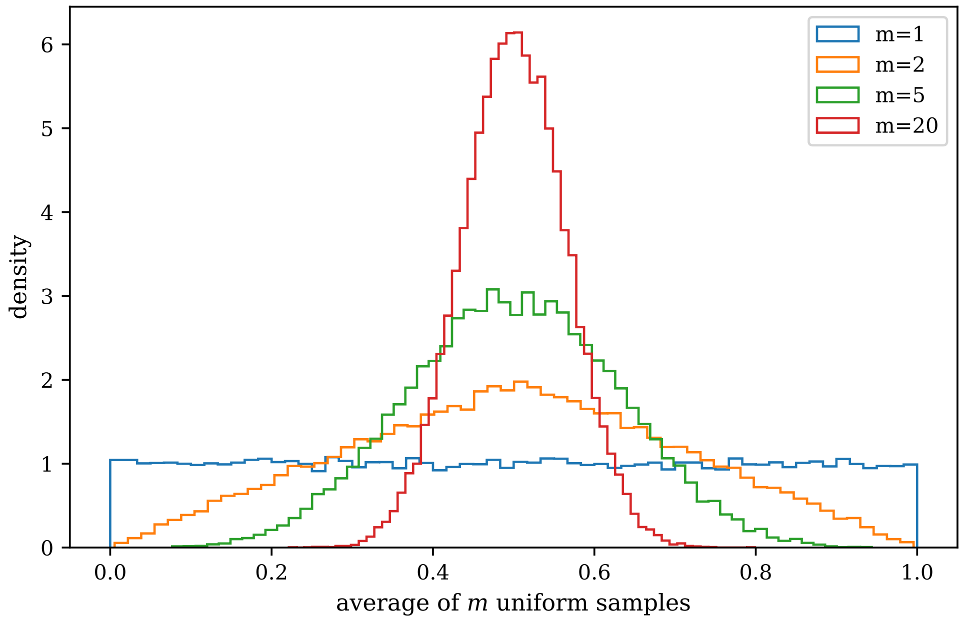

Fig. 12.1 Normalised histograms of 40000 averages of m samples drawn from the uniform distribution on $[0,1)$, for $m=1,2,5$ and 20, with sixty bins each. The parent distribution, $m=1$, is flat at unit density across the whole interval; averaging two samples already folds it into the triangular convolution that peaks at $z=1/2$, and by $m=5$ the shape is indistinguishable from a Gaussian by eye. The width contracts as $1/\sqrt{m}$ and the peak height grows as $\sqrt{m}$, exactly as Eq. (12.11) requires. Computed by `montecarlo.py`.

hardly be less Gaussian: it is flat on $[0,1)$ and drops discontinuously to zero at both ends. A single convolution, $m=2$, already replaces it by the triangle that rises linearly to a density of 2 at $z=1/2$, and a second and third, $m=5$, produce a curve whose peak sits at 3.09 – which is what $1/\sigma\sqrt{2\pi}$ gives for $\sigma=\sqrt{1/60}=0.129$ – and which no longer betrays its origin at all. At $m=20$ the histogram peaks at 6.15 against the predicted $1/\sqrt{2\pi/240}=6.18$ and is confined to roughly $0.38<z<0.62$: the standard deviation has fallen from 0.289 to 0.065, a factor of $\sqrt{20}=4.5$. Two or three samples are already enough for the error bar $\sigma/\sqrt{m}$ to be meaningful, and this is why Monte Carlo error bars can be quoted at all.

There is one caveat that Figure 12.1 also makes visible, and it will matter in Section 12.9. The average of m uniform samples can in principle take any value in $[0,1)$, yet the $m=20$ histogram has no entries whatsoever outside $[0.35,0.65]$. The central limit theorem is a statement about the bulk of the distribution; the extreme tails converge much more slowly, and for a parent distribution with a heavy tail they may not converge at all within any affordable sample. When the quantity being averaged is a local energy with a $1/r_{12}$ singularity in it, the tail is precisely where the trouble lives.

12.3 Correlated data and the true error

Equation (12.12) assumed independent samples. A Markov chain does not produce them: consecutive configurations of a Metropolis walk are similar by construction, and using $\sigma/\sqrt{n}$ then *understates* the error. This is the one mistake that makes a Monte Carlo result look better than it is, and it is worth setting up properly.

Writing the sample variance of mn measurements out in full,

$$\sigma_m^2 = \frac{\sigma^2}{n} + \frac{2}{mn^2} \sum_{\alpha=1}^{m} \sum_{k<l}^{n} \left(x_{\alpha,k} - \langle X_m \rangle\right) \left(x_{\alpha,l} - \langle X_m \rangle\right), \tag{12.13}$$

the first term is what the central limit theorem gives and the second is the covariance, which vanishes only if the measurements are independent. Collecting the covariance into partial sums over the separation d,

$$f_d = \frac{1}{nm} \sum_{\alpha=1}^{m} \sum_{k=1}^{n-d} \left(x_{\alpha,k} - \langle X_m \rangle\right) \left(x_{\alpha,k+d} - \langle X_m \rangle\right), \qquad \kappa_d = \frac{f_d}{\sigma^2}, \tag{12.14}$$

defines the *autocorrelation function* κ_d, which starts at $\kappa_0 = 1$ and decays. The variance becomes

$$\boxed{\sigma_m^2 = \frac{\tau}{n} \sigma^2, \qquad \tau = 1 + 2 \sum_{d=1}^{n-1} \kappa_d} \tag{12.15}$$

with τ the *autocorrelation time*. For independent data $\tau = 1$. Otherwise the effect of correlation is exactly a reduction of the number of measurements,

$$n_{\text{eff}} = \frac{n}{\tau}, \tag{12.16}$$

and the honest error bar is $\sqrt{\tau}$ times the naive one.

How much this matters is easy to underestimate. Sampling a unit Gaussian by Metropolis with a step of 0.5 – a perfectly reasonable-looking choice, with an acceptance rate of 90% – gives $\tau = 54.5$ over 100 000 samples. The naive error is 0.0031 and the true one 0.0231, a factor of seven, and the 100 000 measurements were worth 1836 independent ones.

Blocking.

Computing τ requires the whole time series and a decision about where to truncate the sum, which is awkward. The *blocking* method of Flyvbjerg and Petersen avoids both. Average neighbouring pairs of measurements to produce a shorter series, compute the naive error of that, and repeat. The naive error grows as the blocks get longer and correlations are averaged away, and then plateaus; the plateau is the true error. For the chain above the naive error climbs from 0.0031 through 0.0085 and 0.0181 to a plateau near 0.022, which is the corrected error found from τ. No autocorrelation function, and no truncation decision.

Figure 12.2 puts the two procedures side by side, and the comparison is instructive. The left panel is the raw evidence for the correlation: κ_d starts at unity by construction and decays smoothly, passing 0.5 near $d = 20$ and 0.1 near $d = 70$, so that consecutive measurements from this chain carry almost the same information and even measurements seventy apart are not yet independent. Summing the curve according to Eq. (12.15) gives $\tau = 54.5$. The difficulty of the method is visible in the same panel: beyond $d \approx 180$ the autocorrelation function is buried in noise, wandering to about -0.03 near $d = 240$ before returning. Those fluctuations are pure sampling error, they do not decrease with the lag, and if the sum in Eq. (12.15) is carried out to $d = n - 1$ they accumulate into nonsense. One must therefore truncate, and the truncation point is a judgement.

The right panel shows why the blocking method is preferred in practice. At a block size of one, the estimate is the naive 0.0031 and it is wrong by a factor of seven. As neighbouring measurements are averaged together the correlations inside a block are absorbed into the block mean and the estimate climbs – 0.0083 at a block size of eight, 0.0179 at sixty-four – until at a block size of a few hundred the blocks are longer than the correlation time, become independent of one another, and the curve flattens onto the dashed line at 0.023. That plateau

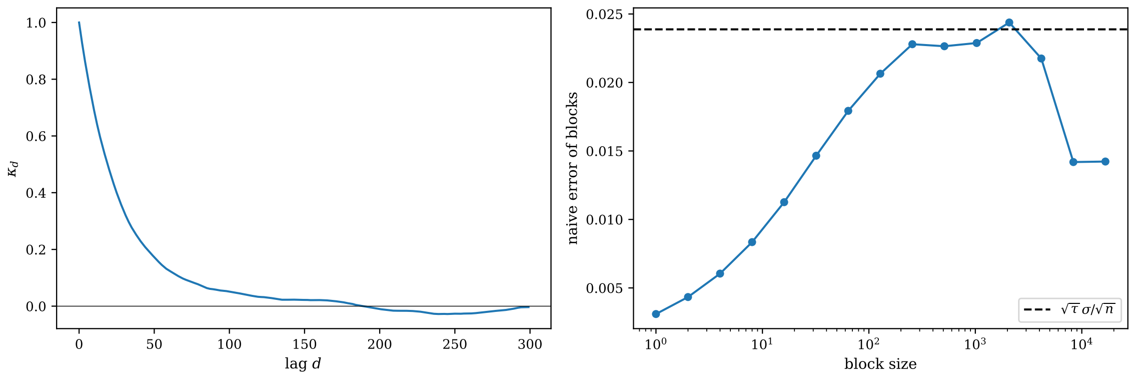

Fig. 12.2 The two ways of extracting an honest error bar, for a Metropolis chain of 100 000 samples of the unit Gaussian with a proposal step of 0.5. Left: the autocorrelation function κ_d of Eq. (12.14) against the lag d, out to $d = 300$, with the zero line drawn in. Right: the naive error $\sigma/\sqrt{n}$ of the blocked series against the block size on a logarithmic abscissa, each point obtained by averaging neighbouring pairs of the previous one; the dashed line is the corrected error $\sqrt{\tau}\,\sigma/\sqrt{n}$ computed from the autocorrelation time $\tau = 54.5$ of the left panel. The blocking curve rises onto that line and stays there for block sizes between about 250 and 2000, and collapses beyond it once too few blocks are left to estimate a variance from. Computed by `montecarlo.py`.

is the answer, and nothing had to be decided to find it. The collapse of the last two points, back to 0.015, is not a failure of the method but the expected one: at a block size of 16 384 only six blocks remain, and a variance estimated from six numbers is itself uncertain by some 30%. The rule for reading such a plot is to take the plateau and to distrust everything to the right of the point where the number of blocks falls below about twenty.

The practical rule. Always report an error obtained either from the autocorrelation time or from blocking. A naive $\sigma/\sqrt{N}$ from a Markov chain is not a small underestimate; in the example above it is wrong by a factor of seven, and in a poorly tuned calculation it can be wrong by orders of magnitude.

12.4 Random numbers

Everything rests on a supply of numbers uniformly distributed on $[0,1)$. Real generators are deterministic algorithms – hence *pseudo*-random – and the classic construction is the linear congruential generator

$$x_{n+1} = (a x_n + c) \bmod M, \tag{12.17}$$

which is fast, has period at most M, and can be very bad. The instructive failure is RANDU, with $a = 65539$, $c = 0$ and $M = 2^{31}$, once shipped with IBM's scientific library. It is uniform, and its consecutive values are uncorrelated at lag one. But it satisfies

$$x_{n+2} = 6x_{n+1} - 9x_n \pmod 1 \tag{12.18}$$

exactly, so every triple of consecutive numbers lies on one of fifteen planes in the unit cube. A one-dimensional integral sampled with RANDU is fine; a three-dimensional one is systematically wrong. Table 12.2 shows the two tests that pass and the one that fails.

Figure 12.3 is the picture that goes with the last column of the table, and the honest thing to say about it is that the defect is not obvious. Both cubes look filled, both look isotropic, and a reader shown them without the labels would be hard pressed to say which generator was the broken one. That is not an accident of the plot. Equation (12.18) confines the RANDU

generator	χ^2, ten bins	lag-one correlation	triples on planes
RANDU	8.46	0.0051	100.0%
PCG64	11.39	−0.0108	0.0%

Table 12.2 Two generators on 20000 numbers. Both are uniform and both are uncorrelated at lag one; only the three-dimensional test separates them. Computed by `montecarlo.py`.

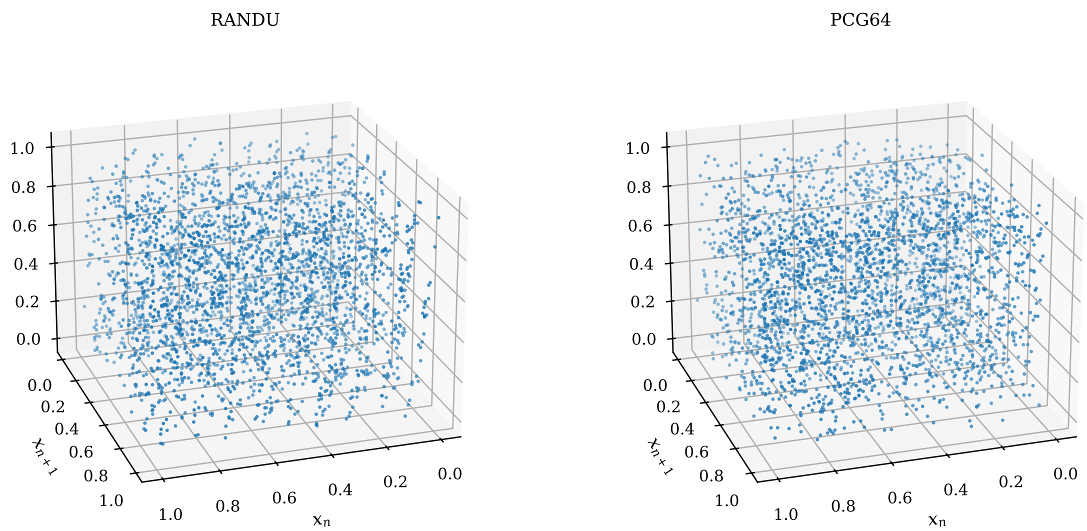

Fig. 12.3 Three thousand consecutive triples (x_n, x_{n+1}, x_{n+2}) from RANDU (left) and from `numpy`'s PCG64 (right), plotted as points in the unit cube and viewed from an elevation of 22° and an azimuth of 72°. Every RANDU point lies on one of the fifteen parallel planes $9x_n - 6x_{n+1} + x_{n+2} = k$, $k = -5, \dots, 9$, which are spaced $1/\sqrt{118} = 0.092$ apart; from the generic direction used here the planes are seen obliquely and the two clouds are hard to tell apart by eye. Computed by `montecarlo.py`.

triples to the fifteen planes $9x_n - 6x_{n+1} + x_{n+2} = k$ with k integer, and those planes are separated by only $1/\sqrt{118} = 0.092$ in the unit cube, so along any line of sight that is not close to the normal $(9, -6, 1)/\sqrt{118}$ they overlap and the structure washes out. Rotate the cube until one looks straight down that normal and the three thousand points collapse onto fifteen sharp lines with nothing in between; from every other direction the failure is invisible. The right-hand panel is what a sound generator does: the PCG64 points fill the cube with no preferred direction at all, and rotating it changes nothing.

The practical moral is stronger than "inspect your random numbers". Three thousand points in three dimensions is far too coarse a test to be trusted by eye, and the only reliable statement in this whole comparison is the 100.0% against 0.0% of Table 12.2, which was obtained by evaluating the linear combination and asking how many triples fall within a tolerance of an integer. A many-body configuration is a point in $3N$ dimensions, and there is no picture of it at all. The lesson is that a generator must be tested in the dimension in which it will be used. We use `numpy`'s PCG64 throughout, always seeded explicitly, so that every number in this chapter can be reproduced exactly.

12.5 Monte Carlo integration and importance sampling

The crudest scheme takes

$$I = \int_0^1 f(x)\,dx = \langle f \rangle \approx \frac{1}{N}\sum_{i=1}^{N} f(x_i), \qquad x_i \sim \mathscr{U}(0,1), \tag{12.19}$$

with the error $\sigma/\sqrt{N}$ from Eq. (12.12). Applied to $\int_0^1 4/(1+x^2)\,dx = \pi$ the product error $\times\sqrt{N}$ stays at 0.64 from $N = 10^3$ to $N = 10^6$, which is the square-root law in the most direct form available.

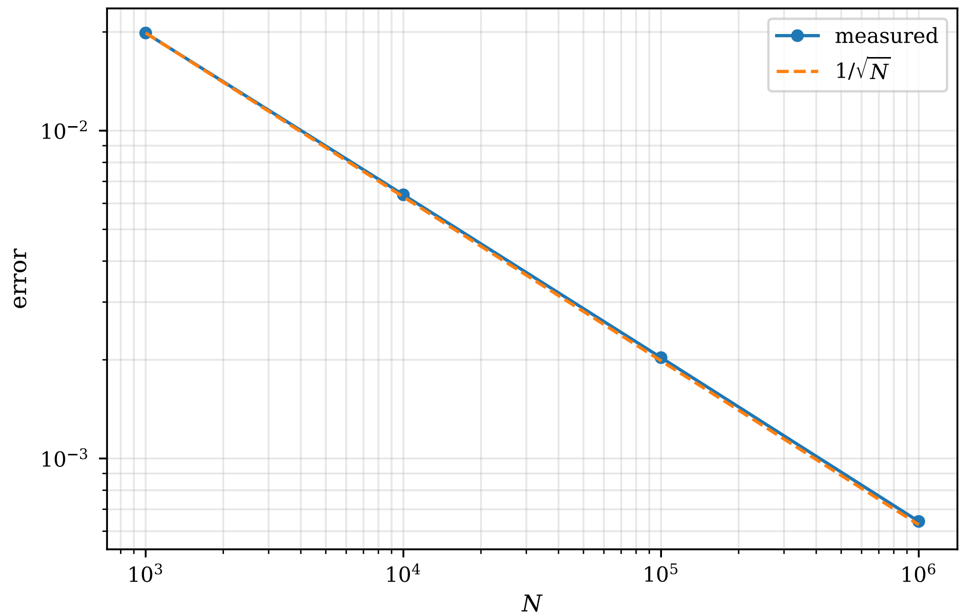

Fig. 12.4 Statistical error of the brute-force Monte Carlo estimate of $\int_0^1 4/(1+x^2)\,dx = \pi$, plotted logarithmically against the number of samples N over four decades. The solid line with circles is the error returned by the calculation itself, $\sigma/\sqrt{N}$ with σ the measured standard deviation of the integrand; the dashed line is the pure $1/\sqrt{N}$ behaviour normalised to the first point. The two are indistinguishable over the whole range, the error falling from 2.0×10^{-2} at $N = 10^3$ to 6.4×10^{-4} at $N = 10^6$. Computed by `montecarlo.py`.

Figure 12.4 draws that statement over four decades, and the measured curve lies on the reference line to within the thickness of the marker at every point. The slope is $-1/2$: a thousandfold increase in the number of samples buys a factor of $\sqrt{1000} = 31.6$ in accuracy, and the error indeed drops from 2.0×10^{-2} to 6.4×10^{-4}. Both faces of Eq. (12.12) are on display. The encouraging one is that the line is straight and that nothing in the argument that produced it mentioned the dimension of the integral; the same plot would be obtained for the six-dimensional integral of Eq. (12.24) below, or for the $3N$-dimensional integral of a many-body expectation value, whereas Simpson's rule with its h^4 error would have needed N^d abscissas to keep the same h. The discouraging one is that the line is straight all the way to $N = 10^6$ with no sign of steepening: there is no regime in which brute force suddenly becomes efficient. Getting one more decimal place from Figure 12.4 means $N = 10^{10}$. Every technique in the rest of the chapter – importance sampling here, the Jastrow factor in Section 12.9 – works by moving the line downwards rather than by tilting it.

Changing variables.

Random numbers arrive uniform on $[0,1)$; integrals rarely do. Conserving probability, $p(y)\,dy = p(x)\,dx = dx$, and integrating,

$$\boxed{x = P(y) = \int_{-\infty}^{y} p(y')\,dy'} \tag{12.20}$$

so that $y = P^{-1}(x)$ generates p from a uniform x. Two examples. The uniform distribution on $[a,b]$ gives $y = a + (b-a)x$. The exponential $p(y) = \lambda e^{-\lambda y}$ has $P(y) = 1 - e^{-\lambda y}$ and therefore

$$y = -\frac{1}{\lambda}\ln(1-x), \tag{12.21}$$

which `montecarlo.py` confirms has mean and variance 1 at $\lambda = 1$. The Gaussian CDF cannot be inverted in closed form, and the standard trick is to do two dimensions at once: with u_1, u_2 uniform,

$$y = \sqrt{-2\ln(1-u_1)}\,\cos(2\pi u_2) \tag{12.22}$$

is normally distributed with zero mean and unit variance. This is the Box-Muller transformation, and it is Eq. (12.7) applied to the polar form of the two-dimensional Gaussian.

Acceptance-rejection.

When even that fails there is von Neumann's method: bound p by M on $[a,b]$, draw x uniformly on $[a,b]$ and s uniformly on $[0,M]$, and accept x if $s < p(x)$. It needs nothing but the ability to evaluate p, and it wastes a fraction $1 - \bar{p}/M$ of the samples. Applied to $\int_0^3 e^x\,dx$ it gives 19.0375 ± 0.0626 against the exact 19.0855.

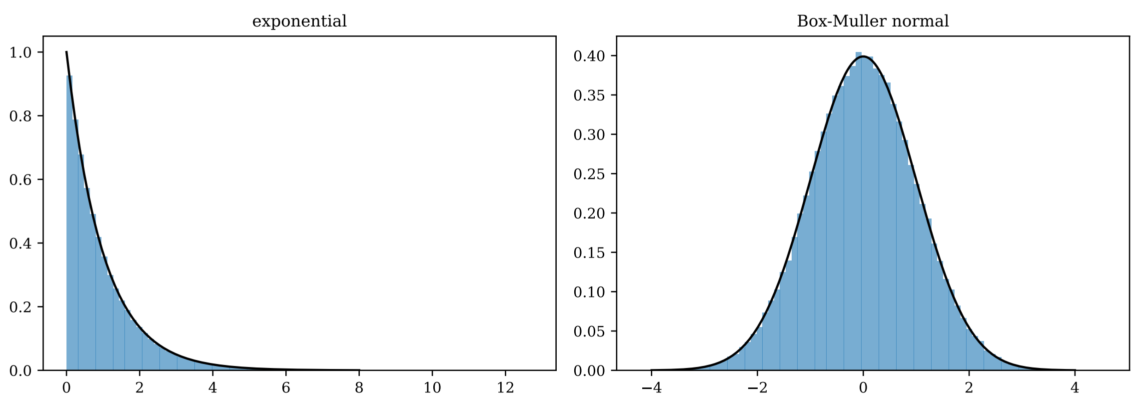

Fig. 12.5 Two hundred thousand samples produced by transforming uniform random numbers, shown as normalised histograms of eighty bins with the target density drawn as a solid black curve. Left: the exponential distribution $p(y) = e^{-y}$ generated by the inverse cumulative distribution (12.21) at $\lambda = 1$. Right: the standard normal distribution generated by the Box-Muller transformation (12.22). The measured moments are mean 1.0000 and variance 1.0000 for the exponential and mean 0.0000 and variance 1.0000 for the normal, in both cases the exact values. Computed by `montecarlo.py`.

Figure 12.5 confirms both transformations against the densities they are supposed to produce. There is nothing approximate about either construction – Eq. (12.7) is an identity, not a fit – and the agreement of the histograms with the analytic curves is limited only by the $\sqrt{n_{\text{bin}}}$ noise of the bin contents, which at 200 000 samples spread over eighty bins is a fraction of a per cent near the peak. It is worth noticing where the two panels stop. The exponential histogram has entries out to $y \approx 13$, which is where one expects the largest of 200 000 exponential variates to fall, since $\text{Prob}(y > \ln 200\,000) = 1/200\,000$ and $\ln 200\,000 = 12.2$; the sampler is reproducing the tail of the distribution and not merely its bulk. The Gaussian panel, by contrast, is empty beyond $|y| \approx 4.5$, and would remain so however long we ran: this is the difference between an exponential tail and a Gaussian one, and it is the reason a Gaussian proposal in Section 12.8 never throws a walker to the other side of the system.

Importance sampling.

The real gain comes from choosing what to sample. If p is a normalised PDF resembling the integrand,

$$I = \int_a^b F(y)\,dy = \int_a^b p(y)\,\frac{F(y)}{p(y)}\,dy \approx \frac{1}{N}\sum_{i=1}^{N}\frac{F(y_i)}{p(y_i)}, \qquad y_i \sim p, \tag{12.23}$$

and the variance is that of F/p rather than of F. If p follows F closely the ratio is nearly constant and the variance nearly vanishes. The conditions on p are that it be normalisable and positive, analytically integrable, and that its integral be invertible so that Eq. (12.20) can be used.

The six-dimensional integral

$$I = \int d^3x\,d^3y\; e^{-x^2-y^2}\,(\boldsymbol{x}-\boldsymbol{y})^2 = 3\pi^3 = 93.018830\ldots \tag{12.24}$$

makes the point. Sampling uniformly in a box wastes almost every point on the exponential tails. Writing $e^{-x^2} = \sqrt{\pi}$ times a normal density of variance $\frac{1}{2}$ and sampling that instead leaves only the polynomial $(\boldsymbol{x}-\boldsymbol{y})^2$ to average. Table 12.3 compares the two.

N	brute force	importance sampling
10^4	81.33 ± 31.36	91.44 ± 0.74
10^5	86.76 ± 10.42	93.08 ± 0.24
10^6	91.21 ± 3.88	92.99 ± 0.08

Table 12.3 The six-dimensional integral (12.24), exact value $3\pi^3 = 93.018830$. At $N = 10^6$ importance sampling is 51 times more accurate, which by the square-root law is a factor of 2600 in computer time. Computed by `montecarlo.py`.

Why quantum mechanics forces this on us.

The expectation value of a Hamiltonian in a trial state can be rewritten as

$$\langle H\rangle = \int dx\,\Psi^*(x)\hat{H}\Psi(x) = \int dx\,|\Psi(x)|^2\,\frac{\hat{H}\Psi(x)}{\Psi(x)} = \int dx\,p(x)\,E_L(x), \tag{12.25}$$

with $p = |\Psi|^2$ the quantum-mechanical probability density and

$$\boxed{E_L(x) = \frac{\hat{H}\Psi(x)}{\Psi(x)}} \tag{12.26}$$

the *local energy*. This is importance sampling with the best possible choice of p, and it has a remarkable property. If Ψ is an exact eigenfunction then E_L is the constant E everywhere, the average is E with *zero variance*, and a single sample suffices. For an approximate Ψ the local energy is smooth near the exact solution and the variance is small. The whole of variational

Monte Carlo is contained in Eq. (12.25), and the variance of E_L is a direct measure of how good the trial function is – often a more sensitive one than the energy itself.

12.6 Random walks and the diffusion equation

Sampling $|\Psi|^2$ for a many-body system cannot be done by inverting a cumulative distribution: we do not know the normalisation, and in $3N$ dimensions we could not invert it if we did. What is done instead is to build a random walk whose stationary distribution is the one we want. This section sets up the machinery; the next turns it into an algorithm.

A walker on a line takes steps $\pm l$ with equal probability. After n steps

$$\langle x \rangle = 0, \qquad \langle x^2 \rangle = n\,l^2, \tag{12.27}$$

so the walker spreads as $\sqrt{t}$ rather than t: this is diffusion, not transport. Simulating 4000 walkers reproduces $\langle x^2 \rangle / n = 1.00$ to two decimals at every n from 25 to 400.

That this really is diffusion can be shown exactly. Fick's law states that the particle flux is proportional to the gradient of the density, $j = -D\,\partial w/\partial x$, and combining it with the continuity equation gives the diffusion equation

$$\frac{\partial w(x,t)}{\partial t} = D\,\frac{\partial^2 w(x,t)}{\partial x^2}. \tag{12.28}$$

Now discretise the walk instead. With $x = il$, $t = n\varepsilon$ and the transition probability

$$W_{ij} = \begin{cases} \frac{1}{2} & |i-j| = 1 \\ 0 & \text{otherwise,} \end{cases} \tag{12.29}$$

the distribution after one step obeys $w_i(t+\varepsilon) = \sum_j W_{ij} w_j(t)$, that is

$$w(x,t+\varepsilon) = \tfrac{1}{2} w(x+l,t) + \tfrac{1}{2} w(x-l,t). \tag{12.30}$$

Subtracting $w(x,t)$ from both sides and multiplying by l^2/ε,

$$\frac{w(x,t+\varepsilon) - w(x,t)}{\varepsilon} = \frac{l^2}{2\varepsilon}\,\frac{w(x+l,t) - 2w(x,t) + w(x-l,t)}{l^2}, \tag{12.31}$$

which is precisely the discretised diffusion equation (12.28) with

$$D = \frac{l^2}{2\varepsilon}. \tag{12.32}$$

A random walk *is* the diffusion equation, and iterating the transition matrix *is* solving it. Starting all walkers at the origin, the exact solution of the discrete problem is the binomial distribution

$$w_i(n\varepsilon) = \frac{1}{2^n} \binom{n}{\frac{1}{2}(n+i)}, \qquad |i| \le n, \tag{12.33}$$

which by the central limit theorem tends to a Gaussian of variance $2Dt$. This connection is the reason diffusion Monte Carlo works at all: the Schrödinger equation in imaginary time is a diffusion equation, and it can therefore be solved by letting walkers diffuse.

Figure 12.6 shows the binomial distribution (12.33) forming. What the reader should take from it is the *rate* at which the distribution broadens rather than its shape. Between 25 and 400 steps the elapsed time grows by a factor of sixteen and the width by a factor of four, which

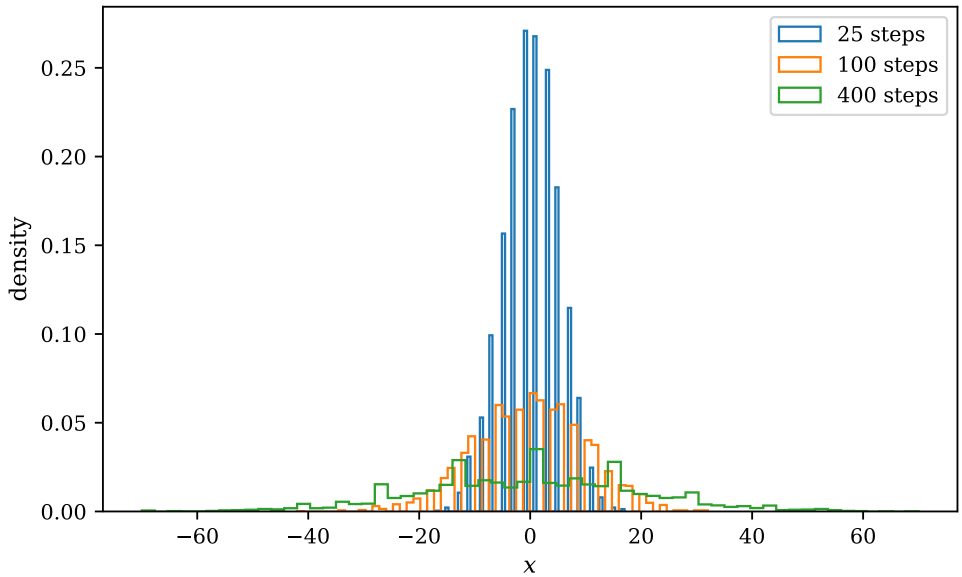

Fig. 12.6 Normalised histograms of the positions of 4000 unbiased walkers on a line after 25, 100 and 400 steps of unit length, sixty bins each. The distributions stay centred on the origin and broaden, with standard deviations 5, 10 and 20 in units of the step length, which is $\sqrt{n}$ as Eq. (12.27) requires; a fourfold increase in the elapsed time widens the distribution by a factor of two, not of four. The picket-fence appearance of the 25-step histogram is the parity of the walk: after an odd number of steps only odd sites can be occupied, so alternate bins are empty. Computed by `montecarlo.py`.

is diffusive spreading and not transport: a walker that has taken four hundred steps of unit length has travelled a path of length 400 but is, typically, only 20 away from where it started. The corresponding entry in the numerical table is $\langle x^2 \rangle / n = 1.00$ at every n, so the coefficient in Eq. (12.27) is unity to the accuracy of 4000 walkers. The shape is the second lesson: the 25-step histogram is visibly a discrete comb, whereas by 400 steps the envelope is a smooth Gaussian of variance 400 – the central limit theorem of Section 12.2 applied to the sum of 400 independent ± 1 steps, which is why Eq. (12.33) tends to $\exp(-x^2/4Dt)$. In a diffusion Monte Carlo calculation this Gaussian is exactly the short-time Green's function that propagates the walkers, and the width $\sqrt{2D\Delta t}$ that Figure 12.6 makes visible is what sets the time step.

Equilibrium and entropy.

Left on an infinite line the walkers spread for ever. On a finite ring they cannot, and what happens instead is worth watching. Define the entropy of the walker distribution,

$$S = -\sum_i w_i \ln w_i, \tag{12.34}$$

which is zero when all walkers sit on one site and $\ln N_{\text{sites}}$ when they are spread evenly. For 4000 walkers on 101 sites with periodic boundaries, S rises from 0.69 after one step through 3.05 after a hundred to 4.55 after three thousand, approaching the uniform value $\ln 101 = 4.6151$ and then staying there.

Figure 12.7 is the whole of the second law on one axis, and the first point on the curve is exact: after one step every walker is at $+1$ or at -1 with probability one half, the distribution has two equally occupied sites, and $S = \ln 2 = 0.6931$. From there the curve is monotone and strongly concave. Half of the total rise is over within the first fifty steps and the remainder

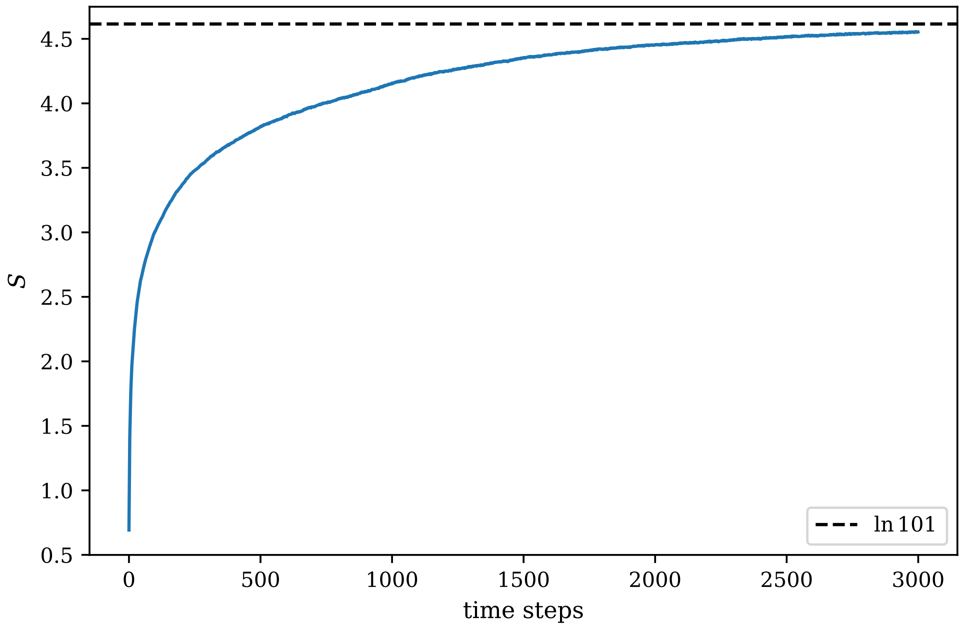

Fig. 12.7 Entropy $S = -\sum_i w_i \ln w_i$ of the distribution of 4000 walkers on a ring of 101 sites with periodic boundaries, against the number of steps. The dashed line is the uniform value $\ln 101 = 4.6151$, the largest entropy the distribution can have. Starting from all walkers at one site the entropy rises steeply – it is $\ln 2 = 0.693$ after a single step, when the walkers are split evenly between the two neighbours of the origin – and then flattens as the ring fills, reaching 4.55 after three thousand steps. Nothing was imposed to make it do so. Computed by `montecarlo.py`.

takes the other two thousand nine hundred and fifty, which is characteristic of diffusive relaxation: filling a ring of 101 sites requires the walkers to travel a distance of order 50, and a diffusing walker needs of order $50^2 = 2500$ steps to do it. The approach is asymptotic and the plateau is reached from below only. There is also a subtler brake on the curve, and it is the same parity that produced the comb in Figure 12.6: a step always changes the parity of the site index, and on a ring with an *odd* number of sites the only way to change sublattice is to wrap all the way round. Until that has happened the walkers occupy every second site and the entropy is capped near $\ln 51 = 3.93$, which is why the curve loiters around 3.8 at five hundred steps before climbing again.

The system has reached a steady state, and nothing in the dynamics put it there. It happened because the walk can reach every site from every other, given enough time. That property is called *ergodicity*, and it is the first of the two conditions the next section will impose.

12.7 Markov chains, detailed balance and the Metropolis algorithm

A *Markov process* is a random walk in which the next state depends only on the present one. Written as a matrix acting on the probability vector,

$$w_i(t+1) = \sum_j W_{ij} w_j(t), \qquad \sum_i W_{ij} = 1, \tag{12.35}$$

and iterating drives w towards the eigenvector of $\boldsymbol{W}$ with eigenvalue one – the steady state. For the four-state example

$$\boldsymbol{W} = \begin{pmatrix} 1/4 & 1/9 & 3/8 & 1/3 \\ 2/4 & 2/9 & 0 & 1/3 \\ 0 & 1/9 & 3/8 & 0 \\ 1/4 & 5/9 & 2/8 & 1/3 \end{pmatrix}, \tag{12.36}$$

starting from $\boldsymbol{w} = (1,0,0,0)$ the iteration converges in 22 steps to

$$\boldsymbol{w}_\infty = (0.244318,\ 0.319602,\ 0.056818,\ 0.379261),$$

which is exactly the eigenvector, computed independently. The starting point is irrelevant: any initial distribution converges to the same steady state.

The master equation and detailed balance.

In continuous time Eq. (12.35) becomes the master equation

$$\frac{dw_i(t)}{dt} = \sum_j \left[W(j \to i)w_j - W(i \to j)w_i\right], \tag{12.37}$$

the rate in balanced against the rate out. At the steady state the left-hand side vanishes, but that alone does not guarantee convergence to the right distribution – cyclic solutions can satisfy it. The stronger condition, which does, is *detailed balance*:

$$\boxed{W(j \to i)\,w_j = W(i \to j)\,w_i \qquad \Longleftrightarrow \qquad \frac{W(j \to i)}{W(i \to j)} = \frac{w_i}{w_j}} \tag{12.38}$$

Every pair of states exchanges probability at equal rates in both directions. Note what has happened: only the *ratio* w_i/w_j appears. The normalisation of the target distribution has disappeared, and for a many-body system that normalisation is exactly what we cannot compute.

The algorithm.

Split the transition probability into a proposal and an acceptance, $W(i \to j) = T(i \to j)A(i \to j)$. If the proposal is symmetric, $T(i \to j) = T(j \to i)$, detailed balance requires

$$\frac{A(j \to i)}{A(i \to j)} = \frac{w_i}{w_j}. \tag{12.39}$$

This fixes only the ratio. To accept as many moves as possible we set the larger of the two to one, giving the Metropolis choice

$$\boxed{A(i \to j) = \min\left(1,\ \frac{w_j}{w_i}\right)} \tag{12.40}$$

A move to a more probable state is always accepted; a move to a less probable one is accepted with probability w_j/w_i, tested against a uniform random number. The algorithm is then:

1. Start from a configuration $x^{(0)}$.
2. Propose x' from a symmetric distribution $T(x'|x^{(k)})$.

3. Accept, $x^{(k+1)} = x'$, with probability $\min(1, w(x')/w(x^{(k)}))$; otherwise set $x^{(k+1)} = x^{(k)}$ and keep the old configuration.
4. Discard an initial burn-in, then accumulate averages.

Rejected moves count: the old configuration is measured again. Omitting it biases the average.

Choosing the step.

The proposal width is the one free parameter and it matters more than it looks. Table 12.4 shows why. A small step is almost always accepted but moves nowhere; a large one is almost always rejected and the walker stands still. Both produce long correlation times and therefore large error bars, and the optimum is in between. The usual rule of thumb is to tune for an acceptance rate near one half.

step	acceptance	τ	$\langle x^2 \rangle$
0.1	97.7%	585.10	0.6226 ± 0.0314
0.5	88.2%	39.37	0.6247 ± 0.0071
2.0	57.3%	4.03	0.6166 ± 0.0031
8.0	16.1%	9.65	0.6183 ± 0.0058
32.0	4.0%	43.51	0.5835 ± 0.0118

Table 12.4 Metropolis sampling of $\exp(-x^2/2 - x^4/10)$, whose normalisation is never computed, with 200 000 samples at each step size. The error bars are corrected for the autocorrelation time by Eq. (12.15). Computed by `montecarlo.py`.

Figure 12.8 turns the table into the shape one should carry around, and the shape is a V. Both branches are easy to understand and they fail for opposite reasons. On the left the proposal is tiny, so the ratio $w(x')/w(x)$ is close to unity, almost every move is accepted – 98% at a step of 0.1 – and the chain nevertheless goes nowhere, because it needs hundreds of accepted moves to cross the distribution. On the right the proposal is enormous compared with the width of the target, almost every proposal lands where w is negligible, and the acceptance rate falls to 4%; the chain now stands still literally rather than figuratively, since 96% of the entries in the time series are copies of their predecessor. In between, at a step of 2.0, the two effects balance and $\tau = 4.0$. The penalty for getting it wrong is not small: by Eq. (12.16) a correlation time of 585 means that only one measurement in 585 was genuinely independent, against one in four at the optimum, so that the run at a step of 0.1 threw away all but a fraction 10^{-2} of its effort. The error bars in the last column of Table 12.4 differ by a factor of ten between the two.

That the minimum is broad is as important as where it lies. Anywhere between a step of 1 and a step of 4 would have given a correlation time within a factor of two of the best, so tuning the step to an acceptance rate of roughly one half is a rule of thumb that costs nothing to follow and rewards no further effort. What one must not do is take a high acceptance rate as evidence that the sampling is going well. It is the most natural diagnostic to look at and it points the wrong way: the worst chain in Figure 12.8 is the one with the highest acceptance rate. The right diagnostic is τ itself, or equivalently a blocking analysis of the kind drawn in Figure 12.2. Section 12.8 shows how to escape the compromise altogether by making the

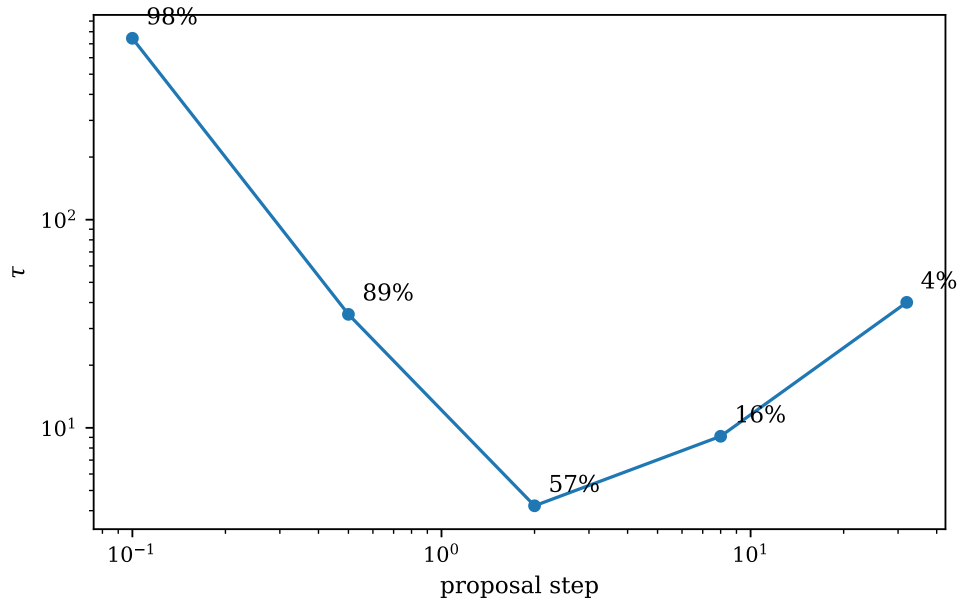

Fig. 12.8 Autocorrelation time τ of a Metropolis chain sampling $\exp(-x^2/2 - x^4/10)$, plotted logarithmically against the width of the symmetric proposal, for the five step sizes of Table 12.4. Each point is annotated with the acceptance rate the chain achieved. The curve is V-shaped with a minimum of $\tau = 4.0$ at a step of 2.0 and an acceptance rate of 57%; a step of 0.1, which is accepted 98% of the time, and a step of 32, which is accepted 4% of the time, both give correlation times an order of magnitude longer. Computed by `montecarlo.py`.

proposal asymmetric and letting it drift towards the regions where w is large, which buys a short correlation time and a high acceptance rate at the same time.

12.8 Fokker-Planck and Langevin equations

A symmetric proposal is wasteful: it proposes moves into regions where the target density is negligible, and they are rejected. The cure is to bias the proposal towards the high-density regions, which requires knowing how a diffusing particle behaves under a force. That is the content of the Fokker-Planck and Langevin equations, and it is the basis of the importance-sampled Monte Carlo of the next chapters.

Consider a transition probability $W(\boldsymbol{x},t|\boldsymbol{x}_0,t_0)$. Composing two successive transitions gives the Einstein-Smoluchowski-Kolmogorov-Chapman relation

$$W(\boldsymbol{x},t|\boldsymbol{x}_0,t_0) = \int d\boldsymbol{x}'\, W(\boldsymbol{x},t|\boldsymbol{x}',t')\, W(\boldsymbol{x}',t'|\boldsymbol{x}_0,t_0), \tag{12.41}$$

which simply says that the particle must be somewhere at the intermediate time. Expanding for a short time step in the moments

$$M_n = \frac{1}{\tau}\int d\xi\, \xi^n\, W(\boldsymbol{x}+\xi,\tau|\boldsymbol{x}) = \frac{\langle[\Delta x(\tau)]^n\rangle}{\tau}, \tag{12.42}$$

and keeping only $n=1$ and $n=2$ – higher moments vanish faster than τ for a diffusive process – gives the *Fokker-Planck equation*

$$\boxed{\frac{\partial W}{\partial t} = -\frac{\partial}{\partial x}(M_1 W) + \frac{1}{2}\frac{\partial^2}{\partial x^2}(M_2 W)} \tag{12.43}$$

The first term is drift and the second diffusion. With $M_2 = 2D$ constant and M_1 a drift velocity the solution is a Gaussian, and it is that Gaussian which will be used as the proposal distribution T.

The same physics from the particle's point of view is the *Langevin equation*. A particle in a fluid feels a systematic drag proportional to its velocity and a random force from the molecular collisions,

$$\frac{d\boldsymbol{v}}{dt} = -\xi\boldsymbol{v} + \boldsymbol{F}(t). \tag{12.44}$$

Requiring that the velocity distribution relax to the equipartition value $\langle v^2\rangle = 3kT/m$ fixes the strength of the random force,

$$\langle \boldsymbol{F}(t)\cdot\boldsymbol{F}(t')\rangle = 6\,\frac{kT}{m}\,\xi\,\delta(t-t'), \tag{12.45}$$

the *fluctuation-dissipation theorem*: the same collisions that damp the motion also drive it, and their magnitudes are not independent. Integrating the Langevin equation gives the mean square displacement, which for long times becomes $\langle(\boldsymbol{r}-\boldsymbol{r}_0)^2\rangle = 6kTt/m\xi$, and comparing with $\langle(\boldsymbol{r}-\boldsymbol{r}_0)^2\rangle = 6Dt$ gives the Einstein relation

$$D = \frac{kT}{m\xi}. \tag{12.46}$$

Where this is going. In a quantum Monte Carlo calculation the role of the drift is played by the *quantum force* $\boldsymbol{F} = 2\nabla\Psi_T/\Psi_T$, which pushes the walkers towards regions where the trial function is large. Proposing moves from the Fokker-Planck Green's function rather than uniformly, and correcting the acceptance ratio for the resulting asymmetry of T – the Metropolis-Hastings algorithm – raises the acceptance rate and shortens the correlation time substantially. The same equation, with the drift supplied by the trial function and the diffusion by the kinetic energy, is what propagates the walkers in diffusion Monte Carlo.

12.9 A first variational Monte Carlo calculation

We can now put all of it to work on the two-electron quantum dot of Chapter 11. Take the trial function

$$\Psi_T(\boldsymbol{r}_1,\boldsymbol{r}_2;\alpha,\beta) = \exp\left(-\frac{\alpha\omega}{2}\left(r_1^2+r_2^2\right)\right)\exp\left(\frac{a\,r_{12}}{1+\beta r_{12}}\right), \tag{12.47}$$

the first factor a scaled oscillator orbital for each electron and the second a Jastrow factor with $a = 1$ fixed by the Coulomb cusp condition for two electrons of opposite spin in two dimensions. The energy is the average of the local energy (12.26) over $|\Psi_T|^2$, sampled by Metropolis.

The check.

Switch the Jastrow factor off and set $\alpha = 1$. The trial function is then the product of the two lowest oscillator orbitals, which is exactly the minimal-basis Hartree-Fock determinant of Chapter 11, and the local energy collapses to

$$E_L = 2\hbar\omega + \frac{1}{r_{12}}. \tag{12.48}$$

The average must therefore be $2\hbar\omega + \sqrt{\pi/2}\sqrt{\hbar\omega} = 3.253314$, the first row of Table 11.2. Two hundred thousand samples give 3.240689 ± 0.009144, which is 1.4 standard errors away. The two chapters agree.

Varying α.

Still without the Jastrow factor, the averages can be done analytically as well. Under $|\Psi_T|^2 \propto \exp(-\alpha\omega(r_1^2+r_2^2))$ one has $\langle r_1^2+r_2^2\rangle = 2/\alpha\omega$, and scaling lengths gives $\langle 1/r_{12}\rangle = \sqrt{\pi/2}\sqrt{\alpha\omega}$, so

$$E(\alpha) = \omega\alpha + \frac{\omega}{\alpha} + \sqrt{\frac{\pi}{2}}\sqrt{\alpha\omega}, \tag{12.49}$$

with a minimum at $\alpha = 0.7631$ where $E = 3.168384$. Every Monte Carlo number can thus be checked against a formula, which is what Table 12.5 does.

α	VMC	exact, Eq. (12.49)	deviation	Var(E_L)
0.7000	3.190122 ± 0.013352	3.177169	$1.0\,\sigma$	6.27
0.7631	3.151440 ± 0.006886	3.168384	$2.5\,\sigma$	2.86
0.8500	3.168811 ± 0.006859	3.181969	$1.9\,\sigma$	2.50
1.0000	3.247816 ± 0.010337	3.253314	$0.5\,\sigma$	4.41

Table 12.5 Variational Monte Carlo for the two-electron dot at $\hbar\omega = 1$ without the Jastrow factor, 200000 samples per row, errors corrected for the autocorrelation time. Computed by `montecarlo.py`.

The agreement is within statistics, but only just, and the reason is instructive. The variance of the local energy is a few units, and it is dominated by the rare configurations in which the two electrons approach and $1/r_{12}$ diverges. A heavy-tailed distribution makes the estimate of the variance itself unstable: repeating the calculation with a different seed can change the numbers in the last column of Table 12.5 by a factor of several, and with them the error bars. This is a mild version of the Cauchy problem of Exercise 12.10, and it is exactly what the Jastrow factor removes – the cusp condition $a = 1$ makes the divergence of $1/r_{12}$ cancel against the kinetic term, so the local energy stays finite as $r_{12} \to 0$.

Switching the Jastrow factor on.

With $a = 1$ enforcing the cusp condition, the local energy stays finite as $r_{12} \to 0$, and Table 12.6 shows what happens.

Two things are worth drawing out. The variance has fallen from about 3 to about 0.002 – three orders of magnitude – and it is telling us something the energy cannot. Both quantities are second order in the error $\delta = \Psi_T - \Psi_0$: writing $E_L - E_0 = (\hat{H} - E_0)\delta/\Psi_T = \mathscr{O}(\delta)$ shows that the variance is $\mathscr{O}(\delta^2)$, and the variational principle makes the energy error $\mathscr{O}(\delta^2)$ as well. The difference is that we know what the variance should be. It is exactly zero for the exact wave function, so it is an *absolute* measure of quality, whereas the energy can only be compared with an E_0 we do not have. For that reason minimising the variance rather than the energy is

α	β	energy	$\mathrm{Var}(E_L)$
0.98	0.30	3.006917 ± 0.001390	0.01775
0.98	0.40	3.000298 ± 0.000365	0.00195
0.98	0.48	3.001957 ± 0.000779	0.00439
0.96	0.44	3.000618 ± 0.000562	0.00220
1.00	0.44	3.001696 ± 0.000775	0.00362
exact (Taut)		3.000000	0
CCSD, 42 orbitals		3.013626	—
Hartree-Fock, 42 orbitals		3.161921	—

Table 12.6 The same calculation with the Jastrow factor, 60000 samples per row. The variance has fallen by three orders of magnitude and the energy sits on the exact value. The last three rows are Taut's analytic result and Tables 11.4 and 11.3. Computed by `montecarlo.py`.

a standard optimisation strategy, and often a more stable one, since the target value is known in advance.

And the energy itself, 3.000298 ± 0.000365, is below the CCSD result of 3.013626 from Chapter 11. That is not because the many-body treatment is better – CCSD is exact for two electrons in its basis. It is because there is no basis here at all. The Jastrow factor depends on r_{12} directly and reproduces the Coulomb cusp exactly, which no finite sum of oscillator products can do. The basis-set error that dominated Chapter 11 has simply been eliminated, and replaced by a statistical error that we know how to estimate and can reduce by running longer.

That trade is the subject of the chapters that follow.

12.10 Exercises

Exercise 1: distributions and their moments

a) Compute the mean and variance of the uniform distribution on $[a,b]$ and of the exponential distribution with rate λ.
b) Use Eq. (12.7) to find the PDF of $Y = X^2$ when X is uniform on $[0,1]$.
c) Show that the Cauchy distribution $p(x) = 1/[\pi(1+x^2)]$ has no finite variance, and say what that does to the central limit theorem.

Answer to a).

For the uniform distribution, $\langle x \rangle = (a+b)/2$ and $\langle x^2 \rangle = (b^3 - a^3)/3(b-a)$, so $\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 = (b-a)^2/12$. On $[0,1]$ this is $1/12$, the value used in Table 12.1. For the exponential, $\langle x \rangle = 1/\lambda$ and $\langle x^2 \rangle = 2/\lambda^2$, so $\sigma^2 = 1/\lambda^2$: mean and standard deviation coincide, which is why radioactive-decay statistics are so noisy.

Answer to b).

With $y = h(x) = x^2$ and $h^{-1}(y) = \sqrt{y}$ on $[0,1]$, $|dh^{-1}/dy| = 1/2\sqrt{y}$, and since $p_X = 1$,

$$p_Y(y) = \frac{1}{2\sqrt{y}}, \qquad 0 \le y \le 1.$$

It is normalised, and it diverges at the origin – squaring a uniform variable piles the values up near zero. This is the mechanism by which the inverse-CDF method works, run backwards.

Answer to c).

$\langle x^2 \rangle = \int x^2/[\pi(1+x^2)]\,dx$ diverges, since the integrand tends to $1/\pi$ at large $|x|$. The central limit theorem does not apply: averages of m Cauchy variables are Cauchy distributed with the *same* width, independently of m, so the error does not fall as $1/\sqrt{m}$ or at all. Heavy tails are not an academic worry – the $1/r_{12}$ tail in Table 12.5 is a mild version of the same problem, and it is what makes those error bars unreliable.

Exercise 2: sampling and importance sampling

a) Derive Eq. (12.21) and explain why $1-x$ may be replaced by x.
b) Verify the Box-Muller transformation (12.22) by transforming the two-dimensional Gaussian to polar coordinates.
c) For $I = \int_0^1 e^{-x}\,dx$ compare the variance of brute-force sampling with importance sampling using $p(x) \propto e^{-x}$ on $[0,1]$.

Answer to a).

The CDF is $P(y) = \int_0^y \lambda e^{-\lambda y'}dy' = 1 - e^{-\lambda y}$. Setting $x = P(y)$ and inverting gives $y = -\ln(1-x)/\lambda$. Since x is uniform on $[0,1)$, so is $1-x$, and the two forms are statistically identical; the version with $1-x$ is safer only because it avoids $\ln 0$ when the generator can return exactly zero.

Answer to b).

For two independent standard normals the joint density is $e^{-(x^2+y^2)/2}/2\pi$. In polar coordinates this is $re^{-r^2/2}dr\,d\theta/2\pi$, which factorises: θ is uniform on $[0,2\pi)$ and r has CDF $1-e^{-r^2/2}$, invertible to $r = \sqrt{-2\ln(1-u_1)}$. Taking $x = r\cos\theta$ with $\theta = 2\pi u_2$ gives Eq. (12.22), and $y = r\sin\theta$ gives a second, independent normal variable for free.

Answer to c).

Brute force averages $f = e^{-x}$ over the uniform distribution, giving $\sigma^2 = \langle e^{-2x}\rangle - \langle e^{-x}\rangle^2 = (1-e^{-2})/2 - (1-e^{-1})^2 \approx 0.0332$. With $p(x) = e^{-x}/(1-e^{-1})$ the ratio $f/p = 1-e^{-1}$ is *constant*, so the variance is exactly zero: the estimator returns the right answer from a single sample. This is the ideal case of Eq. (12.23), and it is the same mechanism that makes the local energy constant for an exact eigenfunction.

Exercise 3: Markov chains and detailed balance

a) Show that the columns of a transition matrix sum to one and that the steady state is the eigenvector with eigenvalue one.
b) Show that detailed balance implies the steady-state condition, but not conversely.
c) Verify that the Metropolis choice (12.40) satisfies detailed balance for a symmetric proposal.
d) What changes if the proposal is *not* symmetric?

Answer to a).

$\sum_i W_{ij} = 1$ says that the walker, starting in j, ends up somewhere with probability one. A steady state satisfies $w = \boldsymbol{W} w$ by definition, which is the eigenvalue equation with eigenvalue one. Such an eigenvalue always exists because $\boldsymbol{W}^T$ has the vector of ones as a right eigenvector with eigenvalue one, and $\boldsymbol{W}$ and $\boldsymbol{W}^T$ share eigenvalues.

Answer to b).

Summing detailed balance $W(j \to i)w_j = W(i \to j)w_i$ over j gives $\sum_j W(j \to i)w_j = w_i \sum_j W(i \to j) = w_i$, which is the steady-state condition. The converse fails: the steady-state condition constrains only the *net* flux into each state, so a cyclic flow $1 \to 2 \to 3 \to 1$ with equal rates satisfies it while violating detailed balance pairwise. Detailed balance is the stronger and the safer requirement.

Answer to c).

With T symmetric, take $w_j > w_i$. Then $A(i \to j) = 1$ and $A(j \to i) = w_i/w_j$, so

$$W(j \to i)w_j = T \frac{w_i}{w_j} w_j = T w_i = T \cdot 1 \cdot w_i = W(i \to j)w_i,$$

which is Eq. (12.38). The case $w_j < w_i$ follows by exchanging the labels, and $w_i = w_j$ is trivial.

Answer to d).

The proposal no longer cancels and the acceptance ratio must compensate for it,

$$A(i \to j) = \min\left(1, \frac{w_j T(j \to i)}{w_i T(i \to j)}\right),$$

which is the Metropolis-Hastings algorithm. This is exactly what is needed when the proposal is drawn from the Fokker-Planck Green's function of Section 12.8, since a drift towards high density is by construction asymmetric.

Exercise 4: errors that are too good to be true

a) Explain why $\sigma/\sqrt{N}$ understates the error of a Markov chain, and in which direction the bias always goes.
b) Show that blocking and the autocorrelation time measure the same thing.
c) A calculation reports $E = 3.0001 \pm 0.0001$ from a Metropolis run with an acceptance rate of 99%. What would you check first?

Answer to a).

Consecutive samples of a Markov chain are positively correlated, so the covariance term in Eq. (12.13) is positive and $\tau = 1 + 2\sum_d \kappa_d > 1$. The true variance is τ times the naive one, so the naive error is always too *small*: neglecting correlations never makes a result look worse than it is, which is precisely why the mistake survives.

Answer to b).

Blocking averages b consecutive measurements. Once $b \gg \tau$ the blocks are effectively independent, so the naive error of the blocked series is the true error. Quantitatively, the variance of a block mean is $\tau\sigma^2/b$ for $b \gg \tau$, so the naive error of n/b such blocks is $\sqrt{\tau\sigma^2/n}$ – Eq. (12.15) again. The plateau in the blocking curve is reached at $b \approx \tau$, so the curve measures the correlation time as a by-product.

Answer to c).

The acceptance rate. Ninety-nine per cent means the proposal step is far too small, the walker is crawling, and the correlation time is enormous – in Table 12.4, an acceptance of 98% gave $\tau = 585$, so the error bar is understated by a factor of $\sqrt{585} \approx 24$. The quoted error is almost certainly wrong by more than an order of magnitude. Check τ, or block the data, and retune the step for an acceptance near one half.

Exercise 5: numerical explorations

a) Reproduce Table 12.2 and plot consecutive triples of RANDU numbers in three dimensions. Find the fifteen planes.
b) Estimate π by the acceptance-rejection method on the unit circle, and confirm that the error falls as $1/\sqrt{N}$.
c) Simulate a two-dimensional random walk and verify $\langle r^2 \rangle = 4Dt$. How does the entropy of the walker distribution behave on a finite lattice with reflecting rather than periodic boundaries?
d) For the trial function (12.47), scan α and β on a grid and locate the minimum of the energy and, separately, the minimum of the variance. Do they coincide?
e) Derive the local energy for Eq. (12.47) analytically and compare with the finite-difference version used in `montecarlo.py`. How large may the step h be before the difference is visible above the statistical error?

f) Repeat the calculation of Table 12.6 at $\hbar\omega = 0.5$ and $\hbar\omega = 0.25$ and compare with Table 11.6. Where is the correlation strongest, and does the simple Jastrow factor still suffice?

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter12/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/statistics.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter12/`.

`montecarlo.py` Random number generator tests (the RANDU planes), brute-force and importance-sampled integration, the central limit theorem, the correlation time and blocking, random walks and entropy, Markov chains and detailed balance, the Metropolis algorithm, and a first variational Monte Carlo run on the chapter-11 dot. Runs in about thirty-five seconds.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 13
Variational Monte Carlo

Chapter 12 assembled the tools – probability distributions, the central limit theorem, the correlation time, Markov chains and the Metropolis algorithm – and ended with a first variational Monte Carlo calculation, run brute force. This chapter is the method itself, worked through properly and taken to the point where it is usable: the variational principle it rests on, how a trial wave function is constructed and why it has the form it does, the local energy and how to differentiate it, the sampling algorithm, and finally importance sampling, which is what turns a working program into an efficient one.

The example throughout is the system of Chapter 11, two electrons in a two-dimensional parabolic trap. It is the right test case for three reasons. It is small enough that everything can be differentiated by hand; it has an analytic answer, $E = 3$ at $\hbar\omega = 1$, so we always know what we are aiming at; and it has already been solved by Hartree-Fock, perturbation theory and coupled cluster in Chapter 11, so the Monte Carlo result can be placed on the same scale as those.

The programs are in `vmc.py`. Everything the sampling needs – the trial function, the local energy and the quantum force – is available in closed form, and each closed form is checked against finite differences before it is used, for a reason that will become clear in Section 13.6.

13.1 The variational principle

Everything rests on one inequality. For any normalisable trial state Ψ_T,

$$\boxed{E[\Psi_T] = \frac{\langle \Psi_T|\hat{H}|\Psi_T\rangle}{\langle \Psi_T|\Psi_T\rangle} \;\geq\; E_0} \tag{13.1}$$

with equality only if Ψ_T is the ground state. The proof is the one from Section 2.7: expand Ψ_T in the exact eigenstates and observe that every excited state can only raise the average.

The strategy is then obvious. Choose a family of trial functions $\Psi_T(\boldsymbol{R};\boldsymbol{\theta})$ depending on a few parameters $\boldsymbol{\theta}$, evaluate $E[\Psi_T]$, and minimise over $\boldsymbol{\theta}$. What makes this hard is the evaluation. For N particles in d dimensions the integral

$$E[\Psi_T] = \frac{\int d\boldsymbol{R}\,\Psi_T^*(\boldsymbol{R})\hat{H}\Psi_T(\boldsymbol{R})}{\int d\boldsymbol{R}\,|\Psi_T(\boldsymbol{R})|^2} \tag{13.2}$$

is over Nd dimensions. For our two electrons in two dimensions that is four, which a grid could still manage; for the six-electron dot it is twelve, and for anything realistic it is hopeless.

Monte Carlo integration converges as $1/\sqrt{M}$ regardless of dimension, which is the whole reason for the method.

What the method can and cannot do.

It is worth being clear about the trade before starting.

- Any expectation value can be computed, for any Hamiltonian, in any number of dimensions, provided we can write down and evaluate a trial wave function. There is no basis set and therefore no basis-set error.
- The energy is a rigorous upper bound, and the bound improves monotonically as the trial function improves.
- But the answer is only as good as the trial function. Unlike coupled cluster or configuration interaction there is no systematic hierarchy to climb; improving Ψ_T is a matter of physical insight.
- And the error is statistical, so an extra digit costs a hundred times the computer time.

13.2 The local energy

The step that makes Eq. (13.2) computable was taken in Section 12.5. Insert Ψ_T/Ψ_T and read the result as an average over a probability density,

$$E[\Psi_T] = \int d\boldsymbol{R}\; \underbrace{\frac{|\Psi_T(\boldsymbol{R})|^2}{\int d\boldsymbol{R}' |\Psi_T(\boldsymbol{R}')|^2}}_{P(\boldsymbol{R})} \; \underbrace{\frac{\hat{H}\Psi_T(\boldsymbol{R})}{\Psi_T(\boldsymbol{R})}}_{E_L(\boldsymbol{R})}, \tag{13.3}$$

where P is the quantum-mechanical probability density and

$$\boxed{E_L(\boldsymbol{R}) = \frac{1}{\Psi_T(\boldsymbol{R})}\hat{H}\Psi_T(\boldsymbol{R})} \tag{13.4}$$

is the *local energy*. The energy is then simply

$$E[\Psi_T] \approx \frac{1}{M}\sum_{i=1}^{M} E_L(\boldsymbol{R}_i), \qquad \boldsymbol{R}_i \sim P, \tag{13.5}$$

an average of a function over samples drawn from P. Two observations make this more than a rewriting.

The zero-variance principle.

If Ψ_T is an exact eigenstate, $\hat{H}\Psi_T = E\Psi_T$, then $E_L(\boldsymbol{R}) = E$ everywhere. The average is E exactly, from a single sample, and the variance is zero. For an approximate Ψ_T the variance

$$\sigma^2 = \langle E_L^2\rangle - \langle E_L\rangle^2 \tag{13.6}$$

is a direct and absolute measure of how far the trial function is from an eigenstate – absolute because we know the target value, zero. This is why σ^2 is quoted alongside the energy in everything that follows, and why minimising the variance is a legitimate alternative to minimising the energy.

The normalisation never appears.

P contains the normalisation integral we cannot compute. But the Metropolis algorithm of Section 12.7 needs only ratios $P(\boldsymbol{R}')/P(\boldsymbol{R}) = |\Psi_T(\boldsymbol{R}')|^2/|\Psi_T(\boldsymbol{R})|^2$, in which the normalisation cancels. That is the second reason the whole scheme works, and it is why a Markov chain is used instead of direct sampling.

13.3 Constructing a trial wave function

A trial wave function is usually built as a product of two pieces,

$$\Psi_T(\boldsymbol{R}) = \Phi(\boldsymbol{R})J(\boldsymbol{R}), \tag{13.7}$$

a mean-field part Φ – a Slater determinant of single-particle orbitals, which enforces antisymmetry – and a symmetric *correlation* or *Jastrow* factor J depending on the interparticle distances. The determinant carries the shell structure; the Jastrow factor carries what a determinant cannot represent, namely that two electrons avoid each other.

The form of J is not guesswork. It is fixed at short distance by a condition on the local energy. As two electrons approach, the Coulomb term $1/r_{12}$ in $\hat{H}$ diverges, and if the local energy is to remain finite the kinetic term must produce a compensating divergence. Isolating the relative motion and keeping only the terms that survive as $r_{12} \to 0$, the requirement E_L finite gives

$$\left(\frac{2}{r_{12}}\frac{d}{dr_{12}} + \frac{2}{r_{12}}\right)\mathscr{R}(r_{12}) = 0 \quad \Longrightarrow \quad \frac{1}{\mathscr{R}}\frac{d\mathscr{R}}{dr_{12}} = \frac{1}{2(l+1)}\bigg|_{r_{12}\to 0}, \tag{13.8}$$

the *cusp condition*. For $l = 0$ and two electrons of opposite spin this integrates to

$$\mathscr{R}(r_{12}) \propto \exp\left(\frac{r_{12}}{2}\right) \qquad \text{(antiparallel spins)}, \tag{13.9}$$

in three dimensions, and to $\exp(r_{12}/4)$ for parallel spins, where the Pauli principle already keeps the electrons apart and less correlation is needed. In two dimensions the corresponding coefficients are 1 and $1/3$. For more than two electrons the condition holds pair by pair.

The exponential in Eq. (13.9) is right at short distance and badly wrong at long distance, where it grows without bound. The standard repair is the *Padé-Jastrow* form, which interpolates between the cusp behaviour and a constant,

$$J = \exp\left(\sum_{i<j}\frac{a_{ij}r_{ij}}{1+\beta r_{ij}}\right), \tag{13.10}$$

with a_{ij} fixed by the cusp condition and β variational. As $r_{ij} \to 0$ the exponent is $a_{ij}r_{ij}$ and the cusp is reproduced; as $r_{ij} \to \infty$ it saturates at a_{ij}/β.

Why this matters later. The construction above is physics put in by hand: we knew to expect a cusp, we knew the exponential was wrong at long range, and we chose a functional form with the right limits. It works well for two electrons and less well as systems grow, because the correlations we have to guess become more complicated. That difficulty is precisely the motivation for the machine-learning wave functions of the later chapters, where the functional form is not guessed but learned.

13.4 The two-electron quantum dot

The system is the one of Chapter 11: two electrons confined to a plane by a parabolic trap, interacting through the bare Coulomb repulsion. In natural units,

$$\hat{H} = \sum_{i=1}^{2} \left(-\tfrac{1}{2}\nabla_i^2 + \tfrac{1}{2}\omega^2 r_i^2\right) + \frac{1}{r_{12}}. \tag{13.11}$$

Separating the motion.

Introducing the centre of mass $\boldsymbol{R} = (\boldsymbol{r}_1 + \boldsymbol{r}_2)/2$ and the relative coordinate $\boldsymbol{r} = \boldsymbol{r}_1 - \boldsymbol{r}_2$, the Hamiltonian separates. The centre-of-mass part is a plain harmonic oscillator and contributes a known constant; the relative part is

$$\hat{H}_r = -\nabla_r^2 + \tfrac{1}{4}\omega^2 r^2 + \frac{1}{r}, \tag{13.12}$$

a one-body problem in the relative coordinate. Taut showed that this has closed-form solutions for a discrete set of frequencies, and at $\hbar\omega = 1$ the total ground-state energy is exactly 3. That number is the target for the rest of the chapter.

The trial function.

Combining the mean-field piece – for two electrons in a spin singlet, simply the product of two lowest oscillator orbitals – with the Padé-Jastrow factor gives

$$\boxed{\Psi_T(\boldsymbol{r}_1, \boldsymbol{r}_2; \alpha, \beta) = \exp\left(-\frac{\alpha\omega}{2}\left(r_1^2 + r_2^2\right)\right) \exp\left(\frac{r_{12}}{1 + \beta r_{12}}\right)} \tag{13.13}$$

Two variational parameters. The first, α, scales the width of the orbital: $\alpha = 1$ is the bare oscillator ground state, and letting it vary allows the orbital to relax in response to the repulsion. The second, β, sets the range over which the correlation factor acts. The numerator of the Jastrow exponent is *not* a parameter – it is fixed at 1 by the cusp condition.

The local energy.

Differentiating Eq. (13.13) twice and dividing by Ψ_T gives, with $d = 1/(1 + \beta r_{12})$,

$$\boxed{\begin{aligned} E_L = \frac{1}{2}\omega^2\left(1-\alpha^2\right)\left(r_1^2+r_2^2\right)+2\alpha\omega+\frac{1}{r_{12}} \\ +d^2\left(\alpha\omega r_{12}-d^2+2\beta d-\frac{1}{r_{12}}\right). \end{aligned}} \tag{13.14}$$

The first line is the result for the pure Gaussian, already met as Eq. (12.49)'s integrand in Chapter 12; the second is everything the Jastrow factor contributes. Note the last term. It carries $-d^2/r_{12}$, which at short distance cancels the $+1/r_{12}$ on the first line, since $d \to 1$ as $r_{12} \to 0$. That cancellation *is* the cusp condition, and Table 13.1 and Figure 13.1 show it happening.

r_{12}	E_L with Jastrow	E_L without
1.0000	3.031237	3.00
0.1000	2.703286	12.00
0.0100	2.611464	102.00
0.0010	2.601159	1002.00
0.0001	2.600116	10002.00

Table 13.1 The local energy for two electrons brought together along a line, $\alpha = 1$, $\beta = 0.4$, $\hbar\omega = 1$. With the correlation factor the local energy converges; without it the Coulomb singularity is unopposed and E_L diverges as $1/r_{12}$. Computed by `vmc.py`.

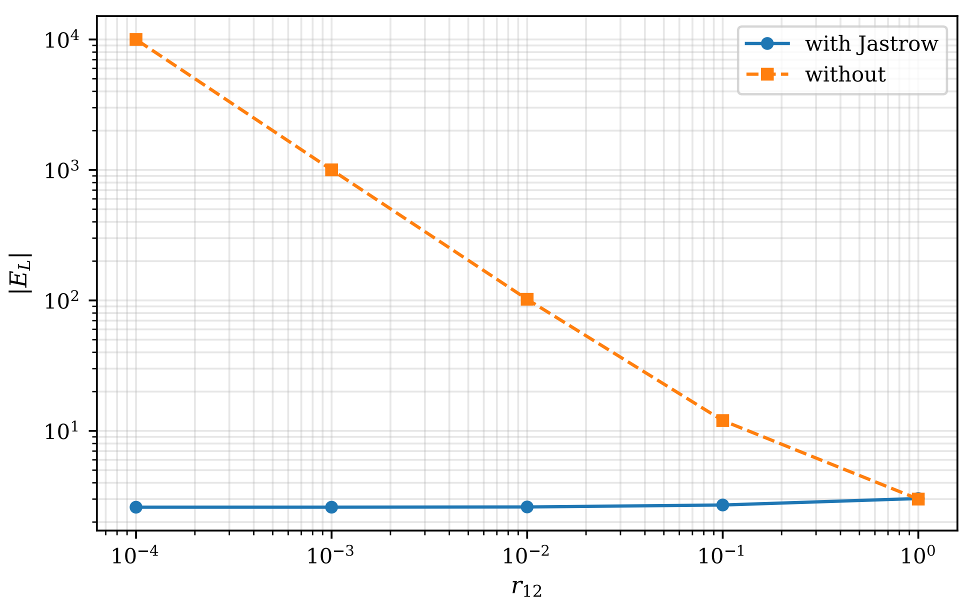

Fig. 13.1 The local energy of two electrons in the two-dimensional dot, evaluated with the two particles placed symmetrically about the origin at a separation r_{12} that is reduced by a decade at a time from 1 to 10^{-4}, at $\alpha = 1$, $\beta = 0.4$ and $\hbar\omega = 1$. Both axes are logarithmic and the ordinate is $|E_L|$. The dashed curve is the trial function without the Jastrow factor, for which $E_L = 2 + 1/r_{12}$ and the Coulomb singularity is unopposed; it is a straight line of slope -1 reaching 10^4 at the smallest separation. The solid curve is the full Padé-Jastrow trial function of Eq. (13.13), whose local energy is bounded and flattens onto the value $2\alpha\omega - 1 + 4\beta = 2.6$. The five points of each curve are the five rows of Table 13.1.

Figure 13.1 is the same five configurations plotted over the five decades they span, and on logarithmic axes the two behaviours separate completely. Without the correlation factor the local energy is $2+1/r_{12}$ and the curve is a straight line of slope -1: every decade by which the electrons are brought closer multiplies $|E_L|$ by ten, so that at $r_{12}=10^{-4}$ a single configuration would contribute 10^4 to an average whose true value is 3. With the Jastrow factor the curve is flat to the eye across the whole range and settles onto 2.600. That plateau is not an artefact of the plotting range but the finite limit of Eq. (13.14). Setting $d \to 1$ and collecting what survives, the two $1/r_{12}$ terms cancel and leave 2β, the term $-d^4$ leaves -1, the term $2\beta d^3$ leaves 2β, and $d^2\alpha\omega r_{12}$ vanishes, so that

$$E_L(r_{12} \to 0) = 2\alpha\omega - 1 + 4\beta, \tag{13.15}$$

which for $\alpha = \omega = 1$ and $\beta = 0.4$ is exactly 2.6. The tabulated 2.600116 at $r_{12} = 10^{-4}$ is that limit approached from above with a correction linear in the separation, and the reader who wants a check on an implementation of Eq. (13.14) has one here that costs a single function call. The modulus on the ordinate is there only because the scale is logarithmic; for this trial function E_L happens to stay positive along the whole line, so nothing is hidden by it.

Notice also that the two curves are indistinguishable at the right-hand edge of the figure. At $r_{12} = 1$ the Jastrow factor changes the local energy by one part in a hundred, 3.031237 against 3.00, and the singularity is still far away; the entire difference between a usable trial function and an unusable one is generated in the last three decades of the plot. Those are configurations the sampler visits rarely, which is exactly what makes them dangerous. It is worth making that quantitative, because it is the reason the cusp is not merely an aesthetic improvement. Sampling $|\Psi_T|^2$ for the uncorrelated Gaussian, the probability of finding the pair at a separation between r_{12} and $r_{12}+dr_{12}$ goes as $r_{12}\,dr_{12}$ in two dimensions, while E_L^2 goes as $1/r_{12}^2$; the second moment $\int r_{12}\,dr_{12}/r_{12}^2$ therefore diverges logarithmically at the origin. The variance defined by Eq. (13.6) does not exist, the sample variance measured in a run creeps upwards without limit as more configurations are collected, and any error bar derived from it is meaningless. With the cusp built in the integrand is bounded, the variance is a genuine finite number, and it is the number quoted in every table below.

This is worth pausing on, because it explains the numbers in Chapter 12. A trial function without the cusp has a local energy with an unbounded tail; rare close encounters then dominate the variance, the variance estimate itself becomes unstable, and error bars stop meaning much. With the cusp built in, the variance for this system falls by three orders of magnitude.

13.5 Brute-force Metropolis sampling

The algorithm is the Metropolis of Section 12.7 applied to $P = |\Psi_T|^2$.

1. Place the particles at random and evaluate Ψ_T.
2. Propose a move of *one* particle, $\boldsymbol{r}_i' = \boldsymbol{r}_i + \Delta \cdot (\boldsymbol{u} - \frac{1}{2})$, with $\boldsymbol{u}$ uniform and Δ the step length.
3. Accept with probability $\min\left(1,\ |\Psi_T(\boldsymbol{R}')|^2/|\Psi_T(\boldsymbol{R})|^2\right)$; if the move is rejected, keep the old configuration.
4. After every particle has been offered a move, accumulate $E_L(\boldsymbol{R})$ and $E_L^2(\boldsymbol{R})$.
5. Discard an initial burn-in, then average.

Two details are easy to get wrong. Moving one particle at a time rather than all of them keeps the acceptance rate up for a given step length. And a rejected move must still be counted – the old configuration is measured again. Dropping rejected steps biases the average, and is one of the most common bugs in a first implementation.

The step length Δ is the one free parameter, and Table 13.2 shows the usual trade-off: too small and the walker crawls, too large and everything is rejected, with the correlation time large at both ends.

Δ	acceptance	energy	error	τ
0.5	90.1%	2.999609	0.001249	24.28
1.0	80.2%	2.999422	0.000643	6.17
2.0	62.2%	3.000750	0.000418	2.80
4.0	35.3%	2.999901	0.000376	2.24
8.0	11.8%	3.000125	0.000701	7.77

Table 13.2 Brute-force Metropolis for the two-electron dot at $\alpha = 0.98$, $\beta = 0.40$, $\hbar\omega = 1$, 30000 cycles. Errors are corrected for the autocorrelation time by Eq. (12.15). Computed by `vmc.py`.

13.6 Importance sampling

The uniform proposal of the previous section is wasteful. It offers moves in every direction with equal probability, including many into regions where $|\Psi_T|^2$ is negligible, and those are rejected. A better proposal would know something about the wave function and bias the move towards where it is large. The machinery for that is the Fokker-Planck and Langevin equations of Section 12.8.

Drift from the Fokker-Planck equation.

A diffusion process with a drift obeys

$$\frac{\partial P}{\partial t} = \sum_i D \frac{\partial}{\partial \boldsymbol{x}_i}\left(\frac{\partial}{\partial \boldsymbol{x}_i} - \boldsymbol{F}_i\right) P(\boldsymbol{x},t), \tag{13.16}$$

with D the diffusion coefficient and $\boldsymbol{F}$ the drift. We want the stationary solution to be our target $P = |\Psi_T|^2$. Setting the left-hand side to zero and requiring every term of the sum to vanish gives

$$\frac{\partial^2 P}{\partial \boldsymbol{x}_i^2} = P\frac{\partial \boldsymbol{F}_i}{\partial \boldsymbol{x}_i} + \boldsymbol{F}_i \frac{\partial P}{\partial \boldsymbol{x}_i}. \tag{13.17}$$

Trying a drift of the form $\boldsymbol{F} = g(\boldsymbol{x})\,\partial P/\partial \boldsymbol{x}$ and demanding that the first- and second-derivative terms cancel forces $g = 1/P$, so that $\boldsymbol{F} = \nabla P/P$. With $P = |\Psi_T|^2$ this is

$$\boxed{\boldsymbol{F} = \frac{2}{\Psi_T}\nabla\Psi_T = 2\nabla \ln \Psi_T} \tag{13.18}$$

the *quantum force*. It points towards regions where the trial function is large, and it is what replaces the aimless uniform proposal.

Proposing a move.

The Langevin equation corresponding to Eq. (13.16) is

$$\frac{\partial \boldsymbol{x}(t)}{\partial t} = D\boldsymbol{F}(\boldsymbol{x}(t)) + \boldsymbol{\eta}, \tag{13.19}$$

with $\boldsymbol{\eta}$ a Gaussian white noise. Integrating it over a short time step Δt by Euler's method gives the proposal

$$\boxed{\boldsymbol{y} = \boldsymbol{x} + D\boldsymbol{F}(\boldsymbol{x})\Delta t + \boldsymbol{\xi}\sqrt{\Delta t}} \tag{13.20}$$

with $\boldsymbol{\xi}$ a standard normal variate. In atomic units $D = \frac{1}{2}$, which comes directly from the $\frac{1}{2}$ in the kinetic energy operator; the time step Δt is a parameter to be tuned, exactly as Δ was.

Correcting for the asymmetry.

The proposal is now *not* symmetric: drifting towards high density makes $\boldsymbol{x} \to \boldsymbol{y}$ and $\boldsymbol{y} \to \boldsymbol{x}$ different. Metropolis is no longer valid and must be replaced by Metropolis-Hastings, which was Exercise 3(d) of Chapter 12. The transition density is the Green's function of Eq. (13.16),

$$G(\boldsymbol{y},\boldsymbol{x},\Delta t) = \frac{1}{(4\pi D\Delta t)^{3N/2}} \exp\left(-\frac{(\boldsymbol{y}-\boldsymbol{x}-D\Delta t\boldsymbol{F}(\boldsymbol{x}))^2}{4D\Delta t} \right), \tag{13.21}$$

and the acceptance ratio becomes

$$\boxed{q(\boldsymbol{y},\boldsymbol{x}) = \frac{G(\boldsymbol{x},\boldsymbol{y},\Delta t)\,|\Psi_T(\boldsymbol{y})|^2}{G(\boldsymbol{y},\boldsymbol{x},\Delta t)\,|\Psi_T(\boldsymbol{x})|^2}} \tag{13.22}$$

The prefactors of G cancel in the ratio, and expanding the exponents leaves a form that needs no exponentials at all,

$$\ln \frac{G(\boldsymbol{x},\boldsymbol{y})}{G(\boldsymbol{y},\boldsymbol{x})} = \frac{1}{2}\left(\boldsymbol{F}(\boldsymbol{x}) + \boldsymbol{F}(\boldsymbol{y})\right) \cdot \left[\frac{D\Delta t}{2}\left(\boldsymbol{F}(\boldsymbol{x}) - \boldsymbol{F}(\boldsymbol{y})\right) - \boldsymbol{y} + \boldsymbol{x} \right], \tag{13.23}$$

which is what the program evaluates.

The quantum force for our trial function.

Differentiating Eq. (13.13),

$$\boldsymbol{F}_1 = -2\alpha\omega\boldsymbol{r}_1 + \frac{2(\boldsymbol{r}_1-\boldsymbol{r}_2)\,d^2}{r_{12}}, \qquad \boldsymbol{F}_2 = -2\alpha\omega\boldsymbol{r}_2 - \frac{2(\boldsymbol{r}_1-\boldsymbol{r}_2)\,d^2}{r_{12}}, \tag{13.24}$$

again with $d = 1/(1+\beta r_{12})$. It is a *sum* of two terms: the first pulls the electron towards the centre of the trap, the second pushes the pair apart.

A wrong quantum force is not a wrong answer. This deserves emphasis, because it is a trap. If Eq. (13.24) is coded incorrectly – and the two terms are easy to combine wrongly – the calculation still converges to the right energy, provided the *same* expression is used both to propose the move and to evaluate the Green's function. Detailed balance holds for any drift; a bad drift merely proposes bad moves. Replacing the sum in Eq. (13.24) by a product, `vmc.py` still gets 3.000256 ± 0.000399, but the acceptance rate falls from 87% to 57% and the correlation time doubles from 1.21 to 2.48. The error is invisible in the answer and shows up only in the efficiency, which is exactly why the quantum force is checked against finite differences before use. A wrong *local energy*, by contrast, gives a wrong number.

13.7 Results

Table 13.3 scans the time step. The comparison with Table 13.2 is the point of the whole chapter. At $\Delta t = 0.5$ the acceptance rate is 88% while the walker is taking steps that a uniform proposal of comparable size would have rejected almost always, and the correlation time is 1.24 against a best of 2.24 for brute force. There is no bias to worry about at large Δt: the Metropolis-Hastings ratio corrects the discretisation of the Langevin equation exactly, so a coarse time step costs accuracy only through the acceptance rate.

Δt	acceptance	energy	error	τ
0.010	100.0%	3.001377	0.001969	52.53
0.050	99.5%	3.001285	0.000768	8.79
0.100	98.7%	3.000852	0.000551	4.54
0.300	93.8%	3.000394	0.000320	1.57
0.500	87.6%	3.000526	0.000280	1.24
1.000	69.6%	3.000962	0.000267	1.17

Table 13.3 Importance-sampled variational Monte Carlo, same system and same number of cycles as Table 13.2. Computed by `vmc.py`.

Figure 13.2 puts the last columns of Tables 13.2 and 13.3 on the same axes and makes the comparison in one picture. In the left panel both samplers show the expected shape but not the same shape. Brute force is a proper valley: τ falls from about 21 at $\Delta = 0.5$ to a minimum of 2.18 at $\Delta = 4$ and climbs again to 7.35 at $\Delta = 8$, because a step much larger than the dot is almost always rejected and the walker then sits still. Importance sampling instead falls steeply and then flattens, from 49.7 at $\Delta t = 0.01$ through 8.5, 5.1 and 1.6 to a plateau of about 1.2 that is already reached at $\Delta t = 0.3$ and does not deteriorate by $\Delta t = 1$. The steep branch at small Δt has nothing to do with rejection – the acceptance rate there is essentially 100% – and everything to do with diffusion: the walker advances a distance of order $\sqrt{2D\Delta t}$ per step, so the number of steps needed to cross the region where $|\Psi_T|^2$ is appreciable grows as $1/\Delta t$. Reducing the time step from 0.05 to 0.01, a factor of five, raises τ from 8.5 to 49.7, a factor of 5.8, which is that law to the accuracy one can expect of a correlation-time estimate. The gap between the two floors, 2.18 against 1.15, is a factor of 1.9 in the effective sample size $n_{\text{eff}} = M/\tau$ and therefore a factor of $\sqrt{1.9} \approx 1.4$ in the error bar, which is the gain quoted in the comparison below.

The right panel is the more instructive of the two, because the acceptance rate is a quantity both samplers possess and both curves can be read against it. They do not share an optimum. Brute force is at its best near 35% acceptance and is nearly three times as expensive by 80%, where $\tau = 5.79$; importance sampling is flat from 70% to 88% and only degrades as the acceptance approaches unity. The often-repeated instruction to tune a Monte Carlo run to

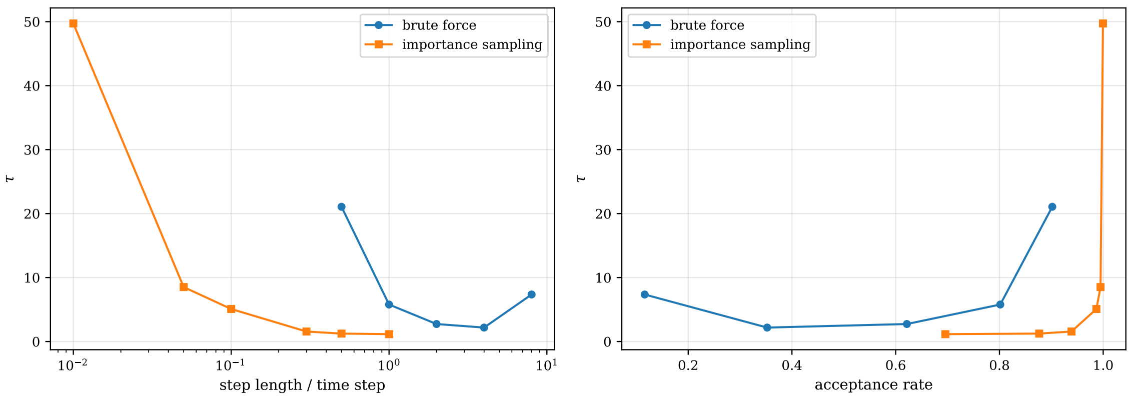

Fig. 13.2 Autocorrelation time τ of the local energy for the two-electron dot at $\alpha = 0.98$, $\beta = 0.40$, $\hbar\omega = 1$, for brute-force Metropolis with step length $\Delta \in \{0.5, 1, 2, 4, 8\}$ (circles) and for importance-sampled Metropolis-Hastings with time step $\Delta t \in \{0.01, 0.05, 0.1, 0.3, 0.5, 1\}$ (squares). Left: τ against the tuning parameter on a logarithmic abscissa; the two abscissae are a step length and a time step and are therefore not commensurable, so what is to be compared is the height of the two floors, 2.18 and 1.15. Right: the same points plotted against the measured acceptance rate, the axis on which the two samplers can be compared, showing that they have quite different optima. The run is an independent one of 20 000 cycles, so the correlation times differ from those of Tables 13.2 and 13.3 in the second digit.

about half acceptance is thus a statement about *symmetric* proposals only, and the reason is visible in the construction of Section 13.6. For a uniform proposal a high acceptance rate can be bought only with a short step, so acceptance and move length are rigidly tied together and the optimum is a compromise between them. The drift breaks that link: the move is aimed at the region where the trial function is large, so it can be long and still be accepted, and at $\Delta t = 0.5$ the walker travels a distance of the order of the width of the orbital and is still accepted seven times in eight.

The near-vertical branch at the right-hand edge of that panel deserves to be read as a warning. The point at 100% acceptance is not a well-tuned sampler but a pathological one, with $\tau = 49.7$ and an error bar seven times the best available for the same cost; the moves are so short that the proposal is barely distinguishable from staying put, and nothing is ever rejected because nothing much is ever attempted. An acceptance rate close to one is a symptom, not a target. The practical conclusion is that acceptance alone is not a sufficient diagnostic and must be monitored together with the correlation time or, better, with the blocking analysis of Section 12.3, which needs no assumption about the shape of the autocorrelation function at all.

Run at their respective best settings on equal numbers of cycles, the two samplers give

sampler	acceptance	energy ± error	τ	n_{eff}
brute force, $\Delta = 4$	35.4%	3.000865 ± 0.000265	2.33	25 788
importance, $\Delta t = 0.5$	87.5%	3.000463 ± 0.000199	1.22	49 017

a factor 1.34 in the error, which by the square-root law is a factor of about 1.8 in computer time. That is a real but unspectacular gain, and it is worth being honest about the scale: for a system this small, with a very good trial function, brute force is already close to optimal. The advantage of importance sampling grows with the number of particles and with the sharpness of the wave function, and it becomes indispensable in diffusion Monte Carlo, where the drift is not an efficiency device but part of the algorithm.

The variational surface.

Table 13.4 scans both parameters, and Figure 13.3 renders the same scan as a pair of colour maps. The energy is remarkably flat near the minimum – the whole central block agrees to within a few parts in 10^4 – while the variance varies by a factor of ten across the same region. This is the practical argument for variance minimisation: the variance has a sharper minimum and a known target value, so it locates the optimum more precisely than the energy does.

	β				
α	0.30	0.36	0.40	0.44	0.50
energy					
0.94	3.01804	3.00780	3.00460	3.00316	3.00357
0.96	3.01232	3.00415	3.00197	3.00127	3.00266
0.98	3.00786	3.00142	3.00043	3.00081	3.00295
1.00	3.00492	3.00033	3.00012	3.00145	3.00445
1.02	3.00294	3.00025	3.00096	3.00297	3.00671
variance					
0.94	0.03241	0.01420	0.00777	0.00478	0.00545
0.96	0.02397	0.00855	0.00378	0.00227	0.00490
0.98	0.01780	0.00521	0.00197	0.00192	0.00640
1.00	0.01360	0.00387	0.00222	0.00356	0.00985
1.02	0.01154	0.00448	0.00434	0.00704	0.01508

Table 13.4 The variational surface, importance sampling with $\Delta t = 0.5$ and 15 000 cycles per point. The energy is flat near its minimum; the variance is not. Computed by `vmc.py`.

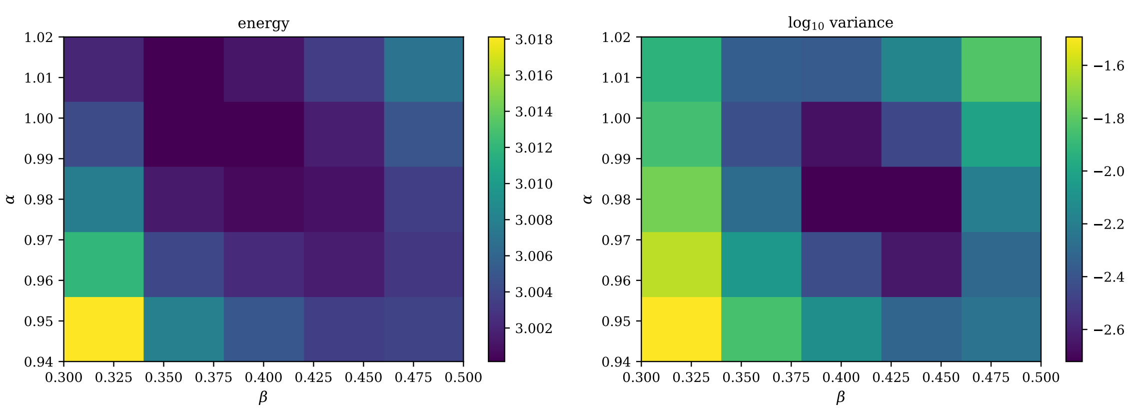

Fig. 13.3 The variational surface of the two-electron dot on the grid $\alpha \in \{0.94, 0.96, 0.98, 1.00, 1.02\}$, $\beta \in \{0.30, 0.36, 0.40, 0.44, 0.50\}$, sampled with importance sampling at $\Delta t = 0.5$. Left: the energy, whose colour bar spans only 3.0001 to 3.018. Right: the base-ten logarithm of the variance, which covers 1.23 decades, a factor of seventeen, over the identical grid. Each of the twenty-five cells in a panel is one scanned pair (α, β), and the axes are drawn so that the extreme parameter values fall on the outer edges of the block rather than at the cell centres. The energy basin is broad and featureless while the variance has a narrow minimum near $\alpha = 0.98$, $\beta \simeq 0.42$, and in both panels the best β drifts downwards as α is raised. Produced by an independent scan of 8 000 cycles per point, so the individual entries differ from Table 13.4 in the last digit or two.

Figure 13.3 makes the asymmetry between the two panels impossible to miss, and it is worth spelling out what is being compared, because the two colour bars are doing very different work. The energy panel has to spread its whole range, 3.0001 to 3.018, over a variation of less than two parts in 10^3, and almost all of that range is consumed by the single cell in the lower left corner at $\alpha = 0.94$, $\beta = 0.30$, where the orbital is too diffuse and the correlation factor too long-ranged at the same time. Everything above and to the right of it is one shade: the nine cells with $\alpha \geq 0.98$ and β between 0.36 and 0.44 span only 2.9×10^{-3} in Table 13.4, and the six best of them only 1.3×10^{-3}, which is two or three statistical errors of a single point in the scan. The energy panel, in other words, cannot locate the optimum, and no amount of squinting at it will help; it can only tell us which corner of the grid to leave. The variance panel, plotted logarithmically because it has to be, carries a factor of seventeen between its extremes and a clearly resolved minimum in the neighbourhood of $\alpha = 0.98$, $\beta \in [0.40, 0.44]$, where Table 13.4 gives 0.00197 and 0.00192. Crossing that row to $\beta = 0.30$ at the same α raises the variance to 0.01780, a factor of nine, while the energy changes by only 7×10^{-3}. That is the practical argument of Section 13.2 made visible: the variance is a sharper instrument because it responds strongly where the energy responds weakly, and because we know its target value is zero whereas the energy's target is what we are trying to determine.

The shape of the minimum is as informative as its depth. Reading the energy panel row by row, the best β slides from 0.44 at $\alpha = 0.94$ and 0.96 to 0.40 at $\alpha = 0.98$ and 1.00 and to 0.36 at $\alpha = 1.02$, and the variance panel makes the same move one step later, from 0.44 to 0.40 between $\alpha = 0.98$ and $\alpha = 1.00$. The dark region in either panel is therefore a trough leaning across the grid rather than a circular basin. The two parameters are anticorrelated, and the physics of that is straightforward. Raising α contracts the Gaussian and pulls both electrons towards the centre of the trap; lowering β lengthens the range over which the Jastrow factor pushes them apart. Either adjustment alone would move the mean separation of the pair, and it is only the combination that keeps the electrons at the distance the Coulomb repulsion and the confinement together prefer. A trial function with two parameters and one physical length scale to fix has a soft direction, and here it is.

That soft direction is a warning about optimisation as much as about scanning. A grid of twenty-five points, each of them a Monte Carlo run in its own right, has bought us a minimum located to about one grid spacing, and the refinement to $\beta = 0.38$ used in Eq. (13.25) required a further scan between the two best columns. With three or four parameters the same strategy is already impossible, and one is forced onto the gradient of Exercise 4(e). Figure 13.3 says what such a gradient method will be up against: along the trough the energy gradient is small compared with its own statistical noise, so steepest descent on the energy will wander, while across the trough it is comparatively steep, so the same method will zigzag. Both are symptoms of an ill-conditioned surface, and both are reasons to minimise the variance instead, or to use a method that accounts for the curvature. Chapter 14 takes that up properly.

The final number.

Two hundred thousand cycles at $\alpha = 1.00$, $\beta = 0.38$ give

$$E_{\mathrm{VMC}} = 3.000603 \pm 0.000120, \qquad \sigma^2 = 0.00258, \qquad \tau = 1.12, \tag{13.25}$$

against the exact 3. The gap of 6×10^{-4} is five standard errors wide, and that is the point rather than a problem. A variational energy is an upper bound, so what is being resolved here is not a statistical discrepancy but the deficiency of a two-parameter ansatz – and it is resolved only because the statistical error has been pushed below it. Removing the residual requires a better trial function, or diffusion Monte Carlo, which projects it out.

Set against Chapter 11, where the best coupled-cluster calculation in a 42-orbital basis gave 3.013626, the variational Monte Carlo error is smaller by a factor of about twenty-three.

The reason is not that the many-body treatment is better – for two electrons CCSD is exact within its basis – but that there is no basis here at all. The Jastrow factor depends on r_{12} directly and satisfies the cusp condition exactly, and no finite sum of oscillator products can do either.

13.8 Summary and what comes next

A variational Monte Carlo program has four parts, and each has been dealt with in turn: a trial wave function whose form is dictated by the physics (Section 13.3); a local energy, differentiated in closed form and verified (Section 13.2); a sampler, which is the Metropolis algorithm applied to $|\Psi_T|^2$ (Section 13.5); and an error estimate that accounts for the correlations a Markov chain introduces (Section 12.3). Importance sampling (Section 13.6) improves the third of these by proposing moves along the quantum force and correcting the asymmetry with Metropolis-Hastings.

Three things are still missing, and they are the subject of what follows. The error analysis here used the autocorrelation time; for production runs the blocking method of Section 12.3 is what one actually uses. The parameters were located by scanning a grid, which does not scale beyond two or three of them; gradient-based optimisation using the derivative of the energy with respect to the parameters is the standard replacement. And the result remains an upper bound set by the quality of Ψ_T. Diffusion Monte Carlo removes that last limitation by letting the trial function evolve in imaginary time towards the true ground state, using the same Fokker-Planck machinery of Section 13.6 – which is why it was worth setting up carefully here.

13.9 Exercises

Exercise 1: the local energy

a) Derive the first line of Eq. (13.14), the local energy of the pure Gaussian trial function.
b) Show that E_L is constant when $\alpha = 1$ and the interaction is switched off, and say what its value is.
c) Show that the $1/r_{12}$ terms in Eq. (13.14) cancel as $r_{12} \to 0$.

Answer to a).

With $\Psi = \exp(-\alpha\omega R^2/2)$ and $R^2 = r_1^2 + r_2^2$, use $\nabla^2\Psi/\Psi = \nabla^2 \ln\Psi + (\nabla \ln\Psi)^2$. In two dimensions $\nabla_i^2(-\alpha\omega r_i^2/2) = -2\alpha\omega$ for each particle, so the first term contributes $-4\alpha\omega$, and $(\nabla \ln\Psi)^2 = \alpha^2\omega^2 R^2$. Hence

$$-\frac{1}{2}\frac{\nabla^2\Psi}{\Psi} = 2\alpha\omega - \frac{1}{2}\alpha^2\omega^2 R^2,$$

and adding the potential $\frac{1}{2}\omega^2 R^2 + 1/r_{12}$ gives $2\alpha\omega + \frac{1}{2}\omega^2(1-\alpha^2)R^2 + 1/r_{12}$.

Answer to b).

Setting $\alpha = 1$ kills the R^2 term and dropping $1/r_{12}$ leaves $E_L = 2\hbar\omega$, independent of position. This is the zero-variance principle in its simplest form: without the interaction the trial function is the exact ground state of $\hat{H}_0$, its energy is $2\hbar\omega$, and a single sample would suffice. All the variance in the interacting problem comes from the Coulomb term and from the imperfection of the Jastrow factor.

Answer to c).

As $r_{12} \to 0$, $d = 1/(1+\beta r_{12}) \to 1$, so the term $-d^2/r_{12}$ tends to $-1/r_{12}$ and cancels the $+1/r_{12}$ on the first line. What survives is finite: expanding to first order, $d^2 \approx 1 - 2\beta r_{12}$, and

$$\frac{1}{r_{12}}\left(1-d^2\right) \longrightarrow 2\beta,$$

so the singular parts cancel and leave a constant. This is Table 13.1 in algebra, and it is why the numerator of the Jastrow exponent had to be exactly 1 and not a free parameter.

Exercise 2: the quantum force and the Green's function

a) Derive Eq. (13.24) from Eq. (13.13).
b) Derive the simplified Green's function ratio Eq. (13.23) from Eq. (13.21).
c) Show that Metropolis-Hastings with the ratio Eq. (13.22) satisfies detailed balance for *any* drift $\boldsymbol{F}$, and explain what goes wrong if $\boldsymbol{F}$ is wrong.

Answer to a).

$\ln\Psi_T = -\frac{1}{2}\alpha\omega(r_1^2+r_2^2) + r_{12}/(1+\beta r_{12})$. The first term gives $\nabla_1 = -\alpha\omega\boldsymbol{r}_1$. For the second, note $\nabla_1 r_{12} = (\boldsymbol{r}_1 - \boldsymbol{r}_2)/r_{12}$ and

$$\frac{d}{dr_{12}}\frac{r_{12}}{1+\beta r_{12}} = \frac{1}{(1+\beta r_{12})^2} = d^2,$$

so $\nabla_1[\,\cdot\,] = (\boldsymbol{r}_1 - \boldsymbol{r}_2)d^2/r_{12}$. Multiplying the sum by two gives Eq. (13.24); $\boldsymbol{F}_2$ follows from $\nabla_2 r_{12} = -(\boldsymbol{r}_1 - \boldsymbol{r}_2)/r_{12}$.

Answer to b).

Write $\boldsymbol{u} = \boldsymbol{y} - \boldsymbol{x}$, $\boldsymbol{a} = D\Delta t\boldsymbol{F}(\boldsymbol{y})$ and $\boldsymbol{b} = D\Delta t\boldsymbol{F}(\boldsymbol{x})$. Then

$$\ln\frac{G(\boldsymbol{x},\boldsymbol{y})}{G(\boldsymbol{y},\boldsymbol{x})} = -\frac{(\boldsymbol{u}+\boldsymbol{a})^2-(\boldsymbol{u}-\boldsymbol{b})^2}{4D\Delta t} = -\frac{2\boldsymbol{u}\cdot(\boldsymbol{a}+\boldsymbol{b})+a^2-b^2}{4D\Delta t}.$$

Substituting back and factorising $a^2-b^2 = (\boldsymbol{a}+\boldsymbol{b})\cdot(\boldsymbol{a}-\boldsymbol{b})$ gives Eq. (13.23). The prefactors of G are identical in numerator and denominator and cancel, which is why the normalisation of the Green's function never has to be computed.

Answer to c).

Detailed balance requires $T(\boldsymbol{x} \to \boldsymbol{y})A(\boldsymbol{x} \to \boldsymbol{y})P(\boldsymbol{x}) = T(\boldsymbol{y} \to \boldsymbol{x})A(\boldsymbol{y} \to \boldsymbol{x})P(\boldsymbol{y})$. With $T = G$ and $A = \min(1,q)$ from Eq. (13.22) this holds identically, whatever $\boldsymbol{F}$ is, because q is by construction the ratio that makes it hold. A wrong $\boldsymbol{F}$ therefore leaves the stationary distribution untouched and the energy correct. What it damages is the acceptance rate: the drift proposes moves the wave function does not like, more of them are rejected, and the correlation time rises. The consistency requirement is that the same $\boldsymbol{F}$ be used in the proposal and in G; using different ones would break detailed balance and give a wrong answer.

Exercise 3: the cusp condition

a) Derive Eq. (13.8) by keeping only the terms in the relative-motion local energy that survive as $r_{12} \to 0$.
b) Why is the coefficient different for parallel and antiparallel spins?
c) Show that the Padé-Jastrow form (13.10) reproduces the cusp at short range and saturates at long range, and say what goes wrong if the bare exponential is used instead.

Answer to a).

In the relative coordinate the kinetic operator contains $d^2/dr^2 + (2/r)\,d/dr$ in two dimensions, and the potential contains $1/r$. For E_L to stay finite as $r \to 0$ the $1/r$ terms must cancel:

$$\frac{2}{r}\frac{\mathscr{R}'}{\mathscr{R}} + \frac{2}{r} = 0 \quad \Longrightarrow \quad \left.\frac{\mathscr{R}'}{\mathscr{R}}\right|_{r\to 0} = \text{constant},$$

which is Eq. (13.8). Everything else – the harmonic term, the energy, the centre-of-mass part – is regular at the origin and cannot participate.

Answer to b).

For parallel spins the spatial wave function is antisymmetric and already vanishes linearly as $r_{12} \to 0$, so the Coulomb singularity is partly screened by the Pauli principle and less is asked of the correlation factor. The angular momentum in the relative motion is $l = 1$ rather than $l = 0$, and the factor $1/2(l+1)$ in Eq. (13.8) halves the coefficient.

Answer to c).

Expanding $ar/(1+\beta r) = ar(1-\beta r+\cdots)$ shows that the exponent is ar to first order, so the logarithmic derivative at the origin is a and the cusp condition is satisfied for any β. As $r \to \infty$ the exponent tends to a/β, a constant, so the factor saturates and the wave function retains the decay of the mean-field part. With the bare $\exp(ar)$ the correlation factor grows without bound and overwhelms the Gaussian at large separation: the trial function is not normalisable in the relevant sense and the electrons are driven apart. This is exactly what Chapter 12 warned about – setting $\beta = 0$ does not switch the Jastrow factor off, it leaves $\exp(r_{12})$.

Exercise 4: numerical explorations

a) Using `vmc.py`, reproduce Tables 13.2 and 13.3. Plot the correlation time against the acceptance rate for both samplers on the same axes. Is there a universal optimal acceptance rate?
b) Replace the analytic local energy by the finite-difference one and compare the cost and the statistical error. How small may the step h be made before round-off dominates?
c) Implement blocking, as in Section 12.3, and compare the error estimate with the one from the correlation time for a run at $\Delta t = 0.01$, where τ is large.
d) Repeat the calculation at $\hbar\omega = 0.5$ and $\hbar\omega = 0.1$. Where is the correlation strongest, and how do the optimal α and β move? Compare with Table 11.6.
e) Compute the derivative of the energy with respect to a variational parameter θ using

$$\frac{\partial E}{\partial \theta} = 2\left(\langle E_L\, \partial_\theta \ln \Psi_T\rangle - \langle E_L\rangle\langle \partial_\theta \ln \Psi_T\rangle\right),$$

and use it to optimise α and β by gradient descent. How many iterations does it take, and how does the cost compare with the grid scan of Table 13.4?
f) Extend the code to six electrons using a Slater determinant of the lowest three oscillator orbitals times the Padé-Jastrow factor, and compare with the coupled-cluster result of Table 11.3.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter13/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/variationalmc.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter13/`.

`vmc.py` The two-electron dot: the analytic local energy and quantum force, both checked against finite differences, brute-force Metropolis and importance-sampled Metropolis–Hastings, and parameter scans. Runs in about a minute.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 14
Optimization and resampling

Chapter 13 left two loose ends. The variational parameters were located by scanning a grid, which worked because there were only two of them; and the error bar was obtained from the autocorrelation time, which is serviceable but not what a production calculation uses. Both are addressed here, and the two are more closely related than they look: the first is about spending computer time cheaply while hunting for a minimum, the second about spending it lavishly once the minimum is found and then reporting honestly what was achieved.

The structure of a real calculation has three stages.

1. **Optimise.** Compute the gradient of the energy with respect to the variational parameters – which turns out to be another Monte Carlo average, costing nothing extra – and follow it downhill. Short, noisy runs suffice: we are locating a minimum, not measuring an energy.
2. **Produce.** With the parameters fixed, run once with as many samples as the budget allows. Millions, and on a cluster, billions.
3. **Resample.** The samples are correlated, so $\sigma/\sqrt{N}$ is wrong. The variance of a correlated mean is a sum over the whole covariance matrix; blocking, the bootstrap and the jackknife are three ways of evaluating it without knowing that matrix.

The middle stage is where most of the chapter goes. We derive the gradient in full, and then set out four families of methods for following it: first-order descent with and without momentum, the quasi-Newton methods of Broyden and BFGS, the derivative-free direction-set method of Powell, and stochastic reconfiguration. The first three are the standard tools of deterministic optimisation and are worth having for their own sake – Broyden's update is the accelerator behind self-consistent field iterations, and L-BFGS is the default optimiser of large-scale numerical analysis. The last is the one built for a stochastic objective, and it is the one that survives when the trial function acquires thousands of parameters.

The programs are in `vmcoptimise.py`, which imports the trial function, local energy and quantum force of Chapter 13 unchanged, and which verifies every algebraic identity of this chapter numerically.

14.1 Why a grid does not scale

Table 13.4 was a 5×5 grid: twenty-five Monte Carlo runs to locate two parameters, and the answer only as good as the mesh. With p parameters and m points along each, the cost is m^p

runs. For $p = 2$ and $m = 5$ that is twenty-five; for $p = 10$ it is ten million; and for the neural-network trial functions of the later chapters, where p runs to thousands, it is not a number worth writing down.

Worse, a grid throws away information. Each run already knows which way is downhill – the samples that produced the energy can produce its derivative too – and a scan discards that and starts again at the next node. What we need is the gradient.

14.2 The gradient of the energy

Let $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_P)$ collect the variational parameters, let $\psi_{\boldsymbol{\theta}}(x)$ be the trial function – x standing for the whole set of many-body coordinates, continuous or discrete – and write

$$\bar{E}_\theta = \frac{d\langle E_L[\theta]\rangle}{d\theta}, \qquad \bar{\Psi}_\theta = \frac{d\psi[\theta]}{d\theta}. \tag{14.1}$$

We shall derive, in this section, that

$$\boxed{\bar{E}_\theta = 2\left(\Big\langle \frac{\bar{\Psi}_\theta}{\psi} E_L\Big\rangle - \Big\langle \frac{\bar{\Psi}_\theta}{\psi}\Big\rangle\langle E_L\rangle\right)} \tag{14.2}$$

This is the central formula of the chapter. It is worth doing the derivation carefully rather than quoting the result, because every step of it recurs: the same manipulation gives the metric of Section 14.6, the derivative of the variance, and the gradient of any other estimator one might want to optimise.

14.2.1 Differentiating the Rayleigh quotient

We take $\psi_{\boldsymbol{\theta}}$ real; the complex case is treated at the end of the section and differs only by a conjugation and a real part. The energy is the Rayleigh quotient

$$E(\boldsymbol{\theta}) = \frac{\langle \psi_{\boldsymbol{\theta}}|\hat{H}|\psi_{\boldsymbol{\theta}}\rangle}{\langle \psi_{\boldsymbol{\theta}}|\psi_{\boldsymbol{\theta}}\rangle} = \frac{\int dx\, \psi_{\boldsymbol{\theta}}(x)\,\left[\hat{H}\psi_{\boldsymbol{\theta}}\right](x)}{\int dx\, \psi_{\boldsymbol{\theta}}(x)^2}, \tag{14.3}$$

and we abbreviate numerator and denominator as

$$A(\boldsymbol{\theta}) = \int dx\, \psi_{\boldsymbol{\theta}}\, \hat{H}\, \psi_{\boldsymbol{\theta}}, \qquad B(\boldsymbol{\theta}) = \int dx\, \psi_{\boldsymbol{\theta}}^2, \tag{14.4}$$

so that $E = A/B$. The quotient rule gives

$$\partial_\theta E = \frac{(\partial_\theta A)\,B - A\,(\partial_\theta B)}{B^2}, \tag{14.5}$$

and the whole derivation consists of evaluating the two derivatives and recognising what comes out.

The denominator is immediate,

$$\partial_\theta B = \int dx\, 2\psi_{\boldsymbol{\theta}}\,(\partial_\theta \psi_{\boldsymbol{\theta}}). \tag{14.6}$$

The numerator has two terms, because θ appears on both sides of $\hat{H}$,

$$\partial_\theta A = \int dx\, \left[(\partial_\theta \psi_{\boldsymbol{\theta}})\, \hat{H} \psi_{\boldsymbol{\theta}} + \psi_{\boldsymbol{\theta}}\, \hat{H}\, (\partial_\theta \psi_{\boldsymbol{\theta}}) \right].$$

Here the hermiticity of $\hat{H}$ enters. Provided the boundary terms generated by the two integrations by parts in the kinetic energy vanish – which they do for any square-integrable trial function, and this is the one assumption of the derivation –

$$\int dx\, \psi_{\boldsymbol{\theta}}\, \hat{H}\, (\partial_\theta \psi_{\boldsymbol{\theta}}) = \int dx\, (\partial_\theta \psi_{\boldsymbol{\theta}})\, \hat{H} \psi_{\boldsymbol{\theta}},$$

so the two terms are equal and

$$\partial_\theta A = 2 \int dx\, (\partial_\theta \psi_{\boldsymbol{\theta}})\, \hat{H} \psi_{\boldsymbol{\theta}}. \tag{14.7}$$

Note what has been gained: the derivative of the *energy* now requires only the derivative of the *wave function*, and never the derivative of $\hat{H}\psi$. That is the difference between an optimisation that costs nothing and one that costs a great deal.

Substituting Eqs. (14.6) and (14.7) into Eq. (14.5) and using $A = EB$ to eliminate A,

$$\partial_\theta E = \frac{2\left(\int dx\, (\partial_\theta \psi_{\boldsymbol{\theta}}) \hat{H} \psi_{\boldsymbol{\theta}}\right) B - EB\left(2 \int dx\, \psi_{\boldsymbol{\theta}}\, \partial_\theta \psi_{\boldsymbol{\theta}}\right)}{B^2},$$

and one factor of B cancels throughout,

$$\partial_\theta E = 2 \left[\frac{\int dx\, (\partial_\theta \psi_{\boldsymbol{\theta}}) \hat{H} \psi_{\boldsymbol{\theta}}}{\int dx\, \psi_{\boldsymbol{\theta}}^2} - E\, \frac{\int dx\, \psi_{\boldsymbol{\theta}}\, \partial_\theta \psi_{\boldsymbol{\theta}}}{\int dx\, \psi_{\boldsymbol{\theta}}^2} \right]. \tag{14.8}$$

The last step is the one that turns an expression into an algorithm. Multiply and divide the first integrand by $\psi_{\boldsymbol{\theta}}^2$,

$$\frac{\int dx\, (\partial_\theta \psi_{\boldsymbol{\theta}}) \hat{H} \psi_{\boldsymbol{\theta}}}{\int dx\, \psi_{\boldsymbol{\theta}}^2} = \frac{\int dx\, \psi_{\boldsymbol{\theta}}^2 \left(\dfrac{\partial_\theta \psi_{\boldsymbol{\theta}}}{\psi_{\boldsymbol{\theta}}} \right) \left(\dfrac{\hat{H} \psi_{\boldsymbol{\theta}}}{\psi_{\boldsymbol{\theta}}} \right)}{\int dx\, \psi_{\boldsymbol{\theta}}^2},$$

and recognise the two brackets. The second is the local energy of Chapter 13,

$$E_L(x; \boldsymbol{\theta}) = \frac{\left[\hat{H} \psi_{\boldsymbol{\theta}}\right](x)}{\psi_{\boldsymbol{\theta}}(x)}, \tag{14.9}$$

and the first is the logarithmic derivative of the trial function with respect to the parameter,

$$O_\theta(x) = \frac{\partial_\theta \psi_{\boldsymbol{\theta}}(x)}{\psi_{\boldsymbol{\theta}}(x)} = \partial_\theta \ln \psi_{\boldsymbol{\theta}}(x). \tag{14.10}$$

What remains outside the brackets is $\psi_{\boldsymbol{\theta}}^2 / \int \psi_{\boldsymbol{\theta}}^2$, which is precisely the normalised sampling density

$$p_{\boldsymbol{\theta}}(x) = \frac{|\psi_{\boldsymbol{\theta}}(x)|^2}{\int dx'\, |\psi_{\boldsymbol{\theta}}(x')|^2}, \qquad \int dx\, p_{\boldsymbol{\theta}}(x) = 1, \tag{14.11}$$

the same density the Metropolis walk of Chapter 13 already generates. Every ratio in Eq. (14.8) is therefore an expectation value with respect to $p_{\boldsymbol{\theta}}$, and with $E = \langle E_L \rangle$ we arrive at Eq. (14.2),

$$\partial_\theta E = 2\left(\langle O_\theta E_L \rangle - \langle O_\theta \rangle \langle E_L \rangle\right) = 2\,\mathrm{Cov}(O_\theta, E_L). \tag{14.12}$$

14.2.2 Three averages, one random walk

Written out, the gradient is made of the three integrals

$$\langle O_\theta E_L \rangle = \int dx\, p_{\boldsymbol{\theta}}(x)\, O_\theta(x)\, E_L(x), \tag{14.13}$$

$$\langle O_\theta \rangle = \int dx\, p_{\boldsymbol{\theta}}(x)\, O_\theta(x), \tag{14.14}$$

$$\langle E_L \rangle = \int dx\, p_{\boldsymbol{\theta}}(x)\, E_L(x). \tag{14.15}$$

The decisive observation is that all three are averages over the *same* density $p_{\boldsymbol{\theta}} \propto |\psi_{\boldsymbol{\theta}}|^2$. One Markov chain, already being run to obtain the energy, delivers all of them; the additional cost is P extra accumulators and the evaluation of P logarithmic derivatives per sample. Given samples $\{x^{(m)}\}_{m=1}^{M}$ drawn from $p_{\boldsymbol{\theta}}$, and writing $O_\theta^{(m)} = O_\theta(x^{(m)})$ and $E_L^{(m)} = E_L(x^{(m)})$, the estimator is

$$\widehat{\partial_\theta E} = 2\left(\frac{1}{M}\sum_{m=1}^{M} O_\theta^{(m)} E_L^{(m)} - \left(\frac{1}{M}\sum_{m=1}^{M} O_\theta^{(m)}\right)\left(\frac{1}{M}\sum_{m=1}^{M} E_L^{(m)}\right)\right). \tag{14.16}$$

It is consistent, and biased at order $1/M$ only through the product of the two separately unbiased means – the usual and harmless bias of a sample covariance.

Three further things about Eq. (14.2) are worth noticing.

First, it is a *covariance*, Eq. (14.12): the gradient is twice the covariance of the local energy with the logarithmic derivative of the trial function. If the local energy and the logarithmic derivative are uncorrelated the gradient vanishes, which is another way of saying that at the minimum the trial function has stopped being able to tell good configurations from bad ones.

Second, it needs only *first* derivatives of $\ln\psi$ with respect to the parameters – no second derivatives, no derivatives of the local energy, and no extra sampling. The same walk that measures the energy accumulates $\langle O_\theta \rangle$ and $\langle O_\theta E_L \rangle$ in the same loop.

Third, and least obviously, it is much better behaved than a finite difference. Estimating $\partial E/\partial\theta$ by running at $\theta \pm h$ and subtracting requires two runs per parameter, and the subtraction cancels most of the signal while the statistical errors add. Equation (14.2) has no cancellation of that kind.

Complex trial functions.

Nothing above is specific to real ψ. If $\psi_{\boldsymbol{\theta}}$ is complex, replace $\psi_{\boldsymbol{\theta}}^2$ by $|\psi_{\boldsymbol{\theta}}|^2$ throughout, note that $O_\theta = \partial_\theta \ln\psi_{\boldsymbol{\theta}}$ is then complex too, and observe that the two terms in $\partial_\theta A$ are now complex conjugates of one another rather than equal. Their sum is twice the real part, and the gradient of the (real) energy becomes

$$\partial_\theta E = 2\,\mathrm{Re}\{\langle O_\theta^* E_L \rangle - \langle O_\theta^* \rangle \langle E_L \rangle\}\,. \tag{14.17}$$

This is the form used for the neural-network wave functions of the later chapters, where the phase carries physical information and cannot be dropped; for the real Slater-Jastrow functions of this chapter and the next, Eq. (14.2) is what we use. The same derivation applies verbatim to excited-state functionals and to any other cost function built as a ratio of quadratic forms in ψ.

For our trial function.

With Ψ_T of Eq. (13.13) the two logarithmic derivatives are elementary,

$$O_\alpha = -\frac{\omega}{2}\left(r_1^2 + r_2^2\right), \qquad O_\beta = -\frac{r_{12}^2}{(1+\beta r_{12})^2}, \tag{14.18}$$

and `vmcoptimise.py` checks both against finite differences, agreeing to 2×10^{-9}. It then checks the assembled gradient Eq. (14.2) against a finite difference of the energy itself; Table 14.1 shows the comparison.

		$\partial E/\partial\alpha$		$\partial E/\partial\beta$	
α	β	Eq. (14.2)	finite difference	Eq. (14.2)	finite difference
0.90	0.30	−0.41171	−0.42413	−0.29347	−0.29289
1.00	0.40	0.02905	0.03823	0.01227	0.01853
1.05	0.50	0.22099	0.21838	0.10431	0.10902

Table 14.1 The analytic gradient against a central difference of the energy with $h = 0.01$, 40000 cycles per run. They agree within the Monte Carlo noise, and the analytic route costs one run instead of four. Computed by `vmcoptimise.py`.

14.3 Descending the surface

With a gradient in hand the simplest scheme is *steepest descent,*

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \eta\, \nabla_{\boldsymbol{\theta}} E(\boldsymbol{\theta}_k), \tag{14.19}$$

with η the learning rate. For a convex function and a small enough η this converges; the difficulty is that “small enough” depends on the curvature, and the curvature is exactly what the gradient does not know. If the surface is a narrow valley – the Hessian badly conditioned – steepest descent zig-zags across it and creeps along it, and the number of iterations grows with the condition number.

Table 14.2 runs it on the quantum dot, starting deliberately far from the minimum. Two features are worth pointing out. The energy falls from 3.1128 to within 2×10^{-3} of the exact answer in about ten steps; and each step uses only 6000 Monte Carlo cycles, which would be far too few to *report* an energy but is ample to decide which way is down. That asymmetry – cheap runs while optimising, one expensive run at the end – is the reason the two halves of this chapter belong together.

Figure 14.1 draws those same twenty iterations as a path in the (β, α) plane, and the first thing to notice is what is absent: there is no zig-zag. With $\eta = 0.05$ the step is far below the stability limit, successive gradients point in very nearly the same direction, and the path is a smooth arc rather than a staircase thrown across a valley. What the figure shows instead is the other half of the steepest-descent pathology, the creep. The first step moves the parameters by 0.045 in α and 0.048 in β; the last five together move them by only 0.003 and 0.008, because the step is proportional to a gradient that has shrunk by a factor of forty-seven along the way. The consequence is legible at the end of the path. The star sits at $\beta = 0.355$, while the minimum – the one the safeguarded methods of Table 14.6 reach – lies near $\beta = 0.397$: α

iteration	α	β	energy	$\lvert\nabla E\rvert$
1	0.85000	0.20000	3.112849	1.321241
2	0.89515	0.24822	3.046038	0.722958
3	0.92182	0.27262	3.029580	0.507078
4	0.94113	0.28906	3.013600	0.366792
6	0.96631	0.31072	3.007260	0.227005
10	0.99126	0.33328	3.002706	0.099140
15	1.00300	0.34749	3.000419	0.045287
20	1.00623	0.35534	3.000180	0.028328
25	1.00660	0.36082	3.001775	0.017045

Table 14.2 Steepest descent from $(\alpha,\beta) = (0.85, 0.20)$ with $\eta = 0.05$ and 6000 cycles per step. The gradient is noisy, which is why the last few energies wander at the level of 10^{-3}; the parameters are nonetheless located well enough. Computed by `vmcoptimise.py`.

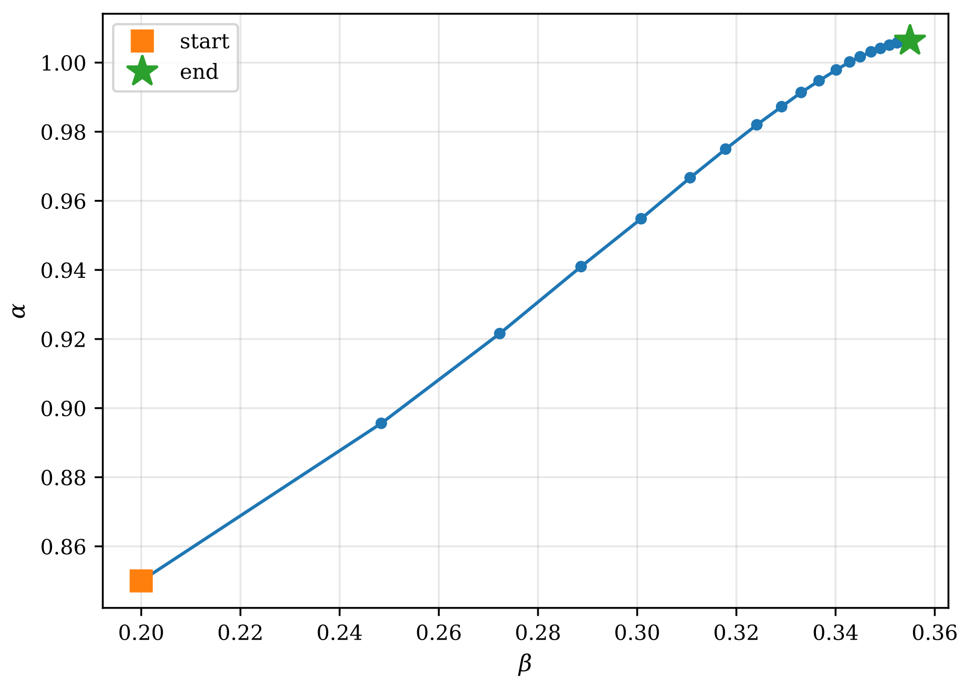

Fig. 14.1 The descent of Table 14.2 drawn as a path in the plane of the two variational parameters, with β horizontal and α vertical, over twenty steps of Eq. (14.19) at a fixed learning rate $\eta = 0.05$. The square marks the deliberately poor starting point $(\alpha,\beta) = (0.85, 0.20)$, the star the point $(1.0062, 0.3553)$ reached after twenty steps, and the circles between them the individual iterates, each obtained from its own short Monte Carlo run. Since the step is $-\eta\nabla E$, the spacing of the circles is a direct picture of the length of the gradient, which falls from 1.32 at the start to 0.028 at the end.

has arrived and stopped to three decimals, and the almost horizontal final stretch is β still creeping towards its answer. The two parameters are not equally stiff, one learning rate has to serve both, and it is necessarily set by the stiff direction while the soft one converges at leisure. The anisotropy is mild here – the metric of Eq. (14.65) has a condition number of 9.7 – and it already costs a factor of several in iterations. The next paragraph makes that statement precise.

Quantifying the zig-zag.

The statement can be made precise for a quadratic, $f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^T\boldsymbol{x}$ with $\boldsymbol{A}$ symmetric positive definite. The error $\boldsymbol{e}_k = \boldsymbol{x}_k - \boldsymbol{x}_*$ obeys $\boldsymbol{e}_{k+1} = (\boldsymbol{I} - \eta\boldsymbol{A})\boldsymbol{e}_k$, so in the eigenbasis of $\boldsymbol{A}$ each component decays as $(1-\eta\lambda_i)^k$. The best single η balances the extremes, $\eta = 2/(\lambda_{\min}+\lambda_{\max})$, and gives the rate

$$\|\boldsymbol{e}_{k+1}\| \leq \frac{\kappa-1}{\kappa+1}\|\boldsymbol{e}_k\|, \qquad \kappa = \frac{\lambda_{\max}}{\lambda_{\min}}, \tag{14.20}$$

with κ the condition number of Section 1.13. The number of iterations therefore grows *linearly* with κ, and it does so even if the step length is chosen optimally at every iteration. This is not a defect of the step size; it is a defect of the direction. Steepest descent knows only the gradient, and the gradient alone cannot tell a narrow valley from a round bowl.

Momentum.

The cheapest repair accumulates a velocity,

$$\boldsymbol{v}_{k+1} = \gamma\boldsymbol{v}_k + \eta\nabla E, \qquad \boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \boldsymbol{v}_{k+1}, \tag{14.21}$$

which damps the zig-zag across a valley and accelerates along it: successive gradients that alternate in sign cancel in $\boldsymbol{v}$, while successive gradients that agree reinforce. With a *stochastic* gradient momentum does something else as well, and arguably more useful: it averages successive estimates, so the noise in each individual one matters less. It is the ancestor of the whole family of adaptive first-order methods – RMSProp, Adam, AdaGrad – that dominate machine learning, all of which are attempts to guess a per-parameter step size without ever forming a matrix.

The three remaining routes all form a matrix, and they differ in where they get it. *Quasi-Newton* methods (Section 14.4) infer the Hessian from successive gradients. *Derivative-free* methods (Section 14.5) build a set of conjugate directions from function values alone. *Stochastic reconfiguration* (Section 14.6) uses not the Hessian of the energy but the metric of the space of wave functions, and it is the one designed for the problem at hand.

14.4 Quasi-Newton methods

Newton's method is the reference against which the rest are measured, so we start there, and we start with root finding rather than minimisation because the algebra is simpler and because the two problems are the same one: at a minimum the gradient vanishes, so minimising f is solving $\nabla f(\boldsymbol{x}) = 0$.

Given an iterate $\boldsymbol{x}_k$ and a vector-valued $\boldsymbol{F} : \mathbb{R}^n \to \mathbb{R}^n$, linearise,

$$\boldsymbol{F}(\boldsymbol{x}_k + \boldsymbol{s}) \approx \boldsymbol{F}(\boldsymbol{x}_k) + \boldsymbol{J}_k\boldsymbol{s}, \qquad \boldsymbol{J}_k = \nabla\boldsymbol{F}(\boldsymbol{x}_k), \tag{14.22}$$

and solve the linearised equation for the step,

$$\boldsymbol{J}_k\boldsymbol{s}_k = -\boldsymbol{F}(\boldsymbol{x}_k), \qquad \boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \boldsymbol{s}_k. \tag{14.23}$$

Newton's method converges quadratically near a simple root, which is as good as it gets, and it has two costs that are often prohibitive. It needs the $n \times n$ Jacobian at every iteration – n^2 derivatives, and for a Monte Carlo objective every one of them carries a statistical error – and it needs that matrix factorised, at $\mathscr{O}(n^3)$. For minimisation the Jacobian is the Hessian $\nabla^2 f$, and the situation is worse: a stochastic Hessian is not merely expensive but is frequently not positive definite, in which case the Newton step points uphill.

Quasi-Newton methods never form either matrix. They build an approximation $\boldsymbol{B}_k \approx \boldsymbol{J}_k$, or an approximation $\boldsymbol{H}_k \approx \boldsymbol{J}_k^{-1}$ to its inverse, out of information that the iteration is generating anyway.

14.4.1 The secant condition

Define the step and the change in the function,

$$\boldsymbol{s}_k = \boldsymbol{x}_{k+1} - \boldsymbol{x}_k, \qquad \boldsymbol{y}_k = \boldsymbol{F}(\boldsymbol{x}_{k+1}) - \boldsymbol{F}(\boldsymbol{x}_k), \tag{14.24}$$

or, for minimisation, $\boldsymbol{y}_k = \nabla f(\boldsymbol{x}_{k+1}) - \nabla f(\boldsymbol{x}_k)$. By the mean value theorem $\boldsymbol{y}_k \approx \boldsymbol{J}(\boldsymbol{x}_k)\boldsymbol{s}_k$, with equality when $\boldsymbol{F}$ is affine, that is when f is quadratic. Quasi-Newton methods take this as a definition and require the new approximation to reproduce it exactly,

$$\boxed{\boldsymbol{B}_{k+1}\boldsymbol{s}_k = \boldsymbol{y}_k} \tag{14.25}$$

the *secant equation*. In one dimension it determines B_{k+1} completely and gives the secant method of elementary numerical analysis: the derivative is replaced by the slope of the chord through the last two points. In n dimensions it is n equations for the n^2 entries of $\boldsymbol{B}_{k+1}$, and the remaining $n^2 - n$ degrees of freedom must be fixed by some other principle. Different choices of that principle give different methods, and all of the useful ones amount to the same instruction: change as little as possible.

14.4.2 Broyden's method as a minimum-change update

Take that instruction literally. We look for the $\boldsymbol{B}_{k+1}$ that satisfies Eq. (14.25) and lies closest to $\boldsymbol{B}_k$ in the Frobenius norm of Section 1.13.1. Writing $\Delta\boldsymbol{B} = \boldsymbol{B}_{k+1} - \boldsymbol{B}_k$ and $\boldsymbol{r}_k = \boldsymbol{y}_k - \boldsymbol{B}_k\boldsymbol{s}_k$ for the *secant residual*, the problem is

$$\min_{\Delta\boldsymbol{B}} \|\Delta\boldsymbol{B}\|_F^2 \qquad \text{subject to} \qquad \Delta\boldsymbol{B}\,\boldsymbol{s}_k = \boldsymbol{r}_k. \tag{14.26}$$

This is a linearly constrained least-squares problem, and it is solved in three lines. Introduce a Lagrange multiplier $\boldsymbol{\lambda} \in \mathbb{R}^n$ and write, using $\|\boldsymbol{X}\|_F^2 = \mathrm{Tr}(\boldsymbol{X}^T\boldsymbol{X})$,

$$\mathscr{L}(\Delta\boldsymbol{B}, \boldsymbol{\lambda}) = \mathrm{Tr}\left(\Delta\boldsymbol{B}^T\Delta\boldsymbol{B}\right) + 2\boldsymbol{\lambda}^T\left(\Delta\boldsymbol{B}\,\boldsymbol{s}_k - \boldsymbol{r}_k\right). \tag{14.27}$$

Stationarity with respect to the matrix $\Delta\boldsymbol{B}$ uses the single derivative $\partial\,\mathrm{Tr}(\boldsymbol{X}^T\boldsymbol{X})/\partial\boldsymbol{X} = 2\boldsymbol{X}$, which is available precisely because the Frobenius norm comes from an inner product,

$$\frac{\partial\mathscr{L}}{\partial\Delta\boldsymbol{B}} = 2\Delta\boldsymbol{B} + 2\boldsymbol{\lambda}\boldsymbol{s}_k^T = 0 \qquad \Longrightarrow \qquad \Delta\boldsymbol{B} = -\boldsymbol{\lambda}\boldsymbol{s}_k^T.$$

The answer is therefore a rank-one matrix before we have imposed anything else, which is the whole content of the method. Imposing the constraint fixes the multiplier,

$$\Delta\boldsymbol{B}\,\boldsymbol{s}_k = -\boldsymbol{\lambda}\left(\boldsymbol{s}_k^T\boldsymbol{s}_k\right) = \boldsymbol{r}_k \qquad \Longrightarrow \qquad \boldsymbol{\lambda} = -\frac{\boldsymbol{r}_k}{\boldsymbol{s}_k^T\boldsymbol{s}_k},$$

and substituting back gives $\Delta \boldsymbol{B} = \boldsymbol{r}_k \boldsymbol{s}_k^T / (\boldsymbol{s}_k^T \boldsymbol{s}_k)$, that is

$$\boxed{\boldsymbol{B}_{k+1} = \boldsymbol{B}_k + \frac{(\boldsymbol{y}_k - \boldsymbol{B}_k \boldsymbol{s}_k)\,\boldsymbol{s}_k^T}{\boldsymbol{s}_k^T \boldsymbol{s}_k}} \tag{14.28}$$

This is the *good Broyden* update.

Three remarks. It is a rank-one outer product, so it changes $\boldsymbol{B}_k$ only in the single direction $\boldsymbol{s}_k$ – along any direction $\boldsymbol{v}$ orthogonal to $\boldsymbol{s}_k$ one has $\Delta \boldsymbol{B}\boldsymbol{v} = 0$, and the old approximation survives untouched. That is the correct behaviour: the iteration has learned about the curvature along $\boldsymbol{s}_k$ and about nothing else, so it should change nothing else. Second, if $\boldsymbol{s}_k^T \boldsymbol{s}_k$ is very small the update is dominated by rounding, and practical codes either skip it or restart from $\boldsymbol{B} = \boldsymbol{I}$. Third, the derivation used no property of $\boldsymbol{F}$ at all, which is why the same formula appears in contexts that have nothing to do with Jacobians.

The inverse update.

Equation (14.28) still leaves a linear system $\boldsymbol{B}_k \boldsymbol{p}_k = -\boldsymbol{F}(\boldsymbol{x}_k)$ to be solved at each step. It is cheaper to approximate the inverse directly. Impose the *inverse secant condition* $\boldsymbol{H}_{k+1}\boldsymbol{y}_k = \boldsymbol{s}_k$ – which is Eq. (14.25) with the roles of $\boldsymbol{s}$ and $\boldsymbol{y}$ exchanged – and repeat the calculation verbatim with $\boldsymbol{q}_k = \boldsymbol{s}_k - \boldsymbol{H}_k \boldsymbol{y}_k$ in place of $\boldsymbol{r}_k$:

$$\boldsymbol{H}_{k+1} = \boldsymbol{H}_k + \frac{(\boldsymbol{s}_k - \boldsymbol{H}_k \boldsymbol{y}_k)\,\boldsymbol{y}_k^T}{\boldsymbol{y}_k^T \boldsymbol{y}_k}. \tag{14.29}$$

The step is now a matrix-vector product, $\boldsymbol{p}_k = -\boldsymbol{H}_k \boldsymbol{F}(\boldsymbol{x}_k)$, at $\mathscr{O}(n^2)$ and with no factorisation at all. There is a third possibility, obtained by applying the Sherman-Morrison formula of Section 2.11 to Eq. (14.28) rather than re-deriving; it gives the exact inverse of the updated $\boldsymbol{B}$ and is what is usually meant by “Broyden’s second” or *bad Broyden* method. The names are unfortunate: neither is bad, they simply minimise different things – the change in $\boldsymbol{B}$, or the change in $\boldsymbol{H}$ – and the first is usually the more robust in practice. Powell damping and trust regions stabilise both.

Many-body connection: Broyden mixing. Every self-consistent field calculation in this book – Hartree-Fock in Chapter 6, Kohn-Sham in Chapter 8 – solves a fixed-point problem $\rho = \mathscr{G}[\rho]$, and simple iteration $\rho \to \mathscr{G}[\rho]$ converges slowly or not at all. Writing the fixed point as $\boldsymbol{F}(\rho) = \mathscr{G}[\rho] - \rho = 0$ and applying Eq. (14.29) is *Broyden mixing*, the standard accelerator in electronic-structure codes. Linear mixing, $\rho_{k+1} = (1-\gamma)\rho_k + \gamma\mathscr{G}[\rho_k]$, is the special case $\boldsymbol{H} = \gamma \boldsymbol{I}$ with no update at all.

14.4.3 From root finding to minimisation: what BFGS must satisfy

For minimisation, $\boldsymbol{F} = \nabla f$ and the matrix we are approximating is the Hessian. That changes the problem, because a Hessian is not an arbitrary matrix. Three properties are now required of $\boldsymbol{B}_{k+1}$:

1. **The secant condition**, $\boldsymbol{B}_{k+1}\boldsymbol{s}_k = \boldsymbol{y}_k$, so that the approximation reproduces the curvature actually observed.
2. **Symmetry**, $\boldsymbol{B}_{k+1} = \boldsymbol{B}_{k+1}^T$, because a Hessian is symmetric and an unsymmetric approximation wastes half its entries on information that cannot exist.
3. **Positive definiteness**, $\boldsymbol{B}_{k+1} \succ 0$, which is what guarantees that $\boldsymbol{p}_k = -\boldsymbol{B}_{k+1}^{-1}\nabla f$ is a descent direction: $\nabla f^T \boldsymbol{p}_k = -\nabla f^T \boldsymbol{B}^{-1}\nabla f < 0$.

Broyden's rank-one update satisfies the first and neither of the other two, which is why it works well as a root finder and badly as a minimiser – as Table 14.4 will show. A rank-one update cannot in general maintain symmetry while satisfying the secant condition; a rank-two update can, and there is a two-parameter family of them. The Broyden family interpolates between the two classical members, DFP and BFGS.

The curvature condition.

Requirement 3 cannot hold unconditionally. Multiplying Eq. (14.25) by $\boldsymbol{s}_k^T$ gives $\boldsymbol{s}_k^T\boldsymbol{B}_{k+1}\boldsymbol{s}_k = \boldsymbol{s}_k^T\boldsymbol{y}_k$, so a positive definite $\boldsymbol{B}_{k+1}$ satisfying the secant condition can exist only if

$$\boxed{\boldsymbol{s}_k^T\boldsymbol{y}_k > 0} \tag{14.30}$$

This is the *curvature condition*, and it says that the directional derivative along $\boldsymbol{s}_k$ has increased – the function curves upwards along the step, as it must if the step is to be informative about a minimum. It holds automatically for a strongly convex f. In general it is enforced by the line search: the second Wolfe condition

$$\nabla f(\boldsymbol{x}_k + \alpha \boldsymbol{p}_k)^T \boldsymbol{p}_k \geq c_2 \nabla f(\boldsymbol{x}_k)^T \boldsymbol{p}_k, \qquad 0 < c_1 < c_2 < 1, \tag{14.31}$$

together with the sufficient-decrease condition

$$f(\boldsymbol{x}_k + \alpha \boldsymbol{p}_k) \leq f(\boldsymbol{x}_k) + c_1 \alpha \nabla f(\boldsymbol{x}_k)^T \boldsymbol{p}_k, \tag{14.32}$$

implies Eq. (14.30) directly, since subtracting $\nabla f(\boldsymbol{x}_k)^T \boldsymbol{p}_k$ from both sides of Eq. (14.31) gives $\boldsymbol{y}_k^T \boldsymbol{p}_k \geq (c_2 - 1)\nabla f(\boldsymbol{x}_k)^T \boldsymbol{p}_k > 0$, and $\boldsymbol{s}_k = \alpha \boldsymbol{p}_k$ with $\alpha > 0$. *The line search is not a convenience; it is what keeps the curvature model well posed.* Remember this when we come to the Monte Carlo case, where a line search is impossible.

14.4.4 The BFGS update and its three properties

It is easiest to write down the update for the inverse and verify it, rather than to re-run the constrained minimisation – although that route works too, minimising a weighted Frobenius norm $\|\boldsymbol{W}^{1/2}(\boldsymbol{H}_{k+1} - \boldsymbol{H}_k)\boldsymbol{W}^{1/2}\|_F$ over symmetric matrices satisfying the inverse secant condition, with $\boldsymbol{W}$ any matrix obeying $\boldsymbol{W}\boldsymbol{s}_k = \boldsymbol{y}_k$. Let $\boldsymbol{H}_k$ be symmetric positive definite, put

$$\rho_k = \frac{1}{\boldsymbol{y}_k^T \boldsymbol{s}_k}, \qquad \boldsymbol{V}_k = \boldsymbol{I} - \rho_k \boldsymbol{s}_k \boldsymbol{y}_k^T, \tag{14.33}$$

and define the *Broyden-Fletcher-Goldfarb-Shanno* update

$$\boxed{\boldsymbol{H}_{k+1} = \boldsymbol{V}_k \boldsymbol{H}_k \boldsymbol{V}_k^T + \rho_k \boldsymbol{s}_k \boldsymbol{s}_k^T} \tag{14.34}$$

We check the three requirements in turn, dropping the subscript k.

The inverse secant condition.

Note first that

$$\boldsymbol{V}^T \boldsymbol{y} = \left(\boldsymbol{I} - \rho \boldsymbol{y}\boldsymbol{s}^T\right)\boldsymbol{y} = \boldsymbol{y} - \rho \boldsymbol{y}\left(\boldsymbol{s}^T \boldsymbol{y}\right) = \boldsymbol{y} - \boldsymbol{y} = 0,$$

because $\rho(\boldsymbol{s}^T \boldsymbol{y}) = 1$ by definition. Hence the first term of Eq. (14.34) annihilates $\boldsymbol{y}$ entirely, and

$$\boldsymbol{H}_+ \boldsymbol{y} = \boldsymbol{V}\boldsymbol{H}\underbrace{\boldsymbol{V}^T \boldsymbol{y}}_{=0} + \rho \boldsymbol{s}\left(\boldsymbol{s}^T \boldsymbol{y}\right) = \boldsymbol{s},$$

which is $\boldsymbol{H}_+ \boldsymbol{y} = \boldsymbol{s}$ as required.

Symmetry.

Immediate: $\boldsymbol{V}\boldsymbol{H}\boldsymbol{V}^T$ is symmetric whenever $\boldsymbol{H}$ is, and $\boldsymbol{s}\boldsymbol{s}^T$ is symmetric. The form Eq. (14.34) was chosen for exactly this reason – it is a congruence transformation plus a symmetric rank-one term.

Positive definiteness.

For any $\boldsymbol{z} \neq 0$,

$$\boldsymbol{z}^T \boldsymbol{H}_+ \boldsymbol{z} = \left(\boldsymbol{V}^T \boldsymbol{z}\right)^T \boldsymbol{H}\left(\boldsymbol{V}^T \boldsymbol{z}\right) + \rho\left(\boldsymbol{s}^T \boldsymbol{z}\right)^2. \tag{14.35}$$

Both terms are non-negative provided $\boldsymbol{H} \succ 0$ and $\rho > 0$, the latter being the curvature condition (14.30). They cannot both vanish. The first vanishes only if $\boldsymbol{V}^T \boldsymbol{z} = 0$, that is $\boldsymbol{z} = \rho\left(\boldsymbol{s}^T \boldsymbol{z}\right)\boldsymbol{y}$, which says $\boldsymbol{z} = c\boldsymbol{y}$ for some $c \neq 0$; and for $\boldsymbol{z} = c\boldsymbol{y}$ the second term is $\rho c^2\left(\boldsymbol{s}^T \boldsymbol{y}\right)^2 = c^2 \boldsymbol{s}^T \boldsymbol{y} > 0$. Hence $\boldsymbol{H}_+ \succ 0$, and by induction every iterate is positive definite if $\boldsymbol{H}_0$ is – $\boldsymbol{H}_0 = \boldsymbol{I}$ being the usual choice, which makes the first BFGS step identical to a steepest-descent step.

The direct form.

Inverting Eq. (14.34) with the Sherman-Morrison-Woodbury formula gives the update for the Hessian approximation itself,

$$\boldsymbol{B}_{k+1} = \boldsymbol{B}_k - \frac{\boldsymbol{B}_k \boldsymbol{s}_k \boldsymbol{s}_k^T \boldsymbol{B}_k}{\boldsymbol{s}_k^T \boldsymbol{B}_k \boldsymbol{s}_k} + \frac{\boldsymbol{y}_k \boldsymbol{y}_k^T}{\boldsymbol{y}_k^T \boldsymbol{s}_k}. \tag{14.36}$$

The structure is easy to read: the second term removes the curvature the old model predicted along $\boldsymbol{s}_k$, and the third installs the curvature that was actually measured. Both are rank one, so BFGS is a rank-two update, which is the minimum needed to keep symmetry.

DFP, and the Broyden family.

Applying Eq. (14.36) to $\boldsymbol{H}$ instead of $\boldsymbol{B}$ – that is, exchanging $\boldsymbol{s} \leftrightarrow \boldsymbol{y}$ and $\boldsymbol{B} \leftrightarrow \boldsymbol{H}$ – gives the older Davidon-Fletcher-Powell update,

$$\boldsymbol{H}_{k+1}^{\mathrm{DFP}} = \boldsymbol{H}_k - \frac{\boldsymbol{H}_k \boldsymbol{y}_k \boldsymbol{y}_k^T \boldsymbol{H}_k}{\boldsymbol{y}_k^T \boldsymbol{H}_k \boldsymbol{y}_k} + \frac{\boldsymbol{s}_k \boldsymbol{s}_k^T}{\boldsymbol{y}_k^T \boldsymbol{s}_k}, \tag{14.37}$$

which satisfies the same three requirements. The convex combination

$$\boldsymbol{H}_{k+1}^{\phi} = (1-\phi)\,\boldsymbol{H}_{k+1}^{\mathrm{BFGS}} + \phi\,\boldsymbol{H}_{k+1}^{\mathrm{DFP}}, \qquad \phi \in [0,1], \tag{14.38}$$

is the *restricted Broyden family*; every member satisfies the secant condition, and on a quadratic with exact line searches every member terminates in at most n steps with $\boldsymbol{H}_n = \boldsymbol{A}^{-1}$ exactly, generating the same iterates as conjugate gradients (Section 1.18). Away from that ideal case they differ, and decades of experience have put $\phi = 0$, BFGS, ahead: DFP is slower to correct a badly scaled initial $\boldsymbol{H}_0$.

Limited memory.

For large n even storing $\boldsymbol{H}_k$ is impossible. *L-BFGS* stores only the last m pairs $(\boldsymbol{s}_j, \boldsymbol{y}_j)$, typically $m \approx 5$–20, and computes $\boldsymbol{H}_k \nabla f$ by a two-loop recursion that applies Eq. (14.34) m times to an initial diagonal $\boldsymbol{H}^{(0)}$, without ever forming a matrix. The cost is $\mathscr{O}(mn)$ in time and memory, and L-BFGS is in consequence the default optimiser for problems with millions of parameters, in quantum chemistry geometry optimisations as much as in machine learning.

Algorithms.

Collecting the pieces gives the two algorithms in the form they are implemented in `vmcoptimise.py`.

Broyden, for $\boldsymbol{F}(\boldsymbol{x}) = 0$. Given $\boldsymbol{x}_0$, $\boldsymbol{H}_0 = \boldsymbol{I}$ and a tolerance ε, repeat for $k = 0, 1, \ldots$:

1. $\boldsymbol{p}_k = -\boldsymbol{H}_k \boldsymbol{F}(\boldsymbol{x}_k)$ and $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \boldsymbol{p}_k$;
2. $\boldsymbol{s}_k = \boldsymbol{x}_{k+1} - \boldsymbol{x}_k$, $\boldsymbol{y}_k = \boldsymbol{F}(\boldsymbol{x}_{k+1}) - \boldsymbol{F}(\boldsymbol{x}_k)$;
3. update $\boldsymbol{H}_{k+1}$ by Eq. (14.29);

until $\|\boldsymbol{F}(\boldsymbol{x}_k)\| \le \varepsilon$.

BFGS, for $\min f$. Given $\boldsymbol{x}_0$ and $\boldsymbol{H}_0 \succ 0$, repeat:

1. search direction $\boldsymbol{p}_k = -\boldsymbol{H}_k \nabla f(\boldsymbol{x}_k)$;
2. line search for α_k satisfying the Wolfe conditions (14.32) and (14.31);
3. $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k$, then $\boldsymbol{s}_k$, $\boldsymbol{y}_k$ as above;
4. if $\boldsymbol{y}_k^T \boldsymbol{s}_k > 0$, update $\boldsymbol{H}_{k+1}$ by Eq. (14.34); otherwise skip the update;

until $\|\nabla f(\boldsymbol{x}_k)\| \le \varepsilon$.

14.4.5 What the methods actually do

Every algebraic claim of the last two subsections is checked numerically in `vmcoptimise.py` on random five-dimensional problems, and Table 14.3 collects the worst discrepancy over two hundred of them. The line worth pausing on is the third: for each Broyden update the program constructs random competitors $\Delta\boldsymbol{B}' = \Delta\boldsymbol{B} + \boldsymbol{Z}\left(\boldsymbol{I} - \boldsymbol{s}\boldsymbol{s}^T/\boldsymbol{s}^T\boldsymbol{s}\right)$, which satisfy the secant condition just as well because the projector annihilates $\boldsymbol{s}$, and confirms that none of them is closer to $\boldsymbol{B}_k$ in the Frobenius norm. The minimum-change property is not a figure of speech.

property	worst residual
$\|\boldsymbol{A}\|_F^2 - \mathrm{Tr}(\boldsymbol{A}^T\boldsymbol{A})$, Eq. (1.9)	1.1×10^{-14}
$\mathrm{vec}(\boldsymbol{A}) \cdot \mathrm{vec}(\boldsymbol{B}) - \mathrm{Tr}(\boldsymbol{A}^T\boldsymbol{B})$, Eq. (1.13)	3.6×10^{-15}
Broyden: $\boldsymbol{B}_+\boldsymbol{s} - \boldsymbol{y}$, Eq. (14.25)	2.0×10^{-15}
Broyden: smallest excess $\|\Delta\boldsymbol{B}'\|_F - \|\Delta\boldsymbol{B}\|_F$	$+5.0 \times 10^{-1}$
BFGS: $\boldsymbol{H}_+\boldsymbol{y} - \boldsymbol{s}$	5.5×10^{-10}
BFGS: $\boldsymbol{H}_+ - \boldsymbol{H}_+^T$	1.2×10^{-10}
BFGS: $\boldsymbol{B}_+\boldsymbol{H}_+ - \boldsymbol{I}$, Eqs. (14.34) and (14.36)	7.3×10^{-8}
BFGS: smallest eigenvalue of $\boldsymbol{H}_+$	$+1.8 \times 10^{-3}$
DFP: $\boldsymbol{H}_+\boldsymbol{y} - \boldsymbol{s}$, Eq. (14.37)	5.4×10^{-14}

Table 14.3 Every property claimed for the two updates, verified over 200 random problems in five dimensions. The two entries carrying a + sign are inequalities that must not be violated: the Broyden update is never beaten by a competitor satisfying the same secant condition, and the BFGS matrix never loses positive definiteness. Computed by `vmcoptimise.py`.

Table 14.4 then puts the methods to work on a six-dimensional quadratic whose Hessian has condition number κ, and on Rosenbrock's banana for contrast. The first column is Eq. (14.20) made visible: even with an exact line search at every step, steepest descent needs a number of iterations proportional to κ. The two rank-two updates are essentially insensitive to it – ten iterations whether the problem is round or stretched by a factor of a thousand – which is the practical meaning of "learning the curvature". Broyden's rank-one update sits in between, and degrades badly at large κ: it satisfies the secant condition but neither symmetry nor positive definiteness, and as a minimiser it is a tool being used for the wrong job.

κ	steepest descent	BFGS	DFP	Broyden	Powell
1	1	1	1	1	2
10	68	11	10	24	2
100	601	9	11	68	2
1000	5538	10	9	1059	5

Table 14.4 Iterations needed to bring $\|\nabla f\|$ below 10^{-6} for $f = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^T\boldsymbol{x}$ in six dimensions, with $\boldsymbol{A}$ diagonal and of condition number κ. Steepest descent uses an exact line search, the quasi-Newton methods a Wolfe line search, and the Powell column counts complete direction-set cycles, each of which contains $n+1$ line minimisations. On Rosenbrock's banana from $(-1.2, 1)$ the corresponding numbers are 35 iterations for BFGS and 19456 for steepest descent. Computed by `vmcoptimise.py`.

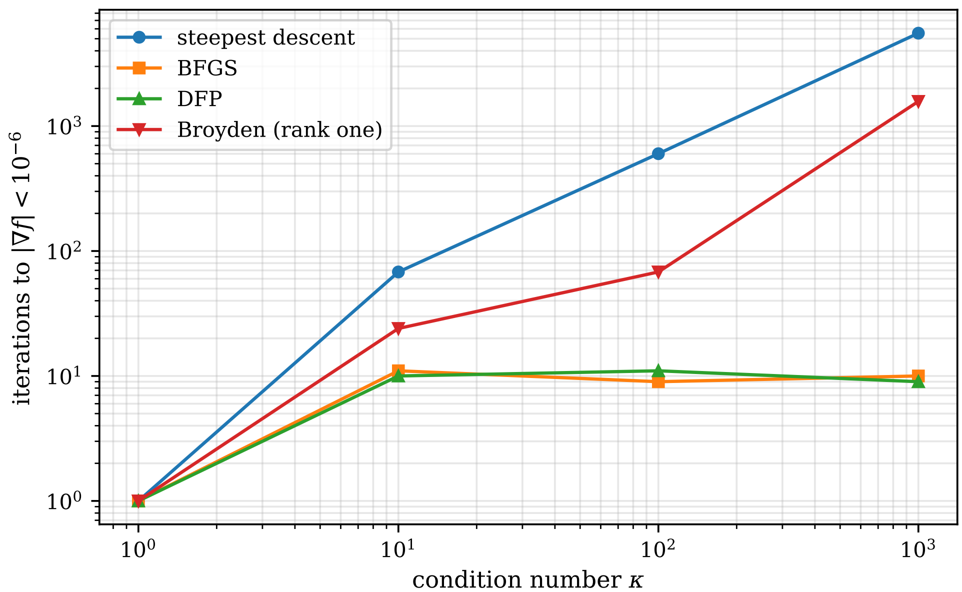

Fig. 14.2 Iterations required to bring $\|\nabla f\|$ below 10^{-6} on the six-dimensional quadratic of Table 14.4, plotted against the condition number κ of its Hessian, both axes logarithmic. Steepest descent with an exact line search (circles) rises along a straight line of unit slope; the two rank-two updates BFGS (squares) and DFP (triangles) are flat at about ten iterations over three decades of κ; Broyden's rank-one update (inverted triangles) lies between them and steepens. All four methods start from the same point and use the same convergence test.

Figure 14.2 plots the first four columns of the table on logarithmic axes, where a scaling law becomes a slope. Steepest descent gives a straight line: 68, 601 and 5538 iterations at $\kappa = 10$, 100 and 1000, which is a factor of 8.8 and then 9.2 per decade of κ, a slope of 0.95. That is Eq. (14.20) measured rather than derived, and since the line search is exact at every step there is nothing about the step length left to blame – the iteration count is proportional to κ because the *direction* is wrong, and it would be wrong in the same way however carefully one moved along it. BFGS and DFP are two flat lines on top of one another at nine to eleven iterations, and stretching the problem by a factor of a thousand costs them nothing measurable. For a six-dimensional quadratic that is exactly what the theory of Section 14.4.4 predicts: after at most $n = 6$ updates the approximation has seen the curvature along six independent directions and $\boldsymbol{H}$ has become $\boldsymbol{A}^{-1}$, after which one step finishes the job whatever κ may be. Broyden's rank-one curve is the interesting one. It is not merely displaced upwards from the rank-two curves but has a different slope, and a slope that grows: a factor of 2.8 from $\kappa = 10$ to $\kappa = 100$, then a factor of 15.6 from 100 to 1000, which is steeper than steepest descent's own scaling. The update is therefore not losing a constant amount of efficiency; it is losing the curvature information itself, and losing it faster the more anisotropic the true Hessian is. That is the price of dropping requirements 2 and 3 of Section 14.4.3, and it is why the same formula that accelerates a self-consistent field iteration so effectively is a poor choice for a minimisation. At $\kappa = 1$ all four methods take a single iteration, which is the sanity check the figure ought to pass: on a round bowl the gradient already points at the minimum.

Rosenbrock's banana makes the same point on a function that is not quadratic at all, and Figure 14.3 shows how BFGS gets down it. The path does three distinct things. From the start at $(-1.2, 1)$ it takes one short step and then plunges almost straight down the wall of the valley to the floor near $x = -0.8$: the Hessian at the starting point has eigenvalues 1.5×10^3 and 24, so the curvature across the valley exceeds that along it by a factor of sixty, and the first step, taken with $\boldsymbol{H}_0 = \boldsymbol{I}$ and therefore along the bare gradient, is dominated accordingly.

It then negotiates the bend at the origin in a sequence of very short steps, one of which, near $x = -0.5$, crosses the floor and is pulled back at the next. And it finally climbs the straight far branch to the minimum at $(1,1)$ in steps that lengthen steadily. The contrast between the short steps in the bend and the long ones on the climb is the whole mechanism on display: where the curvature is changing from step to step the secant information is stale by the time it is used, and BFGS must re-learn the Hessian continuously; where the valley is nearly straight the model is accurate and the line search accepts steps comparable with the distance remaining. Thirty-five iterations suffice in total, against 19456 for steepest descent from the same point, a factor of 550 on a problem in two dimensions. The lesson to carry forward is that this excellent behaviour rests entirely on the differences $\boldsymbol{y}_k$ being meaningful; Section 14.7 shows what happens when they are not.

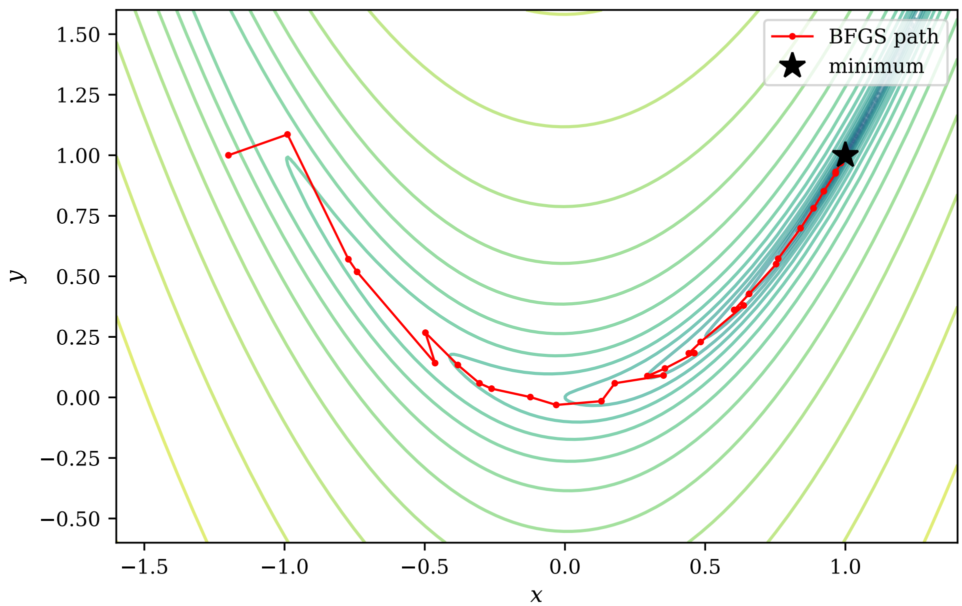

Fig. 14.3 BFGS on Rosenbrock's function $f(x,y) = (1-x)^2 + 100\left(y - x^2\right)^2$, started from $(-1.2, 1)$ and run to $\|\nabla f\| < 10^{-8}$. The background contours are of $\log_{10} f$, so that the crowding of the levels along the parabola $y = x^2$ marks the floor of the curved valley; the marked line is the sequence of iterates and the star the minimum at $(1,1)$, where $f = 0$. The thirty-five steps are short while the path negotiates the bend at the origin and lengthen steadily along the straight climb that follows.

The same program also runs Broyden against Newton on the nonlinear system $x^2 + y^2 - 4 = 0$, $e^x + y - 1 = 0$ from $(1,2)$. Newton reaches $\|\boldsymbol{F}\| < 10^{-10}$ in seven iterations and seven analytic Jacobians; Broyden needs eleven and no Jacobian at all, with the residual falling 5.7×10^{-2}, 5.8×10^{-3}, 4.1×10^{-4}, 4.0×10^{-6}, 6.0×10^{-9} over the last few steps – the superlinear convergence the theory promises, without quadratic convergence and without a single derivative of $\boldsymbol{F}$.

14.5 Derivative-free optimisation and Powell's method

Everything so far has assumed a gradient. Sometimes there is none to be had: the objective may come out of a simulation with no differentiable structure, or be a black box, or – most relevantly here – be so contaminated by Monte Carlo noise that a finite-difference gradient is meaningless. The derivative-free methods are Nelder-Mead's simplex, pattern search, and the direction-set methods of which Powell's is the best known. We describe Powell's because it is the one with a clean theory behind it, and because `scipy.optimize.minimize` offers it as `method="Powell"`.

14.5.1 Line minimisation

The building block is minimisation along a line. Given a point $\boldsymbol{x}$ and a direction $\boldsymbol{d}$, define

$$\phi(\alpha) = f(\boldsymbol{x} + \alpha \boldsymbol{d}) \tag{14.39}$$

and find $\alpha^* = \arg\min_\alpha \phi(\alpha)$; then move, $\boldsymbol{x} \to \boldsymbol{x} + \alpha^* \boldsymbol{d}$. This reduces a p-dimensional problem to a sequence of one-dimensional ones, and the one-dimensional problem is solved by bracketing followed by *golden-section search* or by Brent's method. Golden section maintains a bracket $a < c < d < b$ with $\phi(c)$ and $\phi(d)$ both below the endpoints, and discards one outer segment at each step. Placing the interior points at the golden ratio,

$$\frac{c-a}{b-a} = \frac{b-d}{b-a} = 1 - \frac{1}{\varphi} = 0.381966\ldots, \qquad \varphi = \frac{1+\sqrt{5}}{2}, \tag{14.40}$$

makes the surviving bracket self-similar, so that one of the two interior points can be reused and each iteration costs a single new function evaluation while shrinking the bracket by the factor $1/\varphi = 0.618$. It is slower than Brent's parabolic interpolation and unconditionally robust, which is the trade one usually wants in the innermost loop.

Two warnings. A line minimisation locates the minimum of ϕ to a precision of order $\sqrt{\varepsilon_{\text{mach}}}$ in α, not $\varepsilon_{\text{mach}}$, because ϕ is quadratic there and a quadratic is flat at its minimum. And if f is noisy, golden section will happily converge on a fluctuation.

14.5.2 Conjugate directions

Why choose the directions carefully? Consider the quadratic

$$f(\boldsymbol{x}) = \tfrac{1}{2}\boldsymbol{x}^T \boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}^T \boldsymbol{x} + c, \tag{14.41}$$

and suppose we have just minimised along $\boldsymbol{d}_i$, so that $\boldsymbol{d}_i^T \nabla f = 0$ at the current point. Move along a new direction $\boldsymbol{d}_j$. The gradient changes by $\boldsymbol{A}\boldsymbol{d}_j$ times the step, so the first minimisation is left undisturbed if and only if

$$\boldsymbol{d}_i^T \boldsymbol{A}\, \boldsymbol{d}_j = 0, \tag{14.42}$$

which is the definition of *conjugate* directions, already met in the conjugate gradient method of Section 1.18. A set of n mutually conjugate directions therefore minimises a quadratic exactly in n line minimisations, each one final. Without conjugacy, later minimisations spoil earlier ones and the method creeps down a valley exactly as steepest descent does.

Powell's insight is that conjugate directions can be *generated* from function values alone.

14.5.3 The algorithm

Start with the coordinate directions, $\boldsymbol{d}_1 = (1,0,\dots)$, $\boldsymbol{d}_2 = (0,1,0,\dots)$, and so on. One iteration consists of:

1. Store $\boldsymbol{x}_0$. Minimise along each $\boldsymbol{d}_i$ in turn, $\boldsymbol{x}_i = \boldsymbol{x}_{i-1} + \alpha_i \boldsymbol{d}_i$, recording which direction produced the largest single decrease in f.
2. Form the net displacement of the whole cycle, $\boldsymbol{d}_{\text{new}} = \boldsymbol{x}_n - \boldsymbol{x}_0$, and minimise along it as well.
3. Replace the direction of largest decrease by $\boldsymbol{d}_{\text{new}}$.
4. Repeat until $|f_{\text{new}} - f_{\text{old}}|$, or $\|\boldsymbol{x}_{\text{new}} - \boldsymbol{x}_{\text{old}}\|$, falls below tolerance.

Step 2 is the substance. The net displacement of a full cycle points along the valley – it is the direction in which the accumulated progress was made – and one can show that on a quadratic, successive $\boldsymbol{d}_{\text{new}}$ generated this way are mutually conjugate, so that after n cycles the direction set is a conjugate set and the method terminates. Step 3 is the repair for the obvious danger: naively discarding $\boldsymbol{d}_1$ each time makes the direction set collapse towards linear dependence. Discarding the direction that contributed most removes the one whose contribution $\boldsymbol{d}_{\text{new}}$ has largely absorbed, and keeps the set well conditioned. Table 14.4 shows the consequence: Powell reaches the minimum of the six-dimensional quadratic in two cycles regardless of the condition number, using no derivatives whatever.

When to use it, and when not.

Powell's method is excellent for smooth objectives in moderate dimension where gradients are unavailable – fitting a model to data, optimising a molecular geometry through a black-box energy code, tuning a handful of parameters. It degrades in high dimension, where n line minimisations per cycle become expensive and the direction bookkeeping becomes fragile, and it is *not robust to noise*, because every line minimisation assumes it can resolve the shape of ϕ. Both objections apply with full force to the variational Monte Carlo problem, as the next section shows.

14.6 Stochastic reconfiguration

Stochastic reconfiguration is the method built for this problem, and it comes from a different direction entirely: not from approximating the Hessian of the energy, but from asking what imaginary-time evolution would do and doing as much of it as the variational manifold permits. We follow that derivation, due to Sorella, in full, and only afterwards observe that the answer is a preconditioned gradient descent.

14.6.1 Imaginary time, and why it cannot be followed exactly

The exact route to a ground state is imaginary-time propagation. With

$$|\Psi(\tau)\rangle = e^{-\tau\hat{H}}|\Psi(0)\rangle, \qquad |\Psi(0)\rangle = \sum_n c_n|n\rangle, \qquad \hat{H}|n\rangle = E_n|n\rangle, \tag{14.43}$$

one has $|\Psi(\tau)\rangle = \sum_n c_n e^{-\tau E_n}|n\rangle$, and every excited component decays relative to the ground state by $e^{-\tau(E_n-E_0)}$. Provided $c_0 \neq 0$,

$$|\Psi(\tau)\rangle \xrightarrow[\tau\to\infty]{} c_0 e^{-\tau E_0}|0\rangle. \tag{14.44}$$

This is the same projection that Chapter 15 will implement stochastically as diffusion Monte Carlo. Here we cannot implement it at all, because in variational Monte Carlo we are confined to the manifold

$$\mathscr{M} = \{|\Psi(\boldsymbol{\alpha})\rangle \mid \boldsymbol{\alpha} \in \mathbb{R}^p\} \tag{14.45}$$

of states our ansatz can represent, and $(1-\delta\tau\hat{H})|\Psi(\boldsymbol{\alpha})\rangle$ will in general not lie in $\mathscr{M}$. The idea of stochastic reconfiguration is to take the imaginary-time step anyway and then *project back*: find the parameter change $\delta\boldsymbol{\alpha}$ whose state is as close as possible to the propagated one.

14.6.2 The infinitesimal step

For small $\delta\tau$, $e^{-\delta\tau\hat{H}} = 1-\delta\tau\hat{H}+\mathscr{O}(\delta\tau^2)$, and it is convenient to use the shifted form

$$|\Psi'_{\text{exact}}\rangle = \left[1-\delta\tau\left(\hat{H}-E(\boldsymbol{\alpha})\right)\right]|\Psi(\boldsymbol{\alpha})\rangle, \tag{14.46}$$

which differs from $e^{-\delta\tau\hat{H}}|\Psi\rangle$ only by the overall constant $e^{-\delta\tau E}$, physically irrelevant, and to first order in $\delta\tau$. The reason to subtract E is that

$$\left\langle\Psi\middle|\hat{H}-E\middle|\Psi\right\rangle = 0 \qquad \text{whenever} \qquad E = \frac{\langle\Psi|\hat{H}|\Psi\rangle}{\langle\Psi|\Psi\rangle}, \tag{14.47}$$

so the vector $(\hat{H}-E)|\Psi\rangle$ has no component along $|\Psi\rangle$ itself. The step therefore moves the state *across* the ray it lies on rather than along it: it changes the physical state and not merely its norm. Since the overall normalisation of a wave function carries no information, this is the only part of the step worth reproducing.

14.6.3 The tangent space and the distance functional

On the variational side, expand to first order in the parameter change,

$$|\Psi(\boldsymbol{\alpha}+\delta\boldsymbol{\alpha})\rangle = |\Psi(\boldsymbol{\alpha})\rangle + \sum_k \delta\alpha_k|\Psi_k\rangle + \mathscr{O}(\delta\alpha^2), \qquad |\Psi_k\rangle \equiv \frac{\partial}{\partial\alpha_k}|\Psi(\boldsymbol{\alpha})\rangle. \tag{14.48}$$

The $|\Psi_k\rangle$ span the tangent space of $\mathscr{M}$ at $\boldsymbol{\alpha}$: they are the only directions in Hilbert space the ansatz can move in. Comparing the two candidate states,

$$|\Delta\rangle = |\Psi'_{\text{var}}\rangle - |\Psi'_{\text{exact}}\rangle = \sum_k \delta\alpha_k|\Psi_k\rangle + \delta\tau\left(\hat{H}-E\right)|\Psi\rangle, \tag{14.49}$$

and the reconfiguration is defined by making this mismatch as small as possible,

$$\mathscr{D}(\delta\boldsymbol{\alpha}) = \langle\Delta|\Delta\rangle \qquad \longrightarrow \qquad \text{minimum}. \tag{14.50}$$

Expanding and keeping terms through first order in $\delta\tau$,

$$\mathscr{D} = \sum_{ij} \delta\alpha_i \delta\alpha_j \langle \Psi_i | \Psi_j \rangle + 2\delta\tau \sum_i \delta\alpha_i \,\mathrm{Re}\left\{ \left\langle \Psi_i \middle| \hat{H} - E \middle| \Psi \right\rangle \right\} + \mathscr{O}(\delta\tau^2). \tag{14.51}$$

This is a quadratic form in $\boldsymbol{\delta\alpha}$ with a linear term, so the minimisation is a linear problem. Differentiating,

$$\frac{\partial \mathscr{D}}{\partial(\delta\alpha_i)} = 2\sum_j \langle \Psi_i | \Psi_j \rangle \, \delta\alpha_j + 2\delta\tau \,\mathrm{Re}\left\{ \left\langle \Psi_i \middle| \hat{H} - E \middle| \Psi \right\rangle \right\} = 0, \tag{14.52}$$

that is

$$\sum_j \langle \Psi_i | \Psi_j \rangle \, \delta\alpha_j = -\delta\tau \,\mathrm{Re}\left\{ \left\langle \Psi_i \middle| \hat{H} - E \middle| \Psi \right\rangle \right\}. \tag{14.53}$$

This is the imaginary-time equation projected onto the tangent basis, and everything that follows is a matter of writing its two sides as Monte Carlo averages.

One refinement first. The tangent vectors $|\Psi_k\rangle$ contain a component along $|\Psi\rangle$ itself – a change of parameters generally changes the norm – and that component is redundant. Removing it means working with the normalised state and its derivatives,

$$|\overline{\Psi}\rangle = \frac{|\Psi\rangle}{\sqrt{\langle \Psi | \Psi \rangle}}, \qquad |\overline{\Psi}_k\rangle = \frac{\partial}{\partial \alpha_k} |\overline{\Psi}\rangle, \tag{14.54}$$

in terms of which Eq. (14.53) reads

$$\sum_j \langle \overline{\Psi}_i | \overline{\Psi}_j \rangle \, \delta\alpha_j = -\delta\tau \left\langle \overline{\Psi}_i \middle| \hat{H} - E \middle| \overline{\Psi} \right\rangle. \tag{14.55}$$

14.6.4 The metric and the force as Monte Carlo averages

Write the wave function in configuration space, $\Psi(x)$, and reintroduce the logarithmic derivatives of Eq. (14.10),

$$O_k(x) = \frac{1}{\Psi(x)} \frac{\partial \Psi(x)}{\partial \alpha_k} = \frac{\partial \ln \Psi(x)}{\partial \alpha_k}, \qquad \text{so} \qquad \frac{\partial \Psi(x)}{\partial \alpha_k} = O_k(x)\Psi(x). \tag{14.56}$$

Averages are taken, as always, over $P(x) = |\Psi(x)|^2 / \langle \Psi | \Psi \rangle$. Differentiating the normalised state (14.54) and taking the overlap – the normalisation contributes exactly the subtracted terms – gives

$$\langle \overline{\Psi}_i | \overline{\Psi}_j \rangle = \langle O_i^* O_j \rangle - \langle O_i^* \rangle \langle O_j \rangle, \tag{14.57}$$

which for real wave functions and real parameters is

$$\boxed{S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle} \tag{14.58}$$

the covariance matrix of the logarithmic derivatives. Three properties follow at once from its being a covariance matrix. It is symmetric; it is positive *semi*definite, since $\boldsymbol{v}^T \boldsymbol{S} \boldsymbol{v} = \mathrm{Var}(\sum_k v_k O_k) \geq 0$; and it is singular exactly when some linear combination $\sum_k v_k O_k$ is constant over configuration space, that is when the parameters are locally redundant.

The right-hand side of Eq. (14.55) is the same object we met in Section 14.2. With the local energy $E_{\mathrm{loc}}(x) = [\hat{H}\Psi](x)/\Psi(x)$,

$$g_i = \mathrm{Re}\left\{ \left\langle \overline{\Psi}_i \middle| \hat{H} - E \middle| \overline{\Psi} \right\rangle \right\} = \langle O_i E_{\mathrm{loc}} \rangle - \langle O_i \rangle \langle E_{\mathrm{loc}} \rangle = \tfrac{1}{2} \partial_{\alpha_i} E, \tag{14.59}$$

half the energy gradient of Eq. (14.2). It is called the *generalised force*, and the factor of one half is a convention that disappears into the learning rate.

Assembling, Eq. (14.55) becomes the linear system

$$\sum_j S_{ij}\,\delta\alpha_j = -\delta\tau\, g_i, \qquad \text{that is} \qquad \boldsymbol{S}\,\delta\boldsymbol{\alpha} = -\delta\tau\,\boldsymbol{g}, \tag{14.60}$$

and, formally,

$$\boxed{\boldsymbol{\alpha}_{k+1} = \boldsymbol{\alpha}_k - \eta\,(\boldsymbol{S}+\varepsilon\boldsymbol{I})^{-1}\,\nabla_{\boldsymbol{\alpha}} E} \tag{14.61}$$

with η absorbing $\delta\tau$ and the factor of two. The regularisation ε is discussed below. Note that all four ingredients – $\langle O_i\rangle$, $\langle O_i O_j\rangle$, $\langle E_{\mathrm{loc}}\rangle$ and $\langle O_i E_{\mathrm{loc}}\rangle$ – come from a single random walk, exactly as in Section 14.2.2. Stochastic reconfiguration costs one $p \times p$ accumulator and one linear solve per iteration, and no extra sampling at all.

14.6.5 Why this is the natural gradient

There is a second reading of Eq. (14.61) which explains why it works so much better than Eq. (14.19), and which makes no reference to imaginary time.

The trouble with steepest descent is that it treats parameter space as flat. It takes the step $-\eta\nabla E$, which is the direction of fastest decrease *per unit change of the parameters measured in the Euclidean norm* – and that norm is meaningless. The parameters are labels; what has physical content is the state they produce. Rescale a parameter, and the Euclidean step changes while the physics does not.

The right measure of distance is the distance between the normalised states. For an infinitesimal change,

$$ds^2 = \big\|\,|\overline{\Psi}(\boldsymbol{\alpha}+d\boldsymbol{\alpha})\rangle - |\overline{\Psi}(\boldsymbol{\alpha})\rangle\big\|^2 = \sum_{ij} S_{ij}\,d\alpha_i\,d\alpha_j, \tag{14.62}$$

so $\boldsymbol{S}$ is a Riemannian metric on the variational manifold – the Fubini-Study metric of projective Hilbert space in these coordinates, known in this context as the *quantum geometric tensor* and equal, up to a factor of four, to the quantum Fisher information matrix. Asking for the direction of fastest decrease of E per unit *of this* distance, that is minimising $\nabla E \cdot \delta\boldsymbol{\alpha}$ subject to $\delta\boldsymbol{\alpha}^T\boldsymbol{S}\,\delta\boldsymbol{\alpha} = \text{const}$, gives

$$\delta\boldsymbol{\alpha} \propto -\boldsymbol{S}^{-1}\nabla E, \tag{14.63}$$

which is Eq. (14.61). This is Amari's *natural gradient*, and stochastic reconfiguration is exactly the natural gradient specialised to variational quantum states. Two derivations, one from imaginary-time projection and one from Riemannian geometry, giving the same update: that coincidence is the reason to trust the method.

The practical consequences are the ones the geometry suggests. The step is invariant under reparametrisation, so no hand-tuning of per-parameter learning rates is needed. Redundant parameters – inevitable in a neural network – cost nothing, because $\boldsymbol{S}$ recognises them as flat directions. And the conditioning of the optimisation is set by the geometry of the state manifold rather than by the arbitrary scaling of the coordinates.

14.6.6 Regularisation, and what it costs

$\boldsymbol{S}$ is positive semidefinite, not positive definite, and in a Monte Carlo estimate its small eigenvalues are exactly the ones dominated by noise. Inverting it as it stands amplifies that noise without limit. The standard repair is a diagonal shift,

$$\boldsymbol{S} \longrightarrow \boldsymbol{S}+\varepsilon\boldsymbol{I}, \qquad (\boldsymbol{S}+\varepsilon\boldsymbol{I})\,\delta\boldsymbol{\alpha} = -\delta\tau\,\boldsymbol{g}, \tag{14.64}$$

sometimes with ε multiplying the diagonal of $\boldsymbol{S}$ rather than the identity, and often decreased as the optimisation proceeds. It is exactly Tikhonov regularisation, and it interpolates: at $\varepsilon \to 0$ the update is the natural gradient, at $\varepsilon \to \infty$ it is plain steepest descent with learning rate η/ε.

Table 14.5 shows the crossover for our two-parameter dot. Sampled at the minimum, the metric is

$$\boldsymbol{S} = \begin{pmatrix} 0.6478 & 0.2862 \\ 0.2862 & 0.2260 \end{pmatrix}, \tag{14.65}$$

with eigenvalues 0.0814 and 0.7924 and condition number 9.7. Even here, with two parameters chosen by hand to do different things, the manifold is stretched by an order of magnitude, and α and β move the state in substantially overlapping directions – that is what the large off-diagonal element says. The natural-gradient direction sits 26° away from the bare gradient, and the shift ε rotates it back.

ε	$\lVert(\boldsymbol{S}+\varepsilon\boldsymbol{I})^{-1}\boldsymbol{g}\rVert$	angle to $\boldsymbol{g}$
10^{-6}	0.04654	25.97°
10^{-4}	0.04652	25.95°
10^{-3}	0.04636	25.71°
10^{-2}	0.04492	23.52°
10^{-1}	0.03747	12.52°
1	0.01803	2.16°
10	0.00299	0.23°

Table 14.5 The regularised stochastic-reconfiguration direction at $(\alpha,\beta) = (1.00, 0.40)$, from 40000 cycles. As ε grows the step shortens and rotates towards the bare gradient, which it reaches when ε dominates the eigenvalues of $\boldsymbol{S}$. A useful ε is small enough to leave the geometry intact and large enough to keep the small eigenvalues out of the inverse; here anything below 10^{-2} will do. Computed by `vmcoptimise.py`.

Algorithm.

At each iteration:

1. sample configurations x from $P(x) = |\Psi(x)|^2/\langle\Psi|\Psi\rangle$;
2. accumulate $O_k(x)$ and $E_{\mathrm{loc}}(x)$ in the same loop;
3. form $S_{ij} = \langle O_i O_j\rangle - \langle O_i\rangle\langle O_j\rangle$ and $g_i = \langle O_i E_{\mathrm{loc}}\rangle - \langle O_i\rangle\langle E_{\mathrm{loc}}\rangle$;
4. solve $(\boldsymbol{S}+\varepsilon\boldsymbol{I})\,\delta\boldsymbol{\alpha} = -\delta\tau\,\boldsymbol{g}$;
5. set $\boldsymbol{\alpha} \leftarrow \boldsymbol{\alpha}+\delta\boldsymbol{\alpha}$.

For large p, step 4 is never done by forming $\boldsymbol{S}$: the matrix is $p \times p$ and p may be 10^6. Instead $\boldsymbol{Sv}$ is evaluated as a pair of Monte Carlo averages – $\boldsymbol{Sv} = \langle O(O \cdot v)\rangle - \langle O\rangle\langle O \cdot v\rangle$, which costs one pass over the samples – and the system is solved by conjugate gradients (Section 1.18), or by working in the smaller of the two spaces when the number of samples is less than p. This is what makes stochastic reconfiguration practical for neural-network wave functions.

14.7 The methods compared

Table 14.6 runs six of them on the quantum dot from the same starting point with the same number of samples per step.

method	steps	α	β	final energy
gradient descent	25	1.00639	0.36165	3.001775
with momentum	25	0.99876	0.37802	3.001499
stochastic reconfiguration	25	0.98977	0.39651	3.001494
BFGS, damped + trust radius	25	0.98912	0.39753	3.001491
BFGS, no safeguards	25	1.22282	66.026	3.429908
Powell, derivative free	4	1.06078	0.33372	3.002792

Table 14.6 The same start, $(0.85, 0.20)$, and the same 6000 cycles per step. The Powell row counts direction-set cycles, not gradient steps, and consumed 133 energy evaluations against the 25 of every other row. With two well-scaled parameters the four safeguarded methods all reach the minimum, and the ordering among them is not significant; the last two rows are the ones that carry information. Computed by `vmcoptimise.py`.

The fifth row is the important one. Run BFGS on the Monte Carlo gradient without a trust radius and without a damped update – that is, exactly as one would run it on a deterministic function – and it leaves the physical region entirely, ending at $\beta = 66$ and an energy half a hartree above the start. The reason is the one anticipated in Section 14.4.3: the curvature is inferred from *differences* of gradients, and near the minimum, where the true difference is small, that difference is pure noise. The inferred $\boldsymbol{H}$ then has nothing to do with the Hessian, the curvature condition $\boldsymbol{s}^T\boldsymbol{y} > 0$ fails – thirteen times out of twenty-five in this run – and the ordinary repair, a line search, is unavailable because we cannot reliably test whether the energy has decreased. With Powell's damped update and a trust radius the method is perfectly usable, and it finishes where the others do; but the safeguards are doing the work, not the curvature model.

The last row makes the complementary point. Powell's method needs no gradient, and it converges, but it needs 133 energy evaluations to do what the first-order methods do in 25 – and each of *those* also produced the gradient for free. In the noisy setting the line minimisations resolve fluctuations rather than curvature, and the resulting parameters are the worst of the six.

Why optimisation gets a chapter. For two parameters any of these methods works and the choice hardly matters. The reason to set them up carefully is what comes later: a neural-network trial function has thousands or millions of parameters, no grid is conceivable, the surface is not convex, the gradient is stochastic, and a line search cannot be performed. Of the methods in this chapter, only Eq. (14.2) with a first-order update and Eq. (14.61) survive that list – which is why they are what the neural-quantum-state literature uses. The machine-learning literature knows the first as the REINFORCE or score-function estimator and the second as the natural gradient, and arrived at both independently.

14.8 The production run

Once the parameters are fixed the calculation changes character. There is nothing left to decide, only statistics to accumulate, and that parallelises perfectly: walkers are independent, so an ensemble of them can be advanced with array operations rather than a Python loop. With a thousand walkers taking two thousand steps each, `vmcoptimise.py` collects two million local-energy evaluations in about a second, at an acceptance rate of 87%.

What it returns is the walker-averaged local energy at each step. That series is still autocorrelated *in time* – averaging over independent walkers divides the variance and the autocovariance by the same factor and leaves the normalised autocorrelation function, and therefore τ, unchanged. The error analysis of Chapter 12 is still needed.

14.9 Resampling

Chapter 12 established that a Markov chain gives correlated samples, that $\sigma_m^2 = \tau\sigma^2/n$ rather than σ^2/n, and that the naive error is therefore always too small. It also introduced blocking, informally. A production calculation must report an honest error bar, so we now do the whole thing properly: first the exact variance of a correlated mean, then why that exact expression cannot be used as it stands, and then the three estimators – blocking, the bootstrap and the jackknife – that are used instead.

14.9.1 The exact variance of a correlated mean

Let $\boldsymbol{X} = (X_1,\ldots,X_N)^T$ be the measurements, with $\langle \boldsymbol{X}\rangle = \mu\mathbf{1}$ and $\mathbf{1} = (1,\ldots,1)^T$. The sample mean is a *linear* functional of the data,

$$\bar{X} = \frac{1}{N}\mathbf{1}^T\boldsymbol{X}, \tag{14.66}$$

and for any deterministic $\boldsymbol{a}$ and any random vector, $\mathrm{Var}(\boldsymbol{a}^T\boldsymbol{X}) = \boldsymbol{a}^T\boldsymbol{\Sigma}\boldsymbol{a}$ with

$$\Sigma_{ij} = \mathrm{Cov}(X_i, X_j) = \big\langle (X_i-\mu)(X_j-\mu)\big\rangle. \tag{14.67}$$

Taking $\boldsymbol{a} = \mathbf{1}/N$ gives, with no approximation whatever,

$$\boxed{\mathrm{Var}(\bar{X}) = \frac{1}{N^2}\,\mathbf{1}^T\boldsymbol{\Sigma}\,\mathbf{1} = \frac{1}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N}\mathrm{Cov}(X_i,X_j)} \tag{14.68}$$

This is the whole problem in one line. The familiar σ^2/N is what Eq. (14.68) reduces to when $\boldsymbol{\Sigma}$ is diagonal, and a Markov chain never gives a diagonal $\boldsymbol{\Sigma}$. Everything in this section is an attempt to evaluate the double sum without knowing $\boldsymbol{\Sigma}$.

Stationarity and the Toeplitz structure.

The one structural assumption we make is that the chain is *weakly stationary* after equilibration: $\langle X_i \rangle = \mu$ independent of i, and $\mathrm{Cov}(X_i, X_j)$ depending only on the separation $|i-j|$. Both hold for a Markov chain in its stationary distribution, which is what the burn-in is for. Then

$$\Sigma_{ij} = C(|i-j|), \qquad C(t) = \mathrm{Cov}(X_k, X_{k+t}), \qquad C(0) = \sigma^2, \tag{14.69}$$

so that $\boldsymbol{\Sigma}$ is a *Toeplitz* matrix, constant along its diagonals,

$$\boldsymbol{\Sigma} = \begin{pmatrix} C(0) & C(1) & C(2) & \cdots \\ C(1) & C(0) & C(1) & \cdots \\ C(2) & C(1) & C(0) & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}. \tag{14.70}$$

The N^2 unknown entries have collapsed to N unknown functions $C(t)$, and that is what makes the problem tractable at all.

Collecting the double sum by lag.

Reorganise Eq. (14.68) according to $t = |i-j|$. The diagonal $t = 0$ contributes N terms. For each $t \geq 1$ there are $N-t$ pairs with $j = i+t$ and $N-t$ with $i = j+t$, hence $2(N-t)$ terms. Therefore

$$\mathrm{Var}(\bar{X}) = \frac{1}{N^2}\left[N C(0) + 2\sum_{t=1}^{N-1}(N-t)C(t)\right] = \frac{C(0)}{N} + \frac{2}{N^2}\sum_{t=1}^{N-1}(N-t)C(t). \tag{14.71}$$

With the normalised autocorrelation function

$$\rho(t) = \frac{C(t)}{C(0)} = \frac{C(t)}{\sigma^2}, \qquad \rho(0) = 1, \tag{14.72}$$

this becomes

$$\boxed{\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{N}\left[1 + 2\sum_{t=1}^{N-1}\left(1 - \frac{t}{N}\right)\rho(t)\right]} \tag{14.73}$$

which is *exact* for a stationary series of finite length N. The triangular weight $1 - t/N$ is not a nuisance factor: it is the statement that a separation of t can occur in only $N-t$ of the N positions, so long lags are simply less represented in a finite sample.

The large-N limit and the two conventions.

If the correlations are summable, $\sum_t |\rho(t)| < \infty$, then for large N the weight $1 - t/N \approx 1$ over the lags that contribute and

$$\mathrm{Var}(\bar{X}) \approx \frac{\sigma^2}{N}\tau, \qquad \tau = 1 + 2\sum_{t=1}^{\infty}\rho(t), \tag{14.74}$$

which is Eq. (12.15) of Chapter 12, now derived rather than asserted. Much of the literature, particularly on lattice field theory, instead defines the *integrated autocorrelation time*

$$\tau_{\rm int} = \frac{1}{2} + \sum_{t=1}^{\infty} \rho(t) = \frac{\tau}{2}, \qquad \text{so that} \qquad \mathrm{Var}(\bar{X}) \approx \frac{2\tau_{\rm int}\sigma^2}{N}. \tag{14.75}$$

The factor of two catches everyone once. The two statements are identical; only the bookkeeping differs, and one must check which convention a paper or a code is using before comparing numbers. We use τ throughout.

The interpretation is the same in either convention. Setting $\sigma^2/N_{\rm eff} = \tau\sigma^2/N$ defines

$$N_{\rm eff} = \frac{N}{\tau} = \frac{N}{2\tau_{\rm int}}, \tag{14.76}$$

the number of independent measurements the correlated series is worth. For the production run of Section 14.8, $\tau = 1.27$ and the two thousand walker-averaged steps are worth 1577 independent ones; for the badly tuned run of Table 14.11, $\tau = 13.95$ and four thousand steps are worth fewer than three hundred.

14.9.2 Why the exact formula cannot be used, and why blocking exists

Equation (14.73) is exact, and it is nearly useless as it stands. The reason is that we do not know $C(t)$; we estimate it, as

$$\hat{C}(t) = \frac{1}{N}\sum_{k=1}^{N-t} (X_k - \bar{X})\,(X_{k+t} - \bar{X})\,, \tag{14.77}$$

and each $\hat{C}(t)$ carries a statistical error of order $\sigma^2/\sqrt{N}$ that does *not* shrink as t grows. Adding N of them therefore accumulates an error of order $\sigma^2\sqrt{N}$ against a signal of order $\sigma^2\tau$. The sum over all lags does not give a slightly degraded answer; it gives a meaningless one. Evaluated on the production series, the full quadratic form (14.68) with $\boldsymbol{\Sigma}$ built from $\hat{C}$ returns an error of 1.85×10^{-5}, which is not merely inaccurate but *smaller than the naive* 3.28×10^{-5}, and correspondingly $\tau \to 0$.

The cure is to truncate the sum at some window W, and the two standard rules are to stop at the first t with $\hat{\rho}(t) < 0$, or to use Sokal's self-consistent window, the smallest W with $W \geq c\,\tau_{\rm int}(W)$ and conventionally $c = 5$. Both are defensible and neither is unique. Table 14.7 shows what the answer does as W varies.

This table is the argument for blocking. The autocorrelation route requires a decision that changes the answer by more than the answer's own precision, and provides no principle for making it. Blocking makes the same decision implicitly, and, with Jonsson's stopping rule, makes it by a test rather than by taste.

Figure 14.4 is that argument in one picture. What one wants from such a curve is a plateau: a range of W long enough to have caught all the correlation and short enough not to have accumulated much noise, over which the estimate is flat and from which it may simply be read. There is none. Over three decades the estimate oscillates within about ten per cent of 3.7×10^{-5}, with humps at $W = 2$ and $W = 100$ and dips at $W = 10$ and $W = 500$ that carry no information whatever: each $\hat{C}(t)$ contributes an error of order $\sigma^2/\sqrt{N}$ irrespective of t, so the curve performs a random walk in W whose steps do not shrink as more terms are added. Since the whole correction being estimated is only 13 per cent of the naive value, a wander of ten per cent in the estimate is a wander of most of the correction. Then, at the right-hand edge, the curve falls off a cliff. At $W = N-1$ the sum is exact – no truncation, no choice, nothing left out – and it returns 1.85×10^{-5}, a factor of 1.8 *below* the naive error, with a correlation time that has come out negative. A positively correlated series has been assigned a mean

window W	τ	error
1	1.228	0.00003630
2	1.269	0.00003689
5	1.226	0.00003626
10	1.175	0.00003550
20	1.367	0.00003828
50	1.384	0.00003849
100	1.420	0.00003895
500	1.158	0.00003535
1999	-0.000	0.00001850
blocking	—	0.00004858

Table 14.7 Equation (14.73) evaluated on the production series of $N = 2000$ walker-averaged steps, truncated at various windows W. The first-negative rule gives $W = 2$ and Sokal's rule $W = 4$. Between $W = 1$ and $W = 500$ the answer wanders by ten per cent, and at $W = N - 1$ – the exact sum – it collapses entirely. The last line is what blocking returns, with no window to choose. Computed by `vmcoptimise.py`.

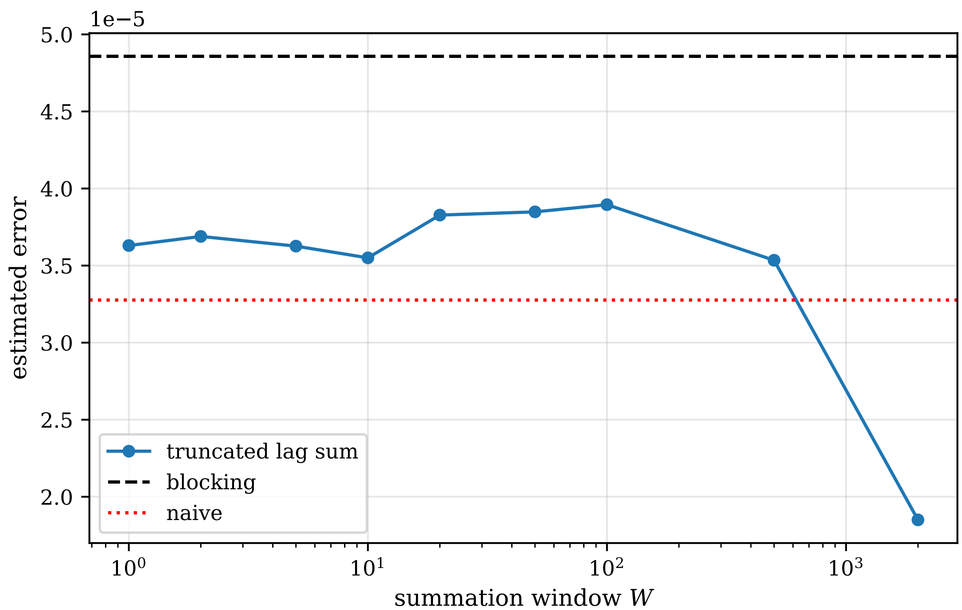

Fig. 14.4 The numbers of Table 14.7 plotted against the summation window: the error on the mean of the production series ($N = 2000$ walker-averaged steps) obtained from the truncated lag sum (14.73), with W logarithmic. The dotted line is the naive $\sigma/\sqrt{N} = 3.28 \times 10^{-5}$ and the dashed line the blocking estimate 4.86×10^{-5}. The truncated sums wander in a band around 3.7×10^{-5} without ever settling on a plateau, and the last point, $W = N - 1$, is the exact sum of Eq. (14.68) and lies below the naive value.

more precise than an independent one would give. That is not an inaccuracy but a failure of sign, and no rule for choosing W protects against it, because the defect is in the estimated covariances and not in where the sum is stopped. The dashed line, by contrast, required no decision at all.

14.9.3 Blocking

Replace the data by the averages of neighbouring pairs,

$$X_i^{(k+1)} = \tfrac{1}{2}\left(X_{2i-1}^{(k)} + X_{2i}^{(k)}\right), \qquad N_k = \frac{N}{2^k}, \tag{14.78}$$

and at each level compute the ordinary, uncorrected estimates

$$s_k^2 = \frac{1}{N_k - 1}\sum_{i=1}^{N_k}\left(X_i^{(k)} - \bar{X}_k\right)^2, \qquad \sigma_k^2 = \frac{s_k^2}{N_k}. \tag{14.79}$$

The mean is unchanged by the transformation, so $\bar{X}_k = \bar{X}$ for every k, and only the error estimate moves.

What blocking does is best seen through the covariance matrix. The transformation is a linear coarse-graining, so $\boldsymbol{\Sigma}^{(k)}$ is again Toeplitz, and each step moves weight from the off-diagonals onto the diagonal: the variance s_k^2 falls, but more slowly than N_k does, and the ratio s_k^2/N_k therefore rises. Once the blocks are long compared with the correlation time the off-diagonals are gone,

$$\boldsymbol{\Sigma}^{(k)} \approx s_k^2 \boldsymbol{I}, \tag{14.80}$$

Eq. (14.68) collapses to the independent-sample formula s_k^2/N_k, and the estimate stops rising. The plateau is the true error. Table 14.8 watches this happen on the badly tuned run, where there is something to see.

k	N_k	s_k^2	$\rho^{(k)}(1)$	$\rho^{(k)}(2)$	$\rho^{(k)}(3)$	$\sqrt{s_k^2/N_k}$
0	4000	4.56×10^{-6}	0.824	0.687	0.580	0.00003377
1	2000	4.17×10^{-6}	0.759	0.540	0.398	0.00004566
2	1000	3.68×10^{-6}	0.631	0.347	0.221	0.00006069
3	500	3.02×10^{-6}	0.464	0.191	0.127	0.00007776
4	250	2.17×10^{-6}	0.352	0.059	−0.024	0.00009309
5	125	1.52×10^{-6}	0.157	0.004	0.039	0.00011023
6	62	9.19×10^{-7}	0.087	−0.113	0.054	0.00012175

Table 14.8 Blocking the badly tuned production run ($\Delta t = 0.02$). The leading correlations $\rho^{(k)}(t)$ of the blocked series fall towards zero, $\boldsymbol{\Sigma}^{(k)}$ becomes diagonal, and the independent-sample estimate $\sqrt{s_k^2/N_k}$ – badly wrong at $k = 0$ – climbs by a factor of nearly four and flattens. Computed by `vmcoptimise.py`.

Figure 14.5 shows four of those levels as pictures of $\boldsymbol{\Sigma}^{(k)}$ rather than as three columns of numbers, and it makes the mechanism hard to miss. At $k = 0$ the corner is uniformly red: neighbours correlate at 0.824, the correlation is still 0.580 three steps away, and the double sum of Eq. (14.68) is therefore dominated by its off-diagonal part – which is precisely why the naive error, which keeps only the diagonal, is wrong by a factor of four here. Two transformations later the red band has narrowed to two or three lags, $\rho^{(2)}(1) = 0.631$; after four it is one lag wide, $\rho^{(4)}(1) = 0.352$; and at $k = 6$ the panel is a bare diagonal on a white field, $\rho^{(6)}(1) = 0.087$. In step with that, the independent-sample estimate $\sqrt{s_k^2/N_k}$ climbs from 3.38×10^{-5} to 1.22×10^{-4}. It is worth being clear about what that climb is: the estimate is not becoming more accurate as the matrix diagonalises, it is being corrected, and the value it

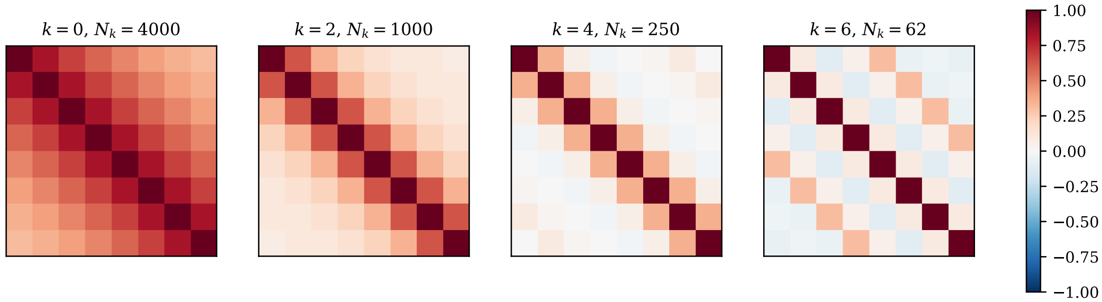

Fig. 14.5 The correlation matrix becoming the identity. Each panel is the leading corner of the correlation matrix $\rho^{(k)}(|i-j|)$ of the blocked series after k blocking transformations of the badly tuned production run of Table 14.8 ($\Delta t = 0.02$, $N = 4000$), drawn as an image on the common scale of the colour bar, red for positive correlation and blue for negative. At $k = 0$ the whole corner is red, the neighbouring-sample correlation being $\rho(1) = 0.824$; by $k = 6$, where the blocks are 64 measurements long and only $N_k = 62$ of them remain, nothing survives but the diagonal, and $\boldsymbol{\Sigma}^{(k)}$ has become the $s_k^2\boldsymbol{I}$ of Eq. (14.80).

climbs towards was true of the original series all along. Blocking changes nothing about the data, only our description of them.

There is a second thing to see, in the last panel. Its off-diagonal cells are not white but faintly mottled, positive and negative at the level of a tenth. With $N_6 = 62$ blocks the sampling error on an estimated correlation is about $1/\sqrt{62} = 0.13$, which is exactly the size of the mottling, so those cells contain nothing at all – and they are the reason one cannot simply block until the picture is perfectly clean. Every transformation halves the number of data points, and beyond the level at which the correlations have gone, all that further blocking does is add variance to the answer. The plateau in $\sqrt{s_k^2/N_k}$ is thus bounded on both sides, by bias below and by noise above, and it is a good deal less obvious in practice than the argument makes it sound.

The same information is conventionally presented as the *blocking curve*, Figure 14.6, which plots $\sqrt{s_k^2/N_k}$ against the block size and invites the reader to find the plateau by eye. It is drawn here for the well-tuned production series, the one on which an honest error bar actually has to be quoted. The curve leaves the naive value 3.28×10^{-5} at block size one, reaches 3.60×10^{-5} by block size four, then *falls* to 3.45×10^{-5} at block size sixteen before rising again to 4.15×10^{-5} at block size sixty-four, and the last two points move in opposite directions. None of that is structure. With $N = 2000$ measurements a block size of 64 leaves 31 blocks, and the relative error of a variance estimated from n blocks is about $1/\sqrt{2n}$, or 13 per cent at $n = 31$ – larger than the entire rise the curve is being asked to display. A plateau read off such a curve is a matter of taste, and two careful people will disagree by more than the correction they are arguing about. The dashed line is Jonsson's answer, 4.86×10^{-5}, above every plotted point: as Table 14.11 records, it is 1.48 times the naive error where the autocorrelation estimate $\sqrt{\tau}$ asks only for 1.13. The rule is deliberately conservative, and on a curve as ambiguous as this one that is the right way to be wrong.

Jonsson's stopping rule.

Reading the plateau by eye is unsatisfactory, and Jonsson turned it into a test: the remaining autocovariance after k transformations, accumulated over the coarser levels and normalised, is a quantity that would be χ^2 distributed if the blocked data were independent, so it is compared with a χ^2 quantile and the first k that passes is used. That is what `vmcoptimise.py`

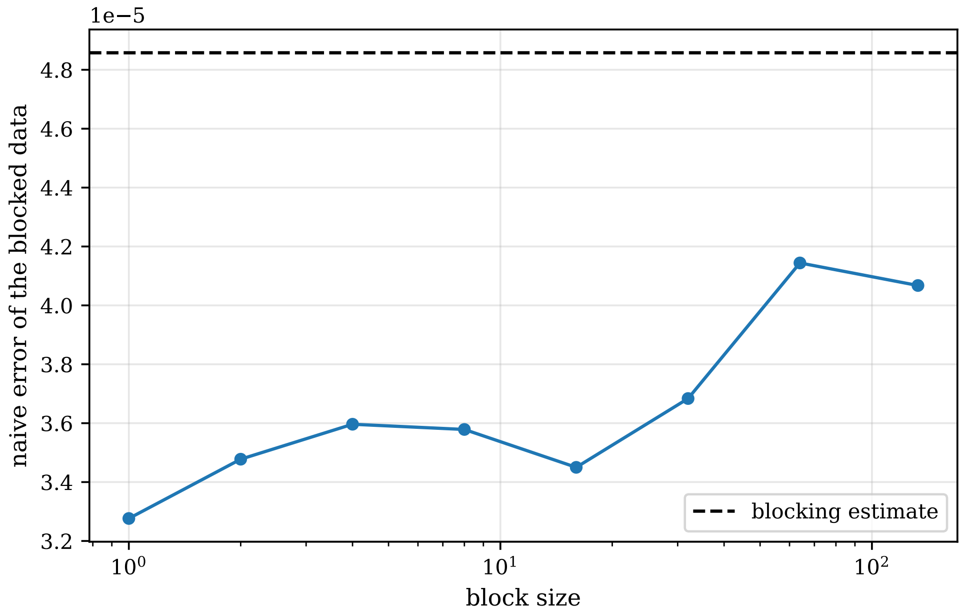

Fig. 14.6 The blocking curve for the well-tuned production series of $N = 2000$ walker-averaged steps: the ordinary uncorrected error $\sqrt{s_k^2/N_k}$ of Eq. (14.79) plotted against the block size 2^k, logarithmically. The dashed line is what the blocking routine returns with Jonsson's stopping rule, 4.86×10^{-5}. The curve is supposed to rise and then flatten; it rises from the naive 3.28×10^{-5} and then wanders, the rightmost points being computed from a few dozen blocks and correspondingly uncertain.

implements, and it makes blocking a method with no free parameters at all – which, after Table 14.7, is not a small thing.

Blocking has one limitation, and it is severe: it estimates the error on a *mean*, and on nothing else. For anything more complicated we need the next two methods.

14.9.4 The bootstrap

Suppose the data are drawn from an unknown distribution F, $X_1, \ldots, X_N \sim F$, and we want the sampling distribution of some estimator $\hat{\theta} = f(X_1, \ldots, X_N)$. If we knew F we could simply repeat the experiment: draw many fresh data sets, compute $\hat{\theta}$ on each, and look at the spread. We do not know F. Efron's observation is that we have something almost as good, the *empirical distribution*

$$\hat{F}_N(x) = \frac{1}{N} \sum_{i=1}^{N} \delta(x - X_i)\,, \tag{14.81}$$

which puts mass $1/N$ on each observed value and which converges to F by the Glivenko-Cantelli theorem. The *plug-in principle* is then to do exactly what we would have done with F, but with $\hat{F}_N$ in its place: treat the observed data set as the population, and generate new data sets by drawing N values from it *with replacement*. Sampling from Eq. (14.81) is precisely resampling with replacement.

The procedure.

For $b = 1,\dots,B$, draw a bootstrap sample $X_1^*,\dots,X_N^*$ and compute $\hat{\theta}_b^* = f(X_1^*,\dots,X_N^*)$. The B values form the *bootstrap distribution*, and everything is read off it:

$$\bar{\theta}^* = \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}_b^*, \tag{14.82}$$

$$\mathrm{Var}_{\mathrm{boot}}(\hat{\theta}) = \frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\theta}_b^* - \bar{\theta}^*\right)^2, \tag{14.83}$$

with $\sigma_{\mathrm{boot}} = \sqrt{\mathrm{Var}_{\mathrm{boot}}}$ the bootstrap standard error. Note that B is limited only by patience: it controls the Monte Carlo error *of the bootstrap itself*, not the information content of the data. B between 10^3 and 10^4 is standard for an error bar; bias estimation and confidence intervals need more.

Why it works.

The justification is a statement about two distributions being close. Under mild regularity conditions $\sqrt{N}(\hat{\theta}-\theta)$ has a limiting distribution as $N \to \infty$, and bootstrap theory shows that $\sqrt{N}(\hat{\theta}^*-\hat{\theta})$, computed under resampling from $\hat{F}_N$ with the data held fixed, has the *same* limit. Hence

$$P^*\left(\hat{\theta}^*-\hat{\theta}\right) \;\approx\; P\left(\hat{\theta}-\theta\right), \tag{14.84}$$

and the variability of the replicates around $\hat{\theta}$, which we can measure, stands in for the variability of $\hat{\theta}$ around the true θ, which we cannot. Nothing was assumed about the shape of F and nothing about the form of f, which is the whole appeal.

Confidence intervals.

Because the bootstrap delivers the entire distribution and not merely its width, an interval can be read off directly. Sorting the replicates, the *percentile interval* at confidence $1-\alpha$ is

$$\left[\hat{\theta}^*_{(\alpha/2)},\, \hat{\theta}^*_{(1-\alpha/2)}\right], \tag{14.85}$$

that is, the 2.5% and 97.5% quantiles for a 95% interval. No normality is assumed anywhere. When the distribution happens to be Gaussian this agrees with $\hat{\theta} \pm 1.96\,\sigma_{\mathrm{boot}}$; when it does not – a positive quantity near zero, a ratio, a fitted parameter at the edge of its range – the percentile interval is the honest one and the symmetric error bar is not.

Bias.

The same replicates estimate bias. With $\mathrm{Bias} = \langle\hat{\theta}\rangle - \theta$, the plug-in estimate is

$$\widehat{\mathrm{Bias}} = \bar{\theta}^* - \hat{\theta}, \qquad \hat{\theta}_{\mathrm{BC}} = \hat{\theta} - \widehat{\mathrm{Bias}} = 2\hat{\theta} - \bar{\theta}^*, \tag{14.86}$$

the second being the bias-corrected estimator. Bias correction should be applied with some care: it removes a systematic error at the cost of adding variance, and if $\widehat{\mathrm{Bias}}$ is itself noisy – which it is unless B is large – the correction can do more harm than good.

Block bootstrap.

There is one assumption in all of the above, and Monte Carlo data violate it: resampling individual points assumes the points are exchangeable, that is, independent. They are not. Drawing single samples with replacement produces data sets that are independent *by construction*, whatever the original looked like, and the bootstrap then returns σ^2/N – the naive answer, with every appearance of having been carefully computed. The repair is to resample *blocks*: cut the series into n contiguous blocks each many correlation times long, and resample whole blocks with replacement. Within a block the correlations are preserved; between blocks there are none left to destroy.

Table 14.9 shows the effect on a deliberately sticky chain with $\tau = 25.3$, for which the honest error should be about $\sqrt{\tau} = 5.0$ times the naive one.

block length	bootstrap error	/ naive
1	0.00010123	0.96
2	0.00014264	1.36
8	0.00023298	2.22
32	0.00038435	3.66
128	0.00047516	4.53
512	0.00055047	5.24
2000	0.00049401	4.71
blocking	0.00058775	5.60

Table 14.9 Block bootstrap on a single chain of 200 000 local energies sampled with a deliberately small time step, $\Delta t = 0.02$, giving $\tau = 25.3$ and a naive error of 1.05×10^{-4}. At block length one the bootstrap reproduces the naive error to two digits; the estimate climbs as the blocks lengthen and plateaus once they exceed the correlation time, which is the blocking plateau in another guise. The last two rows are limited by the number of blocks remaining. Computed by `vmcoptimise.py`.

Figure 14.7 plots those ratios against the three horizontal lines they should be compared with, and there are three things to take from it. The leftmost point is the warning. At block length one the block bootstrap *is* the ordinary bootstrap, and it returns 0.96 times the naive error – a number wrong by a factor of five, produced by two hundred resamplings of two hundred thousand data points, and carrying no internal evidence of being wrong at all. Second, the approach to the honest value is gradual and is governed by the ratio of the block length to τ rather than by the block length itself. At $b = 8$, a third of a correlation time, only 2.2 of the 5.0 has been recovered; at $b = 128$, five correlation times, the ratio is 4.53, which is ninety per cent of $\sqrt{\tau}$; and the curve crosses the $\sqrt{\tau}$ line at about $b = 350$, some fourteen correlation times. The familiar instruction that blocks be "many correlation times long" is exactly this curve: five correlation times get most of the way, twenty get all of it. Third, the curve turns over. At $b = 2000$ the ratio has fallen back to 4.71, not because the estimate has improved but because only a hundred distinct blocks remain in the urn, and a bootstrap resampling a hundred objects cannot resolve the distribution it is trying to build. The two ends of the figure are the two failure modes – correlations destroyed on the left, too few independent objects on the right – and what one wants is the broad shoulder between them. Blocking, which has no block length to choose, arrives at 5.60 without visiting either.

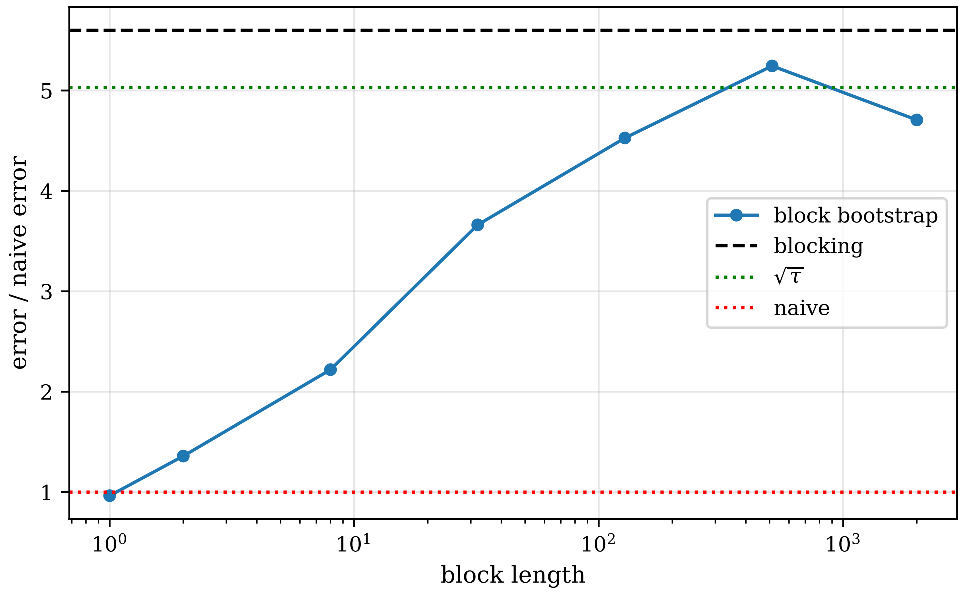

Fig. 14.7 Block bootstrap error on the mean of a deliberately sticky chain of 200 000 local energies ($\Delta t = 0.02$, $\tau = 25.3$) as a function of the length of the resampled block, in units of the naive $\sigma/\sqrt{N} = 1.05 \times 10^{-4}$ and with the block length logarithmic. The horizontal lines mark the naive value itself at 1, the expected $\sqrt{\tau} = 5.03$, and the blocking estimate at 5.60. At block length one the bootstrap returns the naive error to within four per cent; the estimate climbs as the blocks grow past the correlation time and peaks at 5.24 for blocks of 512, falling back thereafter as the number of distinct blocks left to resample from becomes small.

14.9.5 The jackknife

The jackknife predates the bootstrap and is its deterministic cousin. Divide the data into n blocks and let $\hat{\theta}_{(i)}$ be the estimator computed with block i *deleted*, $\bar{\theta}_{(\cdot)}$ the average of the n such values. Then

$$\sigma^2_{\text{jack}} = \frac{n-1}{n} \sum_{i=1}^{n} \left(\hat{\theta}_{(i)} - \bar{\theta}_{(\cdot)}\right)^2, \tag{14.87}$$

and the bias is estimated as

$$\widehat{\text{Bias}}_{\text{jack}} = (n-1)\left(\bar{\theta}_{(\cdot)} - \hat{\theta}\right), \qquad \hat{\theta}_{\text{BC}} = n\hat{\theta} - (n-1)\bar{\theta}_{(\cdot)}. \tag{14.88}$$

The factor $n-1$ in front of a difference of two nearly equal numbers looks alarming, and is exactly right: if the bias is of order $1/m$ in the sample size m, then deleting one block of N/n samples changes it by order $1/(nN)$, and multiplying by $n-1$ recovers the $1/N$ term. The peculiar $(n-1)/n$ in Eq. (14.87) has the same origin: the $\hat{\theta}_{(i)}$ differ from each other far less than independent estimates would, by a factor of about $n-1$, and the prefactor inflates the spread back.

Deleting a whole block rather than a single sample is again what makes the method legitimate on correlated data: the retained data keep their internal correlation structure. For the mean, Eq. (14.87) reduces algebraically to the ordinary standard error of the block means – shown in Exercise 14.12 – so the jackknife earns nothing there. It earns its keep, as does the bootstrap, on estimators that are not means.

14.9.6 Estimators that are not means

The natural non-linear estimator in variational Monte Carlo is the variance of the local energy itself. It is not a diagnostic of secondary interest: by the argument of Exercise 14.12(c), E_L is constant when Ψ_T is an exact eigenstate, so σ^2 measures directly how far the trial function is from one, and variance minimisation is a legitimate alternative optimisation target. Reporting it requires an error bar, and no formula for that error exists.

Table 14.10 takes a single chain of 200000 local energies – a single walker, so that σ^2 really is $\mathrm{Var}(E_L)$ and not the variance of a walker average – cuts it into 200 blocks of 1000, each some eight hundred correlation times long, and applies all three methods to four estimators of increasing non-linearity.

estimator	value	bootstrap	jackknife	blocking	jackknife bias
mean, E	3.000569	0.000105	0.000105	0.000142	$+8.8 \times 10^{-14}$
variance, σ^2	0.002195	0.000013	0.000014	—	-1.1×10^{-8}
σ	0.046851	0.000143	0.000144	—	-3.4×10^{-7}
$\sigma/\lvert E\rvert$	0.015614	0.000048	0.000048	—	-1.1×10^{-7}

Table 14.10 Errors on four estimators from one chain of 200000 local energies, $\tau = 1.25$, resampled in 200 blocks of 1000. For the mean all three methods agree, blocking being the most conservative. For the three below it blocking has nothing to say, and resampling is the only route to an error bar at all. Computed by `vmcoptimise.py`.

Two features of the last column repay attention. The jackknife bias for the mean is 10^{-13}, that is, zero, and it has to be: for a linear estimator every leave-one-out value is an exact rearrangement of the same numbers and there is nothing for the correction to find. Bias is a phenomenon of non-linear estimators. And for the variance the bias can be checked against an exact answer, because the estimator used is the $1/N$ one, biased low by exactly $-\sigma^2/N$:

σ^2, the $1/N$ estimator	0.00219502
exact bias, $-\sigma^2/N$	-1.098×10^{-8}
jackknife estimate	-1.094×10^{-8}
bootstrap estimate, $B = 1000$	-1.995×10^{-7}
jackknife bias-corrected	0.00219503
the $1/(N-1)$ estimator	0.00219503

The jackknife recovers the bias to three digits knowing nothing about what it is estimating, and the correction reproduces the $1/(N-1)$ estimator exactly. The bootstrap estimate is an order of magnitude too large – not because the method is wrong but because a bias of 10^{-8} is being read off $B = 1000$ replicates each carrying an error of 10^{-5}, so what is reported is resampling noise. *Bias estimation needs a far larger B than error estimation*, and the jackknife, being deterministic, does not have the problem at all.

Finally, the percentile interval. On the same chain,

$$E = 3.000569, \qquad \sigma_{\mathrm{boot}} = 0.000105,$$

the 95% percentile interval is $[3.000373, 3.000773]$ against $[3.000362, 3.000775]$ for $\hat{\theta} \pm 1.96\sigma_{\mathrm{boot}}$. They agree, and they should: the mean of two hundred thousand samples is Gaussian by the central limit theorem, and the skewness of the bootstrap replicates is $+0.013$. For σ, for a ratio of averages, or for a fitted parameter the two need not agree, and it is then the percentile interval that should be quoted.

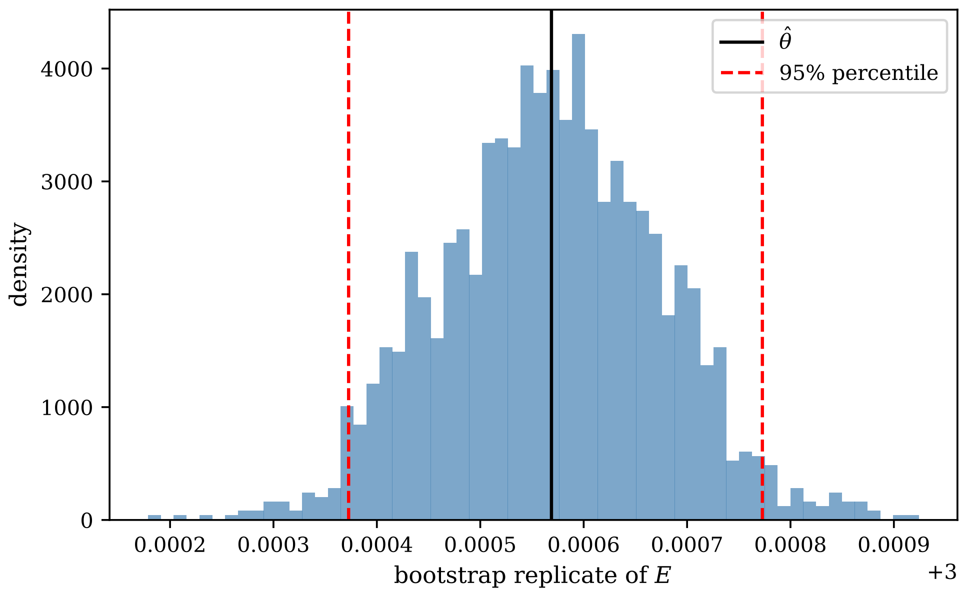

Fig. 14.8 The bootstrap distribution of the mean local energy: a normalised histogram of $B = 2000$ replicates, each the mean of a data set assembled by drawing 200 blocks of 1000 with replacement from the chain of 200 000 local energies of Table 14.10. The solid vertical line is the estimate from the data themselves, $\hat{\theta} = E = 3.000569$, and the two dashed lines are the 2.5% and 97.5% percentiles at 3.000373 and 3.000773, which lie 1.9 standard deviations either side of it. The replicates are symmetric about $\hat{\theta}$, with a skewness of $+0.013$.

Figure 14.8 is the object those two intervals were read off, and it repays a look even in a case where it confirms what we already believed. Its width is the error bar: the standard deviation of the replicates is 1.05×10^{-4}, and the dashed lines stand where a 95% interval should stand. Its shape is the part that no formula supplies, and here the shape is Gaussian – visibly so, and quantitatively so at a skewness of $+0.013$ – because the estimator is a mean of two hundred thousand samples and the central limit theorem has had every opportunity to act. That is why the percentile interval and the symmetric one agree to the last digit that matters. For the three rows below the first in Table 14.10 no such guarantee exists, and for those the histogram is not an illustration of the answer but the answer itself. One further feature deserves notice. The outline of the distribution is smooth near its centre and visibly ragged in the tails, where with $B = 2000$ the extreme bins hold one or two replicates apiece. A standard error depends on the second moment about the centre and is therefore cheap in B; a bias, or a 99% interval, depends on the extremes and is not. That is the same observation as the failure of the bootstrap bias estimate above, seen from the side.

14.9.7 Which to use

- **The mean of a Monte Carlo run.** Blocking with Jonsson’s rule. No free parameters, no window, conservative.

- **Anything else – a variance, a ratio, a fitted parameter, an extrapolated time step.** Block bootstrap or block jackknife. They agree when both apply; the jackknife is deterministic and cheaper, the bootstrap gives the whole distribution and hence confidence intervals.
- **Bias.** The jackknife, or the bootstrap with a large B. Both are worth computing whenever the estimator is non-linear and the sample is small.
- **Never** $\sigma/\sqrt{N}$ from a Markov chain, and never a bootstrap that resamples single points. Both are wrong in the same direction and neither announces itself.

What they give on our data.

Table 14.11 applies all four estimators to two production runs of the same system: one with the well-tuned time step of Chapter 13, and one deliberately badly tuned.

estimator	$\Delta t = 0.5$	$\Delta t = 0.02$
correlation time τ	1.27	13.95
naive $\sigma/\sqrt{N}$	0.00003276	0.00003377
$\sqrt{\tau}\times$ naive	0.00003690	0.00012614
blocking	0.00004858	0.00015405
bootstrap	0.00003486	0.00010611
jackknife	0.00003478	0.00010427
resampled / naive	1.48	4.56

Table 14.11 Error estimates for two production runs of two million and 1.6 million samples. With a well-tuned time step the naive estimate is only modestly wrong; with a badly tuned one it is wrong by a factor of four, and silently. Computed by `vmcoptimise.py`.

Two honest observations. When the sampling is well tuned, τ is close to one and the naive error is not far off – but there is no way to know that without measuring. And the three resampling estimates do not agree exactly: blocking with Jonsson's stopping rule is deliberately conservative and comes out largest. They agree on the order of magnitude, which is what matters, and all three are right in the case where the naive estimate is badly wrong.

14.10 The result

Putting the three stages together – optimise with cheap runs, produce with an expensive one, resample the error – gives

$$E_{\mathrm{VMC}} = 3.000549 \pm 0.000049, \tag{14.89}$$

from two million samples at $\alpha = 1.00$, $\beta = 0.40$. Set against the exact 3 and the results of Chapter 11:

method	energy	error
Hartree-Fock, 42 orbitals	3.161921	1.6×10^{-1}
MP2, 42 orbitals	3.027038	2.7×10^{-2}
CCSD, 42 orbitals	3.013626	1.4×10^{-2}
VMC, this chapter	3.000549(49)	5.5×10^{-4}
exact (Taut)	3.000000	—

The remaining 5.5×10^{-4} is eleven standard errors wide, and it is not statistical: it is what a two-parameter trial function cannot represent. The point of this chapter is precisely that the error bar is now small enough, and trustworthy enough, for that statement to be meaningful. With the naive error of 3.3×10^{-5} one would have quoted a result seventeen standard errors from the exact answer and concluded, wrongly, that something was broken.

14.11 Summary

The gradient of a variational energy with respect to its parameters, Eq. (14.2), is a covariance and costs nothing beyond the energy itself. With it, the grid scan of Chapter 13 is replaced by descent, and the number of parameters stops being the limiting consideration. Four families of descent were set out, and they differ in where they get their curvature. Momentum guesses it from the history of the gradient. Quasi-Newton methods infer it, imposing the secant condition (14.25) and making the smallest change consistent with it – smallest in the Frobenius norm of Section 1.13.1, which is what makes Broyden's update rank one and BFGS the unique symmetric positive definite rank-two update. Powell's method builds conjugate directions from function values alone and needs no gradient at all. And stochastic reconfiguration, Eq. (14.61), does not use the curvature of the energy at all: it uses the metric (14.58) of the space of wave functions, and is simultaneously a projected imaginary-time step and Amari's natural gradient.

Which of them survives contact with a Monte Carlo gradient is the practical lesson. BFGS and Powell are the better methods on a deterministic problem – Table 14.4 is unambiguous about that – and both are fragile here, the first because it differences noisy gradients and the second because it line-minimises a noisy function. Stochastic reconfiguration is built from averages rather than differences and is in consequence far better conditioned statistically, which is why it, and not BFGS, is what scales to thousands of parameters.

Once the parameters are fixed, the production run is pure statistics and parallelises trivially. The error on it is set by Eq. (14.68), $\mathrm{Var}(\bar{X}) = \mathbf{1}^T \boldsymbol{\Sigma} \mathbf{1}/N^2$, which is exact and unusable: the estimated covariances at large lag are noise, and summing them all gives an answer smaller than the naive one. Blocking evades the problem by driving the covariance matrix to a multiple of the identity, and with Jonsson's stopping rule it has no free parameter left; it is what should be quoted for a mean. The block bootstrap and the block jackknife handle everything a mean cannot – a variance, a ratio, a fitted or extrapolated parameter – and give in addition a bias estimate and, for the bootstrap, a confidence interval that assumes no normality. What must never be quoted is $\sigma/\sqrt{N}$, or a bootstrap that resamples single points: they are the same mistake twice, and neither announces itself.

What remains is the trial function. Every result in this chapter is bounded above by what two parameters can express, and no amount of sampling or optimisation will change that. Two directions lead out. Diffusion Monte Carlo removes the dependence on Ψ_T by projecting onto the ground state in imaginary time, using the drift and Green's function of Section 13.6.

And if the difficulty is that we cannot guess a good enough functional form, the alternative is not to guess: to parametrise Ψ_T by a neural network and let Eq. (14.2) and Eq. (14.61) do the rest. Both are taken up in the chapters that follow.

14.12 Exercises

Exercise 1: the energy gradient

a) Rederive Eq. (14.2) by a route different from the one in Section 14.2.1: write $E = \int dx\, p_{\boldsymbol{\theta}}(x) E_L(x;\boldsymbol{\theta})$ and differentiate the sampling density itself.
b) Find the corresponding expression for the derivative of the variance.
c) Show that the gradient vanishes when Ψ_T is an exact eigenstate, for every parameter, and say why that is obvious in advance.

Answer to a).

This is the "score-function" or REINFORCE route, and it is the one the machine-learning literature takes. Differentiating the product,

$$\partial_\theta E = \int dx\, (\partial_\theta p_{\boldsymbol{\theta}})\, E_L + \int dx\, p_{\boldsymbol{\theta}}\, (\partial_\theta E_L)\,.$$

For the first term use the identity $\partial_\theta p = p\,\partial_\theta \ln p$, and note from Eq. (14.11) that $\ln p_{\boldsymbol{\theta}} = 2\ln|\psi_{\boldsymbol{\theta}}| - \ln\int|\psi_{\boldsymbol{\theta}}|^2$, so that

$$\partial_\theta \ln p_{\boldsymbol{\theta}} = 2O_\theta - 2\langle O_\theta\rangle,$$

the subtraction coming from the normalisation. The first term is therefore $2\left(\langle O_\theta E_L\rangle - \langle O_\theta\rangle\langle E_L\rangle\right)$, which is already Eq. (14.2). The second term must therefore vanish, and it does. Differentiating $E_L = (\hat{H}\psi)/\psi$ and using $\partial_\theta\psi = O_\theta\psi$,

$$\partial_\theta E_L = \frac{\hat{H}(O_\theta\psi)}{\psi} - E_L O_\theta,$$

so that

$$\langle\partial_\theta E_L\rangle = \frac{\int dx\,\psi\hat{H}(O_\theta\psi) - \int dx\,\psi^2 E_L O_\theta}{\int dx\,\psi^2} = 0,$$

because hermiticity moves $\hat{H}$ onto the left factor, $\int\psi\hat{H}(O_\theta\psi) = \int(\hat{H}\psi)(O_\theta\psi) = \int\psi^2 E_L O_\theta$. The cancellation is worth understanding: the explicit dependence of the local energy on the parameters contributes nothing to the gradient of its average. That is why no derivative of $\hat{H}\psi$ is ever needed, and it is the same fact that makes E_L a zero-variance estimator at an exact eigenstate.

Answer to b).

With $\sigma^2 = \langle E_L^2\rangle - \langle E_L\rangle^2$ the same manipulation applied to $\langle E_L^2\rangle$ gives

$$\frac{d\sigma^2}{d\theta} = 2\left(\langle O_\theta E_L^2\rangle - \langle O_\theta\rangle\langle E_L^2\rangle\right) + 2\langle\partial_\theta E_L\, E_L\rangle - 2\bar{E}_\theta\langle E_L\rangle,$$

where the middle term appears because E_L itself depends on θ, unlike in a). This is why variance minimisation, although attractive in principle, is more work: it needs $\partial_\theta E_L$, which requires differentiating the Hamiltonian acting on Ψ_T.

Answer to c).

If Ψ_T is an eigenstate then E_L is a constant, so $\langle O_\theta E_L\rangle = E_L\langle O_\theta\rangle = \langle O_\theta\rangle\langle E_L\rangle$ and Eq. (14.2) vanishes identically. It has to: at an exact eigenstate the energy is stationary against *any* variation, parameters included. Note the practical corollary – as the trial function improves, the gradient and the variance vanish together, so a small gradient with a large variance means the optimiser has found a flat direction rather than the answer.

Exercise 2: descent methods

a) For $f(x) = \frac{1}{2}x^T Ax - b^T x$ with A symmetric positive definite, show that steepest descent with the optimal step converges with a rate governed by the condition number of A.
b) Show that the secant condition (14.25) is exactly satisfied by the true Hessian for a quadratic f.
c) Why can a quasi-Newton method behave badly when the gradient is stochastic?

Answer to a).

The gradient is $\nabla f = Ax - b$ and the minimum is at $x_* = A^{-1}b$. Writing $e_k = x_k - x_*$, one line of algebra gives $e_{k+1} = (I - \eta A)e_k$, so in the eigenbasis of A each component decays as $(1 - \eta\lambda_i)^k$. The best uniform η balances the extreme eigenvalues, $\eta = 2/(\lambda_{\min} + \lambda_{\max})$, and gives a rate $(\kappa - 1)/(\kappa + 1)$ with $\kappa = \lambda_{\max}/\lambda_{\min}$. A condition number of 100 therefore needs of order a hundred times more iterations than a well-conditioned one – which is the zig-zag, quantified.

Answer to b).

For f quadratic with Hessian A, $\nabla f(x) = Ax - b$, so $\nabla f(x_{k+1}) - \nabla f(x_k) = A(x_{k+1} - x_k)$, which is Eq. (14.25) with $\boldsymbol{H}_{k+1} = A$. Broyden and BFGS impose this one condition and fix the remaining freedom by asking for the smallest change to $\boldsymbol{H}$ in a suitable norm – BFGS additionally preserving symmetry and positive definiteness, which is what makes it suitable for minimisation rather than root finding.

Answer to c).

The curvature is inferred from *differences* of gradients, and differencing two noisy quantities amplifies the noise. If the statistical error in ∇E is comparable to the change in ∇E between iterations – which happens as soon as the steps get small, that is, near the minimum – the inferred Hessian is dominated by noise and can easily fail to be positive definite, sending the optimiser uphill. The usual remedies are to increase the number of samples as the optimisation proceeds, to fall back on a first-order method near the minimum, or to use stochastic reconfiguration, whose metric (14.58) is an average rather than a difference and is therefore far better conditioned statistically.

Exercise 3: resampling

a) Show that a blocking transformation (14.78) leaves the sample mean unchanged, and that for uncorrelated data it leaves the estimated error unchanged too.
b) Show that the jackknife error for the mean is identical to the blocked standard error.
c) Explain why a bootstrap that resamples individual points rather than blocks reproduces the naive error, and what that implies about using it on Monte Carlo data.
d) Derive Eq. (14.71) by counting: show that the number of pairs (i, j) with $|i-j| = t$ in an $N \times N$ array is N for $t = 0$ and $2(N-t)$ for $t \geq 1$, and that these count to N^2.
e) Show that if $\rho(t) = e^{-t/\xi}$ exactly, then $\tau = \coth(1/2\xi) \approx 2\xi$ for $\xi \gg 1$, and find the finite-N correction from Eq. (14.73).
f) Show that one blocking transformation of a series with autocovariance $C(t)$ gives a series with $C^{(1)}(0) = \frac{1}{2}(C(0)+C(1))$ and $C^{(1)}(t) = \frac{1}{4}(C(2t-1)+2C(2t)+C(2t+1))$ for $t \geq 1$, and hence that τ is halved at each level while $\mathrm{Var}(\bar{X})$ is preserved.

Answer to a).

The mean of the blocked data is $\frac{2}{n}\sum_i \frac{1}{2}(x_{2i-1}+x_{2i}) = \frac{1}{n}\sum_j x_j$, unchanged. For independent data with variance σ^2, each block mean has variance $\sigma^2/2$ and there are $n/2$ of them, so the estimated variance of the mean is $(\sigma^2/2)/(n/2) = \sigma^2/n$: also unchanged. The blocking curve is therefore flat for independent data, and any rise in it is a direct measure of correlation – which is what makes the plateau meaningful.

Answer to b).

With n blocks of means b_i and $\bar{b}$ their average, $\hat{\theta}_{(i)} = (n\bar{b}-b_i)/(n-1)$, so $\hat{\theta}_{(i)} - \bar{\theta} = -(b_i-\bar{b})/(n-1)$. Substituting into Eq. (14.87),

$$\sigma^2_{\text{jack}} = \frac{n-1}{n}\sum_i \frac{(b_i-\bar{b})^2}{(n-1)^2} = \frac{1}{n(n-1)}\sum_i (b_i-\bar{b})^2,$$

which is the ordinary standard error of the n block means. The jackknife adds nothing for a linear estimator, and Table 14.11 shows the two agreeing to three digits.

Answer to c).

Resampling individual points with replacement produces data sets whose elements are independent by construction, whatever the original series looked like. The bootstrap distribution of the mean then has variance σ^2/n – the naive answer – because the correlations have been destroyed by the resampling itself. The implication is that a bootstrap applied naively to Monte Carlo data is not merely inaccurate, it is systematically and confidently wrong, and looks respectable while being so. Block bootstrapping, with a block long compared with τ, is the fix, and Table 14.9 shows exactly this: at block length one the bootstrap returns 0.96 times the naive error, and the honest answer is five times larger.

Answer to f).

Write $Y_i = \frac{1}{2}(X_{2i-1}+X_{2i})$. Then

$$C^{(1)}(0) = \mathrm{Var}(Y_i) = \tfrac{1}{4}\left(C(0) + 2C(1) + C(0)\right) = \tfrac{1}{2}\left(C(0) + C(1)\right),$$

and for $t \geq 1$ the four cross terms of $\mathrm{Cov}(Y_i, Y_{i+t})$ have separations $2t-1$, $2t$, $2t$ and $2t+1$, giving $C^{(1)}(t) = \frac{1}{4}\left(C(2t-1) + 2C(2t) + C(2t+1)\right)$. Summing,

$$\sum_{t\geq 1} C^{(1)}(t) = \tfrac{1}{4}\sum_{t\geq 1}\left(C(2t-1) + 2C(2t) + C(2t+1)\right) = \tfrac{1}{2}\sum_{s\geq 1} C(s) - \tfrac{1}{4}C(1),$$

so that $C^{(1)}(0) + 2\sum_{t\geq 1} C^{(1)}(t) = \frac{1}{2}\left(C(0) + 2\sum_{s\geq 1} C(s)\right)$: the numerator of Eq. (14.74) is exactly halved. Since $N_1 = N/2$ as well, $\mathrm{Var}(\bar{X})$ is unchanged, as it must be – blocking does not alter the data, only our description of them. But $\tau^{(1)} = \tau C(0)/\left(2C^{(1)}(0)\right)$, which for positively correlated data is smaller than τ, and after enough levels reaches one. *Blocking drives τ to unity while holding the true error fixed*, and that is the whole mechanism.

Exercise 4: the Frobenius norm and Broyden's update

a) Show that $\partial\,\mathrm{Tr}(\boldsymbol{X}^T\boldsymbol{X})/\partial\boldsymbol{X} = 2\boldsymbol{X}$, and that $\partial\,\mathrm{Tr}(\boldsymbol{\lambda}^T\boldsymbol{X}\boldsymbol{s})/\partial\boldsymbol{X} = \boldsymbol{\lambda}\boldsymbol{s}^T$. These are the only two matrix derivatives the Broyden derivation uses.
b) Carry out the derivation of Eq. (14.28) in full, including the verification that the stationary point really is a minimum.
c) Show directly that the general solution of the constraint $\Delta\boldsymbol{B}\boldsymbol{s} = \boldsymbol{r}$ is $\Delta\boldsymbol{B} = \boldsymbol{r}\boldsymbol{s}^T/(\boldsymbol{s}^T\boldsymbol{s}) + \boldsymbol{Z}\boldsymbol{P}$ with $\boldsymbol{P} = \boldsymbol{I} - \boldsymbol{s}\boldsymbol{s}^T/(\boldsymbol{s}^T\boldsymbol{s})$ and $\boldsymbol{Z}$ arbitrary, and that the two pieces are orthogonal in the Frobenius inner product. Conclude the minimality without any Lagrange multiplier.
d) Derive the inverse update (14.29) the other way: apply the Sherman-Morrison formula to Eq. (14.28) and show that the result is *not* Eq. (14.29). This is the difference between "good" and "bad" Broyden.

Answer to a).

For the first, $\mathrm{Tr}(\boldsymbol{X}^T\boldsymbol{X}) = \sum_{ij} x_{ij}^2$ by Eq. (1.9), so $\partial/\partial x_{kl}$ gives $2x_{kl}$. For the second, $\boldsymbol{\lambda}^T\boldsymbol{X}\boldsymbol{s} = \sum_{ij}\lambda_i x_{ij} s_j$, whose derivative with respect to x_{kl} is $\lambda_k s_l$, that is $(\boldsymbol{\lambda}\boldsymbol{s}^T)_{kl}$.

Answer to c).

Any $\Delta\boldsymbol{B}$ satisfying the constraint may be written as the particular solution plus a solution of the homogeneous equation $\Delta\boldsymbol{B}\boldsymbol{s} = 0$, and the homogeneous solutions are exactly the matrices of the form $\boldsymbol{Z}\boldsymbol{P}$, since $\boldsymbol{P}\boldsymbol{s} = 0$ and any $\boldsymbol{M}$ with $\boldsymbol{M}\boldsymbol{s} = 0$ satisfies $\boldsymbol{M} = \boldsymbol{M}\boldsymbol{P}$. For the orthogonality, using Eq. (1.11) and the cyclic property of the trace,

$$\left\langle \frac{\boldsymbol{r}\boldsymbol{s}^T}{\boldsymbol{s}^T\boldsymbol{s}}, \boldsymbol{Z}\boldsymbol{P} \right\rangle_F = \frac{1}{\boldsymbol{s}^T\boldsymbol{s}}\mathrm{Tr}\left(\boldsymbol{s}\boldsymbol{r}^T\boldsymbol{Z}\boldsymbol{P}\right) = \frac{1}{\boldsymbol{s}^T\boldsymbol{s}}\boldsymbol{r}^T\boldsymbol{Z}\underbrace{\boldsymbol{P}\boldsymbol{s}}_{=0} = 0.$$

By Pythagoras in the Frobenius inner product, $\|\Delta\boldsymbol{B}\|_F^2 = \|\boldsymbol{r}\boldsymbol{s}^T/\boldsymbol{s}^T\boldsymbol{s}\|_F^2 + \|\boldsymbol{Z}\boldsymbol{P}\|_F^2$, which is minimised by $\boldsymbol{Z}\boldsymbol{P} = 0$. This is the calculation `vmcoptimise.py` checks numerically, and it is the reason the answer had to be rank one: the affine set of admissible updates meets the rank-one matrices $\boldsymbol{r}\boldsymbol{s}^T$ orthogonally.

Exercise 5: properties of the BFGS update

a) Verify Eqs. (14.34) and (14.36) are inverses of one another, using the Sherman-Morrison-Woodbury identity.
b) Show that the second Wolfe condition (14.31) implies the curvature condition (14.30).
c) Show that on a quadratic with Hessian $\boldsymbol{A}$ and exact line searches, BFGS started from $\boldsymbol{H}_0 = \boldsymbol{I}$ generates the same iterates as the conjugate gradient method of Section 1.18, and that $\boldsymbol{H}_n = \boldsymbol{A}^{-1}$.
d) Powell's damped update replaces $\boldsymbol{y}_k$ by $\tilde{\boldsymbol{y}}_k = \phi \boldsymbol{y}_k + (1-\phi)\boldsymbol{B}_k\boldsymbol{s}_k$ with $\phi = 0.8\,\boldsymbol{s}^T\boldsymbol{B}\boldsymbol{s}/(\boldsymbol{s}^T\boldsymbol{B}\boldsymbol{s} - \boldsymbol{s}^T\boldsymbol{y})$ when $\boldsymbol{s}^T\boldsymbol{y} < 0.2\,\boldsymbol{s}^T\boldsymbol{B}\boldsymbol{s}$. Show that this guarantees $\boldsymbol{s}^T\tilde{\boldsymbol{y}} = 0.2\,\boldsymbol{s}^T\boldsymbol{B}\boldsymbol{s} > 0$, so that the update is always well defined.

Answer to b).

Subtract $\nabla f(\boldsymbol{x}_k)^T\boldsymbol{p}_k$ from both sides of Eq. (14.31):

$$\boldsymbol{y}_k^T\boldsymbol{p}_k \geq (c_2 - 1)\,\nabla f(\boldsymbol{x}_k)^T\boldsymbol{p}_k.$$

Since $\boldsymbol{p}_k$ is a descent direction the right-hand side is a negative number times a negative number, hence positive; and $\boldsymbol{s}_k = \alpha_k\boldsymbol{p}_k$ with $\alpha_k > 0$, so $\boldsymbol{s}_k^T\boldsymbol{y}_k = \alpha_k\boldsymbol{y}_k^T\boldsymbol{p}_k > 0$. The line search and the positive definiteness of the Hessian model are thus not two separate concerns but one.

Answer to d).

Write $b = \boldsymbol{s}^T\boldsymbol{B}\boldsymbol{s}$ and $c = \boldsymbol{s}^T\boldsymbol{y}$. Then

$$\boldsymbol{s}^T\tilde{\boldsymbol{y}} = \phi c + (1-\phi)b = b + \phi(c-b) = b + \frac{0.8b}{b-c}(c-b) = b - 0.8b = 0.2\,b,$$

which is positive whenever $\boldsymbol{B} \succ 0$ and $\boldsymbol{s} \neq 0$. The damping therefore interpolates between the measured curvature and the modelled one, using just enough of the model to keep the update legitimate. It is what makes the "damped + trust radius" row of Table 14.6 usable where the unsafeguarded row is not.

Exercise 6: stochastic reconfiguration

a) Derive Eq. (14.57) from Eq. (14.54): differentiate the normalised state and show that the subtracted terms come from the derivative of the normalisation.
b) Show that $\boldsymbol{S}$ is positive semidefinite, and that $\boldsymbol{v}^T\boldsymbol{S}\boldsymbol{v} = 0$ if and only if $\sum_k v_k O_k(x)$ is independent of x. Interpret that condition.
c) Show that if the parameters are rescaled, $\alpha_k \to c_k\alpha_k$, the stochastic-reconfiguration step changes covariantly while the steepest-descent step does not, so that Eq. (14.61) produces the same sequence of *states* in either parametrisation.
d) Show that a redundant parameter – one for which $\partial \ln\Psi/\partial\alpha_k$ is a constant, as with an overall normalisation factor e^{α_0} – contributes a zero row and column to $\boldsymbol{S}$ and a zero component to $\boldsymbol{g}$, so that the regularised update simply leaves it alone.

Answer to b).

For any real $\boldsymbol{v}$, put $A(x) = \sum_k v_k O_k(x)$. Then

$$\boldsymbol{v}^T \boldsymbol{S} \boldsymbol{v} = \sum_{ij} v_i v_j \left(\langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle\right) = \langle A^2 \rangle - \langle A \rangle^2 = \mathrm{Var}(A) \geq 0.$$

The variance vanishes only if $A(x)$ is the same for every configuration in the support of P. Since $O_k = \partial_k \ln \Psi$, that says the combination $\sum_k v_k \partial_k \ln \Psi$ changes $\ln \Psi$ by a constant, that is, changes the state only by an overall factor. A null direction of $\boldsymbol{S}$ is therefore a direction in parameter space along which the physical state does not move at all – a redundancy of the parametrisation – which is exactly why $\boldsymbol{S}$ deserves the name metric and why its null space should be projected out or regularised rather than inverted.

Exercise 7: the bootstrap and the jackknife

a) Show that for the mean the bootstrap variance, Eq. (14.83), converges as $B \to \infty$ to $\frac{1}{N}\left(\frac{N-1}{N}\right) s^2$ with s^2 the sample variance, and hence that the bootstrap is very slightly biased low for a linear estimator. *Hint:* the resampled mean is $\frac{1}{N}\sum_i m_i X_i$ with $(m_1, \ldots, m_N)$ multinomial.
b) Show that the bootstrap bias estimate, Eq. (14.86), vanishes identically for the mean.
c) For the biased variance estimator $\hat{\sigma}^2 = \frac{1}{N}\sum_i (X_i - \bar{X})^2$, show that $\langle \hat{\sigma}^2 \rangle = \sigma^2 (N-1)/N$, so that the bias is exactly $-\sigma^2/N$, and verify that the jackknife formula (14.88) recovers it in the limit of many blocks.
d) The probability that a given sample is absent from one bootstrap replicate is $(1 - 1/N)^N \to e^{-1} = 0.368$. Interpret this, and explain why it makes the bootstrap distribution slightly wider than the jackknife's leave-one-out distribution.
e) Suppose $\hat{\theta}$ is a ratio of averages, $R = \langle A \rangle / \langle B \rangle$. Write down the first-order propagation-of-errors formula for $\mathrm{Var}(R)$ in terms of $\mathrm{Var}(A)$, $\mathrm{Var}(B)$ and $\mathrm{Cov}(A,B)$, and explain when it fails and why the bootstrap does not.

Answer to a).

Under resampling, $\bar{X}^* = \frac{1}{N}\sum_i m_i X_i$ where the counts m_i are multinomial with N trials and probabilities $1/N$, so $\langle m_i \rangle = 1$, $\mathrm{Var}(m_i) = 1 - 1/N$ and $\mathrm{Cov}(m_i, m_j) = -1/N$ for $i \neq j$. With the data held fixed,

$$\mathrm{Var}^*(\bar{X}^*) = \frac{1}{N^2}\left[\left(1 - \frac{1}{N}\right)\sum_i X_i^2 - \frac{1}{N}\sum_{i \neq j} X_i X_j\right] = \frac{1}{N^2}\sum_i (X_i - \bar{X})^2,$$

which is $\frac{N-1}{N} \cdot \frac{s^2}{N}$ with $s^2 = \frac{1}{N-1}\sum_i (X_i - \bar{X})^2$. The bootstrap therefore reproduces the standard error up to the familiar factor $(N-1)/N$, negligible for any N worth analysing. This calculation is also the cleanest proof of Exercise 3(c): nowhere in it does any correlation between the X_i appear, because the multinomial resampling has destroyed it.

Answer to d).

Each replicate omits about 37% of the data and duplicates a comparable fraction, so a bootstrap replicate is a considerably more violent perturbation of the data set than deleting a single point. That is why the bootstrap explores the tails of the sampling distribution – and can therefore produce confidence intervals – while the jackknife, which makes only n small perturbations, is restricted to the local, quadratic information: variance and leading bias. The jackknife is in fact a linear approximation to the bootstrap, and it fails for estimators that are not smooth, the median being the standard counterexample.

Exercise 8: numerical explorations

a) Using `vmcoptimise.py`, reproduce Table 14.2 and plot the path of the optimiser on top of the energy surface of Table 13.4.
b) Study the learning rate. How large may η be before the descent diverges, and how does that limit relate to Eq. (14.20) and to the eigenvalues of the Hessian?
c) Replace the fixed learning rate by a schedule $\eta_k = \eta_0/(1+k/k_0)$ and compare the number of iterations and the scatter of the final parameters.
d) Reproduce Table 14.4 and extend it: how does each method scale with the dimension n at fixed κ? Add L-BFGS with memory $m = 2, 5, 10$ and find where it starts to match full BFGS.
e) Feed the analytic Monte Carlo gradient to `scipy.optimize.minimize` with `method="BFGS"` and compare with `bfgs_descent`. Reduce the number of cycles per gradient evaluation and find the point at which the curvature estimate collapses – you should be able to reproduce the fifth row of Table 14.6, or something worse.
f) Repeat with `method="Powell"` and count function evaluations. How many cycles per energy evaluation are needed before the line minimisations stop chasing noise?
g) Compute the metric $\boldsymbol{S}$ on a grid of (α,β) and plot its condition number. Where on the surface is the natural gradient most different from the bare gradient?
h) Add a third, deliberately redundant parameter to Ψ_T – for instance an overall factor e^{γ} – and confirm both that gradient descent wastes steps on it and that stochastic reconfiguration does not.
i) Verify Exercise 3(c) numerically: bootstrap the production series with a block length of one and confirm that you recover the naive error, then reproduce Table 14.9.
j) Run the production stage at several time steps and confirm that the blocked error, unlike the naive one, is independent of Δt – as it must be, since the physical answer cannot depend on the sampling.
k) Reproduce Table 14.7 and add Sokal's window with $c = 3$, 5 and 10. Which value would you use, and on what grounds? Then generate a synthetic AR(1) series, $X_{k+1} = \phi X_k + \varepsilon_k$, for which $\rho(t) = \phi^t$ and $\tau = (1+\phi)/(1-\phi)$ are known exactly, and find how long a series must be before each rule recovers the true τ to ten per cent.
l) Reproduce Table 14.8 and plot the full correlation matrix $\rho^{(k)}(|i-j|)$ at each level as an image. Watch it become the identity.
m) Compute the error on the variance of the local energy at several values of (α,β) and confirm that σ^2 has its minimum in very nearly the same place as E does, but not exactly. Which minimum would you rather locate, and why might variance minimisation be preferred in practice despite the extra work of Exercise 1(b)?
n) Bootstrap a genuinely awkward estimator: fit $E(\Delta t) = E_0 + a\Delta t$ to the production energies at four time steps and put an error bar on the extrapolated E_0. Compare the percentile interval with the symmetric one.
o) Take the median of the local energies instead of the mean, and compare the bootstrap and jackknife errors. The jackknife should misbehave; explain why with reference to Exercise 7(d).

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter14/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/optimisation.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter14/`.

`vmcoptimise.py` The parameter gradient; gradient descent, momentum and stochastic reconfiguration; the Broyden, BFGS and DFP updates with `check_quasi_newton_algebra`; `broyden_root` and `newton_root`; line searches; `golden_section` and `powell_minimise`; a vectorised many-walker production run; the autocovariance, the summation window and the exact variance of a correlated mean; blocking; and the block bootstrap and jackknife. Runs in about nine minutes; needs `scipy.stats` for the blocking test.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 15
Diffusion Monte Carlo

Chapter 14 ended with a number and a complaint. The number was

$$E_{\mathrm{VMC}} = 3.000549 \pm 0.000049,$$

and the complaint was that the residual 5.5×10^{-4} above Taut's exact 3 is eleven standard errors wide and entirely systematic: it is what a two-parameter trial function cannot represent. Optimising harder does not touch it, and neither does sampling longer. The only way past it, within the variational framework, is a better guess – and guessing is precisely what one would like to stop doing.

This chapter stops doing it. Diffusion Monte Carlo does not evaluate an integral over a trial function; it *projects* onto the ground state, by propagating an arbitrary starting state in imaginary time and letting the excited components die away. The trial function does not disappear – it returns, importantly, as a *guiding* function – but it stops determining the answer and starts determining only the efficiency. For the two-electron dot that is the difference between an energy accurate to 5×10^{-4} and one accurate to 10^{-4} and statistically consistent with the exact result.

The route runs through material already assembled. The connection between random walks and the diffusion equation is Section 12.6; the Fokker-Planck equation, the Langevin equation and the drift are Section 12.8; the quantum force, the Green's function and the Metropolis-Hastings correction are Section 13.6; the optimised parameters are Eq. (14.89); and the blocking analysis that turns a correlated series into an honest error bar is Section 14.9. Diffusion Monte Carlo is what those pieces were being assembled for.

The programs are in `dmc.py`, which imports the trial function, the local energy and the quantum force unchanged from `vmc.py` and the resampling from `vmcoptimise.py`.

15.1 Projection in imaginary time

Replace t by $-i\tau$ in the time-dependent Schrödinger equation. The oscillating phases $e^{-iE_n t}$ become decaying exponentials $e^{-E_n\tau}$, and the equation becomes

$$\boxed{-\frac{\partial \Phi(\boldsymbol{R},\tau)}{\partial \tau} = \left(\hat{H} - E_T\right) \Phi(\boldsymbol{R},\tau)} \tag{15.1}$$

where $\boldsymbol{R} = (\boldsymbol{r}_1, \dots, \boldsymbol{r}_N)$ collects all the coordinates and E_T is a constant energy offset whose purpose will become clear in a moment. The formal solution is

$$\Phi(\boldsymbol{R},\tau) = e^{-\left(\hat{H}-E_T\right)\tau}\,\Phi(\boldsymbol{R},0). \tag{15.2}$$

Now expand the starting state in the exact eigenstates of $\hat{H}$, which we do not know but which exist,

$$\Phi(\boldsymbol{R},0) = \sum_n c_n \Psi_n(\boldsymbol{R}), \qquad \hat{H}\Psi_n = E_n\Psi_n, \tag{15.3}$$

and propagate term by term:

$$\Phi(\boldsymbol{R},\tau) = \sum_n c_n\, e^{-(E_n-E_T)\tau}\,\Psi_n(\boldsymbol{R}) = e^{-(E_0-E_T)\tau}\left[c_0\Psi_0 + \sum_{n>0} c_n e^{-(E_n-E_0)\tau}\Psi_n\right]. \tag{15.4}$$

Every excited component is suppressed relative to the ground state by $e^{-(E_n-E_0)\tau}$, and since $E_n > E_0$ each of those factors goes to zero. As $\tau \to \infty$,

$$\Phi(\boldsymbol{R},\tau) \longrightarrow c_0\, e^{-(E_0-E_T)\tau}\,\Psi_0(\boldsymbol{R}), \tag{15.5}$$

whatever we started from, provided only that $c_0 \neq 0$ – the starting state must not be orthogonal to the ground state.

That is the entire principle. It also explains E_T. The surviving prefactor $e^{-(E_0-E_T)\tau}$ makes the norm of Φ grow without bound if $E_T > E_0$ and collapse to zero if $E_T < E_0$. Tuning E_T so that the norm stays constant makes $E_T = E_0$: the offset that stabilises the propagation *is* the ground-state energy. This is the *growth estimator*, and it is the first of two ways in which the energy will be measured.

Why the excited states are the easy part. The convergence rate is set by the gap $E_1 - E_0$, and the amount of imaginary time needed is of order $1/(E_1 - E_0)$. For the two-electron dot the gap is of order $\hbar\omega = 1$, so a few units of τ suffice. What is *not* controlled by the gap is the starting point: if c_0 is very small, the ground-state signal stays buried under the excited components for a long time. Making c_0 close to one from the outset is one of the jobs of the guiding function introduced in Section 15.3.

15.2 A diffusion equation with a source

Write $\hat{H} = \hat{T} + \hat{V}$ and put Eq. (15.1) in coordinate representation. In the atomic units of Chapter 11, where $\hbar = m = 1$ and the kinetic operator is $-\frac{1}{2}\nabla^2$,

$$\frac{\partial\Phi}{\partial\tau} = \underbrace{D\nabla^2\Phi}_{\text{diffusion}} - \underbrace{\left(V(\boldsymbol{R}) - E_T\right)\Phi}_{\text{rate term}}, \qquad D = \frac{1}{2}. \tag{15.6}$$

Two things are recognisable. The first term is exactly the diffusion equation of Section 12.6, Eq. (12.28), whose solution we already know how to simulate: a random walk with Gaussian steps of variance $2D\Delta\tau$. The second is a first-order rate equation whose solution is an exponential in τ with a position-dependent rate.

The diffusion coefficient $D = \frac{1}{2}$ is not an adjustable property of a physical fluid; it is the coefficient of the Laplacian in the kinetic energy. That the imaginary-time Schrödinger equation looks like a diffusion equation is not an analogy – the two are the same equation with different letters.

The short-time Green's function.

The two terms do not commute, so their effects cannot simply be applied one after the other – except over a short time, where the error is controlled. This is the Trotter factorisation of Section 11.8: for small $\Delta\tau$,

$$e^{-(\hat{T}+\hat{V}-E_T)\Delta\tau} = e^{-\hat{T}\Delta\tau/2}\, e^{-(\hat{V}-E_T)\Delta\tau}\, e^{-\hat{T}\Delta\tau/2} + \mathscr{O}(\Delta\tau^3), \tag{15.7}$$

which in coordinate space gives the *short-time Green's function*

$$G(\boldsymbol{R}' \leftarrow \boldsymbol{R}, \Delta\tau) \simeq \underbrace{\frac{\exp\left[-\dfrac{(\boldsymbol{R}'-\boldsymbol{R})^2}{4D\Delta\tau}\right]}{(4\pi D\Delta\tau)^{3N/2}}}_{G_{\text{diff}}} \times \underbrace{\exp\left[-\Delta\tau\left(\tfrac{1}{2}\left[V(\boldsymbol{R})+V(\boldsymbol{R}')\right]-E_T\right)\right]}_{G_{\text{branch}}}. \tag{15.8}$$

The symmetrised potential in the exponent is the trapezoidal rule for $\int V\,d\tau$ along the step, and it is one order more accurate than evaluating V at either endpoint alone.

Walkers that are born and die.

Equation (15.8) has an immediate stochastic reading. Represent Φ by an ensemble of walkers, points $\boldsymbol{R}$ in the configuration space, whose density is Φ. One time step then consists of

1. **Diffusion.** Displace each walker by a Gaussian random vector of variance $2D\Delta\tau$ in each coordinate. This is G_{diff}, and it is the random walk of Section 12.6.
2. **Branching.** Replace each walker by m copies of itself, where m is an integer with mean G_{branch}. A walker in a region of low potential multiplies; one in a region of high potential dies.

After many steps the density of walkers is Φ, and the trial energy E_T needed to hold the population steady is E_0. There is no wave function anywhere in the algorithm and no basis set: the ground state is represented by where the walkers are.

It works, and then it does not.

For a smooth, bounded potential this bare algorithm is perfectly usable. `dmc.py` applies it to the one-dimensional harmonic oscillator, $V = x^2/2$, with no trial function at all, and the population growth gives

$$E = 0.497322 \pm 0.002792,$$

against the exact $1/2$. Nothing entered but free diffusion and branching.

Now put the quantum dot in. The potential contains $1/r_{12}$, which is unbounded, and the branching weight $\exp[-\Delta\tau(V-E_T)]$ therefore diverges as two walkers approach each other. A single configuration in which the electrons happen to be close acquires an enormous weight, spawns an enormous number of copies, and the entire population descends from it. The variance of the estimator is unbounded and no amount of computer time helps. The same objection applies with more force to the Coulomb attraction in an atom, where the potential is unbounded *below* and the weights diverge faster still.

This is not a technicality to be patched. It is the reason why the algorithm of this section, which is the algorithm as originally proposed, is never the algorithm that is used.

15.3 Importance sampling

The cure is the one that has worked twice already. In Section 12.5 importance sampling replaced a uniform integration density by one shaped like the integrand; in Section 13.6 it replaced a uniform proposal by one drifting along the quantum force. Here it replaces the propagated function itself.

Propagate not Φ but the product

$$\boxed{f(\boldsymbol{R},\tau) = \Psi_T(\boldsymbol{R})\,\Phi(\boldsymbol{R},\tau)} \tag{15.9}$$

where Ψ_T is a fixed, known trial function – from now on called the *guiding function*, because guiding is what it does. Multiplying Eq. (15.1) by Ψ_T and rearranging, using

$$\Psi_T\nabla^2\Phi = \nabla^2 f - 2\nabla\cdot(\Phi\nabla\Psi_T) + \Phi\nabla^2\Psi_T,$$

gives after a short calculation

$$\boxed{\frac{\partial f}{\partial \tau} = D\nabla^2 f - D\nabla\cdot(f\,\boldsymbol{F}(\boldsymbol{R})) - (E_L(\boldsymbol{R}) - E_T)\,f} \tag{15.10}$$

with two quantities that are already familiar:

$$\boldsymbol{F}(\boldsymbol{R}) = \frac{2\nabla\Psi_T(\boldsymbol{R})}{\Psi_T(\boldsymbol{R})}, \qquad E_L(\boldsymbol{R}) = \frac{\hat{H}\Psi_T(\boldsymbol{R})}{\Psi_T(\boldsymbol{R})}. \tag{15.11}$$

The first is the quantum force of Eq. (13.18); the second is the local energy of Eq. (13.4). Both are already coded, already differentiated by hand, and already checked against finite differences in Chapter 13.

Compare Eq. (15.10) with Eq. (15.6). Three things have changed, and all three are improvements.

- A **drift term** has appeared, and it is exactly the Fokker-Planck drift of Eq. (13.16). Walkers are pushed towards regions where Ψ_T is large, which is where the ground state has weight, instead of wandering.
- The **source term contains E_L instead of V**. This is the decisive change. For a trial function satisfying the cusp condition of Section 13.3, E_L is finite everywhere – that is what the cusp condition is *for* – and the branching weights are bounded. In the limit where Ψ_T is the exact ground state, E_L is the constant E_0 and the branching switches off entirely.
- The distribution being sampled is $f = \Psi_T\Phi$ rather than Φ, and it is f, not Φ, that the energy estimator wants.

The mixed estimator.

The last point deserves its own derivation, because it is what makes the method cheap. Once the projection has converged, $\Phi \to \Psi_0$ and $f \to \Psi_T \Psi_0$. Consider

$$\frac{\int d\boldsymbol{R}\, \Psi_T(\boldsymbol{R}) \hat{H} \Psi_0(\boldsymbol{R})}{\int d\boldsymbol{R}\, \Psi_T(\boldsymbol{R}) \Psi_0(\boldsymbol{R})} = \frac{E_0 \int d\boldsymbol{R}\, \Psi_T \Psi_0}{\int d\boldsymbol{R}\, \Psi_T \Psi_0} = E_0, \tag{15.12}$$

using only $\hat{H}\Psi_0 = E_0 \Psi_0$. Now move $\hat{H}$ onto the other factor, which is allowed because $\hat{H}$ is Hermitian, and insert Ψ_T/Ψ_T:

$$\boxed{E_0 = \frac{\int d\boldsymbol{R}\, \Psi_0 \hat{H} \Psi_T}{\int d\boldsymbol{R}\, \Psi_T \Psi_0} = \frac{\int d\boldsymbol{R}\, f(\boldsymbol{R}) E_L(\boldsymbol{R})}{\int d\boldsymbol{R}\, f(\boldsymbol{R})} = \langle E_L \rangle_f} \tag{15.13}$$

This is the *mixed estimator*. It is the same average, of the same function E_L, that variational Monte Carlo computed in Eq. (13.3); the only difference is the distribution it is averaged over, $f = \Psi_T \Psi_0$ here against $|\Psi_T|^2$ there. The code that evaluates the local energy needs no modification whatsoever.

Two remarks are worth making. First, Eq. (15.13) is *exact*: no assumption about the quality of Ψ_T was used, only that it is not orthogonal to Ψ_0. Second, if Ψ_T happened to be the exact ground state then E_L would be the constant E_0 and the estimator would have zero variance. That does not happen, but the closer Ψ_T is, the smaller the variance – the zero-variance principle of Section 13.2 carries over intact.

The mixed estimator is exact only for the energy. For an operator $\hat{A}$ that does not commute with $\hat{H}$ the step taken in Eq. (15.12) fails, and what the simulation returns is $\langle \Psi_T | \hat{A} | \Psi_0 \rangle / \langle \Psi_T | \Psi_0 \rangle$ rather than $\langle \Psi_0 | \hat{A} | \Psi_0 \rangle$. Writing $\Psi_T = \Psi_0 + \delta$ and expanding shows that the error is first order in δ, and that the combination

$$\langle \hat{A} \rangle_{\text{extrap}} = 2\langle \hat{A} \rangle_{\text{mixed}} - \langle \hat{A} \rangle_{\text{VMC}} \tag{15.14}$$

removes that order, leaving an error $\mathscr{O}(\delta^2)$. This *extrapolated estimator* is standard practice for densities, pair correlation functions and anything else that is not the energy, and it is the one place where a diffusion Monte Carlo result still depends visibly on the quality of the guiding function. The unbiased alternative, forward walking, is considerably more expensive.

15.4 The algorithm

The Green's function belonging to Eq. (15.10) is, over a short step, the drifting Gaussian of Eq. (13.21) multiplied by a branching factor built from E_L:

$$G(\boldsymbol{R}' \leftarrow \boldsymbol{R}, \Delta\tau) \simeq \frac{\exp\left[-\dfrac{(\boldsymbol{R}' - \boldsymbol{R} - D\Delta\tau \boldsymbol{F}(\boldsymbol{R}))^2}{4D\Delta\tau}\right]}{(4\pi D\Delta\tau)^{3N/2}} \times \exp\left[-\Delta\tau\left(\frac{E_L(\boldsymbol{R}) + E_L(\boldsymbol{R}')}{2} - E_T\right)\right]. \tag{15.15}$$

One step of the simulation applies the two factors in turn.

Step 1: drift, diffuse, accept or reject.

The first factor is precisely the proposal of Eq. (13.20),

$$\boldsymbol{R}' = \boldsymbol{R} + D\Delta\tau\,\boldsymbol{F}(\boldsymbol{R}) + \boldsymbol{\xi}\sqrt{\Delta\tau},$$

applied one particle at a time exactly as in Section 13.6. Crucially it is followed by the Metropolis-Hastings accept/reject test of Eq. (13.22), with the Green's function ratio of Eq. (13.23).

That accept/reject step was not part of the original algorithm and it is worth being clear about what it does. Without it the walkers sample the drifting Gaussian, which is correct only to first order in $\Delta\tau$. With it they sample $|\Psi_T|^2$ exactly, whatever the time step, because the Metropolis-Hastings test enforces detailed balance with respect to $|\Psi_T|^2$. The remaining time-step error then lives only in the branching factor. Numerically this is worth about an order of magnitude in the time-step bias, which is why every modern implementation includes it.

Step 2: branch.

The second factor is turned into an integer number of copies by the standard device

$$m = \lfloor w + u \rfloor, \qquad w = \exp\left[-\Delta\tau_{\text{eff}}\left(\frac{E_L(\boldsymbol{R}) + E_L(\boldsymbol{R}')}{2} - E_T\right)\right], \qquad u \sim U(0,1), \tag{15.16}$$

which gives $\langle m \rangle = w$ exactly while keeping m an integer. A walker with $m = 0$ is deleted; one with $m = 2$ is duplicated, and the two copies then diffuse independently and decorrelate.

The effective time step.

The $\Delta\tau_{\text{eff}}$ in Eq. (15.16) is not a misprint. A move that the accept/reject test rejects means the walker did not, in fact, diffuse for a full $\Delta\tau$, and reweighting it as though it had is inconsistent. Umrigar, Nightingale and Runge proposed measuring how far the walkers actually moved,

$$\Delta\tau_{\text{eff}} = \Delta\tau\,\frac{\sum_{\text{accepted}}(\Delta\boldsymbol{R})^2}{\sum_{\text{attempted}}(\Delta\boldsymbol{R})^2}, \tag{15.17}$$

and using that in the branching factor. The ratio tends to one as $\Delta\tau \to 0$, so nothing is changed in the limit; at usable time steps it removes a good part of the bias. `dmc.py` implements both, and Section 15.6 compares them.

Step 3: control the population.

Left alone, the number of walkers performs a random walk of its own and will eventually hit zero or exhaust memory. It is stabilised by feeding the population back into the trial energy,

$$E_T = \bar{E} - \frac{\kappa}{\Delta\tau}\ln\frac{N_{\text{walkers}}}{N_{\text{target}}}, \tag{15.18}$$

where $\bar{E}$ is a running average of the mixed estimator. When the population grows, E_T drops and the branching weights fall below one; when it shrinks, the reverse. The feedback strength

κ must be small – `dmc.py` uses 0.05 – because E_T appears in the branching factor, and a violent controller injects its own fluctuations into the simulation.

The price of the feedback is a *population-control bias*: the walkers are no longer independent, since the fate of one affects E_T and hence the fate of all the others. The bias is $\mathscr{O}(1/N_{\text{walkers}})$, and it is checked by varying N_{walkers}, which Section 15.7 does.

Step 4: measure.

The mixed estimator is the average of E_L over the current population, recorded once per step. That series is strongly autocorrelated – more so than in Chapter 14, because branching correlates a walker with its parent as well as with its own past – so the error bar comes from blocking, exactly as in Section 14.9. The growth estimator of Section 15.1 is the average of E_T over the same series, and it is a useful independent check.

Starting the walkers.

The ensemble is initialised by a short variational Monte Carlo run at the same α and β. This is not merely convenient. A good guiding function makes $|\Psi_T|^2$ close to $f = \Psi_T\Psi_0$ already, so the projection has little left to do and the equilibration phase is short – which is exactly the statement that c_0 in Eq. (15.3) is close to one.

What the guiding function is now for. In Chapter 13 Ψ_T *was* the answer, and its deficiencies appeared directly in the energy. Here it does four separate jobs, none of which is to determine the result: it supplies the drift that steers the walkers; it replaces V by E_L in the branching, which is what makes the weights finite; it defines the mixed estimator; and it fixes the nodes, in the fermionic case of Section 15.5. Only the last of these biases the energy. For a nodeless ground state the guiding function affects the error bar and the time-step sensitivity, and nothing else.

15.5 Fermions: the sign problem and the fixed-node approximation

Everything so far assumed that Φ can be read as a density of walkers, which requires it to be positive. For a bosonic ground state that is automatic: the lowest solution of Eq. (15.1) is nodeless. For fermions it is false, and the failure is fundamental rather than technical.

An antisymmetric wave function changes sign, so it has nodes – a surface of one dimension less than the configuration space, on which it vanishes. Meanwhile the imaginary-time propagator projects onto the *lowest* state of the Hamiltonian irrespective of symmetry, and that state is the nodeless bosonic one, which lies below the fermionic ground state. Propagating long enough therefore destroys exactly the state we want:

$$\frac{\text{fermionic signal}}{\text{bosonic noise}} \sim e^{-\left(E_0^{\mathrm{F}}-E_0^{\mathrm{B}}\right)\tau}, \tag{15.19}$$

which decays exponentially in τ and, at fixed statistical accuracy, exponentially in the number of particles. This is the *fermion sign problem*. It is the same obstruction, in different dress, that made the exact diagonalisation of Chapter 5 scale factorially, and no general solution is known.

The practical response is the *fixed-node approximation*. Impose on the projected state the nodes of the guiding function: require

$$\Phi(\boldsymbol{R}) = 0 \quad \text{wherever} \quad \Psi_T(\boldsymbol{R}) = 0, \tag{15.20}$$

so that $f = \Psi_T \Phi \geq 0$ everywhere and the walker interpretation survives. In the algorithm this costs one line: a move that would carry a walker across a node, changing the sign of Ψ_T, is rejected. The nodal surface then partitions configuration space into disjoint pockets, and the simulation solves the Schrödinger equation exactly inside each pocket, with the boundary condition $\Phi = 0$ on the walls.

What this buys is a theorem rather than an approximation of unknown quality. The fixed-node energy is a *variational upper bound* to the exact fermionic ground-state energy,

$$E_{\mathrm{FN}} \;\geq\; E_0^{\mathrm{F}}, \tag{15.21}$$

with equality if and only if the nodes of Ψ_T are the exact ones. The argument is the variational principle of Eq. (13.1) applied within the class of functions vanishing on the imposed surface. The hierarchy is therefore clear: the fixed-node energy is bounded below by the truth and above by the variational energy of the same trial function, and improving the *nodes* – not the amplitude – is the only way to tighten it.

For the system of this chapter none of this applies. Two electrons in a singlet have an antisymmetric spin function, so the spatial part is symmetric, and the trial function

$$\Psi_T(\boldsymbol{r}_1, \boldsymbol{r}_2) = e^{-\alpha\omega(r_1^2+r_2^2)/2}\, e^{r_{12}/(1+\beta r_{12})}$$

of Eq. (13.13) is strictly positive: nodeless. The projection is exact in principle, and the only errors left are the time step, the finite population and the statistics – all three of which can be measured and reduced. This is what makes the two-electron dot the right place to see what diffusion Monte Carlo actually does, uncontaminated by the one approximation that cannot be checked.

15.6 The time step

The short-time Green's function, Eq. (15.15), is exact only as $\Delta\tau \to 0$. Every diffusion Monte Carlo energy therefore carries a systematic error in $\Delta\tau$, and a calculation that quotes one time step has not finished. The procedure is to run at several and extrapolate.

Table 15.1 does this for the quantum dot at the optimised parameters of Chapter 14, $\alpha = 1.00$ and $\beta = 0.40$, with 500 walkers and 4000 steps at each time step.

Three observations, in increasing order of interest.

The acceptance rate stays above 87% even at $\Delta\tau = 0.5$, which is what the drift term buys and is unchanged from Chapter 13. With $\Delta\tau_{\mathrm{eff}}$ switched on, the correction it applies is substantial at the largest time step – the walkers move only 80% of the distance they attempt – and negligible below $\Delta\tau = 0.05$, exactly as it should be.

The correlation time in *steps* grows as the time step shrinks: 1.2 at $\Delta\tau = 0.5$ against 16 at $\Delta\tau = 0.02$. This is not a defect but arithmetic. What decorrelates a walker is imaginary time, and 4000 steps at $\Delta\tau = 0.02$ cover eighty units of τ where the same number of steps at $\Delta\tau = 0.5$ cover two thousand. Halving the time step therefore doubles the number of steps needed for

$\Delta\tau$	energy	acceptance	$\Delta\tau_{\text{eff}}/\Delta\tau$	τ_{corr}
plain $\Delta\tau$ in the branching factor				
0.50	2.999835(50)	87.2%	1.000	1.2
0.30	2.999889(57)	93.6%	1.000	1.7
0.20	2.999959(68)	96.4%	1.000	2.3
0.10	3.000098(89)	98.6%	1.000	4.2
0.05	3.000034(137)	99.5%	1.000	10.1
0.02	3.000142(162)	99.9%	1.000	16.3
extrapolated 3.000090(64), slope -0.00055				
with $\Delta\tau_{\text{eff}}$, Eq. (15.17)				
0.50	2.999927(50)	87.2%	0.805	1.2
0.30	3.000040(58)	93.6%	0.899	1.7
0.20	2.999946(65)	96.4%	0.942	2.3
0.10	3.000155(96)	98.7%	0.978	5.2
0.05	3.000216(115)	99.5%	0.991	7.2
0.02	2.999736(167)	99.9%	0.998	18.3
extrapolated 3.000090(94), slope -0.00031				

Table 15.1 Diffusion Monte Carlo for the two-electron dot at $\alpha = 1.00$, $\beta = 0.40$, with 500 walkers, 4000 steps and 1000 steps of equilibration. Errors on the last digits, from blocking. The extrapolation is a weighted linear fit in $\Delta\tau$. Computed by `dmc.py`.

the same statistics, which is why the error bars in the table grow downwards even though the number of samples is constant. A small time step is not free.

And the bias itself is minute – a few parts in 10^4 across a factor of twenty-five in $\Delta\tau$. That is a statement about the guiding function, not about the method. The time-step error originates in the commutators dropped by the Trotter factorisation, Eq. (15.7), and those vanish when Ψ_T is an eigenfunction. At $\alpha = 1$ the Gaussian part of the trial function is exact, the Jastrow factor satisfies the cusp condition, and the local energy is very nearly constant, so there is almost nothing left for the factorisation to get wrong.

Figure 15.1 plots the two halves of the table against $\Delta\tau$, and it establishes at once the one thing that six-digit numbers in a column do not: how little is going on. The offset printed at the top of the vertical axis is 3, and the entire height of the frame is 7×10^{-4}. Every point in the figure, computed at time steps differing by a factor of twenty-five, lies within 2.7×10^{-4} of the exact answer, and most of them are consistent with it inside their own error bars. Drawn on the scale of the Hartree-Fock error of Chapter 11, 1.6×10^{-1}, the whole of Figure 15.1 would be a single point.

The two straight lines are the weighted fits, and their relationship is the argument for the effective time step. They have different slopes – -5.5×10^{-4} with the plain $\Delta\tau$ against -3.1×10^{-4} with $\Delta\tau_{\text{eff}}$ – and they converge as $\Delta\tau \to 0$, meeting at the common intercept 3.000090. Both statements carry weight. The convergence is a consistency check that the algorithm has not been quietly changed: the ratio $\Delta\tau_{\text{eff}}/\Delta\tau$ tends to one as the time step shrinks, so the two prescriptions are the same algorithm in the limit and are obliged to extrapolate to the same number. That they do so to the last quoted digit, from two sets of points whose slopes differ by almost a factor of two, is evidence that the linear fit is following the systematics and not the noise. The difference in slope is the payoff: at any working time step the residual bias is nearly halved, so that the same bias is tolerated at roughly twice the time step, and since the

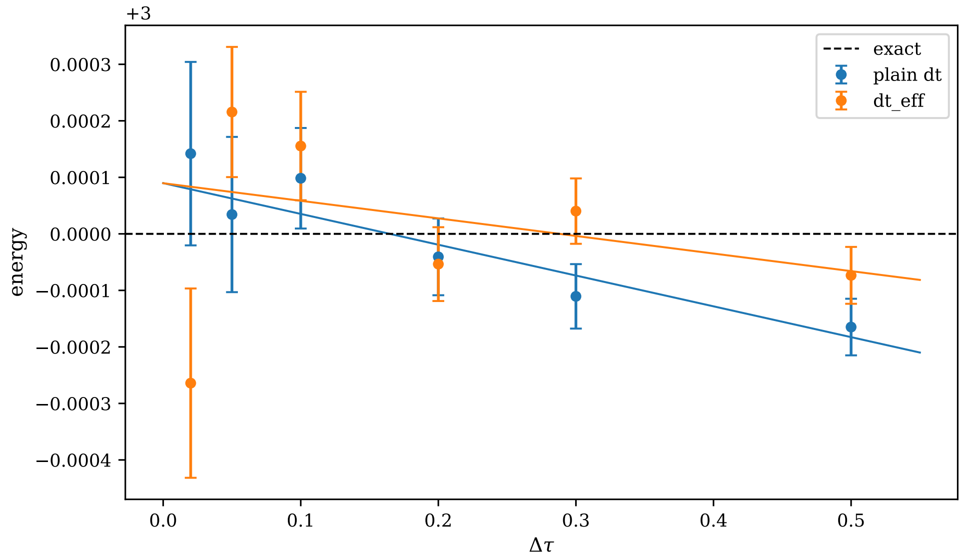

Fig. 15.1 Time-step dependence of the diffusion Monte Carlo energy of the two-electron dot at $\alpha = 1.00$ and $\beta = 0.40$, with 500 walkers and 4000 steps at each time step: the two columns of Table 15.1 plotted against $\Delta\tau$. Circles with blocking error bars are the six time steps from $\Delta\tau = 0.02$ to $\Delta\tau = 0.5$, computed with the plain time step in the branching factor (blue) and with the effective time step of Eq. (15.17) (orange); the straight lines are the corresponding weighted linear fits and the dashed horizontal line is Taut's exact 3. Note the offset printed on the vertical axis: the whole frame spans 7×10^{-4}. The two fits have markedly different slopes, -5.5×10^{-4} and -3.1×10^{-4}, and a common intercept 3.000090.

correlation time in steps grows as $1/\Delta\tau$ that is a factor of two in computer time bought with the one extra line of bookkeeping that Eq. (15.17) requires.

Two further features of the figure repay attention. The first is that both fitted lines cross the exact value at a non-zero time step – near $\Delta\tau \approx 0.16$ for the plain prescription and $\Delta\tau \approx 0.29$ for the effective one. Nothing whatever is special about those time steps. The crossing is merely where the negative slope of the time-step bias happens to cancel the $+9 \times 10^{-5}$ by which the extrapolated result sits above the exact answer, and it is a useful warning to have in a picture: a single diffusion Monte Carlo run that agrees beautifully with a known answer at one particular time step has demonstrated a cancellation, not a convergence. The extrapolation is not decoration. The second feature is the way the error bars grow towards the *left* of the figure, the smallest time steps carrying the largest bars – $\pm 5.0 \times 10^{-5}$ at $\Delta\tau = 0.5$ against $\pm 1.6 \times 10^{-4}$ at $\Delta\tau = 0.02$, a factor of more than three for an identical 4000 steps of sampling, driven by the correlation time growing from 1.2 steps to 16. The whole of the trade-off is thereby visible in one frame: the systematic error is proportional to $\Delta\tau$ and pushes towards the left, the statistical error at fixed effort behaves as $\sqrt{\tau_{\text{corr}}} \sim 1/\sqrt{\Delta\tau}$ and pushes towards the right, and the resolution is not to settle on a compromise but to run at several time steps and let the straight line find the intercept. That intercept, 3.000090(64), carries a smaller error than any individual point in the figure save the two at the largest time steps, which is precisely what a fit through six points ought to deliver.

The next section shows what happens when the guiding function is not so good.

15.7 Results

The chapter's number.

A single run at $\Delta\tau = 0.05$ with 500 walkers gives, in about two seconds,

$$\begin{aligned} \text{mixed estimator} \quad & E = 3.000216 \pm 0.000115, \\ \text{growth estimator} \quad & E = 2.999778 \pm 0.000619, \end{aligned}$$

with a variance of the local energy over the population of 0.0022. The two agree, as they must – one measures the eigenvalue directly, the other the rate at which the population would grow if E_T were frozen – and the mixed estimator has five times the smaller error, which is why it is the one quoted. Extrapolating to zero time step from Table 15.1 gives the result of the chapter,

$$\boxed{E_{\text{DMC}} = 3.000090 \pm 0.000064} \tag{15.22}$$

which sits 1.4 standard errors above the exact 3.

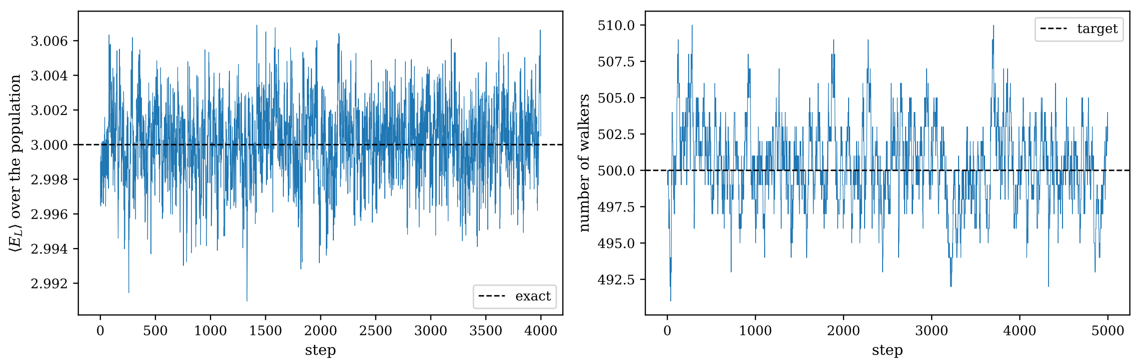

Fig. 15.2 A single importance-sampled diffusion Monte Carlo run for the two-electron dot at $\alpha = 1.00$, $\beta = 0.40$ and $\Delta\tau = 0.05$ with a target population of 500 walkers. Left: the mixed estimator, the average of the local energy over the whole population, recorded once per step over the 4000 measurement steps, against the exact value 3 (dashed). Right: the number of walkers over all 5000 steps, the first 1000 of which are equilibration, against the target of 500 (dashed). Neither series drifts: the energy scatters by some 2×10^{-3} per step about a mean of $3.000216(115)$, and the population is held to within about two per cent of its target by the feedback of Eq. (15.18) with $\kappa = 0.05$.

Figure 15.2 is that run, plotted rather than summarised, and it is worth going through slowly because almost everything the algorithm does is somewhere in it. The left panel is the series whose mean is the mixed estimator and whose blocking analysis produced the error bar. What to notice first is the scale. The whole vertical extent of the panel is 1.6×10^{-2}, more than a hundred times the error bar that four thousand of these points eventually yield, so what one is looking at is overwhelmingly noise with a signal buried in it. The size of that noise is not mysterious. The variance of the local energy over the population is 0.0022, so a single walker reports an energy with a standard deviation of $\sqrt{0.0022} = 0.047$; averaging 500 of them ought to give a per-step scatter of $0.047/\sqrt{500} = 2.1\times10^{-3}$, and that is what the panel shows, with occasional excursions to four or five times it of the kind four thousand samples will always produce. If anything the scatter is a shade larger than the estimate, and that too is expected: two walkers descended from the same parent a few steps ago are not independent reports, so the effective number of walkers behind each point is somewhat below 500. The step from that per-step scatter to the quoted 1.15×10^{-4} is not a factor of $\sqrt{4000}$ – that would give 3.3×10^{-5}, three and a half times too optimistic – but the blocking analysis of Section 14.9, and squaring the discrepancy returns an autocorrelation time of about six steps, which is the τ_{corr} tabulated for this time step in Table 15.1.

The second thing to notice about the left panel is what is missing from it. There is no transient. The series begins at step zero already scattered about its final level, with no visible descent from above and no initial stretch that would have to be discarded. That is the claim of Section 15.4 about starting the walkers, made visible: the ensemble was initialised by a short variational run at the same α and β, so the starting distribution is $|\Psi_T|^2$ while the distribution being converged to is $f = \Psi_T\Psi_0$, and at the optimised parameters those two are so nearly the same function that the projection has almost nothing left to remove. In the language of Eq. (15.3), c_0 is very close to one. The 1000 equilibration steps that precede the panel are fifty units of imaginary time at this time step, against the $1/(E_1 - E_0) \sim 1$ that the gap of the dot demands, and are therefore an extravagance rather than a necessity. The reader who repeats the run at the parameters of Table 15.2 will get a very different-looking panel, an order of magnitude taller and with a settling period one can see.

The right panel is the population, drawn over all 5000 steps so that the equilibration phase is included, and it shows a controller doing its job gently. The excursions are of order ten walkers, two per cent of the target, and they are plainly not corrected within a single step: a population that has drifted up takes some tens of steps to come back down, and that leisureliness is what the small feedback constant $\kappa = 0.05$ means in practice. Over five thousand steps there is no drift at all, which is the statement that E_T has settled at a value for which the branching neither grows nor depletes the ensemble – and that value, by the argument of Section 15.1, is the ground-state energy itself.

This panel is also where one can see why the growth estimator is the poorer of the two. Read Eq. (15.18) backwards. At $\Delta\tau = 0.05$ with $\kappa = 0.05$ the prefactor $\kappa/\Delta\tau$ is exactly one, so a two per cent excursion in the population displaces E_T by $\ln(1.02) \approx 0.02$, ten times the per-step scatter of the mixed estimator in the left panel. The two panels are, in effect, the two estimators side by side, and the right one is visibly the noisier. That the final error bars differ by a factor of five rather than ten is because E_T carries a running average inside it and is thus the smoother of the two series; the ordering, however, is not an accident of this particular run. The mixed estimator measures the eigenvalue directly, configuration by configuration, through Eq. (15.13), whereas the growth estimator infers it from the size of a population that is itself a random variable. The two must agree, and in Figure 15.2 they do; but it is the mixed estimator that is quoted.

A deliberately bad guiding function.

The claim that the guiding function no longer determines the answer ought to be tested rather than asserted. Table 15.2 repeats the calculation at $\alpha = 0.80$ and $\beta = 0.10$, a long way from the variational minimum.

Variationally this trial function is a disaster: 3.3639 against an exact 3, an error of 0.36, worse than Hartree-Fock in a 42-orbital basis. Projected, it gives 3.0011 ± 0.0023 – the right answer. That is the point of the chapter in one line.

The price is visible in the same table. The error bar is twenty-five times larger than in Table 15.1 despite four times the walkers, because the local energy now fluctuates instead of sitting near 3; the correlation times are three times longer; and the slope of the time-step dependence is $+0.249$ against -0.00055, a factor of some five hundred, so that $\Delta\tau = 0.5$ is no longer usable and the extrapolation must be performed from $\Delta\tau \leq 0.1$. A poor guiding function costs computer time and demands a smaller time step. It does not cost accuracy.

$\Delta\tau$	energy	τ_{corr}
0.100	3.025847(966)	8.0
0.050	3.014878(1555)	20.5
0.020	3.002270(2293)	52.8
0.010	3.005206(2222)	50.6
extrapolated	3.001071(2338)	slope +0.249
same Ψ_T, VMC	3.363877(5019)	—
exact	3.000000	—

Table 15.2 The same calculation from a poor guiding function, $\alpha = 0.80$ and $\beta = 0.10$, with 2000 walkers and 3000 steps. The variational energy of this trial function is 0.36 too high; the projected energy is right. Computed by `dmc.py`.

Is it a population-control bias?

The residual in the poor-guide column at a fixed time step is worth one more check, because a systematic error that does not move with $\Delta\tau$ often turns out to move with N_{walkers} instead. Table 15.3 settles it.

walkers	$\alpha = 1.00$, $\beta = 0.40$	$\alpha = 0.80$, $\beta = 0.10$
250	2.999874(131)	3.024045(3611)
500	3.000158(99)	3.021299(2181)
1000	2.999966(62)	3.022742(1509)
2000	2.999934(44)	3.025847(966)

Table 15.3 Dependence on the number of walkers at $\Delta\tau = 0.1$ over 3000 steps. Neither column drifts: the error bars fall as $1/\sqrt{N_{\mathrm{walkers}}}$ and the central values do not move. Computed by `dmc.py`.

Neither central value moves as the population is increased by a factor of eight, while the error bars fall by the expected $\sqrt{8} \approx 2.8$. The population-control bias is therefore below the statistical resolution here, and the residual of Table 15.2 at $\Delta\tau = 0.1$ is time-step error – which the same table then removes by extrapolating.

The dot, by every method in this book.

Collecting the results for two electrons in a two-dimensional parabolic trap at $\hbar\omega = 1$:

method	energy	error	where
Hartree-Fock, 42 orbitals	3.161921	1.6×10^{-1}	Chapter 11
MP2, 42 orbitals	3.027038	2.7×10^{-2}	Chapter 11
CCSD, 42 orbitals	3.013626	1.4×10^{-2}	Chapter 11
VMC, optimised	3.000549(49)	5.5×10^{-4}	Eq. (14.89)
DMC, extrapolated	3.000090(64)	9.0×10^{-5}	Eq. (15.22)
exact (Taut)	3.000000	—	Section 11.7

The two basis-set methods are limited by the basis: a finite sum of oscillator products cannot reproduce the cusp at $r_{12}=0$, and the convergence in the number of orbitals is correspondingly slow. Variational Monte Carlo removes the basis but is limited by the ansatz. Diffusion Monte Carlo removes the ansatz too, and what is left – 9×10^{-5}, consistent with the exact answer at 1.4 standard errors – is statistics, which is to say computer time.

15.8 Summary

Imaginary-time propagation filters out excited states exponentially, Eq. (15.4), and the resulting equation for the ground state is a diffusion equation with a source, Eq. (15.6), which a branching random walk simulates directly. In that bare form the branching weights diverge for any singular potential and the method is unusable for electrons. Importance sampling with a guiding function cures it, Eq. (15.10): the drift becomes the quantum force of Chapter 13, the source becomes the local energy, which the cusp condition keeps finite, and the ground-state energy is the mixed estimator $\langle E_L \rangle_f$ – the same average of the same function as in variational Monte Carlo, taken over a different distribution.

The remaining errors are of four kinds, and it matters that they differ in character. The statistical error falls as $1/\sqrt{N}$ and is measured by blocking. The time-step error is systematic, is removed by extrapolation, and is small when the guiding function is good. The population-control bias falls as $1/N_{\text{walkers}}$ and is checked by varying it. Only the fourth, the fixed-node error, cannot be removed by spending more computer time, and it is absent here because the two-electron singlet is nodeless. For real fermionic systems it is the dominant error, and reducing it means improving the *nodes* of the trial function.

Which points at what is missing. Everything from Chapter 13 onwards has rested on writing down a functional form for Ψ_T and hoping it is close enough – close enough for a useful variational bound in Chapter 13, close enough for a small time-step bias here, and in the fermionic case close enough in its *nodes*, which is a far more delicate requirement than being close in amplitude. The Padé-Jastrow form has two parameters and was chosen by hand from physical arguments about cusps. The alternative is not to choose: to let Ψ_T be a flexible parametrised function – a neural network – and to fix its thousands of parameters with the gradient of Eq. (14.2) and the stochastic reconfiguration of Eq. (14.61), both of which were written for an arbitrary number of parameters precisely so that they would still apply. That is the subject of the chapters that follow.

15.9 Exercises

Exercise 1: the importance-sampled equation

a) Starting from Eq. (15.1) with $\hat{H}=-D\nabla^2+V$, multiply by $\Psi_T(\boldsymbol{R})$ and use the identity

$$\Psi_T\nabla^2\Phi=\nabla^2 f-2\nabla\cdot(\Phi\nabla\Psi_T)+\Phi\nabla^2\Psi_T$$

to derive Eq. (15.10). Identify the step at which the potential V turns into the local energy E_L.

b) Verify that if Ψ_T is the exact ground state then the source term in Eq. (15.10) vanishes identically when $E_T=E_0$. What does the algorithm reduce to in that case, and what is the variance of the energy estimator?

c) Show that the stationary solution of Eq. (15.10) with the source term removed is $f \propto |\Psi_T|^2$, and explain why this is the right check that the drift-diffusion part of the algorithm is the variational Monte Carlo of Chapter 13.
d) Derive the extrapolated estimator, Eq. (15.14), by writing $\Psi_T = \Psi_0 + \delta$, expanding both $\langle \hat{A} \rangle_{\text{mixed}}$ and $\langle \hat{A} \rangle_{\text{VMC}}$ to first order in δ, and eliminating that order.

Exercise 2: branching and the population

a) Show that $m = \lfloor w + u \rfloor$ with u uniform on $[0,1)$ has mean exactly w, and compute its variance. Why is this preferable to rounding w to the nearest integer?
b) `dmc.py` clips m at 3. Explain what this protects against, and argue why for a good guiding function the clip is essentially never reached. Modify the program to count how often it is, at $\alpha = 1.00$, $\beta = 0.40$ and at $\alpha = 0.80$, $\beta = 0.10$.
c) Switch off the population control by setting the feedback κ in Eq. (15.18) to zero, and run the program. Plot the number of walkers against the step number. What kind of stochastic process does it perform, and what happens to it eventually?
d) Raise κ to 1 instead. The population is now held very tightly. Show that the energy is nevertheless *worse*, and explain why in terms of the fluctuations that E_T injects into Eq. (15.16).

Exercise 3: the time step

a) Remove the Metropolis-Hastings accept/reject test from the drift-diffusion step of `dmc.py`, so that every proposed move is taken. Repeat Table 15.1. By how much does the time-step bias grow, and why is the extrapolated value nevertheless still correct?
b) Reproduce Table 15.1 and fit the energies both to $E(\Delta\tau) = E(0) + c_1 \Delta\tau$ and to $E(0) + c_1 \Delta\tau + c_2 \Delta\tau^2$. Compare the two intercepts and their χ^2. Which fit is justified by the data, and what does that say about the order of the Trotter error in Eq. (15.7)?
c) The correlation time in Table 15.1 grows as $\Delta\tau$ shrinks. Show that the product $\tau_{\text{corr}} \Delta\tau$ is roughly constant, and draw the consequence for how the number of steps should be chosen when the time step is reduced.
d) Estimate the total computer time needed to reach a statistical error of 10^{-5} on the extrapolated energy, with the good guiding function and with the poor one. Which is the better investment, and by what factor?

Exercise 4: nodes

a) Explain why the trial function of Eq. (13.13) is nodeless, and why this follows from the two electrons being in a spin singlet rather than being an accident of the functional form.
b) Write down a trial function for the lowest *triplet* state of the same dot, in which the spatial part must be antisymmetric. Where are its nodes? Show that they are fixed by symmetry alone and are therefore exact, so that a fixed-node calculation of this state is exact as well.
c) Implement the triplet in `dmc.py`: multiply the trial function by the antisymmetric factor found in b), reject any move that changes its sign, and run. Compare with the variational energy of the same trial function.
d) For the six-electron dot of Chapter 11 the nodes come from a Slater determinant of oscillator orbitals and are *not* exact. Explain why the fixed-node energy is nevertheless an upper

bound, and why it cannot be improved by increasing the number of walkers, reducing the time step, or running for longer.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter15/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/diffusionmc.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter15/`.

`dmc.py` Importance-sampled diffusion Monte Carlo for the two-electron dot: drift-diffusion with accept/reject, trapezoidal branching, the Umrigar effective time step, population control, the mixed and growth estimators, time-step extrapolation, and a bare branching walk on the one-dimensional oscillator for contrast. Imports `vmc.py` and `vmcoptimise.py`. Runs in about ninety seconds.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 16
Applications to many-body systems

Every method in this book has now been developed, and every one of them has been tested on a system chosen to make it easy. Hartree-Fock, perturbation theory and coupled cluster were applied in Chapter 11 to a quantum dot in a finite oscillator basis, where the two-body matrix elements are analytic. Variational and diffusion Monte Carlo were applied in Chapters 13 to 15 to *two* electrons in the same trap, where the trial function is nodeless and a determinant is not needed at all.

This chapter puts all five on the same problem: six electrons in a two-dimensional parabolic trap. Six is the smallest number for which none of the simplifications survive. The determinant is no longer a formality, the trial function has nodes, the basis-set methods are no longer converged, and there is no exact answer to check against – only the internal consistency of five methods that fail in different ways, and published benchmarks.

Two things have to be built. The first is the Slater-Jastrow trial function and the linear algebra that makes it evaluable, which is the material of Sections 2.11 to 2.14: the determinant ratio from the inverse Slater matrix, the gradient and Laplacian ratios that supply the quantum force and the kinetic energy, and the factorisation $|\boldsymbol{D}| = |\boldsymbol{D}|_{\uparrow}|\boldsymbol{D}|_{\downarrow}$. That material was written in Chapter 2 without a physical application; this is the application.

The second is the spin. Everything in Chapters 13 to 15 sampled positions. Here the assignment of electrons to the two spin blocks is itself a degree of freedom, and we sample it – which turns the Moskowitz-Kalos argument of Section 2.13 from an assertion into a measurement, and which behaves quite differently in the two Monte Carlo methods. That difference is the most interesting thing in the chapter, and it is a trap.

The programs are in `manybodymc.py`, which imports the oscillator orbitals and their derivatives from `slater_update.py`, the blocking analysis from `vmcoptimise.py`, and reproduces `vmc.py` exactly when $N = 2$.

16.1 The system, and what is already known about it

The Hamiltonian is the one of Section 11.1, in the atomic units used there,

$$\hat{H} = \sum_{i=1}^{N} \left(-\frac{1}{2}\nabla_i^2 + \frac{1}{2}\omega^2 r_i^2 \right) + \sum_{i<j} \frac{1}{r_{ij}}, \tag{16.1}$$

with $N = 6$ and $\hbar\omega = 1$ unless stated otherwise. Six electrons fill the two lowest oscillator shells – one orbital in the first, two in the second, each doubly occupied – so the ground state is a closed shell with $S = 0$ and three electrons of each spin. This is what made $N = 6$ one of the magic numbers of Section 11.2.

Chapter 11 has already solved it three ways, in a basis of six oscillator shells:

$$E_{\mathrm{HF}} = 20.72026, \quad E_{\mathrm{MP2}} = 20.30256, \quad E_{\mathrm{CCSD}} = 20.27325. \tag{16.2}$$

None of these is converged. The trouble is the cusp: the exact wave function has a kink at $r_{ij} = 0$, no finite sum of products of oscillator orbitals can reproduce a kink, and the coefficients therefore decay slowly with the number of shells. Lohne *et al.* [6] carried the same CCSD calculation to twenty shells and obtained 20.1737; the 42-orbital number above is a tenth of an atomic unit above it. Whatever else this chapter shows, it is not a fair contest between coupled cluster and Monte Carlo – it is a demonstration that a basis-set calculation must be converged before it is compared with anything.

The benchmark is the fixed-node diffusion Monte Carlo result of the same reference,

$$E_{\mathrm{DMC}} = 20.1597(2), \tag{16.3}$$

against which their best coupled-cluster number, Λ-CCSD(T) in twenty shells, gives 20.1582. Those two agreeing to 1.5×10^{-3} from opposite directions is the strongest statement available about this system, and it is what our own calculation has to reproduce.

16.2 The Slater-Jastrow trial function

The trial function is

$$\boxed{\Psi_T(\boldsymbol{R},\boldsymbol{\sigma}) = |\boldsymbol{D}|_\uparrow |\boldsymbol{D}|_\downarrow \exp\left(\sum_{i<j} \frac{a_{ij}\, r_{ij}}{1+\beta r_{ij}}\right)} \tag{16.4}$$

with $|\boldsymbol{D}|_\uparrow$ and $|\boldsymbol{D}|_\downarrow$ the 3×3 determinants of Eq. (2.62) over the spin-up and spin-down electrons, built from the oscillator orbitals $\phi_{n_x n_y}(\boldsymbol{r};\alpha\omega)$ at the *scaled* frequency $\alpha\omega$. There are exactly two variational parameters, α and β, as in Chapter 13.

For $N = 2$ both determinants are 1×1 and equal to the oscillator ground state, and Eq. (16.4) collapses onto the Padé-Jastrow function of Eq. (13.13). That is worth checking numerically rather than by inspection, and `manybodymc.py` does: for the same configuration and the same parameters, its local energy agrees with that of `vmc.py` to twelve digits.

The cusp coefficients are spin dependent.

The a_{ij} are not variational. They are fixed by requiring E_L to stay finite as two electrons meet, and the requirement is *not* the same for the two spin cases. Repeating the argument of Section 13.3 in d dimensions for a pair whose relative wave function behaves as r^ℓ at contact gives

$$a = \frac{1}{2\ell + d - 1}. \tag{16.5}$$

For an antiparallel pair the spatial function is finite at contact, $\ell = 0$, and $a = 1$ in two dimensions. For a parallel pair the two electrons sit in the same determinant, which vanishes linearly when they coincide, so $\ell = 1$ and

$$a_{\uparrow\uparrow} = a_{\downarrow\downarrow} = \frac{1}{3}, \qquad a_{\uparrow\downarrow} = 1. \tag{16.6}$$

The determinant has already done a third of the work, and the Jastrow factor must not do it again.

Table 16.1 shows what happens if this is forgotten. Two electrons are brought together in a six-electron configuration, once as an antiparallel pair and once as a parallel one, and the third column repeats the parallel case with $a_{\uparrow\uparrow} = 0$ – the mistake of assuming that the determinant handles the parallel case by itself.

r_{ij}	antiparallel, $a = 1$	parallel, $a = \frac{1}{3}$	parallel, $a = 0$
1.000	17.0116	16.7744	17.03
0.300	14.1059	15.1091	17.82
0.100	12.3815	14.4457	23.88
0.030	11.4901	14.3025	47.09
0.010	11.1825	14.2804	113.74
0.003	11.0676	14.2751	347.07

Table 16.1 The local energy as two of six electrons are brought together, with the other four held fixed. With the correct coefficients the local energy converges; with $a_{\uparrow\uparrow} = 0$ it diverges as $1/r_{ij}$, and the variance of the energy is then unbounded. Computed by `manybodymc.py`.

Why this is the interesting parameter. The spin dependence of a_{ij} is what makes the spin assignment a physical variable rather than bookkeeping. If the Jastrow factor were spin blind, exchanging the spins of two electrons would change Ψ_T only through the determinants; with Eq. (16.6) it changes the correlation factor with every other electron as well. That is why the exchange move of Section 16.4 has an acceptance rate worth measuring.

The Jastrow factor, seen.

The correlation factor is usually justified by what it does to the energy. It is more instructive to look at what it does to the electrons. Anticipating the sampling machinery of Section 16.7 by a few pages, we may draw configurations from the square of the trial function twice over, with the same importance-sampled sweep, the same seed and the same 250 walkers: once from $\left| |\boldsymbol{D}|_\uparrow |\boldsymbol{D}|_\downarrow \right|^2$ alone – the bare determinant of oscillator orbitals, obtained by setting every cusp coefficient of Eq. (16.6) to zero – and once from the full Eq. (16.4). Nothing else differs: the same orbitals, the same α, the same trap, the same number of sweeps. From each run we bin the distance of every electron from the centre of the trap and the separation of every one of the fifteen pairs. Figure 16.1 is the result.

Two things happen in Figure 16.1, and they are the same thing seen from two sides. The right-hand panel is the correlation hole itself. The pair distribution of the bare determinant peaks at $r_{ij} \simeq 1.65$ with a height of 0.50; switching the correlation factor on moves the peak out to $r_{ij} \simeq 2.05$ and flattens it to 0.42, and the weight so removed reappears in the tail, where the Jastrow curve is roughly twice the bare one at $r_{ij} = 4$. At small separations the suppression is severe: at $r_{ij} = 0.1$ the bare determinant still puts 0.02 of pair density per unit separation and the correlated function puts about a quarter of that. Neither curve reaches the axis at any finite r_{ij} and neither is flat there – in two dimensions the phase space for a pair at separation r_{ij} vanishes linearly, so $P(r_{ij}) \propto r_{ij}$ near the origin whatever the wave function does, and the interesting comparison is between the two slopes and not with zero.

The size of the effect is fixed by the two parameters and can be read off Eq. (16.4) before any sampling is done. The exponent $a_{ij} r_{ij}/(1 + \beta r_{ij})$ rises from 0 at contact to a saturation

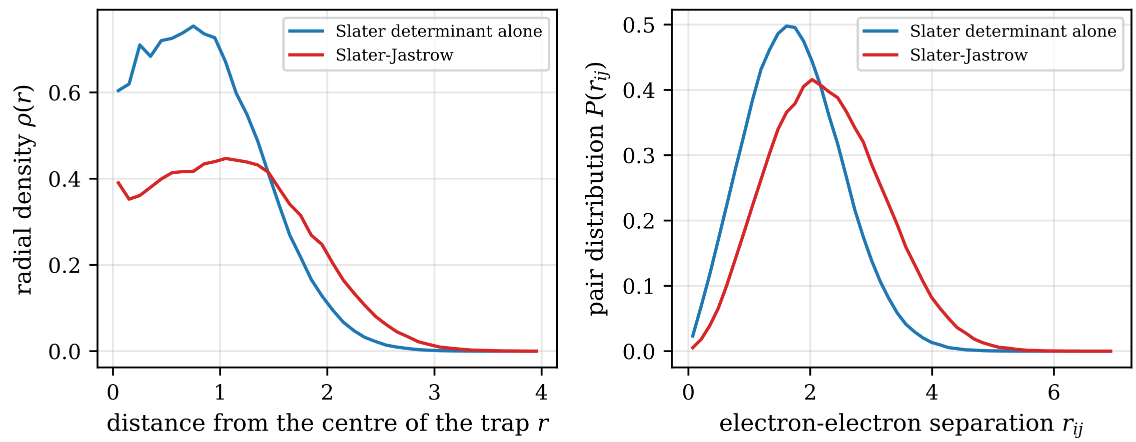

Fig. 16.1 The Jastrow factor digs a correlation hole and swells the dot. Six electrons at $\hbar\omega = 1$, $\alpha = 0.98$ and $\beta = 0.46$, sampled twice with the same importance-sampled sweep and the same seed: from the bare determinant $\left||\boldsymbol{D}|_\uparrow |\boldsymbol{D}|_\downarrow\right|^2$ with all cusp coefficients set to zero (blue) and from the full trial function of Eq. (16.4) (red). Left: the radial density $\rho(r)$, normalised so that $\int \rho\, 2\pi r\, \mathrm{d}r = 6$. Right: the distribution of the fifteen pair separations r_{ij}, normalised to unit area. Both come from 300 sweeps of 250 walkers after 60 sweeps of equilibration, binned every fourth sweep – enough to show the behaviour, and nowhere near enough for a production error bar. Computed by `manybodymc.py`.

value a_{ij}/β at large separation, so the weight $|\Psi_T|^2$ assigns to a pair at contact, relative to the same pair infinitely far apart, is $\exp(-2a_{ij}/\beta)$. At $\beta = 0.46$ that is $e^{-4.35} \approx 1.3 \times 10^{-2}$ for an antiparallel pair and only $e^{-1.45} \approx 0.23$ for a parallel one. The correlation factor suppresses antiparallel contact by nearly two orders of magnitude and parallel contact by a factor of four, and the difference is exactly the factor of three between the two cusp coefficients of Eq. (16.6). The determinant, which already vanishes when two parallel electrons coincide, has dug two thirds of the parallel hole by itself; what remains for the exponential to do is mostly the antiparallel pairs, of which there are nine of the fifteen.

The left-hand panel is the price. Pushing the electrons apart means pushing them up the parabola, and the radial density responds by spreading: the bare determinant peaks at $r \simeq 0.75$ with $\rho \simeq 0.76$, the correlated one at $r \simeq 1.05$ with $\rho \simeq 0.45$, and the two curves cross at $r \simeq 1.45$. That they cross at all is not an accident but a sum rule – both integrate to six electrons, so density removed from the centre has to be found again further out, and it is: beyond $r = 2$ the correlated density is half again as large as the uncorrelated one. The dot is visibly bigger. This is the one-body cost that the two-body gain has to pay for, and the fact that it does pay is the whole content of the correlation energy quoted in Eq. (16.2) and measured again in Section 16.9. The wiggles in the innermost few bins of the blue curve are not physics: the rings there have a small area and a short run puts few counts in them.

16.3 The determinant machinery

A Monte Carlo run makes millions of single-particle moves and needs, at each one, the ratio $\Psi_T(\boldsymbol{R}')/\Psi_T(\boldsymbol{R})$ and then – if the move is accepted – the gradient and Laplacian of $\ln \Psi_T$ for every particle. Section 2.11 showed that all of this follows from the *inverse* Slater matrix. Nothing needs to be rederived here; it needs to be used.

The ratio.

Moving particle i changes one row of one of the two blocks, so by Eq. (2.52)

$$R = \sum_j \phi_j(\boldsymbol{r}_i^{\text{new}})\, d_{ji}^{-1} \tag{16.7}$$

is a single dot product of length $N/2$. The other block is untouched and cancels, which is the saving of Eq. (2.63). If the move is accepted the inverse is repaired by the Sherman-Morrison formulae (2.54). Two hundred consecutive accepted moves in `manybodymc.py` reproduce the brute-force quotient of two determinants to 3×10^{-12}, and $\|\boldsymbol{D}^{-1}\boldsymbol{D}-\mathbf{1}\|$ is still 9×10^{-15} at the end: the inverse does not drift and never has to be rebuilt.

The derivatives.

Equations (2.56) and (2.57) give $\nabla_i|\boldsymbol{D}|/|\boldsymbol{D}|$ and $\nabla_i^2|\boldsymbol{D}|/|\boldsymbol{D}|$ as the same kind of dot product, with ϕ_j replaced by its derivatives. The local energy then follows from

$$\frac{\nabla^2\Psi_T}{\Psi_T} = \frac{\nabla^2|\boldsymbol{D}|}{|\boldsymbol{D}|} + 2\,\frac{\nabla|\boldsymbol{D}|}{|\boldsymbol{D}|}\cdot\nabla U + \nabla^2 U + |\nabla U|^2\,, \tag{16.8}$$

U being the exponent of the Jastrow factor. The determinant and the correlation factor are never differentiated together: only their separate derivatives appear, and each is available in closed form. The quantum force is

$$\boldsymbol{F}_i = 2\left(\frac{\nabla_i|\boldsymbol{D}|}{|\boldsymbol{D}|} + \nabla_i U\right). \tag{16.9}$$

Every one of these is differentiated by hand in `manybodymc.py` and checked against finite differences before anything is sampled – the gradient of $\ln\Psi_T$ to 4×10^{-8} and the local energy to 1×10^{-4}, both at the level of the finite-difference truncation error rather than of the formulae.

A practical note on the scaling. Table 2.3 promises $\mathscr{O}(dN^2)$ instead of $\mathscr{O}(dN^4)$, and that is the reason the algorithm is written this way. For $N=6$, however, the determinants are 3×3, and a vectorised batched factorisation of a thousand such matrices at once is faster in practice than a Python loop carrying Sherman-Morrison updates one walker at a time. `manybodymc.py` therefore contains both: the single-walker class that states the algorithm and verifies it, and a batched evaluator used for the production runs. The asymptotic argument is right and the crossover is simply well above $N=6$.

16.4 Sampling the spins

Section 2.13 argued that for a spin-independent Hamiltonian the full antisymmetric determinant may be replaced by the product $|\boldsymbol{D}|_\uparrow|\boldsymbol{D}|_\downarrow$ at a fixed assignment of electrons to the two blocks, and that the energy is unchanged. Every production code takes that on trust and never touches the assignment again.

We do not have to. Treat the spin configuration $\boldsymbol{\sigma}=(\sigma_1,\ldots,\sigma_N)$, $\sigma_i=\pm1$, as a coordinate alongside the positions, and add a move: pick one spin-up and one spin-down electron at random and exchange their spins. The proposal is symmetric – the reverse move draws the same pair with the same probability – so the ordinary Metropolis test applies,

$$q = \frac{|\Psi_T(\boldsymbol{R},\boldsymbol{\sigma}')|^2}{|\Psi_T(\boldsymbol{R},\boldsymbol{\sigma})|^2}, \tag{16.10}$$

and S_z is conserved exactly. The move is not trivial: the two electrons change determinants, and by Eq. (16.6) their cusp coefficients with every other electron change too. For six electrons at the optimised parameters it is accepted 42% of the time.

What the move buys is discussed in the next section, and it is not what one would expect. What it settles immediately is the Moskowitz-Kalos claim: with the spins sampled the variational energy is 20.2000(18), and with the assignment frozen it is 20.1956(31). The two agree, as promised, and now as a measurement rather than an argument.

16.5 Nodes, and two ways to survive them

Here is the first thing that breaks in going from two electrons to six. The two-electron trial function of Chapter 13 is strictly positive. This one is not: $|\boldsymbol{D}|_\uparrow$ vanishes whenever the three spin-up electrons become collinear, which for the orbitals $(0,0)$, $(1,0)$, $(0,1)$ is exactly the condition that

$$\det\begin{pmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{pmatrix} = 0. \tag{16.11}$$

A walker can approach that surface, and when it does the quantum force $\boldsymbol{F} = 2\nabla \ln \Psi_T$ diverges. The drift step $D\boldsymbol{F}\Delta t$ then hurls the walker across the configuration space, and the local energy it reports on the way is enormous. In a run of three hundred walkers the walker-averaged energy swings between 18.6 and 21.7 and the estimator is worthless – not because of any error in the formulae, all of which have been checked, but because the Langevin proposal of Eq. (13.20) is inappropriate near a singular drift.

There are two cures, and it is instructive that they are independent.

Cut the drift.

Umrigar, Nightingale and Runge [7] replace the drift by

$$\boldsymbol{F} \longrightarrow \boldsymbol{F}\,\frac{-1+\sqrt{1+2a\boldsymbol{F}^2\Delta t}}{a\boldsymbol{F}^2\Delta t}, \tag{16.12}$$

which is the identity for small $\boldsymbol{F}$ and saturates for large, so the drift step never exceeds a fixed size. The essential point is that the *same* modified drift is used in the proposal and in the Green's function of Eq. (13.23). Detailed balance is then untouched and the sampled distribution is still exactly $|\Psi_T|^2$ – which is the lesson of Section 13.6, where a deliberately wrong quantum force was shown to cost efficiency and not correctness. Here that freedom is used on purpose.

Or let the spins move.

A walker that has strayed close to a node has a small $|\Psi_T|$, so by Eq. (16.10) almost any spin exchange is accepted: exchanging one of the three collinear electrons into the other block destroys the near-degeneracy at once. The exchange move is an escape hatch that opens precisely when it is needed, and it requires the walker to move no distance at all.

Table 16.2 runs all four combinations.

drift	spin moves	energy	series range
plain	no	20.2547(498)	18.547 – 21.439
plain	yes	20.2036(18)	20.127 – 20.260
cut	no	20.1956(31)	20.108 – 20.249
cut	yes	20.2000(18)	20.135 – 20.261

Table 16.2 Variational Monte Carlo for six electrons at $\alpha = 1.00$, $\beta = 0.46$, 300 walkers and 400 steps. "Series range" is the smallest and largest walker-averaged energy over the run. Computed by `manybodymc.py`.

The three treated rows agree with each other; the untreated one does not, and its error bar does not warn you. Two further observations are worth making. The spin exchange alone is enough – the second row needs no drift cutoff at all – which says that the pathology really is walkers dwelling near nodes rather than anything in the drift formula. And where both are available the exchange is still the better of the two: its error bar is smaller by a factor between 1.3 and 1.7 across several seeds, which is a factor of two or so in computer time for the same accuracy. Both are used from here on.

16.6 Optimising the trial function

The machinery of Chapter 14 carries over untouched. The energy gradient is still the covariance of Eq. (14.2), and only the parameter derivatives change. That with respect to β is elementary. That with respect to α now runs through the orbitals of the determinant, and is assembled blockwise as

$$\frac{\partial \ln|\boldsymbol{D}|}{\partial \alpha} = \mathrm{Tr}\left(\boldsymbol{D}^{-1}\frac{\partial \boldsymbol{D}}{\partial \alpha}\right), \tag{16.13}$$

which is the same inverse the sampling already maintains. Both derivatives agree with finite differences of $\ln \Psi_T$ to 1×10^{-9}.

Started deliberately far from the minimum, at $\alpha = 0.90$ and $\beta = 0.25$, plain gradient descent with 150 walkers and 120 steps per iteration – short, noisy runs, as Chapter 14 insisted – moves monotonically to $\alpha = 1.00$, $\beta = 0.40$ in fourteen iterations, lowering the energy from 21.008 to 20.221. Refining on a scan puts the minimum at

$$\alpha = 0.98, \qquad \beta = 0.46, \tag{16.14}$$

with the surface flat to within 8×10^{-3} over $\alpha \in [0.96, 1.02]$ and essentially flat in β between 0.44 and 0.48. Figure 16.2 shows the descent itself, once as a sequence of energies and once as a path through the parameter plane.

The left-hand panel of Figure 16.2 makes the point that Chapter 14 insisted on: the energies are estimated from runs far too short to be quoted as results – 150 walkers and 120 steps is a statistical error of some 5×10^{-3} on each point – and yet the sequence they form is monotone to the eye. A descent step is an average of the covariance of Eq. (14.2) over 18 000 samples, and with a learning rate as small as 0.004 the noise in successive gradients cancels faster than the steps accumulate it. Almost all of the improvement is over early: the first six iterations recover 0.72 of the total drop of 0.79, and the last seven together are worth 0.06, which is only a dozen times the error bar on a single point. Continuing beyond fourteen

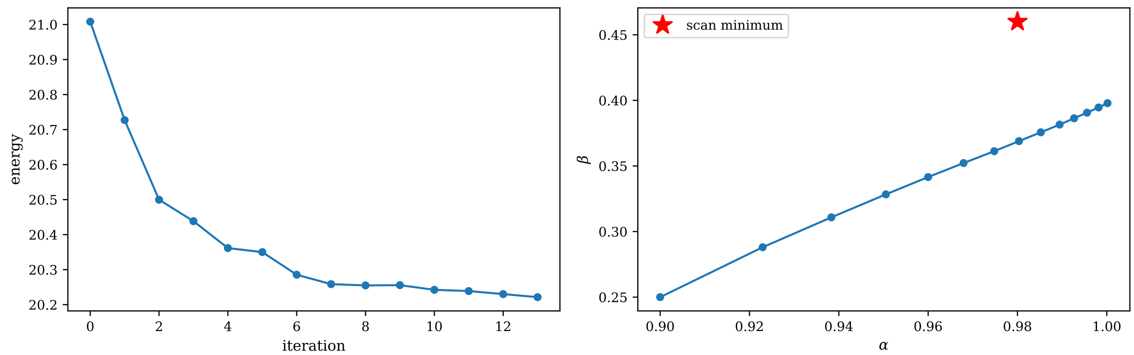

Fig. 16.2 Plain gradient descent on the two parameters of Eq. (16.4) for six electrons at $\hbar\omega = 1$, started deliberately far from the minimum at $\alpha = 0.90$, $\beta = 0.25$ with a learning rate of 0.004, and with the energy and its gradient estimated at each iteration from 150 walkers taking 120 steps. Left: the variational energy against the iteration number, falling monotonically from 21.008 to 20.221 in fourteen iterations. Right: the same fourteen iterations drawn as a path in the (α, β) plane, ending at $\alpha = 1.00$, $\beta \simeq 0.40$; the star marks the minimum $(0.98, 0.46)$ located afterwards by a scan. Computed by `manybodymc.py`.

iterations buys nothing that can be measured at this sample size, which is why the run stops there and a scan takes over.

The right-hand panel says why the scan is needed at all. The path is very nearly a straight line of slope $d\beta/d\alpha \simeq 1.45$, with the points crowding together as it advances: the direction of the gradient hardly changes over the whole descent, and only its magnitude falls. What it does not do is arrive at the star. The descent halts at $\beta \simeq 0.40$, a distance 0.06 short of the scan minimum at $\beta = 0.46$, and stops there not because the gradient in β has changed sign but because the energy has stopped moving by more than the tolerance. That is the flat valley of the paragraph above seen from the inside: over $\beta \in [0.44, 0.48]$ the surface is level to within the statistical error, so the component of the gradient along the valley floor is buried in noise long before the parameter has finished travelling. A steepest-descent method cannot distinguish a flat direction from a converged one, and the honest response is the one taken here – descend until the energy stalls, then map the neighbourhood of the stalling point directly. The residual 0.02 in energy between the end of the descent and the value obtained at the scan minimum in Eq. (16.15) is the size of the mistake one would make by trusting the descent alone.

That α comes out at 1 to within the noise is the physically interesting result. It says that the optimal orbitals in the determinant are the *non-interacting* oscillator eigenfunctions: the Jastrow factor absorbs the effect of the Coulomb interaction on the one-body part so completely that there is nothing left for a scale parameter to do. Harju and co-workers [8] found the same for this system by optimising the single-particle states freely, and noted that it fails for the three-electron dot, where the numbers of spin-up and spin-down electrons differ. Closed shells are forgiving.

16.7 Variational Monte Carlo

At the optimum, six hundred walkers taking nine hundred steps – 540 000 samples, in about half a minute – give

$$E_{\mathrm{VMC}} = 20.19961 \pm 0.00091, \tag{16.15}$$

with an acceptance rate of 97.4%, 41.6% of the spin exchanges accepted, and a correlation time of 2.74 steps. The error is from blocking; the bootstrap gives 0.00065 on the same series, and the two agree on the order of magnitude in the way Section 14.9 led one to expect.

A number of that form conceals the object it was extracted from, which is a time series and not a set of independent draws. Figure 16.3 puts the series back on view. It is a shorter run than the one behind Eq. (16.15) – 120 walkers and 700 sweeps – and it was started with the equilibration deliberately switched off, so that the transient every other run in this chapter discards silently is kept and can be looked at.

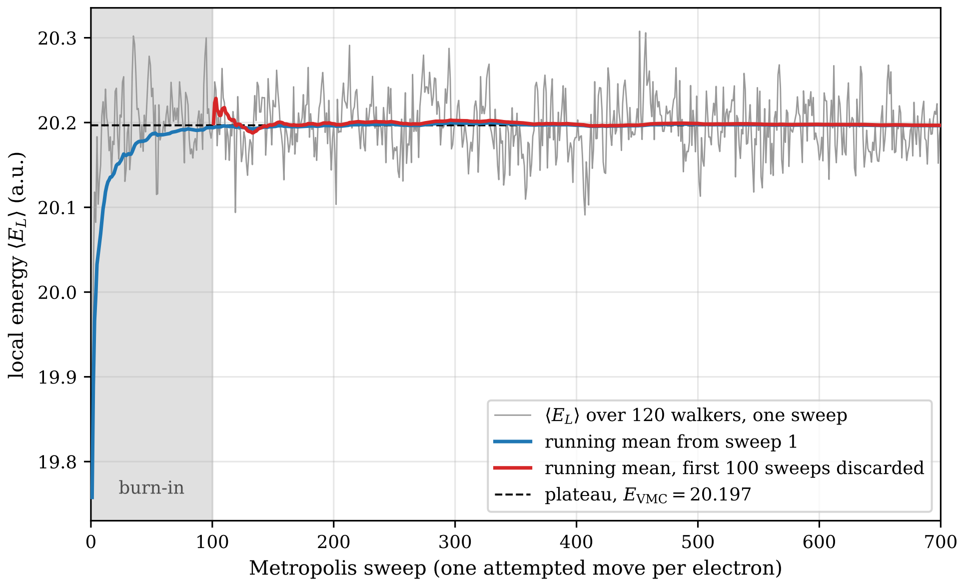

Fig. 16.3 Equilibration and plateau of the variational local energy for six electrons at $\alpha = 0.98$, $\beta = 0.46$, $\hbar\omega = 1$ and a time step of 0.05, with 120 walkers and no equilibration at all. The grey curve is the local energy averaged over the population after each Metropolis sweep, one attempted move per electron; the blue curve is the running mean from the first sweep and the red one the running mean with the first 100 sweeps, shaded, thrown away. The dashed line is the plateau value 20.197 obtained from the retained sweeps by blocking. The walkers start in a Gaussian cloud that is not distributed as $|\Psi_T|^2$, and the first sweep therefore reports an energy half a unit below the plateau. Computed by `manybodymc.py`.

The transient in Figure 16.3 is over in some thirty sweeps. It begins at 19.75, three quarters of the way up in the first ten sweeps, and by sweep 30 the grey curve is fluctuating about the plateau in a way that cannot be told from the rest of the run. The two running means then say what discarding it is worth: the blue one, which keeps the burn-in, is still about one standard error low at sweep 100 and is indistinguishable from the plateau by sweep 400; the red one, which throws the first hundred sweeps away, is within a standard error of its final value almost immediately. That the difference between them is this small is luck rather than virtue. The initial Gaussian cloud happens to be a fair approximation to $|\Psi_T|^2$ for a trap of $\hbar\omega = 1$, and the same picture drawn for a shallow trap, or from a deliberately bad start, would show a transient worth many error bars. The practice of Section 16.8, where the equilibration is measured in imaginary time rather than in steps, exists precisely because that luck cannot be relied on.

The grey curve carries a second lesson, and it is the one that governs every error bar in this chapter. Its scatter about the plateau is roughly ± 0.05, so a naive standard error over the 600 retained sweeps would come out near 2×10^{-3}. That would be wrong, because successive sweeps are not independent: one attempted move per electron changes the configuration very little, and the autocorrelation function of the series gives $\tau = 1 + 2\sum_d \kappa_d \approx 3.9$ sweeps for this run. The true standard error is larger than the naive one by $\sqrt{\tau} \approx 2$, and no amount of staring at the figure would reveal it – an autocorrelated series looks exactly like an independent one drawn with a smaller variance. This is what the blocking analysis of Section 14.9.3 is for, and it is the reason every number in this chapter is quoted through it. The plateau of 20.197 read off this short run is, as it should be, the same as the 20.19961 ± 0.00091 of Eq. (16.15) obtained from six hundred walkers; what the long run buys is not a different answer but an error bar smaller by a factor of five.

This single number already sits 0.074 below the CCSD energy of Eq. (16.2), and 0.040 above the benchmark of Eq. (16.3). Neither comparison is quite what it looks like. The first is a statement about the oscillator basis, not about coupled cluster. The second is the deficiency of a two-parameter trial function, and it is what diffusion Monte Carlo now removes.

16.8 Diffusion Monte Carlo, and a warning

The algorithm is that of Chapter 15, unchanged except that the guiding function is now Eq. (16.4) and that the fixed-node approximation of Section 15.5 is no longer vacuous. The nodal surface is Eq. (16.11), and one thing about it is worth noticing: the Gaussian factors are common to all rows and cancel from the determinant, so the node depends on α only through a uniform dilation of the configuration space. For six electrons the nodal surface is therefore essentially fixed by symmetry, and the fixed-node energy is a well-defined number rather than something that drifts as the trial function is retuned.

The spin move must now be switched off.

This is the trap. The exchange move of Section 16.4 satisfies detailed balance with respect to $|\Psi_T|^2$. That is exactly what variational Monte Carlo samples, and it is exactly *not* what diffusion Monte Carlo samples: the target there is $f = \Psi_T \Phi$ of Eq. (15.9). Applying the exchange at every step inserts a move that pulls the walkers back towards the variational distribution, in competition with the branching that is trying to project them away from it. The result is not a small bias.

spin exchanges	energy
at every step	20.19483(321)
during equilibration only	20.14812(507)
never	20.14980(340)

Table 16.3 Diffusion Monte Carlo at $\Delta\tau = 0.02$ with 200 walkers. Exchanging the spins throughout returns, to within its error bar, the variational energy of Eq. (16.15): the projection has been undone. Computed by `manybodymc.py`.

The remedy is to exchange spins during the variational equilibration, which leaves each walker in a randomised assignment, and then freeze them. The ensemble still averages over assignments – different walkers carry different ones – and nothing interferes with the projection. That is the default in `manybodymc.py`, and Table 16.3 shows it agreeing with the fully frozen calculation.

The time step.

Table 16.4 repeats the extrapolation of Section 15.6, with the equilibration measured in imaginary time rather than in steps – at $\Delta\tau = 0.01$ a fixed number of steps covers a tenth of the imaginary time it does at $\Delta\tau = 0.1$, and a calculation that forgets this reports the time-step error of its own unconverged projection.

$\Delta\tau$	energy	acceptance
0.10	$20.16003(230)$	94.3%
0.05	$20.16028(353)$	97.4%
0.02	$20.15882(327)$	99.2%
0.01	$20.16832(363)$	99.7%
weighted mean	$20.16125(151)$	$\chi^2/\mathrm{dof} = 1.6$

Table 16.4 Diffusion Monte Carlo for six electrons at $\alpha = 0.98$, $\beta = 0.46$, with 250 walkers, 500 steps and an equilibration of five units of imaginary time at every time step. No time-step dependence is resolvable. Computed by `manybodymc.py`.

No time-step dependence survives the error bars, so the weighted mean is the result:

$$\boxed{E_{\mathrm{DMC}} = 20.16125 \pm 0.00151} \tag{16.16}$$

against the published $20.1597(2)$ of Eq. (16.3). The two agree to one standard error.

16.9 The comparison

method	energy	basis	where
Hartree-Fock	20.72026	42 orbitals	Table 11.3
MP2	20.30256	42 orbitals	Table 11.3
CCSD	20.27325	42 orbitals	Table 11.3
CCSD	20.1737	20 shells	Ref. [6]
Λ-CCSD(T)	20.1582	20 shells	Ref. [6]
VMC	$20.19961(91)$	none	Eq. (16.15)
DMC	$20.16125(151)$	none	Eq. (16.16)
DMC, published	$20.1597(2)$	none	Ref. [6]

Table 16.5 Six electrons in a two-dimensional parabolic trap at $\hbar\omega = 1$, in atomic units. The published coupled-cluster results use an effective interaction and a self-consistent Hartree-Fock basis, so they are not the same calculation as the 42-orbital rows, only the same method carried to convergence.

Three things can be read off Table 16.5, and they are different in kind.

The basis is the limitation, not the method. CCSD in 42 orbitals is 0.11 above CCSD in twenty shells, and almost the whole of the apparent gap between coupled cluster and Monte Carlo is that, and not a failure of the cluster expansion. Read the other way: at convergence,

Λ-CCSD(T) and diffusion Monte Carlo – a deterministic expansion in particle-hole excitations and a stochastic projection with a fixed nodal surface, sharing no approximation whatever – agree to 1.5×10^{-3} on a system with no exact solution. That agreement is the real result of this chapter, and neither method could establish it alone.

The Monte Carlo hierarchy behaves as advertised. The variational energy is an upper bound, the diffusion energy improves on it by 0.038, and the diffusion energy is itself an upper bound because the nodes are approximate. Both are basis-free and neither has anything to converge.

The remaining error is nodal. Our 20.16125(151) and the published 20.1597(2) use the same single-determinant nodes and agree; the difference from the truth, whatever it is, is the fixed-node error, and no amount of computer time reaches it. Only better nodes will.

16.10 Dependence on the confinement

Table 11.6 varied the trap frequency and found the correlation energy growing as the trap shallows. Table 16.6 adds the two Monte Carlo columns, at $\alpha = 0.98$ and $\beta = 0.46\sqrt{\omega}$ – the oscillator length is $\omega^{-1/2}$, so that is the only scaling the Jastrow range needs, and a short scan at each frequency confirms it.

$\hbar\omega$	E_{HF}	E_{MP2}	E_{CCSD}	E_{VMC}	E_{DMC}
0.25	7.39179	7.06226	7.03671	7.02274(89)	6.98238(106)
0.50	12.27150	11.89288	11.86345	11.82437(199)	11.78488(116)
1.00	20.72026	20.30256	20.27325	20.19961(91)	20.15960(278)
2.00	35.67233	35.22533	35.19959	35.09581(341)	35.04162(394)
4.00	62.74116	62.27348	62.25276	62.11527(657)	62.05754(472)

Table 16.6 Six electrons at five trap frequencies. The first three columns are the 42-orbital results of Table 11.6; the last two are this chapter, with 600 walkers and $\Delta\tau = 0.05/\omega$. Computed by `manybodymc.py`.

The ordering $E_{\mathrm{HF}} > E_{\mathrm{MP2}} > E_{\mathrm{CCSD}} > E_{\mathrm{VMC}} > E_{\mathrm{DMC}}$ holds at every frequency, and the gap between the coupled-cluster and the Monte Carlo columns widens as ω falls – 0.014 at $\omega = 4$ against 0.054 at $\omega = 0.25$ – because a shallow trap is a more strongly correlated system and the finite basis is worse there.

An honest discrepancy. At $\hbar\omega = 0.5$ the published diffusion Monte Carlo value is 11.7888(2) [6] and the table gives 11.78488(116), which is below it – impossible for two calculations sharing the same nodes, since fixed-node energies are upper bounds. The culprit is population control. Repeating that point with 250 walkers instead of 600 gives 11.77981(144), and with 800 it gives 11.78543(149): the energy climbs as the population grows, which is the standard signature of population-control bias and the standard direction for it. At $\hbar\omega = 1$ the same test finds no movement between 100 and 400 walkers, because the local energy fluctuates less and the branching is gentler. The bias is $\mathcal{O}(1/N_{\mathrm{walkers}})$ and would be removed by running larger populations; it is left visible here because a residual of 4×10^{-3} at the shallowest traps is exactly the kind of thing a table of six-digit numbers is apt to conceal.

16.11 What each method costs

method	scaling	limited by
Hartree-Fock	$\mathcal{O}(n^4)$ per iteration	the basis
MP2	$\mathcal{O}(n^5)$	the basis
CCSD	$\mathcal{O}(n^6)$ per iteration	the basis
Λ-CCSD(T)	$\mathcal{O}(n^7)$	the basis, weakly
VMC	$\mathcal{O}(N^2)$ per sweep, $\mathcal{O}(M^{-1/2})$	the trial function
DMC	as VMC	the nodes of the trial function

Table 16.7 n is the number of single-particle states, N the number of electrons, M the number of Monte Carlo samples. The two families are limited by entirely different things, which is why running both is worth more than running either twice.

The distinction in the last column is the point. A basis-set method has a systematic error that shrinks as n grows and a statistical error of zero; a Monte Carlo method has no basis-set error at all, a statistical error that shrinks as $M^{-1/2}$, and – for diffusion Monte Carlo – one systematic error that no amount of M will touch. When the two agree, as they do at the bottom of Table 16.5, the agreement means something precisely because the failure modes do not overlap.

The costs also scale in the number of *particles* quite differently. Going from six electrons to twenty multiplies the coupled-cluster work by $(n_{20}/n_6)^6$ with a basis that must also grow, while the Monte Carlo work grows as N^2 per sweep with the number of samples fixed by the accuracy wanted. That is why the largest circular dots in the literature are done by diffusion Monte Carlo, and why Ref. [6] needed a renormalised interaction to bring coupled cluster into agreement with it at twenty electrons.

16.12 Summary

Six electrons is where the simplifications end. The trial function needs a determinant, and therefore the inverse Slater matrix of Section 2.11, the derivative ratios of Section 2.12 and the spin factorisation of Section 2.13; the cusp coefficients become spin dependent, 1 for antiparallel pairs and $\frac{1}{3}$ for parallel ones, because the determinant supplies part of the node itself; the wave function acquires a nodal surface, which makes the drift divergent and requires either Umrigar's cutoff or the spin-exchange move to control; and diffusion Monte Carlo acquires a fixed-node error that is now the accuracy limit.

Sampling the spin assignment is exact and useful in variational Monte Carlo, where it confirms the Moskowitz-Kalos factorisation numerically and rescues walkers from nodal regions, and it is wrong in diffusion Monte Carlo, where it preserves $|\Psi_T|^2$ rather than $\Psi_T \Phi$ and undoes the projection almost entirely. The same move, correct in one method and destructive in the other, is a compact reminder of what the two methods actually sample.

The numbers, finally, hang together. A variational calculation with two parameters gives $20.19961(91)$; projecting it gives $20.16125(151)$, in agreement with published diffusion Monte Carlo at one standard error and with converged Λ-CCSD(T) at 1.5×10^{-3}. The 42-orbital coupled cluster of Chapter 11 sits a tenth of an atomic unit above, and that gap is the basis and nothing else.

What no method here can do is improve the nodes. Both Monte Carlo results rest on a single Slater determinant of oscillator orbitals, and Eq. (16.11) – three electrons collinear – is a guess about the ground state that was never optimised and, within this ansatz, cannot be. That is the last of the approximations in this book to remain uncontrolled, and it is the one that motivates what follows: trial functions flexible enough that their nodes are learned rather than assumed.

16.13 Exercises

Exercise 1: the trial function

a) Derive Eq. (16.5) by requiring the local energy of a relative wave function behaving as r^ℓ at contact to stay finite as $r \to 0$ in d dimensions. Evaluate it for $\ell = 0, 1$ in $d = 2$ and $d = 3$ and check the two-dimensional values against Eq. (16.6).
b) Verify by hand that Eq. (16.4) reduces to the two-electron trial function of Eq. (13.13) when $N = 2$, and say which of the two cusp coefficients survives and why.
c) Derive Eq. (16.8) and explain why the determinant and the Jastrow factor never have to be differentiated together.
d) The literature [8] uses two range parameters, one for parallel and one for antiparallel pairs. Implement this in `manybodymc.py` and optimise both with the gradient of Eq. (14.2). How much does the variational energy improve, and does the diffusion Monte Carlo energy change at all?

Exercise 2: determinants in a Monte Carlo loop

a) Starting from Eq. (2.52), show that when the determinant factorises as in Eq. (2.62) a single-particle move leaves one factor untouched, and count the operations saved.
b) Replace the Sherman-Morrison update in `manybodymc.py` by a full recomputation of the inverse at every accepted move and measure the running time for $N = 6$, $N = 12$ and $N = 20$. At which N does the asymptotic argument of Table 2.3 start to pay?
c) Monitor $\|\boldsymbol{D}^{-1}\boldsymbol{D} - \mathbf{1}\|$ over a long run. Under what circumstances does it grow, and what does Section 2.14 suggest doing about it?
d) Show that for six electrons the nodal surface of $|\boldsymbol{D}|_\uparrow$ is the collinearity condition of Eq. (16.11), and that it is unchanged by the scale parameter α. What is the corresponding condition for twelve electrons?

Exercise 3: sampling the spins

a) Show that the exchange proposal of Section 16.4 is symmetric, so that Eq. (16.10) is the correct acceptance ratio, and that it conserves S_z.
b) Set $a_{\uparrow\uparrow} = a_{\uparrow\downarrow}$ in `manybodymc.py` and measure the exchange acceptance rate. Explain the result in terms of what the move can and cannot change.
c) Reproduce Table 16.2. Then record, over a long run, the joint distribution of the local energy and $\ln||\boldsymbol{D}||$, and use it to argue that the walkers with extreme local energies are those close to a node.

d) Reproduce Table 16.3 and explain, from Eq. (15.10), why a move that satisfies detailed balance with respect to $|\Psi_T|^2$ biases a diffusion Monte Carlo calculation. Devise a numerical test that measures the size of the bias as a function of how often the exchange is attempted.

Exercise 4: the comparison

a) Using `quantumdot.py`, extend the Hartree-Fock and CCSD calculations of Chapter 11 to as many shells as your machine allows, and plot the energy against the number of shells. Fit the tail and extrapolate. How close do you get to the 20.1737 of Ref. [6]?
b) Repeat the population-size study of the note in Section 16.10 at $\hbar\omega = 0.25$, and estimate the population-control bias by extrapolating in $1/N_{\text{walkers}}$.
c) Run both Monte Carlo methods for $N = 12$, which fills three shells. Compare with the coupled-cluster results of Ref. [6]. What changes in the cost, and what changes in the quality of the trial function?
d) Table 16.7 claims the two families of method are limited by different things. Design a calculation that would distinguish a basis-set error from a fixed-node error in a system where neither is known independently, and say what it would cost.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter16/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/manybodymc.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter16/`.

`manybodymc.py` Slater–Jastrow variational and diffusion Monte Carlo for N electrons in a two-dimensional dot: Sherman–Morrison ratios and derivative ratios, spin-dependent cusp coefficients, Metropolis sampling of the spin assignment at fixed S_z, the Umrigar drift cutoff, the chapter-14 optimiser, and diffusion Monte Carlo with three spin-move policies. Reduces to `vmc.py` exactly at $N = 2$. Runs in about four minutes.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Part IV
Machine Learning based approaches

Chapter 17
Boltzmann machines and generative models

Every method in this book so far has begun with a guess. Hartree-Fock guesses that the state is a single Slater determinant; coupled cluster guesses an exponential ansatz; the variational Monte Carlo of Chapter 13 guesses a Padé-Jastrow form and two parameters to go with it. The guesses are good ones, built from physics – the cusp condition, antisymmetry, the shape of the potential – and they are the reason the methods work. They are also the reason the methods stop where they do. Chapter 14 ended by observing that its remaining error of 5.5×10^{-4} was not statistical but structural: it was what two parameters could not express.

This chapter takes the opposite approach. Instead of guessing a functional form we write down a family of distributions flexible enough to contain almost anything, and *learn* which member of the family we want. The family is the Boltzmann distribution over a set of visible and hidden variables, the learning is done by gradient descent on a likelihood or on an energy, and the sampling that makes both possible is Gibbs sampling. Nothing in the construction knows any physics at all.

The chapter has three parts.

1. **Generative models in general.** What it means to learn a probability distribution, why energy-based models are the natural family for physicists, and the one formula that drives all of their training: the gradient of the log-likelihood as a difference between an average over the data and an average over the model.
2. **Gibbs sampling.** The second of those averages requires sampling from a distribution known only up to its normalisation. Gibbs sampling does this by drawing one variable at a time from its exact conditional, and we prove that it is correct, show that it is a rejection-free special case of Metropolis-Hastings, and examine when it fails.
3. **Restricted Boltzmann machines and neural quantum states.** The binary-binary and Gaussian-binary machines, with every marginal and conditional derived, and then the step that concerns us most: using the Gaussian-binary marginal as a variational wave function for the quantum dot of Chapters 11 and 13.

The programs are in `rbm.py`, which verifies every closed form derived here against brute-force enumeration and imports the blocking analysis of Chapter 14 unchanged.

17.1 Generative and discriminative models

A *discriminative* model learns a conditional probability $p(y|\boldsymbol{x})$: given an image, which digit is it; given a configuration of spins, is the system above or below its critical temperature. A *generative* model learns the distribution $p(\boldsymbol{x})$ itself, and can therefore be asked not only to classify but to *produce* – to draw a new image, a new spin configuration, a new molecule. The distinction matters for us because a wave function is not a classifier. What we want from a many-particle calculation is $|\Psi(\boldsymbol{x})|^2$, a probability density over configurations, and that is exactly what a generative model provides.

The families in current use are

1. **energy-based models**, of which the Boltzmann machines of this chapter are the canonical example;
2. **variational autoencoders**, treated briefly in Section 17.12;
3. **generative adversarial networks** and **normalizing flows**;
4. **diffusion models**, which generate by reversing a noising process and which are, in their score-based formulation, energy-based models in disguise.

We concentrate on the first because it is the one a physicist already knows. An energy-based model *is* statistical mechanics: a Boltzmann distribution over a set of degrees of freedom, with parameters to be fitted instead of coupling constants to be measured.

There is one structural difference from the supervised learning of a feed-forward network, and it is the difference that organises this whole chapter. A discriminative model computes its gradient by backpropagation, which is exact and cheap. A generative model cannot, because its normalisation constant – the partition function – is a sum over every configuration of the system and is not computable. The gradient must instead be estimated by Monte Carlo, and that is why a chapter on machine learning in this book leans on Chapters 12 and 14 rather than replacing them.

17.2 Bayes' theorem and maximum likelihood

We recall the small amount of probability theory the rest of the chapter needs. Let X and Y be two domains of events, with likelihoods $p(X)$ and $p(Y)$; the probability of a specific event is written $p(X = x_i)$ or simply $p(x_i)$. The union of events obeys

$$p(X \cup Y) = p(X) + p(Y) - p(X \cap Y), \tag{17.1}$$

and the joint probability, or product rule, is

$$p(X,Y) = p(X|Y)\, p(Y) = p(Y|X)\, p(X), \tag{17.2}$$

where $p(X|Y)$ is read as the likelihood of X given Y. For independent events $p(X,Y) = p(X)p(Y)$. Summing the joint over one variable gives the *marginal*,

$$p(X) = \sum_{i=0}^{n-1} p(X, Y = y_i) = \sum_{i=0}^{n-1} p(X|y_i)\, p(y_i), \tag{17.3}$$

and dividing the joint by a marginal gives the *conditional*, provided $p(Y) > 0$,

$$p(X|Y) = \frac{p(X,Y)}{p(Y)}. \tag{17.4}$$

Combining Eqs. (17.2), (17.3) and (17.4) gives Bayes' theorem,

$$\boxed{p(X|Y) = \frac{p(Y|X)\,p(X)}{\sum_{i=0}^{n-1} p(Y|X=x_i)\,p(x_i)}} \tag{17.5}$$

which tells us how the uncertainty in X changes once Y has been observed.

The names of the three factors are worth keeping straight. Evaluated at the observed data Y and regarded as a function of the parameters X, the factor $p(Y|X)$ is the *likelihood function*; it is not normalised as a function of X. The factor $p(X)$ is the *prior*, the left-hand side is the *posterior*, and the denominator is the normalisation, sometimes called the evidence. In this chapter the denominator will be a partition function, and the fact that we cannot compute it will be the central difficulty.

Maximum likelihood.

Suppose the data are independent and identically distributed and come from a model with parameters $\boldsymbol{\Theta}$. *Maximum likelihood estimation* chooses the parameters that make the observed data most probable,

$$\hat{\boldsymbol{\Theta}} = \arg\max_{\boldsymbol{\Theta}} \; p(\boldsymbol{X};\boldsymbol{\Theta}) = \arg\max_{\boldsymbol{\Theta}} \prod_{x_i \in \boldsymbol{X}} p(x_i;\boldsymbol{\Theta}). \tag{17.6}$$

Differentiating a product of many small numbers is both awkward and numerically hopeless – the product underflows long before the data set is exhausted. Because the logarithm is monotonically increasing, maximising $\log p$ gives the same answer as maximising p, and turns the product into a sum. We shall in fact *minimise* the negative log-likelihood, which is the cost function of every model in this chapter.

17.3 Energy-based models

Let $\boldsymbol{X} = \{x_0, \dots, x_{M-1}\}$ be the observed, or *visible*, stochastic variables and $\boldsymbol{H} = \{h_0, \dots, h_{N-1}\}$ a second set of *hidden* variables which we introduce ourselves. Define a joint probability

$$p(x_i, h_j;\boldsymbol{\Theta}) = \frac{f(x_i, h_j;\boldsymbol{\Theta})}{Z(\boldsymbol{\Theta})}, \qquad Z(\boldsymbol{\Theta}) = \sum_{x_i \in \boldsymbol{X}} \sum_{h_j \in \boldsymbol{H}} f(x_i, h_j;\boldsymbol{\Theta}), \tag{17.7}$$

where $f \geq 0$ carries all the parameter dependence and Z is the normalisation. Inspired by statistical mechanics we call Z the *partition function*. The marginals are

$$p(x_i;\boldsymbol{\Theta}) = \frac{\sum_{h_j \in \boldsymbol{H}} f(x_i, h_j;\boldsymbol{\Theta})}{Z(\boldsymbol{\Theta})}, \qquad p(h_j;\boldsymbol{\Theta}) = \frac{\sum_{x_i \in \boldsymbol{X}} f(x_i, h_j;\boldsymbol{\Theta})}{Z(\boldsymbol{\Theta})}. \tag{17.8}$$

A change of notation.

From here on a boldface $\boldsymbol{x}$ denotes a complete *configuration* – one particular assignment of values to all M visible variables at once – and similarly for $\boldsymbol{h}$. The partition function is then a sum over configurations,

$$Z(\boldsymbol{\Theta}) = \sum_{\boldsymbol{x}} \sum_{\boldsymbol{h}} f(\boldsymbol{x}, \boldsymbol{h};\boldsymbol{\Theta}). \tag{17.9}$$

If the variables are binary there are 2^M configurations $\boldsymbol{x}$ and 2^N configurations $\boldsymbol{h}$, and Z is a sum of 2^{M+N} terms. For an image of 28×28 pixels that is 2^{784}, a number with more digits than the universe has atoms. *This is the whole problem.* Everything that follows is an attempt to avoid computing Z.

The optimisation problem.

The hidden variables were introduced for our convenience and we have no interest in their values. What we want to maximise is the probability of the data, obtained by marginalising them away,

$$p(\boldsymbol{X};\boldsymbol{\Theta}) = \prod_{x_i \in \boldsymbol{X}} p(x_i;\boldsymbol{\Theta}) = \frac{1}{Z(\boldsymbol{\Theta})} \prod_{x_i \in \boldsymbol{X}} \left(\sum_{h_j \in \boldsymbol{H}} f(x_i, h_j;\boldsymbol{\Theta}) \right) = \frac{1}{Z(\boldsymbol{\Theta})} \prod_{x_i \in \boldsymbol{X}} f(x_i;\boldsymbol{\Theta}), \tag{17.10}$$

where the last step defines $f(x_i;\boldsymbol{\Theta}) = \sum_{h_j} f(x_i, h_j;\boldsymbol{\Theta})$. Taking the logarithm as promised, the stationarity condition is

$$\nabla_{\boldsymbol{\Theta}} \log p(\boldsymbol{X};\boldsymbol{\Theta}) = \nabla_{\boldsymbol{\Theta}} \left(\sum_{x_i \in \boldsymbol{X}} \log f(x_i;\boldsymbol{\Theta}) \right) - \nabla_{\boldsymbol{\Theta}} \log Z(\boldsymbol{\Theta}) = 0. \tag{17.11}$$

The two terms have names. The first, which involves only the data and a function we have chosen ourselves, is the *positive phase* and is easy. The second is the *negative phase* and is where all the difficulty lives.

The derivative of the partition function.

Although Z itself is out of reach, its logarithmic derivative is not. Write

$$\nabla_{\boldsymbol{\Theta}} \log Z(\boldsymbol{\Theta}) = \frac{\nabla_{\boldsymbol{\Theta}} Z(\boldsymbol{\Theta})}{Z(\boldsymbol{\Theta})} = \frac{\nabla_{\boldsymbol{\Theta}} \sum_{x_i \in \boldsymbol{X}} f(x_i;\boldsymbol{\Theta})}{Z(\boldsymbol{\Theta})}, \tag{17.12}$$

and use $f = \exp(\log f)$, legitimate because $f \geq 0$, so that

$$\nabla_{\boldsymbol{\Theta}} \log Z = \frac{\sum_{x_i \in \boldsymbol{X}} \nabla_{\boldsymbol{\Theta}} \exp\left(\log f(x_i;\boldsymbol{\Theta})\right)}{Z} = \frac{\sum_{x_i \in \boldsymbol{X}} f(x_i;\boldsymbol{\Theta})\, \nabla_{\boldsymbol{\Theta}} \log f(x_i;\boldsymbol{\Theta})}{Z}. \tag{17.13}$$

The ratio $f(x_i;\boldsymbol{\Theta})/Z$ is precisely $p(x_i;\boldsymbol{\Theta})$, so

$$\boxed{\nabla_{\boldsymbol{\Theta}} \log Z(\boldsymbol{\Theta}) = \sum_{x_i \in \boldsymbol{X}} p(x_i;\boldsymbol{\Theta})\, \nabla_{\boldsymbol{\Theta}} \log f(x_i;\boldsymbol{\Theta}) = \langle \nabla_{\boldsymbol{\Theta}} \log f \rangle_{\text{model}}} \tag{17.14}$$

An intractable derivative of an intractable sum has turned into an *expectation value* over the model distribution. Expectation values are what Monte Carlo is for, and the normalisation cancels in the sampling exactly as it does in Chapter 12. The price is that we must sample from $p(\boldsymbol{x};\boldsymbol{\Theta})$, which changes at every optimisation step; that is the subject of Section 17.5.

Introducing the energy.

So far f has been arbitrary. A Boltzmann machine takes

$$p(\boldsymbol{x},\boldsymbol{h};\boldsymbol{\Theta}) = \frac{f(\boldsymbol{x},\boldsymbol{h};\boldsymbol{\Theta})}{Z(\boldsymbol{\Theta})} = \frac{e^{-E(\boldsymbol{x},\boldsymbol{h};\boldsymbol{\Theta})}}{Z(\boldsymbol{\Theta})}, \tag{17.15}$$

with E an *energy function*, and the temperature set to one so that E is dimensionless. For binary variables $x_i, h_j \in \{0,1\}$ the simplest choice that couples the two layers is bilinear,

$$E(\boldsymbol{x},\boldsymbol{h};\boldsymbol{\Theta}) = -\sum_i^M a_i x_i - \sum_j^N b_j h_j - \sum_{i,j}^{M,N} x_i w_{ij} h_j, \tag{17.16}$$

with parameters $\boldsymbol{\Theta} = \{\boldsymbol{a},\boldsymbol{b},\boldsymbol{W}\}$: a visible bias, a hidden bias and a weight matrix. In matrix notation

$$p(\boldsymbol{x},\boldsymbol{h};\boldsymbol{\Theta}) = \frac{\exp\left(\boldsymbol{a}^T\boldsymbol{x} + \boldsymbol{b}^T\boldsymbol{h} + \boldsymbol{x}^T\boldsymbol{W}\boldsymbol{h}\right)}{Z(\boldsymbol{\Theta})}. \tag{17.17}$$

Because the energy is *linear* in every parameter, its derivatives are trivial,

$$\frac{\partial E}{\partial w_{ij}} = -x_i h_j, \qquad \frac{\partial E}{\partial a_i} = -x_i, \qquad \frac{\partial E}{\partial b_j} = -h_j, \tag{17.18}$$

and, as we shall see in Section 17.8, this is the entire reason the training rule is as simple as it is.

17.4 The Kullback-Leibler divergence

Maximum likelihood is one way to say what we want. There is another, which looks different and turns out to be the same statement, and it is worth having both because they suggest different generalisations.

If the goal is to approximate an unknown distribution p by a model q, the natural measure of failure is the *Kullback-Leibler divergence*, also called the relative entropy,

$$\mathrm{KL}(p\|q) = \int_{-\infty}^{\infty} p(\boldsymbol{x}) \log \frac{p(\boldsymbol{x})}{q(\boldsymbol{x})}\, d\boldsymbol{x}. \tag{17.19}$$

It is non-negative, vanishes only when $p = q$ almost everywhere, and is *not* symmetric: $\mathrm{KL}(p\|q) \neq \mathrm{KL}(q\|p)$, so it is not a distance. The asymmetry is not a defect but a choice, and which argument goes first determines whether the model prefers to cover all of the data at the cost of putting mass where there is none, or to fit one mode sharply and ignore the others.

Take p to be the empirical distribution $f(\boldsymbol{x})$ of the training data and q the model $p(\boldsymbol{x}|\boldsymbol{\Theta})$. Then

$$\begin{aligned}
\mathrm{KL}\left(f\|p_{\boldsymbol{\Theta}}\right) &= \int f(\boldsymbol{x}) \log f(\boldsymbol{x})\, d\boldsymbol{x} - \int f(\boldsymbol{x}) \log p(\boldsymbol{x}|\boldsymbol{\Theta})\, d\boldsymbol{x} \\
&= \langle \log f(\boldsymbol{x}) \rangle_{\text{data}} - \langle \log p(\boldsymbol{x}|\boldsymbol{\Theta}) \rangle_{\text{data}} \\
&= \langle \log f(\boldsymbol{x}) \rangle_{\text{data}} + \langle E(\boldsymbol{x}) \rangle_{\text{data}} + \log Z \\
&= \langle \log f(\boldsymbol{x}) \rangle_{\text{data}} + \mathscr{C}_{LL},
\end{aligned} \tag{17.20}$$

where the third line used $\log p = -E - \log Z$ and the last defines the log-likelihood cost function $\mathscr{C}_{LL} = \langle E \rangle_{\text{data}} + \log Z$. The first term is the negative entropy of the data and does not depend on $\boldsymbol{\Theta}$ at all. Hence

$$\boxed{\text{minimising } \mathrm{KL}(f\|p_{\boldsymbol{\Theta}}) \;\equiv\; \text{maximising the log-likelihood}} \tag{17.21}$$

The two criteria differ by a constant, and `rbm.py` exhibits that constant numerically: over a training run the log-likelihood and the KL divergence move in lockstep, separated by exactly the entropy of the data set.

Regularisation.

As in supervised learning, the goal is to do well on data we have not seen. The empirical distribution f is not the true distribution we wish to estimate but only a finite sample of it, so an unregularised model can drive the KL divergence down by memorising the training set. The usual remedies – an L^2 or L^1 penalty on the weights, early stopping, or limiting the number of hidden units – apply here exactly as they do to linear regression, and for the same reason.

17.5 Gibbs sampling

Equation (17.14) reduced training to the evaluation of an expectation value over a distribution known only up to its normalisation. Chapter 12 solved that problem once already, with Markov chains and the Metropolis algorithm: we recall from Eqs. (12.15) and the surrounding discussion that a Markov chain obeying ergodicity and detailed balance converges to a unique stationary distribution, that the Metropolis choice $A_{j\to i} = \min(1, p_i T_{i\to j}/p_j T_{j\to i})$ satisfies detailed balance, and that successive samples are correlated so that error bars require the resampling of Chapter 14. All of that carries over unchanged.

What is new here is a different way of building the chain. Metropolis proposes a move and then decides whether to keep it. Gibbs sampling proposes nothing: it replaces one variable at a time by a fresh draw from its exact conditional distribution, and never rejects.

17.5.1 The update rule

We want to sample from $\pi(x_1,\dots,x_d)$ on a high-dimensional state space, typically known only up to normalisation,

$$\pi(\boldsymbol{x}) = \frac{1}{Z} f(\boldsymbol{x}), \qquad Z = \sum_{\boldsymbol{x}} f(\boldsymbol{x}) \quad \text{or} \quad Z = \int f(\boldsymbol{x})\, d\boldsymbol{x}, \tag{17.22}$$

and in statistical physics $f(\boldsymbol{x}) = e^{-\beta E(\boldsymbol{x})}$. Direct sampling is hard. The *conditional* distributions, however, are often available in closed form:

$$\pi(x_i|\boldsymbol{x}_{-i}) = \frac{\pi(x_i,\boldsymbol{x}_{-i})}{\int \pi(x_i',\boldsymbol{x}_{-i})\, dx_i'}, \tag{17.23}$$

where $\boldsymbol{x}_{-i}$ denotes every coordinate except the ith. Note that the unknown Z cancels between numerator and denominator: *the conditional of a distribution known up to normalisation is known exactly*. That single observation is the whole of Gibbs sampling.

A deterministic *sweep* updates the coordinates in turn,

$$\begin{aligned}
x_1^{(t+1)} &\sim \pi\left(x_1 \,\middle|\, x_2^{(t)}, \ldots, x_d^{(t)}\right), \\
x_2^{(t+1)} &\sim \pi\left(x_2 \,\middle|\, x_1^{(t+1)}, x_3^{(t)}, \ldots, x_d^{(t)}\right), \\
&\vdots \\
x_d^{(t+1)} &\sim \pi\left(x_d \,\middle|\, x_1^{(t+1)}, \ldots, x_{d-1}^{(t+1)}\right),
\end{aligned} \tag{17.24}$$

each draw using the most recently updated values of the other coordinates. This defines a Markov chain. Formally, the update of coordinate i has the transition kernel

$$K_i(\boldsymbol{x}, d\boldsymbol{x}') = \delta_{\boldsymbol{x}_{-i}}(d\boldsymbol{x}'_{-i})\,\pi(dx'_i|\boldsymbol{x}_{-i}), \tag{17.25}$$

which says that every other coordinate is left where it was and coordinate i is resampled exactly. A full sweep is the composition $K = K_d \cdots K_2 K_1$.

17.5.2 Correctness

We must show that π is invariant, $\int \pi(d\boldsymbol{x})K(\boldsymbol{x}, A) = \pi(A)$. Since a composition of kernels each of which leaves π invariant also leaves π invariant, it suffices to prove it for a single coordinate.

Invariance.

Factorise the joint as $\pi(dx_i, d\boldsymbol{x}_{-i}) = \pi(dx_i|\boldsymbol{x}_{-i})\,\pi_{-i}(d\boldsymbol{x}_{-i})$ and integrate the kernel against it,

$$\int \pi(d\boldsymbol{x})\,K_i(\boldsymbol{x}, d\boldsymbol{x}') = \int \pi(dx_i|\boldsymbol{x}_{-i})\,\pi_{-i}(d\boldsymbol{x}_{-i})\,\delta_{\boldsymbol{x}_{-i}}(d\boldsymbol{x}'_{-i})\,\pi(dx'_i|\boldsymbol{x}_{-i}).$$

The integral over x_i acts only on $\pi(dx_i|\boldsymbol{x}_{-i})$ and gives one, because a conditional distribution is normalised. The delta function then sets $\boldsymbol{x}'_{-i} = \boldsymbol{x}_{-i}$, leaving

$$\int \pi(d\boldsymbol{x})\,K_i(\boldsymbol{x}, d\boldsymbol{x}') = \pi(dx'_i|\boldsymbol{x}'_{-i})\,\pi_{-i}(d\boldsymbol{x}'_{-i}) = \pi(d\boldsymbol{x}'), \tag{17.26}$$

which is invariance. The proof is three lines and uses nothing but the definition of a conditional probability.

Detailed balance.

The single-coordinate update satisfies the stronger condition

$$\pi(d\boldsymbol{x})\,K_i(\boldsymbol{x}, d\boldsymbol{x}') = \pi(d\boldsymbol{x}')\,K_i(\boldsymbol{x}', d\boldsymbol{x}). \tag{17.27}$$

If $\boldsymbol{x}_{-i} \neq \boldsymbol{x}'_{-i}$ both sides vanish, since the kernel cannot change the other coordinates. If $\boldsymbol{x}_{-i} = \boldsymbol{x}'_{-i}$, then

$$\pi(\boldsymbol{x})K_i(\boldsymbol{x}, \boldsymbol{x}') = \pi(x_i, \boldsymbol{x}_{-i})\,\pi(x'_i|\boldsymbol{x}_{-i}), \qquad \pi(\boldsymbol{x}')K_i(\boldsymbol{x}', \boldsymbol{x}) = \pi(x'_i, \boldsymbol{x}_{-i})\,\pi(x_i|\boldsymbol{x}_{-i}),$$

and writing $\pi(x_i, \boldsymbol{x}_{-i}) = \pi(x_i|\boldsymbol{x}_{-i})\pi(\boldsymbol{x}_{-i})$ in both makes them manifestly equal. Detailed balance is the equilibrium-flux condition of Chapter 12: the probability current from $\boldsymbol{x}$ to $\boldsymbol{x}'$ is balanced by the reverse current. Each coordinate update is therefore reversible, and so is *random-scan* Gibbs sampling, in which the coordinate to update is chosen at random,

$$K = \sum_{i=1}^{d} p_i K_i, \qquad \pi K = \sum_i p_i \, \pi K_i = \sum_i p_i \pi = \pi. \tag{17.28}$$

The deterministic sweep (17.24) leaves π invariant but is not itself reversible, since it visits the coordinates in a fixed order. That is harmless for correctness and occasionally not harmless in practice, as we shall see.

17.5.3 Gibbs as a rejection-free Metropolis-Hastings method

Metropolis-Hastings proposes $\boldsymbol{x} \to \boldsymbol{x}'$ from a proposal density $q(\boldsymbol{x}'|\boldsymbol{x})$ and accepts with probability

$$\alpha(\boldsymbol{x},\boldsymbol{x}') = \min\left(1, \frac{\pi(\boldsymbol{x}')\, q(\boldsymbol{x}|\boldsymbol{x}')}{\pi(\boldsymbol{x})\, q(\boldsymbol{x}'|\boldsymbol{x})}\right). \tag{17.29}$$

Now make the particular choice of proposing coordinate i from its exact conditional, $q(x_i'|\boldsymbol{x}) = \pi(x_i'|\boldsymbol{x}_{-i})$. For two states differing only in that coordinate,

$$\frac{\pi(\boldsymbol{x}')q(\boldsymbol{x}|\boldsymbol{x}')}{\pi(\boldsymbol{x})q(\boldsymbol{x}'|\boldsymbol{x})} = \frac{\pi(x_i', \boldsymbol{x}_{-i})\, \pi(x_i|\boldsymbol{x}_{-i})}{\pi(x_i, \boldsymbol{x}_{-i})\, \pi(x_i'|\boldsymbol{x}_{-i})} = 1, \tag{17.30}$$

using $\pi(x_i, \boldsymbol{x}_{-i}) = \pi(x_i|\boldsymbol{x}_{-i})\pi(\boldsymbol{x}_{-i})$ twice. The acceptance probability is identically one.

Gibbs sampling is Metropolis-Hastings with the exact conditional as proposal, and therefore never rejects. The two methods are not rivals but the two ends of a spectrum: Metropolis works with any proposal and pays for a poor one in rejections, Gibbs demands a tractable conditional and is rewarded with an acceptance rate of unity.

The practical consequence is worth a number. Table 17.1 runs both on a periodic Ising chain, where the transfer matrix gives the exact energy per spin for comparison. Both reproduce it. But at $\beta = 2$ the Metropolis acceptance has fallen to one per cent – ninety-nine of every hundred proposals are computed and thrown away – while Gibbs keeps every draw.

β	exact e	Gibbs	Metropolis	acceptance
0.25	−0.24492	−0.24565	−0.24554	75.5%
0.50	−0.46212	−0.46266	−0.46204	53.7%
1.00	−0.76396	−0.76679	−0.76577	23.4%
2.00	−0.98782	−0.98998	−0.98905	1.1%

Table 17.1 Energy per spin of a periodic Ising chain of 20 spins, from 20000 sweeps after equilibration. The exact column is the finite-N transfer-matrix result. The last column is the Metropolis acceptance rate; the Gibbs acceptance rate is one by construction. Computed by `rbm.py`.

A cautionary tale about $\Delta E = 0$. Metropolis accepts a move with $\Delta E = 0$ with probability *one*. On the Ising chain such a move is exactly a domain wall hopping by one site, so under a *deterministic* left-to-right sweep every wall moves in lockstep, walls never meet, and the number of walls is conserved: the chain is not ergodic, and the measured energy stays pinned at whatever the random starting configuration had. `rbm.py` reproduces this – an in-order sweep returns $e = 0.00004$ at $\beta = 2$ where the answer is -0.98782 – and the cure is to choose the site at random. Gibbs sampling has no such pathology, because at zero local field it draws the spin from $\mathrm{sig}(0) = 1/2$ rather than flipping it with certainty. Drawing from the conditional and accepting a proposal are not always the same thing.

17.5.4 Examples

A Boltzmann distribution.

For $\pi(\boldsymbol{x}) = e^{-\beta E(\boldsymbol{x})}/Z$ with discrete variables,

$$\pi(x_i|\boldsymbol{x}_{-i}) = \frac{e^{-\beta E(x_i,\boldsymbol{x}_{-i})}}{\sum_{x_i'} e^{-\beta E(x_i',\boldsymbol{x}_{-i})}}, \tag{17.31}$$

a sum over the values of *one* variable rather than over all configurations. Gibbs sampling is thus local equilibrium sampling in a frozen environment: each degree of freedom is brought to thermal equilibrium with its neighbours held fixed, and iterating that brings the whole system to equilibrium.

The Ising model.

With $s_i \in \{-1,+1\}$ and $E(\boldsymbol{s}) = -\sum_{\langle ij\rangle} J_{ij}s_is_j - \sum_i h_is_i$, define the local field at site i,

$$H_i^{\mathrm{loc}} = \sum_{j\neq i} J_{ij}s_j + h_i. \tag{17.32}$$

Then $\pi(s_i|\boldsymbol{s}_{-i}) \propto e^{\beta s_i H_i^{\mathrm{loc}}}$, and normalising over the two values of s_i,

$$\mathrm{Prob}(s_i = +1|\boldsymbol{s}_{-i}) = \frac{1}{1+e^{-2\beta H_i^{\mathrm{loc}}}}, \qquad \mathrm{Prob}(s_i = -1|\boldsymbol{s}_{-i}) = \frac{1}{1+e^{+2\beta H_i^{\mathrm{loc}}}}. \tag{17.33}$$

The logistic function has appeared for the first time, and not by choice: it is what the conditional probability of a two-valued variable *is*, whenever the energy is linear in that variable. We shall meet it again in every RBM in this chapter, and it is the same function that machine learning calls the sigmoid activation.

A bivariate Gaussian.

For

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}\right), \tag{17.34}$$

the conditionals are Gaussian too,

$$X|Y = y \sim \mathcal{N}\left(\mu_X + \rho\frac{\sigma_X}{\sigma_Y}(y-\mu_Y),\ \sigma_X^2(1-\rho^2)\right), \tag{17.35}$$

and symmetrically for $Y|X$. Gibbs sampling therefore alternates exact Gaussian draws, and `rbm.py` checks that two hundred thousand such alternating draws reproduce the means, the variances and the correlation of a direct Cholesky sample of the joint to three decimal places; the comparison is drawn in Figure 17.1. The general result behind Eq. (17.35), for a partitioned multivariate Gaussian with blocks $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$, is

$$p(\boldsymbol{f}|\boldsymbol{g}) = \mathcal{N}\left(\boldsymbol{f};\boldsymbol{a}+\boldsymbol{C}\boldsymbol{B}^{-1}(\boldsymbol{g}-\boldsymbol{b}),\, \boldsymbol{A}-\boldsymbol{C}\boldsymbol{B}^{-1}\boldsymbol{C}^{T}\right). \tag{17.36}$$

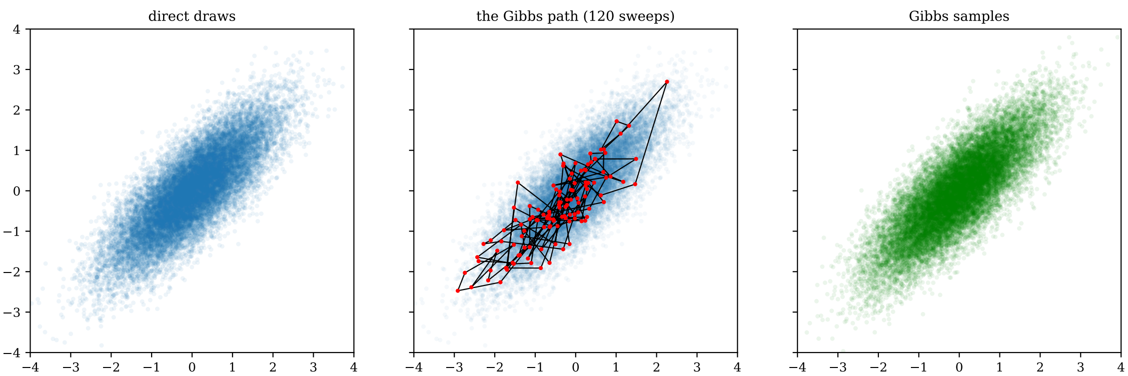

Fig. 17.1 Gibbs sampling of the bivariate Gaussian of Eq. (17.34) with $\mu_X = \mu_Y = 0$, $\sigma_X = \sigma_Y = 1$ and $\rho = 0.8$. Left: twenty thousand independent draws obtained directly from a Cholesky factorisation of the covariance matrix. Middle: the same cloud shown faintly, with the first 120 sweeps of the Gibbs chain drawn on top, one red dot per completed sweep and straight lines joining consecutive sweeps. Right: twenty thousand consecutive Gibbs samples, none of them independent. All three panels share the window $[-4,4]^2$ and an equal aspect ratio, so the clouds may be compared directly. Computed by `rbm.py`.

The first and third panels of Figure 17.1 are indistinguishable by eye, and that is the entire content of the correctness proof of Section 17.5.2 made visible: a chain that never looks at the joint density, only at two one-dimensional conditionals, nevertheless fills the two-dimensional ridge with exactly the right shape, the right orientation and the right density. The numbers behind the picture are that the Gibbs run returns $\langle x\rangle = 0.0014$, $\mathrm{Var}(x) = 0.9977$ and a correlation of 0.8005, against 0.0001, 0.9998 and 0.7994 from the direct draws and 0, 1 and 0.8 exactly. The chain was started at $(3,-3)$, a point the target gives essentially no weight to, with no burn-in discarded; the very first conditional draw of x has mean $\rho y = -2.4$ and the first recorded sweep is already at $(-2.40,-1.74)$, on the ridge. Equilibration here costs one sweep, because the conditional is exact and there is nothing to reject.

The middle panel is where the price is paid. The path is confined to the ridge and wanders up and down it, and although 120 sweeps do carry it from $x = -2.9$ to $x = +2.3$, the excursion happens in many short steps rather than a few long ones. The reason is geometric: a draw of x at fixed y can move the chain only along a horizontal line, and on a ridge of correlation ρ the conditional standard deviation $\sigma_X\sqrt{1-\rho^2} = 0.6$ is much smaller than the unconditional $\sigma_X = 1$. Each coordinate is therefore pinned by the other, and the chain diffuses along the ridge instead of crossing it. At $\rho = 0.8$ this is an inconvenience; at $\rho \to 1$ it is fatal, and Section 17.5.5 measures how fatal.

17.5.5 Convergence, mixing and blocked updates

Invariance guarantees that π is a fixed point, not that we reach it. For convergence we need irreducibility, aperiodicity and positive recurrence, under which

$$\lim_{t\to\infty} \text{Prob}\left(\boldsymbol{X}^{(t)} \in A\right) = \pi(A), \tag{17.37}$$

and the ergodic theorem then lets us replace ensemble averages by time averages, $\frac{1}{N}\sum_t g(\boldsymbol{X}^{(t)}) \to \langle g \rangle_\pi$ almost surely. This is the same statement that justified every Monte Carlo estimate in Chapters 12 to 14.

The samples are correlated, and the analysis is the one already developed at length in Section 14.9.1: with $\rho_g(k)$ the normalised autocorrelation of an observable g, the variance of the mean is inflated by the autocorrelation time and the effective sample size is $N_{\text{eff}} \approx N/\tau$. A useful notion of how long equilibration takes is the *mixing time*

$$t_{\text{mix}}(\varepsilon) = \min\left\{t : \sup_{\boldsymbol{x}} \left\|K^t(\boldsymbol{x},\cdot) - \pi\right\|_{\text{TV}} \le \varepsilon\right\}, \qquad \|\mu - \nu\|_{\text{TV}} = \sup_A |\mu(A) - \nu(A)|. \tag{17.38}$$

Gibbs sampling mixes slowly when the variables are strongly correlated, when the target is multimodal, or near a critical point, where local updates decorrelate the configuration only over diverging times – critical slowing down. The mechanism is easy to see on the bivariate Gaussian: the chain moves parallel to the axes, so on a narrow diagonal ridge it can only creep along in small steps. `rbm.py` measures the correlation time of the x coordinate as a function of ρ and finds $\tau = 1.00$, 1.67, 4.66, 20.32 and 84.45 for $\rho = 0$, 0.5, 0.8, 0.95 and 0.99: at $\rho = 0.99$ a hundred Gibbs sweeps are worth about one independent sample. Figure 17.2 plots them.

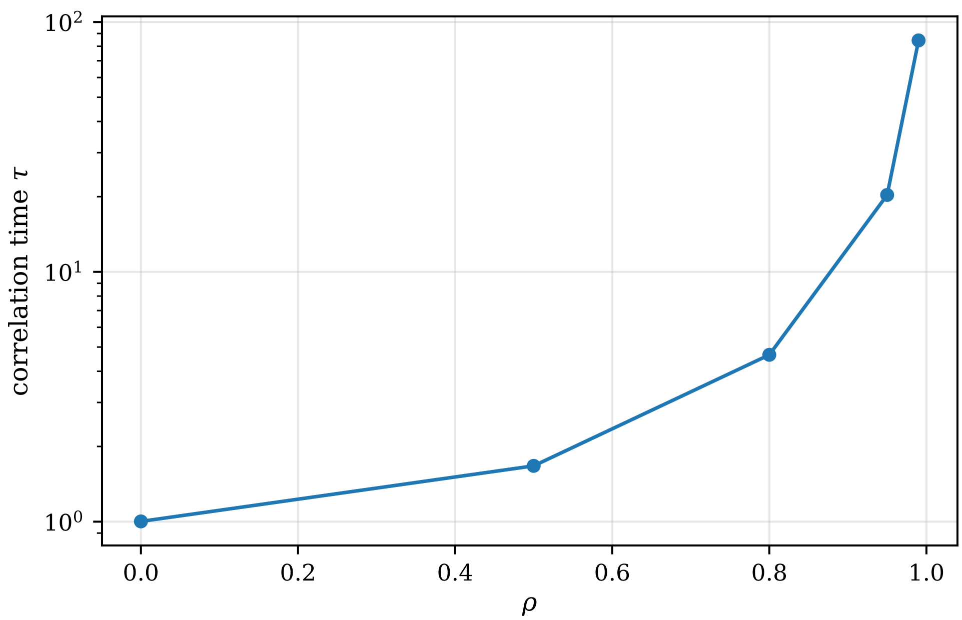

Fig. 17.2 Integrated autocorrelation time τ of the x coordinate of a Gibbs chain on the bivariate Gaussian of Eq. (17.34), as a function of the correlation ρ between the two coordinates, from 40 000 sweeps at each value and with the autocorrelation sum truncated at its first negative term. The vertical scale is logarithmic. The curve rises from $\tau = 1.00$ at $\rho = 0$, where a sweep produces an independent sample, to $\tau = 84$ at $\rho = 0.99$, and the rise steepens rather than flattening as $\rho \to 1$. Computed by `rbm.py`.

The curvature in Figure 17.2 is worth reading carefully, because the vertical axis is already logarithmic and the curve still bends upwards: the growth of τ with ρ is faster than exponential in ρ. For this particular target the answer is known exactly. Sampling x given y and then y given x makes the sequence $x^{(t)}$ an autoregressive process of order one with lag-one autocorrelation ρ^2, so that $\rho_x(k) = \rho^{2k}$ and

$$\tau = 1 + 2\sum_{k=1}^{\infty} \rho^{2k} = \frac{1+\rho^2}{1-\rho^2}, \tag{17.39}$$

which gives 1, 1.67, 4.56, 19.5 and 99.5 at the five values of ρ plotted. The first four agree with the measurements to the accuracy one would expect of a finite chain; the fifth does not, and the discrepancy is instructive rather than alarming. The estimator truncates the sum over lags at the first negative sample autocorrelation, and when τ is large the individual $\rho_x(k)$ are small, noisy and go negative by accident long before the true autocorrelation has decayed. The measured 84.45 is therefore an underestimate of the true 99.5, and the bias is worst exactly where the number matters most. This is the same difficulty that motivated the blocking analysis of Chapter 14, which estimates the inflation of the variance without ever forming $\rho_g(k)$ term by term, and it is the reason blocking rather than a truncated autocorrelation sum is used everywhere else in this book.

The practical statement hidden in Eq. (17.39) is that $\tau \simeq 2/(1-\rho^2) \simeq 1/(1-\rho)$ as $\rho \to 1$: the cost of a single independent sample diverges as the target degenerates onto a line. Nothing about this is special to Gaussians. Any strongly coupled pair of variables – two spins in a domain, two parameters in a stiff fit, the real and imaginary parts of an amplitude near a resonance – produces the same picture, and the cure is always the same one.

Blocked Gibbs.

The remedy is to update a whole *block* of variables at once, drawing from the joint conditional of the block given the rest,

$$\boldsymbol{x}_B^{(t+1)} \sim \pi\left(\boldsymbol{x}_B \,\middle|\, \boldsymbol{x}_{-B}^{(t)}\right). \tag{17.40}$$

Everything proved above goes through unchanged – a block is just a coordinate with more components – while strongly coupled variables now move together. The examples are block spin updates, latent Gaussian blocks in Bayesian models, and, most important for this chapter, the visible and hidden *layers* of a Boltzmann machine. It is precisely the bipartite structure of the restricted machine that makes each layer a legitimate block, and that is the subject of the next section.

Graphical models.

Gibbs sampling is not confined to physics. A Bayesian network is a directed acyclic graph with the factorisation $p(x_1,\ldots,x_d) = \prod_i p(x_i|\mathrm{pa}(x_i))$ over parents, and conditioning on observed evidence generally makes the posterior impossible to sample directly. The conditional of a single node, however, depends only on its *Markov blanket* – its parents, its children and its children's other parents –

$$p(x_i|\boldsymbol{x}_{-i}) = p\left(x_i|\mathrm{MB}(x_i)\right) \propto p\left(x_i|\mathrm{pa}(x_i)\right) \prod_{y\in\mathrm{ch}(x_i)} p\left(y|\mathrm{pa}(y)\right), \tag{17.41}$$

so only local factors are needed. The analogy with a spin system is exact: in both cases a variable interacts only with its neighbourhood, and repeated local conditional updates reconstruct the global equilibrium distribution.

17.6 Boltzmann machines

A Boltzmann machine is an undirected probabilistic graphical model with stochastic units, continuous or discrete. It can equally be read as a stochastic recurrent neural network in which the state of each unit depends on the units it is connected to, the weights measuring the strength of the interaction. A physicist will recognise it immediately as a generalised Ising model with the couplings promoted from constants of nature to parameters to be fitted.

The units are divided into a set of *visible* units $\boldsymbol{x}$, which carry the data, and a set of *hidden* units $\boldsymbol{h}$, whose values are never observed. In full generality the energy contains couplings within each layer as well as between them,

$$E(\boldsymbol{x},\boldsymbol{h}) = -\sum_i a_i x_i - \sum_j b_j h_j - \sum_{i,j} w_{ij} x_i h_j - \sum_{i<i'} u_{ii'} x_i x_{i'} - \sum_{j<j'} v_{jj'} h_j h_{j'}, \tag{17.42}$$

with the joint distribution $p(\boldsymbol{x},\boldsymbol{h}) = e^{-E(\boldsymbol{x},\boldsymbol{h})}/Z$. Machines of this kind are stackable – the hidden layer of one can serve as the visible layer of the next, which is how deep belief networks are built – and they are universal enough to represent any distribution over their visible units given sufficiently many hidden ones.

They are also very hard to train. With intra-layer couplings present, no group of variables is conditionally independent of any other, so the conditional distributions needed for Gibbs sampling do not factorise and every unit must be updated separately, in a chain that mixes slowly.

The restriction.

The *restricted* Boltzmann machine simply deletes the lateral couplings, $u_{ii'} = v_{jj'} = 0$, leaving a bipartite graph in which visible units connect only to hidden units and vice versa,

$$E(\boldsymbol{x},\boldsymbol{h}) = -\sum_i a_i x_i - \sum_j b_j h_j - \sum_{i,j} w_{ij} x_i h_j. \tag{17.43}$$

The consequence is the one property on which everything in the rest of this chapter depends,

$$\boxed{p(\boldsymbol{h}|\boldsymbol{x}) = \prod_j p(h_j|\boldsymbol{x}), \qquad p(\boldsymbol{x}|\boldsymbol{h}) = \prod_i p(x_i|\boldsymbol{h})} \tag{17.44}$$

Given the visible layer the hidden units are conditionally independent of one another, and conversely. A whole layer can therefore be resampled in one step, in parallel, and block Gibbs sampling alternates

$$\boldsymbol{x} \longrightarrow \boldsymbol{h} \longrightarrow \boldsymbol{x} \longrightarrow \boldsymbol{h} \longrightarrow \cdots \tag{17.45}$$

This is why restricted machines are used and unrestricted ones are not. `rbm.py` verifies Eq. (17.44) directly, comparing $p(\boldsymbol{x})\prod_j p(h_j|\boldsymbol{x})$ with the exact joint over all $2^M \times 2^N$ configurations of a machine with $M = 6$ and $N = 4$, and finds agreement to 1.0×10^{-17}.

The layers and the parameters.

- $\boldsymbol{x}$, the **visible layer**: a vector of M nodes. This is both what the machine is shown during training and what we ask it to reconstruct – the pixels of an image, the spins of an Ising configuration, or, in Section 17.10, the coordinates of the particles.
- $\boldsymbol{h}$, the **hidden** or **latent layer**: a vector of N nodes, also called feature detectors. Their purpose is to supply expressive power: complicated correlations among the visible variables are encoded by letting each of them interact *simply* with a set of extra variables, which are then integrated out.
- $\boldsymbol{a}$ (M components), $\boldsymbol{b}$ (N components) and $\boldsymbol{W}$ ($M \times N$): the visible bias, the hidden bias and the weights, the parameters to be learned.

The lower the energy of a configuration $(\boldsymbol{x}, \boldsymbol{h})$, the more probable it is; training adjusts $\boldsymbol{\Theta}$ so as to make the energy landscape fit the problem. The variants of the model differ only in what values the units are allowed to take and, correspondingly, in the form of E. We treat the two that matter here: binary visible and binary hidden units, and continuous Gaussian visible units with binary hidden units.

17.7 The binary-binary restricted Boltzmann machine

The original RBM had binary units in both layers, and was shown to be a universal approximator of discrete distributions; adding hidden units strictly increases its modelling power. The common choice of values is 0 and 1, which we adopt, although ± 1 is sometimes more natural in physics applications. The energy is

$$E_{BB}(\boldsymbol{x}, \boldsymbol{h}; \boldsymbol{\Theta}) = -\sum_{i}^{M} a_i x_i - \sum_{j}^{N} b_j h_j - \sum_{i,j}^{M,N} x_i w_{ij} h_j, \tag{17.46}$$

so that the joint distribution is

$$p_{BB}(\boldsymbol{x}, \boldsymbol{h}; \boldsymbol{\Theta}) = \frac{1}{Z_{BB}} e^{\sum_i^M a_i x_i + \sum_j^N b_j h_j + \sum_{ij}^{M,N} x_i w_{ij} h_j} = \frac{1}{Z_{BB}} e^{\boldsymbol{a}^T \boldsymbol{x} + \boldsymbol{b}^T \boldsymbol{h} + \boldsymbol{x}^T \boldsymbol{W} \boldsymbol{h}}, \tag{17.47}$$

with

$$Z_{BB}(\boldsymbol{\Theta}) = \sum_{\boldsymbol{x}, \boldsymbol{h}} e^{\boldsymbol{a}^T \boldsymbol{x} + \boldsymbol{b}^T \boldsymbol{h} + \boldsymbol{x}^T \boldsymbol{W} \boldsymbol{h}}. \tag{17.48}$$

17.7.1 Marginal probabilities

To find the probability of a configuration of the visible units alone we must sum the joint over every configuration of the hidden layer. This looks like a sum of 2^N terms and is in fact a product of N sums of two terms each, which is the first payoff of the restriction:

$$
\begin{aligned}
p_{BB}(\boldsymbol{x};\boldsymbol{\Theta}) &= \sum_{\boldsymbol{h}} p_{BB}(\boldsymbol{x},\boldsymbol{h};\boldsymbol{\Theta}) = \frac{1}{Z_{BB}} \sum_{\boldsymbol{h}} e^{\boldsymbol{a}^T\boldsymbol{x}+\boldsymbol{b}^T\boldsymbol{h}+\boldsymbol{x}^T\boldsymbol{W}\boldsymbol{h}} \\
&= \frac{1}{Z_{BB}} e^{\boldsymbol{a}^T\boldsymbol{x}} \sum_{\boldsymbol{h}} e^{\sum_j^N \left(b_j+\boldsymbol{x}^T\boldsymbol{w}_{*j}\right)h_j} \\
&= \frac{1}{Z_{BB}} e^{\boldsymbol{a}^T\boldsymbol{x}} \sum_{\boldsymbol{h}} \prod_j^N e^{\left(b_j+\boldsymbol{x}^T\boldsymbol{w}_{*j}\right)h_j} \\
&= \frac{1}{Z_{BB}} e^{\boldsymbol{a}^T\boldsymbol{x}} \left(\sum_{h_1} e^{(b_1+\boldsymbol{x}^T\boldsymbol{w}_{*1})h_1}\right)\left(\sum_{h_2} e^{(b_2+\boldsymbol{x}^T\boldsymbol{w}_{*2})h_2}\right)\cdots\left(\sum_{h_N} e^{(b_N+\boldsymbol{x}^T\boldsymbol{w}_{*N})h_N}\right) \\
&= \frac{1}{Z_{BB}} e^{\boldsymbol{a}^T\boldsymbol{x}} \prod_j^N \sum_{h_j} e^{\left(b_j+\boldsymbol{x}^T\boldsymbol{w}_{*j}\right)h_j} \\
&= \frac{1}{Z_{BB}} e^{\boldsymbol{a}^T\boldsymbol{x}} \prod_j^N \left(1+e^{b_j+\boldsymbol{x}^T\boldsymbol{w}_{*j}}\right).
\end{aligned}
\tag{17.49}
$$

Here $\boldsymbol{w}_{*j}$ is the jth column of $\boldsymbol{W}$, the weights joining every visible unit to hidden unit j. The step from the third to the fifth line is the one that matters: the sum over all 2^N hidden configurations factorises into N independent two-term sums because the energy contains no $h_j h_{j'}$ coupling. Each of those sums is $1+e^{(\cdot)}$ because h_j takes only the values 0 and 1.

It is often convenient to write Eq. (17.49) as an effective energy for the visible layer alone, the *free energy*

$$
p_{BB}(\boldsymbol{x}) = \frac{e^{-F(\boldsymbol{x})}}{Z_{BB}}, \qquad F(\boldsymbol{x}) = -\boldsymbol{a}^T\boldsymbol{x} - \sum_j^N \log\left(1+e^{b_j+\boldsymbol{x}^T\boldsymbol{w}_{*j}}\right), \tag{17.50}
$$

in which the hidden units have been integrated out and their effect appears as a sum of softplus functions. This is the form used in the programs, because it reduces the cost of evaluating $p(\boldsymbol{x})$ from 2^N operations to N.

An identical calculation with the roles of the layers exchanged gives the marginal for the hidden units,

$$
p_{BB}(\boldsymbol{h};\boldsymbol{\Theta}) = \frac{1}{Z_{BB}} e^{\boldsymbol{b}^T\boldsymbol{h}} \prod_i^M \left(1+e^{a_i+\boldsymbol{w}_{i*}^T\boldsymbol{h}}\right), \tag{17.51}
$$

with $\boldsymbol{w}_{i*}$ the ith row of $\boldsymbol{W}$.

17.7.2 Conditional probabilities and the logistic function

The conditionals follow from Bayes' rule, and this is where the sigmoid appears. Dropping the explicit $\boldsymbol{\Theta}$,

$$
\begin{aligned}
p_{BB}(\boldsymbol{h}|\boldsymbol{x}) &= \frac{p_{BB}(\boldsymbol{x},\boldsymbol{h})}{p_{BB}(\boldsymbol{x})} = \frac{\frac{1}{Z_{BB}} e^{\boldsymbol{a}^T\boldsymbol{x}+\boldsymbol{b}^T\boldsymbol{h}+\boldsymbol{x}^T\boldsymbol{W}\boldsymbol{h}}}{\frac{1}{Z_{BB}} e^{\boldsymbol{a}^T\boldsymbol{x}} \prod_j^N \left(1+e^{b_j+\boldsymbol{x}^T\boldsymbol{w}_{*j}}\right)} \\
&= \frac{e^{\sum_j^N \left(b_j+\boldsymbol{x}^T\boldsymbol{w}_{*j}\right)h_j}}{\prod_j^N \left(1+e^{b_j+\boldsymbol{x}^T\boldsymbol{w}_{*j}}\right)} = \prod_j^N \frac{e^{\left(b_j+\boldsymbol{x}^T\boldsymbol{w}_{*j}\right)h_j}}{1+e^{b_j+\boldsymbol{x}^T\boldsymbol{w}_{*j}}} = \prod_j^N p_{BB}(h_j|\boldsymbol{x}).
\end{aligned}
\tag{17.52}
$$

The partition function has cancelled, and so has $e^{\boldsymbol{a}^T\boldsymbol{x}}$: the conditional of the hidden layer is a product of N independent single-unit distributions, which is Eq. (17.44) established by direct calculation. Setting $h_j = 1$ and $h_j = 0$ in turn,

$$\boxed{p_{BB}(h_j = 1|\boldsymbol{x}) = \frac{e^{b_j + \boldsymbol{x}^T \boldsymbol{w}_{*j}}}{1 + e^{b_j + \boldsymbol{x}^T \boldsymbol{w}_{*j}}} = \frac{1}{1 + e^{-\left(b_j + \boldsymbol{x}^T \boldsymbol{w}_{*j}\right)}} = \mathrm{sig}\left(b_j + \boldsymbol{x}^T \boldsymbol{w}_{*j}\right)} \tag{17.53}$$

and $p_{BB}(h_j = 0|\boldsymbol{x}) = 1/(1 + e^{b_j + \boldsymbol{x}^T \boldsymbol{w}_{*j}})$, where

$$\mathrm{sig}(z) = \frac{1}{1 + e^{-z}} \tag{17.54}$$

is the logistic function. By symmetry,

$$p_{BB}(\boldsymbol{x}|\boldsymbol{h}) = \prod_i^M \frac{e^{\left(a_i + \boldsymbol{w}_{i*}^T \boldsymbol{h}\right) x_i}}{1 + e^{a_i + \boldsymbol{w}_{i*}^T \boldsymbol{h}}}, \qquad p_{BB}(x_i = 1|\boldsymbol{h}) = \mathrm{sig}\left(a_i + \boldsymbol{w}_{i*}^T \boldsymbol{h}\right). \tag{17.55}$$

Where the sigmoid comes from. In the machine-learning literature the logistic function is usually introduced as a convenient smooth switch, one choice of "activation function" among many. Equations (17.53) and (17.33) show that it is nothing of the kind. It is the exact conditional probability of a binary variable whose energy is linear in that variable, and the argument $b_j + \boldsymbol{x}^T \boldsymbol{w}_{*j}$ is the local field acting on hidden unit j. A layer of an RBM computing sigmoids is a set of spins coming to thermal equilibrium in the field of the layer below.

Equations (17.53) and (17.55) are all that block Gibbs sampling needs: draw every h_j from its sigmoid, then every x_i from its sigmoid, and repeat. `rbm.py` runs 400 000 such sweeps on a machine with $M = 4$ and $N = 3$ and compares the resulting histogram with the exactly enumerated $p_{BB}(\boldsymbol{x})$; the largest discrepancy over all sixteen visible configurations is 0.001. Figure 17.3 shows the two distributions side by side.

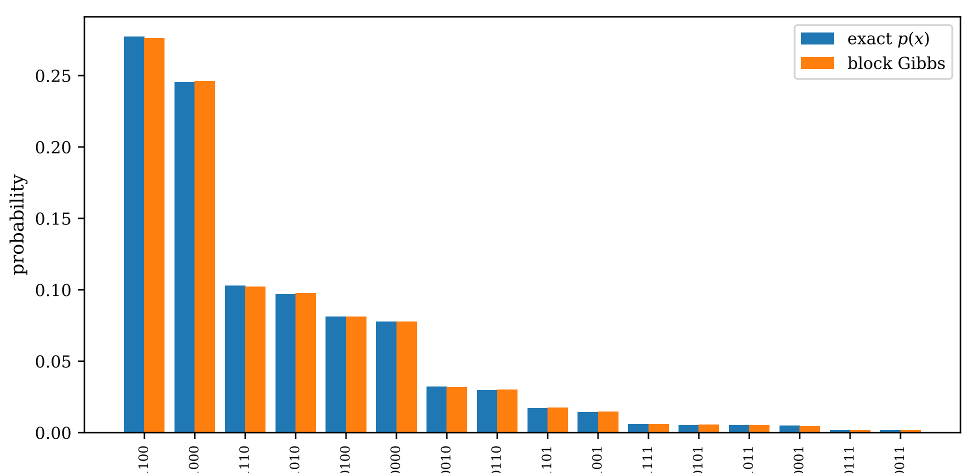

Fig. 17.3 Block Gibbs sampling of a binary-binary restricted Boltzmann machine with $M = 4$ visible and $N = 3$ hidden units and random parameters, tested against its own exact marginal. The blue bars are $p_{BB}(\boldsymbol{x})$ evaluated by summing Eq. (17.49) over all $2^4 \times 2^3$ configurations, the orange bars the relative frequencies of the sixteen visible states in 400 000 block Gibbs sweeps. The states are ordered along the horizontal axis by decreasing exact probability and labelled by their bit strings $x_1x_2x_3x_4$. Computed by `rbm.py`.

The point of Figure 17.3 is not that the two histograms agree – they must, if the algebra of this section is right – but how well and over what range they agree. The machine is far from uniform: the two most probable configurations, 1100 and 1000, carry 0.277 and 0.246 of the total probability between them, while the least probable, 0011, carries 0.001, a ratio of nearly three hundred to one. The sampled bars follow the exact ones across that whole range, including the states in the right-hand tail which the chain visits only a few hundred times in four hundred thousand sweeps. For a probability of 0.28 the binomial standard deviation of a frequency estimated from $N = 400000$ independent draws is $\sqrt{p(1-p)/N} = 7\times 10^{-4}$, so the observed worst deviation of 0.001 is what an *independent* sampler would have produced. Block Gibbs on a small machine is therefore not merely correct but close to perfectly efficient: with the whole hidden layer redrawn in one step and the whole visible layer in the next, one sweep is very nearly one independent sample, and the pathology of Figure 17.2 does not appear. It is worth appreciating why – the visible units are strongly correlated with each other, exactly the situation that ruins single-site Gibbs, but they are *conditionally independent given the hidden layer*, and the bipartite structure lets us update them together. The remedy of Eq. (17.40) is built into the model.

17.8 Training a Boltzmann machine

Denote the parameters collectively by $\boldsymbol{\Theta} = \{a_1,\dots,a_M,b_1,\dots,b_N,w_{11},\dots,w_{MN}\}$. The log-likelihood of the training data is

$$\mathscr{L}(\boldsymbol{\Theta}) = \langle \log p_{\boldsymbol{\Theta}}(\boldsymbol{x})\rangle_{\rm data} = -\langle E(\boldsymbol{x};\boldsymbol{\Theta})\rangle_{\rm data} - \log Z(\boldsymbol{\Theta}), \tag{17.56}$$

where the partition function came out of the average because it does not depend on the data. The cost function is its negative, $\mathscr{C}_{LL} = -\mathscr{L}$, and training is stochastic gradient descent,

$$\boldsymbol{\Theta}_{k+1} = \boldsymbol{\Theta}_k - \eta\,\nabla\mathscr{C}(\boldsymbol{\Theta}_k), \tag{17.57}$$

with the learning rate η and, in practice, one of the adaptive schemes of Section 14.3.

17.8.1 The gradient as a difference of moments

Define, for brevity, the operator

$$O_i(\boldsymbol{x}) = \frac{\partial E(\boldsymbol{x};\Theta_i)}{\partial \Theta_i}. \tag{17.58}$$

The partition function is the generating function of expectation values, and in this case

$$\langle O_i(\boldsymbol{x})\rangle_{\rm model} = \int p(\boldsymbol{x}|\boldsymbol{\Theta})\,\frac{\partial E(\boldsymbol{x};\Theta_i)}{\partial \Theta_i}\,d\boldsymbol{x} = -\frac{\partial \log Z(\Theta_i)}{\partial \Theta_i}, \tag{17.59}$$

which is Eq. (17.14) written for an energy model. Differentiating Eq. (17.56) and substituting,

$$\begin{aligned}\frac{\partial \mathscr{C}_{LL}}{\partial \Theta_i} &= \Big\langle \frac{\partial E(\boldsymbol{x};\Theta_i)}{\partial \Theta_i}\Big\rangle_{\rm data} + \frac{\partial \log Z(\Theta_i)}{\partial \Theta_i}\\ &= \Big\langle \frac{\partial E(\boldsymbol{x};\Theta_i)}{\partial \Theta_i}\Big\rangle_{\rm data} - \Big\langle \frac{\partial E(\boldsymbol{x};\Theta_i)}{\partial \Theta_i}\Big\rangle_{\rm model},\end{aligned} \tag{17.60}$$

that is

$$\boxed{\frac{\partial \mathscr{C}_{LL}}{\partial \Theta_i} = \langle O_i(\boldsymbol{x})\rangle_{\text{data}} - \langle O_i(\boldsymbol{x})\rangle_{\text{model}}} \tag{17.61}$$

The gradient of the log-likelihood is a difference of moments, one computed from the data and one from the model. The data-dependent term is the *positive phase*, the model-dependent term the *negative phase*. Training lowers the energy of configurations near the observed data – increasing their probability – and raises the energy of configurations the model produces but the data do not contain. At the optimum the two moments agree: the model reproduces the relevant statistics of the data exactly, which is a moment-matching condition and, in the language of statistical mechanics, the condition that the fitted couplings reproduce the measured correlation functions.

For the binary-binary machine the derivatives are the elementary ones of Eq. (17.18), so

$$\begin{aligned}
\frac{\partial \mathscr{C}_{LL}}{\partial w_{ij}} &= \langle x_i h_j\rangle_{\text{data}} - \langle x_i h_j\rangle_{\text{model}}, \\
\frac{\partial \mathscr{C}_{LL}}{\partial a_i} &= \langle x_i\rangle_{\text{data}} - \langle x_i\rangle_{\text{model}}, \\
\frac{\partial \mathscr{C}_{LL}}{\partial b_j} &= \langle h_j\rangle_{\text{data}} - \langle h_j\rangle_{\text{model}}.
\end{aligned} \tag{17.62}$$

The positive phase costs nothing: clamp the visible units to a training example, compute $p(h_j = 1|\boldsymbol{x})$ from Eq. (17.53), and average over the data set. The negative phase requires samples from the model itself, and there is no way to obtain them but to run a Markov chain – because Z is intractable, as it has been since Eq. (17.9).

17.8.2 Contrastive divergence

The obvious procedure – run block Gibbs sampling to equilibrium at every gradient step – is far too slow. Hinton's *contrastive divergence* replaces it with something that has no right to work and does:

CD-k. For each training example $\boldsymbol{x}^{(0)}$:

1. compute the positive phase from $p(h_j = 1|\boldsymbol{x}^{(0)}) = \text{sig}(b_j + \boldsymbol{x}^{(0)T}\boldsymbol{w}_{*j})$;
2. run k steps of block Gibbs sampling *started at the data*, $\boldsymbol{x}^{(0)} \to \boldsymbol{h}^{(0)} \to \boldsymbol{x}^{(1)} \to \cdots \to \boldsymbol{x}^{(k)}$;
3. compute the negative phase from $\boldsymbol{x}^{(k)}$ instead of from an equilibrium sample;
4. update $\boldsymbol{\Theta}$ by Eqs. (17.62) and (17.57).

The chain is started at the data rather than at equilibrium and stopped after k steps, often $k = 1$. The result is a *biased* estimate of the gradient – it is not the gradient of the log-likelihood, nor in fact of any function – and yet it moves the parameters in a useful direction, because what it measures is the tendency of the model to drift away from the data in one step.

Table 17.2 makes both halves of that statement concrete on the bars-and-stripes data set: all 3×3 binary images whose rows are constant or whose columns are constant, of which there are 14 out of 512. With only nine visible units the partition function can be summed exactly, so the log-likelihood and the Kullback-Leibler divergence are known at every step – and Eq. (17.21) can be watched holding. The fourteen patterns are shown in Figure 17.4.

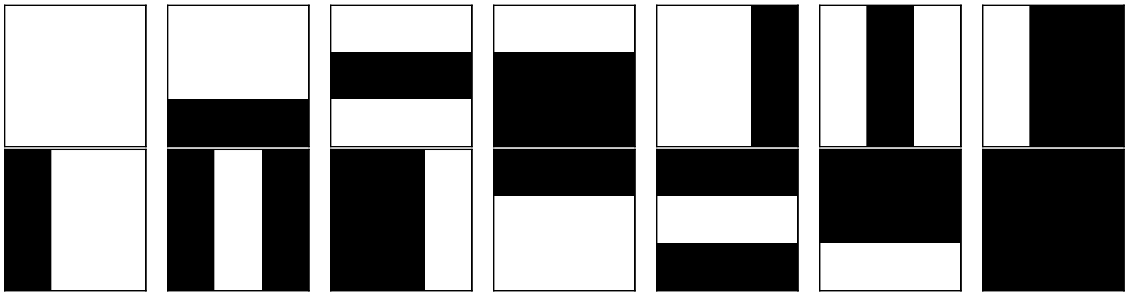

Fig. 17.4 The whole bars-and-stripes data set on a 3×3 grid: the fourteen binary images, out of the $2^9 = 512$ possible ones, in which either every row is constant or every column is constant. Black denotes a pixel value of one and white a value of zero. Reading along the top row and then the bottom, the first eight are the stripe patterns and the remainder the bar patterns; the uniformly white and uniformly black images belong to both families, which is why there are $8+8-2=14$ and not 16. Computed by `rbm.py`.

Small as it is, the data set of Figure 17.4 is a genuine test of a generative model, and it is worth being clear about what makes it one. Every pattern is specified by four bits – three saying which rows, or which columns, are black, and one saying whether the pattern is read horizontally or vertically – so the fourteen images occupy 2.7% of the 512-dimensional configuration space and the model must place its mass on that thin set and nowhere else. The rule cannot be learned pixel by pixel: each individual pixel is on in exactly half the patterns, and each pair of pixels within a row, or within a column, is perfectly correlated, while a pair lying on a diagonal is not correlated at all. A model that captured only the one-body and two-body statistics would therefore assign weight to a great many images that are not in the set. What the hidden units must supply is a higher-order constraint – the exclusive choice between rows and columns – and eight of them turn out to be enough. Notice also that the patterns come in complementary pairs, so the data set is invariant under $x \to 1-x$, a symmetry a well-trained machine ought to reproduce and one the model family can realise exactly: complementing the visible units in Eq. (17.46) is undone, up to an additive constant in the energy, by $\boldsymbol{a} \to -\boldsymbol{a}$, $\boldsymbol{W} \to -\boldsymbol{W}$ and $\boldsymbol{b} \to \boldsymbol{b} + \boldsymbol{W}^T \mathbf{1}$.

epoch	$\mathscr{L}$	$\mathrm{KL}(f\|p)$	k	$\mathscr{L}$	$\mathrm{KL}(f\|p)$
0	-6.2386	3.5995	1	-3.3710	0.7319
2500	-3.6737	1.0346	2	-3.0410	0.4019
5000	-2.9767	0.3377	5	-2.7775	0.1385
10000	-2.8125	0.1734	10	-2.7327	0.0937
20000	-2.7775	0.1385			

Table 17.2 Contrastive divergence on bars and stripes, $M = 9$ visible and $N = 8$ hidden units, learning rate 0.1. Left: training history with $k = 5$. A perfect model would give $\mathscr{L} = \log(1/14) = -2.6391$ and $\mathrm{KL} = 0$. The two columns differ by exactly 2.6391, the entropy of the data, as Eq. (17.20) requires. Right: the same budget of 20000 updates at different k. After training the model places 97.9% of its probability on the fourteen valid patterns, against 2.7% at random. Computed by `rbm.py`.

The machine has learned the rule "rows constant or columns constant" from fourteen examples and eight hidden units, without ever being told it. The right-hand half of the table shows the cost of the approximation: $k = 1$, the cheapest and most biased choice, leaves a residual divergence five times larger than $k = 5$. One Gibbs step from the data is nowhere near the model's equilibrium, so the negative phase is badly estimated – and CD-1 still works, which is the surprising and much-discussed fact about the algorithm.

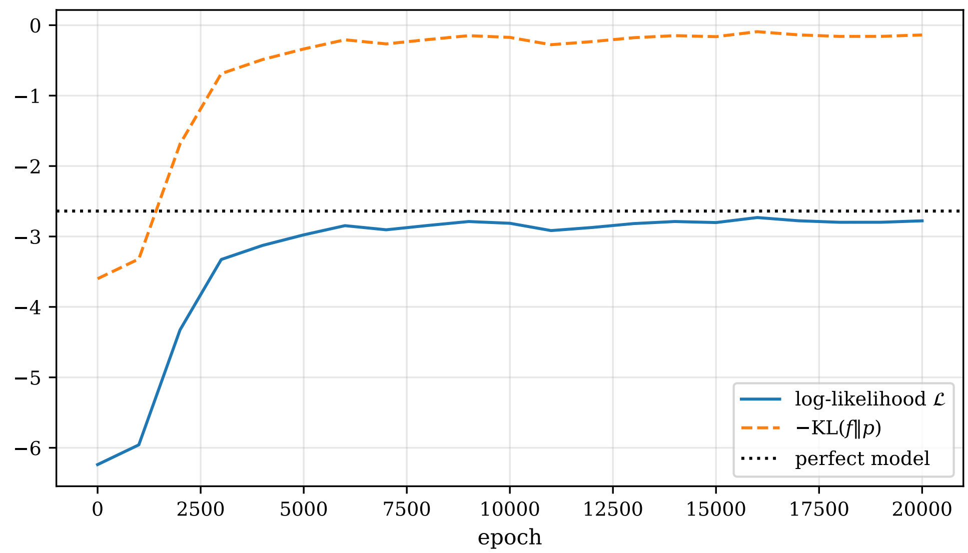

Fig. 17.5 Training history of the run tabulated in the left half of Table 17.2: contrastive divergence with $k = 5$ on bars and stripes, $M = 9$ visible and $N = 8$ hidden units, learning rate 0.1, sampled every thousand epochs. The solid curve is the exact log-likelihood $\mathscr{L}$ of Eq. (17.56), the dashed curve the negative Kullback-Leibler divergence $-\mathrm{KL}(f\|p)$ of Eq. (17.19), both computed by enumerating all 512 visible configurations rather than estimated. The dotted line at $\log(1/14) = -2.6391$ marks the log-likelihood of a model that puts its whole probability uniformly on the fourteen valid patterns. Computed by `rbm.py`.

Figure 17.5 shows the same run as a function of epoch, and what it displays above all is Eq. (17.21) in action. The two curves are not merely similar in shape; they are the same curve translated vertically, and the translation is 2.6391 at every epoch, at the start when the model is hopeless and at the end when it is good. That is the content of Eq. (17.20): the difference between the two criteria is $\langle \log f \rangle_{\text{data}}$, the negative entropy of the empirical distribution, which is fixed by the data set and knows nothing about $\boldsymbol{\Theta}$. Maximising the likelihood and minimising the divergence are not two strategies that happen to converge to the same place – they are the same optimisation, step by step.

The trajectory itself repays a second look. The log-likelihood starts at -6.2386, which is close to $\log(1/512) = -6.238$: a machine with small random weights is very nearly uniform over the whole configuration space, as it should be. It then rises steeply between epochs 1000 and 3000 – almost all of the learning happens there – and by epoch 6000 it has reached -2.82 and stopped improving in any systematic way. What remains is a residual gap of about 0.14 to the dotted line, and a jitter of ± 0.05 about the plateau. Neither is statistical noise in the usual sense, since both curves are computed exactly; they are the signature of the bias in contrastive divergence. The parameters are being pushed around by a gradient that is not the gradient of anything, and they settle into a small orbit near but not at the maximum of the likelihood. Running longer does not help, and Table 17.2 shows that increasing k does: the plateau is set by how far the truncated Gibbs chain is from equilibrium, not by how many updates we are willing to perform. One further caution about reading the figure: the dotted line is the perfect-model value of $\mathscr{L}$, and only the solid curve should be compared with it. The dashed curve crosses it near epoch 1500 for no better reason than that $-\mathrm{KL}$ has a different zero, and rises towards 0 from below, as a divergence must.

17.9 The Gaussian-binary restricted Boltzmann machine

Binary visible units are natural for images and spins and useless for particle coordinates. For continuous data we let the visible units be real-valued and Gaussian while keeping the hidden units binary, which preserves everything that made the binary machine tractable. The energy becomes

$$\begin{aligned} E_{GB}(\boldsymbol{x},\boldsymbol{h};\boldsymbol{\Theta}) &= \sum_i^M \frac{(x_i-a_i)^2}{2\sigma_i^2} - \sum_j^N b_j h_j - \sum_{ij}^{M,N} \frac{x_i w_{ij} h_j}{\sigma_i^2} \\ &= \left\| \frac{\boldsymbol{x}-\boldsymbol{a}}{2\boldsymbol{\sigma}} \right\|^2 - \boldsymbol{b}^T \boldsymbol{h} - \left(\frac{\boldsymbol{x}}{\boldsymbol{\sigma}^2}\right)^T \boldsymbol{W}\boldsymbol{h}. \end{aligned} \tag{17.63}$$

The visible bias a_i has become the centre of a Gaussian rather than a linear coefficient, and a new parameter σ_i has appeared, the width of that Gaussian; it is very often set to one and the same for all units, but we shall see in Section 17.10 that its value is not innocent. The joint distribution is

$$\begin{aligned} p_{GB}(\boldsymbol{x},\boldsymbol{h};\boldsymbol{\Theta}) &= \frac{1}{Z_{GB}} e^{-\left\| \frac{\boldsymbol{x}-\boldsymbol{a}}{2\boldsymbol{\sigma}} \right\|^2 + \boldsymbol{b}^T \boldsymbol{h} + \left(\frac{\boldsymbol{x}}{\boldsymbol{\sigma}^2}\right)^T \boldsymbol{W}\boldsymbol{h}} \\ &= \frac{1}{Z_{GB}} \prod_{ij}^{M,N} e^{-\frac{(x_i-a_i)^2}{2\sigma_i^2} + b_j h_j + \frac{x_i w_{ij} h_j}{\sigma_i^2}}, \end{aligned} \tag{17.64}$$

and the partition function now involves an integral over the visible units and a sum over the hidden ones,

$$Z_{GB} = \int \sum_{\tilde{\boldsymbol{h}}} e^{-\left\| \frac{\tilde{\boldsymbol{x}}-\boldsymbol{a}}{2\boldsymbol{\sigma}} \right\|^2 + \boldsymbol{b}^T \tilde{\boldsymbol{h}} + \left(\frac{\tilde{\boldsymbol{x}}}{\boldsymbol{\sigma}^2}\right)^T \boldsymbol{W}\tilde{\boldsymbol{h}}} d\tilde{\boldsymbol{x}}. \tag{17.65}$$

17.9.1 Marginal probabilities

Marginalising over the binary hidden units is the calculation of Eq. (17.49) again, unchanged in structure because the $\boldsymbol{h}$ dependence is still bilinear,

$$\begin{aligned} p_{GB}(\boldsymbol{x}) &= \sum_{\tilde{\boldsymbol{h}}} p_{GB}(\boldsymbol{x},\tilde{\boldsymbol{h}}) = \frac{1}{Z_{GB}} \sum_{\tilde{\boldsymbol{h}}} e^{-\left\| \frac{\boldsymbol{x}-\boldsymbol{a}}{2\boldsymbol{\sigma}} \right\|^2 + \boldsymbol{b}^T \tilde{\boldsymbol{h}} + \left(\frac{\boldsymbol{x}}{\boldsymbol{\sigma}^2}\right)^T \boldsymbol{W}\tilde{\boldsymbol{h}}} \\ &= \frac{1}{Z_{GB}} e^{-\left\| \frac{\boldsymbol{x}-\boldsymbol{a}}{2\boldsymbol{\sigma}} \right\|^2} \prod_j^N \left(1 + e^{b_j + \left(\frac{\boldsymbol{x}}{\boldsymbol{\sigma}^2}\right)^T \boldsymbol{w}_{*j}} \right). \end{aligned} \tag{17.66}$$

A Gaussian envelope multiplied by N smooth positive factors, one per hidden unit: this is the object that will become a wave function.

Marginalising over the *visible* units is new, because it is the first time we integrate rather than sum. The work is in completing the square:

$$
\begin{aligned}
p_{GB}(\boldsymbol{h}) &= \int p_{GB}(\tilde{\boldsymbol{x}},\boldsymbol{h})\,d\tilde{\boldsymbol{x}} = \frac{1}{Z_{GB}}\int e^{-\left\|\frac{\tilde{\boldsymbol{x}}-\boldsymbol{a}}{2\boldsymbol{\sigma}}\right\|^2+\boldsymbol{b}^T\boldsymbol{h}+\left(\frac{\tilde{\boldsymbol{x}}}{\boldsymbol{\sigma}^2}\right)^T\boldsymbol{W}\boldsymbol{h}}\,d\tilde{\boldsymbol{x}} \\
&= \frac{1}{Z_{GB}}e^{\boldsymbol{b}^T\boldsymbol{h}}\int\prod_i^M e^{-\frac{(\tilde{x}_i-a_i)^2}{2\sigma_i^2}+\frac{\tilde{x}_i\boldsymbol{w}_{i*}^T\boldsymbol{h}}{\sigma_i^2}}\,d\tilde{\boldsymbol{x}} \\
&= \frac{1}{Z_{GB}}e^{\boldsymbol{b}^T\boldsymbol{h}}\prod_i^M\int e^{-\frac{(\tilde{x}_i-a_i)^2-2\tilde{x}_i\boldsymbol{w}_{i*}^T\boldsymbol{h}}{2\sigma_i^2}}\,d\tilde{x}_i \\
&= \frac{1}{Z_{GB}}e^{\boldsymbol{b}^T\boldsymbol{h}}\prod_i^M\int e^{-\frac{\tilde{x}_i^2-2\tilde{x}_i\left(a_i+\boldsymbol{w}_{i*}^T\boldsymbol{h}\right)+a_i^2}{2\sigma_i^2}}\,d\tilde{x}_i \\
&= \frac{1}{Z_{GB}}e^{\boldsymbol{b}^T\boldsymbol{h}}\prod_i^M\int e^{-\frac{\left(\tilde{x}_i-a_i-\boldsymbol{w}_{i*}^T\boldsymbol{h}\right)^2-2a_i\boldsymbol{w}_{i*}^T\boldsymbol{h}-\left(\boldsymbol{w}_{i*}^T\boldsymbol{h}\right)^2}{2\sigma_i^2}}\,d\tilde{x}_i \\
&= \frac{1}{Z_{GB}}e^{\boldsymbol{b}^T\boldsymbol{h}}\prod_i^M e^{\frac{2a_i\boldsymbol{w}_{i*}^T\boldsymbol{h}+\left(\boldsymbol{w}_{i*}^T\boldsymbol{h}\right)^2}{2\sigma_i^2}}\int e^{-\frac{\left(\tilde{x}_i-a_i-\boldsymbol{w}_{i*}^T\boldsymbol{h}\right)^2}{2\sigma_i^2}}\,d\tilde{x}_i \\
&= \frac{1}{Z_{GB}}e^{\boldsymbol{b}^T\boldsymbol{h}}\prod_i^M\sqrt{2\pi\sigma_i^2}\,e^{\frac{2a_i\boldsymbol{w}_{i*}^T\boldsymbol{h}+\left(\boldsymbol{w}_{i*}^T\boldsymbol{h}\right)^2}{2\sigma_i^2}}.
\end{aligned}
\tag{17.67}
$$

Line by line: the product over i comes out of the integral because the exponent is a sum over i with no cross terms, so the M-dimensional integral factorises into M one-dimensional ones; the fourth line expands the square; the fifth completes it, adding and subtracting $(a_i+\boldsymbol{w}_{i*}^T\boldsymbol{h})^2$ so that the a_i^2 cancels; and the last line is the standard Gaussian integral $\int e^{-(u-\mu)^2/2\sigma^2}du = \sqrt{2\pi\sigma^2}$. `rbm.py` checks the result against a numerical quadrature of the joint and finds agreement to 3.8×10^{-14}.

17.9.2 Conditional probabilities

The hidden units behave exactly as before. Dividing Eq. (17.64) by Eq. (17.66), the Gaussian envelope cancels along with Z_{GB},

$$
p_{GB}(\boldsymbol{h}|\boldsymbol{x}) = \prod_j^N \frac{e^{\left(b_j+\left(\frac{\boldsymbol{x}}{\boldsymbol{\sigma}^2}\right)^T\boldsymbol{w}_{*j}\right)h_j}}{1+e^{b_j+\left(\frac{\boldsymbol{x}}{\boldsymbol{\sigma}^2}\right)^T\boldsymbol{w}_{*j}}} = \prod_j^N p_{GB}(h_j|\boldsymbol{x}),
\tag{17.68}
$$

so that

$$
\boxed{p_{GB}(h_j=1|\boldsymbol{x}) = \mathrm{sig}\left(b_j+\left(\frac{\boldsymbol{x}}{\boldsymbol{\sigma}^2}\right)^T\boldsymbol{w}_{*j}\right)}
\tag{17.69}
$$

with $p_{GB}(h_j=0|\boldsymbol{x})$ the complement. The only change from Eq. (17.53) is that the input is scaled by $1/\sigma_i^2$.

The visible units are different, and the form of the answer is what gives the model its name. Dividing the joint by Eq. (17.67),

$$p_{GB}(\boldsymbol{x}|\boldsymbol{h}) = \frac{p_{GB}(\boldsymbol{x},\boldsymbol{h})}{p_{GB}(\boldsymbol{h})} = \frac{\frac{1}{Z_{GB}} e^{-\left\|\frac{\boldsymbol{x}-\boldsymbol{a}}{2\boldsymbol{\sigma}}\right\|^2 + \boldsymbol{b}^T\boldsymbol{h} + \left(\frac{\boldsymbol{x}}{\boldsymbol{\sigma}^2}\right)^T \boldsymbol{W}\boldsymbol{h}}}{\frac{1}{Z_{GB}} e^{\boldsymbol{b}^T\boldsymbol{h}} \prod_i^M \sqrt{2\pi\sigma_i^2}\, e^{\frac{2a_i\boldsymbol{w}_{i*}^T\boldsymbol{h} + \left(\boldsymbol{w}_{i*}^T\boldsymbol{h}\right)^2}{2\sigma_i^2}}}$$
$$= \prod_i^M \frac{1}{\sqrt{2\pi\sigma_i^2}} e^{-\frac{x_i^2 - 2a_i x_i + a_i^2 - 2x_i\boldsymbol{w}_{i*}^T\boldsymbol{h} + 2a_i\boldsymbol{w}_{i*}^T\boldsymbol{h} + \left(\boldsymbol{w}_{i*}^T\boldsymbol{h}\right)^2}{2\sigma_i^2}}$$
$$= \prod_i^M \frac{1}{\sqrt{2\pi\sigma_i^2}} e^{-\frac{\left(x_i - a_i - \boldsymbol{w}_{i*}^T\boldsymbol{h}\right)^2}{2\sigma_i^2}} = \prod_i^M \mathcal{N}\left(x_i \,\middle|\, a_i + \boldsymbol{w}_{i*}^T\boldsymbol{h},\ \sigma_i^2\right), \tag{17.70}$$

the last step recognising that the numerator of the exponent is a perfect square. Hence

$$\boxed{p_{GB}(x_i|\boldsymbol{h}) = \mathcal{N}\left(x_i \,\middle|\, a_i + \boldsymbol{w}_{i*}^T\boldsymbol{h},\ \sigma_i^2\right)} \tag{17.71}$$

a normal distribution with variance σ_i^2 and a mean that the hidden units shift by $\boldsymbol{w}_{i*}^T\boldsymbol{h}$. `rbm.py` compares this density point by point with the ratio of the joint to the marginal and finds agreement to 8.3×10^{-16}.

What the two conditionals say. Given the data, each hidden unit is a biased coin whose bias is a sigmoid of the local field – Eq. (17.69). Given the hidden units, each visible unit is a Gaussian of fixed width whose *centre* the hidden units move – Eq. (17.71). The machine is therefore a mixture of 2^N Gaussians, all of the same width σ_i, with centres $a_i + \boldsymbol{w}_{i*}^T\boldsymbol{h}$ indexed by the 2^N binary configurations of the hidden layer, and with mixing weights given by Eq. (17.67). That is a very flexible family – and note what it cannot do: it cannot change the width. We return to this in Section 17.11.

Figure 17.6 draws the mixture for the smallest case in which there is anything to see, a single visible unit and three hidden ones. The eight components are Gaussians of identical unit width, and all of them peak at $1/\sqrt{2\pi} = 0.399$; what distinguishes them is only their centres $a + \boldsymbol{w}_{1*}^T\boldsymbol{h}$, which for these parameters run from -0.456 when every hidden unit is off to -2.045 when all three are on. The middle weight is -0.004, so switching that unit moves the centre by nothing at all and the eight curves collapse into four visible pairs – a reminder that a hidden unit with a small weight is not a feature but dead weight, which is what an L^2 penalty exploits. The heavy curve, the actual marginal, peaks at $x = -1.80$ with a density of 0.358, lower than any of its components because it is an average of curves that peak in different places.

The quantitative moral is in the width. A mixture of Gaussians of common variance σ^2 with centres $\mu_{\boldsymbol{h}}$ drawn with weights $\pi_{\boldsymbol{h}}$ has variance

$$\mathrm{Var}(x) = \sigma^2 + \mathrm{Var}\left(\mu_{\boldsymbol{h}}\right) \ \geq\ \sigma^2, \tag{17.72}$$

the law of total variance: the spread of the centres adds to the width of the components and can never subtract from it. For the machine plotted here $\mathrm{Var}(x) = 1.255$ against $\sigma^2 = 1$, so the displaced centres have broadened the density by twelve per cent in standard deviation and could have broadened it by an arbitrary amount, but no setting of $\boldsymbol{a}$, $\boldsymbol{b}$ and $\boldsymbol{W}$ could ever have made it narrower than the single curve $\sigma = 1$. The hidden layer buys shape – skewness here, multimodality when the centres are pushed far apart – and it buys extra width, and those are the only two things it buys. When we come to use p_{GB} as a wave function in Section 17.10 the parameter σ will therefore have to be fixed by hand at something at least as small as the true extent of the state, and Eq. (17.87) is that choice.

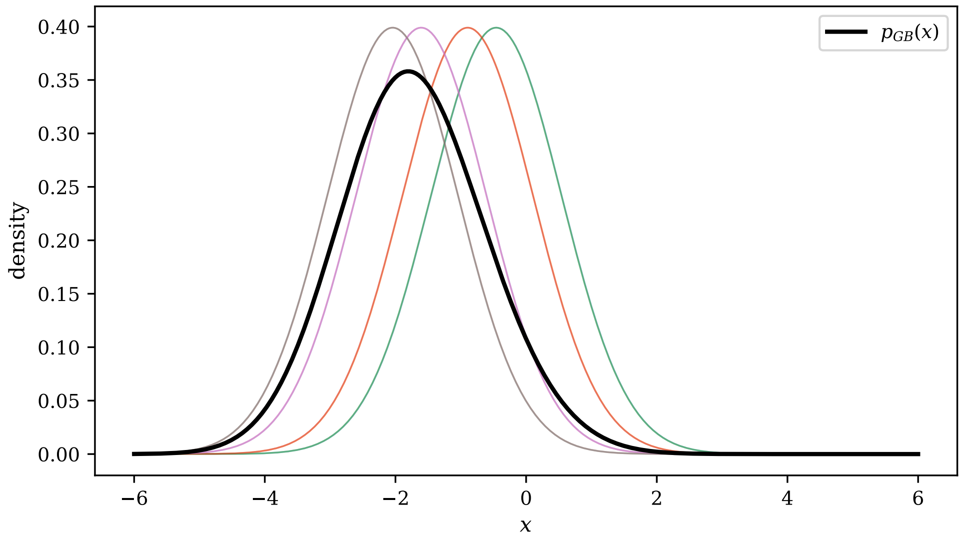

Fig. 17.6 The marginal density $p_{GB}(x)$ of a Gaussian-binary machine with a single visible unit, $M = 1$, three hidden units, $N = 3$, unit width $\sigma = 1$ and random parameters. The thin coloured curves are the $2^N = 8$ mixture components $\mathcal{N}(x|a + \boldsymbol{w}_{1*}^T \boldsymbol{h}, \sigma^2)$ of Eq. (17.71), one for each configuration of the hidden layer, each drawn unweighted so that all of them peak at the same height and only their centres differ; the heavy black curve is the normalised marginal Eq. (17.66) obtained by weighting them according to Eq. (17.67). Only four components are separately visible because one of the three weights came out close to zero, pairing the components off. Computed by `rbm.py`.

Both conditionals are again all that block Gibbs sampling requires: draw the hidden layer from N sigmoids, draw the visible layer from M Gaussians, and repeat. Nothing else in the machinery of Section 17.8 changes; only the derivatives of the energy, which the reader is invited to work out in Exercise 17.14.

17.10 Neural quantum states

We now do the thing this chapter exists for. A wave function is a probability *amplitude* on configuration space; the marginal $p_{GB}(\boldsymbol{x})$ of Eq. (17.66) is a probability *density* on the same space, defined by a handful of parameters and flexible enough to represent essentially arbitrary correlations. Identify the two, and the machinery of Chapters 13 and 14 does the rest.

There are two ways to make the identification, and the difference between them is not cosmetic.

The wave function as the marginal.

The direct choice is

$$\Psi(\boldsymbol{x}) = F_{rbm}(\boldsymbol{x}) = \frac{1}{Z}\sum_{\boldsymbol{h}} e^{-E(\boldsymbol{x},\boldsymbol{h})} = \frac{1}{Z} e^{-\sum_i^M \frac{(x_i - a_i)^2}{2\sigma^2}} \prod_j^N \left(1 + e^{b_j + \sum_i^M \frac{x_i w_{ij}}{\sigma^2}}\right). \tag{17.73}$$

This is the most general option, in the sense that with complex parameters it allows a complex wave function. But it changes the probabilistic interpretation entirely: what was a probability in the RBM framework is now an amplitude, and the graphical-model reading – conditional probabilities, Markov random fields, Gibbs sampling – breaks down, since $|\Psi|^2 = F^2$ is not an RBM distribution.

The wave function as the square root of the marginal.

If we are content with a positive-definite Ψ, as we may be for the ground state of a bosonic or spin-free problem, the better choice is

$$|\Psi(\boldsymbol{x})|^2 = F_{rbm}(\boldsymbol{x}) \qquad \Longrightarrow \qquad \Psi(\boldsymbol{x}) = \sqrt{F_{rbm}(\boldsymbol{x})}, \tag{17.74}$$

because then the sampling density $|\Psi|^2$ *is* an RBM distribution and can be sampled by Gibbs as well as by Metropolis. Carrying the square root through the derivation of Eq. (17.66),

$$\begin{aligned}\Psi(\boldsymbol{x}) &= \frac{1}{\sqrt{Z}}\sqrt{\sum_{\{h_j\}} e^{-E(\boldsymbol{x},\boldsymbol{h})}}\\ &= \frac{1}{\sqrt{Z}} e^{-\sum_i^M \frac{(x_i-a_i)^2}{4\sigma^2}} \sqrt{\prod_j^N \sum_{h_j} e^{b_j h_j + \sum_i^M \frac{x_i w_{ij} h_j}{\sigma^2}}}\\ &= \frac{1}{\sqrt{Z}} e^{-\sum_i^M \frac{(x_i-a_i)^2}{4\sigma^2}} \prod_j^N \sqrt{1+e^{b_j+\sum_i^M \frac{x_i w_{ij}}{\sigma^2}}}.\end{aligned} \tag{17.75}$$

Note the $4\sigma^2$ where Eq. (17.73) has $2\sigma^2$: the square root halves everything in the exponent. We adopt Eq. (17.75) from here on and write $\sigma_i = \sigma$ for all units.

17.10.1 The cost function and its gradient

Here we part company with machine learning. There is no data set and no log-likelihood. The cost function is the *energy of the quantum system*, and the variational principle of Chapter 13 guarantees that minimising it drives the trial state towards the ground state. Everything needed was derived in Chapter 14: with $\boldsymbol{\alpha} = \{a_1,\dots,a_M,b_1,\dots,b_N,w_{11},\dots,w_{MN}\}$ standing for all the RBM parameters, the local energy is $E_L = \Psi^{-1}\hat{H}\Psi$ and the gradient is Eq. (14.2),

$$G_i = \frac{\partial\langle E_L\rangle}{\partial\alpha_i} = 2\left(\left\langle E_L \frac{1}{\Psi}\frac{\partial\Psi}{\partial\alpha_i}\right\rangle - \langle E_L\rangle\left\langle\frac{1}{\Psi}\frac{\partial\Psi}{\partial\alpha_i}\right\rangle\right), \tag{17.76}$$

a covariance of the local energy with the logarithmic derivative of the trial function, and computable from the same random walk that measures the energy. Using $\Psi^{-1}\partial_\alpha\Psi = \partial_\alpha \ln\Psi$, all we need are the derivatives of $\ln\Psi$.

For the direct choice (17.73),

$$\ln\Psi(\boldsymbol{x}) = -\ln Z - \sum_m^M \frac{(x_m-a_m)^2}{2\sigma^2} + \sum_n^N \ln\left(1+e^{b_n+\sum_i^M \frac{x_i w_{in}}{\sigma^2}}\right), \tag{17.77}$$

whence, writing $Q_n = b_n + \sum_i^M x_i w_{in}/\sigma^2$ for the local field on hidden unit n,

$$\frac{\partial\ln\Psi}{\partial a_m} = \frac{x_m-a_m}{\sigma^2}, \qquad \frac{\partial\ln\Psi}{\partial b_n} = \frac{1}{e^{-Q_n}+1}, \qquad \frac{\partial\ln\Psi}{\partial w_{mn}} = \frac{x_m}{\sigma^2\left(e^{-Q_n}+1\right)}. \tag{17.78}$$

The middle expression is $\mathrm{sig}(Q_n)$, which by Eq. (17.69) is exactly $p(h_n = 1|\boldsymbol{x})$: *the derivative of the log wave function with respect to a hidden bias is the probability that the corresponding hidden unit is on.*

For the square-root choice (17.75),

$$\ln\Psi(\boldsymbol{x}) = -\tfrac{1}{2}\ln Z - \sum_m^M \frac{(x_m - a_m)^2}{4\sigma^2} + \tfrac{1}{2}\sum_n^N \ln\left(1 + e^{Q_n}\right), \tag{17.79}$$

and every derivative acquires a factor of one half,

$$\frac{\partial \ln\Psi}{\partial a_m} = \frac{x_m - a_m}{2\sigma^2}, \qquad \frac{\partial \ln\Psi}{\partial b_n} = \frac{1}{2\left(e^{-Q_n} + 1\right)}, \qquad \frac{\partial \ln\Psi}{\partial w_{mn}} = \frac{x_m}{2\sigma^2\left(e^{-Q_n} + 1\right)}. \tag{17.80}$$

17.10.2 The local energy

Take the Hamiltonian of the quantum dot of Chapters 11 and 13, in natural units $\hbar = c = e = m_e = 1$,

$$\hat{H} = \sum_p^P \left(-\tfrac{1}{2}\nabla_p^2 + \tfrac{1}{2}\omega^2 r_p^2\right) + \sum_{p<q} \frac{1}{r_{pq}}, \tag{17.81}$$

with P particles in D dimensions. For a real positive Ψ the identity $\nabla^2\Psi/\Psi = \nabla^2\ln\Psi + (\nabla\ln\Psi)^2$ turns the kinetic term into derivatives of the logarithm, and

$$\begin{aligned} E_L = \frac{1}{\Psi}\hat{H}\Psi &= -\frac{1}{2}\frac{1}{\Psi}\sum_p^P \nabla_p^2\Psi + \frac{1}{2}\omega^2\sum_p^P r_p^2 + \sum_{p<q}\frac{1}{r_{pq}} \\ &= \frac{1}{2}\sum_p^P\sum_d^D\left(-\left(\frac{\partial\ln\Psi}{\partial x_{pd}}\right)^2 - \frac{\partial^2\ln\Psi}{\partial x_{pd}^2} + \omega^2 x_{pd}^2\right) + \sum_{p<q}\frac{1}{r_{pq}}. \end{aligned} \tag{17.82}$$

Letting each visible node of the Boltzmann machine represent one Cartesian coordinate of one particle, so that $M = P \times D$, this reads

$$\boxed{E_L = \frac{1}{2}\sum_m^M\left(-\left(\frac{\partial\ln\Psi}{\partial x_m}\right)^2 - \frac{\partial^2\ln\Psi}{\partial x_m^2} + \omega^2 x_m^2\right) + \sum_{p<q}\frac{1}{r_{pq}}} \tag{17.83}$$

The two derivatives of $\ln\Psi$ with respect to a coordinate follow from Eq. (17.77) by differentiating the softplus twice. For the direct choice,

$$\frac{\partial\ln\Psi}{\partial x_m} = -\frac{x_m - a_m}{\sigma^2} + \frac{1}{\sigma^2}\sum_n^N \frac{w_{mn}}{e^{-Q_n} + 1}, \tag{17.84}$$

$$\frac{\partial^2\ln\Psi}{\partial x_m^2} = -\frac{1}{\sigma^2} + \frac{1}{\sigma^4}\sum_n^N w_{mn}^2 \frac{e^{Q_n}}{\left(e^{Q_n} + 1\right)^2}, \tag{17.85}$$

and for the square-root choice both are halved,

$$\frac{\partial\ln\Psi}{\partial x_m} = -\frac{x_m - a_m}{2\sigma^2} + \frac{1}{2\sigma^2}\sum_n^N \frac{w_{mn}}{e^{-Q_n} + 1}, \qquad \frac{\partial^2\ln\Psi}{\partial x_m^2} = -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_n^N w_{mn}^2 \frac{e^{Q_n}}{\left(e^{Q_n} + 1\right)^2}. \tag{17.86}$$

Note that $e^Q/(1 + e^Q)^2 = \mathrm{sig}(Q)\,(1 - \mathrm{sig}(Q))$, the variance of the corresponding hidden unit, which is the form used in the programs and the reason the second derivative costs no more than the first. The quantum force of Section 13.6, needed for importance-sampled Metropolis-Hastings, is as always $\boldsymbol{F} = 2\nabla \ln \Psi$.

We now have everything: a trial function with $M+N+MN$ parameters, its logarithmic derivatives with respect to coordinates and parameters, a local energy, and a quantum force. `rbm.py` checks all of them against central differences – including the local energy itself against a numerical $\hat{H}\Psi/\Psi$ – with worst discrepancies of 1.2×10^{-11} for the first coordinate derivative, 4.9×10^{-9} for the second and 3.2×10^{-7} for E_L.

17.11 Application: two electrons in a quantum dot

We apply this to the system that has run through Chapters 11, 13, 14 and 16: two electrons in a two-dimensional harmonic trap at $\omega = 1$, for which Taut's exact ground-state energy is 3 a.u. with the Coulomb repulsion and 2 a.u. without it. With $P = 2$ and $D = 2$ the machine has $M = 4$ visible units, and we take $N = 2$, 4 or 8 hidden units, giving between 14 and 44 parameters.

The trial function contains no physics whatever. There is no Slater determinant, no orbital, no single-particle basis; no Padé-Jastrow factor, no cusp condition, and nothing that knows the electrons repel. There is a Gaussian envelope and a set of sigmoidal bumps, and the parameters are found by gradient descent on the energy.

Choosing the width.

One parameter is not learned, and it repays a moment's thought. Setting all the weights to zero in Eq. (17.75) leaves $\Psi \propto \exp\left(-\sum_m x_m^2/4\sigma^2\right)$, while the exact non-interacting ground state is $\exp\left(-\omega\sum_m x_m^2/2\right)$. The two agree when

$$\frac{1}{4\sigma^2} = \frac{\omega}{2} \qquad \Longleftrightarrow \qquad \sigma^2 = \frac{1}{2\omega}, \tag{17.87}$$

which for $\omega = 1$ gives $\sigma = 1/\sqrt{2}$. With that choice the non-interacting problem is *exactly* representable – at zero weights – and the hidden units have nothing to do but describe the correlations. It is the natural division of labour, and it also gives a stringent test: the non-interacting calculation must return $E = 2$ with vanishing variance, or something is wrong. Table 17.3 shows that it does.

N hidden	E	error	$\mathrm{Var}(E_L)$
2	2.000080	0.000014	1.2×10^{-4}
4	2.000049	0.000013	1.1×10^{-4}

Table 17.3 The *non-interacting* two-electron dot, exact energy 2. Sixty gradient-descent steps followed by a production run of 500 walkers and 3000 steps, with the error from the blocking analysis of Chapter 14. The variance of the local energy collapses towards zero, which is the zero-variance principle of Section 14.2 in action: the optimiser has found the exact eigenstate, where E_L is constant. Computed by `rbm.py`.

The interacting dot.

Switching on the Coulomb term gives Table 17.4.

N hidden	E	error	$\mathrm{Var}(E_L)$	$E - E_{\mathrm{exact}}$
2	3.14840	0.00185	3.31	+0.148
4	3.08227	0.00173	3.12	+0.082
8	3.09596	0.00171	2.92	+0.096

Table 17.4 The *interacting* two-electron dot, exact energy 3. Two hundred and forty gradient-descent steps with the learning rate annealed through 0.05, 0.02, 0.01 and 0.005, then a production run of 500 walkers and 3000 steps. Computed by `rbm.py`.

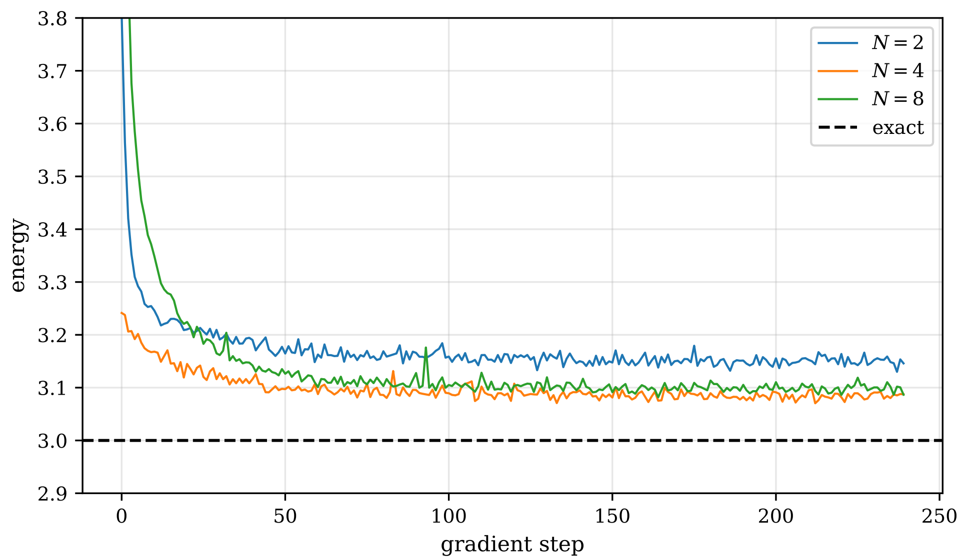

Fig. 17.7 Optimisation of the neural quantum state on the interacting two-electron dot at $\omega = 1$, for $N = 2$, 4 and 8 hidden units. Each curve is the Monte Carlo estimate of $\langle E_L \rangle$ at successive gradient steps, over the 240 steps of the annealing schedule of Table 17.4; the dashed line is Taut's exact energy of 3 a.u. The vertical scale is cut at 2.9 and 3.8, so the first few steps of the $N = 2$ and $N = 8$ runs lie above the frame. Computed by `rbm.py`.

Figure 17.7 shows how those numbers were arrived at, and it is a more informative picture than the table because it separates two questions that the table conflates: whether the optimiser has converged, and whether the ansatz is any good. The answer to the first is plainly yes. All three runs fall out of the frame within twenty or thirty steps, are within 0.05 of their final values by step 50 – while the learning rate is still at its largest value of 0.05 – and spend the remaining two hundred steps fluctuating about a flat line. Annealing the learning rate through 0.02, 0.01 and 0.005 narrows the fluctuations but moves none of the three means: what is left after step 60 is the Monte Carlo noise on an energy estimated from a finite walk, not a systematic descent. Nothing here suggests that a longer run, a better optimiser or a smarter schedule would help.

The answer to the second question is the gap to the dashed line, and it does not close. The $N = 4$ run settles at 3.09 and the $N = 8$ run, with 44 parameters against 24, settles on top of it at 3.10; only $N = 2$, with fourteen parameters, is visibly worse at 3.15. Doubling the hidden layer a second time therefore buys nothing at all, which is the clearest possible statement that the residual 0.09 is not a lack of flexibility. A variational calculation whose error is insensitive to the size of the ansatz has run into a feature of the functional form itself, and Figure 17.8 identifies which one. It is worth contrasting this with the non-interacting case of Table 17.3, where the same optimiser drives the energy to 2.000049 and the variance to 10^{-4}: the machinery is not at fault.

Set beside the other methods applied to the same system:

method	energy	error
Hartree-Fock, 42 orbitals	3.161921	1.6×10^{-1}
RBM, $N=4$ hidden, no physics input	3.082(2)	8.2×10^{-2}
MP2, 42 orbitals	3.027038	2.7×10^{-2}
CCSD, 42 orbitals	3.013626	1.4×10^{-2}
VMC, Padé-Jastrow, two parameters	3.000549(49)	5.5×10^{-4}
exact (Taut)	3.000000	—

Two honest observations.

The first is that the Boltzmann machine beats Hartree-Fock. It was given no single-particle basis, no orbitals and no notion of antisymmetry, and it learned enough of the correlations from the energy alone to do better than a mean-field calculation in a basis of forty-two orbitals. That is a non-trivial thing for fourteen numbers to have achieved.

The second is that it is beaten comfortably by the *two-parameter* trial function of Chapter 13, and by a factor of a hundred and fifty. The reason is visible in the variance column of Table 17.4. Where the Padé-Jastrow function of Eq. (13.13) was constructed to satisfy the cusp condition – so that the $1/r_{12}$ in the Hamiltonian is cancelled exactly by a term from the kinetic energy – the RBM is a Gaussian times a product of smooth sigmoidal factors, and *no choice of its parameters produces a cusp at* $r_{12}=0$. The local energy therefore diverges as $1/r_{12}$ wherever the two electrons meet. The variance of E_L is formally infinite and in practice merely enormous, at three against the 10^{-3} of Chapter 14, and no amount of training removes a divergence the ansatz cannot represent. The energy remains a rigorous variational upper bound; it is simply a mediocre one, and expensive to measure.

Figure 17.8 makes the divergence visible. Each entry in the histogram is already an average over five hundred walkers, so the central limit theorem of Chapter 12 has had every opportunity to render the distribution Gaussian, and it has failed: the histogram is strikingly one-sided, rising almost vertically from nothing at 2.95 to its mode at 3.06 and then falling away in a tail that is still visible at 4.4. The left edge lies 0.13 below the mean and the right tail extends 1.3 above it, a factor of ten in asymmetry after averaging five hundred samples. A distribution with a finite variance would have been symmetric and Gaussian to a good approximation long before that.

The asymmetry is exactly what Eq. (17.83) predicts for a trial function without a cusp. The Coulomb term contributes $+1/r_{12}$ to E_L and nothing in the kinetic term cancels it, so a walker that wanders close to coincidence contributes a large value of *one sign only* – there is no configuration that produces a large negative local energy. Near $r_{12}=0$ the sampling density is finite, and in two dimensions the measure is $r_{12}\,dr_{12}$, so $\langle E_L\rangle$ converges while $\langle E_L^2\rangle \sim \int r_{12}^{-1}dr_{12}$ diverges logarithmically: the mean is well defined, the variance is not. What we measure as $\mathrm{Var}(E_L)\approx 3$ is not a property of the wave function but of how close to coincidence the finite walk happened to come, and it grows slowly with the length of the run. The gap between the mode at 3.06 and the mean at 3.082 is the weight of the tail expressed in atomic units, and it is the same 0.08 that separates the calculation from the exact answer. Two practical consequences follow. The first is that the error bar of 0.0017 in Table 17.4 rests on the blocking analysis of Chapter 14 and should be read with the reservation that blocking assumes a finite variance to begin with. The second is that a cusp in the trial function is not a cosmetic improvement but the difference between an estimator with a variance and one without: the

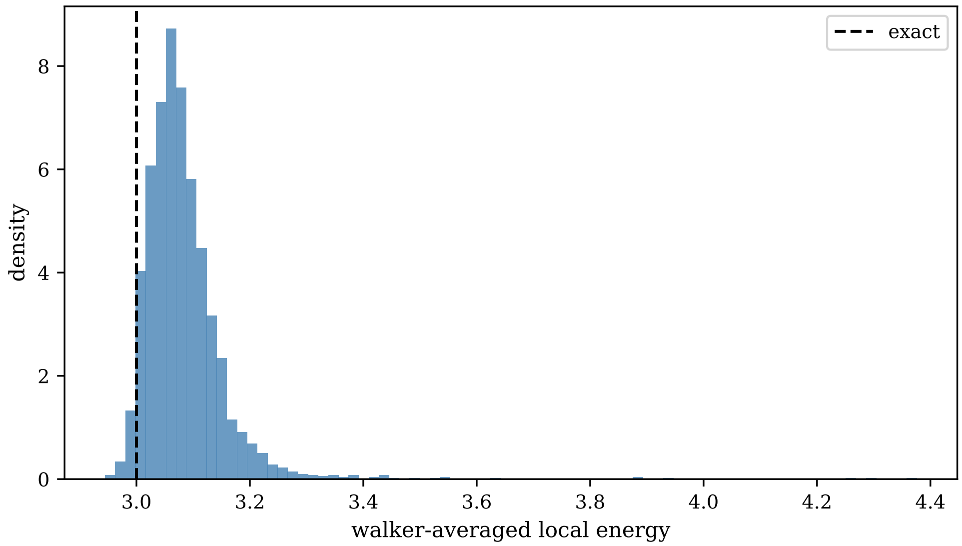

Fig. 17.8 Distribution of the local energy in the production run of the $N = 4$ machine on the interacting dot. Each of the 3000 entries in the histogram is the average of E_L, Eq. (17.83), over the 500 walkers of the ensemble at one Monte Carlo cycle, and the bars are normalised to unit area over eighty bins. The dashed line marks the exact energy 3 a.u. The distribution is sharply cut off just below 3, peaks at 3.06, and carries a tail reaching to 4.4. Computed by `rbm.py`.

Padé-Jastrow function of Chapter 13 produced $\mathrm{Var}(E_L) \sim 10^{-3}$ on the same system, three thousand times smaller, and its histogram would be a spike a few thousandths wide on the scale of this figure.

There is a structural limitation behind this too, already flagged after Eq. (17.71): the Gaussian-binary machine is a mixture of Gaussians all of the same width σ, so the hidden units can move probability around but cannot change the overall extent of the wave function. For the interacting dot the true state is broader than the non-interacting one that Eq. (17.87) fixes, and σ has to be re-tuned by hand or promoted to a variational parameter – Exercise 17.14.

What to conclude. It would be easy to read Table 17.4 as a defeat for machine learning and easy to read it as a triumph, and both readings would be wrong. What it shows is that a generic ansatz with no physics in it reaches, from a standing start, the accuracy of a carefully constructed mean-field theory – and that a small amount of physics, correctly inserted, is still worth more than a large amount of flexibility. The productive response is not to choose between them but to combine them: a neural network multiplying a Slater determinant and a Jastrow factor, so that the cusp and the antisymmetry are exact by construction and the network supplies what is left. That is the subject of Chapter 18.

17.12 Variational autoencoders

We close with the other major family of latent-variable generative models, briefly, because it makes a useful contrast with the Boltzmann machine and because its central identity is the Kullback-Leibler manipulation of Section 17.4 used a second time.

An ordinary *autoencoder* is a pair of neural networks trained together: an encoder $\boldsymbol{z} = g(\boldsymbol{x})$ mapping the input to a low-dimensional latent code, and a decoder $\tilde{\boldsymbol{x}} = f(\boldsymbol{z})$ mapping back, with the reconstruction error $\|\boldsymbol{x} - f(g(\boldsymbol{x}))\|^2$ as the cost. The bottleneck forces the network to discover a compressed representation, and if f and g are linear the optimum is the principal-component subspace, so an autoencoder is a non-linear generalisation of the SVD of Section 1.28. It is *not*, however, a generative model: the encoder produces one point per input, nothing says what the distribution of latent codes is, and a code picked at random decodes to nonsense.

The *variational autoencoder* repairs this by making the latent variable genuinely stochastic. Posit a simple prior $p(\boldsymbol{z})$ – a unit Gaussian – and a decoder that defines $p_{\boldsymbol{\theta}}(\boldsymbol{x}|\boldsymbol{z})$. The model distribution is then

$$p_{\boldsymbol{\theta}}(\boldsymbol{x}) = \int p_{\boldsymbol{\theta}}(\boldsymbol{x}|\boldsymbol{z})\, p(\boldsymbol{z})\, d\boldsymbol{z}, \tag{17.88}$$

which is the same structure as Eq. (17.8) with the hidden units now continuous and the sum an integral. The integral is intractable for the same reason Z was, so training needs the same kind of trick.

Introduce an approximate posterior $q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})$ – the encoder, which outputs a mean and a variance rather than a point – and consider its divergence from the true posterior. Using Bayes' rule $p(\boldsymbol{z}|\boldsymbol{x}) = p(\boldsymbol{x}|\boldsymbol{z})p(\boldsymbol{z})/p(\boldsymbol{x})$,

$$\begin{aligned}
\mathrm{KL}\left(q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) \,\|\, p_{\boldsymbol{\theta}}(\boldsymbol{z}|\boldsymbol{x})\right) &= \left\langle \log q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) - \log p_{\boldsymbol{\theta}}(\boldsymbol{z}|\boldsymbol{x}) \right\rangle_q \\
&= \left\langle \log q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) - \log p_{\boldsymbol{\theta}}(\boldsymbol{x}|\boldsymbol{z}) - \log p(\boldsymbol{z}) \right\rangle_q + \log p_{\boldsymbol{\theta}}(\boldsymbol{x}),
\end{aligned} \tag{17.89}$$

where $\log p_{\boldsymbol{\theta}}(\boldsymbol{x})$ came out of the average because it does not depend on $\boldsymbol{z}$ – the same step that removed $\log Z$ from Eq. (17.56). Rearranging,

$$\boxed{\log p_{\boldsymbol{\theta}}(\boldsymbol{x}) = \underbrace{\left\langle \log p_{\boldsymbol{\theta}}(\boldsymbol{x}|\boldsymbol{z}) \right\rangle_q - \mathrm{KL}\left(q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) \,\|\, p(\boldsymbol{z})\right)}_{\mathscr{L}_{\mathrm{ELBO}}} + \mathrm{KL}\left(q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) \,\|\, p_{\boldsymbol{\theta}}(\boldsymbol{z}|\boldsymbol{x})\right)} \tag{17.90}$$

which is the central equation of variational autoencoders. The last term is non-negative and unknown; dropping it gives a rigorous *lower bound* on the log-likelihood, the *evidence lower bound* or ELBO, and maximising the bound both fits the data and drives the approximate posterior towards the true one. The two terms of the ELBO have clean readings: the first is a reconstruction accuracy, the second a regulariser pulling the latent distribution towards the prior.

Two remarks connect this to the rest of the book. First, the bound is variational in exactly the sense of Chapter 13 – an inequality saturated when the trial object, here $q_{\boldsymbol{\phi}}$, equals the exact one. Second, the gradient of the ELBO with respect to $\boldsymbol{\phi}$ passes through a sampling step, and would therefore need the score-function estimator of Exercise 14.12, which is unbiased but noisy. The *reparametrisation trick* avoids it: write $\boldsymbol{z} = \boldsymbol{\mu}_{\boldsymbol{\phi}}(\boldsymbol{x}) + \boldsymbol{\sigma}_{\boldsymbol{\phi}}(\boldsymbol{x}) \odot \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathscr{N}(0, \boldsymbol{I})$, so that the randomness no longer depends on the parameters and ordinary backpropagation applies.

The contrast with the Boltzmann machine is instructive. Both are latent-variable models; both are trained by a bound or an identity involving a Kullback-Leibler divergence. But the RBM keeps an exact likelihood with an intractable normalisation, and pays for it with Monte Carlo in the negative phase; the VAE keeps a tractable normalisation and pays for it by optimising a bound rather than the likelihood itself. Neither pays nothing.

17.13 Summary

A generative model learns a distribution rather than a decision rule, and for a physicist the natural family is the energy-based one: a Boltzmann distribution $p = e^{-E}/Z$ whose couplings are fitted rather than measured. Its likelihood cannot be differentiated directly, because Z is a sum over every configuration, but its logarithmic derivative is an expectation value, Eq. (17.14), and the gradient of the cost is therefore a *difference of moments*, Eq. (17.61): one average over the data, one over the model. Minimising the negative log-likelihood and minimising the Kullback-Leibler divergence from the data are the same thing, differing by the entropy of the data set.

The model average requires sampling from a distribution known only up to normalisation. Gibbs sampling does this by drawing one variable at a time from its exact conditional – in which the normalisation cancels – and is correct because each such update leaves the target invariant and satisfies detailed balance. It is Metropolis-Hastings with the conditional as proposal, and therefore never rejects, which at low temperature is worth a factor of a hundred in efficiency. Its weakness is that it moves along coordinate axes and mixes slowly when the variables are strongly correlated, which blocked updates alleviate.

The restricted Boltzmann machine makes the conditionals available in closed form by deleting the intra-layer couplings. Its layers are then conditionally independent, Eq. (17.44), each layer can be resampled in one step, and the conditional probability of a binary unit is a logistic function of its local field – not an activation function chosen for convenience but the exact answer. For binary visible units the marginals and conditionals are Eqs. (17.49)–(17.55); for Gaussian visible units they are Eqs. (17.66)–(17.71), the visible conditional being a normal distribution whose mean the hidden units shift. Training is contrastive divergence: a short Gibbs chain started at the data, biased and effective.

Finally, the Gaussian-binary marginal can be used as a variational wave function. With $\Psi = \sqrt{F_{rbm}}$ the sampling density is still an RBM distribution, the derivatives of $\ln\Psi$ are elementary, and the optimisation is the one of Chapter 14 with the RBM parameters in place of α and β. On the two-electron quantum dot the result is $3.082(2)$ against the exact 3: better than Hartree-Fock, worse than a two-parameter Jastrow function, and limited by a cusp the ansatz cannot represent.

What follows from that limitation is the programme of the next chapter. If the difficulty is that a shallow generic ansatz cannot express the short-distance behaviour, there are two ways forward: make the network deeper and more expressive, or build the known physics in and let the network supply only the remainder. Both are pursued in Chapter 18, together with the physics-informed neural networks that impose the Schrödinger equation itself as the training objective.

17.14 Exercises

Exercise 1: the gradient of an energy model

a) Derive Eq. (17.14) in full, and state carefully where the assumption $f \geq 0$ is used.
b) Show that Eq. (17.61) can be written $\partial_{\Theta_i}\mathscr{C}_{LL} = \mathrm{Cov}(O_i, \cdot)$-like differences, and in particular that at the optimum the model reproduces the data moments $\langle x_i \rangle$, $\langle h_j \rangle$ and $\langle x_i h_j \rangle$ exactly.
c) Show that the second derivative of $-\log Z$ with respect to the parameters is minus the covariance matrix of the operators O_i, so that the log-likelihood of an energy model is a concave function of the parameters when the hidden units are absent. Why does the argument fail once hidden units are introduced?
d) Verify Eq. (17.20) line by line, and explain why the first term is independent of $\boldsymbol{\Theta}$.

Answer to b).

For the binary-binary machine $O_{w_{ij}} = \partial E/\partial w_{ij} = -x_i h_j$ by Eq. (17.18), so Eq. (17.61) reads $\partial \mathscr{C}/\partial w_{ij} = \langle x_i h_j \rangle_{\text{data}} - \langle x_i h_j \rangle_{\text{model}}$, and the gradient vanishes exactly when the two agree. The same holds for a_i and b_j with the first moments. A Boltzmann machine at its optimum is therefore the maximum-entropy distribution consistent with the first and second moments of the data – which is the classical statistical-mechanics characterisation of a Boltzmann distribution, arrived at from the other direction.

Answer to c).

Writing $\log Z = \log \sum_{\boldsymbol{x}} e^{-E(\boldsymbol{x};\boldsymbol{\Theta})}$ and differentiating twice,

$$\frac{\partial^2 \log Z}{\partial \Theta_i \partial \Theta_j} = \langle O_i O_j \rangle_{\text{model}} - \langle O_i \rangle_{\text{model}} \langle O_j \rangle_{\text{model}},$$

which is a covariance matrix and therefore positive semidefinite – this is the same computation as the metric $\boldsymbol{S}$ of Section 14.6, and for the same reason. Without hidden units $\langle E \rangle_{\text{data}}$ is linear in $\boldsymbol{\Theta}$, so $\mathscr{C}_{LL}$ is a convex function and the optimisation has no local minima. With hidden units the data term becomes $-\langle \log \sum_{\boldsymbol{h}} e^{-E} \rangle_{\text{data}}$, itself a log-partition function and hence convex, so the cost is a *difference* of two convex functions – and differences of convex functions are not convex. That is the precise origin of the local minima that make Boltzmann machines hard to train.

Exercise 2: Gibbs sampling

a) Prove the invariance (17.26) for a single coordinate, and explain why a composition of invariant kernels is invariant.
b) Prove the detailed-balance relation (17.27), treating the two cases $\boldsymbol{x}_{-i} = \boldsymbol{x}'_{-i}$ and $\boldsymbol{x}_{-i} \neq \boldsymbol{x}'_{-i}$ separately.
c) Show that the deterministic sweep (17.24) leaves $\boldsymbol{\pi}$ invariant but does not in general satisfy detailed balance. Give a three-variable example.
d) Verify Eq. (17.30) and hence that the Metropolis-Hastings acceptance probability for a Gibbs proposal is one.
e) Derive the Ising conditional (17.33) from Eq. (17.31), and show that the Metropolis acceptance ratio $\min(1, e^{-\beta \Delta E})$ for a single spin flip and the Gibbs probability of flipping agree when the local field is large, but not when it vanishes.

Answer to e).

With $\Delta E = 2 s_i H_i^{\text{loc}}$ the Metropolis probability of flipping is $\min\left(1, e^{-2\beta s_i H_i^{\text{loc}}}\right)$, while Gibbs flips with probability $\operatorname{sig}\left(-2\beta s_i H_i^{\text{loc}}\right)$. For $\beta s_i H_i^{\text{loc}} \gg 1$ both are $\approx e^{-2\beta s_i H_i^{\text{loc}}}$ and the methods coincide. At $H_i^{\text{loc}} = 0$, however, Metropolis flips with probability 1 and Gibbs with probability 1/2. That difference is invisible under random-site updating and fatal under a deterministic sweep, as the notebox in Section 17.5.3 shows.

Exercise 3: the non-ergodic sweep

a) Consider the periodic Ising chain at zero field and the Metropolis update with a deterministic left-to-right sweep. Show that a spin whose two neighbours are antiparallel has $\Delta E = 0$ and is therefore always flipped, and that this move translates a domain wall by one site.
b) Show that under such a sweep the number of domain walls is conserved for the moves with $\Delta E \leq 0$, so that walls can never annihilate and the chain is not ergodic.
c) Confirm numerically with `rbm.py` by calling `metropolis_ising_chain` with `random_order=False`, and show that the measured energy is that of the random initial configuration.
d) Show that Gibbs sampling does not suffer from this, and that random-site Metropolis does not either. Which of the two ingredients – the stochastic rule at zero field, or the random site order – is doing the work?

Exercise 4: the binary-binary machine

a) Carry out the marginalisation (17.49) in full, justifying the interchange of sum and product.
b) Derive Eq. (17.51) by the same route.
c) Derive the conditionals (17.53) and (17.55), and verify that they are normalised.
d) Show that with $x_i, h_j \in \{-1,+1\}$ instead of $\{0,1\}$ the conditional becomes $p(h_j = 1|\boldsymbol{x}) = \mathrm{sig}\left(2b_j + 2\boldsymbol{x}^T \boldsymbol{w}_{*j}\right)$, and find the transformation of $(\boldsymbol{a}, \boldsymbol{b}, \boldsymbol{W})$ that maps one convention onto the other.
e) An RBM with N hidden units is a mixture of how many components? Use Eq. (17.50) to argue that the model is a product of experts rather than a mixture, and explain why that distinction matters for its expressive power.

Answer to d).

Substituting $x_i = (1+\tilde{x}_i)/2$ and $h_j = (1+\tilde{h}_j)/2$ with $\tilde{x}, \tilde{h} \in \{-1,1\}$ into Eq. (17.46) and collecting terms gives an energy of the same form with $\tilde{w}_{ij} = w_{ij}/4$, $\tilde{a}_i = a_i/2 + \sum_j w_{ij}/4$ and $\tilde{b}_j = b_j/2 + \sum_i w_{ij}/4$, plus an irrelevant constant. The two conventions therefore describe the same family of distributions, which is why either may be used, but *not* with the same numerical values of the parameters – a point that has caused a good deal of confusion in the literature.

Exercise 5: the Gaussian-binary machine

a) Work out $\partial E_{GB}/\partial a_i$, $\partial E_{GB}/\partial b_j$, $\partial E_{GB}/\partial w_{ij}$ and $\partial E_{GB}/\partial \sigma_i$ from Eq. (17.63), and hence write down the four learning rules corresponding to Eq. (17.62).
b) Carry out the Gaussian integral of Eq. (17.67) in full, showing every step of the completion of the square.
c) Derive Eq. (17.71) and confirm that it is a normalised probability density in x_i.
d) Show that the Gaussian-binary RBM is a mixture of 2^N Gaussians of *equal* width, identify the mixing weights, and hence explain the remark after Eq. (17.71) that the model cannot change the width of the distribution it represents.
e) What goes wrong if σ_i is allowed to become large? And if it becomes small?

Exercise 6: neural quantum states

a) Derive Eq. (17.75) and check the factor of $4\sigma^2$ in the exponent.
b) Derive the parameter derivatives (17.80) and the coordinate derivatives (17.86). Show that the derivative with respect to b_n equals $\frac{1}{2}p(h_n = 1|\boldsymbol{x})$ and interpret this.
c) Derive Eq. (17.82) from the identity $\nabla^2\Psi/\Psi = \nabla^2 \ln\Psi + (\nabla \ln\Psi)^2$, and state where the assumption that Ψ is real and positive is used.
d) Show that $e^Q/(1+e^Q)^2 = \mathrm{sig}(Q)\,(1-\mathrm{sig}(Q))$, and hence that the second derivative in Eq. (17.86) involves the variance of the hidden unit.
e) Show that setting $\boldsymbol{W} = 0$ and $\sigma^2 = 1/(2\omega)$ reproduces the exact non-interacting ground state, and that E_L is then constant and equal to $M\omega/2$. Confirm that this is 2 for the two-electron two-dimensional dot at $\omega = 1$.

Answer to e).

With $\boldsymbol{W} = 0$ and $\boldsymbol{a} = 0$, Eq. (17.79) gives $\ln\Psi = \mathrm{const} - \sum_m x_m^2/(4\sigma^2)$, so $\partial_m \ln\Psi = -x_m/(2\sigma^2)$ and $\partial_m^2 \ln\Psi = -1/(2\sigma^2)$. Substituting into Eq. (17.83) with no interaction term,

$$E_L = \frac{1}{2}\sum_m^M \left(-\frac{x_m^2}{4\sigma^4} + \frac{1}{2\sigma^2} + \omega^2 x_m^2 \right).$$

The x_m^2 terms cancel precisely when $1/(4\sigma^4) = \omega^2$, that is $\sigma^2 = 1/(2\omega)$, leaving $E_L = M\omega/2$, a constant independent of the configuration. For $M = 4$ and $\omega = 1$ this is 2, and the vanishing variance in Table 17.3 is this cancellation happening numerically.

Exercise 7: variational autoencoders

a) Derive Eq. (17.90) from Eq. (17.89), and state why the ELBO is a lower bound and when it is tight.
b) For $q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) = \mathcal{N}(\boldsymbol{\mu}, \mathrm{diag}(\boldsymbol{\sigma}^2))$ and $p(\boldsymbol{z}) = \mathcal{N}(0, \boldsymbol{I})$, show that $\mathrm{KL}(q\|p) = \frac{1}{2}\sum_k \left(\mu_k^2 + \sigma_k^2 - \log\sigma_k^2 - 1\right)$.
c) Explain the reparametrisation trick and why it reduces the variance of the gradient relative to the score-function estimator of Exercise 14.12(a).
d) Compare the roles of the intractable object in a VAE and in an RBM. Which model optimises the likelihood itself, and which a bound?

Exercise 8: numerical explorations

a) Using `rbm.py`, reproduce the correlation times of Gibbs sampling on the bivariate Gaussian as a function of ρ, and compare with a Metropolis random walk on the same target. At which ρ does each become impractical?
b) Implement a blocked Gibbs sampler for the bivariate Gaussian – that is, draw both coordinates at once from the joint – and confirm that the correlation time drops to one for every ρ. Explain why this is cheating, and what the honest analogue is for an RBM.

c) Reproduce Table 17.2 and study the effect of the number of hidden units at fixed k. How many hidden units are needed before the KL divergence stops improving, and how does that compare with the $\log_2 14 \approx 3.8$ bits of information in the data set?
d) Add an L^2 penalty on the weights to the contrastive-divergence update and study the effect on the exact log-likelihood. Does regularisation help on a data set this small?
e) Train the machine on a 4×4 bars-and-stripes set, where the partition function is still enumerable, and check that the conclusions of Table 17.2 survive.
f) Reproduce Table 17.3 and verify Exercise 6(e) numerically: start the optimiser from non-zero weights and watch them be driven to zero.
g) Reproduce Table 17.4. Then plot a histogram of the local energy from the production run and confirm that it has a long tail towards $+\infty$; estimate the exponent of the tail and explain it from the behaviour of E_L as $r_{12} \to 0$.
h) Promote σ to a variational parameter. Derive $\partial \ln \Psi / \partial \sigma$ for Eq. (17.79), add it to the optimiser, and see how much of the 0.082 discrepancy of Table 17.4 it removes.
i) Multiply the RBM trial function by a Padé-Jastrow factor $\exp(r_{12}/(1+\beta r_{12}))$ with β variational, so that the cusp condition is satisfied exactly, and repeat the calculation. You should recover something close to the result of Chapter 14, with a variance smaller by orders of magnitude. This is the simplest example of the hybrid ansatz of Chapter 18.
j) Replace the plain gradient descent of `rbm.py` by the stochastic reconfiguration of Section 14.6, whose metric $S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle$ is built from the same logarithmic derivatives (17.80). With tens of parameters rather than two, the improvement should be considerably more visible than it was in Table 14.6.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter17/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/boltzmannmachines.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter17/`.

`rbm.py` Gibbs sampling checked against a Cholesky draw and a transfer matrix; `BinaryBinaryRBM` and `GaussianBinaryRBM` with `check_identities` verifying every marginal and conditional; contrastive divergence on bars-and-stripes with the exact log-likelihood and Kullback–Leibler divergence; and `NeuralQuantumState`, a Gaussian–binary machine used as a trial wave function for the two-electron dot. Runs in about five minutes.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 18
Neural quantum states and physics-informed neural networks

Chapter 17 ended with a Boltzmann machine that reached 3.082 on a system whose exact answer is 3, and with a diagnosis: the ansatz contained no physics, could not produce a cusp, and paid for it in a local-energy variance of three. The two-parameter Padé-Jastrow function of Chapter 13, which was built from physics, did a hundred and fifty times better. The conclusion drawn there was that the productive move is not to choose between the two but to combine them, and this chapter carries that out.

There are two distinct ideas to keep apart. The first is the *ansatz*: what functional form we allow the wave function to take, and how much of the known physics – antisymmetry, the cusp condition, the symmetries of the Hamiltonian – is built into it exactly rather than learned. The second is the *training objective*: what we minimise. Chapters 13 to 14 minimised the energy, which is a variational principle and therefore returns an upper bound. A physics-informed neural network may instead minimise the residual of the Schrödinger equation itself, which is a different thing, and the difference turns out to matter a great deal.

The chapter is organised around those two ideas.

1. **Physics-informed neural networks** (Sections 18.3–18.4): what a PINN is, the automatic differentiation it rests on, and a careful answer to the question of whether a PINN respects the variational principle. The answer is that it depends entirely on the loss, and the failure mode is demonstrated numerically.
2. **The ansatz** (Sections 18.5–18.6): the Slater-Jastrow-neural-correlator form, and backflow – both the classical analytic version and the learned many-body map, with the proof that antisymmetry survives and a numerical demonstration of what backflow can and cannot do to the nodal surface.
3. **Results** (Sections 18.7–18.8): quantum dots with $N = 2$ to 20 electrons across five orders of magnitude in confinement, benchmarked against diffusion Monte Carlo and exact diagonalisation.

The programs are in `pinn.py`, which implements second-order forward automatic differentiation by hand, demonstrates the variational failure of residual training on a problem with a known exact answer, and optimises a Slater-Jastrow-neural ansatz for the two-electron dot. This chapter is a first pass at a subject that is moving quickly, and it will grow.

18.1 Quantum dots, and why a new method

Semiconductor quantum dots are artificial atoms. The particle number N, the confinement strength ω, the interaction strength and the external fields are all tunable, which makes them a controlled laboratory for interacting fermions in reduced dimensions and puts them at the intersection of condensed matter physics, quantum information and nanotechnology. Even a handful of electrons shows rich many-body behaviour: Wigner molecule formation as the confinement is relaxed, spin ordering, and non-trivial excitation spectra. Coulomb repulsion, exchange and quantum confinement all matter at once, and none of them can be treated as a small perturbation across the whole parameter range.

A second motivation is experimental. Electrons floating on the surface of liquid helium form an exceptionally clean realisation of the same physics: the electrons sit in a vacuum above the liquid, bound by their own image charge, with no lattice, no impurities and correspondingly long coherence times. Electrodes beneath the helium shape the trapping potential, so a double well – and hence a two-electron system whose Coulomb coupling can be tuned continuously – can be built to order. The device physics of such platforms, and the use of machine learning to control them, belong to Chapters 19 and 20; here we take the trapping potential as given and ask only how to solve the resulting many-body problem.

Why the established methods run out.

Each of the methods developed earlier in this book has a regime in which it is the right tool and a regime in which it fails.

- **Exact diagonalisation** (Chapter 5) is limited by the exponential wall to very small N.
- **Quantum Monte Carlo** (Chapters 13 and 15) scales well but carries the fermion sign problem and, in its fixed-node form, an error controlled entirely by the nodal surface of the trial function.
- **Density functional theory** (Chapter 8) is cheap but rests on uncontrolled approximations, and is least reliable exactly where the correlations are strongest.
- **Mean-field and perturbative methods** (Chapters 6 and 9) fail once the correlations are large – which, at small ω, they are.

All of them, in addition, become awkward when one wants to sweep a high-dimensional parameter space or model a realistic device geometry, because each point of the sweep is a fresh calculation. This is the practical opening for machine learning: a wave function represented as a smooth function of the coordinates *and* of the parameters can, in principle, be transferred from one point to the next.

18.2 Two neural paradigms

There are two distinct ways to put a neural network into a many-body calculation, and it is worth being explicit about the difference before developing either.

Method	Core idea	Natural regime
Variational Monte Carlo with a neural ansatz	Minimise $\langle \hat{H} \rangle$ by Monte Carlo sampling of $\lvert\Psi_\theta\rvert^2$	Large N, ground states, lattice systems
Physics-informed neural network	Enforce the equation and its symmetries as residuals in the loss, evaluated at collocation points	Continuous space, complex geometries, dynamics, excited states

Table 18.1 The two ways of using a network. The first replaces the trial function of Chapter 13 and keeps everything else; the second replaces the training objective as well.

The first is what Chapter 17 did with a restricted Boltzmann machine: the network supplies Ψ_θ, the energy is estimated by Monte Carlo, and the parameters are moved by the gradient of Eq. (14.2). Nothing about the variational framework changes.

The second is more radical. A PINN represents the wave function as a continuous function of the coordinates and imposes the Schrödinger equation – together with antisymmetry, normalisation and boundary conditions – as explicit terms in a loss function evaluated on a set of sampled points. It needs no Markov chain, and the wave function it produces can be differentiated and evaluated anywhere, which is what makes excited states, complex geometries and time-dependent problems natural. It also, as Section 18.4 shows, gives up something important unless one is careful.

18.3 Physics-informed neural networks

The general setting is a differential equation. Let $u : \Omega \subset \mathbb{R}^d \to \mathbb{R}$ satisfy

$$\mathscr{N}[u](x) = 0 \quad \text{on } \Omega, \qquad u = g \quad \text{on } \partial\Omega, \tag{18.1}$$

with $\mathscr{N}$ a differential operator. A PINN represents u by a neural network u_θ and minimises

$$\boxed{\mathscr{L}(\theta) = \lambda_{\text{data}} \sum_i \lvert u_\theta(x_i) - y_i\rvert^2 + \lambda_{\text{PDE}} \sum_j \left\lvert \mathscr{N}[u_\theta](\xi_j)\right\rvert^2 + \lambda_{\text{BC}} \sum_k \lvert u_\theta(\zeta_k) - g(\zeta_k)\rvert^2} \tag{18.2}$$

The three terms are a data term over observations $\{x_i, y_i\}$, a *residual* term over *collocation points* ξ_j drawn from the interior of Ω, and a boundary term over points ζ_k on $\partial\Omega$. The weights λ balance them. For a quantum many-body problem there is usually no data at all, the operator is $\mathscr{N}[\Psi] = (\hat{H} - E)\Psi$, and the boundary condition is decay at infinity.

Automatic differentiation.

The residual contains derivatives of the network with respect to its *inputs*, and for the Schrödinger equation second derivatives. These are not obtained by finite differences but by *automatic differentiation*, which evaluates them exactly, to floating-point precision, by propagating derivative information through the same computational graph that evaluates the network. For a network of layers $a^{(k+1)} = \varphi\left(\boldsymbol{W}^{(k)} a^{(k)} + \boldsymbol{b}^{(k)}\right)$ the forward-mode recursion is

$$
\begin{aligned}
\frac{\partial z}{\partial u_k} &= \boldsymbol{W}\frac{\partial a}{\partial u_k}, \qquad \frac{\partial^2 z}{\partial u_k \partial u_l} = \boldsymbol{W}\frac{\partial^2 a}{\partial u_k \partial u_l}, \\
\frac{\partial a'}{\partial u_k} &= \varphi'(z)\frac{\partial z}{\partial u_k}, \\
\frac{\partial^2 a'}{\partial u_k \partial u_l} &= \varphi'(z)\frac{\partial^2 z}{\partial u_k \partial u_l} + \varphi''(z)\frac{\partial z}{\partial u_k}\frac{\partial z}{\partial u_l},
\end{aligned} \tag{18.3}
$$

which for $\varphi = \tanh$, with $t = \tanh z$, reads $\varphi' = 1 - t^2$ and $\varphi'' = -2t(1-t^2)$. `pinn.py` implements exactly this and checks it against central differences, agreeing to 5×10^{-12} for the gradient and 4×10^{-8} for the Hessian – the latter limited by the finite difference, not by the analytic result. Writing the recursion out makes its cost visible: each additional derivative order multiplies the work by the number of inputs, which is why the Laplacian of a network in $3N$ dimensions is expensive and why practical codes sometimes fall back on finite differences for large N.

The activation function is not a free choice.

Equation (18.3) contains φ''. A PINN for a second-order equation therefore requires an activation with a non-trivial second derivative, and the almost universal default of machine learning, $\mathrm{ReLU}(z) = \max(0, z)$, has $\varphi'' = 0$ almost everywhere. It is *unusable* for the Laplacian: the residual it produces is identically wrong. Smooth activations – tanh, GELU, SiLU, softplus – are required, and for equations of order ℓ the activation must be at least ℓ times differentiable in a useful sense.

What generalisation means here.

In ordinary supervised learning, generalisation means accuracy on unseen data. In physics-informed learning it means *consistency with physical laws*, which is a stronger and different requirement, and three mechanisms contribute to it: the algorithmic stability of gradient-based optimisation, the implicit bias of such optimisation towards smooth, low-complexity solutions, and the architectural inductive biases we deliberately build in. The last is the one we control, and the rest of this chapter is largely about using it well.

A practical checklist.

- Prefer weak, variational or energy formulations when the strong form of the equation is unstable.
- Hard-enforce essential boundary conditions by construction rather than penalising them softly.
- Balance the loss terms; an unbalanced λ silently turns the problem into a different one.
- Use smooth activations, and architectures that control the derivatives that appear in the residual.
- Exploit symmetry and equivariance in the architecture – it is free accuracy, and Section 18.6.4 shows it can also be a matter of correctness rather than efficiency.
- Use second-order optimisers: L-BFGS (Section 14.4.4) or stochastic reconfiguration (Section 14.6).
- Use importance or stratified sampling for the collocation points, and space-time protocols for dynamics.

18.4 Does a PINN respect the variational principle?

This is the central question of the chapter, and it has a precise answer: it depends entirely on how the loss is written, and the natural-looking choice is the wrong one.

Recall the variational principle. For any admissible trial state ψ_θ,

$$E[\psi_\theta] = \frac{\langle \psi_\theta | \hat{H} | \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle} \geq E_0, \tag{18.4}$$

with equality only for the exact ground state. *Any* trial state gives an upper bound, which is what makes the number produced by a variational calculation meaningful: it can be compared with another upper bound, and lower is better. Without that guarantee an energy is just a number.

Two training strategies.

A PINN represents ψ_θ by a network, and there are two ways to train it.

1. Variational, or energy-based.

$$\mathscr{L}(\theta) = \frac{\langle \psi_\theta | \hat{H} | \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle}$$

This is Rayleigh-Ritz minimisation, exactly as in Chapter 13. The variational principle is preserved: $E_\theta \geq E_0$ for every θ, at every stage of training.

2. Residual, or equation-based.

$$\mathscr{L}(\theta) = \left\| \left(\hat{H} - E \right) \psi_\theta \right\|^2$$

This enforces the Schrödinger equation directly, and is the natural PINN formulation in the sense of Eq. (18.2). Residual minimisation is *not* energy minimisation.

For the second strategy there is no guarantee whatever that $E_\theta \geq E_0$. The failure modes are: energies below the true ground state, convergence to an incorrect eigenstate – the residual is small at *every* eigenpair, not only the lowest – and the stability problem that a small residual does not imply a small error in the solution.

18.4.1 The failure, demonstrated

The argument is easy to make abstractly and easy to disbelieve, so `pinn.py` makes it concrete on a problem where the exact answer is known to machine precision. Take the quartic oscillator

$$\hat{H} = -\frac{1}{2}\frac{d^2}{dx^2} + \frac{1}{2}x^4, \tag{18.5}$$

whose spectrum follows from diagonalising on a grid: $E_0 = 0.530181$, $E_1 = 1.899833$, $E_2 = 3.727839$. Use the one-parameter family $\psi_a(x) = e^{-ax^2/2}$, which cannot represent the exact state, so that both methods must be wrong – the question being *how*.

The variational route.

With $\langle T\rangle = a/4$, $\langle x^2\rangle = 1/(2a)$ and $\langle x^4\rangle = 3/(4a^2)$,

$$E(a) = \frac{a}{4} + \frac{3}{8a^2}, \tag{18.6}$$

stationary at $a = 3^{1/3} = 1.4422$ and giving $E = 0.540844$. This lies $+0.010663$ *above* E_0, as it must: Eq. (18.6) is a Rayleigh quotient for every a, so every value of it is an upper bound.

The residual route.

A PINN minimises the residual at collocation points sampled over the domain – uniformly, in the simplest and commonest case. Dividing through by ψ, which one does to stop the amplitude collapsing (see below), the residual is the mean squared deviation of the local energy from a free parameter E,

$$\mathscr{L}(a,E) = \frac{1}{J}\sum_{j=1}^{J}\left(E_L(\xi_j;a) - E\right)^2, \qquad E_L(x;a) = \frac{a}{2} - \frac{a^2x^2}{2} + \frac{x^4}{2}. \tag{18.7}$$

For fixed a the optimal E is the *unweighted mean* of the local energy over the collocation points. *That is not a Rayleigh quotient*: the measure is the collocation measure, not $|\psi|^2$, and nothing bounds it below. Table 18.2 shows what happens.

L	a^*	E^*	$E^* - E_0$
2	1.8531	0.238116	−0.292065
3	2.7795	−2.094219	−2.624400
4	3.7059	−9.156153	−9.686334
6	5.5568	−52.812539	−53.342719
variational, same family		0.540844	+0.010663

Table 18.2 Minimising the collocation residual (18.7) for the quartic oscillator on uniformly sampled points in $[-L,L]$, over the same one-parameter family for which the variational calculation gives 0.540844. Every residual-trained energy lies *below* the exact ground state $E_0 = 0.530181$, and the deficit grows without bound as the collocation domain is enlarged. Computed by `pinn.py`.

This is not a pathological construction. It is what one gets by writing down the obvious PINN loss for the Schrödinger equation and sampling collocation points uniformly over a box, which is the default in most PINN tutorials. The reported energy is below the true ground state by an amount that depends on the size of the box.

Figure 18.1 draws those four rows against the size of the box, and the first thing to notice is what has become of the two horizontal reference lines. The exact ground state and the best that a single Gaussian can do differ by 0.010663, and on a vertical axis that has to accommodate −52.8 they are one line thickness apart: the entire variational story – the whole gap between the exact answer and a deliberately inadequate one-parameter ansatz – is invisible at the scale on which the residual error has to be drawn. The residual curve leaves those two lines at the smallest box tested and never returns, and there is no plateau and no hint of one, because the deficit is not converging to anything. For uniformly sampled points the loss (18.7) is the variance of the local energy over $[-L,L]$; using the uniform moments

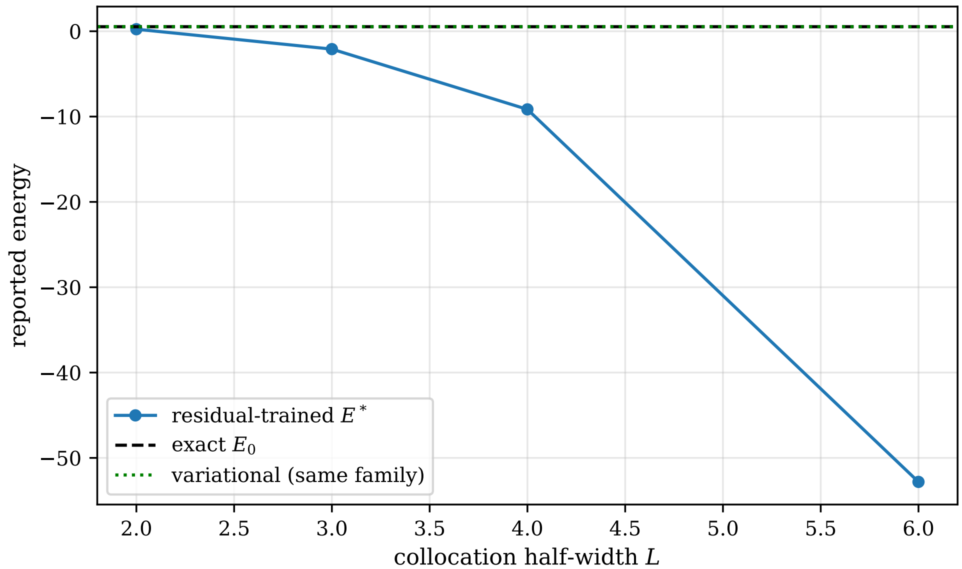

Fig. 18.1 Energy reported by the collocation residual (18.7) for the quartic oscillator, as a function of the half-width L of the box of uniformly sampled collocation points, minimised over the one-parameter Gaussian family $\psi_a = e^{-ax^2/2}$. The circles are the four rows of Table 18.2, the dashed line is the exact ground state $E_0 = 0.530181$ and the dotted line the variational optimum 0.540844 obtained from the very same family. The two reference lines differ by 0.010663 and are indistinguishable on an axis that must reach −53; the residual-trained energy leaves them at the smallest box tested and falls away as L^4 without approaching any limit.

$\overline{x^2} = L^2/3$, $\overline{x^4} = L^4/5$, $\overline{x^6} = L^6/7$ and $\overline{x^8} = L^8/9$, that variance is $a^4L^4/45 - 4a^2L^6/105 + 4L^8/225$, stationary at $a^* = L\sqrt{6/7} = 0.9258\,L$ – which is exactly the linear growth of the second column of Table 18.2 – and the energy reported there is

$$E^*(L) = \sqrt{\frac{3}{14}}\,L - \frac{3L^4}{70}. \tag{18.8}$$

The quartic term is the curve in the figure. It gives −2.08, −9.13 and −52.77 at $L = 3$, 4 and 6 against the tabulated −2.09, −9.16 and −52.81, the small discrepancies being the finite number of collocation points; and it accounts for the observed factor of 5.5 between the last two deficits, close to $(6/4)^4 = 5.06$. The lesson is sharper than "the residual is not variational". The number a residual-trained PINN reports depends on where one decided to put the collocation points, and it can be driven arbitrarily far below the true ground state by the entirely innocent-looking decision to sample over a larger domain. Nothing in the training diagnostics distinguishes the four points in Figure 18.1: each is a converged minimum of a loss that behaved impeccably.

Figure 18.2 shows the same failure from the other direction: the box is held fixed at $L = 3$ and the variational parameter is swept instead. Drawn this way the two objectives can be compared point by point, and the asymmetry between them is stark. The blue curve is a Rayleigh quotient at *every* a and not merely at its minimum, which is the content of Eq. (18.4); it therefore has a floor at E_0 built into it, and the closest it comes is 0.5408, the familiar 0.0107 above. The red curve is $E^*(a) = a/2 - a^2L^2/6 + L^4/10$, a downward-opening parabola in a that is unbounded below, so along this one family there is no lower bound on the reported energy at all. What arrests the optimiser is not the energy but the loss, and the loss arrests it at $a = 2.78$, where the reported energy is −2.1291, some 2.66 below the true ground state. The grey dotted curve makes the incentive quantitative: at the variational optimum $a = 1.4422$

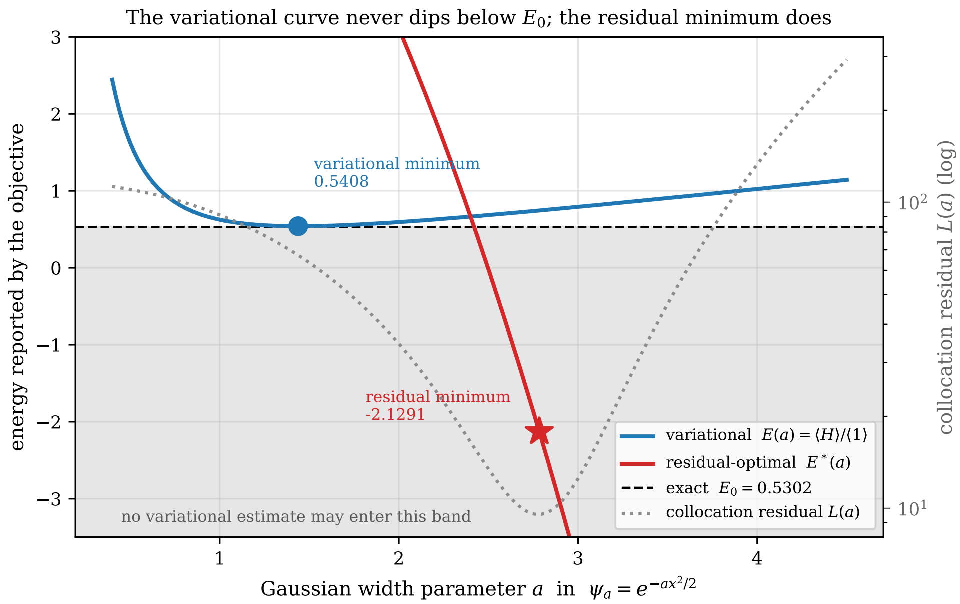

Fig. 18.2 The two objectives plotted against the *same* variational parameter, the width a of $\psi_a = e^{-ax^2/2}$, for the quartic oscillator with collocation points spread uniformly over $[-3,3]$. The blue curve is the Rayleigh quotient $E(a) = a/4 + 3/(8a^2)$ of Eq. (18.6), the red curve the energy $E^*(a)$ that the collocation residual reports, and the grey dotted curve the residual $\mathscr{L}(a)$ itself, read against the logarithmic right-hand axis. The shaded band below the exact $E_0 = 0.530181$ is territory no variational estimate may enter: the blue curve touches its upper edge at $a = 1.44$ and turns back, whereas the red curve crosses it at $a = 2.42$ and keeps going, so that the entirely well-behaved minimum of $\mathscr{L}(a)$ at $a = 2.78$ sits above a reported energy of -2.1291.

the collocation residual is about 67, at $a = 2.78$ it is about 9.5. A residual objective therefore *prefers*, by a factor of seven, the parameter that reports an energy of the wrong sign.

The mechanism is visible in the local energy itself, $E_L(x) = a/2 - a^2x^2/2 + x^4/2$. What the residual asks for is that E_L be as flat as possible across the box in the unweighted least-squares sense, and narrowing the Gaussian buys flatness cheaply: a larger a deepens the middle term until it cancels the quartic over most of the interval, the two balancing at $x = \pm 1.96$ for $a = 2.78$, comfortably inside $[-3,3]$. That the resulting ψ has almost no amplitude where this cancellation is being arranged – $|\psi(1.96)/\psi(0)|^2 = e^{-ax^2} \approx 2\times 10^{-5}$ – is entirely invisible to the loss, because the collocation measure weights the point at $x = 1.96$ exactly as heavily as the point at the origin. Nothing in the objective knows that the measure should have been $|\psi|^2$, and an unweighted mean of the local energy has no reason whatever to lie above E_0. This is the concrete reason why Eq. (18.9) puts the quotient first and the residual second, and why the pipeline of Section 18.7 does not end until the stochastic-reconfiguration tail has restored the bound.

And without normalisation, worse.

The raw residual $\|(\hat{H} - E)\psi_\theta\|^2$, with the network amplitude unconstrained, has a trivial global minimum: $\psi_\theta \equiv 0$, for *any* E whatever. `pinn.py` multiplies the trial function by a constant and recomputes:

amplitude	residual	apparent E
1	1.598×10^{-2}	0.540844
10^{-1}	1.598×10^{-4}	0.540844
10^{-2}	1.598×10^{-6}	0.540844
10^{-3}	1.598×10^{-8}	0.540844

The residual falls as the square of the amplitude while the energy is untouched. An optimiser presented with this loss is rewarded for making the wave function small, which is not physics; and a training curve showing the loss falling by six orders of magnitude looks, to the casual eye, like spectacular convergence.

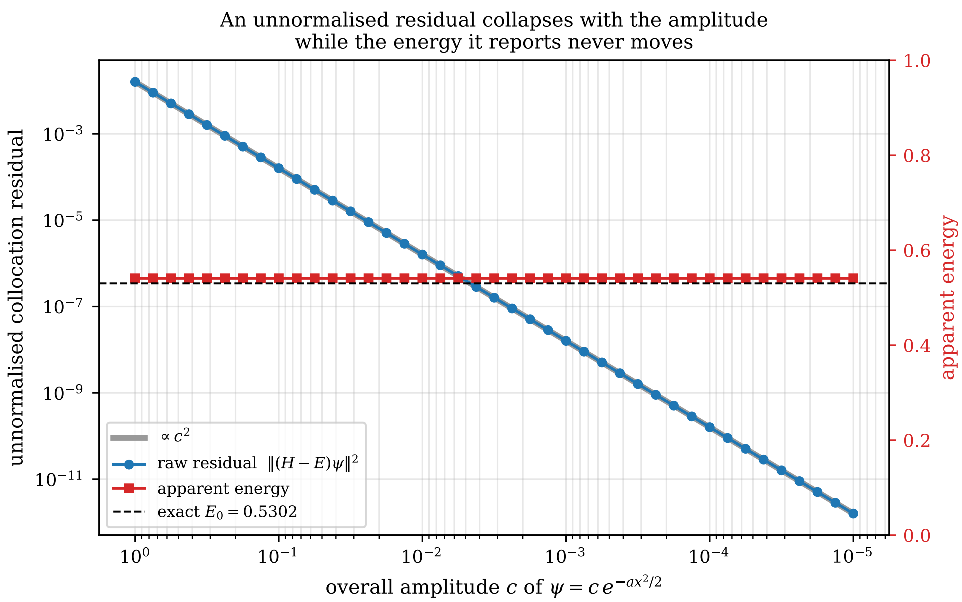

Fig. 18.3 Raw collocation residual $\|(\hat{H}-E)\psi\|^2$ of the unnormalised trial function $\psi = c\,e^{-ax^2/2}$ as the overall amplitude c is reduced from 1 to 10^{-5}, with c decreasing to the right. Blue circles are the residual against the logarithmic left-hand axis and the thick grey line is the guide $\propto c^2$; red squares are the energy the same objective reports, against the linear right-hand axis, with the exact $E_0 = 0.530181$ dashed. The loss falls by ten orders of magnitude, from 1.6×10^{-2} to 1.6×10^{-12}, and sits on the guide line throughout, while the reported energy stays at 0.540844 to machine precision.

Figure 18.3 continues the table down to $c = 10^{-5}$, and the picture is worth more than the four rows above it: ten orders of magnitude in the loss, and not one digit of the energy. The c^2 guide line lies exactly on top of the data, because that is all that is happening – the Hamiltonian is linear, so $(\hat{H}-E)(c\psi) = c(\hat{H}-E)\psi$ and the squared norm carries a factor c^2 out in front, whatever E and whatever ψ. It is worth being clear that nothing here has gone wrong with the physics. The Gaussian used is the variational optimum, and the energy it reports, 0.540844, is a perfectly respectable upper bound sitting the usual 0.0107 above the dashed line. The point is that the loss knows nothing about that number and never moves it: a red curve that is flat to sixteen digits while the blue one descends through ten decades is as clean a statement as one could want that the two are measuring different things.

The moral generalises past this toy. A loss curve is a convergence diagnostic only if the quantity being minimised is invariant under the symmetries of the ansatz, and the overall scale of a wave function is the most trivial such symmetry there is. The energy quotient of Eq. (18.4) is invariant under it identically, the factor c^2 cancelling between numerator and denominator, which is the practical argument for making the energy the primary objective rather than an afterthought. Dividing the residual through by ψ, as was done in Eq. (18.7), removes this particular disease – the local energy does not care about c either – but it should be said plainly that it cures only this one: Figure 18.2 was computed with the residual already divided through, and the bound is still gone.

18.4.2 Practical consequences

Normalisation must be enforced explicitly, or – better – made irrelevant by using the energy quotient, in which it cancels identically.
Symmetry must be built in: fermionic antisymmetry through a Slater determinant, and the spin and spatial symmetries of the Hamiltonian through the architecture. A soft penalty is not the same thing.
Nodal structure is critical for fermions. Incorrect nodes give an incorrect ground state, and no amount of training in the interior of a nodal pocket repairs a node in the wrong place.
Boundary conditions should be enforced by construction where possible. A soft boundary penalty competes with the other loss terms and can be dominated by them.

Hybrid losses.

The practical compromise is

$$\mathscr{L}(\theta) = \frac{\langle \psi_\theta | \hat{H} | \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle} + \lambda\, \mathscr{L}_{\mathrm{BC}} + \mu\, \mathscr{L}_{\mathrm{sym}}, \tag{18.9}$$

with the Rayleigh quotient as the *primary* objective, so that the variational character is preserved, and the residual and symmetry terms as auxiliaries. The recommendation that follows from Table 18.2 is blunt: *do not let the PDE residual be the dominant term if a variational upper bound is wanted*. Where the residual is genuinely useful is as a *pretraining* objective – it converges quickly from a random initialisation, where energy minimisation is slow – after which the energy quotient takes over and restores the bound. That is exactly the pipeline used in Section 18.7.

Variational structure must be built into the loss function. It is not a property of the network, the optimiser or the sampling. A PINN preserves the variational principle when the loss *is* the energy quotient, the normalisation is handled by that quotient, the antisymmetry is exact by construction, and the optimisation ends with a natural-gradient or stochastic-reconfiguration tail. It loses it when the residual dominates the loss, the normalisation is a soft penalty, there is no antisymmetric architecture, or the training is pure gradient descent on a residual.

Property	**Variational PINN**	**Residual PINN**
Loss function	$\langle\psi_\theta\|\hat{H}\|\psi_\theta\rangle/\|\psi_\theta\|^2$	$\|(\hat{H}-E)\psi_\theta\|^2$
Variational bound	**yes**, $E_\theta \geq E_0$	no guarantee
Enforces the equation	implicitly, at the minimum	explicitly
Normalisation	automatic, cancels in the quotient	must be added
Risk of $E < E_0$	none	present
Excited states	needs orthogonality constraints	natural, but which eigenstate is found is not controlled
Connection to QMC	direct (Rayleigh-Ritz)	indirect

Table 18.3 The two formulations compared. Neither column is uniformly better; the point is that they answer different questions, and only the first returns a number that can be compared with another calculation.

18.5 The ansatz: Slater, Jastrow and a neural correlator

Having settled what to minimise, we turn to what to minimise *over*. The form used throughout this chapter is

$$\boxed{\Psi_{\theta,\beta}(\boldsymbol{R}) = \underbrace{\mathrm{SD}(\boldsymbol{R}')}_{\text{antisymmetry}} \exp\Big(\underbrace{U(\boldsymbol{R})}_{\text{analytic pair term}} + \underbrace{W_\theta(\boldsymbol{R})}_{\text{neural correlator}}\Big), \qquad \boldsymbol{R}' = \boldsymbol{R} + \underbrace{\Delta_\beta(\boldsymbol{R})}_{\text{backflow}}} \tag{18.10}$$

with four components, each doing one job:

- $\mathrm{SD}(\boldsymbol{R}')$ – a Slater determinant, giving *exact* fermionic antisymmetry. Not a penalty, not learned: exact.
- $U(\boldsymbol{R})$ – an analytic pair term carrying the cusp conditions, evaluated on the *unscaled* distances r_{ij}.
- $W_\theta(\boldsymbol{R})$ – a compact permutation-invariant neural correlator, evaluated on *trap-scaled* coordinates $\tilde{\boldsymbol{x}}_i = \sqrt{\omega}\,\boldsymbol{x}_i$.
- $\Delta_\beta(\boldsymbol{R})$ – an optional backflow displacement, applied *only* inside the determinant, where it can move the nodes.

The coordinate convention is deliberate and worth dwelling on. The trap sets one length scale, $\ell_{\text{trap}} \sim \omega^{-1/2}$, and the Coulomb interaction sets another; across five orders of magnitude in ω these separate by a factor of order a hundred. Feeding the network trap-scaled coordinates $\sqrt{\omega}\,\boldsymbol{x}_i$ makes its inputs of order unity at every ω, so the same architecture and the same initialisation work throughout the sweep. Evaluating U on physical distances keeps the cusp condition, which is a statement about the bare Coulomb singularity, independent of the trap. Mixing the two up is the commonest reason a network that works at one ω fails at another.

The analytic pair term.

The cusp conditions of Section 13.3 are imposed exactly by

$$U(\boldsymbol{R}) = \sum_{i<j} u_{\sigma_i\sigma_j}(r_{ij}), \qquad u_{\sigma\sigma'}(r) = \gamma_{\sigma\sigma'}\, r\, e^{-r}, \tag{18.11}$$

with spin-dependent coefficients fixed by the dimension d,

$$\gamma_{\uparrow\downarrow} = \frac{1}{d-1}, \qquad \gamma_{\uparrow\uparrow} = \frac{1}{d+1}. \tag{18.12}$$

For $d=2$ these are 1 and $1/3$, which are the coefficients used in Chapter 16; for $d=3$ they are the familiar $1/2$ and $1/4$. The factor e^{-r} makes the term short-ranged, so that it corrects the wave function where the Coulomb singularity lives and leaves the rest to the network.

The neural correlator.

W_θ must be invariant under exchange of identical particles, or the exchange symmetry of the whole ansatz is destroyed. The standard construction is a DeepSets-style sum over permutation-invariant features: one-body features summed over particles, and two-body features summed over pairs,

$$W_\theta(\boldsymbol{R}) = \rho_\theta\Big(\sum_i \phi_\theta(\tilde{\boldsymbol{x}}_i),\, \sum_{i<j} \chi_\theta(\tilde{r}_{ij})\Big), \tag{18.13}$$

with ϕ_θ, χ_θ and ρ_θ small networks. A sum is invariant under any relabelling, so the invariance is exact by construction and does not have to be learned or penalised.

18.5.1 The two-electron dot, and an exact reference

For $N=2$ in a spin singlet the spatial wave function is symmetric, the determinant reduces to a constant, and Eq. (18.10) collapses to something we can compare directly with Chapters 13 and 17. Better still, this is a case with a closed-form answer.

Taut's solution.

Separating into centre-of-mass and relative coordinates, $\boldsymbol{R}_{\rm cm} = (\boldsymbol{r}_1+\boldsymbol{r}_2)/2$ and $\boldsymbol{r} = \boldsymbol{r}_1 - \boldsymbol{r}_2$, the Hamiltonian of Eq. (17.81) becomes

$$\hat{H} = \left[-\tfrac{1}{4}\nabla^2_{\rm cm} + \omega^2 R^2_{\rm cm}\right] + \left[-\nabla^2_r + \tfrac{1}{4}\omega^2 r^2 + \frac{1}{r}\right]. \tag{18.14}$$

The centre-of-mass part is a two-dimensional oscillator with ground state $e^{-\omega R^2_{\rm cm}}$ and energy ω. For the relative part, insert $e^{-\omega r^2/4}(1+ar)$ and collect terms: the $1/r$ contribution cancels only if $a=1$, and matching the constant and linear terms then gives $E_{\rm rel} = 2\omega$ together with the condition $\omega = 1$. Adding the two pieces, and using $|\boldsymbol{r}_1+\boldsymbol{r}_2|^2/4 + |\boldsymbol{r}_1-\boldsymbol{r}_2|^2/4 = (r_1^2+r_2^2)/2$,

$$\boxed{\Psi_{\rm exact}(\boldsymbol{r}_1,\boldsymbol{r}_2) = e^{-\left(r_1^2+r_2^2\right)/2}\,(1+r_{12}), \qquad E=3} \tag{18.15}$$

This is the number quoted as "exact (Taut)" throughout Chapters 11 to 17, now derived. It is also a stringent test of any implementation: the local energy of Eq. (18.15) must be exactly 3 at *every* configuration, and `pinn.py` confirms it over two thousand random points with a spread of 7×10^{-15}.

What the ansatz can and cannot reach.

Write Eq. (18.15) logarithmically:

$$\ln \Psi_{\text{exact}} = -\frac{s}{2} + \ln(1 + r_{12}), \qquad s = r_1^2 + r_2^2. \tag{18.16}$$

Both terms are functions of the two permutation-invariant features (s, r_{12}) – which are exactly the features the neural correlator is given. *The Slater-Jastrow-neural ansatz therefore contains the exact state*, at $\alpha = 1$ and $W = \ln(1+r_{12}) - r_{12}/(1+\beta r_{12})$. The Padé factor of Chapter 13 is a rational approximation to that logarithm, and Figure 18.4 puts the two side by side: they agree to first order at $r_{12} = 0$, which is precisely the statement that both satisfy the cusp condition, and they part company further out. The network has something well defined to learn, and the residual variance measures the optimisation rather than the ansatz.

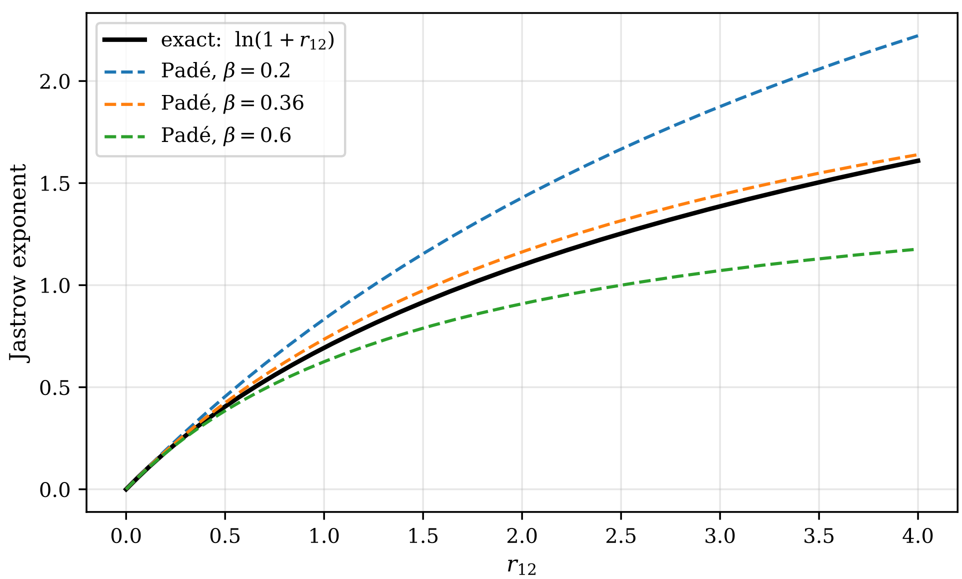

Fig. 18.4 The exponent of the pair-correlation factor for two electrons in a two-dimensional dot at $\omega = 1$, as a function of the interparticle distance r_{12}: the exact $\ln(1+r_{12})$ of Taut's solution, Eq. (18.16), in black, and the Padé form $r_{12}/(1+\beta r_{12})$ of Chapter 13 for three values of β. All four curves leave the origin with unit slope, so every β satisfies the cusp condition; what β controls is how the correlation hole closes further out, and there the rational form cannot follow the logarithm. The vertical gap between the black curve and a coloured one is exactly the correction $W = \ln(1+r_{12}) - r_{12}/(1+\beta r_{12})$ that the neural correlator is asked to supply.

Figure 18.4 repays a careful look, because it shows both how much and how little the cusp condition settles. Every curve in it has unit slope at the origin, and they are visually one curve out to $r_{12} \approx 0.3$; the cusp is a condition on a single derivative at a single point, and it constrains the correlation factor nowhere else. Beyond that the three Padé curves fan out, and they must, because $r_{12}/(1+\beta r_{12})$ saturates at $1/\beta$ while $\ln(1+r_{12})$ grows without bound: for every β the two functions eventually separate, and the only question is where. At $\beta = 0.2$ the separation is early and large, an overshoot of 0.6 by $r_{12} = 4$; at $\beta = 0.6$ the hole closes too fast and the curve falls short by 0.44 at the same distance. The intermediate $\beta = 0.36$ is much the best of the three, tracking the logarithm to within 0.065 across the whole interval, with the worst discrepancy at $r_{12} = 2.16$.

That residual misfit is small in the exponent but the wave function is its exponential, so a gap of 0.065 is a 7% error in amplitude at a separation of about two oscillator lengths – which at $\omega = 1$ is not the tail of the pair distribution but a good part of its bulk. It is also, read the other way, a precise statement of the job the neural correlator has been given: a smooth bounded function of one permutation-invariant variable, zero at the origin by the cusp condition, of one sign over the range that matters, and of depth less than a tenth. That modest a target is what the third row of Table 18.4 buys with its 105 extra parameters, halving the variance and reducing the energy error from 0.0013 to 0.00048. The comparison worth holding beside it is the first row: an unstructured Boltzmann-machine state with no cusp built in is out by 0.082, some sixty times worse, because it must learn the kink at $r_{12} = 0$ as well – and a derivative discontinuity is precisely what a smooth network is worst at.

trial function	parameters	E	$\mathrm{Var}(E_L)$	$E - E_{\text{exact}}$
RBM, no physics (Chapter 17)	44	3.08227(173)	3.12	+0.082
Padé-Jastrow (Chapter 13)	2	3.001323(72)	0.0054	+0.0013
Padé-Jastrow × neural correlator	$2+105$	3.000477(48)	0.0027	+0.00048
exact, Eq. (18.15)	—	3	0	—

Table 18.4 Two electrons in a two-dimensional dot at $\omega = 1$. The hybrid was trained in two phases: the two analytic parameters first, then the correlator switched on from that solution. It halves the variance and lowers the energy by a factor of nearly three. The comparison worth making is with the first row: putting the cusp back is worth more than any amount of extra flexibility – and once it is back, the network improves on what two analytic parameters can do alone. Computed by `pinn.py`.

18.6 Backflow

The remaining component of Eq. (18.10) is the one that acts inside the determinant, and it addresses a limitation that neither the Jastrow factor nor the neural correlator can touch.

The limitation.

In the ordinary Slater-Jastrow ansatz

$$\Psi(\boldsymbol{R}) = e^{J(\boldsymbol{R})} \det[\phi_a(\boldsymbol{r}_i)], \tag{18.17}$$

the Jastrow factor e^J is symmetric and *positive*. It can reshape the wave function anywhere, but it cannot change where the wave function vanishes: the nodes of Ψ are exactly the nodes of the determinant, wherever J puts its weight. Since the nodal surface determines the fixed-node error of diffusion Monte Carlo (Section 15.5) and sets a floor under the variational energy, this is a hard ceiling on the quality of the ansatz. Backflow raises it by modifying the coordinates *before* they enter the determinant.

18.6.1 Standard backflow

Introduce effective, or quasi-particle, coordinates

$$\tilde{\boldsymbol{r}}_i = \boldsymbol{r}_i + \boldsymbol{\xi}_i(\boldsymbol{R}), \qquad \boldsymbol{\xi}_i(\boldsymbol{R}) = \sum_{j \neq i} \eta(r_{ij})\,(\boldsymbol{r}_i - \boldsymbol{r}_j), \qquad r_{ij} = \left|\boldsymbol{r}_i - \boldsymbol{r}_j\right|, \tag{18.18}$$

with η a short-ranged scalar function. Particle i is displaced by a weighted sum of the vectors pointing away from the other particles: an electron is pushed aside by its neighbours, which is the physical picture the name comes from. The form is analytically tractable, easy to interpret, and has few parameters.

Its limitations are equally clear: the functional form is fixed in advance, the structure is strictly pairwise, and there is no access to genuine higher-body correlations. Section 18.6.5 exhibits a case where the restriction is not merely quantitative – a nodal surface that pairwise backflow provably cannot move.

18.6.2 CTNN backflow: a learned many-body map

The conceptual shift is to replace the fixed analytic form by a *learned* map,

$$\tilde{\boldsymbol{r}}_i = \boldsymbol{r}_i + \boldsymbol{F}_{i,\theta}(\boldsymbol{R}), \qquad \text{or collectively} \qquad \tilde{\boldsymbol{R}} = \boldsymbol{F}_\theta(\boldsymbol{R}), \tag{18.19}$$

that is,

$$\underbrace{\text{analytic pairwise deformation}}_{\text{standard}} \quad \longrightarrow \quad \underbrace{\text{learned many-body coordinate transformation}}_{\text{CTNN}}.$$

The generic construction is a graph-neural-network layer: build node features from single particles and edge features from pairs, aggregate the edges into each node, and decode the result into a displacement,

$$\boldsymbol{h}_i = \phi_\theta(\boldsymbol{r}_i), \qquad \boldsymbol{h}_{ij} = \psi_\theta(\boldsymbol{r}_i, \boldsymbol{r}_j), \qquad \boldsymbol{H}_i = \sum_{j \neq i} \boldsymbol{h}_{ij}, \qquad \tilde{\boldsymbol{r}}_i = \boldsymbol{r}_i + \boldsymbol{g}_\theta(\boldsymbol{h}_i, \boldsymbol{H}_i). \tag{18.20}$$

The maps ϕ_θ, ψ_θ and $\boldsymbol{g}_\theta$ are learned, and because $\boldsymbol{g}_\theta$ acts on an *aggregate* of all the edges attached to node i it can represent collective structure that no pairwise sum can. The implementation used in Section 18.7, `CTNNBackflowNet`, carries explicit node features $\boldsymbol{h}_i \in \mathbb{R}^{H_v}$ and edge features $\boldsymbol{h}_{ij} \in \mathbb{R}^{H_e}$ with learned linear transports between the node and edge spaces; the simpler `BackflowNet` is a pairwise message-passing multilayer perceptron whose output is squashed, $s_\beta \tanh(\delta \boldsymbol{x}_i)$, to keep the displacement bounded.

18.6.3 The backflow-modified Slater determinant

Without backflow the Slater matrix is

$$M_{ia}(\boldsymbol{R}) = \phi_a(\boldsymbol{r}_i), \qquad D(\boldsymbol{R}) = \det \boldsymbol{M}(\boldsymbol{R}), \tag{18.21}$$

and each *row* depends only on the coordinate of one particle. With backflow the orbitals are evaluated at the effective coordinates,

feature	standard backflow	CTNN backflow
functional form	analytic	learned
typical structure	pairwise	many-body, hierarchical
parameters	few	many
expressivity	limited	high
interpretability	high	moderate

Table 18.5 The two constructions. In one line: standard backflow is an analytic pairwise correction, CTNN backflow is a learned many-body deformation.

$$\widetilde{M}_{ia}(\boldsymbol{R}) = \phi_a(\tilde{\boldsymbol{r}}_i(\boldsymbol{R})) = \phi_a\big(\boldsymbol{r}_i + \boldsymbol{F}_{i,\theta}(\boldsymbol{R})\big), \qquad D_{\mathrm{BF}}(\boldsymbol{R}) = \det\widetilde{\boldsymbol{M}}(\boldsymbol{R}), \tag{18.22}$$

and every row now depends on the *entire* configuration. The orbitals have become *configuration-dependent quasi-orbitals*. For a small displacement the effect is a gradient correction inside the determinant,

$$\widetilde{M}_{ia} \approx \phi_a(\boldsymbol{r}_i) + \boldsymbol{F}_{i,\theta}(\boldsymbol{R}) \cdot \nabla\phi_a(\boldsymbol{r}_i) + \ldots, \tag{18.23}$$

which is a useful way to see that backflow adds to the determinant precisely the kind of term a larger orbital basis would – but with coefficients that depend on where the other particles are.

18.6.4 Antisymmetry and permutation equivariance

Making every row depend on all the particles looks as though it should destroy the antisymmetry, and it does unless one condition holds.

Claim. If the backflow map is *permutation-equivariant*, that is if relabelling the particles relabels the displacements in the same way,

$$\boldsymbol{F}_{P(i)}(P\boldsymbol{R}) = \boldsymbol{F}_i(\boldsymbol{R}) \quad \text{for every permutation } P,$$

then exchanging particles $i \leftrightarrow j$ exchanges $\tilde{\boldsymbol{r}}_i \leftrightarrow \tilde{\boldsymbol{r}}_j$, hence exchanges two rows of $\widetilde{\boldsymbol{M}}$, hence

$$D_{\mathrm{BF}}(\boldsymbol{R}) \longmapsto -D_{\mathrm{BF}}(\boldsymbol{R}).$$

Antisymmetry is preserved.

The standard form (18.18) is equivariant by inspection: the sum runs over all other particles with a coefficient depending only on the distance, so nothing distinguishes one label from another. The learned form (18.20) is equivariant because $\boldsymbol{H}_i$ is a *sum* over edges, and a sum does not care about order.

The condition is not automatic, and violating it is easy: feed a network the particle index, or sort the particles by some criterion before passing them in, and the equivariance is gone. `pinn.py` exchanges two particles in a four-electron determinant and reports the ratio $D(\text{swapped})/D$:

map	$D(\text{swap})/D$	
no backflow	-1.000000	antisymmetric
equivariant backflow, Eq. (18.18)	-1.000000	antisymmetric
a map privileging particle 0	-452000.27	**not** antisymmetric

The last row is the warning. The object that comes out is not a fermionic wave function at all, whatever energy it may report, and nothing in a training curve will tell you.

18.6.5 What backflow does to the nodes

The purpose of backflow is to move the nodal surface, and it is worth seeing both that it does and where it does not. `pinn.py` drags one particle of a three-electron configuration along a line and locates the first zero of the determinant, with and without a Jastrow factor and with and without backflow.

orbitals	plain	$\times$ Jastrow	$+$ backflow
different Gaussian widths	-0.63242	-0.63242	-0.58680
common Gaussian factor	1.72630	1.72630	1.72630

Table 18.6 Position of the first node along a line as one particle is dragged past the other two. The Jastrow column never moves – which is the whole reason backflow exists. The second row shows that pairwise backflow does not always move it either. Computed by `pinn.py`.

The first row is the expected result: the Jastrow factor leaves the node exactly where it was, and backflow shifts it. The second row is more interesting, and it is a small theorem rather than a numerical accident. When the three orbitals share a common Gaussian factor, $\{1,x,y\}e^{-r^2/2}$, that factor comes out of the determinant and the node is exactly the condition that the three effective positions be *collinear*. But at a collinear configuration every difference vector $\boldsymbol{r}_i - \boldsymbol{r}_j$ already lies along the line, so a displacement built out of those vectors – which is what Eq. (18.18) is – keeps all three points on it. The node cannot move.

Figure 18.5 turns Table 18.6 into a function of the backflow strength, and three features of it deserve comment. The first is the dashed line at zero. It is not a fit and not a small number: the node of $e^J \det \boldsymbol{M}$ is the node of $\det \boldsymbol{M}$ because e^J never vanishes, so the Jastrow column of Table 18.6 is zero as an identity, for both orbital sets and every strength, and the figure simply records that it stays zero everywhere along the axis. The second is the pair of curves that do move. With orbitals of different widths the node responds linearly at small η_0, with a slope of about 0.12, and bends over as the displacement grows, reaching $+0.071$ at $\eta_0 = 1.5$ for $w = 2$; the entry -0.58680 in Table 18.6 is the point at $\eta_0 = 0.6$ on that curve, a shift of $+0.0456$ from the unperturbed -0.63242. Shortening the range to $w = 1$ costs a factor of four, only $+0.016$ at the same strength, and the reason is that $\eta(r_{ij})$ is evaluated at the actual pair separations of the configuration: what matters is not the strength of the backflow but the overlap of its envelope with the distances that are physically present. The short-range curve even dips a little below zero before turning up, an excursion of under 10^{-3} that is at the margin of what the node search resolves.

The third feature is the one the chapter has been building towards. The triangles lie on the dashed line, not near it. There is no strength at which a pairwise backflow displaces the collinear node, because collinearity is preserved by any displacement assembled from the

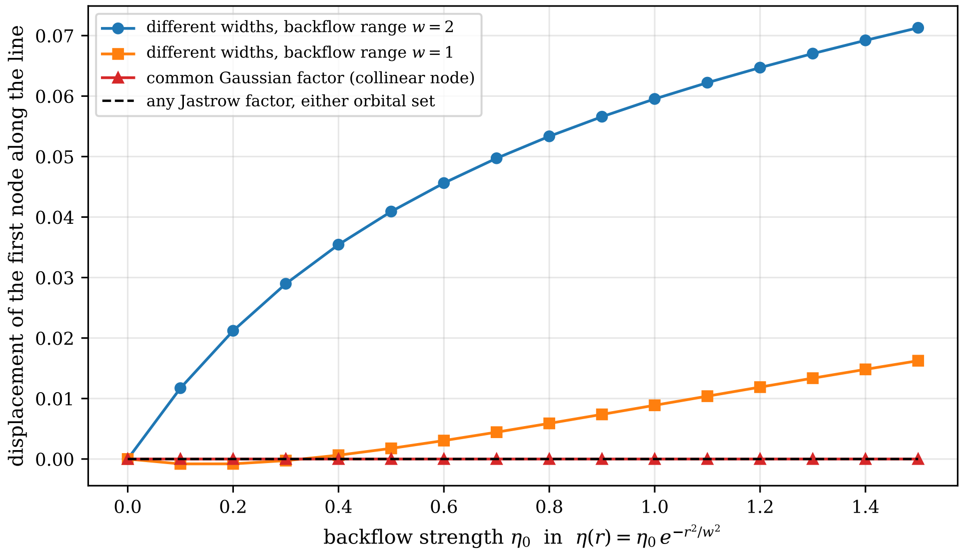

Fig. 18.5 Displacement of the first node of a three-electron determinant along a line, as the strength η_0 of the pairwise backflow of Eq. (18.18) with $\eta(r) = \eta_0 e^{-r^2/w^2}$ is turned up from zero. Circles and squares use three orbitals of different Gaussian widths with backflow ranges $w = 2$ and $w = 1$; triangles use the degenerate set $\{1, x, y\}e^{-r^2/2}$, whose common Gaussian factor leaves the node as the bare condition that the three effective positions be collinear. The dashed line at zero is what a Jastrow factor does to the node, for either orbital set and any strength. The collinear node does not move for any η_0, and it does not move exactly rather than approximately.

difference vectors $\boldsymbol{r}_i - \boldsymbol{r}_j$, and that is a statement about the functional form of Eq. (18.18) rather than about the size of its parameters. Read together, the red and blue curves say that the machinery works and that it has a blind spot, and that the blind spot is not somewhere one can optimise one's way out of. It is exactly the situation in which replacing the analytic form by the learned map of Eq. (18.20) is not a refinement but a change of kind, and it is the microscopic version of what the bold entries of Table 18.7 report at $N = 6$ and above, where the CTNN ansatz beats the pairwise one by roughly an order of magnitude in the percentage error.

Why this is the argument for a learned map. The failure in the second row of Table 18.6 is structural: *no* choice of η in Eq. (18.18) escapes it, because the restriction lies in the functional form and not in its parameters. A map of the form (18.20), whose displacement is decoded from an aggregate rather than assembled from pair vectors, is not so restricted. This is what “limited expressivity” means concretely, and it is why the extra parameters of a CTNN backflow buy something that a better-tuned analytic form cannot.

18.6.6 Derivatives, and the enlarged variational manifold

Backflow is not a preprocessing step applied to the coordinates before the calculation begins. It is an intrinsic part of the trial function, and its parameters are optimised together with the orbital and Jastrow parameters:

$$\theta^\star = \arg\min_\theta \frac{\langle \Psi_\theta | \hat{H} | \Psi_\theta \rangle}{\langle \Psi_\theta | \Psi_\theta \rangle}, \qquad \Psi_\theta(\boldsymbol{R}) = e^{J_\theta(\boldsymbol{R})} \det\left[\phi_a\left(\boldsymbol{r}_i + \boldsymbol{F}_{i,\theta}(\boldsymbol{R})\right)\right]. \tag{18.24}$$

Because θ now includes the map $\boldsymbol{F}$, the variational manifold is larger than the space of Jastrow and orbital coefficients: it contains the learned many-body *geometry* of the effective coordinates. Backflow helps in two ways – it captures additional many-body correlations, and it improves the nodal surface – and both lower $E(\boldsymbol{\theta})$ towards E_0.

The cost appears in the derivatives. Since $\widetilde{M}_{ia} = \phi_a(\tilde{\boldsymbol{r}}_i(\boldsymbol{R}))$ depends on every particle, a spatial derivative requires the chain rule through the map,

$$\nabla_{\boldsymbol{r}_k}\widetilde{M}_{ia} = \sum_{\alpha} \frac{\partial \phi_a}{\partial \tilde{r}_{i\alpha}} \frac{\partial \tilde{r}_{i\alpha}}{\partial \boldsymbol{r}_k}, \tag{18.25}$$

and every quantity built from spatial derivatives – the gradient, the Laplacian, the kinetic energy, the quantum force, and the parameter gradients used for optimisation – acquires a contribution from $\partial\tilde{\boldsymbol{r}}/\partial\boldsymbol{r}$. The Sherman-Morrison update of Section 2.11, which made single-particle moves cheap by exploiting the fact that only one row of the Slater matrix changes, no longer applies: with backflow, moving one particle changes *every* row. This is the main computational price of the construction.

18.7 Training pipeline and results

We now put the pieces together on quantum dots with $N = 2$, 6, 12 and 20 electrons, at confinements $\omega \in \{10^{-3}, 10^{-2}, 0.1, 0.5, 1.0\}$ – five orders of magnitude, from a tightly bound artificial atom to a system on the way to a Wigner molecule. Two ansätze are compared: **PINN+BF**, using the pairwise `BackflowNet`, and **PINN+CTNN**, using the learned `CTNNBackflowNet`.

The pipeline.

1. **Residual pretraining.** The PINN loss enforces the many-body Schrödinger equation at collocation points. This converges quickly from a random initialisation, where energy minimisation is slow, and it does not need a Markov chain.
2. **Stochastic-reconfiguration tail.** The objective switches to the energy quotient and the optimiser to the natural gradient of Section 14.6. This is the phase that *restores the variational upper bound* lost during pretraining, and it is not optional – Table 18.2 is what happens without it.

The architecture is modest: a Jastrow network of two hidden layers of width 64 with GELU activations, taking all pairwise distances and absolute positions and returning a scalar log-amplitude correction; the backflow module, where used, is two layers of width 32. The parameter count runs from about 25 000 at $N = 2$ to about 45 000 at $N = 6$. Training uses Adam at a learning rate of 10^{-4} with a cosine warm-up over 200–400 epochs, batch sizes of 512 to 2048 collocation points depending on N, and gradient clipping at 0.2. For $N > 4$ the Laplacian is evaluated by finite differences with $h = 0.01$ rather than by automatic differentiation, because the $\mathscr{O}(N)$ cost of second-order forward propagation – Eq. (18.3) – becomes the bottleneck.

An internal consistency check.

Before comparing with anyone else's numbers it is worth checking the calculation against itself. For a harmonic trap plus Coulomb interaction the virial theorem reads

$$2\langle T\rangle = 2\langle V_{\mathrm{trap}}\rangle - \langle V_{\mathrm{int}}\rangle, \tag{18.26}$$

which follows from the scaling behaviour of the two potentials – quadratic and inverse-linear – under $\boldsymbol{r} \to \lambda \boldsymbol{r}$. It involves no reference calculation at all, and it is satisfied to sub-percent accuracy across the whole benchmark, worst case $\lesssim 0.4\%$. This is the kind of check that should be reported alongside any variational energy: it tests the physics of the solution rather than its agreement with another method.

18.7.1 Ground-state energies against diffusion Monte Carlo

N	ω	DMC reference	PINN+BF	% err	PINN+CTNN	% err
2	10^{-3}	—	0.0137948(8)	—	0.013778(1)	—
	10^{-2}	0.073839(2)	0.073838(1)	−0.0014	0.0738382(5)	−0.0014
	0.10	0.44079(1)	0.44079(1)	+0.0000	0.440796(4)	−0.0000
	0.50	1.65977(1)	1.65976(2)	−0.0006	1.65976(1)	−0.0006
	1.00	3.00000	2.99999(1)	−0.0003	3.00000(2)	−0.0000
6	10^{-3}	—	0.140913(1)	—	0.1408172(7)	—
	10^{-2}	—	0.69125(2)	—	0.69036(1)	—
	0.10	3.55385(5)	3.5549(1)	+0.0295	**3.55388(5)**	+0.0008
	0.50	11.78484(6)	11.7895(4)	+0.0395	**11.7847(2)**	−0.0000
	1.00	20.15932(8)	20.1610(6)	+0.0083	**20.1585(3)**	−0.0041
12	10^{-3}	—	0.515823(3)	—	0.515365(4)	—
	10^{-2}	—	2.48620(5)	—	2.47363(4)	—
	0.10	12.26984(8)	12.2731(2)	+0.0160	12.2718(1)	—
	0.50	39.1596(1)	39.1786(8)	+0.0485	**39.1604(3)**	+0.0018
	1.00	65.7001(1)	65.717(1)	+0.0257	**65.69556(5)**	−0.0069
20	10^{-3}	—	—	—	1.293033(6)	—
	10^{-2}	—	—	—	6.14645(5)	—
	0.10	29.9779(1)	—	—	29.9888(2)	+0.0364
	0.50	93.8752(1)	—	—	**93.8789(5)**	+0.0040
	1.00	155.8822(1)	—	—	**155.8738(7)**	−0.0024

Table 18.7 Ground-state energies in atomic units for N electrons in a two-dimensional parabolic trap, compared with diffusion Monte Carlo where a reference is available. Percentage errors are relative to the DMC value. Entries in bold are the cases in which PINN+CTNN improves on PINN+BF. For $N = 2$, $\omega = 1$ the DMC reference is Taut's exact 3, Eq. (18.15).

Three observations. At $N = 2$ both ansätze reproduce the reference within statistical error at every ω, which is the calibration the method has to pass. From $N = 6$ upwards the learned backflow is systematically the better of the two at moderate and strong confinement, typically by an order of magnitude in the percentage error – which is the empirical content of Table 18.5. And at $N = 20$ the CTNN ansatz remains accurate to $\lesssim 0.04\%$, so the construction does not degrade with system size over the range tested.

Several entries lie slightly *below* the DMC reference. This is not paradoxical: DMC in its fixed-node form is itself only an upper bound, one limited by the nodal surface of *its* trial function, so a variational calculation with better nodes may legitimately come out lower. It is, however, exactly the situation in which one wants to be certain that the variational property has been preserved – which is why the pipeline ends with the stochastic-reconfiguration tail and why the virial identity is checked.

18.7.2 Energy partitioning and the approach to the Wigner regime

The total energy decomposes as $E = \langle T \rangle + \langle V_{\text{int}} \rangle + \langle V_{\text{trap}} \rangle$, and two dimensionless ratios track how the balance shifts as the trap is relaxed:

$$\Gamma(\omega) = \frac{\langle V_{\text{int}} \rangle}{\langle T \rangle}, \qquad \mathscr{C}(\omega) = \frac{2\langle V_{\text{trap}} \rangle}{\langle V_{\text{int}} \rangle}. \tag{18.27}$$

The first measures interaction dominance and rises as ω falls; the second tends to unity when the trap and Coulomb energies come into the balance characteristic of a classical, crystallised arrangement of charges.

N	$\Gamma_{\omega=1}$	$\mathscr{C}_{\omega=1}$	$\Gamma_{\omega=10^{-3}}$	$\mathscr{C}_{\omega=10^{-3}}$
2	0.93	3.14	8.64	1.226
6	2.41	1.83	24.55	1.074
12	3.61	1.56	41.11	1.049
20	4.66	1.43	57.26	1.038

Table 18.8 Energy partitioning. At $\omega = 1$ and $N = 2$ the kinetic energy still slightly exceeds the interaction energy, $\Gamma < 1$; by $\omega = 10^{-3}$ and $N = 20$ the interaction is nearly sixty times the kinetic energy. The $\mathscr{C}$ column approaches unity from above, and for $N \geq 12$ at $\omega = 10^{-3}$ it is below 1.05: the system satisfies $V_{\text{int}} \approx 2V_{\text{trap}}$ to better than five per cent, which is the quasi-classical balance of a Wigner molecule.

Plotted logarithmically, $\Gamma(\omega)$ rises monotonically and without saturating over the whole range, while $\mathscr{C}(\omega)$ falls towards 1 and flattens; the root-mean-square radius expands by a factor of about 70 between $\omega = 1$ and $\omega = 10^{-3}$. These are quantities that require no reference calculation, and they are the strongest available evidence that the wave function is physically correct in the regime where no DMC benchmark exists.

18.7.3 Reproducibility

A variational energy from a stochastic optimisation is only as trustworthy as its seed dependence. Table 18.9 repeats the calculation from three independent random initialisations against a configuration-interaction reference computed by discrete-variable representation. Every run lies below its CI reference, and the seed-to-seed spread is under 10^{-3} a.u. throughout. Non-MCMC training reaches sub-0.06% accuracy across all N tested, and the loss does not grow with N – a tentative but encouraging observation. The reader should read “every run lies below CI” in the light of the footnote: for $N = 6$ the CI space is severely truncated, so being below it is a weak statement. For $N = 2$ to 4, where the CI is close to converged, it is a strong one.

N	E_{CI}	seed 42	seed 314	seed 901	spread	max $\|\mathrm{err}\|$
2	2.254424	2.253853	2.254049	2.253171	0.00088	0.056%
3	3.637002	3.636503	3.636209	3.636661	0.00029	0.022%
4	5.104449	5.103305	5.103369	5.103568	0.00026	0.022%
6	$8.217032^{\dagger}$	8.213596	8.214344	8.213966	0.00075	0.042%

Table 18.9 Three seeds against a CI benchmark, $\omega = 1$, energies in atomic units. The CI reference uses a two-dimensional discrete variable representation: each of K wells contributes n_{sp} single-particle orbitals from its local oscillator basis, the N-particle product space of n_{sp}^N configurations is sorted by non-interacting energy, and the lowest n_{ci} are retained. †For $N = 6$, $n_{\mathrm{sp}} = 10$ and $n_{\mathrm{ci}} = 500$, which retains only 0.05% of the 10^6 configurations – so the CI number is itself an upper bound, and not a tight one.

18.8 PINNs and variational Monte Carlo: complementary strengths

The two frameworks are not competitors so much as tools with different shapes.

Where a PINN is strong.

- The wave function is a genuine continuous function of the coordinates, which can be evaluated and differentiated anywhere, not merely sampled.
- Excited states, complex geometries and time-dependent problems follow naturally: the loss simply acquires more terms.
- Antisymmetry, normalisation and conservation laws enter as modular, explicit constraints, which makes it easy to see what has been imposed.
- No Markov chain, hence no autocorrelation time and none of the error analysis of Chapter 14.

Where variational Monte Carlo is strong.

- It is the established workhorse for large-N ground states.
- Monte Carlo sampling handles very high-dimensional integrals efficiently.
- It gives variational upper bounds by construction, not by careful loss design.
- Decades of accumulated, well-tested ansatz forms.

The division of labour that emerges is that VMC excels at large-N ground states while PINNs are strong candidates for continuous-space, dynamical, constrained and few-to-intermediate-body problems. The most productive direction is cross-fertilisation, and the pipeline of Section 18.7 is one instance of it: PINN-style residual pretraining to get quickly into the right region, followed by a VMC-style variational tail with stochastic reconfiguration to obtain a number one can defend.

18.9 Summary and open questions

Two things determine what a neural many-body calculation is worth, and they are independent.

The **objective** decides whether the answer means anything. Minimising the energy quotient is Rayleigh-Ritz and returns an upper bound; minimising the residual of the Schrödinger equation does not, and Table 18.2 shows it returning energies below the exact ground state by an amount that grows with the size of the collocation domain. Without a normalisation constraint the residual can be driven to zero by shrinking the wave function. The remedy is to make the energy quotient the primary objective and use the residual, if at all, for pretraining.

The **ansatz** decides how good that bound can be. The construction that works is layered: a Slater determinant for exact antisymmetry, an analytic pair term for the cusp, a permutation-invariant network for everything else, and backflow inside the determinant to reach the nodal surface. Each layer does one job, and the ones that can be made exact are made exact rather than learned. Antisymmetry survives backflow if and only if the map is permutation-equivariant – and, as Section 18.6.5 shows, an analytic pairwise map has nodal surfaces it provably cannot move, which is the concrete argument for a learned many-body one.

On quantum dots from $N = 2$ to $N = 20$ and across five orders of magnitude in confinement, the result is agreement with diffusion Monte Carlo to $\lesssim 0.05\%$, with the learned backflow systematically ahead of the pairwise one from $N = 6$ upwards, and with the virial identity and the energy partitioning confirming physical correctness in the regime where no benchmark exists.

Open questions.

Methodological. Scaling beyond $N = 20$ needs sparse or message-passing architectures to reduce the $\mathscr{O}(N^2)$ cost of the pair features. Transfer learning – pre-train at one ω, fine-tune across the grid – exploits the one structural advantage a continuous representation has over a sampled one. Excited states need orthogonality constraints in the loss; real-time dynamics needs the time variable in it. And the optimiser remains open: the stochastic reconfiguration of Section 14.6 is the current default, but its $\mathscr{O}(p^2)$ metric is itself a bottleneck at large parameter counts.

Physical. The transition to the Wigner regime, and where exactly it should be said to occur. Spin degrees of freedom and magnetic fields. Coupled dots, and transport and conductance calculations. The entanglement structure of the many-body wave function – which connects directly to Chapters 19 and 20.

18.10 Exercises

Exercise 1: automatic differentiation

a) Derive the forward-mode recursion (18.3) for a general activation φ, and specialise it to $\varphi = \tanh$.
b) Show that propagating the full Hessian through a network with d inputs costs $\mathscr{O}(d^2)$ times the cost of the value, and that only the *diagonal* is needed for a Laplacian. How would you exploit that?
c) Explain precisely why ReLU cannot be used in a PINN for a second-order equation, and why the failure is not fixed by using a smooth approximation to ReLU with a very small smoothing parameter.
d) Reverse-mode differentiation costs $\mathscr{O}(1)$ times the forward pass for a scalar output, regardless of the number of inputs. Why is forward mode nonetheless the right choice for the second derivatives in a PINN residual, and at what d does the balance change?

Exercise 2: the variational principle and the residual

a) Prove Eq. (18.4) by expanding ψ_θ in eigenstates of $\hat{H}$.
b) Derive Eq. (18.6) using $\langle T\rangle = a/4$, $\langle x^2\rangle = 1/(2a)$ and $\langle x^4\rangle = 3/(4a^2)$ for the normalised Gaussian, and confirm the stationary point $a = 3^{1/3}$.
c) Show that for a residual weighted by the collocation measure $w(x)$ the optimal E is $\int w\,\psi\hat{H}\psi \,/\int w\,\psi^2$, and that this reduces to the Rayleigh quotient if and only if w is constant *and* the domain is the whole space. Which of the two conditions fails in Table 18.2?
d) Show that minimising $\langle (E_L - E)^2\rangle$ with the average taken over $|\psi|^2$ – that is, variance minimisation – *does* give $E = \langle E_L\rangle$, a Rayleigh quotient, and is therefore variational. What exactly goes wrong when the average is taken over a uniform measure instead?
e) Show that $\|(\hat{H}-E)\psi\|^2$ is minimised over E by $E = \langle E_L\rangle$, and hence that the residual formulation is variational *provided* the norm is the L^2 norm over all space. Reconcile this with Table 18.2.

Answer to c) and e).

These two parts are the same observation from opposite directions, and together they locate the failure precisely. Minimising $\int w(x)\left|(\hat{H}-E)\psi\right|^2 dx$ over E gives

$$E = \frac{\int w\,\psi\hat{H}\psi\,dx}{\int w\,\psi^2\,dx},$$

which is a Rayleigh quotient of $\hat{H}$ – and hence bounded below by E_0 – only when $w \equiv 1$ over all space, since only then is the numerator $\langle\psi|\hat{H}|\psi\rangle$ in the L^2 inner product. In Table 18.2 the residual was divided by ψ^2, which is what one does to prevent the amplitude collapse of Section 18.4.1; that division sets $w = 1/\psi^2$, and the optimal E becomes the mean of E_L over the flat collocation measure, which is not a Rayleigh quotient of anything. The two safeguards – normalise the residual so it cannot be gamed, and keep the measure equal to $|\psi|^2$ so the bound survives – are in direct conflict, and only the energy quotient satisfies both.

Exercise 3: the ansatz

a) Derive the cusp coefficients (18.12) by requiring that the divergence of the kinetic energy as $r_{ij} \to 0$ cancel the Coulomb singularity in d dimensions. Check that $d = 3$ gives the familiar $1/2$ and $1/4$.
b) Verify that $u(r) = \gamma r e^{-r}$ satisfies the cusp condition at the origin, and explain what the factor e^{-r} is for.
c) Show that the DeepSets form (18.13) is exactly invariant under any permutation of the particles, and that it remains so under any choice of the networks ϕ_θ, χ_θ, ρ_θ.
d) Carry out the separation (18.14) in full, insert $e^{-\omega r^2/4}(1+ar)$ into the relative equation, and derive both $a = 1$ and the condition $\omega = 1$. Where does the two-dimensionality enter?
e) Verify from Eq. (18.16) that the exact state satisfies the antiparallel-spin cusp condition with $\gamma_{\uparrow\downarrow} = 1$, and that the Padé factor $r/(1+\beta r)$ does too, for every β. What does β control, if not the cusp?

Answer to e).

Expanding, $\ln(1+r) = r - r^2/2 + \ldots$ and $r/(1+\beta r) = r - \beta r^2 + \ldots$; both have unit slope at the origin, which is the $d = 2$ antiparallel cusp condition $\gamma_{\uparrow\downarrow} = 1/(d-1) = 1$. The cusp fixes only the first derivative at $r = 0$, so it is satisfied for *every* β: what β controls is the rate at which the correlation hole closes further out, and matching the exact $-1/2$ in the second-order coefficient would require $\beta = 1/2$. Optimising the energy in Table 18.4 returns $\beta \approx 0.36$, which is the best compromise the rational form can make across the whole range of r_{12} that is actually sampled – and the discrepancy is exactly what the neural correlator is left to repair.

Exercise 4: backflow

a) Show that the standard backflow (18.18) is permutation-equivariant, and that the aggregated form (18.20) is too.
b) Prove that a permutation-equivariant map preserves the antisymmetry of the determinant, and construct explicitly a map that is not equivariant and a configuration for which the determinant then fails to change sign.
c) Show that a Jastrow factor cannot move the nodes of Ψ, and that this is a consequence of $e^J > 0$ and nothing else.
d) For three particles in two dimensions with orbitals $\{1, x, y\}e^{-r^2/2}$, show that the common Gaussian factors out of the determinant and that the node is exactly the condition that the three positions be collinear. Then show that any displacement of the form $\boldsymbol{\xi}_i = \sum_{j\neq i} \eta(r_{ij})(\boldsymbol{r}_i - \boldsymbol{r}_j)$ maps collinear configurations to collinear configurations, so that the node cannot move. This is the second row of Table 18.6.
e) Construct a permutation-equivariant map that *does* move that node. *Hint:* you will need a displacement that is not a linear combination of the vectors $\boldsymbol{r}_i - \boldsymbol{r}_j$.
f) Explain why the Sherman-Morrison update of Section 2.11 fails once backflow is switched on, and estimate the resulting change in the cost of a single-particle move.

Answer to d).

With $\phi_1 = g(r)$, $\phi_2 = xg(r)$, $\phi_3 = yg(r)$ and $g(r) = e^{-r^2/2}$, the Slater matrix is $M_{ia} = g(r_i)\,(1, x_i, y_i)_a$, so $\det \boldsymbol{M} = g(r_1)g(r_2)g(r_3)\det[(1, x_i, y_i)]$. The Gaussian factors are strictly positive, so the node is the vanishing of the remaining determinant – which is twice the signed area of the triangle with vertices (x_i, y_i), and vanishes exactly when the three points are collinear. Now suppose they are, all lying on a line ℓ. Then every difference vector $\boldsymbol{r}_i - \boldsymbol{r}_j$ is parallel to ℓ, so $\boldsymbol{\xi}_i = \sum_{j\neq i} \eta(r_{ij})(\boldsymbol{r}_i - \boldsymbol{r}_j)$ is parallel to ℓ, and $\tilde{\boldsymbol{r}}_i = \boldsymbol{r}_i + \boldsymbol{\xi}_i$ still lies on ℓ. The three effective positions are collinear, the determinant still vanishes, and the node has not moved. Note what the argument uses: only that the displacement lies in the span of the pair-separation vectors. The restriction is in the functional form, not in η, so no amount of optimisation escapes it.

Exercise 5: numerical explorations

a) Using `pinn.py`, reproduce Table 18.2 and extend it: how does E^* depend on the *distribution* of the collocation points rather than merely their range? Try sampling them from $|\psi_a|^2$ and confirm that the variational bound is restored.

b) Add the residual as an auxiliary term to the energy quotient, Eq. (18.9), and find empirically how large λ may become before the bound is lost.
c) Verify Taut's solution (18.15) numerically for $\omega = 1$, and then show that no a makes $e^{-\omega r^2/4}(1+ar)$ an eigenstate for $\omega \neq 1$. What does this say about the closed-form solutions at the other special frequencies?
d) Reproduce Table 18.4. Then initialise the correlator at the exact residual $W = \ln(1+r_{12}) - r_{12}/(1+\beta r_{12})$ by fitting the network to it directly, and confirm that the variance falls towards zero. How close can you get, and what limits you?
e) Replace the plain gradient descent in `pinn.py` by stochastic reconfiguration (Section 14.6). With a hundred parameters rather than two, the improvement should be much more visible than it was in Table 14.6.
f) Implement the CTNN construction (18.20) with small networks and repeat Exercise 4(e) numerically: show that the learned map moves the collinear node that pairwise backflow cannot.
g) Extend the two-electron calculation to the triplet state, where the spatial wave function is antisymmetric and the determinant is no longer trivial. Check the parallel-spin cusp coefficient $\gamma_{\uparrow\uparrow} = 1/3$ from Eq. (18.12).
h) Compute the virial ratio (18.26) for your optimised two-electron wave function and confirm that it is satisfied to better than a per cent. Then deliberately under-train and watch the virial identity degrade before the energy does – it is the more sensitive diagnostic.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter18/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/pinns.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter18/`.

`pinn.py` `Correlator`, a tanh network carrying its own value, gradient and Hessian by forward-mode automatic differentiation; the quartic-oscillator demonstration that residual training returns energies below the exact ground state; `SlaterJastrowNeural`; `TautExact` as a zero-variance check; and the backflow antisymmetry and collinear-node tests. Runs in about five minutes.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Part V
Quantum Computing

Chapter 19
The quantum Fourier transform and phase estimation

Every method in this book so far has been classical. Several have run into the exponential wall of Section 2.16: the number of Slater determinants that can be built by distributing N particles over n single-particle states is $\binom{n}{N}$, and no amount of cleverness in the algorithm changes that. A quantum computer does not remove the wall; it moves it, by storing a state of M modes in M qubits rather than in 2^M complex numbers. The question is then what can actually be computed with such a register, and this chapter answers it for the one quantity we have wanted throughout: an eigenvalue of a Hamiltonian.

Nothing new is needed from the formalism. Section 2.15 already set out the gate model in the language of Chapter 1 – a state $|\Psi\rangle = |i_1 i_2 \cdots i_N\rangle$ in the computational basis of Section 1.6, evolving by unitary matrices acting on one or two qubits at a time, with measurement given by the projection operators of the same section – and observed that quantum computing is *a restricted form of linear algebra: matrix-vector multiplications with special unitary matrices on exponentially large vectors*. This chapter is that observation carried through to an algorithm. Every ingredient is already in Chapters 1 and 2: the tensor product of Section 1.10, the Pauli matrices of Section 1.7, the unitarity that preserves inner products of Section 1.9, and the spectral decomposition of Section 1.11.

The algorithm is *quantum phase estimation*. Its input is a unitary and one of its eigenstates; its output is the eigenphase, written out in binary in a register of qubits, to a precision that improves *exponentially* with the size of that register. Its engine is the *quantum Fourier transform*, which is the subject of the first half of the chapter and which is worth the space: the QFT is the transformation that converts phase information into computational-basis information, and essentially every quantum algorithm with an exponential speed-up uses it.

The chapter runs in three parts.

1. **The quantum Fourier transform** (Sections 19.1–19.4): the definition, the product form derived twice, the circuit, the gate count, the approximate QFT and the bit-ordering convention that causes more confusion than any other detail in the subject.
2. **Quantum phase estimation** (Sections 19.5–19.5.2): the algorithm, why the inverse QFT is the right final step, the exact and inexact cases, and the precision it achieves.
3. **Hamiltonians** (Sections 19.6–19.9): turning energies into phases, the Trotterisation that makes $e^{-i\hat{H}t}$ a circuit, and worked applications to the Lipkin, pairing, Heisenberg and Hubbard models of Chapter 4.

The programs are in `qpe.py`, a statevector simulator that builds the QFT gate by gate, checks it against the Fourier matrix and against `numpy.fft`, and runs phase estimation on the qubit Hamiltonians of Section 4.7. Every number quoted below comes from it.

19.1 From the discrete Fourier transform to the quantum one

The classical discrete Fourier transform of a vector $x = (x_0, \ldots, x_{N-1})$ is

$$y_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} x_j \, e^{2\pi i jk/N}, \qquad k = 0, \ldots, N-1, \tag{19.1}$$

that is, multiplication by the unitary matrix

$$\boldsymbol{F}_N = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \cdots & \omega^{(N-1)^2} \end{pmatrix}, \qquad \omega = e^{2\pi i/N}. \tag{19.2}$$

The matrix is unitary, $\boldsymbol{F}_N^\dagger \boldsymbol{F}_N = \boldsymbol{I}$, by the orthogonality relation $\sum_j \omega^{j(k-k')} = N\delta_{kk'}$; in the language of Section 1.9 it therefore preserves inner products and norms, which is exactly the condition for a matrix to be an admissible quantum gate. Its columns are an orthonormal basis in the sense of Section 1.8, and $\boldsymbol{F}_N$ is the change of basis between them and the computational basis.

The *quantum* Fourier transform is the same matrix, applied to the amplitudes of a quantum state. For $N = 2^n$ it is the unitary defined on the computational basis $|x\rangle$ of Section 1.6 by

$$\boxed{\mathrm{QFT}_N \, |x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i xk/N} \, |k\rangle \,, \qquad x = 0, \ldots, N-1} \tag{19.3}$$

and extended to a general state $|\psi\rangle = \sum_x c_x |x\rangle$ by linearity, $\mathrm{QFT}_N |\psi\rangle = \sum_x c_x \, \mathrm{QFT}_N |x\rangle$.

Two things should be said at once, because they are the source of both the power and the disappointment of the QFT. The power: the transform acts on $N = 2^n$ amplitudes using, as we shall see, only $\mathscr{O}(n^2)$ gates, against the $\mathscr{O}(N \log N)$ operations of the classical fast Fourier transform – exponentially fewer. The disappointment: the result sits in the *amplitudes* of a quantum state, which cannot be read out. Measuring gives one basis state drawn from $|y_k|^2$, not the vector y. The QFT is therefore useless as a subroutine for computing Fourier transforms of data, and enormously useful as a subroutine inside algorithms that need only one number out of the transform – a period, or, as here, a phase.

19.2 The product form

Everything about the QFT circuit follows from rewriting Eq. (19.3) as a product of one-qubit states. We do it twice: once in general, and once explicitly for two qubits, where every step can be followed by eye.

Binary notation.

Write the integer x in binary as $x = x_1 x_2 \ldots x_n$, meaning

$$x = \sum_{j=1}^{n} x_j 2^{n-j}, \qquad x_j \in \{0,1\}, \tag{19.4}$$

and introduce the *binary fraction*

$$0.x_j x_{j+1} \ldots x_n = \frac{x_j}{2} + \frac{x_{j+1}}{2^2} + \cdots + \frac{x_n}{2^{n-j+1}}. \tag{19.5}$$

This notation is not a convenience: the phases of the QFT factor into exactly such fractions, and the same fractions reappear as the output of phase estimation in Section 19.5.

19.2.1 The general derivation

Start from Eq. (19.3) with $N = 2^n$ and write the *output* label in binary, $k = \sum_{m=1}^{n} k_m 2^{n-m}$. Then

$$\frac{xk}{2^n} = x \sum_{m=1}^{n} \frac{k_m}{2^m}, \qquad \text{so} \qquad e^{2\pi i xk/2^n} = \prod_{m=1}^{n} e^{2\pi i x k_m/2^m}. \tag{19.6}$$

The exponential has factorised into one term per output qubit, and each term depends on k only through the single bit k_m. A sum over all k is a sum over all the bits independently, so it factorises too:

$$\mathrm{QFT}_N |x\rangle = \frac{1}{2^{n/2}} \bigotimes_{m=1}^{n} \left(|0\rangle + e^{2\pi i x/2^m} |1\rangle \right). \tag{19.7}$$

Finally, $e^{2\pi i x/2^m}$ depends only on the fractional part of $x/2^m$, and that fractional part is precisely the binary fraction $0.x_{n-m+1}x_{n-m+2}\ldots x_n$. Hence

$$\boxed{\begin{aligned}\mathrm{QFT}_N |x_1 x_2 \ldots x_n\rangle = \frac{1}{2^{n/2}} \left(|0\rangle + e^{2\pi i 0.x_n} |1\rangle\right) \otimes \left(|0\rangle + e^{2\pi i 0.x_{n-1}x_n} |1\rangle\right) \\ \otimes \cdots \otimes \left(|0\rangle + e^{2\pi i 0.x_1 x_2 \ldots x_n} |1\rangle\right)\end{aligned}} \tag{19.8}$$

Three consequences, all of them important. First, the output is a *product* state in the sense of Section 1.10: the QFT of a computational basis state is completely unentangled, and each qubit carries a phase and nothing else. It is worth pausing on that, because the $\bigotimes$ in Eq. (19.8) is doing real work. A general state of n qubits needs 2^n amplitudes; a product state needs $2n$. The QFT of a basis state is therefore an *exponentially compressible* object, and that is precisely why it can be built with polynomially many gates. Second, the first output qubit depends only on the *last* input bit, the second on the last two, and so on – the output order is reversed relative to the input. Third, each factor can be produced by a Hadamard followed by a few controlled phase rotations, which is the circuit.

19.2.2 Two qubits, explicitly

For $n = 2$, write $|x\rangle = |x_1 x_0\rangle$ with $x = 2x_1 + x_0$, and $k = 2k_1 + k_0$. Then

$$\mathrm{QFT}_4 |x_1 x_0\rangle = \frac{1}{2} \sum_{k_1,k_0 \in \{0,1\}} e^{2\pi i (2x_1+x_0)(2k_1+k_0)/4} |k_1 k_0\rangle, \tag{19.9}$$

and the exponent expands as

$$\frac{(2x_1+x_0)(2k_1+k_0)}{4} = x_1k_1 + \frac{x_1k_0}{2} + \frac{x_0k_1}{2} + \frac{x_0k_0}{4}. \tag{19.10}$$

The first term is an *integer*, so $e^{2\pi i x_1 k_1} = 1$ and it drops out. This is the essential simplification, and it is what makes the QFT efficient: the high-order products of input and output bits contribute nothing at all. Collecting what remains by output bit,

$$e^{2\pi i xk/4} = e^{2\pi i k_1\left(\frac{x_0}{2}\right)}\, e^{2\pi i k_0\left(\frac{x_1}{2}+\frac{x_0}{4}\right)} = e^{2\pi i k_1(0.x_0)}\, e^{2\pi i k_0(0.x_1x_0)}, \tag{19.11}$$

so the sum factorises into

$$\mathrm{QFT}_4\,|x_1x_0\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i 0.x_0}\,|1\rangle\right) \otimes \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i 0.x_1x_0}\,|1\rangle\right), \tag{19.12}$$

which is Eq. (19.8) for $n=2$. The first factor depends only on x_0 and the second on both bits: *the least significant phase appears first*, which is the reversal again, now visible in three lines of algebra.

19.3 The circuit

Two gates suffice, and both are already in hand. The Hadamard

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{19.13}$$

of Section 2.15 creates an equal superposition – it is, in the language of Section 1.7, the operator $(X+Z)/\sqrt{2}$, which is why $H^2 = \mathbb{1}$ – and the phase rotation

$$R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{pmatrix} \tag{19.14}$$

adds one binary digit to the phase. R_k is a rotation about the z axis up to a global phase, $R_k = e^{i\pi/2^k}\exp(-i\pi Z/2^k)$, and $R_1 = Z$, $R_2 = S$, $R_3 = T$ are the standard names. A *controlled* gate applies its 2×2 block only on the subspace where the control qubit is $|1\rangle$; it is the two-qubit construction of Section 2.15, of which the CNOT given there is the case in which the block is X.

Read Eq. (19.8) backwards to see the construction. The last output factor carries the phase $e^{2\pi i 0.x_1x_2\ldots x_n}$. A Hadamard on qubit 1 produces $|0\rangle + e^{2\pi i 0.x_1}\,|1\rangle$ – since H acting on $|x_1\rangle$ gives $|0\rangle + (-1)^{x_1}\,|1\rangle$ and $(-1)^{x_1} = e^{2\pi i 0.x_1}$ – and controlled rotations from qubits $2,3,\ldots,n$ append the digits $x_2,x_3,\ldots,x_n$ one at a time. The same pattern then repeats on qubit 2, and so on.

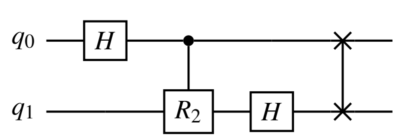

Fig. 19.1 The two-qubit QFT. A Hadamard, one controlled R_2 with $R_2 = \mathrm{diag}(1, e^{i\pi/2})$, a second Hadamard, and a SWAP to undo the reversal of Eq. (19.12).

For $n=3$ the product form reads

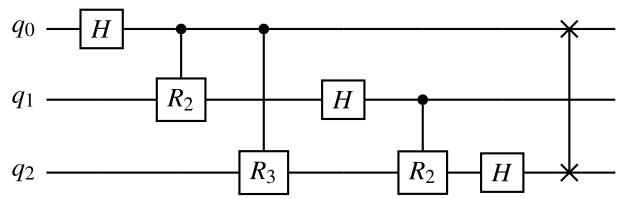

Fig. 19.2 The three-qubit QFT. Hadamard on q_0 then controlled R_2, R_3; Hadamard on q_1 then controlled R_2; Hadamard on q_2; then the reversal. The nested triangular structure is the general pattern.

$$\mathrm{QFT}_8\,|x_1x_2x_3\rangle = \frac{1}{\sqrt{8}}\left(|0\rangle + e^{2\pi i 0.x_3}\,|1\rangle\right)\otimes\left(|0\rangle + e^{2\pi i 0.x_2x_3}\,|1\rangle\right)\otimes\left(|0\rangle + e^{2\pi i 0.x_1x_2x_3}\,|1\rangle\right), \tag{19.15}$$

so the first qubit needs the phase contributions $x_2/2 + x_3/4$, which is exactly what the controlled R_2 and R_3 of Figure 19.2 supply.

The general n-qubit QFT. For $j = 1, \ldots, n$:

1. apply a Hadamard to qubit j;
2. for each later qubit $k > j$, apply a controlled R_{k-j+1} with control k and target j;

then reverse the register with $\lfloor n/2 \rfloor$ SWAP gates.

19.3.1 A worked example

Take $|x\rangle = |101\rangle$, so $x_1 = 1$, $x_2 = 0$, $x_3 = 1$ and $x = 5$. The product form gives

$$\mathrm{QFT}_8\,|101\rangle = \frac{1}{\sqrt{8}}\left(|0\rangle + e^{2\pi i 0.1}\,|1\rangle\right)\otimes\left(|0\rangle + e^{2\pi i 0.01}\,|1\rangle\right)\otimes\left(|0\rangle + e^{2\pi i 0.101}\,|1\rangle\right). \tag{19.16}$$

Evaluating the binary fractions, $0.1 = \frac{1}{2}$, $0.01 = \frac{1}{4}$ and $0.101 = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}$, so

$$e^{2\pi i 0.1} = e^{i\pi} = -1, \qquad e^{2\pi i 0.01} = e^{i\pi/2} = i, \qquad e^{2\pi i 0.101} = e^{5\pi i/4}, \tag{19.17}$$

and

$$\mathrm{QFT}_8\,|101\rangle = \frac{1}{\sqrt{8}}\left(|0\rangle - |1\rangle\right)\otimes\left(|0\rangle + i\,|1\rangle\right)\otimes\left(|0\rangle + e^{5\pi i/4}\,|1\rangle\right). \tag{19.18}$$

Expanding the product gives eight amplitudes, all of modulus $1/\sqrt{8} = 0.353553$, with the phase of $|k\rangle$ equal to $5k/8 \bmod 1$ – which is Eq. (19.3) with $x = 5$, as it must be. `qpe.py` prints the eight amplitudes and confirms both statements, and Figure 19.3 draws them.

Two things in Figure 19.3 are worth dwelling on, and both are properties of the transform rather than of this particular input. The first is that all eight vectors have the same length. A measurement of the transformed register in the computational basis would return each of the eight outcomes with probability $|1/\sqrt{8}|^2 = 1/8$, that is, it would return nothing at all: the entire content of $\mathrm{QFT}_8\,|101\rangle$ sits in the angles, and angles are exactly what a single measurement cannot see. This is the disappointment of Section 19.1 made visual, and it is why the QFT is never the last step of an algorithm. The second is the arrangement of the angles. Reading the labels anticlockwise from $|000\rangle$ gives $0, 5, 2, 7, 4, 1, 6, 3$, which is the sequence $5k \bmod 8$: the input integer has been turned into a *rotation rate*, one eighth of a turn per unit of x per unit of k. Because 5 is coprime to 8 the rate is primitive and the eight vectors are all distinct,

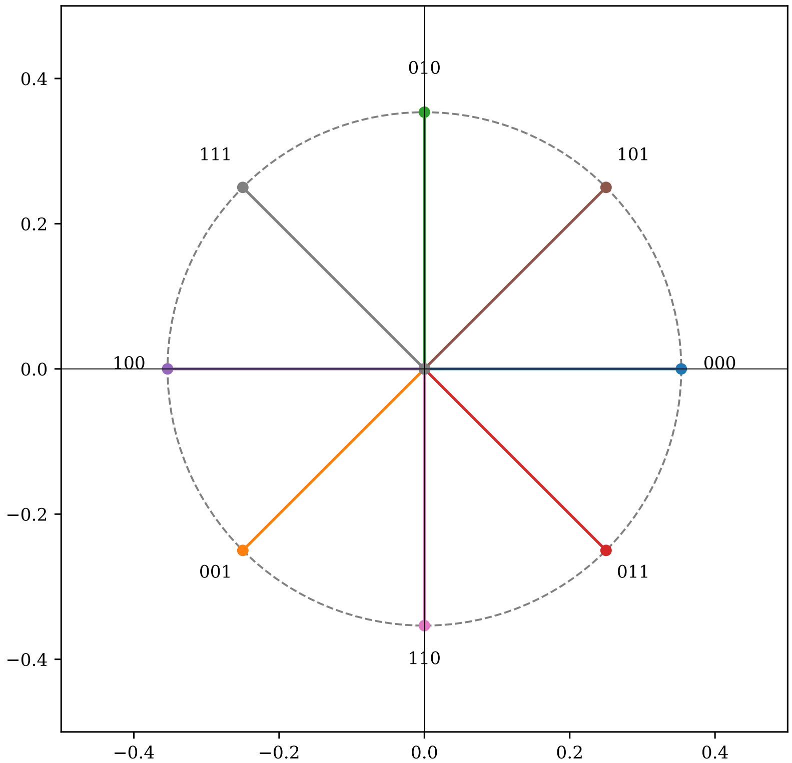

Fig. 19.3 The eight amplitudes of $\mathrm{QFT}_8\,|101\rangle$, drawn as vectors in the complex plane and labelled by the output basis state $|k\rangle$ they belong to. Every one of them has the same modulus $1/\sqrt{8} = 0.3536$, marked by the dashed circle, and the phase of $|k\rangle$ is $2\pi \times (5k/8 \bmod 1)$, which is the Fourier kernel of Eq. (19.3) with $x = 5$. Since 5 and 8 have no common factor the eight vectors occupy the eight eighth-roots of unity exactly once each, reached in the order $0, 5, 2, 7, 4, 1, 6, 3$ as k runs from 0 to 7. Computed by `qpe.py`.

whereas for $x = 4$ the phase $4k/8$ takes only the two values 0 and $\frac{1}{2}$, and the eight vectors would pile up four deep on two positions. That collapse is periodicity, and reading it off is precisely what Shor's algorithm does; here we shall instead read off a single angle, which is phase estimation.

The product form can be traced directly in the picture. Expanding Eq. (19.18), the amplitude of $|k_1k_2k_3\rangle$ is $(-1)^{k_1}\, i^{k_2}\, e^{5\pi i k_3/4}/\sqrt{8}$, so the first bit of the output contributes a half-turn, the second a quarter-turn and the third five-eighths of a turn, independently of one another. The vector labelled $|010\rangle$ therefore stands at 90° and the one labelled $|100\rangle$ at 180°, as the figure shows, and every other position is a sum of these three independent contributions. Three angles, one per qubit, generate all eight: that is the $\bigotimes$ of Eq. (19.8) seen from the outside, and it is the reason the circuit needs only three Hadamards and three controlled rotations rather than an 8×8 matrix multiplication.

The example shows what the QFT *does*: it takes the arithmetic structure of the input – the binary digits of x – and writes it into relative phases. Interference can then act on that structure, which is why the same transform underlies phase estimation, period finding and Shor's algorithm.

19.4 Cost, approximation and conventions

The inverse.

The inverse QFT is obtained by reversing the order of the gates and replacing each R_k by $R_k^\dagger$; the SWAP network is its own inverse. It is therefore no more expensive than the forward transform, and in practice it is the one algorithms use – phase estimation among them.

The gate count.

Counting the circuit of Section 19.3: n Hadamards, $\sum_{j=1}^{n}(n-j) = n(n-1)/2$ controlled rotations, and $\lfloor n/2 \rfloor$ swaps, so

$$\text{gates} = \mathcal{O}(n^2). \tag{19.19}$$

Compare the classical cost of applying $\boldsymbol{F}_N$ directly to a 2^n-dimensional vector, which is $\mathcal{O}(N^2) = \mathcal{O}(4^n)$ operations, or $\mathcal{O}(N \log N)$ with the fast Fourier transform.

n	H	controlled R_k	swaps	direct classical DFT
3	3	3	1	6.4×10^1
6	6	15	3	4.1×10^3
10	10	45	5	1.0×10^6
100	100	4950	50	1.6×10^{60}

Table 19.1 The cost of the QFT circuit against the cost of multiplying by the $2^n \times 2^n$ Fourier matrix. At $n = 100$ the quantum circuit needs about five thousand gates and the classical matrix multiplication needs more operations than there are atoms in the observable universe. Computed by `qpe.py`.

The approximate QFT.

The rotation R_k turns the phase by $2\pi/2^k$, which for large k is tiny. Dropping every controlled rotation with $k - j \geq d$ gives the *approximate* QFT, with $\mathcal{O}(nd)$ gates instead of $\mathcal{O}(n^2)$; choosing $d \sim \log n$ keeps the error bounded and gives $\mathcal{O}(n \log n)$. Table 19.2 shows the trade on six qubits.

d	controlled rotations	largest amplitude error
1	0	2.50×10^{-1}
2	5	2.43×10^{-1}
3	9	1.29×10^{-1}
4	12	3.67×10^{-2}
5	14	1.23×10^{-2}
6	15	0 (exact)

Table 19.2 The approximate QFT on $n = 6$ qubits, applied to $|x\rangle$ with $x = 37$. Each extra level of rotation costs two or three gates and buys about a factor of three in accuracy. Computed by `qpe.py`.

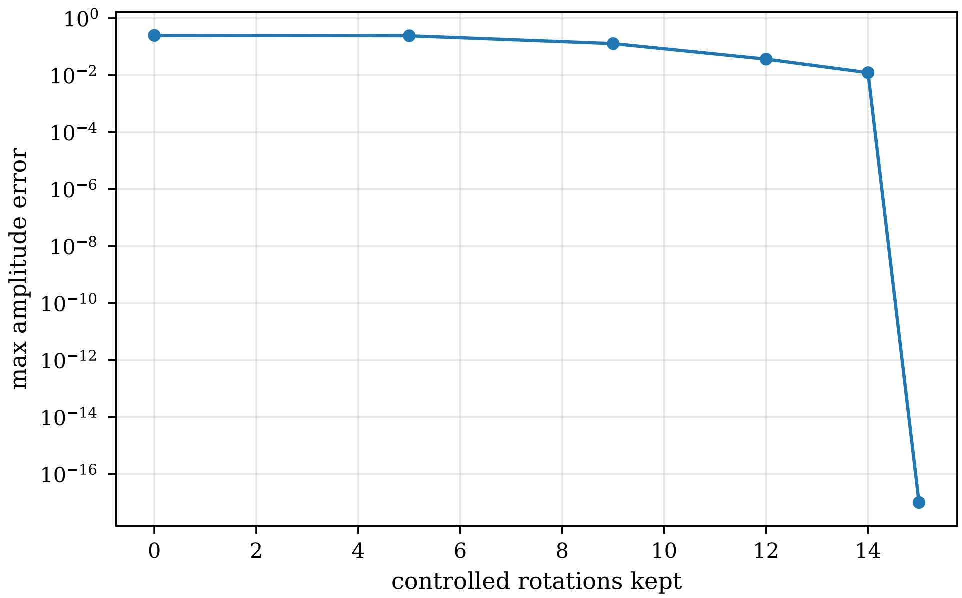

Fig. 19.4 The largest error in any amplitude of the approximate QFT_6 applied to $|x\rangle$ with $x = 37$, plotted against the number of controlled rotations retained. The six points are the six truncation levels $d = 1, \ldots, 6$ of Table 19.2, which keep $0, 5, 9, 12, 14$ and all 15 rotations respectively; the vertical scale is logarithmic and the exact transform, whose error is at the level of machine round-off, is drawn on the 10^{-17} floor of the axis. The curve is nearly flat over its first third and then falls by about a factor of three per level, so that the last and cheapest rotations are the ones that carry the accuracy. Computed by `qpe.py`.

Figure 19.4 plots the same six runs against the quantity one actually pays for, the number of controlled rotations, and the shape it reveals is not the one the asymptotic argument leads one to expect. Keeping the five largest rotations – every controlled R_2, each of which turns the phase by a full half-turn – lowers the largest amplitude error only from 2.50×10^{-1} to 2.43×10^{-1}, a gain of three per cent for a third of the circuit. The error then falls by roughly a factor of three at each further level, while the cost of a level shrinks: level d adds $n - d + 1$ rotations, so the step from $d = 5$ to $d = 6$ is a single controlled R_6, turning the phase by $2\pi/64 \approx 5.6°$, and that one gate takes the error from 1.23×10^{-2} to zero. The marginal value of a controlled rotation therefore *increases* along the ladder, which is the opposite of the truncation's motivation.

There is no contradiction, only a question of scale. The bound of Exercise 19.11(e), $2\pi n 2^{-d}$, equals 37.7×2^{-d} for $n = 6$ and does not drop below unity until $d = 6$: over the whole of Figure 19.4 the estimate is vacuous, and what the figure shows is the pre-asymptotic regime in which every rotation still matters. The prescription $d \sim \log_2 n$ becomes a saving only when n is large enough that $\log_2 n$ is genuinely smaller than n, which six qubits are not; at $n = 100$, the last row of Table 19.1, it replaces the 4950 controlled rotations by some seven hundred, and there it earns its keep. The lesson for a small calculation is simply to keep the whole triangle, and the lesson for a large one is that the gates one may safely discard are the numerous cheap-looking ones at the far end of a *long* ladder, not the last few of a short one.

Bit reversal: the commonest confusion. Equation (19.8) produces the qubits in reversed order, and the final SWAP network exists only to undo that. Many software libraries omit the swaps and document the reversed convention instead, since a relabelling costs nothing classically. The consequence is that a circuit copied from one library into another can be silently wrong. `qpe.py` makes the statement precise by checking that the no-swap output of $|100\rangle$ equals the with-swap output of the bit-reversed input $|001\rangle$, to 10^{-16}. When in doubt, transform $|1\rangle$ and look at where the phase $e^{2\pi i/2}$ ends up.

Verification.

`qpe.py` builds the circuit gate by gate and checks it four ways: the resulting matrix against Eq. (19.2); the output against the product form (19.8) constructed with no circuit at all; the action on a random state against `numpy.fft`; and $\mathrm{QFT}^{-1} \circ \mathrm{QFT} = \mathbb{1}$. The worst discrepancy over all of them is 3.8×10^{-15}. The second check is the one that matters, since the product form is built directly from the binary-fraction factorisation and therefore tests the derivation rather than the implementation's self-consistency.

19.5 Quantum phase estimation

Let U be unitary with an eigenstate $|\psi\rangle$,

$$U\,|\psi\rangle = e^{2\pi i\phi}\,|\psi\rangle\,, \qquad 0 \le \phi < 1. \tag{19.20}$$

The spectral decomposition of Section 1.11 guarantees that such eigenstates exist and form a complete orthonormal set, and that

$$U = \sum_j e^{2\pi i\phi_j}|\psi_j\rangle\langle\psi_j|, \qquad U^k = \sum_j e^{2\pi ik\phi_j}|\psi_j\rangle\langle\psi_j|, \tag{19.21}$$

so that powers of U are obtained by raising the eigenvalues to that power and nothing else – the fact on which the whole algorithm turns. Because U is unitary its eigenvalues have unit modulus, so a single real number ϕ_j per eigenstate carries all the information. Quantum phase estimation determines it.

The algorithm uses two registers: a *control* register of m qubits prepared in $|0\rangle^{\otimes m}$, which will hold the answer, and a *system* register holding $|\psi\rangle$.

Step 1: superposition.

Hadamards on every control qubit give

$$|0\rangle^{\otimes m}\,|\psi\rangle \;\longmapsto\; \frac{1}{2^{m/2}}\sum_{k=0}^{2^m-1}|k\rangle\,|\psi\rangle\,. \tag{19.22}$$

Step 2: controlled powers.

Apply U^{2^j} controlled on control qubit j, for $j = 0, 1, \ldots, m-1$. Since $|\psi\rangle$ is an eigenstate, $U^{2^j}\,|\psi\rangle = e^{2\pi i 2^j\phi}\,|\psi\rangle$, so if the control register is in $|k\rangle$ the total operation applied is U^k and

$$U^k\,|\psi\rangle = e^{2\pi ik\phi}\,|\psi\rangle\,. \tag{19.23}$$

The system register is therefore unchanged, factors out, and the control register is left in

$$\boxed{\frac{1}{2^{m/2}}\sum_{k=0}^{2^m-1} e^{2\pi i k\phi}\,|k\rangle} \tag{19.24}$$

The eigenstate has acted as a catalyst: it is returned untouched, having written its phase into the control register.

Step 3: the inverse QFT.

Compare Eq. (19.24) with Eq. (19.3). If $\phi = m'/2^m$ for an integer m', the control register is *exactly* $\mathrm{QFT}\,|m'\rangle$. Applying the inverse QFT therefore returns $|m'\rangle$, and measuring gives the binary digits of ϕ with certainty:

$$\mathrm{QFT}^{-1}\left(\frac{1}{2^{m/2}}\sum_k e^{2\pi i k\phi}\,|k\rangle\right) = |\phi_1\phi_2\ldots\phi_m\rangle\,, \qquad \phi = 0.\phi_1\phi_2\ldots\phi_m. \tag{19.25}$$

This is why the inverse QFT is the right final step, and it is the whole of the algorithm. Step 2 writes the phase into a Fourier state; step 3 inverts the Fourier transform.

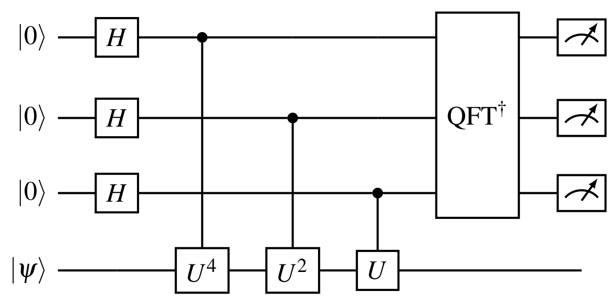

Fig. 19.5 Quantum phase estimation with a three-qubit control register. Hadamards, then controlled powers U^{2^j}, then the inverse QFT, then measurement. The system register emerges in the state it entered.

19.5.1 The bits, one at a time

It is worth seeing the same thing in the product form, because it shows the bits emerging individually. Writing $\phi = 0.\phi_1\phi_2\ldots$, the state (19.24) factorises exactly as Eq. (19.7) did,

$$\frac{1}{2^{m/2}}\left(|0\rangle + e^{2\pi i 2^{m-1}\phi}\,|1\rangle\right)\otimes\left(|0\rangle + e^{2\pi i 2^{m-2}\phi}\,|1\rangle\right)\otimes\cdots\otimes\left(|0\rangle + e^{2\pi i\phi}\,|1\rangle\right). \tag{19.26}$$

Now $2^{m-1}\phi = \phi_1.\phi_2\phi_3\ldots$, and only the fractional part survives in the exponential, so the first factor is $|0\rangle + e^{2\pi i 0.\phi_2\phi_3\cdots}\,|1\rangle$ and so on down the register:

$$\frac{1}{2^{m/2}}\left(|0\rangle + e^{2\pi i 0.\phi_m}\,|1\rangle\right)\otimes\cdots\otimes\left(|0\rangle + e^{2\pi i 0.\phi_1\phi_2\ldots\phi_m}\,|1\rangle\right), \tag{19.27}$$

which is precisely the QFT state (19.8) of the bit string $\phi_1\phi_2\ldots\phi_m$. Each control qubit carries one successive binary suffix of the phase; the inverse QFT decodes them all at once.

19.5.2 Worked example, and the inexact case

Exact.

Let $\phi = 5/8 = 0.101$ and $m = 3$. Then the output is $|101\rangle$ with probability one. `qpe.py` confirms it: probability 1.000000, estimate 0.625000.

Inexact.

If ϕ is not of the form $m'/2^m$ – which, for a Hamiltonian eigenvalue, it never is – the control register is not a computational basis state and the measurement returns a *distribution*, peaked at the best m-bit approximation. The standard result is that the nearest value is obtained with probability at least $4/\pi^2 = 0.405$, whatever ϕ may be, and the probability of landing within one unit of the last bit exceeds $8/\pi^2$. Table 19.3 takes $\phi = 0.3$, which has the non-terminating expansion $0.01001100110011\ldots$, and increases the register.

m	outcome	probability	estimate	error
3	$\|010\rangle$	0.5775	0.250000	5.0×10^{-2}
4	$\|0101\rangle$	0.8756	0.312500	1.3×10^{-2}
6	$\|010011\rangle$	0.8752	0.296875	3.1×10^{-3}
8	$\|01001101\rangle$	0.8751	0.300781	7.8×10^{-4}

Table 19.3 Phase estimation of $\phi = 0.3$. The estimate is the modal outcome read as a binary fraction; the error falls by a factor of two per control qubit. At $m = 4$ the best two outcomes together carry probability 0.931. Computed by `qpe.py`.

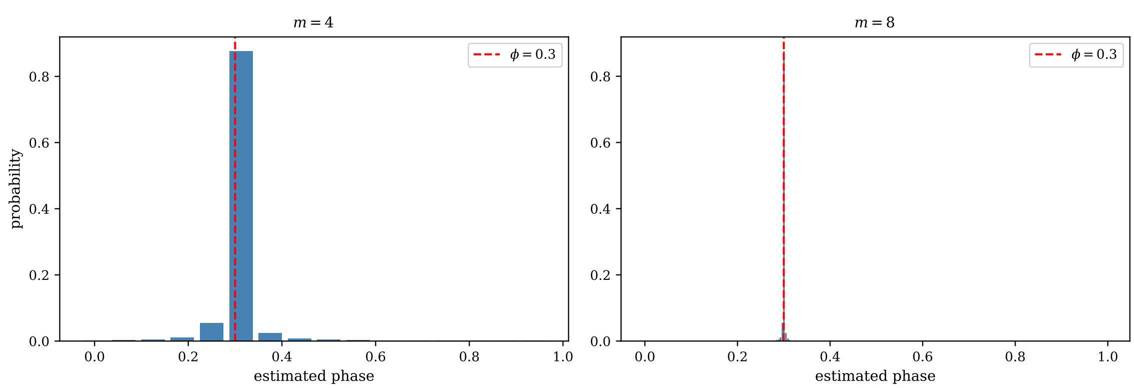

Fig. 19.6 The full output distribution of phase estimation applied to $\phi = 0.3$, with a control register of $m = 4$ qubits (left) and $m = 8$ qubits (right). The bars give the probability of each of the 2^m representable values $m'/2^m$, plotted at that value; the dashed vertical line marks the true phase. The peak carries probability 0.876 at $m = 4$ and 0.875 at $m = 8$: enlarging the register narrows the distribution by a factor of sixteen without making the modal outcome any more likely, since the height is fixed by the Dirichlet kernel of Exercise 19.11(d) and not by m. Computed by `qpe.py`.

Figure 19.6 shows the two extreme rows of Table 19.3 as distributions rather than as modal outcomes, and the comparison makes a point that the table cannot. Going from four control qubits to eight divides the grid spacing by sixteen and the error with it, from 1.3×10^{-2} to

7.8×10^{-4}, yet the peak barely moves in height: 0.8756 against 0.8751. The two numbers are equal for a reason that is arithmetic rather than accidental. The binary expansion of 0.3 is $0.0100110011\ldots$, of period four, so multiplying by 2^4 shifts it by a whole period and leaves the fractional part unchanged; the distance from ϕ to the nearest grid point, measured *in units of the grid*, is therefore $u = 2^m\delta = 0.2$ for $m = 4$ and for $m = 8$ alike. The probability $\sin^2(\pi u)/(\pi u)^2$ then comes out the same both times, and equals $0.34549/0.39478 = 0.875$. It is the offset u, not the size of the register, that decides how likely the best answer is – which is exactly what the $4/\pi^2$ bound says, since $u = \frac{1}{2}$ is its worst case.

The shoulders are worth a look too, because they are the tails of the same kernel and they are not small. In the $m = 4$ panel the outcome one grid step below the peak, $|0100\rangle$ at $\phi = 0.25$, carries 0.055, and the one step above, $|0110\rangle$ at $\phi = 0.375$, carries 0.024; these are $\sin^2(\pi u)/(\pi(u\mp 1))^2$ with $u = 0.2$, and together with the peak they account for the 0.931 quoted in the caption of Table 19.3. The tails fall only as the inverse square of the distance from the peak, so a run of phase estimation has a small but genuinely polynomial chance of landing far from the answer, and the histogram never becomes a delta function however large m is made. This is why the standard statement of the theorem bounds the probability of landing within one unit of the last bit rather than the probability of landing on it, and why the practical recipe is to repeat the circuit and take the median of the estimates rather than to trust a single shot.

With m control qubits the resolution is

$$\Delta\phi \sim \frac{1}{2^m}, \tag{19.28}$$

exponential in the register size. This is the property that makes QPE worth its cost: to gain one more binary digit of an energy one adds one qubit and doubles the number of applications of U, whereas a classical statistical method would need four times as many samples per digit.

19.6 From phases to energies

Everything so far concerned a unitary. Physics gives us a Hamiltonian, and the bridge is time evolution. If $\hat{H}\,|E\rangle = E\,|E\rangle$ then

$$U(t) = e^{-i\hat{H}t} \qquad \Longrightarrow \qquad U(t)\,|E\rangle = e^{-iEt}\,|E\rangle\,, \tag{19.29}$$

so comparing with Eq. (19.20),

$$\boxed{\phi = -\frac{Et}{2\pi} \bmod 1} \tag{19.30}$$

An eigenstate of the Hamiltonian is an eigenstate of $U(t)$, and QPE measures its energy.

Choosing t.

The modulus in Eq. (19.30) is not a formality. Phases are defined only modulo one, so two energies differing by $2\pi/t$ are indistinguishable – aliasing, exactly as in classical signal processing. The time step must therefore satisfy $|E|t < 2\pi$ for every eigenvalue that can appear, which requires a bound on the spectrum. On real hardware the spectrum is precisely what one does not know, and the bound used is the 1-norm of the Pauli coefficients,

$$\Lambda = \sum_{\alpha} |h_\alpha| \;\geq\; \|\hat{H}\|, \qquad t = \frac{2\pi s}{\Lambda}, \quad s < 1, \tag{19.31}$$

available from the Hamiltonian as written. It is a loose bound, and the looseness costs resolution: the useful phases occupy only the fraction $|E|/\Lambda$ of the interval $[0,1)$, so some control qubits are spent resolving empty space. Shifting $\hat{H}$ by a constant, which changes no physics, is the standard way to recover some of them. Choosing t and the offset well is part of designing a QPE calculation, not a detail.

Building $U(t)$.

QPE needs controlled $U^{2^j} = e^{-i\hat{H}t2^j}$, so it needs *Hamiltonian simulation*. This is the application anticipated in the notebox of Section 6.9, and we now use that section rather than repeat it. Writing $\hat{H} = \sum_\alpha h_\alpha P_\alpha$ as a sum of Pauli strings, Eq. (3.138) – which is what Section 4.7 produced for each of our models – each factor $e^{-ih_\alpha P_\alpha \tau}$ is a short circuit: a basis change into the Z eigenbasis of P_α, a ladder of cnots computing the parity, one $R_z(2h_\alpha\tau)$, and the ladder undone. Its depth is set by the *weight* of P_α, which is why the string counts of Table 3.1 and Table 4.7 were worth computing.

The strings do not commute, so assembling them is the Lie-Trotter product formula (6.52) with the Pauli strings in place of H_1 and H_2,

$$e^{-i\hat{H}t} \approx \left(\prod_\alpha e^{-ih_\alpha P_\alpha t/r}\right)^r, \tag{19.32}$$

with the first-order error $\mathscr{O}(t^2/r)$ derived there from the Baker-Campbell-Hausdorff expansion (6.46) and governed by the commutators $[P_\alpha, P_\beta]$. Everything Section 6.9 established carries over unchanged, and two parts of it matter here. The symmetric, or Strang, splitting (6.53) cancels the leading commutator and improves the global error to $\mathscr{O}(t^3/r^2)$, so that a target accuracy ε needs $r = \mathscr{O}(t^{3/2}/\sqrt{\varepsilon})$ steps instead of $\mathscr{O}(t^2/\varepsilon)$ – Exercise 19.11(f) works out what that does to the circuit depth. And strings that *commute* with one another can be exponentiated together with no splitting error at all, so the number of commuting groups, not the number of strings, is the quantity to minimise; that is the last column of Table 19.6.

Higher-order formulas, linear combinations of unitaries and qubitization all improve the scaling further; the point for us is that

$$\text{Hamiltonian simulation} + \text{QPE} \quad\Longrightarrow\quad \text{energy estimation.} \tag{19.33}$$

What the algorithm is, physically.

QPE measures the dynamical phase e^{-iEt} accumulated by an energy eigenstate, using the control register as an interferometer: each control qubit is a which-path degree of freedom, the controlled evolution puts a relative phase between the paths, and the inverse QFT is the recombining beam splitter. It is an interferometric measurement of an energy, carried out to exponential precision.

19.7 Two small Hamiltonians, worked through

Before the many-body models it is worth running the algorithm on two systems small enough to hold in the head. Both are standard exercises in the course that accompanies this book, both are diagonalisable by hand, and both return in Chapter 20, where the variational quantum eigensolver is applied to exactly the same operators. Using the same two Hamiltonians twice is the point: it is the only way to compare two algorithms without also comparing two problems.

19.7.1 One qubit

Take a two-level system with unperturbed energies ε_1 and ε_2 and an interaction λV,

$$\mathscr{H} = \mathscr{H}_0 + \lambda \mathscr{H}_I, \qquad \mathscr{H}_0 = \begin{bmatrix} \varepsilon_1 & 0 \\ 0 & \varepsilon_2 \end{bmatrix}, \qquad \mathscr{H}_I = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix}, \tag{19.34}$$

with $V_{12} = V_{21}$ real. A 2×2 Hermitian matrix has four real parameters and $\{\boldsymbol{I},\boldsymbol{X},\boldsymbol{Y},\boldsymbol{Z}\}$ is a basis for exactly that space, so the rewriting is forced rather than clever. Reading off the coefficients as in Section 4.7,

$$\mathscr{H} = (\mathscr{E} + \lambda c)\boldsymbol{I} + (\Omega + \lambda\omega_z)\boldsymbol{Z} + \lambda\omega_x \boldsymbol{X}, \tag{19.35}$$

with

$$\mathscr{E} = \frac{\varepsilon_1 + \varepsilon_2}{2}, \quad \Omega = \frac{\varepsilon_1 - \varepsilon_2}{2}, \quad c = \frac{V_{11} + V_{22}}{2}, \quad \omega_z = \frac{V_{11} - V_{22}}{2}, \quad \omega_x = V_{12}. \tag{19.36}$$

The eigenvalues follow at once, since $\boldsymbol{I}$ commutes with everything and the remaining piece squares to a multiple of the identity:

$$E_\pm = \mathscr{E} + \lambda c \pm \sqrt{(\Omega + \lambda\omega_z)^2 + (\lambda\omega_x)^2}. \tag{19.37}$$

Every number the algorithm produces can therefore be checked against a closed form, which is what makes this the right place to start. We use the standard choice $\varepsilon_1 = 0$, $\varepsilon_2 = 4$, $V_{11} = -V_{22} = 3$, $V_{12} = V_{21} = 0.2$ and $\lambda = 1$ throughout, for which $E_- = 0.980196$.

The identity term is free, and worth removing.

The coefficient of $\boldsymbol{I}$ in (19.35) is a known constant. It shifts every eigenvalue by the same amount, contributes only a global phase $e^{-i(\mathscr{E}+\lambda c)t}$ to the evolution operator, and is invisible to any measurement. There is no reason to spend the phase register on it, and a good reason not to: the aliasing condition of Section 19.6 requires the *whole* spectrum to fit inside one period, so a large constant offset eats the resolution. Here $\mathscr{E} + \lambda c = 2$, and running QPE on $\mathscr{H}$ directly, with t chosen from the Pauli 1-norm, wraps the phase and returns nonsense. Running it on the traceless part

$$\tilde{\mathscr{H}} = \mathscr{H} - (\mathscr{E} + \lambda c)\boldsymbol{I} = \boldsymbol{Z} + 0.2\boldsymbol{X}, \tag{19.38}$$

whose 1-norm is $\Lambda = 1.2$ and whose ground-state energy is -1.019804, and adding the constant back at the end, is both cheaper and correct. This is standard practice and not a trick of the example: *subtract the identity, estimate the rest, add it back.* Table 19.4 shows the result with $t = 0.9 \times 2\pi/\Lambda = 4.7124$.

Table 19.4 Phase estimation on the one-qubit Hamiltonian of Eq. (19.34), prepared in the exact ground state, with the identity removed and added back. The exact answer is $E_- = 0.980196$. Each extra control qubit halves the grid spacing $2\pi/(t\,2^m)$, and the error follows it down until the true phase lands close to a grid point, where it stalls until the grid refines past it. The success probability is not monotonic: it depends on how near the phase is to a representable value, not on how many qubits are used.

m	outcome	E	$\lvert E-E_-\rvert$	probability
3	110	1.000000	2.0×10^{-2}	0.955
4	1100	1.000000	2.0×10^{-2}	0.828
5	11000	1.000000	2.0×10^{-2}	0.446
6	110001	0.979167	1.0×10^{-3}	0.992
7	1100010	0.979167	1.0×10^{-3}	0.968
8	11000100	0.979167	1.0×10^{-3}	0.878
10	1100001111	0.980469	2.7×10^{-4}	0.864

Two features of the table are worth naming, because both recur in every application. The error is *not* monotone in m in any simple way: it falls in steps, because the estimate is always the nearest point of a grid of spacing $2\pi/(t2^m)$ and a phase that happens to sit near a grid point is resolved well early and then not improved until the grid refines past it. And the success probability wanders between 0.45 and 0.99 without trend. The $4/\pi^2$ bound of Section 19.5.2 guarantees only that the best estimate is likely; it says nothing about which m will be lucky. In practice one repeats the measurement and takes the mode, which costs a constant factor and removes the worry.

19.7.2 Two qubits

The next case up is a 4×4 Hamiltonian of the form met in the pairing problem, with a diagonal of single-particle energies and an interaction that connects the two pairs of states,

$$\mathscr{H} = \begin{bmatrix} \varepsilon_1+V_z & 0 & 0 & V_x \\ 0 & \varepsilon_2-V_z & V_x & 0 \\ 0 & V_x & \varepsilon_3-V_z & 0 \\ V_x & 0 & 0 & \varepsilon_4+V_z \end{bmatrix}. \tag{19.39}$$

The diagonal part is easy to decompose once one notices that the diagonal of *any* two-qubit operator is a combination of exactly the four strings $\boldsymbol{II}$, $\boldsymbol{ZI}$, $\boldsymbol{IZ}$ and $\boldsymbol{ZZ}$ – these are the only Pauli strings that are diagonal, and they are linearly independent, so they span the diagonals. The four coefficients are the four sign-weighted averages

$$\begin{aligned} \varepsilon_{II} &= \tfrac{1}{4}\left(\varepsilon_1+\varepsilon_2+\varepsilon_3+\varepsilon_4\right), \qquad & \varepsilon_{ZI} &= \tfrac{1}{4}\left(\varepsilon_1+\varepsilon_2-\varepsilon_3-\varepsilon_4\right), \\ \varepsilon_{IZ} &= \tfrac{1}{4}\left(\varepsilon_1-\varepsilon_2+\varepsilon_3-\varepsilon_4\right), \qquad & \varepsilon_{ZZ} &= \tfrac{1}{4}\left(\varepsilon_1-\varepsilon_2-\varepsilon_3+\varepsilon_4\right), \end{aligned} \tag{19.40}$$

which are nothing but the inverse of the 4×4 sign matrix built from ±1 patterns of $\boldsymbol{Z}$ eigenvalues – a two-qubit Hadamard-like transform of the diagonal. The off-diagonal V_x, which flips both qubits at once, is $V_x\boldsymbol{X}\otimes\boldsymbol{X}$, and the $\pm V_z$ pattern is another $\boldsymbol{Z}\otimes\boldsymbol{Z}$. Hence

$$\mathscr{H} = \mathscr{H}_0 + \mathscr{H}_I, \tag{19.41}$$

with

$$\begin{aligned}\mathscr{H}_0 &= \varepsilon_{II}\,\boldsymbol{I}\otimes\boldsymbol{I} + \varepsilon_{ZI}\,\boldsymbol{Z}\otimes\boldsymbol{I} + \varepsilon_{IZ}\,\boldsymbol{I}\otimes\boldsymbol{Z} + \varepsilon_{ZZ}\,\boldsymbol{Z}\otimes\boldsymbol{Z},\\ \mathscr{H}_I &= V_z\,\boldsymbol{Z}\otimes\boldsymbol{Z} + V_x\,\boldsymbol{X}\otimes\boldsymbol{X}.\end{aligned} \tag{19.42}$$

Five Pauli strings for a 4×4 matrix, where sixteen were available: the structure of the problem, not the size of the Hilbert space, is what sets the cost. With $\varepsilon = (0, 2.5, 6.5, 7.0)$, $V_z = 0$ and $V_x = 2$ this is

$$\mathscr{H} = 4\,\boldsymbol{II} - 2.75\,\boldsymbol{ZI} - 0.75\,\boldsymbol{IZ} - 0.5\,\boldsymbol{ZZ} + 2\,\boldsymbol{XX},$$

with spectrum $\{-0.531129,\ 1.671573,\ 7.328427,\ 7.531129\}$ and $\Lambda = 6$ after the identity is removed. Table 19.5 collects the runs.

Table 19.5 Phase estimation on the two-qubit Hamiltonian of (19.39), again with the identity removed and $t = 0.9\times 2\pi/\Lambda = 0.9425$. The exact ground-state energy is -0.531129. Chapter 20 solves the same Hamiltonian variationally and reaches machine precision with four parameters, at a cost measured in shots rather than in circuit depth.

m	outcome	E	$\lvert E-E_0\rvert$	probability
3	101	-0.166667	3.6×10^{-1}	0.515
4	1011	-0.583333	5.2×10^{-2}	0.950
5	10110	-0.583333	5.2×10^{-2}	0.810
6	101011	-0.479167	5.2×10^{-2}	0.407
7	1010111	-0.531250	1.2×10^{-4}	1.000
8	10101110	-0.531250	1.2×10^{-4}	1.000
10	1010111000	-0.531250	1.2×10^{-4}	0.999

At $m=7$ the phase happens to be very nearly representable, the distribution collapses onto a single outcome with probability 1.000 to three decimals, and the error drops by nearly three orders of magnitude at once. This is the behaviour the $4/\pi^2$ bound describes: when ϕ is exactly a dyadic rational the algorithm is deterministic, and near-exactness is nearly deterministic. It is also a warning. Nothing about the $m=6$ run announces that its answer is worse than the $m=4$ one; only knowing the exact answer reveals it. Phase estimation returns a number and a probability, not an error bar, and turning the one into the other requires the analysis of Section 19.5.2 rather than inspection of the output.

19.8 The chapter-4 models on a quantum computer

Section 4.7 carried the Lipkin, pairing, Hubbard and Heisenberg models across to qubits, and Table 4.7 counted the resulting Pauli strings. We now feed them to QPE. The cost of the calculation is set by three numbers: how many Pauli strings the Hamiltonian has, what their weight is – the number of non-identity factors, which fixes the length of the cnot ladders – and into how many mutually *commuting* groups they fall, since terms within a group can be exponentiated together with no Trotter error at all.

The table repays reading. *Lipkin* and *Heisenberg* have weight-two Hamiltonians with no Jordan-Wigner tails at all, and for opposite reasons. In the Lipkin model the two levels are degenerate, so the quasispin operators are simple sums of single-particle Paulis,

$$J_z = \tfrac{1}{2}\sum_p Z_p, \qquad J_x = \tfrac{1}{2}\sum_p X_p, \qquad J_y = \tfrac{1}{2}\sum_p Y_p, \tag{19.43}$$

model	qubits	Pauli terms	max weight	commuting groups	E_0
Lipkin, $N=2$	2	4	2	2	-1.41421
Lipkin, $N=4$	4	16	2	4	-4.00000
pairing, 3 levels	3	9	2	4	-0.70530
Heisenberg, 3 sites	3	9	2	4	-0.75000
Heisenberg, 4 sites	4	12	2	2	-2.00000
Hubbard, 2 sites	4	10	3	2	-1.23607

Table 19.6 The qubit Hamiltonians. Parameters: Lipkin $\varepsilon = V = 1$; pairing $g = 1$, $\xi = 1$; Heisenberg $J = 1$, periodic; Hubbard $t = 1$, $U = 2$. The Hubbard ground state -1.23607 is the exact $\frac{1}{2}(U - \sqrt{U^2 + 16t^2}) = 1 - \sqrt{5}$, and the four-site Heisenberg ring gives exactly $-2J$. Computed by `qpe.py`.

and $\hat{H} = \varepsilon J_z + V(J_x^2 - J_y^2) + \dots$ is a sum of one- and two-qubit strings. In the Heisenberg model the degrees of freedom are spins rather than fermions, so operators on different sites commute and there are no parity strings to carry. This is why the Lipkin model is the standard first target for quantum-computing demonstrations, and why spin models were the first systems simulated on hardware.

Hubbard pays the fermionic price: the parity strings of the Jordan-Wigner transformation give weight-three terms even for two sites, and the weight grows with the distance between the modes. *Pairing* in the seniority-zero space sits in between, one qubit per level and weight two, as Section 4.7 showed.

19.8.1 Phase estimation on the models

Table 19.7 runs the algorithm. The system register is prepared in the exact ground state – the question of how one obtains it is Section 19.9 – the time step is chosen by Eq. (19.31) with $s = 0.9$, the evolution operator is exact, and the control register is varied.
The pattern is the expected one and it is worth stating plainly: the accuracy is controlled entirely by the size of the control register, not by the size or difficulty of the physical system. Four qubits give roughly one decimal digit, ten give three, and the Hubbard model is no harder to resolve than the Lipkin model even though its Hamiltonian is nastier. What the difficulty of the Hamiltonian controls is the *cost of each application of U* – and, as we now see, the number of Trotter steps.

19.8.2 The second error: Trotterisation

On hardware $e^{-i\hat{H}t}$ is itself built from gates by Eq. (19.32), and this introduces an error with nothing to do with the control register. Table 19.8 separates the two on the three-site Heisenberg chain, whose nine Pauli strings fall into four commuting groups.

> **Two errors, one budget.** The last column of Table 19.8 falls as $1/r$, which is the first-order product formula's rate, Eq. (6.52). The middle column follows it – and then stops at $r = 8$, because the 8-qubit register cannot resolve anything finer than 2^{-8} of the period. Beyond that point every additional Trotter step is wasted work. The two errors must be *balanced against each other*, not minimised separately, and it is that balance which

model	E_0	t	m	outcome	E_{QPE}	error
Lipkin, $N=2$	-1.414214	2.827	4	1010	-1.388889	2.5×10^{-2}
			6	101001	-1.423611	9.4×10^{-3}
			8	10100011	-1.414931	7.2×10^{-4}
Lipkin, $N=4$	-4.000000	0.707	4	0111	-3.888889	1.1×10^{-1}
			6	011101	-4.027778	2.8×10^{-2}
			8	01110011	-3.993056	6.9×10^{-3}
			10	0111001101	-4.001736	1.7×10^{-3}
pairing, 3 levels	-0.705303	1.331	4	0010	-0.590278	1.2×10^{-1}
			8	00100110	-0.700955	4.4×10^{-3}
			10	0010011001	-0.705566	2.6×10^{-4}
Heisenberg, 3 sites	-0.750000	2.513	4	0101	-0.781250	3.1×10^{-2}
			8	01001101	-0.751953	2.0×10^{-3}
			10	0100110011	-0.749512	4.9×10^{-4}
Hubbard, 2 sites	-1.236068	1.131	4	0100	-1.388889	1.5×10^{-1}
			6	001110	-1.215278	2.1×10^{-2}
			8	00111001	-1.236979	9.1×10^{-4}

Table 19.7 Phase estimation of the ground-state energies of Table 19.6, with the exact ground state as input and the exact $e^{-i\hat{H}t}$. The error falls by roughly a factor of two per control qubit, occasionally plateauing when the modal outcome does not change. Computed by `qpe.py`.

r	E_{QPE}	energy error	$\|U_{\text{Trotter}}-U\|$
1	-0.625000	1.3×10^{-1}	4.5×10^{-1}
2	-0.722656	2.7×10^{-2}	2.1×10^{-1}
4	-0.742188	7.8×10^{-3}	1.0×10^{-1}
8	-0.751953	2.0×10^{-3}	5.0×10^{-2}
16	-0.751953	2.0×10^{-3}	2.5×10^{-2}
32	-0.751953	2.0×10^{-3}	1.3×10^{-2}

Table 19.8 First-order Trotterisation of the three-site Heisenberg chain, $m=8$ control qubits, exact $E_0=-0.75$. The operator error falls cleanly as $1/r$; the energy error follows it down and then stops on the floor set by the control register. Computed by `qpe.py`.

fixes the circuit depth of a real calculation. Getting it wrong in either direction is the commonest way to waste a quantum computer.

Figure 19.7 sets the two errors side by side on logarithmic axes, and the geometry of the two lines is the whole argument. The operator error is a clean straight line of slope -1 – it halves for every doubling of r, from 4.5×10^{-1} down to 1.2×10^{-2}, which is the first-order rate of Eq. (19.32) and nothing else. Extending it, it would meet the dashed 2^{-8} line only near $r\approx115$. The energy error, by contrast, is flat: 1.95×10^{-3} at every r from 1 to 32. That number is not a Trotter error at all. The energy grid of an eight-qubit register at this time step has spacing $2\pi/(t\,2^8)=9.77\times10^{-3}$, the exact $E_0=-0.75$ sits two tenths of a step from the nearest grid point -0.751953, and $0.2\times9.77\times10^{-3}=1.95\times10^{-3}$ is the whole of the discrepancy. The register, not the splitting, is what limits the answer over the entire range plotted.

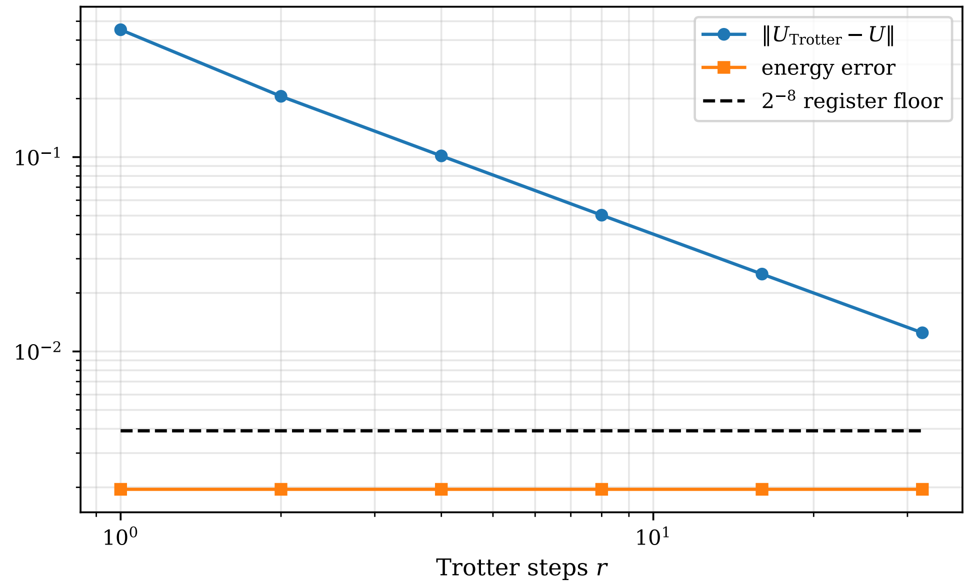

Fig. 19.7 The two error sources of a phase-estimation run on the three-site Heisenberg chain with $J = 1$, $t = 2.5133$ and $m = 8$ control qubits, plotted against the number of first-order Trotter steps r on logarithmic axes. Circles give the largest entry of $U_{\mathrm{Trotter}} - U$ and squares the error of the energy that phase estimation actually reports, with the nine Pauli strings gathered into their four commuting groups; the dashed line is the resolution 2^{-8} of the control register. The operator error is a straight line of slope -1, falling from 4.5×10^{-1} at $r = 1$ to 1.2×10^{-2} at $r = 32$, while the reported energy does not move at all. Computed by `qpe.py`.

Two things follow, and the second is the more interesting. The first is the practical one: on this problem every Trotter step beyond the first is wasted work, and a calculation that spent a hundred-fold increase in circuit depth driving the upper curve down to the dashed line would report exactly the same energy at the end of it. The second is that a large error in the *operator* is not the same thing as a large error in the *eigenvalue* one is measuring. At $r = 1$ the Trotterised evolution operator differs from the true one by 0.45 in its largest entry, which is enormous, and yet the ground-state energy it yields is correct to the resolution of the register. The operator norm is a worst case taken over the whole Hilbert space; phase estimation interrogates a single eigenphase, and the symmetric, four-fold degenerate ground multiplet of the three-site ring is displaced very little by a splitting that scrambles the rest of the spectrum badly. Where the crude splitting does show is in the *sharpness* of the peak rather than in its position, because the input is no longer an exact eigenstate of the operator being powered: the modal outcome carries probability 0.79 at $r = 1$, 0.80 at $r = 4$ and 0.87 at $r = 32$, against 0.875 for exact evolution. Trotterisation has spent itself on repetitions, which is the cheap currency, rather than on accuracy, which is the dear one. There is no guarantee of that, however: a less symmetric ground state, a looser grouping of the Pauli strings or a longer evolution time displaces the peak instead of merely broadening it, and the middle column of Table 19.8 is what that looks like.

Why the Trotter error is fatal here and harmless in unitary coupled cluster.

Chapter 10 met exactly the same splitting. The unitary coupled-cluster operator $e^{\hat{\sigma}(\boldsymbol{t})}$ cannot be applied in one step either, and Eq. (10.58) breaks it into the same kind of product with the same first-order error. Yet Section 10.15.1 found something startling: for the pairing model, *every* Trotter number from $n = 1$ upwards gave the same energy to ten digits. The crudest possible splitting cost nothing at all.

The reason is not that the splitting error was small – holding the amplitudes fixed and re-evaluating showed it falling as $1/n$ in the usual way. It is that the Trotterised UCC ansatz is a *different parametrisation of a similar family of states*, and the amplitudes are *variational*. The optimiser simply finds whichever values are best for the parametrisation it is handed, and absorbs the splitting error into them.

Nothing of the kind is available to phase estimation. There is no parameter to re-optimise: the circuit is fixed, and what it measures is an eigenphase of whatever unitary was actually built. Trotterising U therefore does not perturb the answer around the right one – it changes the operator whose spectrum is being measured, and QPE reports the eigenvalue of the wrong operator to full precision. That is why r appears in Table 19.8 as a genuine, irreducible error source, while it does not appear in the UCC table of Section 10.15.1 at all.

The contrast is worth carrying forward, because it is the difference between the two algorithmic philosophies of this part of the book. A variational method has parameters that absorb the imperfections of its own circuit; a projective method like QPE has none, and every approximation made in constructing the circuit propagates straight into the answer. What one gains in exchange is the exponential precision of Eq. (19.28), which no variational method offers.

19.9 The state you start from

One assumption has been carried silently through the whole chapter: that the system register contains an eigenstate. It need not. The eigenstates are complete, by Section 1.11, so any state may be expanded in them,

$$|\psi_{\text{trial}}\rangle = \sum_j c_j \left|E_j\right\rangle, \qquad c_j = \langle E_j|\psi_{\text{trial}}\rangle, \tag{19.44}$$

and Eq. (19.21) then carries every term through the algorithm independently: the controlled powers act on each $\left|E_j\right\rangle$ with its own phase, and the control register ends *entangled* with the system register,

$$\sum_j c_j \left|\widetilde{\phi}_j\right\rangle \left|E_j\right\rangle. \tag{19.45}$$

Measuring returns E_j with probability $|c_j|^2$ – which is the measurement rule of Section 2.15 applied to Eq. (19.45), and which also collapses the system register onto $\left|E_j\right\rangle$. *Phase estimation is therefore also a state-preparation method*: a successful run leaves the eigenstate behind in the system register, ready to be used.

The overlap requirement. QPE does not average the energies of a superposition, and it does not degrade gracefully: it returns *one* of the eigenvalues, chosen at random with probability $|c_j|^2$, and the run that returns E_0 returns it to the full precision of the register. The overlap does not control the accuracy. It controls how many repetitions are needed.

Table 19.9 makes this concrete on the four-particle Lipkin model, mixing the ground state with the first excited state.

Figure 19.8 is the same four runs drawn as histograms, and it makes the statement of the grey box above almost redundant. Read the four panels from left to right and nothing whatever happens to the *positions* of the peaks: the lower one stands at -3.993 in all four, an error of 6.9×10^{-3} set by the eight-qubit grid, and the upper one at -3.160. What changes is

$\lvert c_0\rvert^2$	$P(E_0$ bits$)$	$P(E_1$ bits$)$	modal E
1.00	0.8751	0.0001	−3.993
0.80	0.7001	0.1965	−3.993
0.60	0.5251	0.3930	−3.993
0.30	0.2625	0.6876	−3.160

Table 19.9 Lipkin, $N = 4$, $m = 8$, with exact $E_0 = -4.000000$ and $E_1 = -3.162278$. The two peaks are the two eigenvalues and their weights are the overlaps. Below $|c_0|^2 \approx 0.5$ the wrong peak becomes the taller one, and a single-shot experiment reporting the modal outcome is actively misleading. Computed by `qpe.py`.

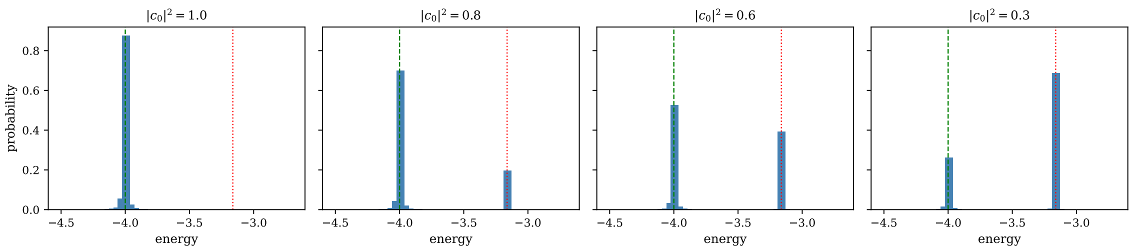

Fig. 19.8 Output histograms of phase estimation on the four-particle Lipkin model with $m = 8$ control qubits, for four trial states of decreasing overlap with the ground state. The horizontal axis is the energy obtained from the measured phase through Eq. (19.30); the dashed green line marks the exact $E_0 = -4.000000$ and the dotted red line the first excited state $E_1 = -3.162278$. From left to right the trial state is the exact ground state and then admixtures with $|c_0|^2 = 0.8,\ 0.6$ and 0.3. The two peaks never move: only their weights change, in proportion to $|c_j|^2$, and no probability whatever appears between them. Computed by `qpe.py`.

only their height, and it changes in exact proportion to the overlap. The lower peak carries $0.8751,\ 0.7001,\ 0.5251$ and 0.2625, which is $|c_0|^2$ times the same Dirichlet factor 0.8751 in every panel; the upper carries $|c_1|^2$ times 0.9825, that eigenvalue happening to sit a little closer to a representable phase. Since $0.9825|c_1|^2$ overtakes $0.8751|c_0|^2$ at $|c_0|^2 = 0.53$, the wrong peak becomes the taller one just above an even mixture, which is what the last two panels straddle.

Compare this with what a variational method does with the same trial states, and the difference between the two philosophies of Chapter 20 and this one is laid out in a single picture. At $|c_0|^2 = 0.6$ the expectation value is $\langle\psi|\hat{H}|\psi\rangle = 0.6 \times (-4.000) + 0.4 \times (-3.162) = -3.665$, and a variational calculation would return precisely that number: a smooth, well-behaved, monotone function of the quality of the state, an upper bound to E_0, and a number that is not an eigenvalue of anything. The third panel of Figure 19.8 has no probability at all at -3.665. Phase estimation never reports the average; it reports -3.993 on 53 per cent of its runs and -3.160 on 39 per cent, each to the full precision of the register, and the run that lands on the ground state is as accurate as the run in the leftmost panel where the input was exact. A poor trial state costs repetitions, not digits. It also means that the histogram carries more than the ground-state energy: both eigenvalues are visible in the second and third panels, with their overlaps as weights, and accumulating the whole distribution is spectroscopy rather than a ground-state calculation.

Two lessons. The honest procedure is to accumulate the whole histogram and identify the peaks, not to report the modal outcome of one run – the histogram *is* the spectrum, weighted by the overlaps, and reading it gives excited states as well as the ground state. And a good approximate eigenstate is valuable even though QPE does not use it variationally, because it is what keeps the number of repetitions bounded.

Where this leads.

That last observation is the reason the two great algorithms of quantum many-body computing fit together rather than compete. The variational quantum eigensolver of Chapter 20 prepares a state by minimising $\langle\psi(\boldsymbol{\theta})|\hat{H}|\psi(\boldsymbol{\theta})\rangle$ over the parameters of a shallow circuit; it tolerates noise, needs no controlled evolution, and gives a variational upper bound, but its accuracy is limited by the ansatz and by the difficulty of the classical optimisation. QPE needs deep coherent circuits and controlled $e^{-i\hat{H}t}$, which puts it out of reach of present hardware, and in exchange gives the eigenvalue itself to exponential precision.

	VQE	**QPE**
Output	a variational upper bound	an eigenvalue, to 2^{-m}
Circuit depth	shallow	deep, coherent
Needs controlled $e^{-i\hat{H}t}$	no	yes
Cost per digit	$\mathcal{O}(\varepsilon^{-2})$ measurements	one more qubit
Limited by	the ansatz and the optimiser	hardware coherence
Noise	tolerant	intolerant
Era	near-term	fault-tolerant

Table 19.10 The two algorithms. The natural workflow uses both: VQE as a state-preparation front end, QPE as the readout.

The combination is the standard long-term plan: use VQE, or any other method in this book, to prepare a state with substantial overlap with the target eigenstate; then use QPE to refine the energy. Chapter 20 takes up the first half.

19.10 Summary

The quantum Fourier transform is the map $|x\rangle \mapsto N^{-1/2}\sum_k e^{2\pi ixk/N}|k\rangle$. Its whole theory follows from one rewriting: the phase factors into binary fractions, Eq. (19.8), so that the output is an unentangled product of one-qubit phase states. That factorisation gives the circuit – Hadamards, controlled R_k, and a final reversal – a gate count of $\mathcal{O}(n^2)$ against $\mathcal{O}(4^n)$ for the matrix it represents, and an approximate version at $\mathcal{O}(n\log n)$. What it does not give is a way to read the transform out: the answer is in the amplitudes.

Quantum phase estimation is what makes that acceptable, because it needs only one number. Controlled powers of U write the eigenphase into a control register as a Fourier state, Eq. (19.24); the inverse QFT decodes it into binary. With m control qubits the resolution is 2^{-m} – exponential in the register size, one more digit per qubit.

For physics the unitary is $e^{-i\hat{H}t}$ and the phase is $-Et/2\pi \bmod 1$. Two consequences follow. The modulus forces a choice of t from a bound on the spectrum, and a loose bound wastes register. And $e^{-i\hat{H}t}$ must be built by Trotterisation, whose error is independent of the register and must be balanced against it, as Table 19.8 shows. Applied to the models of Chapter 4, the algorithm recovers the exact ground-state energies to 10^{-3} with ten control qubits, and the difficulty of the Hamiltonian shows up in the cost of each U, not in the achievable accuracy.

Finally, QPE does not require the exact eigenstate, only overlap with it, and what the overlap buys is repetitions rather than precision. That is the hinge between this chapter and the next.

19.11 Exercises

Exercise 1: the quantum Fourier transform

a) Show that $\boldsymbol{F}_N$ of Eq. (19.2) is unitary, using $\sum_j \omega^{j(k-k')} = N\delta_{kk'}$.
b) Carry out the derivation of Eq. (19.8) in full, justifying the interchange of sum and product and the step in which only the fractional part of $x/2^m$ survives.
c) Repeat the two-qubit calculation of Section 19.2.2 for three qubits, and identify explicitly which products $x_j k_m$ are integers and therefore drop out.
d) Show that the QFT of a computational basis state is always a product state, and that the QFT of a general state need not be. What does that say about where the entanglement in a quantum algorithm comes from?
e) Compute $\mathrm{QFT}_8 |101\rangle$ from Eq. (19.18) by expanding the product, and verify that the amplitude of $|k\rangle$ is $e^{2\pi i 5k/8}/\sqrt{8}$.
f) Show that QFT_N has order four, $\mathrm{QFT}_N^4 = \mathbb{1}$, and find its eigenvalues. *Hint:* the same is true of the classical Fourier transform.

Answer to d).

Equation (19.8) exhibits $\mathrm{QFT}\,|x\rangle$ as a tensor product, so it is unentangled for every basis state $|x\rangle$. Since the QFT is linear and the basis states span the space, this does not extend to superpositions: $\mathrm{QFT}(|x\rangle + |y\rangle)/\sqrt{2}$ is a sum of two product states and is generically entangled. The lesson is that the QFT does not create entanglement out of nothing – it converts entanglement in one basis into entanglement in another, and in phase estimation the entanglement between the control and system registers is created by the *controlled* U operations, not by the transform.

Exercise 2: circuits, cost and conventions

a) Verify that the circuit of Figure 19.1 implements Eq. (19.12), by following $|x_1 x_0\rangle$ through gate by gate for all four inputs.
b) Show that H acting on $|x_j\rangle$ gives $\left(|0\rangle + e^{2\pi i\,0.x_j}|1\rangle\right)/\sqrt{2}$, and hence that the Hadamard is the $k=1$ member of the family R_k combined with a basis change.
c) Count the gates in the general circuit and derive Eq. (19.19). How many gates would the same circuit need without the final swaps, and why do many libraries omit them?
d) Show that reversing the gate order and conjugating each R_k gives the inverse transform.
e) Estimate the error of the approximate QFT when all rotations with $k - j \geq d$ are dropped, and show that it is bounded by $2\pi n 2^{-d}$. Hence justify the choice $d \sim \log n$ and the $\mathscr{O}(n \log n)$ gate count.
f) Using `qpe.py`, confirm the bit-reversal statement: the no-swap output of $|x\rangle$ equals the with-swap output of the bit-reversed input. Then break it deliberately by omitting the swaps in a phase estimation and see what the measured phase becomes.

Exercise 3: phase estimation

a) Derive Eq. (19.24), being explicit about why the system register factors out.
b) Show that Eq. (19.24) is exactly QFT $|m'\rangle$ when $\phi = m'/2^m$, and hence that the inverse QFT gives $|m'\rangle$ with certainty.
c) Derive the product form (19.27) and explain why only the fractional part of $2^{m-1}\phi$ appears.
d) For ϕ not exactly representable, show that the amplitude of the outcome $|m'\rangle$ is

$$\frac{1}{2^m}\sum_{k=0}^{2^m-1} e^{2\pi i k\left(\phi - m'/2^m\right)} = \frac{1}{2^m}\,\frac{1-e^{2\pi i 2^m\delta}}{1-e^{2\pi i\delta}}, \qquad \delta = \phi - \frac{m'}{2^m},$$

and hence that the probability is $\sin^2(\pi 2^m\delta)/\left(2^{2m}\sin^2(\pi\delta)\right)$.
e) Maximise the error $|\delta| \le 2^{-m-1}$ in that expression and show that the probability of the nearest outcome is at least $4/\pi^2$.
f) How many control qubits are needed to obtain ϕ to within 2^{-b} with probability at least $1-\varepsilon$? *Answer:* $m = b + \lceil \log_2(2 + 1/2\varepsilon)\rceil$; derive it.

Answer to d) and e).

The geometric sum is immediate. Taking the modulus squared and using $|1-e^{i\theta}| = 2|\sin(\theta/2)|$ gives the stated probability, a Dirichlet kernel. Its worst case is $|\delta| = 2^{-m-1}$, exactly halfway between two representable values, where $\sin^2(\pi 2^m\delta) = 1$ and $\sin^2(\pi\delta) \le (\pi\delta)^2 = (\pi 2^{-m-1})^2$, so

$$P \;\ge\; \frac{1}{2^{2m}}\cdot\frac{1}{\left(\pi 2^{-m-1}\right)^2} = \frac{4}{\pi^2} \approx 0.405.$$

Note what this does *not* say: it is a bound on a single outcome, and the two neighbouring outcomes together do much better – 0.931 in the $m=4$ row of Table 19.3. In practice one repeats and takes the median, which converts the constant success probability into an exponentially small failure probability.

Exercise 4: energies, aliasing and Trotter

a) Derive Eq. (19.30) and explain the modulus.
b) Two eigenvalues E and $E + 2\pi/t$ give the same phase. Show that choosing $t < 2\pi/\|\hat H\|$ removes the ambiguity, and that shifting $\hat H \to \hat H - \lambda\mathbb{1}$ lets one use the range more efficiently. What is the optimal λ if the spectrum lies in $[E_{\min}, E_{\max}]$?
c) Show that $\Lambda = \sum_\alpha |h_\alpha| \ge \|\hat H\|$, and give an example in which the bound is loose by a factor growing with the system size. What does that cost in control qubits?
d) Write out the circuit for $e^{-ihZ_1Z_2Z_3\tau}$ as two cnots, an $R_z(2h\tau)$ and two more cnots, and generalise to an arbitrary Pauli string. How does the depth scale with the weight?
e) Starting from Eq. (6.46), show that the first-order Trotter error over a time t with r steps is bounded by $\frac{t^2}{2r}\sum_{\alpha<\beta}\left\|[H_\alpha, H_\beta]\right\|$, and hence that grouping commuting terms costs nothing. Compare with the numerical ratios tabulated in Section 6.9.
f) Given a target energy error ε, balance the register error 2^{-m} against the Trotter error $\mathscr{O}(t^2/r)$ and find the optimal (m,r). Repeat for the second-order formula (6.53) and show that the total gate count improves from $\mathscr{O}(\varepsilon^{-1})$ to $\mathscr{O}(\varepsilon^{-1/2})$. Compare with Table 19.8.

g) Section 10.15.1 found that Trotterising the unitary coupled-cluster operator costs nothing once the amplitudes are re-optimised, while Table 19.8 shows the same splitting producing a genuine error in QPE. Explain the difference precisely, and construct a hybrid in which a variational parameter is inserted into the QPE circuit. What goes wrong?

Exercise 5: the models

a) Derive Eq. (19.43) from the definitions of the quasispin operators in Section 4.1, and hence write the Lipkin Hamiltonian as an explicit sum of Pauli strings for $N = 2$ and $N = 4$. Check the term counts in Table 19.6.
b) Show that the Lipkin Hamiltonian conserves the total quasispin J^2, and hence that the qubit Hamiltonian is block diagonal. How would you exploit that to reduce the number of qubits?
c) For the pairing model in the seniority-zero space, derive the closed Pauli form of Section 4.7 and confirm the nine terms of Table 19.6.
d) Show that the Heisenberg Hamiltonian requires no Jordan-Wigner strings, and that the Hubbard Hamiltonian does. For a chain of L sites, how does the maximum Pauli weight grow in each case?
e) Verify that the two-site Hubbard model at half filling has $E_0 = \frac{1}{2}\left(U - \sqrt{U^2 + 16t^2}\right)$, and check the value -1.23607 in Table 19.6.
f) Show that the four-site Heisenberg ring has $E_0 = -2J$ with the normalisation of Section 4.5, and identify the ground state.

Exercise 6: numerical explorations

a) Using `qpe.py`, reproduce Table 19.1 and Table 19.2, and extend the second to $n = 8$ and $n = 10$. At what d does the error fall below 10^{-3}, and how does that d grow with n?
b) Reproduce Table 19.3 and plot the full output distribution for $\phi = 0.3$ at $m = 4$ and $m = 8$. Identify the Dirichlet kernel of Exercise 3(d) in the shape.
c) Verify the $4/\pi^2$ bound numerically by scanning ϕ across a whole interval $[m'/2^m, (m'+1)/2^m]$ and recording the worst case. Where does it occur?
d) Reproduce Table 19.7. Then apply QPE to an *excited* state of the Lipkin model and confirm that the algorithm is indifferent to which eigenstate it is given.
e) Reproduce Table 19.8, then repeat it with the second-order Strang splitting (6.53) and confirm the improved $1/r^2$ scaling – the same factor-of-four-per-doubling seen in the table of Section 6.9. How much does that change the optimal (m, r) of Exercise 4(f)?
f) Reproduce Table 19.9 and extend it: feed QPE a Hartree-Fock state for the pairing model and measure its overlap with the correlated ground state as a function of g. At what coupling does the single-shot answer become unreliable?
g) Use the full histogram rather than the modal outcome to extract *several* eigenvalues at once from a deliberately mixed input state. This is spectroscopy rather than ground-state calculation, and it is one of the things QPE does that no variational method can.
h) Estimate the resources for a calculation you would actually want: the pairing model with ten levels, to chemical accuracy. How many qubits, how many Trotter steps, how many applications of U? Compare with the exact diagonalisation of Chapter 5.

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter19/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/quantumfourier.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter19/`.

`qpe.py` A statevector simulator: gate application by tensor contraction, the quantum Fourier transform built gate by gate and checked against the Fourier matrix, gate counts and the approximate transform, `phase_estimation` for an arbitrary unitary, the model Hamiltonians with their Pauli terms and commuting groups, `choose_time` from the Pauli 1-norm, and exact against Trotterised evolution. Runs in about a minute; needs `scipy.linalg.expm`.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Chapter 20
The variational quantum eigensolver

Chapter 19 solved the eigenvalue problem the way a quantum computer would if quantum computers worked. Phase estimation is exact up to a discretisation we control, its cost in qubits grows logarithmically with the precision demanded, and it comes with a proof. It also needs a circuit whose depth is measured in millions of gates for any Hamiltonian of interest, run coherently, on hardware whose coherence is measured in hundreds. The gap is not a matter of a few years of engineering; it is the reason the algorithm that dominates the literature is a different one.

The variational quantum eigensolver takes the opposite side of every trade. It uses a shallow circuit, tolerates noise, needs no coherent arithmetic, and runs on the machines that exist. In exchange it gives up the proof. What it returns is a variational upper bound on the ground-state energy – exactly the object Chapter 6 obtained from Hartree-Fock and Chapter 13 from Monte Carlo sampling, and with exactly the same epistemic status. The quantum computer is used for one thing only: to hold a trial state that a classical computer could not store, and to report its energy. Everything else – the optimisation, the bookkeeping, the decision about when to stop – happens on a classical machine, in a loop.

This is a hybrid algorithm, and the hybrid character is not a compromise forced by circumstance but the source of its robustness. A noisy circuit prepares a state that is not the one the parameters describe; the energy of that state is still an upper bound on E_0, so the answer degrades rather than breaks. A miscalibrated gate shifts the optimum; the optimiser walks to the shifted optimum and the bound survives. Phase estimation has no such grace: a single phase error in the middle of the controlled evolution corrupts the register and the output is a number with no relation to any energy.

The chapter is organised around what actually has to be built. Section 20.1 sets up the variational principle in the form the algorithm uses it and states what the algorithm can and cannot promise. Section 20.2 constructs the trial states. Section 20.3 is the longest and, in a sense, the most important: it is about the fact that a quantum computer can measure only in one basis, what has to be done to every other observable in consequence, and what goes wrong – silently, and quite badly – when a natural-looking shortcut is taken. Section 20.4 derives the parameter-shift rule, which gives exact gradients from two extra energy evaluations and is one of the genuinely elegant results in the subject. Sections 20.5 and 20.7 apply all of it to the Hamiltonians of Chapter 19, and Section 20.8 weighs the two algorithms against one another with the numbers in hand.

The companion program is `vqe.py`, which reproduces every table below, and the notebook `vqe.ipynb` runs the same material interactively.

20.1 The variational principle, once more

The foundation is the statement proved in Section 2.7 and used in every chapter since. For any Hamiltonian $\hat{H}$ with ground-state energy E_0 and any normalised state $|\psi\rangle$,

$$E[\psi] = \langle\psi|\hat{H}|\psi\rangle \geq E_0, \tag{20.1}$$

with equality if and only if $|\psi\rangle$ is a ground state. The proof is one line in the eigenbasis: writing $|\psi\rangle = \sum_i c_i |\phi_i\rangle$ with $\hat{H}|\phi_i\rangle = E_i |\phi_i\rangle$,

$$\langle\psi|\hat{H}|\psi\rangle = \sum_i |c_i|^2 E_i \geq E_0 \sum_i |c_i|^2 = E_0, \tag{20.2}$$

since every $E_i \geq E_0$ and the coefficients are normalised. Nothing about this is quantum-mechanical in the computational sense; it is a statement about weighted averages.

What changes on a quantum computer is which states are available. Restricting $|\psi\rangle$ to a family $|\psi(\boldsymbol{\theta})\rangle$ labelled by real parameters turns (20.1) into a minimisation,

$$E_{\mathrm{VQE}} = \min_{\boldsymbol{\theta}} \langle\psi(\boldsymbol{\theta})|\hat{H}|\psi(\boldsymbol{\theta})\rangle \geq E_0, \tag{20.3}$$

and that is the whole algorithm. The family is generated by a parametrised circuit,

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})\,|0\rangle^{\otimes n}, \tag{20.4}$$

so the trial states are exactly those a quantum computer can prepare. This is the point of the exercise. A classical variational calculation is limited to states whose amplitudes can be written down; the Slater-Jastrow forms of Chapter 13 and the neural quantum states of Chapter 18 are elaborate ways of parametrising a function of the coordinates that must ultimately be evaluated. A circuit of depth d on n qubits reaches states with 2^n amplitudes, correlated in ways no compact classical parametrisation reproduces, using $\mathcal{O}(nd)$ parameters. The hope of VQE rests entirely on the belief that these states are useful trial functions for physical Hamiltonians. That belief is plausible and largely unproven.

What the bound is worth, and what it is not.

Equation (20.3) says the answer is an upper bound. It does not say the bound is close. If the ansatz cannot represent anything near the ground state, VQE converges beautifully to a number that is simply too high, and the convergence of the optimiser tells us nothing about the quality of the answer. This is the same difficulty that afflicts every variational method in this book – a Hartree-Fock calculation also converges, and also converges to the wrong energy – and the same remedies apply: enlarge the family and watch whether the number moves, compare families of different structure, and check against exact results wherever the system is small enough to permit it. Two variational numbers can be compared and the lower one is better. That is a real and useful guarantee, and it is precisely what phase estimation does not provide.

The loop.

The algorithm as executed is:

1. choose an ansatz $U(\boldsymbol{\theta})$ and initial parameters $\boldsymbol{\theta}^{(0)}$;

2. on the quantum device, prepare $|\psi(\boldsymbol{\theta})\rangle$ and measure enough observables to reconstruct $E(\boldsymbol{\theta}) = \langle\psi(\boldsymbol{\theta})|\hat{H}|\psi(\boldsymbol{\theta})\rangle$;
3. on the classical device, use E (and, if available, its gradient) to propose $\boldsymbol{\theta}^{(k+1)}$;
4. repeat until the energy stops decreasing.

Steps 2 and 3 alternate thousands of times. Step 2 is where the quantum computer earns its place and where almost all the cost lies; the rest of this chapter is largely about how much step 2 costs and how it can go wrong.

20.2 Ansatz

The circuit $U(\boldsymbol{\theta})$ is built from parametrised one-qubit rotations and fixed entangling gates. The rotations are the exponentials of the Pauli matrices introduced in Section 2.15,

$$R_x(\theta) = e^{-i\theta \boldsymbol{X}/2}, \qquad R_y(\theta) = e^{-i\theta \boldsymbol{Y}/2}, \qquad R_z(\theta) = e^{-i\theta \boldsymbol{Z}/2}, \tag{20.5}$$

and since $\boldsymbol{X}^2 = \boldsymbol{Y}^2 = \boldsymbol{Z}^2 = \boldsymbol{I}$ each expands in closed form,

$$R_a(\theta) = \cos\frac{\theta}{2}\,\boldsymbol{I} - i\sin\frac{\theta}{2}\,\boldsymbol{\sigma}_a. \tag{20.6}$$

This innocuous identity is responsible for the parameter-shift rule of Section 20.4: because the exponential closes on two terms, the energy is a trigonometric polynomial in each angle, and trigonometric polynomials can be differentiated by evaluating them at shifted points.

20.2.1 One qubit is a sphere

For a single qubit the general normalised state, up to a phase, is

$$|\psi(\theta,\phi)\rangle = \cos\frac{\theta}{2}\,|0\rangle + e^{i\phi}\sin\frac{\theta}{2}\,|1\rangle\,, \tag{20.7}$$

the standard Bloch-sphere parametrisation of a qubit. Two angles, two real degrees of freedom, and the sphere is covered. A product of two rotations,

$$|\psi(\boldsymbol{\theta})\rangle = R_x(\theta_2)R_y(\theta_1)\,|0\rangle\,, \tag{20.8}$$

therefore reaches every one-qubit state. For a one-qubit Hamiltonian the ansatz is *exact*: the variational minimum is the true ground state, and any error in the result is the optimiser's fault and nobody else's. This is why Section 20.5 begins there. It is also the last case in which anything of the kind is true.

20.2.2 Hardware-efficient ansatz

The workhorse of the near-term literature places a pair of rotations on every qubit and follows them with a fixed layer of entangling gates, repeating the pattern L times:

$$U(\boldsymbol{\theta}) = \prod_{\ell=1}^{L}\left[U_{\text{ent}}\prod_{q=1}^{n} R_z(\theta^z_{\ell q})R_y(\theta^y_{\ell q})\right]. \tag{20.9}$$

With a ladder of cnot gates as U_{ent} this uses $2nL$ parameters and $(n-1)L$ two-qubit gates. It is called hardware-efficient because it is built from whatever gates the machine executes natively and respects whatever connectivity it has, and the name is honest about its virtue and silent about its defect: *nothing in it knows anything about the Hamiltonian*. It does not conserve particle number, does not respect spin, and is not guaranteed to contain any state resembling the ground state. It is a generic family that happens to be cheap.

Two consequences follow, and both appear in the calculations of Section 20.7. The optimisation landscape is riddled with local minima, so restarts from several random initialisations are not a refinement but a necessity. And as n and L grow the landscape flattens: for a sufficiently random circuit the gradient of the energy has zero mean and a variance that decays exponentially in the number of qubits. This is the *barren plateau*, taken up in Section 20.4.3.

20.2.3 Ansätze that know some physics

The alternative is to build the circuit from the physics. The unitary coupled cluster ansatz of Section 10.14 of Chapter 10,

$$|\psi(\boldsymbol{t})\rangle = e^{\hat{T}(\boldsymbol{t})-\hat{T}^{\dagger}(\boldsymbol{t})}\,|\Phi_0\rangle\,, \tag{20.10}$$

starts from a reference determinant and applies particle-hole excitations, so it conserves particle number by construction, has parameters that are recognisable amplitudes, and reduces to the classical coupled-cluster result in the appropriate limit. Its drawback is depth: the exponential must be Trotterised, as in Section 10.15.1, and the resulting circuits are far deeper than (20.9). The trade-off between hardware-efficient shallowness and physically informed depth is the central practical question in ansatz design and is not settled.

Between the two lie adaptive schemes, which grow the ansatz one operator at a time by choosing at each step the excitation with the largest energy gradient. They produce compact, problem-specific circuits at the price of many extra measurements during construction.

20.3 Measurement

Everything so far has been written as though $\langle\psi|\hat{H}|\psi\rangle$ were simply available. It is not. A quantum computer performs exactly one kind of measurement: it reads each qubit in the computational basis and returns a bit. The state is destroyed in the process, so the whole preparation must be repeated for every sample. Turning that into the expectation value of a Hamiltonian requires three things – a decomposition of $\hat{H}$ into measurable pieces, a rotation that brings each piece into the computational basis, and enough repetitions to beat down the statistical noise. The third is the dominant cost of the entire algorithm.

20.3.1 Decomposition into Pauli strings

Any Hermitian operator on n qubits is a real combination of the 4^n Pauli strings, as Section 4.7 established:

$$\hat{H} = \sum_{\alpha} h_\alpha P_\alpha, \qquad P_\alpha \in \{\boldsymbol{I},\boldsymbol{X},\boldsymbol{Y},\boldsymbol{Z}\}^{\otimes n}, \qquad h_\alpha = \frac{1}{2^n}\mathrm{Tr}\left(P_\alpha \hat{H}\right), \tag{20.11}$$

the last equality following from the orthogonality $\mathrm{Tr}(P_\alpha P_\beta) = 2^n\delta_{\alpha\beta}$. Linearity then gives

$$E(\boldsymbol{\theta}) = \sum_{\alpha} h_\alpha \langle \psi(\boldsymbol{\theta}) | P_\alpha | \psi(\boldsymbol{\theta}) \rangle, \tag{20.12}$$

so the problem reduces to estimating one number per Pauli string. The identity string needs no measurement at all – its expectation value is one in every state – which is why it is always split off and added at the end, as in Section 19.7.

What makes this tractable is that physical Hamiltonians have far fewer than 4^n terms. The counts in Table 4.7 are polynomial: the Jordan-Wigner transform of a two-body fermionic Hamiltonian on n modes produces $\mathcal{O}(n^4)$ strings, and lattice models with local interactions produce $\mathcal{O}(n)$.

20.3.2 Rotating into the computational basis

The computational basis is the eigenbasis of $\boldsymbol{Z}$, so a string of $\boldsymbol{Z}$'s and $\boldsymbol{I}$'s can be read directly: measure every qubit, and the eigenvalue of the string on the observed bit string $\boldsymbol{b}$ is

$$\lambda(\boldsymbol{b}) = (-1)^{\sum_{q:P_q \neq \boldsymbol{I}} b_q}, \tag{20.13}$$

the parity of the bits on the qubits the string acts on. Averaging $\lambda(\boldsymbol{b})$ over the shots estimates $\langle P \rangle$.

For $\boldsymbol{X}$ and $\boldsymbol{Y}$ a rotation is needed first. The requirement is a unitary V with $V^\dagger \boldsymbol{Z} V = P$, so that measuring $\boldsymbol{Z}$ after applying $V^\dagger$ is the same as measuring P before. For $\boldsymbol{X}$ the Hadamard does it, since

$$\boldsymbol{HZH} = \boldsymbol{X}, \qquad \boldsymbol{H} = \frac{1}{\sqrt{2}}(\boldsymbol{X}+\boldsymbol{Z}), \tag{20.14}$$

which follows from $\boldsymbol{H}^2 = \boldsymbol{I}$ and the anticommutation of $\boldsymbol{X}$ and $\boldsymbol{Z}$. For $\boldsymbol{Y}$, apply the phase gate $\boldsymbol{S}^\dagger = \mathrm{diag}(1, -i)$ first, which rotates $\boldsymbol{Y}$ into $\boldsymbol{X}$, and then the Hadamard:

$$\boldsymbol{HS}^\dagger \boldsymbol{Y} \boldsymbol{SH} = \boldsymbol{Z}. \tag{20.15}$$

The recipe is worth memorising, because it is the whole of Pauli measurement:

observable	circuit before the computational-basis readout
$\boldsymbol{Z}$	nothing
$\boldsymbol{X}$	$\boldsymbol{H}$
$\boldsymbol{Y}$	$\boldsymbol{S}^\dagger$ then $\boldsymbol{H}$

The rotations act qubit by qubit and independently, so a string such as $\boldsymbol{X} \otimes \boldsymbol{Y} \otimes \boldsymbol{Z}$ requires $\boldsymbol{H}$ on the first qubit, $\boldsymbol{S}^\dagger \boldsymbol{H}$ on the second and nothing on the third, followed by one readout of all three. Figure 20.1 shows the three circuits.

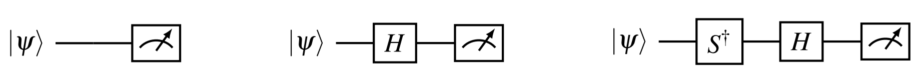

Fig. 20.1 Measuring $\boldsymbol{Z}$, $\boldsymbol{X}$ and $\boldsymbol{Y}$. The meter always reads the computational basis; the observable is selected by what precedes it. These three circuits, applied qubit by qubit, are sufficient for every Pauli string and hence for every Hamiltonian.

Table 20.1 verifies the construction numerically on a random two-qubit state: the value obtained by rotating and averaging parities agrees with $\langle \psi | P | \psi \rangle$ to machine precision.

Table 20.1 The rotation recipe checked against direct evaluation on a random two-qubit state. The left column is what hardware does – apply the rotations of Figure 20.1, read both qubits, average the parity over the outcome distribution. The right column is the matrix element. From `vqe.py`.

string	rotation + readout	$\langle\psi\|P\|\psi\rangle$	difference
$\boldsymbol{ZI}$	0.271768	0.271768	1×10^{-16}
$\boldsymbol{XI}$	−0.353388	−0.353388	2×10^{-16}
$\boldsymbol{YI}$	0.755358	0.755358	2×10^{-16}
$\boldsymbol{XX}$	0.051382	0.051382	8×10^{-17}
$\boldsymbol{YY}$	0.814800	0.814800	6×10^{-16}
$\boldsymbol{ZZ}$	−0.061651	−0.061651	5×10^{-17}
$\boldsymbol{XZ}$	0.344515	0.344515	0
$\boldsymbol{YX}$	0.112842	0.112842	1×10^{-16}

20.3.3 Shot noise

Each shot returns $\lambda(\boldsymbol{b}) = \pm 1$. The estimator of $\langle P\rangle$ from N shots is the sample mean of these values, so it is unbiased, and since $\lambda^2 = 1$ its variance is

$$\mathrm{Var}\left[\langle\hat{P}\rangle\right] = \frac{1-\langle P\rangle^2}{N}, \tag{20.16}$$

giving a standard error $\sqrt{1-\langle P\rangle^2}/\sqrt{N} \le 1/\sqrt{N}$. For the full energy the errors of the individual strings add in quadrature when the strings are measured independently, so

$$\mathrm{Var}\left[\hat{E}\right] = \sum_\alpha \frac{h_\alpha^2\left(1-\langle P_\alpha\rangle^2\right)}{N_\alpha} \;\le\; \frac{1}{N}\left(\sum_\alpha |h_\alpha|\right)^2, \tag{20.17}$$

where the bound assumes a total budget N divided optimally, and $\Lambda = \sum_\alpha |h_\alpha|$ is the same Pauli 1-norm that set the time step in Section 19.6. The same quantity controls the cost of both algorithms, in completely different ways.

The scaling is the point. To reach an accuracy ε,

$$N \sim \frac{\Lambda^2}{\varepsilon^2}, \tag{20.18}$$

so every additional decimal digit costs a factor of one hundred in measurements. Table 20.2 shows this on the two-qubit Hamiltonian: the standard error times $\sqrt{N}$ is constant, as it must be.

Table 20.2 Shot noise in the energy of a fixed trial state of the two-qubit Hamiltonian of Section 19.7.2, over 200 independent repetitions. The third column is the second scaled by $\sqrt{N}$ and is constant to within its own statistical error, confirming Eq. (20.16). The estimator is unbiased, so the last column shows no trend.

shots N	standard error	$\times\sqrt{N}$	bias
10	0.9891	3.13	−0.0078
100	0.2809	2.81	−0.0234
1000	0.0851	2.69	+0.0023
10000	0.0299	2.99	+0.0000

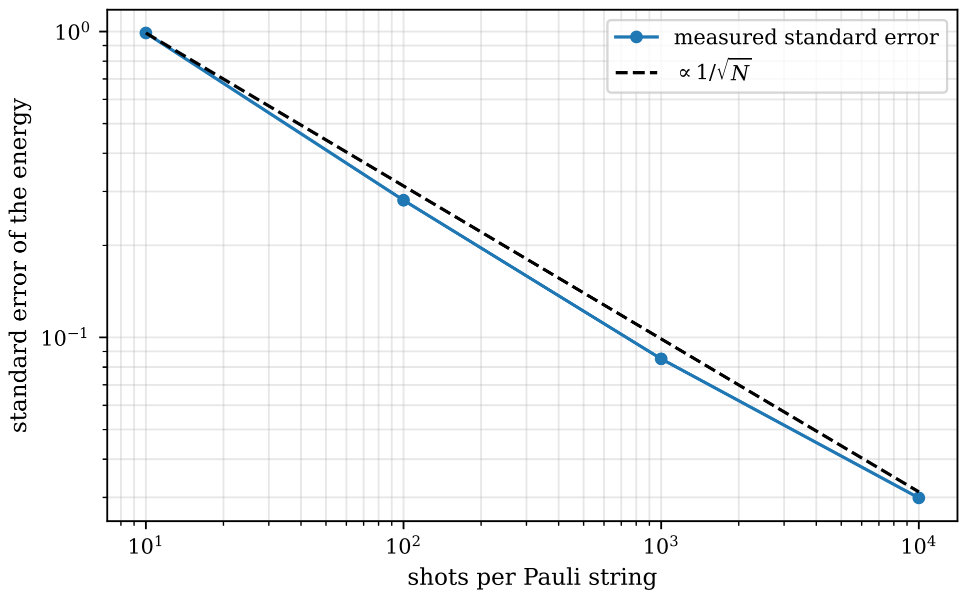

Fig. 20.2 Standard error of the sampled energy of a fixed two-qubit trial state, plotted against the number of shots allocated to each Pauli string on doubly logarithmic axes. Each point is the spread of 200 independent repetitions of the whole measurement procedure, the same data as Table 20.2; the dashed line is the ideal $N^{-1/2}$ law of Eq. (20.16), anchored at the leftmost point and carrying no fitted parameter. Three decades of shots buy a factor of 33 in accuracy, and the straightness of the measured curve is the statement that nothing better is available.

Figure 20.2 plots the same four numbers on logarithmic axes, where the content of Eq. (20.16) becomes a slope. A power law $\varepsilon \propto N^{-p}$ is a straight line of gradient $-p$ on such a plot, and the dashed reference line drawn through the leftmost point with $p = 1/2$ has nothing fitted to it: it is the prediction, and the measured points follow it across three decades. What the eye should take from the figure is how little is bought. Going from ten shots per string to ten thousand – a thousandfold increase in the work done by the quantum device – reduces the standard error from 0.989 to 0.0299, a factor of 33, which is $\sqrt{1000}$ to the accuracy with which the spread of 200 repetitions can itself be determined. The small excursions of the solid curve about the dashed one, of which the largest is the 14% shortfall at $N = 10^3$, are the uncertainty of an estimated standard deviation and not a departure from the law; with 200 repetitions that uncertainty is about 5%, so a 14% excursion is a two-sigma event in four points and entirely unremarkable. It is worth dwelling on how unforgiving the geometry is. There is no regime at the right-hand end of the figure in which the curve steepens, no threshold beyond which additional shots become more productive, and no reason to expect one: the estimator is a sample mean of bounded random variables and the central limit theorem fixes its rate once and for all. Every technique discussed in Section 20.3.4 and every proposal in the literature moves this line *down* – reduces the prefactor Λ – and none of them tilts it. That distinction, between a prefactor one may attack and an exponent one may not, is the whole of the economics of VQE, and it reappears in Section 20.8.2 as the single line of Table 20.8 on which phase estimation is unambiguously the better algorithm.

Two remarks follow. Chemical accuracy, conventionally 1.6 mHartree, on a molecule with $\Lambda \sim 10^3$ Hartree therefore needs of order 10^{11} shots for a single energy evaluation, and an optimisation needs thousands of those. At kilohertz repetition rates this is centuries. Reducing that number – by grouping, by better estimators, by classical shadows – is one of the most active areas of the field, and no proposal so far removes the ε^{-2}. Second, the noise enters the optimisation as well as the answer: the classical optimiser of step 3 is minimising a function it can only evaluate stochastically, which is the situation of Chapter 14 and calls for the same tools.

20.3.4 Grouping commuting strings

Strings that commute and are simultaneously diagonalisable by the same one-qubit rotations can be measured together: one circuit, one set of shots, several expectation values. The simplest and most used version groups strings that are *qubit-wise* commuting – identical non-identity letters wherever both act – since these are diagonalised by the same product of Hadamards and phase gates. Partitioning the strings into as few such groups as possible is a graph-colouring problem, NP-hard in general and easy in practice with a greedy heuristic.

The gain is real and bounded. Grouping reduces the number of distinct circuits, which is the last column of Table 20.7, but it does not reduce the ε^{-2} in (20.18); correlated strings share shots but the variance still falls as $1/N$. Typical reductions for molecular Hamiltonians are an order of magnitude, which matters and does not change the picture.

20.3.5 Why measuring one qubit at a time does not work

Here is a shortcut that looks reasonable and is wrong. Reading all n qubits gives, in particular, the one-qubit statistics $\langle Z_1\rangle,\dots,\langle Z_n\rangle$. It is tempting to build the energy out of those, treating a two-qubit term as a product,

$$\langle \mathrm{Z}_1\mathrm{Z}_2\rangle \overset{?}{=} \langle \mathrm{Z}_1\rangle\langle \mathrm{Z}_2\rangle. \tag{20.19}$$

Some early treatments of VQE do exactly this. It is false, and the way in which it is false deserves care, because the resulting error is not noisy or unstable but quiet and confident.

Theorem 20.1 (Marginals do not determine correlators). *There exist states ρ and ρ' on two qubits with identical single-qubit reduced density matrices, and an observable $\hat{H}$ with* $\mathrm{Tr}(\rho\hat{H}) \neq \mathrm{Tr}(\rho'\hat{H})$. *Consequently the expectation value of $\hat{H}$ is not a function of the single-qubit measurement statistics.*

Proof. Take $\hat{H} = \boldsymbol{Z}\otimes\boldsymbol{Z}$ and the two states

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right), \qquad \left|\Psi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right).$$

Both have maximally mixed reduced density matrices, $\rho_1 = \rho_2 = \frac{1}{2}\boldsymbol{I}$, so *every* single-qubit observable has the same expectation value in the two states – not merely $\langle Z\rangle$, but $\langle X\rangle$ and $\langle Y\rangle$ as well, and hence any function of them. Yet

$$\left\langle \Phi^+\middle|\boldsymbol{Z}\otimes\boldsymbol{Z}\middle|\Phi^+\right\rangle = +1, \qquad \left\langle \Psi^+\middle|\boldsymbol{Z}\otimes\boldsymbol{Z}\middle|\Psi^+\right\rangle = -1.$$

The two numbers differ by 2 while all single-qubit data are identical.

The mechanism is worth stating plainly, because it is not an artefact of a contrived example. The single-qubit statistics of a state are precisely the content of its reduced density matrices, and the reduced density matrices are obtained by tracing out the other qubit – an operation whose entire purpose is to discard correlations. Asking for $\langle Z_1Z_2\rangle$ from $\langle Z_1\rangle$ and $\langle Z_2\rangle$ is asking to recover what was deliberately thrown away. For a product state the factorisation (20.19) happens to hold; for anything entangled it does not, and entanglement is the reason for using a quantum computer at all.

What it does to a VQE calculation.

Take $\hat{H} = \boldsymbol{Z} \otimes \boldsymbol{Z}$ with the one-parameter ansatz

$$|\psi(\theta)\rangle = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle\,. \tag{20.20}$$

Every state in this family has $\langle Z_1 Z_2\rangle = 1$, since both basis states present have even parity, so the exact energy landscape is *flat*:

$$E_{\text{true}}(\theta) = 1 \quad \text{for all } \theta. \tag{20.21}$$

The marginals, on the other hand, are $\langle Z_1\rangle = \langle Z_2\rangle = \cos^2\theta - \sin^2\theta = \cos 2\theta$, so the reconstruction (20.19) gives

$$E_{\text{naive}}(\theta) = \cos^2 2\theta, \tag{20.22}$$

an oscillating function with minima at $\theta = \pi/4$ and $3\pi/4$ where it reports 0 instead of 1. The optimiser sees a smooth landscape with a clear minimum, descends into it, converges, and reports an energy that is wrong by one unit – on a Hamiltonian whose energy does not depend on the parameter at all.

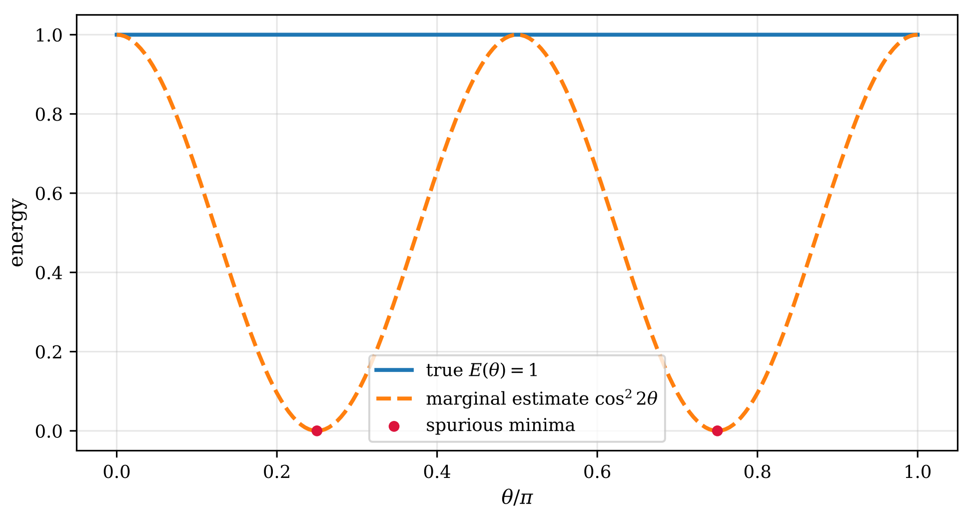

Fig. 20.3 The two energy landscapes of $\hat{H} = \boldsymbol{Z} \otimes \boldsymbol{Z}$ over the one-parameter family $|\psi(\theta)\rangle = \cos\theta\,|00\rangle + \sin\theta\,|11\rangle$, as functions of θ/π. The solid line is the true energy $E_{\text{true}}(\theta) = 1$ of Eq. (20.21), flat because every state of the family has even parity; the dashed line is the marginal reconstruction $E_{\text{naive}}(\theta) = \cos^2 2\theta$ of Eq. (20.22). The two circles mark the spurious minima at $\theta = \pi/4$ and $3\pi/4$, where the reconstruction reports 0 for a quantity whose value is 1.

Figure 20.3 draws the two landscapes on the same axes, and the picture repays more than a glance. The solid line is the truth and it is a horizontal line at $E = 1$; the dashed line is what the marginals report, and it sweeps the entire spectrum of $\boldsymbol{Z} \otimes \boldsymbol{Z}$, which runs from -1 to $+1$, in the space of a quarter period. The error made by the reconstruction is $1 - \cos^2 2\theta = \sin^2 2\theta$, and that quantity is not arbitrary: for the family (20.20) the concurrence is $C(\theta) = |2\cos\theta\sin\theta| = |\sin 2\theta|$, so the error of the factorisation (20.19) is *exactly* C^2, the square of the entanglement of the state on which it was attempted. The two curves touch precisely at $\theta = 0$, $\pi/2$ and π, which are the three points of the family at which the state is a product – $|00\rangle$ and $|11\rangle$ – and they separate most widely at $\theta = \pi/4$ and $3\pi/4$, which are the Bell states $|\Phi^+\rangle$ of the proof of Theorem 20.1. The counterexample is therefore not something adjacent to the calculation; it sits at the bottom of the fictitious well.

What makes this more than a curiosity is the direction of the gradient. The dashed curve has a nonvanishing slope everywhere except at multiples of $\pi/4$, and everywhere on the descending branches that slope points towards the maximally entangled state. An optimiser handed this landscape is not merely misinformed; it is actively driven towards the configuration on which the readout scheme is at its very worst, and the more successfully it optimises the larger the error it accumulates. The convergence is smooth, the reported energy decreases monotonically, and every diagnostic the classical half of the loop has access to says that the run is going well. Only the horizontal solid line, which requires knowing the answer, says otherwise.

Table 20.3 runs the same failure inside a full VQE on the two-qubit Hamiltonian of Section 19.7.2.

Table 20.3 VQE on the Hamiltonian of Section 19.7.2 with joint and with marginal readout, from identical starting parameters. The marginal run converges smoothly to a state lying 0.53 above the ground state, and the number it reports is not the energy of any state. Note in particular that the reported value is *not* constrained to lie above E_0: a quantity that is not $\langle\psi|\hat{H}|\psi\rangle$ for any $|\psi\rangle$ is under no obligation to respect the variational principle, and can equally well come out too low, in which case nothing whatever identifies it as wrong.

	energy
exact ground state	−0.531129
VQE, joint readout	−0.531129
VQE, marginal reconstruction (reported)	0.000024
true energy of the state it prepared	0.000080

And the shortcut buys nothing.

The final irony is that the incorrect procedure is not even cheaper. A single shot of a two-qubit circuit returns *both* bits. The same data that give $\langle Z_1\rangle$ and $\langle Z_2\rangle$ give $\langle Z_1 Z_2\rangle$, at no additional cost: one simply multiplies the two outcomes before averaging instead of averaging first and multiplying afterwards. The rule is therefore easy to state and should never be broken –

measure all qubits, form the parity of each string, then average.

Averaging first and multiplying afterwards is the error, and the order of the two operations is the whole of the difference: $\mathbb{E}[ab] \neq \mathbb{E}[a]\mathbb{E}[b]$ unless a and b are independent, and a quantum state that is worth preparing is one in which they are not.

20.4 Gradients: the parameter-shift rule

The classical half of the loop needs derivatives. On a classical computer one would differentiate the code; here the energy is only accessible through measurement, and the obvious substitute – a finite difference

$$\frac{\partial E}{\partial \theta_j} \approx \frac{E(\theta_j + h) - E(\theta_j - h)}{2h} \tag{20.23}$$

– is a poor one. It has a truncation error $\mathscr{O}(h^2)$ that wants h small, and it subtracts two noisy numbers whose difference is $\mathscr{O}(h)$ while their errors are not, so the relative error of the estimate grows as $1/h$. With shot noise of size σ the total error is minimised at $h \sim (\sigma)^{1/3}$

and is far larger than σ itself. Finite differences on a quantum computer are a bad idea for a reason that has nothing to do with quantum mechanics.

The parameter-shift rule replaces them with something better in every respect: it is exact, uses a shift of $\pi/2$ rather than an infinitesimal, and costs two energy evaluations of exactly the kind already being performed.

20.4.1 The theorem

The rule is a statement about the *generators of the gates*, not about the Hamiltonian. This is worth emphasising at the start, because it is the usual source of confusion. The Hamiltonian enters only through the requirement that $E(\boldsymbol{\theta})$ be measurable at all, which Section 20.3 settled; the differentiability structure comes entirely from how the parameters sit inside the circuit.

Theorem 20.2 (Two-term parameter-shift rule). *Let*

$$E(\theta) = \langle\psi| U^\dagger(\theta)\hat{H}U(\theta)|\psi\rangle, \qquad U(\theta) = e^{-i\theta G}, \tag{20.24}$$

with $\hat{H}$ and G Hermitian and G possessing exactly two distinct eigenvalues $\pm r$, $r > 0$. Then

$$\frac{dE}{d\theta} = r\left[E\left(\theta + \frac{\pi}{4r}\right) - E\left(\theta - \frac{\pi}{4r}\right)\right]. \tag{20.25}$$

Proof. Because G has only the eigenvalues $\pm r$, the projectors onto its two eigenspaces can be written down directly,

$$\Pi_\pm = \frac{1}{2}\left(\boldsymbol{I} \pm \frac{G}{r}\right), \tag{20.26}$$

and satisfy $\Pi_\pm^2 = \Pi_\pm$, $\Pi_+\Pi_- = 0$, $\Pi_+ + \Pi_- = \boldsymbol{I}$ and $G = r(\Pi_+ - \Pi_-)$. The verification is immediate: $G^2 = r^2\boldsymbol{I}$, since a Hermitian operator with eigenvalues $\pm r$ squares to r^2 times the identity, and everything follows from that one relation. The exponential therefore terminates,

$$U(\theta) = e^{-i\theta G} = e^{-i\theta r}\,\Pi_+ + e^{+i\theta r}\,\Pi_-, \tag{20.27}$$

which is the same closing-on-two-terms phenomenon as (20.6). Substituting into (20.24) gives four terms, two of which carry no θ-dependence (those with matching projectors) and two of which carry $e^{\pm 2ir\theta}$. Collecting them, and using that E is real,

$$E(\theta) = A + B\cos(2r\theta) + C\sin(2r\theta), \tag{20.28}$$

with real constants

$$\begin{aligned} A &= \langle\psi|\left(\Pi_+\hat{H}\Pi_+ + \Pi_-\hat{H}\Pi_-\right)|\psi\rangle, \\ B &= 2\Re\,\langle\psi|\,\Pi_+\hat{H}\Pi_-\,|\psi\rangle, \\ C &= 2\Im\,\langle\psi|\,\Pi_+\hat{H}\Pi_-\,|\psi\rangle. \end{aligned} \tag{20.29}$$

The energy is thus a pure sinusoid of frequency $2r$ – three unknowns, no more. Differentiating,

$$E'(\theta) = -2rB\sin(2r\theta) + 2rC\cos(2r\theta). \tag{20.30}$$

Now evaluate (20.28) at $\theta \pm s$ with $s = \pi/(4r)$, so that $2rs = \pi/2$ and the shifted cosines and sines exchange places:

$$E(\theta + s) - E(\theta - s) = 2\left[-B\sin(2r\theta) + C\cos(2r\theta)\right], \tag{20.31}$$

which is exactly $E'(\theta)/r$.

The standard case is $G = \frac{1}{2}P$ with P a Pauli string, so that $r = 1/2$, $s = \pi/2$, and

$$\frac{\partial E}{\partial \theta_j} = \frac{1}{2}\left[E\left(\theta_j + \frac{\pi}{2}\right) - E\left(\theta_j - \frac{\pi}{2}\right)\right], \tag{20.32}$$

all other parameters held fixed. Every rotation in (20.5) and every gate of the form $e^{-i\theta P/2}$ used in unitary coupled cluster qualifies. A gradient with P parameters therefore costs $2P$ energy evaluations, each of which costs the shots of Section 20.3.3 – which is where the real expense of VQE lives.

Why this is not a finite difference.

Equation (20.32) looks like (20.23) with $h = \pi/2$ and it is essential to see why it is a different kind of object. The shift is not small, and there is no truncation error to be traded against noise: the identity is exact for every θ. What makes it work is that (20.28) has only three unknown constants, so three evaluations determine the function completely and two determine its derivative. The rule is interpolation of an exactly known functional form, not approximation of an unknown one. On hardware, where every evaluation carries shot noise, this is worth a great deal: the two terms in (20.32) are typically of order one and well separated, so their difference does not amplify the noise.

Verification.

Two checks, both in `vqe.py`. Fitting A, B, C in (20.28) to three points of a two-qubit, two-layer ansatz and predicting the energy elsewhere reproduces it to 4×10^{-15}, confirming the functional form rather than merely the derivative. And the shift rule agrees with a converged finite difference to 3×10^{-10}, the residual being the finite difference's own error.

20.4.2 When two terms are not enough

The hypothesis of Theorem 20.2 is that G has two eigenvalues, and it is not decorative. Consider

$$G = \frac{1}{2}(\mathbf{Z}_1 + \mathbf{Z}_2), \tag{20.33}$$

with eigenvalues $-1, 0, +1$. What the energy sees are the *differences* of eigenvalues, which are now $0, \pm 1, \pm 2$, so (20.28) is replaced by a two-frequency polynomial

$$E(\theta) = A + B_1 \cos\theta + C_1 \sin\theta + B_2 \cos 2\theta + C_2 \sin 2\theta, \tag{20.34}$$

with five unknowns. Two evaluations cannot determine five constants, and the failure is not graceful. Applying (20.32) to the generator (20.33) at a point where the true derivative is -0.970535 returns 0.000000: the two shifted evaluations coincide exactly, and an optimiser using the result would conclude it had reached a stationary point and halt. A silently wrong gradient is worse than a noisy one.

The remedy is the generalised rule for R equidistant frequencies,

$$\frac{dE}{d\theta} = \sum_{\mu=1}^{2R} \frac{(-1)^{\mu-1}}{4R\sin^2(x_\mu/2)}\, E(\theta + x_\mu), \qquad x_\mu = \frac{(2\mu-1)\pi}{2R}, \tag{20.35}$$

which reduces to (20.32) at $R = 1$ and, at $R = 2$, recovers the derivative above to 5×10^{-10}. The cost is $2R$ evaluations per parameter: a generator with a richer spectrum is not merely awkward, it is proportionately more expensive. This is a good reason to build ansätze from Pauli-string generators, quite apart from their convenience.

20.4.3 Optimisers and barren plateaus

With gradients available, the classical half of the loop is the subject of Chapter 14, and the same considerations apply: plain gradient descent is robust and slow, momentum and adaptive step sizes help, and quasi-Newton methods need care because the gradient is stochastic. Two features are specific to this setting.

Gradient-free methods remain competitive. Simultaneous perturbation stochastic approximation estimates the entire gradient from two energy evaluations, regardless of the number of parameters, by perturbing all parameters at once along a random direction. Against the $2P$ evaluations of the parameter-shift rule this is an enormous saving when P is large, paid for with a much noisier search direction. Which wins depends on how expensive an evaluation is, and for hardware it is often SPSA.

The second feature has no classical analogue. For a sufficiently expressive random circuit the energy landscape flattens exponentially: the gradient has zero mean and variance

$$\mathrm{Var}\left[\frac{\partial E}{\partial \theta_j}\right] \sim 2^{-n}, \tag{20.36}$$

so on 50 qubits the gradient is of order 10^{-15} and cannot be distinguished from zero by any feasible number of shots. This is the *barren plateau*, and the argument behind it is a concentration result: as the circuit approaches a 2-design the expectation values it produces concentrate about their Haar averages, and derivatives of a concentrated quantity are small. The trouble is structural – expressiveness is exactly what causes it – and the escapes proposed all amount to being less generic: shallow circuits, physically motivated ansätze, layerwise training, or initialisations near a known good state. It is the most serious theoretical obstacle to VQE at scale, and it has no counterpart in phase estimation, which does not optimise anything.

20.5 The small Hamiltonians again

We return to the two systems of Section 19.7, now solved variationally. Reusing them is deliberate: the comparison in Section 20.8 is between two algorithms on one problem, not between two problems.

20.5.1 One qubit

For the Hamiltonian (19.34) the ansatz (20.8) covers the Bloch sphere, so the variational family contains the exact ground state and the minimum of (20.3) equals E_- of (19.37) exactly.

Anything left over is the optimiser's residual. Table 20.4 confirms this at three interaction strengths.

Table 20.4 VQE on the one-qubit Hamiltonian, ansatz $R_x(\theta_2)R_y(\theta_1)|0\rangle$, gradient descent with parameter-shift gradients. The ansatz is exact, so the error measures only how far the optimiser has run; it falls without limit as the iteration count is increased, which is not true of any of the tables that follow.

λ	exact E_-	VQE	error
0.0	0.000000	0.000000	1×10^{-30}
0.5	1.490098	1.490140	4×10^{-5}
1.0	0.980196	0.980215	2×10^{-5}

Compare Table 19.4. Phase estimation with ten control qubits reached 2.7×10^{-4}, limited by the resolution $2\pi/(t2^m)$ of its energy grid; VQE reaches 2×10^{-5} and would go further with more iterations, because it is not quantising the answer at all. On a single qubit, with an exact ansatz and no shot noise, the variational method simply wins. Neither of the two conditions in that sentence survives contact with a larger problem.

20.5.2 Two qubits

For the Hamiltonian (19.39) the hardware-efficient ansatz (20.9) is used. The ground state here is a superposition of $|00\rangle$ and $|11\rangle$ – the $\boldsymbol{X}\otimes\boldsymbol{X}$ term couples exactly those – so it is entangled and a single cnot suffices to reach it. Table 20.5 bears that out: one layer, four parameters, and the energy is exact to machine precision. Additional layers add parameters that the optimiser must navigate around without adding anything reachable.

Table 20.5 VQE on the two-qubit Hamiltonian with the hardware-efficient ansatz. One layer already contains the ground state. Every entry lies above the exact value -0.531129, as the variational principle requires; contrast Table 19.5, where phase estimation returned -0.583333 at $m=4$ – a number *below* the true ground-state energy, which no variational method could ever produce and which is a perfectly normal QPE result.

layers	parameters	VQE	error
1	4	-0.531129	1×10^{-16}
2	8	-0.531129	9×10^{-13}
3	12	-0.531129	3×10^{-16}

Figure 20.4 shows how the three rows of Table 20.5 were arrived at, and it is more instructive than the converged numbers. All three runs begin at whatever energy the random initial parameters happen to give – 1.00 for one layer, 1.36 for two and 0.48 for three – and all three fall monotonically onto the dashed line without once passing through it. That is not an accident of this Hamiltonian: every point on every curve is $\langle\psi(\boldsymbol{\theta})|\hat{H}|\psi(\boldsymbol{\theta})\rangle$ for a state the circuit actually prepares, so Eq. (20.1) forbids the crossing. Compared with the oscillating and occasionally sub-E_0 output of phase estimation in Table 19.5, a convergence plot that can only approach its limit from above is a considerable practical comfort, and it is one of the few diagnostics VQE offers for free.

The apparent lesson of the figure – that depth accelerates convergence – is worth resisting, because the horizontal axis is the wrong currency. The three-layer run is within machine precision of E_0 after fewer than ten iterations, the two-layer run after about twenty-five and

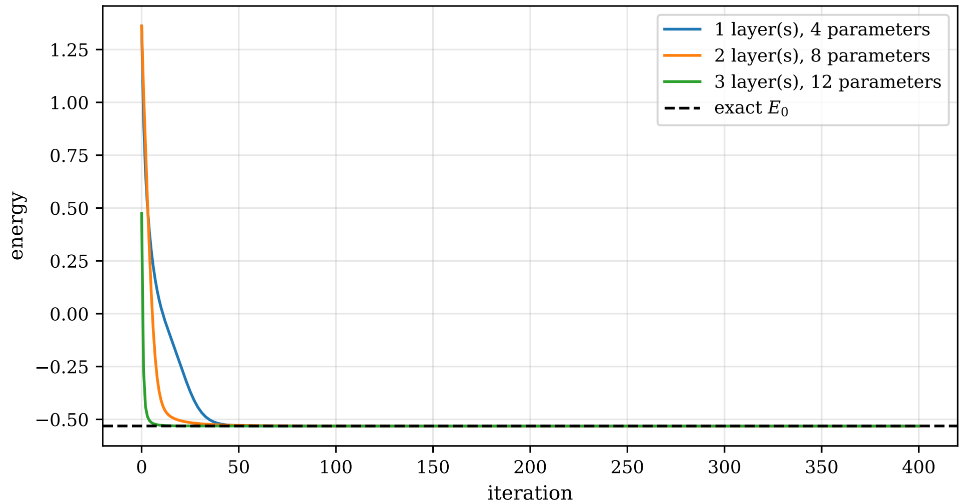

Fig. 20.4 Gradient-descent histories of the three runs of Table 20.5: the energy of the two-qubit Hamiltonian of Section 19.7.2 against the iteration number, for hardware-efficient ansätze of one, two and three layers with 4, 8 and 12 parameters, all started from the same random seed and run for 400 iterations with parameter-shift gradients. The dashed line is the exact ground-state energy $E_0 = -0.531129$. Each curve descends monotonically and none crosses the dashed line, which is the variational principle (20.3) in visual form.

the one-layer run after about forty-five, so in iterations the deepest circuit is four or five times the fastest. But one iteration requires one parameter-shift gradient, which by Eq. (20.32) costs $2P$ energy evaluations, and P is 4, 8 and 12 respectively. Multiplying through, the three runs consume of order 360, 400 and 240 energy evaluations – flat to within a factor of two, and in an order that no longer favours depth in any systematic way. Since every energy evaluation is the several thousand shots of Section 20.3.3, it is the second of these numbers and not the first that a hardware run pays. This is the reason the numerical exercises at the end of the chapter ask for the same plot drawn against the number of measurements rather than the number of iterations; the curves reorder when it is drawn. One further feature is plain in the figure and easy to overlook: for something like seven eighths of the horizontal axis nothing whatever is happening. The optimiser has converged and continues to request gradients, and on hardware every one of those iterations is paid for in shots. A stopping criterion is therefore not a refinement but a substantial fraction of the cost, and shot noise makes it genuinely difficult to set, since an energy that has stopped moving and an energy whose movement is buried in statistical scatter look identical from inside the loop.

The remark in the caption is the single most useful practical difference between the two algorithms. A phase-estimation output can lie on either side of the true energy, and there is no way to tell from the output which side it is on. A VQE output cannot lie below it. Given two runs, the one with the lower energy is the better one – a comparison available for free, without any knowledge of the exact answer.

20.6 The Lipkin model

The Lipkin model of Section 4.1 is the standard testing ground for these methods, and for a good reason that is worth naming: its quasispin structure makes the Hilbert space that matters grow *linearly* with the particle number, so exact answers are available for systems large enough to be interesting. It is also the model on which the choice of qubit encoding can be seen to matter, which is the subject of this section.

20.6.1 Quasispin, and the collapse of the dimension

Chapter 4 introduced the model: N fermions distributed over two N-fold degenerate levels at $\pm\varepsilon/2$, with an interaction that moves pairs between them. In terms of the quasispin operators of Eq. (4.6),

$$J_z = \frac{1}{2}\sum_{p\sigma} \sigma\, a^\dagger_{p\sigma} a_{p\sigma}, \qquad J_\pm = \sum_p a^\dagger_{p\pm} a_{p\mp}, \tag{20.37}$$

which satisfy the $SU(2)$ algebra $[J_z, J_\pm] = \pm J_\pm$, $[J_+, J_-] = 2J_z$, the Hamiltonian is

$$H = \varepsilon J_z + \frac{V}{2}\left(J_+^2 + J_-^2\right) + \frac{W}{2}\left(J_+ J_- + J_- J_+\right). \tag{20.38}$$

The last term simplifies. Using

$$J_+ J_- + J_- J_+ = 2\left(J^2 - J_z^2\right), \tag{20.39}$$

which follows from $J^2 = J_z^2 + \frac{1}{2}(J_+J_- + J_-J_+)$, and restricting to the maximal-quasispin irreducible representation $j = N/2$ in which the ground state lies, so that $J^2 = j(j+1)$ is a constant,

$$H = \varepsilon J_z + \frac{V}{2}\left(J_+^2 + J_-^2\right) - W J_z^2 + W j^2, \tag{20.40}$$

where the constant has been simplified using $N = 2j$: $W j(j+1) - WN/2 = W j^2$. The operator now acts on a space of dimension $2j+1 = N+1$ rather than $\binom{2N}{N}$.

That collapse is what makes the encoding question interesting. An $(N+1)$-dimensional problem needs only $\lceil \log_2(N+1) \rceil$ qubits in principle, but the Hamiltonian is naturally written in terms of operators on N of them. Both routes are used.

20.6.2 The direct mapping

Assign one qubit to each particle, with $|0\rangle$ and $|1\rangle$ the lower and upper level, and realise the quasispin operators as collective spins,

$$J_a = \frac{1}{2}\sum_{i=1}^{N} \sigma_i^a, \qquad a = x, y, z. \tag{20.41}$$

No Jordan-Wigner strings appear, and their absence is not an accident: the transformation of Section 3.16 produces $\boldsymbol{Z}$ tails to enforce fermionic antisymmetry between modes, but the Lipkin interaction moves a particle between the two states of the *same* degenerate index p, and those tails cancel between the creation and annihilation operators. Expanding the squares in (20.40) using $J_\pm = J_x \pm iJ_y$,

$$\frac{V}{2}\left(J_+^2 + J_-^2\right) = V\left(J_x^2 - J_y^2\right) = \frac{V}{2}\sum_{i<k}\left(X_i X_k - Y_i Y_k\right) \tag{20.42}$$

(the $i = k$ terms cancel identically, since $X_i^2 = Y_i^2 = \boldsymbol{I}$), and similarly for the J_z^2 term, one obtains

$$H^{(N\mathrm{q})} = \mathrm{const} + \frac{\varepsilon}{2}\sum_{i=1}^{N} Z_i + \frac{V}{2}\sum_{i<k}\left(X_i X_k - Y_i Y_k\right) + \frac{W}{2}\sum_{i<k} Z_i Z_k. \tag{20.43}$$

Every string has weight at most two. For $N=6$ there are 36 of them on six qubits, all short, all easy to measure, and they fall into a handful of commuting groups.

20.6.3 The compressed mapping

The alternative encodes the $(N+1)$ quasispin states $|j,m\rangle$ directly into $\lceil \log_2(N+1) \rceil$ qubits. For $N=4$ the five states $m=-2,\ldots,2$ fit into three qubits, and writing each operator of (20.40) as a matrix in that basis and decomposing it with (20.11) gives, after some algebra,

$$\begin{aligned} H^{(3q)}_{N=4} = {} & \left(-\varepsilon + \tfrac{5W}{8}\right) III + \tfrac{3W}{8} IIZ + \left(-\tfrac{\varepsilon}{8} + \tfrac{W}{2}\right) IZI + \left(\tfrac{\varepsilon}{8} + \tfrac{W}{2}\right) IZZ \\ & + \left(-\tfrac{\varepsilon}{4} + \tfrac{W}{8}\right) ZII + \left(-\tfrac{\varepsilon}{4} - \tfrac{W}{8}\right) ZIZ - \tfrac{3\varepsilon}{8} ZZI - \tfrac{\varepsilon}{8} ZZZ \\ & + \tfrac{V(3+\sqrt{6})}{4} IXI + \tfrac{V(\sqrt{6}-3)}{4} IXZ + \tfrac{V\sqrt{6}}{4} \left(XXI + XXZ + YYI + YYZ\right) \\ & + \tfrac{V(3+\sqrt{6})}{4} ZXI + \tfrac{V(\sqrt{6}-3)}{4} ZXZ. \end{aligned} \tag{20.44}$$

Three qubits instead of four, at the price of sixteen strings of weight up to three where the direct mapping had short ones.

There is a further cost, and it is not merely bookkeeping. Three qubits carry eight basis states and only five are physical. A hardware-efficient ansatz knows nothing of this and will happily place amplitude on $|101\rangle$, $|110\rangle$ and $|111\rangle$, which correspond to no state of the model at all. The energy of such a leaked state is meaningless, and worse, it need not respect the variational bound with respect to the physical spectrum. The standard remedy is a penalty,

$$H_{\text{pen}} = \lambda P_{\text{unphys}}, \qquad P_{\text{unphys}} = |101\rangle\langle 101| + |110\rangle\langle 110| + |111\rangle\langle 111|, \tag{20.45}$$

with λ large enough to push the unphysical sector above the spectrum of interest and small enough not to wreck the conditioning of the optimisation. Choosing it is an art, and needing to choose it at all is the real cost of compression.

Table 20.6 sets the two side by side.

Table 20.6 Two qubit encodings of the Lipkin model, with $\varepsilon = V = 1$, $W = 0$. Both reproduce the exact quasispin spectrum: the maximal-spin block of the direct qubit Hamiltonian agrees with the $(N+1)$-dimensional matrix (20.40) to machine precision, which is the check that the two describe the same physics. Neither encoding is uniformly better.

N	dim.	qubits (direct)	qubits (compressed)	strings	E_0
4	5	4	3	16, weight ≤ 2	-4.000000
6	7	6	3	36, weight ≤ 2	-8.321118

The trade is the same one that runs through the whole of quantum computing. The direct mapping is exponentially wasteful in qubits – N of them for an $(N+1)$-dimensional problem – and buys, with that waste, a Hamiltonian whose terms are short, local and free of constraints. The compressed mapping is optimal in qubits and pays for it in string length and in the need to police a code space. On today's machines, where qubits are scarce and depth is scarcer, neither consideration dominates the other, and the choice is made case by case.

20.7 The many-body models

We now run VQE on the same four systems Section 19.8 fed to phase estimation: the Lipkin model in its direct encoding, the pairing model in the seniority-zero pair space of Section 4.2, the one-dimensional Heisenberg chain and the Hubbard dimer. The ansatz is the hardware-efficient form (20.9) throughout, deliberately: it carries no information about any of these Hamiltonians, so where it falls short the fault is the ansatz's and not the algorithm's, and it is instructive to see where that happens. Four random restarts are taken at each depth and the best kept – not a refinement but a necessity, as Section 20.2.2 warned.

Table 20.7 VQE on the models of Chapter 4, with exact expectation values and parameter-shift gradients. The last column counts the qubit-wise commuting groups, hence the number of distinct measurement circuits per energy evaluation. Compare Table 19.7: where phase estimation needed a control register and a Trotterised evolution, VQE needs a circuit a few layers deep – and a great many shots.

model	qubits	layers	exact E_0	VQE	error	groups
Lipkin, $N=4$	4	3	-4.00000	-3.99379	6×10^{-3}	4
pairing, 3 levels	3	4	-0.70530	-0.70527	3×10^{-5}	4
Heisenberg, 3	3	4	-0.75000	-0.75000	3×10^{-16}	4
Hubbard, 2 sites	4	2	-1.23607	-1.23607	2×10^{-8}	2

Three of the four are essentially exact. The Lipkin case is not, and the reason is worth following, because it is the characteristic failure mode of the whole approach. The exact ground state of the direct mapping (20.43) lies in the fully symmetric subspace: it is invariant under any permutation of the four qubits, since the Hamiltonian is. The hardware-efficient ansatz with a cnot ladder is not permutation-symmetric – the ladder singles out a direction along the chain – so the symmetric state is reachable only by a conspiracy among the parameters that the optimiser has no particular reason to find. It gets to within 6×10^{-3} and stops – two orders of magnitude worse than the other three rows, on the smallest Hilbert space of the four. An ansatz built from the collective operators (20.41), which is to say one that knows the model has an $SU(2)$ structure, would find the exact answer with a handful of parameters.

This is the general lesson and it is worth stating plainly: *the quality of a VQE result is a property of the ansatz, not of the algorithm*. The optimiser converged in every row of Table 20.7, and the convergence carries no information about whether the answer is right. Only the comparison with the exact column does, and that column is unavailable for any problem worth solving on a quantum computer. Symmetry is the most reliable guide available: an ansatz that conserves the quantities the Hamiltonian conserves searches a smaller space, and searches it better.

What an evaluation costs.

With G commuting groups and S shots per group, one energy costs GS measurements and one parameter-shift gradient costs $2PGS$. For the four-qubit, four-layer row that is $P=32$, $G=4$, so $256S$ measurements per optimisation step, and a few hundred steps at $S=10^4$ comes to 10^9 measurements for a four-qubit problem that a laptop diagonalises instantly. The number is not a reproach – the point of the exercise is what happens at forty qubits, where the laptop stops – but it does locate the bottleneck. VQE is *measurement-bound*. Phase estimation is *depth-bound*. They fail on different hardware, for different reasons, and that is the substance of the comparison we now make.

20.8 Phase estimation and VQE weighed against each other

Two algorithms, the same Hamiltonians, and almost nothing in common. It is worth setting out the comparison carefully, because the literature tends to present it as a choice between a rigorous method one cannot run and a practical method that works, and both halves of that summary are misleading.

20.8.1 What each one promises

Phase estimation solves the eigenvalue problem. Given an eigenstate and enough control qubits it returns the corresponding eigenvalue to any precision, with a probability bounded below by $4/\pi^2$ per run, and the resources grow as $\log(1/\varepsilon)$ in qubits and $1/\varepsilon$ in circuit depth. The guarantee is a theorem. What the theorem is silent about is where the eigenstate came from: the algorithm as stated needs $\left|\psi_j\right\rangle$ as input, and preparing it is as hard as the original problem. In practice one supplies a trial state, and the run succeeds with probability $|\langle\psi_j\rangle\psi_{\text{trial}}|^2$, the overlap of Section 19.9. For a Hartree-Fock reference and a molecule of moderate size that overlap is respectable; for strongly correlated systems it decays exponentially in the particle number, and the guarantee, while still true, becomes vacuous.

VQE promises less and delivers it more reliably. It returns $\min_{\boldsymbol{\theta}} \langle\psi(\boldsymbol{\theta})|\hat{H}|\psi(\boldsymbol{\theta})\rangle$, which is an upper bound on E_0 and nothing more. If the ansatz is good the bound is tight and the method is excellent; if not, it is not, and no feature of the output distinguishes the two cases. Against this: the state is produced rather than assumed, so the state-preparation problem does not appear separately – it has been absorbed into the ansatz, which is where the difficulty now lives.

Neither algorithm has made the many-body problem easy. Phase estimation concentrates the difficulty in state preparation and circuit depth; VQE concentrates it in ansatz design and measurement count. The difficulty is conserved, and recognising where each method has hidden it is the beginning of using either one sensibly.

20.8.2 Resources

Table 20.8 collects the scalings.

The two error scalings are the crux. Phase estimation buys a bit of precision per control qubit – *exponential* accuracy in the register size – while VQE buys it at $\varepsilon \sim N^{-1/2}$, so that each decimal digit costs a hundredfold in shots. Asymptotically this is no contest. The crossover is not asymptotic. At $\varepsilon = 10^{-3}$ with $\Lambda \sim 10^3$, VQE needs of order 10^{12} measurements, which is out of reach but not absurd; phase estimation needs a coherent circuit of order 10^6–10^9 gates, which requires error correction and therefore a machine that does not exist. VQE is not better; it is merely runnable, and it is runnable because its errors are of a kind that can be averaged away rather than a kind that destroys the computation.

20.8.3 The evidence from the small systems

The tables of this chapter and the last are worth reading together.

Table 20.8 Phase estimation and VQE compared. n is the number of system qubits, m the control register, ε the target accuracy, $\Lambda = \sum_\alpha |h_\alpha|$ the Pauli 1-norm, P the number of variational parameters and G the number of commuting groups.

	phase estimation	VQE
qubits	$n+m$, $m = \mathscr{O}(\log 1/\varepsilon)$	n
circuit depth	$\mathscr{O}(\Lambda t/\varepsilon)$, coherent	$\mathscr{O}(nL)$, shallow
repetitions	$\mathscr{O}(1)$ per bit	$\mathscr{O}(\Lambda^2/\varepsilon^2)$ per energy
classical work	none	the whole optimisation
error scaling	$\varepsilon \sim 2^{-m}$	$\varepsilon \sim 1/\sqrt{N}$
input needed	a state with good overlap	an expressive ansatz
output	an eigenvalue, either side of E_0	an upper bound on E_0
noise	fatal	degrades the bound
guarantee	a theorem	the variational principle
bottleneck	depth	measurements

On the one-qubit Hamiltonian, phase estimation with ten control qubits reached 2.7×10^{-4} (Table 19.4) while VQE reached 2×10^{-5} (Table 20.4) and would go further. QPE's accuracy is quantised by its energy grid; VQE's is limited only by the optimiser, in the ideal case where the ansatz is exact and there is no shot noise.

On the two-qubit Hamiltonian, phase estimation at $m = 4$ returned -0.583333 against a true -0.531129 – *below* the ground-state energy. There is nothing wrong with that: QPE estimates a phase, the nearest grid point may lie on either side, and the algorithm makes no claim of one-sidedness. But it means the output cannot be used the way a variational number can. Given two QPE runs at different m there is no principle that says which to prefer; given two VQE runs the lower one is better. Over Table 19.5 the error was also not monotone in m, falling, plateauing, and rising again before collapsing at $m = 7$ – behaviour perfectly consistent with the theory and thoroughly unhelpful as a convergence diagnostic.

On the many-body models, both algorithms got within a few per cent of the exact answers, and for opposite reasons. Phase estimation was limited by the Trotter error and the resolution of a small control register, both of which shrink predictably as resources are added. VQE was limited by the ansatz – the Lipkin row of Table 20.7 – and that does *not* shrink predictably; adding layers to a hardware-efficient circuit eventually makes things worse, not better, as the barren plateau of Section 20.4.3 sets in. This asymmetry is the deepest of the practical differences. QPE's errors are controlled by parameters we can turn. VQE's principal error is controlled by a design choice we can only make well or badly.

20.8.4 Where each fails, and how

The failure modes deserve to be stated side by side, because they are of different characters and the difference matters more than the scalings.

Phase estimation fails *loudly*. Insufficient overlap shows up as a distribution with no dominant peak. Too small a control register shows up as a coarse answer whose resolution is known in advance. Aliasing, as Section 19.7 showed, produces an obviously wrong number the moment the identity term is not removed. Decoherence randomises the register and the output distribution becomes flat. In every case, something in the output announces that something is wrong.

VQE fails *quietly*. A poor ansatz produces a smooth convergence curve and a wrong number. A barren plateau produces a gradient indistinguishable from noise, which an optimiser reads as a converged minimum. The marginal readout of Section 20.3.5 produces a confident answer that is not the energy of any state. Noise shifts the optimum and the optimiser follows it. In each of these the algorithm reports success. The only defence is external: symmetry checks, comparison across ansatz families, extrapolation in the noise, and exact results at small size. Building those checks in is not optional and is not decoration; it is the part of a VQE calculation that does the epistemic work.

20.8.5 What is likely to happen

The two are usually presented as successive eras – VQE now, phase estimation after error correction – and that is probably right as far as it goes. It understates how much they need each other. Phase estimation requires a trial state with good overlap, and the natural source of one is a converged VQE calculation; VQE requires a way to certify its answer, and the natural certificate is a phase-estimation run on the state it produced. The combination is stronger than either: use the shallow circuit to prepare, the deep one to measure. Equation (19.45) is the technical reason this works – QPE leaves the register entangled with the eigenstate components, so a successful run both measures the energy and projects onto the corresponding eigenstate.

The honest summary is that neither algorithm has yet produced a many-body result beyond classical reach, and that the obstacles are different in kind. For phase estimation the obstacle is hardware, and it is being worked on by people building error-corrected qubits. For VQE the obstacles are the barren plateau and the measurement cost, and both are mathematical: no amount of better hardware removes an exponentially small gradient or an ε^{-2}. Which of those is the easier problem is not obvious, and anyone who tells you otherwise is selling something.

20.9 Summary

The variational quantum eigensolver minimises $\langle\psi(\boldsymbol{\theta})|\hat{H}|\psi(\boldsymbol{\theta})\rangle$ over a family of states prepared by a parametrised circuit, using a quantum computer to evaluate the energy and a classical one to move the parameters. What it returns is a variational upper bound, with the same standing as any other in this book.

The energy is obtained by decomposing $\hat{H}$ into Pauli strings, rotating each into the computational basis – nothing for $\boldsymbol{Z}$, $\boldsymbol{H}$ for $\boldsymbol{X}$, $\boldsymbol{S}^\dagger\boldsymbol{H}$ for $\boldsymbol{Y}$ – and averaging parities over many shots. The statistical error falls as $1/\sqrt{N}$ and this, not the circuit, is the dominant cost. Correlators must be measured jointly: reconstructing $\langle Z_1Z_2\rangle$ from $\langle Z_1\rangle\langle Z_2\rangle$ is wrong, costs nothing to avoid, and fails silently.

Gradients come from the parameter-shift rule, which is exact when the gate generators have two eigenvalues, and which is interpolation of a known trigonometric form rather than a finite difference. Generators with richer spectra need $2R$ evaluations, and the two-term rule applied to them can return zero for a nonzero derivative.

Applied to the models of Chapter 4, VQE with a hardware-efficient ansatz reaches the exact energy for the pairing, Heisenberg and Hubbard systems and falls a few per cent short for Lipkin, where the ansatz breaks a symmetry the Hamiltonian respects. That is the characteristic result: the algorithm converges, and the quality of what it converges to is a property of the ansatz.

Against phase estimation, VQE trades an exponential precision guarantee for shallow circuits and noise tolerance. QPE fails loudly and needs a machine that does not exist; VQE fails quietly and runs today. The most plausible use of both is together.

20.10 Exercises

Exercise 1: the variational principle on a circuit

a) Prove Eq. (20.1) and show that equality holds if and only if $|\psi\rangle$ lies in the ground-state eigenspace.
b) Show that the ansatz (20.8) reaches every pure state of one qubit up to a global phase, and find the angles that give the ground state of the Hamiltonian (19.35).
c) The hardware-efficient ansatz (20.9) with L layers on n qubits has $2nL$ parameters, while a general state has $2^{n+1}-2$ real degrees of freedom. At what L do the counts match, and why does matching them not imply that the ansatz is universal?
d) Show that if a noisy device prepares ρ instead of $|\psi\rangle\langle\psi|$, then $\mathrm{Tr}(\rho\hat{H}) \geq E_0$ still holds. Which property of ρ is doing the work?

Exercise 2: measurement

a) Verify $\boldsymbol{HZH} = \boldsymbol{X}$ and $\boldsymbol{HS}^{\dagger}\boldsymbol{YSH} = \boldsymbol{Z}$ by direct multiplication.
b) Write down the measurement circuit for the string $\boldsymbol{X}\otimes\boldsymbol{Y}\otimes\boldsymbol{Z}\otimes\boldsymbol{I}$ and the classical post-processing that turns an outcome bit string into ± 1.
c) Derive Eq. (20.16) and show that the variance vanishes when $|\psi\rangle$ is an eigenstate of P. Explain the result in words.
d) For the two-qubit Hamiltonian of Section 19.7.2, partition the five Pauli strings into qubit-wise commuting groups by hand. How many measurement circuits are needed?
e) Suppose the total shot budget N is to be divided among the strings as N_α. Minimise (20.17) subject to $\sum_\alpha N_\alpha = N$ and show that the optimal allocation is $N_\alpha \propto |h_\alpha|\sqrt{1-\langle P_\alpha\rangle^2}$. Why is this hard to use in practice?

Exercise 3: the marginal trap

a) Compute the reduced density matrices of $|\Phi^+\rangle$ and $|\Psi^+\rangle$ and confirm the claim in the proof of Theorem 20.1: *every* single-qubit observable has the same expectation value in both.
b) Derive (20.21) and (20.22) and plot both over $\theta \in [0,\pi]$.
c) Show that $\langle Z_1 Z_2\rangle = \langle Z_1\rangle\langle Z_2\rangle$ holds for every product state, and construct a non-product state for which it holds accidentally. What does the existence of such states imply about testing for the error empirically?
d) Explain why a single shot of a two-qubit circuit gives $\langle Z_1\rangle$, $\langle Z_2\rangle$ and $\langle Z_1 Z_2\rangle$ at no extra cost, and identify the exact place in the calculation where the incorrect procedure goes wrong. (The answer is the order of two operations.)
e) Run `vqe.py` with `marginal=True` on the Lipkin Hamiltonian and report both the energy the optimiser believes and the true energy of the state it prepared.

Exercise 4: the parameter-shift rule

a) Verify that $G^2 = r^2\boldsymbol{I}$ for a Hermitian G with eigenvalues $\pm r$, and deduce Eqs. (20.26) and (20.27).
b) Fill in the algebra between (20.27) and (20.28), obtaining the coefficients in (20.29).
c) Show that three evaluations of $E(\theta)$ at distinct points determine A, B and C completely, and hence that the shift rule is exact interpolation and not approximation.
d) For $G = \frac{1}{2}(\boldsymbol{Z}_1 + \boldsymbol{Z}_2)$, show that (20.34) is the correct form, and that the two-term rule applied to it returns $\frac{1}{2}[E(\theta+\pi/2) - E(\theta-\pi/2)] = -2C_1 \sin\theta \cdot 0 + \ldots$ – work out exactly which frequency components survive and which cancel, and explain the zero found in Section 20.4.2.
e) Verify (20.35) at $R = 1$ by showing that it reduces to (20.32).
f) Estimate the total number of measurements needed for one gradient of a 20-qubit, 5-layer ansatz on a Hamiltonian with 500 Pauli strings in 30 commuting groups, at 10^4 shots per group.

Exercise 5: the Lipkin encodings

a) Derive (20.39) from $J^2 = J_z^2 + \frac{1}{2}(J_+J_- + J_-J_+)$ and complete the reduction to (20.40).
b) Show that $\frac{V}{2}(J_+^2 + J_-^2) = V(J_x^2 - J_y^2)$ and derive Eq. (20.42), being careful about the $i = k$ terms.
c) Explain why no Jordan-Wigner strings appear in the direct mapping, referring to the structure of the Lipkin interaction.
d) For $N = 4$, write the five states $|j = 2, m\rangle$ as three-qubit basis states and verify two of the coefficients in (20.44).
e) Show numerically that the maximal-spin block of the direct qubit Hamiltonian reproduces the spectrum of (20.40). What are the other blocks physically?
f) Design an ansatz built from the collective operators (20.41) and show that it finds the exact $N = 4$ ground state where the hardware-efficient ansatz of Table 20.7 does not. How many parameters does it need?

Exercise 6: comparing the two algorithms

a) For a target accuracy ε, estimate the number of measurements VQE needs and the circuit depth QPE needs, both as functions of Λ and ε. Where do the two curves cross for $\Lambda = 10$, and for $\Lambda = 10^3$?
b) Table 19.5 contains a QPE estimate lying below the exact ground-state energy. Explain why this is not a bug, and why no VQE run could produce such a number.
c) Both algorithms need something they cannot supply themselves: QPE a state with good overlap, VQE a sufficiently expressive ansatz. Argue that these are the same difficulty in different clothes, and identify one respect in which they are not.
d) Design a hybrid: use VQE to prepare a state, then QPE to measure its energy. What does the combination guarantee that neither does alone, and what is the overlap of the VQE state with the true ground state in terms of quantities the VQE run has already produced?
e) Both algorithms were run above on the same four models. Write a paragraph, of the kind that would appear in a paper, stating what has and has not been demonstrated.

Exercise 7: numerical work

a) Reproduce Table 20.2 and extend it to 10^5 and 10^6 shots. Fit the standard error to aN^{-b} and check $b = 1/2$.
b) Implement SPSA and compare it with parameter-shift gradient descent on the Hubbard dimer, plotting energy against *number of measurements* rather than number of iterations. Which wins, and does the answer depend on the shot count per evaluation?
c) Study the barren plateau: for $n = 2, \dots, 8$ qubits with $L = n$ layers, sample random parameters, compute $\partial E/\partial\theta_1$ by the shift rule, and plot its variance against n on a logarithmic scale. Compare with (20.36).
d) Add depolarising noise to the simulator by mixing the state with the maximally mixed state after each layer, and study how the VQE energy of the two-qubit Hamiltonian degrades. Verify that the result remains an upper bound.
e) Run VQE with shot noise at 10^2, 10^3 and 10^4 shots on the Heisenberg chain and plot the converged energy against the shot count. How many shots are needed before the ansatz error dominates the statistical error?

Programs for this chapter

Every number and every figure of this chapter comes from the programs collected in `BookPrograms/chapter20/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/vqe.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures of this chapter were generated by re-executing that notebook and are collected in `BookFigures/chapter20/`.

`vqe.py` Built on the simulator of `qpe.py`: the model Hamiltonians in Pauli form, `rotate_to_z_basis` and `expectation_sampled` with the three measurement circuits and shot sampling, the marginal-readout counterexample and the deliberate VQE failure it causes, `parameter_shift_gradient` with its trigonometric structure test and its failure for three eigenvalues, the hardware-efficient ansatz and the gradient-descent loop, and both Lipkin encodings. Runs in about six minutes.

All of them run on `numpy` alone unless stated otherwise. Several import a program belonging to another chapter, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebooks show how.

Part VI
Concluding remarks and perspectives

Chapter 21
Concluding remarks and perspectives

Twenty chapters, five families of methods, and one problem. It is worth saying at the outset what that problem is, because everything in this book is a response to it and the responses look more different than they are.

A system of N interacting fermions distributed over n single-particle states has $\binom{n}{N}$ basis states, and the wave function is a vector in a space of that dimension. For six electrons in forty-two orbitals that is about five million; for the twenty-eight electrons of a modest molecule in a hundred orbitals it is 10^{28}, and no computer that will ever be built can hold the vector, let alone diagonalise the matrix. This is the exponential wall of Section 5.5, and it is not a statement about algorithms or hardware. It is a statement about the dimension of a tensor product, which Section 2.16 established before any physics was done.

Everything since has been a way of not storing that vector. The five strategies in this book are genuinely distinct – they fail for different reasons, and where one is helpless another is comfortable – but each is a bargain of the same shape: give up access to the full space, and get back a polynomial algorithm, at the cost of an error that must then be controlled by some other means. What distinguishes the methods is which error they incur and how honestly it can be estimated.

21.1 Five bargains

Truncate the space.

Configuration interaction, Chapter 5, keeps the exact linear structure of the problem and simply throws away most of the basis. Its virtue is that it is systematically improvable in an entirely transparent way – add determinants, get a better answer, and the variational principle guarantees the direction of the improvement. Its defect is that the number kept must grow exponentially to maintain a fixed accuracy as the system grows, so truncated CI is not size extensive: two non-interacting copies of a system do not get twice the energy of one, a failure demonstrated numerically in Table 9.3. Full CI remains the reference against which everything else in this book is measured, and it is available only where the problem is small enough not to need any of the other methods.

Factorise the wave function.

Hartree-Fock (Chapter 6), perturbation theory (Chapter 9) and coupled cluster (Chapter 10) all replace the linear expansion with a structured one. The mean-field solution is a single determinant; perturbation theory is a power series about it; coupled cluster exponentiates the excitation operator, and the exponential is precisely what restores size extensivity, since $e^{T_A+T_B} = e^{T_A}e^{T_B}$ for non-interacting subsystems. These methods are the backbone of quantum chemistry and nuclear structure for good reason: they are polynomial, they are systematically improvable through the excitation hierarchy, and at weak to moderate coupling they are very accurate. Table 10.4 shows coupled-cluster doubles reproducing the pairing model to 3.6×10^{-6} at $g = 0.25$. What they cannot do is strong correlation. The same table shows the perturbation series diverging by $g = 1.5$ and CCD wrong by 0.12 at $g = 2$, where the single-reference picture has stopped meaning anything.

Sample it.

Variational and diffusion Monte Carlo (Chapters 13 and 15) never form the vector at all. They evaluate the trial function pointwise and integrate by sampling, so the cost is set by the dimension of a configuration – linear in N – rather than by the dimension of the Hilbert space. There is no basis-set error, which is a larger advantage than it sounds: Table 16.5 shows almost the whole apparent gap between coupled cluster and Monte Carlo for six electrons to be basis incompleteness rather than any failure of the cluster expansion. The price is paid in two currencies. Statistical error falls as $N^{-1/2}$, so precision is expensive and correlated samples make it more expensive still, which is why Chapter 14 spends as long on resampling as on optimisation. And for fermions the sign problem, and its practical resolution in the fixed-node approximation of Section 15.5, introduces an error that no amount of computer time reduces. It is the only error in the book that cannot be made smaller by running longer.

Compress it.

Chapters 17 and 18 replace the hand-designed trial function with a parametrised one – a Boltzmann machine, a neural network – and let optimisation find the parameters. The appeal is that the functional form is not assumed; the machine is told the energy and nothing else. Table 17.4 makes the point at its sharpest: a restricted Boltzmann machine with four hidden units, given no orbitals, no basis and no notion of antisymmetry, reaches 3.082 for the two-electron dot and so beats Hartree-Fock in forty-two orbitals. Fourteen numbers against a mean-field calculation, and the fourteen numbers win.

The following two rows of the same comparison are the more important lesson. A Padé-Jastrow function with *two* parameters reaches 3.000549, three orders of magnitude better, because those two parameters encode the cusp condition and the network had to discover it from scratch and did not. The resolution, in Table 18.4, is not to choose between them: the analytic form multiplied by a neural correlator reaches 3.000477 with half the variance. *Physics first, then flexibility* is the summary of two chapters, and it generalises well beyond the quantum dot.

Store it natively.

Chapters 19 and 20 take the one route that does not compromise: a register of n qubits holds 2^n amplitudes, so the state is represented exactly and the exponential wall simply is not there. Phase estimation then extracts an eigenvalue to any precision, with a theorem. The compromise reappears somewhere else, as it must. QPE needs a coherent circuit of 10^6 to 10^9 gates and a trial state whose overlap with the target does not vanish exponentially; VQE avoids the depth and pays instead in measurements, $\mathscr{O}(\Lambda^2/\varepsilon^2)$ of them, and in the ansatz problem that Chapter 18 also faces in a different costume. Neither has yet produced a many-body result beyond classical reach.

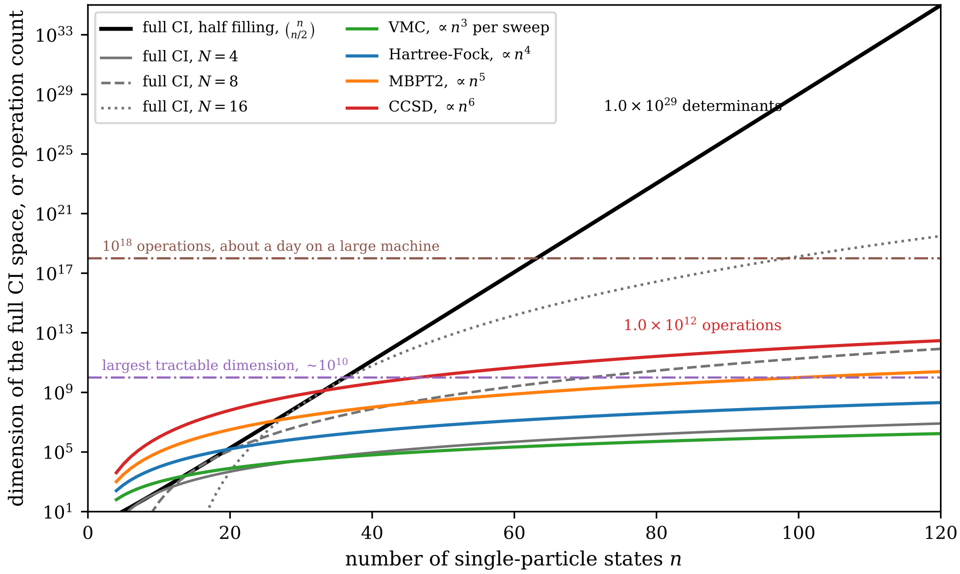

Fig. 21.1 The exponential wall, and what the five bargains buy. The dimension of the full CI space is plotted against the number of single-particle states n on a logarithmic vertical axis, for half filling (the heavy black curve, $\binom{n}{n/2}$) and at fixed particle numbers $N = 4$, 8 and 16 (grey); against these are drawn the operation counts of the polynomial methods, $\mathscr{O}(n^3)$ for a variational Monte Carlo sweep, $\mathscr{O}(n^4)$ for Hartree-Fock, $\mathscr{O}(n^5)$ for MBPT2 and $\mathscr{O}(n^6)$ for CCSD. The two dash-dotted horizontal lines mark a dimension of 10^{10}, about the largest an iterative diagonalisation reaches, and 10^{18} operations, roughly a day on a large machine. Prefactors have been dropped and a dimension is not a floating-point operation, so only the slopes and the gap between the two families of curve carry meaning.

Figure 21.1 puts the wall and the five responses to it on one pair of axes, and it is the picture the rest of this chapter argues from. The vertical scale is logarithmic and spans thirty-four decades, so a curve that climbs steadily across it is not merely large but of a different kind from one that flattens. The heavy black curve is the half-filled full CI dimension. At twenty single-particle states it is 1.8×10^5 and one could hold the vector in memory without thinking about it; at forty it is 1.4×10^{11} and one could not; at a hundred it is 1.0×10^{29}. The same problem treated by CCSD costs of order 10^{12} operations at $n = 100$, which is a long afternoon. Seventeen orders of magnitude separate the two at that point, and because $\binom{n}{n/2} \sim 2^n/\sqrt{n}$ the gap widens by roughly a factor of two for every single-particle state added, while the polynomial curves gain only the ratio $((n+1)/n)^6$, which is a few per cent. Nothing on this plot converges towards anything else.

The horizontal lines are where the picture becomes uncomfortable. A dimension of about 10^{10} is what a Lanczos diagonalisation reaches on the largest machines available, and the half-filled curve passes it at $n \approx 38$ – a system of nineteen particles in thirty-eight orbitals, which is smaller than almost anything one would want to compute. Everything of interest in this book lies above that line. Equally instructive is where the black curve overtakes the CCSD count, at about $n = 33$: below that crossing exact diagonalisation is genuinely the cheaper calculation, which is why the small models of Chapter 4 are solved exactly throughout, and above it the crossing is never reversed. The grey curves at fixed particle number are the quiet warning. They are polynomial – $\binom{n}{N} \sim n^N/N!$ is a polynomial of degree N in the basis size – and they are also, at $N = 16$ and $n = 120$, above 10^{20}. A method can be polynomial and useless in the same breath, which is the reason the exponent in the scaling is quoted as carefully as it is in every chapter of this book.

Read against the five bargains, the figure says what each of them does to the geometry. Truncating the space keeps the exact linear structure and moves down the family of fixed-N curves without leaving it, which is precisely why truncated CI still climbs exponentially as the system grows and why it is not size extensive. Factorising the wave function abandons the black curve altogether for one of the coloured ones, at the price of an error controlled by nothing but the quality of the reference – and Section 21.2.2 below is about how badly that can go. Sampling does not appear on the vertical axis at all in the way the others do: the $\mathscr{O}(n^3)$ curve is the cost of one sweep, and the number of sweeps is a prefactor, so Monte Carlo trades the dimension of the Hilbert space for the dimension of a single configuration and pays in a statistical error falling as $N_{\text{samples}}^{-1/2}$. Compression replaces the vector by a few hundred parameters, which is to say by a curve that is nearly flat. And storing the state natively is the only bargain that does not move off the black curve – the register *is* the black curve, which is the whole of the appeal – so its compromise must reappear elsewhere, and does, as circuit depth and measurement count.

21.2 The evidence

Assertions about method families are cheap. The book has been organised so that they can be checked, by applying as many methods as possible to a small number of systems whose exact answers are known. Two of those systems carry most of the weight.

21.2.1 Two electrons in a quantum dot

The two-dimensional parabolic dot at $\hbar\omega = 1$ has Taut's analytic ground-state energy $E = 3$ exactly, which makes it the ideal proving ground: no method can hide. Table 21.1 collects what every chapter obtained.

Three readings, in increasing order of usefulness.

The obvious one is that the best result is the hybrid at the bottom. The second is that the ordering has almost nothing to do with how elaborate the method is: a two-parameter trial function beats a forty-two-orbital coupled-cluster calculation by an order of magnitude, and a hundred-and-five-parameter neural network improves on it only by a factor of three. The third, and the one worth carrying away, is *why* the basis-set methods stop at 1.4×10^{-2}. They stop because CCSD is exact for two particles, so the entire residual error is the finite basis – and a basis of Gaussians or oscillator functions converges slowly wherever the wave function has a cusp, which is exactly where the physics is. The Monte Carlo methods have no basis and put the cusp in by hand. The comparison is therefore not really between coupled cluster and Monte Carlo at all. It is between a representation that struggles with the short-range correlation hole and one that is built around it.

Figure 21.2 makes that argument quantitative, by doing what Table 21.1 cannot and following the basis-set methods through five oscillator bases from twelve to fifty-six spin-orbitals. Three things are immediate in the left panel. The open FCI circles sit exactly on the CCSD triangles at every basis size, as they must, since for two particles singles and doubles exhaust the excitation space – the cluster truncation is not an approximation here at all, and the whole of the residual is basis incompleteness. The Hartree-Fock line is flat: 3.162691 in twelve orbitals and 3.161909 in fifty-six, a movement of 8×10^{-4} across a fivefold enlargement

method	chapter	parameters	E	error
Hartree-Fock, 42 orbitals	11	—	3.161921	1.6×10^{-1}
RBM, 4 hidden units	17	44	3.08227(173)	8.2×10^{-2}
MP2, 42 orbitals	11	—	3.027038	2.7×10^{-2}
CCSD, 42 orbitals	11	—	3.013626	1.4×10^{-2}
FCI, 42 orbitals	11	—	3.013626	1.4×10^{-2}
Padé-Jastrow VMC	13	2	3.001323(72)	1.3×10^{-3}
Padé-Jastrow $\times$ neural correlator	18	$2 + 105$	3.000477(48)	4.8×10^{-4}
exact (Taut)	18	—	3.000000	—

Table 21.1 Two electrons in a two-dimensional parabolic trap at $\hbar\omega = 1$, in atomic units, treated by six methods drawn from four different parts of this book. CCSD and FCI coincide because for two particles singles and doubles exhaust the excitation space, so the remaining 1.4×10^{-2} is entirely basis incompleteness. The rows are ordered by accuracy, and the ordering is not the ordering of sophistication. The last two rows are taken from Table 18.4 because they are the only pair in the book computed under identical conditions; other production runs of the same two-parameter Padé-Jastrow function appear in Chapter 13 at 3.000463(199) and in Chapter 17 at 3.000549(49). That spread, of about 9×10^{-4} across runs whose quoted errors are two to seven times smaller, is itself worth attention: it is the correlated-sampling effect of Chapter 14 showing through, and a reminder that a stochastic result is only as good as its error estimate.

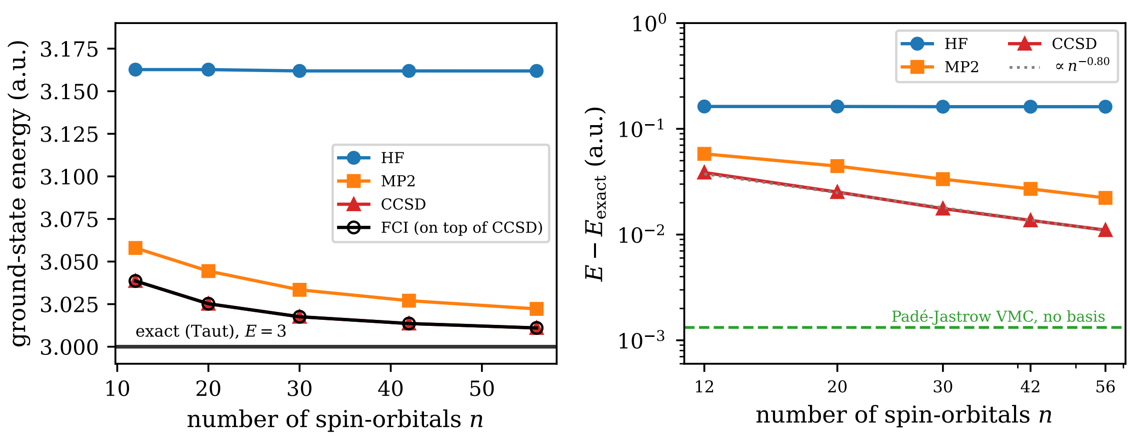

Fig. 21.2 Two electrons in a parabolic dot at $\hbar\omega = 1$: convergence of the basis-set methods towards Taut's exact 3 a.u. Left, the ground-state energy against the number of spin-orbitals n for harmonic-oscillator bases of two to six shells, with the exact value as a horizontal line; the open circles are full CI and lie on top of the CCSD triangles at every point. Right, the same errors on doubly logarithmic axes, so that the rate of convergence is a slope: the dotted line is the fitted $n^{-0.80}$ and the dashed horizontal line is the two-parameter Padé-Jastrow variational Monte Carlo result of Table 21.1, 3.001323(72), which uses no basis and therefore does not move with n.

of the basis, which says as plainly as anything in this book that a mean field does not describe a correlation hole and that giving it more functions does not help. And the correlated curves, which look on this linear scale as though they are settling onto the exact line, are not.

The right panel is why the figure has two panels. On logarithmic axes the CCSD and FCI error is a straight line of slope -0.80, and a straight line of that gradient does not terminate anywhere: it crawls. Going from twelve to fifty-six spin-orbitals – a factor of nearly five in the basis, and a great deal more than that in the cost of transforming the two-body integrals – reduces the error from 3.9×10^{-2} to 1.1×10^{-2}, a factor of three and a half. Extrapolating the same slope, an error of 10^{-3} would require about 1100 spin-orbitals, some thirty-three oscillator shells, for a system of *two* electrons. The 1.4×10^{-2} at which Table 21.1 finds CCSD and FCI is not, then, a value at which the basis-set methods stop; it is simply where one stops paying, and the figure shows that stopping a shell or two later would not have changed the conclusion. The dashed Padé-Jastrow line sits an order of magnitude below the best of the basis-set points and is horizontal because there is no basis to converge: two variational parameters, chosen to satisfy the cusp condition, reach 1.3×10^{-3} where a fifty-six-orbital exact diagonalisation reaches 1.1×10^{-2}. That slope of -0.80 is the price of expanding a non-analytic feature of the wave function in an analytic basis, and it is a property of the representation and not of the many-body method built on top of it – which is the third reading of Table 21.1, drawn as a curve.

21.2.2 The pairing model

Where the dot tests accuracy, the pairing model tests *robustness*, since its coupling g can be turned up until every single-reference method fails. Table 10.4 carried out that experiment. At $g = 0.25$ the methods separate by three orders of magnitude and coupled cluster is best; by $g = 1.5$ the perturbation series has begun to diverge, third-order theory overshooting by 0.089 where second order undershoots by 0.19, which is the characteristic signature of an asymptotic series and not of a numerical accident. At $g = 2$ coupled-cluster doubles is wrong by 0.12 and the RPA, which had been the worst method at weak coupling, is suddenly the best at 0.013 – for the good reason that the RPA is built out of collective pair excitations, which is what the strongly coupled system actually contains.

Five values of g show that something happens; they do not show where it happens. Figure 21.3 therefore repeats the whole of Table 10.4 on a grid of thirty couplings running from $g = 0.05$ out to $g = 3$, using the same routines that produced the table, and the turnover can be watched rather than inferred.

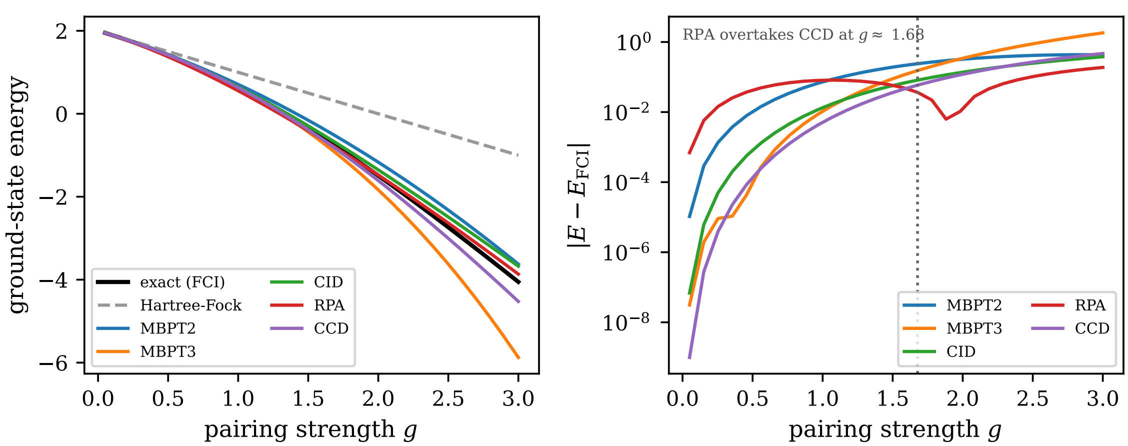

Fig. 21.3 The ordering of the methods is not stable in the coupling: the pairing model with four doubly degenerate levels and four particles, over thirty couplings from $g = 0.05$ to $g = 3$. Left, the ground-state energy itself, with the exact diagonalisation in black and the Hartree-Fock reference as a grey dashed line; right, the absolute error $|E - E_{\mathrm{FCI}}|$ of each approximation on a logarithmic scale, which is the only scale on which the crossings are visible. The dotted vertical line marks $g \approx 1.68$, the coupling beyond which the random-phase approximation – comfortably the worst of the five at weak coupling – is the most accurate method available.

The left panel of Figure 21.3 is there mainly to show that there is nothing to see. Every approximation except Hartree-Fock tracks the exact curve so closely on the scale of the energy that no ranking can be read off at all, while the mean-field line peels away steadily and is wrong by 3.1 at $g = 3$, where it gives -1.0 against an exact -4.1. All the information is in the right panel, and it is stark. At $g = 0.05$ the five methods are spread over five orders of magnitude: coupled-cluster doubles is in error by 9.9×10^{-10}, third-order perturbation theory by 3.1×10^{-8}, and the RPA by 6.9×10^{-4}, so CCD is better than its nearest competitor by a factor of thirty and better than the RPA by six orders of magnitude. Nothing in that picture survives. The CCD and RPA curves cross at $g \approx 1.68$, and from there to the end of the range the ordering is simply reversed: at $g = 3$ the RPA is wrong by 0.19 and coupled cluster by 0.47, so the worst method at weak coupling has become the best by a factor of two and a half, and the best has become worse than second-order perturbation theory. A factor of 7×10^5 in favour of CCD at one end of the axis has turned into a factor of 2.5 against it at the other.

Two smaller features of the right panel are worth pausing over, because both are easy to misread. Third-order perturbation theory is the most accurate of the five in a narrow window around $g \approx 0.36$ and the *least* accurate of the five by $g \approx 2.4$, overshooting by 1.4 at $g = 2.8$ where second order undershoots by 0.43; a correction that alternates in sign and grows in magnitude with the order is the signature of an asymptotic series, and not of a defect in the implementation. And the sharp downward spike in the RPA curve near $g \approx 1.9$ is not a region of unusual accuracy. It is a change of sign: the RPA energy crosses the exact one there, and on a logarithmic plot of $|E - E_{\mathrm{FCI}}|$ a zero of the error is an infinitely deep notch. A method that is right by coincidence at one coupling has learnt nothing that helps it at the next, and the notch should be read as an accident of the plot rather than a property of the RPA.

This is the most instructive single result in the book, and it is worth stating as a principle: *no method is uniformly best, and the ordering at weak coupling tells you nothing about the ordering at strong coupling.* A practitioner who selects a method by reputation rather than by matching it to the physics of the regime will be wrong, and will be wrong in a way that the calculation does not reveal.

21.2.3 Six electrons, and agreement without an exact answer

The two-electron dot has an exact solution and the pairing model can be diagonalised. Chapter 16 confronts the situation that actually obtains in research, where neither is true. Six electrons in the same trap have no analytic solution and a full CI space far too large to diagonalise, so the only available check is the agreement of methods that share no approximation. Table 16.5 provides it: Λ-CCSD(T) converged in twenty shells gives 20.1582, diffusion Monte Carlo with a fixed nodal surface gives $20.16125(151)$, and the two agree to 1.5×10^{-3}.

One is a deterministic expansion in particle-hole excitations, limited by the truncation of the cluster operator and the size of the basis; the other is a stochastic imaginary-time projection, limited by the nodal surface of a trial function and by statistics. Their errors are unrelated. When two such calculations agree, the agreement is evidence in a way that neither result alone could be, and this – rather than any single number – is the standard of proof the field actually works to. It is also the reason a book like this one covers five families of methods rather than the one its author likes best.

21.2.4 On a quantum computer

The last two chapters ran the models of Chapter 4 through both quantum algorithms. Table 19.7 and Table 20.7 report accuracies of 10^{-2} to 10^{-13} on systems of three and four qubits, and the numbers should be read with a clear eye about what they demonstrate. They demonstrate that the algorithms are correctly implemented and that their error mechanisms behave as the theory says. They do not demonstrate any advantage: every system in those tables is diagonalised instantly by the laptop that simulated the quantum computer.

The comparison in Section 20.8 is the substantive part. Phase estimation and VQE fail in opposite ways – depth-bound against measurement-bound, loudly against quietly – and neither failure is the kind that better engineering alone removes. It is worth noticing how closely the VQE ansatz problem resembles the trial-function problem of Chapter 13 and the architecture problem of Chapter 18. The same difficulty appears in three costumes across three parts of this book, which suggests it is the real difficulty.

21.3 What recurs

Reading the twenty chapters together, a small number of ideas keep returning in different notation. They are worth collecting, because a reader who recognises them will find new methods much easier to learn than one who treats each chapter as a separate technique.

The variational principle.

Section 2.7 proved that $\langle\psi|\hat{H}|\psi\rangle \geq E_0$ for any normalised $|\psi\rangle$, in one line, from nothing but the fact that a weighted average of numbers is at least the smallest of them. That line is then the foundation of Hartree-Fock, of truncated CI, of variational Monte Carlo, of the neural quantum states of Chapter 18 and of the variational quantum eigensolver. Five different fields, one inequality. Its practical value is that it makes results *comparable*: two variational numbers can be ranked without knowing the exact answer, and lower is better. Almost nothing else in computational physics offers that, and Section 20.8 showed what its absence costs.

Reference states and what is measured from them.

Hartree-Fock produces a determinant; perturbation theory, coupled cluster, CI and the unitary coupled-cluster ansatz of Chapter 10 all measure corrections *from* it. The particle-hole formalism of Chapter 3 exists to make that bookkeeping automatic. When the reference is good the correction is small and every one of these methods works; when it is not – strong coupling, level crossings, open shells – they all degrade together, because they all share the same assumption. This is why the pairing model at large g defeats four methods at once.

Symmetry as a resource.

The most efficient calculation in the book is not the most sophisticated one; it is the one that exploits a symmetry. Quasispin collapses the Lipkin model from $\binom{2N}{N}$ dimensions to $N+1$, which is why exact answers are available there for systems that would otherwise be impossible. Seniority does the same for the pairing model. Conversely, the clearest failure in Table 20.7 is the Lipkin row, where the hardware-efficient ansatz *breaks* the permutation symmetry the Hamiltonian respects and consequently performs two orders of magnitude worse than on larger problems. An ansatz that conserves what the Hamiltonian conserves searches a smaller space and searches it better. This holds for trial wave functions, for neural architectures and for quantum circuits alike.

Bias against variance.

Every method in the book sits somewhere on this axis. Deterministic methods have no variance and a bias set by their truncation; Monte Carlo methods have no truncation bias and a variance set by the sampling; the fixed-node approximation buys a finite variance with a bias that cannot be removed; shot-based VQE has both. Chapter 14 is in a sense the technical heart of the book, because estimating a variance correctly in the presence of correlation is what makes any stochastic result trustworthy, and because it is the part most often done badly.

The exact answer as an instrument.

Nearly every chapter tests itself against something exactly known – Taut's solution, full diagonalisation, an analytic limit, a published benchmark. This is not pedagogical decoration. It is the only way to establish that an implementation is correct, and the habit of arranging for such a check before trusting a result is probably the single most transferable thing in this book.

21.4 What remains open

It would be dishonest to close a book on the many-body problem by implying that it has been solved. The following are not gaps in the exposition; they are the places where the field itself does not yet know what to do.

The sign problem.

Fermionic Monte Carlo without a fixed nodal surface has a signal-to-noise ratio that decays exponentially in the particle number and the inverse temperature. The fixed-node approximation of Section 15.5 converts this from an intractability into a controlled bias, and the bias is then the accuracy limit of the best available method for large fermionic systems – our $20.16125(151)$ against the published $20.1597(2)$ differ by their nodes and by nothing else. The problem is known to be NP-hard in general, so no universal solution exists; progress consists of better nodal surfaces for particular classes of system, and neural-network nodes are currently the most promising route.

Strong correlation and multireference physics.

Single-reference methods fail where a single determinant is a poor starting point, and this is precisely where the interesting physics lives: superconductivity, Mott transitions, nuclear shape coexistence, transition-metal chemistry. Multireference coupled cluster exists but is neither simple nor uniformly reliable. Table 10.4 is a miniature of the whole difficulty, in a model with two parameters and an exact solution.

Ansatz design.

The same problem, three times over. Chapter 13 asks what trial function to write down; Chapter 18 asks what architecture to choose; Chapter 20 asks what circuit to build. In every case the answer determines the quality of the result, in every case the optimisation converges regardless, and in no case does the output indicate whether the choice was good. There is no theory of ansatz design. There are heuristics – respect the symmetries, put in the known analytic behaviour, prefer physically motivated forms to generic ones – and the evidence of Table 18.4 that they are worth more than raw flexibility.

Barren plateaus and trainability.

Section 20.4.3 described a gradient variance decaying as 2^{-n} for sufficiently expressive circuits. This is a mathematical obstacle, not an engineering one: no improvement in hardware makes an exponentially small gradient measurable. The proposed escapes all amount to being less generic, which trades the trainability problem for the ansatz-design problem. The same tension – expressiveness against trainability – appears in classical neural quantum states, and is not resolved there either.

The measurement cost.

The $\mathcal{O}(\Lambda^2/\varepsilon^2)$ shot count of Section 20.3.3 puts chemical accuracy for a molecule of interest at something like 10^{11} measurements per energy evaluation. Grouping, classical shadows and better estimators have each removed constant factors and an order of magnitude or two. None has removed the ε^{-2}, and none can, since it is the central limit theorem.

Verification without an exact answer.

This is the quiet problem underneath all the others, and it is what Section 21.2.3 was really about. For any system worth computing, the exact answer is unavailable by construction. Every method in this book converges; convergence is not correctness; and the field's actual standard of proof is the agreement of methods with unrelated errors. That standard is sound but expensive, and it is systematically weakened by the tendency to report the best number rather than the spread. A calculation reported with a statistical error bar and no estimate of its systematic error is not a finished calculation.

21.5 Perspectives

The classical methods are not finished.

It is easy to read the last four chapters as an announcement that coupled cluster and Monte Carlo are of historical interest. They are not. Coupled cluster remains the most accurate polynomial-scaling method available for weakly to moderately correlated systems and is used daily to produce results of a quality no quantum computer will approach for many years. Diffusion Monte Carlo, with good nodes, is the reference method for large fermionic systems. The developments most likely to matter in the next decade are improvements to these

– better nodal surfaces, cheaper and more robust multireference schemes, tensor-network representations that make the truncation adaptive – and not their replacement.

Machine learning will be absorbed, not adopted wholesale.

The trajectory of Chapters 17 and 18 is already visible in miniature in Table 18.4. The pure network, given no physics, is beaten by two analytic parameters; the network attached to the correct analytic form beats both. The productive use of these tools is as a flexible correlator on top of a structure that already encodes the symmetries, the cusps and the asymptotics – not as a replacement for understanding. The most substantial results in the field so far, in electronic structure and in nuclear physics, have all had this hybrid character.

Quantum computing is a hardware bet with a mathematical hedge.

The case for phase estimation is a theorem and the obstacle is engineering, which is the good way round: error-corrected machines are being built, and if they arrive at the required scale the algorithm works as advertised. The case for VQE is the reverse – it runs today, and its obstacles are mathematical. The reasonable expectation is that the two combine, VQE or a classical method preparing a state and phase estimation certifying it, and that the first real demonstration of advantage comes from a problem chosen for the hardware rather than a problem chosen for its importance. That is how these things usually go, and it is not a criticism.

Where the methods will meet.

The most interesting near-term work is at the boundaries. Neural-network trial functions inside diffusion Monte Carlo, which attacks the nodal problem with the one tool flexible enough for it. Coupled-cluster amplitudes as the initial parameters of a unitary coupled-cluster circuit, which uses fifty years of classical development to solve the quantum ansatz problem. Machine-learned functionals in density functional theory, which addresses the one approximation in Chapter 8 that has never had a systematic route to improvement. In each case the gain comes from a method with a structural strength being used to repair a method with a matching weakness, and a reader who has followed this book has the background to see which weaknesses match which strengths.

What to take from this book.

Three things, in order of durability. The formalism – second quantisation, Wick's theorem, the particle-hole picture – because it is the language every method in every chapter is written in, and it will outlast all of them. The habit of checking against something exactly known, because it is what separates a calculation from a computation. And the judgement, which is what Table 10.4 was really teaching, that method selection is a physical question and not a technical one: what matters is whether a method's assumptions match the physics of the regime, and every method in this book is excellent somewhere and useless somewhere else.

The many-body problem is not going to be solved. It is going to be approximated, in an ever-growing number of ways, each honest about what it gives up. Knowing what has been given up is the whole of the skill.

21.6 Projects

These are open-ended, in the manner of a term project rather than an exercise. Each requires code from more than one chapter.

1) **Extend Table 21.1.** Add rows for diffusion Monte Carlo (Chapter 15) and for a Gaussian-binary RBM used as a correlator rather than as the whole wave function (Chapter 17). Which of the two closes more of the remaining 4.8×10^{-4}, and at what cost in samples?
2) **Map the crossover.** Reproduce Table 10.4 and extend it to $g = 3$ and $g = 4$, adding variational Monte Carlo with a seniority-zero trial function. Identify the coupling at which each method is overtaken, and explain each crossing in terms of what the method assumes. Which method would you use at $g = 10$, and how would you defend the choice without access to the FCI answer?
3) **Symmetry-projected ansatz.** The Lipkin row of Table 20.7 fails because the circuit breaks permutation symmetry. Build an ansatz from the collective operators $J_a = \frac{1}{2}\sum_i \sigma_i^a$, verify that it reaches the exact $N = 4$ ground state, and count the parameters. Then do the same for a classical neural quantum state: enforce permutation equivariance and measure what it buys.
4) **The nodal surface.** Take the six-electron dot of Chapter 16 and replace the single-determinant nodes with a backflow-transformed determinant (Section 18.6). Measure the change in the fixed-node diffusion energy and compare with the 1.5×10^{-3} discrepancy between Λ-CCSD(T) and DMC in Table 16.5. Can you account for it?
5) **An honest error bar.** Choose any result in this book reported with a statistical error and produce a defensible estimate of its *systematic* error as well – basis, time step, nodal surface, ansatz, whichever apply. Write the result as a research paper would have to. This is harder than it looks, and that is the point of the exercise.
6) **Two methods, one system, unrelated errors.** Pick a system from Chapter 4 large enough that you cannot diagonalise it, compute its ground-state energy by two methods from different parts of this book, and argue from the agreement or disagreement what the true value is. State what would change your mind.

Programs for this chapter

This chapter has no programs of its own. Its figures are produced by the notebook `LectureNotes/conclusions.ipynb`, which draws on the programs of the earlier chapters — `BookPrograms/chapter05/fci.py`, `BookPrograms/chapter09/mbpt.py`, `BookPrograms/chapter07/rpa.py`, `BookPrograms/chapter10/coupledcluster.py` and `BookPrograms/chapter11/quantumdot.py` — and puts them side by side. The figures are collected in `BookFigures/chapter21/`.

Appendix A
Projects

The exercises at the end of each chapter have answers. The projects collected here do not, or not entirely. Each takes one of the programs of the book as its starting point and asks for it to be extended in the direction of a research paper, so that the reader ends up reproducing – on a system small enough that every step can be checked – a method that is used in current work on real nuclei, molecules or materials. They are meant as term projects: several weeks of work, a written report, and a conclusion that the reader has to defend rather than look up. The open-ended projects of Section 21.6 asked for methods from different chapters to be set against each other; the projects here go deeper into one method at a time.

Every project comes with a program in `BookPrograms/appendixA/` and a companion notebook in `LectureNotes/`, which carry the reference solution to the first tasks and the scaffolding for the remaining ones. The reader is expected to read them, run them, and then leave them behind.

A.1 Project 1: the finite amplitude method for the random-phase approximation

A.1.1 Background and aim

Chapter 7 solved the random-phase approximation by building the matrices A and B of Eqs. (7.11) and (7.17) and diagonalising the non-Hermitian matrix (7.16). For the model of that chapter this is trivial: sixteen particle-hole pairs, a 32×32 matrix. For a real nucleus described by an energy-density functional on a three-dimensional mesh it is not. The number of particle-hole pairs runs into the millions, the residual interaction $\partial h/\partial \rho$ of a density-dependent functional has dozens of terms, and coding every one of them without error is, in the words of the paper this project is built on, “very tedious and difficult”.

Nakatsukasa, Inakura and Yabana [9] proposed a way around both difficulties, which they called the *finite amplitude method*. The observation is that the RPA never needs A and B themselves; it needs their action on a vector of amplitudes, and that action is an induced mean field, $\delta h = (\partial h/\partial \rho)\,\delta\rho$. The induced field can be obtained numerically, by evaluating the ordinary mean-field routine $h[\rho]$ of a static Hartree-Fock code for a density perturbed by a small, finite amplitude and dividing the difference by that amplitude. The only extension the static code needs is the ability to build h from independent bra and ket orbitals, because the perturbed density is not Hermitian. The linear-response equations at a fixed frequency are then solved iteratively, and the strength function follows. The method has since become the standard way of doing RPA and quasiparticle RPA with modern functionals, in the coordinate-space Skyrme calculations of Ref. [10], in the quasiparticle extension of Ref. [11], in the harmonic-oscillator-basis Skyrme-QRPA of Ref. [12], and in the contour-integration technique of Ref. [13] which extracts discrete modes from it.

The paper demonstrates the method on the deformed nucleus ^{20}Ne, with the schematic BKN functional [14] on a mesh of 8217 points. That calculation requires a coordinate-space Hartree-Fock code, which this book does not provide and which is the last, open-ended task below. The central claims of the paper, however, are claims about the *method*, and every one of them can be reproduced on the pairing plus particle-hole model of Chapter 7: that the finite-difference induced field reproduces the explicit residual interaction; that the iterative solution reproduces the matrix RPA "in the accuracy of three to four digits"; how that accuracy depends on the finite amplitude η and the smoothing Γ; how the cost of the iteration depends on the frequency; and how the strength function is turned into discrete modes, transition strengths and sum rules. The model offers one thing the nucleus cannot: the exact response, for comparison.

The aim of the project is therefore to extend the program `rpa.py` of Chapter 7 into a finite-amplitude solver, to verify it against the matrix RPA of that chapter, to study its numerics in the manner of the paper, and then to take it in one of two directions – towards the quasiparticle RPA with its Nambu-Goldstone mode, or towards a coordinate-space mean field on which the calculation of the paper itself could be repeated.

A.1.2 The linear-response equation

The starting point is the time-dependent Hartree-Fock equation for the one-body density matrix, met in Section 7.6 in its equation-of-motion form. With a weak external one-body field $\hat{V}_{\text{ext}}(t)$ the density obeys

$$i\frac{d}{dt}\rho(t) = [h[\rho(t)] + V_{\text{ext}}(t),\, \rho(t)]\,, \tag{A.1}$$

where $h[\rho] = \partial E/\partial\rho$ is the mean-field Hamiltonian, the same object that the self-consistent iteration of Chapter 6 diagonalises. Writing $\rho(t) = \rho_0 + \delta\rho(t)$ about the static solution ρ_0, using $[h_0, \rho_0] = 0$ with $h_0 = h[\rho_0]$, and keeping first order in the perturbation gives

$$i\frac{d}{dt}\delta\rho = [h_0, \delta\rho] + [V_{\text{ext}} + \delta h,\, \rho_0]\,, \qquad \delta h = \frac{\partial h}{\partial\rho}\cdot\delta\rho. \tag{A.2}$$

The induced field δh is the residual interaction acting on the density fluctuation; it is what turns the independent-particle response into the RPA. For a field with a single frequency, $V_{\text{ext}}(t) = \eta\, F e^{-i\omega t} + \eta^* F^\dagger e^{i\omega t}$ with η a small parameter, the density and the induced field respond at the same frequency, and Eq. (A.2) becomes

$$\omega\,\delta\rho(\omega) = [h_0, \delta\rho(\omega)] + [F + \delta h(\omega),\, \rho_0]\,. \tag{A.3}$$

Because $\rho(t)^2 = \rho(t)$, the fluctuation has only particle-hole and hole-particle matrix elements in the basis that diagonalises h_0, and the two are independent once ω is fixed: $\delta\rho(\omega)$ is not Hermitian. This is why it takes two sets of amplitudes to describe it,

$$\delta\rho(\omega) = \sum_i \Big(|X_i(\omega)\rangle\langle\phi_i| + |\phi_i\rangle\langle Y_i(\omega)|\Big), \qquad |X_i\rangle = \sum_m |\phi_m\rangle X_{mi}, \quad |Y_i\rangle = \sum_m |\phi_m\rangle Y^*_{mi}, \tag{A.4}$$

with i occupied and m unoccupied as in Chapter 7. Taking the particle-hole and the hole-particle matrix elements of Eq. (A.3) gives

$$(\omega - \varepsilon_m + \varepsilon_i) X_{mi}(\omega) - \delta h_{mi}(\omega) = F_{mi}, \tag{A.5}$$

$$(-\omega - \varepsilon_m + \varepsilon_i) Y_{mi}(\omega) - \delta h_{im}(\omega) = F_{im}, \tag{A.6}$$

where $F_{mi} = \langle \phi_m | \hat{F} | \phi_i \rangle$ and $\delta h_{mi}(\omega) = \langle \phi_m | \delta h(\omega) | \phi_i \rangle$. These are the equations the whole project revolves around. If the induced field is expanded explicitly, $\delta h_{mi} = \sum_{nj} [\bar{v}_{mjin} X_{nj} + \bar{v}_{mnij} Y_{nj}]$, the coefficients are exactly the interaction parts of the A and B matrices of Eqs. (7.11) and (7.17), and Eqs. (A.5)–(A.6) collapse into the matrix form

$$\left[\begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} - \omega \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] \begin{pmatrix} X(\omega) \\ Y(\omega) \end{pmatrix} = - \begin{pmatrix} f \\ g \end{pmatrix}, \qquad f_{mi} = F_{mi}, \quad g_{mi} = F_{im}. \tag{A.7}$$

This is the linear-response version of the RPA eigenvalue problem (7.16): the same matrix, with a source term on the right and the frequency prescribed rather than solved for. At $\omega = \omega_\nu$ and $F = 0$ the two coincide.

What one wants from the amplitudes is the response of an observable. For the field $\hat{F}$ itself, first-order time-dependent perturbation theory gives

$$S(F;\omega) = \sum_i \Big(\langle \phi_i | \hat{F}^\dagger | X_i(\omega) \rangle + \langle Y_i(\omega) | \hat{F}^\dagger | \phi_i \rangle \Big) = \sum_\nu \left(\frac{|\langle \nu | \hat{F} | 0 \rangle|^2}{\omega - \omega_\nu} - \frac{|\langle \nu | \hat{F}^\dagger | 0 \rangle|^2}{\omega + \omega_\nu} \right), \tag{A.8}$$

where the second form follows from expanding the solution of Eq. (A.7) in the RPA eigenmodes, normalised as in Eq. (7.18), and $\langle \nu | \hat{F} | 0 \rangle = \sum_{mi} (X^{\nu *}_{mi} F_{mi} + Y^{\nu *}_{mi} F_{im})$ is the RPA transition amplitude. The strength function is thus a meromorphic function of ω with a simple pole at every RPA root, and giving the frequency a small imaginary part, $\omega \to \omega + i\Gamma/2$, turns the poles into Lorentzians of width Γ:

$$\frac{dB(F;\omega)}{d\omega} = \sum_\nu |\langle \nu | \hat{F} | 0 \rangle|^2 \delta(\omega - \omega_\nu) \;\longrightarrow\; -\frac{1}{\pi} \operatorname{Im} S\big(F; \omega + \tfrac{i}{2}\Gamma\big). \tag{A.9}$$

This is the quantity plotted in every figure of the paper. Nothing so far is specific to the finite amplitude method; Eq. (A.7) with the A and B of `rpa.py` is the “conventional RPA” of the paper’s Fig. 1(b), and it is the reference against which everything else is measured.

A.1.3 The finite amplitude method

The residual interaction enters Eqs. (A.5)–(A.6) only through the induced field $\delta h(\omega)$, and δh is, by construction, the first-order change of $h[\rho]$ under the change $\delta\rho$ of the density. The finite amplitude method takes this literally. Given amplitudes X and Y, one forms the perturbed orbitals

$$|\psi_i\rangle = |\phi_i\rangle + \eta |X_i\rangle, \qquad \langle \psi_i' | = \langle \phi_i | + \eta \langle Y_i |, \tag{A.10}$$

with η a small real number, builds the density $\rho_\eta = \sum_i |\psi_i\rangle\langle\psi_i'|$ from them, and evaluates

$$\delta h(\omega) = \frac{1}{\eta} \Big(h\big[\langle \psi' |, |\psi\rangle\big] - h_0 \Big). \tag{A.11}$$

The density ρ_η is not Hermitian, which is why the bra and ket orbitals must be independent, and why the mean-field routine of a static Hartree-Fock code – written for $\rho = \sum_i |\phi_i\rangle\langle\phi_i|$ – has to be generalised to accept them. That generalisation is the whole of the programming effort: the residual interaction is never written down.

Two remarks on accuracy, both of which the project asks the reader to verify. First, the perturbed density is

$$\rho_\eta = \rho_0 + \eta\,\delta\rho + \eta^2 \sum_i |X_i\rangle\langle Y_i|, \tag{A.12}$$

so even for a Hamiltonian that is exactly linear in the density, as the model of Chapter 7 is, the forward difference (A.11) carries an error of order η from the quadratic term. A density-dependent functional adds its own $O(\eta)$ terms. A central difference, $(h[+\eta]-h[-\eta])/2\eta$, removes the quadratic term at the cost of a second evaluation. Second, the subtraction in Eq. (A.11) loses digits as η shrinks: the round-off error grows as $\varepsilon_{\mathrm{mach}}/\eta$. The paper chooses $\eta = 10^{-5}$, normalised to the size of the amplitudes, and a convergence tolerance of 10^{-5} on the residual of Eqs. (A.5)–(A.6); the two numbers are not independent, and a solver asked for a residual smaller than the consistency of the finite difference will not converge.

Since δh depends linearly on the amplitudes, Eqs. (A.5)–(A.6) are a linear system for (X,Y) whose matrix is never formed but whose action on a vector costs one evaluation of h. This is precisely the situation the Krylov methods of Chapter 1 were designed for; because ω is complex and the system non-Hermitian, the paper uses the bi-conjugate gradient method, and the program uses GMRES and Bi-CGSTAB from `scipy`. The simplest alternative, the plain iteration

$$X_{mi} \leftarrow \frac{F_{mi} + \delta h_{mi}[X,Y]}{\omega - \varepsilon_m + \varepsilon_i}, \qquad Y_{mi} \leftarrow -\frac{F_{im} + \delta h_{im}[X,Y]}{\omega + \varepsilon_m - \varepsilon_i}, \tag{A.13}$$

with linear mixing, is worth implementing first, because its failures are instructive: it converges only if the residual interaction is small compared with $|\omega - \varepsilon_m + \varepsilon_i|$, and it therefore slows down near every pole and for every small Γ.

A.1.4 *Moments, poles and the correlation energy from the strength function*

A strength function on a grid of real frequencies is the natural output of the method, but it is not the only one. Because $S(F;\omega)$ is meromorphic, Cauchy's theorem turns its contour integrals into sums over the enclosed poles [13],

$$\frac{1}{2\pi i}\oint_C S(F;\omega)\,d\omega = \sum_{\nu\in C} |\langle \nu|\hat F|0\rangle|^2, \qquad \frac{1}{2\pi i}\oint_C \omega\, S(F;\omega)\,d\omega = \sum_{\nu\in C} \omega_\nu |\langle \nu|\hat F|0\rangle|^2. \tag{A.14}$$

A circle enclosing a single root returns its energy as the ratio of the two integrals and its transition strength as the first; a circle enclosing every positive root returns the non-energy-weighted and energy-weighted sum rules m_0 and m_1. The trapezoidal rule on a circle converges exponentially, so a few dozen finite-amplitude solutions at complex frequencies – with no smoothing at all – replace the diagonalisation. The energy-weighted sum rule is also a check on self-consistency: Thouless' theorem [15] states that for an RPA built on the self-consistent Hartree-Fock state

$$m_1 = \sum_\nu \omega_\nu |\langle \nu|\hat F|0\rangle|^2 = \tfrac{1}{2}\langle \mathrm{HF}|\left[\hat F^\dagger,\left[\hat H,\hat F\right]\right]|\mathrm{HF}\rangle, \tag{A.15}$$

a double commutator that the `build_AB` routine of Chapter 7 evaluates directly.

Even the correlation energy of Section 7.8, $E_{\mathrm{corr}}^{\mathrm{RPA}} = \frac{1}{2}\left(\sum_\nu \omega_\nu - \mathrm{Tr}A\right)$, is available without the matrix A. For the unit field $\hat F_K = \hat a_m^\dagger \hat a_i$ the residues of $S(F_K;\omega)$ are $|X_K^\nu|^2$ at $+\omega_\nu$ and $-|Y_K^\nu|^2$ at $-\omega_\nu$, so with a circle C_+ around the positive roots and its mirror image C_- the normalisation (7.18) gives

$$\sum_\nu \omega_\nu = \sum_K \left[\frac{1}{2\pi i}\oint_{C_+} \omega\, S(F_K;\omega)\,d\omega - \frac{1}{2\pi i}\oint_{C_-} \omega\, S(F_K;\omega)\,d\omega\right], \tag{A.16}$$

while $A_{KK} = \varepsilon_m - \varepsilon_i + \delta h_{mi}[X = e_K, Y = 0]$ is one induced field per pair.

A.1.5 What the program provides

The program `fam.py` implements all of the above on top of `rpa.py`, and the notebook `fam.ipynb` runs it. Its structure mirrors the logic of the paper. A class `MeanField` plays the static Hartree-Fock code: it holds t and $\bar{v}$, builds $h[\rho] = t + \sum_{cd} \bar{v}_{acbd}\rho_{dc}$ for any ρ, and offers the one extension `h_braket(bra, ket)` for independent orbitals. A class `MatrixResponse` solves Eq. (A.7) with the A and B of Chapter 7 and also diagonalises them, so that both forms of Eq. (A.8) are available. A class `FiniteAmplitudeMethod` evaluates Eq. (A.11) with a forward or a central difference (or, as a check, exactly), and solves Eqs. (A.5)–(A.6) by GMRES, by Bi-CGSTAB or by the plain iteration (A.13). A class `ExactResponse` diagonalises the N-particle Hamiltonian in the Fock space of `rpa.py` and returns the exact strength function. Two external fields are provided: the level operator $\hat{Q}_0 = \sum_{p\sigma}(p-1)\hat{n}_{p\sigma} = \hat{H}_0/\xi$, diagonal and pair-conserving, the discrete analogue of a monopole field, and the hopping operator $\hat{Q}_1 = \sum_{p\sigma}\left(\hat{a}^\dagger_{p+1,\sigma}\hat{a}_{p\sigma} + \text{h.c.}\right)$, which connects neighbouring levels one spin at a time and therefore breaks pairs, the analogue of a dipole field. All results below are for $L = N = 4$, $g = 1$ and $f = 0.5$, the most strongly coupled case of Chapter 7, where the residual interaction lowers the lowest particle-hole energy from 1.811 to the RPA root 1.644.

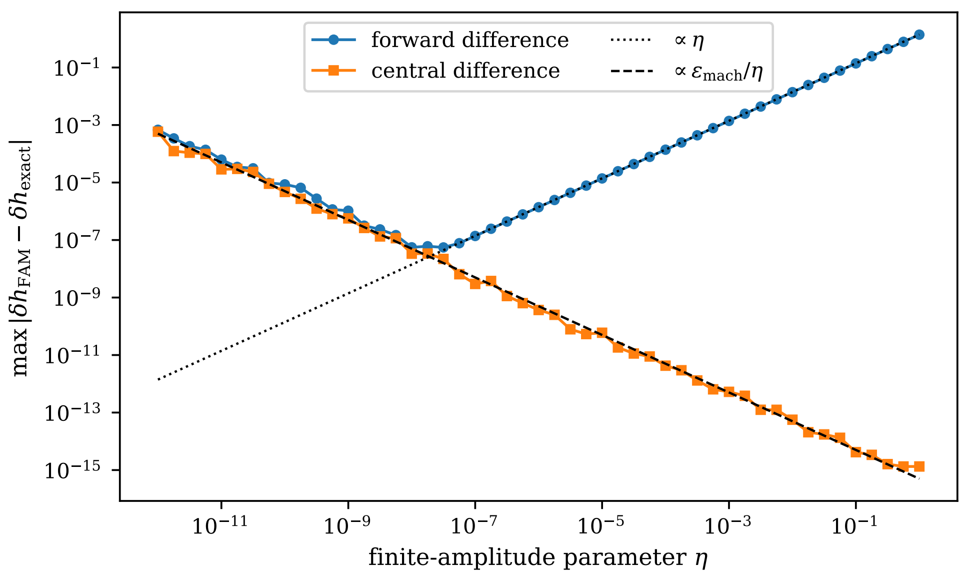

Fig. A.1 Error of the finite-difference induced field, $\max|\delta h_{\text{FAM}} - \delta h_{\text{exact}}|$, against the finite amplitude η for a random complex amplitude vector, for the forward difference (A.11) and for the central difference. The dotted line is proportional to η, the dashed one to $\varepsilon_{\text{mach}}/\eta$. The forward difference is limited by the quadratic term of Eq. (A.12) and reaches at best 10^{-7}; the central difference is exact for this Hamiltonian, which is linear in the density, and only round-off remains.

Figure A.1 is the first thing to look at. The finite-difference induced field is compared with the exact $(\partial h/\partial\rho)\,\delta\rho$, which for the model is $(A-\Delta)X + BY$ with the matrices of Chapter 7. The forward difference has the $O(\eta)$ error predicted by Eq. (A.12) down to $\eta \approx 10^{-8}$ and the $\varepsilon_{\text{mach}}/\eta$ round-off below; the central difference follows the round-off line all the way. With the paper's $\eta = 10^{-5}$ the forward scheme is accurate to about 10^{-5}, which is the origin of its "three to four digits".

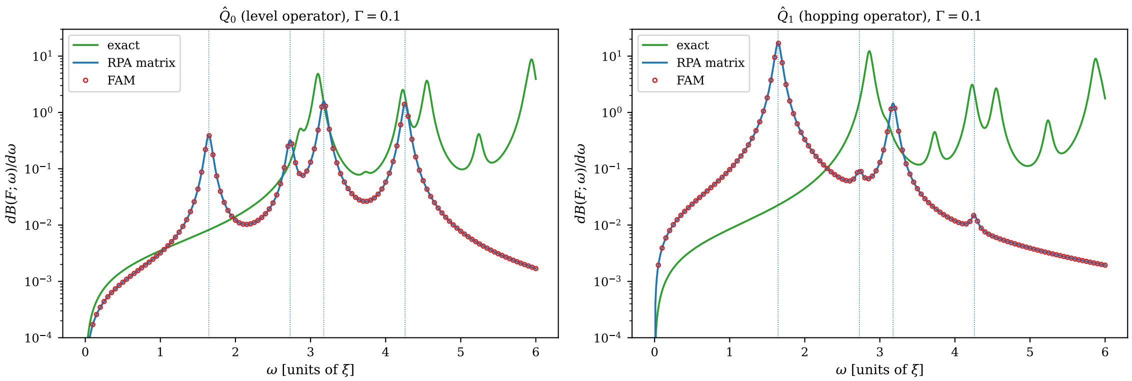

Fig. A.2 Strength functions $dB(F;\omega)/d\omega$ of the pairing plus particle-hole model for $g=1$, $f=0.5$ and a smoothing $\Gamma=0.1$, for the level operator $\hat{Q}_0$ (left) and the hopping operator $\hat{Q}_1$ (right). The full lines are the exact response from the diagonalisation of the $N=4$ system and the matrix RPA of Eq. (A.7); the open circles are the finite amplitude method with a forward difference, $\eta = 10^{-5}$, solved by GMRES to a relative residual of 10^{-6}. The dotted vertical lines mark the four distinct RPA roots, each four-fold degenerate. This is the content of Fig. 1 of Ref. [9], on a system where the exact answer is also known.

Figure A.2 is the counterpart of Fig. 1 of the paper. The finite-amplitude points lie on the matrix-RPA curve at every frequency and for both operators; the program prints the differences, which are of order 10^{-6} relative, and shows that the central difference brings them down to 10^{-11}. The exact response is a different matter, and this is where the model earns its keep. At $g=1$, $f=0.5$ the exact ground state is strongly correlated, its strength sits higher in energy and is spread over many states, and its non-energy-weighted sum m_0 is six times the RPA value for $\hat{Q}_0$. The finite amplitude method reproduces the RPA, no more and no less; whether the RPA is adequate is a question about the quasiboson approximation of Section 7.6, not about the solver, and the model answers it without mercy.

Figure A.3 shows Eq. (A.14) at work. Sixteen finite-amplitude solutions on a circle around the lowest root return its energy to six digits with the forward difference and to eleven with the central one, and its transition strength alongside. The program also evaluates the two sum rules by a circle enclosing every positive root, finds $m_1 = 1.98184$ for $\hat{Q}_0$ and 5.15401 for $\hat{Q}_1$ in agreement with the double commutator (A.15) to six digits, and assembles the correlation energy from Eq. (A.16): $\sum_\nu \omega_\nu = 47.22842$, $\mathrm{Tr}A = 50.15812$, $E_{\mathrm{corr}} = -1.46485$, which is what `rpa.py` obtains by diagonalisation. Not one matrix element of A or B was used.

Figure A.4 addresses the cost. With 2×16 unknowns a Krylov solver converges in eight to sixteen steps whatever the frequency and whatever Γ, so the model cannot reproduce the growth with ω that the paper reports for its millions of unknowns. What it does reproduce is the mechanism behind the paper's other observation, that a smaller Γ costs more iterations: the plain iteration is governed by the residual interaction divided by $|\omega - \varepsilon_m + \varepsilon_i|$, and it slows down as the frequency approaches a pole and as Γ shrinks, failing altogether in windows around each root that widen as $\Gamma \to 0$.

A.1.6 Tasks

The first five tasks have reference solutions in the program; the reader should nevertheless write their own version before reading it. The last two are open.

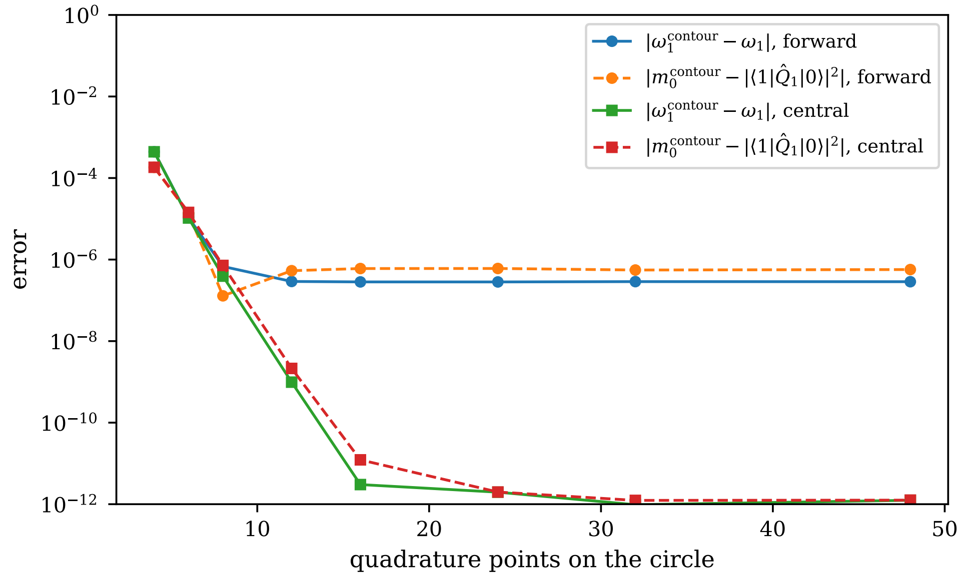

Fig. A.3 Locating the lowest RPA root by contour integration (A.14) of the finite-amplitude strength function of $\hat{Q}_1$ on a circle of radius 0.3 around it, as a function of the number of trapezoidal quadrature points. The full lines show the error of the energy ω_1 obtained as the ratio of the two moments, the dashed lines the error of the transition strength; circles are the forward difference with $\eta = 10^{-5}$, squares the central difference. The convergence is exponential, and the floor is set by the accuracy of the finite difference.

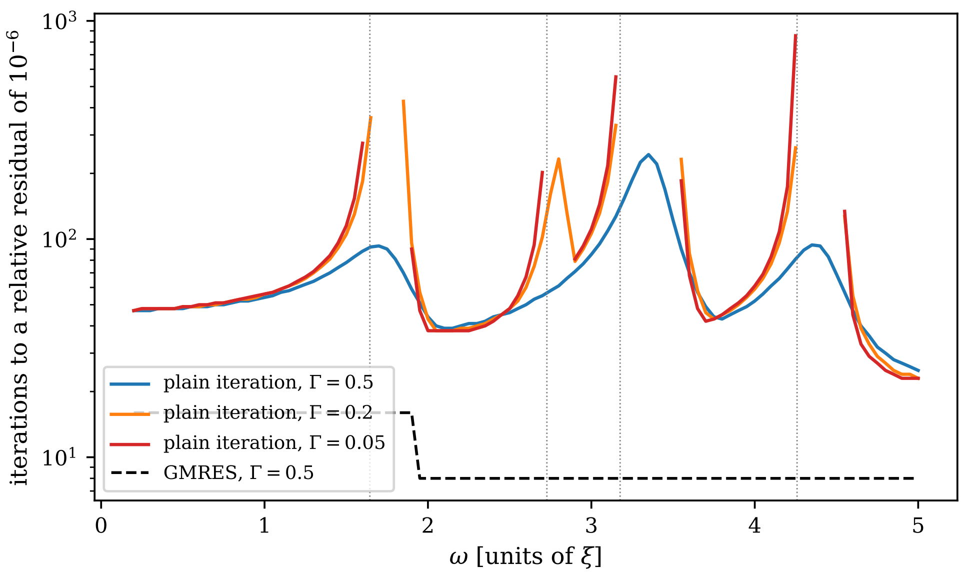

Fig. A.4 Number of iterations needed to solve Eqs. (A.5)–(A.6) for $\hat{Q}_1$ to a relative residual of 10^{-6}, as a function of the real part of the frequency, for the plain iteration (A.13) with mixing 0.5 at three values of the smoothing Γ, and for GMRES. Gaps in the curves are frequencies at which the plain iteration diverges; the dotted lines are the RPA roots.

a) **Linear response with the matrices of Chapter 7.** Take A and B from `rpa.tda_rpa` and solve Eq. (A.7) for a complex ω. Verify that the strength function obtained from the solution equals the spectral sum in Eq. (A.8) built from the RPA eigenvectors – and note that this requires the eigenvectors of a degenerate root to be orthogonalised in the indefinite metric of Eq. (7.18), which a library eigensolver does not do. Check Thouless' theorem (A.15) for both fields, and check that the exact ground state satisfies the same sum rule with the exact double commutator. Plot $dB/d\omega$ for several Γ and explain what a Lorentzian smoothing does to a sum rule evaluated by integrating over a finite frequency window.

b) **The induced field by finite differences.** Extend the mean-field routine to independent bra and ket orbitals and implement Eq. (A.11). Derive Eq. (A.12) and reproduce Fig. A.1, including the two asymptotes. Then add a term to the mean field that is *not* linear in the density – a three-body contact term contributes $\frac{3}{16}t_3\rho^2$ to the BKN functional, and its analogue in the model is a term proportional to ρ_{aa}^2 on the diagonal of h – and show that the central difference is then accurate to $O(\eta^2)$ rather than exact. Which η would you choose, and does the answer depend on the scheme?

c) **The finite amplitude method against the matrix RPA.** Solve Eqs. (A.5)–(A.6) with a Krylov solver and reproduce Fig. A.2. Report the agreement with task a) as a function of η, of the scheme and of the solver tolerance, and explain why the tolerance cannot usefully be smaller than the consistency of the finite difference. Repeat at $g = 0.5$, $f = 0.1$, where the exact and the RPA responses are closer, and comment on what the comparison with the exact response says about the quasiboson approximation at each coupling.

d) **The cost of the solution.** Implement the plain iteration (A.13) and reproduce Fig. A.4. Write the iteration as $z \leftarrow (1-\alpha)z + \alpha D^{-1}(r + Kz)$, identify D and K, and show that it converges if and only if the spectral radius of $(1-\alpha)I + \alpha D^{-1}K$ is below one. Compute that spectral radius from the matrices of task a) along the frequency axis and show that it predicts the gaps in the figure. Then compare with the iteration counts of GMRES and Bi-CGSTAB and explain, from Chapter 1, why the latter are bounded by the number of unknowns.

e) **Poles and sum rules without diagonalisation.** Implement Eq. (A.14) and reproduce Fig. A.3. Use it to follow the lowest root and its $\hat{Q}_1$ strength as functions of f at fixed $g = 1$, from $f = 0$ to $f = 1$, using only finite-amplitude solutions, and compare with Section 7.9. Then evaluate the correlation energy from Eq. (A.16) and reproduce the value quoted above. How does the number of h-evaluations needed for E_{corr} scale with the number of particle-hole pairs, and how does that compare with forming and diagonalising the RPA matrix?

f) **Open: the quasiparticle RPA and the Nambu-Goldstone mode.** Section 7.11.1 showed that the BCS reference breaks particle-number conservation and that the number operator generates an exact zero mode of the QRPA. Section II D of Ref. [9] gives a prescription for removing such a mode from a calculated transition density, based on the symmetry generator $\hat{P}$ and its canonically conjugate operator $\hat{R}$ with $[\hat{R},\hat{P}] = i$. Set up the linear response of the cranked QRPA (7.33) of `rpa.py` to the pair fields $\hat{P}^\dagger = \sum_p \hat{a}_{p+}^\dagger \hat{a}_{p-}^\dagger$ and $\hat{N}$, exhibit the Nambu-Goldstone contribution as a peak near $\omega = 0$ whose height diverges as $\Gamma \to 0$, construct the conjugate operator $\hat{R}$ – the gauge angle, whose amplitudes follow from the Thouless-Valatin equation $[\hat{H}',\hat{R}] = -i\hat{P}/\mathscr{M}$, understood in the sense of the double commutators of Section 7.4, with $\mathscr{M}$ the inertia – and apply the prescription of the paper. Verify that the removal does not touch the physical modes. Then generalise the finite amplitude method itself to the quasiparticle case following Ref. [11], where the mean-field routine must return both h and the pairing field Δ for independent bra and ket quasiparticle states, and reproduce your matrix-QRPA results with it.

g) **Open: a coordinate-space mean field.** The paper's own calculation needs $h[\rho]$ on a spatial mesh. Write a spherical Hartree-Fock code for the BKN functional [14], Eq. (54) of

Ref. [9], on a radial grid – the spin-isospin degeneracy of four makes ^{16}O a problem with two radial orbitals, $0s$ and $0p$ – and solve it for the ground state. Extend the mean-field routine to independent bra and ket orbitals, implement the finite amplitude method for the isoscalar monopole field r^2, and compute the strength function. Check it against the energy-weighted sum rule, which for r^2 is $2\hbar^2 A\langle r^2\rangle/m$, and against a matrix RPA in the basis of the radial eigenfunctions. Only then consider the deformed ^{20}Ne of the paper on a three-dimensional mesh, where the Nambu-Goldstone modes of translation and rotation appear and the prescription of task f) becomes indispensable. This is a research project, and Refs. [10, 12, 13] are the literature to enter it with.

Programs for this appendix

Every number and every figure of this appendix comes from the programs collected in `BookPrograms/appendixA/`. Each of them runs as a script and prints the tables reproduced in the text, and each is also imported by the executable companion, the notebook `LectureNotes/fam.ipynb`, so that the numbers on this page and the numbers on the Jupyter-book site come from the same code. The figures were generated by re-executing that notebook and are collected in `BookFigures/appendixA/`.

`fam.py` The finite amplitude method for the pairing plus particle-hole model, built on `BookPrograms/chapter07/rpa.py`: the mean field for independent bra and ket orbitals, the matrix linear response from the A and B of Chapter 7, the finite-difference induced field with forward, central and exact schemes, Krylov and plain iterative solvers, the exact response from the diagonalisation of the N-particle system, the energy-weighted sum rule as a double commutator, and the contour integrals for poles, moments and the correlation energy. Runs in about fifteen seconds; needs `scipy.linalg`, `scipy.sparse` and `scipy.sparse.linalg`.

All of them run on `numpy` alone unless stated otherwise. The program imports `rpa.py` of Chapter 7, so the whole of `BookPrograms/` should be placed on the search path rather than a single directory; the notebook shows how.

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